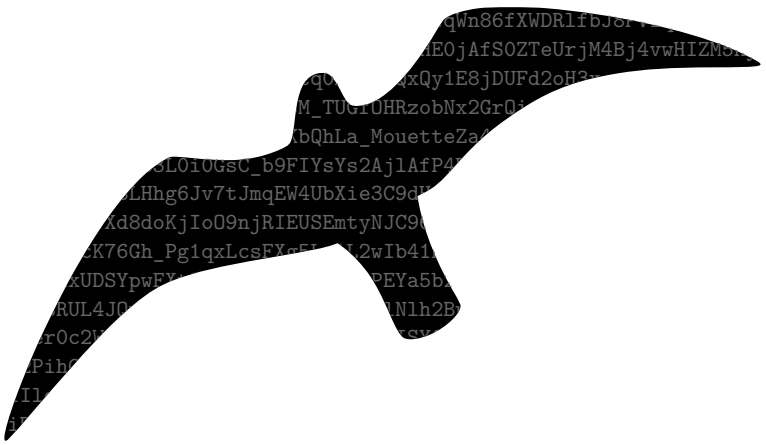


The Black Seagull

2018-04-13

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7058 – TOTS 1C-Management of the Engineering Department 1945

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8532 – Ex basic refresher and assessment 1962

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8536 – Ex inspection and maintenance 1990

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8539 – Exi verification and documentation 2004

8540 – Exi installation and inspection 2008

0001 – Personal Safety

Version 7.0 (1166)

Accidents happen for a reason. Based on what you have seen in this module, what would you say is the cause of most accidents at sea? Select the ONE best answer.

- **The actions that people take, the so-called 'human element'**
- Faulty procedures
- Lack of training
- We cannot avoid danger at sea

According to most authorities, what is the cause of most accidents at sea? Select the ONE correct answer.

- **The human element**
- Flags of convenience
- Lack of training
- Reductions in manning levels

According to the International Chamber of Shipping, what are the THREE key components to developing an effective safety culture? Select the THREE correct answers.

- **Commitment from top management**
- **Measuring current performance and behaviour**
- **Modifying behaviour**
- Finding out who is to blame for accidents

According to this module, where does the ship's safety culture begin? Select the ONE correct answer.

- **The safety and familiarisation tour when joining the ship**
- With regular Safety Management Team meetings
- With the correct procedures being written
- With the ship's SMS

According to this module, why is the Galley a potentially dangerous place? Select the ONE correct answer.

- **Because there is a greater risk of slips trips and falls, and there are many other additional hazards.**
- Because galley staff are less aware of safety
- Because galley staff are not trained about safety in the same way as deck and engine room staff
- Because galley staff do not usually wear the correct PPE

Anyone joining a vessel that they are not familiar with should be given a safety equipment and familiarisation tour. According to this module, where does this requirement come from? Select the ONE best answer

- **The ISM Code and STCW**
- MLC guidelines
- SIRE and OCIMF requirements
- The International Chamber of Shipping (ICS)

Anyone joining a vessel that they are not familiar with should be given a safety equipment and familiarisation tour. Which of the following should this include? Select any answers that you think apply.

- **Alarm signals and Muster stations**
- **Any security duties that they have**
- **Firefighting equipment**
- **Lifesaving appliances**

How can accidents still happen, even when the correct procedures are followed? Select the ONE best answer.

- **If there are any 'holes' or weaknesses in procedures, accidents can still happen in some circumstances**
- Accidents can never happen if all the procedures are followed
- It is not worth trusting in procedures, you should always decide what is best yourself
- There are often a lot of very poor procedures around

If you need to use a chemical cleaning agent, where can you find complete information about any potential hazards and precautions? Select the ONE best answer.

- **The Safety Data Sheet for the product**
- In the ship's SMS procedures
- No special precautions are needed if the correct PPE is worn
- On the label of the product

In a lockout-tagout system, what is the main purpose of the lock and tag? Select the ONE correct answer.

- **To prevent anyone else making equipment you are working on dangerous by trying to re-start or re-energise it.**
- So that the duty officer knows you are working on equipment
- To give you permission to work on equipment
- To hold valves in the closed position, or isolators and switches in the open position

What operations does the ISM Code cover? Select the ONE best answer.

- **Operations which could affect safety or cause pollution**
- Only operations conducted at the management level
- Only operations that affect safety
- Operations by ship management companies

When working on electrical equipment, we need to have it isolated. What does this mean? Select the ONE correct answer.

- **We have created a physical separation between the electricity supply and the item to be worked on**
- The equipment has a tag on it, where possible
- The equipment is switched off
- We have roped off the area around the equipment

Which of the following are evidence of a POOR safety culture onboard a ship? Select any options that you think apply.

- **No security procedures at the gangway**
- **People not paying attention to safety briefings**
- Other people wearing PPE such as hard hats, when you cannot see any risks
- Wearing sports shoes in a recreational area of the accommodation

Why do people need familiarisation training every time they join a new ship? Select the ONE correct answer.

- **Because the content of familiarisation training will change depending on the type of ship and their job on board.**
- Because it is checked during SIRE inspections
- Because they are liable to have forgotten how to launch lifeboats and operate rescue equipment.
- If they are experienced seafarers, they do not need such a tour

Why should 'near miss incidents' be reported? Select the ONE best answer.

- **To avoid accidents and injuries in the future**
- It is best not to report them; it makes the company look like they are having too many incidents
- To comply with the regulations
- To find the person who was to blame

0002 – Ship General Safety

Version 5.0 (5016)

Annex IV of MARPOL deals with which one of the following types of pollution?

- **Sewage**
- Air
- Garbage
- Oil

Annex V of MARPOL deals with which one of the following types of pollution?

- **Garbage**
- Air
- Oil
- Sewage

Annex VI of MARPOL deals with which one of the following types of pollution?

- **Air**
- Garbage
- Oil
- Sewage

Corrosion in Bulk Carriers can be caused by:

- **Severe local corrosion can result from mechanical damage to tank coatings caused by cargo grabs**
- Mechanical damage to tank coatings caused by cargo grabs tends to cause slow corrosion of the whole tank surface
- Tank coatings are self repairing
- Tank coatings are strong enough to withstand any mechanical damage from cargo work

Guidelines for Ballast Water Management are given by which one of the following?

- **The IMO Convention on Ballast Water Management**
- Classification society rules
- MARPOL
- SOLAS

If the thickness of the hull is reduced by corrosion then:

- **The stress on the hull will be increased**
- The bending moments on the hull will be increased
- The shear force and bending moments on the hull will be increased
- The shear force on the hull will be increased

Regulations give three options for ensuring that GMDSS equipment is always available for use in an emergency. How many of these options must a ship use if it operates near to shore in what are known as sea areas A1 and A2?

- **One**
- All three
- It depends on classification society rules
- Two

Regulations give three options for ensuring that GMDSS equipment is always available for use in an emergency. How many of these options must a ship use if it operates well away from shore in what are known as sea areas A3 and A4?

- **Two**
- All three
- It depends on classification society rules
- One

Rules on how a vessel is built are contained in

- **SOLAS, Classification Society Rules and other publications**
- Classification Society Rules only
- SOLAS and MARPOL only
- SOLAS only

Shear force and bending moments on a ship's hull may be caused by:

- **Cargo, ballast and heavy weather**
- Cargo and ballast Only
- Cargo loading only
- Corrosion

The ISM Code is part of:

- **SOLAS**
- ISGOTT
- MARPOL
- STCW

The ISM Code means that Ships must have a Safety Management System which requires that:

- **The ship must carry written instructions for the operation of safety and pollution prevention equipment**
- Only certificated officers can operate safety and pollution prevention equipment
- Ships staff must be able to operate equipment without any reference to written procedures
- The Master must chair all safety committees.

The ISM Code was introduced by:

- **The International Maritime Organisation (IMO)**
- The International Association of Classification Societies (IACS)
- The International Standards Organisation (ISO)
- The US Coastguard (USCG)

The stresses on a Ship's hull are caused by:

- **Shear force and bending moments**
- Only bending moments
- Only buoyancy forces
- Only shear force

When repairs to the hull are required, which one of the following statements is the most accurate?

- **Only the flag State administration or their nominated surveyor can approve repair work**
- If only a small amount of steelwork is to be replaced, ships staff can approve repair work
- If only a small amount of steelwork is to be replaced, the Company Superintendant can approve repair work
- Only the flag State administration can approve repair work

Which one of the following statements is correct when a vessel is said to be 'hogging'

- **The fore and aft parts of the vessel are more heavily loaded than amidships**
- The bow is regularly covered with water in heavy seas
- The fore and aft parts of the vessel are more lightly loaded than amidships
- The ship develops a very slow rolling motion

Which one of the following statements is correct when a vessel is said to be 'sagging'

- **The amidships of the vessel are more heavily loaded than fore and aft parts**
- Bending moments have become excessive
- The bow is regularly covered with water in heavy seas
- The ship develops a very rapid rolling motion

Which one of the following statements is correct?

- **Safety devices designed to prevent the accidental release of free fall lifeboats must be engaged during maintenance operations**
- Accidental release of free fall lifeboats during maintenance operations is not possible
- No maintenance is required for free fall lifeboats or their release mechanisms
- There have never been any accidents with free fall lifeboats during maintenance operations

Which one of the following statements is correct?

- **Procedures for pumping operations should be available to the crew and be followed**
- Only certificated officers can carry out pumping operations
- Ship's staff should carry out pumping operations without any reference to written procedures
- The Chief Engineer must authorize all pumping operations

Which one of the following statements is most accurate?

- **The emergency fire pump must be located outside the engine room.**
- The emergency fire pump can be located anywhere, provided it has a power supply from outside of the engine room
- The emergency fire pump must be located as close to the engine room as possible.
- The emergency fire pump must be located inside the engine room.

Which one of the following statements is most accurate?

- **The main reason for lagging of exhausts in the engine room is to reduce the fire risk**
- The main reason for lagging of exhausts in the engine room is to increase engine efficiency
- The main reason for lagging of exhausts in the engine room is to reduce the risk of injury to the crew during maintenance
- The main reason for lagging of exhausts in the engine room is to try and keep the engine room from being too hot

Which one of the following statements is most accurate?

- **Self closing levers on fuel oil sounding pipes must always be left free to operate and the screw cap should also fitted to the pipe**
- Self closing levers on fuel oil sounding pipes can be temporarily removed when taking soundings
- Self closing levers on fuel oil sounding pipes can be tied open provided the screw cap is fitted to the pipe
- Self closing levers on fuel oil sounding pipes must can be temporarily tied open when taking soundings

Which one of the following statements is most accurate?

- **Steering gear must be tested within 12 hours of departure, and emergency steering drill should be held once every 3 months**
- Steering gear must be tested within 24 hours of departure, and emergency steering drill should be held once every 3 months
- Steering gear must be tested within 24 hours of departure, and emergency steering drill should be held once every 6 months
- Steering gear must be tested within 48 hours of departure, and emergency steering drill should be held once every 3 months

Which one of the following statements is most accurate?

- **Lifeboat release systems can be dangerous unless operated correctly**
- Lifeboat release systems are designed to be maintenance free
- Lifeboat release systems are designed to be safe in any circumstances
- There have never been any accidents with lifeboat release systems

Which one of the following statements is most accurate?

- **Watertight doors which are manually operated should always be fully secured, with all fastenings tightened**
- Once clear of restricted waters, and in good weather, watertight doors which are manually operated need only be secured with one fastening tightened
- Once clear of restricted waters, watertight doors which are manually operated can be left open
- Once clear of restricted waters, watertight doors which are manually operated need only be secured with one fastening tightened

Which one of the following statements is most accurate?

- **Any defects with the seals on watertight or weathertight doors must be repaired immediately**
- Any defects with the seals on watertight doors must be repaired immediately, but the seals on weathertight doors are not so important
- Any defects with the seals on watertight or weathertight doors should be repaired at the next dry-dock
- Weathertight doors do not require seals

Which one of the following statements is most accurate?

- **STCW requirements for bridge watch ratings include steering, communication skills, keeping a lookout, and a basic knowledge of GMDSS operations**
- STCW only covers requirements for officers
- STCW only requires bridge watch ratings to hold a steering ticket
- STCW states that the competences required for a rating are entirely at the discretion of the Master given the nature of the voyage

Which one of the following statements is most accurate?

- **STCW requires navigational ratings at the support level to have knowledge of emergency duties and alarm signals, pyrotechnic distress signals, satellite EPIRBs and SARTs.**
- Navigational ratings at the support level would never be called upon to operate EPIRBs or SARTs in any circumstances
- STCW requires navigational ratings at the support level to have total responsibility for pyrotechnic distress signals, satellite EPIRBs and SARTs.
- STCW rules prevent navigational ratings at the support level from having any duties in relation to alarm signals, pyrotechnic distress signals, satellite EPIRBs and SARTs.

Which one of the following statements is most accurate?

- **Voyage planning is an essential form of risk management and should ensure that the navigating officer has all the necessary up to date information**
- Voyage planning can be done on a watch to watch basis
- Voyage planning is only necessary in restricted waters
- Voyage planning is the job of a nominated officer and need not involve anyone else at any stage

Which one of the following statements is most accurate?

- **All deck officers who have bridge watch keeping duties must hold a GMDSS (Global Maritime Distress and Safety System) operators certificate**
- The Master and at least one other deck officer who has bridge watch keeping duties must hold a GMDSS (Global Maritime Distress and Safety System) operators certificate
- The Master and at least two other deck officers who have bridge watch keeping duties must hold a GMDSS (Global Maritime Distress and Safety System) operators certificate
- There is no requirement for anyone to hold a GMDSS (Global Maritime Distress and Safety System) operators certificate if such equipment is automated

Which one of the following statements is most accurate?

- **Single hull bulk carriers have had more structural failures than oil tankers but have caused less damage to the environment**
- Single hull bulk carriers have had about the same number of structural failures as any other type of vessel
- Single hull bulk carriers have had fewer structural failures than oil tankers and have had less effect on the environment
- Single hull bulk carriers have had more structural failures than oil tankers and have caused more damage to the environment

Which one of the following statements is most accurate?

- **Oily water separators require regular maintenance to work properly**
- Oily water separators are designed to be maintenance free
- Oily water separators only need maintenance in dry-dock
- Regular flushing with sea water is the only maintenance required for an oily water separator

Which one of the following statements is most accurate?

- **Without regular maintenance and correct operation shipboard incinerators can cause air pollution**
- Flushing through with diesel will mean that incinerators are maintenance free
- Shipboard incinerators are designed to be maintenance free
- Shipboard incinerators only need to be maintained at dry dock

Which one of the following statements is most accurate?

- **The main and emergency switchboards are connected to each other but will be in separate compartments**
- The emergency switchboard is only used in emergencies
- The main and emergency switchboards are connected together and usually in the same compartment
- The main and emergency switchboards are not connected to each other in any circumstances

Which one of the following statements is most accurate?

- **Vessel propulsion usually depends on the main engine, steering gear, electrical supply system and auxiliary systems**
- Main propulsion is completely independent of other systems
- Vessel propulsion relies only on the main engine
- Vessel propulsion relies only on the main engine, propeller shaft and propeller

Which one of the following statements is most correct?

- **Watertight doors should usually be kept closed at sea and some may be required to be permanently closed at sea**
- Operation of watertight doors is at the discretion of the officer in charge
- The position of watertight doors only matters in restricted waters
- Watertight doors should usually be kept open at sea

With respect to corrosion of the hull, which one of the following statements is the most accurate?

- **The thickness of the hull must be accurately measured when corrosion is found**
- An experienced surveyor can rely on a visual assessment of corrosion
- Any evidence of corrosion means that steelwork must be replaced
- Ships are designed with generous allowances for corrosion

With respect to the introduction of requirements for tankers and bulk carriers to have a double hull, which one of the following statements is most accurate?

- **Double hulls increase vessel safety and reduce the risk of pollution but make it difficult to check for corrosion in void spaces**
- Double hulls have proved to be more likely to fail in service
- Double hulls increase vessel safety and reduce the risk of pollution without any difficulties
- The added weight of double hulls has led to stability problems

With respect to the introduction of requirements for tankers and bulk carriers to have a double hull, which one of the following statements is most accurate?

- **The double bottom and void spaces must be coated and regularly checked for corrosion**
- If the double bottom only carries oil it does not need to be coated
- Since void spaces are empty, they do not need to be coated
- The double bottom and void spaces do not form part of the load carrying structure

With respect to the need to exchange ballast water, which one of the following statements is most accurate?

- **Ballast water can contain marine life harmful to the local environment and pathogens which can cause diseases such as cholera**
- Any harmful marine life in ballast water cannot survive more than three weeks in a ballast tank
- Ballast water can contain marine life but is otherwise harmless
- Ballast water can normally be considered inert

0004 – SOPEP

Version 4.0 (4209)

For which one of the following would your SOPEP require a pollution report to be made?

- **The oily water separator overboard discharge contained oil above 15 ppm**
- Oil leaked into the drip tray when disconnecting one of the cargo lines at the ship's manifold
- Oil leaked into the drip tray when disconnecting the diesel oil bunker line
- The oily water separator pump broke down

Which of the following best describes the main aim of the OPRC Convention?

- **To improve co-operation between governments when oil pollution needs to be cleaned up**
- To ensure that oil polluters pay for pollution clean-up costs
- To ensure that oil pollution is detected by a network of spotter planes
- To improve co-operation between governments air/sea rescue services

Which of the following best describes the main purpose of a SMPEP?

- **To ensure that chemical pollution is reported to the relevant authorities quickly and effectively**
- To ensure that cargo handling training is provided to crews of chemical tankers
- To ensure that chemical pollution is reported to the harbour master
- To ensure that records of chemicals are kept onboard for use during a pollution investigation

Which of the following best describes the main purpose of a SOPEP?

- **To ensure that oil pollution is reported to the relevant authorities quickly and effectively**
- To ensure that oil pollution is prevented by trained crews
- To ensure that oil pollution is reported to the harbour master
- To ensure that records of oil are kept onboard for use during a pollution investigation

Which of the following best describes the meaning of 'mitigating activities' that will be covered by your SOPEP/SMPEP?

- **Activities that prevent, or reduce the amount of, pollution**
- Activities that involve cleaning up noxious liquids on deck
- Activities that involve monitoring the situation after a casualty
- Activities that prevent, or reduce the ship owner's liability for, pollution clean-up costs

Which of the following best describes what you would expect to be covered in the Preamble to a SOPEP or SMPEP?

- **It is a general overview of the plan and introduces the basic concept of the plan**
- It contains detailed reporting procedures for oil spills.
- It is a guide to the route to be taken when doing a deck watch during cargo or bunkering operations in port
- It is a list of books that must be studied before the plan can be understood

Which of the following best describes when a SOPEP pollution report must be made, despite the fact that no oil has entered the water?

- **When something happens on the ship that makes it probable that oil may enter the water soon after**
- When the ship is rolling more than 30 degrees from vertical during a storm
- When there is a serious oil leak into the cargo pump room bilges
- When there is a serious oil leak into the engine room bilges

Which of the following best describes when instructions in a SOPEP or SMPEP can be ignored?

- **When there is another pollution response plan which takes priority**
- Never, the SOPEP/SMPEP must be followed every time
- When the ship is in a US port
- When the ship is in port

Which of the following best describes when your SOPEP should NOT be used during an oil pollution emergency?

- **When local laws require a different response procedure**
- When oil has spilled on deck but not into the sea
- When the oil pollution emergency occurs inside harbour limits
- When the ship is Not Under Command (NUC)

Which of the following best describes where the master can get up to date information on the contents of a bunker tank that has been found to be leaking through the hull?

- **The location of the information will normally be identified in the non mandatory part of the SOPEP/SMPEP**
- Ask the seaman who takes the morning soundings
- Find the duty engineer
- Try to find the chief engineer

Which of the following best describes where you should send pollution reports if you are at sea?

- **To the nearest coastal State at the contact point given in the IMO list attached to your SOPEP/SMPEP**
- To the harbour master of your last port
- To the harbour master of your next port
- To your flag State at the contact point given in the IMO list attached to your SOPEP/SMPEP

Which of the following best describes where you should send pollution reports if you are inside a harbour?

- **To the contact point listed in your SOPEP/SMPEP or obtained when arriving at the port**
- To the harbour master of the port
- To the nearest coastal State at the contact point given in the IMO list attached to your SOPEP/SMPEP
- To your agent in the port

Which of the following best describes who decides what has to be covered in a SOPEP or SMPEP?

- **The IMO content guidelines should be followed**
- SOPEPs/SMPEPs are written by IMO approved specialist companies
- The marine superintendent decides after having looked at the ship's trading pattern
- The technical superintendent can put anything he wants into it

Which of the following best describes who else you should send pollution reports to, after the local authorities have been informed?

- **To the ship interest contacts listed in your SOPEP/SMPEP**
- To the company's Designated Person Ashore, flag State, class society and insurance company
- To the ship's superintendent and insurance company
- To the ship's superintendent, class society and insurance company

Which of the following best describes why SOPEP/SMPEPs often have a section on 'public affairs' in them?

- **To set out company policy on talking to the media during an emergency**
- To give company policy on crew behaviour during shore leave
- To prevent crew from talking to the media
- To set out company policy on talking to the public while on cargo watch

Which of the following best describes why SOPEP/SMPEPs often have instructions on the taking of oil samples in them?

- **To set out company policy on taking samples during a pollution emergency, in case of court proceedings**
- To make sure that oil samples are kept for a minimum of 3 years
- To set out company policy on taking oil samples instead of opening machinery for inspection, so reducing machinery failures
- To set out safety precautions to prevent injury while taking oil samples

Which of the following best describes why a scenario involving 'loss of tank environmental control' is required to be covered by a SMPEP?

- **The cargo may reach a critical condition, leading to serious damage to the ship and a major leak**
- Duplicate environmental control systems are required
- The air pressure on the cargo can cause the tank sides to bend and leak
- The cargo may boil off into the air

Which of the following best describes why a special form is included in your SOPEP or SMPEP for use in making pollution reports?

- **It ensures that all relevant details are reported and can be understood by persons who do not speak English as their first language**
- It ensures that the superintendent gets all the relevant details of the incident
- The form is in a special emergency e-mail format which is given priority during sending
- The form is specially designed to be faxed ashore

Which of the following best describes why cargo contamination is required to be covered by a SMPEP?

- **Cargo contamination could lead to a hazardous condition**
- Contaminated cargo may not be accepted ashore in the event of a tank overflow
- Contaminated cargo may not be accepted ashore in the event of excessive list
- The cargo contamination could be a result of seawater leaking into the tank

Which of the following best describes why it is necessary to carry out regular 'SOPEP drills'?

- **The plan is of little value if not familiar to the crew**
- It is a MARPOL requirement to hold SOPEP drills every 3 months
- It is an IMO requirement to hold SOPEP drills every month
- It is an STCW requirement to hold SOPEP drills every 6 months

Which of the following best describes why the SOPEP/SMPEP is required to cover containment system failure and hull failure separately?

- **Oil leaks from hull failure will go straight into the water but failed containment systems may allow water pollution to be avoided**
- Containment system failure is only relevant to noxious liquids cargoes but hull failure effects all ships
- Containment system failure is only relevant to ships with deck tanks but hull failure affects all ships
- Containment system failure would never lead to pollution of the water and it is not covered in a SOPEP

Which of the following best describes why the 'Intervention Convention' may be mentioned in a SOPEP or SMPEP?

- **It allows coastal States to dictate actions during a marine pollution incident**
- It allows coastal States to make the ship owner pay for any pollution caused
- It allows for ships to make an unscheduled port stop if they have a pollution emergency
- SOPEPs/SMPEPs are a requirement of the Intervention Convention

Which of the following best describes why your SOPEP/SMPEP contains a section on national and local co-ordination?

- **Co-ordination is necessary to keep the effects of a pollution incident small**
- The coastal State will organise the clean-up of pollution and co-ordinate the contractors' bills for the ship owner to pay
- The local harbour authority will organise the clean-up of pollution and co-ordinate the contractors' bills for the coastal State to pay
- The ship has to organise the clean-up of pollution and co-ordinate with the coastal State to ensure the right contractors are used

Which of the following best describes why your SOPEP/SMPEP may contain a section on salvage?

- **To provide guidance to the master on when salvage assistance should be sought for their ship**
- A section on salvage is mandatory in a SOPEP
- To provide guidance to the master on when salvage assistance should be given to another ship
- To provide the contact details of salvage tugs worldwide

Which of the following is the best reason why casualties causing 'hazardous vapour release' are required to be covered by your SOPEP/SMPEP?

- **Vapour release may lead to a fire and explosion followed by major oil pollution**
- The function of the SMPEP is to minimise air pollution
- The vapour released may lead to the main engine failing due to lack of air
- Vapour release will rapidly fall down, leading to oil pollution of the water

Which one of the following best describes which ships need a carry a SOPEP onboard?

- **Oil tankers over 150 GT and all other ships over 400 GT**
- All oil tankers
- All oil tankers and all other ships over 400 GT
- Only ships over 400 GT

Which one of the following best describes which ships need to carry a SMPEP onboard?

- **Ships that carry noxious liquid cargoes that are over 150 GT**
- All ships that carry noxious liquid cargoes
- All ships that carry noxious liquids of any type
- Ships that carry noxious liquid cargoes that are over 400 GT

Which one of the following describes a 'mitigating activity' that may be found in a SOPEP/SMPEP?

- **Use absorbent materials to remove oil from the water**
- Increase the frequency of deck rounds during bunkering
- Mix any spilled oil and water to form an emulsion
- Stop the bunker barge when the ship's fuel tank is 90% full

Which one of the following is a safety precaution that might be required by your ship's SMPEP after a small spillage of noxious liquid onto deck?

- **Protective clothing is to be worn during spill clean-up**
- All noxious liquid spills are to be left to evaporate away naturally
- All noxious liquid spills are to be washed over the side as quickly as possible
- Sports training shoes should be worn to prevent slips

Which one of the following is a way that you can minimise pollution after a bunker tank has overflowed?

- **Transfer some oil from the over full tank to an empty one**
- Close the tank vent
- Open the tank vent
- Shut the tank filling valve

Which one of the following is an important consideration, after your ship has run aground, that should be covered by your SOPEP/SMPEP?

- **Transfer of oil from a leaking tank may cause the ship to break up**
- Grounding on sand may cause less damage than grounding on rock
- The EPIRB will broadcast the ship's position to shore
- The GPS uses WGS 84 so chart positions must be corrected before marking

Which one of the following is an operational spill that may be covered by your SOPEP/SMPEP?

- **Steady loss of oil from the stern tube header tank**
- A can of oil knocked over in the deck workshop
- A leaking hydraulic pipe on the steering gear ram
- Automatic shutdown of the oily water separator by the bilge monitor

Which one of the following is the correct full name for a SMPEP?

- **Shipboard Marine Pollution Emergency Plan**
- Ship and Mariner Prevention Evaluation Policy
- Ships and Marine Pollution Emergency Plan
- Solvent and Methane Pollution Evaluation Policy

Which one of the following is the correct full name for a SOPEP?

- **Shipboard Oil Pollution Emergency Plan**
- Sea Oil Prevention and Evaluation Policy
- Ship Only Pollution Emergency Plan
- Ship Only Pollution Emergency Preparedness

0005 – ISM Code

Version 6.0 (1651)

According to the I.S.M. Code, what is the definition of a "non-conformity"? Select the ONE correct answer.

- **A situation where objective evidence indicates the non-fulfilment of a specific requirement stated by the S.M.S.**
- A situation where an auditor thinks that a requirement in the S.M.S. has not been met
- A situation where someone has an accident on-board
- A situation where the ship fails to get to its destination on time

According to the ISM Code, which of the following are functional requirements of the Safety Management System? Select any answers that you think apply.

- **Defined levels of authority and lines of communication between shore and shipboard personnel**
- **Procedures to prepare for, and respond to, emergency situations**
- Defined means of communication between shore and ship
- Reporting of maintenance

According to this module, what is the cause of most accidents on-board ships? Select the ONE correct answer

- **Recognised safe working practices or company procedures have not been followed**
- Collisions with jetties and other ships
- Fatigue
- Unsafe manning levels

According to this module, what is the definition of a "hazardous occurrence"? Select the ONE best answer.

- **An undesired event which could have resulted in an accident (a near-miss)**
- An occasion when hazardous materials are spilt
- An occasion when hazardous materials are used
- Any undesired event

According to this module, why do experienced seafarers still have accidents? Select the ONE correct answer.

- **Over-confidence can lead to carelessness**
- They are slow to learn new work practices
- They are unlucky
- They do the most dangerous work

From the options below, which is the most important to check on a work permit issued to you? Select the ONE best answer.

- **The time for which it is valid**
- The name of the ship
- The number of the permit
- The S.M.S. procedure that requires the permit

How can the Designated Person (DP) quickly change an unsafe situation onboard your ship? Select the ONE best answer.

- **The DP can bring critical issues to the top management's attention without delay**
- The DP can contact the Class Society quickly for advice
- The DP is an experienced Chief Engineer
- The DP is an experienced Master

How often does the I.S.M. Code require the company to carry out an internal audit of the S.M.S.? Select the ONE correct answer.

- **At least once a year**
- Every 5 years
- Every 6 months
- Twice in 5 years

Inspectors may ask about a ship's "critical equipment". What does the I.S.M. Code require the company to do about such equipment? Select the ONE correct answer.

- **Promote the reliability of such equipment or systems, including regular testing of equipment not in continuous use**
- Ensure that it is less than 10 years old
- Ensure that it is listed on the Class Society survey schedule
- Ensure that it is only used when essential

Once the S.M.S. is verified and working effectively, what document is issued to the ship? Select the ONE correct answer.

- **The Safety Management Certificate**
- The Document of Compliance
- The Document of Conformance
- The I.S.M. Certificate

The procedure for checking the fire-fighting appliances does not include the new Breathing Apparatus (BA) that has been supplied. What should you do about this? Select the ONE correct option.

- **Send a "Suggested modifications and improvement to the S.M.S." report to the Designated Person**
- Don't touch the BA until the company sends instructions for it
- Follow the old procedure, as far as possible, when checking the BA
- Write your own procedure for checking the BA

Under the I.S.M. Code, a company has to have an 'S.M.S.'. What is an 'S.M.S.'? Select the ONE correct answer.

- **A Safety Management System**
- A Ship Management System
- A Ship Messaging Service
- A Shore Manager for Safety

What is the full title of the I.S.M. Code? Select the ONE correct answer.

- **The International Management Code for the Safe Operation of Ships and for Pollution Prevention**
- The International Management Code for the Safe Operation of Ships
- The International Ship Management Code
- The International SOLAS and MARPOL Code for the Safe Operation of Ships

What is the purpose of the many checklists found on-board ships? Select the ONE correct answer.

- **They are written memos to accomplish tasks safely and without causing pollution**
- They are all filled in and sent to the company
- They are there to impress Port State Control Officers
- They are used in place of S.M.S. procedures

Which I.S.M. Code requirement is met by carrying out regular safety and anti-pollution drills? Select the ONE best answer.

- **To prepare for, and respond to, emergency situations**
- To comply with STCW
- To have a procedure for reporting non-conformities
- To have defined levels of authority between personnel

Which IMO convention is the I.S.M. Code part of? Select the ONE correct answer.

- **SOLAS**
- It is not a part of any convention. It is a stand-alone code of practice
- MARPOL
- STCW

Which ONE of the following best describes the main things that the I.S.M. Code requires the company to do to improve safety?

- **To assess all identified risks to its ships, personnel and the environment, and to establish appropriate safeguards**
- To employ good engineers
- To maintain ships according to Class Society standards
- To use work permits

Which ONE of the following best describes what the I.S.M. Code requires shipping companies to do?

- **Establish procedures, plans and instructions to prevent human error**
- Employ extra staff to become I.S.M. auditors
- Improve communication between ship and office with satellite phones
- Write procedures to prevent the company being liable for its crews

Which ONE of the following will the company check as part of its management review of the S.M.S.? Select the ONE correct answer.

- **The Masters' Reviews**
- Insurance premiums
- The manning levels of each vessel
- Whether each ship is making a profit

Which of the following best describes one of the main goals of a Safety Management System? Select the ONE best answer.

- **To establish standardised company procedures**
- To provide a tracking system
- To provide a training schedule
- To provide the ship with a 24 hour contact number for the office

Which of the following best describes why the Safety Management System is continually reviewed? Select the ONE best answer.

- **To ensure that it remains realistic and adequate to meet its objectives**
- Because it is part of all routine audits
- To fulfil obligations to MARPOL
- To see that the pages are up-to-date

Which of the following is the main reason why the I.S.M. Code was made mandatory? Select the ONE correct answer.

- **To improve safety and prevent pollution**
- To get lower insurance premiums
- To satisfy the US Coast Guard
- To save money for the company

Who can contact the Designated Person with their safety concerns? Select the ONE correct answer.

- **All crewmembers**
- Senior officers only
- The company's shore staff only
- The Master only

Who can suggest changes to the S.M.S.? Select the ONE best answer.

- **Any Company employee**
- Any Senior Officers
- The Designated Person
- The Master

Why are checklists often provided for use during emergency situations? Select the ONE best answer.

- **To ensure that important tasks are not missed during stressful situations**
- So they can be sent to the company
- So they can be sent to the insurance company as evidence
- To make sure emergency drills are realistic

Why are some procedures not kept within the S.M.S. Manual? Select the ONE correct answer.

- **MARPOL and SOLAS have specific requirements for plans and procedures that may be kept separate from the S.M.S.**
- The S.M.S. Manual is kept on a computer
- They are not used on all of the company's ships
- They are old procedures from before the I.S.M. Code was introduced

Why does the I.S.M. Code require hazardous occurrences to be reported to the company? Select the ONE correct answer.

- **So that procedures can be changed to avoid a future accident**
- So that safety bonuses can be calculated
- So the knowledge of the crew can be checked
- The I.S.M. Code only requires accidents to be reported to the company

Why does the S.M.S. include procedures for what happens in the company's shore-side office? Select the ONE correct answer

- **Decisions taken ashore can directly affect operations at sea**
- Crew members need to know what happens in the personnel department
- It is easier to cover the ship and office together
- Office staff need the S.M.S. to give them authority over the Master

Why does the S.M.S. require permits for some types of work? Select the ONE correct answer.

- **To help plan any work and confirm it will be safe**
- So the technical superintendent can control all of the work on-board
- To ensure that work is done to Class Society standards
- To please vetting inspectors

Why should accidents be reported to the Designated Person (DP)? Select the ONE best answer.

- **To prevent them from happening again**
- Because the company has an obligation to keep statistics
- To be able to settle insurance claims
- To find someone to blame

You are told to do a job you have never done before. What should you do? Select the ONE best answer.

- **Check the S.M.S. to see if there is any procedure describing how to do it**
- Ask your friend if they know how to do it
- Do the job very carefully the first time
- Refuse to do the job

You find a folder full of used Hot Work Permits on the bridge. What should you do with the folder? Select the ONE best answer.

- **Check the S.M.S. to see how long permits have to be kept as evidence of following the S.M.S. procedure for Hot Work**
- Keep any permits issued this month, and throw away the rest
- Take the permit binder to your cabin to study them
- Throw away any permits that have expired

0006 – OPA 90

Version 3.0 (2441)

After a grounding in OPA 90 waters, what extra assistance can a ship get to minimise the risk of fuel oil leaking into the sea?

- **Suitable equipment for lightening the ship**
- Engine spare parts
- Firefighting assistance
- Tidal information

Drag the following actions into their list of priority after an oil spill on your ship.

- 1. **Prevent spilled oil entering the water**
- 2. **Stabilise the situation**
- 3. **Reduce the flow of the spill**
- 4. **Clean up the oil**

Following an oil spill, you will be required to 'mitigate' the pollution. Which of the following best describes the meaning of 'mitigate'?

- **To reduce the impact of**
- To clean up
- To move to another place
- To stop

How can an oil tanker sail in OPA 90 waters without an approved Vessel Response Plan (VRP)? Pick the answer you think is best.

- **It is on 'innocent passage'**
- It has been in service less than 3 months and its VRP has been submitted but not yet approved
- It is only making one port call in the United States
- It is only visiting an offshore oil terminal

How does a ship's OPA 90 Vessel Response Plan (VRP) affect its ability to discharge in a US terminal where it has never been before?

- **The ship can only operate in areas which are covered by its VRP**
- Ships can discharge non oil cargoes without restrictions, but not oil cargoes
- The ship can only discharge in new terminals if permission has been granted by the US Coast Guard
- The ship can only load in new terminals, unless permission has been granted by the US Coast Guard

How long are you allowed to take to test all parts of your ship's Vessel Response Plan according to OPA 90?

- **All parts within 3 years**
- All parts within a calendar year
- During a major exercise lasting no less than 3 hours
- During a major exercise lasting no more than 12 hours

How long do you have to keep records of OPA 90 drills and exercises?

- **3 years**
- 1 year
- 5 years
- For ever

How long must you keep records of OPA 90 training onboard?

- **3 years**
- 1 year
- 5 years
- Until that crew member has gone on leave

How often must the ship review an OPA 90 Vessel Response Plan?

- **At least once per year**
- At least once every 5 years
- Every 3 months
- When a change is required

How often must you conduct Qualified Individual notification drills while your ship is in OPA 90 waters?

- **Every 3 months**
- Every 12 months
- Every 6 months
- Every month

If you carry out a remote assessment and consultation exercise while in OPA 90 waters how long should it take to get advice back from shore?

- **Less than 1 hour**
- Less than 15 minutes
- Less than 3 hours
- Less than 6 hours

Ships over 300 GT which are trading to US ports need to have a COFR. What does COFR stand for?

- **Certificate of Financial Responsibility**
- Certificate of Final Rescue
- Contract for Oil Fuel Removal
- Contract for Oil Fuel Responsibility

What assistance can a ship covered by OPA 90 get to minimise the risk of oil pollution during a fire onboard?

- **Boarding by trained shoreside firefighters**
- No assistance until the ship is berthed
- Radio advice from the local Coast Guard Commander
- Radio advice from the local shoreside fire brigade chief

What can a Federal On Scene Coordinator do if they are not satisfied with your contractor's response to an oil spill from a ship in OPA 90 waters?

- **They can demand changes to the pre-planned response in exceptional circumstances**
- They can change your Qualified Individual if they do not like them
- They can replace your contractor with another one if they do not like them
- They will gather evidence for a prosecution

What is the longest period that an OPA 90 Vessel Response Plan can go before needing to be re-approved by the US Coast Guard?

- **5 years**
- 10 years
- 3 years
- Until there is an important change made to it

What must be done during a Qualified Individual notification drill? Pick the answer you think is best.

- **Contact the Qualified Individual as required by the VRP and get confirmation that they responded**
- Contact the Qualified Individual by email or fax so you can save them as proof that the drill was carried out
- Contact the Qualified Individual in any way possible
- Contact the Qualified Individual using all of the ways required by the VRP

Which of the following best describes the area covered by OPA 90 regulations?

- **The inland waters of the United States, and the seas within its 200 mile Exclusive Economic Zone**
- The inland waters of the United States, and the seas up to 12 miles out
- The seas up to 12 miles from the coasts of the United States and its island territories
- The seas up to 200 miles from the coast of the mainland United States

Which of the following best describes the main job of a Federal On Scene Coordinator during an oil spill in OPA 90 waters?

- **They will monitor your QI's management of the clean up**
- They will control every aspect of the clean up
- They will gather evidence for a prosecution
- They will translate messages between the crew and the US Coast Guard

Which of the following best describes the maximum penalty for a ship trading to a US port without a Certificate of Financial Responsibility?

- **The ship may be seized and sold**
- A fine of $30,000
- A fine of $30,000 per visit
- A fine of $30,000 per week

Which of the following best describes the role of a US Coast Guard Strike Team during a noxious liquid spill from a ship?

- **They can assist with the clean up, if called upon by the On Scene Coordinator**
- They will carry out the clean up
- They will control every aspect of the clean up
- They will gather evidence for a prosecution

Which of the following best describes the role of your Qualified Individual (QI) in meeting OPA 90 requirements?

- **They have full authority to arrange whatever is needed to prevent an oil spill, or clean one up**
- They have passed the IMO's Qualified Individual examination
- They have passed the US Coast Guard's Qualified Individual examination
- They have the operator's authority to sign a Lloyd's Open Form salvage contract

Which of the following best describes where the Oil Spill Liability Trust Fund gets its money since OPA 90 came into force?

- **From an oil shipment tax and from fines for oil pollution**
- From a tax on fuels for transportation
- From an oil shipment tax
- From fines for oil pollution

Which of the following describes one of the main problems experienced when dealing with the oil spilled from the Exxon Valdez?

- **Government and industry response plans proved to be unable to handle the spill**
- The Alaskan government ran out of money
- The ship capsized
- The ship exploded and caught fire

Which of the following is the best way to find out your ship's minimum stock of oil clean up materials for OPA 90 purposes?

- **Look in the ship's Vessel Response Plan**
- Ask the Designated Person Ashore for a list
- Get a copy of OPA 90 from the internet
- Get a copy of OPA 90 from the ship's agent

Which of the following should be notified first of a probable oil spill from a ship in OPA 90 waters?

- **The National Response Center in Washington DC**
- The local harbourmaster
- Your Designated Person Ashore
- Your Qualified Individual

Which of the following was one of the important findings in the report into the Exxon Valdez accident?

- **Prevention is the first line of defence**
- All oil tankers should be fitted with inert gas systems
- Oil shipments from Alaskan ports should be banned
- Oil tankers should be restricted in size

Which of the following would be the best choice to deal with media enquiries after an oil spill from a ship in OPA 90 waters?

- **Someone appointed by the ship operator**
- The Federal On Scene Coordinator
- The master
- The ship's safety officer

Which one of the following could be the result of you being found 'grossly negligent' during an oil pollution incident within OPA 90 waters?

- **Your company could face unlimited liability, putting it out of business**
- You could be given a reward by the Department of Justice
- Your maximum fine could be doubled
- Your maximum prison sentence could be tripled

Which one of the following describes what action should be taken after an oil spill during bunkering in the USA has been cleaned up?

- **The effectiveness of the Vessel Response Plan should be reviewed**
- The chief engineer should be disciplined
- The next emergency response drill should cover bunker spills
- The next spill emergency response drill need not be carried out

Which one of the following happened to the Exxon Valdez, causing her to spill her cargo of oil?

- **She ran aground**
- She capsized
- She caught fire
- She had exploded and caught fire

Which one of the following is a recognised sign that oil pollution is serious enough for you to notify to the authorities while your ship is in OPA 90 waters?

- **A visible oil sheen on the water**
- A 1 millimetre oil film on the water
- A 3 millimetre oil film on the water
- You need to see a stream of oil flowing into the water from your ship

Which one of the following is a rule for oil tankers that was introduced by OPA 90?

- **They need to be built with double hulls**
- They need extra crew
- They need high capacity cargo pumps
- They need to have GPS and electronic charts

Which one of the following shipping accidents led to the introduction of the Oil Pollution Act of 1990 to the USA?

- **Exxon Valdez**
- Erika
- Prestige
- Torrey Canyon

Which one of the following would you expect to contact your Oil Spill Response Organisation (OSRO) after your ship had an oil spill in OPA 90 waters?

- **Your Qualified Individual**
- The Designated Person Ashore (DPA)
- The ship's master
- The US Coast Guard

Which ships are required by OPA 90 to have a Vessel Response Plan? Pick the answer you think is best.

- **All tankers and other ships over 400 GT**
- All ships over 300 GT
- All ships over 400 GT
- Tankers over 150 GT and other ships over 400 GT

Which ships need to report any oil pollution seen to the National Response Center in Washington? Pick the answer you think is best.

- **All ships in US waters**
- All ships in Washington State waters
- All ships over 300 GT
- All tankers and other ships over 400 GT

Why can your OPA 90 Vessel Response Plan list several Salvage and Marine Fire Fighting service providers? Pick the answer you think is best.

- **Assistance must be provided within specified times so the providers must be near if there is an incident**
- Each US State uses a different provider
- The cheapest provider will be chosen when the incident happens
- US competition rules require that more than one provider be used

Why did the report into the Exxon Valdez recommend that OPA 90 require an effective response to oil pollution? Pick the answer you think is best.

- **It recognised that while oil is transported some spills are inevitable**
- It decided that insurance would not cover the costs of cleaning up spills
- It recognised that there were not enough double hull tankers available to serve the US oil market
- It saw that there were no response plans before Exxon Valdez

Why is it necessary to comply with rules which are extra to OPA 90 when visiting some States in the USA? Pick the answer you think is best.

- **The US legal system allows states to make extra laws, provided they do not contradict federal laws**
- Insurance in some States is more expensive than others
- Some States don't like foreign ships in their ports
- The water in some States is used for drinking

Why, under OPA 90, can the nearest salvage tug capable of towing your ship be stationed further away than the nearest oil skimming machine? Pick the answer you think is best.

- **Small pollution discharges must be dealt with in shorter times than worst case discharges**
- An oil skimmer must be carried onboard each ship
- Salvage tugs of the right power are not always available in the US
- The ship may be in water that is not deep enough for the tug

0007 – Inert Gas Generator

Version 2.0 (2713)

A water column type pressure vacuum breaker device may be fitted to the deck main of an inert gas system. Which of the statements given in the options is true?

- **This type of device is usually designed to operate before the deck water seal reaches its limit.**
- This type of device is intended to protect individual tanks against over pressure or under vacuum.
- This type of device is intended to protect individual tanks against over pressure or under vacuum.
- This type of device is usually designed to operate only after the deck water seal fails.

Considering an oil tanker fitted with an independent inert gas generator system. Which of the items identified in the options is most likely to be operating continuously when the cargo tanks are full of cargo?

- **Deck seal pump**
- Capacity control valve
- Combustion air blower
- Scrubber water pump

Considering the ullage space of a cargo tank containing petroleum based oil. Which of the given statements best describes the atmosphere of the ullage space?

- **It will contain a mixture of atmospheric air and hydrocarbon gases with an oxygen content of less than 21% by volume.**
- It will contain a layer of hydrocarbon gases and a layer of atmospheric air since the two cannot mix.
- It will contain a mixture of atmospheric air and hydrocarbon gases with an oxygen content equal to approximately 21% by volume.
- It will contain a mixture of atmospheric air and hydrocarbon gases with an oxygen content greater than 21% by volume.

During loaded passage the inert gas pressure in the cargo tanks can vary due to changes in ambient temperature. Which of the given options would be the preferred way to maintain the correct pressure on a vessel fitted with an independent inert gas generator?

- **Intermittent operation of the generator in automatic mode to top up the pressure.**
- Continuous operation of the generator in automatic mode at a very low capacity setting using the automatic atmospheric vent to maintain the pressure.
- Continuous operation of the generator in manual mode to maintain the pressure with manual control of the atmospheric vent valve.
- Intermittent operation of the generator in manual mode to top up the pressure

During start up sequence of an independent inert gas generator it is necessary to allow time for a pre-purge of the furnace whether in manual or automatic mode. What is the recommended minimum time for such a pre-purge operation?

- **3 minutes**
- 1 minute
- 30 seconds
- 5 seconds

During the operation of an independent inert gas generator, a problem develops with the scrubber unit effluent valves such that the water level in the scrubber unit reaches alarm condition. In addition to the alarm sounding what other action is most likely to result from this condition?

- **Immediate shut down of the inert gas generator system.**
- Only the main burner would shut down
- The deck main valve would close and the automatic vent would open but the generator would continue operating.
- The scrubber pump would immediately stop but the generator would continue.

For an independent inert gas generator with automatic control which one of the parameters in the given options is most likely to have a controller set point that can be adjusted by the operator?

- **The pressure of the inert gas in the deck main.**
- The pressure of the seawater to the deck seal.
- The temperature of the scrubber effluent.
- The viscosity of the fuel to the burner unit.

High efficiency burners are often used in independent inert gas generators to ensure good combustion and that high quality inert gas is produced. Monitoring the gases from the combustion process can give an indication of the efficiency of the combustion. Which of the given options indicates good combustion?

- **Minimum carbon monoxide content**
- Minimum carbon dioxide content
- Minimum nitrogen oxide content
- Minimum oxygen content

How is the quantity of inert gas from an independent inert gas generator normally monitored?

- **By the pressure in the deck main.**
- By the blower electrical load.
- By the oxygen content at the scrubber outlet.
- By the temperature at the scrubber outlet.

In the event of a multiple alarm condition occurring during operation of an independent inert gas generator what will be the normal arrangement for the alarm display?

- **All alarms will be listed in order of priority.**
- All alarms will be listed in the order they occurred.
- Only the last alarm to occur will be shown.
- Only the most important alarm to occur will be shown.

In the event that the oxygen content of the inert gas supply from an inert gas generator exceeded the value of the set point an alarm would sound. Which one of the given options is also the most likely to occur in such a situation?

- **The automatic vent valve would open and the deck main valve would close.**
- The generator would immediately shut down.
- The main burner would shut down but the blower would continue to run.
- The scrubber pump would automatically stop.

Independent inert gas generator systems fitted to gas carriers often have refrigerated type cooler dryer units fitted to control the humidity of the inert gas. Which condition of the gas is used to control entry of the gas into the drier section of such a unit?

- **The temperature of the inert gas**
- The humidity of the inert gas
- The oxygen content of the inert gas
- The pressure of the inert gas

Independent, oil fired, inert gas generators are usually operated on distillate fuel rather than residual fuel. What is the main reason for this?

- **Operating on distillate fuel normally produces a cleaner exhaust gas with a lower oxygen content due to relatively good combustion.**
- Distillate fuel is more readily available than residual fuel
- More exhaust gas is available when burning distillate fuel.
- The inert gas plant does not require a scrubber unit since distillate fuels are have a zero sulphur content.

On which one of the vessel types given in the options would a nitrogen generator most likely to be fitted for the purpose of providing inert gas to the cargo system and associated spaces?

- **Chemical tanker**
- Container vessel
- Crude oil carrier
- Steam ship used as a heavy fuel oil carrier

SOLAS stipulates the maximum content of oxygen by volume that an inert gas system, including an independent inert gas generator type, must be capable of supplying. Which of the given options is this maximum value?

- **5%**
- 11%
- 3%
- 8%

Select the best option from those given to complete the following statement in relation to pressure vacuum breakers as fitted to the distribution system of some independent inert gas generator systems.

The correct operation of a liquid column type pressure vacuum breaker is dependent on . . .

- **. . . the correct specific gravity of the liquid and the liquid column height**
- . . . the correct operation of the automatic vent to atmosphere valve
- . . . the correct operation of the deck water seal
- . . . the correct operation of the mechanical non-return device in the deck main

Some oil fired, independent inert gas generators incorporate capacity control to regulate the rate of delivery of the inert gas to the distribution system. Which of the given options is the primary element regulated by such a control feature?

- **The combustion air blower load.**
- The fuel pump pressure.
- The fuel rate to the main burner.
- The fuel recirculation valve.

The combustion air blowers of an inert gas generator may be used to supply fresh air during venting and gas freeing operations following purging of cargo tanks and lines. What precautions must be taken when carrying out such operations with an independent inert gas generator?

- **Ensure all none purged tanks are properly isolated from the inert gas main.**
- Ensure that the capacity control setting is at a minimum
- Ensure that the inert gas generator fuel pump is running on recirculation throughout the operation.
- Ensure the atmospheric vent valve is fully opened.

What is generally taken as the minimum amount of oxygen in a mixture of atmospheric air and hydrocarbon gases that gives a safe margin against the risk of combustion?

- **8%**
- 16%
- 20%
- 5%

What is the SOLAS requirement regarding non-return devices in the inert gas main for inert gas systems on oil tankers?

- **There must be at least two devices one of which should be a water seal type.**
- As long as there is a water seal no other devices are required
- There must be at least two neither of which should be a water seal type.
- There must be at least two water seal type devices fitted.

What is the main reason for controlling the humidity of the inert gas produced by an independent inert gas generator when fitted to a gas or chemical tanker?

- **To avoid compatibility problems with the cargo.**
- To avoid contamination of the deck seal water.
- To avoid corrosion problems in the cargo pipelines.
- To avoid corrosion problems in the inert gas deck main.

What is the main risk associated with dilution of a normally inerted cargo oil tank atmosphere with fresh air?

- **During the process the atmosphere may be within the explosive range for some of the time.**
- During the process the atmosphere may become over rich for some of the time.
- During the process the atmosphere may become too lean for some of the time.
- During the process the oxygen content may exceed the maximum limit for a flammable mixture.

When considering an independent, oil fired inert gas generator how is the required flow of inert gas normally generated?

- **The combustion air fans provide sufficient pressure to ensure the necessary inert gas flow.**
- Additional booster fans are fitted in the deck main.
- Cargo discharge rates are kept high to create a partial vacuum in the tanks which helps to draw in the inert gas.
- Main inert gas fans are fitted after the scrubber unit to increase the flow of inert gas.

When considering an independent, oil fired inert gas generator suitable for service onboard an oil tanker what is the SOLAS requirement regarding the composition of the tank atmosphere to be maintained?

- **A maximum of 8% oxygen content**
- A maximum of 1% oxygen content
- A maximum of 11% oxygen content
- A maximum of 5% oxygen content

When considering an independent, oil fired inert gas generator what is the main purpose of the pilot burner?

- **It is used to provide an ignition source for the main burner.**
- It is used to establish the correct oxygen content of the inert gas before the main burner starts.
- It is used to preheat the combustion air to make it easier to ignite the main burner.
- It is used to preheat the furnace.

When considering an independent, oil fired inert gas generator which incorporates a sea water scrubber unit. How is the water content of the inert gas normally kept to a minimum if the generator is used onboard an oil tanker?

- **A demister unit is fitted before the scrubber inert gas outlet.**
- The inert gas is passed through a refrigerated drying unit before it enters the deck water seal.
- The inert gas is passed through a series of coalescer filters after the scrubber unit.
- The water flow through the scrubber unit is kept to a minimum.

When considering the automatic operation of an independent inert gas generator the start up sequence normally has delays built in to ensure correct timing of the sequence. For example there will be a delay in the sequence to ensure sufficient furnace purging time. How can these delays be ensured in manual mode?

- **By the use of timed interlocks which operate even in the manual mode.**
- The delays are only an extra safety precaution so it doesn't matter if they missed out.
- The operator is relied upon to ensure the correct timing of the sequence.
- There is no need for any delays if the operator is in attendance during start up.

When considering the water supply to the deck water seal for an inert gas system which of the following statements is true?

- **There should be at least one independent pump for continuously supplying water to the deck seal.**
- A water supply to the deck seal is only required during cargo operations
- The deck seal must be supplied directly from the engine room sea water main and auxiliary systems.
- The water supply can be taken directly from the fire main.

When gas freeing a normally inerted cargo oil tank what should be the first step in the operation?

- **Purge with inert gas to bring the atmosphere to below the critical dilution line.**
- Purge with fresh air to bring the atmosphere to below the critical dilution line.
- Purge with fresh air to bring the atmosphere to below the lower explosive limit.
- Purge with inert gas to bring the atmosphere to below the upper explosive limit.

When shutting down an independent inert gas generator there is a correct sequence to the operation. Put the given options in the order that they should occur when an automatic shut down of the generator is initiated. Ignore the fact there will be delays built into the sequence.

- Automatic vent to atmosphere open followed by inert gas deck main closed **Firstly**
- Combustion air blower and scrubber pump stop. **Finally**
- Fuel pump stop **Thirdly**
- Main burner off and fuel recirculating **Secondly**

Which is the best test gas to use when checking and adjusting the zero setting of an oxygen analyser as fitted to an independent inert gas generator system?

- **Nitrogen**
- Carbon dioxide
- Carbon monoxide
- Hydrogen

Which of the given options describes the lower explosive limit in relation to the atmosphere of a cargo oil tank atmosphere?

- **The least proportion of hydrocarbon gases in a mixture with atmospheric air which is flammable.**
- The least proportion of oxygen in a mixture of hydrocarbon gases and atmospheric air which is flammable.
- The lowest amount of oxygen which will support a flame when mixed with hydrocarbon gases.
- The lowest point in a cargo tank ullage space where the mixture of hydrocarbon gases and air is flammable

Which of the given options is the main reason for ensuring that the air cone of the burner unit for an independent inert gas generator is properly positioned and maintained?

- **If not it will result in poor combustion**
- If not it will result in a high oxygen content of the inert gas
- If not it will result in a low carbon monoxide content of the inert gas
- If not it will result in overpressure in the inert gas main

Which of the given options represents the approximate percentage content by volume of nitrogen in atmospheric air?

- **79%**
- 21%
- 50%
- 70%

Which of the given ranges of oxygen content by volume is likely to be achieved when supplying inert gas from an independent inert gas generator operating on distillate fuel?

- **0.5% to 3%**
- 3% to 5%
- 5% to 8%
- Less than 0.5%

Which one of the alarm conditions given in the options would result in shutdown of the burner unit and the blower continuing to run for a cooling down period when considering the operation of an independent inert gas generator?

- **Scrubber pump low discharge pressure.**
- Combustion air blower low load
- Deck inert gas main high pressure.
- High oxygen content of inert gas

Which one of the given options is the most likely type of fixed gas measuring device to be fitted for monitoring and control of the quality of the inert gas produced by an independent inert gas generator?

- **A paramagnetic type oxygen analyser.**
- A carbon dioxide monitor.
- A carbon monoxide monitor.
- A heated filament combustible gas analyser.

Which term is used to identify the operation whereby fresh air is introduced into a cargo tank in order to remove toxic, flammable and inert gases and to increase the oxygen content of the tank to 21% by volume?

- **Gas freeing**
- Diluting
- Inerting
- Purging

Which term is used to identify the operation whereby inert gas is introduced into a cargo tank in order to reduce the oxygen content throughout the atmosphere of the tank to 8% or less by volume?

- **Inerting.**
- Gas freeing
- Purging
- Topping up

While operating an independent inert gas generator during cargo discharge it is noted that the automatic vent valve is continuously 50% open. Which of the given options would be the best adjustment to make in order to minimise the gas being lost to atmosphere whilst maintaining the inert gas main pressure?

- **Reduce the system capacity control setting.**
- Increase the cargo discharge rate
- Increase the set point setting for the deck pressure control.
- Reduce the fuel pressure to the generator burner.

With reference to the adjustment of the set points for the process controllers and the alarm for an independent inert gas generator which one of the statements given in the options is the most accurate?

- **There is usually some in built security feature to limit access to these settings.**
- Any of the settings can be adjusted by anyone operating the generator
- Normally only the alarm settings can be adjusted on this type of inert gas generator by ship's staff.
- The settings are factory set and cannot normally be adjusted.

0008 – Flue Gas Plant

Version 2.1 (2237)

According to the Fire Safety Systems Code what is the maximum oxygen content permitted in the cargo tanks of an oil tanker?

- **8% by volume**
- 11% by volume
- 21% by volume
- 3% by volume

According to the Fire Safety Systems Code which one of the following would be considered a safe amount of oxygen in the output from an inert gas plant?

- **Less than 5% oxygen**
- Less than 11 % oxygen
- Less than 16 % oxygen
- Less than 21% oxygen

According to the instructions in this module, which of these options would be carried out first when starting up a flue gas plant?

- **Open the boiler uptake valve**
- Open the blower discharge valve
- Open the gas regulating valve
- Start the blower

If the inert gas at the blower outlet gets too hot a high temperature alarm is given. What will happen if the temperature of the gas continues to rise?

- **The inert gas blowers will stop and the gas regulating valve closes**
- The backup scrubber water pump will be started automatically
- The inert gas will be vented to air until the temperature drops
- The temperature of the inert gas in the cargo tanks should be monitored every 15 minutes

In an oil cargo tank in which there is a mixture of hydrocarbon gas and air, which of the following is the best definition of the "Lower flammable limit" (LFL)?

- **It is the concentration of hydrocarbon vapour below which there is insufficient gas in the mixture to support combustion**
- It is the point where the concentration of inert gas becomes too high to support combustion
- It is the point where the concentration of inert gas becomes too low to prevent combustion
- It is the point where the oxygen content becomes too high to support combustion

In an oil cargo tank in which there is a mixture of hydrocarbon gas and air, which of the following is the best definition of the "upper flammable limit" (UFL)?

- **It is the concentration of hydrocarbon gas above which there is insufficient oxygen to support and propagate combustion**
- It is the concentration of a hydrocarbon gas which is just enough to support combustion
- It is the point where the concentration of inert gas becomes high enough to prevent combustion
- It is the point where the oxygen content becomes too high to support combustion

In relation to oil tanker cargo spaces which one of the following is the best definition of the expression "purging"?

- **To blow inert gas into a tank to reduce the oxygen or hydrocarbon gas level**
- To replace the inert gas in a cargo tank with hydrocarbon gas
- To replace the inert gas in a cargo tank with oil cargo or ballast
- To replace the inert gas in a tank with fresh air

In some types of inert gas plants using flue gas, air is supplied to the boiler uptake valves. What is the reason for this?

- **To ensure no flue gas leaks pass the uptake valves when the plant is shut down**
- To cool the scrubber tower when the plant is shut down
- To prevent condensation in the inert gas plant when it is shut down
- To prevent inert gas passing into the boiler uptake when the plant is running

In which one of the following IMO publications would you find the requirements for the design of a new inert gas system?

- **International Code for Fire Safety Systems (FSS Code)**
- ISM Code
- MARPOL Annex I
- SOLAS Chapter IV

Many oil tankers are fitted with an inert gas system supplying their cargo tanks. What is the main purpose of introducing inert gas into a cargo tank?

- **To create a non-explosive atmosphere in the tank**
- To make the tank safe for entry
- To prevent evaporation of the cargo
- To reduce the generation of static electricity in the tank

On an oil tanker how long must the deck water seal supply be left on after the inert gas system is shut down?

- **It must be left running continuously to maintain an adequate seal**
- It can be stopped as soon as the inert gas plant has been shut down
- It is left running for 30 minutes after the inert gas plant is shut down
- It is left running for two hours after shut down of the inert gas plant

On inert gas plants when the scrubber water flow drops below a certain rate an alarm is given. What will happen if the flow rate drops further?

- **The inert gas blowers will stop and the gas regulating valve closes**
- The boiler uptake valve will close
- The deck seal sea water pump will automatically switch to supply the scrubber
- The pressure control system will reduce the plant output

On some inert gas plants using flue gas the speed of the blowers is variable. Which of the following is the best explanation for fitting variable speed blowers?

- **The pressure of the gas delivered to deck can be controlled**
- The electrical power used by the blowers can be reduced
- The speed can be reduced during blower washing
- The speed can be varied to prevent bearing damage

Some oil tankers are fitted with one large inert gas blower and one smaller blower. Which of the following is the best explanation for fitting a small blower?

- **It can be used when topping up the gas pressure at sea**
- It is for use when gas freeing
- It keeps electrical loads lower during discharges
- The total blower capacity can be reduced

The power supply to a flue gas type inert gas plant is normally left on, even when the plant is shut down. Which of the following best describes the reason for this?

- **Some alarm functions must be operational even when the plant is not**
- The blowers will automatically be turned every 6 hours
- The oxygen level must be continuously monitored
- The oxygen level must be continuously monitored

What is the Fire Safety Systems Code requirement regarding non return devices in the inert gas main for inert gas systems on oil tankers?

- **There must be at least two devices one of which should be a water seal type**
- As long as there is a water seal no other devices are required
- There must be at least two neither of which should be a water seal type
- There must be at least two water seal type devices fitted

What is the main risk associated with dilution of a normally inerted cargo oil tank atmosphere with fresh air?

- **During the process the atmosphere may be within the flammable range for some of the time**
- During the process the atmosphere may become over rich for some of the time
- During the process the atmosphere may become too lean for some of the time
- During the process the oxygen content may exceed the maximum limit for a flammable mixture

What is the purpose of the deck water seal fitted to an oil tanker's inert gas system?

- **To prevent backflow of hazardous vapours into the gas safe areas**
- To clean the inert gas before it enters the cargo tanks
- To cool the inert gas before it enters the cargo tanks
- To prevent inert gas entering the tanks during loading operations

What is the purpose of the pressure/vacuum breaker fitted to an oil tanker's inert gas system?

- **To relieve excess pressure or vacuum in the inert gas main**
- To control the pressure of inert gas supplied to the cargo tanks
- To improve cargo pump performance in case the inert gas blower cannot keep up with the cargo discharge rate
- To prevent backflow of cargo vapours into the inert gas scrubber

When an oil tanker has a flue gas type inert gas plant which is also used for gas freeing of cargo tanks where is the air normally drawn from?

- **The blower fresh air inlet**
- A gas free tank
- The deck seal vent
- The flue of a shut down boiler

When entering any part of the inert gas system to carry out maintenance or inspections, which of the following is the main danger to personnel?

- **Oxygen deficiency**
- Acidic attack
- Bacterial infection
- Gas flammability

When opened up the scrubber unit of a flue gas type inert gas plant is found to contain a mattress of polypropylene mesh. Which of the following is the best explanation for this mesh being there?

- **It is fitted to the scrubber to dry the gas**
- It acts as a wet filter inside the scrubber
- It has fallen from the blower inlet into the scrubber
- The lining of the scrubber has come off

When the inert gas plant on an oil tanker suffers an automatic shutdown which one of the options below is true of the seawater pumps and blowers?

- **Only the seawater pumps are left running**
- All pumps and blowers are left running
- All pumps and blowers are left running, the boiler uptake valve shuts the gas off
- All pumps and blowers are shut down

Which of the following best describes the action to be taken by the watchkeeper as soon as the blower in a flue gas type inert gas plant is stopped during shut down of the plant?

- **The blower washing water should be turned on**
- The blower drains should be shut
- The blower washing water should be turned off
- The scrubber water should be turned off

Which of the following best describes the most likely cause of low deck main pressure while a flue gas type inert gas plant is running?

- **Cargo discharge rate in excess of the blower output**
- Deck seal empty
- Low boiler load
- P/V valve seized shut

Which of the following best describes the most likely cause of the boiler soot blowers failing to work while a flue gas type inert gas plant is running?

- **The interlock is operating to prevent excessive soot entering the inert gas**
- The cargo pump turbines will reduce steam pressure too far
- The exhaust gas is diverted away from the tubes during inert gas production
- The exhaust gas pressure will be too low for effective soot blowing

Which of the following best describes the most likely initial consequence of low boiler load when a flue gas type inert gas plant is running?

- **A high oxygen level**
- A low deck main pressure
- High gas temperature at the blower outlet
- Increased inert gas recirculation

Which of the following best describes the reason for the fixed oxygen meter being fitted close to its sampling point?

- **To reduce the time between the sample being taken and analysed**
- The oxygen level falls as it passes along the sample line
- The oxygen level rises as it passes along the sample line
- To protect it from heat

Which of the following is the best definition for 'inert gas', as used to protect the cargo spaces of oil tankers?

- **It is a gas containing insufficient oxygen to support the combustion of hydrocarbons**
- It is a gas containing no oxygen
- It is a gas containing too much hydrocarbon vapour to support combustion
- It is any gas containing less than 21 % by volume of oxygen

Which of the following is the best explanation of the purpose of the scrubber unit in a flue gas type inert gas plant?

- **To cool the gas while removing soot particles and sulphur dioxide**
- To cool the gas
- To prevent flue gas entering the plant while it is shut down
- To remove moisture from the flue gas before being sent to the tanks

Which of the following is the main purpose of filling the free space above the oil cargo onboard a tanker with inert gas?

- **To keep the tank atmosphere outside the flammable range**
- To cool the tank
- To prevent contamination of the cargo
- To reduce evaporation of the cargo

Which of the following is the typical oxygen level in the exhaust gas that can be achieved from good combustion in a ship's boiler?

- **4 to 5%**
- 0.1 to 0.5%
- 7 to 8%
- 9 to 10%

Which of the following operations is carried out last when shutting down a typical flue gas plant?

- **Shut down the scrubber water pump**
- Close the boiler uptake valve
- Close the gas regulating valve
- Open the ventilation valve to atmosphere

Which one of the following is a maintenance problem that can be seen during an external examination of the blower of an inert gas plant?

- **Cracking**
- Epoxy coating damage
- Impeller corrosion
- Soot build up

Which one of the following is likely to happen if the oxygen content of the inert gas being supplied to the cargo spaces of an oil tanker rises above 8 percent and the 'high-high' oxygen alarm sounds?

- **The gas regulating valve will automatically close, and the vent or recirculation valve will open**
- The boiler burners will shut down
- The gas regulating valve and the boiler uptake valves will automatically close
- The inert gas blowers will stop and the plant is shut down

Which one of the following is the generally agreed reason for the rise in explosions experienced in the 1960s and 1970s, as oil tankers became larger?

- **Static electricity**
- High sulphur crude
- High voltage electrical systems
- Low sulphur crude

Which one of the following is the minimum required capacity of the blowers fitted to an oil tanker's inert gas plant, as a proportion of the ship's maximum cargo discharge rate?

- **125% in total**
- 100% in total
- 100% per blower
- 110% in total

Which one of the following would be the best gas to use to zero the fixed oxygen analyser monitoring an inert gas plant?

- **Nitrogen**
- 50/50 Methane/Butane mix
- Butane
- Methane

Which one of the following would you expect to happen on an oil tanker if the pressure in the inert gas main falls to about 50 mm Water Gauge during cargo discharge operations?

- **The cargo pumps shutdown automatically**
- The inert gas blowers shutdown automatically
- The inert gas recirculating valve automatically opens fully
- The P/V valve will open fully

Which term is used to identify the operation whereby inert gas is introduced into a cargo tank in order to reduce the oxygen content throughout the atmosphere of the tank to 8% or less by volume?

- **Inerting**
- Gas freeing
- Purging
- Topping up

0009 – Fuel Oil System

Version 2.1 (2235)

Biodiesel is being proposed as an alternative fuel for use in marine diesel engines. Which one of the options best describes biodiesel that can be used in an unmodified diesel engine.?

- **A blend of petroleum based diesel fuel and less than 20% bio fuel.**
- A blend of petroleum based fuel and water.
- A pure product from the processing of vegetable crops.
- A pure product of hydrocarbon based oil refining

Combustion of fuel oil can be considered to be an energy conversion process. Which of the following options best describes the conversion?

- **Chemical to heat energy**
- Chemical to mechanical energy
- Heat to chemical energy
- Heat to mechanical energy

Dual fuel diesel engines which use gas and fuel oil are used on some ships. Select the option which best describes one of the methods of operation of the fuel systems for these engines

- **Low pressure gas is used as main fuel with fuel oil used to ignite it**
- High pressure gas is added to the fuel oil and both injected into the cylinder together
- High pressure gas is used as fuel when fuel oil is not available
- Low pressure gas and fuel oil are injected into the cylinder separately

Fuel oil viscosity control is considered necessary for diesel engines operating on heavy fuel oil. Which one of the following best describes the principle of operation of a Viscotherm unit?

- **Measurement of differential pressure in a capillary tube**
- Measurement of differential pressure between inlet and outlet to the unit
- Measurement of outlet pressure of the unit pump
- Measurement of the differential temperature across the unit

Fuels are sometimes blended onboard the ship prior to use. Which one of the following best describes why this process carried out?

- **To control the fuel viscosity at the engine inlet**
- To control the density at the purifier inlet
- To control the temperature at the engine exhaust
- To control the temperature at the engine inlet

In fuel combustion what is meant by the term “stoichiometric mixture”?

- **A fuel air mixture which has the ideal minimum amount of air required for complete combustion of the fuel**
- A fuel air mixture that will not burn
- A fuel air mixture which has less than the ideal minimum amount of air required for complete combustion of the fuel
- A fuel air mixture which has more than the ideal minimum amount of air required for complete combustion of the fuel

Many boiler burner units include an air register. What is the purpose of an air register?

- **It controls the quantity of combustion air entering the furnace**
- It controls the direction and movement of the combustion air entering the furnace
- It measures the quantity of combustion air entering the furnace
- It prevents backflow of exhaust gas from the furnace

Many diesel engine high pressure fuel pumps have plungers with helical profiles cut into them. How do these helical profile type plungers control the timing and quantity of fuel delivered to the cylinder.

- **The leading edge controls the start of injection and the trailing edge controls the end of injection**
- The leading edge controls both the start and end of injection
- The trailing edge controls both the start and end of injection
- The trailing edge controls the start of injection and the leading edge controls the end of injection

Some boiler use rotary cup burners instead of jet nozzle types. Which one of the following options best describes how rotary cup burners differ from jet nozzle types?

- **Rotary cup burners operate on much lower fuel pressures than simple jet nozzle types**
- Rotary cup burners normally use steam atomisation
- Rotary cup burners operate with simple on/off control
- Rotary cup burners rely on electrostatic charge for atomisation of the fuelatomization of the fuel.

Some diesel engine manufacturers have introduced the use of fuel and water emulsions as part of their exhaust emissions control measures. How do these emulsions help to reduce air pollutants in exhaust gas emissions?

- **They lower the maximum cylinder temperature and reduce NOx levels**
- The oxygen in the water improves combustion of carbon and reduces carbon monoxide levels
- The water washes out sulphur and reduces SOx levels
- They dissolve the ash components of the fuel and reduce particulate levels

Some diesel engines feature a twin injection system designed to improve fuel injection performance. Select the option which best describes this feature.

- **A constant amount of fuel is injected by the first injector at all loads while the amount through the second one varies**
- A variable amount of fuel is injected by the first injector while the amount through the second one is constant at all loads
- An equal amount of fuel passes through each injector in sequence
- An equal amount of fuel passes through each injector in simultaneously

Some diesel engines have fuel injection systems which incorporate pilot injection. What is the main purpose of using pilot injection?

- **To reduce ignition delay**
- To increase ignition delay
- To increase maximum cylinder pressure
- To reduce maximum cylinder pressure

Some diesel engines operate on a fuel mixture which is blended onboard. Which one of the following is a means to control the make up of the blended fuel?

- **Preset viscosity and temperature with varying ratio of blend components.**
- Preset temperature and blend component ratio and varying viscosity
- Preset viscosity and temperature with varying pump pressure
- Preset viscosity, temperature and blend component ratio

Some machines use fuel directly to provide the energy required for operation. Which of the following machines would you include in this group?

- **Diesel engine**
- Air compressor
- Electric motor
- Steam turbine

The term "atomisation" is often used in relation to diesel engine fuel injection. Which of the following options best defines the term "atomisation"?

- **The break up of a liquid fuel stream into fine droplets**
- The distribution of fuel in a diesel engine cylinder
- The formation of a combustible fuel air mixture
- The mixing of fuel and air in a diesel engine cylinder

The vessel you are sailing on, which has one settling tank and one service tank in the fuel system, is due to operate in a SECA. When would you need to start to change over to low sulphur fuel if you are currently operating on normal heavy fuel?

- **At the time before entry into the SECA stated in the ship's fuel changeover procedure**
- In time to consume the contents of the in service system
- In time to consume the contents of the service tank and the in service system
- Just as you enter the SECA

Under Marpol Annex VI legislation which one of the following best describes the difference between operating an engine inside a SECA compared to one outside a SECA ?

- **Fuel with a lower sulphur content must normally be used in a SECA**
- Fuel with a higher sulphur content must normally be used in a SECA
- Fuel with a lower nitrogen content must normally be used in a SECA
- Fuel with a lower viscosity must normally be used in a SECA1

What are the products from the complete combustion of hydrogen?

- **Water vapour and heat**
- Carbon dioxide and heat
- NOx emissions
- SOx emissions

What is meant by the term direct injection in relation to diesel engine fuel injection systems?

- **A system in which fuel is injected into the combustion chamber**
- A system in which fuel is admitted with the inlet air
- A system in which fuel is injected into a pre-combustion chamber
- A system in which fuel is taken straight from the settling tanks to the engine

What is the most common method used to allow the position of fuel cams to be individually adjusted on the camshaft?

- **The majority of fuel cams are mounted on hydraulic sleeve couplings**
- The majority of fuel cams are bolted to flanges
- The majority of fuel cams are mounted on splines
- The majority of fuel cams are of split design

What is the normal source of ignition in most diesel engines?

- **Heat from the compressed air charge in the cylinder**
- A spark igniter
- Heat from a glow plug
- Heat from the fuel

When diesel engines are operated on heavy fuel oil the fuel is heated up to achieve the correct injection viscosity. How is the fuel system prevented from gassing up when operating this way?

- **By maintaining a high enough pressure in the low pressure system to prevent vapour formation**
- By increasing the viscosity control setting to limit the fuel temperature
- By operating the system with an open vent
- By setting a temperature limit of below 100° C

Which of the following options best describes the term "self ignition temperature"?

- **The minimum temperature at which a combustible mixture of fuel vapour and air will ignite without an external ignition source**
- Another term for the flash point of a fuel
- The minimum temperature at which a combustible mixture of fuel vapour and air can be ignited with an external ignition source
- The minimum temperature at which combustion will continue when the source of ignition is removed

Which one of the following best describes the purpose of the spring in a fuel injection valve?

- **To give positive seating of the valve to prevent back flow of gas from cylinder into the fuel system**
- To allow adjustment of the valve closing period.
- To allow recirculation of fuel through the valve
- To limit the lift of the valve needle

Which one of the following best describes why a low pressure alarm is often fitted in the low pressure part of a diesel engine heavy fuel oil system?

- **To minimise the risk of engine fuel starvation**
- To indicate when distillate fuel has reached the pump supply manifold
- To indicate when the level in the fuel oil service tank is low.
- To indicate when the service tank valve is not fully open.

Which one of the following best explains why high pressure fuel pipes fitted to many diesel engines are often double skinned?

- **So that any leakage is contained**
- So that if the inner skin fails operation can continue using the outer skin
- To prevent engine room personnel from contacting the hot inner skin
- To prevent heat loss by insulating the inner skin

Which one of the following boiler operating parameters would normally give automatic shut down of the boiler in event of alarm condition?

- **Flame failure**
- Exhaust high oxygen level
- High fuel temperature
- High steam temperature

Which one of the following elements of diesel engine exhaust emissions are considered as atmospheric pollutants?

- **Carbon dioxide**
- Heat
- H_2O
- Nitrogen

Which one of the following features would you expect to find with an inline multi cylinder fuel pump block arrangement?

- **A single rack to control all of the fuel pumps simultaneously.**
- A separate camshaft for each cylinder
- High pressure fuel pipes without bends
- Positioning on the engine at crankshaft level

Which one of the following gases would you normally associate with the combustion of hydrocarbon fuel containing sulphur in a diesel engine?

- **Sulphur dioxide, SO2**
- Hydrogen sulphide, H2S
- Sulphur carbide, SC3
- Sulphur oxide, SO

Which one of the following options best describes how combustion in a boiler furnace is different to that in a diesel engine cylinder?

- **Boiler combustion is a continuous process.**
- Boiler combustion is controlled by measuring the exhaust temperature
- Boiler combustion is started from cold using distillate fuel
- Boiler combustion requires more excess air

Which one of the following options best describes the gas used as a source of fuel on some LNG carriers?

- **It is a low pressure gas, having evaporated from the cargo tanks**
- It is a high pressure liquid, and stored in bunker tanks within the cargo area
- It is a high pressure liquid, having been pumped from the cargo tanks
- It is a low pressure liquid, after processing by the reliquefaction plant

Which one of the following options best describes when you would normally expect to find a thermal oil heating unit fitted onboard a ship?

- **When the ship operates on heavy fuel but has no steam boiler**
- When the ship has steam propulsion
- When the ship only has 4 stroke diesel engines
- When there is only one steam boiler

Which one of the following options best describes why incinerators need a fuel oil system?

- **To establish a flame prior to burning waste oil.**
- To allow the incinerator to be operated when there is no waste oil, sludge or waste material.
- To inject fuel into the sludge before they are pumped to the furnace
- To soak solid waste material before burning

Which one of the following options best describes why recirculation type fuel injectors are sometimes fitted to diesel engines?

- **A separate fuel valve cooling system is not required**
- Removes the need for the vessel to carry distillate fuel
- Removes the need to have viscosity control
- They allow engine to be run on heavy fuel oil

Which one of the following options is a requirement of a diesel engine fuel injection pump?

- **Ensure the correct quantity of fuel injected**
- Ensure the correct quantity of air admitted
- Ensure the fuel at the right temperature to ensure combustion
- Ensure the fuel is at the viscosity to ensure combustion

Which one of the following options is an acceptable method of checking the timing of diesel engine high pressure fuel pumps?

- **Spill cut off method**
- Matching witness marks on pump casing and camshaft cover
- Measuring fuel pump discharge pressure
- Testing fuel injector lift pressure

Which one of the following options is considered to be a product of incomplete combustion of fuel in a diesel engine cylinder?

- **Carbon monoxide**
- Carbon dioxide
- Nitrogen
- Water vapour

Which one of the following problems is likely to occur if the correct procedure for changing over from heavy fuel to distillate fuel prior to stopping a diesel engine is not followed?

- **Gassing up due to distillate fuel component overheating**
- SECA limits may not be met
- The engine may not stop
- Viscosity control may remain in manual mode

Which one of the following problems is most often associated with blended marine fuels?

- **Instability**
- High density
- High viscosity
- Low flash point

Which one of the following statements is true?

- **Marine diesel oil is a mixture of distillate and a small amount of residual or heavy fuel**
- Marine diesel oil is a mixture of distillate and a large amount of residual or heavy fuel
- Marine diesel oil is a pure distillate fuel
- Marine diesel oil is a residual fuel

Why is the exhaust from an oil fired boiler a suitable source of inert gas to use to inert cargo tanks on an oil tanker?

- **The boiler exhaust normally contains less than 5% oxygen**
- The boiler exhaust gas is normally clean and contaminant free
- The boiler exhaust normally contains about 30% oxygen
- The boiler exhaust normally contains no oxygen

Why is variable injection timing, VIT, incorporated in some diesel engine fuel systems?

- **To increase thermal efficiency and reduce specific fuel consumption**
- To increase the maximum load of the engine
- To increase the minimum compression pressure
- To reduce the maximum cylinder pressure

“Turbulence” is the term used to describe the movement of the charge air in a diesel engine cylinder. Which one of the following options best describes a feature which helps in generating turbulence.

- **The geometry of air inlet ports and valves**
- The combustion chamber volume
- The length of the charge air receiver
- The piston crown material

0010 – Marine Fuel Handling and Pre-Treatment

Version 2.2 (1849)

A fuel oil tank is fitted with a sounding pipe which is used from within the engine room. Which one of the following fittings given must be fitted?

- **Self closing cock with test cock.**
- Flame trap.
- Non return float valve.
- Tank high level alarm.

Back flushing filters are sometimes fitted in the onboard fuel oil treatment system. Which one of the following options best describes what the unit is normally back flushed with?

- **Cleaned fuel.**
- Air.
- Diesel oil.
- Water.

Centrifuges are often used in pairs to give combined cleaning effect on fuel oil. Which of the options given is the preferred arrangement for these combinations?

- **Purifier and clarifier in series.**
- Purifier and clarifier in parallel.
- Two clarifiers in series.
- Two purifiers in parallel.

Chemical additives and stabilisers are sometimes used as part of the fuel oil treatment. Which one of the following best describes how fuel stabilisers are intended to work?

- **Preventing the fuel from ageing.**
- By introducing chemicals which ignite more easily.
- Changing the fuel molecules.
- Forming stable emulsions.

Edge filters are used for removal of very small particles. Which one of the following best describes where you would be likely to find edge filters fitted in the fuel oil system?

- **Before fuel injection equipment.**
- At purifier inlet connections.
- At the fuel tank outlet connection.
- Before the viscosity controller.

Filters are often fitted as part of the fuel oil cleaning arrangement. Which one of the following best describes why duplex (double) filter units are normally fitted?

- **To allow cleaning without interrupting the fuel supply.**
- To allow automatic cleaning of the elements.
- To double the cleaning effect.
- To have filters with different gauge mesh in operation at the same time.

Fuel oil treatment systems sometimes include ultrasonic devices. Which one of the following best describes how these devices improve the condition of the fuel?

- **They generate high energy vibrations which break down contaminant particles and heavy molecules.**
- They generate electrical pulses which break down contaminant particles and heavy molecules.
- They generate magnetic fields which break down contaminant particles and heavy molecules.
- They grind down contaminant particles and heavy molecules.

In which one of the following would you expect to find the bunkering procedures for the ship that you are sailing on?

- **Safety Management Manual**
- MARPOL
- Planned Maintenance System (PMS)
- SOLAS

MARPOL Annex I deals with the prevention of which of the following forms of marine pollution from ships?

- **Pollution by oil**
- Pollution by garbage
- Pollution by noxious liquid substances in bulk
- Pollution by sewage

MARPOL Annex I requires new ships with a total bunker capacity greater than a certain limit to have fuel oil tanks positioned so that no part of the tank is formed by the outer hull plating. What is the limit?

- **600 m^3**
- 150 m^3
- 2500 m^3
- 5000 m^3

MARPOL Annex VI deals with the prevention of which one of the following forms of pollution from ships?

- **Air pollution due to exhaust emissions.**
- Environmental pollution by garbage.
- Environmental pollution by harmful substances carried in packaged form.
- Marine pollution by oil.

Mechanical homogenisers are sometimes fitted as part of the fuel oil treatment arrangement. Which one of the following options given best describes the effect of an homogeniser on fuel oil?

- **Grinds solid particles down in size.**
- Increases the density.
- Reduces the density.
- Separate solids and water from the fuel.

Safe operation and pollution prevention procedures should be contained in the ships Safety Management Manual as part of the company's Safety Management System. Which legislation requires this?

- **ISM Code**
- COLREGS
- FSS Code
- MARPOL

Ships are required to take a bunker oil sample for MARPOL Annex VI purposes. Which of the following options gives the time that this sample must be retained onboard?

- **At least 12 months.**
- At least 3 months.
- At least 3 years.
- Until all of the bunkers have been used.

Some centrifuges used for oil have a solenoid valve in the water outlet which is controlled by a water sensor in the oil outlet. What is this type of centrifuge used for?

- **Treating high density residual fuel oils.**
- Treating distillate fuel oils.
- Treating fuels with minimum water content.
- Treating lubricating oil.

Suction strainers are fitted to most fuel oil treatment systems. Which one of the following best describes the main purpose of a suction strainer?

- **To protect the pumps from large solid particles.**
- To improve the flow rate to the centrifuges.
- To protect the pumps from any water in the fuel.
- To remove all of the solids in the fuel.

Taking samples of oil when bunkering is necessary for a number of reasons. Which one of the following is the recommended manner of obtaining a representative sample?

- **Fill a sample container at ship's manifold by a continuous drip method.**
- Directly from the supplier from shoreside/barge tanks.
- Fill a sample container at ship's manifold at the end of bunkering.
- Fill a sample container at ship's manifold at the start of bunkering.

The majority of ships are require to have a Shipboard Oil Pollution Emergency Plan (SOPEP) while some others are required to have a Shipboard Marine Pollution Emergency Plan (SMPEP).

Which of the following best describes the kinds of ship that require an SMPEP?

- **Ships carrying MARPOL Annex II cargoes**
- Bunker barges
- Gas carriers
- Ships carrying MARPOL Annex I cargoes

The number of fuel oil tanks in a ship is decided upon by considering a number of factors. Which one of the options is one of the most important factors which would be taken into consideration?

- **Segregation of fuel types**
- Cargo carrying capacity.
- Fuel transfer pump capabilities.
- Tank vent positions.

There are various health and environmental hazards associated with handling and using fuel oil, including bunkering operations. Where would you expect to find the information relating to these hazards?

- **The Material Safety Data Sheets (MSDS).**
- The bunker receipt.
- The engine manufacturers fuel specification.
- The ISO 8217 fuel standards tables for residual and distillate fuel.

Water/fuel oil emulsions are being used in some diesel engines. Which one of the following best describes the purpose of these emulsions?

- **To reduce harmful exhaust emissions.**
- To help to clean the fuel system.
- To help to lubricate the engine cylinders.
- To reduce the fuel consumption.

When a centrifuge is set up as a fuel oil clarifier it normally only separates solids from the dirty fuel. How is the set up of a clarifier different to that for a purifier?

- **The water outlet is closed off by the top cover of the disc stack.**
- The disc stack has no distribution holes.
- The gravity discs with the largest diameter holes are fitted.
- The water outlet is closed off by a solenoid valve.

When centrifuges are set up as fuel oil purifiers they often have gravity discs or gravity plugs fitted. How does fitting different sizes of these allow fuels of different densities to be treated?

- **They control the position of the interface.**
- They alter the flow rate through the purifier.
- They alter the height of the disc stack.
- They alter the position of the oil outlet.

When checking the quantity of liquid in a tank either a sounding or ullage may be taken to calculate the amount. Which one of the following best describes the term “ullage”?

- **The distance from the top of the sounding pipe to the surface of the oil.**
- The depth of the oil in the tank.
- The distance from the surface of the oil to the top of the tank.
- The reading obtained from a remote contents gauge.

When used in relation to fuel oil, the term instability refers to which one of the following situations?

- **Possibility that fuel components will separate during storage.**
- Possibility that the fuel will burn unevenly during combustion.
- Possibility that the fuel will require additional heating for transfer.
- The fuel is a blend of two or more grades.

Which of the following options best completes the following statement?

The main reason for heating fuel oil before it enters a purifier is to . . .

- **increase the density difference between fuel and water.**
- decrease the density difference between fuel and water.
- decrease the viscosity difference between fuel and water.
- increase the viscosity difference between fuel and water.

Which of the following options best describes the term "incompatible" when applied to fuel oil?

- **The fuel become unstable and form sludges when mixed with another fuel.**
- The components of a fuel separate out over time.
- The fuel does not match the ISO 8217 specification.
- The fuel is not suitable for the machinery onboard.

Which of the following options describes a feature required where sight glasses are used to show fuel oil tank contents?

- **Self closing cocks or valves.**
- Screw caps.
- Seamless tubular glass.
- Test cocks.

Which of the following options gives the maximum temperature allowed for the heating medium in a fuel oil tank?

- **220° Celsius.**
- 100° Celsius.
- 150° Celsius.
- 180° Celsius.

Which of the following options gives the minimum flashpoint for marine fuel oils?

- **60° C**
- 100° C
- 40° C
- 80° C

Which of the following options would you describe as being a double bottom tank?

- **A tank formed between the hull bottom plating and the inner floor plating.**
- A tank where the hull plating is double thickness.
- A tank which can be used for two different purposes.
- A tank with a void space in the bottom.

Which of the options best completes the following statement?

It is recommended that fuel oils from different supplies should not be mixed together in a tank when bunkering in case they . . .

- **are incompatible**
- are contaminated
- are unstable
- contain water

Which one of the fittings listed would you expect to find on a fuel oil tank air vent positioned on an open weather deck?

- **Vent head with float type shut off valve or similar.**
- Remote closing valve.
- Self closing cock.
- Test cock.

Which one of the following best describes how water is removed from fuel oil when static filter modules are used instead of centrifuges?

- **They contain coalescer elements.**
- They contain edge filter elements.
- They contain fine mesh elements.
- They contain paper elements.

Which one of the following best describes the difference made to a ship's fuel oil requirements if it enters a SOx Emission Control Area, SECA?

- **The maximum sulphur content of any fuel burnt will be lower.**
- Combustion changes will cause an increase in fuel consumption.
- The minimum nitrogen content of any fuel burnt will be higher.
- The minimum sulphur content of any fuel burnt will be higher.

Which one of the following describes the normal way of removing accumulated water from a fuel oil settling tank?

- **Regular draining using the drain cock.**
- During manual cleaning.
- With a centrifuge.
- With a water pump.

Which one of the following is the best method for onboard testing of fuel oil density?

- **A hydrometer.**
- A falling ball viscometer.
- Flostick.
- Spot test.

Which one of the following options best describes the conditions under which microbial contamination of distillate fuel is most likely to occur?

- **Water is present in the fuel.**
- Oxygen is not present in the fuel.
- Oxygen is present in the fuel.
- The fuel contains sulphur.

Which one of the following options describes the aim of legislative changes that now require many ships to have fuel oil tanks sited away from the ship's side and bottom?

- **Reduce pollution in the event of collision or grounding.**
- Increase the amount of fuel that can be carried.
- Increase the ballast capacity of ships.
- Reduce the amount of cargo that can be carried.

Which one of the following options describes when remote closing valves are required to be fitted in fuel oil tank outlet lines?

- **When the fuel oil tank is in the same space as the machinery it supplies.**
- Only required for fuel oil service tanks.
- Only required for fuel oil settling tanks.
- Required for all fuel oil tanks.

Which one of the following options is a fitting that will always be found on fuel oil service and settling tanks?

- **Self closing drain valve or cock.**
- High and low level alarms.
- Remote reading contents gauge.
- Sight glass.

Which one of the following options must be fitted on an air vent for a fuel oil tank?

- **Flame screen.**
- Filter.
- Pressure indicator.
- Self closing cock.

Which one of the following would be the normal way to remove microbial contamination from fuel oil?

- **Treatment with biocides.**
- Regular draining of tanks.
- Treatment with combustion improvers.
- Treatment with fuel stabilisers.

0012 – Tank I

Version 2.0 (1263)

A Material Safety Data Sheet contains specific information relating to a substance. Which regulations cover the requirement for MSDS to be available for cargo oil and bunker oil carried onboard an oil tanker?

- **IMO Resolution Marine Safety Circular (MSC).**
- The International Convention for Safety of the Life at Sea, (SOLAS).
- The International Labour Convention (ILO).
- The MARPOL annexe IV Regulations.

After an oil tanker has completed loading operations and the cargo documents have been cleared, the ship's Master will be given permission to leave the port. What is the term normally used for this permission?

- **Port clearance.**
- Customs clearance.
- Port dues paid, vessel free to depart.
- Vessel's departure permission.

All ballast handling must be recorded in dedicated forms. Where are these forms usually found onboard a crude oil tanker?

- **In the Water Ballast Management Plan.**
- In the company's TQM system.
- In the OPA 90 VRP.
- In the SOPEP.

All oil tankers are required to complete a pre-arrival check list. How are negative answers in the pre-arrival check list handled?

- **They should be reported to the Ship Owner/Operator.**
- The Master calls the Chief Officer and Chief Eng. for a meeting.
- The negative answer doesn't need to be reported it is just entered in the weekly work program.
- They should be reported to the owner's agent at discharge port.

All tankers have an approved Loading Manual on board. Which body approves the Loading Manual?

- **The classification Society to which the ship is classed.**
- The National Maritime Authority.
- The ship owner's/Operator's Technical Department.
- The ship's building yard.

Ballast water carried on board oil tankers is governed by stringent regulations. Which one of the given options includes the regulations covering ballast water for crude oil tankers?

- **IMO Resolutions, MARPOL.**
- International Convention on Load Lines for Ships.
- Oil Companies International Marine Forum (OCIMF) guidelines.
- The International Convention for the Safety of Life at Sea.

Because of its nature, operation of oil tankers poses a risk to personnel and the environment. Which of the given options is the main source for guidelines and recommendations for safe operation of oil tankers?

- **International Safety Guide for Oil Tankers and Terminals (ISGOTT)**
- IMO MARPOL Annex I
- The International Convention on Safety of Life at Sea (SOLAS)
- The Oil Companies International Marine Forum (OCIMF)

Crude oil tankers often load the next cargo on top of the residue or slops from the previous cargo Who decides if the vessel is to Load on Top of the slops?

- **The charterers.**
- The cargo owner.
- The cargo receiver.
- The ship owner or operator.

During cargo loading and discharge of an oil tanker the cargo hoses or loading arms must be connected. Is connection of cargo hoses (arms) part of the vessel's risk assessment?

- **Yes, it is always part of the vessel's risk assessment.**
- No, it is covered by the terminal risk assessment.
- Yes, but only when the ship's own hoses are being used.
- Yes, but only when visiting a new port.

It is required that the condition of the ballast tanks on a crude oil tanker are recorded following inspection. Where are the results of these inspections normally recorded?

- **In the vessel's Planned Maintenance System (PMS).**
- In the Chief Engineer's files.
- In the Chief Officer's files.
- In the owner's or operator's Vetting department.

On large crude oil tankers, why is it good practice to have the vessel as close to even keel as possible and with zero list when topping off the tanks?

- **In order to reach the correct quantity in the tanks.**
- In order to assist the port authorities to check the mid ship loading marks.
- In order to read the draught marks correct.
- The SOLAS regulations require no list when a vessel leaves port.

Once the vessel has been cleared by the quarantine and customs authorities, the independent cargo surveyors may board the vessel. Who are the cargo surveyors representing?

- **The surveyors are representing the shippers and cargo owners or charterers.**
- The cargo surveyors are representing the National Customs Office.
- The cargo surveyors are representing the ship owner and the cargo owner.
- The cargo surveyors are representing the vessel's P & I Club.

Prior to commencement of loading an oil tanker the independent cargo surveyors carry out a tank survey. What do they survey and calculate?

- **The on board quantity, OBQ.**
- The total oil and water remaining on board.
- The total slop and bunker on board.
- The vessel's experience factor.

Some crude oil loading ports operate an early departure procedure, EDP. Why is EDP used in some ports?

- **In order to make a quick turn-around and not cause delay at the terminal.**
- Because of a comprehensive and time-consuming handling of cargo documents.
- The charterers want to check the figures before the Bills of Lading are issued.
- Too expensive to have the vessel waiting for the cargo documents.

The cargo figures for the official Bill of Lading are determined following completion of loading operations for a crude oil tanker. Which figures are the figures for the Bill of Lading initially based on?

- **The terminal (shore) figures.**
- The figures given by the cargo owners.
- The ship's figures calculated by the cargo surveyors.
- The ship's figures calculated by the Chief Officer.

The duty deck officer is required to complete all details of cargo operations in the 'time sheet'. Is the 'time sheet' normally considered to be part of the official cargo documentation?

- **Yes, the time sheet is considered to be part of the official documentation.**
- No, it is only for the ship owner's or operator's internal use.
- No, it is only required by the agent.
- No, it is only used for calculating crew overtime.

The laden voyage is often used as an opportunity to carry out internal inspection of ballast tanks of a crude oil tanker. What is the normal frequency for inspection of ballast tanks on a crude oil tanker?

- **Annually.**
- 24 months.
- 36 months.
- Every 6 months.

The requirements for reporting of oil spills, and situations that may lead to an oil spill are strictly regulated. Where, onboard ship, are the details of these reporting procedures normally found?

- **In the Shipboard Oil Pollution Emergency Plan (SOPEP).**
- In the MARPOL Annex IV regulations.
- In the Oil Companies International Marine Forum (OCIMF) Guidelines.
- In the vessel's Damage Stability Manual.

What are the alternatives if the charterers of a crude oil tanker do not wish to Load on Top of the cargo residues or slops from the previous cargo? Select all of the correct options from those given.

- **Segregate the slop and no freight will be paid to the ship owner on the deadweight used.**
- **The charterers may require the slop to be discharged prior to loading.**
- **The ship owner/operator may sell the residues and de-slop the vessel prior to loading.**
- Distribute the slop into all cargo tanks prior arrival loading port.

What is the basis for making a loading plan on an oil tanker?

- **The loading order.**
- The loading rate.
- The reception facilities in the discharge port.
- The vessel's total loading capacity.

When the Master of a VLCC receives the following instruction: "After completion discharge at US Gulf, proceed to Fujairah for order". What kind of order is this?

- **Voyage order.**
- Basis for a complete passage plan.
- Loading order.
- Preliminary loading order.

When the Master on a ULCC receives following instruction: "After completion discharge at US Gulf, proceed to Bonny River, Nigeria, and load full cargo of Bonny Light crude. Disport intention US Gulf or NW Europe". Who issues this order? (List all alternatives)

- **The charterers.**
- **The operator**
- **The ship owner.**
- The cargo owners
- The shippers.

When the Master on a VLCC receives following instruction: "After completion discharge at Rotterdam, proceed to AG via Suez, intention full cargo Ras Tanura/Ju'aymah for NW Europe incl. Med.Sea. Pls. advise bunker requirements ex. barge Fujairah". What kind of order is this?

- **Voyage order.**
- Basis for a complete voyage passage plan.
- Loading order.
- Preliminary loading order.

When the Master on a VLCC receives following instruction: "On completion bunkering at Fujairah, proceed to Kharg Island and load full cargo Iranian Light crude. Disport Antifer, France. Pls. advise cargo intake". What kind of order is this?

- **Loading order.**
- Loading - and discharge order.
- Preliminary loading order.
- Voyage order.

When the vessel is securely moored alongside the discharge terminal, who should be the first to board the vessel?

- **The quarantine and customs officers.**
- The loading master and terminal representatives.
- The local agent.
- The terminal's security officer.

Where oil loading ports operate an early departure procedure, EDP, Who must agree to accept to operate with procedure prior to arrival at the load port?

- **The ship owner or operator and the charterers.**
- The Master.
- The Protection & Indemnity (P&I) Club.
- The shippers.

Who is responsible for developing and implementing an International Safety Management System on board?

- **The ship owner or operator.**
- The classification society to which the ship is classed.
- The Master.
- The National Maritime Authority.

Who would normally sign all of the cargo documentation prior to departure once a crude oil tanker is finished loading?

- **The ship's Master and the shippers.**
- The chief officer and the loading master.
- The ship's Master and the cargo surveyors.
- The ship's Master and the agent.

Why is it less important if crude oil tankers to receive a late loading order compared to chemical, gas and product tankers?

- **Crude oil tankers have less tank preparations prior to loading.**
- Crude oil tankers always carry the same cargo grade.
- Crude oil tankers always Load on Top
- Crude oil tankers have less ROB.

0013 – Tank II

Version 2.0 (2709)

A discharge order may change during the vessel's voyage. What is the usual reason for such a change to the discharge orders of an oil tanker?

- **The cargo is sold during the voyage.**
- Congestion at discharge port.
- The cargo receivers are not able to pay for the cargo.
- The cargo receivers do not accept the cargo quality specifications.

According to the MARPOL Annex I regulations, the cargo tanks of a crude oil tanker need to be crude oil washed for sludge control. What are the requirements in relation to crude oil washing of cargo tanks?

- **No tank need to be crude oil washed more than once in every four months.**
- All tanks to be crude oil washed annually.
- One quarter of the total number of tanks to be crude oil washed every discharge.
- The slop tanks and 4 other tanks to be crude oil washed every discharge.

Crude oil washing was made mandatory for new tankers by the 1978 Protocol to the MARPOL Convention. What was the main reason for this requirement?

- **To prevent oil pollution.**
- To avoid the need for sludge removal by hand.
- To increase the amount of cargo discharged.
- To reduce the need for water washing.

During discharge and COW operations of a crude oil tanker it is important to monitor the Inert Gas (IG) quality and pressure. What is the procedure if the IG system fails during such operation?

- **Stop the discharge and COW.**
- Continue the operation and rectify the IG failure.
- Continue the operation as the P/V valves will open if the tanks go into an under-pressure condition.
- Stop the COW operation and reduce the discharge rate.

Oil tankers are required to complete pre-arrival check list before entering port. How are negative answers in the pre-arrival check list handled?

- **Reported to the Ship Owner/Operator.**
- Reported to the owner's agent at discharge port.
- The Master calls the Chief Officer and Chief Eng. for a meeting.
- The negative answer is entered in the weekly work program.

Once an oil tanker is cleared by the quarantine and customs authorities the independent cargo surveyors can normally board the vessel. Who do the cargo surveyors usually represent?

- **The surveyors are representing the cargo receivers and cargo owners or charterers.**
- The cargo surveyors are representing the National Customs Office.
- The cargo surveyors are representing the ship owner and cargo owner.
- The cargo surveyors are representing the vessel's P & I Club.

Once an oil tanker is securely moored alongside the discharge terminal, who is normally the first to board the vessel?

- **Quarantine and customs officers.**
- The loading Master and terminal representatives.
- The local agent.
- The terminal's security officer.

Operating oil tankers poses a risk to personnel and the environment. Which of the given options is considered as the main source of information on guidelines and recommendations for safe operation of oil tankers?

- **International Safety Guide for Oil Tankers and Terminals (ISGOTT)**
- IMO MARPOL Annex I
- The International Convention on Safety of Life at Sea (SOLAS)
- The Oil Companies International Marine Forum (OCIMF)

Prior to arrival at a discharge port, a crude oil tanker will require permission to carry out COW operations during discharge. Who would normally need to give permission to carry out COW during discharge operations?

- **The port authority and terminal prior to arrival.**
- The cargo receiver gives permission as he is focusing on the best out-turn of the cargo.
- The charterer.
- The Ship Owner/Operator.

The IG produced on the crude oil tankers will normally contain some oxygen. What are the requirements for the Oxygen content in the IG during discharge, cow and tank cleaning?

- **5% in the IG supply and 8% in the tank atmosphere.**
- 3% in the IG supply and 5% in the tank atmosphere.
- 5% in the IG supply and 5% in the tank atmosphere.
- 8% in the IG supply and 15% in the tank atmosphere.

The Ship – Shore meeting deals with all aspects related to the discharge operation of an oil tanker. What are the main issues discussed in this meeting? Select all of the relevant options.

- **The discharge plan.**
- **The environment protection.**
- **The safety and security.**
- **The ship/shore communication.**
- The Class Annual Survey.
- The crew change.

The cargo tanks on a crude oil tanker are required to be inspected internally. How often must internal inspection of the cargo tanks normally be carried out?

- **At intervals not exceeding 30 months.**
- At intervals not exceeding 12 months.
- During the annual survey by Class.
- Prior to each dry docking.

The discharge plan of an oil tanker will normally include ballast operations. According to IMO regulations all ballast handling is required to be recorded. Where are ballast handling operations normally recorded onboard an oil tanker?

- **In the Ballast Water Management Plan's record forms.**
- In the Deck Log Book.
- In the Oil Record Book Part II
- In the Pump Log.

What is it that independent cargo surveyors normally measure and calculate prior to commencement of discharge operations by an oil tanker?

- **The cargo figures on arrival.**
- The Onboard quantity (O.B.Q.).
- The remaining onboard quantity (R.O.B.).
- The total ship figures on arrival.

What was the size limitation when the MARPOL Annex I made COW mandatory for new crude oil tankers?

- **Tankers of 20.000 tons deadweight and above.**
- Tankers of 10.000 tons deadweight and above.
- Tankers of 50.000 tons deadweight and above.
- Tankers of 70.000 tons deadweight and above.

When carrying out COW operations onboard a crude oil tanker the regulations require that certain check lists are used. Where onboard are these check lists available and how many of them are required?

- **In the COW Manual and 3 different check lists are required.**
- In the company's ISM/TQM system and 3 different check lists are required.
- In the International Safety Guide for Oil Tankers and Terminals (ISGOTT) and 3 check lists are required.
- In the SOPEP Manual and 2 check lists are required.

When entering a cargo tank, or enclosed space, it is by regulations required to record such entry. Where should the entry into the tank be recorded?

- **The Deck Log Book.**
- In the PMS tank inspection records.
- The Oil Record Book Part I.
- The Oil Record Book Part II.

When the Loading Master orders the discharge to start at slow rate, a certain amount of cargo has to be discharged before full rate is ordered. What is this discharge sequence called?

- **Line displacement.**
- Flow control.
- Line test.
- Shore clearing.

When the Master of a VLCC receives following instruction: "After completion loading at Ras Tanura, proceed to dis.port NW Europe incl. Med.Sea and Scandinavia, via Cape". What kind of order is this?

- **Voyage order.**
- Basis for a complete voyage plan.
- Discharge order.
- Preliminary discharge order.

When the Master receives instruction to proceed to Rotterdam and discharge all cargo at a port such as Europoort Maatschap Tankfarm, who would normally issue this instruction?

- **The ship owner/operator or charterers.**
- The cargo owner.
- The cargo receiver at Europoort.
- The charterer's agent in Rotterdam.
- The owner's agent in Rotterdam.

When the discharge of an oil tanker is completed, the cargo surveyors start surveying the cargo tanks. What action would be expected to be taken if an unacceptable quantity of cargo oil remains on board?

- **The cargo surveyors issue a Letter of Protest on behalf of the cargo receivers and charterers.**
- The cargo surveyors request stripping of the tanks to be resumed.
- The cargo surveyors request the terminal to assist with a new tank survey.
- The vessel will be detained until all cargo oil is discharged.

Which activity is considered to be the most critical on an oil tanker?

- **Gas freeing and entering a cargo tank.**
- Crude oil washing (COW).
- Hot work in the engine room.
- Tank cleaning.

0014 – Vessel Struct. Cond.

Version 3.0 (1447)

For a loaded, simply supported beam which one of the following statements about compressive and tensile forces is most accurate?

- **The load side is under compression and the opposite side is under tension**
- Both the load side and the opposite side are under compression
- Both the load side and the opposite side are under tension
- The load side is under tension and the opposite side is under compression

For a loaded, simply supported beam with uneven loading, which one of the following statements about the loading of the supports is most accurate?

- **If the loading is uneven, the support nearest to the load has the greatest force**
- Even if the loading is uneven, each support carries equal force
- Even if the loading is uneven, the reactions at each support will be zero.
- If the loading is uneven, the support furthest from the load has the greatest force

For a loaded, simply supported symmetrical beam which one of the following statements about the resulting bending stresses is most accurate?

- **If the beam is symmetrical, the bending stresses are highest at the outside surfaces and zero along the neutral**
- If the beam is symmetrical, the bending stresses are the same throughout any cross-section of the beam
- If the beam is symmetrical, the bending stresses are zero at the outside surfaces and highest along the neutral
- If the beam is symmetrical, the bending stresses are zero throughout any cross-section of the beam

For an evenly loaded, simply supported beam which one of the following statements about the bending stress is most accurate?

- **The bending stress is greatest half way along the length of the beam**
- The bending stress is greatest at each end
- The bending stress is zero half way along the length of the beam
- The ends of the beam cannot move

When the distance between wave crests is equal to the length of the ship's hull, which one of the following statements is most accurate?

- **When the wave crests are fore and aft, they will make the ship tend to sag**
- The smallest bending moment caused by the wave motion will be when either the peak or the trough of the wave is at mid-ships
- There will be no overall effect on the hull
- When the wave crests are fore and aft, they will make the ship tend to hog

Which one of the following statements about bending of the hull is most accurate?

- **When the hull is hogging, the deck will be in tension whilst the keel will be in compression**
- When the hull is hogging, the deck will be in compression whilst the keel will be in tension
- When the hull is sagging, the deck and keel will be under tension
- When the hull is sagging, the deck will be under tension whilst the keel will be in compression

Which one of the following statements about corrosion is most accurate?

- **Corrosion can greatly reduce the strength of the ship**
- Corrosion of the hull is relatively unimportant, as there is an allowance for this at the design stage
- No permanent loss of hull strength occurs because of operation in heavy weather
- The strength of the hull always decreases rapidly in service

Which one of the following statements about forces on the hull is most accurate?

- **Shear forces on the hull will tend to be high if one hold is fully loaded and the next is empty**
- Shear forces on the hull will be tend to be high if all holds are evenly loaded
- Shear forces on the hull will tend to be high if all holds are empty
- Shear forces on the hull will tend to be low if one hold is fully loaded and the next is empty

Which one of the following statements about forces on the ship's hull is most accurate?

- **Panting is the in and out movement of the shell plating due to the variation in water pressure on the shell plating.**
- Panting occurs when the vessel is lifted by the sea and then slams back down on the water
- Pounding damage is most likely at mid-ships
- Pounding is most likely to occur when the ship is fully loaded

Which one of the following statements about hull inspections is most accurate?

- **Ship's crew should regularly inspect the cargo holds, hatch covers and ballast tanks with a view to detecting damage and defects**
- Ship's crew are not allowed to inspect the cargo holds, hatch covers and ballast tanks
- Ship's crew have insufficient training to inspect the cargo holds, hatch covers and ballast tanks with a view to detecting damage and defects.
- There is no need to detect damage and defects between surveys

Which one of the following statements about hull strength is most accurate?

- **A ship regularly operating in heavy weather is subject to many stress reversals which lower the strength of the hull**
- A ship regularly operating in heavy weather will remain as strong as when new if properly maintained
- Fatigue cracking is never related to corrosion
- Small Fatigue cracks will normally be okay until the next dry-dock or survey

Which one of the following statements about sloshing of fluid in tanks is most accurate?

- **Excessive forces from fluids sloshing in tanks can be prevented by not having them between 20% and 90% full**
- Excessive forces from fluids sloshing in tanks can be prevented by having them between 20% and 90% full
- Excessive forces from fluids sloshing in tanks can be prevented by having them only half full
- Excessive forces from fluids sloshing in tanks can only be prevented by having them completely full

Which one of the following statements about the loads on the hull is most accurate?

- **Reducing the dynamic loads on the hull by slowing down or altering course in heavy weather is effective in both increasing safety and maintaining the strength of the vessel**
- Little can be done in heavy weather to reduce the dynamic loads on the hull
- Reducing the dynamic loads on the hull by slowing down or altering course in heavy weather should only be considered in an emergency for safety reasons
- Static loads on the hull can be reduced by slowing down or altering course

Which one of the following statements about the strength of the hull is most accurate?

- **Factors such as operation in rough weather and damage from cargo work can greatly reduce the strength of the hull**
- Incidental damage from cargo work has no effect on the strength of the hull
- The strength of the ship always decreases rapidly in service
- The strength of the ship never changes much after the building stage

Which one of these IMO publications requires that Bulk Carriers have a loading plan?

- **SOLAS**
- COLREG
- Load line
- MARPOL

With reference to crack formation in the hull structure, which one of the following statements is most accurate?

- **Cracks can occur due to fatigue, corrosion, or sharp changes of section**
- Cracks in hatch coamings are not a significant problem
- Cracks occur when the material can no longer withstand compressive forces
- Small cracks often occur and are not a significant problem

With reference to cracking of the hull structure, which one of the following statements is most accurate?

- **Cracks occur when the material can no longer withstand tensile forces**
- Corrosion does not affect whether or not cracks occur
- Cracks occur when the material can no longer withstand compressive forces
- Fatigue does not affect whether or not cracks occur

With reference to looking for defects in the hull structure around hatch openings, which one of the following statements is most accurate?

- **The hatch corners and hatch coaming brackets need close attention as they are prone to cracking**
- Hatch openings do not form part of the structure and therefore do not need close inspection
- The deck plating mid way along the breadth of the hatch needs close attention as it is prone to cracking
- The deck plating mid way along the length of the hatch needs close attention as it is prone to cracking

With reference to looking for defects in the hull structure, which one of the following statements is most accurate?

- **Indentations in the side and floor plating of Bulk carriers from cargo work can lead to corrosion and cracking**
- Double bottom tanks are not usually subject to corrosion
- Indentations in the side and floor plating of Bulk carriers from cargo work are usually of no great importance
- Large hatch openings on container and bulk vessels do not form part of the structure and therefore do not need close inspection

With reference to shear forces on a ship's hull, which one of the following statements is most accurate?

- **Shear forces cause further stress on the hull in addition to that caused by bending**
- Shear forces are not significant compared to the stress on the hull caused by bending
- Shear forces cancel out some of the stress on the hull caused by bending
- Shear stress always opposes the stress on the hull caused by bending

With reference to static and dynamic forces acting on the hull, which one of the following statements is most accurate?

- **High static forces, for instance due to poor cargo loading, mean that the vessel is less safe when operating in heavy weather**
- Dynamic loads on the vessel are insignificant compared to the weight of cargo and ballast
- Provided the maximum static loading for the hull has not been exceeded, the vessel should be safe in any weather conditions
- Static forces on the vessel are insignificant compared to sea forces

With reference to stiffening of side shell plating, which one of the following statements is most accurate?

- **The middle and ends of stiffeners and girders will be subject to the highest bending stress**
- Bending stress is equal along the entire length of stiffeners and girders
- The ends of stiffeners and girders will be subject to zero bending stress
- The middle and ends of stiffeners and girders will be subject to zero bending stress

With reference to the forces acting on the hull, which one of the following statements is most accurate?

- **Panting, pounding, and sloshing of liquids in tanks are examples of dynamic forces**
- Dynamic forces are always small compared to static forces
- Static forces tend to cancel out dynamic forces
- The weight of fuel, cargo, fresh water and ballast are examples of dynamic forces

With reference to the hull structure, which one of the following statements is most accurate?

- **Resistance to buckling depends on the length and cross section of the stiffener or girder**
- Buckling can occur in tension or compression
- Resistance to buckling depends only on the cross section of the stiffener or girder
- Resistance to buckling depends only on the length of the stiffener or girder

With reference to the hull structure, which one of the following statements is most accurate?

- **Fatigue is the reduction in strength of a material due to repeated cycles of stress**
- Fatigue cracking is most likely when the ship is new
- Fatigue is the reduction in strength of a material due to constant high load
- If protective coatings are maintained, fatigue will never affect a ship

With reference to the stress distribution along the length of a ship's hull subjected to bending, which one of the following statements is most accurate?

- **If the ship is evenly loaded, the bending stresses are at a maximum at mid-length**
- If the ship is evenly loaded, the bending stresses are at a maximum fore and aft
- If the ship is evenly loaded, the bending stresses are at a minimum at mid-length
- If the ship is evenly loaded, the bending stresses are equal along the length of the ship

With reference to the stress distribution of a ship's hull subjected to bending, which one of the following statements is most accurate?

- **The bending stresses will be maximum at the deck and bottom plating and reduce towards the neutral axis at approximately the mid-point of the side plating**
- The bending stresses will be equal at the deck, bottom plating and the neutral axis at approximately the mid-point of the side plating
- The bending stresses will be maximum at the deck plating and reduce to zero at the bottom plating
- The bending stresses will be zero at the deck and bottom plating and increase towards the neutral axis at approximately the mid-point of the side plating

With reference to water on the deck of a ship, which one of the following statements is most accurate?

- **Sea water coming over the deck will cause downward forces on the ship's structure and is most likely to affect the fore end of the vessel**
- Sea water coming over the deck causes additional buoyancy forces on the ship's structure
- Sea water coming over the deck has little overall effect on large vessels
- Sea water coming over the deck will cause downward forces on the ship's structure and is most likely to affect the aft end of the vessel

With regard to looking for defects in the hull structure, which one of the following statements is most accurate?

- **Connections between parts of the structure, or changes of section, are common areas to find defects**
- Connections between parts of the structure are so strong that they rarely fail
- Loss of protective coating above welds is fairly common and of no immediate concern
- Small cracks in shell plating above the waterline are of no concern

With regard to shear forces acting on the hull, which one of the following statements is most accurate?

- **Shear force is created when there are two parallel, opposite forces, like gravity and buoyancy**
- Shear force is created when there are two parallel forces in the same direction, like gravity and buoyancy
- Shear force refers to the horizontal force at the aft part of the vessel
- Shear force refers to the horizontal force at the forward part of the vessel

With regard to the hulls of Ro-Ro's and car ferries, which one of the following statements is most accurate?

- **The bow visor and inner door are considered the most vital areas of the hull for inspection on car ferries and Ro-Ro vessels**
- The bottom plating is considered the most vital areas of the hull for inspection on car ferries and Ro-Ro vessels
- The bow visor and inner door are above the waterline and therefore less vital than other areas of the hull for inspection on car ferries and Ro-Ro vessels
- The deck plating is considered the most vital areas of the hull for inspection on car ferries and Ro-Ro vessels

With respect to bending stresses on the ship's hull, which one of the following statements is most accurate?

- **Bending stresses will normally be highest at mid length and at the deck and bottom plating levels**
- Bending stresses will normally be highest at mid length and half way between the deck and bottom plating levels
- Bending stresses will normally be highest fore and aft and at the deck and bottom plating levels
- Bending stresses will normally be highest in way of the engine room and half way between the deck and bottom plating levels

With respect to flat bar cross sections of stiffeners used for the ship's hull, which one of the following statements is most accurate?

- **Flat bar is relatively poor at resisting buckling**
- Flat bar is difficult to apply a protective coating to
- Flat bar is relatively expensive to make
- Flat bar is relatively good at resisting buckling

With respect to framing types for the ship's hull, which one of the following statements is most accurate?

- **Combination framing uses both transverse and longitudinal framing**
- Combination framing is required on all new oil tankers
- Combination framing uses longitudinal frames and box girders
- Combination framing uses transverse frames and box girders

With respect to rolling causing stress on the ship's hull, which one of the following statements is most accurate?

- **Rolling mostly causes stresses in the transverse bulkheads**
- Rolling mostly causes stresses at the aft part of the ship
- Rolling mostly causes stresses at the forward part of the ship
- Rolling mostly causes stresses in the longitudinal stiffeners

With respect to the cross sections of stiffeners used for the ship's hull, which one of the following statements is most accurate?

- **T-profile performs better in most situations**
- Angle bar performs better in most situations
- Flat bar performs better in most situations
- It is difficult to apply protective coatings to Bulb profiles

With respect to the ship's hull, which one of the following statements is most accurate?

- **The main purpose of the longitudinal stiffeners is to resist bending**
- Longitudinal stiffeners are only used in longer vessels
- Longitudinal stiffeners are only used in shorter vessels
- The main purpose of the transverse frames is to support the deck structure

With respect to the ship's hull, which one of the following statements is most accurate?

- **Longitudinal framing tends to be used for longer vessels**
- Longitudinal and transverse framing should never be used on the same ship
- Longitudinal framing cannot resist bending stresses
- Transverse framing tends to be used for longer vessels

0015 – Corrosion Protection I

Version 2.0 (0)

Oxygen is usually present where corrosion takes place. What would happen to steel in an area where no oxygen exists?

- **A different form of corrosion could take place**
- It will be protected against corrosion.
- It will not corrode
- Will corrode in the normal way

What are the most important parameters to be observed during application of a convertible coating system?

- **The over-coating interval being a product of time and steel temperature to be observed**
- The absolute number of days stated on over-coating intervals to be observed
- The over-coating interval being a product of time and air temperature to be observed
- To observe the weather condition during the period.

What are the most important things to monitor when inspecting coating conditions?

- **Inspect both intact and corroded areas and also inspect angles, corners, behind bars etc.**
- Inspect a typical area where you can access easily
- Inspect suspected areas only.
- Inspect the areas only where coating has broken down

What determines the rate at which the corrosion process of steel proceeds?

- **The conductivity of the electrolyte, ambient temperature, amount of oxygen present and the relative sizes of cathodic and anodic areas?**
- The amount of oxygen present and the ambient temperature?
- The Conductivity of the electrolyte.
- The relative sizes of cathodic and anodic areas and the conductivity of the electrolyte?

What is the key role that you carry out when evaluating steel condition?

- **You need to be able to assess the condition and be able to report that condition accurately (so that decisions can be made based on your report.)**
- You are there to approve the work done.
- You are there to decide the next repair action
- You are there to understand what has happened to the steel

What is the main single factor, which affects the cost of preparation and application to difficult vessel areas?

- **The cost of access to an area**
- The cost of the paint
- The specialised labour required
- The Weather condition

What should be observed during application of convertible coating systems in cold weather?

- **That sufficient curing time, above the minimum curing temperature of the coating, is allowed prior to application of the next coat.**
- That sufficient curing time is allowed prior to application of the next coat
- That the surface is dry
- The correct ambient temperature.

What type of corrosion on a coated deck-area would you most worry about?

- **3% scattered corrosion**
- 1% localised pitting corrosion
- 6% localised corrosion
- Localised damage corrosion

Would cargo remnants, e.g. coal mixed with water affect the corrosion rate?

- **The water in contact with the sulphur in the coal will form an active electrolyte, which will tend to accelerate the corrosion process.**
- There will be no corrosion.
- This would have no effect if the water present is fresh water
- This would have none or little effect on the corrosion rate.

0016 – Corrosion Protection II

Version 2.0 (0)

A large area of exposed deck is to be blasted to an agreed standard of SA2 before application of an epoxy paint system. In the vicinity of some pipes where difficult access means that the blasting is not able to remove all the corrosion. What do you do?

- **Discuss the various options with the contractor.**
- Accept the blasted areas and go ahead with painting.
- Reject the whole blasted areas until blasting is redone to the correct SA2 standard.

During a drydock paint application why only check dft's on spot blasted areas, how do you check the antifouling thickness?

- **Spot blasted areas are the only areas coated from the steel surface upwards - you control the antifouling thickness by controlling the distribution.**
- You can also check dft's on non blasted areas
- You check the wet film of each coat, at various parts of the hull.

Paint is applied to a deck in the morning. Each morning the air temperature is 7°C. In early afternoon the temperature of the air rises to 23°C. You are told that the maximum overcoating time for the paint coating is 36 hours at 20°C and 48 hours at 10°C. In this case, what maximum overcoating time do you allow?

- **You consult the paint suppliers representative**
- 24 hours
- 36 hours
- 48 hours
- You consult the Technical data sheet for the paint

The main contaminants in a Marine environment are SALT, CORROSION, MOISTURE, OIL & GREASE, DUST AND HULL FOULING.

What is the most efficient method of removing HULL FOULING?

- **High Pressure Water Washing.**
- Fan Jet Cleaning.
- Low Pressure High Volume Water Washing.
- Scraping to remove the fouling.

The wet film of paint, which is applied, bears a direct relationship to the final dry film thickness of the coating. Taking wet film measurements can be very important. Why is it important to take these measurements as quickly as possible?

- **To obtain as close a reading to the actual wet film as possible.**
- So that the painter is disturbed as little as possible.
- To avoid the fresh coat softening the coats underneath leading to possible wrong readings.

What is the minimum number of degrees centigrade the steel temperature must be above the dew point before paint application?

- **3°C**
- 1°C
- 2°C
- 4°C

What piece of equipment would be the most effective in removing corrosion from difficult corners or angled areas?

- **Needle gun**
- Chipping hammer
- Disc sander
- Scraper

When using Airless Spray equipment what are the 3 most important safety rules to observe?

- **Always wear the correct personal protective equipment**
- **Do not point the gun at anyone or pull trigger when bare hands or fingers are close to the spray outlet tip**
- **Do not snag the paint lines or drag them over sharp edges**
- Do not operate the equipment at excessively high paint pressures
- Do not use the airless spray unit when it is dirty

Which are the 3 factors which influence the length of time from finishing an antifouling coating to the flooding of the dry-dock?

- **Drying time**
- **Temperature**
- **Ventilation**
- Correct film thickness
- Number of coats applied

Which are the 3 factors, which help contribute to correct performance of a paint system.

- **Proper Application of Paint**
- **Proper Surface Preparation**
- **The Paint itself**
- That clean Airless spray equipment is used
- The amount of Time and Money spent on the paint and the application
- The Correct Intervals between coats of paint
- The Correct Specification

Which of the following statements best describes one of the main advantages of Hydroblasting with fresh water as opposed to Dry Abrasive Blasting?

- **As well as removing corrosion products it also removes salt contamination.**
- Only requires air power for operation.
- Produces a Surface Profile to aid adhesion of coatings to the surface.
- The prepared surfaces will flash rust.

Why is fresh water washing important before commencing paint application?

- **To remove soluble salts**
- To remove corrosion
- To remove dust & debris
- To remove oil & grease

Why should the use of long lines be avoided, if possible, during spray application?

- **Risk of pressure drops**
- Risk of blockages
- Risk of paint curing in the line
- Safety risks due to increased length

0017 – Steering Gear Operation

Version 5.0 (3653)

According to SOLAS, what special steering gear requirement must tankers above 10,000 tonnes meet that other ships do not require?

- **There must be oil leak detection, and steering capability must be regained within 45 seconds following the loss of one power actuating system**
- Steering capability must be regained within 60 seconds following an oil leak
- There must always be full duplication of components up to the rudder stock
- There must be two or more identical power units

According to the SOLAS rules the steering gear should be capable of being recharged with hydraulic oil from a fixed storage tank. What is the main reason for this?

- **To ensure that the steering gear can be rapidly recharged with clean hydraulic oil in the event of a leak from the system**
- Because a steady leakage of oil is to be expected from some steering gears and having a fixed storage tank reduces the time taken for topping up
- To cut down the fire risk and reduce the risk of spillages
- To ensure that the correct grade of oil is always used

According to the SOLAS rules, when must ships fitted with more than one steering gear power unit have at least two of these units running?

- **In all cases, in areas where navigation demands special caution**
- In areas where navigation demands special caution, unless each individual power unit can perform as a main steering gear
- In areas where navigation demands special caution, unless the system is arranged to change over automatically within 45 seconds of a failure to either power unit
- When entering or leaving port, or at the discretion of the Master

For an electro-hydraulic steering gear, if the stand by pump fails to start following a failure of the running pump, which one of the following best describes the likely cause?

- **The pump has not been set to standby or its power supply has been switched off**
- The starting current has caused overload
- There has been a significant loss of hydraulic fluid
- There is a valve shut somewhere in the system

For an electro-hydraulic steering gear, if there is a lot of movement of the wheel in hand steering mode before the steering gear responds, which one of the following best describes the likely causes?

- **Air in the hydraulic control system**
- Excessive wear in the hydraulic pumps
- Loss of one phase of the electrical supply to the pump motor
- Overheating of the hydraulic fluid

If there are no alarms, but the ship stops responding correctly to helm orders, which one of the following best describes the likely causes?

- **A failure in the remote rudder angle indication system**
- A leaking safety valve
- An electrical power failure in the control system
- An electrical power failure in the power actuating system

In the event of an oil leak on a steering gear fitted with an automatic change-over system, at what point should the system be automatically split into two?

- **When a main low level alarm operates on either hydraulic oil tank**
- At the same time as one of the pumps is automatically stopped
- The system will be split into two in the event of any alarm sounding
- When the final low level alarm on either hydraulic oil tank operates

Requirements for steering gears are given in which one of the following publications?

- **SOLAS**
- COLREG
- ISM Code
- MARPOL

The steering mode in which the rudder angle is pre-set manually and kept in position automatically, is called?

- **Follow up mode**
- Auto pilot mode
- Hand steering mode
- Non follow up mode

What is the most likely cause of the hydraulic lock alarm occurring on a steering gear system?

- **A failure in the control system such as a directional valve sticking open**
- A failure of one of the hydraulic pumps
- Failing to purge the system of air
- Having both hydraulic pumps running at the same time

What special requirement for steering gears is applied to any ship over 70,000 tonnes?

- **The main steering gear must have two or more identical power units**
- One of the steering gear motors must be supplied from the emergency switchboard
- There must be automatic change-over in the event of hydraulic leakage
- There must be oil leakage detection

What types of valves are actuated in order to move the rudder when local steering is being carried out?

- **Directional valves**
- Local connecting valves
- One way valves
- Safety valves

When a heavy sea strikes the rudder, which one of these statements best describes what will happen?

- **If the force is large enough, the hydraulic pressure in the steering gear will be high enough to operate the safety valve, and the rudder will move**
- The rudder is free to move, but the auto-pilot will bring the ship back on course
- The rudder is hydraulically locked in position and will not move
- The shape and lightweight design of the rudder means that wave forces have no effect

Where a steering gear is fitted with oil leak detection what would be the first alarm in the event of loss of oil?

- **Low level alarm in the hydraulic fluid tank**
- Hydraulic system power failure
- Hydraulic system separation
- Low level alarm in the hydraulic fluid tank together with stand by pump auto-start

Where are remote rudder angle indicators usually fitted?

- **At the bridge steering console and on the bridge wings**
- At the bridge steering console only
- In the cargo control room
- In the steering gear compartment only

Where is the changeover to local steering carried out?

- **The steering compartment**
- The bridge
- The engine control room or the bridge
- The steering compartment, but it must be acknowledged by the bridge before it can take effect

Which is the only one of the following NOT known to have caused moisture to have contaminated the hydraulic oil used for the steering gear?

- **A leaking rudderstock gland**
- Air in the hydraulic control system containing moisture
- Oil for topping up stored in open containers in the steering compartment
- Water collecting on top of hydraulic oil drums stored on deck

Which is the only one of the following that is NOT likely to cause overheating of a hydraulic steering gear?

- **Leakage of hydraulic fluid**
- Air in the main hydraulic system
- High ambient temperatures and blocked oil coolers
- Pumps running on no-load for long periods of time

Which of the following statements best describes the importance of the purity of the hydraulic oil used in steering gear?

- **Even small solid particles will cause wear, whilst moisture will cause corrosion**
- Provided the oil remains incompressible, the steering gear will continue to work correctly
- Provided the oil viscosity remains within acceptable limits, the steering gear will continue to work correctly
- Steering gear power actuators are very robust and therefore a degree of oil contamination can be tolerated

Which one of the following is NOT caused by air in a steering gear hydraulic system?

- **Rudder tends to return to mid-ships**
- Corrosion
- Erratic movement of the rudder
- Overheating

Which one of the following statements about steering gear is NOT correct?

- **A main steering gear must be able to move the rudder from 30° on either side to 30° on the other side in less than 28 seconds**
- A main steering gear must be able to move the rudder from 35° on either side to 35° on the other side
- An auxiliary steering gear must be able to move the rudder from 15° on either side to 15° on the other side
- Ships with more than one rudder do not need an auxiliary steering gear if they are navigable with one rudder

Which one of the following statements best describes the actions would you take if there is a leak of hydraulic fluid from a steering gear which does not have an automatic change-over system?

- **Isolate the leaking part of the steering gear by following the procedures on the emergency steering instructions**
- Immediately switch off all power units
- Immediately switch over to another power unit
- Top up the system oil until repairs can be made

Which one of the following statements best describes the arrangement of steering gear alarms?

- **Steering gear alarms can only be acknowledged on the bridge, but are indicated elsewhere**
- Steering gear alarms are only indicated and acknowledged in the steering compartment
- Steering gear alarms are only indicated and acknowledged on the bridge
- Steering gear alarms can be acknowledged on the bridge, or elsewhere

Which one of the following statements best describes the checks carried out during emergency steering drills?

- **All emergency arrangements must be tested including operation from the steering compartment and communication with the bridge.**
- Checks previously carried out prior to leaving port do not need to be repeated, but hand steering from the bridge must be carried out.
- In addition to the checks carried out prior to leaving port, hand steering from the bridge must be tested.
- The checks carried out are the same as those required prior to leaving port, but are more realistic because the ship is moving through the water.

Which one of the following statements gives one of the steering gear requirements for passenger ships?

- **Each power unit must be capable of moving the rudder from 35° on either side to 30° on the other side in not more than 28 seconds**
- The requirements for each steering gear remains the same, even if there is more than one rudder
- There must be oil leakage detection fitted
- There must be two identical power units which together can move the rudder from 35° on either side to 30° on the other side in not more than 28 seconds

Which one of the phrase given in the options most accurately completes the following statement? A leaking safety valve on a hydraulic steering system is most likely to cause which one of the following:

- **Slow rudder movement**
- A time delay before the rudder first starts to move
- Excessive hydraulic oil consumption
- Leakage of oil into drip trays in the steering flat

Which one of the phrase given in the options most accurately completes the following statement? According to the SOLAS rules, if a ship has more than one rudder and can be manoeuvred with only one rudder, then:

- **Only one main steering gear needs to be fitted to each rudder**
- An additional main or auxiliary steering gear is only required on each rudder for passenger ships of over 70,000 tonnes
- An additional main or auxiliary steering gear is only required on each rudder for tankers, chemical tankers and gas carriers above 10,000 tonnes
- Each rudder must be fitted with a main and auxiliary steering gear

Which one of the phrase given in the options most accurately completes the following statement? The safety valve on a steering gear hydraulic system is most likely to operate because:

- **A heavy sea has hit the rudder**
- An attempt has been made to put too much helm on, too quickly
- It has been opened as part of the normal operation of the steering gear
- The output pressure from the hydraulic pumps is too high

Which one of the phrase given in the options most accurately completes the following statement? The ships steering gear must normally be checked and tested by the ship's crew:

- **Within 12 hours before departure**
- Within 24 hours before departure
- Within 48 hours before departure
- Within 72 hours before departure

Which one of these statements about rudder carrier bearings is true?

- **A separate rudder carrier bearing is only required on ram type steering gears**
- A rudder carrier bearing is not required on rotary vane type steering gears
- A rudder carrier bearing is only required on ram type steering gears
- A rudder carrier bearing is only required on rotary vane type steering gears

Which one of these statements about the force developed by a hydraulic system is true?

- **Force equals pressure times area**
- Force equals pressure divided by area
- Force equals pressure plus area
- Pressure equals force times area

Which one of these statements about types of steering gear is true?

- **Higher pressures are needed for the ram type**
- A greater angle of possible rudder movement is available with ram type steering gears
- Higher pressures are needed for the rotary vane type
- The ram type takes up less space than the rotary vane type

Which one of these statements best describes the effect of using two steering gear hydraulic pumps instead of one?

- **The speed of rudder movement is greatly increased**
- The force on the rudder is greatly increased
- The speed of rudder movement and the force on the rudder are greatly increased
- There is little effect on the speed of movement or the force on the rudder, but reliability is increased

0018 – Cooling Systems

Version 2.0 (3071)

A transducer can be used in a cooling water temperature control system. What do you understand by the term transducer?

- **A device that measures a physical characteristic and generates different physical output.**
- A device to boost the controller output signal.
- A device to compare measured value and set point.
- A device to measure control valve travel.

At which point does the jacket water normally enter the engine?

- **At the bottom of the liner.**
- At the cylinder cover.
- At the top of the liner.
- At the turbocharger.

At which point in a diesel engine jacket cooling water system is the temperature normally measured for temperature control purposes?

- **At the engine outlet.**
- At the engine inlet.
- At the header tank.
- At the outlet from the heat exchanger.

Bore cooling is a common design feature for the main components of a diesel engine. Why is bore cooling used?

- **To allow intense cooling while maintaining component strength.**
- To make the components lighter.
- To prevent corrosion
- To prevent scale formation.

Cavitation erosion leading to corrosion can be a problem with some diesel engine components. Which one of the following are the most likely to suffer this form of damage?

- **Water side of cylinder liners.**
- Combustion side of cylinder liners.
- Exhaust valves.
- Piston crowns.

Central cooling systems are widely used onboard modern ships. What is the main reason for using such a system?

- **To minimize the amount of seawater pipework required.**
- To allow all of the pipes to be the same length.
- To have the equipment in the middle of the engineroom to minimize damage in the event of collision.
- To minimize the number of heat exchangers required.

Chromate based corrosion inhibitors are not allowed as the cooling water treatment in some situations. In which of the following must alternatives be used?

- **When the system includes a fresh water generator for producing potable water.**
- In a central cooling system.
- When the system doesn't include a fresh water generator.
- When the system includes a fresh water generator for producing boiler feed water.

Differential pressure cells are often fitted to diesel engine jacket cooling water systems. What is the main purpose for fitting them in such a system?

- **To monitor flow rate in the system.**
- To measure pressure drop across filters.
- To measure pressure rise across the circulating pump.
- To measure the level in the header tank.

Direct air cooling is used for some diesel engines. Which of the following duties are likely to have diesel engines which are directly air cooled?

- **Emergency electrical generator.**
- Main electrical generator.
- Medium speed main propulsion.
- Slow speed main propulsion.

How are the main and bottom end bearings cooled in a diesel engine?

- **Heat is carried away by lubricating oil.**
- Air cooled.
- Freshwater cooled.
- Seawater cooled.

How are the pistons cooled on a four stroke trunk piston type diesel engine? There may be more than one correct option.

- **By lubricating oil splashed up from the crankcase.**
- **By passing lubricating oil up the connecting rod and into the piston crown.**
- By passing jacket water through them.
- With cylinder lubricating oil.

How is expansion of the tube stack accommodated for in a shell and tube type heat exchanger?

- **Only one tube plate is clamped, the other is free to move longitudinally in the shell.**
- Non expanding materials are used for the tubes.
- The heat exchanger is only fixed at one end.
- The tubes are loosely fitted in the tube plate.

How is the flow normally monitored in a piston cooling system?

- **Sight glass and flow alarm in each return line.**
- Sight glass and flow alarm in each inlet line
- Sight glass and flow alarm in pump outlet.
- Sight glass and flow alarm in the pump inlet.

How is the temperature of the LT circuit controlled in a central cooling system?

- **By using a 3 way diverter valve to direct flow through the central coolers.**
- By using a 3 way diverter valve to recirculate the seawater.
- By using a 3 way mixing valve to mix water from the HT and LT circuit.
- By using a 3 way mixing valve to mix water from the LT circuit with seawater.

How should the jacket water flow through the deaerator?

- **With a rotational flow through the baffles.**
- From the center outwards through the baffles.
- Vertically downwards through the baffles.
- Vertically upwards through the baffles.

In some heat exchangers the tubes are fitted with fins. What is the purpose of these fins?

- **To extend the heat transfer surface.**
- To direct the fluid flow.
- To slow down the fluid flow.
- To strengthen the tubes.

Many shell and tube heat exchangers are described as multi-pass types where one of the fluids passes through the heat exchanger a number of times between entry and final exit. How are these multi-passes created?

- **By suitable division of the end covers.**
- By using a number of tube stacks.
- By using diverter valves
- By using recirculation valves

Only part of the heat released during combustion is converted to useful work. What happens to the rest of it? Select all applicable options.

- **It passes to exhaust**
- **It passes to the cooling water.**
- **It passes to the surrounding air.**
- It is used to heat the fuel.

Problems can occur when operating a diesel engine that is not adequately cooled. Select, from the options given, all those items that can be affected?

- **Component clearances.**
- **Component material properties.**
- **Lubricating oil condition.**
- **Pump seals and o-rings.**

The piston rings and liner wall heat up during the operation of the diesel engine. Apart from the combustion process what else causes these components to heat up?

- **Friction**
- Convection
- Cylinder lubricating oil
- Jacket water

Three way valves used in diesel engine cooling water systems can be arranged as either diverter or mixing valves. What is the normal port arrangement for a mixing valve?

- **Two inlet and one outlet port.**
- One inlet and two outlet ports.
- One inlet, one outlet and a bypass port.
- Three inlet ports.

Wax element control valves are often used in diesel engine cooling water systems. How is the valve actuated?

- **Expansion and contraction of the wax element moves the valve.**
- The temperature of the wax is measured by a sensor which outputs a signal to a controller.
- Water flow past the wax element moves the valve.
- Water pressure on the wax element moves the valve.

What do you understand by the term primary fluid when referring to diesel engine cooling systems?

- **The coolant that passes through the engine and is cooled in a heat exchanger.**
- The coolant that is used to preheat the engine.
- The coolant that passes through the heat exchanger to cool another fluid.
- The fluid that is bled through the fuel injectors.

What form of treatment is a seawater cooling system normally treated with to avoid fouling by marine growth?

- **Electrochlorination**
- Chromate based inhibitors
- Nitrite – borate based corrosion inhibitors
- Zinc anodes

What is the approximate value for the specific heat capacity of freshwater?

- **4.2 kJ/kgK**
- 1 kg/m^3
- 42 kJ/kgK
- 420 kJ/kgK

What is the main problem associated with using water to cool diesel engine pistons?

- **Risk of leakage into the crankcase.**
- Cocktail shaker effect cannot be achieved.
- Risk of cavitation erosion.
- Specific heat capacity too low.

What is the main reason for cooling the charge air supplied by the turbocharger?

- **Increase the density of the air entering the cylinder.**
- Heat up the seawater to feed the fresh water generator.
- Reduce the exhaust gas temperature.
- Reduce the temperature of the air entering the cylinder.

What is the main reason for including a fresh water generator in the main engine jacket water cooling circuit?

- **To increase the overall plant efficiency.**
- To allow a smaller jacket water cooler to be fitted.
- To help cool the jacket water.
- To reduce the number of pumps fitted.

What is the most common method used for cooling fuel injection valves on large modern heavy fuel oil burning 2 stroke diesel engines?

- **Recirculation of heavy fuel oil.**
- Direct circulation of diesel oil.
- Direct circulation of freshwater.
- Direct circulation of seawater.

What is the most common type of pump used in a diesel engine jacket water system?

- **Single stage centrifugal pump**
- Multi stage centrifugal pump
- Positive displacement pump
- Reciprocating pump

What is the purpose of a controller in a cooling water temperature control system?

- **To compare the measured value with the set point and generate an output signal for the control valve actuator.**
- To control the flow of the water in the system.
- To indicate the control valve position.
- To measure the temperature of the cooling water system.

What is the purpose of the diffuser ring fitted to some centrifugal pumps?

- **To convert the high velocity kinetic energy of the fluid to pressure energy.**
- It is fitted instead of an impeller.
- To accelerate the fluid in the pump.
- To help distribute the fluid inside the pump.

When describing heat exchangers what does the term parallel flow mean?

- **Both fluids flow in the same direction through the heat exchanger.**
- Both fluids flow through the tubes.
- Both fluids have horizontal flow paths.
- Both fluids have vertical flow paths.

When would a split range control system normally be used in a diesel engine cooling system?

- **When both heating and cooling of the system are required.**
- When the system is used for main and auxiliary engines.
- When the system is used for two engines.
- When two jacket water coolers are fitted.

Where would you expect to find zinc anodes to be fitted in a diesel engine cooling water system?

- **Inside the end covers of seawater cooled shell and tube type heat exchangers.**
- On a plate type heat exchanger using titanium plates.
- On the air side of charge air coolers.
- On the freshwater side of a jacket water cooler.

Where would you find detailed information relating to health and safety and environmental hazards associated with cooling water system treatment chemicals?

- **From the material safety data sheets.**
- From the delivery note.
- In the engine manual.
- On the packaging or container for the treatment.

Which of the following equipment is normally cooled by the LT circuit in a central cooling system? There may be more than one correct option.

- **Air compressors.**
- Diesel engine cylinder liners and cylinder covers.
- Diesel engine lubricating oil systems.
- Turbochargers.

Which of the following is normally included in the HT circuit of a central cooling system? There may be more than one correct option.

- **Auxiliary diesel engine cylinder liners and cylinder covers.**
- **Fresh water generator.**
- **Main diesel engine cylinder liners and cylinder covers.**
- Charge air coolers.

Which of the options given describes the desired water quality for a diesel engine cooling water system?

- **Slightly alkaline with a low level of total dissolved solids.**
- Slightly acidic with a high level of total dissolved solids.
- Slightly acidic with a low level of total dissolved solids.
- Slightly alkaline with a high level of total dissolved solids.

Why are the plates for plate type heat exchangers often made from stainless steel or titanium? More than one option could be correct.

- **Good resistance to corrosion**
- **Good resistance to erosion.**
- **They are strong even though they are thin.**
- **They can be made thin to promote heat transfer.**

Why is a header tank fitted in a jacket cooling water system? There may be more than one correct answer.

- **To accommodate expansion in the system.**
- **To allow air and vapour to exit the system.**
- **To provide a positive suction head for the circulating pump.**
- To allow the system to be drained.

Why is it necessary to preheat a diesel engine before the first start? Select all applicable options?

- **To help ignite the fuel.**
- **To minimise thermal stresses.**
- To help cool the auxiliary engines.
- To stop any water leaks at the seal rings.

Why is seawater rarely used as the primary coolant for a diesel engine jacket water system?

- **There is a high risk of fouling and corrosion.**
- It has a very low specific heat capacity.
- It is too cold.
- It may cause cracks in the components.

Why is silicate usually added to nitrite – borate based cooling water treatment?

- **To protect aluminium components in the system.**
- To protect against microbial problems.
- To protect steel and iron components in the system
- To protect the pump seals.

Why would a U tube manometer normally be fitted to a diesel engine charge air cooler?

- **To measure the pressure drop across the cooler.**
- To measure the atmospheric pressure in the engine room.
- To measure the output pressure of the turbocharger.
- To measure the pressure in the scavenge air manifold.

0019 – Remote Ctrl. Sys. (AC4)

Version 2.0 (2656)

Digital Governor Unit receives speed command through

- **Speed Set Lever**
- ECR Telegraph

During normal operation from Bridge the Main Engine can be stopped by . . . . . .

- **Bridge Telegraph**
- ECR telegraph
- Engine room telegraph
- Speed set lever

Engine Room Panel is subdivided into how many fields?

- **10**
- 11
- 4
- 8

From where does Autochief 4 takes RPM command information?

- **bridge telegraph lever.**
- speedset lever.

How does the bridge unit transfer command to ECR unit?

- **serial communication.**
- parallel communication.

If you are maneuvering from Engine Control Room, will the telegraph order be recorded at the Bridge OPU?

- **yes**
- no

In AC4 remote control system; manoeuvering of ME can be carried out from

- **from all three locations**
- bridge
- engine Control Room
- engine room panel

Name the field that contains “Rough Sea Mode” function?

- **other functions field**
- cancel function field
- command position field
- miscellaneous field

Pneumatic Valve Cabinet controls

- **all three functions of Main Engine**
- reversing of Main Engine
- starting of Main Engine
- stopping of Main Engine

Slow turning Diode will be illuminated when Engine has been stopped for . . . . . .

- **more than 30 minutes**
- more than 15 minutes
- more than 20 minutes
- more than 40 minute

System Warning & System Fail indication will be activated when . . . . . .

- **the RPM Detector system has failed**
- the Engine failed to reverse
- the Engine fails to start
- the Safety System Unit is malfunctioning

The bridge panel is subdivided into how many fields?

- **11**
- 10
- 4
- 8

The pick-up for measuring RPM is located . . .

- **at the Engine Fly Wheel**
- at the engine Panel
- in the Control Room
- on the Bridge

The reversing of Main Engine is done through a set of . . . . . .

- **Pneumatic valves**
- Hydraulic valve
- Mechanical linkage
- Solenoid valves

Try to place the different units beside their correct description.

This is the Bridge-panel which enables the operator to carry out Operations and gives information about different status in system. The panel is also used for commissioning.



This is the Engine Control Room-panel, which is a display and control panel It enables the operator to carry out certain operations and gives information about different status in the system .The panel is also used for commissioning



What are the functions of "Cancel" buttons?

- **all the three functions**
- to cancel Engine Shut down function
- to cancel Speed Limitation
- to cancel Slow down function

What component is installed at all three locations in AC4 remote control unit?

- **a telegraph panel**
- order printer unit
- safety system unit
- speed-set lever.

What is this?

- **this is the Emergency Control El-box used for informing the operator about safety actions and direction of rotation as well as for cancelling shut-downs and emergency stop.**
- this is the electric Actuator used for regulating the supply of fuel to the main engine.
- this is the local Telegraph Unit used for receiving and acknowledge of manoeuvre orders from bridge when the engine is controlled from local stand.

Which one of the following statement are true?

- **the fourth generation of remote control system for low speed main engines made by New Sulzer Diesel or MAN B&W.**
- the fourth generation of signal control system for high speed main engines made by New Sulzer Diesel or MAN B&W.
- the third generation of signal control system for highspeed main engines made by New Sulzer Diesel or MAN B&W.

0020 – Search and Rescue

Version 5.0 (2342)

A dotted line is seen on a radar display when a signal is received back from a SART. Which ONE of these options best identifies the location of the SART?

- **At the beginning of the 12 dots**
- At the 6th dot
- At the end of the 12 dots
- In the general direction of the bearing of the dots

A master decides NOT to provide assistance to a vessel in distress. What must they do next? Select the option that is the most important.

- **Record their decision in the ship's log and tell the search and rescue authorities**
- Advise the owner, charterer and flag State
- Record their decision in the ship's log and tell the ship owner
- Report to the RCC

During a search and rescue mission, what is the best position for an effective lookout during darkness? Select the ONE best option.

- **Forward and close to the water**
- Close to the galley to keep warm and fed
- Close to the ship's medical facility
- On the bridge

During a search and rescue mission, what is the best position for an effective lookout during daylight? Select the ONE best option.

- **High up on the ship**
- As close to the waterline as possible
- In a warm mess room with portholes
- In the forecastle store

From what direction should a burning ship be approached? Select the best option.

- **From the windward side**
- From ahead
- From astern
- From the leeward side

How can the visual methods of communication for use in search and rescue be checked? Select the TWO best answers.

- **A copy of SOLAS 1, Life saving signals and rescue methods is carried on the bridge of every ship**
- **All SAR facilities should carry a copy of the International Code of Signals**
- Contact the OSC and ask for confirmation
- Contact the SMC and ask for confirmation

If there have been deaths during an SAR operation, what should be done with the personal belongings of the deceased? Select the best answer.

- **Put them in a labelled container for transport with the deceased**
- Dump them overboard
- Gather all such effects together and declare them at the next port
- Incinerate them

In relation to search and rescue, what do the initials 'OSC' represent?

- **On-Scene Co-ordinator**
- Offshore Search Co-ordinator
- On-Scene Controller
- Overall Search Controller

In relation to search and rescue, what do the initials 'RCC' represent?

- **Rescue Co-ordination Centre**
- Radar Control Centre
- Regional Co-ordination Centre
- Rescue and Casualty Centre

In relation to search and rescue, what do the initials 'SMC' represent?

- **Search and Rescue Mission Co-ordinator**
- Safety Management Controller
- Search and Rescue Management Centre
- Search Management Controller

Is the following statement true or false?

"The international regulations for preventing collisions at sea still apply during SAR operations"

- **True**
- False

Is the following statement true or false?

'You have to acknowledge and investigate every DSC alert'

- **True**
- False

Select the ONE option which most accurately completes the following statement.

"An EPIRB should not be transferred from one ship to another unless . . .

- **. . . its registration details are changed.**
- . . . it is tested and works satisfactorily
- . . . the new ship's master agrees and new batteries are fitted.
- . . . the nearest rescue co-ordination centre is notified in writing.

There has been a death during an SAR operation. Where is the best place to store the dead body? Select the ONE best answer.

- **Properly wrapped, in a refrigerator**
- It should be put back into the sea
- Properly wrapped, in a lifeboat
- Properly wrapped, in the steering gear flat

What ONE of the following most accurately identifies the radio frequencies that will be used at the scene of a search and rescue?

- **The SMC will usually select SAR dedicated frequencies for use on scene**
- Channel 16 will always be the single SAR frequency used at the scene
- Channel 70 will always be the single SAR frequency used at the scene
- The Rescue ship will usually select the best SAR frequency to use at the scene

What information should be recorded by the OSC? Select the THREE correct options.

- **Actions taken**
- **Areas searched**
- **Names of other ships engaged in the operation**
- Number of crew on each assisting ship

What is a 'place of safety' for the survivors of a rescue? Select the best answer.

- **A place where the survivor's life is no longer threatened and basic human needs can be met**
- A place where the survivors are no longer in immediate danger
- Any rescue craft such as a life-raft, lifeboat or helicopter
- Once ashore in a member State of the SAR Convention

What is the main role of the OSC with respect to communications? Select the best option

- **Maintain communication with all SAR facilities**
- Deploy a SART to assist in homing
- Fix the position of the casualty using an EPIRB
- Recover and switch off the EPIRB

What is usually the most effective search method when ships and aircraft are involved? Select the best option.

- **A 'creeping line' search**
- A 'parallel track' search
- A 'track line' search
- An 'expanding square' search

What should a recue ship do if its radar breaks down during a search? Select the best option.

- **Drop astern of the other ships and inform the OSC**
- Maintain course and speed, but inform the MSC
- Maintain course and speed, but inform the OSC
- Post extra lookouts

When does a NAVTEX receiver monitor the transmission of Marine Safety Information? Select the best answer.

- **Automatically**
- At set time periods
- Only when entering the correct NAVAREA
- When requested

When is a 'sector search' most effective? Select the best option.

- **When the position of the search object is accurately known**
- When a datum marker can be used as a reference
- When the position of the search object is not accurately known
- When the search pattern radius is larger than 20 nautical miles

When is a 'track line search' most effective? Select the best option.

- **When a ship or aircraft has disappeared without a trace**
- When a direct communication between the search object and the search facility is available
- When the position of the search object is accurately known
- When the position of the search object is not accurately known

When should an initial 'SITREP' be transmitted? Select the best option.

- **As soon as details of an incident make it clear that SAR involvement is necessary**
- As soon as all the facts are known
- When required for the media
- When the situation is under control

When should distress messages be cancelled? Select the best option.

- **When a false alert has been sent**
- When the rescuing craft decide
- When the situation seems to be under control
- When weather conditions have improved

When would a 'parallel track' search be used? Select the best option.

- **When searching a large area**
- When searching a small area
- When searching during the night
- When searching in rough weather

When would a master NOT have to respond to a distress signal? Select the ONE answer that is the most accurate.

- **When the master considers it unsafe or unreasonable to do so**
- When the Company says not to, because of the vessel's charter
- When the master hears other ships responding to the distress signal
- When the master is already late on his ETA

When your ship is talking to a craft in distress, what information should you give to them? Select the THREE correct answers.

- **The distressed crafts true bearing and distance from your ship**
- **Your own ships identity, call sign, name and position**
- **Your own ships speed and ETA to the distressed craft**
- Your own ships original course, speed and destination

Which ONE of the following describes the function of an AIS SART?

- **To show ID and position of the casualty on equipment displaying AIS information**
- To allow two way communications for survival craft
- To indicate a ship in distress on radar by 12 dots
- To indicate distress by alerting satellites

Which ONE of the following most accurately completes the following statement?

"During a parallel track search . . .

- **. . . all ships proceed with the same speed"**
- . . . all ships proceed with a different speed"
- . . . all ships proceed with their own maximum safe speed"
- . . . all ships proceed with their own minimum safe speed"

Which ONE of the following most accurately completes the following statement?

"Communication between an aircraft and a ship is . . .

- **. . . difficult because of different radio bands.**
- . . . easy and straight forward.
- . . . not allowed.
- . . . not possible.

Which ONE of the following most accurately completes the following statement?

"The first ship arriving at the scene of a SAR incident would normally . . .

- **. . . take on OSC duties.**
- . . . take on ACO duties.
- . . . take on SMC duties.
- . . . take on SRU duties.

Which ONE of the following pieces of information is always included in a 406 MHz EPIRB signal?

- **The identification of the vessel or aircraft in distress**
- The flag State of the vessel
- The last known course and speed of the vessel
- The nature of the emergency

Which ONE of the following signals is NOT a distress signal?

- **A flare showing a white light**
- A flare showing a red light
- A square flag with a ball above or below it
- An orange smoke float

Which ONE of the following types of equipment would display an activated radar SART?

- **X-band radar**
- DSC equipment
- Inmarsat equipment
- S-band radar

Which ONE of the following volumes of IAMSAR must be carried on board?

- **Volume three**
- All volumes
- Volume one
- Volume two

Which ONE of the following would be the best person to speak to the media after an SAR incident?

- **Someone with experience of talking to the media, and authorised by the company**
- A representative of the ship's crew
- Any deck officer on duty during the rescue operation
- The master

Which ONE of these is the most important factor when using an expanded square search? Select the best option.

- **Accurate navigation**
- Constant speed
- Fixed ROT
- Fixed turn radius

Which ONE of these search patterns creates a particular problem in restricted visibility?

- **Parallel track search**
- Creeping line search
- Expanding square search
- Track line search

Which THREE of the following pieces of information are immediately available to a ground control station when a 406 MHz EPIRB signal is first picked up?

- **That an EPIRB has been activated and that this should be taken as a sign of distress**
- **The approximate location of the beacon**
- **The identification of the vessel or aircraft in distress**
- That the vessel or aircraft has been abandoned

Which TWO of the following responsibilities does a rescue vessel have to survivors following a search and rescue mission?

- **To deliver them to a place of safety**
- **To provide initial medical care, food and clothing**
- To give them some money
- To take them to their destination

Which person would usually guide a SAR operation from start to finish? Select the best option

- **SMC**
- OSC
- RCC
- SRR

Who is responsible for the co-ordination of communications at the scene of a rescue? Select the best option.

- **OSC**
- NRC
- SAR
- SMC

Who is usually responsible for providing the search action plan? Select the best option.

- **SMC**
- NRC
- OSC
- SAR

Why must all distress signals be entered into the ship's log book? Select the best option.

- **Because it is a legal requirement**
- Because it is a company requirement
- Because it is usually part of the master's standing orders
- Because it will help any claims for financial compensation

0021 – GMDSS

Version 4.0 (2047)

Direct or space wave (line of sight) propagation in the VHF and UHF bands mainly involve

- **AIS, SART, EPIRB, Inmarsat Fleet communications**
- HF communications only
- MF communications only
- Navtex broadcasts

For what period of time should an EPIRB be able to operate on its batteries to meet the regulations relating to GMDSS?

- **48 hours**
- 24 hours
- 36 hours
- 96 hours

How should a satellite antenna be mounted for approval in the GMDSS?

- **With free line of sight in every direction**
- As high as possible
- As near the transmitter as possible
- Midships

In relation to GMDSS what does the acronym MSI stand for?

- **Maritime Safety Information**
- Maritime Security Information
- Maritime Service Information
- Maritime Surface Information

In relation to GMDSS what does the acronym RCC stand for?

- **Rescue Coordination Centre**
- Research Coordinating Centre
- Resource Coordinating Centre
- Retransmitting Coordinating Centre

In which sea area are vessels required to carry either INMARSAT or HF equipment or a combination of both under GMDSS?

- **A3**
- A3 and A4
- A4
- All sea areas

On which VHF DSC channel should all GMDSS equipped ships keep watch?

- **Channel 70**
- 2187.5 kHz
- Channel 13
- Channel 16

What does the acronym GMDSS stand for in relation to communications onboard ships?

- **Global Maritime Distress and Safety System.**
- Global Maritime Digital Safety System.
- Global Maritime Disaster Search System.
- Global Maritime Distress Surface Search.

What is Tropospherical Propagation?

- **It is the propagation of a space wave directly or after reflecting from earth's surface to the troposphere surface of the earth.**
- It is the propagation of a ground wave directly after reflecting from earth's surface to the stratosphere of the earth.
- It is the propagation of a relay wave directly or after reflecting from earth's surface to the stratosphere surface of the earth.
- It is the propagation of a sky wave directly after reflecting from earth's surface to the ionosphere surface of the earth.

What is the coverage of Inmarsat's geostationary satellites in relation to GMDSS?

- **Global except the polar areas.**
- Global including the polar areas.
- The Atlantic, Indian and Pacific oceans and the polar areas.
- The Atlantic, Indian and Pacific oceans.

What is the most important factor for the range of a VHF radio station?

- **The height of the antenna**
- The output power
- The power supply
- The size of the antenna

What is the most important factor for the range of an HF radio station?

- **The frequency**
- The height of the antenna
- The output power
- The size of the antenna

What is the name of the international convention that regulates GMDSS?

- **SOLAS**
- GMDSS
- MMSI
- STCW

What is the propagation path taken by the radio signal on the VHF Channel 16?

- **Direct wave (line of sight)**
- Ground wave
- Sky wave
- Space wave

What is the two-digit 'Special Access Code' or 'Short Access Code' or 'SAC' designated for maritime safety services when requesting for medical advice under the context of GMDSS?

- **32**
- 38
- 39
- 40

What is the two-digit 'Special Access Code' or 'Short Access Code' or 'SAC' designated for maritime safety services when requesting for search and rescue (SAR) assistance under the context of GMDSS?

- **39**
- 40
- 41
- 42

What should be the very first step for a GMDSS radio operator to take when initiating a distress priority message via Inmarsat Fleet terminal?

- **Pressing the "Distress Button" or "Distress Hot Key" on the equipment**
- Contacting the CES operator and informing about the distress condition
- Contacting the CES operator by using the radiotelephone distress procedure
- Dialling the correct code on the telephone remote unit

Which VHF channel should be used for bridge-to-bridge communications concerning safety of navigation and manoeuvring in the GMDSS?

- **Channel 13**
- Channel 08
- Channel 16
- Channel 70

Which is the distress and safety frequency for MF DSC?

- **2187.5 kHz**
- 2182 kHz
- 8291 kHz
- 8414.5 kHz

Which one of the certificates given in the options is required to be held by an operator on a GMDSS ship trading in all sea areas?

- **GOC**
- DSC
- MOC
- ROC

Which one of the following EPIRB transmits a distress alert that is received and relayed by Inmarsat satellites globally?

- **406 MHz EPIRB**
- Class-A EPIRB
- Class-B EPIRB
- Class-E EPIRB

Which one of the following INMARSAT services has the exceptional features of 'Prioritisation' and 'Pre-emption' in the Global Maritime and Safety System (GMDSS)?

- **Inmarsat-Fleet 77**
- Inmarsat-Fleet 33
- Inmarsat-Fleet 55
- Inmarsat-Mini M

Which one of the given options best completes the following statement. "In relation to GMDSS, F1B is used for . . .

- **. . . TLX radiotelex"**
- . . . distress communications on 2182 kHz"
- . . . SSB radiotelephony"
- . . . VHF telephony"

Which one of the given options best completes the following statement. "In relation to GMDSS, F3E is used for . . .

- **. . . VHF telephony"**
- . . . distress communications on 2182 kHz"
- . . . SSB radiotelephony"
- . . . TLX radiotelex"

Which one of the given options best completes the following statement. "In relation to GMDSS, J3E is used for . . .

- **. . . SSB radiotelephony"**
- . . . distress communications on 2182 kHz"
- . . . TLX radiotelex"
- . . . VHF telephony"

Which one of the given options best completes the following statement. "In relation to GMDSS, sea area A1 means an area . . .

- **. . . within coverage of at least one Coast Station with VHF DSC and telephony"**
- . . . coastal water in sight of a Coast Station"
- . . . within 150 nautical miles from the coast"
- . . . within 30 nautical miles from the coast"

Which one of the given options best completes the following statement. "In relation to GMDSS, sea area A2 means an area . . .

- **. . . within coverage of at least one coast station with MF DSC and telephony"**
- . . . coastal water in sight of a Coast Station"
- . . . within 150 nautical miles from the coast"
- . . . within Radar coverage of at least one Coast Earth Station"

Which one of the given options best completes the following statement. "In relation to GMDSS, sea area A3 means an area . . .

- **. . . within coverage of an Inmarsat geostationary satellite"**
- . . . in one of the three major oceans"
- . . . that is further out than 150 nautical miles from the coast"
- . . . that is global"

Which one of the given options best completes the following statement. "In relation to GMDSS, sea area A4 means an area . . .

- **. . . which is outside areas A1, A2 and A3"**
- . . . that is global"
- . . . which is outside the ordinary ship traffic routes"
- . . . with no ordinary radio communications"

Which one of the given options best completes the following statement. "One of the most important advantages with GMDSS is that . . .

- **. . . a ship in distress should always be able to send a ship-to-shore distress alert"**
- . . . a ship can always communicate via radiotelex"
- . . . a ship can always communicate via satellite"
- . . . every ship has the same equipment"

Which one of the given options best completes the following statement. "Ship-to-ship distress alerting is primarily based on . . .

- **. . . digital selective calling in the VHF and MF bands"**
- . . . mobile telephones"
- . . . satellite communications"
- . . . VHF telephony"

Which one of the phrases given in the options most accurately completes the following statement? Among the services provided by Inmarsat-Fleet 77 in the context of Global Maritime Distress and Safety System (GMDSS), the foremost is . . .

- **. . . global mobile voice and flexible data communications services**
- . . . high speed fax only
- . . . store and forward messaging only
- . . . telex services

Which one of the phrases given in the options most accurately completes the following statement? The Inmarsat system communicates through . . .

- **. . . space wave**
- . . . a combination of ground wave and sky wave
- . . . ground wave
- . . . sky wave

Which one of the phrases given in the options most accurately completes the following statement? The propagation path used by HF signals, from the SSB radio transceiver to achieve long-range communications, is . . .

- **. . . the sky wave**
- . . . direct wave (line of sight)
- . . . relay wave
- . . . the ground wave

Which one of the phrases given in the options most accurately completes the following statement? The two-digit codes that are designated for maritime safety and general services are referred as . . .

- **. . . Special Access Codes or Short Access Codes**
- . . . General Access Codes
- . . . Routine Access Codes
- . . . Safety Access Codes

Which one of the phrases given in the options most accurately completes the following statement? When selecting a HF band for a particular range, the one important advice is . . .

- **to select a higher frequency in the band by day and a lower frequency in the night**
- to select a higher frequency in the band by day and night
- to select a lower frequency in the band by day and a higher frequency in the night
- to select a lower frequency in the band by day and night

Which one of the systems given in the options most accurately completes the following statement? In the Global Maritime Distress and Safety System (GMDSS), the ISDN (Integrated Services Digital Network) and MPDS (Mobile Packet Data Service) data communication facility is available using . . .

- **Inmarsat-Fleet 77**
- Inmarsat-A
- Inmarsat-B
- Inmarsat-C

Which piece of equipment on the bridge is used for locating a SART?

- **3 cm (9 GHz) Radar**
- 10 cm (3 GHz) Radar
- Radar Direction Finder
- Radio Direction Finder

Why is channel 16 a simplex channel under the GMDSS context?

- **Every ship within coverage should be able to hear and communicate**
- It is cheaper
- Not every country provides duplex service
- Simplex is less sensitive to interference than duplex

Why should the SART be mounted as high as possible in the survival craft in relation to GMDSS?

- **To increase the range**
- The flashlight should be easily discovered by a rescuer
- To protect the people in the survival craft from radiation
- To protect the SART from sea spray

You have received the 9-digit MMSI of a ship in distress. In which ITU publication can you find the name of the ship?

- **List of call signs and numerical identities**
- List of radio determination and special service stations
- List of ship names
- List of ship stations

0022 – Maritime English, Ship Familiarisation

Version 1.2 (28)

Any action taken to avoid collision shall . . .

- **Be a positive action, made in ample time and using good seamanship**
- **Be carefully checked until the other vessel is finally past and clear**
- **Result in passing at a safe distance**

Choose the correct option.
In ABC of First Aid, B stands for . . . . . .

- **Breathing**
- Bandage
- Bleeding
- Breastline

Choose the correct option.
In ABC of first aid, A stands for . . . . . .

- **Air Way**
- Alleways
- Always
- Arteries

Choose the correct option.
OSC stands for . . . . . .

- **On scene co-ordinator**
- On scene commander
- On ship captain
- Over ship commander

Choose the correct option?
ETA stands for . . . . . .

- **Estimated time of arrival**
- Estimated time of anchoring
- Extra terrestrial alien
- Extra time allowance

Choose the correct option?
RORO stands for . . . . . .

- **Roll on roll off**
- Rhumbline on river ornico
- Ride on ride out
- Roll out roll over

Identify the flag and its meaning.

- **B - BRAVO; I am taking in, discharging, or carrying dangerous cargo.**
- A – ALFA; I have a diver down; keep well clear at slow speed.
- D – DELTA; Keep clear of me; I am manoeuvring with difficulty.
- E – ECHO; I am altering my course to starboard.

Is the statement correct? Reef Knot is used to tie two ropes of equal size or thickness together.

- **True**
- False

Is the statement correct? The sheet bend is used to join two ropes of different size together.

- **True**
- False

Look at the image and choose what it indicates ?

- **Gas tanker**
- Bulk carrier
- Container ship
- Passenger ship

Look at the image and choose what it indicates?

- **Bulbous bow**
- Poop deck
- Porthole
- Rudder

Look carefully at the navigational lights shown by this vessel and click on the right option what these lights indicate.

- **Vessel constrained by her draft**
- Vessel aground
- Vessel not under command
- Vessel restricted in her ability to manoeuver

SAFETY Onboard is to ensure that all work procedures and operations are planned keeping in mind the following:

- **The protection of personnel**
- **The protection of property**
- **The protection of the environment or (Protection from pollution)**

Safe speed is defined as that speed where . . .

- **Vessel can take proper and effective action to avoid collision**
- Vessel can be stopped within range of visibility
- Vessel is navigating at slower speed than vessels around
- Vessel is proceeding at manoeuvring speed

Select the most appropriate phrase to indicate that eye of first mooring line has been laid around the bollard on jetty?

- **First line ashore**
- All fast
- First fast
- Mooring on jetty

To indicate a new danger, a duplicate mark is provided besides the normal original mark. Is the statement correct?

- **True**
- False

What is the Maximum possible Latitude?

- **90 degrees north**
- 180 degrees north
- 270 degrees north
- 360 degrees north

What is the Maximum possible Longitude?

- **180 degrees east**
- 270 degrees east
- 360 degrees east
- 90 degrees east

What is the meaning of Flag Hotel depicted above.

- **I have a pilot onboard.**
- I am in process of embarking a pilot.
- I require a pilot.
- Identification signal given by ship to pilot boat.

What is the most common name for ships, which carry oil in bulk?

- **Product tankers**
- Ferries
- General cargo ships
- RoRo

What is the most widely used general emergency alarm onboard ships?

- **Signal of seven short blasts, followed by one long blast**
- Signal of one short blast, one long blast repeated seven times
- Signal of seven long blasts followed by one short blast
- Signal of three long blasts

What is the primary function of echo sounder?

- **To provide depth of water underkeel**
- To provide course of ship
- To provide position of ship
- To provide speed of ship

What should you do when you hear a General Emergency Alarm?

- **Proceed to your designated muster station**
- Contact bridge
- Evacuate the premises
- Shout Fire

When a man falls overboard, we should immediately throw the nearest lifebuoy overboard. Is the statement correct?

- **True**
- False

When a man falls overboard, which of the following is not the required immediate action?

- **Keeping sickbay ready**
- Shouting 'man overboard'
- Sounding general alarm
- Throwing a lifebuoy

Where on ship would you find a Fire Blanket?

- **Galley**
- Bridge.
- Engine control room
- Radio room

Which IMO Convention deals exclusively with the Environmental Pollution?

- **MARPOL**
- ISM
- SOLAS
- STCW

Which Publication gives detailed guidelines about conduct of various parties during Search and Rescue?

- **IAMSAR Manual**
- IMO SAR Manual
- MARPOL
- SOLAS

Which is the correct type of stopper to be used for wire ropes?

- **Chain Stopper**
- Nylon Stopper
- Rope Stopper
- Wire Stopper

Which is the recommended colour of the Foam Type Fire Extinguishers?

- **Yellow**
- Black
- Blue
- Red

Which list details the duties and responsibilities in case of emergency onboard? The statement is:

- **Muster list**
- Crew list
- Daily work list
- Hot work checklist

Which of the following is the correct abbreviations for two most common standard sizes of containers?

- **TEU and FEU**
- DWT and FWA
- TFT and FFT
- TPC and FTC

Who has the overall responsibility of the engine room and its maintenance?

- **Chief Engineer**
- First Engineer
- Master
- Second Engineer

0024 – Operating Auxilary Diesel Engines

Version 2.0 (3054)

A tandem type gear pump is fitted in the rocker arm lubricating oil system of the Ulstein Bergen diesel engine covered in this module. Which one of the given options is the recommended inspection/maintenance routine that should be performed on the pump?

- **Check the tell-tale holes for leakage once a month**
- Check the lub oil level once a week
- Lubricate once a month

An actuator is fitted in the governing system of the Ulstein Bergen diesel engine covered in this module. How often should the linkage between this actuator and the fuel pumps be inspected for wear, loosening and lubrication?

- **Daily**
- Every month
- Every second week
- Every week

At what load should the engine be operating at when measuring maximum combustion pressure on an Ulstein Bergen diesel engine according to the manufacturer's recommendations?

- **100%**
- 0%
- 50%
- 75%

Before starting the Ulstein Bergen diesel engine covered in this module certain conditions must be met. Select the option which best describes the minimum recommended conditions that must be met for cold starting on MDO?

- **Jacket cooling water above 0°C.**
- Jacket cooling water above 50°C and nozzle oil temperature above 80°C.
- Jacket cooling water above 50°C.

Before starting the Ulstein Bergen diesel engine covered in this module certain conditions must be met. Select the option which best describes the minimum recommended conditions that must be met for warm starting on MDO?

- **Jacket cooling water above 50°C.**
- Jacket cooling water above 0°C.
- Nozzle oil temperature above 80°C.

Before starting the Ulstein Bergen diesel engine covered in this module certain conditions must be met. What are the conditions to be met before starting the engine on HFO?

- **Jacket cooling water above 50°C, nozzle oil temperature above 80°C, and fuel oil viscosity at injection as given in UBE documentation.**
- Jacket cooling water above 50°C and nozzle oil temperature above 80°C.
- Jacket cooling water above 50°C.

Certain operating conditions are specified when water washing the turbine side of the turbo charger for the Ulstein Bergen diesel engine covered in this module. Select the main criteria to be used from the options given?

- **Engine operating with a charge air pressure of at least 0.3 bar**
- Engine operating on at least 25% load
- Engine operating on at least 50% load
- Engine operating with a charge air pressure of at least 1.3 bar

From which end of the engine does cylinder number designation begin for Ulstein Bergen diesel engines?

- **From the pump end**
- From the flywheel end

How many overspeed trips are fitted to the Ulstein Bergen diesel engine covered in this module?

- **Two**
- Four
- One
- Three

How often should the filter element in the rocker arm lubricating oil system of the Ulstein Bergen diesel engine covered in this module be cleaned?

- **Daily**
- At least once a month
- At least once a week
- At least once every second week

How often should the filter housing in the rocker arm lubricating oil system of the Ulstein Bergen diesel engine covered in this module be cleaned?

- **At least once a week.**
- At least once a month.
- At least once every second week.
- Daily

How often should the fuel oil filter elements in the fuel oil system of the Ulstein Bergen diesel engine covered in this module be checked?

- **Every month**
- Daily
- Every second week
- Every week

How often should the maximum combustion pressure be measured for the Ulstein Bergen diesel engine covered in this module?

- **At least every 3 months.**
- At least once a month.
- At least once a week.
- At least once every second week.

How often should the strainer in the rocker arm lubricating oil system of the Ulstein Bergen diesel engine covered in this module be cleaned?

- **At least once a week**
- At least once a month
- At least once every second week
- Daily

In this module a standstill period, when the engine is stopped, is defined as either short or long. Which of the given options is used to define a short period of standstill?

- **A period of less than 24 hours**
- A period of less than 12 hours
- A period of less than 48 hours
- A period of less than 96 hours

Is it possible to clean the centrifugal filter while the Ulstein Bergen diesel engine covered in this module is running?

- **Yes, but the valve to the centrifugal filter must be closed.**
- No, the centrifugal filter can only be cleaned with the engine stopped.
- No, the engine must have been stopped for at least 24 hours.
- Yes, but the valve to the centrifugal filter must be closed and the lubricating oil pressure should be reduced to 4 bar.

It is important when re-assembling the main lubricating oil filter for the Ulstein Bergen engine covered in this module that the required steps are carried out in the correct sequence. Place the five steps given in the options in the correct sequence.

- 1. **Check all gaskets to ensure there is no damage.**
- 2. **Replace strainer, cover and drain plug.**
- 3. **Fill the filter and ventilate the air from the filter.**
- 4. **Put the cock into the correct in service position.**
- 5. **Check for leakage under normal running conditions.**

It is recommended in the module that only one of the lubricating oil centrifugal filters fitted to the Ulstein Bergen diesel engine be dismantled at any one time. Which of the given options is the main reason for this recommendation?

- **To avoid the risk of mixing or interchanging the filter parts.**
- Because the filtration will be impaired.
- Because the lubricating oil pressure is too high.
- Because there are no bypass possibilities.

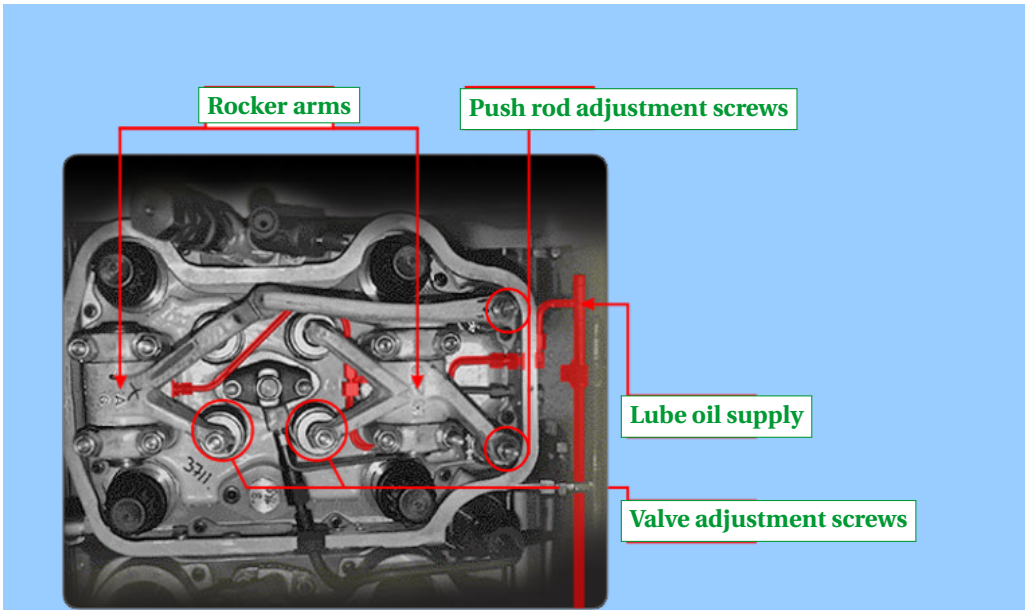
Place the given text in the correct order for the procedure for cleaning the main engine lubricating oil sump as stated in the module.

- 1. **Shut down the engine.**
- 2. **Drain the oil system while the oil is still warm.**
- 3. **Remove and clean the strainers.**
- 4. **Clean the oil sump and the crankcase thoroughly with rags.**
- 5. **Replace the strainers.**
- 6. **Clean the main filter housing and the centrifugal filter.**
- 7. **Replace main filter elements, if it has not been recently done.**

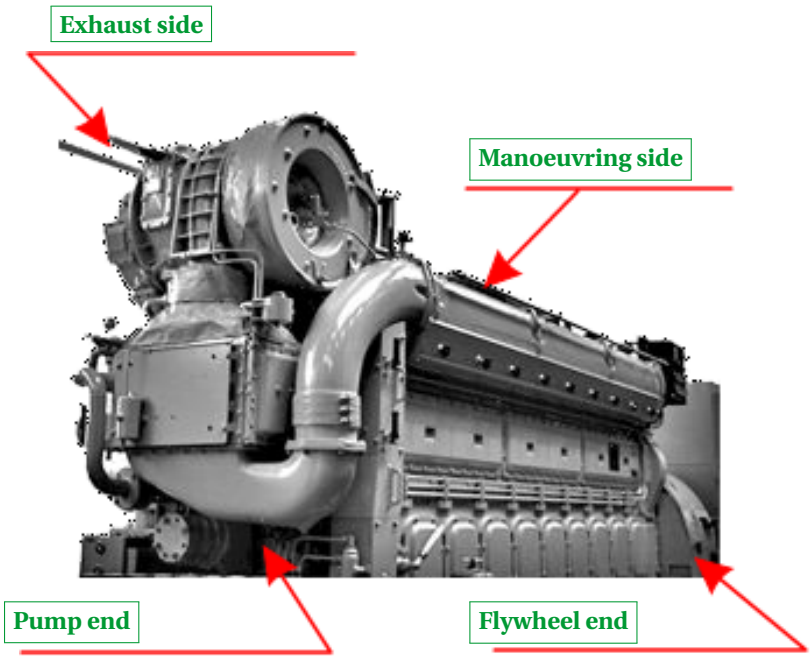
Place the given text in the correct order to give the procedure for cleaning the strainer in the rocker arm lubricating oil system as given in the module

- 1. **Unscrew the drain plug.**
- 2. **Pull out the strainer element and clean it**
- 3. **Remove sludge in the strainer housing**
- 4. **Reinstall the strainer element**
- 5. **Reinstall the plug**
- 6. **Watch the lubricating oil pressure until the reading is normal and the low pressure alarm is off.**

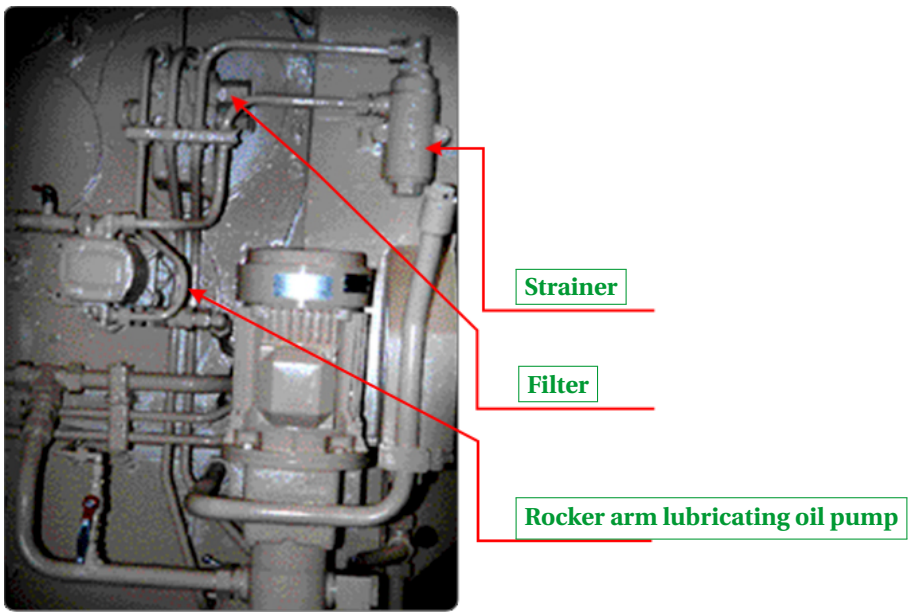
Place the labels to correctly identify the parts of the engine cylinder cover as indicated in the picture.



Place the labels to correctly identify the reference points of the engine as indicated in the picture.



Some of the components of the rocker arm lubricating oil system are mounted on the pump end or free end of the Ulstein Bergen diesel engine. Place the labels to correctly identify these components as indicated in the picture.



The Ulstein Bergen diesel engine described in this module is fitted with a control air module. Which of the statements given in the options best describes the main functions of the control air module fitted?

- **It regulates the humidity and pressure of the control air and also filters it.**
- It regulates the humidity and temperature of the control air and also filters it.
- It regulates the pressure, temperature and humidity of the control air.
- It regulates the temperature and pressure of the control air and also filters it.

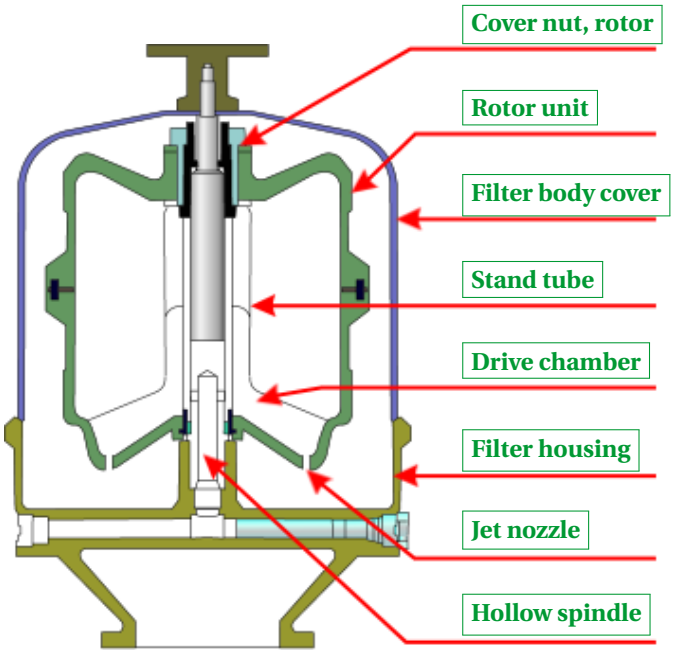
The Ulstein Bergen diesel engine described in this module is fitted with a nozzle temperature control system for the fuel injection valves. How often should the filter in the system be cleaned?

- **Every week**
- Daily
- Every month
- Every second week
- Every third month

The filter cock in the rocker arm lubricating oil system of the Ulstein Bergen diesel engine covered in this module has two marked positions, D and F. Which of the marks should correspond with the mark on the filter housing when the filter is in the in-service operating position?

- **F**
- D

The given sketch shows the centrifugal filter for the main lubricating oil system of the Ulstein Bergen diesel engine covered in this module. Place the labels to correctly identify the components of the centrifugal filter.



The main lubricating oil system of the Ulstein Bergen diesel engine covered in this module includes a centrifugal filter. How often should this centrifugal filter be cleaned?

- **At least every month.**
- At least every second week.
- At least once a week.
- Once a day.

The main lubricating oil system of the Ulstein Bergen diesel engine covered in this module is fitted with a centrifugal filter. Does all the lubricating oil flow go through the centrifugal filter during normal operation?

- **No, only 10% of the flow**
- No, only 30% of the flow
- No, only 50% of the flow
- Yes

The main lubricating oil system of the Ulstein Bergen diesel engine covered in this module is fitted with a centrifugal filter. How often should the centrifugal filter be checked for excessive vibrations according to the recommendations given in this module?

- **At least once a week.**
- At least every month.
- At least every second week.
- Once a day.

The sketches show three positions for the main lubricating oil duplex filter for the Ulstein Bergen diesel engine covered in this module. Which of the shown positions would the filter cock be in to bypass the left hand filter unit so that maintenance can be carried out on the filter?

- **A**
- B
- C

There are certain recommended operating conditions for the Ulstein Bergen diesel engine covered by this module. Is it recommended to operate the engine on HFO at less than 10% load?

- **No, this is not recommended**
- Yes, but only for 24 hours before running on 50% load for 15 minutes.
- Yes, but only for 24 hours before running on MDO for 1 hour.
- Yes, but only for 24 hours before running on MDO on at least 50% load for 1 hour

What is the function of the actuator fitted in the governing system of the Ulstein Bergen diesel engine covered in this module?

- **It transforms electrical signals to mechanical movement.**
- It transfers pneumatic air signal to mechanical movement.
- It transfers signals to the alarm and monitoring system.

What is the function of the change-over valve in the fuel oil system of the Ulstein Bergen diesel engine covered in this module?

- **To change between HFO and MDO.**
- To change the direction of the fuel flow.

What is the main purpose of the heater module fitted in the jacket cooling water system of the Ulstein Bergen diesel engine covered in this module?

- **To keep the jacket cooling water temperature high enough when the engine is stopped to ensure a warm start up can be performed.**
- To keep the jacket cooling water temperature high enough to heat the charge air in the charge air cooler during low load operation.
- To prevent freezing of jacket cooling water when temperatures get below 0°C during a long period of standstill.

What is the maximum level of water that should be in the jacket water system expansion tank for the Ulstein Bergen diesel engine covered in this module?

- **2/3 of tank volume**
- 1/2 of tank volume
- 1/3 of tank volume.
- 1/4 of tank volume.

What is the purpose of the regulating valve in the jacket water system for the Ulstein Bergen diesel engine described in this module?

- **To adjust the jacket water temperature.**
- To adjust the jacket water flow rate.
- To adjust the jacket water pressure.
- To shut off the jacket water flow when performing maintenance operations.

What is the recommended frequency given in this module for waterwashing the compressor side of the turbocharger?

- **Daily**
- Every month
- Every second week
- Weekly

Which end of the engine is the reference end when deciding the direction of rotation of an Ulstein Bergen inline engine?

- **The flywheel end**
- The pump end

Which of the following combinations best describes the health risks that are generally associated with diesel engine lubricating oil as described in this module?

- **Intoxication and irritation of the skin and eyes**
- Caustic burns and skin irritation
- Intoxication and caustic burns
- Intoxication, caustic burns and skin irritation

Which of the given options is the fluid used in the cleaning system for the fuel oil injection pumps as described for the Ulstein Bergen diesel engine in this module?

- **Marine Diesel oil.**
- The lubricating oil used in the main lubricating oil system
- Turbine lubricating oil.

Which of the sketches shows the correct operation for a fuel oil injection valve nozzle?

- **B**
- A
- C

0026 – Voyage Planning

Version 2.0 (0)

Can RNC charts be used for voyage planning?

- **Yes, when used in conjunction with paper charts**
- Never
- Yes, when being used in the ECDIS mode
- Yes, when the equipment is approved by the flag state

Can the Officer of the watch be the sole lookout?

- **Only during day light hours when visibility is good and traffic etc. make it safe to do so**
- Always during day light hours
- Never
- Only during periods of good visibility

Can voyage planning be executed on an ECDIS only?

- **Yes, only when equipment complies and is operated as approved by the Flag State**
- Never
- Yes, only if approved and operating as specified by the ship owner
- Yes, only when the ECDIS uses UK Admiralty Raster Charts

Do most piracy attacks take place in territorial or international waters?

- **Most attacks take place in territorial waters**
- Most attacks take place in international waters
- Number of attacks are approximately 50/50% in international and territorial waters
- The most serious attacks take place in international waters

How can a ship be kept "head up" into the weather, with the rudder jammed in hard over position?

- **Careful ahead manoeuvring of the main engine**
- By applying astern power
- By applying emergency full astern power
- Not possible

How can you reduce ship's downwind progress in deep water when the main engine is incapacitated?

- **By lowering the anchor to about 4 shackles**
- By effectively steering the vessel
- By lowering the anchor to the water surface
- Not possible

How should effective bridge communications be conducted?

- **In clear and precise messages delivered in standard maritime English**
- All messages should be in the language of the majority of the bridge team
- Any instructions should be in the native language of the pilot
- Responses to any orders are sufficient indication of understanding

If the Officer of the watch must deviate from the voyage plan, he should?

- **Inform the Master**
- Check the new plan is safe and no further action
- Make an appropriate note in the log book and no further action
- No special action is necessary

In coastal waters GPS positions should be checked by?

- **By Visual, radar and any other available means**
- Radar observation only
- Visual bearings of Buoys and sea marks only
- Visual observations only

In which geographical waters are piracy attacks most frequent?

- **Indonesian waters**
- Brazilian waters
- Malaysian waters
- Open sea

Ship to ship communication during SAR should be executed by?

- **VHF and MF**
- MF only
- Satellite only
- VHF only

Should all on-board activities be planned and any risks assessed?

- **Yes, always**
- Not always necessary
- Only the important ones
- When it is obvious there may be dangerous circumstances

Voyage planning consists of how many main stages?

- **4 (Appraisal; Planning; Execution and Monitoring)**
- 1 (Planning)
- 2 (Planning and Monitoring)
- 3 (Planning; Execution and Monitoring)

What are the main purposes of VTS?

- **To provide detailed navigational and safety information for ships in the area**
- To evaluate ships using the port or coastal region
- To provide only local knowledge and instructions for all ships in the area
- To provide only tidal and weather information for ships participating in VTS

What can good voyage planning provide?

- **A reduction of stress, improved voyage efficiency and additional safety**
- A safe passage throughout the voyage
- Reduction of involvement in everyday navigational activities by the Master
- Sufficient information at hand without reference to chart and publications

What documentation and charts should be used for a voyage?

- **Only up-to-date charts and publications fully corrected**
- British Admiralty charts and documents
- Flag state documents and charts
- Only charts using WGS84 datum and appropriate publications

What is an effective way to continually monitor coastal progress within the voyage plan

- **By Parallel Indexing on the radar**
- By GPS positions
- By use of transit bearings
- By visual bearings

What is important to check when transferring a position from GPS to a chart?

- **Make sure that the chart and the GPS use the same datum or make any necessary corrections to the GPS readout**
- Make sure the chart has updated corrections, before plotting the GPS readout
- Plotting the position carefully using the GPS readout
- Reading the position correctly before plotting on the chart

What is most important when transferring route plans to other navigational systems?

- **Both systems use same chart datum**
- Both systems are approved by the classification society
- Both systems are made by the same manufacturer
- Both systems work according to specification

What is the accepted international definition of Piracy?

- **An illegal act committed on the high seas**
- The act of boarding a vessel by unwanted persons
- The act of removing items from a ship at sea, by persons using force
- The interference of a voyage by unwanted persons on the high seas

What is the fundamental purpose of Voyage Planning?

- **To provide adequate preparation and support to the bridge team for the intended voyage**
- To comply with statutory requirements
- To comply with STCW95 requirements
- To provide general information about the intended voyage

What is the shortest distance between two points?

- **Great circle**
- Composite great circle
- Rhumb line
- Straight line on Mercator chart

What items should be checked when approaching the destination port after a long sea passage?

- **All navigational, propulsion, communication and mooring equipment**
- Astern movement on engines only
- Hand steering and engine manoeuvring effectiveness only
- Mooring equipment

What should a shipping or management company provide for the bridge team to successfully complete the voyage?

- **Clearly defined bridge procedures for planning and execution of voyage plan**
- A complete management procedures guide
- A copy of the STCW code of practice
- Complete authority to the master to allow him/her to establish correct bridge procedures

What unexpected weather occurrence should always affect the existing voyage plan?

- **A TRS**
- A frontal depression
- A severe temperature change
- An anticyclone

When should operational checks on navigation equipment be done?

- **Regularly and when preparing for sea and entering port**
- No set times, just as often as possible
- When errors are indicated
- When time allows

When should parallel indexing be used on the passage?

- **When ever there is suitable headlands or marks to provide referencing**
- Never
- When navigating in a river or port entrance only
- When navigating on the coast only

When should the Officer of the Watch be able to hand over the watch to his relief?

- **When the relief understands the present situation, is fully capable and all existing manoeuvres have been completed**
- As soon as the relief arrives on the bridge
- When the OOW is satisfied his relief is understanding the situation
- When the relief is aware of present situation and has gained his/her night vision

When should voyage planning be completed?

- **Prior to sailing**
- Before the pilot leaves at the departure port
- During the sailing
- During the voyage

When steering by autopilot at sea, how often should manual steering be checked?

- **At least once a watch**
- Every day
  When time allows
- Every day
- No checking is necessary

When the pilot is on board, is he/she in charge of the vessel?

- **The Master is still responsible, but the pilot aids the navigation with local knowledge**
- The harbour authority is responsible for the vessel
- The OOW is fully responsible
- The pilot takes over the responsibility of the vessel

Which of the following items would NOT be included in the Radio Log Book?

- **Reception of weather forecasts**
- Problems of interference
- Ships position
- Summary of communications regarding distress and urgency

Which of the following items would probably NOT be included in the Passage Plan?

- **Occurrence of a TRS**
- Distance along each course line (leg)
- Tidal flow currents
- Wheel over positions

Who is ultimately responsible for the voyage plan?

- **The Master**
- The navigation officer
- The Officer of the Watch
- The person delegated to do the planning

0027 – Ballast Management

Version 2.0 (830)

According to the Ballast Water Management Convention which one of the following events must be recorded in the Ballast Water Record Book?

- **Circulation or treating of ballast water**
- Bunker figures
- Distance to waypoint when taking on ballast
- Transfer of ballast to correct a list

Can a coastal or port State which has signed up to the Ballast Water Management Convention implement stricter rules than laid down by the IMO without prior notice?

- **Yes – but only if an emergency or epidemic situation is declared**
- No - six months notice must be given before stricter rules are enforced
- No – the standards in the Convention must be used at all times
- Yes - port States can make any rules they want without giving any prior notice

How many times the volume of a ballast tank would you expect to have to pump in order to meet the Ballast Water Management Convention requirement of a 95% change of the water if using the flow-through method of ballast exchange?

- **3**
- 1
- 2
- 4

If you are on a ship which meets the requirements of the Ballast Water Management Convention which one of the following best describes how you can find out where the ballast water sampling points are located?

- **In the Ballast Water Management Plan**
- In the Safety Management System
- In the Trim and Stability Booklet
- On the Damage Control Plan

Previously, the main way of dealing with organisms and pathogens in ballast was by exchanging the water during the voyage. Which one of the following best explains why the Ballast Water Management Convention requires ships to install ballast treatment systems?

- **The technology is now available to treat ballast to a higher standard than obtained by exchange**
- Ballast treatment is generally a cheaper option than ballast exchange
- Ballast treatment systems are simpler to operate than ballast exchange methods are
- The ISPS Code has been amended to encourage the use of treatment systems so as to prevent alien species getting on board

Ships have been loading and discharging ballast water for many years. Which one of the following best describes why invasive species like Mitten Crabs HAVE NOT spread all around the world?

- **Alien species can only invade where environmental conditions in their new surroundings are similar to those where they were brought from**
- Alien species only invade cold water environments
- Alien species only invade warm water environments
- Mitten Crabs do not like cold water

When were the problems caused by invasive aquatic species first brought to the attention of the IMO by a member State?

- **The 1980s**
- 2004
- The 1920s
- The 1990s

Where would you find a detailed description of the ballast water management procedures for a ship which complies with the Ballast Water Management Convention?

- **In the Ballast Water Management Plan**
- In the captain's safe
- In the Safety Management System
- In the SOLAS training binder

Which one of the following best describes the Ballast Water Management Convention requirements for ships which do not have treatment systems and are on routes between ports where they are never far enough from land to conduct ballast exchange?

- **Ships can carry out exchanges within designated coastal exchange areas which do not meet the normal Convention requirements, if provided**
- Coastal States must build ballast water reception facilities by 2016 or ships will need to divert their route to meet Convention exchange requirements
- Ships must divert their route to meet Convention exchange requirements
- Ships must fit treatment systems by the end of 2012

Which one of the following best describes the Ballast Water Management Convention requirements for ships which intend to carry out ballast water exchange?

- **Exchange should be performed in water at least 200 metres deep and whenever possible at least 200 miles from land**
- Exchange should be performed in water at least 50 metres deep and whenever possible at least 50 miles from land
- Exchange should be performed only in designated coastal exchange zones
- Exchange should be performed while steaming at a minimum of 7 knots while at least 200 miles from land

Which one of the following best describes the Ballast Water Management Convention requirements for ships which only trade within a limited area?

- **Ships which only load and discharge ballast water within a common ecosystem can be exempted from the requirement to fit a ballast treatment system**
- Offshore supply vessels or tugs which only operate from one port can be exempted from the requirement to use ballast reception facilities
- Ships which only load and discharge ballast water within a common ecosystem only need to install a Class B ballast treatment system
- Ships which only trade within a limited area do not need to carry ballast water

Which one of the following best describes the Ballast Water Management Convention requirements for ships which take water into a ballast tank as a result of a grounding?

- **The flag State and port State must be contacted before the water is pumped out**
- The water will need to be discharged overboard as soon as possible
- The water will need to be pumped to a ballast water reception facility
- The water will need to be treated before being discharged overboard

Which one of the following best describes the Ballast Water Management Convention requirements for the control systems of approved ballast treatment systems?

- **The control system must record critical data for the last 2 years in a form that can be displayed or printed for inspectors to see**
- The control system must be serviced by the manufacturers at least once per year
- The control system must display information in the working language of the ship
- The control system must print out the date and time when of any of the 5 required alarms sounded

Which one of the following best describes the main function of the Ballast Water Management Plan required by the Ballast Water Management Convention?

- **To ensure that ballast treatment is carried out effectively and safely**
- To ensure that ballast exchange is carried out safely
- To ensure that ballast is loaded and discharged safely
- To ensure that ballast loaded in one port is not allowed to contaminate ballast from another location

Which one of the following best describes the main reason why ships fitted with static filters as part of their ballast water treatment system use them when loading ballast but often bypass them while discharging ballast?

- **The organisms filtered out during discharge need to be collected and disposed of properly**
- Static filters are unreliable and used only when essential
- Static filters tend to clog with paint and rust when used during discharge
- The elements inside static filters are expensive and used as little as possible

Which one of the following best describes the main reason why ships normally use another treatment technology in their ballast water treatment system in addition to static filtration?

- **Static filtration is unlikely to remove pathogens from the ballast so a second treatment technology is needed**
- Another treatment technology is fitted so that ballast can be loaded or discharged using gravity instead of the ballast pumps
- Static filters are unreliable so a back up treatment is needed to be safe
- The running costs of static filters means they are used as a backup treatment only

Which one of the following describes the Ballast Water Management Convention requirements for keeping ballast water records?

- **Two years records must be available on board, with records for the preceding 3 years being kept under the control of the company**
- Records must be available on board for 5 years
- Records must be available on board until the ship is scrapped
- Two years records must be available on board or kept under the control of the company

Which one of the following describes the best way to find out how to take on ballast on a ship which complies with the Ballast Water Management Convention?

- **Read the Ballast Water Management Plan**
- Complete Seagull CBT No 27
- Read the manufacturers manual for the ballast water treatment system
- Read the Stability Book

Which one of the following describes the best way to find out if a ship with an International Ballast Water Management Certificate has been granted any exemptions?

- **Any exemptions must be recorded in the Ballast Water Record Book**
- Any exemptions will be in the ship's certificate file
- Any exemptions will be listed on the IMO website
- Any exemptions will be listed on the ship's Classification Certificate

Which one of the following is a technology used in ballast water treatment systems to separate liquids and solids?

- **Coagulation**
- Cavitation
- Centripetalism
- Cohabitation

Which one of the following options best describes how ballast water can be deoxygenated within a ballast water treatment system?

- **By creating a vacuum above the water or by adding an inert gas with a low oxygen level**
- By breaking the oxygen from water using ultrasonic waves
- By creating a vacuum above the water using a vacuum pump
- By injecting chemicals that react with and use up the oxygen in the water

Which one of the following options best describes how the Ballast Water Management Convention aims to prevent harmful organisms and pathogens building up within sediment inside ballast tanks?

- **The volume of sediment inside the ballast tanks has to be monitored at regular intervals and cleaned when necessary**
- Ballast tanks are to be cleaned of sediment at every dry docking
- Ballast tanks are to be fitted with tank washing guns
- Sediment filters are to be fitted to ballast intake sea connections

Which one of the following options best describes how the IMO has addressed safety concerns about the use of "active substances" as part of a ballast water treatment system?

- **The risks to the crew and the marine environment from active substances must be assessed before they are approved for use**
- Chemical data sheets are required by MARPOL
- Chemical data sheets are required by SOLAS
- They are to be handled in accordance with MARPOL Annex II

Which one of the following options best describes one of the aspects of training required for officers and crew engaged in ballast water exchange, according to the Ballast Water Management Convention?

- **The method of ensuring that sounding pipes are clear, and that air pipes and their non-return devices are in good order**
- Ballast pump maintenance requirements
- Methods of proving that ballast pipes are clear
- Tank coatings

Which one of the following options best describes the basic principle of a "hydrocyclone", as used in some ballast water treatment systems?

- **Water flow is used to create a vortex in which clean water flows around and upwards with the denser particles falling downwards**
- A hydrocyclone is injected into a static filter to clean and extend the life of the filter elements
- The rotor in the hydrocyclone throws heavier particles outwards so the clean water can flow downwards
- Water flow is used to throw heavy particles outwards and through a filter mesh

Which one of the following options best describes the basic principle of using ultra violet (UV) light as a ballast water treatment?

- **It works by damaging the DNA of the living organisms as they pass by the light, so that cells cannot reproduce or regenerate**
- It works by raising the temperature of organisms so that they cannot survive
- The light produces a chemical in the water which is toxic to aquatic organisms
- The light reacts with the proteins in cell walls causing them to bend and break

Which one of the following options best describes the basic principle of "flocculation", as used in some ballast water treatment systems?

- **Flocculants are added to the ballast water to increase the size or density of the particles so they are easier to separate**
- Flocculants are added to the ballast water to kill any organisms or pathogens
- Flocculants are magnetic and cling to rust particles so they are easier to separate
- Flocculation is the act of passing water through a fabric, which removes solids

Which one of the following options best describes the basic principle used in advanced oxidation technology as a ballast water treatment?

- **It works by light producing free radicals, which destroy the cell membranes of the living organisms**
- Advanced chemical preparations are used to actively target organisms
- It works by raising the temperature of organisms so that they cannot survive
- The light reacts with the proteins in cell walls causing them to bend and break

Which one of the following options best describes the main action of non-oxidising biocides when used in a ballast water treatment system?

- **Non-oxidizing biocides kill by disrupting the metabolic or reproductive processes of organisms**
- Non-oxidizing biocides kill by emitting radioactivity
- Non-oxidizing biocides kill by increasing the oxygen level in the water
- Non-oxidizing biocides kill by removing the oxygen from the water

Which one of the following options best describes the main advantage claimed for using electrochemical activated water as a ballast water treatment?

- **Low electrical demand and no need to supply or handle dangerous chemicals**
- After treatment is not needed
- It protects ballast tanks against corrosion
- It works at low temperatures, like those found in the Arctic

Which one of the following options best describes the main advantage of using oxidising biocides as a ballast water treatment?

- **Treatment systems using oxidising biocides are comparatively simple and require little electrical power**
- Oxidising biocides are very safe to handle and store
- Oxidising biocides create no harmful by-product
- The running costs of oxidising biocides treatment systems are very low

Which one of the following options best describes the main aim of the Ballast Water Management Convention?

- **To prevent the movement of harmful aquatic organisms and pathogens**
- To prevent the movement of pathogens
- To prevent the pollution of the sea by coastal muds
- To reduce the use of water for ballast on ships

Which one of the following options best describes the main way or ways that cavitation is created within a ballast water treatment system?

- **It is produced by injecting gas into the water or by using ultrasonic waves**
- It is produced by electrical discharges from a ring of electrodes
- It is produced by spinning a set of small propellers within the water flow
- It is produced by spinning the water around in a tight circle

Which one of the following options best describes the precaution that all ships, whether they are using ballast exchange or treatment, can take to try and minimise the chances of transporting an alien species to an area where it may be invasive?

- **Try to take on ballast where the number of organisms in the water is low**
- Try to heat the ballast using sunshine or waste heat from engines
- Try to keep ballast pump sea strainers clean
- Try to take on ballast only when inside ports

Which one of the following options best describes who onboard a ship needs to have knowledge of the ship's Ballast Water Management Plan, according to the Ballast Water Management Convention?

- **All personnel involved in ballast handling, appropriate to their duties**
- All deck and engineer officers and ratings
- All officers must be familiar with the plan
- All personnel onboard

Which one of the following options contains the names of two of the methods for ballast water exchange at sea described in the Ballast Water Management Convention?

- **Sequential and Flow-through**
- Dilution and Overfilled
- Flow-through and Mechanical
- Sequential and Mechanical

Which one of the following options is a technique that may be used as part of a ballast water treatment system approved under the procedures in the Ballast Water Management Convention?

- **Electrochemical Activated Water**
- Activated charcoal
- Aging
- Oxygenation

Which one of the following options is the best estimate of the number of different species believed to be currently carried in ballast tanks around the world each day?

- **Around 7,000**
- Around 70
- Around 70,000
- Around 700

Which one of the following options would be the best way to find out who the Ballast Water Management Officer is on your ship and what their responsibilities are?

- **Look in the ship's Ballast Water Management Plan**
- Look at the ship's muster list
- Look in the ship's emergency response plan
- Look in the ship's SOPEP

0028 – P & I Insurance

Version 2.0 (1189)

By definition, what type of organisations are P&I Associations?

- **Non-profit making, mutual societies.**
- International commercial enterprises.
- Public, limited-liability companies.
- Registered banking subsidiaries.

How do P&I Associations secure their economic position?

- **From a system of mutual support.**
- By minimizing insurance settlements through clever interpretation of insurance rules.
- By pursuing strong counter claims through the courts.
- From levying high premiums on ship owners.

How does a ship owner or charterer become a Member of a P&I Association?

- **By insuring a vessel with that particular Association.**
- By applying to the Association's Membership Committee.
- By producing a valid Certificate of Fitness from an approved Classificiation Society.
- By proposal from an existing member and a vote at the Annual General Meeting of Members.

In respect of an act of piracy, what claims will be met by P&I insurance? (Select all appropriate answers)

- **crew substitution.**
- **illness.**
- **loss of life.**
- **loss of personal effects.**
- **personal injury.**
- **repatriation.**
- an incident later assessed as an act of terrorism.

In respect of collision between two vessels, a number of factors are not usually covered by Hull and Machinery insurance and are met instead through P&I insurance. (Select those items which are covered in this way).

- **Death.**
- **Oil pollution.**
- **Personal injury.**
- Damage to property.
- Loss of hire.

In respect of goods which are intended to be, or being, or which have been carried in the entered ship, which of the following claims are covered by P&I insurance? (Select all appropriate answers)

- **Damage.**
- **Loss.**
- **Short delivery.**
- **Theft.**
- Damage whilst in independent storage.
- Loss of commercial value during transit.

There is an outbreak of food poisoning on board, which affects every crew member in one way or another. The vessel diverts and spends three days at anchor off an unscheduled port. Seven crew members are hospitalised and four of those eventually repatriated. Which of the following would be covered by P&I insurance? (Select all appropriate answers)

- **Disinfection expenses.**
- **Extraordinary expenses associated with the outbreak.**
- **Hospital costs.**
- **Quarantine costs.**
- **Repatriation costs.**
- Costs of local press representatives covering the incident.
- Refund of provisioning bills.

What are the two elements of the insurance premium that a ship owner will pay to a P&I Association? (Select all appropriate answers)

- **Supplementary contributions.**
- **The advance premium.**
- Conditional payments.
- Limited payments.
- The mandatory premium.
- The policy premium.
- The statutory premium.

What is a supplementary contribution?

- **A contribution to be paid after the end of the policy year, based on the performance to the whole membership of the P&I Association.**
- A contribution to be paid after the end of the policy year and dependant on the performance of the individual vessel/owner.
- A mandatory contribution to be paid after the end of the policy year, applicable to Members who have made very large claims.

What is an 'Advance Premium'?

- **A sum agreed between the owner and the Association for a specified term of insurance.**
- A fixed sum paid for marine P&I insurance, taken from an international scale and based on gross tonnage.
- A supplementary premium paid before a specific voyage to insure risks that are not covered under the general policy.
- A variable sum, depending on the performance of all Members considered together, paid at the end of a policy year.

What name is generally given to appointed representatives of P&I Associations in ports around the world?

- **Correspondents.**
- Agents.
- Defendants.
- Handlers.
- Members.
- Overseas Inspectors.

What was it, from around 1850 onwards, that brought about the creation of P&I insurance?

- **An increase in third party and contractual liabilities of shipowners.**
- A significant increase in international commercial trade.
- An increase in ship and cargo losses due to maritime warfare.
- The size of the world maritime fleet reached 1 million tons.

What, basically, is P&I insurance?

- **Third Party liability insurance.**
- Cargo insurance.
- Comprehensive hull insurance.
- Crew insurance.
- Underwritten insurance cover for all maritime claims.

When should the Master release the cargo to the consignee(s) without production of the Bills of Lading?

- **When explicit instructions have been given to that effect by the ship owner/operator to the Master.**
- Never.
- When explicit instructions have been given to that effect by the receivers to the Master.
- When the shipper and receiver are the same person or company.

Which of the following are covered by P&I insurance? (Select all appropriate answers)

- **Costs of compulsorily raising of wreck.**
- **Damage to fixed and/or floating objects.**
- **Damage to other ships.**
- **Damage without contact.**
- **Indemnity.**
- **Pollution.**
- Acts of terrorism.
- Structural damage to the vessel resulting from grounding.

0029 – Hull and Machinery

Version 3.0 (881)

As explained in this module, for which of the following reasons should the Hull and Machinery Insurer be involved in all stages of an insurable incident? (Select all applicable answers).

- **Because damage may not be immediately known.**
- **Because repairs may be deferred.**
- Because all associated costs must be settled within twelve months.
- Because it is a legal requirement.
- Because the Insurer may refuse to pay out unless they are fully involved.

From the list below, select the type of claims covered by Hull and Machinery insurance (Select all applicable answers).

- **Liability in connection with fixed or floating objects.**
- **Loss or damage to the vessel.**
- Liabilities to third parties.
- Loss caused by ordinary wear and tear.
- Loss of income.

In all cases of loss or damage covered by a vessel's Hull and Machinery policy, what is considered necessary to be able to compile a repair specification and cost estimate? (Select the most applicable answer)

- **A survey sufficient to gain all necessary information.**
- A meeting of representatives of all insurers.
- Full dry-docking.
- The appointment of a Claims Leader or Independent Adjuster.

In documenting repair and yard costs for the purposes of an insurance claim, which of the following are items to be included? (Select all applicable answers)

- **agent's general account**
- **bunker and other consumables during the period off hire and under repair**
- **fees and charges paid to Surveyors**
- **port, pilot and tug costs**
- **vouchers for any special payments to crew**
- loss of hire costs
- repainting the accommodation exterior, which you took the opportunity to do whilst the ship was being repaired.

What are the three general "stages" of an incident from an insurance point of view and as described in this module?

- **The incident.**
- **The loss or damage.**
- **The repairs.**
- Pollution.
- The payment.

What is an "adjustment" in the context of marine insurance and as outlined in this module?

- **A detailed summary document, prepared by an adjuster, listing all items related to a claim,**
- A tax refund, based on loss of earnings due to an incident.
- The difference between the sum claimed by an owner and that agreed by the Claims Leader, minus any deductibles.
- The increase in premium paid by an Owner, following a major claim.
- The sum allocated by the Claims Leader or Independent Adjuster to be proportionately shared by participating insurers.

What is the essential difference in the way that insurance claims are handled and processed between the Scandinavian system and the British / American system?

- **The Scandinavian system uses a Claims Leader to organize and supervise all claims related to an incident, while the British / American system uses an Independent Adjuster.**
- The Claims Leader in the Scandinavian system is paid out of the claim settlement, while the Independent Adjuster in the British / American system is jointly paid for by the various insurers.
- The Scandinavian system is independent of national law, while the British or American systems is governed and regulated by the legal systems in those countries.
- The Scandinavian system makes interim payments as a clim is being processed, while the British / American system pays a single lump sum on final settlement.

What kind of special deductible is sometimes agreed to?

- **One applying to machinery claims.**
- One accounting for currency variations.
- One applying to personal injury claims.
- One covering war risk claims.

Where will primary procedures and reporting forms relating to a ship's involvement in an "incident" be found?

- **In the company Safety Management System.**
- From the embassy at the next port of call.
- In the manufacturer's manuals.
- On the Classification Society website.
- They can be requested from the Agent at the next port of call.

Which of the following are obligations on the insured party under the Hull and Machinery policy? (Select all applicable answers).

- **To minimise any loss or damage.**
- **To report, without delay, any damage or deficiencies, which may lead to a possible claim under the policy.**
- Not to become directly involved in any incident resulting in loss or damage, but to document its extent fully for the purposes of submitting a claim.
- To ensure that any claims are submitted within 6 months of an incident.
- To maximize sums posted in any claims for loss or damage.

Which of the following is another term for the Hull and Machinery insurer?

- **The Underwriter.**
- The Adjuster.
- The Claims Leader.
- The Under-Assessor.
- The Undertaker.

Which of the following might legitimately be granted access to a vessel in her first port of call following involvement in an incident that results in an insurance claim? (Select all applicable answers)

- **Flag and Port State representatives.**
- **Lawyer representing Hull and Machinery insurers / Owners.**
- **Surveyor representing Classification Society.**
- **Surveyor representing Hull and Machinery insurers.**
- **Surveyor representing other party's insurers.**
- **Surveyor representing P&I Association.**
- Media personnel limited to local coverage.
- Media Personnel specifically engaged by international TV stations.

Who commonly collects the settlement from the various insurers?

- **The Insurance Broker.**
- The Admiralty Court.
- The Claims Leader.
- The Independent Adjuster.
- The Owner (directly).

Your vessel has been involved in a collision. Damage is limited and she is able to proceed to the next port. Once alongside, a surveyor comes up the gangway and introduces himself as representing the owners of the other ship involved in the collision. How should this person be treated?

- **With courtesy, but he should be given only limited information.**
- He should be allowed on board but told nothing and not allowed to take any photographs.
- He should not be allowed on board under any circumstances.
- The Agent should be asked to summon the police and have him removed.
- With courtesy and he should be given any information he requests.

0030 – Marine Lubricants

- High ring/liner wear rates: Abrasive wear. **Excessive cat fines in fuel.**
- High ring/liner wear rates: Adhesive wear **Insufficient oil to maintain an adequate film**
- High ring/liner wear rates: Corrosive wear. **High sulphur fuel.**

Lubricant base oils generally exhibit boiling points above:

- **300 degrees C.**
- 100 degrees C.
- 150 degrees C.
- 200 degrees C.

Marine lube oil extracted from Naphthenic Base oil is suitable as

- **Industrial process oils**
- **Refrigerator oils**
- Engine oils
- Turbine oils

Match the following

- Boundary lubrication **Lubrication in which a considerable degree of contact occurs between the mating surfaces**
- Extreme pressure lubrication **Solid lubricants when in extreme conditions of operation.**
- Fluid lubrication **Liquid lubricants reduce friction by completely separating the moving surfaces**

Match the following

- BN decrease **Sulphuric acids, oxidation acids.**
- Flash point **Fuel Contamination**
- Viscosity increase **Oxidation, insolubles build-up.**
- Water **Combustion, coolant leaks.**

Match the following

- Fluoroesters and fluorocarbons **Inert fluids with excellent chemical stability, low volatility and high autoignition tempera**
- Phosphate esters **Good fire-resistant lubricants with natural anti-wear properties.**
- Polyglycols **copolymers of ethylene- and propylene oxides**
- Silicones and silicate esters **Possess extremely high viscosity indices, excellent low temperature fluidity.**

Mineral oils are generally complex mixture of Hydrocarbons. What is the range of presence of carbon atoms in one molecule of hydrocarbon in a Lubricant base oils?

- **20 -50**
- 50- 70

Name the properties that are exhibited by Aromatic hydrocarbons?

- **Poor chemical stability**
- high pour points
- high viscosity indices

Name the two properties of Marine Lube oil extracted from paraffin base oil?

- **High viscosity index**
- **Low volatility/evaporation loss**
- High pour point
- High thermal stability/resistance to cracking

Solvent extraction is a:

- **physical process removing aromatics**
- atmospheric distillation process
- hybrid process
- vacuum distillation process

Synthetic hydrocarbons are:

- **polyolefins, hydrocracked mineral oils and alkylates**
- fluoroalkylether and silicate esters
- organic esters
- polyethers and polyalkylene glycols

Synthetic oils are manufactured by

- **Polymerisation of selected monomers**
- **Synthesised by chemical processes using a variety of raw materials.**
- By Hybrid process
- Hydro-catalytic Process

The main advantages of Synthetic oil over mineral oil are

- **Longer Oil life**
- **The properties required by the equipments can be accurately met**
- Low cost factor
- Low water absorption

The refined lubricant base oil properties are greatly influenced by:

- **crude oil composition or origin**
- chemical treatment
- refining process
- storage and treatment

This is an example of . . .

- **Elastohydrodynamic lubrication**
- Boundary Lubrication
- Extreme pressure lubrication
- Hydrodynamic lubrication

What is EP (Extreme Pressure) oil?

- **Additive forming a protective chemical surface film**
- High maximum pressure cylinder oil.
- High pressure hydraulic oil
- Standard gear oil

What is mixed lubrication?

- **When formation of a fluid film is partly established**
- When changing type of lubricating oil from time to time
- When mixing mineral and synthetic oils
- When using grease and lubricating oil together

What is the "NLGI" number of a grease indicating?

- **How fluid or non-fluid the grease is**
- How water resistant the grease is
- The grease compatibility
- The grease expected life time.

What is typical BN level for a trunk piston engine operating on residual fuel oil?

- **BN 30 - BN 40**
- BN 40 - BN 50
- BN 5 - BN 10
- BN 70 - BN 100

What type of refining process will greatly reduce the influence Crude oil Origin?

- **Severely hydroprocessed**
- Solvent refining

0031 – Pusnes ETS

Version 4.0 (1220)

Deployment, Emergency procedure: Which statement is correct?

- **After breaking the weak link in the securing device the tugboat should let the wire run out by its own weight.**
- Both the pick-up gear and the towing wire must be in the water before the tugboat can pick it up.
- The tugboat shall always use full engine power to pull out the towing wire.
- When the tugboat has secured the connection, it can start pulling out the towing wire, preferable at full engine power to break the weak link.

How often is it necessary to mount the air motor and unreel the entire length of the towing wire and check for any damage?

- **In the 2-year Maintenance**
- In the 12th months Maintenance
- In the 6th months Maintenance
- In the daily Maintenance

How often must the centrifugal brake be inspected?

- **In the 6th months Maintenance**
- In the 12th months Maintenance
- In the 2-year Maintenance
- In the daily Maintenance

How often shall the weak link and the turnbuckle be checked?

- **In the 6th months Maintenance**
- In the 12th months Maintenance
- In the 2-year Maintenance
- In the daily Maintenance

Move the names of the three types of ETS to their correspondent picture, and click the Continue-button when finished.



AFT ETS below deck



Fore ETS

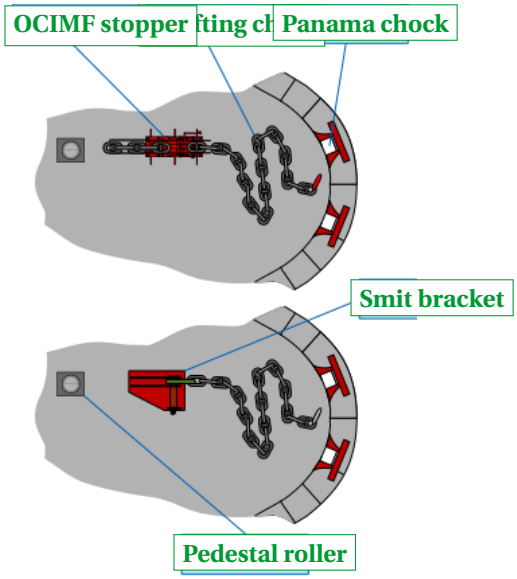


AFT ETS on deck

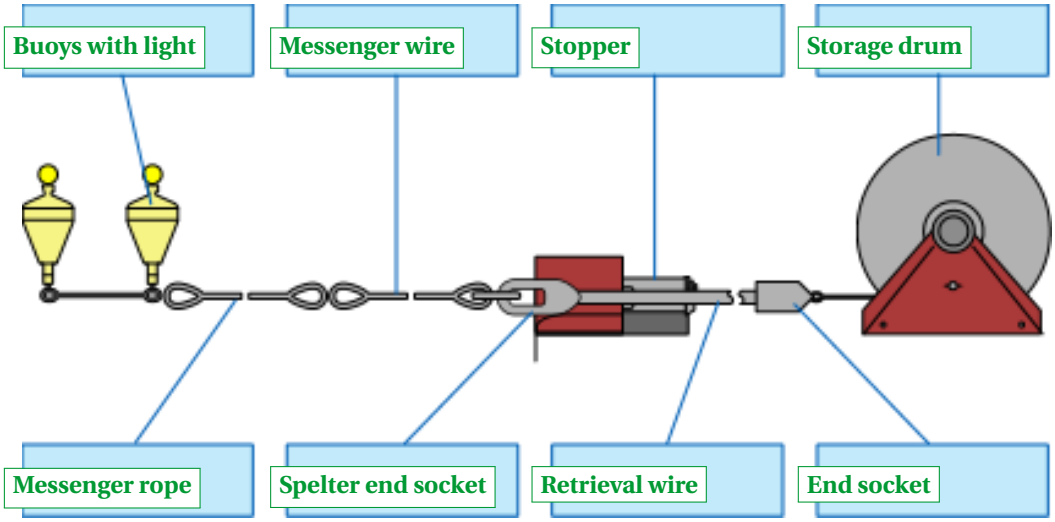
Packing of pick-up gear container procedure: How do you check that the light bulbs is flashing?

- **Hold the buoys in an upright position for a few seconds.**
- Move the buoys up then down rapidly again a few times.
- Turn the boys with the bulbs pointing down for a few seconds and up again.
- Turn the light bulb off then on again for a few seconds.

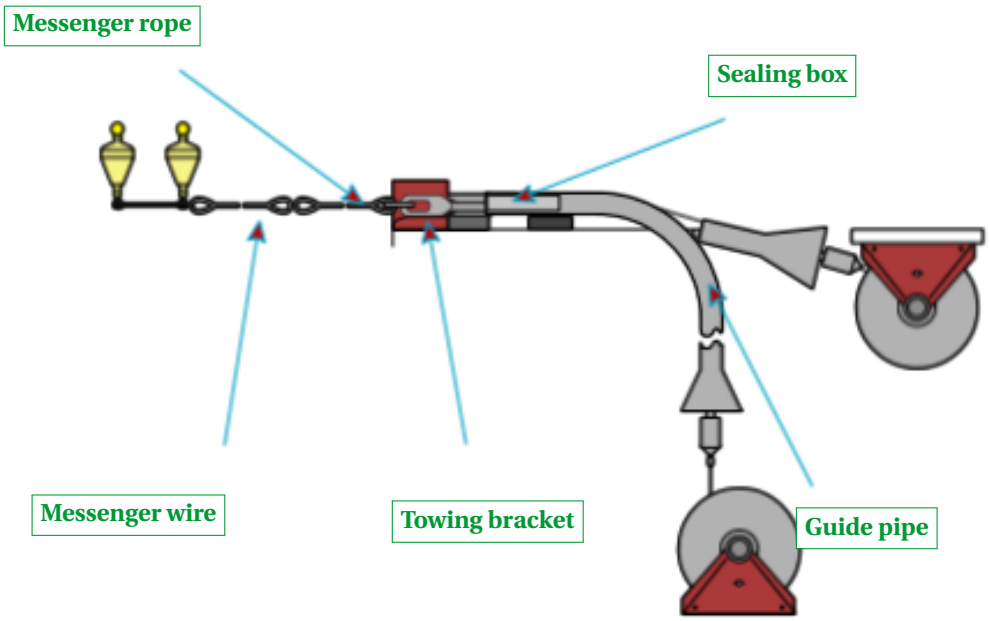
Place the part names in the correspondent box, and click the Continue-button when finished.



Place the part names in the correspondent box, and click the Continue-button when finished.



Place the part names in the correspondent box, and click the Continue-button when finished.



Test Procedure: Which statement is correct when you are following the test procedure?

- **Remove the weak link at the storage drum, before you go to the pick-up gear container and throw the buoys in the water.**
- Go to the pick-up gear container and throw the buoys in the water, before removing the weak link at the storage drum.
- In the test procedure, the wire will not run out by its own weight.
- You must never feed the messenger rope and wire into the sea.

The ETS 100D is designed for the following ship type?

- **Tankers between 20 and 50.000 tdw.**
- Tankers above 60.000 tdw.
- Tankers below 30.000 tdw.
- Tankers between 30 and 60.000 tdw.

What is the correct thing to do when the light bulbs are not flashing when holding the buoys in an upright position?

- **Replace the batteries and/or the electronic unit with bulb.**
- First replace the bulb then replace the batteries.
- Replace the batteries and/or the lamp.
- Replace the entire top unit.

What is the maximum allowed deployment time for the forward ets system?

- **1 hour**
- 1.5 hours
- 15 minutes
- 30 minutes

What is the maximum allowed deployment time of the pick-up gear for one man? (Aft System)

- **15 minutes**
- 10 minutes
- 20 minutes
- 5 minutes

When shall the three batteries in each buoy be replaced?

- **In the 12th months Maintenance**
- In the 2-year Maintenance
- In the 6th months Maintenance
- In the daily Maintenance

0032 – Cargo Properties

Version 3.0 (980)

A liquid at different temperatures is made to flow through identical orifice passages. The one with the easiest flow will have liquid at:

- **The highest temperature**
- The average temperature
- The lowest temperature

During a cargo voyage, the inert gas pressure in the tank will decrease as a result of:

- **Inert gas being absorbed by the liquid (diffusion).**
- Increasing viscosity level during the voyage.
- Leaks which always are present due to the pipeline system, gauges etc.

Hydrocarbons belongs to a chemical group named:

- **Organic Substances**
- Alkanes
- Carbohydrates

Light products are more volatile than heavier products. This means that light products:

- **Have lower viscosity**
- Have higher flash point
- Have higher viscosity

On a fully refrigerated ship, gas would be transported:

- **Close to its atmospheric boiling point**
- In high viscosity tank domes
- In insulated pressure tanks to keep the gas cold

The O2 meter shows an oxygen level of 18.3 % in a tank that you are to inspect. This means that:

- **You cannot enter the tank because the body requires air with the content of 21% oxygen for normal breathing.**
- You can enter the tank safely, because the body only requires air with the content of 17.8% oxygen.
- You cannot enter the tank because the body requires air with the content of 22.8% oxygen for normal breathing.

The flammable range in air of ethylene is 3% - 32%. This means that:

- **Ethylene vapour in proportion 3% - 32% to air, is flammable.**
- Ethylene vapour is flammable when there is 3% - 32% air present in the mixture.
- Ethylene vapour is flammable when there is 3% - 32% oxygen present in the mixture.

The gases present in crude oil are termed:

- **Natural Gas Liquids (NGL)**
- Inert Gas
- Liquefied Petroleum Gas (LPG)

The molecular structure below is named:

- **Benzene**
- Carbohydrate
- Cyclohexane

Threshold Limit Value is:

- **A measure of the toxicity level in the atmosphere**
- A measure of the weight of a substance (kg/dm3)
- The maximal partial pressure (%)

What information should always be to find on the datasheet of a substance?

- **Information regarding safe handling of the product.**
- Production date and level of flammability.
- The price and quality of the product.

0033 – Stability

Version 4.0 (3288)

A ship floating on an even keel and upright has a weight loaded on the port side of the deck and towards the stern of the vessel. What will be the final condition of the ship after loading? Select the ONE correct answer.

- **Listed to port and trimmed by the stern**
- Heeled to port and trimmed by the head
- Heeled to port and trimmed by the stern
- Upright and trimmed by the stern

A ship with a large Transverse GM will have which ONE of the following rolling periods? Select the ONE correct answer.

- **Fast and to small angles of heel**
- A moderately comfortable period with medium angles of heel
- Slow period of roll either side of an angle of loll.
- Very slow and to large angles of heel.

A tank on a vessel can hold 100 tonnes of FW when full. How many tonnes of liquid (Relative density 0.5) could it take when full? Select the ONE correct answer.

- **50 tonnes**
- 100 tonnes
- 150 tonnes
- 200 tonnes.

A weight of 200 tonnes is loaded amidships on the deck at a height of 7 metres above the keel on a ship of 3800 tonnes and KG 5.0 metres. What is the final KG of the vessel after loading?

- **5.1 metres**
- 4.9 metres
- 5.0 metres
- 5.2 metres

Add the appropriate missing words into the following sentences:

- The Centre of Gravity G will move **parallel to** a moved weight already on board
- The Centre of Gravity G will move directly **away from** a weight discharged
- The Centre of Gravity G will move directly **towards** a weight added.

At what point during the dry docking of a ship does the upthrust from buoyancy become zero? Select the ONE correct answer.

- **When the water drops below the keel**
- Never
- When the vessel settles on the block overall.
- When the vessel touches the blocks aft

How can a ship's officer change the value of the Transverse Metacentric Height? Select the ONE correct answer.

- **By loading or discharging cargo and in other ways changing the positions of both G and M**
- By loading or discharging cargo and changing the position of G only
- By vertically shifting cargo already on board and changing the position of both G and M
- Only by loading cargo and changing the position of Centre of Gravity "G" of the ship.

How can the adverse effects of Free Surface on a ship be reduced by a ship's officer? Select the TWO best answers.

- **The draining of the liquid from any slack tanks**
- **The pressing up of the liquid within any slack tanks**
- Loading more ballast and increasing the displacement
- Transferring ballast from high tank to a low tank

If a vessel is rolling heavily in a seaway and has a negative GM and a small Angle of Loll, describe how the vessel will roll. Select the ONE best answer.

- **Roll unevenly flopping over between the Angle of loll on each side**
- Roll between the angle of loll on each side
- Roll evenly each side of the angle of loll
- Roll evenly either side of the upright position

The buoyancy forces upwards act at what points on the hull?

- **All over the underwater volume**
- Along the keel
- At the Centre of Buoyancy B.
- At the Centre of Gravity G

The condition of Equilibrium for a floating vessel is which of the following conditions? Select the ONE best answer.

- **When G is vertically below B with equal and opposite forces of displacement and buoyancy**
- When the ship is upright, the Fore and Aft draughts are steady and the GM is positive
- When the vessel completes loading or discharging cargo
- When the vessel is upright and on an even keel and GM is positive

The ships own derrick is used to discharge a heavy weight from the centreline of deck to the jetty. At what instant does the most dangerous situation occur with respect to the ships effective stability? Select the ONE correct answer.

- **When the weight is swung out over the jetty and starts to be lowered**
- When the weight is being swung over towards the jetty
- When the weight is landed on the jetty
- When the weight is lifted off the deck.

What are the units of Relative Density? Select the ONE correct answer.

- **Has no units**
- Cubic metre per tonne
- Tonnes
- Tonnes per cubic metre

What does the term "Block Coefficient" mean? Select the ONE correct answer.

- **The ratio of the ships underwater shape to a rectangular block of the same extreme dimensions**
- The efficiency of the derrick topping lift used to load cargo.
- The ratio of the ship's Waterplane area to a rectangle of the same extreme dimensions
- The ratio of the vessel's even keel, underwater shape to the actual underwater shape when trimmed

What does the term "Layer Correction" mean? Select the ONE correct answer.

- **It is the correction between the Draught Amidships and the True Mean Draught**
- It is a correction to apply to the lightship draught to allow for ballast already on board
- It is another term for the Dock Water Allowance
- It is the vertical separation between two parallel waterlines.

What is lever which returns the vessel to the upright? Select the ONE correct answer.

- **GZ**
- BM
- GM
- KM

What is minimum value of the GM allowed under the international Load Line Rules? Select the ONE correct answer.

- **0.15 m**
- 0.25m
- 1.5 m
- 2.0 m

What is the Curve of Transverse Statical Stability? Select the ONE best answer.

- **A graph showing the GZ values for different angles of heel**
- A curve showing the stability of a vessel at different displacements
- A graph showing the value of GM at different displacements
- A graph showing the values of KM for different angles of heel

What is the change in the apparent loss of GM due to free surface effect when two transverse bulkheads are introduced into the tank holding the liquid?

- **Nothing**
- 1/2
- 1/4
- 1/9

What is the danger of having a large bunker tank only partially filled with bunkers? Select the ONE best answer.

- **Free Surface Effect and virtual loss of GM**
- Excessive corrosion of the tank sides
- Insufficient bunkers to complete the voyage
- Structural damage due to liquid movement in the tank

What is the meaning of the term "Ship's Displacement"? Select the ONE correct answer.

- **It is the weight of the ship and everything onboard**
- It is the lightship weight of a ship only.
- It is the underwater volume of a ship.
- It is the weight of cargo on board a ship

What is the purpose of the "Inclining Experiment"? Select the ONE correct answer.

- **To determine the lightship KG of a vessel**
- To determine the angle of inclination when a known weight is shifted a known transverse distance across the vessel.
- To determine the GG_1 when shifting a known weight across the deck
- To determine the lightship displacement of a vessel

What is the relationship between GM and the righting ability of the ship? Select the ONE correct answer.

- **GZ = GM x sin (Angle of Heel)**
- GM is the righting lever returning the vessel upright
- GZ = GM x tan (Angle of Heel)
- The Curve of Statical Stability

What is the "Angle of Loll"? Select the ONE correct answer.

- **The angle of equilibrium for a ship with a negative upright GM**
- The angle of heel experienced in a steady beam wind
- The maximum angle to which a vessel inclines when discharging a heavy weight using the ship's derrick.
- The trim angle measured at the after perpendicular

Which ONE of the following would be the specific point about which a ship will roll in a seaway? Select the ONE correct answer.

- **The Centre of Flotation, CF**
- The Centre of Buoyancy, B.
- The Centre of Gravity, G
- The Centreline, CL

Which THREE of the following items of information can be determined from examining the Curve of Statical Stability? Select the THREE correct options.

- **Angle of Maximum Stability**
- **The Initial value of Transverse GM**
- **The maximum value of the Righting Lever**
- The Righting Moment of the vessel at a specific angle of heel

Which TWO of the following cause a small positive GM at the start of a voyage to become dangerous during the passage? Select the TWO correct answers.

- **Consumption of bunkers in a low tank**
- **Partially filled tanks giving free surface effect**
- Consumption of bunkers in a high tank
- Distribution of weights due to bad stowage

Which of the following initial conditions onboard are required to conduct an inclining experiment? Select each correct option.

- **Minimum bunkers, water and ballast**
- **Minimum of personnel onboard to conduct experiment**
- **No cargo on board**
- **No free liquid surface in any tanks**

Why can the draught amidships not be used to extract the value of displacement from the displacement table? Select the ONE correct answer.

- **The draught amidships is not the true mean draught**
- The arithmetical mean of the end draughts should be used to determine correct displacement.
- The displacement is not dependent on the draught
- The draught amidships is not the mean of the draughts at the forward and after perpendiculars

Why is a vessel entering dry dock normally slightly trimmed by the stern? Select the ONE best answer.

- **To position the vessel easily over the blocks**
- To allow better manoeuvring when entering into the dry dock
- To allow drainage of water off the decks
- To minimise the loss of stability

0034 – AC4, Digital Governor System

Version 2.0 (1532)

.Which of the factors does not cause governor limits to be changed during service

- **Engine Room temperatures**
- Fuel Oil Quality Variations
- Hull Fouling
- Wear of Fuel Pumps

The RPM Command function does not do which of the following

- **Manual setting of fuel supply reference and speed setting Reference**
- Commanding Input signal from remote control system or control room lever
- Control Engine Acceleration or deceleration
- Critical Speed Discrimination

The RPM reference values are

- **Computed for all Engine Running Conditions**
- Computed manually by operator
- Computed only for some Engine Running Conditions
- Stored in memory for later use

The Servomotor of the fuel actuator has a brake which acts to keep fuel lever in

- **Current position**
- Mid Position
- New Position
- Non fail safe position

What are the Actuator main functions?

- **Positioning the actuator (and fuel-rack) according to commanded value**
- **Sense the actuator position command from the Regulator**
- Displaying Data Values
- Limiting speed of fuel-rack
- Output for fuel-rack (actuator) indication (option)
- Repeated testing of System Failures

What are the Regulator main functions?

- **Output and limit the command signal to the fuel actuating function**
- **Speed Measurement and Filtering**
- **Speed Reference Computation**
- Displaying Data Values
- Off-line Testing
- On-line Testing
- Programming user-dependent parameters
- Selection of alternative mode operations

Which Data is Adjustable?

- **RPM Limiter level**
- Measured RPM
- RPM Command
- Scav. Air Pressure

Which Data is Readable?

- **Scavenge Air Pressure**
- Dead Band
- Fuel Limiter level
- RPM limiter level

0035 – Inspections

Version 3.0 (39)

A standard pre-arrival procedure should include:

- **A good wipe down of oily surfaces**
- A briefing in unsafe work practices
- A meeting on increased fire hazards
- Re-assignment of emergency duties

A vetting is:

- **A risk management process used by oil & chemical companies**
- A filing system
- A hazard recognition process
- Required by statutory regulations

An Oxygen & Acetylene cutting torch must be fitted with:

- **Flashback suppressors**
- A proper size torch head
- Analogue gauges
- Digital gauges

Drills aboard ship must:

- **Demonstrate crew competence**
- Comply with ISGOTT
- Satisfy vetting inspectors
- Thorough confuse the crew

During a fire drill, the crew must ensure that:

- **Water pressure is established, electrical power & ventilation to the area is secured**
- The main engine is properly shut down
- The SMS is placed in the lifeboat
- The steering is in non-follow-up mode

Emergency systems to be tested at regular intervals includes:

- **Fire flaps & quick closing valves**
- **The emergency fire pump**
- **The emergency generator**
- The Main Engine

Europe and Asia joined in identifying substandard ships through:

- **The Paris MOU & The Tokyo MOU**
- The Hong Kong Stock Exchange
- The London MOU & Singapore MOU
- The United Nations MOU

If a serious material defect is known, the Master should:

- **Advise the technical manager to notify the Classification Society**
- Call port state authorities
- Call the flag state
- Hide it from inspectors

In the workshop, a waste bin is considered unsafe if:

- **It is uncovered**
- It is empty
- It is in a corner
- It is made of tin

Mobile phones:

- **Should never be used on the deck of a petroleum tanker**
- Are explosion proof
- Provide safe communications in hazardous atmospheres
- Should be of an approved type.

One of the most common deficiencies in vetting inspections is:

- **Inadequate passage planning**
- Crew apathy
- Failure to properly tend notice of readiness
- Lack of document control

One way to demonstrate knowledge of the dangers posed by cargo is:

- **Posting the MSDS**
- Posting a Gas Free Certificate
- Posting he Document of Compliance
- Posting the Safety Equipment Certificate

Passage planning on tankers must include:

- **Berth to sea buoy and sea buoy to berth**
- **Parallel Indexing Information**
- **The methods & frequency of position fixing**
- Radar Maintenance

Prior to a boat drill, the engine should:

- **Be warmed up to ensure smooth operation when under observation by inspectors**
- Have a safety check
- Have the oil changed
- Have the sea suction disconnected

Several years ago, the USCG introduced risk management methods that include:

- **Targeting Substandard Vessels**
- Checking Fire Extinguisher Manufacturers
- Pressure testing Nitrogen Bottles
- Requiring Life Jacket Drills

Standard Operating Procedures are a major component of:

- **A Safety Management System**
- Charterers requirements
- Marpol rules
- The IMDG Code

Standing Orders need not be signed by navigating officers:

- **False**
- True

Statutory Regulations include:

- **SOLAS Regulations**
- IMO SMCP
- MARPOL Requirements
- The IMDG Code

Tanker crews face additional examination from:

- **Major Oil and Chemical companies**
- Classification Surveyors
- Green Peace
- The P & I Club

The inspector should not:

- **Impose any test that could jeopardize safety**
- Drink coffee
- Make a thorough exam
- Report findings

The poor condition of fire mains and hydrants or the absence of hoses or extinguishers would point to:

- **A need for closer inspection of fire safety equipment**
- A need to examine the hospital
- The ISM Code
- The third mate

Too many general deficiencies would warrant:

- **Inspection of logs & machinery records**
- An abbreviated inspection
- Giving control to sea staff
- Taking control away from sea staff

Work on steam systems does not require a permit to work:

- **False**
- True

0036 – Medical First Aid

Version 2.2 (4263)

A bandage is always applied to an open wound.

- **False**
- True

A casualty with burns has less need for fluid.

- **False**
- True

A chemical burn from fluid must be bandaged firmly to prevent fluid from covering greater parts of the skin.

- **False**
- True

A fractured spine will always imply unconsciousness.

- **False**
- True

A good splint should prevent adequate blood circulation and compress nerves so the casualty will not feel any pain.

- **False**
- True

A haematoma can be caused by head injury.

- **True**
- False

A large dose of muscle relaxing medicaments can cause respiratory problems due to the reduced activity in the respiratory muscles.

- **True**
- False

A limb with an external bleeding should be lowered to maintain circulation in order to supply the cells with oxygen and prevent cell death.

- **False**
- True

A patient who breathes will always have a pulse rate?

- **True**
- False

A patient who is shouting has an open airway.

- **True**
- False

A patient with a major burn is liable to develop circulation failure.

- **True**
- False

A person suffering from a moderate to hard blow to the head should always be under close observation in the early hours after the accident, considering possible development of edeme in the head.

- **True**
- False

A person under influence of alcohol will suffer from frost injuries quicker than a sober person.

- **True**
- False

A seriously hypothermic patient should as soon as possible be placed in a warm bath (>50°C) to regain normal body heat.

- **False**
- True

A spine/neck injury will always have priority over every other injury due to the risk of paralysis.

- **False**
- True

A sudden fall in blood pressure does not normally lead to circulation failure.

- **False**
- True

A triangular bandage is always used in its triangular shape.

- **False**
- True

Abdominal thrust is applied to remove foreign objects in the airway.

- **True**
- False

All fires will burn easier and faster in an oxygen-rich environment. Therefore, do not use oxygen bottles near an open fire. Smoking is not allowed.

- **True**
- False

An internal abdominal bleeding is not dangerous as it will stop bleeding by itself due to the increased pressure in the abdomen.

- **False**
- True

An unconscious patient suffering from a major spine injury should not be placed in recovery position as this may increase the damage and lead to paralysis.

- **False**
- True

Any major bleeding must be stopped before treating a wound.

- **True**
- False

Bandages must be sterile before use.

- **False**
- True

Bleeding will always occur together with a fracture that breaks the skin.

- **True**
- False

Breathing is controlled by the respiratory centre in the brain.

- **True**
- False

Cardiac arrest leads to circulation failure.

- **True**
- False

Cardiac arrest will always lead to unconsciousness.

- **True**
- False

Cooling does not reduce pain in a sprain.

- **False**
- True

Cooling of a third degree burn should be terminated after 15 minutes.

- **False**
- True

Damage to the spinal cord will always result in immediate neurological malfunction.

- **True**
- False

Do not touch anything that is sticking to, or embedded in, the eyeball, or over the coloured part of the eye, when treating a person with a foreign body in the eye.

- **True**
- False

Dressings should always be larger than the wound they cover.

- **True**
- False

Flush away any residual chemical on the skin with plenty of cold water. Be sure that the water does not flush away from the casualty's body.

- **False**
- True

Food poisoning may be caused by eating food that is contaminated by bacteria or by toxins produced by bacteria that were already in the food.

- **True**
- False

Grease and oil must never be used in connection with oxygen bottles, couplings and valves.

- **True**
- False

Head injury will lead to circulation failure.

- **False**
- True

Heat cramps seldom occurs after excessive sweating precipitated by strenuous exercise.

- **False**
- True

Heat-stroke can occur suddenly, causing unconsciousness within minutes. This may be signalled by the casualty feeling uneasy and ill

- **True**
- False

If a patient has heat exhaustion, the symptoms are weakness, vertigo, headache and nausea. Initially the skin is cool and clammy.

- **True**
- False

If there are signs of tendon damage, the tendon must be sutured by a surgeon before closing the wound. If so, put on a dressing until qualified treatment can be administered.

- **True**
- False

If you are unable to give a patient with an increased circulation failure due to bleeding, intravenous treatment, he should promptly be given plenty to drink to replace lost fluid.

- **False**
- True

Improvised splints can only be made of solid materials.

- **False**
- True

Inspiration is mainly caused by muscle activity of the Diaphragm.

- **True**
- False

Internal bleeding will never lead to circulation failure.

- **False**
- True

It is the increased pressure in the head that is fatal, not the bleeding itself.

- **True**
- False

It may be difficult to tell if there is a fracture or a sprain.

- **True**
- False

Moderate and deep hypothermia cause very unstable condition, and the patient must be inactivated and treated very carefully.

- **True**
- False

Most adults will endure a loss of blood of 1 litre.

- **True**
- False

None of the tissue cells will die, only damaged when the casualty has a crush injury.

- **False**
- True

Normal resting rate of pulse in a healthy adult is about 2-3 beats per second.

- **False**
- True

Once a bandage is in place it should be left alone.

- **False**
- True

One of the main tasks of the blood is to transport oxygen to body's cells.

- **True**
- False

Open weave roller bandages are used to support joints

- **False**
- True

Oxygen bottles should be handled with care and should be removed if they are exposed to heat.

- **True**
- False

Oxygen is transported to the cells by blood vessels called veins.

- **False**
- True

Oxygen treatment is most important with injury to the head/neck/spine.

- **True**
- False

Parts of the body can be used as splints.

- **True**
- False

R-I-C-E treatment has no effect if postponed more than 4 hours after the injury took place.

- **False**
- True

Respiration while resting is normal if the frequency increases to more than 20 ventilations per minute.

- **False**
- True

Shock and circulation failure are equivalent.

- **True**
- False

Spillage of corrosive chemicals is poisonous and can also result in burns.

- **True**
- False

Symptoms of serious head injury are unconsciousness, low pulse rate and high blood pressure.

- **True**
- False

The brain of hypothermic patients will consume less oxygen than usual.

- **True**
- False

The main goal of splinting is to render a pulling strain on the fractured bone ends, in order to prevent them from touching each other.

- **False**
- True

The motor function and circulation distal to the fracture should always be assessed.

- **True**
- False

The position and direction of entry and exit wounds will alert you to the likely extent of hidden injury, and to the degree of shock that may ensue.

- **True**
- False

The red blood cells are responsible for the transportation of oxygen.

- **True**
- False

The respiration system of a person suffering from flame burns is very exposed to damage.

- **True**
- False

The respiration system, above all, reacts to carbon dioxide levels in the blood.

- **True**
- False

The walls in the blood vessels are completely resistant to diffusion.

- **False**
- True

There are at least four distinct signs of fracture.

- **True**
- False

There is nothing more you can do to prevent food poisoning, than ensure that frozen poultry and meat are fully defrosted before it is cooked. Cook meat, poultry, fish, and eggs thoughly to kill harmful bacteria. Never keep food lukewarm for long periods; bacteria can multiply without obvious signs of spoilage.

- **False**
- True

To protect a burn injury we should never remove the patients clothing.

- **False**
- True

To render correct first aid one should be certain that there is a fracture before treating the patient.

- **False**
- True

Visible shortening of a limb is not necessarily a sign of a fracture.

- **False**
- True

We have 3 circulatory systems for blood transportation.

- **False**
- True

What is the most important treatment concerning a foreign body in the eye?

- **Rinsing of the eye**
- Local anaesthetic
- Wring the eye-lid to prevent blinking

What is the most serious damage to the tissue concerning damage by pressure?

- **Increasing pressure and developing oedema.**
- Dirt in the wound
- Pain and discolouring

When an ambulance or more qualified personnel arrives at the scene, there is no need to tell them about the poisonous chemicals inhaled since they will find out themselves.

- **False**
- True

When dealing with a heat stroke your aims are to lower the casualty's body temperature as quickly as possible, and to arrange removal of the casualty to hospital. Do NOT use iced water as they may constrict blood vessels.

- **True**
- False

When performing C-P-R it is advisable to elevate the patient's legs to secure sufficient blood to the chest and head.

- **True**
- False

When there is an electrical burn there is a great danger of internal injuries.

- **True**
- False

When treating an injury obtained by falling, it is of great importance to check the motor function in the extremities.

- **True**
- False

When we deal with food poisoning your aims are to encourage the casualty to rest, to seek medical advice or aid, and to give the casualty plenty of bland fluids to drink.

- **True**
- False

You can threat a person with heat exhaustion by giving him salt and water.

- **True**
- False

You should always wash your hands before preparing food. Wear protective gloves or waterproof plasters if you have cuts on your hands.

- **True**
- False

You should not supply oxygen when resuscitating, as this will only disturb the treatment. It is however important to do so immediately after the resuscitation has ended.

- **False**
- True

You should only use double nose catheters when treating two patients on one oxygen bottle.

- **False**
- True

You should terminate resuscitation after 15 minutes.

- **False**
- True

Your aims are to maintain the airway, breathing and circulation, to remove any contaminated clothing, to identify the poison and to obtain medical aid.

- **True**
- False

0040 – Maritime English, Pilot on the Bridge

Version 2.0 (298)

Click on the letter to which this phrase applies

- **B**
- A
- C
- D

Click on the letter to which this phrase applies

- **C**
- A
- B
- D

Click on the letter to which this phrase applies

- **D**
- A
- B
- C

Click on the phrase you hear

- **Ease to 5**
- East to 5
- Easy 5

Click on the phrase you hear

- **Fifteen minutes to the berth, Captain**
- Fifty minutes to the berth, Captain
- Fifty minutes to unberth, Captain

Click on the phrase you hear

- **Lower away the springs please**
- Lower and weigh the springs please
- Lower in way of the springs please

Click on the phrase you hear

- **First line ashore**
- First line in-shore
- First line is short
- First line to shore

Click on the phrase you hear

- **Hard-a-starboard**
- Half to starboard

Click on the phrase you hear

- **Keep the springs slack please**
- Keep the springs back please
- Keep the springs on deck please

Click on the phrase you hear

- **We have a heaving line ashore aft**
- We have a leading line ashore aft
- We'll heave a line on shore aft

Click on the phrase you hear

- **Dead slow astern**
- Go slow astern
- Let's go astern

Click on the phrase you hear

- **In position - make fast**
- In collision - main part
- In position - make a start
- In position - make aft

Click on the phrase you hear

- **That's 'All fast' Pilot**
- That's a start, Pilot
- That's a fast pilot

Click on the phrase you hear

- **All fast, port side to**
- All far ports to do
- All fast, fourth side to

Click on the phrase you hear

- **Are you going off to shore or by boat?**
- Are you going often to shore by boat?

Click on the phrase you hear

- **Checklists completed**
- Check is completed

Click on the phrase you hear

- **Pilot information card**
- Pilot uniform issue card

Click on the phrase you hear

- **We will have two tugs, Captain**
- We will have to talk, Captain
- We will have two tracks, Captain

Click on the phrase you hear

- **We can single up, Captain**
- We can sign them up, Captain
- We can signal up, Captain

Click on the phrase you hear

- **Singled up aft**
- Signals apart
- Signals up aft

Click on the phrase you hear

- **We have clearance from VTS to proceed**
- We have a clear run from Vlissingen to the sea
- We have clear ice from Vlissingen to outside
- We have clearance from VTS to the sea

Click on the phrase you hear

- **Thruster slow to starboard**
- Thrust dead slow to starboard
- Thruster more to starboard

Click on the phrase you hear

- **Gone and clear aft**
- Go and clear aft
- Going to clear aft

Click on the phrase you hear

- **Stop thruster**
- Stock the store
- Stop the stern

Click on the phrase you hear

- **Starboard ten**
- Standard ten
- Start port ten

Click on the phrase you hear

- **Midships**
- Red ships

Click on the phrase you hear

- **Steady as she goes**
- State it as she goes
- Status as she goes
- Steer it as she goes

Click on the phrase you hear

- **Course 306**
- Course is 006

Click on the phrase you hear

- **Let's see if there's any inbound traffic at the river**
- Let's see if there's any in port traffic at the river
- Let's see if there's any traffic around at the river
- Let's see if there's any inboard traffic at the river

Click on the phrase you hear

- **That might slow us down a touch**
- That might slow us down a bunch
- That might slow us down so much

Click on the phrase you hear

- **The mooring arrangement will be the same on the next berth.**
- The mooring or range might be the same on the next berth.

Click on the phrase you hear

- **4, 2 and 2 each end**
- 4 and 2 to each end

Click on the phrase you hear

- **You'll be running springs first**
- You'll be running springs last
- You'll be running sprinklers

Drag and drop the pictures onto the matching labels

Rudder indicator on hard-to-port	
Rudder indicator on midships	
Rudder indicator on port 5	
Rudder indicator on starboard 15	

Drag and drop the pictures onto the matching labels

Mooring boat	
Pilot boat	
Tug	

Drag and drop the pictures onto the matching labels

Rudder indicator	
Telegraph	
Wheel	

Whistle (or horn)	

Drag and drop the pictures onto the matching labels

Echo sounder	
Gyro compass	
Radar	
VHF radio	

Drag and drop the pictures onto the matching labels

First line ashore!	
In position, make fast!	
Pilot on board!	
Tug fast forward!	

Drag and drop the pictures onto the matching labels

Bow thruster control	
Heaving Line	
Pilot ladder	
Portable radio	

Drag and drop the pictures onto the matching labels

3rd Officer	
Captain	
Helmsman	
Pilot	

How were the headlines sent ashore on the 2nd berth?

- **They were sent ashore by mooring boat**
- They were sent ashore by heaving line
- They were sent ashore via the tug
- They were sent direct to shore

In what direction is a ship said to be moving when it is going forward?

- **Ahead**
- Afront
- Astern
- Return

Select the correct phrase

- **It was about forty minutes to the next berth**
- It were about forty minutes to the next berth

Select the correct phrase

- **Some lines were sent ashore by mooring boat**
- Some lines was sent ashore by mooring boat

Select the correct phrase

- **The Helmsman put the wheel to midships**
- The Helmsman put the wheel for midships
- The Helmsman put the wheel in midships

Select the correct phrase

- **The Captain stated that the ship was in position**
- The Captain stated that the ship was for position
- The Captain stated that the ship was on position
- The Captain stated that the ship was up position

Select the correct phrase

- **The Captain and the Pilot were on the bridgewing**
- The Captain and the Pilot was on the bridgewing

Select the correct phrase

The Chief Officer informed the Captain that everything was . . . . . forward

- **gone and clear**
- going too clear
- gone in clear
- gone to clear

Select the correct word

Before the ship moved off the berth, the crew . . . the tugs fast forward and aft.

- **made**
- maid
- mate

Select the correct word

The 2nd Mate . . . the Pilot Ladder

- **arranged**
- arrange
- arranging

Select the correct word

The 3rd Officer . . . the Information Card to the Pilot.

- **gave**
- give
- gived
- giving

Select the correct word

The Captain asked the 3rd Mate if everything . . . ready

- **was**
- were

Select the correct word

The Captain told the 3rd Mate to go down and . . . the Pilot

- **meet**
- meat
- meeting
- met

Select the correct word

The Pilot told the Captain that there would be . . . tugs

- **two**
- to
- too

Select the correct word

The radar was . . . on the three mile range

- **set**
- sat
- send
- sent

Select the correct word

The rudder is . . . 15 degrees to starboard

- **on**
- by
- in

Select the correct word

The telegraph . . . dead slow astern

- **shows**
- show
- showing

What did the 3rd Officer show to the Pilot?

- **Pilot information card**
- Port advisory card
- Ship's advisory information
- The Bell Book

What mooring arrangement did the ship have once she was all secured?

- **4 + 2 + 2 forward and aft**
- 4 + 2 + 1 forward
- 4 + 2 + 1 forward and aft
- 4 + 2 forward and aft

When mooring ropes are put on the jetty, where are they said to go?

- **Ashore**
- Aligned
- Beached
- To ground
- To port

When the Captain wants to release the last ropes from the poop deck, what order does he give?

- **He gives the order "Let go aft!"**
- He gives the order "Let go forward!"
- He gives the order "Pick up the lines aft!"
- He gives the order "Slack away aft!"

When the Captain wants to secure a tug forward, what order does he give?

- **Make fast the tug forward!**
- Make fast the tug aft!
- Pick up the tug forward!
- Tug's line forward!

Which lines were used to secure the tugs to the ship?

- **The tugs were secured using the tug's Lines**
- The tugs were secured using breastlines
- The tugs were secured using headlines and sternlines
- The tugs were secured using ship's lines

Which word does NOT fit?

The 3rd Officer . . . that the Pilot was on board

- **tell**
- announced
- reported
- stated

Which word does NOT fit?

The Captain and Officer . . . on the radio

- **told**
- spoke
- talked

Which word does NOT fit?

The Captain spoke . . . the Pilot

- **by**
- to
- with

Which word does NOT fit?

The Officer . . . the Captain that all was secure

- **said**
- advised
- notified
- told

Which word does NOT fit?

The Pilot . . . down the ladder

- **ascended**
- climbed
- went

Who was first on the bridge?

- **The 3rd Officer**
- The Captain
- The Helmsman
- The Pilot

0041 – Generator

Version 2.1 (5419)

How many % of nominal voltage is required before main circuit breaker can be connected to main bus bar?

- **70%**
- 50%
- 80%
- 90%

How many degrees are there between the phases in a 3-phase generator?

- **120 degrees**
- 180 degrees
- 60 degrees
- 90 degrees

If we have a generator with 4 poles, how many rpm does it need to give 60Hz?

- **1800 rpm**
- 1200 rpm
- 900 rpm

Normally the power factor on the generator is . . . ?

- **0,8-0,9**
- 0,6-0,7
- 0,9-1,0

What does it mean if the synchronising lamp rotation is counter-clockwise?

- **Generator frequency is to slow relative to main bus**
- Generator frequency is to fast relative to main bus
- The voltage is wrong, but the frequency is correct

What is the minimum requirement of isolation resistance for a 440V system?

- **440 kilo Ohm**
- 220 kilo Ohm
- 600 kilo Ohm
- No requirement at all

What will happen if Load dependent start/stop of generator sets is in auto mode?

- **It will start/stop the generator sets in response to varying load**
- It will start but never stop the generator sets in response to varying load
- It will start/stop the generator sets in response to fuel consumption
- It will stop but never start the generator sets in response to varying load

Where will you find the instruments for synchronising the generator?

- **On the main switch board**
- On the circuit breaker
- On the diesel engine
- On the generator

0043 – Bilge water separator

Version 1.1 (2)

Are you allowed to discharge oily water from the oily water separator and into the sea when the ship is at anchor?

- **No, the ship has to be proceeding en route**
- Yes, if the oil content is less than 5 ppm.

Due to residues from heavy fuel the bilge water should have a temperature of . . . . . before it is pumped from the bilge water tank.

- **50-55 °C**
- 10-30 °C
- 95-100 °C

For how long must the oil record book be kept onboard after the last entry has been made?

- **3 years**
- 1 year
- 10 years
- 5 years

IMO regulations demands that the accuracy of the bilge alarm should be within . . . . . ?

- **+/- 5 ppm.**
- +/- 1 ppm.
- +/- 100 ppm.
- +/- 15 ppm.

Is it possible to separate an emulsion by settling

- **No**
- Yes

Oily mixtures discharged into the sea after passing through the bilge water separator must never have an oil content exceeding:

- **15 parts per million**
- 100 parts per million
- 5 parts per million
- 50 parts per million

The suction pipe to the bilge water separator should be placed so that the suction is . . . . than the heating coils.

- **Higher**
- Lower

What is done to avoid anyone tampering with the bilge alarm?

- **Anything but essential operation requires the breaking of a seal**
- Anything but essential operation will give an alarm

What is the main weakness of a bilge water separator of gravity type?

- **It doesn't separate emulsified bilge water**
- It cannot operate in heavy weather
- It requires heating of the bilge water

What is the maximum response time between detection of to high oil content out of the bilge water separator and operation of an automatic stopping device preventing overboard discharge?

- **20 seconds**
- 15 seconds
- 5 seconds
- 60 seconds

What is used to calibrate the bilge alarm?

- **Clean water**
- A sample from bilge water tank
- A sample from sludge tank
- Diesel oil

What regulation controls the discharge of oil in to the sea?

- **MARPOL**
- ISPS Code
- SOLAS
- STCW

When should the accuracy of the bilge alarm be checked?

- **At renewal of IOPP Certificate**
- Every 6 months
- Every time the bilge water separator is used
- Once a week

Which of the following areas are defined as special areas with strict controls on discharges of oily wastes?

- **The Mediterranean Sea area**
- **The Red Sea area**
- The Indian Ocean
- The South China Sea area

Which of the following shall be recorded in oil record book part 1?

- **Bunkering of fuel or bulk lubricating oil**
- **Collection and disposal of oil residues**
- **Condition of oil discharge monitoring and control system**
- Bunkering of fresh water
- Incineration of garbage
- Main engine fuel oil consumption

Which persons shall sign in the oil record book?

- **Master**
- **Officer in charge of the operation**
- 2nd engineer
- Chief engineer
- Chief officer
- Duty engineer

0045 – Container Lashings - Basic

Version 2.0 (2959)

As a container is being lowered into position on board a container ship, it swings in the fore and aft plane and the actuating wire for the semi-automatic base twistlock catches under the container.

Select the ONE best action to take in these circumstances.

- **Lift and re-land the container correctly, freeing the wire as you do so**
- Cut the wire and free the twistlock later with a spanner
- Make a note to tell the terminal representative at the discharge port
- Try and pull the wire free, attaching it to the gantry hook if necessary

The number of lashing rods used and positions, in which they are fitted on container stacks, are determined by a number of factors.

Select the ONE correct pair of factors.

- **The stack weight and the weight of each container in the stack**
- The container size and the weight of each container in the stack
- The container weight and the weight of each container in the column
- The container weight and the weight of each container in the row

What are intermediate stackers used for?

Select the ONE best answer.

- **Securing containers inside cargo holds and container stacks into composite units**
- Securing containers of different height
- Securing containers of different length
- Securing containers on deck

What disadvantage does an under deck lashing arrangement that uses pressure elements, have?

Select the ONE best answer.

- **Each tier of containers in the same block needs to be of the same height and all the containers in the same block must be for same port of discharge**
- All the containers in the same block must be for same port of discharge but they all must be of one size only
- If an individual pressure element breaks the whole block of containers is affected
- If over three units high, containers in the same block need to be secured with extra locking pins

What is the main advantage in using automatic locking cones?

Select the ONE best answer.

- **Some units combine the function of both the semi-automatic twistlocks (SAT) and the midlock and as such operating of cones during cargo operations are fully automatic**
- Dedicated operating rods are supplied for use in conjunction with the cones in case of their automatic function failure
- Their flangeless design leads to a higher centre of gravity
- Their flangeless design leads to a higher centre of gravity as well as large GM

What is the main advantage in using double and quadruple stacking cones?

Select the ONE best answer.

- **They secure loaded containers into a single block**
- No inspection or maintenance is required
- They can be used even where inter-connected stacks are for discharge at different ports
- They can be used where adjacent stacks are of different heights

What is the main advantage of a lashing bridge as used in securing and lashing systems on container ships?

Select the ONE most accurate answer.

- **It allows anchoring points for each stack to be moved higher up the stack resulting in reduction of tipping moments.**
- 20 foot and 40 foot containers can be secured to it when they are required to be loaded in front of the ship's bridge.
- It allows for lashing the ship's navigating bridge with the containers loaded in front and behind it.
- Ship's torsional forces are considerably reduced as the lashing bridge transfers all the container load to the ship's navigating bridge.

What is the main advantage of a semi-automatic twistlock?

Select the ONE best answer.

- **Most of the models are fully compliant with OSHA requirements can be attached and removed from the container corner casting by a stevedore stationed on the dock**
- It is one of the heaviest twistlocks in its class and hence more durable
- It relocks itself automatically if accidentally opened
- Most models are fully compliant with OSHA requirements and hence require no routine maintenance

What is the main advantage of using removable cell guides in relation to container ships?

Select the ONE correct answer.

- **Removable container guides allow multipurpose freighters, reefer vessels, etc., to carry containers in their regular or insulated holds without any risk of damage to the holds themselves**
- There is no risk of damage to the cargo in the containers or in the cargo holds
- They can be adjusted for the carriage of containers of different lengths within the same column
- They can be removed so as to accommodate carriage of high density bulk cargoes

What is the main disadvantage of using double and quadruple stacking cones?

Select the ONE best answer.

- **They can only be used where adjacent stacks are of the same height and where inter-connected stacks are for discharge at the same port**
- Frequent inspection and excessive levels of maintenance are required
- They cannot be used to secure loaded containers into a single block
- They cannot be used where adjacent stacks are of same heights and where inter-connected stacks are welded inside the holds

What was the main limitation of the lashing equipment manufacturer's manual on 'first generation' container vessels?

Select the ONE correct answer.

- **An ideal stow, referred to as the 'homogenous stack', was assumed.**
- Calm weather conditions were assumed during the voyage.
- Manuals were available only in the 'English' language.
- No lashing equipment description was provided.

What was the main reason that the stacking cones method became redundant on 'first generation' container vessels?

Select the ONE correct answer.

- **Increased utilisation of containers of different heights and the introduction of 'high-cube' containers**
- Recommendations from the International Maritime Organisation (IMO).
- Recommendations in the cargo securing manual (CSM).
- Recommendations in the Safety of Life at Sea (SOLAS) Convention.

Which ONE of the following is normally carried in type IMO5, ISO tank containers?

- **Only liquefied petroleum gases.**
- All dangerous bulk materials.
- All inflammable liquids.
- Refrigerated liquefied cargoes of any type.

Which ONE of these is a disadvantage of a lashing bridge as used in securing and lashing systems on container ships?

- **Only one end of 20 foot containers can be secured to it due to the limited capacity of the conventional lashing system.**
- 45 foot containers cannot be secured to it due to their exceptional weight and size.
- It can only be fitted athwartships, across 20 foot container bays due to the limited capacity depending on the container ship's size.
- Only one end of 40 foot containers can be secured to it when empty.

Which ONE of these is a prime cause of excessive lashing tension in lashing rods?

- **Over-tightening of lashing rods using large spanners**
- Inadequate lubrication of lashing equipment
- Tightening lashing rods manually by hand
- Tightening turn buckles to the end of their threads

Which ONE of these is another name for a slewing eye?

- **Lashing bridge**
- Cell guide
- D-ring
- Pad eye

Which ONE of these is the best description of a lashing bridge?

- **It is a substantial athwartships steel structure used for the better lashing of second and third tier containers.**
- It is a movable under-deck cell guide, which facilitates the stowage of containers of different lengths within the same hold.
- It is a substantial longitudinal steel structure used for the lashing of second and third tier containers.
- It is an alternative term for a colour-coded steel plate, welded to the ship's structure, onto which lashing rods and turnbuckles are secured.

Which ONE of these is the definition of a lashing bridge as used in securing and lashing systems on container ships?

- **It is a substantial steel structure running athwartships between each forty foot container bay that allows the second and third tiers of containers to be secured to the bridge using lashing rods and turnbuckles.**
- It is a substantial steel structure for storing loose turnbuckles and lashing rods when they are not in use onboard ships in ballast passage.
- It is a substantial steel structure running athwartships for safe passage of personnel during adverse weather conditions.
- It is a substantial steel structure running athwartships to lash second and third tier containers for transferring excess container loads to the vessel.

Which ONE of these statements is correct?

- **The compressive forces act on a container's corner posts and fittings as a result of tilting of stacks and vertical accelerations.**
- Container collapse can never result from compressive forces generated by exceeding the allowable loads on corner posts.
- The risk of container failure due to compressive forces can be minimised by exceeding the weight of the upper containers in a stack.
- The risk of container failure due to compressive forces can be minimised by over-tensioning of the lashing devices.

Which ONE of these statements is the most accurate?

- **Containers tend to pivot on the bottom edge and may eventually overturn when subjected to toppling forces caused by rolling motion of the ship due to extreme heavy weather conditions.**
- Containers tend to lift from the bottom edge which may disconnect the corner castings or twistlocks when subject to toppling forces caused by extreme heavy weather conditions in port.
- Containers tend to stay in position as they naturally counteract toppling forces caused by extreme rolling motion of the ship due to extreme heavy weather conditions.
- Containers tend to stay upright when fully loaded even when subjected to the toppling forces caused by extreme heavy weather conditions resulting in smooth rolling motion while in port.

Which ONE of these statements is the most accurate?

- **Convertible stowage frames were developed in order to accommodate containers of different dimensions securely, without any appreciable delay, on feeder ships, multipurpose freighters and container ships in certain regions.**
- Loaded containers cannot be locked in place by means of securing pins when using convertible stowage frames.
- The closure rails, fitted on both sides of the guide rails by means of screw connections, cannot be removed once containers have been loaded when using convertible stowage frames.
- The frames could not be secured at the sides by pins, which engage in bushes that are let into the wing bulkheads of the container ship.

Which of the statements regarding the combined static and dynamic loads acting on container securing systems of 'second generation' container vessels is true?

Select the ONE most accurate answer.

- **The combined static and dynamic loads acting on a securing system during adverse weather were not considered and there were no uniform standards for container lashing systems.**
- The combined static and dynamic loads acting on a securing system during adverse weather were considered and presumed to be correct.
- The containers were correctly stowed with the assumption that the lashing recommendations in the cargo securing manual (CSM) took static and dynamic loads into account.
- There were no cargo securing manuals for non-cellular container ships so the static and dynamic loads were not recognised.

Which of these contains the names of two types of fixed ISO container fittings?

Select the ONE correct answer.

- **Raised and sliding ISO twistlock foundations**
- Automatic and semi-automatic twistlocks
- Midlocks and deadlocks
- Sliding and non-sliding ISO twistlock foundations

Which of these does the section on 'stack weights', in a cargo securing manual, contain information about?

Select the ONE correct answer.

- **This section shows the maximum hatch cover, pedestal and tank top loading and lashing system for 20' containers without lashing bridge**
- This section shows how to use load-bearing base twistlocks, bridge fittings, lashing bridges, single and double stacking cones in conjunction with lashing bars or lashing chains
- This section shows the maximum number of stacks that can be loaded on bulk carrier ships with respect to their size and types when not loading break bulk cargo
- This section shows the maximum number of stacks that can be loaded on cargo ships with respect to their size and types along with loading of general cargo

Which of these identifies the main difference between raised and sliding ISO twistlock foundations?

Select the ONE most accurate answer.

- **Dovetail foundations need to be welded onto the deck or hatch pontoon to accommodate sliding ISO twistlock foundations but not for raised types.**
- Sliding ISO twistlock foundations are considered by classification societies to be less prone to stresses that raised ISO foundations.
- Sliding ISO twistlock foundations must be welded into specially designed sockets welded onto the deck or hatch pontoon but raised types do not.
- With sliding ISO twistlock foundations, there is no problem in using left or right-handed twistlocks whereas there is with raised types.

Which of these identifies the primary resultant effects on container stacks of the vessel 'rolling'?

Select the ONE correct answer.

- **Racking force is the primary resultant transverse effect caused by a vessel's rolling motion that tends to distort a container's end or its side frames, resulting from static and dynamic forces parallel to the deck.**
- Racking force is the primary resultant effect caused by a vessel's rolling motion that virtually damages the container ship itself and is the result of static and dynamic forces inside the hull.
- Racking force is the primary resultant effect caused due to the vessel's rolling motion. It is an arthward force that tends to distort a container's contents temporarily, resulting from static and dynamic forces perpendicular to the deck.
- Racking force is the primary resultant longitudinal effect caused by a vessel's rolling motion that tends to damage a container and its cargo, resulting in spilling of cargo on the deck.

Which of these is a recognized potential deficiency of dovetail foundations when securing containers?

Select the ONE most appropriate answer.

- **The chamfered edges are particularly susceptible to damage by compression and to flaring outwards when in use**
- The chamfered edges are particularly susceptible to damage by flaring outwards in bad weather conditions even when not in use
- The versatility of a stowage arrangement is seriously reduced if dovetail foundations are incorporated
- When damaged, sliding fixtures cannot be properly handled and stored

Which of these is the most common and preferred stowage and securing arrangement used on board modern cellular container vessels?

Select the ONE most accurate answer.

- **Cell guides in holds and stacked stowage and lashing with twistlocks, lashing bars and turnbuckles on deck of the cellular container vessel.**
- Block stowage and stabilization in holds and cell guides on deck of a general cargo-cum-container vessel.
- Cell guides in holds and on poop decks of a general cargo ship or a bulk carrier.
- Stacked stowage and securing with twistlocks, lashing rods and turnbuckles in holds and block stowage and securing with stacking cones and bridge fittings on deck of a general cargo ship.

Which of these precautions should be taken when using raised ISO twistlock foundations?

Select the ONE most accurate answer.

- **The foundations should be regularly inspected to ensure that they are not cracked as they are susceptible to stresses**
- It should be ensured that twistlocks are inserted with the locking bar/toggle facing inwards from the container's corner castings
- Markings and labels should clearly indicate that the use of both left or right-handed twistlocks is permitted
- The foundations do not require close-up inspection at regular intervals as they do not get cracked despite long usage

Which of these should be contained in the section on 'Equipment Operating Instructions', in a cargo securing manual?

Select the ONE correct answer.

- **The 'Equipment Operating Instructions' should include instructions on the use of twistlocks, actuator poles, turnbuckles and midlock stackers and should illustrate lashing operations using short and long rods**
- The 'Equipment Operating Instructions' should include instructions on the use of compressors and generators for reefer cargoes and illustrates their detailed operating instructions
- The 'Equipment Operating Instructions' should include instructions on the use of various types of container handling equipment required for loading and unloading of containers securely on break bulk carriers
- The 'Equipment Operating Instructions' should include instructions on the use of various types of container handling equipment required for loading and unloading of containers securely on general cargo ships

Which of these statements correctly applies to a 'hard' open-top container in relation to container ships?

Select the ONE correct statement.

- **A 'hard' open top container is one that is fitted with a removable steel top and can be loaded from the top or side door with the cross-bar swung out of the container.**
- They are loaded with a tarpaulin sheet covering its doors and fitted all around the container.
- They can be over-stowed by any other type of containers on cargo ships.
- They have a lower centre of gravity than the usual 2590mm (8 feet and 6 inches) high containers.

Which of these statements normally applies to cargo securing manuals in relation to a container ship?

Select the ONE correct statement.

- **They contain ship-specific information about different container securing arrangements each having its own permissible stack and tier weight limits**
- They contain general information about loading, stowage and securing containers as approved by the local port State administration
- They contain general information about manually securing of different types of containers only
- They contain ship-specific information about securing of containerised cargo on deck that varies from ship to ship

Which of these statements only correctly applies to the high-cube dry cargo containers?

Select the ONE correct statement.

- **The high-cube dry cargo containers are generally used for the carriage of non-perishable general cargoes and come in lengths of 20', 40', 45' and 48'.**
- They are generally used for the carriage of general break-bulk cargoes on cargo ships and bulk carriers because of their open ventilation.
- They can be stowed under-deck without any height limitations on cargo ships and bulk carriers because of their open ventilation.
- They have a lower centre of gravity than the usual 2590mm (8'6") high containers.

'A cargo securing manual (CSM) must be approved by . . .

Select the ONE phrase that most accurately completes the above statement.

- **. . . a classification society on behalf of a flag State.'**
- . . . the classification society in consultation with the ship owner.'
- . . . the ship yard in consultation with the shipowner and the naval architect.'
- . . . the shipowner in consultation with the classification society.'

0046 – Auxiliary boiler plant

Version 2.0 (3717)

A burner should never be ignited on a hot furnace wall because

- **There could be danger of explosion**
- It would increase maintenance time on equipment
- the burner may not work properly

After the economizer the feed water is entering:

- **A perforated feed pipe inside the upper secondary drum.**
- The desuperheater control valve.
- The suction side of the feed water pump.

Before firing the boiler, the oil is heated and re-circulated to

- **so that it is at the correct viscosity at the burners**
- to avoid wastage of oil
- to reach every part of the circulating system

For how long should you be firing the boiler before you reach working pressure?

- **Depending on the size and design of the actual boiler.**
- 15-30 minutes.
- Continuously.

For how long should you be purging before you fire the boiler?

- **2-4 minutes.**
- 10-15 minutes.
- 10-15 seconds.

How do we maintain a constant steam pressure?

- **By controlling the amount of heat supplied to the boiler water.**
- By adjusting the safety valves.
- By dumping steam from the boiler.

How is the circulation of the water inside the boiler obtained?

- **By the boiler design including placement of burners and direction of forced draft.**
- A separate circulating pump of the centrifugal type placed inside the boiler insulating wall.
- The feed water pump is circulating the water in the boiler.

How is the water in the primary steam system circulated?

- **By the difference in density of steam and water.**
- By a circulating pump outside the boiler.
- By the feed water pressure.

If the water level is too high during firing you should?

- **Blow off some water through the foam blowing valve.**
- Stop the feed water pump.
- Use the bottom blowing valve.

Is the boiler trip system an integrated part of the boiler control system?

- **It has to be a separate system.**
- It can be a part of the boiler control system.
- Some of the signals are common for the two systems.

It is very important to have steam flow through superheater because close to the burners the temperatures can be

- **Close to 2000 deg C**
- Close to 200 deg C
- Close to 400 degrees C
- varying widely

On Boilers with High Steam production, combined with low water content, the level control system is based on

- **water level, steam consumption and feed water supply**
- water level and feed water supply only
- Water level and steam consumption only

The Burner Management system will fire and shut off the burners in sequence to

- **adjust the boiler load**
- to control the air/fuel ratio
- to prevent breakdown of burners
- To reduce fuel consumption

The Density of water is:

- **30 times that of steam**
- 10 times that of steam
- 20 times that of steam
- equal to that of steam

The vent valve on boiler must be kept open when boiler is depressurized

- **To prevent vacuum formation inside boiler shell**
- to expel air
- To give visible indication of boiler being shut down

What is a dual pressure boiler?

- **Steam from a primary system is heating the secondary system.**
- You can select steam pressure by changing set point during operation.
- You have two different sets of burners.

What is a good indication on when to start soot blowing?

- **When the exhaust temperature after the air preheater has increased approximately 10° C.**
- When the exhaust gas from the funnel is getting grey to black in color.
- When the superheater outlet temperature has decreased approximately 50° C.

What is the function of a "high/low selector"?

- **To select fuel/air lead/lag during reducing/increasing boiler load.**
- Selecting high or low boiler load.
- Selecting high or low boiler water level automatically.

What is the purpose of the deaerator?

- **To remove air and other non-condensable gasses from the feed water.**
- To control the combustion airflow.
- To control the feed water pump.

What is typical steam consumption for a soot blower of the multi dual nozzle type?

- **2 kg/s**
- 100 kg/s
- 300 kg/s

When the water is heated at constant pressure which of the following happens?

- **The temperature will increase until we reach the saturation temperature**
- Plastic deformation of the tubes will occur
- The behaviour of the heated water is unpredictable.
- The temperature will not increase unless additional heating is applied

Which of the following is not an advantage of Dual Pressure boilers?

- **The temperature will rise above the evaporation temperature**
- Feed water is not contaminated
- No deaerator required in Primary system
- The Heating Surface will always be cooled by water flow

Which of the following statements is not correct?

- **Some of the Energy content from the steam is converted to mechanical energy at the turbine**
- All of the energy content of steam is converted to mechanical energy at the turbine
- The steam-water cycle is a closed loop cycle with some losses
- Water is converted to Steam in the Boiler by burning Oil

Why are we often using a cascade control system in combustion control applications?

- **To achieve the best possible process stability.**
- Make it easy to change between diesel and heavy fuel.
- To obtain system backup.

Why is it good practice to blow through the tubing to pressure and level gauges?

- **To check that the tubes and connections are not blocked.**
- To check that the instrumentation is working properly.
- To remove any condensate inside the tubings.

Why shall the combustion air blowers be running for a while after you have closed the oil to the burners?

- **To remove all gasses from the furnace.**
- To help cooling down the boiler as fast as possible.
- To let the blowers cool down.

Why should the condensate temperature be as high as possible?

- **To minimize the energy loss in the condenser.**
- To correspond to the economizer design criteria.
- To increase the vacuum in the condenser.

Will the secondary steam drum pressure be above or below the primary steam drum pressure?

- **Below.**
- Above.
- The same pressure.

0047 – Fuel Combustion Efficiency

Version 2.0 (1032)

A large increase in cylinder pressure occurs

- **When there is a large delay between beginning of injection and actual ignition**
- When there is a small delay between beginning of injection and actual ignition
- When there is no delay between beginning of injection and actual ignition

A too late combustion leads to?

- **Increased exhaust gas temperature**
- Decreased exhaust gas temperature
- No change in exhaust gas temperature
- Reduced soot formation and cleaner turbo chargers

Fuel must be atomized and distributed finely through the combustion chamber

- **To increase the surface area available for ignition**
- To increase the pumping volume
- To prevent fuel accumulation

How will a large content of "Cat fines" in the fuel influence the injection timing?

- **Abrasive wear of pumps and valves will lead to late injection**
- Cat fines will have no influence on the injection and combustion
- Cat fines will improve the atomisation and thus the combustion
- Cat fines will improve the fuel quality and lead to early injection

The combustion process starts when

- **When there is an increase in cylinder pressure**
- The exhaust valve is closed
- The fuel pump starts working

The cylinder liner temperature is lower than normal when?

- **You have an early injection of fuel**
- Increased RPM
- You have a late injection of fuel
- You have a poor atomisation of the fuel

The maximum cylinder pressure will be higher than normal when?

- **You have an early injection of fuel**
- The fuel valve opening pressure is too high
- You have a late injection of fuel
- You have a leaking fuel valve

The maximum cylinder pressure will be lower than normal when?

- **You have a late injection of fuel**
- The cylinder liner temperature is below normal
- The mean indicated pressure is higher than normal
- You have an early injection of fuel

The opening of the Exhaust valve in MAN B&W MC engines is by

- **Hydraulic oil pressure**
- Air spring pressure
- Fuel Oil pressure

The purpose of VIT is

- **To optimize fuel economy at part load**
- To optimize fuel economy at full load
- To reduce wear on fuel pumps

The purpose of suction valve is

- **To avoid additional opening of injection valves**
- To permit inlet to fuel pump
- To prevent sticking of fuel pump plunger

What do we mean by the expression after burning?

- **When the combustion continues after the injection has ended**
- Some fuel will continue to burn until the exhaust gas has passed the exhaust boiler
- The fuel is ignited twice to obtain a more clean combustion
- The fuel valves are leaking and the combustion continue

What do we mean by the fuels cetane number?

- **The cetane number is related to the ignition quality of the fuel**
- The cetane number is related to the combustion temperature
- The cetane number is related to the cylinder oil
- The cetane number is related to the pump index

What is controlled combustion?

- **Vaporising and reaction with oxygen is executed at the same rate as injection**
- Changing pump index according to fuel quality
- Checking the exhaust gas for possible soot content
- Using the VIT system continuously during combustion

What is the advantage of using three injection valves on each cylinder?

- **A more uniform temperature distribution around the combustion space**
- Each injector is smaller and easier to handle
- You can inject more fuel to each cylinder
- You will have longer maintenance intervals and lower cost

What is the approximate opening pressure for the fuel injection valves on a MAN B&W MC90 engine?

- **420 bar**
- 350 bar
- 480 bar
- 500 bar

What is the effect of high viscosity on after burning?

- **The after burning will last longer due to increased droplet size**
- Fuel pump capacity will be reduced and give less after burning
- The after burning will be reduced due to better atomisation of fuel
- Viscosity has no effect on after burning

When the fuel injector is closing a pressure wave will start travelling in the fuel pipes at a velocity in order of 1300-1500 m/s. If this is continuing what problem will it cause?

- **The fuel injectors will normally start dripping**
- The fuel pipes will be damaged and start leaking
- The fuel pump suction valve will start leaking
- The fuel pump will supply a variable amount of fuel to the injector

You have received a fuel with poor ignition quality, which of the following would you do?

- **Collective adjustment towards more early injection**
- Change the Pump index
- Collective adjustment towards more late injection
- Mix the fuel with additives

0048 – Gas Measurement

Version 2.0 (2820)

A sampling pump for an oxygen analyser with a capacity of 100ml per stroke connected to a 10 metre sampling line with a capacity of 25 ml per metre is being used to obtain a sample of a cargo tanks atmosphere. How many strokes of the pump are required to ensure that a truly representative sample is obtained in practice?

- **4**
- 10
- 2
- 25

How is a time weighted average reading obtained from a diffusion type detector tube that has been used over a working period of eight hours?

- **Divide the total reading on the tube by 8.**
- Half the reading on the tube to get a mid point reading.
- Multiply the total reading on the tube by 8.
- Take a direct total reading from the tube.

How is the correct volume of the sample controlled when using short term gas detector tubes with a manually operated hand pump?

- **By using the indicated number of complete pump strokes for that particular tube.**
- By always operating the pump for the same period of time for each tube.
- By always using a fixed number of complete strokes of the pump for each tube.
- By always using a single complete stroke of the pump for each tube.

If the oxygen content in an enclosed space falls below 21% during entry by personnel what action should be taken?

- **Stop working and leave the space immediately**
- Complete the work as quickly as possible
- Increase the ventilation and continue working
- Stop the work and put on breathing apparatus before carrying on

If the oxygen content of an enclosed space is less than 20% under what conditions should entry be made by personnel?

- **In an emergency only using suitable breathing equipment**
- For a short period only using breathing equipment
- Only in pairs using breathing equipment
- Using breathing equipment

It is generally recommended that non metallic sampling lines and sensors are used when obtaining gas samples from cargo tanks. What additional precautions are required when taking samples during inerting of a cargo tank?

- **The inerting operation should be stopped for at least 30 minutes before introducing sampling equipment into the tank.**
- It is safe to take samples during inerting.
- Samples should never be taken from a partly inerted tank.
- The inerting operation should be stopped while samples are obtained.

Many of the oxygen analysers used onboard ship use paramagnetic type sensors. What is meant by the term 'Paramagnetic'?

- **That a material is temporarily magnetized when exposed to a magnetic field.**
- That a material is non magnetic under any circumstances.
- That a material is permanently magnetized when exposed to a magnetic field.
- That a material is repelled by a magnetic field.

Metal oxide semiconductor type sensors are increasingly being used in personal gas monitors. What is the operating principle of this type of sensor?

- **Change of electrical resistance as oxide adsorbs the target gas.**
- Absorption of specific wavelengths of light.
- Combustion of target gas on heated filament.
- Conduction of heat from oxide surface by target gas.

What are the minimum checks that should be carried out before a gas detecting instrument is used to check the contents of an enclosed space? Select all correct options.

- **Check battery condition**
- **Check calibration**
- **Check condition of sample pump and lines and any filters**
- **Check for general condition**

What are the two main reasons for measuring the oxygen content of an enclosed space onboard a ship?

- **To check if the oxygen content is low enough to avoid flammable mixtures.**
- **To ensure the oxygen content is sufficient to allow safe entry.**
- To check the amount of hydrocarbon gas present.
- To check the amount of toxic gases present.

What facility is normally provided for making a zero adjustment to a paramagnetic oxygen analyser?

- **Alter relative position of optical system.**
- Adjust the electrical circuit.
- Change the rating of the light source lamp.
- Reposition the meter needle to zero.

What is meant by the term 'bump test' in relation to gas detecting instruments?

- **Checking the reading of the instrument against a known concentration of test gas.**
- Adjusting the span of the instrument.
- Adjusting the zero setting in fresh air.
- Knocking the instrument to see if the meter needle is sticking.

What is the effect on the oxygen content of the atmosphere of an enclosed space if hydrocarbon gases are present?

- **Oxygen content is reduced**
- No effect
- Oxygen content is increased
- The effect cannot be determined

What is the main advantage of using a personal gas monitor rather than a long term gas detector tube when working in a potentially hazardous space?

- **The personal monitor will give an alarm if the exposure limits are exceeded.**
- The personal monitor can be used to check total exposure throughout the working period.
- The personal monitor can be used to obtain a time weighted average.
- The personal monitor can check for any hazardous gas.

What is the most likely reason for the presence of high levels of NOx and SOx gases in the atmosphere of a cargo tank on an oil tanker?

- **Inert gas from the combustion of fuel oil has been used to inert the tank.**
- A cargo of heavy fuel oil with a high sulphur content has been carried in the tank.
- Sour crude oil has been stored in the tank.
- The tank has been fully gas freed following discharge of cargo.

What is the normally accepted value for the lower explosive limit, LEL, for hydrocarbon gas in a cargo tank?

- **1% by volume**
- 10% by volume
- 21% by volume
- 8% by volume

What is the normally accepted value for the maximum oxygen content in an inerted cargo tank?

- **8% by volume**
- 1% by volume
- 21% by volume
- 5% by volume

What is the normally accepted value for the upper explosive limit, UEL, for hydrocarbon gas in a cargo tank?

- **10% by volume**
- 1% by volume
- 21% by volume
- 8% by volume

What is the purpose of gas freeing a cargo tank?

- **To replace the contents with fresh air**
- To remove the hydrocarbon gases only
- To remove the inert gas only
- To replace the contents with inert gas

What is the time period that is normally used when stating threshold limit values for the short term exposure limit, TLV – STEL, for toxic gases?

- **15 minutes**
- 1 hour
- 5 minutes
- 8 hours

What is the time period that is normally used when stating threshold limit values for the time weighted average for exposure, TLV – TWA, for toxic gases?

- **8 hours**
- 1 hours
- 15 minutes
- 24 hours

What might indicate that a galvanic cell type oxygen measuring instrument requires a replacement sensor cell?

- **Unstable readings during calibration**
- Meter indicates maximum reading during test
- Reading less than 8% oxygen when testing an inerted cargo tank atmosphere
- Reading of 20.9% when testing fresh air

What precautions should be taken prior to use when using an oxygen analyser that has automatic calibration?

- **The analyser should be switched on in fresh air**
- No precautions are necessary
- The analyser should be switched on 30 minutes before use
- The analyser should be switched on with the sensor placed in the space to be tested

What reading will be obtained if a explosimeter, with a range of 0 – 100% LEL, is used to check the hydrocarbon gas content in an atmosphere containing a concentration of hydrocarbon gas which is three times the LEL.

- **The meter needle will move to full scale deflection and then fall close to zero.**
- The meter needle will move to full scale deflection and remain there.
- The meter reading will oscillate between full scale deflection and zero.
- The meter reading will stay at zero.

What should be used as the reference gas when carrying out the zero setting of a combustible or hydrocarbon gas detecting instrument?

- **Fresh air**
- 2% butane in nitrogen
- Inert gas
- Pure oxygen

What should be used for the span adjustment of a galvanic cell type oxygen analyser which is to be used for checking low concentrations of oxygen?

- **A test gas with 2% oxygen and 98% nitrogen**
- A test gas with 2% butane and 98% nitrogen
- Fresh air
- Inert gas with 8% or less oxygen content

What should be used when checking the span during calibration of a combustible gas or hydrocarbon gas detector?

- **A known concentration of butane in nitrogen.**
- A known concentration of butane in air.
- A known concentration of butane in inert gas.
- Fresh air.

What simple test can be carried out to check for leaks in a bulb type aspirator pump for a gas detecting instrument?

- **Nip the inlet line after squeezing the bulb to see if it stays deflated**
- Nip the outlet line after squeezing the bulb to see if it stays deflated
- Pressure test the pump with gas from a span test canister
- Submerge the pump in liquid and check for bubbles as it is operated

What two types of sensor are normally used in portable oxygen analysers onboard ship?

- **Galvanic cell**
- **Paramagnetic**
- Catalytic filaments
- Infra red absorption

What work related recommendations are covered by IMO Resolution A864(20)?

- **The recommendations for entering an enclosed space onboard a ship**
- Those for crude oil washing a cargo tank
- Those for inerting a cargo tank onboard a tanker
- Those for operating an inert gas plant onboard a tanker

What would be the effect, if any, on the meter reading of a catalytic filament type combustible gas indicator if the oxygen content of the sample was less than 11%?

- **The reading would be unreliable.**
- The reading would be zero.
- The reading would show full scale deflection.
- There would be no effect and an accurate reading would be obtained.

What would be the likely outcome if the combustion control of an independent inert gas generator, supplying inert gas to a cargo tank system, was incorrectly set resulting in poor combustion?

- **Increased level of carbon monoxide in the inert gas.**
- Increased level of hydrogen sulphide gas in the inert gas.
- Increased level of NOx and SOx in the inert gas.
- Increased level of oxygen in the inert gas.

What would be the most common source of hydrogen sulphide gas being present in onboard spaces of a ship?

- **Sour crude or petroleum products from sour crude in tanks.**
- Combustion of distillate fuel oil in the inert gas generator.
- Incomplete combustion of fuel oil in a boiler furnace.
- Low sulphur marine fuel oil being used in a diesel engine.

When carrying out a span check of a combustible or hydrocarbon gas detector the sample gas should normally be taken from a gas bag rather than direct from the gas canister? Why should this procedure be followed?

- **To avoid excess pressure or flow in the instrument.**
- Connection of the gas bag to the instrument is easier.
- To control the quantity of test gas used.
- To reduce the temperature of the test gas by expanding it into the bag.

When carrying out checks of the atmosphere in a cargo tank following gas freeing how many levels in the tank should samples be taken to ensure there are no variations in the tank atmosphere?

- **At least three levels**
- At least two levels
- One level is sufficient
- Samples should be taken every metre depth in the tank.

When obtaining a representative sample of the atmosphere of an enclosed space for purposes of determining the oxygen content what action should be taken with the ventilation of the space?

- **It should be stopped for 10 minutes before sampling**
- It should be increased for 10 minutes before sampling
- It should be reduced for 10 minutes before sampling
- Normal ventilation should be maintained during sampling

Which gas would be most likely to affect the reading from a refractive index type hydrocarbon gas measuring instrument?

- **Carbon dioxide**
- Hydrogen sulphide
- Nitrogen
- Oxygen

Which material is normally used to remove dirt and moisture from a gas sample prior to entry into a combustible gas indicator?

- **Cotton wool**
- Cardboard
- Paper
- Soda lime

Which material is recommended for the removal of carbon dioxide from a gas sample when measuring the hydrocarbon gas content of an inerted cargo tank atmosphere with a refractive index or infrared absorption type detector?

- **Sodalime**
- Cotton wool
- Gauze
- Silica gel

Which of the following toxicity threshold limit exposure terms would normally have the lowest numerical value?

- **TLV - TWA**
- TLV
- TLV - C
- TLV - STEL

Which term is normally used to describe the property of a gas that has a long term harmful effect on a living organism as a result of repeated or long term exposure to even low concentrations of the gas?

- **Chronic toxicity**
- Acute toxicity
- Long toxicity
- Low toxicity

Which two of the following gases are likely to result in poisoning of the sensor element in a catalytic filament type combustible gas detector?

- **Carbon monoxide**
- **Hydrogen sulphide**
- Carbon dioxide
- Oxygen

Which two types of combustible/hydrocarbon gas detectors rely on resistance change with temperature for their operation?

- **Catalytic filament type**
- **Non-catalytic heated filament type**
- Infra red absorption type
- Refractive index type

Which type of hydrocarbon gas detecting instrument uses thermal conductivity of the gas as the means of measuring gas content in a sample?

- **Non-catalytic heated filament type**
- Catalytic filament type
- Infra red absorption type
- Refractive index type

0049 – Radar Observation and plotting

Version 2.0 (0)

Can the frequency of radar pulses transmitted per second affect the radar's maximum range?

- **Yes.**
- No.
- Only at short range.
- Only during rain, snow etc

Does the Collision Regulations give any preference to ships equipped with radar?

- **No.**
- Only during reduced visibility.
- Only in good visibility.
- Yes.

Does the ships trim affect radar minimum range?

- **Only at short distance.**
- **Yes, it may introduce obstructions and blind sectors.**
- No.
- Only at long distance.

How can the Beam Width Distortion be minimized?

- **Reduce the setting of the Gain control.**
- Adjust centre position.
- Adjust Heading Marker.
- Adjust the setting of the Brilliance control.

How can the horizontal extent of a "Blind Sector" be accurately determined?

- **By turning the ship through 360 degrees and determining when a strong echo disappears and reappears on the radar screen.**
- By determining the direction of indirect echoes.
- By looking at the sectors on the radar display with no sea clutter.
- By visually looking for on-board obstructions.

How does radar antenna height influence on sea clutter?

- **High antenna height results in more sea clutter displayed on the screen.**
- Antenna height has no effect on the amount of sea clutter displayed.
- High antenna height results in less sea clutter being displayed.
- Low antenna height results in more sea clutter being displayed.

How is the amplifier protected from the transmitted pulse?

- **By the Transmitter/Receiver switch.**
- By having separate transmission and receiver scanners.
- By timing the pulse transmissions to precise times.
- Not necessary.

IMO resolution 477 applies to all ships built after?

- **1984**
- 1972
- 1982
- 1992

Is it a requirement to have the HM-suppress button spring loaded?

- **Yes, it is a IMO requirement.**
- HM switch must not be spring loaded.
- HM-switch may be spring loaded.
- No requirement.

Marine radar bearing accuracy is generally?

- **Good, but not as good as the range rings.**
- Bad.
- Excellent.
- Not very good.

Marine radar range accuracy is generally?

- **Good, but the range rings are better than the VRM.**
- Excellent , but better with the VRM than the fixed rings.
- Poor.
- Variable.

The action taken to avoid collision in reduced visibility should be:

- **Positive, in ample time and avoiding a series of small alterations.**
- Always by a large alteration of course to starboard.
- Always by a reduction of speed.
- Sometime by no alterations as own ship may be the “Stand-on” vessel.

The actions required by the Collision Regulations can be ignored when:

- **Never.**
- In a traffic separation Lane.
- In clear visibility.
- When at anchor.

The radar must be able to operate in relative wind speeds up to?

- **100 knots**
- 50 knots
- 75 knots
- No requirement

The reflection of radar pulses is similar to that of?

- **Light waves.**
- Ice.
- Snow.
- Sound waves.

We have normal transmission of radar waves when the radar horizon is?

- **10% longer than the visible horizon.**
- 25% longer than the visible horizon.
- 50% longer than the visible horizon.
- The same as the visible horizon when at the same vertical height.

What is known as "Side Lobes" with reference to marine radar?

- **Unwanted lobes of energy transmitted outside the main radar transmission beam.**
- The horizontal extensions of the radar scanner.
- The separate transmissions eventually being returned to the scanner after bouncing back and forth between the sides of own ship and another vessel.
- The unwanted affects on magnetic compass, when being close to radar equipment .

What is sometimes known as a Coded Racon?

- **A racon which displays a Morse code on the radar screen.**
- A racon only working at night.
- A racon which automatically activate it self.
- A racon which has the transmitter sending a signal continuously.

What is the correct setting for the "Gain" control?

- **A little noise (speckled effect) remaining visible on the screen.**
- A noise (speckled effect) visible across the whole screen.
- No noise (speckled effect) visible on the screen.
- No sea clutter remaining visible on the screen.

What is the correct type of radar display to be selected when at sea?

- **Always use a compass stabilized display either True Motion or Relative Motion to suit experience and environment.**
- Always a North-up Stabilized Relative Motion display.
- Always a North-up Stabilized True Motion display.
- Always use Ship's Head-up unstabilised display.

What is the effect of Sub-refraction?

- **Distance to the radar horizon is reduced.**
- Distance to the radar horizon is increased.
- Distance to the radar horizon not effected.
- Possible radar black-out.

What is the effect of error in own ships speed, when completing a plot?

- **Error in the targets course and speed.**
- Error in calculated CPA and TCPA.
- Error in calculated target aspect and CPA.
- Error in CPA, TCPA and target course.

What is the effect of gyro error?

- **All bearings will be wrong.**
- All ranges will be wrong.
- Bearings and ranges will be wrong.
- None of the three alternatives.

What is the fundamental principle of a marine radar?

- **Detection of other objects outside own ship.**
- Determine the course of other ships.
- Provide an early warning of ships on collision course.
- Provide early warning of ships.

What is the main content of Rule 19 in the Collision Regulations?

- **The conduct of vessels in restricted visibility.**
- The conduct of vessels in any any visibility.
- The conduct of vessels in sight of one another.
- The necessity of keeping a proper lookout.

What is the main disadvantage of true plotting?

- **Collision risk (CPA) can only be found by completing the plot.**
- Can be done on ordinary piece of paper.
- It provides immediate collision risk information.
- The chart scale is not always suitable to be used for true plotting.

What is the main function of the antenna?

- **To transmit the radar signals and receive echoes from objects.**
- To provide information about echo range bearing.
- To receive radar signals from objects outside own vessel.
- To transmit radar signals.

What is the main function of the receiver?

- **Amplify the incoming signals.**
- Amplify the outgoing signals.
- Display signals of interest.
- Reduce receiver noise.

What is the main purpose of plotting?

- **Obtain information about whether danger of collision exists, CPA, TCPA, target course and speed.**
- Obtain CPA and TCPA only.
- Obtain target course and speed only.
- Obtain target-calculated aspect only.

What is the main purpose of the parallel index lines?

- **To assist the navigator in maintaining the desired course over the ground . . .**
- To assist the navigator in finding the ships set and drift.
- To assist the navigator in maintaining the ships heading.
- To assist the navigator in maintaining the ships steered course.

What is the maximum radar "warm-up" time?

- **4 min**
- 1 min
- 2 min
- 3 min

What is the meaning of "radar Bearing Discrimination"?

- **The radar ability to display as separate spots on the screen, close targets on the same range.**
- The radar ability to display as separate spots on the screen, close targets on the same bearing.
- The radar ability to pick up large targets.
- The radar ability to pick up small targets.

What is the meaning of "radar's Range Discrimination"?

- **The radar ability to display as separate spots on the screen, close targets on the same bearing.**
- The radar ability to provide accurate bearings of small echoes
- The radar ability to provide accurate ranges of small echoes.
- The radars ability to display as separate spots on the screen, close targets on the same range.

What is the minimum display diameter required on ships bigger than 1600 tons but less than 10 000 tons?

- **250 mm**
- 340 mm
- 360 mm
- 380 mm

What is the minimum number of range scales required?

- **5**
- 2
- 3
- 6

What is the minimum radar display diameter for ships of 10 000 tons and upwards?

- **16 inch**
- 12 inch
- 20 inch
- 9 inch

What is the purpose of a radar reflector?

- **Making small objects more radar visible.**
- Making large echoes smaller.
- Making objects less radar visible.
- Making small targets more visible to the eye.

What is the purpose of determining the Aspect of a target vessel ?

- **Provide the same information as could be seen visually out of the wheelhouse window.**
- Avoid a collision risk developing.
- Determine the probable reflective properties of the target.
- Provide the relative bearing of the target.

What is the purpose of the Anti Sea Clutter controls?

- **Reduces the affects of the echo returns from the sea waves.**
- Reduces the amount of echoes from the sea waves.
- Reduces the height of the sea waves.
- Reduces the strength of the transmitted signal.

What is the purpose of the Gain control?

- **To adjust the amount of amplification of the echoes.**
- Adjust antenna sensitivity.
- Adjust the brightness of the screen.
- Adjust the strength of the transmitted signal.

What is the purpose of the Heading Line (HL) -suppression control

- **To switch off the HL temporarily in order to see under the line on the screen.**
- To adjust the position of the HL in azimuth.
- To switch off the HL permanently in order to see under the line on the screen.
- To switch on the HL.

What is the purpose of the VRM control?

- **To measure distance of a target on the screen accurately from own ship.**
- To measure bearing of a target on the screen accurately.
- To measure range and bearing of a target on the screen accurately.
- To measure the length of all the target trails on the screen.

What is the quickest method of deciding whether collision danger exists or not?

- **By taking several bearings of the target.**
- By relative plotting.
- By taking several range measurements to the target.
- By true plotting.

What is the relation between the scanner's horizontal width (size) and horizontal beam width?

- **Large scanner horizontal width, narrow horizontal beam width.**
- Large scanner horizontal width, wide horizontal beam width.
- No relation.
- Small scanner horizontal width, narrow horizontal beam width.

What is the required accuracy of the heading marker?

- **+/- 1.0°**
- +/- 0.5°
- +/- 1.5°
- No requirement

What radar range should be selected to suit all conditions?

- **The range most suitable to the proximity of targets and land, but with frequent checks of higher and lower ranges.**
- The 12 mile range with frequent changes to higher ranges to determine distant targets.
- The 12 miles range.
- The 6 mile range with frequent observations of the 12 mile range.

When completing a full Relative plot (North up), what input information is required?

- **A series of target ranges and bearings and own ship's course and speed through the water.**
- A series of target ranges and bearings and own ship's course and speed over the ground.
- A series Ranges and bearings only.
- The CPA and TCPA.

When does a radar display the identity signal of a racon?

- **When the wavelength of the racon is similar to the radar in use.**
- Always when using a S- band radar.
- Always when using all marine radars.
- No radar can activate a racon.

Which course should be fed in to the radar with a stabilised radar picture?

- **True course steered by own ship.**
- Magnetic Compass course steered.
- Master Gyro course steered.
- True Course made good over the ground.

Which of the following provides the complete answer, when related to the effect of using incorrect bearings in plotting?

- **CPA, TCPA, Course and Speed of the target are effected.**
- Calculation of CPA is effected.
- Calculation of target course and speed are effected.
- Calculation of TCPA is effected.

Which of the radar controls can be left at the setting determined when the set was first switched on?

- **None.**
- Pulse Length.
- Sea Clutter.
- Tuning control.

Which radar condition can be associated with a temperature inversion?

- **Ducting.**
- Radar black-out.
- Sub-refraction.
- Super refraction.

Why should radar still be used during the day in clear visibility?

- **To improve confidence and competence in the use of that specific radar.**
- Because radar is the only method to determine any risk of collision from other ships.
- To avoid looking out of the wheelhouse windows.
- To remove the necessity of keeping a good lookout.

0050 – ARPA Theory

Version 2.0 (0)

Comparing manual and automatic acquisition, which is most sensitive?

- **Manual acquisition**
- Automatic acquisition
- Not relevant
- Same sensitivity

Does current and drift affect the CPA/TCPA calculations on a true motion, true vector presentation?

- **No**
- Only a little bit
- Yes
- Yes, very seriously

Does incorrect speed input affect the true vectors on an ARPA display?

- **Yes**
- No
- Only a little
- Only if the input is speed over ground

How can the true course and speed of a target be found when the display is showing relative vectors?

- **By looking at the digital readout of target information**
- By comparing target vector with own ships vector
- By examining the vectors displayed
- Can only be found by changing to true vectors

How can true vectors be easily identified as the vectors on display?

- **Own ship has a vector**
- CPA and TCPA limits are broken and alarm is activated
- Own ship has no vector
- The vectors are the same direction as history dots

If the input gyro course is in error and fed to the ARPA, what can be expected?

- **Errors in other ships calculated course and speeds**
- Error in other ships course only
- Error in other ships speed only
- No problem

Ship A is navigating in an area of restricted visibility and has detected, by radar alone, a crossing ship, B, on her port bow. Which ship is the "Stand-On" vessel?

- **None**
- Ship A
- Ship A, as long the ARPA clearly shows that ship B is crossing from port
- Ship B

The plotting process can be divided into four separate operations, which?

- **Detection - Selection - Tracking - Plotting**
- Detection - Selection - Observation - Tracking
- Observation - Plotting - Tracking - Cancellation
- Selection - Observation - Plotting - Detection

To which condition of visibility does Rule no. 5, Lookout, apply?

- **Any condition of visibility**
- Good visibility at day-time
- Night-time only
- Restricted visibility only

What do the sides in the relative plotting triangle represent?

- **Targets true course and speed - Own ship true course and speed - Targets relative course and speed**
- Targets true course - Own ship true course - Targets relative course
- Targets true course and speed - Own ship relative course and speed - Targets relative course and speed
- Targets true speed - Own ship true speed - Targets relative speed

What does manual acquisition mean?

- **The operator must select and acquire each target separately**
- The operator must repeatedly place the cursor on each target and press “acquire”
- The operator selects an area of interest and targets inside the area are acquired
- The operator selects the limits of CPA and TCPA and targets breaking these limits are automatically acquired

What does the lost target alarm tell you?

- **A previously tracked target is no longer tracked**
- A previously tracked target has moved out of range and is no longer tracked
- Radar failure
- To many targets are tracked

What does the target ship’s aspect mean?

- **The relative bearing of own ship measured at the target ship**
- The relative bearing of the target ship measured at own ship
- The true bearing of own ship measured at the target ship
- The true bearing of target ship measured at own ship

What influence does an error in speed and/or gyro input have on the calculated CPA and TCPA?

- **None**
- Both CPA and TCPA will be wrong
- The CPA will be wrong but TCPA will be correct
- The TCPA will be wrong but CPA will be correct

What information is immediately apparent from the "relative plot" of another vessel?

- **CPA**
- Own ships course over ground
- Targets true course and speed
- TCPA

What information is immediately apparent from the "true plot" of another vessel?

- **Targets true course and speed**
- CPA
- Own ships course and speed
- TCPA

What is the correct speed input to an ARPA used for traffic surveillance?

- **Speed through water**
- Speed from doppler log
- Speed from GPS
- Speed over ground

What is the meaning of this symbol?

- **The target has been manually acquired and is in acquisition state**
- Dangerous target
- Lost target
- The target has been automatically acquired and is in acquisition state

What is the meaning of this symbol?

- **Dangerous target**
- Acquired target
- Lost target
- Selected target

What is the meaning of this symbol?

- **Lost target**
- Auto acquired target
- Dangerous target
- Manual acquired target

What is the purpose of a Guard ring on an ARPA?

- **It provides automatic target acquisition**
- It alarms to operator of a close CPA
- It avoids the need to continually monitor the radar display
- It provides warning of another target on the screen

What is the purpose of digitising the radar echoes?

- **The ARPA tracking system only work with digital signals**
- Increase accuracy of system
- No special purpose
- Reduce number of radar echoes

What is the result if there is no heading input to the radar?

- **An un-stabilised Head-Up presentation will be the only one available**
- A stabilised Head-Up presentation will be the only option
- An un-stabilised Course-Up presentation will be the only available mode
- The North-Up presentation will still be available

What is the result of "Target Swop"?

- **A previously tracked target is no longer tracked**
- A target has lost its vector and activates the lost target alarm
- A target will no longer be visible on the screen
- The operator can swap over vectors between ships

Which ARPA presentation provides the information "closest to what you see through the bridge windows?

- **True Motion, Course Up, True Vector**
- Relative Motion, North Up, Relative vector
- True Motion, Course Up, Relative Vector
- True Motion, North Up, True Vector

Which sensor input is required for the ARPA to be able to calculate and display true vectors?

- **Course and speed**
- Course
- None
- Speed

Why at different times is there a change in the size of the tracking gate?

- **There has been a change in course or speed**
- Target swop has occurred
- The gate is continually changing
- The tracking gate size is always the same

You have radars installed on your ship, are you required to use them?

- **Yes**
- No
- Only at night-time
- Only in restricted visibility

0051 – Introduction to Navigation

Version 2.0 (238)

A light is given with the following characteristics: Fl W (3+1) 12s 25m 12M. Is this:

- **A white light flashing for 12 seconds and having a range of 12 Nautical Miles?**
- A red light flashing for 12 seconds, having a range of 25 Nautical Miles?
- A red light flashing for 3 seconds then occulting once and having a range of 12 Nautical Miles?
- A white light flashing for 3 seconds then 1 second up to a total of 12 seconds and having a range of 25 Nautical Miles?

How do you find the Nautical Mile scale on the chart?

- **By measuring one minute of Latitude**
- By measuring one degree of Longitude
- By measuring one minute of Longitude
- By measuring the cosine of one degree of Latitude

How do you take a running fix?

- **By taking bearings from a landmark at time intervals and measuring the distance covered in the intervals.**
- By measuring the angle between two fixed landmarks and finding the sine of the angle.
- By taking bearings from a landmark at time intervals and finding the sine of the angles.
- By taking the relative bearings from 2 sets of landmarks and plotting the difference.

How long would 15 minutes of longitude be at a latitude of 60 degrees North?

- **7.5 NM**
- 15 NM
- 30 NM
- 60 NM

How long would one degree of latitude be at a latitude of 30 degrees North?

- **60 NM**
- 22.5 NM
- 30 NM
- 45 NM

If there were suddenly an immediate danger for both the vessel and its crew, would you send a MAYDAY on:

- **Channel 16 with 25 Watts output?**
- Channel 06 with 25 Watts output?
- Channel 06 with Dual Watch?
- Channel 16 with 1 Watt output?

If you sight the sun's meridian in the Southern Hemisphere in August, would you then:

- **Subtract the elevation from 90 degrees and subtract the declination?**
- Subtract 90 degrees from the elevation and add the declination?
- Subtract 90 degrees from the elevation and subtract the declination?
- Subtract the elevation from 90 degrees and add the declination?

If you spot a Racon on the radar and try to visually identify its light, would you look for:

- **It doesn't matter**
- Probably a quick flashing red light?
- Probably a sector light?
- Probably a white isophase light?

If your timepiece is correct to within 1 minute of an hour, what would be your margin of error in estimating your Longitude at Latitude 60 degrees North?

- **Within 7.5 NM**
- Within 0.5 NM
- Within 15 NM
- Within 6 NM

In the unlikely event that a distress situation allows you to send only a very short message, after MAYDAY would you first say:

- **Your callsign and position?**
- The nature of the distress situation?
- The number of persons on board?
- Your name and callsign?

On a route from Santander, Spain, (43 36 N, 005 38 W) to Brest, France (48 15 N, 004 46 W) would a Great Circle navigation bring you:

- **About the same as the Rhumb Line navigation?**
- Exactly the same as the Rhumb Line navigation?
- To the East of the Rhumb Line navigation?
- To the West of the Rhumb Line navigation?

The average time between two High Waters is 12 hours and 25 minutes because this is the time it takes for:

- **The moon to be at the opposite side of the earth**
- The moon to be at its apogee
- The sun and the moon to be on the same side
- The sun to be at the opposite side of the moon

What can you find from a single aeronautical RDF

- **A bearing**
- A hyperbolic LOP
- A range
- Your speed

What does a Racon help with?

- **Identifying a seamark**
- Identifying a vessel
- Measuring the distance
- Measuring the relative speed

What does the Doppler log measure?

- **The Speed Over Ground**
- The depth of water
- The Speed Made Good
- The speed through the Water

What is a Gnomonic chart?

- **A chart where Great Circle lines can be drawn as a straight line.**
- A chart giving true bearings.
- A chart where Rhumb Lines can be drawn as a straight line.
- Another type of Mercator projection.

What is the easiest way to find the latitude? By measuring the elevation of:

- **The Pole star**
- The moon
- The sun
- Venus

What is the tidal Height?

- **The water above the Lowest Astronomical Tide**
- The range of the tide
- The water above or under the Mean Level (ML)
- Twice the tide amplitude

What made finding the longitude by lunar transit inaccurate?

- **The lack of an accurate almanac**
- The lack of a reliable chronometer
- The lack of a sextant
- The ship's motion

What was the Selective Availability?

- **A reduction in the system's accuracy**
- A method of finding a more accurate position
- A range measuring device
- The selection of satellites available over the horizon

What was the advantage of the Omega system over its predecessors?

- **It has worldwide coverage**
- It is cheap to install and to use
- It is more accurate
- It works in any weather conditions

What would be the colour of an IALA starboard hand lateral buoy, as seen when sailing toward land, in Europe?

- **Green**
- Black and yellow stripes
- Red
- Red and green stripes

Which Beaufort Force is defined as a gale?

- **Force 8**
- Force 10
- Force 7
- Force 9

Which instrument uses a plumb-line?

- **The quadrant**
- The astrolabe
- The back-staff
- The cross-staff

Who has right of way in a Traffic Separation Scheme?

- **The vessel coming from the starboard side**
- The inbound vessel
- The outbound vessel
- The vessel coming from the port side

Who ran the Decca system?

- **A British company**
- An independent U.S. company
- The IMO
- The U.S. Department of Defence

Would you take the morning sun's azimuth in order to:

- **Find the true North?**
- Find a Line of Position (LOP)?
- Find the sun's declination?
- Find your latitude?

0053 – HAZMAT - IMDG Code, Basic

Version 11.0 (6434)

According to the IMDG Code, what are “labels”?

- **Notices that are fixed to the outside surfaces of small packages to indicate the hazards associated with the substances in the packages.**
- Large notices that are fixed to the outside surfaces of cargo transport units to provide a warning that the contents of the unit are dangerous goods and present risks.
- Large signs identifying dangerous goods.
- The unique identifying numbers assigned to dangerous substances, materials or articles.

According to the IMDG Code, what are “placards”?

- **Large labels that are fixed to the outside surfaces of cargo transport units to indicate the hazards associated with the substances contained in the unit.**
- Large signs identifying dangerous goods.
- Notices that are fixed to the outside surfaces of small packages to provide a warning that the contents of the package are dangerous and present risks.
- The unique identifying numbers assigned to dangerous substances, materials or articles.

According to the IMDG Code, what is the correct way to store drums containing dangerous goods?

- **Upright.**
- In a pyramid pattern, not more than three tiers high.
- On pallets and shrink-wrapped, in batches of eight drums.
- On their sides.

According to the IMDG Code, where should placards be located on a freight container containing dangerous goods?

- **One on each end and one on each side.**
- One on each end and one visible from directly above.
- One on each end only.
- One on each side only.

According to the IMDG Code, which one of the options identifies the item or items that road vehicles which are used for the transport of dangerous goods must have?

- **Suitable supports and securing attachments which provide a safe and stable platform during transportation.**
- A yellow flashing light.
- Motion sensors which activate a remote alarm if the vehicle breaks loose during transportation by sea.
- Suitable bracing arrangements, such that they can be attached to other vehicles and create a more rigid unit during transportation.

Class 2 (Gases) has three divisions. Two of these are “Flammable” and “Non-flammable, non-toxic”. Which of the following is the third division of this Class?

- **Toxic.**
- Corrosive.
- Inflammable.
- Reactive.

Column 16b of the Dangerous Goods List shows the requirements for segregation. What is segregation in terms of carriage of dangerous goods onboard ship?

- **The way that dangerous goods are separated by location on board.**
- Stowage describes the location of dangerous goods horizontally on board a ship, while segregation indicates their vertical separation.
- The requirements for loading and discharging dangerous goods.
- The system used in plans showing the stowage of dangerous goods.

For which one of the following can guidance be found in the Supplement to the IMDG Code?

- **The safe use of pesticides in ships.**
- Anti-piracy measures.
- Reporting procedures for arrival in quarantined ports.
- The safe handling of animals on board cargo ships and livestock carriers.

If a mixture is above its upper flammable limit, how is the atmosphere described?

- **Too rich.**
- Inflammable.
- Saturated.
- Too lean.

If you discover a leaking tank container on the weather deck, which of the following actions should you take first?

- **Report it immediately to the Duty Officer.**
- Examine the tank container and try and close off any leaking valves.
- Make sure the Chief Officer knows about it before arrival, so that no vehicles drive through the spillage.
- Wash it away immediately.

In addition to the IMDG Code, which other book should be referred to for guidance on security matters?

- **The International Ship and Port Facility Security (ISPS) Code.**
- MARPOL 73/78 (as amended).
- The International Maritime Security (IMS) Code.
- The Seafarers Security Handbook.

In which of these can you find the Dangerous Goods List?

- **Volume 2 of the IMDG Code.**
- MARPOL Annex II
- The Supplement to the IMDG Code.
- Volume 1 of the IMDG Code.

In which part of the diamond-shaped shipping label for dangerous goods would you find the class and division numbers?

- **The lower part.**
- The left-hand part.
- The right-hand part
- The upper part.

The proper shipping name and UN Number of a substance must be listed on the Dangerous Goods Form. This is so that the substance can be readily identified during transport, especially in the event of an emergency. But which one of the following is also a reason for this?

- **So that the Master has the necessary information to hand if he has to report a pollution incident involving this substance.**
- To assist in stowage of the package.
- To avoid being detained by the authorities for improper documentation when entering port.
- To demonstrate compliance with section 6.5 of the ISPS Code.

What does this symbol indicate in relation to packages used for dangerous goods?

- **The way in which a package should be stored.**
- How many packages may be stored on top of this package.
- The lid of the package.
- Where the vents are on the package.

What is meant by the term salvage packaging in relation to the IMDG Code?

- **Special packaging into which damaged, defective, leaking or non-conforming packages or dangerous goods that have spilled or leaked are placed for the purposes of transport for recovery or disposal.**
- Packaging in which goods recovered from the sea are held, pending investigation by the appropriate authority.
- Packaging used only at the direction of a contracted environmental protection response company.
- Packaging which is manufactured from recycled material.

When should all placards, orange panels, marks and signs be removed from a cargo transport unit?

- **As soon as the dangerous goods or their residues are discharged.**
- As soon as the cargo transport unit is discharged from the ship.
- Once the cargo transport unit has been loaded and sealed.
- Placards, orange panels, marks and signs should never be removed from a cargo transport unit.

Where dangerous goods have been loaded into a road vehicle, what is the name of the form that must be supplied by the person responsible for that operation (assuming the information has NOT been incorporated into the Dangerous Goods Form)?

- **Container / vehicle packing certificate.**
- Modular operation transport certificate (MOTC)
- Packing note.
- Vehicle manifest.

Which group of dangerous goods is generally understood to be the most dangerous for marine transportation?

- **Organic peroxides.**
- Flammable liquids.
- Infectious substances.
- Toxic gases.

Which of the following is another name for "smell"?

- **Odour.**
- Odeum.
- Odonate.
- Odorant.

Which one of the following is a correct statement?

- **The hazard posed by a substance can arise from a dangerous reaction of the substance or article with other substances in the surrounding environment.**
- A dangerous reaction between a substance or article and other substances in the surrounding environment is only possible if those goods are packaged in a container constructed of soft, non-water-resistant material.
- Substances in the surrounding environment can have no dangerous impact on packaged dangerous goods.
- The hazard posed by a substance is no higher onboard ship than it is ashore.

Which one of the following is the most accurate definition of the term "incompatible goods" in relation to the IMDG Code?

- **Those which when stowed together result in a hazardous situation in the event of spillage, leakage or any other accident.**
- Goods of different classes, as listed in the Dangerous Goods List.
- This is just another way of saying "dangerous goods".
- Those which when stowed together with foodstuffs present a risk to health through the leakage of poisonous vapours.

Which one of the following would you consult for guidance on the fire fighting procedures to follow in the event of an incident involving dangerous goods?

- **The EmS Guide.**
- MARPOL 73/78.
- The International Marine Fire Fighting Directory.
- The MFAG.

Which one of the options best describes a multi-element gas container in relation to the IMDG Code?

- **A special arrangement of cylinders and tubes which has a manifold and is assembled within a framework and fitted with service and structural equipment.**
- A gas container which is divided into individual compartments and is capable of simultaneous storage of different gases.
- A number of independent gas cylinders being transported under a single shipment and with the same UN Number.
- A portable tank which is divided by a number of internal bulkheads and is capable of being emptied via a single drain valve.

Which one of the options describes a requirement of the IMDG Code in respect of the testing of packaging?

- **All packagings must be of a successfully tested design.**
- Each individual packaging must undergo testing by the manufacturer before use.
- Packaging will be subject to testing in the event of leakage or other failure.
- The marking on every package must indicate when it was last tested.

Which one of the options identifies the first part of any marking on a packaging as required by the IMDG Code?

- **The UN packaging symbol.**
- The manufacturer's mark.
- The mark of the authorizing State.
- The UN Number.

Which one of the options identifies where empty, uncleaned cylinders, which have previously contained gas of a toxic nature, be safely stowed, according to the IMDG Code?

- **Only on deck.**
- Either on deck or under deck – there is no restriction on empty cylinders.
- Empty gas cylinders must be cleaned before they can be accepted for carriage by sea.
- Only under deck.

Which one of these statements defines a "marine pollutant"?

- **A substance which is subject to the provisions of Annex III of MARPOL.**
- A substance that discolours the water.
- A substance which requires the highest form of protection during transportation and is regulated accordingly by the International Transportation Authority.
- A substance which, because of its tendency to degrade in seafood, or because of its low hazard potential to the aquatic environment is subject to the provisions of SOLAS and transported accordingly.

Which packing group is assigned to substances presenting a low level of danger?

- **Packing Group III.**
- Packing Group I.
- Packing Group II.
- Packing Group L.

Which two of the following are standard tests performed on representative samples of packagings?

- **Drop test.**
- **Stacking test.**
- Crash test.
- Rigidity test.

Which two of the following can the term "cargo transport unit" be applied to?

- **Portable tanks.**
- **Road transport tank.**
- Aviation transport containers.
- International livestock container.

Who is primarily responsible for completing the documents relating to the shipping of dangerous goods (unless the Code specifically requires otherwise)?

- **The Shipper.**
- The Loading Master at the port of loading.
- The Master.
- The Receiver.

Within the IMDG Code, the term “marking” refers to which one of the following?

- **Identification of the substance contained in a package.**
- Damage done to a package of dangerous goods while onboard ship.
- Identification of the risks and hazards presented by substances contained in a package.
- Identification of the risks and hazards presented by substances contained in cargo transport units.

0054 – COW

Version 2.0 (699)

According to IMO the Manual has to be approved by the Administration. With regard to general practice who is giving the final approval of the Manual?

- **The classification society.**
- The coast guard.
- The IMO.
- The port state.

Are we allowed to perform COW at sea?

- **Yes, between discharge ports.**
- No, we are not allowed to perform COW at sea.
- There are no restrictions regarding COW at sea.
- Yes, on ballast passages.

From how many or which one of the cargo pumps must it be possible to take "bleed-off" from main(s) for COW supply?

- **From any one of the cargo pumps.**
- From 50% of the cargo pumps.
- From a particular cargo pump.
- From at least two of the cargo pumps.

How many per cent of the cargo tanks to be crude oil washed on rotational basis for sludge control during a 4 months period?

- **Approximately 25 per cent.**
- Approximately 10 per cent
- Approximately 30 per cent
- Approximately 50 per cent

How to avoid excessive electrostatic generation in the washing process due to the presence of water in the crude oil washing fluid?

- **By discharge at least 1 metre from the tank before use as a source of washing fluid.**
- By heating the cargo in the tank before use as a source of washing fluid.
- By recirculating cargo in the tank before use as a source of washing fluid.
- Increase the inert gas pressure in the tank before use as a source of washing fluid.

How to avoid vapour emission during filling of departure ballast (non SBT vessel)?

- **By simultaneous ballasting and cargo discharge.**
- By minimise the amount of departure ballast.
- By reduce the IG pressure.

Water that is put into a tank which has been crude oil washed but not water washed shall be regarded as:

- **Dirty ballast.**
- Arrival ballast.
- Clean ballast.
- Segregated ballast.

What are the requirements regarding experience for persons who are intended to have particular responsibilities as defined in the Manual?

- **They shall have at least 6 months experience on oil tankers where they should have been involved in COW operation.**
- They shall have at least 12 months experience on oil tankers where they should have been involved in COW operation.
- They shall have at least 9 months experience on oil tankers where they should have been involved in COW operation.
- With regard to experience there is no such requirement.

What is IMO's requirement regarding the oxygen level in the tank before and during it is crude oil washed?

- **The oxygen level must not exceed 8 % by volume.**
- The oxygen level must not exceed 10 % by volume.
- The oxygen level must not exceed 12 % by volume.
- The oxygen level must not exceed 5 % by volume.

What is the difference between SBT-system and CBT-system?

- **CBT-system can utilise cargo pumps/lines for ballast handling.**
- CBT-system must have separate ballast pumps
- SBT-system can utilise cargo pumps/lines for ballast handling.
- There is no difference.

What is the dissolving effect dependent on?

- **The composition of the oil.**
- **The temperature of the oil.**
- **The type of oil.**

What is the maximum quantity of oil you are permitted to discharge into the sea from an "existing tanker"?

- **1/15,000 of the total quantity of the cargo.**
- 1/20,000 of the total quantity of the cargo.
- 1/30,000 of the total quantity of the cargo.
- There is no such restriction.

What is the minimum distance from the nearest land when you are pumping departure ballast overboard?

- **50 nautical miles.**
- 10 nautical miles.
- 100 nautical miles.
- 30 nautical miles.

What is the purpose of the small diameter line?

- **To discharge ashore the drainings form pumps and lines.**
- To be utilised when decanting the slop tank
- To strip the pump room.
- To supply driving media to the eductor.

What is the requirement regarding arrangements for hand dipping the cargo tanks after COW?

- **Arrangements in the aftermost portion of the tank and in 3 other suitable locations.**
- Arrangements in each corner of the tank.
- Arrangements in the aftermost portion of the tank and in 4 other suitable locations.
- Arrangements in the middle of the tank

What is the requirement regarding capacity of inert gas plant?

- **At least 125% of the maximum discharge capacity.**
- At least 115% of the maximum discharge capacity.
- Same as the discharging capacity.
- There is no such requirement.

What is the requirement regarding hydrant valves on the COW line?

- **They must be blanked off.**
- No requirement.
- They must be disconnected.
- They must be fitted with a special cap.

What is the requirement regarding stripping system capacity when washing bottom of the cargo tanks?

- **The stripping capacity must be 25% more than the total throughput of washing fluid.**
- The stripping capacity must be 10% more than the total throughput of washing fluid.
- The stripping capacity must be 50% more than the total throughput of washing fluid.
- There is no requirement to stripping capacity.

What is the requirement to the number of drive units on vessels equipped with tank washing machines where the drive units are detachable?

- **That no drive unit has to be shifted more than twice.**
- No particular requirement.
- That no drive unit has to be shifted more than 3 times.
- That no drive unit has to be shifted more than once.

What kind of action must be taken in the event of total failure of the inert gas system during discharge/COW?

- **Stop the discharge/COW.**
- Call the Chief Engineer.
- Stop the COW and reduce the discharge rate.

What kind of action to be taken if an alternation affecting the crude oil washing system is made?

- **Revise the Manual.**
- Advise IMO.
- Advise port state.
- Advise the coast guard.

What requirements must the person in charge of COW operation meet with regard to the COW Manual?

- **Be fully knowledgeable of the content of the Manual.**
- Be familiar with the Manual.
- Been instructed in the content of the Manual.
- Have detailed knowledge of the Manual.

Where in the tank shall the oxygen level be determined before crude oil washing?

- **1 metre from the deck and in the middle region of the ullage space.**
- 1 metre from the deck and in the bottom of the tank.
- 2 metre from the deck and in the middle region of the ullage space.
- In the middle of the tank.

Where shall the verification of submerged washing machines be recorded?

- **In the Supplement 2 to the Oil Record Book.**
- In the deck logbook.
- In the engine logbook.
- In the PMS (planned maintenance system).

Where should you record the crude oil washing after completion?

- **Oil Record Book.**
- Cow Manual.
- Inert plant manual.
- Pumping Log.

Who is signing the Ship/Shore Safety Checklist?

- **Ship and Shore Representative.**
- Chief officer.
- The Captain.
- The surveyor.

Why is COW more efficient than water washing?

- **Because of the dissolving property in crude.**
- Because of the 'splash-back-effect'
- Because tank corrosion is reduced.
- Because the cargo is circulated through fixed washing machines.

Why was COW made compulsory?

- **To reduce pollution.**
- To improve cargo outturn.
- To prevent corrosion in cargo tanks.
- To reduce amount of sludge in cargo tanks

0055 – ODME

Version 3.0 (99)

Are you allowed to discharge oil while at anchor more than 50 nautical miles from shore.

- **No**
- Yes

For how long must you save the Oil Record Book onboard?

- **3 years**
- 1 year
- 3 months
- 5 years

For how long must you save the printouts from the ODME?

- **3 years**
- 3 months
- 5 years
- Until the end of the voyage

How far from the nearest land do you have to be before you can discharge any oily waste or residue into the sea?

- **50 nautical miles**
- 150 nautical miles
- 30 nautical miles
- 80 nautical miles

What is the maximum instantaneous rate at which oil can be discharged into the sea?

- **30 litres per nautical mile**
- 10 litres per nautical mile
- 100 litres per nautical mile
- 50 litres per nautical mile

What medium is used to calibrate the ODME?

- **Fresh water**
- Crude Oil
- Gasoline
- Sea water

What regulation controls the discharge of oil in to the sea?

- **MARPOL**
- ISM Code
- SOLAS
- STCW

Which of the following items shall be recorded in the oil record book?
Click all that applies.

- **Ballasting of cargo tanks**
- **Discharge of slop water to a shore facility**
- **Failure on the ODME**
- **Maintenance of the ODME**
- Discharge of ballast from segregated ballast tanks
- Failure on a cargo pump

Which of the sea areas below are defined as special areas in MARPOL?
Click all that applies.

- **The Baltic Sea**
- **The Black Sea**
- **The Mediterranean Sea**
- **The Red Sea**
- The Atlantic Ocean
- The Pacific Ocean

Which water can be discharged into the sea without using the ODME?

- **Ballast water from segregated ballast tanks (SBT)**
- Clean ballast water from cargo tanks
- Dirty ballast water from cargo tanks
- Water from slop tanks

Who must sign for each completed operation in the Oil Record Book?

- **The Officer in charge of the operation**
- The Captain shall sign for each operation
- The duty rating
- The Loading Master

0056 – Assessor Training

Version 2.0 (2293)

As an assessor you have been asked to evaluate the competence of an individual with respect to the STCW95 function "Controlling the operation of the ship and care for persons on board at the operational level". Recommend a method for demonstrating competence of practical skills?

- **In-service demonstration**
- Oral tests
- Project assignments
- Written tests

Before a seafarer is allocated to any duty he/she should:

- **Be given proper familiarisation with the duty**
- Get a good night sleep
- Go through a dedicated computer based training module for the actual duty
- Pass a relevant onshore training course

How many competence elements are defined by NMD in the "Assess Candidate Performance" table?

- **6**
- 2
- 4
- 8

How many function areas are defined in the STCW95 Convention?

- **7**
- 4
- 5
- 6

How many responsibility levels are defined in the STCW Code?

- **3**
- 1
- 2
- 4

In STCW '95 standards of competence is defined for?

- **Competence for master, officers, ratings and other personnel, related to duty and safety functions**
- Officers professional certificates only
- Only for officers and rating on watch
- The in-service assessor

The ISM Code implies that the ship's complement is able to operate their ship in a safe manner and without polluting the environment. Who has the obligation to fulfil this implication?

- **The shipping company**
- The flag state
- The individual seafarer
- The port state

The ISM Code is part of:

- **SOLAS**
- High Speed Craft Code
- MARPOL
- STCW

The ISM code states that the shipping company has to:

- **Establish and maintain procedures for identifying any training, which may be required**
- Develop a training plan for training supervisor and on board training instructors
- Develop career plans for all categories of ship personnel
- Develop career plans for senior officers

The Safety Management System requires that the shipping company develops and implements:

- **A system to report, analyse and follow up non-conformities, accidents and hazardous situations**
- A system for main engine maintenance
- A system for managing unplanned maintenance tasks
- A system for voyage planning

The competence elements for an assessor are described by?

- **Knowledge, understanding and proficiency, methods for demonstrating competence and criteria for evaluating competence**
- Criteria for demonstrating competence only
- Knowledge and proficiency requirements only
- Knowledge, understanding and proficiency requirements only

The four letters in STCW Convention stands for?

- **Standards for Training, Certification and Watch keeping**
- Seafarers' Training, Certification and Watch keeping
- Ship owners' Tracking Control World-wide
- Ship owners' Training, Certification and Watch keeping

The tables defining standards of competence have:

- **4 columns named**
  - competence
  - knowledge, understanding and proficiency
  - methods for demonstrating competence
  - criteria for evaluating competence
- 3 columns named:
  - competence
  - knowledge, understanding and proficiency
  - criteria for demonstrating competence
- 4 columns named:
  - competence
  - knowledge and understanding
  - proficiency
  - methods for demonstrating competence
- 4 columns named:
  - function
  - competence
  - knowledge, understanding and proficiency
  - competence evaluation

The training supervisor has asked you to do an in-service assessment of proficiency regarding the competence element "Operate life-saving appliances" which of the following methods would you recommend?

- **Observation of a designed test case**
- Observation during normal work situations
- Written test
- Written test and interview

The training supervisor has asked you to do an in-service assessment of the basic understanding of the competence element "Operate emergency equipment and apply emergency procedures". Which of the listed methods would you recommend for this task?

- **Observation in designed test situations**
- Computer based multiple choice questions
- Interview
- Observation during normal work situations

What are interviews useful for?

- **Assessment of knowledge, understanding and attitudes**
- Assessment of practical skills
- Checking ability to use bridge instrumentation
- Checking ability to write reports

What are the names of the responsibility levels specified in the STCW Code?

- **Management, operational and support**
- Deck, engine and catering
- Management, operational and assistance
- Senior officers, junior officers and ratings

What is meant by competence within the STCW-95 framework

- **The ability to perform activities to the standards required in employment**
- The knowledge needed for performing a work task
- The level of understanding of complex work tasks
- The motivation of the seafarer when coping with a difficult task

What is meant by the expression "Standard of competence"?

- **The level of proficiency to be achieved for the proper performance of functions on board ships.**
- The level of knowledge needed to understand how a ship should be operated.
- The level of understanding needed to demonstrate ability to cope with specific work tasks on board.
- The lower performance limit for all types of shipboard personnel.

What is the main task of Training Supervisor?

- **Co-ordinate and facilitate onboard and onshore training relating to certification requirements.**
- Check that all crewmembers have valid certificates.
- Fight for money for training activities.
- Train and retrain instructors and assessors.

What is the strength of computer-based assessment tools for in-service assessment?

- **Scoring will always be objective**
- Attitudes and motivation can be scored using such tools
- Such tools are designed for performance assessment
- There will not be any feedback

What is the strength of "observations of work activities" as assessment method?

- **Provides high quality evidence of competence**
- Can be used in emergency situations
- Competence can easily be demonstrated across the full range of operational conditions
- Easy to evaluate

Where are the standards of competence for Training Supervisor, Instructor and Assessor specified?

- **The standards of competence are vaguely defined in A-I/6.**
- The standards are not defined anywhere in the STCW Code.
- The standards of competence are defined in part B.
- The standards of competence are defined in the Convention articles.

Which of the listed methods are the most objective for assessing the knowledge component in the competence element "Organise and manage the provision of medical care on board" in the function "Controlling the ship/care for persons on board at the management level"?

- **Interviews**
- Computer based assessment
- Observations
- Projects and assignments

Which of the listed methods would you recommend for assessing the understanding of the competence element "Organise and manage the crew" in the function "Controlling the ship/care for persons on board at the management level"?

- **Observation and Interviews**
- Computer based assessment
- Projects and assignments
- Written test

You are planning an in-service assessment session. The competence element to be assessed is: "Ensure compliance with pollution prevention requirements" from the function "Controlling the operation/care for persons on board". Which assessment method would you recommend?

- **Observation in normal operation**
- Interview
- Observation of a designed test case
- Written exercise

0057 – Doppler log

Version 4.0 (153)

Aerated water can affect a Doppler log. Which one of the options is the most likely to produce aerated water?

- **The wake caused by the other ships' propellers**
- Passage through a shoal of fish
- The Doppler log losing "Bottom track"
- Varying density layers caused by water temperature differences

Select the option which correctly completes the following statement.

'If undisturbed reverberation measurement in shallow water is not possible, the system automatically switches to . . .

- **. . . bottom track.'**
- . . . docking mode.'
- . . . heading mode.'
- . . . water track.'

Select, from the options given, six factors that are likely to affect a Doppler log system.

- **Aerated water**
- **Density differences in water**
- **Loss of bottom track**
- **Loss of signal or weak signal**
- **Loss of water track**
- **Orientation of the transducer**
- Clock error
- DGPS-speed error

Which mode is the most suitable to use if the primary function of the connected ARPA is traffic surveillance?

- **Water track**
- Bottom track
- Track mode

Which one of the following statement depends on the "Janus configuration"?

- **Small errors due to trim, roll and pitch**
- High sound energy
- Small errors due to salinity and density
- Small errors due to sound velocity

Which one of the modes given in the options should be used when the Doppler log is connected to an ARPA-radar which is primarily used for traffic surveillance (according to IMO-rules)?

- **Water track**
- Bottom track
- Docking mode
- Sailing mode

Which one of the modes given in the options should be used when the Doppler log is connected to an autopilot with track keeping functions?

- **Bottom track**
- Docking mode
- Sailing mode
- Water track

Which one of the options identifies when the Doppler log would normally be switched off?

- **During dry-docking**
- At sea, during daylight hours
- The Doppler log should never be switched off
- When loading cargo

Which one of the options is the most likely reason that a Doppler log's operation might be disturbed during docking operations?

- **Due to aerated water from the backwash from own propeller**
- Due to a weak signal (normal in harbour areas)
- Due to interference from the GPS-speed
- Due to the orientation of the transducer

Which one of the options is the type of measurement used by the Doppler log?

- **Frequency measurement**
- Hyperbola measurement
- NMEA-0183 measurement
- Time measurement

0058 – COLREG

Version 4.0 (5265)

A bearing is taken of an approaching vessel and shortly afterwards another observation is made of the same vessel. The bearing is changing slowly, but under what circumstances (as defined in the Rules) could there still be a risk of a collision in this situation?

- **All of the suggested answers**
- When approaching a very large vessel
- When approaching a vessel at close range
- When approaching a vessel engaged in towing.

A power driven vessel is sighted coming towards own ship from 45 degrees on the port side and the bearing is not changing and it is getting closer and appears to be taking no action. You have sounded five or more short and rapid blasts and it still appears to be taking no action. As "stand-on" vessel can you take action under the rules at this time?

- **Your vessel may take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to you that the vessel required to keep out of the way is not taking appropriate action.**
- You must take substantial action altering your course to the port and give two short blasts.
- You must take substantial action and give one short blast and change your course to the starboard.
- You must take substantial action and give two short blasts and change your course to the starboard.

A power driven vessel is sighted coming towards own ship from 45 degrees on the port side at a distance of five nautical miles and the bearing is not changing. What is the initial action you should take?

- **There is a risk of close quarter situation, but the other ship is the "give way" ship so I should maintain my course and speed, but monitor the situation.**
- There is a risk of a close quarter situation and therefore both vessels should give way to the other.
- There is a risk of a close quarter situation, but I am the "give way" vessel and therefore I shall keep well clear of the other ship.
- There is no risk of a close quarter situation and therefore I should maintain my speed and course.

A power driven vessel is sighted coming towards own ship from 45 degrees on the port side at a distance of two nautical miles and the bearing is still not changing. What is the first action you should take?

- **There is a close quarter situation developing and the other vessel is not keeping clear therefore I would sound at least 5 short and rapid blasts to indicate my concern and monitor the situation.**
- There is a close quarter situation developing and the other vessel is the "keep clear" vessel, therefore I should maintain my speed and course.
- There is a close quarter situation developing and under the Rules both vessels should give way to the other.
- There is a close quarter situation developing, but I am the "give way" vessel and therefore I shall keep well clear of the other ship.

A vessel is observed approaching from the port side showing two black balls. A series of bearings indicate that a close quarter situation is likely to occur. As Officer of the Watch, indicate what action should be taken within the meaning of the Rules.

- **Take early and substantial action to keep well clear.**
- Give two short blasts on the whistle and alter course to port.
- It would be expected that the vessel will be slowing down and therefore no action should be necessary
- Maintain your course and speed.

A vessel is observed approaching from the port side showing two red lights and also a green side light. A series of bearings indicate that a close quarter situation is likely to occur. As Officer of the Watch, indicate what action should be taken within the meaning of the Rules.

- **Take early and substantial action to keep well clear.**
- Give two short blasts on the whistle and alter course to port.
- It would be expected that the vessel will be slowing down and therefore no action should be necessary
- Maintain your course and speed.

A vessel is proceeding along a narrow channel and hears the following signal "two long blasts and two short blasts on a whistle" from astern. How should the vessel respond?

- **The response should be one long, one short, one long and one short blast on the whistle, if in agreement to be overtaken**
- The response is one long blast on the whistle if not in agreement to be overtaken
- The response should be one long, one short, one long and one short blast on the whistle.
- The response should be two long blasts and two short blasts on the whistle if in agreement to overtaken on the port side.

A vessel is seen showing a black ball in the fore part of the vessel. What does this day time signal indicate?

- **It is a ship at anchor.**
- It is a cable-laying vessel.
- It is a ship that it is showing that it is "under way" but not making any way through the water.
- It is a vessel that has run aground.

A vessel is seen showing a rigid replica of international code flag A. What does this day time signal indicate?

- **The flag is a signal shown by certain vessels involved in a diving operation**
- The boat has engine failure and is calling for help.
- The flag has no particular meaning under these Rules.
- The flag is a signal that the vessel has a pilot on board.

A vessel is seen showing a white light in the fore part of the vessel. What does this night time signal indicate?

- **It is a ship at anchor, under 50 metres in length.**
- It is a cable-laying vessel.
- It is a ship that it is showing that it is "under way" but not making any way through the water.
- It is a vessel that has run aground under 100 metres in length.

A vessel is seen showing three black balls in a triangle at the head of the fore mast of the vessel and another single black ball forward. What does this day time signal indicate?

- **A ship involved in mine clearance operation at anchor.**
- A ship involved in compass adjustment.
- A ship involved in underwater operation.
- A ship that is dredging where the dredging equipment extending more than 150 metres in the direction of the single black ball.

A vessel is seen showing three black balls. What does this day time signal indicate?

- **It is the signal for a ship that has run aground.**
- It is the ship at anchor.
- It is the signal for a ship that is taking provision.
- It is the signal of a ship involved in an underwater operation.

A vessel is seen showing three green lights in a triangle at the head of the fore mast of the vessel and another single white light forward. What does this night time signal indicate?

- **A ship involved in mine clearance operation at anchor.**
- A ship involved in compass adjustment.
- A ship involved in underwater operation.
- A ship that is dredging where the dredging equipment extending more than 150 metres in the direction of the single black ball.

A vessel is seen showing three lights in a vertical line; the top and bottom ones are red and the middle one is white in the fore part of the vessel. What does this night time signal indicate?

- **It is a vessel restricted in her ability to manoeuvre, underway but not making way.**
- It is a ship at anchor.
- It is a ship dredging.
- It is a ship engaged in a mine clearance operation.

A vessel is seen showing three shapes in a vertical line; the top and bottom ones are black balls and the middle one is a black diamond in the fore part of the vessel. What does this day time signal indicate?

- **It is a vessel restricted in her ability to manoeuvre.**
- It is a ship at anchor.
- It is a ship dredging.
- It is a ship engaged in a mine clearance operation.

A vessel is seen showing two black balls on one side; two black diamonds on the other side of the vessel and also three shapes in a vertical line, the upper and lower black balls and the middle one a diamond. What does this day time signal indicate?

- **It is a vessel engaged in dredging or underwater operations which is restricting its ability to manoeuvre - showing the side where the obstruction exists and the side it is safe to pass.**
- It is a vessel engaged in mine clearance operations, which is restricting its ability to manoeuvre - showing the side where the obstruction exists.
- It is power driven vessel not under command due to damaged steering gear which is restricting its ability to manoeuvre.
- It is vessel engaged in towing which is restricting its ability to manoeuvre - showing the side where the tow exists.

A vessel is seen showing two black balls on one side; two black diamonds on the other side of the vessel and also three shapes in a vertical line, the upper and lower black balls and the middle one a diamond. Which is the correct side to pass this vessel?.

- **The side which displays the two diamond shapes is the side it is safe to pass.**
- It doesn't matter on which side a vessel passes the vessel as long as it is done at a slow safe speed.
- Other vessels should pass well clear of either side and avoid passing within 100 metres of the other vessel .
- The side which displays the two balls is the side it is safe to pass.

A vessel is seen showing two black cones points together and another cone point upwards on one side of the vessel. What does this day time signal indicate?

- **It is a vessel engaged in fishing (other than trawling) where outlying gear extends more than 150 metres horizontally from the vessel.**
- It is a vessel engaged in fishing (other than trawling) where outlying gear extends more than 100 metres horizontally from the vessel.
- It is a vessel engaged in trawling where outlying gear extends less than 100 metres horizontally from the vessel.
- It is a vessel engaged in trawling where outlying gear extends more than 100 metres horizontally from the vessel.

A vessel is seen showing two black cones points together. What does this day time signal indicate?

- **It is a vessel engaged in any type of fishing**
- It is a vessel engaged in fishing, but not trawling
- It is a vessel engaged in trawling only
- It vessel engaged in underwater operations.

A vessel is seen showing two red lights in a vertical line and also a white light forward and another white light aft. What does this night time signal indicate?

- **It is the signal for a ship that has run aground.**
- It is the ship at anchor.
- It is the signal for a ship that is taking provision.
- It is the signal of a ship involved in an underwater operation.

A vessel is seen showing two red lights on one side; two green lights on the other side of the vessel and also three lights in a vertical line, the upper and lower red and the middle one white. What does this night time signal indicate?

- **It is a vessel engaged in dredging or underwater operations which is restricted in its ability to manoeuvre, showing the side where the obstruction exists and the side it is safe to pass – under way but not making way.**
- It is a vessel engaged in mine clearance operations, which is restricting its ability to manoeuvre, showing the side where the obstruction exists – under way but not making way.
- It is power driven vessel not under command due to damaged steering gear which is restricting its ability to manoeuvre - – under way but not making way.
- It is vessel engaged in towing which is restricting its ability to manoeuvre, showing the side where the tow exists – under way but not making way.

A vessel is seen showing two red lights on one side; two green lights on the other side of the vessel and also three lights in a vertical line, the upper and lower red and the middle one white. Which is the correct side to pass this vessel?.

- **The side which displays the two green lights is the side it is safe to pass.**
- It doesn't matter on which side a vessel passes the vessel as long as it is done at a slow safe speed.
- Other vessels should pass well clear of either side and avoid passing within 100 metres of the other vessel .
- The side which displays the two red lights is the side it is safe to pass.

Another vessel is observed dead ahead steaming in the opposite direction. What action would the Officer of the Watch expect from the other vessel?

- **Expect the other vessel to alter its course to the starboard.**
- Expect the other vessel to alter its course to the port.
- Expect the other vessel to maintain his course and speed.
- Expect the other vessel to slow down

Are there any particular rules for sailing vessels and vessels less than 20 metres in length, when they are proceeding in a narrow channel?

- **They should not impede the passage of a vessel that can only safely navigate within a narrow channel of fairway.**
- They should always keep to the port side in a narrow channel.
- They should always stop and allow larger vessels to pass in the narrow channel.
- They should not pass a vessel that can only safely navigate within a narrow channel of fairway.

As Officer of the Watch you sight another power driven vessel on the port side approaching your vessel on a collision course. Which of the following answers is the correct action under the Rules if the other vessel appears to be taking no action to avoid a close quarter situation?

- **You give a signal to the approaching vessel consisting of at least five short and rapid blasts combined with a light signal and keep monitoring the situation.**
- You give a signal to the approaching vessel consisting of one short blast combined with a light signal and altering your course to the starboard.
- You give a signal to the approaching vessel consisting of two short blasts combined with a light signal and altering your course to the port.
- You slow down and let the other vessel pass ahead

As Officer of the Watch you sight another power driven vessel on the port side approaching your vessel on a collision course. Which of the following answers is the correct initial responsibility under the Rules?

- **Your vessel should maintain its speed and course.**
- Your vessel is obliged to keep out of the way of the other vessel therefore you alter your course to the port.
- Your vessel is obliged to keep out of the way of the other vessel therefore you alter your course to the starboard.
- Your vessel is obliged to keep out of the way of the other vessel therefore you should slow down and aloow the other vessel to pass ahead.

As Officer of the Watch you sight another power driven vessel on the starboard side approaching your vessel on a collision course. Which of the following answers is the correct initial action to be taken under the Rules?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed
- Make an early and substantial alteration of course to starboard.
- Sound two short blasts on the whistle and alter course to port

As Officer of the watch of a power driven vessel, what should you take into consideration when entering a narrow channel?

- **Proceed at a safe speed, keeping to the starboard side of the fairway**
- Be prepared to reduce your speed and keep to the starboard side of the fairway.
- Proceed with great care and keep to the port side of the fairway.
- Reduce speed to a speed just to maintain steerage.

The visibility is about one mile and as Officer of the Watch you visually see a vessel dead ahead on a reciprocal course coming out of the fog towards you. What should be the action to take?

- **Give one short blast and make a substatial alteration of course to starboard**
- Alter course to starboard and continue sounding the fog signal, but do not sound one short blast as this is for clear visibility only.
- Maintain your course and speed.
- Stop your vessel and change the fog signal to two long blasts every two minutes until the vessel has past.

Two ships are seen close to each other and both are showing a black diamond in the fore part of the vessel. What does this day time signal indicate?

- **One ship is towing the other and the length of the tow is more than 200 metres.**
- Both ships are engaged in towing another vessel or craft where the length of tow in both cases is under 200 metres.
- One ship is towing the other.
- The ships are trawling together as a single unit.

Under what circumstances should a vessel consider itself to be an overtaking vessel within the meaning of the Rules?

- **A vessel is overtaking when coming up with another vessel from a direction more than 22.5 degrees abaft her beam or if in any doubt she shall assume that she is.**
- When you are faster than the other vessel.
- When you are overtaking the other vessel from a direction more than 20 degrees abaft her beam.
- When you are overtaking the other vessel from abaft her beam.

What fog signal should be sounded by a power driven vessel underway and making way through the water, when in reduced visibility?

- **One long blast on the whistle every two minutes.**
- Four short blasts every two minutes.
- One long and two short blasts on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

What fog signal should be sounded by a power driven vessel underway but stopped and making no way through the water, when in reduced visibility?

- **Two long blasts on the whistle every two minutes.**
- Four short blasts every two minutes.
- One long and two short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.

What is the clearance distance required by a ship when navigating in the vicinity of a vessel engaged in a mine clearance operation?

- **1000 metres**
- 1500 metres
- 2000 metres
- 500 metres

What is the fog signal for a 150 metre long power driven vessel when at anchor in reduced visibility?

- **Sound the bell for about 5 seconds and in the after end of the vessel immediately after the bell, a gong for about 5 seconds every minute.**
- Sound the bell for 5 seconds and after that, sound a gong for 5 seconds every minute.
- Sound the bell for about 10 seconds and after that sound the gong for 10 seconds in the aft part of the vessel every two minutes.
- Sound the bell for about 5 seconds and after that sound the gong for about 5 seconds in the aft part of the vessel every two minutes.

What is the fog signal for a power driven vessel engaged in pushing another vessel ahead, but not as a composite unit?

- **One long blast followed by two short blasts every two minutes.**
- One long blast every two minute.
- Two short blasts and one long every one minute.
- Two short blasts followed by one long blast every two minutes.

What is the fog signal for a power driven vessel engaged in towing another vessel?

- **One long blast followed by two short blasts every two minutes.**
- One long blast every two minute.
- Two short blasts and one long every one minute.
- Two short blasts followed by one long blast every two minutes.

What is the fog signal to be sounded by a 150 metre long power driven vessel when aground in reduced visibility?

- **Every minute a ringing of the bell for about 5 seconds and in addition three distinct strokes before and after the ringing of the bell and at the after end of the vessel immediately after the bell, a gong for about 5 seconds.**
- Sound the bell for about 10 seconds, plus three distinct strokes before and after the bell and in the after end of the vessel the gong for about 10 seconds every minute.
- Sound the bell for about 5 seconds, plus three distinct strokes before and after the bell and in the after end of the vessel the gong for about 5 seconds every 2 minutes.
- Sound three blasts on the whistle, namely one long blast followed by two short blasts every two minutes

What is understood by the term “prolonged blast”?

- **A blast of from four to six seconds in duration.**
- A blast of about one second in duration.
- A blast of about three seconds in duration.
- A blast of about two seconds in duration.

What is understood by the term “short blast”?

- **A blast of about one second in duration.**
- A blast of about four seconds in duration.
- A blast of about three seconds in duration.
- A blast of about two seconds in duration.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when altering course to port?"

- **Two short blasts**
- A minimum of five short blasts.
- One short blast.
- Three short blasts.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when altering course to starboard?"

- **One short blast.**
- A minimum of five short blasts.
- Three short blasts.
- Two short blasts.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when going astern on the engines?"

- **Three short blasts on the whistle**
- Five short and rapid blasts on the whistle.
- One short blast on the whistle.
- Two short blasts on the whistle.

Which fog signal should a power driven pilot vessel under way and making way through the water on pilotage duty, sound in reduced visibility?

- **One long blast on the whistle every two minutes and may sound an identy signal of four short blasts.**
- An identity signal of four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes and an identy signal of four short blasts.

Which fog signal should a power driven vessel constrained by her draught, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a sailing vessel sound in reduced visibility?

- **One long blast followed by two short blasts on the whistle every two minutes**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel engaged in fishing, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel engaged in towing or pushing another vessel, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel not under command, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel restricted in her ability to manoeuvre, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which of the following statements is correct within the Rules?

- **You should join a traffic separation system at as small angle to the general direction of traffic flow as practicable.**
- It is important that all vessels in a traffic separation system keep the same speed.
- You are not allowed to fish in a traffic separation area.
- You can enter a traffic separation only at the end points.

Which vessels are allowed to use an inshore traffic lane of a Traffic Separation Scheme?

- **All of the suggested options**
- A vessel shorter than 20 metres
- Sailing vessels
- Vessels engaged in fishing

You are Officer of the Watch and see another vessel displaying two cones points together and a cone point up on one side, this vessel is sighted dead ahead and the bearing is not altering. What actions should be taken?

- **Take early and substantial action to keep well clear of the other vessel.**
- Sound not less than five short and rapid blasts on the whistle.
- The other vessel should give way to your vessel.
- You give it right of way and alter your course to best keep to your track.

You are Officer of the Watch and you are approaching a bend in a narrow channel, where you can not see round the bend. What action should you take?

- **You should give one long blast on the whistle.**
- You should give one short blast on the whistle.
- You should give three short blasts on the whistle.
- You should give two short blasts on the whistle.

You are Officer of the Watch of a power driven vessel and your vessel is proceeding in fog, when you hear the fog signal of another power driven somewhere ahead of you. The signals seem to be getting closer, but nothing is detected on the radar. How should you proceed within the meaning of these rules?

- **Reduce the speed to lowest manoeuvring speed and if necessary take all the way off and proceed with extreme caution.**
- Slow down the speed and proceed with caution.
- Slow down the speed, keep a sharp lookout and change the fog signal to two long blasts.
- Stop the vessel and sound one long blast followed by two short blasts

You are Officer of the Watch of a power driven vessel when your vessel encounters fog during your watch, what are the important initial actions that should be taken under these Rules?

- **All of the suggested answers**
- Call the Master
- Ring Standby on the engines and reduce the speed to a safe speed as soon as possible.
- Start sounding the appropriate fog signal.

You are Officer of the Watch on a power driven vessel in a Traffic Separation Scheme and sight a fishing vessel engaged in fishing inside the Traffic Separation Scheme? Who has the responsibility to keep clear?

- **A vessel engaged in fishing shall not impede the passage of any vessel following a traffic lane.**
- A fishing vessel is not allowed to fish within the Traffic separation Scheme and should be forced to leave the area.
- A power driven vessel should keep clear of a vessel engaged in fishing
- Your vessel is in the Traffic Separation Scheme and therefore does not need to keep clear of other vessels.

You are Officer of the Watch on a vessel meeting another vessel in clear visibility on a collision course and you are the "give way" vessel. Under the Rules what action should you take to avoid a close quarter situation?

- **You should take early and substantial action to keep well clear.**
- You should avoid a collision by altering course to port and giving two short blasts, under these Rules.
- You should avoid a collision by sounding one short blast and altering course to starboard.
- You take whatever action is necessary to avoid a collision.

You are Officer of the Watch on a vessel when you observe another two vessels crossing from port and both of these vessels are showing a black diamond and one is also showing three shapes, the highest and lowest are black balls and the middle one a black diamond. What would be the most suitable initial action to be taken by your vessel under these Rules, if the bearing of the nearest vessel is only changing slowly?

- **Slow down your vessel and allow the two vessels to pass ahead**
- Maintain your course and speed as the vessels are crossing from port.
- Maintain your course and speed until you are close to the other vessel, in case it makes an alteration of course.
- Sound one short blast and make a series of small alterations of course to starboard.

You are Officer of the Watch on a vessel when you observe another two vessels crossing from port and both of these vessels are showing a black diamond. What would be the correct initial action to be taken by your vessel under these Rules, if the bearing of the nearest vessel is only changing slowly?

- **Maintain your course and speed and continually monitor the situation.**
- Slow down your vessel and allow the two vessels to pass ahead
- Sound one short blast and alter your course to the starboard
- Sound two short blasts and alter course to port.

You are Officer of the Watch on a vessel when you observe another vessel under full sail showing a black cone apex downwards and on a constant bearing of about 50 degrees on your port bow. What would be the correct initial action to be taken by your vessel?

- **Maintain your course and speed**
- Alter your course to the port and sound two short blast on the whistle.
- Alter your course to the starboard and sound one short blast on the whistle.
- Reduce your speed until the other vessel has passed ahead.

You are Officer of the Watch proceeding in fog and hear one long blast every two minutes coming from ahead by a vessel approaching, which you have been tracking on your ARPA and know that there a risk of a close quarter situation. What would be considered the best action to take?

- **You should take avoiding action in ample time avoiding an alteration to port.**
- You alter your course 15 degrees to the starboard.
- You reduce your speed to a minimum manoeuvring speed, navigate with extreme caution and keep your engines ready to stop
- You should make a series of small alterations to starboard.

You are Officer of the Watch proceeding on your voyage in bad visibility; you have plotted a target on the radar screen and see that it is a ship on a collision course. What are the recommended actions to be taken in this situation?

- **All of the suggested answers**
- Avoid an alteration of course towards a vessel abeam or abaft the beam.
- Avoid an alteration to port for a vessel forward of the beam, other than for a vessel being overtaken
- Take action in ample time and avoid a series of small alterations.

You are bridge Officer of the Watch and observe another vessel approaching from the port side across your course line and showing a red light over a white light and another white light on its starboard side and also a green side light. What would be the correct action to be taken by your vessel, if a close quarter situation is developing?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed for a crossing vessel from port
- Sound one short blast and alter course to port.
- Sound two short blasts and alter course to starboard

You are bridge Officer of the Watch and observe another vessel approaching from the port side across your course line and showing two black cones points together and a cone point upwards on its starboard side. What would be the correct action to be taken by your vessel, if a close quarter situation is developing?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed for a crossing vessel from port
- Sound one short blast and alter course to port.
- Sound two short blasts and alter course to starboard

You are bridge Officer of the Watch in poor visibility and plot another vessel on the radar approaching your ship from the port side so as to involve a close quarter situation. Which of the following is the most suitable action required under these Rules?

- **Make an early and substantial alteration of course to starboard**
- Make a small reduction in speed to allow the other vessel to pass ahead.
- Make an early and substantial alteration of course to port.
- Manitain your course and speed as the “stand-on” vessel

You are officer of the watch and you are approaching a bend in a narrow channel, where you can not see round the bend and you sound one long blast on the whistle. What action would you expect an approaching vessel on the other side of the bend to answer to your signal?

- **It should answer with one long blast on the whistle.**
- It should answer with one long one short one long and one short blast on the whistle.
- It should answer with one short blast on the whistle.
- It should answer with two short blasts on the whistle.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What is the meaning of this signal?

- **It is a ship that is intending to pass you on your starboard side where you need to take some action to allow this manoeuvre.**
- It is a pilot vessel coming up from the starn
- It is a ship that is intending to pass you on your port side.
- It is a ship that is intending to pass you on your starboard side.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What should you give in response to this signal, if you feel doubtful of the safety of their intended manoeuvre?

- **At least five short and rapid blasts.**
- One long blast.
- One long one short one long and one short blast.
- One short blast.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What should you give in response to this signal?

- **If in agreement to being overtaken - one long, one short, one long and one short blast and take the appropriate action to allow passing.**
- If in agreement to being overtaken - one long, one short, one long and one short blast.
- One long blast.
- One short blast.

You are on a power driven vessel approaching the vessel illustrated (showing one black ball), directly ahead on your course line. What is the correct action to be taken by your vessel?

- **Taking into account any navigational hazards, take early and substantial action to keep well clear**
- Slow down and allow the other vessel to pass clear of your vessel.
- Taking into account any navigational hazards, you should adjust your course to avoid passing ahead of the other vessel.
- Taking into acount any navigational hazards alter course to starboard.

You are on a power driven vessel approaching the vessel illustrated (showing two white lights), directly ahead on your course line. What is the correct action to be taken by your vessel?

- **Taking into account any navigational hazards, take early and substantial action to keep well clear**
- Slow down and allow the other vessel to pass clear of your vessel.
- Taking into account any navigational hazards alter course to starboard.
- Taking into account any navigational hazards, you should adjust your course to avoid passing ahead of the other vessel.

You are on a vessel approaching a Traffic Separation Scheme and wish to enter a port within the inshore traffic zone on the side of the Scheme which is used by vessels coming in the opposite direction. What is the correct approach to safely gain entry into the destination port in compliance with the Rules?

- **Take the "inshore traffic zone" and keep out of the main lanes of the scheme.**
- Enter the scheme in the correct main lane and at the appropriate position alter hard over and cross the other lane on a course at right angles to the main traffic flow.
- Enter the scheme into the correct main lane and cross over the scheme at as small an angle as possible to the general direction of traffic flow.
- Take the correct main lane until the end of the scheme and then return down the other lane until arriving at the appropriate position to exit for the destination.

Your vessel is a deep draught vessel and is approaching the entrance to a narrow channel. Which of the following answers provide the most appropriate instructions on how you should proceed?

- **Reduce to a safe speed, keep a sharp lookout and keep to the starboard side of the fairway**
- Engines on standby ready to reduce your speed when required and stay to the starboard side of the fairway.
- Reduce the speed and keep in the centre of the channel for maximum depth
- Reduce your speed and keep a sharp lookout.

Your vessel is approaching a bend in a narrow channel with an obstructed view ahead. How should you proceed?

- **All of the suggested answers**
- Keep to the starboard side of the fairway.
- Proceed at a safe speed.
- Sound one long blast.

0059 – DGPS

Version 3.0 (1361)

How does the Differential GPS, improve the accuracy of the output position

- **An error of position message is provided by a differential reference station and sent to the ship**
- A combination of signals from the Navstar GPS satellites are sent to the ship's receiver and compared to provide a more accurate position output
- The position from the Navstar GPS produced by the receiver is compared with a eLoran signal to improve the accuracy of the ship's position
- The two satellite signals L1 and L2 are received by the ship and the difference between the two signals produces a correction factor to the position output

In the Navstar GPS system there are normally 24 active satellites, how many different carrier frequencies are used for transmissions by all of the GPS satellites?

- **Each satellite uses 2 carrier frequencies.**
- Each group of satellites uses the same frequency, therefore 6 carrier frequencies used in total.
- The same single carrier frequency is used by all satellites.
- There is a different frequency used by each satellite therefore 24 carrier frequencies used in total.

In the Navstar GPS system, satellites orbit the earth at an altitude of approximately what distance?

- **20200 km**
- 10000 km
- 15000 km
- 5000 km

In the Navstar GPS system, satellites orbit the earth in approximately?

- **12 hours**
- 24 hours
- 3 hours
- 6 hours

It is commonly recognised that the Navstar GPS system consists of three main segments, what are these segments?

- **Space, Control and User segment**
- Latitude, Longitude and Time
- Position, course and speed
- Satellite, Signal and Receiver

It is recognized that Global Navigation Satellite Systems (GNSS) transmit signals that can be used to provide three different services, what are these?

- **Position, Navigation and Timing**
- Latitude, Longitude and Altitude
- Position, course and speed
- Position, HDOP and Datum

What action should be taken by the Officer of the Watch when the Navstar GPS receiver is set to provide an output position of latitude and longitude using the datum of WGS 84, which is different to the datum on the paper chart?

- **Apply the corrections marked on the chart before plotting the latitude and longitude.**
- Change the datum used by the GPS receiver to the same datum as used on the chart.
- Plot the output latitude and longitude and insert warning note on the chart of possible inaccuracies
- There will be very little difference between the two datums and it can be ignored

What are the dangers of the Officer of the Watch being totally reliant on the Navstar GPS?

- **It is always dangerous to be totally reliant on one navigational system, because no single navigational system is 100% reliable**
- It is always dangerous to be totally reliant on one navigational system, because no system is accurate enough
- It is always dangerous to be totally reliant on the Navstar GPS, because it has periods when satellites are not visible to the ship and can't provide a position
- It is only dangerous because that system receiver may break down and be unavailable

What types of Global Navigation Satellite Systems (GNSS) receivers are used at sea for navigational purposes

- **Navstar GPS; GLONASS; and combined Navstar GPS and GLONASS, while other GNSS are under development**
- Combined Navstar GPS and Decca navigator receivers
- Only Navstar GPS
- Only the Global Positioning System

When considering GPS what is the main function of RAIM?

- **To assess the integrity of the GPS signals**
- To calculate ship's COG and SOG
- To improve the DGPS position accuracy
- To select the satellites with best configuration

When considering GPS, a 3D position has three unknown parameters; latitude, longitude and altitude. Why are signals from four satellites required to find an accurate 3D position?

- **Because of the receiver clock error.**
- Because of the ephemeris error .
- Because of the ionospheric delay.
- Signals from four satellites are not required, three is enough since there are only three unknown parameters.

When considering GPS, how does the elevation mask angle influence the number of available satellites for position fixing?

- **Normally, the higher the elevation mask angle is set, the fewer satellites are available.**
- The higher the elevation mask angle is set, the more satellites are available.
- The lower the elevation mask angle is set, the fewer satellites are available.
- There is no difference in the number of satellites available when the elevation mask angle is changed.

When considering GPS, what is the result of receiving multipath signals?

- **System accuracy decreases.**
- Signals are never deflected into multipath signals with GPS.
- System accuracy increases.
- System accuracy stabilizes.

When considering Navstar GPS receiver, what is the minimum number of satellites that must be available to calculate a 3 Dimensional (Latitude, Longitude and Altitude) position?

- **4**
- 2
- 3
- 5

When considering Navstar GPS, what is the meaning of pseudo-range?

- **Initial calculated distance between the satellite and the receiver**
- Position of a satellite
- Relative distance between satellites
- Relative position between two satellites

When considering Navstar GPS, what is the name of the common error affecting the orbital satellite position information?

- **Ephemeris error**
- Fluxgate error
- Ionosphere error
- Kalman error

When considering Navstar GPS, what is the safest operating mode to select?

- **Three dimensional**
- DR mode
- Two dimensional
- WGS 84

When considering Navstar GPS, which DOP is most important for a maritime user when checking GPS accuracy?

- **HDOP**
- PDOP
- TDOP
- VDOP

When considering Navstar GPS, which coordinate system (datum) is recommended to be used at all times?

- **WGS-84**
- ED-50
- NAD-27
- NAD-83

When considering a Navstar GPS receiver the output course and speed provided by the receiver provides what information?

- **The course and speed over the ground**
- The course and speed through the water
- The course steered and the log speed
- The gyro course and the Doppler speed

When considering a Navstar GPS receiver, what control should be operated when a member of the crew falls over the side of the ship?

- **The "Man overboard" button**
- The Officer of the Watch should record the Latitude and Longitude position given on the GPS display as soon as possible
- The officer of the watch should sound four long blasts on the whistle as soon as possible
- There is no special controls on the GPS available for this type of incident

When considering a Navstar GPS receiver, what is the minimum number of satellites that must be available to calculate a 2 Dimensional (latitude and Longitude) position?

- **3**
- 2
- 4
- 5

When considering a basic GPS, which of the following statements is true?.

- **If own ship speed is low, the GPS calculated speed over the ground is of low accuracy.**
- If own ship speed is high the GPS calculated speed over the ground is of low accuracy.
- If own ship speed is low, the GPS calculated speed over the ground is not calculated.
- If own ship speed is low, the GPS calculated speed over the ground is of high accuracy.

With reference the Navstar GPS outputs, what does a low HDOP figure indicate?

- **A low HDOP figure indicates a probable accurate position, provided by broad angles of intersections between the position lines from the satellites in use**
- A low HDOP figure indicates a poor inaccurate position because of small angles of intersections between the position lines from the satellites in use
- A low HDOP figure indicates that there are only a few satellites visible and suitable for use
- A low HDOP figure indicates that there is a Clock error discrepancy which needs to be resolved before accurate positions will be obtained

With reference to Navstar GPS system used in conjunction with a Differential Reference Station, on what is the accuracy of the resultant position mainly dependant?

- **The distance between the observer and the reference station**
- The number of satellites available
- The price of the equipment
- The relative angle between the satellites

With reference to Navstar GPS system, what happens to the GPS radio signal speed when it passes through the ionosphere?

- **The speed decreases.**
- It only passes through the ionosphere when the angle of elevation of the satellite is small.
- The speed increases.
- The speed remains unchanged

With reference to Navstar GPS, how many ground based control stations are used to monitor the satellites

- **The Master control station and 4 regional control stations**
- The Master Control Station in USA and six regional tracking stations
- There is only the Tracking Station in US which tacks all satellites every 24 hours during which times they will all be visible to that location
- Three control stations

With reference to Navstar GPS, why is the initial calculated range of the satellite call "pseudo-range"?

- **The initial range is calculated uses an assumption to make the calculation**
- The pseudo-range is only one of several ranges from a satellite required to plot a position of latitude and longitude
- The range calculated is not the information that is required for a position, but it is used to determine the course and speed of the ship.
- The range is called pseudo-range to ensure it is not confused with the radar range

With reference to a standard ship's Navstar GPS receiver, how would the probable accuracy of the output position be indicated

- **HDOP**
- Probability in cables
- Probability in decimals of a nautical mile
- Probability in Metres

With reference to the Navstar GPS system, how are the inaccuracies of position due to " clock error" removed?

- **By making use of the signal from one more satellite than the minimum required**
- By applying a correction to the acquired position
- By correcting the receiver clock to the same time as the satellite clocks
- By correcting the satellite clock to the same time as the receiver clocks

With reference to the Navstar GPS system, which statement is true?

- **GPS provides good global coverage.**
- GPS provides good coverage between latitudes 55°N and 55°S.
- GPS provides good coverage on the northern hemisphere but poor on the southern hemisphere.
- The coverage provided by GPS is exactly the same anywhere on earth.

With reference to the Navstar GPS, what are the effects on the ship's position resulting from an incident of "spoofing"

- **Spoofing can deliberately produce a feasible, but incorrect position**
- Spoofing is the deliberate downgrading of the Navstar GPS signals resulting in less accurate positions
- Spoofing may increase the amount of ionospheric interference and reduce the accuracy of the GPS position
- Spoofing may result in the GPS not being able to provide any form of position.

With reference to the Navstar GPS, what is the result of the height of the GPS antennae being incorrectly entered into the GPS receiver

- **When in the 2 dimensional mode of operation the GPS position may be less accurate**
- The antennae height being incorrectly entered will only affect the GPS output of altitude
- The antennae height input will never make any difference to the GPS positional output
- When in the 3 dimensional mode of operation the GPS position of latitude and longitude may be less accurate

With reference to the Navstar GPS, what is the significance of the "pseudo-random code" of the satellite transmissions

- **The signal is so complicated that it almost looks like random electrical noise**
- The satellite send a false signal prior to the correct signal to reduce the possibility of spoofing
- The transmitted signal from the satellite changes content randomly to avoid being duplicated.
- The transmitted signal from the satellite changes frequency randomly to reduce the ionospheric interference

With reference to the Navstar GPS, what is "selective availability" and does it affect the GPS position accuracy at this time (2011)

- **Selective Availability is when the GPS accuracy is intentionally degraded to a maximum of 100 meters, 95% of the time. It ceased to be operational in the year 2000**
- Selective Availability is the automatic system where the GPS receiver selects the available satellites which are above the horizon and is continuously operational
- Selective Availability is the available access to a Differential GPS reference station and is controlled by the operation management of the station
- Selective Availability is the method used by a combined GPS/GLONASS receiver to select which satellites to use to obtain a position and is continuously operational

0060 – Navigation in Ice

Version 1.1 (0)

An ice berg has a minimum height above sea level of?

- **5 meters**
- 10 meters
- 2 meters
- 7 meters

Anchoring in ice covered waters are?

- **Not recommended**
- No problem
- Only to be done in deep water
- Unsafe

Another name of first year ice is?

- **White ice**
- Blue ice
- Cream ice
- Green ice

Backing in ice?

- **Is a dangerous manoeuvre**
- Is a safe manoeuvre
- Is a safe manoeuvre with the rudder hard a port
- Is a safe manoeuvre with the rudder to starboard

Berthing in ice covered water is normally?

- **A long process**
- A problem
- Not a problem
- To be avoided

Close range hazard detection should be done?

- **Visually and by radar**
- By GPS
- By Radar
- Visually

Detection of low icebergs on radar should include?

- **Observation of a radar shadow behind the target**
- Careful observation of all small radar targets
- Observation of targets close to other ships
- Observation only of large targets

How can you reduce the risk of hull damage when executing "ramming"?

- **Use minimum speed**
- Rudder amidships before ramming
- Stop engine before ramming
- Use maximum speed

How many primary groups of steel is used in ship construction?

- **2**
- 1
- 3
- 4

Ice bergs normally move with the:

- **Current**
- The resultant of wind and current
- Wind
- Wind and current does not influence the movement of ice bergs

If possible, an iceberg should be passed?

- **On the windward side**
- Ahead
- Astern
- On the leeward side

If the speed of the ship is increased from 8 to 12 knots, the force of impact has?

- **More than doubled**
- Decreased
- Increased 4 times
- Increased a little

In connection with icebreaker operations, what does code letter "A" mean?

- **Go ahead**
- Anchor
- Go astern
- Stop

In connection with icebreaker operations, what does code letter "E" mean?

- **I am altering course to starboard**
- I am altering course to port
- Increase distance
- Remember to listen to the radio

In low temperature, what is important to do when hot fluid is loaded?

- **Start with very low loading rate**
- Start with highest possible loading rate
- Stop loading
- Wait until the temperature rise

In regards to radar imaging, ice is defined as:

- **A poor radar target**
- A good radar target
- A very good radar target
- Not being visible on radar

In traditional ice breaker design the distance between main frames is approximately?

- **40 cm**
- 20 cm
- 30 cm
- 60 cm

New ice is recognized by its characteristic?

- **Matt appearance**
- Brown appearance
- Glossy appearance
- Greenish appearance

Old ice is more than?

- **More than one year old**
- More than five year old
- More than three year old
- More than two year old

Pancake ice consists of circular pieces of ice with a diameter of?

- **30 – 300 cm**
- 10 –30 cm
- 30 –100 cm
- 50 – 150 cm

The accuracy of charts in the Arctic can be expected to be?

- **Poor**
- Better than normal
- Good
- Same as the rest of the globe

The first principle of successful navigation in ice is?

- **Avoid stopping**
- Avoid going astern
- Avoid night operations
- Avoid slow speed

The stern design of an icebreaker is mainly controlled by?

- **The number of propellers**
- The design draft of the icebreaker
- The engine power
- The number of rudders

The thickness of young ice is normally between?

- **10 and 30 cm**
- 10 and 15 cm
- 30 and 40 cm
- 5 and 10 cm

Under some atmospheric conditions white lights may take on

- **A reddish hue**
- A bluish hue
- A greyish hue
- A yellowish hue

What is an "Ice horn"?

- **Wedge shaped structure to help protect the rudder when going astern**
- A ice formation normally found in Arctic waters
- A tasty drink
- A very high ice berg

What is the effect of low temperature on steel?

- **Increase brittleness**
- Decrease brittleness
- Little effect
- No effect

What is the meaning of "Tripping" when used in connection with Arctic ship design?

- **Collapse of a frame against the side shell**
- A high speed engine stall
- A major generator failure
- A ship designed according to Arctic shipping rules

What is the purpose of "rudder stops"?

- **To protect the rudder and steering gear from damage during backing**
- To be able to operate with only one steering gear operational
- To increase the rudder stall angle
- To reduce steering gear power consumption

What is "nilas"

- **A thin elastic crust of ice**
- A massive piece of ice
- Rafted ice
- Ridged ice

What rudder position is recommended to obtain minimum loads on the steering gear when going astern?

- **Amidships**
- Hard a port
- Hard a starboard
- No special position

What type of chart projection is not suitable for use in high latitudes?

- **Mercator**
- Grid
- Lambert
- Polar grid

When a ship is expected to enter areas with very cold air, what should be done with tanks and lines?

- **Stripping all lines and tanks for water**
- Filling all tanks and lines with water
- Insulate all possible tanks and lines
- Nothing

When an icebreaker is breaking a track through large heavy floes at slow speed, the track can be expected to be how wide?

- **30 – 40 % wider than the icebreaker beam**
- 10 – 20 % wider than the icebreaker beam
- 50 – 60 % wider than the icebreaker beam
- 70 – 80 % wider than the icebreaker beam

When can the tactics for execution of a voyage start?

- **Once the planning of the passage is completed**
- After approval from the owner
- After approval from the ship owner an charter
- After careful route selection

When entering ice, the speed should be?

- **Very low**
- High
- Manoeuvring speed
- Slow

When in ice, one important basic rule is?

- **Keep moving**
- Stop moving
- Use high speed
- Work against the ice movement

When operating in ice, course changes should be executed when?

- **If possible, in open water**
- As soon as possible
- At dead slow speed
- At full speed

When the concentration of drift ice is higher than 7/10 the term drift ice may be replaced by the term?

- **Pack ice**
- Floe
- Ice berg
- Nilas

When will towing normally be provided by the icebreaker?

- **In an emergency**
- Never
- On Owners request
- When requested by escorted ships Master

Which materials are most used in propeller blades for icebreakers and ice breaking cargo ships?

- **Stainless steel and nickel-aluminium bronze**
- Carbon fibre
- High quality steel
- Stainless steel and carbon fibre

Which ships produce waste?

- **All ships**
- Dry cargo ships
- Passenger ships
- Tankers

Which type of glacial ice is most difficult to detect?

- **Growlers**
- Bergy bits
- Icebergs
- Pack ice

0061 – Stability 2

Version 2.0 (1965)

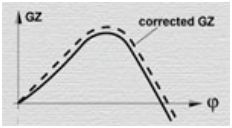
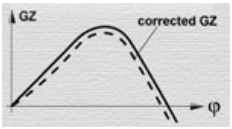
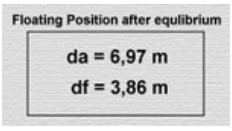
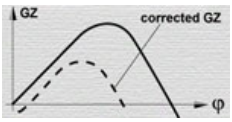
A ship has VCG = 9,80 m above keel. The metacentre position above keel varies according to the hydrostatic table shown here. What is the maximum permissible draught for the ship to fulfil the stability requirements?

- **7,0 m**
- 5,5 m
- 6,0 m
- 6,5 m

Click on the different watertight compartments and analyse the triangles.
How many damaged compartments does the following ship survive?

- **2 compartments**
- 1 compartment
- 3 compartments
- 4 compartments

Drag and drop the correct description of the situation on the images below:

Double bottom damage, amid ship	
Symmetrical three compartment damage in the fore ship.	
Symmetrical two compartment damage in the aft ship.	
Two compartment, unsymmetrical damage in the aft ship.	

Drag and drop the different terms to their corresponding boxes:

- A water line used in determining the number of water tight compartments in the ship. **Subdivision load line**
- An assumed line 3 inches or 76 mm below bulkhead deck of the ship. **Margin line**
- Ingress of water through internal openings to compartments assumed to be intact. **Progressive flooding**
- means that in any sea condition water will not penetrate into the ship **Weathertight**
- The positive range of the righting lever curve, after damage, with external heeling levers taken into account. **Residual stability**

In the added weight method, the free communication effect in damaged condition:

- **Reduces the GM and decreases the dynamic stability by sinus correction of the GZ curve**
- Gives the ship decreased GM when the damage is symmetrical.
- Reduces the GM and decreases the dynamic stability by cosinus correction of the GZ curve
- Reduces the GM and increase the dynamic stability of the ship.

What is the minimum required GM for a ship with the following GM(min) curve and draught 6.3 meters?

- **1.7m**
- 1.3m
- 2.0m
- 2.5m

What is the minimum required residual dynamic stability for a damaged passenger ship at equilibrium?

- **0.015 metre-radians**
- 0.05 metre-radians
- 0.05 metres
- 0.15 metres

What is the minimum required righting lever for a damaged passenger ship at final state of flooding?

- **0.10 metres**
- 0,10 metre-radians
- 0.05 metres-radians
- 0.15 metres-radians

What is the minimum required righting lever for a damaged passenger ship at intermediate state of flooding?

- **0.05 metres**
- 0.05 metre-radians
- 0.10 metre-radians
- 0.15 metres

What is usually the effect on G when the ship is damaged?

- **It is unchanged**
- It first rises then lowers.
- It lowers
- It rises

Which of the following statement (s) are true?

- **A damage control plan shows the boundaries of watertight compartments and openings therein.**
- **Cross flooding arrangements reduce the heeling angle in damaged condition.**
- Cross flooding arrangements stop the water from reaching the opposite side of the damaged compartment.
- Valves of scuppers that go through the damaged compartment should be opened.

Which of the following statements are true, with respect to change of stability parameters for a damaged ship?

- **M moves, B moves and GM increases or decreases.**
- B moves and M remains at its original place.
- M moves, B moves and GM is always constant.
- M moves, but B remains at its original place.

Which statement is true about the metacentric radius?

- **The distance between B and metacentre, depending on the displacement and the moment of inertia of the waterplane area.**
- It is the distance between the keel and the metacentre, depending on the speed and the location of G.
- The distance between G and the metacentre, depending on the displacement and the location of G.
- The distance between the keel and G depending on the waterline area and the location of G.

Which statement is true about volume permeability?
(Select all applicable answers)

- **If the volume permeability is 0%, no water will enter the compartment in case of damage.**
- **If the volume permeability is 100%, the damaged compartment below the waterline will be completely filled with water in case of damage.**
- If the volume permeability is 100%, no water will enter the compartment in case of damage.
- The volume permeability in machinery spaces is 50% according to SOLAS.

Which statements are true concerning the added weight method:
(Select all applicable answers)

- **It is a suitable damage stability calculation method for manual calculations.**
- **The free surface effect from the water in the damaged compartment has influence on residual stability even though the damage is symmetrical.**
- You can disregard the change in trim caused by the damage when calculating the new GZ-curve.
- You have no use of the hydrostatic tables and cross curves for the ship due to increased displacement . . .

Which statements are true concerning the loss of Buoyancy method:
(Select all applicable answers)

- **The damaged compartment is considered non-existing.**
- **The method is suitable for computer programming.**
- The damaged compartment is still considered intact.
- The original KN-values can be used for calculating GZ.

0063 – Mooring

Version 3.0 (63)

According to which criteria are the cable lifter units designed?

- **The brake force and cable diameter**
- The band brake diameter and drive motor force
- The brake force and band brake diameter
- The cable diameter and drive motor force

Automatic winches may be used on all winch lines at the same time.

- **False**
- True

Choose the statements you think are correct according to safe mooring practice.

- **Rope should never be used as a stopper on wires**
- **Two or more lines leading in the same direction should always be of the same material**
- **Wear goggles if you operate the windlass**
- A safety helmet without chinstraps is allowed
- New ropes have the same elongation as used ropes
- Wire stoppers can be used on synthetic fibre ropes

Choose the statements you think are correct in connection with oil sampling contamination control

- **Allow a quantity of oil to flush before collecting a sample**
- **Oil is extracted through fully open ball valves from high pressure line**
- **Take at least two samples from each sampling place in the system**
- Fill up the sampling bottle to 90 %
- The sampling flow should not exceed 10 % of the main line flow.

Does the angle between the line and the horizontal have anything to do with the number of ropes to be used?

- **Yes, smaller angle means less rope to be used**
- NO
- No, only when wire rope is used
- Yes, greater angle means less rope to be used

How is the anchor cable length marked?

- **White paint**
- Black paint
- Blue paint
- Red paint

How many complete rope turns are recommended around the warping drum?

- **3-4**
- 1-2
- 5-6
- 7-8

If one strand on the 8-strand fibre rope is broken, how much strength remains in the fibre rope?

- **80%**
- 25%
- 50%
- 65%

If we have different types of mooring lines leading in the same direction, which type of rope, mentioned below, will take most of the load?

- **Steel wire rope**
- Manila rope
- Nylon rope
- Polypropylene rope

Snapback is a serious danger when we are handling . . . ?

- **Synthetic ropes**
- Both wire and synthetic ropes
- Only nylon ropes
- Wire ropes

To identify a fibre material rope you can check if it floats. Which of the following ropes float if you put it into water?

- **Polyethylene and Polypropylene**
- All types of ropes can float.
- Nylon and Polyester
- Steel wire rope

What does STS mean?

- **Two ships moored alongside each other for the purpose of cargo transfer**
- Ordinary mooring
- The ship is moored in position using both anchors forward and stern is secured to buoys located around the stern
- The ship is only moored to one point

What is the operating parameter for the mooring drums brakes?

- **60% of the MBL**
- 40% of the MBL
- 50% of the MBL
- 70% of the MBL

What is the purpose of a Pedestal Roller?

- **To change the direction of lead of a mooring line**
- To facilitate towing
- To secure a chain
- To secure the eye of a mooring line

What is the purpose of the brake test?

- **It is to find the correct tightening torque**
- Is to find the highest necessary tightening torque
- It is to find the maximum tightening torque
- To update the ship records.

What kind of damage is this?

- **Bend**
- Birdcage
- Kink
- Mechanical damage

What should the operator of an undivided drum be aware of?

- **The difficulty to spool and stow the wire drum satisfactory**
- Always have sufficient number of turns on the tension drum
- Of standing instructions given by duty officer.
- The danger of personnel injury when the rope is transferred on a split drum

What will happen if you have more than one layer of wire on the tension drum on a split-drum winch?

- **It will decrease the brake holding capacity and the pulling power**
- It will increase only the brake holding capacity
- It will increase the brake holding capacity and the pulling power
- The brake holding capacity will be the same

Which of the following statements are correct?

- **Avoid placing wire and synthetic rope on the same bollard**
- **Do not drag the rope over sharp edges**
- **Exposure of sunlight can damage the rope**
- A new mooring rope should be uncoiled from the inner end
- Chafting at chocks and fairleads has no effect on the rope

Which of the ships have the greatest current force acting on them?

- **Vessel D**
- Vessel A
- Vessel B
- Vessel C

Who is in charge of the mooring operation?

- **Master**
- Duty officer
- Harbour master
- Pilot

0064 – Electronic Chart Display

Version 2.0 (0)

A chart made by using the Lambert projection can be considered to present?

- **Correct areas**
- Correct angles
- Correct area and angles
- None of the above answers are correct

A geoid refers to?

- **The actual size and shape of the earth.**
- A mathematical model of the earth's surface
- A satellite derived model of the earth's surface
- A theoretical model of the earth's surface

A raster chart is a chart that is?

- **Used both in RCDS and ECDIS**
- Only used in a ECDIS
- Only used in a RCDS
- Only used together with paper charts

A typical ECDIS "base display" shows the following information?

- **Coastline, sub sea dangers and traffic systems**
- Coastline, danger areas and soundings
- Coastline, seamarks and traffic separation schemes
- Information according to operators choice

According to IMO, what type of charts should be used on ECDIS?

- **A vector chart**
- A computerised chart
- A raster chart
- A scanned paper chart

An ECDIS consists of?

- **ENC and ECDIE**
- ENC and GPS
- ENC and RCDS
- RCDS and GPS

Before using a pre-programmed ECDIS route, what should you do?

- **Check all route data thoroughly**
- Check all way points
- Check log and gyro input
- Check own ships position

Can a raster chart be displayed in head-up mode on today's ECDIS?

- **No**
- Only charts with small scale
- Only during route planning
- Yes

Can raster charts be "customised"?

- **No**
- Only when used on a RCDS
- Only when used on an ECDIS
- Yes

Can raster charts provide navigational warnings?

- **No**
- Only when used on a RCDS
- Only when used on an ECDIS
- Yes

Chart projections are classified according to the type of projection surface used. How many commonly used surfaces are there?

- **3**
- 1
- 2
- 4

Does STCW specify special training in use of ECDIS?

- **No**
- Only for ships carrying dangerous cargo
- Only for ships over 10 000 DWT
- Yes

ECDIS is an abbreviation for?

- **Electronic Chart Display and Information System**
- Electronic Chart Data System
- Electronic Chart Display
- Electronic Chart Display System

ECS is a system that?

- **Does not meet SOLAS requirements**
- Meets SOLAS requirements
- Uses radar information
- Will meet SOLAS requirements after Y-2000

HMI is an abbreviation for?

- **Human Machine Interface**
- High Maritime Intelligence
- Human Maritime Interference
- Hydrographic Maritime Institute

How can the ships course change be described in detail?

- **By specifying ships turn radius**
- By specifying ships present and future course
- By specifying ships present and future heading
- The ships course change can not be described in detail

How is an ellipsoid formed?

- **By rotating an ellipse about its minor axis**
- By moving a circle about its east/west axis
- By moving a circle about its north/south axis
- By rotating an ellipse about its larger axis

How many stages are there in the planning of a safe voyage?

- **Four**
- Eight
- Six
- Two

If own ships position is erroneous on an ECDIS system, what is the result?

- **All positions, directions and courses taken from the ECDIS are erroneous**
- All other positions on the chart are erroneous
- The direction to other positions in the chart is faulty
- The distances to other positions on the chart is faulty

In worst case, how much can the difference be expected to be between local datum and satellite datum?

- **Up to several nautical miles**
- Up to 100 metres
- Up to 1000 metres
- Up to 500 metres

The ECDIS system can operate in how many modes?

- **Two**
- Five
- Four
- Three

The accuracy of data on a raster chart is exactly the same as?

- **The paper chart**
- The computerised chart
- The Mercator chart
- The Transverse Mercator Chart

The first satellite datum was based on:

- **American NNSS**
- American DGPS
- American GPS
- Russian Glonass

The surface of an ellipsoid is?

- **Regular**
- Regular in direction E/W
- Regular in direction N/S
- Uneven

The worlds first satellite based datum was named?

- **WGS-72**
- GGS-72
- WGS
- WGS-84

The "flattening" of the earth is approximately?

- **1/300**
- 0
- 1/400
- 1/900

Vector charts can?

- **Can be displayed with any direction up**
- Only be displayed Course up
- Only be displayed Head up
- Only be displayed with North up

What does a navigator who uses an ECDIS require?

- **Good navigational knowledge and a professional job attitude**
- Good computer knowledge
- Good system knowledge
- Nothing special

What is a raster chart?

- **A raster chart is a digital facsimile of the paper chart**
- A raster chart is a chart made by giving analogue values to each and every object on the chart
- A raster chart is a chart made by giving digital values to each and every object on the chart
- A raster chart is a computerised chart

What is the IMO definition of ECDIS?

- **A system that display hydrographic information in order to assist the safe navigation.**
- A system which provides information based on all available data.
- A system which provides information based on GPS.
- The display equipment used to display ENC

What is the definition of a leg?

- **A line connecting two turn-points**
- A line connecting the start and end position of a route
- A line in the direction of the next way-point
- A line in the direction of the previous way-point

What is the definition of a waypoint?

- **A waypoint is a position where the ships course is to be changed**
- A way-point is a position on a leg
- A way-point is the destination of a route
- A way-point is the starting point of a voyage

What is the definition of an Electronic Navigation Chart (ENC)?

- **ENC is a subset of ECDB**
- ENC is a subset of ECDIS
- ENC is a subset of ECS
- ENC is a subset of RCDS

What is the effect of sailing in "Course mode"?

- **Ships course will not be corrected for off-set, but the course will be adjusted towards the nearest way point**
- Ships course will automatically be corrected for off-set
- Ships course will be great circle
- Ships course will be rhumb line

What is the main element of geographical data?

- **Description of positions**
- Description of objects
- Geometric data
- Hydrographic data

What is the main purpose of the pan function?

- **Move the current chart on the screen**
- Change chart scale
- Deselect an area of no interest
- Reduce displayed information

What is the main purpose of the zoom function?

- **To select appropriate chart scale to the actual situation**
- To deselect an area of no interest
- To reselect a previous area of special interest
- To select an area of special interest

What is the most important thing to know about computerised systems?

- **Their limitations**
- Their features
- Their functions
- Their technical capacity

What is the name of the international standard for exchange of maritime digital electronic chart information?

- **S-57**
- S-52
- S-75
- S-84

What is the number of the IMO resolution concerning ECDIS?

- **IMO res. A 817(19)**
- IMO res. A 572(14)
- IMO res. A 728(17)
- IMO res. A 827(19)

What is the purpose of the "off track" limit?

- **Assist in controlling the ship track from planned track**
- Display warning limits on ECDIS
- Give warning about possible danger
- Give warning about possible grounding

What is the purpose of voyage planning?

- **To support the bridge team**
- To meet company regulations
- To satisfy IMO regulations
- To satisfy STCW regulations

What is the reference datum for GPS?

- **WGS 84**
- GGS-72
- WGS
- WGS-72

What is the worst known difference between a local datum and WGS84?

- **7 NM**
- 1 NM
- 3 NM
- 5 NM

When own ships position input to ECDIS is wrong, what is the result?

- **Positions, range and bearings taken on the ECDIS will be wrong.**
- ECDIS will automatically be switched off
- ECDIS will give warning
- Nothing

When the route planning process is completed, what should then be done?

- **Validate and check all route data**
- Checking basic route data
- Make a safety copy of the route data
- Store the route in a safe way

Where can you find information about standard terms used in connection with electronic charts?

- **IMO resolution A. 817(19)**
- IHO resolution A.888 (21)
- IMO resolution A.472 (XII)
- IMO resolution A.572 (14)

Which of the following chart is considered "intelligent"?

- **The vector chart**
- The computerised chart
- The paper chart
- The raster chart

0066 – Basic Ship Handling

Version 3.0 (603)

A conventional rudder can be expected to be effective up to?

- **40 - 45 degrees rudder angle**
- 20 - 25 degrees rudder angle
- 50 - 55 degrees rudder angle
- 70 - 75 degrees rudder angle

A ship's manoeuvring characteristics are directly related to?

- **Hull form**
- Engine power
- L/B ratio
- Rudder type

A ships ability to manoeuvre depends on?

- **Steering, stopping and turning ability**
- Steering and stopping ability
- Steering and turning ability
- Turning and stopping ability

An experienced shiphandler is expected to?

- **Make a ship do what they want when they want**
- Execute shiphandling quickly
- Make a ship do what it is expected to do
- Take good care of the machinery during the shiphandling operation

For most ships, how far is the "wheel over point" from the start point of the turn?

- **1.0 - 1.5 ship lengths**
- 0.5 - 1.0 ship length
- 1.5 - 2.0 ship lengths
- 2.0 - 2.5 ship lengths

Forces acting upon a ship during manoeuvring are in this course divided into how many groups?

- **3 groups**
- 2 groups
- 4 groups
- 5 groups

If a rudder has its entire area aft of the rudder stock, then it is?

- **Unbalanced**
- Balanced
- Counter balanced
- Semi-balanced

The Twin Shilling rudders can rotate through a total arc of?

- **145 degrees**
- 110 degrees
- 45 degrees
- 70 degrees

The change of pitch of the CP propeller is usually done by?

- **Hydraulic power**
- Electrical power
- Manually
- Mechanical power

The helicopter pilot is normally situated on?

- **Starboard side of the helicopter cockpit**
- Centre of the helicopter cockpit
- In the back of the helicopter cockpit
- Port side of the helicopter cockpit

What effect has shallow water on ships turning radius?

- **The turning radius is increased**
- The ship is not able to turn at all
- The turning radius is reduced
- There is no effect on the turning radius

What influence to-days ship designs the most?

- **Commercial interests**
- Environmental interests
- National interests
- Seafarer interests

What is shiphandling?

- **Using forces under control to manage forces not under control**
- Operating lines and anchors in a proper way
- Operating the main engine and thrusters in a proper way
- Operating the main engine in a proper way

What is the advantage of the "Single turn"?

- **It is the fastest way to turn the ship around**
- It brings the ship back on an opposite course
- It stops the ship quickly
- The speed is reduced very little

What is the advantage of the “Williamson turn”?

- **It brings the ship back on an opposite course**
- It is the fastest way to turn the ship around
- It stops the ship quickly
- The speed is reduced very little

What is the angle of the Twin Shilling rudders when full astern power is ordered?

- **105 degrees**
- 145 degrees
- 70 degrees
- 95 degrees

What is the approximate emergency stopping distance for a 250 000 tdw VLCC travelling at 16 knots?

- **3 Nautical miles**
- 1 Nautical miles
- 5 Nautical miles
- 7 Nautical miles

What is the first bridge response in an "immediate action" man overboard situation?

- **Rudder hard over to the side of the casualty**
- Order stand-by in engine
- Prepare a pick-up boat
- Sound the general alarm

What is the main purpose of the “combinator” in a CPP system?

- **Adjust engine RPM against propeller pitch**
- Improve propeller efficiency
- Reduce engine load
- Reduce fuel consumption

What is the main purpose of the “flap rudder”?

- **To ensure smoother water flow over the rudder**
- To increase rudder reliability
- To obtain cheaper rudder operation cost
- To reduce rudder size

What kind of propeller are most ships equipped with?

- **Fixed propeller**
- Constant speed propeller
- Ducted propeller
- Variable pitch propeller

What will happen with the pivot point when your ship is making headway?

- **It will move forward**
- It will be in the centre of the vessel
- It will move astern
- It will move down to the bottom of the vessel

What will happen with water when it is forced to pass under the hull in shallow waters?

- **The speed of the water will increase and the pressure will decrease**
- The speed of the water will decrease and the pressure will decrease
- The speed of the water will decrease and the pressure will increase
- The speed of the water will increase and the pressure will increase

Where is the safest place to learn Shiphandling?

- **On a shiphandling simulator**
- On a model ship
- On a small ship with a powerful engine
- On a small ship with little engine power available

Which alternative below lists three forces under control?

- **Main engine – Thruster – Anchor**
- Hydrodynamic forces – Rudder – Moorings
- Propeller – Thruster – Fouling
- Rudder – Inertia Momentum – Current

Which of the following methods is best to obtain precision track keeping in a turn?

- **Using preset turning radius**
- Using autopilot
- Using fixed rate of turn
- Using manual steering

Which of the following rudder types gives the smallest turning diameter?

- **T-rudder**
- Flap rudder
- Normal rudder
- Rotor rudder

Which propulsion system tends to be the least expensive to operate?

- **Diesel engines**
- Gas turbines
- Steam turbines
- Turbines

0068 – IMSBC Code

Version 3.0 (7320)

A bulk cargo may shift within the cargo hold and cause the vessel to list.

Which one of the following circumstances is this most likely to happen under?

- **The moisture content of the cargo is the same or higher than the flow moisture point of the cargo**
- The cargo is very dry and the hold is cubically full
- The flow moisture point of the cargo is high
- The transportable moisture limit is six parts per hundred

A bulk carrier has badly maintained hatch cover seals which allow seawater to leak into the cargo space in bad weather.

Which one of the following is considered to be the main danger of this situation when considering a non liquefying cargo?

- **The cargo space will fill with water and reserve buoyancy will be lost**
- The cargo may self-heat
- The cargo will be commercially damaged
- The cargo will produce flammable gas

A cargo of coal appears to be self heating on voyage.

Which one of the following actions should the ship's Master take in such circumstances?

- **Follow guidance in IMO Bulk Cargo Code and measure carbon monoxide levels**
- Continue on voyage
- Discharge cargo over side
- Open hatch covers and spray large quantities of water onto the cargo

A cargo of crushed coal, particle size 1mm and a flow moisture point of 20 is to be loaded. The coal had a moisture content of 17 measured 12 days previously but has lain uncovered on the quayside in heavy rain since.

Which one of the following actions should the Master of the vessel take?

- **Request a new test certificate for the moisture content as it should not be more than 7 days old and then decided whether to load**
- Load only part of the cargo
- Load the cargo
- Refuse to load the cargo and sail

A number of deaths have occurred in cargo spaces that have been used for cargoes such as grain and steel scrap (swarf).

Which one of following is thought to have been absorbed by the cargo, from the cargo space atmosphere, so causing these deaths?

- **Oxygen**
- Carbon dioxide
- Hydrogen
- Nitrogen

A vessel is to load 100,000 tonnes of coal with a stowage factor (SF) of 1.50 m^3/t.

Which one of the following will be the required volumetric cargo space to load this cargo, measured in cubic metres, m^3?

- **150,000 m^3**
- 1,500 m^3
- 15,000 m^3
- 20,000 m^3
- 200,000 m^3

A vessel is to load 100,000 tonnes of iron ore with a stowage factor (SF) of 0.50 m^3/t.

Which one of the given options is the required stowage volume for this cargo?

- **50,000 m^3**
- 40,000 m^3
- 76,000 m^3
- 80,000 m^3

A vessel is to load 20,000 tonnes of iron ore with a stowage factor (SF) of 0.30 m^3/t. The vessel has four cargo holds each with a volume of 6,000 m^3 and a maximum allowed loading of 7,000 tonnes in each hold.

Which one of the following statements is correct in relation to the above?

- **All four holds will each be approximately 25% full by volume**
- All four holds will be completely filled with cargo
- Each cargo hold will be approximately 50% full by volume
- Two holds will be empty and two holds will each be 75% full by weight

In relation to the classes given in the IMSBC Code, is the following statement true or false?

Two bulk cargoes, a Class 5.1 dangerous good and a Class 6.2, requiring a category 3 segregation, must have at least 3 metres separating them.

- **False**
- True

In relation to the groups given in the IMSBC Code, is the following statement true or false?

Group C contains a list of materials which are neither liable to liquefy nor to possess chemical hazards.

- **True**
- False

Is the following statement true or false?

A bulk cargo with low angle of repose has a high likelihood of shifting.

- **True**
- False

Which one of the bulk cargo types given in the options is more likely to have a liquefaction risk?

- **Heavy metal ores**
- Grains such as maize
- Sand and small stones
- Scrap steel

Which one of the bulk cargoes given in the options is most likely to be subject to liquefaction?

- **Ore concentrates**
- Grains
- Steel

Which one of the following conditions is likely to occur when a bulk carrier is heavily loaded in the mid section and lightly loaded at the ends?

- **Sagged**
- Hogged
- Unstressed

Which one of the following groups of gases should the atmosphere of an enclosed space, for example a cargo hold, be checked for, before entry is permitted?

- **Toxic gases, oxygen and flammable gas**
- Argon, nitrogen and flammable gas
- Oxygen and carbon dioxide
- Oxygen, toxic gases and nitrogen.

Which one of the following identifies how a bulk cargo should generally be loaded in order to reduce the likelihood of the cargo shifting when the ship is rolling heavily in bad weather?

- **Trimmed and levelled near to the horizontal**
- Secured with timber
- Stowed against hold bulkheads
- Stowed with a high peak at centre of hold

Which one of the following identifies the action that should be taken with bulk cargoes that may be subject to liquefying?

- **They should only be loaded and carried if the transposable moisture limit is less than 90% of the flow moisture point**
- They may be carried without any restriction
- They should be allowed to dry out completely before loading
- They should never be loaded on a ship

Which one of the following identifies the main gases that are likely to be produced by a coal cargo?

- **Carbon monoxide and Methane**
- Hydrogen, Oxygen and Carbon monoxide
- Oxygen and Methane

Which one of the following identifies what is recommended should be monitored on a bulk carrier that is carrying a cargo of coal?

- **Temperatures, Carbon monoxide and Methane levels**
- Carbon monoxide levels only
- Methane levels prior departure only
- Oxygen levels only

Which one of the following is considered to be the main danger to a bulk carrier in relation to seawater leakage into a cargo space?

- **The cargo is more likely to liquefy**
- The cargo is likely to be damaged commercially
- There is an increased risk of steelwork corrosion

Which one of the following is the main reason why bulk cargo should be trimmed level, or nearly level?

- **To avoid the dangers associated with cargo shifting**
- To avoid loading an excessive quantity of cargo
- To avoid oxygen depletion

Which one of the following is the minimum oxygen content that is required before entry into any enclosed space, such as a cargo hold, should be allowed?

- **21%**
- 10%
- 30%
- 5%

Which one of the following is the most likely result of moisture migration in a bulk cargo?

- **A flow state existing on the cargo's surface**
- Build up of toxic gases
- Oxygen depletion
- Self-heating of the cargo

Which one of the following procedures should be followed when a bulk cargo of coal is being carried at sea?

- **All hot work and smoking should be prohibited, otherwise normal deck procedures**
- Access to deck area only for safety inspections
- Deck area out of use to all crew
- Normal deck safety procedures

Which one of the following statements is the most accurate with regard to actions to be carried out before entry into an enclosed space, such as cargo space, is made?

- **A risk assessment and permit to work must be completed**
- Informing the bridge and the engineroom is the only precaution required
- Log book entries only need to be made
- No specific procedures need to be followed

Which one of the following statements is the most likely to be true, in relation to a cargo with a high angle of repose of 45°?

- **The cargo pile will be stable**
- The cargo pile may produces gases
- The cargo pile will be very unstable
- The cargo pile will liquefy quickly

Which one of the following statements is the most likely to be true, in relation to a cargo with a low angle of repose of 32°?

- **The cargo pile may easily shift**
- The cargo pile will be stable
- The cargo pile will be very high
- The cargo pile will self-heat

Which one of the following will be the likely result of a bulk carrier being loaded with a high-density cargo such as iron ore?

- **The vessel will have a high GM and fast roll period**
- The vessel will have a high GM and slow roll period
- The vessel will have a low GM and a fast roll period
- The vessel will have low GM and a slow roll period

Which two of the following apply to a vessel carrying a coal cargo when at sea?

- **No hot work to be carried out and no smoking where gases from cargo holds may be present.**
- **Normal work to be carried out on deck**
- All access to deck prohibited.
- All work on deck to be stopped

0069 – Basic Refrigeration Theory

Version 2.0 (3)

0 Kelvin is equal to . . . ?

- **–273,16 ˚Celsius**
- + 273,16 ˚Celsius
- 0 Fahrenheit
- 0 ˚Celsius

During automatic operation of a refrigeration system the starting and stopping of the compressor is usually achieved by which of the following?

- **Low pressure cut out.**
- Manual on/off switch.
- Temperature in the cold room.
- Thermostatic switch.

From the following options select those which may cause the refrigeration system high pressure cut out to operate and stop the compressor?

- **Loss of condenser coolant.**
- **Overcharge of refrigerant.**
- Ice on the evaporator coils.
- Oil carry over.

If a running system regularly requires charge of refrigerant, what should be done?

- **Find the leak and rectify the problem before the system is charged.**
- Charge the system, find the leak and rectify the problem.
- Charge the system, modern refrigerants are cheap and do not harm the environment.
- Replace the lost refrigerant with nitrogen.

Put in the correct words.
According to the second law of thermodynamics heat can only transfer naturally from a region of high temperature to a region of . . . . . .unless mechanical work is done on the system

- **low temperature**
- equal temperature
- high pressure

The closed cycle refrigeration system works on the principle of which of the following?

- **Vapour compression and recirculation of the refrigerant.**
- Rapid expansion and loss of refrigerant.
- Rapid expansion and recirculation of the refrigerant.
- Vapour compression with loss of refrigerant.

The ideal temperature of a freeze room is:

- **-18°C to -25°C**
- -26°C to -30°C
- -5°C to -10°C
- 0°C to -5°C

The operation of the automatic expansion valve is based on?

- **Constant evaporator pressure.**
- Constant evaporator pressure and temperature.
- Constant evaporator temperature.
- Constant flow of refrigeration medium.

The operation of the thermostatic expansion valve is based on?

- **Constant degree of superheat in the vapour at the evaporator.**
- Constant evaporator pressure.
- Constant flow of refrigeration medium.
- Constant temperature in the condenser.

The term "specific enthalpy" of a fluid, measured in kJ/kg, is used to quantify the energy per unit mass due to . . . . . ?

- **the sum of the internal energy plus the product of the pressure and volume.**
- the internal energy.
- the sum of the internal energy plus the volume.
- the sum of the pressure plus the volume.

To be able to evaporate a medium at –20˚C and to condense the same medium at +30˚C we have to . . .

- **increase the pressure**
- decrease the pressure

What are the disadvantages of ammonia as a refrigerant?

- **It is toxic and inflammable.**
- It can only be used on very large refrigeration systems.
- It cannot be used in refrigeration systems with screw compressor.
- It is very expensive.

What are the main components in a closed refrigeration cycle?

- **Compressor, condenser, expansion valve and evaporator.**
- Compressor, condenser and evaporator.
- Compressor, condenser, expansion valve and receiver.
- Compressor, oil separator, expansion valve and evaporator.

What is absolute pressure?

- **Atmospheric pressure plus gauge pressure.**
- It is equal to atmospheric pressure.
- It is the same as 100% vacuum.
- It is the same as gauge pressure.

What is the disadvantage of the manually operated expansion valve?

- **It is unresponsive to changes in load.**
- It can only be used with modern refrigerants.
- It cannot be used with modern refrigerants.
- It responds to quickly to changes in load.

What is the main problem with Carbon dioxide as a refrigerant?

- **Extremely high working pressure.**
- Extremely high cost.
- Extremely inflammable.
- Extremely low working pressure.

What is the saturation temperature?

- **The temperature at which a fluid boils or condenses, it varies with pressure.**
- It is the point where the ice starts to melt.
- It is the same as critical point for a gas.
- The temperature at which a fluid boils or condenses, it is independent of pressure.

What will be an adequate allowance for oil circulation in a refrigeration system?

- **One litre of oil for every twenty kilo of refrigerant charge.**
- One litre of oil for every kilo of refrigerant charge.
- One litre of oil for every thirty kilo of refrigerant charge.
- One litre of oil for every two kilo of refrigerant charge.

When the desired cold room temperature is reached in a refrigeration system operating on the direct expansion principle, which of the following shuts off the evaporator for the room?

- **Solenoid valve operated by thermostatic switch.**
- High pressure cut out.
- Low pressure cut out.
- Thermostatically controlled expansion valve.

Which of the following components are parts of the high-pressure side in a refrigeration system?

- **Condenser**
- **Liquid line**
- **Receiver**
- Evaporator
- Suction line

Which of the following does the term latent heat describe when referring to a refrigerant?

- **Heat required to cause evaporation at constant boiling temperature and pressure.**
- The heat required to raise the liquid temperature to boiling point.
- The heat required to superheat the refrigerant gas.
- The temperature.

Which of the following indicates refrigerant undercharge?

- **Compressor running continuously.**
- **Low head pressure.**
- **Vapour bubbles in the liquid line sight glass.**
- High head pressure.
- High receiver level.
- Low temperature in the refrigerated room.

0070 – Food Handling

Version 1.0 (0)

A steward reaches into a sink of soapy water to release the plug and cuts his hand on a knife lying in the bottom. Which of the following policies would have been most effective in avoiding this accident, while still following a practical and hygienic approach to the cleanliness of the equipment?

- **Every person washes their own knives.**
- Detergent is not used when washing knives.
- Heavy-duty kitchen gloves are used when washing knives and other utensils.
- There is a dedicated sink for washing knives.
- There is a dedicated time for washing knives.

According to regulation 9 to Annex V of MARPOL 73/78 for every ship of 12 metres or more in length placards must be displayed notifying passengers and crew of the disposal requirements for garbage. In what language, or languages, must these placards be written?

- **In the official language of the ship's Flag State and also in English or French for ships travelling to ports or offshore terminals of other nations.**
- In English and in the common language of the majority of the ship's crew.
- In English and in the language, or languages, of the passengers, if carried.
- In the common language of the ship's crew.

After cooking, at what minimum temperature must rice be maintained during service?

- **63°C.**
- 10°-12°C.
- 41°C.
- 83°C.

An electrically-powered heat counter breaks down. What should the catering staff do about it?

- **They should report the defect through the proper channels, isolate the equipment from the power supply and place a safety notice on the unit.**
- The catering staff are fully responsible at all times for any repairs necessary to galley equipment and should therefore make it serviceable again themselves at the earliest opportunity.
- They should immediately order a replacement unit for delivery in the next port.
- They should run an extension lead from a socket elsewhere in the galley and try to get the counter to run somehow from this alternative source.

At what recommended temperature should frozen fish be stored?

- **-20°C.**
- -1°C.
- -2°-3°C.
- -50°C.

How long is it estimated that it will take a tin can to degrade in the sea?

- **100 years.**
- 1 year.
- 200-500 years.
- 3-14 months.

How long must a Garbage Record Book be kept on board?

- **Two years from the date of the last entry.**
- Five years from the date of the first entry.
- Five years from the date of the last entry.
- One year from the date of the last entry.
- Until the vessel is sold, changes flag or is scrapped.

How many portions of fruit and vegetables are recommended per day for the average adult?

- **5**
- 10
- 3
- At least 1
- Not more than 7

How should wooden cutting boards be arranged after washing?

- **They should be stacked in such a way that air circulates over all surfaces during drying.**
- They should be placed in a dishwasher, which is programmed on the “accelerated drying” cycle.
- They should be rested on the galley range with the plate on minimum heat, until dry to the touch.
- They should be roughly wiped with a drying cloth and stacked one on top of another.
- Wooden cutting boards should only ever be scraped, sprayed with a sanitizer spray and wiped dry; they should never be washed with water.

How should you pick up a knife from a bench?

- **By the handle, with the point of the blade facing away from you.**
- By the blade, but only gripping the back and keeping clear of the cutting edge.
- By the handle, with the point of the blade facing towards you.
- It doesn’t matter really and depends how it was left by the last person who used it.
- Slide it to the edge and then pick it up by the handle.

In most colour coding systems for kitchen knives, which colour is used to indicate that the knife is to be used only for the preparation of raw meat?

- **Red.**
- Black.
- Blue.
- Orange.
- White.
- Yellow.

In order to meet the definition of "comminuted waste" under Annex V to MARPOL 73/78, what size of mesh must the particles be able to pass through?

- **25 mm.**
- 10 mm.
- 100 mm.
- 30 mm.
- 50 mm.

In terms of food temperature, what is the "danger zone"?

- **Between 5°C and 63°C.**
- Above 63°C.
- Below 5°C.
- Between 37°C and 63°C.
- Between 5°C and 37°C.

Is smoking permitted in the galley? (Select the most correct answer).

- **No.**
- No if the vessel is in port, yes if it is at sea.
- Yes, but only between the hours of 2000 and 0600.
- Yes, but only when the serving of food is complete.
- Yes.

Is staphylococcal food poisoning spread directly from person-to-person? (Select the most correct answer).

- **No.**
- Generally no, although there is a remote possibility if the average temperature in the galley does not drop below 22.7°C throughout the working day.
- Yes, but only where the carrier has suffered from an infectious food poisoning virus within the previous 3 months.
- Yes.

It is recommended that personnel engaged in food handling should avoid using perfume and scented anti-perspirant body sprays. Why is this?

- **Because there is a possibility that the scent can transfer to food during handling.**
- Because the heat of the galley area will increase the evaporation of the scent and mask the aroma from the food.
- Because the heat of the galley will react with the alcohol in the perfume and intoxicate the wearer.
- Because the heat of the galley will react with the perfume and induce itching.

Some fresh prawns have been peeled in preparation for a cocktail dish. Some fresh salad has been cleaned and prepared to accompany it. The dish is not required for another four hours. What will you do with these ingredients in the meantime?

- **The prawns and the salad will be placed in separate containers, covered and refrigerated until required.**
- The ingredients will be placed together in a dish, covered and refrigerated until required.
- The prawns and salad will be placed in separate dishes, covered in tin foil and left on the workbench until required.
- The prawns will be refrigerated and the salad left in cold water to keep it fresh.

The United States Centers for Disease Control and Prevention run a program designed to protect passenger and crew health by minimizing the risk of gastrointestinal illness aboard cruise ships. What is it called?

- **The Vessel Sanitation Program.**
- The Cruise Industry Best Practise in Passenger Health Scheme.
- The Cruise Ship Health Monitoring System.
- The Homeland Security (Food Terrorism) Program.

The cook on this ship had been under pressure from the Catering Superintendent and the Captain regarding the amount of food wasted over the month and the overtime hours of the catering department. In order to save both time and food, he decided to cook a large portion of rice, keep it warm for service during the evening meal, then refrigerate the leftover rice and reheat the next day for the lunchtime meal. Is what he did acceptable? (Select the most correct answer).

- **No, what he did is extremely dangerous and could have led to food poisoning throughout the crew.**
- No, but the chance of anything going wrong was minimal as the rice was not directly mixed with any other food during reheating and the refrigeration would have ensured that the rice remained fresh.
- Yes, but he would have to ensure that the rice was reheated to a temperature of at least 43°C.
- Yes, it certainly saved both on preparation time and the amount of food wasted and, being only rice, posed no health risk to the crew.

Under Annex V of MARPOL 73/78, at what distance from shore may a vessel inside a Special Area dispose of inert ash, comminuted (slurry) into the sea?

- **Disposal to sea is prohibited.**
- >12 miles.
- >25 miles.
- >3 miles.

Under Annex V of MARPOL 73/78, at what distance from shore may a vessel inside a Special Area dispose of paper, rags and glass bottles that have been ground down or comminuted into the sea?

- **Disposal to sea is prohibited.**
- >12 miles.
- >25 miles.
- >3 miles.

What are the three principles on which many Garbage Management Plans are based?

- **Disposal.**
- **Recycling.**
- **Reduction at source.**
- Compatibility of packagings.
- Consolidation of waste types.
- Incineration potential.

What is a "butcher's steel"?

- **A hardened-steel tool, with closely packed grooves providing an abrasive surface and used for sharpening knives.**
- A galvanized steel hook used for hanging large cuts of meat.
- A retaining bar, fitted to the galley range in bad weather, to prevent movement of the pots and pans.
- A special short bladed knife used for scoring joints of meat prior to seasoning and basting.

What is the best policy to follow, in order to prevent personnel from other departments introducing germs and other forms of contamination into the galley?

- **Prohibit unauthorized persons from the galley and storeroom areas at all times.**
- Only permit other persons to enter between the hours of 2000 and 0600.
- Post a notice outside the galley access door requesting personnel to knock before entering, so that you can check them prior to granting permission.
- Require all visitors to the galley to pass through a designated decontamination area on entering.

What is the estimated average required daily calorie intake for an active male?

- **2900**
- 1940
- 2150
- 3790
- 4100
- 5250

What is the most common form of food poisoning worldwide?

- **Salmonella enteritidis.**
- Eschenchia coli.
- Norwalk virus.
- Staphylococcal enterotoxin.

What kind of saw is used to cut bone that is deep and well covered by meat?

- **A bow saw.**
- A bone saw.
- A gullet saw.
- A rake saw.
- A tendon saw.
- A tenon saw.

What precautions should you take before washing down the galley at the end of the day?

- **All power to the galley range and other electrical equipment should be switched off and isolated.**
- All electrical equipment should be switched on to ensure that they dry off in the shortest possible time.
- Check that the water supply is clean and fresh to minimize the risk of electrical shock.
- Remove all electrical equipment from the galley, or cover it in plastic sheeting.

What should be on the deck immediately outside the door to a walk-in refrigerator or freezer space?

- **Non-slip matting.**
- An area marked with red diagonal lines.
- An area marked with yellow diagonal lines.
- Disposable brown paper sheeting.
- Fire-resistant coir matting.
- Plastic sheeting.

Which are the two recommended methods of cooling stock after preparation?

- **By standing deep pots in a sink of cold water.**
- **By transferring to wide, shallow pans.**
- By cooling overnight on a galley bench.
- By placing in a refrigerator immediately after cooking.

Which international regulations deal specifically with the prevention of pollution by garbage from ships?

- **Annex V, MARPOL 73/78**
- Annex I, ISPS Code.
- Annex VII, MARPOL 73/78
- Chapter II, SOLAS 1974
- Chapter VI, SOLAS 1974.
- Section III, ISPS Code.

Which of the following are acknowledged as common sources of staphylococcal food poisoning? (Select all applicable answers).

- **Food that is not cooked thoroughly.**
- **Food that is not kept properly hot or cold as required.**
- **Foods that are kept at slightly elevated temperatures after preparation.**
- **Foods that require considerable handling during preparation.**
- **Handling of foodstuffs with bare hands, after touching the face or mouth.**
- Raw salad leaves originating from the African continent.
- Uncooked fish which has been defrosted in a refrigerator.
- Waxes citrus fruit.

Which of the following are designated as "Special Areas" under Annex V of MARPOL 73/78?

- **The Antarctic Area.**
- **The Baltic Sea area.**
- **The Gulfs area.**
- **The Mediterranean Sea.**
- **The North Sea.**
- **The wider Caribbean Region.**
- The Bay of Biscay.
- The South China Sea.

Which of the following are forms of food poisoning?

- **Norwalk virus.**
- **Salmonella enteritidis.**
- **Staphylococcal enterotoxin.**
- Allergic rhinitis.
- Chronic dyspraxia.
- Cytomegalovirus infection.
- Hypothermia.
- Type "A" diabetes.

Which of the following are precautions necessary before using electrical equipment in the galley? (Select all applicable answers).

- **Check that all guards are in position.**
- **Confirm that the equipment has been properly assembled and set-up.**
- **Confirm that the machine has been properly cleaned before use.**
- **Ensure that you are properly and safely dressed.**
- **Ensure that your colleagues are aware that you are about start to use the equipment.**
- **Establish that the equipment is in good working order.**
- **Verify that the machine is suitable for the intended task.**

Which of the following categories of garbage are prohibited from disposal to sea under Annex V to MARPOL 73/78?

- **Paper, rags, glass, etc, comminuted or ground, vessel within a Special Area.**
- **Plastic, including synthetic ropes, fishing nets and plastic bags, vessel inside a Special Area.**
- **Plastic, including synthetic ropes, fishing nets and plastic bags, vessel outside a Special Area.**
- Ash (inert) comminuted (slurry), vessel outside a Special Area.
- Food waste, comminuted or ground, vessel outside a Special Area.

Which of the following conditions must be applied to left-over cooked meat from an evening meal if it is to be re-used in a pie dish for the following mid-day meal? (Select all applicable answers).

- **The cooked meat must be finely chopped or minced to allow reheating to at least 70°C as quickly as possible.**
- **The pie containing the re-used meat must only be heated once.**
- **The previously cooked meat must not be mixed with newly prepared raw meat.**
- Once cooked initially, the pie may be safely reheated up to three times in 24 hours.
- The cooked meat can be refrigerated and then taken out again up to two hours before serving, mixed with hot vegetables and sauce and safely left standing at room temperature prior to final re-heating just before service.
- The cooked meat must be layered in the pie and heated to at least 63°C.

Which of the following food or food constituents are considered to be healthy when consumed on a regular basis?

- **Fat-free or low-fat produce**
- **Fruit**
- **Low levels of sugar**
- **Vegetables**
- Cholesterol
- High levels of alcohol
- High levels of sodium
- Saturated fats

Which of the following food or food constituents are considered to be unhealthy when consumed on a regular basis?

- **Cholesterol**
- **Fatty acids**
- **High levels of alcohol**
- **High levels of sodium**
- **Saturated fats**
- Fruit
- Low-fat produce
- Vegetables

Which of the following is the recommended form of footwear for the galley?

- **B.**
- A.
- C.
- D.

Which of the following items of equipment should be available with the hand-washing facilities in the galley? (Select all applicable answers).

- **A bacterial liquid soap dispenser;**
- **A hot air dryer or disposable towel dispenser.**
- A light with a waterproof switch.
- A shaving mirror.
- Lightweight latex rubber gloves.

Which of the types of fire extinguisher shown here would you routinely find in a galley and storeroom area? (Select all applicable answers).

- **Carbon dioxide extinguisher**
- **Dry powder extinguisher**
- Foam extinguisher
- Water extinguisher

You are an assistant cook working on a large ferry. You have a boil on your wrist which has become infected. What must you do? (Select all applicable answers).

- **Avoid handling foodstuffs until the condition passes.**
- **Obtain medical assistance from the responsible Officer.**
- **Report the fact to your superior immediately.**
- Cover the boil with a small flesh-coloured band-aid and take two paracetamol.
- Don't tell anyone and try and work through it.
- Wear long cotton gloves when handling any raw food.

You are taking stores in port. A ship's officer and a port official approach and say they wish to look inside some of the packages. Under which Code are they required to carry out such inspections?

- **The International Ship and Port Facility Security (ISPS) Code.**
- MARPOL 73/78.
- The International Health and Safety (Ships Provisions) Regulations.
- The World Health Organization Foodstuffs Monitoring Guidelines.

You are the Cook on a cargo vessel. On which of the following occasions would you wash your hands? (Select all applicable answers).

- **After blowing your nose.**
- **After handling raw meat, poultry, fish and vegetables.**
- **After handling refuse or food waste.**
- **After using the toilet.**
- **Before commencing work.**
- **Before preparing cold foods which will not require further cooking.**

You have just collected provisions from the freezers and the dry store room and are returning up the stairway to the galley. You accidentally lose your balance and fall, dropping the tray and banging your knee against the edge of the step, causing minor bleeding. Which of the following statements describes what you should now do?

- **I will report the accident to my superior and ensure that an entry is made in the galley accident logbook in compliance with the company safety policy. I will obtain basic first aid treatment. I will ensure that the foods I was carrying are inspected for damage and any defective foods discarded.**
- I will report the accident to my superior just in case there are any questions at a later stage. I will get a bandage from the first aid kit and wrap it round my knee. I will pick up all the dropped food, give the fresh and frozen produce a quick wash and place it in the galley for preparation.
- The accident is minor and did not actually take place in the galley. It is really my own fault, because I was carrying too much on the tray and it is all frankly a bit embarrassing, so I will say nothing and just carry on working.

You have just joined a large oil tanker as cook for the first time. What will you do in respect of the alarms and other safety devices fitted to the walk-in refrigerators and freezers ? (Select all applicable answers).

- **Ask the departing cook to demonstrate any devices I am not confident and competent with.**
- **Check the Safety Manual for procedures applicable to the devices.**
- **Check when the devices were last tested and that the test proved them in satisfactory condition.**
- **Establish where the devices are located.**
- **Identify which devices are fitted.**
- Make a mental note to have a look at the alarms within the first few days.
- Not sign-on until the ship's Safety Officer has demonstrated their use.

You have just made up a batch of chocolate éclairs, containing fresh cream. At what temperature must they be stored until required for service?

- **5°-7°C.**
- -3°C
- 0°C.
- 10°C.
- 1°-2°C.

0072 – Basic Hydraulics

Version 1.2 (81)

By reducing the flow, the torque will:

- **Be the same**
- Be higher
- Be reduced

Counterbalance valve is used:

- **To support a load and prevent "creeping"**
- It is the same as a check valve
- To unload the delivery pump

Energy not used to move the load in a hydraulic system turns into what?

- **Heat**
- Noise
- Pressure

How much pressure will a force of 2000 N create in a cylinder with a piston area of 0,0025 m2?

- **800 kPa**
- 1600 kPa
- 400 kPa

Internal leaks create:

- **Heat**
- Faster movement of the actuator
- Higher pressure in the system

Is hydraulic oil compressible?

- **Aprox. ½ % pr. 7 MPa**
- 10% pr 5 MPa
- No

Movement of the actuator depends on:

- **Flow**
- Pressure

Pascal's law says:

- **Force = pressure x area**
- Force = pressure + area
- Force = pressure / area

Rate of flow determines:

- **Speed**
- Pressure
- Torque

This is a:

- **Variable delivery, bi directional pump**
- Variable delivery, bi directional motor
- Variable delivery, unidirectional pump

This is a:

- **4 port, 3 position v/v with float centre**
- 4 port, 3 position v/v with tandem centre
- 5 port, 3 position v/v with closed centre

This is a:

- **Relief valve**
- Pressure reducing valve
- Unloading valve

What creates pressure in a hydraulic system?

- **The resistance to flow**
- The directional valve
- The pump

What is the most common reason for failures in hydraulic systems?

- **Contamination of the oil**
- The use of the wrong type of oil
- Water in the oil

What is the most reliable indication of a faulty hydraulic pump?

- **Excessive case drain flow**
- Low oil temperature
- Low system pressure

What type of gas should be used to charge an accumulator?

- **Nitrogen**
- Compressed air
- Oxygen

When do we use a pilot operated check valve?

- **To prevent creeping**
- To prevent overload
- To reduce load

Which statement is not true?

- **Hydraulics cannot be used in hazardous condition**
- Hydraulics has a simple design, few parts and is easy to operate
- Hydraulics produce constant torque with variable speed

Which statement is true?

- **Hydraulic components have a high power to weight ratio.**
- Difficult to control speed of actuator
- Hydraulic components need external lubrication.

Which type of pump is most common in hydraulic systems?

- **Positive displacement pump**
- Non-positive displacement pump

0073 – Crisis Management

Version 2.0 (147)

A company must update its Safety Management System and Manual:

- **whenever regulations change**
- at least once every 12 months
- at least once every 18 months
- at least once every 6 months

A moderate level of stress provides which useful function in an emergency?

- **To prepare for peak performance that will improve the chances of survival**
- To immobilize the individual until he has decided upon the best course of action
- To slow down the heart rate to conserve energy
- To slow down the thought processes to ensure a thorough assessment of the threat

After a drill, when is the most effective time to conduct a debriefing?

- **Immediately after the drill.**
- Debriefings should only be held when necessary.
- The day after the drill, to allow crew members to reflect on their actions.
- Within 24 hours after the drill.

Describe how Captain Smith demonstrated for Titanic crewmembers the principle of leading by example:

- **Throughout the crisis, he was calm, decisive, and approachable.**
- He wasn't to blame for the collision because he wasn't on the bridge.
- His order to board women and children first was admirable.
- His rank and his years of service were enough to ensure respect.

During the Morro Castle crisis, crewmembers showed a complete failure to deal with the emergency. What caused them to abdicate their safety roles?

- **Lack of leadership**
- Inadequate fire-fighting provisions
- Inadequate fire-fighting training
- No lifeboat drills for passengers

Effective drills are Relevant, Realistic and . . . ?

- **Unpredictable.**
- Exciting.
- Familiar.
- Interesting.

In an emergency, if passengers receive clear, direct communications, what result can be expected?

- **Passenger cooperation is increased.**
- Messages don't have to be repeated.
- Passenger stress is increased.
- Passengers are more likely to panic.

In an emergency, why is it important that the emergency teams give constant feedback to the bridge?

- **It gives the master a good basis for decision making.**
- The bridge team has to keep the head office up to date.
- The cruise director has to decide what to tell the passengers.
- The master needs to know when to abandon ship.

In preparing for emergency response, which are the two most important elements?

- **Emergency plans and drills**
- The design and layout of the ship
- The ISM Code and the Safety Management System
- The safety requirements of the USCG and the company

Name one advantage of drills that are unpredictable?

- **It increases the crew's ability to handle changes and uncertainty.**
- It is more interesting for the captain.
- It keeps the crew from getting bored during a drill.
- It makes the crew talk about the drill after it has been complete.

Select four factors that affect the situational awareness of the crew

- **Experience.**
- **Knowledge.**
- **Safety awareness.**
- **Training.**
- Cultural background.
- General intelligence.
- Maturity.

Studies of shipboard emergencies show that many passengers do not read or remember safety information. As an officer, how can you best compensate for this fact?

- **In an emergency, repeatedly tell passengers what's happening, what's being done, and what they should do.**
- Ensure that printed safety information onboard is of the highest quality.
- Post large, graphic safety signs conspicuously throughout the ship.
- Prior to departure, provide passengers with a good safety video.

The ISM code requires every company to develop a Safety Management System. For what kind of emergencies should the SMS system include plans and procedures?

- **All imaginable emergencies.**
- Fire, collision and grounding.
- Fire, collision, grounding and oil pollution.
- Fire, collision, grounding and piracy.

The Safety Management Manual contains checklists that are:

- **clear, concise summaries of critical action items in the emergency plan**
- clear, concise summaries of all action items in the emergency plan
- detailed descriptions of action items most easily forgotten in emergencies
- to be used when crewmembers feel especially stressed in emergencies

There are three recognized states of stress; Resistance, Exhaustion and . . . ?

- **Alarm.**
- A sense of hopelessness.
- Apathy.
- Panic.

What is the best strategy for reducing passenger stress in an emergency?

- **Give passengers information as soon as possible.**
- Inform the passengers only if evacuation becomes necessary.
- Tell passengers something that will make them feel secure.
- Try to resolve the emergency without notifying the passengers.

What is the most common reaction to emergencies among passengers?

- **Confusion.**
- Being rational.
- Panic.
- Shock.

What is the most important factor in the master's ability to make good decisions in an emergency?

- **Feedback from the crew**
- Following safety regulations
- Input from company personnel
- Planning of emergency procedures

What should you do if a passenger is unable to understand emergency instructions due to language problems?

- **If possible, try to find a passenger or crew member who can translate.**
- Ignore the passenger and assume that he or she will follow the others.
- Speak louder and hope that the passenger will understand.
- The passenger will understand English if you speak slowly enough.

Where can you find plans and procedures for responding to emergencies?

- **In the Safety Management Manual.**
- Ask the captain.
- On the internet.
- Posted on the wall in the crew mess.

Which is the most important factor in the safety awareness of the ships crew?

- **The level of emergency-preparedness**
- The quality of emergency drills
- The safety awareness of the senior officers
- The shipboard training provided

Which of the following approaches is more likely to encourage feedback from the passengers?

- **The safety awareness of the crew and the delivery of safety information are the best ways to encourage passengers to report unusual situations.**
- Passengers should be encouraged to report unusual situations to an officer because the crew are focused on performing their routine duties.
- Passengers should not be encouraged to report unusual situations because this can cause them to doubt the safety of the ship.
- The crew should calm passengers by explaining that their safety concerns are unfounded.

Which of the following is most likely to influence changes to emergency plans and procedures?

- **The design and layout of the ship**
- MARPOL 73/78
- The ISM Code
- The safety requirements of the USCG

Which statement about stress is correct?

- **Individuals in a shipboard emergency experience different levels of stress.**
- An example of stress is a shipboard fire.
- Stress interferes with an individual's ability to respond to a threat.
- Stress is an unusual reaction to a threatening situation.

Why is it important that crew members are safety-conscious and alert?

- **They can notice and report something wrong before any detection system.**
- It gives the passengers a good impression of the on board safety.
- It makes them more alert to the passenger's needs and therefore improves the passenger service.
- To ensure that the crew is ready for a drill at any time.

0074 – Sultzer Medium Speed Diesel Engine

Version 2.0 (894)

Select the correct option to accurately complete the following statement according to the information provided in this module. During slow turning of the main engine, the indicator cocks . . .

- **. . . must be fully open throughout the operation.**
- . . . may be either open or closed.
- . . . must be kept shut throughout the operation.
- . . . need only be partially opened.

Select the correct option to accurately complete the following statement according to the information provided in this module. The main engine can be started after a long period of shutdown . . .

- **. . . only after performing basic checks listed in this module.**
- . . . almost immediately
- . . . only after performing advanced checks and critical measurements
- . . . only with the permission of the head office ashore

Select the correct option to accurately complete the following statement according to the information provided in this module. The turning of the cylinder lubricators before start up

- **. . . is to ensure pre-lubrication and assess the force required to pump the cylinder lubricating oil and check its flow.**
- . . . is merely a part of routine.
- . . . makes no difference at all.
- . . . wastes unnecessary lubricating oil and must be avoided.

Select the correct option to accurately complete the following statement according to the information provided in this module. When paralleling two main engines the rpm of the incoming engine should be . . .

- **. . . slightly higher than the running engine.**
- . . . fluctuating between higher and lower rpm than the running engine.
- . . . slightly lower than the running engine.
- . . . the same rpm as the running engine.

Select the correct option to accurately complete the following statement according to the information provided in this module. When two of the Sulzer 12ZAV40S engines are running in parallel . . .

- **. . . one of the speed governors controls the fuel setting for both engines, ordering the same fuel setting for each engines.**
- . . . one of the speed governors controls the fuel setting for both engines, ordering an increased fuel setting for the first connected engine.
- . . . each speed governor controls the fuel setting for its own engine.
- . . . one of the speed governors controls the fuel setting for both engines, ordering an increased fuel setting for the second connected engine.

Select, from the given options the one that completes the following statement correctly, based on the information given in this module. The options for starting the auxiliary engines are . . .

- **. . . from the SELMA computer and a manual start facility.**
- . . . from any remote location such as the cargo control room.
- . . . from the engine room only.
- . . . from wheelhouse and control room.

Select, from the statements given in the options, the one which is most accurate according to the information provided in this module.

- **Some of the main engine auxiliary pumps and machines are normally running continuously.**
- All of the main engine auxiliary pumps and machinery normally run continuously.
- All of the main engine auxiliary pumps and machines are started and stopped using Pressostat control.
- All of the main engine auxiliary pumps and machines are started using a group start system.

Which of the given options is the highest voltage available onboard the vessel studied in this module?

- **660 V**
- 110V
- 220V
- 440V

Which of the given options is the power rating for each of the main engines featured in this module?

- **7100KW**
- 4000KW
- 5000 KW
- 6000 KW

Which of the manufacturers given in the options made the separators used in the ship studied in this module?

- **Alfa Laval**
- De Laval
- Mitsubishi
- Westfalia

0078 – Pumps and Pumping Operations

Version 4.0 (966)

A pump is to be used to pump fresh water at a temperature where the vapour pressure of the water is equivalent to the pressure caused by 0.2 m head of fresh water, and atmospheric pressure is equivalent to 10.6 m head of fresh water. The maximum suction lift is 6.0 m, and the pipeline losses on the suction side are equivalent to a head of 3.0 m.
What will the pump NPSH available be?

- **1.4 m**
- 0.3 m
- 0.7 m
- 1.1 m

If a pump had available NPSH of 0.5m would a NPSH requirement of 0.48 be suitable for this purpose?

- **No**
- Yes

Is the following statement true or false: 'No types of positive displacement pump can change their delivery rate unless the speed of the drive motor is changed'.

- **False**
- True

Is the following statement true or false: 'When operating pumps in parallel, if we use three pumps we will get three times the flow'

- **False.**
- True.

Is the following statement true or false: in centrifugal pump series operation the flow through all of the pumps is equal.

- **True**
- False

Is this pump a single acting pump or a double acting pump?

- **Double acting**
- Single acting

On which side of point A will cavitation occur?

- **Side B**
- Side A

Under what system conditions would operating pumps in parallel achieve very little benefit? Select the ONE best answer.

- **If the piping system is undersized or some component in the system acts as a bottleneck**
- If the piping system is oversized, causing a lack of discharge pressure
- If the pumps are of different size.

What is the main advantage of starting a centrifugal pump with the delivery valve closed? Select the ONE best answer.

- **The starting load on the motor will be minimum.**
- The pump will be self-priming.
- There will be less vibration.
- There will be no surge of pressure in the pipeline, causing the relief valve to open.

What is the main cause of cavitation of the fluid in a centrifugal pump? Select the ONE best answer.

- **The vapour pressure in the suction pipe falls below the vapour pressure of the liquid at a given temperature.**
- The viscosity of the fluid is too high, the extra power absorbed being converted into heat.
- Too high a speed of impeller rotation creating adverse heat.

What is the main reason for operating pumps in parallel? Select the ONE best answer.

- **It allows a larger range of flow than for a single arrangement.**
- A greater discharge pressure can be achieved.
- The combined electrical load on the pumps is lower than an equivalent single pump.

What is the main reason for operating pumps in series? Select the ONE best answer.

- **So that commercially available pumps (i.e. small) can be used and to avod extremely high discharge pressures which would be necessary with only one pump.**
- To avoid cavitation in high-pressure systems.
- To share the power absorbed using many small motors instead of one large motor, which is more energy efficient.

What operating principle do positive displacement pumps rely on? Select the ONE best answer.

- **Changes of volume of the pump chambers.**
- Compressing the fluid and then re-expanding it.
- High speeds giving the fluid Kinetic energy.
- Speeding up the fluid and then slowing it down.

What will happen if the NPSHA is less than the NPSHR? Select the ONE best answer.

- **The pump will suffer cavitation damage.**
- The pump needs to run faster.
- The pump will not work at all.
- The pump will overheat.

Which ONE of the following statements is correct?

- **Positive displacement pumps can require priming**
- Positive displacement pumps always require priming
- Positive displacement pumps are always self priming

Which one of the following does not directly affect the efficiency of a pump? Select the ONE best answer.

- **NPSHA**
- Pump flowrate
- Pump Losses
- The discharge head.

Which one of the following statements best describes the energy transformation within a centrifugal pump? Select the ONE best answer.

- **The impeller increases the liquid velocity, increase the fluid's kinetic energy. This kinetic energy is then converted into pressure by reducing the velocity in the volute casing.**
- Decreasing the velocity in the impeller decreases the kinetic energy, decreasing the kinetic energy whilst increasing the velocity of the fluid in the volute casing increasing its pressure.
- Pressure is converted into kinetic energy by the impeller; this kinetic energy is converted to an increase in velocity by the volute casing.

Why are positive displacement pumps well suited to fuel transfer duties? Select the ONE best answer.

- **They are self priming and can handle some amount of vapours and gases.**
- They can deliver high pressure.
- They have very small clearances.
- They require no maintenance.

Why do positive displacement pumps require a relief valve? Select the ONE best answer.

- **Because the pump will always try and displace a constant volume of fluid, it can generate dangerously high pressures against a closed discharge valve.**
- A high pressure could overload the motor.
- It allows priming liquid to return to the suction.
- It can regulate the delivery pressure.

Why is the axial flow pump ideal for trimming and heeling duties? Select the ONE best answer.

- **It is reversible and has a high capacity flow.**
- It has excellent suction lift.
- Its discharge pressure is increased by the speed of the ship.

0082 – Crowd Management

Version 2.0 (183)

Approximately how many of the passengers will immediately accept that there is an emergency?

- **10%.**
- 25%.
- 50%.
- 75%.

Give clear, direct instructions in an emergency, and show passengers exactly what they need to do. What is the most important reason for doing this?

- **75% of passengers are confused, and feel helpless.**
- 25% of passengers go into shock.
- 3% of passengers have little or no ship safety knowledge.
- 3% of passengers panic.

How can you expect a passenger in shock to behave?

- **They will become paralyzed by fear and not react at all.**
- They will act hysterical.
- They will be able to focus and look after themselves.
- They will wait for others to lead and tell them what to do.

How can you expect a passenger who is confused to behave?

- **They will wait for others to lead and tell them what to do.**
- They will act hysterical.
- They will be able to focus and look after themselves.
- They will become paralyzed by fear and not react at all.

How can you recognize that a person is panicking?

- **They will lose control and act hysterical.**
- They will act like nothing is wrong.
- They will be able to focus on what they need to do.
- They will sit still and act apathetic.

How is body language generally communicated?

- **the way you stand, gesture, and walk**
- looking at someone directly in the eyes
- the tone of your voice
- touching other people

How should a crew member treat a passenger who is panicking?

- **Immediately take control of them. They need others to look after them.**
- Guide them in a gentle manner.
- Leave them alone, they will eventually start looking after themselves.
- Shout at them until they calm down.

If the lighting fails, what is a technique for keeping a line of moving passengers in contact with one another?

- **Have each person place a hand on the shoulder of the person in front.**
- Have everyone hold hands, forming a chain.
- Have one crewmember lead and one follow the line of passengers.
- Tie a rope around everyone.

If you are looking for passengers who may be trapped or hiding, what is the best command to shout?

- **Is anyone here? Call out if you need my help.**
- Don't stay here.
- Go to mustering stations immediately.
- Take warm clothes and lifejackets.

In a non-verbal demonstration, which action provides good communication?

- **Exaggerate your movements.**
- Avoid eye contact.
- Do not say the instructions verbally.
- Do the procedure as you would without an audience.

In addition to the type of ship and nature of the voyage, which factor in general does the passenger safety knowledge depend on?

- **When during the voyage an emergency occurs.**
- Information given on board.
- The information given in advance of the voyage.
- The safety knowledge of the crew.

In an emergency, passengers are most likely to be familiar with the ship's layout and be able to find escape routes, if they are:

- **Frequent travellers on a 3-hour domestic ferry trip, and they are in the passenger lounge at the time of the emergency**
- First-time travellers on an overnight ferry trip, and they are in their cabins at the time of the emergency
- On an 8-day cruise, and they are in their cabins when the emergency occurs on the first day of the cruise
- On an 8-day cruise, and they are in their cabins when the emergency occurs on the sixth day of the cruise

In an emergency, should passengers be allowed to return to their cabins to get their pets after having mustered at their station?

- **If lives are not endangered, the master may allow crew to get pets.**
- If passengers wish to collect pets or belongings, they should always be allowed to do so.
- No, pets are not considered important in an emergency.
- Pets will be collected in the "warn and sweep" procedure.

In an emergency, why is it important for the crew to use positive commands when communicating with passengers?

- **Passengers are more likely to hear the correct instructions and understand what to do.**
- A positive approach makes it less likely that passengers panic.
- Passengers often do not listen to negative commands.
- To keep the passenger's moods up

In order to speak clearly and effectively you should:

- **open your mouth, pronounce your consonants, and shape your sounds well**
- gesture with your arms
- never open your mouth too wide
- stand tall with shoulders squared

In which category of passengers do you think a woman who shows you a police badge might belong?

- **30% who are rational**
- 25% who go into shock
- 3% who panic
- 75% who are confused and helpless

Personal life-saving equipment available to passengers and crew include:

- **immersion suits**
- painter lines
- self-contained breathing apparatus
- TPAs

The benefits of emergency response plans are only realized if:

- **the crew carries out the plans**
- escape routes are used
- lifejackets are put on
- survival craft are launched

To Sweep the Ship in an emergency means that crewmembers are:

- **searching for passengers in cabins or other hidden places onboard**
- directing passenger crowds to muster and embarkation stations
- giving safety announcements from the bridge
- sounding alarm bells from the bridge

To which position(s) should you direct the first passengers into the lifeboat?

- **at either end of the boat**
- across from the entrance to the boat
- anywhere passengers want to sit
- right beside the entrance to the boat

What is NOT typical passenger behavior during an emergency?

- **Being alert and searching for information.**
- Moving to the upper side of a listing ship.
- Slow reactions.
- Staying in an area where they feel safe.
- Trouble understanding directions.

What is important to remember when giving passengers information in an emergency?

- **Only give the passengers information you know is correct.**
- As long as the passengers are at the assembly station, the amount of information they need is limited.
- Tell the passengers everything you hear, no matter the source.
- Try to only give them good news, even if you have to lie.

What is one reason for repeating a command to passengers in an emergency?

- **More people will have a chance of hearing the command.**
- You can establish your presence.
- You don't have to be as clear in what you say.
- You won't have to say the command more than twice.

What is the main reason for crewmembers to practice search and rescue operations on ships in an emergency?

- **Passengers seek safety in their cabins, or in other places onboard, where they think they can escape danger.**
- Passengers may not hear the alarm bells, and they may not know that they should proceed to mustering stations.
- Passengers may not hear the public address system, and they may not know that they should proceed to mustering stations.
- Passengers seek safety in mustering stations, or other common areas, where they think they can escape danger.

What is the main reason that passenger baggage creates problems for crowd control?

- **Passenger baggage clutters escape routes, and embarkation stations, and slows down an evacuation.**
- If passengers are already in muster stations, you will have to allow them to return to cabins to retrieve baggage.
- Muster stations are crowded, and passenger baggage takes up space.
- There isn't room in the life rafts, and crewmembers will have to tell passengers to leave baggage behind.

What is the most important reason for separating a panicking passenger from the group?

- **Panic can spread and affect other passengers.**
- It is easier to calm the person down in private.
- You don't want other passengers to hear if you have to become very strict with the panicking person.
- You want to protect the panicking person as much as possible.

What is your most appropriate action before demonstrating the donning of a lifejacket?

- **Stand in a high spot where everyone can see you.**
- Do up the strap.
- Pull the strap tight.
- Put on the lifejacket.

When doing a "warn and sweep" procedure; how can you ensure that a passenger cabin is empty?

- **Check under desks, beds, in closets and other places where a person can hide.**
- It's enough to knock on the door – if no-one opens, the cabin is empty.
- Open the door and ask if anyone are there - if there is no answer, the cabin is empty.
- Open the door and have a quick look – if you cannot see anyone, the cabin is empty.

When people fear for the safety of their friends or relatives, which of the following actions would you take to help calm them?

- **Assure them that the crew is sweeping the entire ship for passengers.**
- Give them a lifejacket to put on.
- Inform them about the nature of the emergency.
- Prevent them from looking for their loved ones.

Which action is the responsibility of Crew Members C or D, in addition to removing passengers from the chute?

- **Count the passengers.**
- Call up "Go!" to Crewmember B at the opening of the chute.
- Load the liferaft.
- Tell passengers to put on their hats or sunglasses if removed.

Which of the following is NOT an important tool for crowd control?

- **Shouting.**
- Body language.
- Establishing presence and control.
- Non-verbal skills.
- Verbal skills .

Which of the following statements is correct when discussing warm clothing and passengers?

- **Warm clothing will protect passengers from hypothermia.**
- After being mustered, passengers will return to their cabins for warm clothing.
- The crew will pass out warm clothing for passengers.
- Warm clothing will protect passengers from drowning.

Why is it a good idea to set passengers to look after other passengers in an emergency?

- **It allows the crew to focus on crowd control and other duties.**
- It allows the passenger to focus on other things than the emergency.
- Passengers often trust other passengers more than they trust crew members.
- Text: It gives the passenger something to keep them occupied.

You are standing on the floating platform and helping to load the life raft. When can you safely let go of the passenger's arm?

- **When most of the passenger's weight is transferred into the life raft.**
- After you walk into the life raft with the passenger.
- When someone inside the life raft grabs the passenger's arm.
- When the passenger says it's okay to let go.

You have ensured that a wheelchair contains no equipment essential to the passenger's survival. How should you help this person reach the embarkation station?

- **Remove the person from the wheelchair and assign people to carry the person.**
- Have several people carry the person in the wheelchair up or down the stairs.
- Use the elevator.
- Use the wheelchair in corridors but carry the person down stairs without the wheelchair.

You should use short and simple words, rather than long, complex words, when giving emergency instructions. What is the main reason for this?

- **People are more likely to understand you.**
- Short words take less time to say and you don't have much time.
- You won't have to repeat the instruction.
- You won't have to think too hard about what you are saying.

0083 – Automation

Version 2.0 (0)

Can too high viscosity cause damages to the fuel injection system?

- **It can damage parts of the fuel pump and their drive, it can cause the relief valves to spray off, and it can affect the atomisation of the fuel.**
- It can only affect the atomisation.
- It cannot cause any damages.

If an input value of a temperature controller connected to a heater is above the set point, this will result in

- **A reduction of the output signal.**
- A narrowing of the proportional band.
- A stabilization of the output signal above the proportional band.
- An increase of the output signal.

Select the components you mean are the main components in a steam viscosity system.

- **Controller (proportional, proportional + integral or proportional + integral + differential).**
- **Differential pressure converter.**
- **Flow rate regulator.**
- **Measuring tube built into the fuel line, and provided with a thermometer and manometer.**
- **Steam regulating valve.**
- **Viscosensor.**
- Fuel filter.
- Separator.

What are the highest viscosity modern diesel engines are capable of burning?

- **Approximately 700 cSt, at 50° C.**
- Approximately 1000 cSt, at 50° C.
- Approximately 1200 cSt, at 50° C.
- Approximately 500 cSt, at 50° C.

What determines the oil's viscosity?

- **Temperature and composition.**
- Only the temperature.
- The oil's content of cat fines.
- The oil's content of sediments.

What do we mean with a "180 grade fuel"?

- **Viscosity equal to or less than 180 cSt at 50 deg C.**
- Viscosity equal to or less than 180 cSt at 70 deg C.
- Viscosity equal to or more than 180 cSt at 50 deg C.
- Viscosity equal to or more than180 cSt at 70 deg C.

What gives the best separating effect?

- **Low throughput (m3/hr) low viscosity.**
- High throughput (m3/hr) high viscosity.
- High throughput (m3/hr) low viscosity.
- Low throughput (m3/hr) high viscosity.

What is the most effective way of cleaning liquid fuels from solids and water?

- **Centrifugal separators.**
- Additives.
- Filter.
- Settling tanks.

What is the working principle of the viscosity controller based on?

- **The drop in pressure that occurs in a capillary tube.**
- The increase in pressure that occurs in a capillary tube.
- The oil's speed through a capillary tube.
- The temperature of the oil.

What makes a Proportional-Integral-Derivative (PID) control different from a PI control.

- **A "derivative" action increases the reaction time of the control system.**
- The control switches before the signal overshoots the set point value.
- The proportional band is narrower.
- There is no difference between a PID and a PI.

What type of unit is used when referring to 'dynamic' viscosity?

- **centipoise.**
- centiparsecs.
- milliparsel-seconds.
- zahn.

Which statement is true?

- **The proportional feedback control can reduce error response to disturbances but that it still allows a non-zero steady-state error.**
- The proportional feedback can raise error response.
- The proportional feedback control can reduce error response to disturbances, and keep it to a steady correct value.
- The proportional feedback control cannot reduce any errors.

With the described viscosity sensor, the differential pressure is transferred to . . .

- **An indicator and the viscosity controller.**
- A magnetic coupling between viscosity sensor motor and gear pump.
- A resilient stainless steel housing and a stainless steel damping capillary.
- A thermometer.

0087 – Personal Survival and Survival Craft

Version 6.0 (1229)

A passenger is sailing with a baby. What should you do when they arrive for the pre-sailing abandon ship drill? Select the ONE best answer.

- **Find a buoyant cot to put the baby in**
- Give the passenger an adult lifejacket and some extension cords
- Give the passenger an adult lifejacket and tell them to hold the baby tightly
- Put the passenger and baby into the lifeboat first

How can a freefall lifeboat be launched if the ship is stern-to a reef? Select the ONE best answer.

- **Using the auxiliary davit**
- As normal, the lifeboat only draws 40 cm of water
- As normal, with the engine running astern
- It cannot be launched without hitting the reef

How can you maximise the range of a SART after you have abandoned ship? Select the ONE correct answer.

- **Fix it high on the survival craft**
- Fit the special battery
- Fix it high on the ship
- Press the boost button

How can you prevent someone who has fallen overboard from being lost at sea? Select the ONE best answer.

- **Keep them in sight**
- Fire a parachute flare to alert other vessels
- Stop the ship immediately
- Turn around and steer a reciprocal course

How long can a portable survival craft radio be expected to work for in an emergency situation? Select the ONE correct answer.

- **8 hours**
- 2 hours
- 24 hours
- 48 hours

How might a dedicated rescue boat be different to a lifeboat? Select the ONE best answer.

- **It can be of rigid or inflatable construction**
- It is always of inflatable construction
- It will be red, instead of orange
- It will have an engine

On joining a new ship, how would you find out which Muster Station you have been given? Select the ONE best answer.

- **Look at the Muster List or emergency plan**
- Look around the deck
- Look at the Safety Plan
- Look on the back of the cabin door

Solid lifejackets cover more of your chest than your back. Why is this? Select the ONE best answer.

- **So that you float with your face up and your mouth clear of the water**
- So that they are easier to put on
- So that they keep your heart warm
- So that you can swim more easily

Some ships are fitted with an M.E.S.' as part of their equipment. What does "M.E.S." stand for? Select the ONE correct answer.

- **Marine Evacuation System**
- Marine Effluent System
- Master Escape System
- Multi-person Escape Slide

The lifeboat on a gas tanker has an air supply inside. How long would you expect the air supply to last? Select the ONE best answer.

- **10 minutes**
- 1 hour
- 24 hours
- 30 minutes

The other crew have abandoned ship in a lifeboat because of a fire, but you are trapped on the fo'csle. There is no ladder nearby. How can you get into the lifeboat? Select the ONE best answer.

- **Climb down a fire hose**
- Jump into the water
- Jump onto the lifeboat roof
- Wait for a helicopter

What does this symbol mean? Select the ONE correct answer.

- **The position of an embarkation ladder**
- The pilot boarding position
- The position where life-rafts are lowered
- The position where lifeboats are lowered

What does this symbol mean? Select the ONE correct answer.

- **An evacuation slide**
- A life-raft
- A mechanical descent unit
- The position of the swimming pool

What does this symbol mean? Select the ONE correct answer.

- **Fasten seat belts**
- Fasten lifejackets
- Fasten twice
- Immersion Suit – Extra Large

What equipment is available to deal with a lifeboat engine fire? Select the ONE best answer.

- **A fire extinguisher**
- A bailer, to put water over it
- A bucket of sand
- A fixed CO2 system in the engine compartment

What is the best way to send a distress message from a sinking ship? Select the ONE correct answer.

- **Use the main GMDSS transmitter**
- Use a parachute flare
- Use an air band radio
- Use the EPIRB

What is the main purpose of a Thermal Protective Aid (T.P.A.)? Select the ONE correct answer.

- **To keep someone warm**
- To allow you to abandon ship without getting wet
- To cool someone down
- To cover a Partially Enclosed lifeboat

What is the standard IMO general alarm signal? Select the ONE correct answer.

- **Seven short blasts followed by one long blast on the ship's whistle or siren, with the same signal sounding simultaneously with the alarm bells**
- A spoken command over the public address system
- Continuous ringing of the alarm bell
- Intermittent ringing of the alarm bell

What is "hypothermia"? Select the ONE correct answer.

- **A low body temperature**
- A fear of cold weather
- A high body temperature
- An injection against colds and flu

What would you use an "air band" radio for? Select the ONE correct answer.

- **To talk to a search aircraft**
- To transmit ashore via a satellite
- To transmit further than a UHF radio
- To transmit further than a VHF radio

Where can you get more information about a type of life-raft you have not seen before? Select the ONE correct answer.

- **In the L.S.A. Training Manual**
- Ask the storeman
- In the safety management manual
- On the Safety Plan

Where should a Totally Enclosed lifeboat be boarded? Select the ONE best answer.

- **At the stowed position**
- After being lowered enough to release the securing gripes
- At boat deck level
- At main deck level

Where should you be to launch a freefall lifeboat from? Select the ONE best answer.

- **Inside the lifeboat**
- At the top of the davit, near the winch
- On the bridge
- On the deck, next to the davit

Where would you normally find a Partially Enclosed lifeboat? Select the ONE best answer.

- **At the side of a passenger ship**
- At the side of a cargo ship
- At the stern of a cargo ship
- At the stern of a passenger ship

Which TWO of these statements about inflatable life-rafts are true?

- **Some life-rafts cannot inflate upside down**
- **Sometimes life-rafts inflate upside down**
- An upside down life-raft can only be righted using the rescue boat and a rope
- An upside down life-raft cannot be righted

Which of the following can be used to reduce the risks to the lifeboat crew when recovering a lifeboat after a man overboard rescue? Select the ONE best answer.

- **Foul weather recovery strops**
- Nylon falls
- Use a freefall lifeboat
- Use a neck brace

Which of these can be seen furthest away from you? Select the ONE correct answer.

- **A parachute flare**
- A buoyant smoke float
- A hand flare
- A rocket line

Which one of these lifeboat types would you expect to find on a crude oil tanker? Select the ONE best answer.

- **Totally enclosed, with air supply and water spray**
- Freefall, with air supply
- Freefall, with water spray
- Totally or partially enclosed, with air supply and water spray

Why might you have to learn two sets of emergency duties? Select the ONE best answer.

- **So you can act as reserve for someone who cannot do their duties**
- Because the master is tough
- Because your duties might not be clear
- So you can apply for promotion

You are in the water on a foggy night. How can your lifejacket help you to be found? Select the ONE best answer.

- **Blow the whistle**
- It is brightly coloured
- Reflective tape will show your position
- The light will show your position

You are in the water wearing a SOLAS inflatable lifejacket. It is losing pressure. What can you do? Select the ONE best answer.

- **Blow into the top up tube**
- Activate the 2nd gas bottle
- Nothing
- Pull the red toggle

You are launching a "throw-over" life-raft. Drag these steps into the correct order to do this.

- 1 **Secure the painter to a strong point**
- 2 **Undo the life-raft lashing**
- 3 **Throw the life-raft overboard**
- 4 **Pull sharply on the painter**

You have abandoned a burning ship in a lifeboat. Which piece of equipment is used to keep the lifeboat near to where the ship is? Select the ONE best answer.

- **A sea anchor**
- A painter
- An anchor
- Oars

You have abandoned ship in a life-raft. How can you increase the range of your radar reflector? Select the ONE best answer.

- **Rig it as high as possible**
- Flatten it out
- Open all the canopy doors
- Tie it to the grab ropes outside

You have abandoned ship in a life-raft. Which of these would you do when a rescue helicopter wants to winch you up? Select the ONE best answer.

- **Flatten the raft canopy**
- Fire a parachute flare
- Take off your Immersion Suit
- Take off your lifejacket

You have abandoned ship in a lifeboat. What should you do now? Select the ONE best answer.

- **Get all the survival craft together, clear of the ship**
- Open all the doors to get air
- Spread out and search for help
- Use the engine to head for the nearest land

You have abandoned ship in a lifeboat. Which of these increases your risk of hypothermia? Select the ONE best answer.

- **Wet clothes**
- Distance from land
- Drinking too much water
- Lack of water

0088 – Human Relations

Version 5.0 (2124)

Accident investigations have often found that someone in a junior rank has not spoken out even though they knew that something was wrong. According to this module, what can often be the reason for this behaviour? Select the ONE correct answer

- **In some cultures, respect for senior persons is a barrier to speaking out even when things are going badly wrong**
- Juniors ranks should never question the decisions of their superiors
- Many company procedures do not allow senior ranks to be challenged
- Some junior ranks don't care what happens, as long as they don't get the blame

According to the International Chamber of Shipping, what are the THREE key components to developing an effective safety culture? Select the THREE correct answers.

- **Commitment from top management**
- **Measuring current performance and behaviour**
- **Modifying behaviour**
- Finding out who is to blame for accidents

According to this module, and the IMO, what percentage of accidents are caused by 'the human element'? Select the ONE correct answer.

- **80%**
- 100%
- 20%
- 50%

According to this module, how can respect be gained at sea? Select the ONE best answer.

- **By a good attitude to work and by respecting others**
- By being nice to everyone and doing what they want
- By virtue of rank and giving firm orders
- Respect should be an automatic thing

According to this module, how much of our communication is by non-verbal means? Select the ONE correct answer.

- **More than two thirds**
- About one fifth
- Half
- None

According to this module, what does human relations management theory tell us? Select the ONE correct answer.

- **People want to be part of a supportive team**
- Most people are happy just to take orders and be led
- That an authoritative management style is best
- That many people just want to be left alone

If the bridge team do not all speak a common language, what does SOLAS say they should do? Select the ONE correct answer.

- **Use English for bridge-to-bridge and bridge-to-shore safety communications as well as for communications on board between the pilot and bridge team**
- Nominate one of the team to translate between the different languages
- Use the majority language for bridge-to-bridge and bridge-to-shore safety communications as well as for communications on board between the pilot and bridge team
- Use the Master's language for bridge-to-bridge and bridge-to-shore safety communications as well as for communications on board between the pilot and bridge team

Onboard ship, what should the attitude to practical jokes be? Select the ONE best answer

- **They should not be done**
- People should not be at sea if they cannot take a joke
- They are a good thing, they help to make the trip more fun
- They are okay, as long as they are not dangerous

What do we mean by a 'Stop the job' policy? Select the ONE best answer

- **That anyone can stop work being done when it seems unsafe, or when they do not understand the scope of the work**
- That anyone can stop work being done for any reason
- That anyone can stop work being done if they think it is a waste of their time
- That anyone can stop work being done if they think someone else should be doing it

What does the IMO term 'a common language' mean? Select the ONE correct answer.

- **A language used in everyday work situations, that everyone onboard a ship can understand**
- A language that is spoken by the navigating officers and the harbour authorities
- A very basic form of English that everyone onboard must know
- The English language

What is the IMO's 'SMCP'? Select the ONE correct answer.

- **Standard Marine Communication Phrases in simple English**
- Search and Rescue Marine Contact Person
- Search and Rescue Maritime Contact Protocols
- Standard Marine Communication Phrases in six different languages

What is the best way to try and avoid accidents due to the 'human element'? Select the ONE best answer.

- **By following procedures and communicating effectively with others**
- By following procedures in every situation
- By relying totally on automated systems
- By strong leadership and not listening to other people

What is the name of the convention which tries to reduce fatigue by limiting hours of work onboard ship? Select the ONE best answer.

- **The Maritime Labour Convention (MLC)**
- SOLAS
- STCW
- The Working Time Directive (WTD)

When giving orders to someone, what is the best way to make sure they have been understood? Select the ONE best answer.

- **Get them to repeat or explain the order back to you**
- Look for non-verbal clues, like them nodding in agreement
- Repeat what you have said at least twice
- Talk loudly and slowly

When should the IMO's 'SMCP' be used? Select the ONE correct answer.

- **Whenever there are language difficulties, to avoid misunderstandings which could cause accidents**
- At all times when two ships are communicating
- During Search And Rescue (SAR) operations
- Only in emergencies, when there is no shared language

Which of the following are evidence of a POOR safety culture onboard a ship? Select any options that you think apply.

- **No security procedures at the gangway**
- **People not paying attention to safety briefings**
- Other people wearing PPE such as ear defenders, when you cannot see any risks
- Wearing sports shoes in a recreational area of the accommodation

Why are good relationships among the crew important? Select the ONE best answer.

- **Good relationships lead to a better work performance and a positive atmosphere**
- Good relationships prevent any complaints to port State inspectors
- Good relationships reduce complaints to the office ashore
- Good relationships will be useful in an emergency

0089 – Medical Care

Version 2.0 (0)

A person has the following symptoms: 1. Radiating pain to the neck, left arm and abdomen 2. Nausea 3. Breathlessness. What does these symptoms indicate?

- **Ischaemic heart disease**
- Cetoacidosis
- Pulmonary embolus
- Sepsis

A person who may have become a carrier of the disease should not be allowed to prepare or handle food or utensils until cleared for such duty by a doctor.

- **True**
- False

Acute abdominal condition should be treated with:

- **Give fluid intravenously**
- **Relieve pain with painkiller, start with Paracetamol**
- Give fluid by month
- Give food by mouth
- Give stronger drugs (Morphine) at once

Even a small piece of the afterbirth left inside the mother can be dangerous.

- **True**
- False

Gingivitis is inflammation of the gums

- **True**
- False

If the throat is covered with a grey layer, that is a sign for:

- **Diphtheria**
- Lepra
- Plague
- Typhoid

In acute situations, how often should you monitor the primary parameters and record them?

- **Every 15 minutes**
- Every 30 minutes
- Every 5 minutes
- Every hour

In case of more severe pain, as in burns, fractures, acute abdominal pain etc, it might be necessary to employ more potent drugs such as morphine.

- **True**
- False

In patients with suspected fractures, it is essential that you do not stabilise the limb/spine prior to transportation.

- **False**
- True

In temperate climatic condition, how much fluid should you give a person suffering from dehydration?

- **2 litre**
- 1 litre
- 3 litre
- As much as the person wants to drink.

On what injury/injuries should ointment be used together with gauzed dressings?

- **On burns only**
- On strains and burns
- On wounds and burns
- On wounds only

Stewards or one of the persons in the galley department should be selected as the nurse onboard.

- **False**
- True

The medical record should be available to the patient, on his request.

- **True**
- False

The patient room temperature should be?

- **21-24 Degree Celsius**
- 18-21 Degree Celsius
- 20 Degree Celsius
- 23 Degree Celsius

The weight of the patient is not necessary to inform the doctor about, only the age and sex of the patient.

- **False**
- True

What causes myocardial infarction?

- **Rupture of plaques in the coronary vessels.**
- Embolus traveling from the veins in the legs.
- High glucose level in the blood.
- Spasm (cramp) in the coronary vessels and heart wall.

What is ICE an abbreviation for?

- **Ice – Compression – Elevation**
- Ice – Compression – Examination
- Immobilize – Compression – Examination
- Immobilize – Conscious – Elevation

What is the most common sexually transmitted disease in the western world?

- **Chlamydia and Gonorrhea**
- Hepatitis A, B and D
- HIV and AIDS
- Syphilis

What is the most important measure taken by nurses to prevent spread of disease between patients?

- **Hand hygiene**
- Mouth mask
- Patients in single rooms
- Use of special clothing

What kind of wound is shown on the picture?

- **Incisions**
- Abrasion
- Lacerations
- Puncture wound

What language should the medical record be written in?

- **English**
- Both English and the language spoken onboard
- The language required by the Company
- The language spoken onboard

What should you give a person suffering from malnutrition the first two days?

- **Sugar and water or milk or soup**
- As much food as the person wants
- Small amount of normal food
- Water only

What sort of infection is Malaria?

- **Tropical parasite infection**
- Tropical bacterial infection
- Tropical fungal infection
- Tropical viral infection

What vaccine should not be trusted completely?

- **Tuberculosis**
- Mumps
- Tetanus
- Yellow fever

When moving and transporting a patient, primary attention should be paid to the basal functions – ABC. What does ABC stand for?

- **Airway – Breathing - Circulation**
- Airway – Bleeding - Circulation
- Airway – Bleeding - Consciousness
- Airway – Breathing - Consciousness

Which of the following diseases may be treated with penicillin?
Mark the answers you think are correct.

- **Gonorrhoea**
- **Syphilis**
- Chlamydia
- Hepatitis A
- Hepatitis B and C
- HIV

Which of the following procedures are correct to follow after a death onboard a ship?

- **If there is to be a post-mortem examination, the body should be put in a body bag and kept in a refrigerator or cold store**
- **Make a brief description of each article and note any initials or names on the garments**
- **Use a camera to take photographs, which might illustrate how the death occurred**
- Do not strip the body for all clothing

Which of the following statement are correct?

- **Do not pull on or cut the umbilical cord**
- **Do not pull on the baby's head.**
- **Lift the baby away from the birth canal.**
- Once the widest part of the baby's head is visible, make sure that the mother carry through a big push.
- Pull on the baby's shoulders.
- Smack the baby if he/she doesn't cry.

Which statement is correct?

- **It is correct to apply some pressure on a strain wound in addition to elevating the injury.**
- After sewing, the stitches should be left for at least 14 days to be sure there is a strong healing.
- Burns should be treated with care to sterility, but these are seldom infected because the plasma that drains from them contains anti-bacterial matters.
- Internal bleedings are difficult to localise, but they are seldom life threatening, because the hollow cavities in the body are too small to contain a serious bleeding.

You should try to delay the birth, if the ship can reach the shore within some hours.

- **False**
- True

0091 – Incinerators

Version 3.0 (330)

At which chamber temperature can the sluice door be opened and feeding of garbage can start? Select the ONE correct answer.

- **750°C**
- 1200°C
- 200°C
- 600°C

Shortly after the combustion chamber, the temperature of the flue gas should not exceed . . . . . .?

- **350°C**
- 1200°C
- 200°C
- 650°C

The temperature in the combustion chamber can vary between 850° C and . . . . . ° C.

- **1200**
- 1400
- 2000
- 900

The maximum suction height from the tank to the D.O. burner is . . . . . ?

- **3.5m**
- 12m
- 6.5m
- 9m

The suction from the flue gas fan creates a . . . . . pressure in the combustion chamber.

- **Negative**
- Atmospheric
- Positive

What can you throw overboard within 3 miles of the nearest land & in all inland waters? Select the ONE correct answer.

- **You cannot throw anything at all overboard.**
- The only thing you can't throw overboard is plastics.
- You can throw floating lining and packing materials overboard.
- You can throw food and glass overboard.

What should the diesel oil pressure be at the diesel oil burner? Select the ONE correct answer.

- **16 bar**
- 0.4 bar
- 1 bar
- 60 bar

When burning sludge, the sludge will be heated up to which temperature? Select the ONE correct answer.

- **85°C**
- 145°C
- 45°C
- As long as it is more than 0°C it doesn't matter.

Where is it legal to dispose of plastic? Select the ONE correct answer.

- **Only at shore facilities.**
- Outside 25 miles offshore.
- Within 12 – 25 miles offshore.
- Within 3 – 12 miles offshore.

Which key should be pressed if you want to see the history alarm log? Select the ONE correct answer.

- **[LIST]**
- [ACK]
- [INFO]
- [PREV]

0092 – Introduction to MARPOL

Version 3.0 (3762)

A ship has sewage stored in a tank. Which one of the following best describes the MARPOL Annex IV requirements that must be met before the ship is allowed to discharge the tank contents overboard?

- **The ship must be away from land and pump the sewage at a controlled rate**
- The ship must be away from land and moving at a minimum of 12 knots
- The ship must be outside port limits
- The ship must be outside port limits and pump the sewage at a controlled rate

Select the option which is the reason why MARPOL Annex I requires most crude oil tankers to have cargo oil piping designed to minimise oil retention.

- **To reduce the amount of oil in the slop tank**
- To reduce the corrosion in the pipes
- To reduce the pollution when the cargo tank is ballasted
- To reduce the risk of capsize while in ballast

Some MARPOL annexes require ships to be surveyed and certificated. Which one of the following is the most likely reason for this?

- **Compliance with some MARPOL rules cannot easily be proven while the ship is in service**
- It allows IMO to count the number of compliant ships
- It is an excuse for flag states to visit their ships
- It produces fees for class societies

The MARPOL convention requires ships to report pollution incidents. Which one of the following is the main reason for this?

- **So that action can be taken to clean up the pollution**
- So that insurance companies can put up premiums
- So that ships can be blacklisted by ports
- So that someone can be prosecuted

Under MARPOL Annex III what operational discharges of harmful substances in packaged form are allowed?

- **Discharge of leakages might be allowed by the flag state**
- Damaged packages can be discharged overboard
- None
- None, unless via the bilge separator

Which one of the following best describes how MARPOL Annex I reduces the amount of oil in ballast water discharges from oil tankers?

- **Ballast water is normally kept segregated from the cargo tanks**
- Ballast water is kept in fore and aft peak tanks only
- Ballast water must be discharged via the slop tank
- Tankers are designed not to need water ballast

Which one of the following best describes how MARPOL Annex III attempts to reduce accidental loss of harmful substances in packaged form into the sea?

- **Packages are required to be properly stowed and secured**
- Packages are required to be inside a shipping container
- Packages are required to be made of durable plastic
- Packages are required to be stowed below deck only

Which one of the following best describes how MARPOL Annex VI is intended to reduce leakages of Ozone Depleting Substances (ODS)?

- **By requiring an ozone depleting substance record book**
- By banning some of the ozone depleting substances from 2012
- By increasing the cost of ozone depleting substances
- By requiring a SOx Emissions Compliance Plan

Which one of the following best describes how coastal states can find out if a sinking ship is carrying MARPOL Annex III cargo?

- **A manifest listing the harmful substances must be left ashore**
- A red flag is to be flown while harmful substances are on board
- The ship's AIS will broadcast the list of harmful substances on board
- The SMPEP will require the Mayday broadcast to say that harmful substances are on board

Which one of the following best describes the main goal of MARPOL Annex III?

- **To reduce the harm to the environment from harmful substances in packaged form**
- To make the clean up of harmful substances in packaged form in the sea easier
- To reduce the amount of dumping into the sea
- To reduce the amount of harmful substances in the sea

Which one of the following best describes the main goal of MARPOL Annex IV?

- **To reduce the harm to the environment from sewage from ships**
- To ensure that all sewage on ships is treated to break it down
- To minimise the harm to the environment after an accidental discharge of sewage
- To reduce the amount of sewage produced by ships

Which one of the following best describes the main goal of MARPOL Annex V?

- **To reduce the harm to the environment from garbage from ships**
- To ban discharges containing harmful substances
- To ban the discharge of garbage into the sea
- To reduce the harm to the environment from sewage from ships

Which one of the following best describes the main goal of MARPOL Annex VI?

- **To reduce the harm to the environment from air pollution from ships**
- To ban the emissions of air pollutants from ships
- To capture emissions of air pollutants from ships
- To reduce the harm to the environment from oil pollution from ships

Which one of the following best describes the main purpose of the SOPEP required by MARPOL Annex I?

- **To describe the actions to be taken to reduce the discharge of oil following an accident**
- To act as a telephone directory
- To describe the actions needed to evacuate the ship while surrounded by oil
- To record the actions taken following an accidental discharge of oil

Which one of the following best describes the means required by MARPOL Annex IV to reduce accidental discharges of sewage?

- **Holding tanks must have a means to indicate visually the amount of its contents**
- Holding tanks must be in a protective location away from the hull
- Sewage comminuting and disinfecting systems must have a disinfectant low level alarm
- Sewage treatment plants must be fitted with an alarm on the outlet pipe

Which one of the following best describes the purpose of the garbage management plan that MARPOL Annex V requires some ships to carry?

- **It gives instructions for the collection, processing and storage of garbage before its disposal**
- It gives instructions for the disposal of garbage at sea only
- It only gives instructions for garbage recycling
- It only gives the actions to be taken following accidental discharge of garbage

Which one of the following best describes the purpose of the placards required on some ships by MARPOL Annex V?

- **They show the garbage disposal requirements.**
- They show the garbage collection areas
- They show the garbage disposal positions
- They show the penalties for illegal garbage discharges

Which one of the following best describes the purpose of the slop tank that MARPOL Annex I requires oil tankers to be fitted with?

- **To ensure tank washings can settle before the water is discharged overboard**
- To heat oil for crude oil washing
- To heat water for tank washing
- To store oil for crude oil washing

Which one of the following best describes the records of noxious liquid operations that ships have to maintain to satisfy MARPOL Annex II?

- **A cargo record book must be completed after any transfer of cargo or ballasting of cargo tanks**
- A record book must be completed only after cargo operations
- A record book must be completed only after cargo tank washing operations
- The oil record book Part II must be completed after cargo operations

Which one of the following best describes the requirement in MARPOL Annex VI intended to minimise accidental emissions from incinerators?

- **The operator is to be trained and have an operating manual**
- The incinerator is to be supplied with an operating manual
- The incinerator is to have an annual service
- The operator is to have been on the manufacturer's training course

Which one of the following best describes the requirement in MARPOL Annex VI intended to minimise accidental emissions of Sulphur Oxides (SOx) inside Sulphur Emission Control Areas (SECAs)?

- **Requiring a written fuel changeover procedure**
- Requiring a bunker sample be kept on board
- Requiring separate high and low sulphur fuel day tanks
- Requiring sulphur meters in exhaust pipes

Which one of the following best describes the requirement in MARPOL Annex VI intended to minimise operational emissions of CO2?

- **All ships are required to have a Ship Energy Efficiency Management Plan**
- All ships will have to reduce speed by 10%
- New ships will have to go slower than old ships
- Only new ships will be required to have a Ship Energy Efficiency Management Plan

Which one of the following best describes the requirement in MARPOL Annex VI intended to minimise operational emissions of Ozone Depleting Substances (ODS)?

- **Emissions during maintenance are banned and gases must be recaptured for recycling**
- All types of ODS must be removed from the ship by 2012
- Equipment containing ODS must be maintenance free
- Fire extinguishers must not be serviced

Which one of the following best describes the requirement in MARPOL Annex VI intended to minimise operational emissions of Sulphur Oxides (SOx)?

- **The sulphur content of fuel is limited**
- Bunker samples must be taken and kept ashore
- New diesel engines have to meet Tier II SOx limits
- Sulphur scrubbers are required on all new engines

Which one of the following best describes the requirements of MARPOL Annex V for plastic garbage?

- **It must be discharged ashore, even after incineration**
- It can only be discharged outside a special area
- It must be discharged ashore or incinerated
- It must be incinerated at high temperatures

Which one of the following best describes the way that sewage can damage the marine environment?

- **It can contain bacteria, parasites and viruses**
- It can block out the sunlight
- It can cover the seabed
- It can increase oxygen levels in the water

Which one of the following best describes what MARPOL Annex II requires to minimise the amount of noxious liquids when cleaning cargo tanks?

- **Tanks have to be stripped down to small volumes of cargo**
- Tanks are not allowed to be washed with water
- Tanks have to be emptied using ventilation fans
- Tanks have to be washed with the minimum of water

Which one of the following best matches the MARPOL Annex II definition of a "noxious liquid substance"?

- **Any substance listed as pollution category X, Y or Z in the International Bulk Chemical Code**
- Any oil not listed in MARPOL Annex I
- Any substance listed in the IMDG Code
- Any substance which could kill a human within 12 hours

Which one of the following describes the most common way that the substances regulated by MARPOL Annex III cause harm to the marine environment?

- **They are poisonous**
- They cannot be seen
- They cover the sea bed
- They spread across the surface of the water

Which one of the following documents is required by MARPOL Annex II in order to minimise the harm caused by an accidental discharge of noxious liquids?

- **A Shipboard Marine Pollution Emergency Plan**
- A Noxious Liquids Cargo Record Book
- A Procedures and Arrangements Manual
- A Safety Management System

Which one of the following explains why MARPOL Annex I requires ships to have a sludge tank?

- **It is required to hold oil separated off by the bilge separator, sludge from purifiers and other waste oil**
- It is only required to hold the oil separated off by the bilge separator
- It is only required to store bilge water until it can be discharged overboard
- It is required only when ships have oil purifiers

Which one of the following is a kind of environmental damage caused by air pollutants regulated by MARPOL Annex VI?

- **Acid rain**
- Drought
- Heavy rain
- Lightning strikes

Which one of the following is the main goal of MARPOL Annex I?

- **To reduce the harm to the environment from oil pollution from ships**
- To minimise the harm to seafarers from oil on ships
- To minimise the number of oil tanker voyages
- To reduce the harm to the environment from sewage pollution from ships

Which one of the following is the most likely way in which a noxious liquid substance could harm marine plants?

- **By blocking the sunlight**
- By changing the water colour
- By making the leaves slippery
- By making the rocks slippery

Which one of the following kinds of garbage does MARPOL Annex V allow to be discharged overboard while the ship is inside a special area?

- **Food**
- Cargo residues
- Paper
- Wood

Which one of the following most closely matches the definition of 'garbage' in MARPOL Annex V?

- **All kinds of food, domestic and operational waste**
- All kinds of food and domestic waste
- All solid waste produced by the ship
- Solid waste and packages containing liquid waste produced by the ship

Which one of the following most closely matches the definition of 'oil' in MARPOL Annex I?

- **Any oil from petroleum**
- Any oil that burns in air
- Any oil that floats on water
- Unrefined (crude) oil

Which one of the following most closely matches the definition of 'sewage' in MARPOL Annex IV?

- **Drainage from animals, medical premises, toilets and urinals**
- Drainage from sinks, baths and toilets and urinals
- Grey water
- Solids from toilets

Which one of the following options best describes how MARPOL Annex I controls the amount of oil discharged into the sea from the cargo tank areas of oil tankers?

- **The rate of discharge of oil is limited**
- No oil can be discharged into the sea from cargo tank areas
- Permits must be bought to allow oil to be discharged
- There is a fixed limit on the total amount of oil discharged per year

Which one of the given options best explains why MARPOL Annex I requires most new tankers to be built with double hulls?

- **To reduce the chances of oil leaking into the sea following collision or grounding**
- To ensure that ships have enough ballast capacity to cope with storms
- To increase port fees from tankers
- To reduce instability caused by the free surface effect on cargo

Which one of the given options would be the best action to take following an incident where leaking oil polluted the water around your ship?

- **Contain the oil with booms**
- Set fire to the oil
- Spray oil dispersant as soon as possible
- Throw rags on the oil

0093 – Wall Wash

Version 2.0 (0)

(Select all applicable answers) A "failed" wall wash can result from:

- **Dirty equipment.**
- **Poor technique.**
- **Traces of previous cargo.**
- Using a clean plastic spray bottle.

(Select all applicable answers) The basic routine laboratory tests that we run onboard include:

- **Appearance.**
- **Hydrocarbons.**
- Gas Chromatography.
- Ultra-Violet Spectrum.

(Select all applicable) Wall wash results should be recorded for the following reasons:

- **It enables progress to be tracked during tank cleaning.**
- **It often figures as an item in Vetting Inspections.**
- It is an ASTM requirement.
- It is in all Captain's Standing Orders.

(Select as applicable) The two kinds of detectable odour in a wall wash sample are:

- **Characteristic.**
- **Residual.**
- Distinctive.
- Nominal.

A drying-oven Non-Volatile Matter test cannot be effectively done on board a ship because:

- **A sufficiently stable and vibration-free platform cannot be provided for the required measurements.**
- Ship's laboratories cannot be maintained at constant room temperature.
- Spares are difficult to obtain in many parts of the world.
- The equipment works on a different voltage.

A wall wash should not be taken from a hot bulkhead because:

- **Too much Methanol will evaporate.**
- Any contaminants will evaporate.
- Any contaminants will stick to the surface.
- The Officer (or Surveyor) will burn his hands.

A wall wash should not be taken from a wet bulkhead because:

- **Surface tension and dilution will lead to a less efficient wall wash.**
- All contaminants will collect at the bottom of the bulkhead.
- Any contaminants will stick to the surface.
- The Methanol will evaporate on contact with the moisture on the bulkhead.

All cargoes loaded into Stainless Steel tanks require a wall wash standard of tank cleanliness (True or False):

- **False.**
- True.

An Inhibitor in a last cargo will almost certainly influence the result in this routine shipboard laboratory test:

- **Permanganate Fade Time.**
- Appearance.
- Hydrocarbons.
- Odour.

Another name for the Hydrocarbon test is "Water Misidentification" (true or False):

- **False.**
- True.

Blue graduation marks on the side of a test tube may lead to:

- **Misinterpretation of low levels of Hydrocarbons.**
- Increased levels of Hydrocarbons.
- Misinterpretation of high levels of Chlorides.
- Poor analysis of Permanganate Fade Time.

Heavy smokers may, in particular, have a problem accurately conducting this routine laboratory test:

- **Odour.**
- Appearance.
- Chlorides.
- Colour.

How many drops of Silver Nitrate solution are added to the sample in a Chloride test?

- **5.**
- 1.
- 10.
- 2.

If Bleach products have been used in a tank cleaning process and not properly rinsed away, the following is most likely to be noted in a wall wash analysis:

- **Chlorides.**
- Colour.
- Hydrocarbons.
- Suspended Matter.

If you encounter a new wall wash or analysis procedure, you should:

- **Forward full details to Company Operations Department.**
- Advise the Ship's Superintendent.
- Ignore it.
- Inform the Purchasing Department.

In a Hydrocarbons test, the presence of slight Hydrocarbons is indicated by:

- **A translucent bluish tint.**
- A milky white discoloration.
- A pale yellow haze.
- A silver sheen on the surface of the sample.

In a Permanganate Fade Time test using Methanol, at what temperature should the samples be maintained?

- **15°C.**
- 10°C.
- 25°C.
- Not more than 10°C.

In a Permanganate Fade Time test, how long should the samples be allowed to "stabilise" before the Permanganate solution is added?

- **At least 10 minutes.**
- 2 minutes.
- Approx 45 minutes.
- Maximum 30 minutes.

In a Permanganate Fade Time test, samples should be checked against the standard at the following interval:

- **Every 10 minutes.**
- After 30 minutes and every 10 minutes thereafter.
- Continuously.
- Every 30 minutes.

Inhibitor in a previous cargo commonly affects Permanganate Fade Time test results (True or False):

- **True.**
- False.

Maintaining the samples at a constant temperature is a necessary part of the Permanganate Fade Time test procedure. True or False?:

- **True.**
- False.

Of the four problems listed below and assuming all are included in a customer specification, which would be the priority?

- **Moderate Hydrocarbons.**
- Appearance (some suspended matter).
- Heavy Chlorides.
- Very slight Odour.

Orange or red particles noted in a wall wash sample are an indication of the presence of:

- **Rust particles.**
- Chlorides.
- Hydrocarbons.
- Undissolved Potassium Permanganate.

Permanganate Fade Time test samples should start off with the following colour:

- **A rich, deep purple.**
- A bluish tint.
- A pale, watery yellow.
- Completely clear.

Permanganate Fade Time tests are run for:

- **As long as a customer's specification requires.**
- 1 hour.
- 40 minutes.
- As long as the Chief Officer chooses.

Regular marker pens should not be used to identify sample jars because:

- **Methanol may dissolve the marker ink.**
- Methanol will turn the marker ink sticky and it may get transferred to the wall wash funnel.
- Under the regulations, coloured marker pens are prohibited.
- You may need to remark them at a later stage.

Soapy bubbles noted on the top of a wall wash sample in a test tube are a likely indication of:

- **Residues from a detergent based cleaning chemical.**
- Acetone.
- Residues from an oil based cleaning agent.
- Residues of a water based cargo.

The Ultra-Violet Spectrum test is run to detect:

- **Low levels of dissolved impurities.**
- Acid.
- High levels of solid particle impurities.
- Hydrocarbons.

The dish used in a shore laboratory Non-Volatile Matter test is commonly made from:

- **Platinum.**
- Potassium.
- Stainless steel.
- Teflon coated metal.

The following amount of Potassium Permanganate Solution is added to the sample in the Permanganate Fade Time test:

- **2 ml.**
- 1 ml.
- 10 ml.
- 5 ml.

The following important information should be prominently posted in the ship's laboratory:

- **Material Safety Data Sheets for any chemicals in use.**
- A calendar.
- Cleaning Chemical Index.
- The Deck Watchkeeping Schedule.

The letters "G.C." in the shore laboratory test stand for:

- **Gas Chromatography.**
- Gas Condition.
- General Chromatography.
- General Condition.

The most apparent effect of samples left standing in test tubes is:

- **Accelerated discoloration of the glass.**
- Chemical reaction with any Hydrocarbons present.
- Permanent etching of the glass.
- Very slight inaccuracies in certain laboratory tests.

The most likely source of Hydrocarbons detected in a wall wash sample is:

- **Traces of previous cargo.**
- Dust in the laboratory Methanol.
- Equipment failure.
- Polluted air drawn into the tank.

The over-riding principle behind a wall wash and the subsequent analysis is:

- **Cleanliness.**
- Agreement between the Chief Officer and the Surveyor.
- Selecting a good part of the tank, in order to ensure a "pass" result.
- Speed.

The purpose of a wall wash is to determine:

- **Whether a tank meets customer cleanliness requirements.**
- If a tank is dry.
- If a tank is safe to enter.
- Whether a tank meets company maintenance requirements.

The ratio and components used in the preparation of a Permanganate Solution are:

- **0.1g Permanganate crystals to 0.5 ltr Deionised Water.**
- 0.1g Permanganate crystals to 0.5 ltr Methanol.
- 0.2g Permanganate crystals to 0.5 ltr Deionised Water.
- 0.5g Permanganate crystals to 0.5 ltr Deionised Water.

The sealed reference standard used in a Permanganate Fade Time test is:

- **A Platinum Cobalt 500 colour standard.**
- A Potassium Chloride 600 colour standard.
- A Potassium Cobalt 530 colour standard.
- An API 500 Colour standard.

The significant difference with a Whatman Filter Paper test is that it:

- **Commonly includes a sample taken from the tank bottom.**
- Involves more samples per tank.
- It is more difficult to pass.
- It is only performed in Japan.

The two components in a Hydrocarbon test "blank" standard are:

- **Methanol and Deionised Water.**
- Methanol and Silver Nitrate Solution.
- Wall wash sample and Deionised Water.
- Wall wash sample and Methanol.

The wall wash procedures in this programme, and those used by the majority of shore laboratories, are built around industry standards issued by the following organisation:

- **The American Society for Testing and Materials.**
- Det Norske Veritas.
- System International.
- The American Petroleum Institute.

To assess the Appearance of a wall wash sample, we:

- **Examine the sample in good light and compare to a sample of "good" Methanol.**
- Add Silver Nitrate and look for cloudiness.
- Heat a small sample of the wall wash and check for any residues remaining.
- Send the sample ashore for laboratory analysis.

To determine the presence of Chlorides in a wall wash sample, we add:

- **5 drops of 10% Silver Nitrate Solution.**
- 2 mg of Potassium Permanganate crystals.
- 2 ml of 20% Potassium Permanganate Solution.
- 5 ml of 3% Hydrochloric Acid.

Under normal handling conditions, a Permanganate Solution will last for approximately:

- **3-5 days.**
- 1 month.
- 1 year.
- Indefinitely.

Wall wash samples will not really be affected by an occasional drop of sweat (True or False):

- **False.**
- True.

When performing a Hydrocarbons test, the sample should be viewed:

- **Looking down through the tube against a black background.**
- Looking down through the tube against a white background.
- Looking sideways through the tube against a black background.
- Looking sideways through the tube against a good light.

Where possible, a wall wash stream should be run for:

- **Minimum 1 metre.**
- 2 metres.
- At least 0.5 metre.
- Maximum 1 metre.

0094 – Flexinert Gas Generator, Standard

Version 3.0 (8)

At start-up by utilising the flexinert mode, the correct start-up sequence is:

- **Choose flexinert mode, then start scrubber pump, then start the blowers immediately after.**
- Choose flexinert mode last.
- Start the blowers first, then the scrubber pump immediately after, then choose flexinert mode.
- Start the scrubber pump first, then the blowers, then choose flexinert mode.

How do you inspect the combustion chamber of the inert gas generator?

- **By opening the front cover.**
- By dismantling the burner from the scrubber.
- By opening the bottom cover.
- By opening the inspection hatch.

How do you look up a spare part?

- **Open the Operating and Maintenance Manual, look in parts list, then find component number.**
- Open the Operating and Maintenance Manual, go to mailing list and ask for help.
- Open the Operating and Maintenance Manual, look for sales department and ask for component number.
- The operating and maintenance manual and contact maker.

How do you look up spare parts from a vendor manual?

- **Open the Operating and Maintenance Manual, go to relevant Vendor manual, then find component number.**
- Open the operating and maintenance manual, and contact relevant vendor.
- Open the Operating and Maintenance Manual, go to Ch. 7, Parts list, then find component number.
- Open the Operating and Maintenance Manual, go to parts list, then find component number.

Routine maintenance: How often and how should demister inside scrubber be cleaned?

- **3 monthly with sea water.**
- 3 monthly with fresh water.
- Monthly with fresh water.
- Weekly with fresh water.

Routine maintenance: How often should calibration of stationary oxygen analyser be carried out:

- **Weekly**
- 3 monthly
- Daily
- Monthly

Routine maintenance: How often should cleaning of fuel filters be carried out:

- **3 monthly.**
- Monthly.
- Weekly.
- Yearly.

Safety on deck: Before opening a cargo tank hatch cover, it is important to check:

- **That the cargo tank pressure is zero**
- Check safety gear
- That cigarettes are distinguished
- That the master has been informed

The drain valve of the deck water seal shall be:

- **Closed.**
- Open.
- Partially closed.
- Partly open.

The heavy fuel control/re-circulation valve should be pre-set to:

- **3 bar.**
- 1 bar.
- 10 bar.
- 5 bar.

The pressure/vacuum - breaker should be filled up with:

- **Fresh water and antifreeze.**
- Fresh water.
- Salt water and anti freeze.
- Salt water.

What are the modes of operation for this inert gas system:

- **Flexinert mode/Generator mode/Fresh air mode.**
- Flue gas mode/fresh air mode.
- Flue gas mode/Generator mode/Fresh air mode.
- Generator mode/Fresh air mode.

What is Inert gas?

- **It is gas containing insufficient oxygen to support the combustion of hydrocarbons.**
- it is a gas containing more than 21% Oxygen.
- It is a gas containing no oxygen.
- it is gas containing less than 21 % by volume of oxygen.

What is the purpose of the Venturi scrubber?

- **To clean the flue gas before it enters the blowers.**
- To clean the flue gas /air mixture before it enters blower.
- To clean the flue gas after it enters the blowers.
- To clean the inert gas from the Inert gas generator.

When is the Flexinert mode applied:

- **When the auxiliary engine exhaust gas is of acceptable quality and quantity.**
- When all tanks are to be gas-freed before the ship enters dry dock.
- When auxiliary engine gas is not available.
- When exhaust gas is not available.

Where is the inert gas outlet of the scrubber located?

- **At the top of the unit.**
- Above the sea water nozzles.
- At the bottom of the unit.
- Below the sea water nozzles.

Which document contains all specific plant performance data?

- **The Package data sheet.**
- The Alarm list.
- The material data sheet.
- The supplier document list.

0096 – Cargo Compatibility

Version 2.0 (0)

(Select as applicable) The IBC Code requires that cargo sample lockers are

- **Cell-divided, to avoid bottles shifting at sea.**
- **Equipped with adequate ventilation arrangements.**
- **Made of resistant material suitable for the cargoes carried.**

A blank square in the Compatibility Chart indicates which of the following?

- **Cargoes of the corresponding Groups are compatible, subject to any specific Exceptions listed in Appendix I.**
- Cargoes of the corresponding Groups are always compatible.
- Cargoes of the corresponding Groups are always incompatible.
- Cargoes of the corresponding Groups are incompatible, subject to any specific Exceptions listed in Appendix I.

A cargo which has unusual carriage requirements or potentially has compatibility problems may be assigned which of the following Group Numbers in Table II "Grouping of Cargoes"?

- **0**
- 1
- 13
- 5

Across the top of the Compatibility Chart, we find which of the following:

- **Cargo Groups.**
- **Reactive Groups.**
- Cargo Groups and Reactive Groups.
- Compatibility Groups.

An "X" in a square in the Compatibility Chart indicates which of the following?

- **Cargoes of the corresponding Groups are incompatible, subject to any specific Exceptions listed in Appendix I.**
- Cargoes of the corresponding Groups are always compatible.
- Cargoes of the corresponding Groups are always incompatible.
- Cargoes of the corresponding Groups are compatible, subject to any specific Exceptions listed in Appendix I.

Aromatic Amines are listed under which of the following Chemical Group numbers?

- **9.**
- 10.
- 7.
- 8.

Can Acrylic Acid be safely stowed adjacent to Tall Oil?

- **Yes.**
- No.

Can Acrylonitrile be stowed adjacent to 1,1,1 Trichloroethane?

- **Yes.**
- No.

Can Aniline be stowed adjacent to Diphenylmethane discarnate?

- **No.**
- Yes.

Can Butyl Alcohol be safely stowed adjacent to Propylamine?

- **No.**
- Yes.

Can Dicyclopentadiene be stowed adjacent to Nonyl Phenol?

- **Yes.**
- No.

Can Ethylene Glycol be stowed adjacent to Epichlorohydrin?

- **Yes.**
- No.

Can Heptene be safely stowed adjacent to Formic Acid?

- **Yes.**
- No.

Can Methyl Iso Butyl Ketone be safely stowed adjacent to Nitric Acid?

- **No.**
- Yes.

Propionic Acid is found in which Cargo Group:

- **4 Organic Acids**
- 0 Unassigned Cargoes
- 1 Non-Oxidizing Mineral Acids
- 2 Sulphuric Acids

The Compatibility Chart is officially part of which of the following US Regulations?

- **The Code of Federal Regulations.**
- MARPOL 73/78.
- The Florida State Contingency Plan.
- The Noxious Liquid Substances Rules (1988).

The USCG-approved format for results of a reactivity test is found in:

- **Appendix IV to the Compatibility Chart**
- The Cargo Handling Manual
- The Condensed Chemical Dictionary
- The ICS Tanker Safety Guide (Chemicals)

To find data on Dowanol DB, which of the following cargo names would you would look up?

- **Diethylene Glycol Butyl Ether.**
- Diethylene Glycol Ethyl Ether.
- Diethylene Glycol Iso Ether.
- Diethylene Glycol Methyl Ether.

To find data on MEK, which of the following cargo names would you look up?

- **Methyl Ethyl Ketone.**
- Meta Ethanol Korpanol.
- Methyl Ethyl Kersosene.
- Mono Ethyl Ketone.

To find data on n-Propyl Carbinol, which of the following cargo names would you look up?

- **n-Butyl Alcohol.**
- 2-Propyl Alcohol.
- Propanol.
- Propylene Carbolic Alcohol.

Where specific cargoes are to be carried in a stowage that does not comply to published USCG Compatibility criteria but for which tests have been carried out to determine reactivity, which of the following conditions must be in effect?

- **A copy of the USCG Letter of Approval, or a copy of the test method and results in the approved format, must be on the vessel prior to commencement of loading.**
- A copy of the USCG Letter of Approval must be received by the testing laboratory.
- A copy of the USCG Letter of Approval, or a copy of the test method and results in the approved format, must be lodged with the Owners.
- The ship's Certificate of Fitness must be endorsed within one year.

Which of the following Chemical Group numbers is assigned to 2-Ethyl Hexyl Acrylate?

- **14.**
- 18.
- 24.
- 7.

Which of the following Chemical Group numbers is assigned to Diethyl Glycol Butyl Ether Acetate?

- **34.**
- 32.
- 33.
- 35.

Which of the following Chemical Group numbers is assigned to Propylene Oxide?

- **16.**
- 11.
- 15.
- 9.

Which of the following Chemical Groups numbers is assigned to Tetrahydrofuran?

- **41.**
- 21.
- 31.
- 39.

Which of the following conclusions applies to the stowage of Acetone Cyanohydrin adjacent to cargoes of Group 22 (Caprolactam Solution)?

- **Incompatible (Exception).**
- Compatible (Exception).
- Compatible.
- Incompatible.

Which of the following conclusions applies to the stowage of Acetone adjacent to Epoxy Resin?

- **Compatible.**
- Compatible (Exception).
- Incompatible (Exception).
- Incompatible.

Which of the following conclusions applies to the stowage of Acrylonitrile adjacent to Triethanolamine?

- **Compatible (Exception).**
- Compatible.
- Incompatible (Exception).
- Incompatible.

Which of the following conclusions applies to the stowage of Allyl Alcohol adjacent to Palm Kernel Oil (Fatty Acid Methyl Esther)?

- **Compatible.**
- Compatible (Exception).
- Not Compatible (Exception).
- Not Compatible.

Which of the following conclusions applies to the stowage of Caustic Soda Solution adjacent to Epichlorohydrin?

- **Incompatible.**
- Compatible (Exception).
- Compatible.
- Incompatible (Exception).

Which of the following conclusions applies to the stowage of Cyclohexanone adjacent to Toluene Diisocyanate?

- **Not Compatible (Exception).**
- Compatible (Exception).
- Compatible.
- Not Compatible.

Which of the following conclusions applies to the stowage of Dimethylamine adjacent to Propylene Oxide?

- **Incompatible.**
- Compatible (Exception).
- Compatible.
- Incompatible (Exception).

Which of the following conclusions applies to the stowage of Ethylene Dichloride adjacent to Ethylenediamine?

- **Incompatible (Exception).**
- Compatible (Exception).
- Compatible.
- Incompatible.

Which of the following conclusions applies to the stowage of Ethylenediamine adjacent to Glycerine?

- **Compatible (Exception).**
- Compatible.
- Not Compatible (Exception).
- Not Compatible.

Which of the following conclusions applies to the stowage of Methyl Alcohol adjacent to Iso-Propyl Acetate?

- **Compatible.**
- Compatible (Exception).
- Incompatible (Exception).
- Incompatible.

Which of the following conclusions applies to the stowage of Methyl Ethyl Ketone adjacent to Phenol?

- **Compatible.**
- Compatible (Exception).
- Incompatible (Exception).
- Incompatible.

Which of the following conclusions applies to the stowage of Methyl Tertiary Butyl Ether adjacent to Phosphoric Acid?

- **Incompatible (Exception).**
- Compatible (Exception).
- Compatible.
- Incompatible.

Which of the following conclusions applies to the stowage of Phenol adjacent to Coconut Oil, Fatty Acid?

- **Compatible.**
- Compatible (Exception).
- Incompatible (Exception).
- Incompatible.

Which of the following conclusions applies to the stowage of Propylene Glycol adjacent to Methyl Acrylate?

- **Compatible.**
- Compatible (Exception).
- Not Compatible (Exception).
- Not Compatible.

Which of the following conclusions applies to the stowage of Sodium Silicate Solution adjacent to Nitric Acid (70% or less)?

- **Incompatible (Exception).**
- Compatible (Exception).
- Compatible.
- Incompatible.

Which of the following conclusions applies to the stowage of Trichloroethylene adjacent to cargoes of Group 5 (Caustics)?

- **Incompatible (Exception).**
- Compatible (Exception).
- Compatible.
- Incompatible.

Which of the following conclusions applies to the stowage of Vinyl Acetate adjacent to Ammonium Nitrate, Urea Solution (containing Ammonia)?

- **Incompatible.**
- Compatible (Exception).
- Compatible.
- Incompatible (Exception).

Which of the following is an alternative name for 2-Ethyl -1- Butanol?

- **Ethyl Butanol.**
- 1,2- Methyl Ethanol.
- 2,1- Ethyl Alcohol
- Butyl Etanol.

Which of the following is an alternative name for Ethylidene Dichloride?

- **1,1-Dichloroethane.**
- 1,3 – Ethylene Chloride.
- Chlorothene.
- Ethyl Dichloride.

Which of the following is not permitted as a barrier between two hazardously reactive cargoes?

- **An integral stainless steel bulkhead.**
- A cofferdam.
- A tank containing a compatible cargo.
- An empty tank.

Which two of the following statements are correct descriptions of Exceptions as described and listed in Appendix I to the Compatibility Chart?

- **Cargoes which have been tested and found to be compatible and may be stowed in adjacent tanks, although most products with the same respective Chemical Group numbers are dangerously reactive.**
- **Cargoes which have been tested and found to be dangerously reactive and may not be stowed in adjacent tanks, although most products with the same respective Chemical Group numbers are compatible.**
- Cargoes which have been tested and found to be highly reactive, although products in the same Chemical Groups are usually deemed moderately reactive.
- Cargoes which have been tested and found to be only occasionally reactive, although products in the same Chemical Groups are usually deemed consistently reactive.

"The check according to the USCG Compatibility Chart is the only check actually necessary during stowage planning". True or False?

- **False.**
- True.

0097 – Ship Energy Efficiency

Version 4.0 (2497)

About how much of the energy supplied to a marine engine is lost?
Select the ONE correct answer.

- **about 50%**
- about 20%
- about 5%
- about 75%

According to this module, approximately how much of the cost of running a cargo vessel is due to fuel consumption?
Select the ONE best answer.

- **50%**
- 10%
- 20%
- 80%

According to this module, how much fuel can be saved by fitting a bulbous bow to some ships?
Select the ONE best answer.

- **up to 20%**
- up to 40%
- up to 5%
- up to 60%

According to this module, how often should a traditional antifouling paint be applied?
Select the ONE best option.

- **once a year**
- every three years
- every two years
- twice a year

According to this module, what is a typical hull roughness value for a new-build ship?
Select the ONE correct answer.

- **100 microns**
- 10 microns
- 5 microns
- 500 microns

According to this module, what is a typical roughness value for a newly polished propeller?
Select the ONE best answer.

- **10 microns**
- 100 microns
- 5 microns
- 500 microns

According to this module, what is the optimal interval for self-polishing antifouling application?
Select the ONE best answer.

- **24 months**
- 12 months
- 18 months
- 6 months

According to this module, what speed loss would justify an average vessel docking?
Select the ONE best answer.

- **0.5 knot**
- 0.1 knot
- 1 knot
- 2 knots

An upgraded propeller gives a 10% decrease in fuel consumption and engine modifications also give a 10% decrease in fuel consumption. Calculate the overall fuel savings.

- **19%**
- 10%
- 15%
- 20%

Choose the most fuel efficient option.

- **large propeller and a slow revolving shaft**
- large propeller and a fast revolving shaft
- small propeller and a fast revolving shaft
- small propeller and a slow revolving shaft

For a variable pitch propeller vessel what is the best pitch and propeller speed combination for optimum fuel consumption?
Select the ONE correct answer.

- **high pitch, low RPM**
- high pitch, fast RPM
- low pitch, fast RPM
- low pitch, low RPM

For a vessel with a variable pitch propeller, what is the best way to adjust the speed of the ship?
Select the ONE best answer.

- **adjust RPM**
- adjust load.
- adjust pitch

If there is low frequency vibration of the engine at low speed, what is the probable location of the problem?
Select the ONE best answer.

- **The main propulsion shaft**
- One of the engine cylinders
- The main gearing
- The turbocharger bearings

In general terms, what is the most fuel efficient shape for a vessel?
Select the ONE best answer.

- **long and slim vessel**
- long and shallow vessel
- long and wide vessel
- short and wide vessel

What does SEEMP stand for?
Select the ONE correct answer.

- **Ship Energy Efficiency Management Plan**
- Safety Executive Energy Management Plan
- Ship Energy Efficiency Market Place
- Ship European Employment Management Plan

What is the main reason for de-rating the engine and changing the propeller?
Select the ONE best answer.

- **A change in the service speed of the vessel**
- increasing the lifetime of the vessel
- the vessel is getting old
- vibration problems

Which ONE of the given options is the minimal instrumentation necessary to estimate fuel consumption?
Select the ONE correct option.

- **fuel gauge and speed log**
- fuel gauge and Doppler log
- fuel gauge and exhaust thermometer
- fuel gauge and radio navigation instrument

Which ONE of these would benefit the most from a nozzle (tunnel) propeller?
Select the ONE best answer.

- **A slow speed ship with a high propeller load**
- A fast coastal ships
- A long distance cargo ship
- A vessel with a propeller which rotates at high speed

Which THREE of these measures increases the efficiency of steam heating?
Select the THREE correct answers.

- **Make sure that pipe insulation is fitted and in good condition**
- **Make sure that the steam leaves the heaters as condensate.**
- **Prevent leakage of steam from the pipework system.**
- Make sure that the steam is still dry when it leaves the heaters.

Which of these vessels would NOT benefit from using self-polishing antifouling?
Select the ONE best answer.

- **ice operating vessels**
- non-ferrous hull vessel
- riverboats
- slow moving tug boats

0098 – Fresh Water Generator

Version 3.0 (1321)

After which ONE of these stages should the hot water inlet and outlet valves be opened during start-up?

- **After there is a 90% vacuum**
- After closing the hot water bypass valve
- After opening the ejector pump overboard valve
- After there is a 30% vacuum

From which ONE of these is the feed-water taken in a fresh water generator?

- **The seawater outlet from the condenser**
- The jacket water cooling system
- The seawater inlet to the condenser
- The seawater inlet to the jacket water cooler

How is the fresh water produced in a fresh water generator delivered to the storage tank?

- **Using the attached fresh water pump**
- Using a fresh water ejector
- Using a stand-alone fresh water pump
- Using gravity

In which order does the feed water pass through the fresh water generator, after pre-heating? Drag the items into the correct order.

- 1 **Evaporator**
- 2 **Demister**
- 3 **Condenser**
- 4 **Salinometer**

What is the FIRST step when stopping the fresh water generator? Select ONE answer.

- **Stop the hot water supply to the evaporator, after opening the bypass valve**
- Stop the ejector pump
- Stop the feed water supply
- Stop the fresh water discharge pump

What is the LAST step when stopping the fresh water generator? Select ONE answer.

- **Close the overboard valve for the combined air/brine ejector**
- Open the air valve
- Stop the ejector pump
- Stop the feed water supply

What should the pressure be before the ejector when starting the fresh water generator? Select ONE answer.

- **0.28 - 0.30 MPa**
- 0.06 - 0.08 MPa
- 0.20 - 0.25 MPa
- Over 1.0 MPa

When should the air valve on the top of the fresh water generator (FWG) be open? Select ONE answer.

- **While the FWG is stopped**
- During vacuum creation at start-up
- Never
- Never – except during maintenance

Which ONE of these actions would be best to increase the flow of jacket water to the fresh water generator while it is running?

- **Adjust the by-pass valve until the desired inlet and outlet temperatures are reached**
- Open the jacket water inlet valve fully
- Open the jacket water outlet valve fully
- Start a second jacket water circulating pump

Which ONE of these best describes the function of the sight glass on a fresh water generator?

- **It enables the brine level to be seen**
- It enables the evaporated steam to be seen
- It enables the flow of condensate to be seen
- It enables the flow of feed water to be seen

Which ONE of these best describes the process of "vacuum distillation"?

- **Seawater is boiled at low pressure and temperature before being condensed into fresh water**
- Fresh water is boiled at low pressure and temperature before being condensed into alcohol
- Fresh water is boiled at low pressure but high temperature before being condensed into alcohol
- Fresh water is boiled at low pressure but high vacuum

Which ONE of these best describes the purpose of a fresh water generator is?

- **To convert seawater into fresh water, for drinking and for use in ship's systems**
- To convert fresh water into DC electricity
- To convert seawater into AC electricity
- To convert seawater into drinking water

Which ONE of these best describes the purpose of the air valve, fitted to the fresh water generator

- **It allows the vacuum to be broken when shutting down**
- It allows air to be removed, so that a vacuum can be created
- It allows vapour samples to be taken for analysis
- It prevents overpressure in the unit

Which ONE of these best describes what is supplied by the fresh water generator ejector pump?

- **Condenser cooling, feed-water for evaporation and driving water for the brine/air ejector**
- Only the driving water for brine/air ejector
- Only the feed-water for evaporation and driving water for brine/air ejector
- Only the feed-water for the evaporator only

Which ONE of these best describes what takes the heating medium into the fresh water generator?

- **It is pumped around by the jacket water circulating pump**
- A thermostatic valve controls the flow of heating medium
- An integral hot water circulating pump is fitted
- The steam pressure drives

Which ONE of these best describes why fresh water disinfection equipment is often fitted with fresh water generators?

- **Because boiling water at low temperatures does not kill all harmful bacteria**
- Because the titanium heat exchangers leak poison
- To improve the taste of drinking water
- To prevent corrosion inside the unit

Which ONE of these best describes why fresh water generators are more reliable than they used to be?

- **They use seawater resistant materials, rather than coatings**
- They have fewer moving parts
- They use a different working principle to older designs
- They use steam for heating instead of jacket water

Which ONE of these best describes why fresh water generators can be used in an unmanned engine room?

- **They are designed with continuous monitoring of the fresh water quality**
- All of their systems have remote controls
- They are built with no local indicators or gauges
- They have no moving parts and therefore do not go wrong

Which ONE of these best describes why modern fresh water generators are simple to install?

- **They can be assembled in place**
- They can be folded for transportation
- They have fewer moving parts
- They use steam for heating instead of jacket water

Which ONE of these best describes why re-hardening filters are sometimes fitted with fresh water generators?

- **To improve the taste of drinking water**
- To prevent corrosion inside the unit
- To prevent corrosion of the cooling circuit
- To prevent corrosion of the hot water circuit

Which ONE of these best describes why spring loaded valves might be fitted in the fresh water discharge line?

- **They prevent air being sucked in**
- They dump the fresh water when the salinity is too high
- They dump the fresh water when the tank is full
- They enable the reciprocating discharge pump to work

Which ONE of these best describes why two similar fresh water generators might have a different number of plates in their heat exchangers?

- **They have been designed to operate with different heating or cooling water temperatures**
- One is designed for greater condenser throughput, to improve ejector performance.
- They have been designed to fit different spaces onboard
- They use different vacuum levels

Which ONE of these creates the vacuum inside the fresh water generator?

- **The water powered ejector**
- The compressed air powered venturi
- The fresh water pump
- The rotary vacuum pump

Which ONE of these describes why the sea suction for fresh water generators is often separate from other suctions?

- **So it is away from any anti-fouling chemical injection systems**
- Because it is installed by the fresh water generator manufacturer, not the shipyard
- Because the suction line will be made of titanium
- So it never takes water from the low suction

Which ONE of these does the vapour generated in the evaporator pass through?

- **The demister and condenser**
- The brine pool, the demister and the condenser
- The condenser only
- The demister only

Which ONE of these explains why the ejector has a suction from the lower part of a fresh water generator?

- **To prevent the brine level from rising too high**
- o help remove salt water from the vapour
- To keep the ejector wet
- To remove the incondensable gases

Which ONE of these explains why we create a vacuum inside a fresh water generator?

- **To decrease the boiling point of the feed water**
- To draw in the feed-water
- To help discharge the fresh water output
- To increase the boiling point of the feed water

Which ONE of these happens to the feed water when it enters a running fresh water generator?

- **It partially evaporates**
- It all condenses
- It all evaporates
- It partially condenses

Which ONE of these is the jacket water cooler?

- **Item C**
- Item A
- Item B
- Item D

Which ONE of these statements about the diagram is true

- **D is the distiller shell**
- A is the distiller shell
- C is the demister
- D is made of titanium

Which ONE of these statements about the diagram is true?

- **A is the condenser inlet and B is the condenser outlet**
- A is the feed water inlet and D is the condenser outlet
- C is the condenser inlet and B is the brine outlet
- C is the condenser inlet and D is the condenser outlet

Which ONE of these statements about the diagram is true?

- **B is the steam heater**
- C is the fresh water generator
- C is the steam heater
- D is the steam heater

Which ONE of these vacuums will result in an evaporation temperature of 40 - 50 °C in a fresh water generator?

- **90 – 95 %**
- 60 – 70 %.
- 70 – 80 %.
- 95 - 100 %.

Which TWO of these are potential causes of high salinity in the water output from a fresh water generator?

- **A high brine level**
- **A leaky condenser**
- A dirty evaporator
- A low brine level

Which TWO of these are potential causes of loss of vacuum in a fresh water generator?

- **The ejector inlet pressure is too low**
- **The ejector nozzle is worn**
- The jacket water is too cold
- The jacket water is too hot

Which TWO of these are potential causes of low fresh water output from a fresh water generator?

- **A loss of vacuum**
- **The jacket water flow is too low**
- The brine level is too low
- The jacket water is too hot

Which TWO of these help remove drops of seawater from the evaporated water?

- **The deflector plate**
- **The demister**
- The condenser
- The incondensable gas extraction line

Which of these is the minimum heating medium temperature above which an anti-scale chemical dosing system is recommended?

- **75°C**
- 50°C
- 85°C
- 95°C

0099 – Low Temperature Insulation on Gas Carriers

Version 2.0 (0)

How often are inspections of void spaces normally carried out?

- **Once a year.**
- Every 6 months.
- Once every 5 years.
- Whenever the need arises.

If closed moulds are used for in-situ insulation, the filling volume is not to exceed . . .

- **200 l pr. filling.**
- 120 l pr. filling.
- 300 l pr. filling.
- 60 l pr. filling.

Is it necessary to refill cavities that are filled with perlit?

- **Yes, within one year after the vessel has been taken into service, and then at regular intervals.**
- No, it is note necessary to refill.
- Only if a tank has been damaged.
- Only if the perlit has been soaked with water.

The cellular plastics insulation contracts approximately . . .

- **2.5 times more than aluminium and 5 times more than Nickel Steel at - 163 deg C.**
- 0.25 times more than aluminium and 2 times more than Nickel Steel at - 163 deg C.
- 0.5 times more than aluminium and 2 times more than Nickel Steel at - 163 deg C.
- equal to aluminium and Nickel Steel at - 163 deg C.

What are some of the indicators of thermal loss

- **Ice or humidity on the surface of the insulation.**
- Bumps or irregular surface on the insulation.
- Discoloration of the vapour barrier.

What is important to keep in mind about U-profiles that are used on pipes supports?

- **That they must not be fastened against the pipe.**
- That they must be made of a flexible material to protect against stress.
- That they should be fastened tightly around the pipe.
- That they should only allow movement on the lateral axis.

What is the atmospheric content of oxygen that is required for entry?

- **21%**
- 12%
- 14%
- 24.5%

What is the maximum allowable LEL content for permitting entry when entering void spaces?

- **1%**
- 12%
- 2.8%
- 9%

What is the most common type of insulation material onboard gas tankers today?

- **Cellular plastics.**
- Cellular glass.
- Cork.
- Mineral wool.

What must you check before personnel are allowed to enter void spaces?

- **A rescue plan has been made.**
- **The level of oxygen and contaminants are within safe limits.**
- **The space is vented while personnel remain inside.**

What should you do if the elastic bellow between the hatch opening and tank on a cylindrical cargo tank should crack?

- **Replace it.**
- Patch it with approved mastic filler.
- Remove it.
- Repair it with an elastic filler and seal it off with a vapour barrier.

When insulating straight line pipes it is important that each layer is . . .

- **staggered at 180 degrees.**
- parallel to the previous layer.
- secured with externally mounted straps or pop rivets.
- separated by a moisture barrier.

Where will you find information about emergency procedures for a specific cargo

- **On the cargo data sheet.**
- On the cargo tank's labelling .
- On the ship's cargo manifest.
- On the ship's emergency plan.

Which of the following has highest diffusion resistance?

- **Polyurethane foam.**
- Air.
- Mineral wool.
- Perlit.

Which of the following is NOT a requirement of insulation used on pipes and containers

- **It should prevent gas leakages.**
- It should prevent heat transfer.
- It should prevent water leakage.
- It should provide external protection of the insulation.

Which of the following properties are needed to fulfil the requirements set for vapour- and physical protection?

- **Prevent moisture transport.**
- **Provide adequate fire protection.**
- **Provide necessary flexibility.**
- **Provide physical strength.**
- Be explosion proof.
- Insulate.

Which of these is the most common way to reduce tank pressure on a LNG tanker?

- **To use the gas as fuel.**
- To discharge the gas into the atmosphere.
- To let the pressure escape through the safety valves.
- To transfer the pressure over to another tank.

Which type of material is normally used in the construction of a cargo tank designed for transporting LPG at atmospheric pressure?

- **Carbon-Manganese steel.**
- 9% Nickel steel.
- Aluminium.
- Stainless steel.

Why do we insulate cargo tanks and other components?

- **To prevent heat flow into the cargo, and to protect the ship's structure.**
- To prevent heat flow in the cargo, and to protect the ship's bunker tanks.
- To protect the crewmembers.
- To save fuel.

Why does the formation of ice along the hatch frame on the deck tank require reinsulation?

- **It can cause corrosion.**
- It can cause loss of cargo
- It can cause the cooling plants to overload.
- It can cause the hull structure to become brittle.

0101 – IALA Buoyage

Version 1.2 (2560)

As prescribed by the IALA Maritime Buoyage System, if you are proceeding down a channel AGAINST the Conventional Direction of Buoyage and you come across a yellow Special mark with a can shape, how will you pass it?

- **Leave it to starboard**
- Leave it to port
- Pass either side – it is a Special mark and there is safe water all round
- Refer to the Sailing Directions for further instructions

As prescribed by the IALA Maritime Buoyage System, you are proceeding down a channel AGAINST the Conventional Direction of Buoyage and you come across a West Cardinal mark. How will you pass it and why?

- **I will pass to the West of it. Cardinal marks are not used in conjunction with Direction of Buoyage, so this has no bearing on my decision**
- I will pass to the East of it, as going against the Direction of Buoyage, my passage is reversed
- It does not matter which side I pass it. A Cardinal buoy in a channel has clear water all round it.
- There must be a mistake. I should not encounter a Cardinal mark in or near a channel. I will drop anchor and investigate further

As prescribed in the IALA Maritime Buoyage System, what rhythm may the light of this mark display?

- **Any that is not used for Cardinal, Isolated Danger and Safe Water marks**
- Any that is not used for a red or green light under the IALA Maritime Buoyage System
- Isophase
- VQ(6)+LFl

As prescribed in the IALA Maritime Buoyage System, where would you find a spherical-shaped buoy?

- **In the middle of a channel**
- On the port side of a channel
- On the starboard side of a channel

Click on the countries and continents whose waters form Region A under the IALA Maritime Buoyage System:

- **Australia and New Zealand**
- **Europe**
- **Most of Asia**
- **Parts of Africa**
- Central America
- Japan
- Korea
- The Philippines

Click on the countries and continents whose waters form Region B under the IALA Maritime Buoyage System:

- **Central America**
- **Japan**
- **Korea**
- **North America**
- **South America**
- **The Philippines**
- Australia and New Zealand
- Europe

Click on the light pattern below that may be found on a "preferred channel to starboard" mark in region A of the IALA Maritime Buoyage System:

- **Fl(2+1) R**
- Fl(2+1) G
- Iso R
- Q.R

Click on the light pattern below that may be found on a "preferred channel to starboard" mark in region B of the IALA Maritime Buoyage System:

- **Fl(2+1) G**
- Fl(2+1) R
- Iso G
- Q.G

Going with the Conventional Direction of Buoyage, on which side of a channel are can-shaped marks located in the IALA Maritime Buoyage System:

- **Port hand**
- Can shaped marks are not used to mark channels in the IALA System
- Starboard hand
- They can be found on both sides of a channel

How are Lateral marks of the IALA Maritime Buoyage System modified to show preferred channels?

- **Horizontal red or green stripes are added to standard Lateral marks**
- The buoys are marked with a topmark of the opposite colour and display a light which flashes twice in the primary colour, followed by once in the secondary colour
- The buoys are painted with red and green vertical stripes
- The letters "PC" are added to the buoy as an identification mark

How many flashes per minute will a "Quick Flash" light display, as precribed by the IALA Maritime Buoyage System?

- **Between 50 and 79**
- Between 100 and 129
- Between 30 and 49
- Between 80 and 99

If numbered, do port hand marks in IALA Region B generally carry odd numbers or even numbers?

- **Odd**
- Even

In addition to the use of buoys, what additional warning may be given by a mark of the IALA Maritime Buoyage System to indicate the presence of a "New Danger"?

- **A racon signal, emitting the Morse Code letter "D"**
- A blue strobe light fitted onto, or directly above the marked hazard
- A racon signal, emitting the Morse Code letters "ND"
- An automatic sound signal of the International Code Signal "U", meaning "You are standing into danger"

Sometimes seen as an alternative mark to those prescribed in the IALA Maritime Buoyage System, what is a nun buoy?

- **A cylindrical or can shaped buoy with a triangular top, which may be cut-off or slightly rounded, used on the starboard side of channels in US waters**
- A buoy with a triangular top used on the port side of channels in European waters
- A fairway or landfall buoy fitted with a built-in automatic fog signal
- An IALA-approved beacon with a black conical topmark

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **By leaving it to starboard**
- By leaving it to port
- To the South of it
- To the West of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the West of it**
- By leaving it to port
- By leaving it to starboard
- To the East of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the North of it**
- By leaving it to port
- By leaving it to starboard
- To the South of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the South of it**
- By leaving it to port
- By leaving it to starboard
- To the North of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the East of it**
- By leaving it to port
- By leaving it to starboard
- To the West of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the West of it**
- By leaving it to port
- By leaving it to starboard
- To the East of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the North of it**
- By leaving it to port
- By leaving it to starboard
- To the South of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the South of it**
- By leaving it to port
- By leaving it to starboard
- To the North of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **To the East of it**
- By leaving it to port
- By leaving it to starboard
- To the West of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

This buoy is part of the IALA Maritime Buoyage System. How would you pass it?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

This mark is part of the IALA Maritime Buoyage System. Where does the navigable water lie around it?

- **All round it**
- To the East of it
- To the South of it
- Upstream when proceeding against the Conventional Direction of Buoyage

This mark is part of the IALA Maritime Buoyage System. Where does the navigable water lie around it?

- **All round it**
- Downstream when proceeding with the Conventional Direction of Buoyage
- To the North of it
- To the West of it

This mark is part of the IALA Maritime Buoyage System. Where does the navigable water lie around it?

- **All round it**
- To the East of it
- To the South of it
- Upstream when proceeding against the Conventional Direction of Buoyage

This mark is part of the IALA Maritime Buoyage System. Where does the navigable water lie around it?

- **All round it**
- Downstream when proceeding with the Conventional Direction of Buoyage
- To the East of it
- To the North of it

This mark is part of the IALA Maritime Buoyage System. Where does the navigable water lie around it?

- **All round it**
- Downstream when proceeding with the Conventional Direction of Buoyage
- To the North of it
- To the West of it

Under the IALA Maritime Buoyage System and when proceeding with the Conventional Direction of Buoyage, how is the point at which two channels join marked?

- **No special provision is made to mark the point at which two channels join**
- By a beacon, red with a horizontal black band, topmark black sphere and exhibiting a light Fl(3) R 10s
- By the appropriate preferred channel marks
- It depends on whether you are in Region A or Region B, as the buoys used will be different colours

What colour are Isolated Danger Marks, as prescribed by the IALA Maritime Buoyage System?

- **Black and red**
- Green
- Red and white
- Yellow

What colour is used on port hand Lateral marks in Region A of the IALA Maritime Buoyage System?

- **Red**
- Green
- Red and white vertical stripes.
- Yellow over black

What colour is used on port hand Lateral marks in Region B of the IALA Maritime Buoyage System?

- **Green**
- Black with a red band
- Red
- Yellow over black

What colour is used on starboard hand Lateral marks in Region A of the IALA Maritime Buoyage System?

- **Green**
- Black over yellow
- Blue and yellow vertical stripes
- Red

What colour is used on starboard hand Lateral marks in Region B of the IALA Maritime Buoyage System?

- **Red**
- Black
- Green
- Green with a red band

What is the term used in the IALA Maritime Buoyage System to identify the feature that Cardinal marks are placed around?

- **The Point of Interest**
- The Danger Point
- The Point of Reference
- The Significant Hazard

What name is given to a system where an AIS signal projects the position of a buoy artificially, for the purpose of navigational safety, without a buoy being physically present in the corresponding position?

- **Virtual buoyage**
- Electronic buoyage
- Replacement buoyage
- Simulated buoyage

What name is given to the coloured strips attached to buoys which "beam back" when a light is played on them at night?

- **Retroreflectors**
- Autobeamers
- Reflective actuators
- Reprodiffusers

What topmark (if fitted) is used on a North Cardinal mark of the IALA Maritime Buoyage System?

- **Two black cones, points up**
- Two black cones, bases together
- Two black cones, points down
- Two black cones, points together

What topmark (if fitted) is used on a Safe Water mark of the IALA Maritime Buoyage System?

- **A single red sphere**
- A yellow cone
- Two black cones, points together
- Two black spheres

What topmark (if fitted) is used on a South Cardinal mark of the IALA Maritime Buoyage System?

- **Two black cones, points down**
- Two black cones, bases together
- Two black cones, points together
- Two black cones, points up

What topmark (if fitted) is used on a Special mark of the IALA Maritime Buoyage System?

- **A yellow X**
- A yellow cone
- Two black cones, bases upwards
- Two black cones, points together

What topmark (if fitted) is used on a West Cardinal mark of the IALA Maritime Buoyage System?

- **Two black cones, points together**
- Two black cones, bases together
- Two black cones, points down
- Two black cones, points up

What topmark (if fitted) is used on an East Cardinal mark of the IALA Maritime Buoyage System?

- **Two black cones, bases together**
- Two black cones, points down
- Two black cones, points together
- Two black cones, points up

What topmark (if fitted) is used on an Isolated Danger mark of the IALA Maritime Buoyage System?

- **Two black spheres**
- A single red sphere
- Two black cones, bases together
- Two black cones, points together

What type of buoy may be used as an alternative to the conventional conical, pillar and spar shapes of the IALA Maritime Buoyage System, to mark the starboard side of a channel in U.S. waters?

- **A nun buoy**
- A truncated pillar buoy
- A tun buoy
- An elongated spar buoy

When establishing the Conventional Direction of Buoyage, as defined by the IALA Maritime Buoyage System, which two of the following factors are considered?

- **The general direction of buoyage**
- **The local direction of buoyage**
- The prevailing direction of wind and tide
- The Traffic Management System established by the Flag State Authority

Wherever possible, which way does the General Direction of Buoyage run, as described in the IALA Maritime Buoyage System?

- **Clockwise around continents**
- Always upstream, towards a port
- Anti-clockwise around continents
- Predominantly North-South

Which of the symbols displayed represents a green conical buoy of the IALA Maritime Buoyage System on a British Admiralty chart?

- **A**
- B
- C
- D

Which two of the following are types of mark incorporated in the IALA Maritime Buoyage System?

- **Cardinal**
- **Lateral**
- Local
- Orbital

You are in IALA Region A and proceeding up a channel with the Conventional Direction of Buoyage. The buoys are numbered. How would those on the starboard side generally be marked?

- **With successive odd numbers, following the Conventional Direction of Buoyage**
- With successive even numbers, following the Conventional Direction of Buoyage

You are in IALA Region B and proceeding up a channel with the Conventional Direction of Buoyage. The buoys are numbered. How would those on the starboard side generally be marked?

- **With successive even numbers, following the Conventional Direction of Buoyage**
- With successive odd numbers, following the Conventional Direction of Buoyage

You are in Region A of the IALA Maritime Buoyage System and proceeding down a channel AGAINST the Conventional Direction of Buoyage. You come across a green conical-shaped buoy - how will you pass it?

- **Leave it to port**
- Leave it to starboard
- Pass either side – it is a Special mark and there is safe water all round
- Refer to the Sailing Directions for further instructions

You are in Region A of the IALA Maritime Buoyage System and proceeding down a channel AGAINST the Conventional Direction of Buoyage. You come across a red can-shaped buoy - how will you pass it?

- **Leave it to starboard**
- Leave it to port
- Pass either side – it is a Special mark and there is safe water all round
- Refer to the Sailing Directions for further instructions

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the North of it
- To the West of it

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the South of it

You are in Region A of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

You are in Region B of the IALA Maritime Buoyage System and proceeding down a channel AGAINST the Conventional Direction of Buoyage. You come across a green can-shaped buoy - how will you pass it?

- **Leave it to starboard**
- Leave it to port
- Pass either side – it is a Special mark and there is safe water all round
- Refer to the Sailing Directions for further instructions

You are in Region B of the IALA Maritime Buoyage System and proceeding down a channel AGAINST the Conventional Direction of Buoyage. You come across a red conical-shaped buoy - how will you pass it?

- **Leave it to port**
- Leave it to starboard
- Pass either side – it is a Special mark and there is safe water all round
- Refer to the Sailing Directions for further instructions

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the North of it

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to port**
- By leaving it to starboard
- To the East of it
- To the West of it

You are in Region B of the IALA Maritime Buoyage System. How would you pass this buoy?

- **By leaving it to starboard**
- By leaving it to port
- To the East of it
- To the West of it

0102 – Safe Cargo Handling - FRAMO

Version 1.1 (0)

After blowing lines to shore and closing the manifold valve, what is it recommended to do before stripping?

- **Relieve any residual pressure in the deck cargo line.**
- Close the shore manifold valve.
- Purge the pump cofferdam.
- Visually check the sump.

After purging a pump, from where will any liquid resulting from leakage be collected?

- **The exhaust trap drain valve.**
- The air/nitrogen outlet tap.
- The relief valve.
- The stripping valve.

After the discharge of a cargo of acid and following the purging of the cofferdam, which of the following procedures is recommended?

- **Circulate fresh water through the cofferdam.**
- Carry out an analysis of the hydraulic oil supply.
- Perform a full overhaul on the pump.
- Repeat the purging with a higher pressure to break away any blockages.

At what opening pressure is the relief valve set?

- **3 - 3.5 bar.**
- 0.3 - 0.35 bar.
- 1 - 1.5 bar.
- 5 bar.

Cargo that has leaked into the cofferdam can have come from which of the following places? (Select all applicable answers).

- **Cracks in the cofferdam check pipe.**
- **Cracks in the pipestack.**
- **Flange face seals.**
- **Shaft seals.**
- Cracks in the cofferdam purge pipe.
- The exhaust trap.
- The stripping valve.

Contaminated hydraulic oil can lead to which of the following?

- **Excessive downtime.**
- **Increased maintenance costs.**
- **Reduced system efficiency.**
- Lower hydraulic oil system temperature.
- Reduced leakage detection during purging.

During start up of loading, which of the following precautionary measures will avoid pressure shocks in the pipeline system? (Select all applicable answers):

- **Open valves in a controlled manner.**
- **Start the loading slowly.**
- Increase flow rate to maximum as quickly as possible.
- Open as many valves simultaneously as possible.
- Open cargo valves quickly.

How are the bearings on the shaft lubricated?

- **By the low pressure hydraulic return oil.**
- By an oil sump in the cargo pump casing.
- By grease added manually during pump overhaul.
- By the high pressure hydraulic oil supply.

How do Framo recommend that a number of cargo pumps are run, when required to discharge a homogenous cargo up a common line?

- **Run a higher, yet practical number of pumps in parallel at reduced hydraulic pressure, rather than a smaller number of pumps at maximum hydraulic pressure.**
- Run a smaller, yet practical number of pumps at maximum hydraulic pressure, rather than a higher number of pumps at reduced hydraulic pressure.

How does purging assist the ship's personnel? (Select all applicable answers).

- **Condition monitoring.**
- **Leak detection.**
- **Planning maintenance.**
- Complying with SOLAS regulations.
- Consuming excess spare parts.

How might washing water be tested during tank cleaning to determine if acid residues have been successfully removed?

- **It can be tested with litmus paper.**
- A colour test may be run in the ship's laboratory.
- A sample may be sent ashore for a G.C. test in a shore laboratory.
- It can be filtered through carbon paper.

Hydraulic oil that has leaked into the cofferdam can have come from which of the following places? (Select all applicable answers).

- **Cracks in the pipestack.**
- **Flange face seals.**
- **Mechanical seals.**
- Cracks in the cofferdam check pipe.
- Cracks in the cofferdam purge pipe.
- The high pressure hydraulic supply line.
- The relief valve.

If a tank that previously contained Sulphuric Acid cannot be cleaned immediately after discharge, what should be done with that tank?

- **It should be kept dry.**
- Enough water should be put in the sump to submerse the impeller.
- It should be circulated with a caustic solution for a minimum of 2 cycles with the tank cleaning machines.
- It should be thoroughly ventilated.

If four tanks, of identical size and filled to the same ullage, are to be discharged at the same time up a common shore line, what is the recommended procedure to follow when controlling the overall discharge rate?

- **Keep all the pump controllers in the maximum position and regulate the main hydraulic system pressure.**
- Keep the main hydraulic pressure at maximum and run all cargo pumps at 35 bar less than the main setting
- Keep the main hydraulic system pressure at maximum and regulate individual pump pressures as required.

In order to prevent damage to a stainless steel tank, what will be required after salt water washing?

- **Rinsing down with fresh water.**
- Purging the pump cofferdam.
- Recirculation with an acid-based cleaning chemical.
- Testing the bulkhead with reagent chemical.

In which two ways can a Framo cargo pump be started manually?

- **Local start-up at the top cover plate on deck.**
- **Remote start-up at the Framo pump control panel.**
- Remote start-up at the manifold.
- Remote start-up from the engine room.

Once the cargo pump has been run initially and the pressure is then adjusted, to what level is it increased?

- **Until it is above the cargo manifold pressure.**
- 110 bar.
- By an additional 50% above the initial level.
- To a level at least 75% of the final planned discharge pressure.

Phosphoric Acid sediment can damage which of the following parts of a cargo pump in particular?

- **The seals.**
- The cofferdam purge pipe flange gasket.
- The hydraulic motor.
- The impeller.

Rotation of the impeller causes the liquid to flow towards the periphery of the pump casing through which of the following forces:

- **Centrifugal.**
- Centripetal.
- Gravitational.
- Pneumatic.

Under which of the following circumstances should the emergency stop be activated? (Select all applicable answers).

- **Critical pressure pulsation in the system (hunting).**
- **Critical vibration on the power packs.**
- **Damage to hydraulic pressure pipes or hoses that leads to leakage.**
- **Damage to the cargo lines.**
- **Uncontrolled spill/leakage during discharging.**

What is the backstop unit?

- **A device that prevents the impeller running in reverse when loading is carried out through the pump.**
- A device that allows the impeller to be rotated manually from deck to clear a blockage.
- A non-return valve on the hydraulic oil supply line.
- A tool which provides easy access to the cargo seals during maintenance.

What is the maximum recommended duration for which "live" steam should be applied during tank cleaning, in order to avoid possible damage to the cargo seals?

- **10 minutes.**
- 1 hour.
- 2 minutes.
- 30 minutes.
- Stainless steel and live steam are incompatible and contact must therefore be avoided.

What is the purpose of the cofferdam space?

- **It forms a small void space between the hydraulic oil system and the cargo, capable of being checked at intervals for any possible leakage from either side.**
- It is an auxiliary system built into the cargo pump for the purposes of cleaning the mechanical seals.
- It provides a solid barrier between the hydraulic oil system and the cargo, thereby preventing any possible leakage between the two.

What is the recommended operating range for the temperature of the hydraulic oil?

- **Between 30°C and 55°C.**
- Between 10°C and 25°C.
- Between 50°C and 75°C.
- Between 80°C and 120°C.

What is the recommended supply pressure to have available for clearing the deck cargo line?

- **6-7 bar.**
- 10-12 bar.
- 3 bar.
- More than 15 bar.

What is the standard opening time for an actuator controlled cargo line valve?

- **Approximately 20 seconds.**
- Approximately 3 seconds.
- Exactly 10 seconds.
- More than 30 seconds.

What is the term applied to the formation and collapse of bubbles around the impeller, creating a partial vacuum, causing the pump to surge and heat to be generated, resulting from uncontrolled running of the pump with cargoes such as latex or molasses?

- **Cavitation.**
- Cathodation.
- Causation.
- Centrifugation.
- Gravitation.

What maintenance procedure must be carried after cargo discharge and stripping operations are complete?

- **Purge the pump cofferdams.**
- Pressure test the cargo lines
- Remove the plugs and open the valves on the stripping lines.
- Test the cargo valves for full and correct operation.

What must be tended while foot samples are being loaded?

- **The manifold valve.**
- The cargo pump discharge valve.
- The Framo pump control panel.
- The hydraulic system pressure controller.

What name is given to the process by which the final residues of cargo are removed from a tank and its associated piping?

- **Stripping.**
- Clearing.
- Reducing.
- Straining.

What piece of portable equipment is used to safely release any excess pressure before a blowing hose is disconnected?

- **3-way valve.**
- Quick coupling.
- Snap-on connection.
- Triple release valve.

What term is applied to the testing process used to check for any leakage past the seals in the cargo pump?

- **Purging.**
- Displacing.
- Purifying.
- Stripping.

When a cargo pump is started, at what hydraulic pressure should it initially be run and for how long before it is increased?

- **40-50 bar for a minimum of 1 minute.**
- 120 bar for a minimum of 1 minute.
- 15-20 bar for approximately 3-5 minutes.
- 75-100 bar for approximately 2 minutes.

When discharging a number of tanks simultaneously, how would you adjust the rate from an individual tank?

- **By adjusting the individual pump hydraulic pressure.**
- By regulating the main system hydraulic pressure.
- By throttling the cargo pump discharge valve.
- By throttling the manifold valve.

When preparing for the carriage of fuel oil, which of the following liquids should be added to the pump cofferdam?

- **Diesel oil.**
- Di-octyl phthalate.
- Fresh water.
- Hot fresh water.

When preparing for the carriage of molasses, which of the following liquids is filled to the pump cofferdam?

- **Fresh water.**
- Acetone.
- Di-octyl phthalate.
- Diesel oil.
- Mineral spirit.
- Salt water.

When running at full rate, what should be the balance between the hydraulic system pressure and the hydraulic pressure at the pumps

- **The system pressure should be no more than 20 bar above the highest pump pressure.**
- The hydraulic system pressure should be set to the combined total requirement of all the pumps that are to be run in the current operation.
- The system pressure must be exactly the same as the highest consumer pressure.
- The system regulator automatically adjusts the balance to the maximum discharge performance.

When using the local control valve on the top cover plate to operate the cargo pump, in what position should the control handle on the Framo control panel be?

- **At maximum.**
- At minimum.
- It doesn't matter when the local control valve is being used.

Where is the mechanical seal located?

- **Between the low pressure return oil side and the cofferdam.**
- Between the cofferdam and the cargo.
- Between the high pressure supply oil and the cofferdam.
- Between the impeller and the pump shaft.

Where would you find the results of a vessel's stripping test?

- **The Procedures and Arrangements Manual.**
- The Cargo Record Book.
- The Certificate of Registry.
- The MARPOL Regulations.

Which direction would you turn the local control valve on the top cover plate of the cargo pump to open it?

- **Counter clockwise.**
- Clockwise.

Which of the features listed below is an indication that a pump is losing suction? (Select all applicable answers):

- **A distinctive surging noise from the pump.**
- **Noticeable vibration from the pump.**
- **The needle on the pressure gauge observed to be "hunting".**
- A distinct increase in manifold pressure.
- A noticeable reduction in discharge rate
- A rapid and sustained increase in rate as the last few tonnes are discharged.

Which of the following are functions of the double cargo lip seal?

- **To prevent any leakage of hydraulic oil in the cofferdam from leaking on further into the cargo tank.**
- **To prevent leakage of cargo into the cofferdam.**
- To prevent any leakage of cargo in the cofferdam from leaking on further into the hydraulic oil system.
- To prevent leakage of hydraulic oil into the cofferdam.

Which of the following are typical signs of worn wear rings? (Select all applicable answers).

- **Heavy spraying during stripping.**
- **Slow discharge rate.**
- Increased discharge rate.
- Increased hydraulic oil consumption.
- Raised cargo temperature as the discharge operation approaches completion.

Which of the following cargoes is the most corrosive towards stainless steel?

- **Sulphuric Acid.**
- Acetic Acid.
- Acetic Anhydride.
- Methyl Methacrylate Monomer.
- Propylene Oxide.

Which of the following factors is considered an advantage of a closed loop hydraulic system?

- **The size of the hydraulic tank.**
- The air release procedure.
- The amount of piping.
- The number of components.

Which of the following is the only reason for closing the service valves on the hydraulic supply lines?

- **For maintenance work on the pump or lines.**
- At the start-up of cargo operations.
- On completion of cargo operations.
- When no cargo operations are taking place.

Which of the following precautions would be carried out in preparation for the loading of a cargo of toluene diisocyanate?

- **Filling the cofferdam with di octyl phthalate.**
- Filling the cofferdam with fresh water.
- Plugging the cofferdam check pipe inlet and purging pipe outlet.
- Replacing the cargo seals in the pump.

Which of the following statements is a definition of parallel pumping?

- **Running a number of cargo pumps together to discharge a homogenous cargo from several tanks simultaneously up a common line.**
- Adjusting the hydraulic pressures of a number of cargo pumps so that the tanks complete discharge at the same time.
- Running a group of cargo pumps together by running each up independently, but to the same hydraulic pressure.
- Running adjacent cargo pumps at the same time.

Which of the following substances can easily damage stainless steel surfaces during short-term exposure if improperly handled? (Select all applicable answers):

- **Acid/water mixtures.**
- **Bleach.**
- **Salt water.**
- Fresh water.
- Fuel oil.
- Homogenous cargo.

Which of these functions is performed by a Jockey pump in an open loop hydraulic system?

- **Maintaining the hydraulic oil pressure above the cargo static pressure when the cargo system is not in operation.**
- **Providing cooling/lubrication to the main hydraulic pumps.**
- Maintaining the hydraulic oil temperature above the minimum required operating temperature.
- Preventing back-flow within the hydraulic system.

Which two items from the list below must you consider before opening a lid to check visually if the sump is empty?

- **Safety precautions.**
- **Terminal regulations.**
- If the surveyor is on deck.
- The MARPOL category of the cargo.

Which two items of equipment, located in the hydraulic control room, should be checked as running prior to starting discharge operations?

- **The cooling water pump.**
- **The ventilation fan.**
- The emergency switchboard control board.
- The hydraulic oil heater isolation unit.

Why are wear rings fitted around the base of a pump?

- **To prevent cargo from escaping back into the cargo tank and direct it instead through the discharge line.**
- To hold the base of the cargo pump securely in place and prevent vibration.
- To hold the cargo seal in position at the base of the cofferdam.
- To reverse the direction of flow within the volute casing.

Why is a closed loop hydraulic system considered safer in an operation which involves filling the pump cofferdam with liquid?

- **Because the feed pump maintains a small positive pressure in the circuit when the cargo system is not in use, thereby avoiding the possibility of oil contamination.**
- Because the feed pump absorbs any excess pressure in the hydraulic system, thereby avoiding the possibility of mechanical seal failure.
- Because the header tank in a closed loop system is of a smaller capacity and therefore exerts a lower gravitational pressure on the system.
- Because the pipework in a closed loop circuit is of a smaller diameter and it therefore runs at a a lower system pressure.

Why is it necessary to use hot water in the cofferdam when carrying Phenol?

- **Due to the high melting point of Phenol.**
- Because Phenol is reactive with cold water.
- Because the Code of Carriage of Dangerous Liquid Chemicals at Sea (1987) requires that hot water must be used.
- Due to the low temperature of the hydraulic oil.

Why, with certain cargoes, it is an advantage to fill the pump cofferdam with liquid:

- **Cargo leakages will not solidify and block the cofferdam.**
- **The working lifetime of the seals can be extended.**
- A higher cargo discharge rate can be achieved.
- Purging intervals will be reduced.
- The hydraulic oil system can be run at a higher pressure.

0103 – INS - Integrated Navigation System

Version 1.1 (0)

A good INS system consists of how many main parts?

- **4**
- 2
- 6
- 8

How many earth based monitor stations are in use by the GPS system?

- **4**
- 2
- 3
- 6

Is ECDIS considered to be a primary or secondary INS sub system?

- **Primary**
- ECDIS is not a part of INS
- Not primary, not secondary
- Secondary

The ECDIS consist of how many main components?

- **2**
- 4
- 6
- 8

The information provided by UAIS can be divided in how many groups?

- **4**
- 2
- 3
- 5

What is considered to be the most important part of an INS system?

- **The INS operator**
- The INS back-up systems
- The INS communication system
- The INS computer system

What is the main difference between ECS and ECDIS?

- **ECDIS must be approved by IMO and IHO**
- ECS is only approved by IHO
- ECS/ECDIS is only approved by IMO
- No difference

What is the main purpose of ECDIS?

- **Assist in safe navigation of the vessel**
- Automatically display own ships position in the chart
- Provide navigation information in an easy way
- Provide "anti-grounding" warning in due time

What is the main purpose of IBS?

- **Increasing safe and efficient ship management**
- Increasing safe and efficient navigation and traffic surveillance
- Increasing safe and efficient navigation of the ship
- Increasing safe and efficient traffic surveillance and management

What is the main purpose of INS?

- **Increasing safe and efficient navigation and traffic surveillance**
- Increasing safe and efficient navigation of the ship
- Increasing safe and efficient ship management
- Increasing safe and efficient traffic surveillance and management

What is the main purpose of the operational procedures?

- **Ensuring that the whole system performs satisfactorily**
- Ensuring that technical failures are detected and displayed
- Meeting classification requirements
- Meeting IMO requirements

What is the meaning of proper setting of warning limits?

- **Optimal setting of all warning parameters**
- Optimal setting of important parameters
- Optimal setting of less important parameters
- Proper settings of warning limits are not possible

What is the meaning of "Total system monitoring"

- **Monitoring of technical and human performance**
- Monitoring of all primary technical systems on board the ship
- Monitoring of all technical equipment on the bridge
- Monitoring of both navigation and engine systems

What is the meaning of "adaptive autopilot"?

- **The autopilot is self-adjusting**
- The autopilot can execute pre-programmed turns
- The autopilot is computerized
- The autopilot is not computerized

What is the most important ARPA function?

- **Ability to detect collision danger at an early stage**
- Ability to automatically acquire targets within a selected area
- Ability to detect grounding warning at an early stage
- Ability to give important information in close quarter situations

What is the most important part of INS?

- **The human operator**
- ARPA
- GPS
- The computer system

What is the most important thing to know about the INS system?

- **System limitations**
- How to run test programs and use fault finding procedures
- Main system functions and features
- Warning settings and limits

What is the purpose of Differentional GPS?

- **Improve GPS position accuracy**
- Check the standard GPS position
- Provide course and speed through water
- Provide GPS positions and DR-positions

What is the purpose of INS?

- **Enable more accurate navigation and sailing**
- Enable more economically ship operation
- Increasing safe and efficient ship management
- Obtain more accurate information about important ship parameters

What kind of system is the conning display?

- **Information system**
- Navigation system
- Secondary system
- Warning system

When should "Track mode" be used?

- **In restricted waters**
- Always
- At regular intervals in order to check how it works
- In open sea areas

Which system is considered “Secondary” in INS?

- **CCTV**
- ARPA
- DGPS
- ECDIS

Why does the adaptive autopilot need speed through water information?

- **In order to calculate correct rudder commands**
- In order to calculate correct rate of turn parameters
- In order to calculate course changes as geometrical turns
- In order to calculate speed over ground

0104 – Use of Nitrogen (Chemical tanker)

Version 2.0 (0)

(Allowing for regional variations) regular nitrogen cylinders are commonly:

- **40-50 litres capacity at 180-200 bar pressure.**
- 100 litres capacity at 175-200 bar pressure.
- 200 litres at 50 bar pressure.
- 70-75 litres capacity at 125-130 bar pressure.

Approximately how long will it take to deliver 5000 cubic meters of nitrogen gas through a 100mm (4") pipeline at 5.2 bar (75 psi)?

- **15 minutes.**
- 30 minutes.
- 45 minutes.
- 60 minutes.

Approximately how much nitrogen is there in the air we breathe?

- **80%.**
- 30%.
- 50%.
- 67%.

By injecting nitrogen into the shore pipeline during the loading process, which of the following advantages may be obtained?

- **The requirement to pre-inert is much reduced.**
- Loading can take place at a higher rate.
- The need to monitor nitrogen levels after loading is eliminated.
- The need to pre-inert is avoided entirely.

Chemicals tankers operating predominantly in which of the following trades routinely use oil-fired inert gas (assuming they are fitted with the appropriate generating equipment):

- **Gasolines and Light Gas Oils.**
- Acids.
- Heavy Oils.
- Propylene Oxide.

Click in the boxes next to mediums commonly used in drying cargo tanks:

- **De-humidified air.**
- **Nitrogen.**
- Denatured air.
- Deodorized air.

Click on the boxes next to the two main reasons behind the use of nitrogen on chemical tankers:

- **Preserving cargo quality.**
- **Preventing conditions under which a hazardous reaction might occur.**
- Nitrogen is cheaper to produce than other mediums.
- Preserving cargo quantity.

Click on the boxes to indicate the possible purposes of additives:

- **Antioxidant**
- **Inhibitor.**
- **Stabiliser.**
- Coagulant.
- Coolant.
- Destabiliser.

Click on the boxes to indicate the two types of shipboard nitrogen generator:

- **Membrane separator.**
- **Pressure-swing adsorption.**
- Comparative isolator.
- Pressure separator.

If a ship enters a channel, where the water is relatively cold and still, what may occur in the cargo tanks?

- **Sweating of the bulkheads.**
- Desorption in the upper area of the wing tanks.
- Evaporation on the bulkheads.
- Steam emission.

If an amount of nitrogen equivalent to 3.5 times the volume of a cargo tank was injected into that space (by dilution), what level of oxygen might it be reasonable to expect at that time?

- **2%**
- 1%
- 5%
- 8%

If, during loading of liquid nitrogen, drops of the liquid are allowed to contact the steel deck, which of the following risks is created?

- **Localised steel fracture.**
- A toxic atmosphere will develop.
- Extensive steel distortion.
- Localised steel delamination.

Indicate two types of liquid pad which may be encountered in the carriage of certain specific liquid chemicals:

- **Base oil.**
- **Fresh water.**
- Liquid nitrogen.
- Sulphuric acid.

Inert gas drawn from an in-line unit in the main or auxiliary boiler exhaust, cannot normally be used in chemical tanker operations because:

- **It will be laden with hydrocarbons and soot particles that would have a significant negative impact on the cargo quality.**
- MARPOL Annex II prohibits tank washing direct to sea after inerting with such gases.
- SOLAS requires this type of generator to be fitted only on oil tankers.
- The gas will be above the LEL.

Inert gases create which of the following?

- **An oxygen deficient atmosphere.**
- An oxygen compliant atmosphere.
- An oxygen enhanced atmosphere.
- An oxygen rich atmosphere.

Nitrogen from cylinders will need to pass through which of the following for safe use on deck?

- **A reduction valve.**
- A cooler.
- A de-humidifier unit.
- A reactivation valve.

Oil tankers use an inert atmosphere specifically to prevent:

- **Fire or explosion in a tank containing a hydrocarbon gas/air mixture.**
- Corrosion of the tank surfaces.
- Loss of cargo quantity during a loaded passage.
- Reduction of cargo quality during a loaded passage.

Putrefaction is a term applied to which of the following terms?

- **The decomposition of vegetable oils.**
- The blanketing of low-flash cargoes.
- The dissolution of Caustic Soda.
- The pyrolyzation of light refraction oils.

Symptoms of skin damage resulting from contact with liquid nitrogen may resemble which of the following conditions?

- **Frostbite**
- Alaplecia.
- Dengi fever.
- Melenoma.

The exhaust gas from a nitrogen generator exhaust outlet can typically contain which of the following oxygen levels?

- **35-40%**
- 20.9%
- In excess of 60%
- Less than 10%

The maintenance of a nitrogen pad is also sometimes referred to as:

- **Blanketing.**
- Blending.
- Blowing.
- Purging.

The storage temperature of liquid nitrogen is approximately:

- **-196°C**
- -42°C
- -96°C
- 196°C

The term "Asphyxiates" is applied to:

- **All inert gases.**
- All cargoes with a high evaporation rate.
- All oxygen dependent inhibitors.
- Inerting techniques on chemical tankers.

The traditional manual dew point meter employs which of the following substances?

- **Acetone.**
- **Dry ice.**
- Cryogenic ice.
- Methanol.
- Toluene.
- Xylene.

The two parts of the rotating bed in a de-humidifier unit are referred to as:

- **The process sector.**
- **The reactivation sector.**
- The elimination sector.
- The primary sector.
- The redistribution sector.

Two of the terms below indicate methods by which a pipeline may be cleared to a ship. Which are they?

- **Blow through.**
- **Pigging.**
- Blow down.
- Filling.

Unconsciousness resulting from entering into an inerted space without breathing apparatus can occur earliest in which of the following times?

- **20 seconds.**
- 1 minute.
- 2 minutes.
- 45 seconds.

Under normal circumstances, gas moves through a pipeline more easily than liquid. True or false?

- **True.**
- False.

What additional process can be applied to de-humidified air to make it more efficient in drying tanks:

- **Heating.**
- Cooling.
- Depressurizing.
- Filtering.

What colour identification strip is commonly used to mark nitrogen lines?

- **Signal Yellow.**
- International orange.
- Powder blue with a 10cm (4”) green stripe.
- Signal Red.

What is “dew point”?

- **The temperature at which a gas is saturated with respect to a condensable component.**
- The temperature at which a condensable component is saturated with respect to a gas.
- The temperature at which a saturated gas becomes inambient.
- The temperature at which dry-ice melts in acetone.

What should be reached before commencing any operation involving nitrogen?

- **The full agreement on, and understanding of all aspects of the operation by all parties involved.**
- Agreement on completion time.
- Agreement on who will operate the shore valves.
- Joint calculation of the minimum nitrogen quantity required.

What size of cargo tank would cause special inerting requirements (as listed under the 2000 Amendments to the SOLAS 1974 Protocol) to be implemented on a chemical tanker carrying low-flash products?

- **3000 cubic meters**
- 1000 cubic meters
- 2000 cubic meters
- 5000 cubic meters

What term is applied to the discharge of rail cars by the application of nitrogen?

- **Pressing off.**
- Pulling off.
- Pushing off.
- Putting off.

When maintaining a nitrogen pad on passage, a daily application will involve:

- **Low pressure and low volume.**
- High pressure and high volume.
- High pressure and high volume.
- Low pressure and high volume.

When purging a cargo tank with nitrogen, the critical relationship is:

- **Rate of nitrogen flow in versus capacity of venting system out.**
- Liquid flow in versus liquid flow out.
- Rate of nitrogen supply measured against cargo pump capacity.
- Tank capacity measured against nitrogen supply rate.

Which International Chamber of Shipping publication contains specific details on the use of nitrogen on chemical tankers?

- **The Tanker Safety Guide - Chemicals.**
- Inert Gas Systems.
- SOLAS 1974.
- The Nitrogen Handling Manual.

Which method of atmosphere replacement is generally practised on chemical tankers?

- **Dilution.**
- Dislocation.
- Displacement.
- Dissolution.

Which of the following cargoes is an example of one loaded in a tank where the surrounding empty voids and ballast tanks will also be inerted?

- **Propylene oxide.**
- Polyethylene glycol.
- Propyl alcohol.
- Propylene glycol monomethyl ether.

Which of the following chemicals has an inhibitor added for marine transportation:

- **Ethyl acrylate.**
- Caustic soda solution.
- Iso-propyl alcohol.
- Sulphuric acid.

Which of the following is a true statement?

- **Prolonged blowing of the cargo hose used for loading a cargo containing an oxygen-dependant additive with nitrogen may lead to a significant drop in the efficiency of that additive.**
- Extended blowing of a cargo hose will lead to the destabilisation of an additive.
- Polymerization will result from prolonged blowing of a cargo hose with air.
- The specification of a colour sensitive cargo will be adversely affected if the cargo hose is blown with nitrogen after discharge.

Which of the following items are required to be indicated on a Certificate of Inhibitor?

- **The date added and the effective duration.**
- **The dependency of the additive.**
- **The type of inhibitor.**
- The manufacturer of the additive.

Which of the following items have to be entered on an enclosed space entry permit? (Select all applicable)

- **Minimum permitted oxygen level required for entry.**
- **Observed gas concentration values for tanks which previously contained toxic cargoes.**
- **Signature of Master or responsible officer.**
- **Validity of the permit.**

Which of the following items must a vessel legally have on board prior to loading a cargo of Styrene Monomer?

- **A Certificate of Inhibitor.**
- A Certificate of Analysis.
- A Certificate of Purity.
- A Certificate of Test.

Which of the following statements is an accepted definition of purging?

- **The filling of a cargo tank and associated piping systems, or the spaces surrounding a cargo tank, with a gas or vapour which will not support combustion and which will not react with the cargo, and maintaining that condition.**
- Pre-drying.
- The filling of a cargo tank and associated piping systems, or the spaces surrounding a cargo tank, with a saturated gas or latent vapour which will not induce hydrolysis, and maintaining that condition.
- The gas-freeing of a cargo tank, to a level at which the hydrogen sulphide content, as measured, is less than 35 ppm.

Which regulations detail the requirements for the control of tank atmospheres in compartments containing chemicals?

- **The IBC Code.**
- ISGOTT.
- SOLAS 1974.
- The ICS Tanker Safety Guide (Chemicals).

Why is a hose cleared prior to disconnection? Click on the boxes next to the relevant reasons.

- **Pollution avoidance.**
- **Safety of personnel.**
- Compliance with SOLAS regulations.
- Ship's personnel cannot undertake these tasks as per company regulations.

You are discharging vegetable oils to rail cars on shore. What additional precautions is it prudent to adopt?

- **Blowing down of ship's lines in the event of a protracted delay between groups of rail cars.**
- Drawing up written procedures to cover the discharge and have them signed by a shore representative prior to commencement of dicharge.
- Keeping a fresh water supply adjacent to the shore gantry to wash down any spillage.
- Positioning extra personnel on the ship's manifold to ensure continuous communication.

You are the officer on watch and are about to undertake a line clearing operation from shore. What must you ensure is available in the required tank?

- **Adequate space to accommodate the remaining product without exceeding maximum filling limits.**
- Adequate pressure to ensure good flow in the vapour return system.
- An inert atmosphere.
- Independent portable ullaging equipment.

You are to receive cargo from rail cars. Which of the following risks should you be specifically aware of?

- **As a tank car finishes, nitrogen may pass into the ship's tanks, resulting in an overflow or structural damage if not controlled.**
- As a tank car commences, the ship's tank will be over-pressured and structural damage will result.
- Once the maximum operating pressure is achieved in the rail car, its safety valves will lift.
- The pressure build-up in the ship's tank may result in a reduced discharge rate from the rail cars.

You have just completed loading 450 tonnes of Caustic Soda to two tanks on your vessel and will be trading to NW Europe with an ETA Rotterdam of December 6th. What would be a prudent operational measure to carry out?

- **Additional line blowing.**
- Additional ballasting.
- Additional sampling.
- Additional venting.

You have just loaded a cargo of Toluene. What particular hazard must you be aware of in terms of line clearing?

- **Toluene is a static accumulator.**
- Toluene has a high melting point.
- Toluene is a high viscosity cargo.
- Toluene needs to be blanketed.

0109 – AIS - Automatic Identification System

Version 2.0 (0)

Can AIS with MKD operate as a standalone system?

- **Yes**
- No
- Only in ship to ship mode
- Only in ship to shore mode

How is the AIS technical status checked?

- **Automatically**
- During annual service
- Manually
- No check is necessary

How is the voyage related information updated?

- **Manually**
- Automatically
- No update is necessary
- Semi automatically

How much can the AIS broadcast mode be overloaded without any significant operational effect?

- **400 - 500 %**
- 200 -300 %
- 50 -100 %
- No overload is accepted

How often is the static information updated?

- **every 6 minutes**
- every 3 minutes
- every 12 second
- manual update when necessary.

In case of AIS overload, what can you expect?

- **Targets far away drop out**
- AIS will shut down
- AIS will “freeze”
- Close targets drop

Seen from an operational point of view, how will you describe the AIS?

- **AIS is an automatic system**
- AIS is a manual system
- AIS is a semi automatic system
- AIS is both manual and automatic

The AIS transmitted information is divided into three main groups, which?

- **Dynamic data, Static data and Voyage related data**
- Dynamic data, Destination data and Cargo data
- Position data, Identity data and Dynamic data
- Ship data, Speed Data and Course data

The AIS transponder is designed to transmit information?

- **Automatically**
- AIS is not transmitting at all
- Manually
- Semi automatically

The dynamic information update rate depends on?

- **ship speed and ROT**
- ships size
- type of cargo
- type of ship

The lost target symbol is shown where?

- **In the last known target position**
- At the bottom of the screen
- At the top of the screen
- In operator defined position

The selected target symbol represents?

- **A manually selected target**
- A semi automatically selected target
- A target lost by the AIS
- An automatically selected target

What are the criteria for a dangerous target in AIS?

- **A target calculated to violate preset CPA/TCPA limits**
- A target bigger than own ship
- A target carrying dangerous cargo
- A target faster than own ship

What can happen in case of AIS overload?

- **Targets far away will drop out**
- The system will freeze
- The system will give an alarm
- The system will shut down

What happens when an AIS sensor fails?

- **AIS display "not available"**
- AIS is automatically switched off
- Nothing
- The AIS calculate the missing data

What is important to remember when using AIS for collision avoidance?

- **AIS may not give a complete picture of the traffic situation**
- AIS is more accurate than ARPA
- AIS is not allowed to be used for collision avoidance
- AIS is not as accurate as ARPA

What is the IMO minimum display requirement for AIS?

- **AIS information on MKD**
- AIS information on ECDIS
- AIS information on ARPA
- AIS information on Radar

What is the accuracy of the AIS information?

- **The accuracy is as good as the accuracy of the transmitted information**
- The accuracy can be compared with the accuracy of radar
- The accuracy is good
- The accuracy is not so good

What is the required minimum ship reporting capacity for AIS according to IMO performance standards?

- **2000**
- 1000
- 4000
- 4500

What kind of system is AIS?

- **Transponder system**
- ECDIS system
- Navigation system
- Radar system

When is static information entered into the AIS?

- **During installation**
- After the voyage is completed
- At sea
- At the AIS factory

When should the AIS transponder normally be activated?

- **Always**
- In coastal waters
- In open waters
- When the OOW think it is necessary

When using AIS the OOW should be aware of?

- **AIS information may not be correct**
- AIS information is always correct
- AIS information is automatically switched off if not correct
- AIS information need to be corrected before use

Which of the following is defined as dynamic information?

- **Ship's speed over ground**
- Ship's call sign and name
- Ship's destination and ETA
- Ship's route plan

Which of the following is defined as static information?

- **MMSI**
- Navigational status
- Route plan
- Safety related messages

0110 – AC4 - SULZER - DENIS 1

Version 2.0 (0)

Can DENIS be used together with controlled pitch propeller?

- **Yes.**
- No.

Can the main engine be operated with a malfunctioning governor?

- **Yes, from the engine local manoeuvring stand.**
- No, the main engine cannot be operated with a malfunctioning governor.
- Yes, from the bridge.
- Yes, from the engine control room.

Can the main engine be operated with faulty rpm sensors?

- **Yes, from the engine local manoeuvring stand.**
- Yes, from the bridge.
- Yes, from the engine control room.

Is it possible to cancel a shutdown from bridge.

- **Yes.**
- No.

Is it possible to cancel a shutdown function?

- **Yes but not the overspeed shutdown function.**
- No shutdown functions can be cancelled.
- Yes, all shutdown function can be cancelled.

Is it possible to cancel a slowdown from the bridge?

- **Yes but only when on bridge control.**
- No.
- Yes, always.

The safety cut-out device must immediately stop the engine when:

- **Engine over speed, Emergency stop actuated, pressure drop in the air spring of the exhaust valves.**
- Emergency stop actuated, Engine over speed, High exhaust temp.
- High exhaust temp., Fuel leak, Low cooling water temp.

There are two types of processor cards, what is the difference between them?

- **One is 5 MHz the other is 8 MHz.**
- One is 12 V DC the other is 24 V DC.
- There is no difference.

What happens if both RPM systems have failed?

- **Start block.**
- Shutdown.
- Slowdown.

What is DENIS?

- **An interface between the engine and the remote control system.**
- The emergency system.
- The governor.

What is an adaptor card?

- **Interface card for input/output signals.**
- It is used to simulate input/output signals.
- The same as a processor card.

What is reading the RPM signals in the AC-4 and SSU?

- **The units have RPM cards for this purpose.**
- One adaptor card in each unit.
- There are no RPM signals to the AC-4 and SSU.

What is the input voltage to the AC-4 and SSU units?

- **24 V.**
- 12 V.
- 220 V.
- 440 V.

What is the main purpose of the load program?

- **Preventing the engine from thermal overload.**
- Preventing the engine from overspeeding.
- Rapid increase of rpm in order to reach charter service speed.

What is the purpose of the safety system?

- **Monitor shutdowns and slowdown functions.**
- Inhibit start.
- Prevent the engine from overheating.
- Prevent the engine from overspeeding.

What voltages are generated by the power supply?

- **5 V DC, +15 V DC and -15 V DC.**
- 110 V AC and 220 V AC.
- 5 V AC and 15 V AC.

Where is the safety cut-out device mounted?

- **Fuel pump.**
- Governor.
- Turbocharger.

Which of the following triggers the 3 start Attempts Failed indicator?

- **Third and last start attempt has finished unsuccessfully.**
- Fourth start attempt begins.
- It is always activated until engine starts.
- Only active when shut down.

Which of the following triggers the Start Failure alarm?

- **3 start attempts failed.**
- Low control air pressure.
- Low start air pressure.
- Turning gear engaged.

Which of the following triggers the Start Interlock alarm?

- **Turning gear engaged.**
- Low fuel temp.
- Low-level lub oil.
- Scavenging press. high.

Which of the following will trigger the Crash Astern command?

- **The function is automatic and will be activated if the manoeuvre lever on the bridge is moved to astern when the engine speed is above brake air limit.**
- Call the duty engineer.
- Push the "Crash Astern" button.

0111 – Heavy Weather Damage - Container Vessel

Version 2.0 (1434)

By 31 December 1997, which ships were required to carry an approved CSM, under SOLAS '74? (Select all applicable answers)

- **All new container ships.**
- **All ships of 500 gt and upwards engaged in the carriage of cargoes other than solid and liquid bulk cargoes.**
- All new ships excluding new passenger ships.
- All ships of 500 gt and upwards.
- All ships.

During heavy weather, what could be the approximate induced transverse wind force acting on a five-tier stack of containers?

- **18 tonnes.**
- 1.8 tonnes.
- 36 tonnes.
- 4.2 tonnes.

During pitching, when will pressure changes on the forward container stacks reach their maximum values?

- **When the bows reverse their direction of motion.**
- When a wave crest is amidships.
- When the bows are rising onto a wave crest.
- When the bows fall into a wave trough.

From the formula below, which is correct? (Select all applicable answers)

- **The closer a container is stowed to the base line, the less will it experience transverse accelerations.**
- **The wider the ship's beam, the less will it experience transverse accelerations (at).**
- Increasing the metacentric height (GM) will reduce the ship's roll period.
- Reducing the metacentric height (GM), will result in greater transverse accelerations (at) experienced by the ship.

From the formula below, which statements are correct? (Select all applicable answers)

- **The roll period Tr is inversely proportional to the square root of the metacentric height (GM).**
- **Transverse accelerations are directly proportional to the distance from the base line (Z).**
- **Transverse accelerations are directly proportional to the metacentric height (GM).**
- The wider the ship's beam, the greater will be the transverse accelerations.
- Transverse accelerations (at) are directly proportional to the ship's breadth (B).

How do the accelerations always affect a container lashing system?

- **By inducing forces into the entire lashing system.**
- By causing containers to slip on their base twistlocks.
- By setting up vibrations throughout the system.
- By slackening the turnbuckles.

How would you properly grease turnbuckles?

- **By first cleaning dirty or rusty spindle threads and then applying grease to them.**
- By frequently greasing the hooked ends.
- By frequently greasing the spindles.

How would you properly grease twistlocks? (Select all applicable answers)

- **By injecting grease into the shaft by a specially designed shaft nozzle.**
- **By injecting grease to the wire handling by inserting the shaft handle into the housing through the handle guide.**
- By frequently dismantling and greasing the inner twistlock springs.
- By frequently greasing the twistlock cones.
- By frequently greasing the twistlock handles.

How would you recognise the possible onset of Parametric rolling? (Select all applicable answers)

- **The ship experiences seas from astern;**
- **The ship rolls once for every two waves encountered;**
- **The ship starts developing large roll angles within a very short time;**
- The ship is pitched down by the bow when upright.

If you were on a large container ship with flared bows and a flat, wide stern experiencing seas in near-gale conditions, what action would you take to avoid the possible onset of Parametric rolling?

- **Maintain sea speed and steer a course preferably with the seas approaching from close to the beam, so that roll and pitch motions do not synchronise.**
- Head your ship into the seas.
- Reduce speed to the minimum revolutions to which the ship will respond, under the circumstances.
- Run with the seas from astern.

In accordance with what should the lashing and securing of containers be carried out?

- **The CSM.**
- OSHA regulations.
- The International Convention for Safe Containers, 1972 (CSC).
- The manufacturer's lashing equipment manual.
- The stability booklet.

Once you have calculated the transverse forces acting on a particular container, what would you do with this information? (Select all applicable answers)

- **Compare the information with the permissible forces given by the loading computer.**
- **Compare the information with the permissible transverse forces of the container's Classification Society.**
- Compare the value obtained with the Ship Classification Society's permissible transverse forces.
- Keep it on file for future reference.
- Present the information to attending surveyors in the event of an investigation for that particular voyage.

To what limits do lashing equipment manufacturers design their equipment?

- **To those imposed by Classification Societies.**
- To IMO's requirements.
- To maximum roll amplitudes between 22 and 30 degrees.
- To minimum standards laid down by leading manufacturers.

Under OSHA regulations, what inherent problems are associated with a container spreader suspended from a single point? (Select all applicable answers)

- **Containers cannot be lifted off their stacks in a vertical or relatively stable manner.**
- **It does not allow the spreader to be lifted vertically at the four corner castings.**
- **It precludes the use of semi-automatic twistlocks due to the suspension not being vertical at the four lifting points.**
- It precludes the use of fully automatic twistlocks due to the suspension not being vertical at the four lifting points.

What action(s) would contribute to the prevention of lashing failure? (Select all applicable answers)

- **Checking container lashings during cargo operations;**
- **Inspecting base twistlocks regularly for wear-down and/or cracks.**
- Securing turnbuckles to pad eyes that have slight cracks around their welded seams.
- Using only wires to secure containers on after bays, due to a shortage of lashing rods;

What are some of the basic causes for container stows to collapse? (Select all applicable answers)

- **Carrying heavier over lighter containers;**
- **Incorrect application of the 'wind' lashings;**
- **The use of left and right-hand twistlocks leading to the mistaken notion that the containers are 'locked';**
- Forty five foot long containers stowed outboard.
- Out of gauge containers such as high cube boxes stowed together on a centre bay;

What are the disadvantages of securing containers to a Lashing Bridge? (Select all applicable answers)

- **The lashing bridge becomes less effective as the longitudinal distance between it and the container increases.**
- **The lashing bridge becomes less effective when stowing forty-foot containers in bays arranged for alternative forty-five foot or forty-eight foot containers.**
- Corner-post tension and compression reactions are reduced.
- The lashings resist the overturning of containers.

What are the limitations of CSMs? (Select all applicable answers)

- **They are often not updated in respect of new items or models of container lashing gear supplied to a vessel.**
- **They do not cover all information provided by the loading computer.**
- **They do not cover all permutations and eventualities to which a ship may be exposed.**
- **They provide stowage and securing instructions for pre-approved load conditions only.**
- The information contained differs greatly from the practical aspect of ship operations.

What are the responsibilities of a ship owner in ensuring that containers can safely be carried on board? (Select all applicable answers)

- **The owner must ensure that the items supplied are of the required strength.**
- **The owner must provide a comprehensive stowage and securing manual that is understood by the ship's staff.**
- **The owner must provide and maintain adequate supplies of container securing equipment.**
- The owner must appoint a class surveyor to inspect newly supplied items.
- The owner must maintain a register ashore, of newly supplied items.

What causes longitudinal accelerations? (Select all applicable answers).

- **A ship influenced by stern seas.**
- **A ship meeting head seas.**
- **The rate of change of velocity.**
- The rate of change of velocity during heaving.
- The rate of change of velocity during pitching.
- The rate of change of velocity during rolling.

What causes vertical accelerations?

- **Heaving, Pitching and Rolling.**
- Only Heaving and Pitching.
- Only Heaving.
- Pitching, Rolling and Surging.

What contributes to the safe carriage of containers? (Select all applicable answers)

- **Never exceeding stack weight(s).**
- **Using twistlocks whenever possible.**
- Carrying containers more than two tiers high below deck, in the absence of cell guides.
- In the absence of a CSM, lashing containers as on previous voyages.
- Leaving the inspection of items of lashing equipment upto the Class surveyor;
- Placing dunnage along the bottom of the lowest stack of containers, on non-cellular ships, to prevent slippage.

What dangers are associated with attempting to steer out of following seas? (Select all applicable answers)

- **Course alterations may result in rolling.**
- **Large course alterations are required.**
- **Large roll angles may quickly develop, setting up forces in the container lashing system.**
- The large roll angles will lead to Parametric Rolling.

What does the integrity of a container lashing system depend upon?

- **The weakest component of that system.**
- The extent to which lashings rods are tightened.
- The number of Bridge Fittings securing the containers.
- The strongest item of lashing equipment.

What information do loading computers provide us with? (Select all applicable answers)

- **Alarms when stack weight exceeds the designed maximum stack weight.**
- **Examples of the forces that can be expected in various stowage positions on board.**
- **Facilities for automatic monitoring of individual stack weights.**
- **Heeling moments caused by wind loads.**
- Container inspection dates.
- Survey reports.

What would you look for when inspecting bridge fittings? (Select all applicable answers)

- **Buckling.**
- **Proper lubrication of the item.**
- **Seizing.**
- Elongation.

What would you look for when inspecting deck sockets? (Select all applicable answers)

- **Clogging of drain holes.**
- **Cracking in welded seams.**
- **Signs of pitting.**
- Elongation of sockets.

What would you look for when inspecting dovetail foundations? (Select all applicable answers)

- **Curling of chamfered edges.**
- **Distortion of chamfered edges.**
- **Ease of inserting twistlocks.**
- **Wastage of chamfered edges.**
- Cracks in welded seams.

What would you look for when inspecting lashing plates? (Select all applicable answers)

- **Cracks in welded seams.**
- **Pitting.**
- **Plate distortion.**
- **Reduction of plate thickness.**
- Freedom of movement.

What would you look for when inspecting lashing rods? (Select all applicable answers)

- **Signs of pitting.**
- **Straightness of rods.**
- Rod compression.
- Stretching of rods.

What would you look for when inspecting turnbuckles? (Select all applicable answers)

- **Bending.**
- **Seizing or cracking.**
- **Signs of pitting.**
- Turnbuckle compression.

What would you look for when inspecting twistlocks? (Select all applicable answers)

- **Freedom of cone movement when operated by the handle.**
- **The bases of twistlocks are not cracked.**
- **'Sacrificial exchangeable wear plates', where fitted, have not reached the predefined level of wear and tear.**
- The spring holding the twistlock cones in a closed position is non-resilient.

What would you look for when inspecting 'D' rings? (Select all applicable answers)

- **Cracks in welded seams.**
- **Deterioration of the retaining sleeve.**
- **Wastage of the main shaft.**
- Compression.
- Elongation.

When will a ship experience 'Resonance'? (Select all applicable answers)

- **When it encounters beam seas.**
- **When it encounters one wave during one roll cycle.**
- When it encounters head seas.
- When it encounters two waves during one roll cycle.

When will parametric rolling occur? (Select all applicable answers)

- **when a ship is running with the seas from astern;**
- **when a ship's low speed results in large roll amplitudes.**
- **when roll and pitch motions occur in phase with each other;**
- when a ship's natural period of roll is equal to approximately the wave encounter period;

When will parametric rolling occur? (Select all applicable answers)

- **when a ship is meeting seas fine on the port bow;**
- **when successive wavelengths are of the order between 0.8 and twice the ship's length;**
- **when wave heights exceed a critical level.**
- when a ship's natural period of roll is approximately half the wave encounter period;

Where are the statutory requirements for the carriage of CSMs found? (Select all applicable answers)

- **In chapters six and seven of SOLAS.**
- In chapter four and five of SOLAS.
- In the company's standing instructions.
- In the ship's stability booklet.

Where would you find tools to assist in calculating wind pressure? (Select all applicable answers)

- **In the CSM.**
- **In the loading computer program.**
- In SOLAS.
- In the IMO code on intact stability.
- In the Occupational Safety and Health Administration (OSHA) Regulations.
- In the stability booklet.

With respect of Classification Societies, what is the concern regarding container lashing equipment? (Select all applicable answers)

- **Despite their respective rules and regulations, the industry still lacks a standard approach to the securing of containers aboard ships.**
- **Rules applied by one Classification Society differ from another.**
- **The risk level associated with container stowage in accordance with the various class rules is inconsistent.**
- Having been approved by Class, the information contained in the CSM is still unreliable.
- The information contained in the CSM can be completely relied upon.

With respect to OSHA requirements when can stevedores work on top of containers? (Select all applicable answers)

- **When a stevedore standing on deck, is trying to unlock a container stowed in a high tier by means of a long lashing pole.**
- **When bridge clamps are to be installed.**
- **When stevedores wear appropriate safety gear.**
- When the duty officer considers it safe for stevedores to do so.
- When weather conditions permit.

With respect to container lashings, what could happen in heavy weather if the extreme pitch angle occurred simultaneously with heaving? (Select all applicable answers)

- **Turnbuckles could fail.**
- **Twistlock failure could occur.**
- Hatch pontoon cleats could fail.
- Lashing bridges could collapse.

With respect to lashing bars and turnbuckles, which of the following statements is true? (Select all applicable answers)

- **American Bureau of Shipping (ABS) allows a SWL of 293 kN.**
- **Lloyd's Register (LR), Germanischer Lloyd (GL), Det Norske Veritas (DnV), and Bureau Veritas (BV) allow a SWL of 245 kN.**
- All Classification Societies allow a SWL of 245 kN.
- All Classification Societies allow a SWL of 293 kN.

0112 – Zero Damage

Version 1.1 (0)

A carrying instruction should always contain:

- **Delivery air temperature, maximum CO2 level, speed of circulation fans and shock treatment instructions if so required.**
- Delivery air temperature, maximum CO2 level, dew point, humidity and speed of circulation fans.
- Delivery air temperature, return air temperature, seawater salinity and humidity.
- Return air temperature, set point, humidity, and shock treatment instructions if so required.

A pallet has been dropped on the quay, and is badly damaged. Do you sign for this on the Zero Damage Exceptions Checklist?

- **No, the pallet was in good condition when it left the vessel.**
- Yes, and I include a remark about time and date.
- Yes, the pallet is now in damaged condition, so I have to sign for it.

At the discharge port the custody of the cargo is handed over from the vessel to the terminal. Where should the quality check be done once the pre-slung pallets have been discharged?

- **In the hold before handling the cargo.**
- In the terminal, where the cargo can be properly scanned.
- On the quay, after lifting the cargo out of the vessel.

Before opening the hatches, the tweendecks have to be blown clear of debris because:

- **Debris contaminates the cargo.**
- Debris gets stuck in the gratings so cold air is not possible.
- Debris is toxic to the consumers.
- Debris makes the gratings slippery.

Drag each photo from the cluster, watch carefully and then drop each photo onto the best matching damage code.

DC A -
Empty and/or missing cartons.



DC B -
Damage to packing materials and/or cartons.



DC D -
Pallet base damaged.



DC G -
Wet, mouldy, greasy or oil-stained cartons.



Drag each photo from the cluster, watch carefully and then drop each photo onto the best matching damage code.

DC C -
Damage to content of cartons.



DC F -
Dirty or stained cartons.



DC H -
Chafed, scuffed or scratched cartons.



DC J -
Compression damage to cartons.



Drag each photo from the cluster, watch carefully and then drop each photo onto the best matching damage code.

DC A -
Empty and/or missing cartons.



DC B -
Damage to packing materials and/or cartons.



DC E -
Pallets leaning, collapsed or dismantled.



DC G -
Wet, mouldy, greasy or oil-stained cartons.



How do you determine the apparent good order & condition of the cargo?

- **Use common sense, look, smell, and inspect.**
- Discuss the condition of the cargo with the port captain.
- Inspect the pallets like a fruit specialist would do.
- Look and inspect the fruit inside the cartons.

How often and when must temperature reports be sent to the office?

- **On a daily basis, starting after closing the first cooling section in the first load port.**
- On a daily basis, but only if temperature problems are encountered.
- On a daily basis, starting at departure from the first load port.
- On Monday, Wednesday and Friday, immediately after departure from the load port.

If the condition of the cargo leads to an amber light according to the traffic light system, the terminal should:

- **Repair and/or replace the cargo or record its damage.**
- Accept the cargo without listing the cargo on the damage sheet.
- Mark the cargo with an amber sticker.
- Set the cargo aside and load it only when nobody sees it.

In which case does the T-bar have the highest Maximum Securing Load?

- **With an angle of 45 degrees and 2 lashings applied.**
- With an angle of 30 degrees and 1 lashing applied.
- With an angle of 30 degrees and 2 lashing applied.
- With an angle of 60 degrees and 2 lashings applied.

Indicate the equipment that stevedores use to load the cargo safely and without damage.

- **Pallet spreader**
- **Pallets jacks**
- **Slipsheets**
- **Walking boards**
- Airbags
- Hammer
- Scanner
- T-bars

Indicate the equipment that stevedores use to load the cargo safely and without damage.

- **Pallet spreader**
- **Pallets jacks**
- **Slipsheets**
- **Walking boards**
- Airbags
- Hammer
- Scanner
- T-bars

Mark the 3 main reasons to have a pre-discharge meeting:

- **To clarify the local Zero Damage procedure.**
- **To discuss the use of Zero Damage equipment during discharge.**
- **To make the relevant partners known to each other and organise communication.**
- The pre-discharge meeting is not needed.
- To discuss the lunch breaks.
- To find out whether the terminal will do a proper quality check.
- To go through the Vessel Checklist.
- To have a nice cup of soup before the work starts.

Mark the 3 main reasons to have a pre-loading meeting:

- **To clarify the local Zero Damage procedure.**
- **To discuss the stowage plan and how the cargo will be loaded.**
- **To make the partners involved known to each other and organise communication.**
- The pre-loading meeting is not needed.
- To discuss the lunch breaks.
- To find out whether the terminal did a proper quality check.
- To go through the Vessel Checklist.
- To have a nice cup of coffee before the work starts.

Mark the five Checklist items to prepare the vessel in the pre-loading stage:

- **Check if all gratings are intact.**
- **Check if the bilges and scupper drains are clean and clear.**
- **Check if the delivery- and return air sensors are calibrated and working.**
- **Check if the refrigeration system is free of leakage.**
- **Check if there are sufficient airbags on board.**
- Check if a bunker sample kit has been ordered.
- Check if the vessel has sufficient stores on board.
- Check the navigation planning to the loading port.

Take this example through the 1st step of the 4-step load questions. Is the cargo in apparent good order and condition?

- **Yes, the cargo is in apparent good order and condition.**
- No, the cargo is not in apparent good order and condition.

Take this example through the 1st step of the 4-step load questions. Is the cargo in apparent good order and condition?

- **No, the cargo is not in apparent good order and condition.**
- Yes, the cargo is in apparent good order and condition.

Take this example through the 2nd step of the 4-step load questions. Is it likely that the cargo quality will meet the Receiver's demand?

- **No, the quality of this cargo is not good and must be rejected.**
- Yes, the cargo quality will meet the Receiver's demand.

Take this example through the 2nd step of the 4-step load questions. Is it likely that the cargo quality will meet the Receiver's demand?

- **Yes, the quality of this cargo will meet the Receiver's demand**
- No, the quality of this cargo is not good and must be rejected.

Take this example through the 3rd step of the 4-step load questions. Will it be possible to load the cargo safely, fast and easily?

- **No, it will not be possible to load the cargo safely, fast and easily.**
- Yes, it will be possible to load the cargo safely, fast and easily.

Take this example through the 3rd step of the 4-step load questions. Will it be possible to load the cargo safely, fast and easily?

- **No, it will not be possible to load the cargo safely, fast and easily.**
- Yes, it will be possible to load the cargo safely, fast and easily.

Take this example through the 4th step of the 4-step load questions. Will the cargo negatively affect the condition of other cargo?

- **Yes, the cargo will negatively affect the condition of other cargo.**
- No, the cargo will not negatively affect the condition of other cargo.

Take this example through the 4th step of the 4-step load questions. Will the cargo negatively affect the condition of other cargo?

- **No, the cargo will not negatively affect the condition of other cargo.**
- Yes, the cargo will negatively affect the condition of other cargo.

The 5 reasons for writing down the Supply Chain Procedure are:

- **To be able to measure damages and take corrective action.**
- **To clarify who is responsible for what.**
- **To define at which point the responsibility is handed over to the next partner in the chain.**
- **To know who must pay when there is damage.**
- **To motivate all partners in the supply chain to strive for quality.**

- To collect signatures.
- To have a thick document.
- To motivate the consumption of fresh fruit

What is a Supply Chain Procedure?

- **A document, which lists the tasks, duties and authorities of the partners who are involved in the supply chain**
- A document, which describes the local Zero Damage procedure in the port.
- A document, which describes the responsibility of the various partners, but it is not a document that you can rely upon.
- A document, which has to be issued before the cargo can be transported from its origin to its destination.

What is the definition of Checkpoint Vessel-In?

- **In the vessel after all handling by the stevedores at the final point of stow in the hold.**
- From in the terminal, by handling of the stevedores into the hold.
- From in the terminal, via the quay, into the hold.
- From the quay, via the stevedores' cranes into the hold.

What is the main purpose of the Vessel Checklist:

- **It's a guideline to prepare the vessel for the different stages in the supply chain.**
- It's a guideline on how to handle, stow, load and tally the cargo.
- It's a tool for the master on how to prepare the vessel before loading.

What is the main purpose of the rain directives?

- **To protect the cargo and avoid delays in loading or discharge during light rain.**
- To close the hatches immediately when it starts to rain.
- To discharge the cargo in the right sequence when it rains
- To protect the cargo in the hatch square when it rains.

What is the most important rule to keep in mind when you use slings?

- **Always use slings of the same type, SWL and length.**
- Always minimum 24 pallets must be pre-slung in the hatchsquare.
- Always use clean slings.
- Always use slings to lash the cargo.

When different temperatures are required in different compartments it is important to write down the different temperatures per compartment in the stowage plan:

- **Yes**
- No
- Not relevant

When the terminal checks the condition of the cargo according to the traffic light system, the red light will be given to:

- **Cargo that is not fit for transportation.**
- A pallet with a missing carton.
- Cargo that is damaged but fit for transportation.
- Cargo without damage or defects

Which 3 parties do you have to contact in case of a major cargo damage?

- **The operator**
- **The owner or manager**
- **The P&I club**
- The General Support department
- The shipper
- The stevedore
- The tally clerk
- The terminal

Which 3 rules do you have to keep in mind when you place airbags?

- **Do not place the airbags between the sideboards and the pallets.**
- **Do not place the airbags in the shape of the letter "L".**
- **The maximum gap to be filled by an airbag is 30 cm but preferably less.**
- Avoid as much as possible airbags in the longitudinal direction of the stowage.
- That out of 15 airbags one will deflate during the voyage.
- Use as many airbags as possible to fill up the cargo hold.

Which of the following statements is correct?

- **Rejected cargo has to be set-aside in a protected cargo shed area on the quay.**
- Rejected cargo has to be loaded anyway into the cargo hold.
- Rejected cargo has to go back to the receiver.
- Rejected cargo has to go straight back to the terminal.

Which picture gives the preferred broken stowage in view of lashing.

- **Drawing A with pallets fore and aft.**
- Drawing B with pallets port and sb side.
- Drawing C with pallets fore and aft and port and sb side.
- Drawing D with pallets fore and port and sb side.

Which rain scenario do you have to use when it rains? It depends on:

- **Whether the vessel has spar decks or individual cooling decks.**
- What kind of vessel you sail on and which discharge port you sail to.
- Whether the vessel has the right equipment like tarpaulins, rain covers and so on.

Which statement describes the correct loading sequence?

- **Start with section A and E, then section B and D and finally section C.**
- Start with section A, then sections B, C and D and finally section E.
- Start with section B and D, then section A and E and finally section C.
- Start with section E, then sections B, C and D and finally section A.

Which statement is correct:

- **The vessel has to comply with the speed instructions, but there should be no compromise to the Zero Damage condition of the cargo.**
- The vessel always has to comply with the speed instructions and does not have to comply with the Zero Damage instructions.
- The vessel always has to comply with the speed instructions to make sure to arrive on time at the discharge port, no matter what.
- The vessel has to comply with the speed instructions and never stop or change course or change speed.

Which statement is correct:

- **The operator is there to assist you, and can be contacted at all times when necessary.**
- The operator is not to be contacted by the vessel after regular office hours.
- The operator is there to assist you, and can be contacted normally during office hours only.

Which statement is correct?

- **Walking boards must be available on arrival in the load port and are compulsory for stevedores and crew.**
- Walking board must be available after opening the hatches and are compulsory for the stevedores.
- Walking boards must be available just before loading the pre-slung pallets and are compulsory for the crew.
- Walking boards must be left on top of the cargo after completion of loading the hatch square.

Who is responsible for this kind of damage?

- **The stevedores.**
- The captain and the stevedores.
- The captain.

Why do we have Checkpoints in the supply chain?

- **To check the cargo for damage and establish a handover of responsibility for the cargo from one partner to the next.**
- To allow the owner of the cargo to examine his goods.
- To check that the cargo is going in the right direction in the supply chain.
- To check the cargo for damage before it goes out of the terminal.

Why do you have to stow the pallets parallel to the centre-line and not parallel to the sides of the cargo hold?

- **Otherwise you do not get a square for loading the pre-slung pallets.**
- Otherwise the discharge of the pallets will be very difficult.
- Otherwise you do not get a maximum cargo intake.
- Otherwise you will not get enough pallets in the cargo hold.

Why is it so important to clean and maintain the cargo holds?

- **To ensure that food products can be safely loaded in the cargo holds.**
- To avoid the occurrence of stevedore damage in the cargo holds.
- To ensure that a nice party can be organised in the cargo holds.
- To maximise the cargo intake in the cargo holds.

Why is the Checkpoint Vessel-In so important for the vessel?

- **As the vessel takes responsibility for the cargo at this point.**
- As it is important that the cargo documents are correct.
- As the terminal will check the cargo quality at this point.
- As this is a part of the Zero Damage policy of the shipping line.

Why must there always be a deck officer on deck during loading?

- **To monitor, control and enforce the Zero Damage principles.**
- To monitor and control if damage occurs in the cargo holds.
- To monitor and control if it starts to rain.
- To monitor, control and enforce the use of the Vessel Checklist.

With an agreed Supply Chain Procedure in place, the responsibility for damages is always clear, because:

- **The Supply Chain Procedure clearly describes where the responsibility for the cargo is handed over from one partner to the next one in the chain.**
- The Supply Chain Procedure clearly describes that all responsibility for the cargo is with the ship and its crew.
- The Supply Chain Procedure clearly describes which damages have to be recorded.

0115 – Security awareness

Version 6.0 (1721)

According to this module, what should happen if a ship is in port when a bomb warning is received? Select the ONE correct answer.

- **The Master, the Ship Security Officer and the Port Security Officials will discuss the situation and decide what action to take**
- The Master must be informed and they should immediately evacuate all personnel from the ship
- The person receiving the threat should immediately sound the general alarm, and then inform the Master
- The Ship Security Officer must contact the Company Security Officer immediately, and they will decide what action to take

Are weapons of any type allowed to be carried on-board ship? Select the ONE correct answer.

- **The management company of the ship sets the policies for what type of weapons are allowed on-board the ship and this must be clearly stated in the Ship Security Plan**
- No firearms or explosives of any type are allowed on-board ship
- Persons may carry firearms for which they have a licence, subject to local legislation
- The Master may sanction licensed users to carry firearms, subject to local legislation

During a bomb search, which of the following is the most important principle to follow?

- **Do not touch any suspicious packages**
- Know exactly what a bomb looks like
- Throw any suspicious items overboard
- Try to reach behind bulkheads to find a bomb

Under the ISPS Code, which of the following are likely to be 'restricted areas'? Select ANY answers that you think apply.

- **The Bridge**
- **The Cargo control room**
- **The Engine control room**
- The Mess room

What is a Continuous Synopsis Record? Select the ONE correct answer.

- **A record of the vessel's history**
- A plan for the continuous maintenance of security equipment
- A plan which includes all the security measures on-board
- A record of all security incidents

What is a 'Declaration of Security'? Select the ONE correct answer.

- **An agreement (checklist) between a ship and either a port facility or another ship, specifying the security measures each will implement**
- A declaration from the ship, stating the ship's security level
- A document between the port and the cargo owner, stating the security measures required
- An agreement (checklist) jointly completed by the Ship Security Officer and the U.S. Coast Guard

What percentage of the ship's stores need to be inspected at Security Level 3? Select the ONE correct answer.

- **100%**
- 25 - 50%
- 5 - 20%
- 50 - 80%

When should the identities of all visitors trying to board the ship be checked? Select the ONE correct answer.

- **At Security Levels 1, 2 & 3**
- Only at Security Level 3
- Only at Security Levels 2 & 3
- Only if they look suspicious

Which Security Level requires the highest security alert? Select the ONE correct answer.

- **Security Level 3**
- Security Level 1
- Security Level 2
- Security Level 4

Which TWO persons check the Declaration of Security? Select the TWO correct answers.

- **The Port Facility Security Officer**
- **The Ship Security Officer**
- The Company Security Officer
- The local Coast Guard Officer

Who is responsible for issuing an International Ship Security Certificate? Select the ONE correct answer.

- **The flag State**
- The insurance company
- The port State
- US Coast Guard

Who is responsible for the development of the Ship Security Plan? Select the ONE correct answer.

- **The Company Security Officer**
- The Classification Society
- The Port Facility Security Officer
- The Ship Security Officer

Who is responsible for the implementation and maintenance of the Ship Security Plan? Select the ONE best answer.

- **The Company Security Officer and the Ship Security Officer**
- The Company Security Officer only
- The Ship Security Officer and the Master only
- The Ship Security Officer only

Who is responsible for the regular security inspections of the ship? Select the ONE correct answer.

- **The Ship Security Officer**
- The Chief Officer
- The Master
- The Port Facility Security Officer

Who is responsible for the regular security inspections of the ship? Select the ONE correct answer.

- **The Ship Security Officer**
- The Classification Society
- The Company Security Officer
- The Port Facility Security Officer

Your ship is entering an area known for pirate activity. Which ONE of the actions listed below is recommended?

- **To transit with maximum safe speed**
- To confine all ship's personnel to one room on-board
- To transit at night time only
- To turn off all lights, including navigation lights

0116 – AC4 - MAN B and W

Version 2.0 (0)

Can the main engine be operated with a malfunctioning governor?

- **Yes, from the engine manoeuvring stand.**
- No, the main engine can not be operated with a malfunctioning governor.
- Yes, from the bridge.
- Yes, from the engine control room.

Can the main engine be operated with faulty rpm sensors?

- **Yes, from the engine manoeuvring stand.**
- Yes, from the bridge.
- Yes, from the engine control room.

Is it possible to cancel a shut down from bridge?

- **Yes**
- No

Is it possible to cancel a shutdown function?

- **Yes but not the over speed shutdown function.**
- No shutdown functions can be cancelled.
- Yes, all shutdown function can be cancelled.

Is it possible to cancel a slowdown from the bridge?

- **Yes but only when bridge control.**
- No
- Yes, always.

There are two types of processor cards, NN 791.10 and NN 791.12. What is the difference between them?

- **791.10 are 5MHz and 791.12 are 8MHz.**
- 791.10 are 12 volt and 791.12 are 24 volt.
- There is no difference.

What does “handle match” mean?

- **Bridge handle matches engine speed and transfer of command position can take place.**
- Bridge and Engine control room handles are in same position.
- Load program is activated.

What happens if both RPM systems have failed?

- **Start block.**
- Shutdown.
- Slowdown.

What is an adaptor card?

- **Interface card for input/output signals.**
- It is used to simulate input/output signals.
- The same as a processor card.

What is reading the RPM signals in the AC-4 and SSU unit?

- **The units have RPM cards for this purpose.**
- One adaptor card in each unit.
- There are no RPM signals to the AC-4 and SSU unit.

What is the input voltage to the AC-4 and SSU units?

- **24V**
- 220V
- 440V

What is the main purpose of the load program?

- **Preventing the engine from thermal overload.**
- Preventing the engine from overspeeding.
- Rapid increase of rpm in order to reach charter service speed.

What is the purpose of the safety system?

- **Monitor shutdowns and slowdown functions.**
- Inhibit start.
- Prevent the engine from overheating.
- Prevent the engine from overspeeding.

What voltages are generated by the internal power supply?

- **5V DC. +15V DC and –15V DC.**
- 110V AC and 220V AC.
- 5V AC and 15V AC.

Which of the following functions is NOT integrated in the Bridge Panel?

- **Displaying ship position.**
- **Displaying ship speed.**
- **Physical operation of the main engine.**
- Displaying process alarm status.
- Displaying process signal values.
- Displaying status for the remote control system.
- Interfacing Ahead/Astern signals from the bridge lever.
- Interfacing bridge lever rpm command signal.

Which of the following trigger the: 3 fail starts indicator?

- **Third and last start attempt has finished unsuccessful.**
- Fourth start attempt begins.
- It is always activated until engine starts.
- Only active when shut down.

Which of the following will trigger the Crash Astern command?

- **The function is automatic and will be activated if the manoeuvre lever on the bridge is moved from full ahead to full astern.**
- Call the duty engineer.
- Push a “Crash Astern” button.

0117 – Integrated Survey Programme - ISP

Version 2.0 (0)

Below you have find an example of coating condition in a ballast tank. Provided this condition is representative throughout the tank what would be the correct rating according to your judgement?

- **Good.**
- Fair.
- Poor.

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- **Fair.**
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Below you have find an example of coating condition in a ballast tank. Provided this condition is representative throughout the tank what would be the correct rating according to your judgement?

- **Poor.**
- Fair.
- Good.

Coating is applied in ballast tanks for protection of steel against corrosion. Which type of coating is not considered by the class to be an efficient corrosion protection system?

- **Soft coating.**
- Hard coating.
- Semi hard coating.

During inspection of the aft peak, where are cracks unlikely to be found?

- **Fatigue cracks at the top of tanks.**
- Indentation due to bottom slamming.
- Vibration crack at connection between floor and hull plating.

For a simply supported beam with a concentrated load at middle of the span, the maximum bending moment is higher:

- **At middle of span.**
- At the end supports.
- At the quarter length of span.

For efficient cathodic protection of ballast tanks which of the following does not apply:

- **Anodes are to be painted.**
- Anodes are to be properly attached.
- Anodes are to be submerged in water.
- Consumed anodes are to be replaced.
- Sufficient number of anodes are to be provided.

How does the class notify deficiencies?

- **Condition of Class (CC) for owner in case of minor deficiency.**
- Memo to the owner (MO) covering the deficiency.
- Memo to the surveyor.
- Write a report only.

In case securing equipments on board a Container ship is worn beyond use and new equipment has to be ordered. In that connection, please mark which of the below statement that is correct.

- **The equipment should be certified by the classification society.**
- The equipment should be as specified in the container securing manual.
- The equipment should be of similar type and size.

In connection with loading of containers, the correct information about the weight and location of containers are crucial for ship's stability and strength, as well as the stowing and securing of them. Where would you look for information on how to provide proper securing of the containers?

- **In the Container securing manual.**
- In the ISM system.
- In the loading manual.

In the bottom plating in a ballast tank, the thickness diminution may be due to, general corrosion , pitting corrosion, grooving corrosion or erosion corrosion. Where would you look for pitting corrosion?

- **On horizontal flanges.**
- In way of cut-outs for drainage.
- Under suction heads for ballast line.

It is very important to be familiar with the technical terminology when dealing with container ships. Is a floor a part of the:

- **Double bottom structure**
- Deck structure
- Ship

Maintenance of loose stowing and securing equipment is very important for the safety of the cargo during transportation overseas, such as (please indicate which is incorrect).

- **Cell guides.**
- Lashing bars.
- Manual twistlocks.
- Semi-automatic twistlocks.
- Sliding base sockets.
- Turn buckles.

On a container ship, which part of the ship represents the web of hull girder?

- **The longitudinal bulkheads and the ship sides.**
- The longitudinal bulkheads.
- The transverse bulkheads.

Pitting corrosion is most commonly found on horizontal surface in:

- **Ballast tanks.**
- Cargo holds.
- Heavy fuel oil tanks.

Regarding inspection of the cargo holds, where is the least likely location for finding cracks?

- **At the cell guides.**
- At corner of stringer decks.
- At hatch corner.

Reporting in PIMS of the results from the inspection is very important Checklist and Narrative reports. Consider you have carried out an inspection in a ballast tank. The tank was found in order except for an indention in the ship side which has caused a slight buckling of two side longitudinals. How would you report this?

- **Fill in the checklist, take some photo and make a narrative report where you describe the indent and location.**
- Fill in the checklist and take some photos.
- Fill in the checklist.

Similar to Classification Society, the Owner also has acceptance criteria to be used when deficiencies are found. These acceptance criteria should normally be somewhat stricter than the class. Where do you find Owner's acceptance criteria?

- **In the PIMS manual on board.**
- In the ISM system, Reg. 10.
- In the working guidance by chief officer.

Some areas of minimal attention with respect to coating, during inspection of ballast tanks are:

- **On the web of stiffeners.**
- Edge part of stiffeners.
- Inside edge of cut-outs.
- Welding beads.

The class society has acceptance criteria for different types of deficiencies, such as:
Which two alternatives are not correct?

- **Buckling.**
- **Cracks.**
- Coating breakdown.
- Corrosion.
- Indentations.

The coating condition in the ballast tanks may be rated as: Indicate which of these coating conditions require annual follow up of the ballast tank in question by the class surveyor.

- **Poor.**
- Fair.
- Good.

The fore ship may suffer damage due to impact from the sea. Which structural element will then suffer the most significant damage?

- **The webframes and girders.**
- The shell plating.
- The stiffeners from the shell plating.

The function of a loaded beam is to transfer loads towards the end supports (in shear). Which part of beam has this function?

- **The web.**
- Both web and flange.
- The flange.

The hull girder of a container ship is as other ships subject to tension and compression due to the loads from the lightweight and cargo, and the counteracting buoyancy loads. Since the container ship normally operates in a hogging loading condition, where would you expect tension stresses in the hull girder?

- **In the deck area**
- In the bottom area

The pressure testing to prove strength and tightness is not required for which of the following items:

- **Cargo holds.**
- Piping systems.
- Tank boundaries.

What is assumed by the class for retention of class?

- **Owner/manager to ensure adequate maintenance, handling and manning of the ship.**
- Owner to ensure adequate maintenance.
- Owner to ensure that the vessel is presented for surveys.
- Owner/manager to ensure adequate handling and manning of the ship.

What is the Integrated Survey Programme, ISP?

- **A hull inspection and maintenance programme for container ships.**
- A maintenance programme for cargo ships.
- A maintenance system for machinery on container ships.
- A survey program, which is integrated in the maintenance system onboard.

What is the classification in the SiO phase?

- **To verify that the standard of the ship is acceptable through a system periodical surveys.**
- To make sure that the vessel is well maintained.
- To verify that the standard of ship in operation (SiO) acceptable after conversion.
- Verify that the standard of the ship is acceptable after damage repairs.

What is the condition for the 5-year validity of the class Certificate?

- **That annual surveys are carried out with satisfactory result.**
- That all condition of class (cc) are carried out.
- That survey is carried out every 5 years with satisfactory result.
- That surveys are carried out every 2,5 years and with satisfactory result.

When doing inspections of the upper deck area, where would you not find cracks immediately?

- **At the middle of hatch end coaming.**
- At the hatch coaming stays.
- At the hatch end coaming brackets.
- At the lashing bridges connection to the deck.

When looking for cracks in the ship structure like for instance the bottom structure, would you then first focus on:

- **At the connection between stiffeners and girders.**
- At the connection between plate and stiffeners .
- The middle of the plate.

Where would you look for cracks in the side ballast tanks in the first place?

- **At ship side longitudinals connection to the web frames below waterline.**
- At the longitudinal bulkhead longitudinal connection to the web frames.
- At the tank top longitudinals connection to the web frames.

Which is outside the scope of Planned Inspection and Maintenance System (PIMS)?

- **Dry docking/bottom survey.**
- Piping in ballast tanks.
- Survey of ballast tanks with Fair coating.
- Survey of container fittings.

Which is required from the crew on board for practising of ISP?

- **Carry out inspections and report in accordance with the maintenance system, which has been accepted by the class.**
- Carry out inspections and submit report to the class.
- Carry out inspections in accordance with the maintenance system on board, which has been accepted by the class.
- Carry out inspections when deemed necessary.

Which of the following items is not mandatory in the PIMS:

- **Safety Equipment.**
- Cargo holds.
- Deck and hatches.
- Piping system outside machinery space.
- Tanks.

Which of the following precautions while pressure testing of a ballast tank is the most important?

- **Hatch covers and/or manhole covers should be kept open during pressure testing.**
- Filling should be to the top of tank.
- Venting done through the air pipes.

0118 – Steering Gear Construction and Maintenance

Version 5.0 (1813)

SOLAS requires the fitting of visible and audible low level alarms for every steering gear hydraulic fluid reservoir. Which one of the following identifies the locations where it states that these alarms must be fitted?

- **On the bridge and in the machinery space, where they can be seen**
- In the steering compartment and in the engine control room, where they can be seen
- On the bridge and in the steering compartment, where they can be seen
- On the bridge, in the steering compartment and in the engine control room, where they can be seen

Select from the following the one feature that the steering gear of all ships greater than 70,000 tonnes displacement must have that not every other ship requires according to SOLAS?

- **Two or more identical power units**
- A main and auxiliary steering gear
- A means of quickly restoring steering in the event of a single hydraulic failure in the hydraulic system.
- Duplication of all components up to the rudderstock

Select from the following the type of rudder that is most commonly used on ferries and smaller ships?

- **Spade rudder, because of its good manoeuvrability**
- All three types are equally used
- Flap rudder, because of its very good manoeuvrability
- Semi-spade rudder, because of the reduced steering gear torque required

Select the one statement that best completes the statement "The maximum required working torque of a steering gear is determined by . . .

- **. . . the torque necessary to turn the rudder to the maximum required angle at the ship's full speed and maximum draft"**
- . . . the effective torque at the relief valve opening pressure"
- . . . the friction at the bearings and other moving parts"
- . . . the resistance it meets in service, so there is no limit"

Select the phrase which most accurately completes the following statement. "According to the SOLAS rules the rudder speed should be at least 35-0-30 degrees in a maximum of 28 seconds for . . .

- **. . . the main steering gear of all ships, other than those with special requirements, such as those with non-conventional steering arrangements**
- . . . only cargo ships above 10,000 DWT or passenger ships carrying more than 50 passengers
- . . . only oil and chemical tankers and gas carriers
- . . . the main and auxiliary steering gear of all ships, other than those with special requirements, such as those with non-conventional steering arrangements

What is the main purpose of a rapson slide when fitted to a ram type steering gear?

- **The purpose of the rapson slide is to change the linear motion of the rams into circular motion of the tiller**
- The purpose of the rapson slide is to carry the weight of the rudder
- The purpose of the rapson slide is to keep the rams correctly aligned to prevent sideways force on the ram seals
- The purpose of the rapson slide is to limit the maximum movement of the rudder

When a heavy sea strikes the rudder, which one of these statements best describes what will happen?

- **If the force is large enough, the hydraulic pressure in the steering gear will be high enough to operate the relief valve, and the rudder will move**
- The rudder is free to move, but the auto-pilot will bring the ship back on course
- The rudder is hydraulically locked in position and will not move
- The shape and lightweight design of the rudder means that wave forces have no effect

Which of the following best describes how the tiller or a rotary vane steering gear is connected to the rudderstock?

- **The tiller or rotary vane can be connected to the rudder stock by being hydraulically forced onto a taper, by keys, bolts or a combination of these methods**
- The tiller or rotary vane is usually connected to the rudder stock by a nut which pushes it against a stepped shaft
- The tiller or rotary vane must always be connected to the rudder stock by being forced onto a taper
- The tiller or rotary vane must always be keyed to the rudder stock and held in place by a nut

Which of the following is the amount of hydraulic oil that must be held in the fixed storage tank connected by piping to the steering gear according to SOLAS?

- **Enough to recharge one power actuating system**
- Enough to make up expected leakage for the voyage
- Enough to recharge all power actuating systems
- Enough to recharge at least two power actuating systems

Which of the following statements best describes the typical maintenance program for changing the hydraulic oil used in steering gear?

- **Even small solid particles will cause wear, whilst moisture will cause corrosion so the oil needs to be changed regularly**
- Provided the oil remains incompressible oil changes are not required
- Provided the oil viscosity remains within acceptable limits, the oil does not need changing
- Steering gear power actuators are very robust and oil should never need to be changed

Which one of the following are the alarms that should be fitted to an electro-hydraulic steering gear according to the SOLAS rules?

- **Power failure for the motors, power failure for the control system, failure of any phase of a three phase electrical supply, electrical overload, and low level alarms for each hydraulic fluid reservoir.**
- Power failure for the motors, power failure for the control system, and low level alarms for each hydraulic fluid reservoir
- Power failure for the motors, power failure for the control system, electrical short circuit, electrical overload, and low level alarms for each hydraulic fluid reservoir
- Power failure for the motors, power failure for the control system, failure of any phase of a three phase electrical supply, electrical short circuit, and low level alarms for each hydraulic fluid reservoir

Which one of the following are the clearances on a ram type steering gear that need to be checked at dry-dock?

- **Jumping clearance and carrier bearing wear-down clearance**
- Internal carrier bearing clearances
- Ram to cylinder clearances
- Swivel block to guide bar clearance

Which one of the following best describes where the forces on the rudder arrangement come from?

- **Propeller, waves, weight of components and side forces**
- Propeller and drag
- Side forces, propeller and waves
- Weight of ship, propeller, waves and side force

Which one of the following factors do you think will NOT affect the lifetime of the bearings in a rotary vane steering gear actuator?

- **The relief valve pressure setting**
- Cleanliness of the hydraulic oil
- Operation in heavy seas
- The alignment of rudder stock with the steering gear installation

Which one of the following identifies where the changeover to local steering is carried out?

- **The steering compartment**
- The bridge
- The engine control room or the bridge
- The steering compartment, but it must be acknowledged by the bridge before it can take effect

Which one of the following is NOT said to be an advantage of ram type steering gears compared to rotary vane type steering gears?

- **Ram type steering gears require lower operating pressures**
- On ram type steering gears, the torque increases with rudder angle, whereas for rotary vane types it is constant
- Ram type steering gears are easier to maintain
- Ram type steering gears are of simple construction

Which one of the following is a special steering gear requirement that passenger ships must meet that not all other ships require according to SOLAS?

- **They, must be able to move the rudder from 35° on either side to 30° on the other side in not more than 28 seconds with one power unit out of operation**
- For a single failure, steering capability must be regained within 45 seconds
- There must be oil leakage detection.
- They must have a main and auxiliary steering gear

Which one of the following is most likely to cause the hydraulic lock alarm to operate on a steering gear system?

- **A failure in the control system such as a directional valve sticking open**
- A failure of one of the hydraulic pumps
- Failing to purge the system of air
- Having both hydraulic pumps running at the same time

Which one of the following is most likely to cause the safety valve on a steering gear hydraulic system to operate?

- **A heavy sea has hit the rudder**
- An attempt has been made to put too much helm on, too quickly
- It has been opened as part of the normal operation of the steering gear
- The output pressure from the hydraulic pumps is too high

Which one of the following is the main advantage of running two steering gear motors?

- **The speed of rudder movement will be doubled**
- It reduces the wear on individual steering gear pumps
- The available rudder torque will be doubled
- There is less chance of overheating the hydraulic fluid in hot weather

Which one of the following is the main purpose of the relief valve on a hydraulic steering gear?

- **To protect the actuator against too high an oil pressure**
- To protect against leakage between the pressure chambers
- To protect the operator
- To regulate the return flow of oil from the steering gear

Which one of the following is the most important factor in reducing wear in a hydraulic steering gear system?

- **Ensuring the cleanliness of the hydraulic oil is maintained**
- Limiting the maximum helm angle when using the auto-pilot
- Using both steering gear motors as often as possible
- Using only one steering gear motor as often as possible

Which one of the following options best describes the main things that affect the torque needed to turn the rudder?

- **Size and type of rudder and the ship's speed**
- Size and shape of the hull
- The maximum displacement of the ship and the prevailing weather
- The type of steering gear fitted

Which one of the following statements best describes the difference in SOLAS requirements for the steering gears of single rudder tankers above and below 100,000 tonnes?

- **Below 100,000 tonnes, non duplicated hydraulic rudder actuators may be installed, provided they meet IMO guidelines. Above 100,000 tonnes, there must always be full duplication.**
- Below 100,000 tonnes, an automatic change-over for oil leakage is optional. Above 100,000 tonnes, an automatic change-over for oil leakage is always required.
- Below 100,000 tonnes, oil leakage detection is optional. Above 100,000 tonnes, oil leakage detection is always required.
- Below 100,000 tonnes, the power units can be different sizes. Above 100,000 tonnes, two or more identical power units are required.

Which one of the following statements best describes the importance of the purity of the hydraulic oil used in steering gear?

- **Even small solid particles will cause wear, whilst moisture will cause corrosion**
- Provided the oil remains incompressible, the steering gear will continue to work correctly
- Provided the oil viscosity remains within acceptable limits, the steering gear will continue to work correctly
- Steering gear power actuators are very robust and therefore a degree of oil contamination can be tolerated

Which one of the following statements best describes the pressure to which the steering gear relief valves are set?

- **Slightly lower than design pressure**
- At 25% above the design pressure
- At the maximum working pressure
- Slightly lower than the maximum working pressure

Which one of the following statements best describes the purpose of the upper radial bearing in a rotary vane steering gear, and how it is lubricated?

- **It takes up the radial loads generated by horizontal forces on the rudder and is lubricated by the steering gear system oil**
- It takes up the radial loads generated by horizontal forces on the rudder and is separately lubricated by oil or grease
- It takes up the radial loads generated by the weight of the rudder and is lubricated by the steering gear system oil
- It takes up the radial loads generated by the weight of the rudder and is separately lubricated by oil or grease

Which one of the following statements best describes where SOLAS says that you must be able to disconnect the bridge control of the steering gear?

- **From within the steering gear compartment**
- From the bridge
- From the emergency headquarters and the bridge
- From the engine control room

Which one of the following statements best describes why SOLAS requires hand rails, gratings and non-slip surfaces to be fitted in the steering compartment?

- **So that there is safe access to the steering gear immediately for repairs, even in heavy weather.**
- Because it is usual for steering gears to leak oil
- To ensure that the ship owner will be not be liable for any injuries due to slips.
- To save time when the steering gear is opened up for inspection in dry dock

Which one of the following statements correctly describes the main purpose of the rudder carrier bearing?

- **To support the weight of the rudder and any downward forces on the rudder caused by waves**
- To align the rudderstock and take up any radial forces caused by waves
- To support the weight of the rudder when the steering gear is being maintained in dry-dock
- To take up any forces caused by the weight of the rudder, and upwards or downward forces on the rudder caused by waves

Which one of these statements about power supplies to electric and electro-hydraulic system for steering gears is correct?

- **The steering gear motors are fitted with alarms for overload, but to avoid loss of steering will only trip in the event of an electrical short circuit**
- Because the second steering gear motor will start automatically, the alarms and trips fitted are decided by the shipbuilder
- The steering gear motors are fitted with alarms for overload, loss of one phase from a 3-phase supply and electrical short circuit, but to avoid loss of steering will not trip in the event of any of these conditions
- To protect the steering gear motors, they are fitted with trips for overload, loss of one phase from a 3-phase supply and electrical short circuit

Which one of these statements about rudder carrier bearings for rotary vane steering gears is most accurate?

- **Rotary vane steering gears may be fitted with external rudder carrier bearings, but in many designs the rudder carrier bearing is part of the rotary vane unit**
- Rotary vane steering gears are always fitted with external rudder carrier bearings
- Rotary vane steering gears cannot be fitted with external rudder carrier bearings
- Rotary vane steering gears do not need rudder carrier bearings

Which one of these statements about rudder carrier bearings is true?

- **A separate rudder carrier bearing is only required on ram type steering gears**
- A rudder carrier bearing is not required on rotary vane type steering gears
- A rudder carrier bearing is only required on ram type steering gears
- A rudder carrier bearing is only required on rotary vane type steering gears

Which one of these statements about the force developed by a hydraulic system is true?

- **Force equals pressure times area**
- Force equals pressure divided by area
- Force equals pressure plus area
- Pressure equals force times area

Which one of these statements about types of steering gear is true?

- **Higher pressures are needed for the ram type**
- A greater angle of possible rudder movement is available with ram type steering gears
- Higher pressures are needed for the rotary vane type
- The ram type takes up less space than the rotary vane type

Which one of these statements best describes the main effect of using two steering gear hydraulic pumps instead of one?

- **The speed of rudder movement is greatly increased**
- The force on the rudder is greatly increased
- The speed of rudder movement and the force on the rudder are greatly increased
- There is little effect on the speed of movement or the force on the rudder, but reliability is increased

0121 – Ship Security Officer Training System

Version 3.0 (296)

A Ship Security Assessment is integral to the creation of your ship's security plan.

- **True.**
- False.

A contractor is hired to install new navigation equipment onboard your ship while it's berthed. For a period of time he's left unsupervised and photographs schematics of the ship that he finds rolled up and stored in the corner of a nearby office. Later, from home, he hacks into the network and prints off information about the ship's security procedures. Which of these information security measures would have prevented his unauthorized access?

- **Secure area, passwords, a firewall and a secure network.**
- Firewall, protective markings, vetting and a secure network.
- Protective markings, reference checks, and passwords.
- Secure area, passwords, a firewall and protective markings.

All training programs have a beginning, a middle and an end.

- **True.**
- False.

Assign men to search men and women to search women unless a device such as a metal detector is used.

- **True.**
- False.

Brainstorming potential motives for security incidents against your ship helps you to create possible threat scenarios.

- **True.**
- False.

During a profiling interview, the service man you are questioning starts to act suspiciously. He is:

- **Nervous, sweaty and trying to avoid eye contact.**
- Friendly and cooperative.
- Professional, courteous and forthcoming.
- Talkative and good-humored.

Eavesdropping and phone taps are examples of which threat to information security?

- **Espionage.**
- Sabotage.
- Subversion.
- Terrorism.

For a search to be effective it must be:

- **Centrally controlled.**
- Conducted by personnel with limited knowledge of the ship's layout.
- Inclusive of all personnel.
- Organized haphazardly.

How many different learning styles are there?

- **4.**
- 1.
- 2.
- 3.

If drugs are discovered onboard your ship:

- **Ensure the witness to the discovery signs your incident report.**
- Disembark crew and passengers as quickly as possible at the next port of call so the authorities can conduct their investigation.
- Notify the authorities after you arrive at the next port of call.
- Write a report a few days after the event and describe everything that occurred.

If drugs or suspected drugs are found onboard your ship, follow the five C's. Confirm, Clear, Cordon, Control and:

- **Check.**
- Change.
- Cheer.
- Chuck.

In which of these circumstances can your ship request a DoS?

- **Your ship is conducting activities with a port or ship that is not required to implement an approved security plan.**
- There is a heightened security risk for your ship and a port facility.
- Your ship has added a new port to its list of ports of call.
- Your ship is operating at a lower security level than the ship or port it is interfacing with.

Information about your ship's security arrangements and procedures is stored electronically. Which of these measures will help safeguard it from potential threats?

- **Passwords.**
- Encoded email messages.
- Protective markings.
- Work history verification.

One of the categories of information that must be included in your SSP is:

- **Security arrangements.**
- Ship security survey.
- Threat scenarios.
- Weaknesses in security measures.

Only search vehicle compartments where you suspect objects may be hidden.

- **False.**
- True.

Part way through a profiling interview with a supplier, the package the individual is carrying arouses your suspicion. What do you do?

- **Discreetly inform someone of your suspicions so he or she can get assistance.**
- Call for help on the radio.
- Confront the individual and demand that he open the package.
- Take the package and open it.

Records relating to breaches of security and changes in security level must be secured against unauthorized access and available for review by contracting governments.

- **True.**
- False.

Searches are often triggered by:

- **An increase in security level by the Flag State.**
- Lost baggage.
- News stories of stowaways.
- The receipt of a shipment of damaged stores.

Select from the options given the one piece of information which is considered the most sensitive from a ship's security aspect?

- **Voyage itinerary with details of departure and arrival times.**
- HR policies.
- Planned maintenance schedule.
- Training records.

Ship Security Plans all look the same.

- **False.**
- True.

Successful training programs build on learners past experience, are delivered with enthusiasm, present relevant information in a logical sequence and invite participation through varied activities.

- **True.**
- False.

The Declaration of Security:

- **Addresses the security requirements shared between ships or between a port facility and a ship.**
- Details a ship's security measures.
- Identifies the security responsibilities of shipboard personnel.
- States the reporting procedures to government contact points.

The DoS addresses the responsibility for the security of the water around the ship and the verification of increased threat levels.

- **True.**
- False.

The International Maritime Bureau Piracy Reporting Centre attributes the increased numbers of hijackings to:

- **The greater involvement in piracy of well-organized and armed crime networks.**
- Easy access to military weapons.
- Higher crime rates around the world.
- More crew involvement.

The best way to ensure you get the equipment you need is to:

- **Determine your needs and do some research.**
- Buy the most expensive equipment on the market.
- Buy the newest models.
- Buy what everyone else is buying.

The best way to identify an IED is to:

- **Recognize its components.**
- Measure the size of the object.
- Shake the object.
- Weight the object.

The best way to prevent stowaways from boarding your ship is to:

- **Seal spaces that are not in use while in port, and perform a search of the ship before leaving.**
- Conduct a Nominated Officers search.
- Conduct routine, but irregular searches of the ship.
- Search the ship when you arrive at port and again just after leaving.

The first step in completing an SSA is to:

- **Create a list of potential motives for security incidents against your ship.**
- Develop an onboard security survey checklist.
- Identify the key shipboard operations, systems, areas and personnel that are critical to protect.
- List the existing security measures.

The middle portion of a training program takes the most time to plan. Which of these items will assist you in planning and delivering this portion of the program?

- **Session plans.**
- A description of the content to be covered.
- A list of the page references to regulations and other documents that will be referred to in the program.
- An equipment list.

The most effective way to combat the threat of drug smuggling is to:

- **Combining routine, but irregular searches of the ship with spontaneous targeted searches.**
- Organize crew into pairs and conduct weekly searches of the ship.
- Perform routine, but irregular searches using teams of two or more personnel from the same department.
- Perform spontaneous targeted searches using teams of two or more personnel from the same department.

The person who takes a bomb threat call has an opportunity to extract information about the bomb from the caller.

- **True.**
- False.

The process of learning involves doing, reflecting, linking and:

- **Planning.**
- Acting.
- Contemplating.
- Explaining.

The purpose of profiling is to:

- **Get beneath the outer shell of an individual to obtain a more complete picture.**
- Categorize people based on their nationality and ethnicity.
- Identify different personality types.
- Make judgements about people based on their appearance.

This device can detect a package of drugs inside a large delivery of ship's stores. What is it?

- **Pallet scanner.**
- Access control system.
- Security container.
- Vapour detector.

This is an example of a good learning objective: Given a mock-up of an explosive device, identify the components of the device with 100% accuracy.

- **True.**
- False.

Using x-ray equipment, perform a baggage search to locate a small hunting knife. Which instructional technique is appropriate for this objective?

- **Demonstration.**
- Brainstorming.
- Presentation.
- Questioning.

What are the three tasks of a profiler?

- **Check documentation, ask questions, observe body language and behaviour.**
- Check documentation, conduct physical searches, act intimidating.
- Check documentation, conduct physical searches, ask questions.
- Conduct physical searches, observe body language and behaviour, act intimidating.

What do crew, visitors and contractors all have in common?

- **They all have opportunity to smuggle drugs.**
- They're all employees of the ship.
- They're all onboard the ship for the duration of its voyage.
- They're all trained in ship emergency procedures.

What is an IED?

- **A homemade bomb.**
- A type of gun.
- An explosive.
- An incendiary device.

What is the purpose of the Ship Security Plan?

- **Protect your ship from risks posed by security threats or incidents.**
- Detail the duties of shipboard personnel assigned security responsibilities.
- Outline specific measures for your ship to move from security level 1 to 2 and from 2 to 3.
- Outline the organizational structure for the ship.

What measures should be taken when handling mail?

- **X-ray or scan parcels before opening them.**
- Open any mail that looks suspicious.
- Tear open letters with your hands.
- Wear rubber gloves and protective clothing when opening suspicious looking mail.

What type of bomb search should you conduct to avoid panic when the credibility of the threat is in doubt and you don't want to disrupt ship business?

- **Nominated officers search.**
- Crew search.
- External search team.
- Known hiding spot search.

When handling drugs:

- **Wear skin protection and a facemask.**
- Briefly inhale powders, fumes or vapours to verify the substance is a drug.
- Shut off the ventilation in the room where the drugs are being stored to ensure any vapours or fumes do not spread to other parts of the ship.
- Test out the potency of the drug.

Which information should be included in your search plan?

- **Areas to the searched and personnel to be involved in the search.**
- Areas to be searched.
- Known hiding spots to be searched.
- Personnel to be involved in the search.

Which of these actions should crew take if a suspicious object that may be a bomb is located during a search?

- **Confirm with their search partner that the object found is suspicious.**
- Cover the suspicious object with a box or blanket.
- Place the object in a desk drawer or file cabinet.
- Throw the object overboard.

Which of these actions should you take if your ship is successfully boarded by armed pirates?

- **Assure your captors that you're not planning an attack to overthrow them.**
- Fight back.
- Keep quiet and ignore any questions you're asked.
- Scream in fear and refuse to cooperate.

Which of these are indications that a parcel is suspicious?

- **The parcel is addressed to no one in particular, arrives unexpectedly and seems heavy for its size.**
- The parcel came from Joe in the safety department and arrived quicker than expected.
- The parcel is wrapped in clean, brown paper and is addressed to a crewmember from his wife.
- The parcel smells like chocolate.

Which of these are indicators that an individual may be a drug smuggler?

- **The individual wears bulky or out of season clothing.**
- The individual is disinterested in the ship's cargo.
- The individual makes small local purchases.
- The individual spends his or her free time hanging out with others.

Which of these measures would you implement to prevent drugs from being smuggled onboard your ship while it's berthed?

- **Maintain restricted areas.**
- Check a portion of the bags and packages brought onboard.
- Eliminate fore and aft deck watch at night.
- Search some of the packages, spares and stores received.

Which of these precautionary measures can reduce the threat of piracy, if implemented?

- **Sail at full speed.**
- Plan to arrive at port at night.
- Stay at least 15nm away from the shore.
- Turn off all of the ship's lights at night.

Which of these procedures should crew follow when conducting a search?

- **Stand quietly in various spots around the space, listening for unusual sounds like the tick tock of a clock mechanism.**
- Conduct random or spot checks of logical threat areas.
- Search the perimeter of the space up to a height of 5 feet.
- Wait in the area they've finished searching until they're contacted for a status update.

Which of these statements about drug smugglers modes of operating is true?

- **Drugs smuggled by an organized conspiracy are usually concealed in a primary ship system such as the engine room or in a tank, void or compartment.**
- An organized conspiracy usually smuggles a small amount of drugs.
- Drugs hidden by individual entrepreneurs are usually difficult to detect.
- Individual entrepreneurs usually smuggle large quantities of drugs.

Which of these statements about drug smuggling is true?

- **The risks to ships are not restricted to specific areas or trading routes.**
- Drug smuggling is only a problem in certain ports, so only ships sailing in those ports need to implement preventative measures.
- Drugs are difficult to conceal onboard a ship.
- The preventative measures you incorporate into your ship's security plan should be exhaustive, regardless of the level of threat identified by your ship's security assessment.

Which of these statements about profiling is correct?

- **Verify that answers to questions match up with what's already known about the person being questioned.**
- Check the ID of all visitors and some crew prior to boarding.
- Pay less attention to body language and behaviour and more attention to answers to questions.
- Use only open-ended questions.

Which of these tasks is performed as part of the SSA?

- **Assess the likelihood and potential consequences of security incidents.**
- Assign security duties to ship personnel.
- Implement measures to address weaknesses in ship security.
- Train shipboard personnel in their security duties.

Which of these weapons are widely available and commonly used by criminals and terrorists to further their aims?

- **Gun.**
- Incendiary device.
- Liquid explosive.
- Vehicle borne device.

Which of this information must be included in a piracy attack alert?

- **Your ship's name and call sign.**
- The number of crew onboard.
- The number of pirates/highjackers.
- The type of weapons being carried by the pirates/highjackers.

Which statement about IED's is true?

- **All IED's have four common components.**
- All IED's look alike.
- IED's are difficult to manufacture.
- IED's don't cause serious damage.

Which statement about the use of profiling onboard a ship is true?

- **A random selection process must be established.**
- Detection equipment can be used in place of profiling.
- Only visitors can be profiled.
- Profilers need at least half an hour to gather the information they need.

Which type of equipment can be used to detect explosives?

- **Particulate detector.**
- CCTV.
- Metal detector.
- Water cannon.

Which type of profiling is used to obtain information about rival companies and their employees?

- **Industrial.**
- Commercial.
- Criminal.
- General.

Who is responsible for completing the DoS on behalf of the ship?

- **Ship Security Officer.**
- Chief Engineer.
- Chief Officer.
- Company Security Officer .

Who is responsible for ensuring that your ship's security plan meets the requirements of the ISPS Code?

- **Company Security Officer.**
- Flag State Administration.
- Recognized Security Organization.
- Ship Security Officer.

Who is responsible for ensuring your ship completes a security assessment?

- **Company Security Officer.**
- Flag State Administration.
- Recognized Security Organization.
- Ship Security Officer.

You can reduce the need for security guards in certain areas by installing:

- **Anti-intruder devices.**
- Metal detectors.
- Vapour detectors.
- Water cannons.

Your ship is berthed at a port that is frequently used for drug smuggling. Which of these measures is it appropriate to implement?

- **Limit the number of access points to the ship.**
- Check the identity of everyone boarding the ship.
- Secure unattended spaces that visitors have access to.
- Suspend cargo-handling operations.

Your ship security plan must include procedures for responding to security threats, auditing security activities and interfacing with the port facility.

- **True.**
- False.

You've just completed a training drill. Which of these guidelines should you follow when giving feedback?

- **Ask to give feedback or wait to be asked.**
- Give lots of information and examples.
- Listen to your hunches and give appropriate feedback.
- Take a one-hour break for lunch and give feedback after you resume.

0122 – ISO 14001 Environmental Management

Version 6.0 (1651)

Carbon di-oxide will be produced by burning hydrocarbon based fuels. What is the main environmental problem that this causes? Select the ONE best answer.

- **Greenhouse effect (global warming)**
- Acid rain
- Damage to plant life

Do all companies have to follow ISO 14001? Select the ONE correct answer.

- **No, use of the ISO 14001 standard is on a voluntary basis.**
- No, but all shipping companies must use ISO 14001.
- Yes, if their activities might cause pollution.

ISO 14001 uses the term 'compliance obligations'. What does this mean? Select the ONE best answer.

- **Legal requirements that an organization has to comply with and other requirements that an organization has to, or chooses to, comply with.**
- The IMO requirements that an organization has to comply with.
- The legal requirements that an organization has to comply with.
- The MARPOL requirements that an organization has to comply with.

ISO 14001 uses the term 'documented information'. What does this mean? Select the ONE best answer.

- **Any information required to be controlled and maintained by an organization and the medium on which it is contained.**
- Digital information required to be controlled and maintained by an organization.
- Written information required to be controlled and maintained by an organization.

ISO 14001 uses the term 'environmental impact'. What does this mean? Select the ONE best answer.

- **Change to the environment, whether good or bad, resulting from the company's activities.**
- Adverse change to the environment, resulting from the company's activities.
- Positive changes to the environment, resulting from the company's activities.

ISO 14001 uses the term 'life-cycle'. What does this mean? Select the ONE best answer.

- **All the stages of a product or service from getting the raw materials to their final disposal.**
- Any part of a product or service that can be re-cycled.
- The active working stage of the product or service

If engine room bilge water cannot be processed through an oily water separator, what should be done with it? Select ANY answers that you think apply.

- **It should be discharged to a shore reception facility.**
- It can be discharged overboard, if not in a special area, en-route and more than 50 miles from the nearest land or ice shelf.
- It should be allowed to settle in a storage tank, which can then be lowered by pumping out overboard, keeping a watch for any oil in the overboard discharge.

In sulphur emission control areas, we must burn low sulphur fuels. What is the reason for this? Select the ONE best answer.

- **To reduce the production of oxides of sulphur, which cause acid rain.**
- To prevent the formation of smoke or coloured exhaust.
- To reduce the formation of soot and particulate matter, which are bad for health.

In the cycle of continuous improvement used in ISO14001, what does 'PDCA' represent?

- **Plan, Do, Check and Act**
- Plan, Do, Correct and Adjust

Inside a special area, and if more than 12 nautical miles from land and en-route, what types of garbage can be discharged overboard? Select ANY answers that you think apply

- **Ground food waste**
- Food waste, cardboard, paper and similar bio-degradable materials.
- Food waste, either ground or not ground.
- Food waste, if ground and placed in plastic bags.

Is it true that a company having ISO 14001 certification must have a documented environmental policy?

- **Yes**
- No

Outside a special area, and if more than 12 nautical miles from land and en-route, what types of garbage can be discharged overboard? Select ANY answers that you think apply

- **Food waste, either ground or not ground.**
- Food waste, cardboard, paper and similar bio-degradable materials.
- Food waste, cardboard, paper, wood, and incinerator ashes.
- Food waste, if ground and placed in plastic bags.

Oxides of Sulphur will be produced by burning fuels containing sulphur. What is the main environmental problem that this causes? Select the ONE best answer.

- **Acid rain**
- Damage to the Ozone layer
- Greenhouse effect (global warming)

What do the initials 'ISO' stand for?

- **International Organization for Standardization**
- International Standards Office

What does ISO 14001 include in the term 'the environment'? Select the ONE best answer.

- **All plant and animal life, air, water, land, natural resources, humans and their interrelationships.**
- All plant and animal life, air, water, and land which could be directly affected by shipping.
- Marine plant and animal life, air, water, and land which could be directly affected by shipping.

What is ISO 14001? Select the ONE best answer.

- **An International standard for environmental management.**
- An IMO regulation about pollution from ships.
- Standards and guidelines for vessels and ship managing companies about safety and protection of the environment.
- The International Ship Operator's list of approved garbage reception facilities in ports.

What is the best strategy for dealing with pollution? Select the ONE best answer.

- **Prevention**
- Dilution
- Dispersal
- Local clean up

Where can we dispose of plastic?

- **At a shore reception facility only.**
- At a shore facility and at sea only outside special areas.
- At a shore facility and at sea when more than 12 miles from land.

Why is it important that refrigerant gasses are not released to the atmosphere? Select the ONE best answer.

- **Refrigerant gasses, particularly CFCs and HCFCs can cause depletion of the ozone layer.**
- They are flammable.
- They are very expensive.

Why should ship's sewage be collected in a storage or treatment tank? Select the ONE best answer.

- **It is a threat to life in coastal areas and in the sea, and needs to be treated or have its discharge controlled.**
- It looks unpleasant in harbour areas or if washed up on the beach.
- There are local regulations in place in a few ports and harbours.

0123 – Risk assessment and management

Version 4.0 (2321)

Check the boxes on the factors that can be categorized as human errors

- **Carelessness.**
- **Ignorance to the hazards of a job**
- **Lack of understanding or skill in how to work safely around hazards**
- **Recklessness**
- **Task vs. safety focus**
- **"It won't happen to me" attitude**
- Faulty equipment
- Weather conditions

Check which of the following actions that can be used to reduce the risk in a job

- **Change the frequency of performing the job**
- **Change the physical conditions**
- **Figure out a different way to do the job**
- **Identify better tools to do the job**
- **Modify the job procedure**
- **Wear personal protective equipment**
- Let someone else do the job

How many percent of all accidents can be related to human error?

- **80%**
- 50%
- 70%
- 90%

If a risk is not acceptable you can

- **Reduce the likelihood and/or consequence**
- Continue with your work but be very careful
- Get more people to help with the job
- Leave the job to someone else

Risk estimation is?

- **Rating likelihood and consequence**
- A system to see the consequences
- Calculating how many times an accident has occurred
- Identifying accidents

What do you do with a "Risk Profile Matrix"?

- **Check if a risk is acceptable**
- Check how many times you can do a dangerous job without accidents
- Reduce the consequences of an accident
- Reduce the risks of an accident

What is a definition of " likelihood"?

- **How often a risk can lead to an accident**
- How much I like a risk
- How often an accident can lead to a risk
- How often I will do a specific work

What is a definition of "consequence"?

- **The outcome of an accident**
- How likely the consequences are
- How many times an accident occur
- The reduction of likelihood

What is the first step in a risk assessment?

- **Risk identification**
- Consequence estimation
- Likelihood estimation
- Risk evaluation

What should you do before performing any job onboard?

- **Follow procedures and apply informal risk management**
- Ask your superior
- Check how many persons you can get to help you
- You do not have to follow procedures if you have done a thorough risk analysis

0124 – Greasing Mooring Winches and Windlasses

Version 1.2 (0)

Clogged grease channels are indicated by:

- **Grease that only comes out on one side of the bearing.**
- A watery consistency of grease around the bearing.
- Dirty grease around the bearing.
- Excess grease spilling out of the grease nipple.

Failing to establish a clear signal for rotating the winch during greasing can result in:

- **Loss of a finger**
- Back injury
- First degree burn
- Sprained ankle

Greasing must be performed using a pneumatic grease gun.

- **False**
- True

How often should enclosed bearings be greased?

- **Monthly**
- Daily
- Every two months
- Weekly

If the winch is not greased to keep it in good working order incidents can occur that:

- **Cost thousands of dollars in insurance claims.**
- Increase revenue generation.
- Require shipping companies to design self-greasing mooring winches.
- Require shipping companies to purchase safer mooring winches.

In addition to coveralls, goggles, and a helmet, which type of personal protective equipment should be worn when greasing?

- **Leather-palm working gloves**
- Face shield
- Respirator
- Safety harness

Which of these actions should be performed to ensure greasing is effective?

- **Use kerosene to flush out clogged greasing channels.**
- Apply grease for five minutes.
- Keep the winch secure, so it does not move.
- Monitor the bearing to ensure grease is only coming out on one side of the bearing.

Why is it important to immediately clean up any spilled grease on the winch platform?

- **Reduce potential for slipping or falling from the platform.**
- Keep the deck clean for shore inspections.
- Prevent fire.
- Prevent harmful fumes from spreading.

Why is it important to keep the winch greased?

- **Maintain the winch in good working order.**
- Ensure the vessel is accepted for servicing.
- Prevent accidents.
- Prevent unnecessary equipment replacements.

Why is it important to perform greasing after prolonged exposure to sea spray?

- **Grease can be washed away.**
- Grease can dry up.
- Sea salt can mix with the grease and make it ineffective.
- Too much water can change the consistency of the grease and make it ineffective.

0125 – Hot work

Version 2.0 (2256)

A broken hand rail at the bottom of a ladder in a cargo tank is to be cropped and a replacement length welded in place. Which of the following "permits to work" will be required before this job can be started? (Select all applicable answers)

- **A "Hot Work Permit".**
- **An "Enclosed Space Entry" permit.**
- A permit to "Work over the Side".
- A "Permit to Work Aloft".
- No permit will be required.

A hot work modification is to be carried out in the middle of the after transverse bulkhead of an engine room spare gear store. Which of the following could be at risk of damage directly on the other side of the bulkhead in the fitter's workshop? (Select all applicable answers)

- **Paint.**
- **Welding electrodes, contained in a plastic wallet, stored on the wooden shelves.**
- **Wooden shelving.**
- Electrical cables on the opposite transverse bulkhead of the fitter's workshop.
- Spare small bore air and hydraulic hoses hanging in racks on the adjacent longitudinal bulkhead in the fitter's workshop.
- The electric light fittings on the deck head in the fitter's workshop.

A hot work repair is required on a pipeline on the deck of a tanker. The first available flange at which the pipeline can be disconnected and isolated is a long way aft, just in front of the pumproom. What recommended precaution will you follow while the repair is being carried out?

- **Continuously through-ventilating the pipeline with fresh air and monitoring the exhausting air for hydrocarbon vapour.**
- Flushing the pipeline continuously with fresh water.
- Periodically blowing through the pipeline with inert gas and monitoring for oxygen content of at least 15%.

A hot work repair is to take place on a bracket in a forward fuel tank. The tank has been cleaned and gas-freed but how should the heating coils in the tank be treated before that hot work begins?

- **They should be flushed or blown through with steam and then tested to ensure they are clear of hydrocarbons.**
- They should be continuously blown through with fresh air while the hot work repair is taking place.
- They should be continuously blown through with inert gas while the hot work repair is taking place.
- They should be disconnected and removed from the tank.

A level monitor is to be installed in a slop tank on a chemical tanker that is not inerted. Hot work is required. How should the space be treated before that work starts? (Select all applicable answers)

- **The tank should be thoroughly cleaned and gas freed to hot work standard.**
- It is not permitted to perform hot work in a dedicated slop tank, unless the vessel is in dry dock.
- The tank should be inerted.
- The tank should be pre-washed to remove the majority of residues.

A retaining bracket is to be re-welded to the bottom of a bulkhead in a storeroom off the engine room. Which of the following areas will need to be checked for the presence of combustible materials before any hot work is commenced? (Select all applicable answers)

- **Shelving above and immediately to one side of the hot work location.**
- **The space directly below the work site.**
- **The space on the other side of the bulkhead.**
- **The storeroom where the repair is to take place.**
- The complete machinery space.
- The Engine Control Room.

A section of pipe has been disconnected and taken aft to a dedicated gas safe area on the poop deck behind the accommodation block. Is it necessary to perform a Risk Assessment?

- **Yes.**
- No.

A what minimum distance is a hot work repair permitted from the bulkhead of a bunker tank, assuming it has not been cleaned and tested for hot work?

- **500 mm.**
- 2 m.
- 30 m.
- 30 mm.

At which of the following times should hot work be prohibited on a tanker? (Select all applicable answers)

- **During ballast operations.**
- **During cargo operations.**
- **During purging or inerting.**
- **During tank cleaning.**
- **When gas freeing.**
- At a lay-by berth where no cargo operations are to take place.
- At anchor.
- At night.

Before any hot work task can take place, what is required?

- **Careful assessment of all related hazards, and the probability of their occurrence, to determine the severity of the risks involved.**
- A quick check of the intended hot work site to make sure everything seems safe.
- Only an order from the Master or Chief Engineer.
- The completion of the record in the “Hot Work file”, so that the vessel will pass its audits.

Fire safety precautions that may be taken in preparation for hot work might include which of the following? (Select all applicable answers)

- **Arranging for the testing of the atmosphere at the hot work site and in adjacent spaces.**
- **Briefing personnel on action to be taken in the event of an emergency.**
- **Bringing extinguishers and other equipment to a stand-by location.**
- **Dressing personnel in fire fighting gear to be fully ready to move immediately in the event of an emergency.**
- **Inerting adjacent spaces, where appropriate.**
- **Rigging fire curtains and blankets to contain welding sparks and molten slag.**
- **Starting the main or emergency fire pump and running and charging fire hoses.**
- **"Damping down" or performing boundary cooling in adjacent spaces.**

For hot work to take place in the machinery space, what is the minimum distance that it can take place from a bunker tank containing fuel oil?

- **At least 500 mm.**
- At least 1000 mm.
- At least 600 mm.
- No welding can take place in a machinery space where any of the fuel storage tanks has not been cleaned to hot work standard and inspected and tested to that effect.

How can a welder, working in a confined and restricted area and in close contact with the ship's structure, be additionally protected against the risk of electric shock through touching the hull or frames?

- **By providing dry insulating mats or boards.**
- By flushing down the area with fresh water before welding commences.
- By wearing a cotton boiler suit.
- By wearing knee length boots.

How might the tank lining (coating) on the bottom of a tank be protected from damage due to falling molten slag? (Select all applicable answers)

- **By putting a "heel" of water in the bottom of the tank.**
- **By rigging fire resistant blankets.**
- By inerting the tank.
- By rinsing the bottom with Odorless Mineral Spirit before commencing hot work.

If hot work is to be undertaken on a bracket under a tanker's catwalk, at a height of 900mm above the tank deck, what options are available for preparing cargo tanks within a 30m radius of the hot work location to the required safety standards? The vessel is at sea. (Select all applicable answers)

- **They should be cleaned and gas freed, with the hydrocarbon vapour content reduced to not more than 1% LFL and maintained at that level.**
- **They should be completely filled with water.**
- **They should be emptied, purged and the hydrocarbon vapour content reduced to not more than 2% by volume and inerted.**
- There is no requirement to prepare such compartments in any way for hot work at this height above the deck.
- They should be stripped and inerted to not more than 8% oxygen content by volume.

It is intended to burn some old brackets off a bulkhead in the after end of the machinery space. The bulkhead to a lubricating oil storage tank is 480 mm away from the closest bracket. Can the repair take place without further precautionary measures being taken?

- **No.**
- Yes.

On this tanker, a hot work repair is to be carried out at a height of 760mm above the tank deck. The ship is not inerted. Cargo tanks within a specified distance, including those diagonally positioned, should either have been cleaned and gas freed to hot work standard, or completely filled with water. What is that specified distance?

- **30 metres.**
- 100 metres.
- 25 metres.
- 50 metres.

On this tanker, a hot work repair is to take place in cargo tank 2P. Tank 3S is diagonally across the point at which longitudinal and transverse tank bulkheads of the two tanks meet. Is tank 3S considered to be adjacent for the purposes of hot work requirements?

- **Yes.**
- No.

On this tanker, preparations have been made for a damaged ladder in cargo tank 4 centre to be cut away and replaced. The tank has been thoroughly cleaned and gas freed. The vessel is at sea and all other compartments are inerted. How should the inert gas line to 4 centre be treated, before the repair can take place?

- **It should be purged with inert gas to not more than 2% by volume of hydrocarbon vapour and isolated.**
- It is not necessary to do anything with the inert gas line to 4 port.
- It must be flushed through and gas freed, with the hydrocarbon vapour content reduced to not more than 1% LFL and maintained at that level.
- It should be purged with inert gas to not more than 8% by volume of hydrocarbon vapour and isolated.

On this tanker, preparations have been made for a damaged section of heating coil in cargo tank 2 port to be cut away and replaced. The tank has been thoroughly cleaned, gas freed and vented. All enclosed space entry procedures are in place. The vessel is at sea and all other compartments are inerted, but still contain residues. What options are there for preparing the adjacent tanks, before this repair can safely take place? (Select all applicable answers)

- **They must be cleaned and gas freed, with the hydrocarbon vapour content reduced to not more than 1% LFL and maintained at that level.**
- **They must be completely filled with water.**
- **They must be emptied, purged and the hydrocarbon vapour content reduced to not more than 2% by volume and inerted.**
- It is not necessary to do anything with adjacent spaces.
- They must be cleaned but it is not necessary to gas free these spaces.

The ship is alongside, but is not working cargo. The vessel's SMS allows welding in the engine room workshop without a permit, considering it a designated hot work space. The terminal regulations do not permit hot work alongside. The Chief Engineer wants to repair a small crack on a section of pipeline that has been disconnected and taken down below. Can the crack be welded? (Select the most correct answer)

- **No (even though the SMS makes provision for it, terminal regulations take precedence).**
- Strictly speaking no, but it might be possible during the night.
- Yes (even though terminal regulations prohibit it, the SMS takes precedence on board).
- Yes, but the terminal must be informed.

The ship is to carry out some steel replacement in and around the forecastle head, following storm damage. Shoreside contractors are to be brought on board for the hot work. Why must the Risk Assessment for the intended work take special account of these contractors?

- **Because they will be exposed to hazards not normally encountered on shore and they may have little or no maritime experience.**
- Because port regulations require that any Risk Assessment requires contractors to be specifically noted.
- Because SOLAS 1974 requires that special provision be made for shoreside contractors working on board vessels afloat.
- Because such workers are classified as "high risk" under the port regulations.

What approach does the ISM Code take towards safety?

- **It is based on Risk Assessment and Risk Management.**
- It contains highly detailed procedures, covering all aspects of shipboard operation.
- It requires extensive checklists to be completed at every stage.
- It seeks to identify persons responsible in the event of an incident.

What is the definition of "cold work"?

- **Work which cannot create a source of ignition.**
- An atmosphere in which no flammable vapours are present.
- Work that creates a cold spark.
- Work that creates a spark in water.
- Work that will still create a spark in temperatures of less than 15.6°C.
- Work which creates a source of ignition.

What is the maximum concentration of hydrocarbon gas permitted in the atmosphere in a space (not a designated space) before hot work can take place?

- **Less than 1% LFL.**
- Less than 1% UFL.
- Less than 2% Oxygen content.
- Less than 3% LFL.
- Less than 8% Oxygen content.

When a ship is in dry-dock, which of the following primarily governs the management of all safety matters?

- **The shipyard SMS.**
- SOLAS.
- The national regulations of the country in which the dry-dock is located.
- The Port Authority Safety Regulations.
- The shipboard SMS.

When is a tank, compartment or container considered to be gas free?

- **When sufficient fresh air has been introduced into it to lower the level of any flammable, toxic, or inert gas to that required for a specific purpose.**
- When a Senior Officer declares it to be fit for entry.
- When sufficient fresh air has been introduced into it to raise the level of oxygen present to that necessary to sustain human life, that is above 17.9%.
- When sufficient inert gas has been introduced into it to prohibit the possibility of combustion in the presence of a source of ignition.

Where portable lights are required to adequately illuminate a hot work site, how should they be used?

- **They should be clamped or otherwise secured in position, not hand-held, with leads kept clear of the working area.**
- Additional illumination from portable lights is not required; there is enough light from the hot work activity itself.
- They should be held by the assistant at a safe distance from the hot work, with leads kept clear of the working area.
- They should be suspended by natural fibre rope at a minimum distance of 2.5 metres from the hot work activity.

Wherever possible and for safety, which of the following is the most preferable location for performing a hot work repair on a cargo line valve?

- **In the designated space, having been disconnected and removed from the tank deck.**
- In a gas-free cargo tank.
- In position.
- On the poop deck, behind the accommodation, having been disconnected and removed from the tank deck.

Which of the following is considered as the best location on board for hot work?

- **A designated space, such as the engine room workshop.**
- A dedicated area, such as behind the accommodation block.
- Anywhere behind the tank deck area.
- Anywhere within the machinery space.
- At the actual location of the required repair.
- Inside a gas free compartment.
- There is no area on a tanker which is better than any other for hot work; it is equally dangerous all over the ship.

Which of the following is the ISGOTT definition of a "Responsible Officer" (or "Responsible Person")?

- **A person appointed by the employer or the Master of the ship and empowered to take all decisions relating to a specific task, having the necessary knowledge and experience for that purpose.**
- A person appointed by a terminal and empowered to take all decisions relating to a vessel alongside, having the necessary knowledge and experience for that purpose.
- A seafarer with a Certificate of Competency.
- An Officer who has been trained how to complete standard reporting forms.

Which of the following is the definition of a "Risk Assessment"?

- **A thorough and careful analysis of what, in the range of operations, could cause harm, so that decisions can be made as to whether enough precautions have been taken, or whether more should be done to prevent harm.**
- A brief visual check of the work site carried out immediately before the work is started, to ensure that the area is safe and clean.
- An investigation into the circumstances which have led to an incident, such as an explosion that occurred during a welding repair.
- The full and detailed briefing of all team members involved in a work task, to ensure that everyone is aware of their roles and responsibilities.

Which of the following regulates hot work outside the designated space?

- **The SMS by means of a Permit to Work system.**
- Hot work outside the designated space is not regulated.
- Instructions in the Chief Engineer's Standing Orders.
- The SMS through the company Risk Analysis Programme.

Who is responsible for ensuring that adequate arrangements are provided for contractors or work gangs, temporarily employed on board, to understand and comply with all relevant safe working practices?

- **The Master.**
- The Contractor.
- The Shore supervisor.
- The Superintendent.

0127 – Enclosed Space Entry

Version 3.0 (300)

A cleaning and repair job is to be done in a tank. The tank is heavily framed, of an awkward shape, with several distinct sections and built over and around a number of other storage tanks. Access lids and ventilation pipes are all at the forward end of the tank. How can it be ensured that the conditions in the area of the repair are fully checked? Select the ONE correct answer.

- **Assess and plan for an initial entry, carry out further assessment within the space and then plan the entry for the repair on the basis of the additional information obtained.**
- Assess and plan as far as possible and then brief those entering the space for the repair to use their judgement when they get to the after end of the tank.
- Enter the tank and take a quick look round to get an overall picture of the general conditions inside the space and plan the subsequent entry and repair job on that basis.
- Look back in the records and find a report from a similar repair in the past. If it was more-or-less the same situation, base all planning on that.

A crew member doing some cleaning in the bottom a cargo tank reports smelling fumes. What should now happen? Select all answers that you think apply.

- **Any additional controls deemed necessary by the Responsible Person should be put in place before work resumes.**
- **Permits should be withdrawn and only re-issued with any appropriate revisions.**
- **The safety of the situation should be re-assessed.**
- **The work should be stopped and everyone should leave the space immediately.**
- The crew member should be issued with a filter mask and instructed to continue with the cleaning.
- The Responsible Person should immediately enter the tank to check the situation for himself.

A crew-member is to be sent into an enclosed space to remove remaining sludge deposits. Fumes are known to be present. Which of the following would be considered a safe procedure for them to follow? Select all answers that you think apply.

- **Using a self-contained breathing apparatus.**
- **Using an externally-generated air supply.**
- If a mechanical fan is running, he needs do nothing special – the atmosphere should be safe enough to breathe naturally.
- Wearing a filter mask.

A forecastle store containing paint and chemicals is normally continuously ventilated but the area has suffered an electrical failure and the ventilation fan has been stopped overnight. How would you now consider that storeroom? Select the ONE correct answer.

- **As dangerous. It must be treated as an enclosed space for the purpose of entry procedures.**
- It should be alright for entry but the door should be left open for a few minutes before anyone enters.
- So long as the electrical fault has been corrected, it will be OK to enter without additional precautions.
- There is no problem. It is only a storeroom and has been regularly used up until last night.

A ship has a minimum safe complement of four personal oxygen monitors. Two are to be sent ashore for service in the next port but there is a programme of ballast tank inspections scheduled in that port. What could be done to ensure that the ship's personnel continue to work under safe conditions? Select all answers that you think apply.

- **Arrange temporary replacements until the ship's monitors are returned.**
- **Change the schedule of entry operations, so that any person going inside has a monitor at all times.**
- Issue instructions that the remaining meters should be continually transferred between tanks to enable at least some checking of the atmospheres where personnel are working inside the enclosed spaces.
- Stop enclosed space entry altogether as two monitors is not legally enough.

A tank that contained a cargo of vegetable oil has been rough-washed with salt water. It has then been allowed to lie untouched for two days. Which of the following toxic substances is likely to have been released into the atmosphere within the tank over this period? Select the ONE correct answer.

- **Hydrogen sulphide.**
- Ferrous oxide.
- Nitrogen dioxide.
- Nitrous oxide.

According to IMO requirements, what is the minimum steady reading that must be obtained on an oxygen analyser before unprotected entry into an enclosed space can be permitted? Select the ONE correct answer.

- **21%**
- 16%
- 17.5%
- 22.5%
- 25%

Breathing apparatus should be inspected and tested periodically, as per the company maintenance schedule. When else should it be tested for correct operation? Select the ONE correct answer.

- **Before and after use.**
- 2-4 days prior to arrival in port.
- After leaving port.
- When it has been in use for any routine or emergency task where it may have been submerged in water or exposed to water spray.
- When it has been in use in an enclosed space for more than 1 hour.

Can a combustible gas detector be used to detect a gas other than the standard gas on which it is calibrated? Select the ONE best answer.

- **Yes, but only for a limited range of similar gases and reference must be made to the calibration curves supplied with the meter.**
- No.
- Yes, but the operator should be aware that toxic gases with a vapour density of greater than 1.2 may produce erroneous readings.
- Yes.

Can an explosimeter detect the presence of a flammable vapour in a space filled with inert gas? Select the ONE best answer.

- **Some types can, and some types cannot. So this has to be checked.**
- No.
- Yes.

Cargo tank 3 centre has been cleaned and is scheduled for inspection and mopping. You are the Responsible Person. When will you test the atmosphere in the tank? Select all answers that you think apply.

- **At regular and frequent intervals whilst entry continues.**
- **Before anyone enters the space.**
- Once every two to three hours during the validity period of the entry permit.
- Upon completion of the inspection and as the team exits from the tank.

For enclosed space entry, which one of the following describes the "Responsible Person"? Select the ONE correct answer.

- **A person authorised to permit entry into an enclosed space and having sufficient knowledge of the procedures to be established and complied with on board, in order to ensure that the space is safe for entry.**
- A Junior Officer appointed by the Safety Officer to report all activity related to a specific task within an enclosed space, having the required understanding and knowledge for that purpose.
- A Senior Officer, appointed by the Master, who can further delegate responsibility for enclosed space entry to a selected Junior Officer or Rating.
- A Superintendent appointed by the company to oversee all safety-related activities on board the ship, capable of delivering specialised training and empowering him to report information relating to enclosed space entry to the management office.

If a small crack on a bracket in the middle of a bulkhead between two tanks is to be welded, what atmosphere tests should be performed before starting the work? Select all answers that you think apply.

- **The atmosphere in the tank to be entered should be tested for flammability and oxygen content.**
- **The atmospheres in the adjacent spaces should be tested for flammability and oxygen content.**
- The atmosphere in all cargo spaces should be tested for oxygen content.
- The atmosphere in all empty ballast tanks should be tested for flammability and oxygen content.

If the atmosphere in a compartment is considered to be suspect or unsafe, then it should only be entered for which of the following purposes? Select all answers that you think apply.

- **For essential operations.**
- **For the safety of life, or of the ship.**
- **If it is essential for testing purposes, that being not possible from a remote location.**
- For regular cleaning and maintenance under uncontrolled conditions.
- If it is the fastest and most convenient way to prepare the space for cargo.

In relation to entry into an enclosed space, what is "approved equipment"? Select the ONE correct answer.

- **Equipment of a design tested and approved by an appropriate authority for use in a specified hazardous environment.**
- Apparatus that has been supplied by an approved safety company.
- Filter masks and replacement cartridges issued directly by the Chief Officer.
- Monitors and gas test equipment that have been examined and tested within the previous three months.

In safety terms, why might you suspend cargo operations that involve spaces adjacent to a tank intended for entry? Select all answers that you think apply.

- **Because a valve on common piping might be opened in error, flooding the entered compartment.**
- **Because poisonous fumes expelled from those spaces might be drawn in by mechanical ventilators on the entered space.**
- Because safety regulations require all operations personnel to assist in any enclosed space entry procedure.
- Because the charter party contract allows for such interruptions without penalty to the shipowner.

Some cargoes are "carcinogenic". What does this mean? Select the ONE correct answer.

- **They can generate cancer.**
- They can induce indigestion and nausea.
- They may cause permanent blindness.
- They will cause 2nd degree burns.
- They will trigger an allergic reaction to inorganic cargo vapours.

What are the employer's legal obligations in respect of the provision of training on entry into enclosed spaces? Select the ONE correct answer.

- **They must provide the necessary training, instruction and information to employees, in order to ensure that the regulations pertaining to entry into enclosed spaces are complied with.**
- The Employer is not actually legally required to provide any training at all.
- They must provide general safety manuals, CD's and videos that could include some basic guidelines on entry into enclosed spaces.
- They must provide training, instruction and information on entry into enclosed and confined spaces only if a request comes from the Safety Committee of a ship through the Master.

What does the term "lock out and tag out" refer to? Select the ONE correct answer.

- **The securing and labelling of equipment against inadvertent use.**
- Restrictions on personnel against entering certain compartments and spaces on the vessel without the appropriate personal security tag.
- The part of a risk analysis where the Competent Person identifies the valves and switches to be secured by placing a red tag on them.
- The removal of locking devices and withdrawal of advisory notices from valves and switches that had earlier been secured against inadvertent use during an enclosed space entry operation.

What is the first step in any potentially hazardous work task on board. Select the ONE correct answer.

- **Risk assessment.**
- A planning meeting to discuss how to carry out the repair.
- Filling out the permits to work.
- Reporting.
- Taking equipment required for the job into the space.

What is the main hazard associated with entry into an enclosed space? Select the ONE best answer.

- **The potentially hazardous atmosphere inside.**
- The possible loss of communication with those outside the space.
- The Responsible Person's signature missing from the entry permit.
- Unsafe access arrangements that could lead to a slip, trip or fall.

What is the normal oxygen content in fresh air (by volume)? Select the ONE correct answer.

- **20.9%**
- 16.7%
- 17.5%
- 18.2%
- 22.6%
- 23%

When is an enclosed space considered safe and fit for unprotected entry? Select the ONE correct answer.

- **Only when it has been tested and certified as "gas free" by a competent person and approved for entry.**
- When cleaning has finished and the compartment has been rinsed with fresh water.
- When it smells OK from the deck hatch.
- When somebody else has already gone inside without a gas mask.

Which ONE of the following is a correct statement?

- **Personal monitors only provide a warning against oxygen deficiency and hydrocarbon concentrations when there is a change in conditions in the immediate vicinity of the wearer.**
- Personal monitors are an acceptable alternative to a remote analyser in determining if an enclosed space is safe to enter in an unprotected condition.
- Personal monitors are now replacing remote reading meters as the standard equipment for determining the oxygen deficiency and hydrocarbon concentration within the atmospheres contained in enclosed spaces prior to entry.
- Personal monitors provide a warning against oxygen deficiency and hydrocarbon concentrations throughout an enclosed space in which the wearer is working.

Which of the following are standard procedures to observe prior to entry into an enclosed space? Select all answers that you think apply.

- **A Competent Person makes an assessment of the space.**
- **A Responsible Person is appointed to take charge.**
- **An entry permit is issued.**
- **Appropriate safety and operational procedures are put in place.**
- **The atmosphere is tested.**
- **The potential hazards are identified.**
- **The space is secured for entry.**
- All tags are removed, thus indicating that there is no restriction on entry.
- The senior rating enters the space first, to carry out a preliminary visual inspection before any work starts.

Which of the following is the definition of a gas free compartment? Select the ONE best answer.

- **One that is in a suitable and safe condition, such that sufficient fresh air has been introduced into it to lower the level of any flammable, toxic or inert gas to that required for a specified purpose.**
- One that is believed to be in an acceptable condition suitable for an intended task, based on the visual inspection by a designated responsible person.
- One that is listed on a permit which confirms that the atmosphere contained therein has flammability, toxicity and oxygen readings above the corresponding minimum recommended levels listed in the Safety Management System.
- One which the Chief Officer has verbally confirmed to be safe for entry based on his experience and knowledge.

Which one of the following is an acceptable definition of an enclosed space? Select the ONE correct answer.

- **A space which has limited openings for entry and exit, or inadequate ventilation or is not designed for continuous worker occupancy**
- A space or void enclosed by bulkheads and an overhead which, by its design and access points, reduces the hazard exposure of any person entering, or working within that space.
- Any space which is enclosed by bulkheads and decks.

Who is permitted to test the atmosphere in an enclosed space for the purposes of authorising entry? Select the ONE correct answer.

- **Only a person trained in the use of the equipment and listed as an Approved Operator.**
- Anyone who can operate the test equipment and is available at the time entry is required.
- Only Deck Officers
- Only the person who will be entering the space first.

Whose overall duty is it, to ensure that the plan for entry into an enclosed space is followed? Select the ONE best answer.

- **The Responsible Person.**
- The 2nd Engineer.
- The Attendant.
- The Competent Person
- The person in charge of the team that will enter the space

You are assigned as the attendant at the entrance to 3 starboard double bottom ballast tank, inside which the Chief Officer and Cadet are carrying out an inspection. What do you understand you are permitted to do? Select the ONE best answer.

- **I cannot be assigned to other duties while I am attendant and must remain at the entrance to the space while it is occupied.**
- I must stay at the entrance until it is time for lunch.
- I was working in the starboard lifeboat before the Chief Officer called me for the attendant's duty. If the 3rd Officer tells me to go back to the starboard lifeboat, then I should go, because that was my original job for the day.
- Provided I have radio contact with the Cadet inside the tank, it is alright to leave the entrance and carry on working elsewhere on the deck.

You are assisting outside an enclosed space. A maintenance crew are inside and are using air-line breathing apparatus, supplied from a compressor located on deck close to the manhole cover access. The compressor has an emergency supply in a cylinder connected via a manifold. The compressor suddenly stops. You switch over to the emergency supply straight away, but what should the team inside the space do? Select the ONE correct answer.

- **They should immediately stop work and exit the space.**
- Each person inside should take a deep breath, remove their face mask and make their way to the exit.
- They should check the local pressure gauge and estimate how much longer they can remain inside the space.
- They should finish their assigned work as quickly as possible.

You are engaged in cleaning out an oil sludge storage tank that has been previously tested and approved for unprotected entry and for which a valid enclosed space entry permit exists. While you are working, you begin to smell cargo vapours. Which of the procedures listed below should be followed? Select all answers that you think apply.

- **Further tests should be made.**
- **Immediately exit the tank.**
- **The appropriate tag should be placed on the access prohibiting further entry.**
- **The enclosed space entry permit should be cancelled.**
- **The Responsible Person should be notified.**
- All personnel in the space should put on filter masks and continue working.
- All personnel inside should exit and take a short break while the vapours are dispersed.
- An additional portable ventilation fan should be connected.
- The attendant should be sent to get the Duty Engineer.

You are in charge of a team that must enter an enclosed space to rescue a man who has collapsed inside. What is your main priority? Select the one best answer.

- **To ensure that adequate precautions have been taken to protect the rescue team, thereby preventing additional casualties.**
- To collect and prepare an appropriate range of safety equipment at the entrance to the space, so that whatever is needed can be passed inside without further delay.
- To ensure an assistant records all times and other details for the purposes of completing the report required under the Safety Management System.
- To get to the casualty as soon as possible and evacuate them to fresh air, even if it means entering without full safety equipment.

You are the Responsible Person for a cleaning job in a cargo tank. Why must you check around inside the space once cleaning is completed? Select the ONE best answer.

- **To check that no rags, flashlights, lines, clothing or other equipment has been left inside, where it could later be drawn into a pump or valve and damage or block it with serious consequences.**
- Because I have to make sure that all small gear that has been used in the operation is returned to the locker.
- Because if anything like that is found during tank or hold inspection, the surveyor might report it.

You are the Responsible Person for a maintenance inspection in a ship's side ballast tank. The after access manhole has been removed but the compartment is still very dark. How might you safely improve the situation? Select all answers that you think apply.

- **Install temporary lighting.**
- **Issue flashlights.**
- **Remove additional manhole covers and lids.**
- Issue a flare for the entry team to place and light in the bottom of the space.
- Switch off lighting in other areas.
- Tell the team leader to strike a match if he cannot see too well.

You are the Responsible Person for a tank entry operation, where those working inside are being supplied with air from a compressor in the machinery space. Who will you advise of this arrangement and why? There is one correct answer for 'who,' and one correct answer for 'why'. Select the TWO correct answers.

- **So that those in charge of the equipment understand its current purpose and do not shut the compressor down until the work is completed.**
- **The Engineer on Watch.**
- So that the outlet delivery pressure to deck can be reduced to the combined breathing rate of the seamen in the tank.
- So that the running hours on the compressor can be noted.
- The Master.
- The Officer on Watch on the bridge.

You are the attendant at the entrance hatch to a tank where three men, wearing self-contained breathing apparatus, are working inside. You hear the low level audible alarm whistle from one of the sets. What are you going to do? Select the ONE correct answer.

- **Look for the user moving immediately towards the exit and assist them once they emerge.**
- Immediately sound the emergency alarm.
- Lower a full cylinder into the space for the user and wait to pull the empty one out, so it can be re-charged
- Take another fully-charged set inside for the user.
- Try and establish which set it is that is sounding the alarm, by shouting to, or otherwise communicating with, those inside.

You are the attendant at the entrance to an enclosed cargo hold. Why must you "tally", or check the crew off as they exit the space after finishing their work inside? Select the ONE correct answer.

- **To see if anyone is missing and therefore still inside the space.**
- Because the attendant is responsible for all the equipment inside the hold and I need to know the number of flashlights, personal meters and safety helmets that have been used.
- Because the attendant is the responsible person for a hold entry and should check his squad are all present and correct before he dismisses them.
- Because the Chief Officer's instructions always list that as one of the duties of the attendant.
- To record the names for the Chief Officer's work records and overtime sheets.

You are the designated attendant at the top of a tank. Three crew members are inside overhauling a faulty pump. You see one man collapse and hear the other two shouting. What are you going to do? Select the ONE correct answer.

- **I will sound the alarm immediately, report the situation to the Responsible Person and assist the rescue team once they have arrived.**
- I will go inside and pull the collapsed man out, because getting him into fresh air quickly could be critical.
- I will immediately put a breathing apparatus on and go inside the space and see if I can help the collapsed man.
- I will immediately run to the bridge, raise the alarm and then wait there until the Master arrives to take charge.
- I will sound the alarm, then enter the enclosed space and supervise preparations to evacuate the casualty, because the rescue team will bring the stretcher as soon as they are assembled.

You are to assist in "securing" a cargo tank after a cleaning squad have been inside. All personnel and small gear are accounted for and the space is not to be re-entered. Another crew member is returning the safety and working equipment to storage and the Responsible Person is taking care of notifications, documentation and the removal of isolation-tags and notices in the control room. What should you do at or near the tank itself? Select all answers that you think apply.

- **Close the lids**
- **Remove green "safe to enter" tag**
- **Stop the ventilation fan**
- **Withdraw lines and hoses, etc**
- Start pump
- Switch on deck lights

You are to be part of a two-man team to enter an enclosed lower deck in the forecastle head. You are to wear self-contained breathing apparatus and the Responsible Person has decided that a lifeline is to be used. What must you ensure you can do with that lifeline? Select the ONE best answer.

- **That I can easily detach it, if it becomes tangled on the steps or other fittings in the forecastle.**
- I should check the label on the lifeline to ensure that the breaking strain is adequate to support the weight of my colleague, together with the breathing apparatus and other equipment he is wearing.
- I should estimate the distance that the team is to enter the space and ensure that we carry twice that amount of line in with us, so that if it becomes tangled at the bottom, we can still get back to the entrance.
- I should verify that the colour coding on the end of the lifeline conforms to the requirements of the company Safety Management System, indicating that it is suitable for use in a storeroom area.

You are to enter an enclosed space as part of a small team to undertake some cleaning. How will you ensure that the operation continues in a safe manner? Select the ONE correct answer.

- **All of us entering the space, as well as the attendant at the entrance, must remain alert and watch for any changes in circumstances that may render the space unsafe.**
- One team member should be assigned to perform attendant duties immediately inside the space.
- We must check the Enclosed Space Entry permit to see that the Responsible Person has approved the space for a long enough period of time to allow us to complete the cleaning.
- We should talk to each other the whole time, so that we will always know that everyone is OK.

You are to test the atmosphere in a tank prior to entry for cleaning and maintenance purposes. Where will you take the readings? Select the ONE correct answer.

- **At multiple levels and in different sections of the space, through the various lids and vents, to increase the possibility of detecting pockets of gas that might remain in less well ventilated areas.**
- At the bottom of the space, immediately below the access hatch, because toxic vapours are all heavier than air and will sink to the bottom and this is the most likely location for registering any dangerous readings.
- At the top of the space, immediately inside the access hatch, because toxic vapours are all lighter than air and will rise to the top and this is the most likely location for registering any dangerous readings.

You are to test the atmosphere in a tank prior to entry. If at all, how will you adjust the ventilation? Select the ONE correct answer.

- **I will stop it about 10 minutes before testing and not re-start it until testing is complete.**
- I will close over all the lids and secure the ventilator or pressure/vacuum valve closed, so as to create an over-pressure within the space.
- I will have it running at as close to full pressure as possible throughout testing.

- I will just test it as I find it, without touching any ventilation that is running.
- I will reduce it to minimum around 30 minutes before testing and only speed it up again once testing is complete.

0130 – LRM - IHI GTC

Version 1.2 (0)

According to MSC/Circ.1093, which of the following is required to be onboard for use in all operations involving the inspection, maintenance, adjustment and resetting of the lifeboat and associated equipment, such as davits or release gear?

- **A full set of maintenance manuals.**
- A specialized tool-kit supplied by the manufacturer.
- A training video or CBT supplied by the manufacturer and approved by the I.M.O.
- Original certificates of competency in lifeboat maintenance for all personnel involved in inspections and maintenance of such equipment.

According to SOLAS 1974, regulation III/9, where do posters or signs illustrating the procedures for operating survival craft have to be located?

- **In the vicinity of survival craft and their launching controls.**
- In all passenger cabin alleyways.
- On the bridge.
- On the inside of the forward hatches of all enclosed lifeboats.

All inspections, other than weekly and monthly and any servicing and repair, other than routine, should be conducted by which of the following persons?

- **A manufacturer's representative or a person appropriately trained and certified by the manufacturer for the work to be done.**
- A Company Superintendent.
- A senior officer or a fitter suitable qualified and holding a certificate in lifeboat maintenance.
- The Master.

An "on-load release mechanism" is defined as one:

- **. . . designed to be operated when the boat has been lowered and is waterborne, whether or not a load is being applied to it by the action of the ship or the sea.**
- one that can be released at any time from any height when the ship has an adverse list of up to 16 degrees.
- . . . designed to be operated when the boat is hanging in the falls at height of 2-4 metres above the water.
- . . . designed to be operated when the load applied to them is completely removed – that is, when the boat has been lowered and is waterborne.
- . . . that can be released at any time from any height.

By SOLAS, how often does a lifeboat (other than those of free-fall design or those on short international voyages and for whom the Administration has approved alternative arrange-ments) have to be launched and manoeuvred in the water by its assigned operating crew?

- **At least once every three months.**
- At least once a month.
- At least once a year.
- Weekly.

During resetting of the release mechanism, the hook is manually returned to the upright position. Which of the following will also happen as a result of this action?

- **The locking plate will lift back into place.**
- The remote release handle will return to the closed position.
- The second hook will automatically reset.
- The spring-loaded stopper lever will disengage.

How is the D-shaped cam pin turned back into position under the locking plate during resetting?

- **By manually lifting the cam reset lever.**
- By pushing the remote release handle back into position.
- By re-engaging the stopper lever.
- By the hook pressing against the locking plate.

In the case study into the accident on board the Philippine-flagged bulk carrier "Washington Trader", the Australian marine accident investigators found the reset lever extremely stiff. What reason did they suggest for this?

- **Lack of use.**
- Absence of a manufacturer's certificate.
- Lack of understanding of general evacuation procedures.
- Poor illumination of the area.
- The lever was in a poor location.

In the case study of the accident on board the Turkish-flagged bulk carrier "Gulser Ana" in 2001, which of the following deficiencies were listed by the authorities, as contributing to the accident with the lifeboat?

- **Lack of information on the equipment.**
- **Poor maintenance on the hook.**
- **Poor understanding by the crew.**
- **The locking pin missing.**

In which two of the following regulations is the legislation relating to life saving equipment on board a merchant vessel found?

- **SOLAS.**
- **The International Life Saving Appliances Code.**
- The Company Management Manual.
- The ISM Code
- The Lifeboat Manufacturer's Operating Handbook.

Indicate which of the following requirements apply to muster and embarkation areas, according to SOLAS 1974, regulation III/11:

- **They shall be adequately illuminated by lighting powered from the emergency electrical equipment.**
- **They shall be readily accessible from accommodation and work areas.**
- **They shall be so arranged as to allow stretcher cases to be placed in the survival craft.**
- All such areas must have labels indicating where certain responsible personnel must stand.

Into which Code were many of the technical items originally found in SOLAS 1974 transferred in 1996?

- **The Life-Saving Appliances Code.**
- MARPOL
- MSC/Circ.1093.
- S.T.C.W.

Prior to recovering a lifeboat back on board, it is essential to:

- **. . . check that the hook arrangement has properly reset and that the falls are fully engaged.**
- . . . check that everybody is sitting down.
- . . . ensure that the engine is in neutral.
- . . . establish contact on VHF channel 16 between the ship and the lifeboat.
- . . . maintain a minimum of five knots forward speed.

SOLAS requires that, wherever possible, instructions for on-board maintenance of life-saving appliances should include which of the following:
(Select as applicable)

- **A checklist for use when carrying out the inspections.**
- **A diagram of lubrication points.**
- **A list of replaceable parts.**
- **A list of sources of spare parts.**
- **A log for records of inspections and maintenance.**
- **A schedule of periodic maintenance.**
- **Maintenance and repair instructions.**

The British Marine Accident Investigation Branch safety study into accidents involving lifeboat launch, recovery and maintenance highlighted a number of additional significant contributing factors. Which of the following were included in that list?

- **Crew confidence.**
- **Failure of release mechanisms due to involuntary release and complicated design.**
- **Failure to properly reset the mechanism.**
- **Incorrect or inadequate procedures used in training.**
- **Poor understanding of the operating principles by the crew.**

The two main components of this release mechanism are:

- **The hooks.**
- **The remote release handle.**
- The falls.
- The securing gripes.

Weekly and monthly inspections and routine maintenance on lifeboats and associated equipment should be conducted under the direct supervision of which of the following persons?

- **A senior officer.**
- A junior officer assigned to safety duties.
- A manufacturer's representative.
- The Master.

What safety features are fitted to this mechanism to guard against accidental release? (Select all applicable answers)

- **A safety pin.**
- **A spring-loaded stopper lever.**
- A hydrostatic release.
- The locking plates.

When there have been crew changes, special requirements apply for the first abandon ship drill, as detailed in SOLAS 1974, regulation III/19. Which of the following statements applies?

- **If more than 25% of the crew have not participated in an abandon ship drill on that particular ship within the previous month, a drill must be held within 24 hours of leaving port.**
- If any crew members have been changed, a muster should be held before the ship leaves port.
- If any new crew members are not in possession of a valid survival craft training certificate from an approved training facility, then a special abandon ship drill must be held for them within 24 hours of boarding the ship.
- If more than 50% of the crew have not participated in an abandon ship drill on any ship within the previous 3 months, a drill must be held within 3 days of leaving the port.

Where are details of specific responsibilities assigned to crew members during the launch of the lifeboat indicated?

- **On the Muster List.**
- In the seafarer's employment contract.
- In the ship's Personnel Manual.
- On the Bridge WatchKeeping Schedule.
- On the main notice board.

Where would you find specific information on the operation and maintenance of the lifeboats and davits fitted on board your ship?

- **The ship's SOLAS Training Manual.**
- The Company Management Manual.
- The International Guide to Life-Saving Appliances on Merchant Vessels.
- The SOLAS Convention 1974.

Which component directly holds the hook in the closed position?

- **The locking plate.**
- The D-shaped cam pin.
- The safety pin.
- The spring-loaded stopper lever.

Which face of the D-shaped cam pin is in contact with the underside of the locking plate when the mechanism is in the closed position?

- **The curved face.**
- The flat face.

Which of the following additional items must be done after a weekly or monthly inspection of life-saving appliances?

- **A report of inspection must be made in the log-book.**
- A fire drill must be held.
- The equipment must be treated with preservative.
- The lifeboat must be lowered with a complete launching crew embarked.
- The Superintendent must be notified.

Which of the following additional items should be checked during an inspection of the lifeboat release gear?

- **Associated power systems.**
- **Limit switches.**
- **The davit structure.**
- **Wires and sheaves.**

Which of the following is a correct statement?

- **While resetting the release mechanism, it is essential to visually confirm that the spring-loaded stopper lever has returned to the locked position and the locking pin has been replaced**
- While launching the lifeboat, it is essential that the coxswain is seated at the forward end of the boat.
- While re-stowing the lifeboat, it is essential to confirm visually that the remote release handle has rotated in the correct direction.
- While resetting the release mechanism, it is essential to confirm that all personnel are clear of the boat.

Which of the following topics must be included in the vessel's SOLAS Training Manual:

- **Boarding, launching and clearing the survival craft and rescue boats.**
- **Methods of launching from within the survival craft.**
- **Recovery of survival craft and rescue boats, including stowage and securing.**
- **Release of the launching appliances.**

Which two controls need to be pulled by the coxswain to start the release of the lifeboat?

- **The remote release handle.**
- **The spring-loaded stopper lever.**
- The cam reset lever.
- The D-shaped cam pin.

Which two factors from the list below are commonly held to be most important in ensuring the safe and successful release of a lifeboat?

- **The efficiency of the crew in operating the launching equipment, as established and maintained by the regime of training on board.**
- **The working condition of the launching equipment, as determined by the standard of maintenance on that equipment.**
- The international safety rating mark assigned to the lifeboat by the Classification Society.
- The quality of technical drawings and manufacturer's manuals relating to the lifeboat release mechanism.

Why have several national authorities released safety circulars in recent years, focusing on maintenance and operation of lifeboat release mechanisms?

- **Because there continue to be serious incidents related to the lifeboat release mechanisms.**
- Because ship's Captains have mailed too many "Experience Feedback" forms about lifeboat release mechanisms.
- Because they feel it necessary to copy releases from the I.M.O.
- Because they have been advised to do so by the I.M.O.

Why is it important to follow a good maintenance routine on the lifeboat release mechanism?

- **So that the mechanism will always operate immediately and successfully whenever required.**
- Because it gives the crew a regular task on weekends.
- Because the Chief Engineer must be able to fill in the information in the Planned Maintenance System.
- So that the Remote Release Handle will not jam during a Superintendent's Inspection.

Why is it important to train at launching the lifeboat?

- **So that the crew become thoroughly familiar with the launching mechanism.**
- Because the Captain must be able to prove that he is following company procedures.
- Because there is an instruction to do so in the planned maintenance system.
- In order that launching can be done successfully in the dark.

"On-load release gear shall be overhauled and subjected to an operational test under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of persons and equipment". At which of the following intervals must this be done?

- **At least once every five years.**
- 3-monthly.
- Annually.
- Weekly.

"Survival craft, rescue boats and launching appliances shall be visually inspected to ensure that they are ready for use."
At which interval does SOLAS require this procedure to be carried out?

- **Weekly.**
- Annually.
- Daily.
- Monthly.

0131 – LRM - MILLS -TITAN-

Version 1.2 (0)

According to MSC/Circ.1093, which of the following is required to be onboard for use in all operations involving the inspection, maintenance, adjustment and re-setting of the lifeboat and associated equipment, such as davits or release gear?

- **A full set of maintenance manuals.**
- A specialized tool-kit supplied by the manufacturer.
- A training video or CBT supplied by the manufacturer and approved by the I.M.O.
- Original certificates of competency in lifeboat maintenance for all personnel involved in inspections and maintenance of such equipment.

All inspections, other than weekly and monthly and any servicing and repair, other than routine, should be conducted by which of the following persons?

- **A manufacturer's representative or a person appropriately trained and certified by the manufacturer for the work to be done.**
- A Company Superintendent.
- A senior officer or a fitter suitable qualified and holding a certificate in lifeboat maintenance.
- The Master.

An "on-load release mechanism" is defined as one:

- **. . . designed to be operated when the boat has been lowered and is waterborne, whether or not a load is being applied to it by the action of the ship or the sea.**
- . . . designed to be operated when the boat is hanging in the falls at height of 3-4 metres above the water.
- . . . designed to be operated when the load applied to them is completely removed – that is, when the boat has been lowered and is waterborne.
- . . . that can be released at any time from any height.
- . . . where the lifeboat will be launched from a ship underway.

By SOLAS, how often does a lifeboat (other than those of free-fall design or those on short international voyages and for whom the Administration has approved alternative arrange-ments) have to be launched and manoeuvred in the water by its assigned operating crew?

- **At least once every three months.**
- At least once a month.
- At least once a year.
- Weekly.

How is the D-shaped cam pin rotated back, locking the tail of the hook in preparation for recovery of the lifeboat?

- **By lifting the cam reset lever.**
- By adjusting the control cable.
- By ensuring that the remote release lever engages in the locking groove.
- Under the action of the interlock.

In the case study into the accident on board the Philippine-flagged bulk carrier "Washington Trader", the Australian marine accident investigators found the reset lever extremely stiff. What reason did they suggest for this?

- **Lack of use.**
- Absence of a manufacturer's certificate.
- Lack of understanding.
- Poor design.
- Poor illumination.

In the case study of the accident on board the Turkish-flagged bulk carrier "Gulsar Ana" in 2001, which of the following deficiencies were listed by the authorities, as contributing to the accident with the lifeboat?

- **Lack of information on the equipment.**
- **Poor maintenance on the hook.**
- **Poor understanding by the crew.**
- **The locking pin missing.**

In which two of the following regulations is the legislation relating to life saving equipment on board a merchant vessel found?

- **SOLAS.**
- **The International Life Saving Appliances Code.**
- MARPOL.
- The Lifeboat Manufacturer's Operating Handbook.
- The Ship Safety Manual.

Into which Code were many of the technical items originally found in SOLAS 1974 transferred in 1996?

- **The Life-Saving Appliances Code.**
- MARPOL
- MSC/Circ.1093.
- S.T.C.W.

Prior to recovering a lifeboat back on board, it is essential to:

- **. . . check that the hook arrangement has properly re-set and that the falls are fully engaged.**
- . . . check that everybody is sitting down.
- . . . ensure that the engine is in neutral.
- . . . establish contact on VHF channel 16 between the ship and the lifeboat.
- . . . maintain a minimum of five knots forward speed.

SOLAS requires that, wherever possible, instructions for on-board maintenance of life-saving appliances should include which of the following:
(Select as applicable)

- **A checklist for use when carrying out the inspections.**
- **A diagram of lubrication points.**
- **A list of replaceable parts.**
- **A list of sources of spare parts.**
- **A log for records of inspections and maintenance.**
- **A schedule of periodic maintenance.**
- **Maintenance and repair instructions.**

The emergency release procedure for this mechanism involves:

- **Smashing the glass panel and lifting the interlock indicator.**
- Removing the safety pin and pulling the release lever.
- Removing the top cover and screwing down the control handle.
- Smashing the glass panel and pulling the control cable.

The fall link is prevented from accidentally slipping out from under the hook by:

- **A weighted trip bar.**
- An audible alarm at the Coxswain's position.
- The D-shaped cam pin.
- The hydrostatic release.

The function of the hydrostatic release is to:

- **Prevent accidental operation of the remote release handle.**
- Lock the tail end of the hook in the closed position.
- Maintain the lever in the green sector.
- Rotate the D-shaped cam pin.

The two main components of this release mechanism are:

- **The hooks.**
- **The remote release handle.**
- The falls.
- The securing gripes.

Weekly and monthly inspections and routine maintenance on lifeboats and associated equipment should be conducted under the direct supervision of which of the following persons?

- **A senior officer.**
- A junior officer assigned to safety duties.
- A manufacturer's representative.
- The Master.

Where are details of specific responsibilities assigned to crew members during the launch of the lifeboat indicated?

- **On the Muster List.**
- In the seafarer's employment contract.
- In the ship's Personnel Manual.
- On the Bridge WatchKeeping Schedule.
- On the main notice board.

Where should the lifeboat be located when an operational test on the on-load release mechanism is to be performed?

- **Partly in the water, such that the mass of the boat is substantially supported by the falls and the hydrostatic interlock system has not operated.**
- Between 4 and 5 metres above the level of the water.
- Fully in the water, free of all load on the falls and the hydrostatic interlock system having successfully operated.
- Fully secured in the davits, with additional retention devices applied.

Where would you find specific information on the operation and maintenance of the lifeboats and davits fitted on board your ship?

- **The ship's SOLAS Training Manual.**
- The International Guide to Life-Saving Appliances on Merchant Vessels.
- The Ship's Safety Management Manual.
- The SOLAS Convention 1974.

Which component secures the hook in the closed position?

- **The D-shaped cam pin.**
- The locking groove.
- The remote release handle.
- The safety pin.

Which of the following additional items must be done after a weekly or monthly inspection of life-saving appliances?

- **A report of inspection must be made in the log-book.**
- A fire drill must be held.
- The equipment must be treated with preservative.
- The lifeboat must be lowered with a complete launching crew embarked.
- The Superintendent must be notified.

Which of the following additional items should be checked during an inspection of the lifeboat release gear?

- **Associated power systems.**
- **Limit switches.**
- **The davit structure.**
- **Wires and sheaves.**

Which of the following components is activated by the diaphragm in the hydrostatic release?

- **The interlock.**
- The D-shaped cam pin.
- The emergency alarm.
- The remote release handle.

Which of the following is a correct statement?

- **While resetting the release mechanism, it is essential to confirm visually that the D-shaped cam pin and remote release handle are in the correct positions.**
- When resetting the release mechanism, it is essential to confirm visually that the coxswain is seated at the control position.
- While launching the lifeboat, it is essential to confirm that the D-shaped cam pin has not released.
- While re-stowing the lifeboat, it is essential to confirm visually that the remote release handle has rotated in the correct direction.

Which of the following topics must be included in the vessel's SOLAS Training Manual:

- **Boarding, launching and clearing the survival craft and rescue boats.**
- **Methods of launching from within the survival craft.**
- **Recovery of survival craft and rescue boats, including stowage and securing.**
- **Release of the launching appliances.**

Which two factors from the list below are commonly held to be most important in ensuring the safe and successful release of a lifeboat?

- **The efficiency of the crew in operating the launching equipment, as established and maintained by the regime of training on board.**
- **The working condition of the launching equipment, as determined by the standard of maintenance on that equipment.**
- The initial cost of the lifeboat.
- The standard of diagrams relating to the lifeboat release mechanism as displayed in the vessel's SOLAS Training Manual.

Why have several national authorities released safety circulars in recent years, focusing on maintenance and operation of lifeboat release mechanisms?

- **Because there continue to be serious incidents related to the lifeboat release mechanisms.**
- Because ship's Captains have mailed too many "Experience Feedback" forms about lifeboat release mechanisms.
- Because they feel it necessary to copy releases from the I.M.O.
- Because they have been advised to do so by the I.M.O.

Why is it important to follow a good maintenance routine on the lifeboat release mechanism?

- **So that the mechanism will always operate immediately and successfully whenever required.**
- Because it gives the crew a regular task on weekends.
- Because the Chief Engineer must be able to fill in the information in the Planned Maintenance System.
- So that the Remote Release Handle will not jam during a Superintendent's Inspection.

Why is it important to train at launching the lifeboat?

- **So that the crew become thoroughly familiar with the launching mechanism.**
- Because the Captain must be able to prove that he is following company procedures.
- Because there is an instruction to do so in the planned maintenance system.
- In order that launching can be done successfully in the dark.

"On-load release gear shall be overhauled and subjected to an operational test under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of persons and equipment". At which of the following intervals must this be done?

- **At least once every five years.**
- 3-monthly.
- Annually.
- Weekly.

"Survival craft, rescue boats and launching appliances shall be visually inspected to ensure that they are ready for use."
At which interval does SOLAS require this procedure to be carried out?

- **Weekly.**
- Annually.
- Daily.
- Monthly.

0132 – LRM - SHIGI S2

Version 1.2 (0)

According to MSC/Circ.1093, which of the following is required to be onboard for use in all operations involving the inspection, maintenance, adjustment and re-setting of the lifeboat and associated equipment, such as davits or release gear?

- **A full set of maintenance manuals.**
- A specialized tool-kit supplied by the manufacturer.
- A training video or CBT supplied by the manufacturer and approved by the I.M.O.
- Original certificates of competency in lifeboat maintenance for all personnel involved in inspections and maintenance of such equipment.

All inspections, other than weekly and monthly and any servicing and repair, other than routine, should be conducted by which of the following persons?

- **A manufacturer's representative or a person appropriately trained and certified by the manufacturer for the work to be done.**
- A Company Superintendent.
- A senior officer or a fitter suitable qualified and holding a certificate in lifeboat maintenance.
- The Master.

An "on-load release mechanism" is defined as one:

- **. . . designed to be operated when the boat has been lowered and is waterborne, whether or not a load is being applied to it by the action of the ship or the sea.**
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- . . . that can be released at any time from any height.
- . . . where the lifeboat will be launched from a ship underway.

By SOLAS, how often does a lifeboat (other than those of free-fall design or those on short international voyages and for whom the Administration has approved alternative arrange-ments) have to be launched and manoeuvred in the water by its assigned operating crew?

- **At least once every three months.**
- At least once a month.
- At least once a year.
- Weekly.

How are the pistons extended in preparation for recovery of the lifeboat?

- **By returning the remote release handle to the closed position.**
- By depressing the manual override rod and replacing the top cover.
- By re-setting the safety locking bar.
- By removing the top cover and lifting the manual override rod.

In the case study into the accident on board the Philippine-flagged bulk carrier "Washington Trader", the Australian marine accident investigators found the reset lever extremely stiff. What reason did they suggest for this?

- **Lack of use.**
- Absence of a manufacturer's certificate.
- Lack of understanding.
- Poor design.
- Poor illumination.

In the case study of the accident on board the Turkish-flagged bulk carrier "Gulsar Ana" in 2001, which of the following deficiencies were listed by the authorities, as contributing to the accident with the lifeboat?

- **Lack of information on the equipment.**
- **Poor maintenance on the hook.**
- **Poor understanding by the crew.**
- **The locking pin missing.**

In what position is the piston, when it is securing the tail of the hook in the closed position?

- **Extended.**
- Withdrawn.

In which two of the following regulations is the legislation relating to life saving equipment on board a merchant vessel found?

- **SOLAS.**
- **The International Life Saving Appliances Code.**
- MARPOL.
- The Lifeboat Manufacturer's Operating Handbook.
- The Ship Safety Manual.

Into which Code were many of the technical items originally found in SOLAS 1974 transferred in 1996?

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- MARPOL
- MSC/Circ.1093.
- S.T.C.W.

Prior to recovering a lifeboat back on board, it is essential to:

- **. . . check that the hook arrangement has properly re-set and that the falls are fully engaged.**
- . . . check that everybody is sitting down.
- . . . ensure that the engine is in neutral.
- . . . establish contact on VHF channel 16 between the ship and the lifeboat.
- . . . maintain a minimum of five knots forward speed.

SOLAS requires that, wherever possible, instructions for on-board maintenance of life-saving appliances should include which of the following:
(Select as applicable)

- **A checklist for use when carrying out the inspections.**
- **A diagram of lubrication points.**
- **A list of replaceable parts.**
- **A list of sources of spare parts.**
- **A log for records of inspections and maintenance.**
- **A schedule of periodic maintenance.**
- **Maintenance and repair instructions.**

The emergency release procedure for this mechanism involves:

- **Removing the top cover and lifting the override rod.**
- Removing the safety pin and pulling the release lever.
- Smashing the glass panel and lifting the interlock indicator.
- Smashing the glass panel and pulling the control cable.

The fall link is prevented from accidentally slipping out from under the hook by:

- **A weighted trip bar.**
- An audible alarm at the Coxswain's position.
- The D-shaped cam pin.
- The hydrostatic release.

The function of the hydrostatic release is to:

- **Prevent accidental operation of the remote release handle.**
- Extend the piston.
- Hold the safety pin in place.
- Lock the tail end of the hook in the closed position.

The two main components of this release mechanism are:

- **The hooks.**
- **The remote release handle.**
- The falls.
- The securing gripes.

Weekly and monthly inspections and routine maintenance on lifeboats and associated equipment should be conducted under the direct supervision of which of the following persons?

- **A senior officer.**
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- A manufacturer's representative.
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Where are details of specific responsibilities assigned to crew members during the launch of the lifeboat indicated?

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- In the seafarer's employment contract.
- In the ship's Personnel Manual.
- On the Bridge WatchKeeping Schedule.
- On the main notice board.

Where should the lifeboat be located when an operational test on the on-load release mechanism is to be performed?

- **Partly in the water, such that the mass of the boat is substantially supported by the falls and the hydrostatic interlock system has not operated.**
- Between 4 and 5 metres above the level of the water.
- Fully in the water, free of all load on the falls and the hydrostatic interlock system having successfully operated.
- Fully secured in the davits, with additional retention devices applied.

Where would you find specific information on the operation and maintenance of the lifeboats and davits fitted on board your ship?

- **The ship's SOLAS Training Manual.**
- The International Guide to Life-Saving Appliances on Merchant Vessels.
- The Ship's Safety Management Manual.
- The SOLAS Convention 1974.

Which component secures the hook in the closed position?

- **The piston.**
- The locking groove.
- The release handle.
- The safety pin.

Which of the following additional items must be done after a weekly or monthly inspection of life-saving appliances?

- **A report of inspection must be made in the log-book.**
- A fire drill must be held.
- The equipment must be treated with preservative.
- The lifeboat must be lowered with a complete launching crew embarked.
- The Superintendent must be notified.

Which of the following additional items should be checked during an inspection of the lifeboat release gear?

- **Associated power systems.**
- **Limit switches.**
- **The davit structure.**
- **Wires and sheaves.**

Which of the following components is activated by the diaphragm in the hydrostatic release?

- **The safety locking bar.**
- The after hook.
- The piston.
- The remote release handle.

Which of the following is a correct statement?

- **While resetting the release mechanism, it is essential to confirm visually that the piston and remote release handle are in the correct positions.**
- While launching the lifeboat, it is essential that the coxswain is seated at the forward control position.
- While re-stowing the lifeboat, it is essential to confirm visually that the remote release handle has rotated in the correct direction.
- While resetting the release mechanism, it is essential to confirm visually that all personnel are clear of the lifeboat.

Which of the following topics must be included in the vessel's SOLAS Training Manual:

- **Boarding, launching and clearing the survival craft and rescue boats.**
- **Methods of launching from within the survival craft.**
- **Recovery of survival craft and rescue boats, including stowage and securing.**
- **Release of the launching appliances.**

Which two factors from the list below are commonly held to be most important in ensuring the safe and successful release of a lifeboat?

- **The efficiency of the crew in operating the launching equipment, as established and maintained by the regime of training on board.**
- **The working condition of the launching equipment, as determined by the standard of maintenance on that equipment.**
- The initial cost of the lifeboat.
- The standard of diagrams relating to the lifeboat release mechanism as displayed in the vessel's SOLAS Training Manual.

Why have several national authorities released safety circulars in recent years, focusing on maintenance and operation of lifeboat release mechanisms?

- **Because there continue to be serious incidents related to the lifeboat release mechanisms.**
- Because ship's Captains have mailed too many "Experience Feedback" forms about lifeboat release mechanisms.
- Because they feel it necessary to copy releases from the I.M.O.
- Because they have been advised to do so by the I.M.O.

Why is it important to follow a good maintenance routine on the lifeboat release mechanism?

- **So that the mechanism will always operate immediately and successfully whenever required.**
- Because it gives the crew a regular task on weekends.
- Because the Chief Engineer must be able to fill in the information in the Planned Maintenance System.
- So that the Remote Release Handle will not jam during a Superintendent's Inspection.

Why is it important to train at launching the lifeboat?

- **So that the crew become thoroughly familiar with the launching mechanism.**
- Because the Captain must be able to prove that he is following company procedures.
- Because there is an instruction to do so in the planned maintenance system.
- In order that launching can be done successfully in the dark.

"On-load release gear shall be overhauled and subjected to an operational test under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of persons and equipment". At which of the following intervals must this be done?

- **At least once every five years.**
- 3-monthly.
- Annually.
- Weekly.

"Survival craft, rescue boats and launching appliances shall be visually inspected to ensure that they are ready for use."
At which interval does SOLAS require this procedure to be carried out?

- **Weekly.**
- Annually.
- Daily.
- Monthly.

0133 – LRM - FASSMER

Version 1.2 (0)

According to MSC/Circ.1093, which of the following is required to be onboard for use in all operations involving the inspection, maintenance, adjustment and re-setting of the lifeboat and associated equipment, such as davits or release gear?

- **A full set of maintenance manuals.**
- A specialized tool-kit supplied by the manufacturer.
- A training video or CBT supplied by the manufacturer and approved by the I.M.O.
- Original certificates of competency in lifeboat maintenance for all personnel involved in inspections and maintenance of such equipment.

All inspections, other than weekly and monthly and any servicing and repair, other than routine, should be conducted by which of the following persons?

- **A manufacturer's representative or a person appropriately trained and certified by the manufacturer for the work to be done.**
- A Company Superintendent.
- A senior officer or a fitter suitable qualified and holding a certificate in lifeboat maintenance.
- The Master.

An "on-load release mechanism" is defined as one:

- **. . . designed to be operated when the boat has been lowered and is waterborne, whether or not a load is being applied to it by the action of the ship or the sea.**
- . . . designed to be operated when the boat is hanging in the falls at height of 3-4 metres above the water.
- . . . designed to be operated when the load applied to them is completely removed – that is, when the boat has been lowered and is waterborne.
- . . . that can be released at any time from any height.
- . . . where the lifeboat will be launched from a ship underway.

By SOLAS, how often does a lifeboat (other than those of free-fall design or those on short international voyages and for whom the Administration has approved alternative arrange-ments) have to be launched and manoeuvred in the water by its assigned operating crew?

- **At least once every three months.**
- At least once a month.
- At least once a year.
- Weekly.

How is the D-shaped cam pin rotated back, locking the tail of the hook in preparation for recovery of the lifeboat?

- **By pulling on the control cable.**
- By lifting the cam reset lever.
- By returning the remote release handle to the closed position.
- Under the action of the interlock.

In the case study into the accident on board the Philippine-flagged bulk carrier "Washington Trader", the Australian marine accident investigators found the reset lever extremely stiff. What reason did they suggest for this?

- **Lack of use.**
- Absence of a manufacturer's certificate.
- Lack of understanding.
- Poor design.
- Poor illumination.

In the case study of the accident on board the Turkish-flagged bulk carrier "Gulsar Ana" in 2001, which of the following deficiencies were listed by the authorities, as contributing to the accident with the lifeboat?

- **Lack of information on the equipment.**
- **Poor maintenance on the hook.**
- **Poor understanding by the crew.**
- **The locking pin missing.**

In which two of the following regulations is the legislation relating to life saving equipment on board a merchant vessel found?

- **SOLAS.**
- **The International Life Saving Appliances Code.**
- MARPOL.
- The Lifeboat Manufacturer's Operating Handbook.
- The Ship Safety Manual.

Into which Code were many of the technical items originally found in SOLAS 1974 transferred in 1996?

- **The Life-Saving Appliances Code.**
- MARPOL
- MSC/Circ.1093.
- S.T.C.W.

Prior to recovering a lifeboat back on board, it is essential to:

- **. . . check that the hook arrangement has properly re-set and that the falls are fully engaged.**
- . . . check that everybody is sitting down.
- . . . ensure that the engine is in neutral.
- . . . establish contact on VHF channel 16 between the ship and the lifeboat.
- . . . maintain a minimum of five knots forward speed.

SOLAS requires that, wherever possible, instructions for on-board maintenance of life-saving appliances should include which of the following:
(Select as applicable)

- **A checklist for use when carrying out the inspections.**
- **A diagram of lubrication points.**
- **A list of replaceable parts.**
- **A list of sources of spare parts.**
- **A log for records of inspections and maintenance.**
- **A schedule of periodic maintenance.**
- **Maintenance and repair instructions.**

The emergency release procedure for this mechanism involves:

- **Smashing the glass panel and removing the pin from the shackle.**
- Removing the safety pin and pulling the release lever.
- Removing the top cover and lifting the override rod.
- Smashing the glass panel and pulling the control cable.

The fall link is prevented from accidentally slipping out from under the hook by:

- **A weighted trip bar.**
- An audible alarm at the Coxswain's position.
- The D-shaped cam pin.
- The hydrostatic release.

The function of the hydrostatic release is to:

- **Prevent accidental operation of the remote release handle.**
- Hold the safety pin in place.
- Lock the tail end of the hook in the closed position.
- Rotate the D-shaped cam pin.

The two main components of this release mechanism are:

- **The hooks.**
- **The remote release handle.**
- The falls.
- The securing gripes.

Weekly and monthly inspections and routine maintenance on lifeboats and associated equipment should be conducted under the direct supervision of which of the following persons?

- **A senior officer.**
- A junior officer assigned to safety duties.
- A manufacturer's representative.
- The Master.

Where are details of specific responsibilities assigned to crew members during the launch of the lifeboat indicated?

- **On the Muster List.**
- In the seafarer's employment contract.
- In the ship's Personnel Manual.
- On the Bridge WatchKeeping Schedule.
- On the main notice board.

Where should the lifeboat be located when an operational test on the on-load release mechanism is to be performed?

- **Partly in the water, such that the mass of the boat is substantially supported by the falls and the hydrostatic interlock system has not operated.**
- Between 4 and 5 metres above the level of the water.
- Fully in the water, free of all load on the falls and the hydrostatic interlock system having successfully operated.
- Fully secured in the davits, with additional retention devices applied.

Where would you find specific information on the operation and maintenance of the lifeboats and davits fitted on board your ship?

- **The ship's SOLAS Training Manual.**
- The International Guide to Life-Saving Appliances on Merchant Vessels.
- The Ship's Safety Management Manual.
- The SOLAS Convention 1974.

Which component holds the hook in the "locked" position?

- **The D-shaped cam pin.**
- The link.
- The remote release handle.
- The safety pin.

Which face of the D-shaped cam pin should be in contact with the tail of the hook when it is in the closed position?

- **The curved face.**
- The flat face.

Which of the following additional items must be done after a weekly or monthly inspection of life-saving appliances?

- **A report of inspection must be made in the log-book.**
- A fire drill must be held.
- The equipment must be treated with preservative.
- The lifeboat must be lowered with a complete launching crew embarked.
- The Superintendent must be notified.

Which of the following additional items should be checked during an inspection of the lifeboat release gear?

- **Associated power systems.**
- **Limit switches.**
- **The davit structure.**
- **Wires and sheaves.**

Which of the following components is activated by the diaphragm in the hydrostatic release?

- **The clamp.**
- The after hook.
- The D-shaped cam pin.
- The remote release handle.

Which of the following is a correct statement?

- **While resetting the release mechanism, it is essential to confirm that the D-shaped cam pin and the remote release handle are in the correct positions.**
- While launching the lifeboat, it is essential to confirm that the D-shaped cam pin has not released.
- While re-stowing the lifeboat, it is essential that the Coxswain is seated at the forward control position.
- While resetting the release mechanism, it is essential to confirm visually that all personnel are clear of the lifeboat.

Which of the following topics must be included in the vessel's SOLAS Training Manual:

- **Boarding, launching and clearing the survival craft and rescue boats.**
- **Methods of launching from within the survival craft.**
- **Recovery of survival craft and rescue boats, including stowage and securing.**
- **Release of the launching appliances.**

Which two factors from the list below are commonly held to be most important in ensuring the safe and successful release of a lifeboat?

- **The efficiency of the crew in operating the launching equipment, as established and maintained by the regime of training on board.**
- **The working condition of the launching equipment, as determined by the standard of maintenance on that equipment.**
- The initial cost of the lifeboat.
- The standard of diagrams relating to the lifeboat release mechanism as displayed in the vessel's SOLAS Training Manual.

Why have several national authorities released safety circulars in recent years, focusing on maintenance and operation of lifeboat release mechanisms?

- **Because there continue to be serious incidents related to the lifeboat release mechanisms.**
- Because ship's Captains have mailed too many "Experience Feedback" forms about lifeboat release mechanisms.
- Because they feel it necessary to copy releases from the I.M.O.
- Because they have been advised to do so by the I.M.O.

Why is it important to follow a good maintenance routine on the lifeboat release mechanism?

- **So that the mechanism will always operate immediately and successfully whenever required.**
- Because it gives the crew a regular task on weekends.
- Because the Chief Engineer must be able to fill in the information in the Planned Maintenance System.
- So that the Remote Release Handle will not jam during a Superintendent's Inspection.

Why is it important to train at launching the lifeboat?

- **So that the crew become thoroughly familiar with the launching mechanism.**
- Because the Captain must be able to prove that he is following company procedures.
- Because there is an instruction to do so in the planned maintenance system.
- In order that launching can be done successfully in the dark.

"On-load release gear shall be overhauled and subjected to an operational test under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of persons and equipment". At which of the following intervals must this be done?

- **At least once every five years.**
- 3-monthly.
- Annually.
- Weekly.

"Survival craft, rescue boats and launching appliances shall be visually inspected to ensure that they are ready for use."
At which interval does SOLAS require this procedure to be carried out?

- **Weekly.**
- Annually.
- Daily.
- Monthly.

0134 – LRM - HARDING FF

Version 1.2 (0)

According to MSC/Circ.1093, which of the following is required to be onboard for use in all operations involving the inspection, maintenance, adjustment and resetting of the lifeboat and associated equipment, such as davits or release gear?

- **A full set of maintenance manuals.**
- A specialized tool-kit supplied by the manufacturer.
- A training video or CBT supplied by the manufacturer and approved by the I.M.O.
- Original certificates of competency in lifeboat maintenance for all personnel involved in inspections and maintenance of such equipment.

According to SOLAS 1974, regulation III/9, where do posters or signs illustrating the procedures for operating survival craft have to be located?

- **In the vicinity of survival craft and their launching controls.**
- In all passenger cabin alleyways.
- On the bridge.
- On the inside of the forward hatches of all enclosed lifeboats.

All inspections, other than weekly and monthly and any servicing and repair, other than routine, should be conducted by which of the following persons?

- **A manufacturer's representative or a person appropriately trained and certified by the manufacturer for the work to be done.**
- A Company Superintendent.
- A senior officer or a fitter suitable qualified and holding a certificate in lifeboat maintenance.
- The Master.

Before launching a free-fall lifeboat, which of the following items is it essential to check:

- **That additional securing straps have been released and locking pins removed.**
- **That all personnel are safely strapped in to their seats.**
- **That there is a clear launch path.**
- **That there is adequate depth of water.**

By SOLAS, how often does a free-fall lifeboat normally have to be launched by free-fall mode and manoeuvred in the water by its assigned operating crew?

- **At least once very three months.**
- At least once a month.
- At least once a year.
- At least once every six months.

How is the hook held in the closed position?

- **By a hinged catch.**
- By a locking pin.
- By the remote release handle.
- By the weight of the lifeboat alone.

If the remote release handle fails to operate properly, what is the emergency procedure?

- **The glass panel in the casing of the hydraulic cylinder must be smashed and the screw arrangement turned, to build up the pressure in the cylinder and extends the piston to turn the hinged catch.**
- Pressurize the hydraulic cylinder by connecting the emergency gas supply.
- The locking pin must be withdrawn from the safety bolt and the hinged catch manually forced clear, allowing the hook to open.
- Use the emergency handle as a crowbar to force open the hook.

In preparation for recovery of the lifeboat, the pressure in the hydraulic cylinder must be relieved. How is this done?

- **By opening the bypass valve next to the remote release handle.**
- By applying a downward pressure on the securing hook, causing the hinged catch to rotate and forcing the piston back into the housing.
- By closing the bypass valve and operating the remote release handle until the pressure dissipates and the handle action goes slack.
- By manually rotating the end screw on the cylinder in an anti-clockwise direction.

In the case study into the accident on board the Philippine-flagged bulk carrier "Washington Trader", the Australian marine accident investigators found the reset lever extremely stiff. What reason did they suggest for this?

- **Lack of use.**
- Absence of a manufacturer's certificate.
- Lack of understanding of general evacuation procedures.
- Poor illumination of the area.
- The lever was in a poor location.

In which direction must the end screw be rotated to pressurize the hydraulic cylinder during the emergency release procedure?

- **Clockwise.**
- Anti-clockwise.

In which two of the following regulations is the legislation relating to life saving equipment on board a merchant vessel found?

- **SOLAS.**
- **The International Life Saving Appliances Code.**
- The Company Management Manual.
- The ISM Code
- The Lifeboat Manufacturer's Operating Handbook.

Indicate which of the following requirements apply to muster and embarkation areas, according to SOLAS 1974, regulation III/11:

- **They shall be adequately illuminated by lighting powered by the emergency electrical equipment.**
- **They shall be readily accessible from accommodation and work areas.**
- **They shall be so arranged as to allow stretcher cases to be placed in the survival craft.**
- All such areas must have labels indicating where certain responsible personnel must stand.

Into which Code were many of the technical items originally found in SOLAS 1974 transferred in 1996?

- **The Life-Saving Appliances Code.**
- MARPOL
- MSC/Circ.1093.
- STCW

Is it necessary for a crew-member to remain outside this type of lifeboat during release?

- **No.**
- Yes.

Once the hinged catch is rotated clear during release, what actually opens the hook?

- **The weight of the lifeboat alone.**
- Extension of the hydraulic piston.
- Increasing pressure within the hydraulic cylinder.
- Rotation of the hydraulic release bypass valve by the coxswain.

Prior to recovery of the lifeboat after a drill, the pressure in the hydraulic cylinder is relieved. Why is this an essential requirement?

- **In order to prevent damage to the piston, which might cause a failure of the release mechanism in a real emergency situation.**
- It acts as an additional safeguard, in case the locking pins have not been properly replaced.
- So that all components are restored back to their original positions ready for the next drill.
- So that the hinged catch rotates back to the locked position.

SOLAS requires that, wherever possible, instructions for on-board maintenance of life-saving appliances should include which of the following: (Select as applicable)

- **A checklist for use when carrying out the inspections.**
- **A diagram of lubrication points.**
- **A list of replaceable parts.**
- **A list of sources of spare parts.**
- **A log for records of inspections and maintenance.**
- **A schedule of periodic maintenance.**
- **Maintenance and repair instructions.**

The British Marine Accident Investigation Branch safety study into accidents involving lifeboat launch, recovery and maintenance highlighted a number of additional significant contributing factors. Which of the following were included in that list?

- **Crew confidence.**
- **Failure of release mechanisms due to involuntary release and complicated design.**
- **Failure to properly reset the mechanism.**
- **Incorrect or inadequate procedures used in training.**
- **Poor understanding of the operating principles by the crew.**

There are two safety pins that need to be removed as the lifeboat is prepared for launching. Where are they located?

- **In the locking bolt of the hook casing.**
- **In the remote release handle casing.**
- In the bypass valve of the hydraulic cylinder.
- In the davit frame link.
- In the hinged catch.
- Under the glass panel in the hydraulic cylinder casing.

Weekly and monthly inspections and routine maintenance on lifeboats and associated equipment should be conducted under the direct supervision of which of the following persons?

- **A senior officer.**
- A junior officer assigned to safety duties.
- A manufacturer's representative.
- The Master.

What kind of release hook does this arrangement employ?

- **A single point release hook.**
- A dual function release hook.
- A manual interlock release point.
- A multiple point release facility.

When there have been crew changes, special requirements apply for the first abandon ship drill, as detailed in SOLAS 1974, regulation III/19. Which of the following statements applies?

- **If more than 25% of the crew have not participated in an abandon ship drill on that particular ship within the previous month, a drill must be held within 24 hours of leaving port.**
- If any crew members have been changed, a muster should be held before the ship leaves port.
- If any new crew members are not in possession of a valid survival craft training certificate from an approved training facility, then a special abandon ship drill must be held for them within 24 hours of boarding the ship.
- If more than 50% of the crew have not participated in an abandon ship drill on any ship within the previous 3 months, a drill must be held within 3 days of leaving the port.

Where are details of specific responsibilities assigned to crew members during the launch of the lifeboat indicated?

- **On the Muster List.**
- In the seafarer's employment contract.
- In the ship's Personnel Manual.
- On the Bridge WatchKeeping Schedule.
- On the main notice board.

Where it is impracticable to launch a free-fall lifeboat by the free-fall mode, alternative arrangements are acceptable under SOLAS 1974, regulation III/19. These are:

- **The lifeboat may be lowered into the water by winch instead, provided that the lifeboat is launched by free-fall mode with its assigned operating crew aboard and manoeuvred in the water at least once every six months.**
- The lifeboat may always be lowered into the water by winch instead, provided that the frequency of such drills is increased to once every two months.
- The lifeboat may be lowered into the water by winch instead, provided that an entry is made in the log book explaining why launching by the free-fall mode could not be done.
- The lifeboat may be lowered into the water by winch instead, provided that the lifeboat is launched by free-fall mode at least once a year.

Where would you find specific information on the operation and maintenance of the lifeboats and davits fitted on board your ship?

- **The ship's SOLAS Training Manual.**
- The Company Management Manual.
- The International Guide to Life-Saving Appliances on Merchant Vessels.
- The SOLAS Convention 1974.

Which of the following additional items must be done after a weekly or monthly inspection of life-saving appliances?

- **A report of inspection must be made in the log-book.**
- A fire drill must be held.
- The equipment must be treated with preservative.
- The lifeboat must be lowered with a complete launching crew embarked.
- The Superintendent must be notified.

Which of the following additional items should be checked during an inspection of the lifeboat release gear?

- **Associated power systems.**
- **Limit switches.**
- **The davit structure.**
- **Wires and sheaves.**

Which of the following topics must be included in the vessel's SOLAS Training Manual:

- **Boarding, launching and clearing the survival craft and rescue boats.**
- **Methods of launching from within the survival craft.**
- **Recovery of survival craft and rescue boats, including stowage and securing.**
- **Release of the launching appliances.**

Which two factors from the list below are commonly held to be most important in ensuring the safe and successful release of a lifeboat?

- **The efficiency of the crew in operating the launching equipment, as established and maintained by the regime of training on board.**
- **The working condition of the launching equipment, as determined by the standard of maintenance on that equipment.**
- The international safety rating mark assigned to the lifeboat by the Classification Society.
- The quality of technical drawings and manufacturer's manuals relating to the lifeboat release mechanism.

Why have several national authorities released safety circulars in recent years, focusing on maintenance and operation of lifeboat release mechanisms?

- **Because there continue to be serious incidents related to the lifeboat release mechanisms.**
- Because ship's Captains have mailed too many "Experience Feedback" forms about lifeboat release mechanisms.
- Because they feel it necessary to copy releases from the I.M.O.
- Because they have been advised to do so by the I.M.O.

Why is it important to follow a good maintenance routine on the lifeboat release mechanism?

- **So that the mechanism will always operate immediately and successfully whenever required.**
- Because it gives the crew a regular task on weekends.
- Because the Chief Engineer must be able to fill in the information in the Planned Maintenance System.
- So that the Remote Release Handle will not jam during a Superintendent's Inspection.

Why is it important to train at launching the lifeboat?

- **So that the crew become thoroughly familiar with the launching mechanism.**
- Because the Captain must be able to prove that he is following company procedures.
- Because there is an instruction to do so in the planned maintenance system.
- In order that launching can be done successfully in the dark.

"Survival craft, rescue boats and launching appliances shall be visually inspected to ensure that they are ready for use."
At which interval does SOLAS require this procedure to be carried out?

- **Weekly.**
- Annually.
- Daily.
- Monthly.

0139 – Awareness of Lifeboat Release and Retrieval Systems

Version 1.0 (1614)

During lowering and raising of a side-launched lifeboat, which of the following precautions must be taken by operating crewmembers to avoid injury by sudden movement of the lifeboat? Select the ONE correct answer.

- **Ensuring seat harnesses are tightly and securely fastened**
- Standing up and hold on to some firm structure
- Wearing a lifejacket and a survival suit
- Wearing eye protection

During retrieval, why is it necessary to stop raising the lifeboat when just clear of the water? Select the ONE best answer.

- **To recheck that the lifeboat release and retrieval system is properly reset and all interlocks are in place**
- To brief the operating crew
- To check the painter is properly secure.
- To give time for the operating crew to store equipment.

How does the locking pin fall preventer device prevent accidental release of a side-launched lifeboat? Select the ONE correct answer.

- **It prevents the moveable hook component pivoting**
- It locks the operating system in place
- It secures the hook assembly to the fall suspension link
- It stops the hook locking part moving if the hook load is too high.

How must a manufacturer ensure that a lifeboat release and retrieval system fitted to a lifeboat is safe and reliable? Select the ONE correct answer.

- **A lifeboat release and retrieval system must go through strict testing and evaluation to make sure it meets the standard required under SOLAS**
- The design needs only to have a mechanical device to prevent accidental release.
- The design only needs to be made of non-rusting materials
- The manufacturer needs only to check that it works properly before leaving the factory

How often are the hooks of a lifeboat release and retrieval system tested to ensure they release when the lifeboat is loaded to 1.1 times its fully loaded weight? Select the ONE correct answer.

- **Every five years**
- Monthly
- Weekly
- Yearly

If the hook assembly is badly damaged, under what circumstances can a side-launched lifeboat be launched and retrieved? Select the ONE correct answer.

- **The lifeboat cannot be launched and retrieved until it has been properly repaired and tested by a suitably qualified technician authorized by the manufacturer**
- It can be launched and retrieved but the maximum number of persons it is certified to carry must be reduced by half
- It can be launched and retrieved during a drill with a fall preventer device
- It can be launched and retrieved only after a risk assessment has been completed following temporary repairs by the chief engineer.

In a lifeboat release and retrieval system, what is the part that prevents the moveable hook component pivoting until it is moved by the operating system? Select the ONE correct answer.

- **Hook locking part**
- Hydrostatic safety interlock
- Locking pin
- Release handle in locked position

In which of the following situations should a side-launched lifeboat be released on-load? Select the ONE correct answer?

- **When abandoning ship in rough seas.**
- After the lifeboat release and retrieval system has been repaired by the ship's staff
- During a drill when permission is obtained from the ship's superintendent
- During a lifeboat drill when the lifeboat is being lowered into the water

In which of the following situations would it be necessary to manually raise and hold the hydrostatic safety interlock of a lifeboat release and retrieval system? Select the ONE correct answer.

- **When it is necessary to launch a side-launched lifeboat off-load and the hydrostatic safety interlock fails to unlock the release handle.**
- To launch a side launched lifeboat off-load in calm seas
- To reconnect the lifeboat to the hooks prior to retrieval
- When the operating system safety pin interlock is stuck and cannot be removed

What is the correct order for the actions the coxswain must carry out when releasing a side-launched lifeboat off-load? Select the ONE correct answer?

- **Check the hydrostatic interlock has lifted, remove the safety pin and pull back the release handle.**
- Check the hydrostatic interlock has lifted, pull back the release handle and remove the safety pin.
- Pull back the release handle, check the hydrostatic interlock has lifted and remove the safety pin.
- Remove the safety pin, check the hydrostatic interlock has lifted, pull back the release handle.

What is the hook load when a side-launched lifeboat is launched off-load? Select the ONE correct answer.

- **Zero**
- Half the weight of the lifeboat
- The weight of the lifeboat
- Twice the weight of the lifeboat

What is the name of the part that connects the release handle to the hook locking parts in a lifeboat release and retrieval system? Select the ONE correct answer.

- **Hook control cable**
- Lockable hook component.
- Moveable hook component
- Removable safety pin

What is the purpose of an interlock in the operating system of a lifeboat release and retrieval system? Select the ONE correct answer.

- **To prevent accidental pulling of the release handle**
- To lock the lifeboat access to prevent unauthorized use of the operating system
- To lock the moveable hook component in place
- To stop the hook locking part from moving

When is it best to remove a fall preventer device during launching a side-launched lifeboat? Select the ONE correct answer.

- **When the lifeboat is less than one meter above the water**
- After operating the release handle
- As soon as the fall block is clear of the davit
- When the lifeboat is four meters above the water

When lowering a side-launched lifeboat, when is the hook load approximately zero? Select the ONE correct answer.

- **When the lifeboat is stationary and floating in the water**
- As soon as it clears the lifeboat davits
- As soon as the lifeboat keel touches the water
- If the boat starts swinging

Where can detailed maintenance requirements for lifeboat release and retrieval systems be found? Select the ONE best answer.

- **In the manufacturer's maintenance manual**
- In the shipboard management system manual
- In the SOLAS life Saving appliances training manual
- On the maintenance instructions that are posted in the lifeboat

Which of the following could be an indicator of poor maintenance on a lifeboat release and retrieval system? Select the ONE best answer.

- **The release handle is very stiff to operate**
- Grease found on greasing points and linkages
- Release handle cannot able to be opened due safety pin in place
- Spanner found lying in the lifeboat bilges

Which of the following is a control measure that directly prevents accidental release of a side-launched lifeboat? Select the ONE correct answer?

- Fitting a fall preventer device
- Posting operating instructions in the lifeboat
- Rigging a lifeboat painter
- Wearing a safety helmet

Who carries out the annual thorough inspection on a lifeboat release and retrieval system? Select the ONE correct answer.

- **The manufacturer's representative or contractor authorized by them**
- Any contractor who has experience working on lifeboats and davit systems
- The Chief Engineer
- The Chief Officer as long as they have read the manufacturer's maintenance manual

Why must the handle of a fall preventer device locking pin be painted with a colour different from the hook assembly of a side launched lifeboat? Select the ONE correct answer.

- **To make it apparent to the crew that it must be removed before launching a side-launched lifeboat.**
- So that is easy to find if misplaced or lost.
- So that it can be fitted quickly in an emergency
- So that it is not fitted the wrong way round.

0145 – SMCP - Distress Urgency and Safety

Version 1.1 (0)

Choose the correct option:
MF frequency for distress communication by radio telephony is:

- **2182 k Hz**
- 2187 k Hz
- Channel 16
- Channel 71

Choose the correct option:
OSC stands for . . .

- **On Scene Co-ordinator**
- On Scene Caller
- On Scene Crew
- Over Seas Call

Choose the correct option:
RT signal to announce a distress message is . . .

- **Mayday**
- Main day
- My Day
- Pan-Pan

Choose the correct option:
SART stands for . . .

- **Search & Rescue Transponder**
- Search & Rescue Transmitter
- Search & Rescue Tug
- Search & Respond Transponder

Choose the correct option:
Signal to announce a safety message is . . .

- **SECURITE**
- MAYDAY
- PAN PAN
- SAFETY

Choose the correct option:
Signal to announce an urgency message is . . .

- **PAN PAN PAN PAN PAN PAN**
- MAYDAY
- PAN PAN
- PAN PAN PAN

Choose the correct spelling option for SHIP:

- **Sierra Hotel India Papa**
- Sam Hotel Indigo Papa
- Sierra Hot India Papa
- Sierra Hotel Indian Papa

Choose the correct spelling option for TUG:

- **Tango Uniform Golf**
- Tang Uniform Golf
- Tango Uniform Girl
- Tango Union Golf

Listen and click on the correct option.

- **Will you abandon vessel?**
- Will you abandon weasel?
- Will you act on vessel?
- Will you action vessel?

Listen and click on the correct option.

- **Watchkeepers to the assembly stations**
- keep watch to the assembly stations
- Watchkeepers to the assembly
- Watchkeepers to the stations

Listen and click on the correct option.

- **Close watertight doors and report**
- Close doors and report
- Close water in doors and report
- Tight the water doors and report

Listen and click on the correct option:

- **Instruction**
- Information
- Inspection
- Intention

Listen and click on the correct option:

- **Fire spreading to purifier room**
- Fire in to purifier room
- Fire spreading to pure room
- Fire spreading to room

Listen and click on the correct option:

- **Is danger imminent?**
- Is danger important?
- Is danger intention?
- Is dangerous imminent?

Listen and click on the correct option:

- **Was the distress alert acknowledged?**
- Was the alert acknowledged?
- Was the distress acknowledged?
- Was the distress alert acclaimed?

Listen and click on the correct option:

- **Repeat**
- Repel
- Report
- Request

Listen and click on the correct option:

- **What is state of tide?**
- What is standard of tide?
- What is state of time?
- What is statue of tide?

Listen and click on the correct option:

- **Operate the general emergency alarm**
- Open emergency alarm
- Open the general alarm
- Operate the alarm

Listen and click on the correct option:

- **What kind of assistance is required?**
- What is assistance required?
- What is kind of assistance?
- What kind of assistant is required?

Listen and click on the correct option:

- **Is person in water located?**
- Is people in water located?
- Is person in water looked?
- Is person in west located?

Listen and click on the correct option:

- **Let go rescue boat**
- Let go resting boat
- Let good rescue boat
- Let us go to rescue boat

Listen and click on the correct option:

- **Can you proceed to distress position?**
- Can you proceed to distress ?
- Can you proceed to position?
- Can you process to distress position?

Listen and click on the correct option:

- **Is your SART transmitting?**
- Is your CART transmitting?
- Is your SAR transmitting?
- Is your START transmitting?

Listen and click on the correct option:

- **Navigate with caution**
- Navigate with caption
- Navigate with care
- Navigate with caught on

Listen and click on the correct option:

- **SECURITE**
- Say cure
- SECURE
- Secure it

Listen and click on the correct option:

- **Pressure on fire mains**
- Press fire mains
- Pressure on fire menu
- Purser on fire mains

Select the Most Appropriate Phrase to indicate . . .
Silence has been imposed to avoid interference to distress traffic.

- **Seelonce Mayday**
- Quiet Please
- Seelonce Distress
- Silence

Select the most appropriate phrase to indicate 9.8 m (Select all applicable answers):

- **Nine decimal eight meters**
- **Nine point eight meters**
- Nine meters eighty centimeters

Select the most appropriate phrase to indicate:
A person falls into sea from the ship.

- **Man Overboard**
- Man on the board
- Man out of the board
- Man over the board

Select the most appropriate phrase to indicate:
All personnel are out of engine room.

- **engine room evacuated**
- all empty of engine room
- engine out
- engine room empty

Select the most appropriate phrase to indicate:
Question about vessel being underway again after being aground.

- **Has vessel refloated?**
- Has vessel aweighed?
- Has vessel come off aground?
- Has vessel reflected?

Select the most appropriate phrase to indicate:
Rescue team is ready.

- **Rescue team standing by**
- Rescue team standing buy
- Rescue team standing bye
- Rescue team understanding

Select the most appropriate phrase to indicate:
Vessel has touched bottom and is stuck on seabed.

- **We are aground**
- We am aground
- We on ground
- We sit aground

Select the most appropriate phrase to indicate:
Vessel is drifting after main engine breakdown.

- **I am not under command**
- I am drifting under command
- I am not commanding
- I was under the command

Select the most appropriate phrase to indicate:
Vessel's arrival time at distress position.

- **ETA**
- EAT
- V Arrival T
- VAT

Select the most appropriate phrase to indicate: Information requested can not be obtained.

- **No information**
- No advice
- No comments
- Not available

True or false. Is the expression correct?
"Decimal" sign can also be spoken as "Point".

- **True**
- False

True or false. Is the expression correct?
Message marker, if used, is spoken after the full phrase has been spoken.

- **False**
- True

True or false?
In open Sea, channel 16 can be used to communicate with other ships for long time.

- **False**
- True

True or false?
In the boat, if someone is very thirsty, he should be allowed to drink seawater.

- **False**
- True

True or false?
MAYDAY is spoken two times, once before and once after the distress message.

- **False**
- True

True or false?
MRCC stands for maritime rescue coordination centre.

- **True**
- False

True or false?
Merchant ships must help other ships in distress if they are in position to do so.

- **True**
- False

True or false? Is the expression correct?
Man has fallen out of board.

- **False**
- True

True or false? Is the expression correct?
To announce a safety message, word SECURITE is repeated three times.

- **True**
- False

True or false? Is the expression correct?
Word "PAN PAN" is spoken three times to announce a urgency message.

- **True**
- False

0146 – SMCP - Navigation and Cargo Handling

Version 3.0 (58)

Choose the correct option for filling in the blanks.
Never smoke . . . . . bed

- **In**
- Above
- At
- Over

Choose the correct option for filling in the blanks.
We also have radiocontact . . . . . other ships / radio coast stations.

- **With**
- At
- By
- For

Choose the correct option for filling in the blanks.
Passengers are requested . . . . . read all notes and leaflets concerning safety regulations.

- **To**
- At
- For
- In

Choose the correct option for filling in the blanks.
Please follow the instructions given . . . . . the officers and crew.

- **By**
- For
- From
- With

Choose the correct option.
COW stands for . . . . .

- **Crude Oil washing**
- Coal Ore Washing
- Code Of Work
- Crude Oil Water

Choose the correct option.
Hard a port means:

- **Helm to 35 degrees port**
- Alter course to port by 35 degrees
- Helm to 28 degrees port
- Helm to 30 degrees port

Choose the correct option.
SWL stands for . . . . . .

- **Safe Working Load**
- Safe Weather Load
- Safety Worker Load
- Ship Working Load

Choose the correct option.
TEU stands for . . .

- **Twenty Foot Equivalent Unit**
- Twenty Equivalent Unit
- Twenty Foot Equal Unit
- Twenty Ton Equivalent Unit

Choose the correct option.
VTS stands for . . .

- **Vessel Traffic Services**
- Vessel Traffic separation
- VHF Traffic Services
- Visible Traffic Services

Choose the correct option:
ETA stands for . . . . . .

- **Estimated Time of Arrival**
- Equivalent Twenty Area
- Expected Time of Anchor
- Extra Terrestrial Alien

Choose the correct spelling option for SHIP.

- **Sierra Hotel India Papa**
- Sam Hotel Indigo Papa
- Sierra Hot India Papa
- Sierra Hotel Indian Papa

Choose the correct spelling option for TUG.

- **Tango Uniform Golf**
- Tang Uniform Golf
- Tango Uniform Girl
- Tango Union Golf

Listen and click on the correct option.

- **I am constrained by draught**
- I am coming by draught
- I am constrained by deratting
- I am contradicted by draught

Listen and click on the correct option.

- **Make a lee on your port side**
- Make a bee on your port side
- Make a leap on your port side
- Make a lee on your poster

Listen and click on the correct option.

- **steady as she goes**
- steadily as she goes
- steady as ship goes
- straight as she goes

Listen and click on the correct option.

- **Repeat**
- Repell
- Report
- Request

Listen and click on the correct option.

- **pick up the slack on the springs**
- pick back slack on the springs
- pick up back pack on the springs
- pick upon slack on the springs

Listen and click on the correct option.

- **Are the holds clean, dry and free of smell?**
- Are the holds clean and dry?
- Are the holds clean and free of smell?
- Are the holds clean, direct and forward of smell?

Listen and click on the correct option.

- **The hold is in order**
- The hold is odourfree
- The hold is the other
- This hold is on otherside

Listen and click on the correct option.

- **Backpressure for stripping is 2 bars**
- Backpack for stripping is 2 bars
- Backpressure for ship is 2 bars
- Bad pressure for stripping is 2 bars

Listen and click on the correct option.

- **Complete the stowage plan**
- Complete the mortgage plan
- Complete the shortage plan
- Complete the stopover plan

Listen and click on the correct option.

- **Check the lashings and securings everyday**
- Check that lashings and securings are there
- Check the lashings and security everyday
- Check the lashings now and securings everyday

Listen and click on the correct option.

- **What is the handling capacity of the Container Crane?**
- What is the capacity of the Container Crane?
- What is the handling capacity of the Container ?
- What is the handling capacity of the Crane?

Listen and click on the correct option.

- **All passengers must attend this drill**
- All passengers must attend this day
- All people must attend this drill
- All personal managers must attend this drill

Listen and click on the correct option.

- **Take your lifejackets and blankets with you**
- Take your lifeguard and blankets with you
- Take your lifejackets and brackets for you
- Take your liferafts and blankets you

Listen and click on the correct option.

- **All escape routes are clear**
- All and every routes are clear
- All escape reports are clear
- All escape rollers are clear

Listen and click on the correct option.

- **I will complete the manoeuvre**
- I will compete the manoeuvre
- I will complete the man over
- I will contemplate the manoeuvre

Listen and click on the correct option.

- **Instruction**
- Information
- Inspection
- Intention

Listen and click on the correct option.

- **What was your last port of call?**
- What was your last port of card?
- What was your last position of call?
- What was your last post of call?

Select the most appropriate phrase to indicate 9.8 m (Select all applicable answers):

- **Nine decimal eight meters**
- **Nine point eight meters**
- Nine meters eighty centimeters

Select the most appropriate phrase to indicate . . .
All mooring lines are tied and secure.

- **All made fast**
- All lines ready
- All over
- Everything Tight

Select the most appropriate phrase to indicate . . .
Maximum weight a crane can lift safely.

- **Safe working load**
- High safe limit
- Maximum weight of crane
- Safe weather load

Select the most appropriate phrase to indicate . . .
Question about height of main deck above water level.

- **What is your freeboard?**
- What is your above water height?
- What is your air draught?
- What is your main deck height?

Select the most appropriate phrase to indicate . . .
Vessel's heading is not changing as desired despite giving large helm orders.

- **She is not answering the wheel**
- Heading no change with wheel
- She do nothing when wheel
- Wheel no change

Select the most appropriate phrase to indicate . . .
Watch has been handed over to the reliever.

- **You have the watch now**
- All is yours now
- You are in charge now
- You watching now

Select the most appropriate phrase to indicate . . .
While berthing, first mooring line is on the bollard on quay.

- **First line ashore**
- Fast line ashore
- First line at port
- First line on ship

Select the most appropriate phrase to indicate . . .
You want to drop anchor.

- **Let go anchor**
- Anchor go
- Leave anchor
- Let fall the anchor

Select the most appropriate phrase to indicate . . . Information requested can not be obtained.

- **No information**
- No advice
- No comments
- Not available

True or false. Is the expression correct?
"Decimal" sign can also be spoken as "Point".

- **True**
- False

True or false. Is the expression correct?
Message marker, if used, is spoken after the full phrase has been spoken.

- **False**
- True

True or false?
International regulations require all passengers to be assembled in a drill which has to take place within 24 hours of departure.

- **True**
- False

What does the load master want to know?

- **Maximum quantity of oil which the vessel can discharge in an hour**
- Discharging pressure if the vessel discharges maximum cargo
- Maximum quantity of oil which a vessel can discharge
- Price of cargo if the vessel discharges maximum cargo.

What does the terminal want to know from chief officer?

- **Are the cargo hoses connected?**
- Are the cargo holes connected?
- Are the cars connected?
- Are the cars going connected?

0147 – Maritime English, Introduction

Version 1.0 (0)

What is shown in the picture?

- **Valve**
- Anchor
- Funnel
- Winch

What is shown in the picture?

- **Stair**
- Light
- Radar
- Torch

What is shown in the picture?

- **Lifeboat**
- Lifebuoy
- Liferaft
- Tug boat

What is shown in the picture?

- **Spanner**
- Gloves
- Hammer
- Helmet

What is shown in the picture?

- **Rudder**
- Bollards
- Fairlead
- Railing

What is shown in the picture?

- **Scupper**
- Cylinder
- Manhole
- Overalls

What is shown in the picture?

- **Windlass**
- Gangway
- Shackle
- Wheel

What is shown in the picture?

- **Paint brush**
- Paint roller
- Paint scraper
- Wire brush

What is shown in the picture?

- **Life jacket**
- Breathing apparatus
- Immersion suit
- Safety harness

What is shown in the picture?

- **Propeller**
- Radar mast
- Satellite transponder
- Walkie-talkie

What is shown in the picture?

- **Fire hose**
- Binoculars
- Fire extinguisher
- Hawse pipe

What is shown in the picture?

- **Safety goggles**
- Ear defenders
- Pilot ladder
- Rat guard

What is shown in the picture?

- **Flag**
- Buoy
- Mast
- Pipe

What is shown in the picture?

- **Chain**
- Alarm
- Rope
- Wire

What is shown in the picture?

- **Hatch**
- Chart
- Crane
- EPIRB

What phrase do you hear?

- **Where is the ship's fire plan?**
- Where is the fresh water connection?
- Where is the schedule for the crew?

What phrase do you hear?

- **Have you received a visitor badge?**
- Are all holds ready for inspection?
- Have you deballasted the aftpeak tank?

What phrase do you hear?

- **How do you operate this fire extinguisher?**
- How do you dispose of oily rags?
- Who is in charge of sludge tanks?

What phrase do you hear?

- **Can I take samples from the cargo tanks?**
- Can I see your identity card?
- In which tank will you take bunkers?

What phrase do you hear?

- **When did you last check the mooring lines?**
- When did you last test the SART?
- Where are the additional lifejackets kept?

What phrase do you hear?

- **What is the working language of this vessel?**
- What are your work hours as per STCW?
- What is the colour of smoke from the funnel?

What phrase do you hear?

- **When will you use the foam type fire extinguisher?**
- When will you use the CO2 type fire extinguisher?
- When will you use the DCP fire extinguisher?

What phrase do you hear?

- **What are your duties in the event we need to abandon vessel?**
- What are you duties in the event of main engine failure?
- What are your duties in the event of fire onboard?

What phrase do you hear?

- **Is the maintenance plan for the fire protection system in place and effective?**
- Is the condition of the radio reserve source of energy, including charger unit, satisfactory?
- Is the muster list correctly compiled, maintained and crew members aware of their duties?

What phrase do you hear?

- **Where can I smoke?**
- Where is the bunker manifold?
- Where is the galley?

What phrase do you hear?

- **What is the name of the Designated Person for the company?**
- What are your duties in case of oil pollution from vessel?
- What is the alarm limit for the oily water separator?

What phrase do you hear?

- **What is the abandon ship signal?**
- What is the emergency signal?
- What is the man overboard signal?

What phrase do you hear?

- **Where is the engine control room?**
- Where is the Chief Engineer?
- Where is the incinerator located?

What word do you hear?

- **Ship**
- Sea
- Sick
- Sign

What word do you hear?

- **Safety**
- Scale
- Spaces
- State

What word do you hear?

- **Damage**
- Danger
- Doctor
- Draught

What word do you hear?

- **Rescue**
- Restow
- Result
- Rising

What word do you hear?

- **Current**
- Command
- Control
- Correct

What word do you hear?

- **Change**
- Channel
- Charted
- General

What word do you hear?

- **Search**
- Service
- Speed
- Station

What word do you hear?

- **Lane**
- Line
- Rain
- Tank

What word do you hear?

- **Moderate**
- Immediately
- Manoeuvring
- Measures

What word do you hear?

- **Transfer**
- Attention
- Transmit
- Tropical

What word do you hear?

- **Radio**
- Other
- Radar
- Range

What word do you hear?

- **Pilotage**
- Pipeline
- Polluter
- Pressure

What word do you hear?

- **Continue**
- Carefully
- Collision
- Condition

What word do you hear?

- **Dangerous**
- Disabled
- Distress
- Equipment

What word do you hear?

- **Spillage**
- Stability
- Starboard
- Survivors

What word do you hear?

- **Explosion**
- Emergency
- Engineer
- Expected

What word do you hear?

- **Instructions**
- Assistance
- Information
- Permission

0149 – Incident Investigation

Version 2.0 (1217)

A Junior Engineer is injured during the replacement of a main engine cylinder liner. An investigator, examining the period immediately before the accident, may look for two distinct types of actions (referred to in this training program). What are they?

- **Commission**
- **Omission**
- Prevention
- Solution

Acts and practises identified in an incident are compared against a "standard". What is that standard?

- **The applicable operating procedure**
- The appropriate start-up routine
- The relevant maintenance schedule
- The required shut-down method

An investigator arrives on board to conduct an inquiry into an incident involving some damaged containers, several of which have collapsed and fallen forward onto the back of the forecastle deck. Which of the following will he initially seek to identify?

- **The transfer of energy**
- How the greatest damage was caused
- The sequence of events leading to the incident
- Where the biggest impact has taken place

As described in this training program, an accident is generally caused by contact with which two of the following?

- **A source of energy**
- **A substance above the threshold limit of the body concerned**
- A body capable of generating a disproportionately high level of energy
- A body within which all energy has been contained during an impact

Basic causes are generally divided into which two of the following categories?

- **Job factors**
- **Personal factors**
- Fundamental causes
- Substandard conditions

Following an accident onboard, there is an enquiry from a lawyer on the ship's satellite telephone, whilst at anchor awaiting port entry. Who should the Duty Officer direct the call to?

- **The Master**
- The Agent
- The Ship's Safety Officer
- The Ship's Security Officer

If accidents are investigated and the information obtained is constructively fed back into the system, which of the following might reasonably be anticipated?

- **A reduction, or even elimination of such accidents in the future**
- A more efficient system will be developed for the purposes of allocating blame
- A reduction in personal injuries, matched by a corresponding increase in incidents involving material damage
- An increased likelihood of the repetition of such accidents in the future

If an officer is instructed to investigate an incident which involved a broken wire on a mooring winch and he is supporting his inquiry with photographs, how should he try to photograph the wire?

- **In place, still on the winch drum**
- Ashore, having been removed from the vessel
- He should not take photographs of the broken wire
- In the forecastle store, having been removed from the winch drum.

If control is assessed as inadequate, which two of the following factors may be representative of inadequate performance standards?

- **They are assessed as not high or demanding enough**
- **They are not specific enough**
- The standards state exactly what is required
- There are too few system activities found

Immediate causes are generally divided into which two of the following categories?

- **Substandard acts**
- **Substandard conditions**
- Job factors
- Underlying factors

In respect of incident investigation, complacency, incompetence, negligence, recklessness and lack of communication are all examples of what?

- **Fallability**
- Bad luck
- Infallibility
- Pre-existing management failure

In terms of investigation, which of the following may be considered as the results or effects of an accident?

- **The loss experienced at the end of a sequence of events**
- Serious material damage, but not any physical injury
- The elements in a sequence of events prior to the point of contact
- Visible costs in material replacement and repair only

Inadequate compliance is normally identified by which two of the following circumstances?

- **Evidence that there was inadequate knowledge or understanding of a standard which resulted in it not being observed.**
- **Evidence that there was knowledge of a declared standard and that it was wilfully disregarded**
- Proof that the injured party had undergone a training course
- The assumption that personnel involved in an incident have not listened to instructions

The Chief Officer has been appointed to investigate an incident in the engine room workshop. He is planning to interview a number of crewmembers. Which two of the following should he do, prior to commencing the interviews? (Select all applicable answers)

- **Analyse the evidence already secured**
- **Identify evidence still to be secured**
- Compile questions which will deliver evidence to support his ideas of what happened
- Work out beforehand who is lying and how to trap them in the interview

The Superintendent is going on board to conduct an investigation into an accident in which a crewmember was seriously injured during the stowage of an accommodation ladder. Which four of the following individuals or groups should he particularly try to interview?

- **Any eye witnesses**
- **Any person who can contribute to the understanding of the events leading up to, and the prevailing conditions at the time of, the accident**
- **Any person who can provide additional evidence, relevant to the investigation**
- **The person in charge of the operation at the time of the accident**
- A representative from the manufacturer of the accommodation ladder
- All crewmembers

The maximum safe working load of a crane has been exceeded. When the operator attempted the lift, the retaining bolts at the foot of the crane mounting sheared and the crane toppled over. Which three of the following may be considered as evidence of "fallibility" in this situation?

- **Incompetence**
- **Lack of monitoring**
- **Negligence**
- Inadequate design

The ship's Safety Officer is conducting an on-board investigation into a minor incident involving a fall on a lifeboat access platform. A Seaman who was working nearby at the time is being interviewed and is having difficulty in remembering some of the details. Which two techniques might the Safety Officer use to help him remember?

- **He might talk through the incident with the Seaman**
- **He might work through a list of potentially relevant items, asking the Seaman to include or reject them**
- He should ask the Seaman to say anything, even if he is not absolutely sure of the facts, so that he can include some testimony from him in the investigation report
- He should get firm with the Seaman and tell him how important it is that he remembers

What are the three common reasons for lack of control?

- **An inadequate system**
- **Inadequate compliance with standards**
- **Inadequate standards**
- Insufficient investment in training
- Lack of personnel
- Poor communication

What broad rule is often referred to by safety investigators when explaining the selective investigation of reported incidents?

- **The 80 / 20 rule**
- Bernhard's principle of Selective Accident Investigation
- The 20 / 20 rule
- The Accident Pyramid

What is it that starts the process of cause and effect and ultimately results in the "loss" evidenced in the incident?

- **Lack of control**
- Improper risk assessment
- Intentional abuse
- Lack of training

When an accident occurs on board that may have an effect on the safety of the crew and on the environment, an investigator will afterwards examine the way the situation developed afterwards. In which two ways is it generally understood that a situation will go?

- **It may be recovered**
- **It may continue to deteriorate**
- It will be considered to have no significant impact on the environment
- The investigation will focus on the facts at the time; subsequent risk to the safety of the crew or the environment are not considered relevant to the investigation

When we refer to a "blame culture", what kind of culture do we consider typically exists?

- **One where people look no further than "human error"**
- One in which the level of responsibility of each person involved in an incident is allocated as a percentage
- One where investigators seek to blame as many people as possible
- One which seeks to avoid apportioning any degree of blame

Which four of the following are possible consequences of accidents, incidents and near-misses?

- **Adverse publicity**
- **Lost time**
- **Material damage**
- **Personal injury**
- Enhanced reputation
- Increased business

Which maritime Code requires shipping companies to provide for safe practices in ship operation and a safe working environment, establish safeguards against all identified risks and to continuously improve safety management skills of personnel ashore and aboard ships, including preparing for emergencies related both to safety and environmental protection?

- **The International Safety Management (ISM) Code**
- The International Convention for the Prevention of Pollution from Ships (MARPOL)
- The International Convention for the Safety of Life at Sea (SOLAS)
- The International Ship and Port Facility Security (ISPS) Code

Which of the following best describes the form that the narrative section of an Incident Investigation Report should follow?

- **A simple, orderly description of the events leading up to the accident, incident or near miss, the circumstances of the accident, incident or near miss itself, the consequences and the post-incident actions, written in a neutral way, without opinion or judgment.**
- A complex and detailed account of the accident, incident or near miss only, reflecting, so far as possible, the investigator's understanding of the circumstances and including recommendations for the attention of management.
- A random description of the events preceding an incident, or near miss and the corrective actions necessary to prevent a recurrence, written in a neutral way, but including the investigator's summary assessment.

Which of the following best explains the purpose of an investigation into an accident, incident or near miss on board ship?

- **The establishment of facts and the determination of corrective action, aimed at the prevention of a recurrence of the incident. It is not to determine the allocation of blame**
- Primarily the allocation of blame, but also the identification of relevant contributory factors. It may suggest corrective action
- The completion of the necessary forms, so that an incident is on record and an auditor can see that a system for selectively documenting certain incidents exists
- The drawing up of corrective action, including disciplinary procedures where appropriate. Applicable contributory factors may not necessarily be identified

Which of the following is the definition of an "impartial" witness, within the context of this program?

- **One who has no personal interest and nothing to gain, or lose**
- One who has a problem with authority and feels threatened by the interview
- One with a personal interest in the outcome of the inquiry
- One with whom there is a language-based communication problem

Which of the following is the definition of "circumstantial evidence", as understood by this module?

- **Something that may reasonably be said to be relevant to an incident**
- Accounts told by witnesses
- Information obtained through a third party
- Something found in physical objects, which cannot often be disputed

Which of the following may be categorized under inadequate tools?

- **Inadequate orientation or briefing**
- **Inadequate updating or refresher training**
- **Lack of experience**
- Excessive frustration
- Inadequate support
- Lack of incentives

Which of the following may be categorized under lack of knowledge?

- **Inadequate orientation or briefing**
- **Inadequate updating or refresher training**
- **Lack of experience**
- Excessive frustration
- Inadequate support
- Lack of incentives

Which safety initiative, launched by the Oil Companies International Marine Forum in 1993, takes the form of a risk-assessment tool for charterers, ship operators, terminal operators and government bodies and includes verifying if an accident and incident investigation policy exists within a company?

- **The SIRE Programme**
- The International Risk Assessment Guide for Ship and Terminal Operators
- The IPPA Initiative
- The Ship Operator Voluntary Audit Scheme

Which three of the following are common interview techniques, as explained in this program?

- **Conversational approach**
- **Free recall**
- **Question and answer**
- Numerical assessment
- Trick questions

Which three of the following are examples of possible "hidden" costs resulting from an accident or incident?

- **Increased insurance premiums**
- **Lost time**
- **Reduced capacity**
- Injuries
- Material damage
- Pollution

Which three of the following are examples of post-contact controls?

- **Emergency evacuation plans**
- **First Aid**
- **Search and Rescue operations**
- Briefing of the ship's crew prior to the operation
- Creation of a Contingency Plan
- Preparation of fire fighting equipment for possible use

Which three of the following are examples of "latent factors" that may be found to contribute to an accident?

- **Design**
- **Faulty maintenance**
- **Incorrect installation**
- Substandard practises
- Unsafe acts
- Updated procedures

Which three of the following may be an indication that entries in an engine-room log-book have been added, or amended, at a later stage?

- **An entry between others in a different handwriting style**
- **An entry between others using a different pen**
- **An entry made by someone who was not on watch at the time**
- An entry found to have been made by the Chief Engineer

Which two of the following are examples of abnormal behaviour?

- **Commencing the overhaul of a ballast valve, without first checking that it has been properly isolated and the appropriate warning notices put in place in the control room**
- **Operating a manifold crane on which no training and familiarization has been delivered**
- Carrying out a risk assessment and completing a "Permit to work aloft" form before repairing a floodlight on the front of the accommodation
- Wearing footwear with steel-toecaps during storing

Which two of the following are examples of substandard conditions?

- **Congested or restricted work area**
- **Inadequate or defective warning systems**
- Operating equipment at an improper or unsafe speed
- Working under the influence of alcohol or drugs

Which two of the following are examples of substandard practises?

- **Failure to use personal protective equipment**
- **Over-riding or bypassing safety devices**
- Exposure to excessive high or low temperatures
- Missing guards or barriers

Which two of the following are examples of transfer of energy by contact?

- **Damage to spare parts in storage cabinet sustained when work bench broke loose in the electrician's workshop during heavy weather**
- **Steward opens locker door and is hit by falling crockery as ship rolls**
- Failure of electrical control panel resulting in bilge pump not starting
- Wear and tear on the shaft of a sledge hammer, resulting in the head working loose

Which two of the following factors may be evidence of a failing loss control element in a Safety Management System?

- **Inaccurate and out-of-date activities are observed**
- **There are too few system activities**
- Performance standards are specifically defined
- Training standards are too high and demanding

Which two of the following will commonly be amongst an investigator's aims when planning a series of interviews as part of the inquiry procedure?

- **To establish and confirm the sequence of events**
- **To understand how the witness was involved in the incident**
- To determine if the witness is to blame for the incident
- To determine if the witnesses have "arranged a story" between them

Within the context of this training module, which of the following is the definition of a "near-miss"?

- **An unintended or unwanted event, or uncontrolled development in the course of an operation, that almost resulted in unintended harm or damage, or potential for legal action, or is not consistent with acceptable standards**
- An intended or desired event, or controlled sequence of events, in the course of an operation, that was not accounted for in the applicable risk assessment
- An occurrence in which personnel are found to have been operating in compliance with the International Safety Management Code
- An operation, or phase in an operation, in which personnel, the ship, its equipment or cargo, or the environment, almost, but not quite, comply with all applicable regulations and procedures

Within the context of this training module, which of the following is the definition of an "incident"?

- **An uncontrolled development in the course of an operation which has resulted in unintended harm or damage, or the potential for legal action, or is not consistent with acceptable standards**
- A planned event that has resulted in deliberate damage
- A situation where personnel have been injured as a result of failure to comply with the operating requirements of the company's Safety Management System
- An unforeseen event that has resulted in unintended harm or damage

0150 – Maritime English, Superintendent Inspection

Version 2.0 (430)

Click on the phrase you hear.

- **Burner atomizers.**
- Burn a atomizers.
- Burn on atomizers.
- Burning atomizers.

Click on the phrase you hear.

- **Safety valves.**
- Safe valves.
- Say tee valves.

Click on the phrase you hear.

- **Water washing.**
- Water wall spin.
- Water wash in.

Click on the phrase you hear.

- **Funnel top.**
- Fan on top.
- Funnel stop.

Click on the phrase you hear.

- **Boiler dosing chemicals.**
- Boiler dose and chemicals.
- Boiling dose or chemicals.

Click on the phrase you hear.

- **Exhaust gas economiser.**
- Exhaust economiser.
- Exhausted economiser.

Click on the phrase you hear.

- **I'll look at the engine room today.**
- I'll look at the engine in two days.

Click on the phrase you hear.

- **The port boiler uptake.**
- The port boiler upper drain.
- They'll sort the boiler uptake.

Click on the phrase you hear.

- **Fridge compressor.**
- Quick compressor.
- This compressor.

Click on the phrase you hear.

- **Diesel generator crankcase vents.**
- Diesel generating crankcase vents.
- Generate crankcase vents.

Click on the phrase you hear.

- **Diesel generator.**
- These will generate.
- This old generator.

Click on the phrase you hear.

- **Within specification.**
- Winter specification.
- With thin specification.

Click on the phrase you hear.

- **Are you testing them regularly?**
- Are you checking them regularly?
- Run the test on them regularly.

Click on the phrase you hear.

- **We should get them properly balanced.**
- We should get them properly back rinsed.
- With soot, get them properly balanced.

Click on the phrase you hear.

- **We have two complete sets.**
- We have incomplete sets.
- We have to complete tests.

Click on the phrase you hear.

- **It needs a full overhaul.**
- It needs it done overall.
- It's been on for overhaul.

Click on the phrase you hear.

- **When was it last tested?**
- When it was last tested.
- When was it last pressed in?

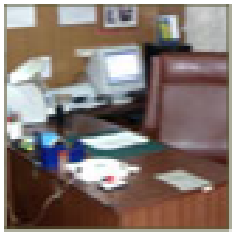
Click on the phrase you hear.

- **We'll have it open on arrival.**
- We'll have an opening trail run.
- You'll have to open up for arrival.

Drag and drop the correct picture onto each label

Boiler	
Purifier	
Steering gear	
Turbocharger	

Drag and drop the correct picture onto each label

Chief Engineer's office	

Spare gear store room	
Steering flat	
Workshop	

Drag and drop the correct picture onto each label

Condenser	
Cooler	
Incinerator	

Purifier



On which pumps had there been a high consumption of mechanical seals in the last few months?

- **The main engine fuel booster pumps.**
- The domestic fresh water pumps.
- The fire and general service pumps.
- The fuel oil transfer pumps.
- The piston cooling pumps.

Select the correct phrase.

Are you . . . the glasses regularly?

- **blowing through**
- blow through
- blowed through

Select the correct word.

Have you . . . any water contamination . . . the lub oil system?

- **had . . . in**
- been . . . in
- done . . . on
- had . . . at
- have . . . in
- took . . . in

Select the correct word.

I'll have it . . . .

- **measured**
- measure
- measuring

Select the correct word.

The Chief Engineer said he had . . . sets of spares.

- **two**
- to
- too

Select the correct word.

The bearings . . . noisy.

- **were**
- am
- be
- is
- where

Select the correct word.

The density . . . 0.983.

- **was**
- are
- were

Select the correct word.

The exhaust temperatures . . . a bit high.

- **are**
- is

Select the correct word.

The inventory . . . up to date.

- **is**
- are

Select the correct word.

The running hours . . . about 14,000 . . . the moment.

- **are . . . at**
- are . . . by
- is . . . for
- is . . . on

Select the correct word.

The ship will dry-dock . . . Singapore.

- **in**
- for
- on

Select the correct word.

There will be some steel replacement . . . the bottom.

- **on**
- at
- for

Select the correct word.

We . . . seven millimetres of . . . on number four.

- **measured . . . wear**
- measure . . . worn
- measured . . . worn
- measured . . . where
- measuring . . . wear
- measuring . . . worn

Select the correct word.

We have just . . . a delivery.

- **received**
- receipt
- receive
- receiving

Select the correct word.

When . . . the heat exchanger last opened up, Chief?

- **was**
- were

Select the correct word.

Where did you . . . your present bunkers?

- **take**
- got
- make
- took

Select the letter to which this phrase applies.

- **A**
- B
- C
- D

Select the letter to which this phrase applies.

- **C**
- A
- B
- D

Select the letter to which this phrase applies.

- **A**
- B
- C
- D

Select the letter to which this phrase applies.

- **D**
- A
- B
- C

Select the letter to which this phrase applies.

- **B**
- A
- C
- D

Select the letter to which this phrase applies.

- **A**
- B
- C
- D

Select the letter to which this phrase applies.

- **B**
- A
- C
- D

Select the letter to which this phrase applies.

- **C**
- A
- B
- D

The Chief Engineer reported a high chloride level in the starboard boiler. What did he say was identified as the cause of the problem?

- **A small leak in the cargo pump condenser.**
- A leak in a heating coil in the port aft bunker tank.
- A pinhole in a tube in the boiler.
- Contamination from the hotwell.

The Chief Engineer was awaiting delivery of a spare light unit. Where was it for?

- **The cargo alarms.**
- The main engine over-speed alarm.
- The telephone system.
- The walk-in refrigerators.

What do we call this?

- **The uptake.**
- The intake.
- The out-take.
- The outflow.
- The upblow.

What is the name of the important piece of equipment that is fitted alongside the bilge pump?

- **The oily water separator.**
- The oil in water separator.
- The oil mist and water separator.
- The oiling water separator.
- The oily water overboard meter.

What is this?

- **The workshop.**
- The engine control room.
- The spare gear store room.
- The steering flat.

What is this?

- **The incinerator.**
- The Electrician's workshop / store.
- The purifier room.
- The spare gear store room.

What is this?

- **A turbocharger.**
- A compressor.
- A diesel alternator.
- A purifier.

What is this?

- **The main sea water pump.**
- The boiler feed pump.
- The domestic fresh water pump.
- The piston cooling pump.

What is this?

- **An engine room fan.**
- A boiler forced draught fan.
- An air receiver.
- The service air compressor.

What is this?

- **The jacket water cooler.**
- The cargo pump condenser.
- The fridge condenser.

What is this?

- **An expansion piece.**
- A compression piece.
- A contraction piece.
- A reduction piece.

What is this?

- **A mechanical seal.**
- A contraction sail.
- A mechanical sole.
- A reduction sale.

What is this?

- **A gasket.**
- A basket
- A gas mask
- A gas ring

What is this?

- **A safety valve.**
- A safety releak valve.
- A safety release valve.
- A safety relive valve.

What specific item did the Superintendent ask the Electrician about when they were discussing the main switchboard?

- **The main breakers.**
- The auxiliary breakers.
- The main switches.
- The shore connection.

When he and the Superintendent were standing beside the settling and storage tanks, the Chief Engineer said he was getting something measured up. What was it?

- **The sections of broken lagging that needed replacement.**
- The depths of the tanks for replacement sounding pipes.
- The size of the identification plates on the tanks for the new IMO safety labels.
- The steel pipe for the new vent to the funnel top.

Where is this?

- **The funnel top.**
- The AC room.
- The purifier room.
- The steering flat.

Which of the following has been written correctly?

- **The air receiver will be surveyed in dry-dock.**
- The manufacturer's representative will servicing the turbocharger.
- The rotors will be landed to be clean.
- There is no problems with the steering gear.

Which of the following has been written correctly?

- **Have you had any contamination in the system?**
- Do you thinking cleaning will be enough?
- The manufacturer's representative will servicing the turbocharger.
- They is tested every three months.

Which of the following has not been written correctly?

- **Alternater.**
- Compressor.
- Condenser.
- Purifier.

Which of the following has not been written correctly?

- **Deisel.**
- Analysis.
- Sample.
- Specification.

Which of the following has not been written correctly?

- **Rotar.**
- Inventory.
- Motor.
- Requisition.

Which of the following has not been written correctly?

- **Overhual.**
- Replace.
- Sample.
- Spares.

Which of the following has not been written correctly?

- **Have you get enough spares?**
- Have you had any problems with this?
- I am here to do my six monthly inspection.
- I'll put it on the list.

Which of the following has not been written correctly?

- **How many tubes was you plug?**
- It may be due to vibration at certain revolutions.
- Where is this motor from?
- You have a small leak.

Which word does not apply?

Have you got . . . spares to satisfy Class minimum requirements, Chief?

- **insufficient**
- adequate
- enough
- sufficient

Which word does not apply?

The bearings were . . . in the port motor.

- **charged**
- changed
- renewed
- replaced

Which word does not apply?

The turbocharger will be . . . in dry-dock.

- **resisted**
- overhauled
- repaired
- serviced

Which word does not apply?

We need to look at any . . . equipment that might be needed.

- **above**
- additional
- extra
- further

Which word does not apply?

We will be . . . a new air compressor in dry-dock.

- **dismantling**
- fitting
- installing

Who is this?

- **The Electrician.**
- The 2nd Engineer.
- The Chief Engineer.
- The Manufacturer's Representative.

Why did the Superintendent instruct the Chief Engineer to have the heavy oil bunker tank be empty and clean on arrival in dry-dock?

- **Because there was going to be some steel replacement on the bottom of the tank.**
- Because it is a dry-dock regulation that all bunker tanks are empty and clean.
- Because it will be used for temporary fresh water storage while the ship is in dry-dock.
- Because the tank will be inspected and surveyed in dry-dock.

0151 – HAZMAT - IMDG Code, Advanced

Version 11.0 (7633)

A cargo transport unit contains 50 drums, each with the UN number 2967 displayed on it. Which number should be displayed in the lower half of the diamond on the placard on the front, rear and both sides of the unit?

- **8**
- 2967
- 3
- 50

A fibreboard package containing 4 metal cans of Caesium Nitrate, individual weight 1kg each, transported under the provisions of chapter 3.4 of the Code, will bear which of the following markings?

- **D**
- A
- B
- C

A marking label coloured green will refer to dangerous goods of which class?

- **Class 2.2 - Non-flammable, non-toxic gases.**
- Class 2.1 - Flammable gases.
- Class 4.1 - Flammable solids.
- Those listed with the UN Environmental Committee approval.

A "Dangerous Goods Declaration" is not required to be provided by the consignor when shipping a package marked "Dangerous Goods of Class 3– Limited Quantity". Which one of the following is a correct assessment of this statement?

- **False.**
- True, but only when contained in rigid containers of not more than 0.3 m^3 capacity, contained within an inner packaging.
- True, but only when the total quantity does not exceed 5 litres.
- True.

According to the IMDG Code, a portable tank, with capacity 3000 litres or more, which is loaded with dangerous goods and which is intended for transport by sea, must display the placards as described by which one of the following statements?

- **On each side and on each end of the tank.**
- On each side and on the back end of the tank.
- On each side only.
- On the back end only.

According to the guidelines in the IMDG Code, what is the maximum mean temperature that should be used in calculations to ensure that the filling of an IBC does not exceed 98% of its water capacity?

- **50°C.**
- 100°C.
- 15.6°C.
- 60°C.

As defined within the IMDG Code, the term "labelling" refers to which one of the following?

- **Identification of the risks and hazards presented by substances contained in a package.**
- Identification of the consignor of the article.
- Identification of the risks and hazards presented by substances contained in cargo transport units.
- Identification of the substance contained in a package.

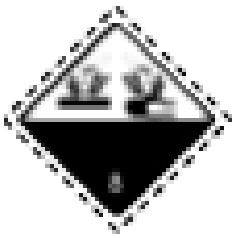
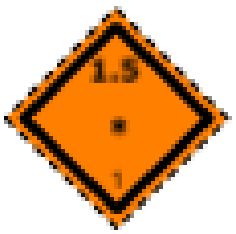
As defined within the IMDG Code, the term "marking" refers to which one of the following?

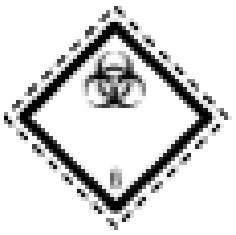
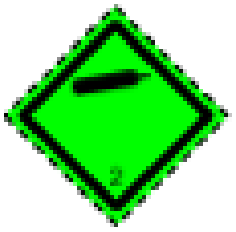
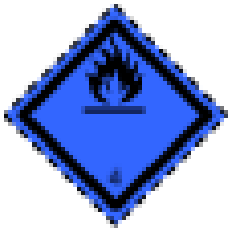
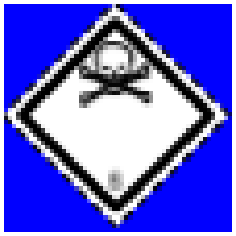
- **Identification of the substance contained in a package.**
- Identification of the consignor of the article.
- Identification of the risks and hazards presented by substances contained in a package.
- Identification of the risks and hazards presented by substances contained in cargo transport units.

Drag and drop the alpha-numeric codes as found in the Dangerous Goods List onto the applicable packing Instructions and Special Packing Provisions:

Packing Instructions for Intermediate Bulk Containers.	IBC
Packing Instructions for large packagings.	LP
Packing Instructions for packagings other than IBCs and large packagings.	P
Special Packing Provisions for Intermediate Bulk Containers.	B
Special Packing Provisions for large packagings.	L
Special Packing Provisions for packagings other than IBCs and large packagings.	PP

Drag and drop the correct class descriptions onto the applicable classes labels:

Corrosive Substances.	
Explosives, division 1.5.	

Flammable Gases.	
Flammable Solids.	
Infectious Substances.	
Non-flammable, non-toxic gases.	
Substances which, on contact with water, emit flammable gases.	
Toxic Substances.	

Drag and drop the substance names to match the correct provisions and descriptions:

- Flexible IBCs shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner. **Magnesium Fluorosilicate**
- Homogeneous mixtures containing not more than 35% by mass and at least 65% of inert substance are not subject to this Code unless criteria of other classes are met. **Azodicarbonamide**
- Marine Pollutant; Packing Group 1; Not permitted for transport under Limited Quantity provisions; **Sodium Cyanide Solution**

- Stowage Category A stowage, stowage code SW1,
  Protected from sources of heat. Handling code H2: Keep as cool as reasonably practicable **Butyric Acid**

Drag and drop the substance names to match the correct provisions and descriptions:

- EmS Fire Schedule E, Spillage Schedule D. **PHENYL MERCAPTAN**
- In contact with acid, evolves silane, a spontaneously flammable gas. **CALCIUM MANGANESE SILICON**
- Stow "separated from" class 5.1, acids and alkalis. **AZODICARBONAMIDE**
- Toxic if swallowed, by skin contact or by dust inhalation. **LEAD NITRATE**

Emergency information on shipments of dangerous goods shall be available at all times and may be provided in which three of the following methods, permitted under the IMDG Code?

- **By keeping separate books such as the EmS Guide and MFAG available at all times.**
- **By making the appropriate entries in the manifest.**
- **On a separate document, such as a Material Safety Data Sheet (MSDS).**
- By providing the ship with internet access details for the USCG Dangerous Substances Transportation Database.
- By stamping the contact telephone number of the responsible emergency service on the package.

How would you expect to find a package that previously contained a consignment of UN Number 2067 marked? It is in an uncleaned condition and there are still heavy residual vapours inside the package.

- **AMMONIUM NITRATE BASED FERTILIZER UN 2067**
- Empty after AMMONIUM NITRATE.
- Empty after UN 2067.
- Empty packages are not required to be marked.

Packages of UN 0477 and UN 2008 are intended for transport by sea. Which one of the following applies?

- **Must be separated by a complete compartment or hold from each other.**
- Must be separated longitudinally by an intervening complete compartment or hold from each other.
- Must be stowed away from each other.
- No specific segregation is required.

Packages of UN 2805 and UN 1120 are intended for transport by sea. Which of the following applies?

- **Must be separated from each other.**
- Must be separated by a complete compartment or hold from each other.
- Must be separated longitudinally by an intervening complete compartment or hold from each other.
- Must be stowed away from each other.

Proper Shipping Names and UN Numbers are used in dangerous goods marking and documentation for which two primary purposes?

- **To ensure that the substance, material or article can be readily identified during transport, especially in the event of an emergency.**
- **To provide the necessary information for the Master to be able to comply with reporting requirements in the event of a pollution incident.**
- To complete the additional security documents when entering a port subject to a level "orange" security alert.
- To demonstrate compliance with MARPOL 73/78 when subject of a Port State Inspection.

The Medical First Aid Guide for Use in Accidents Involving Dangerous Goods provides advice intended for which one of the following?

- **The initial management of chemical poisoning/toxic effects and diagnosis within the limits of the facilities available at sea.**
- The accurate transfer of information relating to a spilled cargo to the appropriate shoreside medical and environmental agencies.
- The comprehensive and total management of chemical poisoning and diagnosis to full recovery, with assistance from shore as required.
- The proper documentation to the flag state authority in the event of a fatality involving dangerous goods on a vessel.

The classes used by the UN for the IMDG Code are based on which ONE of the following?

- **The type of hazard that Dangerous Goods present.**
- The alphabetical order of the list of commonly shipped Dangerous Goods.
- The results of accredited laboratory tests of different chemical groups.
- The UN Numbers of Dangerous Goods.

To what specific source of safety information, as indicated in this program, would you refer in the case of a fire involving a substance identified with the UN number 2257?

- **EmS schedule F-G.**
- Dangerous Goods List, column 12.
- EmS schedule S-N.
- MFAG Table 17.

To what specific source of safety information, as indicated in this program, would you refer in the case of a spillage of a substance identified with the UN number 1230?

- **EmS schedule S-D.**
- Dangerous Goods List, column 16a.
- EmS schedule F-E.
- MFAG Table 16.

To which explosives compatibility Class are "Articles containing a secondary detonating explosive substance, without means of initiation, with a propelling charge (other than one containing a flammable liquid or gel or hypergolic liquids)" assigned?

- **E**
- B
- C
- G

Unless otherwise provided for in the Code, a package containing dangerous goods of UN Number 2196 should be marked as which one of the following?

- **TUNGSTEN HEXAFLUORIDE UN 2196**
- TUNGSTEN HEXAFLUORIDE
- TUNGSTEN HEXAFLUORIDE CLASS 8
- UN 2196

What are the Proper Shipping Name and UN Number for the substance Bromomethane?

- **METHYL BROMIDE with not more than 2.0% chloropicrin UN 1062.**
- BROMOCHLOROMETHANE UN 1887.
- BROMODIPHENYLMETHANE, UN 1770
- BROMOMETHANE , UN 1062.

What does the Code say about the transportation of CHLOROPRENE, STABILIZED as a "limited quantity"?

- **The entry in column 7(a) of the Dangerous Goods List is "0" and therefore indicates that this substance cannot be transported as a limited quantity shipment.**
- The entry in column 16(b) of the Dangerous Goods List has code "SW2" which means that the substance should be stowed clear of living quarters. A limited quantity can therefore be carried, provided it is stowed more than 24 metres away from the accommodation.
- The entry in column 7(b) of the Dangerous Goods List is "E0" and therefore indicates that there is no restriction on this substance being transported as a limited quantity shipment.
- The entry in column 8 of the Dangerous Goods List refers the user to a Packing Instruction 0, which explains how the substance may be transported under "Limited Quantity" requirements.'

What is the maximum permitted temperature on the outer surface of the shell of a portable tank, or its thermal insulation, during transportation?

- **70°C.**
- 100°C.
- 45°C.
- 60°C.

What material would a drum, the marking on which began with be constructed from?

- **Plastics.**
- Aluminium.
- Metal, other than steel or aluminium.
- Steel.

What medical warnings are found in the Dangerous Goods List for 3-TRIFLUOROMETHYLANILINE?

- **Toxic if swallowed, by skin contact or by inhalation.**
- Causes burns to skin, eyes and mucous membranes.
- Corrosive if swallowed.
- High rate of skin decomposition on prolonged contact.

What pressure relief provisions apply to the carriage of TRIETHYLENETETRAMINE in portable tanks?

- **Normal provisions for pressure relief devices apply.**
- Such a tank is required to have a relief device comprising a bursting disc preceding a spring-loaded pressure relief device.
- Such a tank is required to have a relief device constructed of Polycarbonate resin.
- Tanks containing this product are not fitted with a pressure relief device.

What special packing provisions would apply to an IBC, of construction code 11A, used for the transportation of OSMIUM TETROXIDE?

- **Such an IBC shall be carried in a closed cargo transport unit or in a freight container or vehicle, which shall have rigid sides or fences at least to the height of the IBC.**
- IBCs shall not be used for the transportation of this substance.
- Such an IBC shall be sift-proof and water-resistant or shall be fitted with a sift-proof or water-resistant liner.
- Such an IBC shall meet the performance group II performance level.

When a Cargo Transport Unit is loaded with a mixture of dangerous goods and non-hazardous materials, how should the dangerous goods consignments be loaded?

- **Wherever possible, adjacent to the doors, with labels and markings visible.**
- Away from all doors or openings to prevent falling out when the unit is opened.
- On the bottom of the unit, to ensure no damage of other cargo occurs in the event of leakage.
- On top of the non-hazardous material, so as to be readily accessible for priority unloading.

Where in the IMDG Code do you find the provisions relating to temperature control of cargo transport units?

- **Part 7. Provisions Concerning Transport Operations**
- Column 9 of the Dangerous Goods List.
- Part 2.4.3: Substances liable to Spontaneous Combustion.
- Part 5.3.2.2 “elevated temperature substances”.

Where will a dangerous substance, material or article not mentioned specifically by name in the Dangerous Goods List, appear in the IMDG Code?

- **In the List of Generic and N.O.S. Proper Shipping Names.**
- In Packing Provision P200.
- In the List of Class 9 Substances, Materials and Articles.
- It will not appear anywhere in the Code.

Where would you stow a package of dangerous goods, subject to the provisions of Stowage Category D, on a cargo ship, of length 81 metres overall, carrying 12 passengers?

- **ON DECK ONLY.**
- Either ON DECK or UNDER DECK.
- In a locked cargo compartment.
- Such stowage is PROHIBITED.

Which four of the following statements describe characteristics of a gas?

- **Easily compressed.**
- **Having a very low density when compared to the other main states of matter.**
- **Having a very low viscosity when compared to the other main states of matter.**
- **Will readily expand under applied heat.**
- Having an indefinite mass.
- Resistant to mixing and poor at filling a space uniformly.

Which four of the following types of container are covered by the general term "Cargo Transport Unit", as listed in the Code?

- **Portable tanks.**
- **Railway freight wagon.**
- **Railway transport tank.**
- **Road freight vehicles.**
- LASH barge.
- Military armoured vehicles.

Which one of the following code prefixes, as used in the Dangerous Goods List, is applied to Special Packing Instructions for Packagings other than IBC's and Large Packagings?

- **PP**
- B
- L
- LP

Which one of the following is a definition of the control temperature of a dangerous substance as applicable to the IMDG Code?

- **The maximum temperature at which certain substances, such as organic peroxides and self-reactive and related substances can be safely transported during a prolonged period of time.**
- The lowest temperature at which decomposition may occur for a substance in a packaging as used in transport.
- The overall average temperature drawn from daily observations on passage.
- The temperature at which emergency procedures shall be implemented.

Which one of the following is a design feature of all Intermediate Bulk Containers?

- **They must be suitable for handling by mechanical means.**
- They must be capable of withstanding a drop test of 12 metres.
- They must be fitted with a minimum of three pressure-relief valves.
- They must be individually approved by the Flag State authority.

Which one of the following is a valid example of a mark for a packaging containing substances of class 6.2?

- **A**
- B
- C
- D

Which one of the following is the definition of a "consignee", as indicated in the IMDG Code?

- **Any person, organization or Government that is entitled to take delivery of a consignment.**
- A member of a competent authority charged with the administration of the transportation of dangerous goods.
- A shipment of dangerous goods.
- Any person who prepares a shipment for transport.

Which one of the following is used to denote a Marine Pollutant in the Dangerous Goods List?

- **P**
- MP
- P(M)
- PP

Which one of the following statements is correct?

- **The hazard posed by a substance can arise from a dangerous reaction of the substance or article with other substances in the surrounding environment.**
- A dangerous reaction between a substance or article and other substances in the surrounding environment is only possible if those goods are packaged in a container constructed of soft and non water-resistant material.
- Substances in the surrounding environment can have no dangerous impact on packaged dangerous goods.

Which one of the following statements most accurately describes salvage packaging in relation to the IMDG Code?

- **Special packaging into which damaged, defective, leaking or non-conforming packages or dangerous goods that have spilled or leaked are placed for the purposes of transport for recovery or disposal.**
- Packaging in which goods recovered from the sea are held, pending investigation by the appropriate authority.
- Packaging which has been reconditioned after previous use.
- Packaging which is manufactured from recycled material.

Which one of the following symbols would you expect to find on a thermal tank containing liquid of a high temperature?

- **A**
- B
- C
- D

Which one of the options does the mark shown here indicate when found on a package?

- **The package contains dangerous goods in Excepted Quantities**
- The package contains 2.2 Kg of Class 1 explosives
- The package contains a marine pollutant subject to MARPOL Annex 2
- The package contains dangerous goods meeting European Union regulations

Which one of the three Packing Groups is assigned to substances presenting the lowest level of danger?

- **Packing Group III.**
- Packing Group I.
- Packing Group II.

Which one of these is the UN Packaging Symbol?

- **C**
- A
- B
- D

Which one of these labels would be attached to a package containing explosives?

- **A**
- B
- C
- D
- E

Which three of the following sections are contained in the Supplement to the IMDG Code? (Select all applicable answers)

- **Emergency Response Procedures for Ships carrying Dangerous Goods.**
- **The International Code for the Safe Carriage of Packaged Irradiated Nuclear Fuel, Plutonium and High-level Radioactive Wastes on board Ships.**
- **The Medical First Aid Guide for Use in Accidents involving Dangerous Goods.**
- The Dangerous Goods List.
- The List of Generic and N.O.S. Proper Shipping Names.

Which two of the following statements apply to packagings in direct contact with dangerous goods?

- **They shall not be affected or significantly weakened by the dangerous goods.**
- **They shall not cause a dangerous effect in the dangerous goods.**
- They shall be constructed of non-porous material.
- They will absorb a rise in surface temperature of not more than 20°C as a result of such contact.
- They will be deliver a standard of protection equivalent to, or better than, a class 2 (sift-proof and water-resistant) liner.
- They will not exhibit any visible distortion directly - caused by such contact.

0152 – Hatch Cover Maintenance - Bulk Carrier

Version 1.0 (3651)

After defective seals, what is one of the largest causes of serious cargo damage resulting from ingress of water, as highlighted in this module?

- **Blockage of the drain channels, non-return valves or socks by cargo residues**
- Blocked bilges
- Distorted cargo hatches
- Heavy weather

Article III, rule 1 of the Hague-Visby Rules requires the carrier to fulfil which of the following conditions? (Select all applicable answers)

- **Make the holds, refrigerating and cool chambers, and all other parts of the ship in which goods are carried, fit and safe for their reception, carriage and preservation**
- **Make the ship seaworthy**
- **Properly man, equip and supply the ship**
- Ensure that the inside surfaces of hatch covers are maintained in condition suitable for the safe carriage of foodstuffs and animal feed at all times
- Equip and supply the ship to the minimum requirement, as defined by ICAS

As quoted in this program, which of the features listed below, render all hatch covers potentially dangerous? (Select all applicable answers)

- **They are manoeuvred by heavily-loaded wire arrangements, hydraulic cylinders or chains and often make use of gravity, particularly when closing**
- **They move quite quickly, very often with no audible or visible alarms to give warning of their movement**
- Roller wheels frequently jump off the trackways, causing unexpected sideways movement
- Securing devices are considered unnecessary by many crews and changes in draught during loading can cause them to move rapidly and unexpectedly
- They are controlled by wires and chains, which could part at any time if the weight of the hatch comes on them unexpectedly
- They are generally slow moving, which makes their movement difficult to detect

Before attempting to open the hatch covers, what should happen to all cross-joint cleats and wedges?

- **They should be disengaged**
- They should be engaged
- They should be examined for signs of wear or failure
- They should be greased

Defective hatch covers are particularly associated with vessels of which of the following age brackets?

- **10-20 years**
- 0-3 years
- 25-30 years
- 5-10 years

How does a side-rolling, chain-drive "piggy-back" hatch cover work?

- **A dumb panel with no wheels is lifted by hydraulic lift cylinders fitted to the coaming, a rolling panel is moved underneath it and the dumb panel is lowered onto the rolling panel, which is then driven to one side as the hatch is opened**
- A special flexible hinge arrangement allows the first hatch cover to turn through 180 degrees and fold back on top of the second panel; these two then rotate together, coming to rest on top of the third panel, and so on
- The hatches are opened by an arrangement of wires or chains, which stack the hatches one on top of the other at one end of the hatch coaming
- The panels are lifted one on top of the other at one end of the hatch coaming by means of a travelling gantry crane

How is a chalk test carried out?

- **Chalk is spread on the compression bar and the hatch is then closed; when it is re-opened, the seal is examined for the absence of chalk at any point, which indicates a lack of contact**
- Chalk is spread on the compression bar and the hatch is then closed and sealed; water is applied by hose; when the hatch is re-opened, any leak will be indicated by blue discoloration of the chalk where water has penetrated
- Chalk is spread on the compression bar and the hatch is then closed; when it is re-opened, the seal is examined for the presence of chalk at any point, which indicates a lack of contact
- Chalk is spread on the hatch cover seal and the hatch is then closed; when it is re-opened, the seal is examined for traces of cargo in the chalk, which indicates a leak

If coaming movements are expected to be significant, what type of seal will the designer commonly consider fitting?

- **A sliding rubber seal**
- A double lip seal
- A hollow section seal
- A seal comprising of a sponge core with outer skin
- A solid seal
- An inflatable seal

If it is assumed that the covers have been properly installed, which of the following lead to the majority of the problems of accelerated wear of hatch seals? (Select all applicable answers)

- **Large vertical deflections of hatch cover corners, due to solar heating of the top plating, or when cargo is loaded on top**
- **Unexpectedly large relative movement between hatch cover and coaming**
- Blocked drain channels or non-return valves
- Improperly sited locators
- Loads coming from inside the hatch covers, such as those due to sloshing water ballast
- Over-compression caused by corrosion or breakdown of the steel-to-steel contact

If personnel are in, or could be in the vicinity of a hatch cover, what should you do before you open or close it?

- **Use all available means of communication to notify them that a hatch cover movement is about to take place**
- Ask them to leave the area
- Nothing – it is for them to take due care
- Walk all around the hatch cover, so that you can complete the "inspection" section of the safety checklist

If there has been water ingress on passage, how will this normally be seen?

- **It will usually follow the pattern of the cross-joint and coaming seals**
- It will be at the sides of the cargo, having run down the sides of the hold
- It will be on the centreline
- It will be under the most forward panel on a hatch

In a side-opening rack and pinion hatch cover arrangement, what is the function of the "pot lifts"?

- **To raise a short section of rail to bring the rolling wheels up to the running level**
- To balance the hatch cover on the compression bar
- To deliver power to the rolling wheels
- To drive the continuous wire connecting (and opening) both panels
- To lower the longitudinal section of guide rail on either side of the hatch and take the rolling wheels down to the running level
- To maintain the seal compression

In addition to those items that would normally be covered at an annual survey, which of the following is checked at a 5-year special survey? (Select all applicable answers)

- **Thickness measurements of plate and structural stiffeners**
- **Watertightness, as established by a hose test or equivalent**
- Class approval of all maintenance records
- Measurement of any clearances between the hatch cover wheels and the trackway when the hatch is in the last 10% of movement before reaching the stops

In an ultrasonic test, what is the severity of a leak referenced to?

- **The open-hatch reading**
- The calibration reading obtained in the surveyor's laboratory
- The closed-hatch reading
- The reading obtained on the opposite side of the hatch

In determining when a hatch cover seal has reached the point when it needs to be replaced, what general "rule of thumb" is commonly applied?

- **When permanent set reaches half the design compression**
- After not less than 9 months and not more than 15 months service
- When a groove is seen in the seal
- When impact damage or localized corrosion is found on the top of the compression bar
- When it appears as a recommendation following a Class survey
- When permanent set reaches full design compression

In order to understand the basic principle behind hatch cover design, how must the hatch be considered?

- **As a dynamic (mobile) object on the coaming**
- As a rigid object on the coaming
- As a static object on the coaming
- As an open area continually exposed to salt water

In terms of the safety of hatch covers, which of the following are appropriate considerations? (Select all applicable answers)

- **All fittings used for routine or emergency operation must be of adequate strength**
- **They must be able to be fully secured in the open position and released without endangering personnel**
- **They must be able to be operated without endangering personnel up to the maximum angles of heel and trim which are specified**
- Facilities must be in place for hose testing to take place at not more than 30 day intervals
- Their vertical edges must be marked with luminous paint
- Wires and chains used for opening and closing must be tested under full load at 6-monthly intervals
- X-ray analysis must take place of any suspected or actual welding failures, under the supervision of a Class surveyor

In terms of the sealing of a hatch cover, what are the design requirements? (Select all applicable answers)

- **Cargo must be kept inside the holds**
- **Seal material must be compatible with cargo carried**
- **The seal must be easily maintained**
- **The seal must be resilient or elastic enough to make up for hull and coaming deformations**
- **The seal must be suitable for expected weather conditions**
- **Water must be kept out of the ships holds**
- The seal must be maintenance-free
- The seal must be rigid enough to resist any compression

In which of the following operational conditions would a seal with a sponge core with an outer skin made of nitrile rubber commonly be used?

- **For oil cargoes on an OBO vessel**
- On side-rolling cover fitted to a bulk-carrier
- On vessels with large open hatchways, where coaming movement will be a problem
- Where there are very heavy hatch covers

In which of the following operational conditions would a solid rubber seal commonly be used? (Select all applicable answers)

- **Where the vessel trades in Arctic conditions**
- **Where there are very heavy hatch covers**
- For oil cargoes on an OBO vessel
- On vessels with large open hatchways, where coaming movement will be a problem

In which of the following operational conditions would an extruded hollow section seal commonly be used?

- **On side-rolling cover fitted to a bulk-carrier**
- For oil cargoes on an OBO vessel
- On vessels with large open hatchways, where coaming movement will be a problem
- Where the vessel trades in Arctic conditions

Industry statistics clearly indicate a high proportion of reported cargoworthiness defects as being caused by which of the following areas of operation?

- **Hatch covers, their seals and securing arrangements**
- Cargo containers
- Cargo handling cranes, wires and control systems
- Negligence in securing the ship for sea
- Overloading
- Untypical weather conditions

It is February 3rd. You have just loaded a cargo of gravel in Trondheim, Northern Norway. You anticipate that, if you close the cargo hatches, the rubber hatch seal may stick to the compression bar. What can you do to prevent this?

- **Spread a non-organic grease or commercial glycerine on the compression bar prior to closure**
- Apply whale grease to both the compression bar and the hatch seal surface 15 minutes before closing the hatch
- Heat the rubber hatch seal with steam immediately prior to closure
- Replace the hatch seal with special reinforced compound sealing material

What controls the geometry of seal compression on hatch covers?

- **The steel-to-steel contact**
- The age of the vessel
- The dimensions of the hatch cover
- The length of the chains or wires
- The thickness of the compression bar
- The weight of the hatch cover

What general rule does this program suggest designers of hatch covers, those who maintain them and those who operate them should direct their efforts towards?

- **Defending the thin line where the compression bar and the seal rubber meet**
- Ensuring seals must be resilient or elastic enough to make up for hull and coaming deformations
- Minimizing the time delays due to opening and closing of hatch covers
- That intervals between replacement of hatch seals are maximised

What is the class requirement for a hose test?

- **Nozzle of 15–18mm in diameter, water pressure of about 2 bar, the jet of water aimed not more than 1metre from, and directly at, every part of the sealing areas, moved at not more than 1 metre per 3 seconds, where the hatch covers are closed and cleated.**
- Nozzle of 10-15mm in diameter, water pressure of about 3 bar, the jet of water aimed not more than 1metre from, and directly at, every part of the sealing areas, moved at not more than 1 metre per 10 seconds, where the hatch covers are closed but not cleated.
- Nozzle of 20–24mm in diameter, water pressure of about 1.5 bar, the jet of water aimed not more than 1.5metres from, and directly at, every part of the sealing areas, moved at not more than 1 metre per 5 seconds on the cross-joints and 1 metre per 3 seconds on the peripheral joints, where the hatch covers are closed and cleated.

What is the function of the non-return valves fitted to the drain pipes at the corners of the coamings?

- **To permit water to drain freely onto the deck but stop water from the deck entering the hold**
- To allow water to drain from, and allow air into, the drain pipes
- To permit air to enter the hold and allow water to drain freely onto the deck
- To stop water from, but allow air into, the hold

What is the minimum length of any straight piece of rubber seal inserted during a repair, as recommended in this module?

- **500 mm**
- 100 mm
- 1000 mm
- 250 mm
- 300 mm
- 750 mm

What is the name given to the restraining devices fitted to hatch covers to absorb longitudinal and athwartships forces that will be experienced due to the motion of the ship when at sea?

- **Locators**
- Containers
- Cross-joints
- Implanters
- Indicators
- Wedges

What is "permanent set"?

- **The groove in the hatch seal face, resulting from the loss of elasticity**
- The distance from the outer edge of a rubber hatch cover seal to the middle of the groove created by the penetration of the compression bar, measured in mm and averaged from readings taken on all four sides of the hatch
- The exact angle at which the hatch cover lies when closed
- The exact position in which the hatch cover rests when closed
- The group of hatch seals which are in continuous contact with the coaming compression bar when the hatch is closed
- The limit of good steel on the compression bar down to which steel deterioration may be considered acceptable

What normally determines the size of lift-away (pontoon) covers on container ships?

- **That they are capable of being lifted by shore-side container gantry cranes**
- That the longitudinal length of any one hatch cover does not exceed 0.25 of the registered length of the vessel on which it is fitted and the width is not more than 0.85 of the moulded breadth.
- That they are capable of supporting a minimum four-stack of 40' containers
- That they are fitted with an adequate number of anti-lift bolts to resist anticipated upward forces generated by stacks of containers as identified in the ship's cargo handling manual

What technique do automatic cleats employ to hold the hatch firmly in place once closed?

- **Wedges on the hatch cover engaging mechanically in matching wedges on the coaming**
- They are applied and tightened using hydraulic cylinders to provide the force
- They are dropped into place and then tightened up with a spanner
- They slide across like a wedge bolts

What type of hatch cover is frequently fitted with an indirect cross-joint cleating arrangement, where wedge-shaped battening devices attached to the covers engage in coaming sockets, causing the panels to be forced together as they are lowered onto the coaming?

- **Side-rolling covers**
- Four panel folding covers
- Pontoon covers
- Stacking covers

What will an inspection prior to discharge primarily establish?

- **Whether the measures taken in sealing the hatch after loading were successful**
- Whether the checklist completed after loading was correctly filled out
- Whether the hatch was closed properly
- Whether the non-return valve was clear
- Whether there is any damage to the hatch cover panels
- Whether there is any damage to the trackway
- Whether there was unusually heavy weather on passage

When considering the security of a hatch cover arrangement, what measures must be incorporated in the design specification? (Select all applicable answers)

- **Cleats need to be resilient to allow for small horizontal and vertical coaming movements at sea**
- **There must be means fitted to prevent the hatch cover sliding longitudinally or athwartships when at sea**
- **There must be means of keeping the hatch cover in contact with the coaming**
- There must be balanced sets of wheels on each panel, to ensure even movement during closing
- There must be no more than two pairs of peripheral cleats and six cross-joint cleats per panel

When considering the strength of a hatch cover, what kind of loads must it be able to withstand? (Select all applicable answers)

- **Loads coming from cargo loaded on top of hatch covers**
- **Loads coming from inside the hatch covers, such as those due to sloshing water ballast**
- **Loads coming from outside the hatch, such as seawater over decks and hatches**
- Loads produced by adjacent holds, when loaded
- Loads produced through the ship's hull (varying with draft)
- Structural loads induced by deck penetrations for cranes, derricks, etc

When de-ballasting is about to start, why is it necessary to have the vents on water-ballast hatches open?

- **If they are closed, the tank may collapse under vacuum**
- Because it is a requirement on the safety checklist
- Because it makes cleaning easier
- If they are closed, they may be forgotten when the ship goes to sea

Which of the following are among the critical hatch cover spares to be carried on board, as recommended in this program? (Select all applicable answers)

- **Cleats**
- **Running wheels and bearings**
- **Seals**
- Chain links
- Lengths of hose for temporary drain “socks”
- Luminous paint for marking hatch cover sides
- Wire ferrules and hard eyes

Which of the following are common patterns of cross-joint cleat? (Select all applicable answers)

- **Manual**
- **Quick-acting**
- **Screw**
- Automatic
- Hydraulically-actuated

Which of the following are common patterns of peripheral cleat? (Select all applicable answers)

- **Automatic**
- **Hydraulically-actuated**
- **Manual**
- **Quick-acting**
- Cross-joint wedge
- Screw

Which of the following is a description of the function of periphery cleats around a hatch cover?

- **They provide vertical restraint, stopping the panel lifting, while permitting some relative movement between the coaming and the cover**
- They provide horizontal restraint, control the cross-joint seal compression and ensure the edges of the hatch cover panels at the cross joint remain firmly pressed or held together and that the cross joint seal rubber remains properly compressed
- They provide transverse restraint, restricting the horizontal movement of the panel, ensuring that the compression bar remains within the groove in the hatch seal

Which of the following is checked at an annual survey? (Select all applicable answers)

- **Evidence of any significant changes to the hatch cover system since the last survey**
- **The condition of seals, compression bars, tracks, cleats, stoppers and the panel structures**
- Documentary evidence of monthly inspections by ship's personnel
- That class approval has been given for any and all repairs to the hatch covers and attendant structures and fittings
- Thickness measurements of plate and structural stiffeners
- Watertightness, as established by a hose test or equivalent

Which of the following is the definition of "steel-to-steel geometry", as explained in this module?

- **The hatch cover is designed such that when it rests, with steel-to-steel contact between the side plate or the support pads and the coaming, the geometric relationship between the compression bar and the seal rubber is exactly right**
- The hatch coamings are designed such that when the hatch cover rests on the seal, with continuous and even contact between the compression bar and the seal rubber along the length of the hold opening perimeter, the geometric relationship between the support pads and the coaming is exactly right
- The tolerances during manufacture of hatch covers are less than 0.01% of the side plate thickness

Which of the following is the definition of "weathertightness", as given in this program?

- **That in any sea condition water will not penetrate into the ship**
- As identified in the applicable reading drawn from the IACS empirical scale for hatch cover thickness, based on distance from the bow of the ship
- That a minimal amount of water will penetrate if the hatch is impacted by excessively heavy weather conditions
- That hatch covers are installed in such a way as to reduce the ability of moisture to penetrate the seal by a combination of design measures that minimise the effects of sea water impacting on the hatch panels
- That in normal sea conditions, any water ingress past the hatch seals can be suitably controlled without the application of additional defensive measures

Which of the following may be considered as key design elements of hatch design? (Select all applicable answers)

- **Safety**
- **Sealing**
- **Security**
- **Strength**
- **Support**
- Flexibility
- Minimal expansion and contraction due to changes in temperature
- Resistance to bulk chemical cargoes

Which of the following must there be a comprehensive record of, in order to be able to demonstrate "due diligence"?

- **All associated classification survey histories**
- **Inspections**
- **Regular maintenance**
- **Repairs**
- **Testing**
- Communications with the vessel's superintendent
- Personnel on board at the time of any cargo-related incidents or structural failures

Who performs the statutory annual survey on the hatch covers?

- **A classification society surveyor**
- A cargo surveyor
- A port-state control inspector
- The chief engineer
- The company safety manager
- The master
- The superintendent

Why is a chalk test considered to have limited value in establishing the watertightness of a cargo hatch cover?

- **Because it gives no indication of the amount of seal compression that exists when the ship is actually underway in a seaway**
- Because compression of the seal is marginal
- Because it does not show where the rubber comes into contact with the compression bar
- Because it is carried out before any cargo is loaded
- Because it is out carried out at sea

You are carrying out maintenance on number 3 hatch cover. The permanent set has reached half the design compression in places and there are signs of wear. You have cut out two metres of bad seal along the port side. What length of rubber will you actually cut to insert in its place?

- **2.06 m**
- 1.94 m
- 1.97 m
- 2 m
- 2.03 m

You are onboard a bulk-carrier, in ballast, at sea. You are to clean and examine the inner faces of the side-rolling hatch covers on no 2 hold. What particular safety measure is required?

- **Safety fencing, consisting of portable stanchions with an upper and lower safety wire, should be in place**
- All personnel not directly involved in the inspection should leave the deck
- Ensure the covers can move freely at any time
- Nothing particular is required
- Personnel should be instructed not to climb up onto the hatch cover at any time

You have just replaced the seal on number 2 hatch. What should you now do?

- **Measure the seal compression and record it**
- Compare the appearance of the new seal to that of the old seal
- Measure the groove in the seal in the forward corners of the hatch
- Secure the lid as tightly as possible and leave it closed until arrival; in the next port

0155 – Stowaways, Migrants and Refugees

Version 1.0 (2904)

A group of stowaways is discovered in a container on board. There are three males and one female. There are no other females on board. What is the recommended policy on searching the female stowaway?

- **Carry out a minimal, non-contact search and then isolate the stowaway until proper assistance can be obtained from shore.**
- Ask the stowaway if she minds being searched first, make a written record of her response and ask her to initial the entry.
- Carry out a full search anyway but do not make any record of the procedure.
- Get the stowaway to change into new clothing, examine that which has been removed. Consider that adequate and take no further action.

A regular cargo ship is to pick up a group of 16 refugees, who have been adrift in a small boat in the South China Sea. It will be some time before the ship reaches her next port. Which of the following resources may be seriously impacted by their presence on board?

- **Available living space.**
- **Food stocks.**
- **Medicines.**
- **Water reserves.**
- Communication facilities.
- Fuel supplies.
- The ship's speed.

A ship arrives in port. There are concerns about stowaway activity in the area. How should the master regard the level of security risk that has been declared by the port authority?

- **He should assess the consequent risk to the vessel of the level and adjust his own procedures if necessary.**
- He should accept the security risk level declared by the Port Facility Security Officer .
- He should ignore the security risk level declared by the Port Facility Security Officer and automatically increase the on-board level to the maximum as required by the Ship Security Plan . . .

A stowaway is discovered on board. He is very sick when found. What practical steps may be taken by master to safeguard the health and safety of others on board? (Select all applicable answers).

- **Contact control, such as gloves, masks and protective eye-glasses should be employed when searching the stowaway.**
- **The hiding place in which the stowaway had been hiding should be disinfected.**
- **The stowaway should be isolated.**
- **The stowaway should be supplied with clothing and the clothing he was wearing when discovered should be incinerated.**
- **The stowaway should be supplied with his own bedding.**
- **The stowaway should be supplied with his own crockery and cutlery.**
- The stowaway should be hosed down before he is taken into the accommodation.
- The stowaway should be supplied with medicines to treat himself, but nobody should touch him and he should not be moved until he is well.

A stowaway is discovered on board. The vessel is at sea, on passage and in international waters. What legal rights does the stowaway have?

- **He has a right to expect provision to be made for his security, general health, welfare and safety until disembarkation.**
- He has a right to asylum in the first port of call.
- He has a right to communication facilities on board, to advise other parties in his country of origin and the United Nations Refugee Administration of his safety and ongoing treatment on board.
- He has a right to employment on the vessel.
- He has no rights.

According to industry representatives, what is the estimated cost of stowaways to the shipping industry and their P&I Clubs?

- **Around US$ 20,000,000 a year**
- An average of US$ 50,000 per ship of 500 grt or over.
- Approx. € 3.6 million per year.
- US$ 10,000 per incident

Amendments to which IMO Convention first made proposals related to crew and passenger lists and specific recommendations concerning the administration and handling of stowaways?

- **The Convention on Facilitation of International Maritime Traffic (1965 FAL Convention).**
- The 1957 International Convention relating to Stowaways.
- The 1979 International Convention on Maritime Search and Rescue.
- The First United Nations Convention on the Law of the Sea (UNCLOS 1956).
- The International Convention for the Safety of Life at Sea (SOLAS 1974).

As part of gangway security, a record of persons coming on board should be maintained. Why will this prove of value as the ship prepares to leave?

- **Because a check can be made that every person who boarded the vessel, has subsequently returned to shore.**
- If the Port Security Officer visits the ship at this time, he will immediately see that security has been maintained throughout.
- It will confirm whether all crew members are on board.
- So that the foreman stevedore can tally the hours his men have worked and present a timesheet to the chief officer.

Does the master have the legal right to require a stowaway to work? (Select the most correct answer).

- **No.**
- No, unless written permission is obtained from the embassy or consulate of the stowaway's country of origin.
- Yes, but only if the stowaway agrees.
- Yes.

For which of the following can a ship be fined? (Select all applicable answers)

- **Allowing a stowaway to escape.**
- **Arriving in port with a stowaway on board.**
- Arriving in port with over the permitted number of stowaways.
- Arriving in port without a complete medical declaration for a stowaway.
- Failing to arrange for a stowaway to secure an entry visa.
- Failing to indicate an assembly point for a stowaway on the ship's Muster List.

If a person is discovered hiding in a lifeboat during the vessel's pre-departure search, how will he normally be considered within the law?

- **As a trespasser.**
- As a dangerous criminal.
- As a migrant worker.
- As a simple nuisance.
- As an illegal entrant.
- As having alien status.

If a stowaway will not cooperate and give his identity, how might the master establish it?

- **He should search the stowaway's clothing and belongings. Some clue as to his identity may be discovered.**
- He should contact the agent in the port of departure and request records of known stowaways.
- The master should encourage the crew to befriend the stowaway and see if they can get any information from him.
- The stowaway must be physically forced to divulge information.

In general, which of the following situations will result in a higher fine for a vessel and her operators?

- **The vessel arrives in port with a stowaway on board. The stowaway is declared to the authorities. During a busy and complex cargo operation on the first night alongside, the stowaway disappears.**
- The vessel has a standard complement of 23 persons, as advised on the Crew List mailed to the ship's agent in the port of destination 3 days before arrival. The next day, a stowaway is discovered in the vessel's paint locker. He is detained and presented to the authorities on arrival.

Is it recommended by the IMO "Guidelines on the allocation of responsibilities to seek the successful resolution of stowaway cases" that a ship's master, having discovered stowaways on board his ship at sea, should divert from his planned voyage to disembark them? (Select the most correct answer)

- **No, unless he receives confirmation that repatriation has been arranged with sufficient documentation and that permission has given for disembarkation, or unless there are extenuating security or compassionate reasons.**
- No, he must always maintain his planned voyage under any circumstances.
- Yes, but only if he receives verbal confirmation from the ship's agent that it will be possible to arrange disembarkation of the stowaway.
- Yes, stowaways should always be disembarked at the first opportunity, irrespective of geographical location of the port in question, or the potential safety risks to the stowaway.

Security patrols will check that all appropriate spaces are locked. What additional purpose do they serve?

- **They are a visual deterrent to potential stowaways.**
- They enable cargo operations to be performed in safety.
- They ensure compliance with statutory port security regulations.
- They keep the watchkeepers alert.

Specialist equipment for detecting concealed persons may work by measuring which of the following? (Select all applicable answers).

- **Body heat.**
- **Elevated Carbon Dioxide (CO_2) levels.**
- **The presence of a heart-beat.**
- Body odour.
- Gamma-rays.
- Reduced Carbon Monoxide levels.

The charterer may be liable for some or all of the costs associated with stowaways. Under what circumstances could this be the case?

- **Special provision may be written into the charter party, indicating that the charterer accepts responsibility for costs when stowaways have hidden in the charterer's goods and boarded the vessel in that way.**
- Charterers are always legally responsible for all costs associated with stowaways on vessels contracted to them under international law.
- Charterers are always totally exempt from liability for costs associated with stowaways under international law.
- Special provision may be incorporated in the Mate's Receipt, indicating that costs associated with stowaways will be determined by arbitration in the flag State of the chartered vessel.

The ship is approaching a small boat, which is believed to contain a group of refugees, with a view to rescuing them and bringing them on board. What, at all times, must at all times be the main priority of the master?

- **The safety of his own ship and her crew.**
- Minimizing the time required for the rescue operation and the costs to the shipowner.
- The recovery of the boat and the refugees' belongings.
- The rescue of the refugees on the boat.

What action is a master, who is in receipt of information that persons are in distress and is in a position to be able to provide assistance, required to take?

- **He is bound to proceed with all speed to their assistance, if possible informing them or the search and rescue services that the ship is doing so.**
- He is required to proceed to the general area at an appropriate speed and then report to the controlling rescue coordination centre.
- He must inform the charterer of the situation and await further instructions.
- He must inform the shipowner of the situation and await further instructions.

What is the essential difference between the way in which those seeking asylum and those considered as refugees will be handled by the immigration authorities?

- **Asylum is usually dealt with on a case-by-case basis, while modern refugee law is written around large-scale movements of population.**
- Political asylum can only be applied if the number of applicants in any one group is less than 12. If greater than this number, then refugee law is applied.
- The authorities will seek to prove membership of an oppressed political movement or religious organization before considering asylum claims; if this cannot be done, they will consider application for entry based on the more general international refugee law.

What is the significant legal difference between migrants and refugees that are rescued at sea and stowaways that are found on board?

- **Rescued migrants and refugees board a ship with the full knowledge and permission of the master, but stowaways board a ship without such permission and consent, intending to remain on board undetected, at least until the vessel is at sea.**
- Refugees and migrants always carry valid and proper identification documents, while stowaways always have false papers.
- Refugees and migrants will in general be genuine, law-abiding and cooperative, while stowaways may often be violent or engaged in criminal activity.
- Refugees attempt to conceal themselves and escape detection, while stowaways openly seek official approval from the authorities.

What practical measures can be taken to prevent persons boarding illegally and stowing away? (Select all applicable answers).

- **Access control.**
- **Patrols.**
- **Pre-departure searches.**
- **Securing doors and spaces.**
- Daylight only cargo operations.
- Posting of notices prohibiting access.
- The use of luminous barrier tape around restricted areas.

What significant impact did the introduction of the ISPS Code have on statistics on stowaway activity since 2004?

- **In ports which have successfully implemented the appropriate measures, it has been more difficult for potential stowaways to gain access to a ship.**
- Records have revealed that, since this date, stowaway activity has been confined to North Africa, the Far East and Central America.
- Since that time, more comprehensive statistics have been possible due to the reporting requirements under the Code.
- The average cost per port call expended on shore security services has reduced progressively since this date.

Whenever a stowaway is discovered on board, who is responsible for the proper and accurate reporting of the relevant details to the authorities, the owner, charterer and other interested parties?

- **The master.**
- No reporting is required.
- The stowaway.
- The UNHCR.

Which medical condition may these people be suffering from? Drag and drop the likely condition onto the corresponding set of circumstances.

- These migrants were in a small boat. Their water supply had run out six days before they were found. **Dehydration.**
- These people were on a raft, without shade, for several days in a tropical area. **Sunburn.**
- These refugees had been in an open boat in rough weather. Temperatures have approached freezing at times and it has been raining continuously. **Hypothermia.**

Which of the following does the United Nations Convention on the Law of the Sea require the master of a ship to do?

- **He is "to render assistance to any person found at sea in danger of being lost".**
- He is directed "to assist any person who, at any port or place in the vicinity thereof, secludes himself in a ship without the consent of the shipowner or the master or any other person in charge of the ship and is discovered on board after the ship has left that port or place."
- He is "to establish, operate and maintain such search and rescue facilities as are deemed practicable and necessary to provide a means of locating and rescuing persons at sea."
- "If requested by a Rescue Coordination Centre (RCC), he is bound to proceed with all speed to their assistance, if possible informing them or the search and rescue services that the ship is doing so."

Which of the following is a correct statement?

- **A person cannot automatically obtain official refugee status.**
- A displaced person has automatic right to refugee status under international refugee law.
- A refugee has automatic right to political asylum.
- An economic migrant has automatic access to full social rights in his country of choice.

Which of the following is a principle, defined in the 1979 IMO International Convention on Maritime Search and Rescue that must be applied to the rescue of persons in distress at sea?

- **Rescue must take place regardless of the nationality or status of such a person or the circumstances in which that person is found.**
- Rescue should only take place if the master has a guarantee from the appropriate authorities that persons taken on board can be landed at the next port.
- The master has the ultimate authority to decide whether he will divert and assist on the basis of the commercial impact of his action.
- The master must carefully consider whether his own vessel will be able to provide enough food, water, accommodation, medicines, etc. for an indefinite period before deciding whether to rescue persons in distress.

Which of the following is the correct definition of a "displaced person" as recorded in this module?

- **A person who as a result of the actions of the authorities, has been deported from, or has been forced to leave the country of his or her nationality or former habitual residence.**
- A person relocating in an attempt to obtain a better standard of living.
- A person who has left his or her country of origin, has applied for recognition as a refugee in another country and is awaiting a decision on their application
- A person who is discovered in the process of travelling, other than in his or her country of residency with documentation that gives a reasonable and valid explanation for their journey.
- A person who is found in a place that is not his or her country of citizenship or residency, trading or otherwise operating a business for profit and not registered for taxation purposes in that country.
- A person who lives temporarily or permanently in a country where he or she was not born, and has acquired some significant social ties to this country

Which of the following is the definition of a "place of safety" by the IMO Guidelines on the Treatment of Persons Rescued at Sea?

- **A place where a survivor's safety of life is no longer threatened and where their basic human needs can be met.**
- A location where medical treatment is available and a refugee will be housed and fed, awaiting processing by the immigration authorities.
- A port recognized by the UNHCR.
- A rescue ship.
- A state where a refugee's claim for asylum will be recognized.
- Solid land.

Which of the following is the definition of an "asylum seeker" as described in this program?

- **A refugee, trying to reach a country where asylum may be granted, or economic migrants claiming asylum, so as not to be repatriated.**
- A migrant, demanding to travel to another country which is not his own, with the deliberate intention of registering for social benefits in that country.
- A person hoping to avoid the attention of the immigration authorities and hoping to enter a country undetected.
- A person who is attempting to escape war or civil unrest, or has experienced political or religious persecution.

Which of the following is the definition of an "attempted stowaway", as described by the FAL Convention?

- **A person who is secreted on a ship, or in cargo which is subsequently loaded on the ship, without the consent of the shipowner or the master or any other responsible person, and who is detected on board the ship before it has departed from the port.**
- A person who boards a vessel for the purposes of work but fails to return to shore by the time of departure and remains on board as the vessel puts to sea.
- A person who, at any port or place in the vicinity thereof, secludes himself in a ship without the consent of the shipowner or the master or any other person in charge of the ship and who is on board after the ship has left that port or place.

Which of the following is the definition of an "economic migrant" as described in this program?

- **A person moving simply in an attempt to obtain a better standard of living.**
- A migrant, demanding to travel to another country which is not his own, with the deliberate intention of registering for social benefits in that country.
- A person who has been forced to leave his or her native place.
- A person who is attempting to escape war or civil unrest, or has experienced political or religious persecution.
- A refugee, trying to reach a country where asylum may be granted, or economic migrants claiming asylum, so as not to be repatriated.

Which of the following maritime codes is generally believed to have led to a reduction in the number of stowaways in many areas recent years?

- **The ISPS Code.**
- SOLAS
- The BC Code
- The IMDG Code.
- The Life Saving Appliance Code.
- The Maritime Inspection and Testing Code.

Which of the following may be among those factors taken into account when planning the rescue of persons in distress at sea? (Select all applicable answers).

- **The available light.**
- **The condition of the craft from which they are being recovered.**
- **The number of persons being rescued.**
- **The physical condition of those being rescued.**
- **The prevailing weather conditions.**
- **The resources available on board . . .**
- **The type and manoeuvring characteristics of the rescuing vessel.**
- The potential financial loss if the vessel stops to pick up the refugees.

Which of the following sea areas, as listed in this module, has recorded large-scale, sea-borne migrant and refugee activity since 1990? (Select all applicable answers)

- **The Eastern Atlantic Ocean between the Canary Islands and South-west Spain.**
- **The Gulf of Aden between Somalia and Yemen.**
- **The Mediterranean Sea between North Africa and southern Europe.**
- The Black Sea.
- The Indian Ocean between Tanzania and the Seychelles.
- The Tasman Strait.

Why should a stowaway be detained in secure quarters on board? (Select all applicable answers).

- **For the security of the ship.**
- **For the stowaway's own safety**
- **Medical grounds.**
- **To lessen the chance of stowaway escaping.**
- **To lessen the chance of theft or other criminal activity.**
- **To limit contact with the crew.**
- So that there is no need to include them in safety drills.
- To comply with ISPS Code requirements.

You are at sea and discover a stowaway on board. The man has no documents or other proof of identity. How is it likely that the immigration authorities will consider him when your ship arrives in port?

- **As an illegal entrant.**
- As a convicted criminal.
- As a crew member.
- As a refugee.
- As a tresspasser.
- As an economic migrant.

You are conducting a pre-departure stowaway search. You find three men hiding in the rope locker? What do you do?

- **You report your finding to the Responsible Officer immediately. You do not attempt to physically restrain the men. You secure the door to prevent their escape. You stand by for assistance from the shore authorities.**
- You immediately report your discovery to the foreman stevedore and negotiate a fee for him to arrange the removal of the men to shore. You record the appropriate details in the ship's logbook.
- You seal the locker hatch with a chain and padlock and report your discovery to the Responsible Officer at the end of the search.
- You summon other crew members and physically remove the men to the ship's office where you keep them under heavy guard. You secure the rope locker hatch to prevent any further stowaways entering.

You arrive in port with a stowaway on board. He is not making any claim for asylum. Before they consider permitting disembarkation, what will immigration authorities invariably demand the stowaway produces?

- **Valid travel documents.**
- A birth certificate (or certified copy).
- A letter of confirmation of identity from the ship's agent.
- A letter of guarantee from an embassy or consulate.
- A letter of invitation from a friend in that country.
- A medical certificate showing he is HIV-negative.
- Valid records of yellow fever and typhoid inoculations.

You discover three male stowaways on board. They are young, in good health and willing to work. The ship is on her last passage before entering dry-dock and there is a lot of preparation work in progress. Why should the master not put the stowaways to work? (Select all applicable answers).

- **Because it is illegal under international law.**
- **Because the safe operation of the ship could be at risk by using untrained labour.**
- **Because the ship's work routine and team set-up may be disrupted.**
- **Because there may be health risks by allowing the stowaways to share work facilities with the crew.**
- **Because they may claim against the shipowner for wages.**
- **If one of the stowaways is injured in the course of work on board, he may claim against the shipowner for compensation.**
- Because the stowaways then become eligible to be entered on the crew list and are entitled to the protection of the flag State.

You have departed Dar Es Salaam in East Africa and are bound for a European discharge port. You discover a stowaway on board. Should he be forced into making a claim for asylum?

- **No.**
- Yes.

Your ship is approaching port. You are assigned to secure all storage lockers, machinery rooms, deckhouses and pumprooms not required to be open for cargo handling or for safety reasons. What should you do before sealing each door?

- **Follow Enclosed Space Entry procedures and check that no ship's personnel are working inside.**
- All emergency lighting within the space should be tested.
- Check that the space has been ventilated as required.
- Check that there is a back up set of keys to open the padlock, if this is the method of securing.
- Post warning notices, in English, the language of the crew and the local language outside all locked doors.

0156 – Piracy and armed robbery 1

Version 5.0 (2703)

By what name are the larger vessels that are used by pirates to extend their operations out into ocean areas generally known? Select ONE of the given options.

- **Mother ships.**
- Control ships.
- Father ships.
- Support Vessels.

If a merchant vessel is successfully hijacked by a small group of pirates, what can normally be anticipated within a short while afterwards? Select ONE of the given options.

- **More pirates will board.**
- A mother-ship, if used, will make fast to the vessel.
- The crew will immediately be removed from the vessel.
- The pirates will give the Master a letter claiming ownership of his vessel, to comply with international law.

If a pirate attack is happening on a vessel, which kind of message would the vessel broadcast on the VHF radio? Select the ONE correct answer?

- **Mayday**
- Pan Pan
- Sécurité

If a small craft suspected of being involved in pirate activity, is manoeuvring dangerously directly ahead of the vessel, what should the Master do? Select ONE of the given options.

- **Maintain speed and zig-zag manoeuvre.**
- Manoeuvre according to reports from the forecastle lookout.
- Slow down and manoeuvre to avoid the craft.
- Slow down, but warn the engine room to be ready to increase speed again immediately.

If a vessel is attacked within territorial waters, what might it NOT be able to depend upon?
Select one of the given options.

- **Assistance from warships of an international counter-piracy force operating in adjacent international waters.**
- Any other vessel picking up her emergency messages.
- Being able to out-run the pirates.
- That the pirates will be less prepared, being closer to shore.

International law recognises that piracy takes place on the high seas. Which term is generally applied to similar aggressive acts that take place on a vessel within a state's territorial waters?

- **Armed robbery.**
- Attempted seizure.
- Regional piracy.
- Territorial piracy.

It is recommended that the accommodation doors are secured against entry during transit of a known piracy area. Which ONE of the options specifies the door that is generally suggested to be used as the designated access for patrol and lookout purposes?

- **A bridge deck or wheelhouse door.**
- An internal door, leading to the machinery space.
- The aftermost door, leading onto the poop deck.
- The door closest to the crew mess room.

Pirates generally observe a potential target before attacking. Which of the following will they typically look for on a merchant ship as signs of weakness? Select THREE of the options.

- **Lack of preparation and alertness.**
- **Slow response when approached.**
- **Slow speed and low freeboard.**
- Faulty radio and radar equipment.
- High speed and low freeboard.
- Shallow draught and high freeboard.

Pirates have boarded a vessel and managed to reach the bridge, where they are threatening a crew member with injury. Select ONE of the options to best describe the action the vessel Master should take.

- **Cease any further resistance and attempt to calm the situation.**
- Advise the pirates that they must release the CM before you will have anything to do with them. Ignore any instructions they give while they still hold the CM.
- Continue to resist the pirates and attempt to expel them from the bridge.
- Offer them money and hope that they will leave quickly.

Select ONE of the options that best describes the primary role of an on-board security team.

- **To assist and advise the Master on security for a high risk area.**
- To shoot pirates.
- To talk with pirate gang leaders and make sure the vessel is not attacked.
- To tell the crew what to do if pirates get on board.

Select ONE of the options which best describes the value of a citadel on a vessel, sailing in an area with a naval presence, if pirates board the vessel.

- **It can buy time for the naval task force to reach the vessel.**
- The Master will be able to communicate directly with the task force commander from the citadel and get advice.
- The Master will be able to coordinate and control all defence measures against the pirates from this one location.
- With all ship's personnel in the citadel, the pirates will not be able to sail the vessel or control the engines and will give up and leave.

Select ONE of the options which best identifies the product that is recommended to be applied to a window to prevent glass splinters flying around if it is hit by a bullet.

- **Shatter-resistant, adhesive film.**
- A layer of clear varnish on the inside.
- A layer of clear varnish on the outside.
- Adhesive tape, arranged in an "X" pattern.

Select ONE option that best describes the kind of signal that the Ship Security Alert System sends.

- **A covert, silent ship-to-shore signal, alerting the Company, the Flag State authorities and closest national authorities.**
- A loud, pre-arranged Tannoy signal, which is duplicated on the ship's whistle.
- A pre-configured message on Digital Selective Calling frequencies, indicating the ship's name, her position and the nature of the emergency.
- An open speech, pre-recorded verbal message announcing the pirate attack.

Select ONE option that best describes the nature of nitro-glycerine explosive.

- **Nitro-glycerine is a clear, very dangerous liquid.**
- Nitro-glycerine is a clear liquid needing a detonator to make it explosive.
- Nitro-glycerine is a clear liquid safe to handle without a detonator.
- Nitro-glycerine is a solid material.

Select ONE option that best describes the nature of plastic explosive.

- **Plastic explosive is a soft, pliable material.**
- Plastic explosive is a clear liquid that is very unstable.
- Plastic explosive is a hard, brittle material.
- Plastic explosive is always a bright orange colour for identification.

Select ONE option that describes the likely course of action that pirates who wish to negotiate a ransom for a hi-jacked vessel will take.

- **They will take the vessel to a pirate held port.**
- They will keep the vessel at sea.
- They will land the crew and sail the vessel to a different location.
- They will take the vessel to its home port.

Select ONE option to correctly identify the kind of wire that should be arranged around the main-deck rails or bulwarks to prevent boarding.

- **Razor wire.**
- Barbed wire.
- High-tensile steel mesh netting.
- Rusty wire.

Select ONE option to explain why pirate boats sometimes broadcast false radio distress messages.

- **To try to get a merchant vessel to reduce speed in response.**
- As a coded communication to other pirates.
- Because the pirates get excited and activate the alarm by mistake.
- To frighten a merchant vessel before attacking it.

Select ONE term to best describe the successive counter-piracy defensive arrangements supporting a merchant vessel.

- **Layered Defence.**
- Best Management Defence.
- Concentric Support Measures.
- Ringed Defence.

Select the ONE option that describes what personnel on the open deck, such as lookouts, should do immediately if the pirates get close and start firing.

- **Stay out of the line of fire and retreat inside the accommodation.**
- Crouch down and make a report to the bridge.
- Make their way across to the other side of the vessel and continue to monitor developments when possible.
- Prepare fire hoses to direct at any pirates who attempt to board the vessel.

The primary information on preparing the vessel against attacks by pirates is located in which of the following sources?
Select ONE of the options.

- **The Ship Security Plan.**
- Alerts from the Piracy reporting Centre.
- NAVTEX broadcasts.
- The ISPS Code.

The reduction in the number of successful pirate attacks is attributed to the international counter piracy measures and which one of the following factors?

- **Vessels being better prepared.**
- Falling competence levels amongst the pirates.
- Increased media coverage.
- Vessels generally being faster.

There is good evidence to show that pirates will avoid attacking which ONE of the following vessels?

- **A vessel that is well prepared against attack.**
- A new ship.
- A ship that has a satellite communications dome.
- A vessel with a bulbous bow.

To avoid injury from weapons fired at a vessel by pirates, what is the main basic guideline to be followed?
Select ONE of the given options.

- **The crew should remain under cover.**
- The crew on deck should run to the side which is not being fired on.
- The crew should all be issued with steel helmets.
- The crew should display a sign saying ‘explosive material on board’.

What speed should a vessel maintain in order to put pirates off from attacking? Select the ONE correct answer.

- **15 knots.**
- 10 knots.
- 25 knots.
- 8 knots.

What type of vessel do pirate boats sometimes mix with as a form of disguise, particularly when operating near to the coast? Select ONE of the given options.

- **Local fishing boats.**
- Naval patrol boats.
- Pilot boats.
- Yachts.

When the shipping company contracts with a Private Maritime Security Company and sends a team onboard, who will direct the vessel's crew in their safety and security duties? Select ONE of the options.

- **The Master and ship's officers.**
- The Company Security Officer.
- The Master and Security Team Leader will decide between them.
- The Security Team Leader.

Which ONE of the following is a particular feature of piracy activity off West Africa?

- **Oil cargo and bunker theft.**
- Taking vessels and crews hostage and demanding ransom.
- The use of high-tech monitoring systems to identify targets well ahead of interception.
- Violence associated with an inability to communicate with the pirates in a common language.

Which ONE of the following options will NOT normally be part of an onboard security team's duties?

- **Negotiating with pirate gang leaders for a safe passage through their waters.**
- Assisting the crew in constructing barriers as per the BMP guidance document.
- Conducting an onsite security risk assessment of the vessel.
- Running anti-piracy drills.

Which ONE of the following organisations broadcasts regular "live" piracy alerts and bulletins?

- **The International Maritime Bureau Piracy Reporting Centre.**
- INTERBULK
- The International Maritime Organization (IMO).
- The United Nations News Agency.

Which ONE of the given expressions best describes the tiered system of defensive measures for a merchant vessel to defend itself against pirate attack?

- **Layered defence.**
- Limited defence.
- Logical defence.
- Logistical defence.

Which ONE of the options best describes the areas for which extra vigilant monitoring by look-outs is recommended for?

- **Blind and shadow sectors behind the vessel.**
- Directly abeam, either side of the vessel.
- The quadrant ahead of and to starboard of the vessel.
- Under the bow, directly ahead of the vessel.

Which is the first (outer) layer in the "Layered defence" strategy for a merchant vessel to defend itself against pirate activity?

- **Avoidance of known high risk areas.**
- Defeat the boarding.
- Delay the entry.
- Deny access to the accommodation.

Which item is considered to be the most effective item of a military or naval anti-piracy task force? Select the ONE correct answer.

- **The helicopter.**
- The high-speed rigid inflatable boat (RIB).
- The international intelligence network.
- The laser-guided missile.

Which of the following are examples of navigational procedures that have been implemented in known piracy areas to better protect merchant ships? Select TWO of the given options.

- **The creation of "safety corridors".**
- **"Group Transit" merchant ship convoys.**
- The monitoring of the movement of small boats by helicopters.
- The total ban on all radio communications between merchant ships.
- The use of special light signals to be displayed by merchant ships.

Which person is responsible for establishing and monitoring the general fleet-wide anti-piracy procedures? Select the ONE correct answer.

- **The Company Security Officer.**
- Shipping companies do not have fleet-wide policies; Each ship is responsible for establishing its own procedures.
- The Company Personnel Manager.
- The Port Security Officer at the departure port.

Who is responsible for the conduct of anti-piracy lookouts during the transit of a high-risk area? Select ONE of the given options.

- **The Officer of the Watch (in accordance with the Master's Orders and Company Procedures).**
- Each individual is responsible for his own conduct when in a piracy area.
- The Security Team Leader (as per the contract between the Shipping Company and the Private Marine Security Company).
- The Ship's Safety Officer (as per the ship's Safety Management System).

Why can the Master and Watchkeeping Officers not rely on radar to alert them to the approach of pirates in small boats? Select TWO of the given options.

- **Because the boats are often made of wood and present poor radar targets.**
- **There may be blind spots in the radar coverage.**
- Because pirate boats are fitted with devices that interfere with the radar picture.
- Because the Master and Officers cannot watch the radar screens all the time.
- Because the pirate boats will move too fast for the radar to track them.

Why should crew members not attempt to manually direct fire hoses at pirate boats?
Select TWO of the given options.

- **Because their actions may attract gunfire and they are exposed.**
- **Because they may not get back into the secure accommodation in time.**
- Because a sudden increase in water pressure may knock them off their feet and therefore reduce the security status of that area of the deck.
- Because all available personnel should be used in extra look-out positions.

0157 – TMSA 2

Version 2.0 (2090)

Although best practices and operating experience is shared through TMSA, who is able to see a TMSA submission?

- **Only those identified by the operator submitting the TMSA .**
- Any member company of OCIMF.
- Any operator taking part in the TMSA programme.
- Any prospective charterer or body who OCIMF recognise.

During a TMSA self assessment it might not be possible to reach a certain stage until certain changes have been made. Which one of the following statements best describes the course action that should be taken?

- **Enter the TMSA now without waiting for the changes to be made.**
- Do not enter the TMSA until all the changes have been made.
- Enter a TMSA as though the changes had already been made.
- Submit a partially complete TMSA.

Following the submission of a TMSA, which one of the following statements best describes the next step?

- **Conduct a gap analysis to identify the elements and stages which should be prioritised.**
- Allow a gap of at least one year before starting the next improvement cycle.
- Wait for feedback from senior management.
- Wait to hear from OCIMF.

If actions are required during TMSA, but there are not the resources to carry them out, which one of the following statements best describes what should be done?

- **The issue should be referred to senior management.**
- It should be recognised that this stage cannot be reached.
- The issue should be referred to OCIMF.
- The KPI should be marked as "Not Applicable".

If there is any doubt about the meaning or interpretation of any terms or phrases in TMSA, which one of the following statements best describes the action to be taken?

- **Consult the Glossary in part 4 of TMSA.**
- Consult the meaning of that term in SOLAS.
- Contact OCIMF for assistance.
- Refer the matter to senior management.

In TMSA, the steps in the continuous improvement cycle are called:

- **"Plan" -"Act"- "Measure"- "Improve".**
- "Plan" -"Act"- "Measure"- "Do".
- "Plan" -"Do"- "Measure"- "Review".
- "Think" -"Act"- "Review"- "Improve".

The abbreviation "TMSA" stands for which one of the following:

- **Tanker Management and Self Assessment**
- Tanker Management and Safety Assessment
- Tanker Management and Statutory Assessment
- Tanker Manning and Self Assessment

The best performance of a management system in TMSA is known as:

- **Stage 4**
- Best Practice
- ISM compliant
- Stage 1

What is the name of the regulations that govern the safe management and operation of ships and pollution prevention?

- **ISM Code**
- MARPOL
- SOLAS
- TMSA

Which one of the following is not a feature of an effective management system?

- **Total reliance is placed on procedures.**
- Clear goals are set by senior management.
- Hazards and risks are systematically identified.
- Measurable indicators are used to judge the system's effectiveness.

Which one of the following is not a valid reason to declare a key performance indicator in TMSA "not applicable"?

- **When senior staff think it might be too difficult.**
- When local regulations prevent it being applied.
- When the company is very small and cannot achieve it.
- When the nature of the trading pattern prevents it.

Which one of the following is not one of the 12 elements of TMSA?

- **Budget management**
- Environmental management
- Navigational safety
- Safety management.

Which one of the following is not the purpose of the best practice guidance within TMSA?

- **It must be followed exactly at all times.**
- It enables good operating procedures to be shared within the industry.
- It enables lessons learned from incidents to be shared within the industry.
- It provides recommendations when drawing up new procedures.

Which one of the following parts of TMSA are 'key performance indicators' specifically linked to?

- **Each stage of an aim**
- Each objective
- Each overall aim
- Part Two

Which one of the following statements best describes TMSA?

- **A voluntary self assessment tool to help operators improve their management systems.**
- A mandatory assessment system put in place by ISM.
- A mandatory assessment system put in place by OCIMF.
- A voluntary self assessment tool aimed at compliance with the ISM Code.

Which one of the following statements best describes a key performance indicator (KPI) within TMSA?

- **A clear value or standard against which performance can be measured.**
- A clear value or standard required for ISM compliance.
- A clear value or standard required for ISM or MARPOL compliance.
- A general guideline for performance measurement.

Which one of the following statements best describes how senior management or OCIMF can validate a TMSA submission?

- **There should be referenced documentary evidence available.**
- As it is a self assessment, this will not be necessary.
- By questioning those involved.
- By seeing which KPI's have been ticked off.

Which one of the following statements best describes how to determine which stage of an aim has been reached?

- **By careful consideration of the KPI's for that stage.**
- By careful consideration of the best practices for that stage.
- By careful consideration of the guidance notes for that stage.
- By careful consideration of the last ISM audit.

Which one of the following statements best describes what are known as the 'elements' within TMSA?

- **They set out performance measures and best practices for the twelve essential parts required in a management system.**
- They exactly match the requirements of the ISM Code.
- They set out twelve basic rules of management that must be followed.
- Two or more elements are grouped together to form an Aim in TMSA.

Which one of the following statements best describes what the emphasis of TMSA should be?

- **Achieving long-term improvements.**
- Applying a quick fix.
- Cost saving.
- Regulatory Compliance.

Which one of these organisations is TMSA obtained from?

- **OCIMF**
- IMO
- INTERTANKO
- OPEC

0158 – Maritime English, Report Writing

Version 1.0 (3830)

A cook slipped and fell in the galley. What should not be stated in a report covering this accident? (Select all applicable answers.)

- **In my opinion the cook lost his balance.**
- **In my opinion the cook slipped.**
- The cook slipped and fell.
- The cook slipped because he was wearing improper footwear.
- The cook slipped.

A cook slipped and hurt himself in the galley. What information would not be required in a report? (Select all applicable answers.)

- **He had been preparing breakfast prior to the accident.**
- **The second cook had been assisting with preparing breakfast.**
- He also hurt his head whilst falling.
- He slipped.
- He was wearing improper footwear.

A ship has lost its bow. Why would it be necessary to state that upon departure from the load port, there remained 200 tonnes of ballast in the fore peak tank? (Select all applicable answers.)

- **Due to the circumstances of the accident, it was a pertinent fact.**
- Because it is a mandatory reporting requirement.
- Because that was the status of the vessel's tanks upon departure from the load port.

Although designed for verbal communications, what can you refer to for report writing purposes?

- **The GMDSS handbook.**
- Standard Marine Communication Phrases (SMCP).
- The Bridge Procedures guide.
- Webster's dictionary.

Based upon what should you make recommendations in your report?

- **The cause of damage.**
- Past experience.
- The extent of damage.
- The nature of damage.
- The opinions of the ship's safety committee.
- Witness accounts.

Before starting to write a report what should you decide upon? (Select all applicable answers.)

- **Collecting witness statements.**
- **Whether the report covers an accident or incident, a near miss, hazard or a non-conformance.**
- The report audience.
- Whether or not photographs were taken.

Captain Pitroda often confuses the word "weather" with "whether". At which stage of the report writing process should he identify and correct this mistake?

- **Editing**
- Drafting
- Pre-writing
- Revising

Captain Pollock usually writes long and complicated sentences in his report. At which stage of the report writing process should he simplify the language in his report?

- **Revising**
- Drafting
- Editing
- Pre-writing

Chief Engineer Melville needs to include photographs of the damaged turbocharger, taken by the Second Engineer, into his report. At which stage of the report writing process should he collect this evidence?

- **Pre-writing**
- Drafting
- Editing
- Revising

How should ship positions be reported? (Select all applicable answers.)

- **As a bearing from a charted object.**
- **In terms of latitude and longitude.**
- As a bearing from another ship.
- By describing the area in which the ship is.
- By the ship's distance from the nearest port.

How should your reports be written?

- **In the correct chronological order.**
- **In the past tense.**
- In order of the importance of events.
- In the present tense.

In a ship damage report, why is it necessary to clarify that there were two second mates on board? (Select all applicable answers.)

- **Because if not clarified, someone reading the report might assume there was only one second mate on board.**
- **Because if not clarified, the reported actions of one second mate may be confused with the actions of the other.**
- **Because this might need clarification by investigators in future.**
- Because the duties of each second mate are different.

In which stage of writing process is it advisable to put all your thoughts on paper without worrying about punctuation and spellings?

- **Drafting**
- Editing
- Pre-writing
- Revising

Prior to an incident, a verbal agreement had been made between an oil terminal and a vessel. Why would it be necessary to state this fact in your report? (Select all applicable answers.)

- **Because it was a normal practice carried out by this specific vessel.**
- **Because the verbal agreement constituted a non-conformity.**
- **To emphasise that nothing was agreed in writing**
- Because both the vessel and terminal were bound by this verbal agreement.

The sudden operational failure of which of the following could result in a hazardous situation on board? (Select all applicable answers.)

- **Inert gas scrubber during cargo operations**
- **Navigation lights at night**
- **Steering gear or associated pumps**
- Emergency fire pump during a port state inspection
- Emergency generator during routine testing

What action is required where there is a major non-conformance?

- **Immediate corrective action**
- A log entry
- An emergency on-board safety meeting
- Future corrective action

What are non-conformances?

- **Departures from standard shipboard procedures**
- **Departures from the requirements specified in a ship's SMS**
- Compromising situations in which the outcome could have resulted in an accident or incident
- Compromising situations in which the outcome resulted in an accident or incident
- Unintentional and undesired events

What do you understand by, "Lagging and other debris were found strewn around the main engine top plates?" (Select all applicable answers.)

- **Lagging and other debris were found all around the main engine top plates.**
- **Lagging and other debris were found lying around the main engine top plates.**
- **Lagging and other debris were found scattered around the main engine top plates.**
- Lagging and other debris were found to have damaged the main engine top plates.

What do you wish to convey by saying, "The vessel was extremely sluggish"?

- **The vessel was extremely slow-moving.**
- The vessel did not respond to rudder movements.
- The vessel listing heavily.
- The vessel was rolling heavily.

What does the word 'there' indicate? (Select all applicable answers.)

- **A location, such as the engine room or accommodation.**
- Something belonging to more than one person.
- They are.

What is an accident? (Select all applicable answers.)

- **An event that causes damage to cargo.**
- **An event that causes injury.**
- An event that does not, necessarily, lead to environmental pollution.
- An intentional event.

What is an incident? (Select all applicable answers.)

- **An accidental event inconsistent with an acceptable set of standards**
- **An accidental event that often happens due to certain developments spinning out of control during an operation**
- An accidental event that does not cause harm or damage
- An event consistent with an acceptable set of standards

What should accompany your reports? (Select all applicable answers.)

- **Hand sketches.**
- **Photographs.**
- **Witness statements.**
- Crew list.
- Ship's particulars.
- The ship's logbook.

What would be the correct response to: 'What is your present maximum draught?'

- **My present maximum draught is nine point three meters.**
- My present draught is six point three meters forward and seven meters aft.
- My ship is on an even keel.
- My ship is trimmed zero point five metres by the stern.

What, apart from an unforeseen circumstance, could cause an event to spin out of control?

- **A non-conformity**
- A hazard
- An accident
- An incident

Which of the following are incidents? (Select all applicable answers.)

- **A fire in the main engine scavenge space**
- **An oil spill**
- A ship damaging the harbour wall
- A ship losing its bow
- An error in navigation

Which of the following makes it mandatory for ships to report non-conformities, accidents, and hazardous occurrences?

- **The ISM Code**
- Company's instructions
- MARPOL 73/78
- SOLAS.74

Which of these are major non-conformances? (Select all applicable answers.)

- **By-passing the oily water separator.**
- **Incorrect entries in the oil record book.**
- **Not carrying out risk assessments, where required.**
- **Not issuing 'hot work' permits, where required.**
- Non-existent records of safety meetings on board.
- Poor records of safety meetings on board.

Which of these can be regarded as a report recommendation?

- **Securing points should be routinely inspected against an appropriate checklist.**
- Fire-fighting equipment had not been regularly inspected.
- Fire-fighting equipment had not been regularly tested.
- Had engine-room ratings not been working in the engine room, the consequences of this incident could have been a major engine-room fire.

Which of these events can be categorised as a 'near-miss'?

- **A ship pushed off her berth with a cargo arm attached, but without incident.**
- A general lack of shipboard safety standards.
- A rising flood tide.
- Tugs not being available to keep a ship safely moored to her berth.

Which of these is not a hazard? (Select all applicable answers.)

- **Crew injury**
- **Ship collisions**
- A ship exposed to risk
- A ship's personnel exposed to risk
- The outcome of a situation that could result in an accident
- The outcome of a situation that could result in an incident

Which of these is the correct form of response to the question: "From what direction is your vessel approaching?" (Select all applicable answers.)

- **"My vessel is approaching from the eastern lighthouse."**
- **"My vessel is approaching from the South."**
- "From Manila."
- "My vessel is approaching from Manila."

Which of these is the correct form of response to the question: "What is your position?" (Select all applicable answers.)

- **"My position is fifteen degrees north, ninety five degrees west."**
- **"My position is three point six nautical miles east of Small Island".**
- "At the harbour limit."
- "In the anchorage".

Which of these sentences is correct?

- **The crew were in their mess room at the time of the incident.**
- The crew were in there mess room at the time of the incident.
- The crew were in they're mess room at the time of the incident.

Which of these statements does not describe an event?

- **Associated alarm lights**
- **Lubricating oil and boiler feed water**
- Activation of associated alarms
- Overflow of boiler feed water
- Overflow of lubricating oil
- Surging of main engine turbochargers

Which of these statements is correct?

- **The ship passed through the straits of Gibraltar.**
- The ship passed threw the straits of Gibraltar.
- The ship passed throughout the straits of Gibraltar.
- The ship through the straits of Gibraltar.

Which of these statements is correct?

- **The crew searched for the missing seaman in vain.**
- The crew searched in vane for the missing seaman.
- The crew searched in vein for the missing seaman.

Which of these statements is correct?

- **'Vein' refers to a part of the body through which blood flows.**
- 'Vain' refers to a part of the body through which blood flows.
- 'Vane' refers to a part of the body through which blood flows.
- 'Vene' refers to a part of the body through which blood flows

Which stage of the report writing process is appropriate for checking spellings and correcting punctuations?

- **Editing**
- Drafting
- Pre-writing
- Revising

Why are many near-misses not reported? (Select all applicable answers.)

- **Because an individual's competency might be questioned.**
- **Because near-misses come close to things going wrong when, in fact, they don't.**
- **Because of the misconception that if something didn't go wrong, there is no need to report it.**
- Because a near-miss situation cannot be used as a learning tool.
- Because there is no need to report a near-miss.

Why are photographs an important tool with regard to report writing? (Select all applicable answers.)

- **Because photographs capture minor details that you may not have noticed or recorded.**
- **Because they may serve as evidence in a court of law.**
- It is a statutory requirement to add photographs to a report.
- It is standard practice to add photographs to a report.

Why is a ship that loses its bow termed an accident? (Select all applicable answers.)

- **Because it is an undesired event.**
- **Because it leads to disastrous consequences.**
- **Because it might occur during an operational phase of the ship.**
- **Because it might result from an incident.**

Why is it mandatory to report a 'near miss'?

- **Because it is an ISM Code requirement.**
- Because it is a company requirement.
- Because it is a flag-state requirement.
- Because it is a port-state control requirement.

Why is it necessary for narrative reports to be accurately written? (Select all applicable answers.)

- **So that a person who was not involved with the occurrence can understand clearly what the report coveys.**
- **So that there is a clear record of events that occurred.**
- So that a person's report writing skills can be improved upon.
- So that there are no conflicting statements regarding dates or times of certain important events.

Why would an oil spill be termed an incident?

- **Because the spillage would have been inconsistent with an acceptable set of standards.**
- Because the spillage would have been consistent with an acceptable set of standards.
- Because the spillage would not have been controlled.
- Because the spillage would not have been foreseen.

Why would an outbreak of fire in a main engine scavenge space be termed an incident? (Select all applicable answers.)

- **Because the fire would have occurred from developments whilst operating the main engine.**
- **Because the fire would not have been controlled.**
- Because the fire would not have been foreseen.
- Because the fire would not have occurred from developments whilst operating the main engine.

Why would it be necessary to report that a Master makes no mention to authorities of (a) damage to a vessel such as loss of its bow or (b) oil pollution incident? (Select all applicable answers.)

- **Because the information could be required in follow-up investigations.**
- **To provide facts to the report audience.**
- To comply with a company's reporting procedures.
- To comply with the ISM Code requirements.

Why would it be necessary to report that despite a ballast pump taking suction from a forepeak tank, the level of water in the tank remained constant? (Select all applicable answers.)

- **Because it would indicate that the forepeak might be open to the sea.**
- **Because it would indicate that there was a problem with the forepeak.**
- Because it was logged in the ship's logbook
- Because the ballast pump suddenly loses suction

Why would it be necessary to report upon a Master's instructions to abandon ship? (Select all applicable answers.)

- **Because the instructions would be in accordance with safety procedures.**
- **Because this is important information that could be useful in any follow-up investigations**
- Because it is part of a report format
- Because this type of information is required under the ISM Code.

0160 – Fire Fighting - Basic

Version 5.0 (1465)

How does carbon dioxide extinguish a fire in an enclosed space?

- **By depriving the fire of oxygen.**
- By cooling down the fire.
- By cooling the air around the fire.
- By reacting chemically with the fire.

How does foam remove oxygen from a burning oil fire?

- **By smothering it.**
- By changing it.
- By cooling it.
- By mixing with it.

How is heat normally removed from a class A fire?

- **By using water or foam.**
- By using CO2 or foam.
- By using dry powder or CO2.
- By using dry powder.

If there is a fire onboard, who will stop ventilation, close fire flaps and start the emergency fire pump if required?

- **Those allocated to do this on the muster and emergency list.**
- Fire team 1, working under the guidance of the chief officer.
- Fire team 3, working under the guidance of the bosun.
- The bridge team, who know what needs to be done.

Is the following statement true or false?
'Always attempt to extinguish a fire before sounding the alarm'.

- **False**
- True

On a vessel, which plan lets you see where fire extinguishers are located?

- **The fire control plan.**
- The emergency plan.
- The general arrangement plan.
- The loading plan.

On a vessel's muster list and emergency instructions, how many of the crew are named?

- **All crew members.**
- Only the deck crew.
- Only the deck officers.
- Only the engineer officers.

What are the four parts of the fire tetrahedron?

- **Fuel, heat, oxygen and a chemical reaction.**
- Fuel, heat, flammable gas and a chemical reaction.
- Fuel, heat, oxygen and a violent reaction.
- Fuel, smoke, oxygen and a chemical reaction.

What information is found on a fire control plan?

- **The location of the fire-fighting appliances.**
- How the Master can be reached in case of emergency.
- How to respond when a fire is discovered.
- How to use the fire-fighting appliances correctly.

What is one characteristic of a metal fire?

- **It is very hot.**
- It is relatively cool.
- It is very common.
- It is very easy to start.

What should you do if you detect a fire?

- **Sound the alarm, and then try to extinguish the fire.**
- First try to extinguish the fire, and then activate the alarm.
- Immediately muster at your fire station.
- Report to the bridge.

Which fire extinguisher is most suitable for putting out fires involving metal?

- **Special powder.**
- Carbon dioxide.
- Dry Powder.
- Water.

Which of the words describes what happens to a fire when oxygen is removed from it?

- **Smothering.**
- Damping.
- Quenching.
- Stimulating.

Which type of fire can water mist be used to quench?

- **Any type apart from D.**
- Class A and D only.
- Class B and C only.
- Class F only.

Which type of fire extinguisher is best suited to putting out an oil fire in the galley?

- **A suitable foam.**
- Dry Powder.
- Water - applied by a jet.
- Water - applied by a spray.

Which type of fire is carbon dioxide especially suitable for fighting?

- **One where electricity may be present.**
- Class A, ordinary combustible materials.
- Class B, oil and other flammable liquids.
- Class D, a metal fire.

Who is likely to be in charge of the firefighting operation if there is a fire in accommodation?

- **The chief officer.**
- The chief engineer.
- The master.
- The second officer.

0161 – Fire Fighting - Advanced

Version 2.2 (760)

Dry distillation can cause which of the following?

- **Fire gases**
- Breathing equipment failure
- Flash burns
- Structural failure

If someone does not report to their mustering station during a fire what should be done?

- **Conduct a search.**
- Advise the fire fighting teams to keep an eye out for them but continue extinguishing operations.
- Find someone to take over their duties at that station.
- Use the PA system to ask them to report to their mustering station.

In the event of a fire investigation what should the crew do?

- **Aid in the collection of evidence.**
- Clean away debris from the site.
- Make sure all equipment that was used is put back in the proper place.
- Make sure that any problem that could have caused the fire is already fixed.

In the event of a fire involving dangerous goods onboard, where can you find information on the chemical reactions taking place?

- **On its data card.**
- In the cargo manifest.
- In the hazardous materials catalogue.
- On stickers placed on the cargo's container.

It is important that messages to smoke divers are made:

- **short and precise.**
- before they enter the fire zone.
- on VHF ch. 06.
- through hand signals in noisy environments.

One litre of water equals approximately how many litres of steam or mist?

- **1700**
- 1200
- 1400
- 1900

What is the first phase of onboard fire fighting tactics?

- **Alert**
- Evaluate
- Extinguish
- Rescue

What is the first thing that should be done upon detecting a fire?

- **Notify the bridge.**
- Initiate the fixed fire fighting systems.
- Meet up at your muster station.
- Send a distress call.

What is the most common cause of engine room fires?

- **Oil leaks and hot surfaces.**
- Cigarettes coming in contact with flammable waste.
- Faulty electrical equipment.
- Overload on the engine.

What is the safest and most effective way to apply water in a fire fighting operation onboard a vessel?

- **Fog**
- Direct spray
- Flooding
- Sprinkle

What kind of fires should be reported?

- **All types of fires.**
- Any fire involving damage to equipment.
- Any fires involving death or injury.
- Large fires requiring aid from local authorities.

What should fire-fighters always be equipped with?

- **Communications equipment**
- Axes
- GMDSS
- Reflective metallic suits

When discovering an injured person what is the first thing you should check for?

- **Hazards**
- For broken bones
- Their breathing
- Their pulse

When writing a report, which one of the following questions should be asked when interviewing witnesses?

- **Was equipment working properly.**
- **Was the area well maintained.**
- **Were proper routines followed.**

Which extinguishing agent must you absolutely not use in the event of a fire involving magnesium?

- **Water**
- CO2
- Halon
- Powder

Which of the following must smoke divers be equipped with?

- **UHF radio**
- First Aid kit
- Heat measuring monitors
- Infra-red viewing equipment

Which of the following should be installed in areas with a high risk of fire?

- **A fixed extinguishing installation.**
- A fire locker containing protective clothing.
- A fire pump.
- A PA system.

Which of the following should not be done when fighting a fire?

- **Opening the ventilation.**
- Cooling down bulkheads.
- Removing gas bottles from the area.
- Shutting down electrical power.

Which of the following will the local authorities take care of?

- **Requisitioning the necessary resources.**
- Contacting the company's main office.
- Informing the media.

Who bares the overall responsibility for a fire fighting operation onboard a vessel until help arrives?

- **The Master**
- The Chief Engineer
- The First Officer
- The Local Harbour Master

Who will usually carry out statutory surveys and certifications?

- **The ship's classification society.**
- The IMO.
- The ships company inspectors.
- The ships home port authorities.

Why is it important to report fires?

- **To prevent future fires.**
- It is needed for insurance claims.
- It is required by the IMO.
- Local fire authorities involved in the operation require it.

0162 – Separators, ALCAP Separation principles

Version 3.0 (899)

Based on what you have learned so far, which of the following statements are correct?

- **An increase in viscosity will reduce separation efficiency.**
- Free water and solids can be removed from oil in a clarifier.
- Sealing and displacement water is not added to a purifier bowl.
- Too low a separating temperature will not cause the interface to move into the disc stack.

Centrifugal separation has advantage over gravitational separation because

- **Centrifugal force is many times greater than gravitational force**
- Separators are easier to install than large settling tanks
- Tank heating systems are complex

During a partial discharge, the bowl opens for

- **a few milliseconds only**
- a few minutes only
- a few seconds only

How are the solids removed from a Solid Bowl separator?

- **By hand**
- Through the drain holes
- Through the solids discharge ports

How long is the bowl open during a partial discharge?

- **A few milliseconds**
- A few minutes
- A few seconds

Identify from where the V/V Block is drawing Oil ?

- **Storage Tanks**
- Service Tank
- Settling Tank

Identify the blinking part

- **Re-circulation Valve**
- Clarifier
- Purifier

Identify the blinking part

- **Purifier**
- Clarifier
- Flow Controller

Identify the blinking part

- **Clarifier**
- Service Tank
- Settling Tank

Identify the blinking part

- **Service Tank**
- Recirculation Valves
- Settling Tank

Identify the blinking part

- **Settling Tank**
- Pre-Heater
- Service Tanks

Identify the blinking part

- **Suction Strainer**
- Feed Pumps
- Pre-Heater

Identify the blinking part

- **Feed Pumps**
- Flow Control Valve
- Pre-heater

Identify the blinking part

- **Pre-heater**
- Clarifier
- Purifier

Identify the blinking part

- **Flow Control Valve**
- Pre-heater
- Pumps

Identify the blinking part?

- **Transfer Pump**
- Pre-Heater
- Settling Tank

If the backpressure in the oil discharge line is increased, which of the statements is correct?

- **Oil level in paring chamber moves inwards**
- Nothing happens.
- Oil level in paring chamber moves outwards
- Water level in paring chamber moves inwards

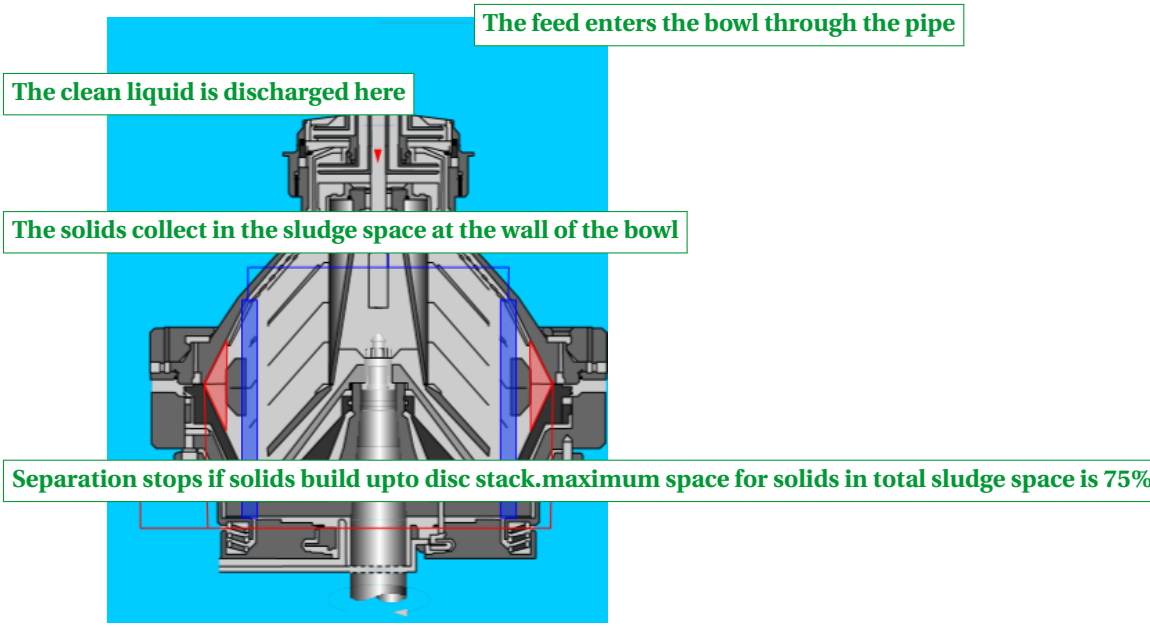
If the density of the oil being supplied to the purifier has increased, would you have to change the gravity disc?

- **Not necessarily, but a check should be made**
- No, never
- Yes, always

Increasing viscosity of a liquid makes it

- **less pumpable and more difficult to flow**
- more pumpable and easier to flow
- viscosity has no relationship with flow or pumpability

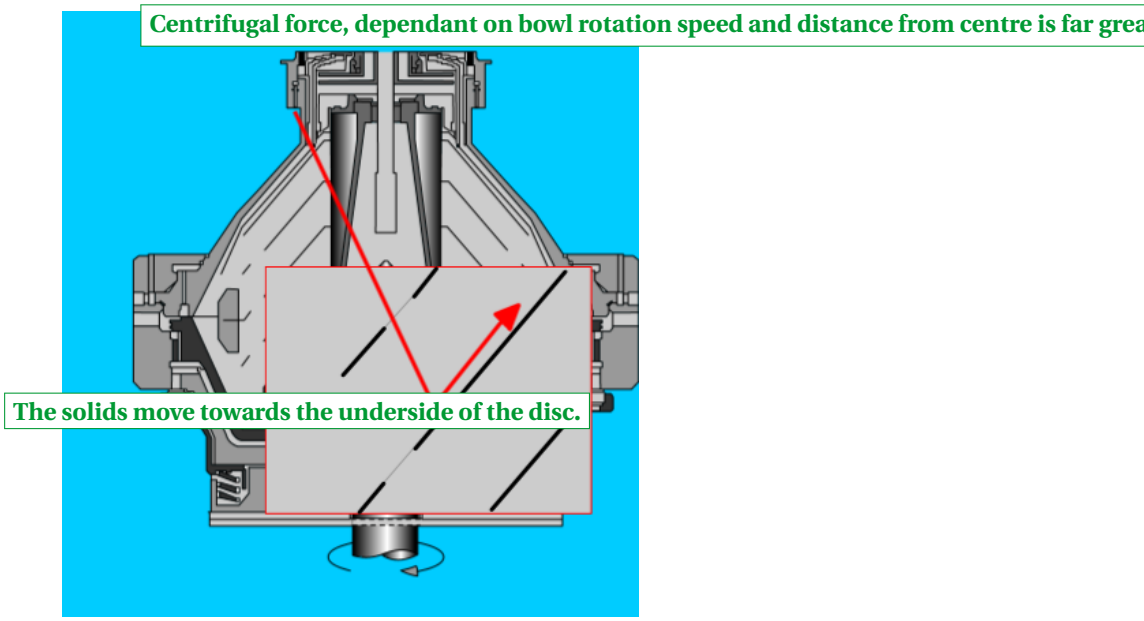
Match the following:-



So, can we increase the speed of separation?

- **Yes**
- No

The liquid flow between the discs moves the solids as indicated by the arrow, but they are also affected by a centrifugal force.



The purpose of a purifier is to separate

- **Liquids of different densities**
- Liquids and solids
- Solids of different densities

The purpose of the paring disc is

- **to pump away the purified oil and water away from the bowl**
- to enclose the bowl from the top end
- to pump purified oil and water into the bowl

The recirculation valve is positioned close to purifier

- **to lessen effect of temperature fluctuation when valve operates**
- for ease of installation
- for ease of maintenance

The time required for particles to separate in sedimentation depends on

- **Speed and size**
- Colour and temperature
- Form and shape

To achieve the best possible results, the interface should be positioned

- **as far out as possible, but no further than the top disc**
- as close as possible and within the top disc
- wherever it can be positioned.

What is a Solid Bowl separator suitable for?

- **Processing liquids with small solids volume**
- Processing liquids with large solids volume
- Processing liquids with negligible solids volume

What is the purpose of the oil paring disc?

- **To discharge the oil under pressure.**
- To discharge the water under pressure.
- To keep the interface in the correct position.
- To pour off the water from the oil.

Which of the following cause the interface to move outwards?

- **Increase in Oil Density**
- Decrease in Feed Flow
- Decrease in Oil Density
- Increase of Feed Temperature

Why is a heater with PI (proportional and integral) required?

- **To keep steady control of temperature**
- To allow for temperature fluctuation
- To avoid manual temperature control

With Decreasing back pressure, the inner level of liquid in the paring chamber

- **Moves outwards**
- Moves inwards
- Stays static

Your Clarifier is NOT cleaning the liquid as it should. Click the alternative you think would improve cleaning.

- **Reduce the feed**
- Increase temperature of process liquid
- Increase the feed
- Reduce temperature of process liquid

0163 – Separators, The ALCAP system

Version 4.0 (2399)

Capacitance is defined as

- **Ratio of change in charge in a system to the change in its electric potential.**
- The capacity to conduct electricity in a fluid.
- The capacity to hold charge.

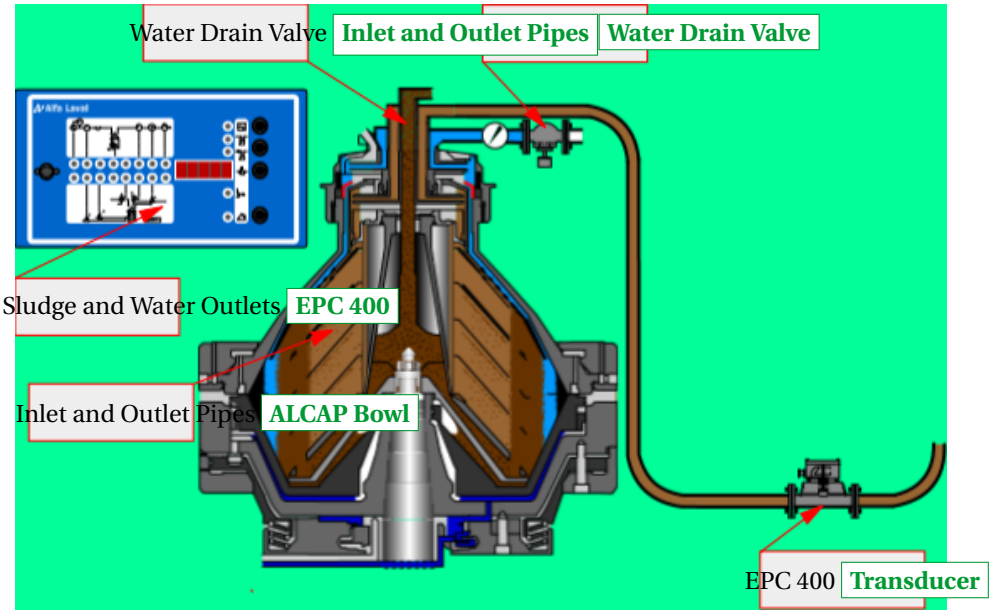
Cat Fines are

- **Aluminium and Silicon catalytic fines which can cause abrasive wear.**
- Fine hair from cats which contaminate Fuel Oil.
- Unknown particles whose role in fuel is not known.

Clarifiers needed modification to ALCAP design because

- **Entrained water could enter from a number of sources.**
- New Technology was preferred.
- The size of the machine could be reduced.

Drag (move the cursor to the part name, press left mouse button and keep it pressed until it is in correct position) the part names to the correct position.



Essentially, an ALCAP Purifier is

- **A Clarifier that removes water.**
- A new product offering from Alfa Laval to diversify product range.
- A totally different machine from Purifiers and Clarifiers.

How is water removed from the ALCAP fuel oil separator bowl?

- **By ejecting it with solids at discharge or by opening the water drain valve.**
- The ALCAP is a clarifier and a clarifier does not remove water.
- The ALCAP separator is for removing catalytic fines only.
- Water is only removed when necessary, if there is no water in the fuel, there is no need to remove it.

How many gravity discs are fitted in the ALCAP separator?

- **None, but a flow control disc is fitted.**
- It depends on the density of the fuel.
- One, but there are many in a set.
- Only one at a time.
- The size is determined from the table in the instruction manual.

Is the measured capacitance of oil higher or lower than that of water?

- **Capacitance of oil is lower than that of water.**
- Capacitance depends on the type of oil and water.
- The dielectric constant of water is 4.
- Water has a lower dielectric constant.

One of the main reasons for the ALCAP purifier was

- **A Clarifier gave better performance than a well adjusted purifier.**
- Results showed purifiers performed better than clarifiers.
- The New technology had to be used as a lot of investment was made.

The EPC 400 controller is a sophisticated piece of microelectronics, which of the following statements best describes components and functions to be controlled?

- **Fail safe actions of valve actuation and motor stop facility.**
- Control of opening and closing water and the positioning of the interface for optimum separation.
- Temperature control and correct positioning of the interface.
- To keep a good check on the separator so the operator can sit in the control room and smoke!
- To monitor the transducer value and to take action if the level gets too high.

The Flow control valve

- **Allows a reduced flow of water from the water outlet when the drain valve is opened.**
- Allows unrestricted flow of water.
- Has no role in control of water flow.

The Function of the EPC 400 is

- **To receive input parameter signals and take controlling action.**
- To increase the complexity of the installation.
- To provide automatic back up for manual operation.

The transducer function is

- **To sense the water content in the oil by difference in dielectric capacity.**
- To amplify flow signal.
- To measure the current flow in the oil.

What are the letters ALCAP understood to stand for?

- **Alfa Laval Clarifier and Purifier.**
- Alfa Lavals' Clarifiers Are Perfect!
- All Capacity Separator.
- Don't know.

What were the key issues which led to the development of the ALCAP separator?

- **The oil crisis of 1973 and 1978, along with the increasing demand for distillates.**
- Laboratory tests and oil price increases.
- The need to use only one separator and more carbon in the residue.
- The stepping up of the refining process and the interface instability.
- Too many catalytic fines and the need for cheaper fuel.

Which of the following are the most important for optimum cleaning efficiency in a centrifugal separator?

- **The oil water interface to be positioned outside the disc stack but inside the top disc.**
- Fuel with a low density and viscosity.
- Making sure that the purifier does not loose its water seal.
- Stopping all the water leaks into the fuel system.
- Using a purifier and a clarifier in series.

Which of the following statements is most accurate?

- **Particle separation is best in an ALCAP separator because the interface is always outside the outer edge of the top disc.**
- The clarifier will always be better at removing particles from heavy fuel oil.
- The purifier which was commissioned by the Alfa laval engineer 2 months ago gives best particle separation.
- The series operation of purifier and clarifier gives best particle removal.

Which of the following statements is most favourable with regard to achieving the cleanest possible oil?

- **An ALCAP separator with accurate temperature control.**
- A clarifier used for lubricating oil cleaning where analysis of the oil has revealed some water in the oil.
- A Purifier and clarifier in series where there is never a seal alarm.
- A purifier where the gravity disc was checked by the Alfa Laval representative two weeks ago.
- A single purifier where the temperature is fluctuating by 10 degrees C.

Why can the interface position be such a problem in the purifier bowl?

- **It is very easily affected by changes in density, viscosity temperature and flow.**
- If the interface is in the disc stack the separation result will be bad.
- It can be in the wrong place by fitting the wrong size gravity disc.
- It is easy to get the interface in the wrong position in a clarifier.
- To get a good night sleep the engineer should ensure the interface is well into the disc stack.

Why was a Clarifier type design preferred over Purifier in designing ALCAP?

- **Clarifiers were seen to be better at particle removal.**
- Clarifiers were easier to manufacture.
- It was the decision of Alfa Laval Engineers.

0164 – Separators, ALCAP Fuel oil system

Version 3.0 (9239)

A screw pump is a good solution to the feed pump required for the separator because

- **It does not separate the particles in the oil before reaching the separator.**
- It costs less than a reciprocating pump.
- It is more energy efficient.

Loss of Water Pressure to the Valve block will

- **Make system malfunction.**
- Water pressure loss can be accommodated for some time.
- Will not affect the system.

One of the main functions of the three way re-circulation valve is

- **To recirculate oil away from the separator in the event of an abnormal situation.**
- To increase the complexity of the system.
- To provide different paths for the oil to travel.

The back pressure setting is

- **Midway between delivery height pressure and pressure required to overflow bowl.**
- Midway between feed pressure and pressure required to overflow bowl.
- Not governed by delivery pressure or by overflow pressure.

The oil entering the separator should be preheated and of lower viscosity because

- **It ensures better separation.**
- It reduces the speed of separation.
- It is necessary to run the preheater.

The oil paring discs for conventional and ALCAP separators are different in the following ways?

- **The edge of the ALCAP oil paring disc has 2 lips to make a seal and prevent air getting drawn in.**
- ALCAP oil paring discs have regulating holes.
- ALCAP paring discs are larger to cope with more viscous oils.
- There is no difference, they are inter-changeable.

What is the purpose of these holes?

- **To allow movement to minimise over-heating.**
- So there is always a flow through the small hole in the flow control disc to stop it getting blocked.
- To allow air to be vented from the oil paring chamber.
- To prevent build up of pressure when the drain valve is closed.

When might you adjust parameter 20?

- **When commissioning a new installation or fitting a new EPC card.**
- Parameter 20 relates to a separator type and should not be changed.
- When using a fuel oil separator to clean lube oil.

Which group of parameters may have to be adjusted for a different oil type?

- **Process parameters**
- All of them
- Installation parameters
- Service parameters
- Timer parameters

Which of the following is most accurate?

- **The transducer monitors changes in the condition of the oil leaving the bowl.**
- The ALCAP Separator has to have a back-pressure of 1.5 - 2.0 bar to run on auto.
- The back pressure regulating valve is automatic and part of the ALCAP principle.
- The recirculation valve always changes position for an abnormal situation.

Which of the following statements best describe the purpose of parameters?

- **To adjust the EPC 400 for different separator types and installations, to ensure optimum performance.**
- It makes adjustments very easy.
- Part of the fashion in moving to more sophisticated electronics.
- To adjust the EPC 400 to the correct variables for each installation to ensure optimal performance.

Which statement best describes why a flow control disc can replace a set of gravity discs?

- **With the ALCAP we are not so concerned with the position of the interface, it can be outside the edge of the top disc. It is not necessary to change gravity discs to ensure correct interface position.**
- An ALCAP separator is a clarifier, and a clarifier does not have a gravity disc.
- By keeping interface outside the disc stack, the water seal is less important, only one flow control disc required.
- Normally the ALCAP separator runs as a clarifier, and when draining water, it is a purifier with a restricted broken water seal.
- The small hole enables water to be drained off when the drain valve is open.
- With the ALCAP the chance of having the interface into the disc stack is reduced because of the small hole in the flow control disc.

Why is water added to the ALCAP fuel separator bowl after each discharge?

- **By keeping the sludge moist hard sludge is prevented, and a good discharge results.**
- This puts a precise quantity of water into the oil, which can be measured by the transducer.
- To check that the transducer is still working the reference has to change.
- To displace the oil and minimise on oil losses.

0165 – Separators, ALCAP Lube oil system

Version 3.0 (3201)

How do you know if the conventional self cleaning Lube oil purifier is working correctly?

- **By careful selection of gravity disc after ensuring process variables; temperature, viscosity and flow are correct and stable.**
- **By ensuring the interface is stable in the best position.**
- By routine analysis of the oil.
- There are no alarms.

How do you know if the conventional self cleaning Lube oil purifier is working correctly?

- **By ensuring the interface is stable in the best position and by careful selection of gravity disc after ensuring process variables; temperature, viscosity and flow are correct and stable.**
- By routine analysis of the oil.
- There are no alarms.

More detergent and dispersant additives are used in Lube Oils today

- **To keep the engine clean and prevent accumulation of sludge particles**
- To change the specific gravity of the Lube Oil to acceptable limits
- To increase the anti-friction property of Lube Oil

The ALCAP Lube Oil system contains which components?

- **Separator (LOPX), Transducer, Recirculation Valve, EPC 400 and Water Valve Block.**
- Transducer, Drain Valve, Water Valve Block, EPC 400 and FOPX Separator.
- Water Valve Block, LOPX Separator, Separator Sam's adjustment tool and EPC 400.

The ALCAP Lube Oil system contains which components?

- **Separator (LOPX), Transducer, Recirculation Valve, EPC 400 and Water Valve Block.**
- Recirculation Valve, Transducer, Drain Valve, EPC 400 and LOPX Separator.
- Transducer, Drain Valve, Water Valve Block, EPC 400 and FOPX Separator.
- Water Valve Block, LOPX Separator, Separator Sam's adjustment tool and EPC 400.

The ALCAP purifier for L.O. is

- **Is basically a clarifier but uses water for oil displacement and sludge conditioning**
- Is a clarifier with a gravity disc
- Is exactly like a conventional Clarifier

The conditioning water is added through the pipe which is in the middle of the feed tube

- **To prevent the turbulent contact between the conditioning water and Oil.**
- This is the most suitable design for the bowl.
- To give the flow of water the least resistance.

The function of the back pressure valve is

- **To set the Pressure between the Minimum positive pressure and the Maximum pressure at which oil overflows**
- To set the pressure at which the Purifier operates
- To set the pressure of the water block

Which of the following statements is correct?

- **The dispersant additives in lube oil break up the contaminants into small particles and keep them in suspension.**
- A smaller volume means more dirt will be going into the lube oil.
- From Stokes Law particle size does not affect separation.
- Lube oil additives assist the separator to remove water from the oil.

Which of the following statements is correct?

- **The dispersant additives in lube oil break up the contaminants into small particles and keep them in suspension.**
- A smaller volume means more dirt will be going into the lube oil.
- From Stokes Law particle size does not affect separation.
- Lube oil additives assist the separator to remove water from the oil.

Which of the following statements is correct?

- **The inlet for displacement and conditioning water for LOPX is different than that for FOPX 605 - 613.**
- The LOPX removes water by discharging with sludge and draining from the water outlet.
- There are two paring discs in the outlet of the LOPX separator.
- Water is added to the middle of the FOPX bowl.

Which of the following statements is true

- **In case of ALCAP FO Purifier, Displacement is added until "trigger point" is reached. In LO purifier ALCAP design this is not the case**
- In both FO and LO ALCAP design, displacement water is added in the same way
- The addition of displacement water in ALCAP design is optional

Why is a predetermined amount of conditioning water added to the bowl?

- **To ensure the sludge removed from the lube oil does not get hard, and an excessive amount of separated sludge will cause water to be detected by the transducer.**
- Because Separator SAM said so!
- In case there is no water in the oil the transducer will need it to take a reference.
- The ALCAP lube oil separator is a clarifier and does not need water in the bowl.
- To displace the oil inwards to economise on oil losses.

Why is a predetermined amount of conditioning water added to the bowl?

- **To ensure the sludge removed from the lube oil does not get hard, and an excessive amount of separated sludge will cause water to be detected by the transducer.**
- In case there is no water in the oil the transducer will need it to take a reference.
- The ALCAP lube oil separator is a clarifier and does not need water in the bow
- To displace the oil inwards to economise on oil losses.

0166 – Separators, ALCAP Monitoring and control

Version 4.0 (2409)

Before disconnecting any cables to the EPC 400 it should be isolated by?

- **Switching off the electrical supply to the EPC.**
- Switching off the EPC with the fuse switch, located in the top left hand corner of the box.
- Switching off the motor starter.
- The EPC 400 only runs on 48 volts, it is not necessary to isolate.

The EPC 400 controller is a sophisticated piece of microelectronics, which of the following statements best describes components and functions to be controlled?

- **Fail safe actions of valve actuation and motor stop facility.**
- Control of opening and closing water and the positioning of the interface for optimum separation.
- Temperature control and correct positioning of the interface.
- To keep a good check on the separator so the operator can sit in the control room and smoke!
- To monitor the transducer value and to take action if the level gets too high.

The fuses on the front circuit board are numbered from the top as?

- **F2, F1 and F3**
- F1, F2 and F3
- F2, F3 and F1
- F3, F1 and F2
- F3, F2 and F1

What happens if you keep the alarm accept button pressed for more than 5 seconds?

- **A 'Lamp test' of all LED's.**
- 32 previous alarms are displayed.
- A test of the alarm output to the central control room.
- It is possible to display extra process conditions.

Which of the following statements are correct?

- **The emergency stop for the separator should be used in the event of heavy vibrations. It provides a similar function to the vibration switch.**
- **The 'FIRE' stop (from some remote location), usually cuts the power to the motor. The power to the EPC 400 must be supplied from the motor starter, for the process to be stopped in this way.**
- In the event of a burst or leaking pipe, the 3-way valve will always change to re-circulation.
- In the event of having a burst pipe, the emergency stop should be used.
- The emergency stop for the separator takes care of every emergency situation.

Which of the following statements is correct?

- **The EPC 400 monitors the transducer and co-ordinates the ALCAP functions with the operating sequence.**
- The EPC 400 can be adjusted to suit different operating voltages.
- The EPC 400 has 2 printed circuit cards to monitor ALCAP functions.
- There are 4 different generations of EPC 400.

Which of the following statements is correct?

- **By changing P90 to 1, timer countdown is displayed.**
- P1 is not used for changing the time between discharges.
- Process parameters are in the group 50 to 89.
- To change P37, C1 must be first set to 12.

0167 – Tanker Training System - Familiarisation

Version 4.0 (594)

A fully refrigerated gas carrier is designed for excess pressure in the cargo tank below . . . . ?

- **0,7 bar**
- 0,3 bar
- 11 bar
- 2,5 bar

Can a mix of propane and air be ignited if the percent of propane are 20 percent by volume?

- **No**
- Yes

In sour crude oil you will find Hydrogen sulphide (H2S), is H2S a poisonous gas?

- **Yes already at a concentration of 300 ppm it is deadly.**
- No.
- Only if the concentration is 2% or more.

Is an "Independent tank" a part of the hull structure?

- **No**
- Yes

Oil tankers of 70.000 dwt and above must have how many slop tanks?

- **2**
- 1
- 3
- 4

What do we mean by "Load on top"?

- **New cargo loaded on top of old residues.**
- Load cargo into the ships bunker tanks.
- Load the cargo tanks to more than 98% full.

What is IMO's definition of condensed gases?

- **It is liquids with vapour pressure above 2,8 bar at 37,8oC**
- It is liquids with vapour pressure above 3,2 bar at 37,8oC
- It is liquids with vapour pressure lower 0,7 bar at 37,8oC
- It is liquids with vapour pressure lower 2,8 bar at 37,8oC

What is the lowest percent of oxygen that have to be present before fire and explosion can take place even though both hydrocarbon gas and ignition sources are present?

- **10.8 % by volume**
- 12,3 % by volume
- 14 % by volume
- 6,5 % by volume

What is the maximum allowed oxygen content in a cargo tank?

- **8%**
- 10%
- 17%
- 5%

What must be the minimum oxygen content measured in a cargo tank before a person can enter the tank?

- **21%**
- 14%
- 20%
- As long as it is more than 12% oxygen it is safe to enter the tank.

Where do you find design standards for new ships (oil tankers)?

- **MARPOL**
- IGC Code
- SOLAS
- STCW

Which hydrocarbon compound has the formula C3H8?

- **Propane**
- Methane
- n-Butane
- n-Octane

0168 – Gas Tanker Training System - Advanced

Version 3.0 (864)

A Chlorine carrier must be equipped with a chlorine absorption plant connected to the cargo tanks and cargo lines. What is the gas detectors alarm setting?

- **5 ppm**
- 10 ppm
- 15 ppm
- 20 ppm

A fully refrigerated gas carriers, equipped with membrane tanks with MARVS less than 0.7 bars and cargo temperature less than – 10° C. What type of barrier is required?

- **Complete secondary barrier.**
- No barrier required.
- No secondary barrier required.
- Partial secondary barrier.

After arms and communication plugs are connected, the ESD is tested from?

- **Ship and shore**
- No testing required
- Ship
- Shore

Cargo hoses have to be tested at a pressure of 1,5 times the maximum working pressure. How often do we have to carry out the pressure test?

- **After 12 months**
- After 3 months
- After 6 months
- After each discharging operation

Cascade cooling plant is basically a direct cargo cooling plant where the cargo is condensed against a secondary refrigerant. Which type of secondary refrigerant is most commonly used for cooling of the cargo condenser?

- **Freon 22**
- Ethylene
- Freon 12
- Seawater

Enthalpy is an expression of how much energy is tied up in one kilo of a substance. A substance's total energy consists of;

- **External and internal energy**
- External energy
- Internal energy

Fully refrigerated gas carriers are designed for excess tank pressure less than 0.7 bars. What tank types are they built with?

- **All mentioned**
- Independent tank type A or B
- Membrane tanks
- Prismatic or spherical tanks.

How do we control the cargo tank pressure when loading a gas carrier?

- **When loading we control the cargo tank pressure with cargo compressors, vapour return to shore or the loading rate.**
- When loading we are not able to control the cargo tank pressure.
- When loading we control the cargo tank pressure with vapour return to shore or the loading rate.
- When loading we control the cargo tank pressure with cargo compressors or the loading rate.

How many carbon atoms are there in solid materials?

- **Above 16 carbon atoms**
- From 5 to 16 carbon atoms
- I don't know
- Up to 4 carbon atoms

How many electrons are there totally in a carbon atom?

- **6**
- 4
- 7
- 8

How many elements are arranged the periodic table?

- **109**
- 101
- 106
- 112

How many groups is the cargo tanks divided into?

- **4**
- 3
- 5
- 6

How safe is LNG compared to other substances handled in ports and land-based facilities?

- **LNG does not burn because it contains no oxygen. It is not toxic and can only burn within the narrow range of a 5 – 15 percent gas–to-air mixture**
- LNG is toxic and can cause a threat to human life
- LNG itself will cause a danger because it is flammable and can explode
- The vapour of LNG is heavier than air and will create a danger of fire on ground.

Isomer compounds is substances with;

- **Different characteristic**
- Different boiling point
- Different formula
- Same characteristic

Name three international conventions that cover the international shipping trades.

- **The three international conventions that cover the shipping trade are SOLAS 74/78, STCW 78/95 and Marpol 73/78.**
- The three international conventions that cover the shipping trade are SOLAS 74, MARPOL 73/78 and STCW 78.
- The three international conventions that cover the shipping trade SOLAS 74, SOLAS 78 and STCW 78.
- There is not any international conventions hat cover the shipping trade.

On all pressure vessels above 20 m3 that is used to safety valves. What do we mean with MARVS?

- **Maximum allowed relieve valve setting**
- Maritime Relieve valve
- Maximum relief valve setting storage of liquefied gases there is a requirement for
- Minimum allowed relieve valve setting

One of the purposes to MARPOL Annex I is to:

- **Prevent operational oil pollution by providing ships with discharge criteria and other guidelines to follow.**
- Ensure that the ships certificates are valid.
- Prevent sea-going incidents involving tankers.
- The tanker is en-route and meets certain discharge conditions.

Organic chemistry mostly deals with chemical compounds containing carbon.

- **Correct**
- Wrong

Propane has a boiling point at -42.8° C at atmospheric pressure. Can we load fully refrigerated propane on a fully pressurised gas carrier?

- **Yes, if we heat the cargo to more than -10° C**
- No
- Only if the cargo tanks are insulated
- We are not allowed to load propane on a fully refrigerated gas carrier

Referring to IMO IGC Code all gas carriers mentioned in chapter 19 is given one of the following descriptions for tanks:
Which tank type can carry all cargoes mentioned in chapter 19?

- **1G**
- 2G
- 2PG
- 3G

The ABC method stands for air, breathing and circulation. If a person has stopped breathing, how many respirations and heart compressions are to be given if there are two rescuers?

- **2 respirations and 30 compressions**
- 2 respirations and 20 compressions
- 2-3 respirations and 10 compressions
- 3 respirations and 15 compressions

The Compressed Natural Gas (CNG) or Pressurised Natural Gas (PNG) technology offers interesting possibilities for handling of associated gas and for exploitation of marginal fields. What is the advantage of this system?

- **A fleet of CNG ships will serve as both storage and transport vehicles and can discharge directly into the land based gas grid via an offshore discharge terminal, and offshore platform or offshore buoys.**
- **LNG storage and regasification at the discharge location will not be necessary.**
- **The system will not require a gas liquefaction plant or LNG storage tanks.**
- There will be no risk of sloshing damages inside the cargo tanks.

The IMO gas carrier code defines liquefied gases as gases with vapour pressure higher than 2.8 bar at which temperature?

- **37,8° C**
- 27,8° C
- 35.8° C
- 43,8° C

The Reid Vapour Pressure (RVP) used for measuring the volatility of petroleum liquids is conducted at 37.8 ° C (100° F).
What RVP has crude oil?

- **0,1 to 0,8 kg/cm2**
- 0,1 to 0,5 kg/cm2
- 0,1 to 0,6 kg/cm2
- 0.1 to 0,7 kg/cm2

The Threshold Limit Value (TLV) refers to as parts to million (ppm) by volume of gas in air. What does the time weighted average of ppm indicates?

- **The maximum exposure to a toxic substance per 8 hours working day**
- An average of exposure to a toxic substance per 8 hours working day
- An average of exposure to a toxic substance per day
- The maximum exposure to a toxic substance per day

The periodic system is built on the principle that the electrons in the outer shell determine the chemical properties of a material. What is the maximum number of electrons in the outer shell?

- **8**
- 12
- 5
- 7

There are three gas codes issued by IMO – which are they:

- **Code for construction and equipment of ships carrying liquefied gases in bulk.**
- **Code for existing ships carrying Liquefied gases in bulk.**
- **IGC code: International code for construction and equipment of ships carrying liquefied gas in bulk.**
- IBC code: International Bulk Chemical Code.

What IMO conventions take care of the human safety at sea?

- **It is the SOLAS conventions that take care of the human safety at sea.**
- It is the MARPOL conventions that take care of the human safety at sea.
- It is the STCW 78/95 that takes care of the human safety at sea.
- There are not any conventions that take care of the human safety at sea.

What are actual cargoes for fully pressurised gas carriers without cooling capacity?

- **LPG and some chemical gases**
- LNG
- LPG, LEG, Ammonia and some chemical gases
- NGL

What are saturated hydrocarbons?

- **Saturated hydrocarbons are colourless and odourless liquids and they are chemically non-reactive.**
- Saturated hydrocarbons are chemically non-reactive liquids and they are transported under high pressure.
- Saturated hydrocarbons are chemically reactive liquids with strong colour.
- Saturated hydrocarbons are colourless and odourless liquids and they are transported under high pressure.

What are the most common products in NGL?

- **Ethane, LPG, Pentane and heavier fractions of hydrocarbons or a mixture of these.**
- Ethane, propane, butane and nitrogen.
- Propane, butane and a mixture of these.
- Propane, methane and a mixture of these.

What does the IGC code primarily deals with?

- **The IGC code primarily deals with gas carriers design and equipment.**
- The IGC code primarily deals with gas carriers cargo equipment.
- The IGC code primarily deals with gas carriers design.
- The IGC code primarily deals with gas carriers safety equipment.

What does the cargo's data sheet tell us about the character of the cargo?

- **Information about health hazards, fire, explosion, chemical data, reaction data and physical data.**
- **The condition of material in freight.**
- **The emergency procedure for a cargo fire or cargo spill.**
- Information about the transportation.

What is IMO's requirement to maximum temperature for fully pressurised cargo tanks at ambient temperature?

- **45° C**
- 40° C
- 43° C
- 50° C

What is LEG (Liquefied Ethylene Gas) utilised for?

- **As raw material for plastic and synthetic fibres**
- As fuel for electric power plants
- As raw material for the fertiliser industry
- For energy purposes and in the petro-chemical industry

What is characteristic for Alkanes?

- **Alkanes have only single bonds between the atoms.**
- Alkanes has double or triple bonds between the atoms.
- Alkanes has single or triple bonds between the atoms.
- Alkanes has triple bonds between the atoms.

What is extremely important to remember when measuring hydrocarbon gas in an inerted atmosphere using the Riken model 17HC?

- **To reduce the readings by 2,5%**
- The indicator should not be used in an inerted atmosphere
- To increase the reading by 1%
- To use a special inert gas filter.

What is likely to happen if a randomly chosen hose is used?

- **The hose will most likely absorb hydrocarbon gases.**
- Nothing in particular
- The hose won't fit the indicator inlet ports.
- The indicator will malfunction

What is the atmospheric content of oxygen that is required before tank entry?

- **21%**
- 17 %
- 19 %
- 20,5 %

What is the boiling temperature of propane at atmospheric pressure?

- **43° C**
- - 28° C
- - 32° C
- - 45° C

What is the intension of the Safety Meeting?

- **The intension of this meeting is to inform and train the crew in safety matters and to get the crew to participate in discussion and make own suggestion for improvement.**
- The intension of this meeting is to assign planned work to the crew, and to insure that checklists and permits have been filled in before the work start.
- The intension of this meeting is to gather the crew that's not on duty to get to know each other better, which will affect the safety onboard.
- To inform the crew of safety information.

What is the maximum allowable filling limit on cargo tanks on gas carriers?

- **Maximum allowable filling limitation cargo tanks are 98% on gas carriers.**
- Maximum allowable filling limitation cargo tanks are 100% on gas carriers.
- Maximum allowable filling limitation cargo tanks are 90% on gas carriers.
- Maximum allowable filling limitation cargo tanks are 95% on gas carriers.

What is the maximum size of a cargo tank on a Chlorine carrier?

- **600 m^3**
- 1000 m^3
- 1200 m^3
- 400 m^3

What is the most common way of handling the LNG boil off gas?

- **It is used as fuel for the boilers**
- It is liquefied and directed back to the cargo tank.
- It is used as fuel for the main engine
- The boil off is let to the atmosphere

What is the name of UN maritime division?

- **The UN maritime division is IMO International Maritime Organisations.**
- The UN maritime division is ISO International Shipping Organisation.
- The UN maritime division is UNM United Nations Maritime.
- UN does not have any maritime division.

What is the normal relief valve setting on a fully pressurised LPG carrier?

- **A relief vale setting at 18 bars**
- A relief vale setting at 15 bars
- The minimum relief valve setting must not be less than 12 bars
- The relief valve setting equal to cargo temperature at 45° C at ambient temperature.

What is the number one job for the ship management team when fire break out?

- **The ship management team must organise the fire squads and then the fire squads have to search for missing personal.**
- The ship management team and the crew must evacuate the ship.
- The ship management team must call the nearest fire brigade and police station.
- The ship management team must fight the fire and then call the fire squads.

What is the purpose of an automatic trip system onboard, and how does it function?

- **Automatic trip is to avoid tanks and pressurised vessels becoming overloaded,the trip function is often initialised by a float switch.**
- Automatic trip is to avoid over pressuring of cargo tanks and other pressurised vessels, the trip function is often initialised by a float switch.
- Automatic trip is to avoid tanks becoming over loaded and the trip function is often initialised by a float switch.
- Automatic trip is to avoid tanks being cooled to much, the trip function is often initialised by a float switch or a temperature switch.

What is the purpose of the ship/shore safety check list?

- **To ensure the safety of personnel, ship and terminal**
- To avoid the avoidance of cargo contamination
- To ensure the safety of the personnel.
- To ensure the ship is safe moored

What is the typical size of a LNG carrier?

- **125000 - 145000 m^3**
- 12000 - 50000 m^3
- 200000 - 250000 m^3
- 75000 - 80000 m^3

What kind of span gas is used to adjust the Servomex OA 262 to the 21% mark?

- **Clean air**
- Nitrogen
- Oxygen free nitrogen
- Pure oxygen

What kind of span gas is used to zero adjust the Servomex OA 262?

- **Oxygen free nitrogen**
- Butane
- Oxygen
- Propane

What purpose has the discharge booster pump on a gas tanker?

- **To increase the manifold pressure**
- To compensate for friction losses in the delivery line
- To increase the discharge rate
- To maintain the same temperature on the cargo

What temperature do we calculate on when loading clean cargoes?

- **We calculate the liquid weight at the actual cargo temperature.**
- We calculate the liquid weight at 15° C.
- We calculate the liquid weight at 25° C.
- We calculate the liquid weight at the ambient temperature.

What temperature do we calculate on when using the ASTM-IP tables?

- **We calculate the liquid weight at 15° C.**
- We calculate the liquid weight at 25° C.
- We calculate the liquid weight at the actual cargo temperature.
- We calculate the liquid weight at the ambient temperature.

What type of compressors is used as cargo compressors onboard gas carriers?

- **Piston, screw and centrifugal type compressors**
- Oil free screw and piston compressors
- Three stages piston compressors
- Three stages screw compressors

When did the transportation of fully refrigerated liquefied gases start?

- **In mid – 1960's.**
- In 1912.
- In the beginning of the 1930's.
- In the beginning of the 1960's.

Where do we find information about what cargo the vessel is allowed to carry?

- **Certificate of Fitness for Carriage of Liquefied Gas in Bulk.**
- Safety Construction Certificate.
- Safety Equipment Certificate.
- Solas.

Which are the three most common energy forms?

- **The three most common energy forms are mechanical energy, chemical energy and electrical energy.**
- The three most common energy forms are temperature, pressure and electrical energy.
- The three most common energy forms are mass energy, physical energy and electrical energy.
- The three most common energy forms are speed energy, electrical energy and rolling energy.

Which are the three most common used valves for cargo handling equipment on gas carriers?

- **Ball valves**
- **Butterfly valves**
- **Seat valves**
- Gate valves

Which certificate is issued in accordance with the IGC code?

- **It is the Certificate of Fitness that is issued in accordance with the IGC code.**
- It is the Cargo ship Safety Construction Certificate that is issued in accordance with the IGC code.
- It is the Cargo ship Safety Equipment Certificate that is issued in accordance with the IGC code.
- There is no Certificate issued in accordance with the IGC code.

Which is the most utilised pump type for cargo discharge on a gas tanker?

- **An electrical operated submerged centrifugal pump.**
- Booster pump.
- Displacement pump.
- Hydraulic driven centrifugal pump.

Which of the following indicators is based on the paramagnetic principle?

- **Servomex OA 262**
- Draeger detector tube
- Riken model 17HC
- Riken Model GP-204

Which of the following materials has the highest diffusion resistance?

- **Freon 22**
- Glass
- Mineral wool
- Stationary air

Why is it important that you become familiar with the vessel, the equipment and your own duties onboard as soon as possible?

- **In an emergency situation you need to know the position and use of different safety – and fire fighting equipment, and what to do.**
- In an emergency situation it is important for you to know where to find the safety officer for reporting.
- It is important for you to know so you can find your way without someone guiding you all the time.
- To be able to carry out safe working onboard.

Why must some chemical gases be carried on gas carriers as specified by the IMO's gas code?

- **Because of the gases boiling point at atmospheric pressure and special requirements for temperature control.**
- Because their vapour pressure does not exceed 2,8 bars at 37,8° C.
- Because their vapour pressure is above 2,8 bar at 37,8° C.
- Because they are toxic.

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A system for continuous monitoring of the concentration of hydrocarbon gases in the cargo pump-rooms shall be fitted. What is the max pre-set level ?

- **No higher than 10% LFL**
- No higher than 3% LFL
- No higher than 5% LFL
- No higher than 8% LFL

Any discharge into the sea of oil or oily mixtures from the cargo area of an oil tanker shall be prohibited except when . . .

- **The tanker is more than 50 nautical miles from nearest land**
- The tanker is more than 100 nautical miles from nearest land
- The tanker is more than 20 nautical miles from nearest land
- The tanker is more than 30 nautical miles from nearest land

**As a general guidance to the suitability of an oil for crude oil washing the following criteria should be used:
For aromatic crude oils whose kinematic viscosity is the temperature controlling characteristic, the kinematic viscosity of the oil used for COW should not exceed ??? centistokes at the oil wash medium temperature.**

- **60 centistokes**
- 40 centistokes
- 50 centistokes
- 65 centistokes

**As a general guidance to the suitability of an oil for crude oil washing the following criteria should be used:
For paraffinic crude oils whose pour point temperature is the controlling characteristic, the temperature of the cargo to be used for COW should exceed its cloud point temperature by at least ???? if excessive sludging is present and should only be used once in a "closed cycle" washing programme.**

- **10o C**
- 15o C
- 20o C
- 5o C

Before any portable gas indicators are brought to the measuring spot, what is very important to do with these analysers first?

- **To do a full calibration.**
- Just check the filters.
- Just renew the batteries.
- Zero adjust them.

Cargo hoses in service should have a documented inspection at least annually to confirm their suitability for continued use. What is the normal testing pressure to check for leakage?

- **1.5 times the Rated Working Pressure (RWP)**
- 1.0 times the Rated Working Pressure (RWP)
- 2.0 times the Rated Working Pressure (RWP)
- 2.5 times the Rated Working Pressure (RWP)

For what purpose do we use inert gas onboard oil carriers?

- **We use inert gas to replace the oxygen in the cargo tanks to maintain a neutral atmosphere.**
- We use inert gas for tank stripping when discharging
- We use inert gas to clean cargo lines after discharging
- We use inert gas to held pressure in the cargo tanks when we are discharging.

From where on board can you find out the different types of protection equipment regarding where placed, how much, how many ans.?

- **The ship's safety plan.**
- In cargo control room.
- On the bridge
- Stated in your cabin.

In order to avoid excessive electrostatic generation in the washing process during COW how many meters do you have to discharge from a cargo tank before you can use the content as source of washing fluid?

- **1 m**
- 0.5 m
- 1.5 m
- 2 m

In which Annex of MARPOL do we find regulation for the prevention of pollution by garbage from ships?

- **In Annex V**
- In Annex III
- In Annex IV
- In Annex VI

In which Annex of MARPOL do we find regulation for the prevention of pollution by oil?

- **In Annex I**
- In Annex II
- In Annex III
- In Annex V

In which way may intake of poisoning material occur?

- **By inhaling**
- **Skin penetrating and skin absorbing**
- **Swallowing**

Near by the access to shore a certain plan is to be kept in case of a emergency situation. What kind of plan?

- **Safety plan**
- Discharging plan
- Emergency plan
- Loading plan

The fire control draft on board is called the «safety plan» and shall be posted onboard. In port, a copy of this plan shall in addition be available from somewhere else. Where shall this copy be available?

- **At the gangway**
- Handed over to the surveyors
- In the engine control room
- On the terminal

What do you call the method used for bone-soft part injures?

- **ICE-method.**
- ABC-method.
- First Aid -method.
- REHAB-method.

What does OBO means?

- **Ore Bulk Oil.**
- Oil Bulk Oil.
- Only Basic Oil.
- Only Bulk Oil.

What does VLCC means?

- **Very large crude carrier.**
- Very large combination carrier.
- Very large common carrier.
- Very large crude combination carrier.

What does the abbreviation TLV means?

- **Threshold Limit Value**
- Time limitation value
- Total level value
- Total limit value

What is it called when it is possible to ignite the vapour above the oil?

- **Flash point**
- Boiling pointy
- Ignition point
- Pour point

What is the Permissible Exposure Limit (PEL) of hydrogen sulphide expressed as a Time Weighted Average (TWA)?

- **10ppm**
- 20 ppm
- 30ppm
- 5 ppm

What is the common expression for all chemical compounds that includes carbon and hydrogen?

- **Hydrocarbons**
- Alkanes
- Alkenes
- Arenes

What is the flammable range (%) for Propane?

- **2.2 – 9.5**
- 0.2 – 0.9
- 7.8 – 1.5
- 8.5 – 1.9

What is the main quencher for an oil tanker?

- **Foam**
- Carbonic acid
- Powder
- Water

What is the maximum oil content allowed in the arrival ballast water?

- **15ppm.**
- 0
- 10ppm.
- 5ppm.

What is the maximum oil content in the ballast/washing water allowed to be pumped over board during a voyage?

- **30 litres pr. nautical mile**
- 20 litres pr. nautical mile
- 40 litres pr. nautical mile
- 60 litres pr. nautical mile

What is the maximum volume percent oxygen allowed in the tank during COW?

- **Maximum volume percent oxygen shall not exceed 8 percent**
- Maximum volume percent oxygen shall be the same as LEL for actual cargo
- Maximum volume percent oxygen shall not exceed 1 percent
- Maximum volume percent oxygen shall not exceed 10,8 percent

What is the meaning of LEL?

- **Lower explosive limit.**
- Limited explosive level.
- Lower explosive level.
- Lower exposure level.

What is the meaning of LOT?

- **Load on Top**
- Load over Top
- Loaded oil Total
- Loss of Tugs

What is the meaning of UEL?

- **Upper explosive limit**
- Upper evaporation level
- Upper explosion level
- Upper exposure level

What is the meaning of the abbreviation ABC due to first aid?

- **Air, Breathing, Circulation.**
- Abandon, Balance, Circulation.
- Air, Breath, Concentration.
- Air, Burning, Critic.

What is the name of the device used for supervising all the ballast water to be pumped over board?

- **Oil Detection Monitoring Equipment (ODME).**
- Ballast Handling Monitor (BHM).
- Ballast Monitor (BM).
- Ballast Supervising Monitor (BSM).

What is the normal clearance of the main suction (bellmouth) from the tank bottom to the stripping suction in the cargo tank?

- **10 cm**
- 15 cm
- 20 cm
- 5 cm

What is the proper name of CH4?

- **Methane**
- Ethane
- Pentane
- Propane

What is the requirement to capacity of stripping system during bottom COW of the cargo tanks?

- **1.25 times the total through-put of all tank cleaning machines**
- 1.15 times the total through-put of all tank cleaning machines
- 1.35 times the total through-put of all tank cleaning machines
- 1.50 times the total through-put of all tank cleaning machines

What is the requirement to capacity of the inert gas plant?

- **At least 125% of the maximum discharge capacity**
- At least 110% of the maximum discharge capacity
- At least 115% of the maximum discharge capacity
- At least 150% of the maximum discharge capacity

What is the requirement to number of drive units where the drive unit is not integral with the cleaning machine?

- **No drive unit need to be moved more than twice from its original position**
- No drive unit need to be moved more than 1 time from its original position
- No drive unit need to be moved more than 3 times from its original position
- No drive unit need to be moved more than r times from its original position

What is the total quantity of the particular cargo you can discharge into the sea from a new tanker?

- **1/30,000 of the total quantity of the particular cargo**
- 1/15,000 of the total quantity of the particular cargo
- 1/20,000 of the total quantity of the particular cargo
- 1/45,000 of the total quantity of the particular cargo

What is the usual activation point for high level alarms in cargo tanks?

- **95% of tank capacity**
- 90% of tank capacity
- 92% of tank capacity
- 98% of tank capacity

What kind of cargo can be carried with an O/O ship?

- **Either Ore or Oil**
- Only Oil
- Only Ore
- Ore and Oil simultaneously

What kind of extinguishing-remedy would you choose to put out electrical fire?

- **Dry extinguishing remedy**
- Combination of powder and water
- Foam
- Water

What kind of fixed extinguishing plant is installed in an oil tankers engine room and pump room.

- **CO_2 plant**
- Foam plant
- Powder plant
- Water spray plant

When re-inerting a cargo tank before commence air venting, what is the maximum oxygen content in the supplied inert gas?

- **5% by volume.**
- 10% by volume.
- 6% by volume.
- 8% by volume.

Where do we find regulation regarding venting system on tankers?

- **SOLAS**
- ISGOTT
- MARPOL
- OILPOL

Where venting is by high-velocity discharge valves, what is the minimum distance above the cargo tank deck?

- **2 m**
- 10 m
- 3 m
- 5m

Which oxygen content shall we measure before entering a tank or space after venting?

- **The oxygen content shall be measured to 21% by volume**
- The oxygen content shall be measured to 10,8% by volume
- The oxygen content shall be measured to 19% by volume
- We don't need to measure the oxygen content

You are discovering a small leak of oil at the manifold during discharging. What is you first action?

- **Call immediately the person in charge.**
- Call the terminal.
- Do not bother.
- Try to stop the leak.

You are going to enter a tank to lift out some sediments. What kind of permits are needed to be issued before entering?

- **Enclosed Space Entry Permit and Cold Work Permit**
- Enclosed Space Entry Permit and Hot Work Permit.
- Enclosed Space Entry Permit only
- Hot Work Permit and Cold Work Permit.

You find the element carbon in only two different natural conditions. Which?

- **Graphite and diamond**
- Diamond and titanium
- Graphite and vanadium
- Vanadium and silver

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An atom consists of what?

- **Protons, neutrons and electrons**
- Electron shells
- Protons and electrons
- Protons and neutrons

Are you allowed to carry cargoes subject to the Code in the fore or aft peak tanks?

- **No**
- Yes
- Yes, if the tank has separate ventilation
- Yes, if the tank is inerted

Enthalpy is an expression for what?

- **For a substance's internal energy**
- For a substance's external energy
- For a substance's freezing point
- For a substance's melting temperature

For the purpose of Annex II of MARPOL how many categories are noxious liquid substances divided into?

- **4 categories**
- 3 categories
- 5 categories
- 7 categories

Is it compulsory to have permanent arrangements for hoisting injured persons in the cargo pump-room ?

- **Yes**
- No
- Only if the vertical distance is more than 10 meters
- Only if the vertical distance is more than 5 meters

Phenol is a key product in the chemical industry. What is the main source of Phenol?

- **Benzene derivative, cumene and acetone**
- Methanol and Methyl Alcohol
- Styrene Monomer
- Xylenes and Tpolene

The different types of cargo pumps are divided in two main groups, what groups?

- **Displacement- and kinetic pumps**
- Axial- and ejector pumps
- Piston- and gear pumps
- Rotating- and screw pumps

Under which pollution category is Acrylic acid listed?

- **Pollution category Y**
- Pollution category X
- Pollution category OS
- Pollution category Z

What are organochlorines?

- **Substances containing chemically combined chlorine and carbon**
- Substances containing chemically combined chlorine and bromine
- Substances containing chemically combined chlorine and oregano
- Substances containing chemically combined chlorine and oxygen

What are the requirements for carrying products with flashpoint less than 60° C?

- **To be carried in vessels of oil tanker standard**
- To be carried in vessels of chemical tanker standard
- To be carried in vessels of heavy duty standard
- To be carried in vessels of phosphoric acid standard

What are the simplest hydrocarbon compounds ?

- **Alkanes**
- Alkynes
- Arenes
- Ketones

What is evaporation?

- **When a liquid changes to gas**
- When a gas changes to liquid
- When a liquid condensates
- When a liquid freezes

What is the LFL and UFL for methane?

- **5.3% and 14.0%**
- 2.8% and 6.9%
- 4.5% and 12.2%
- 8.2% and 16.4%

What is the TLV for nitrogen monoxide?

- **25 ppm**
- 10 ppm
- 2 ppm
- 35 ppm

What is the max design pressure on gravity tank?

- **Not greater than 0.07 MPa**
- Not greater than 0.05 MPa
- Not greater than 0.09 MPa
- Not greater than 1.0 MPa

What is the maximum quantity of type 1 chemicals to be transported in each tank?

- **1,250 m3**
- 1,400 m3
- 2,100 m3
- 3,000 m3

What is the meaning of the abbreviation COF?

- **Certificate of Fitness**
- Certificate of Finance
- Certificate of Flashpoint
- Certificate of Freeboard

What is the minimum capacity of the cargo tank venting system?

- **At least 125% of maximum loading capacity**
- At least 150% of maximum loading capacity
- Minimum 110% of maximum loading rate
- No such requirement

What is the minimum number of portable vapour (toxic or flammable) detection instruments required on board chemical tankers fitted with fixed vapour-detection system?

- **1**
- 0
- 2
- 3

What is the name of the United Nations organisation dealing with maritime matters?

- **IMO**
- IGC
- ILO
- IRA

What is the normal concentration of Carbon Monoxide (CO) in inert gas?

- **0.01%**
- 0.10%
- 0.25%
- 1.10

What is the “main system” in the cargo handling system?

- **The piping and valves**
- The cargo pumps
- The tank cleaning machines
- The ventilation system

Where do we find International regulations for the control of marine pollution by cargoes from chemical tankers?

- **In Annex II of MARPOL and in the IBC Code**
- In chapter 18 of the Bulk Chemical Code
- In MARPOL Annex I and in the IGC Code
- In SOLAS

Where do we find cargoes subject to the requirements of the “expanded” International Bulk Chemical Code ?

- **In Chapter 17 of the Code**
- In Chapter 18 of the Code
- In MARPOL Annex I
- The IGC Code

Where do we find requirements regarding types of gauging for individual products?

- **In the table of chapter 17 (IBC Code)**
- In MARPOL Annex II (Regulation 4)
- In SOLAS
- In the table of chapter 15 (IBC Code)

Where do you find list of cargoes not subject to the requirements of the “expanded” International Bulk Chemical Code”?

- **Chapter 18 of the Code**
- Chapter 17 of the Code
- In MARPOL Annex I
- In SOLAS

Which international convention is dealing with pollution prevention on chemical tankers?

- **MARPOL Annex II**
- ISGOTT
- MARPOL 73/78, Annex I
- STCW 95

Which is simplest member of the olefin family?

- **Ethylene**
- Butadiene
- Propylene
- Vinyl chloride

Which of the Codes applies to chemical vessels constructed after 1 July 1986?

- **The IBC Code**
- The IMDG Code
- The BCH Code
- The IGC Code

Which one of the conventions is dealing with safety and fire prevention?

- **SOLAS**
- MARPOL
- OPA 90
- STCW 95

Which one of the pumps listed below has no self priming capacity ?

- **Centrifugal pump**
- Piston pump
- Reciprocal pump
- Screw pump

Which three stages of illness may be distinguished in a typical case of poisoning?

- **The latent stage, the active stage and the late stage**
- The early stage, the critical stage and the hazard stage
- The ingestion stage, the inhalation stage and the exposure stage
- The symptoms stage, the drowsiness stage and the mental stage

Who shall verify that inhibited cargo is correctly inhibited before loading?

- **The Master and the shore representative**
- The Chief Officer and Loading Master
- The Master and the Safety Officer of the Port
- The Officer on duty and Port Captain

With regard to Annex II of MARPOL what is the definition of "liquid substances"?

- **Those having vapour pressure not exceeding 0.28 MPa absolute at a temperature of 37.8o C**
- All liquid cargoes
- Those having vapour pressure not exceeding 0.28 MPa absolute at a temperature of 39.7o C
- Those having vapour pressure not exceeding 0.38 MPa absolute at a temperature of 37.8o C

With regard to the IMO Bulk Chemical Code what is the definition of "new ship"?

- **A ship constructed on or after 1 July 1986**
- A ship constructed before 1 January 1984
- A ship constructed before 1 May 1983
- A ship constructed on or after 1 July 1984

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Version 3.0 (705)

(Other than a vessel en route to or from a port, installation, structure or pilot station within an inshore traffic zone or for the purposes of avoiding immediate danger) what length of vessel is permitted to use an inshore traffic zone, even if adjacent to a traffic separation scheme?

- **Less than 20 metres**
- Between 7 and 12 metres
- Less than 12 metres
- Less than 50 metres

A vessel is permitted to engage in fishing within a traffic lane but what obligations does that vessel have to other vessels proceeding within that traffic lane?

- **The fishing vessel shall not impede the passage of any vessel following a traffic lane**
- The fishing vessel shall not use outlying gear horizontally over a distance of more than 150 metres
- The fishing vessel shall only manoeuvre in accordance with the applicable special rules made by the appropriate authority for the roadsteads, channels and fairways connected to the traffic lane

According to Rule 15 "Crossing Situation", which vessel shall keep out of the way?

- **The vessel which has the other on her own starboard side**
- The vessel which has the other on her own port side

According to Rule 16 "Action by Give-way Vessel" in what way must a vessel take action to keep out of the way?

- **She shall, so far as possible, take early and substantial action to keep well clear**
- She shall avoid altering course to port for a vessel on her own port side
- She shall slacken her speed and then take all way off by stopping or reversing her means of propulsion
- She shall sound one short blast and make a broad alteration of course to starboard

According to Rule 17, if a stand-on vessel finds herself so close that collision cannot be avoided by the action of the give-way vessel on its own, what must that stand-on vessel do?

- **She shall take such action as will best aid to avoid collision**
- She must continue to maintain her course and speed in compliance with Rule 17 "Action by Stand-on Vessel"
- She must immediately stop and take all way off and in any event avoid passing ahead of the other vessel
- She shall sound one short blast and make a broad alteration of course to starboard

According to Rule 17, what may a stand-on vessel do, once it becomes apparent to her that a vessel required to keep out of the way is not taking appropriate action in compliance with the Rules?

- **She may take action to avoid collision by her manoeuvre alone**
- She may disregard the Rules of the Road from that point on
- She must attempt to contact the other vessel by VHF
- She should take action to avoid collision by alteration of her course and/or speed

According to Rule 18 "Responsibilities between Vessels", how should a "Vessel Constrained by her Draught" navigate?

- **With particular caution having full regard to her special condition**
- If there is sufficient sea-room, she should avoid impeding the passage of other vessels navigating in or near any deep-water routes
- In compliance with any special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected to the high seas and navigable by seagoing vessels
- She should give way only to dredgers and sailing vessels

According to Rule 20, when must the Rules referring to lights be complied with?

- **From sunrise to sunset in restricted visibility**
- **From sunset to sunrise**
- From sunrise to sunset

According to Rule 8 "Action to Avoid Collision", what action should you take if you need more time to assess a developing close-quarters situation?

- **Slacken your speed or take all way off by stopping or reversing your means of propulsion**
- Attempt to make contact with the other vessel by VHF
- Sound one short blast and make a broad alteration of course to starboard
- Use the signalling lamp to transmit the code letter "U", thereby indicating to the other vessel that you request they slacken their speed

**Assuming it is deemed necessary to your passage plan,
how should you plan to cross a traffic separation scheme?**

- **Track 1**
- Track 2
- Track 3

At what distance must outlying gear extend from a fishing vessel for it to be obliged to display an all-round white light or a cone apex upwards in the direction of the gear?

- **150 metres**
- 100 metres
- 1000 metres
- 500 metres
- 5000 metres

At what minimum safe distance should you pass this vessel?

- **1000 metres**
- 1 mile
- 100 metres
- 2000 metres
- 500 metres

Can Traffic Separation Schemes created by a Government or State be adopted by the International Maritime Organization for the purpose of the COLREGs?

- **Yes**
- No

How do the COLREGs define "Restricted Visibility"?

- **Any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes**
- Any condition in which visibility is reduced to less than a ship's length by fog, mist, falling snow, heavy rainstorms, dust, smog or other similar causes
- Fog, mist, heavy precipitation and sandstorms
- Rain, drizzle, hail or snow falling heavily enough to restrict visibility

How do the COLREGs explain the term "Underway"?

- **A vessel that is not at anchor, or made fast to the shore, or aground**
- A vessel that is entitles to exhibit sidelights and a sternlight, or by day the equivalent shapes where they can best be seen
- A vessel that is making way through the water

How do the Collision Regulations say that a "head-on situation" can be recognised?

- **The masthead lights of the other vessel would be in a line or nearly in a line and/or both sidelights can be seen and by day the corresponding aspect of the other vessel**
- The masthead lights of the other vessel would be in a line or nearly in a line but only one sidelight can be seen and by day the corresponding outline of the other vessel
- The sidelights of the other vessel would be in a line or nearly in a line and/or both masthead lights can be seen and by day the corresponding aspect of the other vessel
- The sternlight of the other vessel would be visible and the masts would be in a line or almost in a line

How do the Regulations say a vessel should proceed in terms of "speed"?

- **Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions**
- Every vessel shall at all times proceed at the speed listed in a vessel's Passage Plan and authorized by the Master
- Every vessel shall when possible proceed at a speed adapted to the prevailing circumstances and conditions, so as to minimize the deviation from her charter speed and schedule

How do the Rules describe "any action to avoid collision"?

- **Any action to avoid collision shall be taken in accordance with the Rules of this Part and shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship**
- Any action to avoid collision shall be made in accordance with any special rules made by the Government of any State with respect to close-quarters situations in waters within their jurisdiction
- Any action to avoid collision shall be taken in accordance with special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected to the high seas and navigable by seagoing vessels

How do the Rules describe "any alteration of course and/or speed to avoid collision"?

- **Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided**
- Any alteration of course and/or speed to avoid collision shall be made in accordance with special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected to the high seas and navigable by seagoing vessels; a series of alterations that may not be readily apparent to another vessel should be avoided
- Any alterations of course and/or speed to avoid collision shall, if possible in the prevailing circumstances and conditions, be such as to minimize any deviation from the track indicated in the vessel's passage plan

How does Rule 3 "Definitions" describe a "Sailing Vessel"?

- **Any vessel under sail provided that propelling machinery, if fitted, is not being used**
- Any vessel fitted with sails as the principal form of propulsion power
- Every vessel registered with an authorized sailing club, society or similar organization

How does Rule 3 "Definitions" describe a "Vessel Constrained by her Draught?

- **A vessel which because of her draught in relation to the available depth and width of navigable water is severely restricted in her ability to deviate from the course she is following**
- A vessel which because of her draught in relation to the surrounding depth and length of the navigable channel in which she is proceeding is likely to be restricted in her ability to deviate from the course she is following
- A vessel which because she has a deeper draught than the vessels around her has right of way

How does Rule 3 "Definitions" describe a "Vessel Not Under Command?

- **A vessel which through some exceptional circumstance is unable to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel**
- A vessel which from the nature of her work is restricted in her ability to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel
- A vessel which through a listed technical clause applied by the relevant Classification Society is exempt from manoeuvring in compliance with the International Collision Regulations

How does Rule 3 "Definitions" describe a "Vessel Restricted in her Ability to Manoeuvre?

- **A vessel which from the nature of her work is restricted in her ability to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel**
- A vessel which through a listed technical clause applied by the relevant Classification Society is exempt from manoeuvring in compliance with the International Collision Regulations
- A vessel which through some exceptional circumstance is unable to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel

How effective must any action taken to avoid collision be, according to Rule 8 "Action to Avoid Collision"?

- **Such as to result in passing at a safe distance**
- Adequate enough to permit both vessels to continue to observe successfully by radar until all danger is past and clear
- It must result in a minimum closest approach of at least 0.5 mile
- Such that a close-quarters situation is also avoided

How far does the gear extend horizontally from this fishing vessel?

- **More than 150 metres**
- At least 250 metres
- Less than 100 metres
- Nor more than 1000 metres
- There is no gear extending

How many forward masthead lights must be displayed in a vertical line forward on a power-driven vessel engaged in towing, when the length of the tow, measured from the stern of the towing vessel to the after end of the tow, exceeds 200 metres?

- **Three**
- Five
- One
- Two

How should a vessel maintain a look-out, as identified in Rule 5: "Look-out"?

- **By sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision**
- By all possible means, including radar, ARPA, VHF and AIS to establish if there is a risk of collision
- By visual and aural means, as well as by any other applicable electronic means possible in the prevailing circumstances and conditions, so as to make a reasonable appraisal of the situation and decide if there might be a risk of collision

How should you normally join a traffic separation scheme?

- **At the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable**
- At a junction in the lane, but when joining or leaving from either side shall do so at as large an angle to the general direction of traffic flow as practicable
- At the termination of the lane, but when joining or leaving from either side shall do so whenever a suitable break in the general flow of traffic presents itself
- Only from the side of a traffic lane and whenever a suitable break in the general flow of traffic presents itself.

Identify the vessel

- **Vessel engaged in fishing, length unknown**
- Power-driven vessel underway, restricted in her ability to manoeuvre
- Vessel constrained by her draught, underway
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the cone shapes

Identify the vessel

- **Vessel engaged in fishing (trawling), length unknown, not making way through the water**
- Vessel engaged in fishing other than trawling, less than 50 metres in length, making way through the water
- Vessel engaged in fishing, at anchor
- Vessel engaged in pilotage duty

Identify the vessel

- **Vessel engaged in fishing (trawling), less than 50 metres in length, making way through the water**
- Vessel engaged in fishing other than trawling, less than 50 metres in length, making way through the water
- Vessel engaged in fishing, at anchor
- Vessel engaged in pilotage duty

Identify the vessel

- **Vessel engaged in fishing (trawling), length unknown, making way through the water**
- Vessel engaged in fishing other than trawling, less than 50 metres in length, not making way through the water
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the all-round white light

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water**
- Power-driven vessel underway, restricted in her ability to manoeuvre
- Vessel engaged in fishing other than trawling, less than 50 metres in length, making way through the water
- Vessel engaged in fishing, at anchor

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water**
- Vessel engaged in fishing other than trawling, less than 50 metres in length, making way through the water
- Vessel engaged in fishing, hauling nets
- Vessel engaged in fishing, shooting nets

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, not making way through the water**
- Vessel engaged in fishing other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water
- Vessel engaged in fishing other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water, outlying gear extending more than 150 metres horizontally in the direction of the all-round white light
- Vessel engaged in towing, length of tow unknown

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, shooting nets**
- Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, hauling nets
- Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water, outlying gear extending more than 150 metres horizontally in the direction of the all-round white lights

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, shooting nets**
- Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, hauling nets
- Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water, outlying gear extending more than 150 metres horizontally in the direction of the all-round white lights

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, shooting nets**
- Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, shooting nets, at anchor
- Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water, outlying gear extending more than 150 metres horizontally in the direction of the all-round white lights

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, hauling nets**
- Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, shooting nets
- Vessel engaged in fishing (trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, nets fast on an obstruction

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, making way through the water, hauling nets**
- Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, making way through the water, shooting nets
- Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, nets fast on an obstruction

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 20 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction**
- Vessel engaged in fishing (trawling), more than 20 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, not making way through the water, shooting nets
- Vessel engaged in fishing other than trawling, more than 20 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction**
- Vessel engaged in fishing other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction

Identify the vessel

- **Vessel engaged in fishing (trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction**
- Vessel engaged in fishing other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, not making way through the water, nets fast on an obstruction

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), length unknown, at anchor**
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the diamond
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the diamond
- Vessel restricted in her ability to manoeuvre, engaged in mine clearance operations, at anchor

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), length unknown, at anchor**
- Power-driven vessel underway, restricted in her ability to manoeuvre
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the diamond
- Vessel restricted in her ability to manoeuvre, engaged in mine clearance operations, at anchor

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), more than 50 metres in length, or less than 50 metres in length and opting to show the lights for and aft, at anchor**
- Power-driven vessel underway, restricted in her ability to manoeuvre
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the diamond
- Vessel restricted in her ability to manoeuvre, engaged in mine clearance operations, at anchor

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), length unknown, at anchor**
- Power-driven vessel underway, restricted in her ability to manoeuvre
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally in the direction of the all-round white light
- Vessel in distress

Identify the vessel

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, length unknown, obstruction to her port side**
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, length unknown, obstruction to her starboard side

Identify the vessel

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, length unknown, obstruction to her starboard side**
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, length unknown, obstruction to her port side

Identify the vessel

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light, obstruction to her starboard side**
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light, obstruction to her port side
- Vessel engaged in towing alongside, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light, objects being towed to both sides, restricted in their ability to deviate from their course

Identify the vessel

- **Vessel engaged in diving operations**
- Vessel engaged in fishing, less than 12 metres in length
- Vessel engaged in pilotage duty
- Vessel in distress

Identify the vessel

- **Vessel engaged in mine clearance operations, underway**
- Vessel engaged in fishing, outlying gear extending more than 150 metres horizontally on both sides of the vessel
- Vessel engaged in pilotage duty
- Vessel in distress

Identify the vessel

- **Vessel engaged in mine clearance operations**
- Vessel engaged in helicopter operations, not making way
- Vessel engaged in underwater operations, at anchor
- Vessel engaged on convoy duties, making way through the water
- Warship engaged on manoeuvres as part of a fleet

Identify the vessel

- **Vessel engaged in mine clearance operations, underway, less than 50 metres in length**
- Vessel engaged in dredging or underwater operations, underway, no obstruction exists on either side of the vessel
- Vessel engaged in mine clearance operations, underway, more than 50 metres in length
- Vessel engaged on convoy duties, making way through the water

Identify the vessel

- **Vessel engaged in mine clearance operations, underway, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light**
- Vessel engaged in mine clearance operations, underway, less than 20 metres in length
- Vessel engaged in towing, more than 50 metres in length, tow under 200 metres in length
- Vessel engaged on convoy duties, making way through the water

Identify the vessel

- **Vessel constrained by her draught, underway, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light**
- Vessel constrained by her draught, underway, less than 50 metres in length
- Vessel not under command, making way through the water
- Vessel restricted in her ability to manoeuvre, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water

Identify the vessel

- **Vessel constrained by her draught, underway, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light**
- Vessel constrained by her draught, underway, less than 50 metres in length
- Vessel not under command, making way through the water
- Vessel restricted in her ability to manoeuvre, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water

Identify the vessel

- **Vessel constrained by her draught, underway, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light**
- Vessel constrained by her draught, underway, less than 50 metres in length
- Vessel not under command, making way through the water
- Vessel restricted in her ability to manoeuvre, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water

Identify the vessel

- **Vessel constrained by her draught, underway**
- Vessel constrained by her draught, at anchor
- Vessel not under command, at anchor
- Vessel restricted in her ability to manoeuvre, at anchor

Identify the vessel

- **Vessel engaged in pilotage duty, underway**
- Sailing vessel underway
- Vessel engaged in fishing other than trawling
- Vessel not under command, at anchor

Identify the vessel

- **Vessel engaged in pilotage duty, underway**
- Sailing vessel underway
- Vessel engaged in fishing other than trawling
- Vessel not under command, at anchor

Identify the vessel

- **Vessel engaged in pilotage duty, at anchor, exhibiting identification signal**
- Sailing vessel underway, using propelling machinery
- Vessel engaged in pilotage duty, underway, exhibiting identification signal
- Vessel not under command, at anchor, exhibiting warning signal to approaching vessels

Identify the vessel

- **Vessel at anchor**
- Vessel displaying signal requesting a pilot
- Vessel engaged in underwater operations
- Vessel not under command

Identify the vessel

- **Vessel at anchor, length unknown, head-on**
- **Vessel at anchor, length unknown, stern-on**
- **Vessel at anchor, less than 50 metres in length, displaying single all-round white light**
- Vessel aground

Identify the vessel

- **Vessel at anchor, seen from starboard side**
- Vessel aground
- Vessel at anchor, seen from port side
- Vessel engaged in towing alongside, stern-on

Identify the vessel

- **Vessel aground, seen from port side**
- Vessel aground, seen from starboard side
- Vessel engaged in towing alongside, stern-on
- Vessel not under command

Identify the vessel

- **Vessel aground**
- Vessel engaged in fishing, hauling nets
- Vessel not under command, at anchor
- Vessel restricted in her ability to manoeuvre

Identify the vessel

- **Vessel in distress**
- Vessel engaged in underwater operations, bout to deploy divers
- Vessel not under command, at anchor, giving warning signal to approaching vessels
- Vessel requesting pilot

Identify the vessel

- **Vessel in distress**
- Pleasure craft landing passengers by dinghy
- Vessel engaged in diving operations
- Vessel engaged in pilotage duty

Identify the vessel

- **Vessel in distress**
- Vessel engaged in fishing by trawl lines
- Vessel engaged in pilotage duty
- Vessel engaged in underwater operations

Identify the vessel

- **Vessel in distress**
- Aircraft in distress
- Vessel engaged in underwater operations, about to deploy sonar
- Vessel requesting pilot

Identify the vessel

- **Vessel in distress**
- Aircraft in distress
- Port signal to indicate “Fairway closed”
- Vessel engaged in underwater operations, about to deploy divers

Identify the vessel

- **Vessel in distress**
- Vessel aground
- Vessel engaged in underwater operations, divers down
- Vessel on sea trials

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), not making way through the water**
- Vessel constrained by her draught
- Vessel engaged in fishing other than trawling, making way through the water, hauling nets
- Vessel not under command, at anchor

Identify the vessel

- **Submarine, underway**
- Aircraft carrier, engaged in flying on and off aircraft
- Vessel engaged in fishing with purse seine gear
- WIG craft

Identify the vessel

- **Power-driven vessel, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light, engaged in towing where tow exceeds than 200 metres in length, being an inconspicuous, partly submerged vessel or object, restricted in her ability to manoeuvre**
- Power-driven vessel, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light, engaged in towing where tow is under 200 metres in length, being an inconspicuous, partly submerged vessel or object, restricted in her ability to manoeuvre

Identify the vessel

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, length unknown, at anchor, no obstructions exist on either side of the vessel**
- Power-driven vessel, less than 50 metres in length, engaged in towing alongside to both sides, tug and tow restricted in their ability to deviate from their course
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, length unknown, not making way, obstructions on both sides of the vessel

Identify the vessel

- **Power-driven vessel underway, less than 50 metres in length**
- Power-driven vessel underway, more than 50 metres in length
- Power-driven vessel, at anchor
- Vessel engaged in towing, less than 50 metres in length

Identify the vessel

- **Power-driven vessel underway, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light**
- Power-driven vessel underway, less than 50 metres in length
- Vessel at anchor, over 100 metres in length

Identify the vessel

- **Air-cushioned vessel operating in the non-displacement mode, less than 50 metres in length**
- Power-driven vessel underway, less than 50 metres in length
- Vessel engaged in fishing, using purse-seine gear, starboard side
- Wing-in-Ground Vessel taking-off, landing or in flight near the surface

Identify the vessel

- **Air-cushioned vessel operating in the non-displacement mode, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, underway**
- Air-cushioned vessel operating in the non-displacement mode, less than 50 metres in length, underway
- Vessel engaged in fishing, using purse-seine gear
- Wing-in-Ground Vessel taking-off, landing or in flight near the surface

Identify the vessel

- **Power-driven vessel underway, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, restricted in her ability to manoeuvre**
- Power-driven vessel engaged in towing, less than 50 metres in length, length of tow under 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Power-driven vessel underway, more than 50 metres in length, constrained by her draught
- Vessel engaged in fishing other than trawling, at anchor, hauling nets

Identify the vessel

- **Wing-in-Ground Vessel, less than 50 metres in length, taking-off, landing or in flight near the surface**
- Air-cushioned vessel operating in the non-displacement mode, less than 50 metres in length
- Sailing vessel underway
- Vessel engaged in fishing other than trawling, not making way through the water, nets fast on an obstruction

Identify the vessel

- **Wing-in-Ground Vessel, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, taking-off, landing or in flight near the surface**
- Air-cushioned vessel operating in the non-displacement mode, less than 50 metres in length
- Power-driven vessel underway, engaged in underwater operations
- Vessel engaged in fishing other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, not making way through the water, nets fast on an obstruction

Identify the vessel

- **Vessel engaged in fishing other than trawling, length unknown, making way through the water, where there is outlying gear extending more than 150 metres horizontally in the direction of the additional all-round white light**
- Pilot vessel on station, showing the additional identification white signal light
- Vessel engaged in fishing other than trawling, more than 20 metres in length, not making way through the water, hauling nets
- Vessel engaged in trawling, length unknown, not making way through the water, where there is outlying gear extending more than 500 metres horizontally in the direction of the additional all-round white light

Identify the vessel

- **Sailing vessel underway**
- Pilot vessel underway
- Sailing vessel, being propelled by machinery
- Vessel engaged in fishing, other than trawling

Identify the vessel

- **Sailing vessel underway**
- Pilot vessel underway
- Sailing vessel at anchor
- Vessel engaged in fishing, gear extending 150 metres horizontally in the direction of the white light

Identify the vessel

- **Sailing vessel underway**
- Pilot vessel underway
- Vessel engaged in fishing, gear extending 150 metres horizontally in the direction of the white light

Identify the vessel

- **Sailing vessel underway, less than 20 metres in length**
- Sailing vessel making way, less than 50 metres in length
- Sailing vessel, length unknown, at anchor
- Vessel engaged in fishing, other than trawling, less than 12 metres in length

Identify the vessel

- **Sailing vessel underway**
- Pilot vessel at anchor
- Power-driven vessel underway but not making way through the water
- Sailing vessel underway, but not making way through the water

Identify the vessel

- **Vessel engaged in fishing with purse seine gear, length unknown, making way through the water and hampered by its gear**
- Sailing vessel, propelled by machinery
- Vessel engaged in fishing, other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, shooting nets
- Vessel engaged in pilotage duty, deploying pilot launch

Identify the vessel

- **Vessel engaged in fishing with purse seine gear, length unknown, making way through the water and hampered by its gear**
- Sailing vessel, propelled by machinery
- Vessel engaged in fishing, other than trawling, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, at anchor, shooting nets
- Vessel engaged in pilotage duty, deploying pilot launch

Identify the vessel

- **Vessel not under command, length unknown, making way through the water**
- Sailing vessel underway
- Vessel constrained by her draught, length unknown, underway
- Vessel engaged in pilotage duty

Identify the vessel

- **Vessel not under command, length unknown, not making way through the water**
- Sailing vessel underway
- Vessel not under command, length unknown, making way through the water
- Vessel restricted in her ability to manoeuvre, length unknown, underway

Identify the vessel

- **Vessel not under command**
- Pilot vessel, at anchor
- Vessel constrained by her draught
- Vessel restricted in her ability to manoeuvre

Identify the vessel

- **Vessel not under command, making way through the water**
- Vessel constrained by her draught
- Vessel engaged in pilotage duty
- Vessel restricted in her ability to manoeuvre, making way through the water

Identify the vessel

- **Vessel restricted in her ability to manoeuvre, (other than a vessel engaged in mine clearance operations), less than 50 metres in length, making way through the water**
- Power-driven vessel, less than 50 metres in length, engaged in towing where length of tow is under 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Vessel constrained by her draught, underway
- Vessel restricted in her ability to manoeuvre, (other than a vessel engaged in mine clearance operations), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water

Identify the vessel

- **Vessel restricted in her ability to manoeuvre, (other than a vessel engaged in mine clearance operations), less than 50 metres in length, making way through the water**
- Power-driven vessel, less than 50 metres in length, engaged in towing where length of tow is under 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Vessel constrained by her draught, underway
- Vessel restricted in her ability to manoeuvre, (other than a vessel engaged in mine clearance operations), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water

Identify the vessel

- **Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, making way through the water**
- Power-driven vessel, less than 50 metres in length, engaged in towing where length of tow is under 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Vessel constrained by her draught, underway
- Vessel restricted in her ability to manoeuvre (other than a vessel engaged in mine clearance operations), length unknown, making way through the water

Identify the vessel(s)

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, less than 50 metres in length, obstruction to her port side**
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, less than 50 metres in length, obstruction to her starboard side
- Vessel engaged in towing alongside, less than 50 metres in length, objects being towed to both sides, restricted in their ability to deviate from their course

Identify the vessel(s)

- **Power-driven vessel engaged in towing, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light and tow, where length of tow under 200 metres, not restricted in their ability to deviate from their course**
- Power-driven vessel engaged in towing, less than 50 metres in length, length of tow exceeds 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Vessel engaged in fishing other than trawling,
more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, shooting nets

Identify the vessel(s)

- **Power-driven vessel, length unknown, engaged in towing where length of tow exceeds 200 metres, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel, length unknown, engaged in towing a dracone, less than 25 metres in breadth, towing vessel and tow restricted in their ability to deviate from their course
- Power-driven vessel, restricted in her ability to manoeuvre, at anchor
- Power-driven vessel, restricted in her ability to manoeuvre, engaged in underwater operations

Identify the vessel(s)

- **Power-driven vessel, engaged in towing alongside, length unknown and vessel being towed, length of tow unknown, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel, less than 50 metres in length, engaged in pushing ahead, lit as a composite unit, restricted in their ability to manoeuvre
- Power-driven vessel, more than 50 metres in length, engaged in towing alongside and vessel being towed, length of tow unknown, towing vessel and tow restricted in their ability to manoeuvre

Identify the vessel(s)

- **Power-driven vessel engaged in towing alongside, length unknown and vessel being towed, length of tow unknown, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel, less than 50 metres in length, engaged in pushing ahead, lit as a composite unit, restricted in their ability to manoeuvre
- Power-driven vessel, more than 50 metres in length, engaged in towing alongside and vessel being towed, length of tow unknown, towing vessel and tow restricted in their ability to manoeuvre

Identify the vessel(s) (select all applicable answers)

- **Power-driven vessel, length unknown, engaged in towing a dracone, less than 25 metres in breadth, towing vessel and tow restricted in their ability to deviate from their course**
- **Power-driven vessel, length unknown, engaged in towing and vessel being towed, length of tow unknown, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel, engaged in towing, where the length of tow exceeds 200 metres,
- Vessel engaged in fishing, using purse-seine gear, restricted in her ability to manoeuvre and hampered by its gear

Identify the vessel(s) (select all possible answers)

- **Power-driven vessel underway, more than 50 metres in length, or less than 50 metres in length and showing the a second masthead light**
- **Vessel engaged in towing, less than 50 metres in length, length of tow under 200 metres in length**
- Power-driven vessel underway, less than 50 metres in length
- Vessel engaged in towing, less than 50 metres in length, length of tow over 200 metres in length

Identify the vessel(s) (select all applicable answers)

- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, length unknown, at anchor, obstruction to the left side as observed**
- **Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, length unknown, not making way, obstruction to the left side as observed**
- Power-driven vessel, less than 50 metres in length, engaged in towing alongside to port, tug and tow restricted in their ability to deviate from their course
- Vessel engaged in dredging or underwater operations, restricted in her ability to manoeuvre, making way through the water, stern-on, obstruction to her port side

Identify the vessel(s) (select all possible answers)

- **Power-driven vessel engaged in towing, less than 50 metres in length, length of tow exceeds 200 metres, towing vessel and tow restricted in their ability to deviate from their course**
- **Power-driven vessel engaged in towing, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, length of tow under 200 metres, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel underway, more than 50 metres in length, restricted in her ability to manoeuvre
- Vessel engaged in fishing, shooting nets, restricted in her ability to manoeuvre, at anchor

Identify the vessel(s) (select all possible answers)

- **Vessel aground, head-on**
- **Vessel aground, stern-on**
- **Vessel not under command, length unknown, making way through the water, stern on**
- Vessel restricted in her ability to manoeuvre, making way through the water, stern-on

Identify the vessels

- **Vessels engaged in fishing (pair-trawling), more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, making way through the water**
- Vessel engaged in fishing with purse-seine equipment, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, or less than 20 metres in length and also showing the additional signals voluntarily, making way through the water

Identify the vessels

- **Power-driven vessel engaged in towing, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light and tow, where length of tow exceeds 200 metres, towing vessel and tow restricted in their ability to deviate from their course**
- Power-driven vessel engaged in towing, less than 50 metres in length, where length of tow exceeds 200 metres, towing vessel and tow restricted in their ability to deviate from their course
- Power-driven vessel engaged in towing, more than 50 metres in length, or less than 50 metres in length and showing a second masthead light, length of tow under 200 metres, towing vessel and tow restricted in their ability to deviate from their course

Identify the vessels

- **Vessels engaged in fishing (pair-trawling), more than 20 metres but less than 50 metres in length, or less than 20 metres in length and showing the additional signals voluntarily, making way through the water, hauling nets**
- Power-driven vessel engaged in towing, where each vessel is illuminating the other for the purposes of safety
- Vessels engaged in fishing (pair-trawling), more than 50 metres in length, making way through the water, hauling nets
- Vessels engaged in fishing (pair-trawling), more than 50 metres, making way through the water, shooting nets

If whistles are fitted on a vessel at a distance apart of more than 100 metres, how will manoeuvring and warning signals be given?

- **On one whistle only**
- On both whistles simultaneously
- On the forward whistle first, then the after whistle immediately afterwards

In order for a Wing-in-Ground craft to exhibit a high intensity all-round flashing red light in compliance with the Rules, what must she be doing?

- **Taking-off, landing and in flight near the surface**
- Loading or boarding passengers
- Navigating as a power-driven vessel underway, but restricted in her ability to manoeuvre
- Refuelling

In terms of "Risk of Collision", what do the Rules say about "scanty radar information"?

- **Assumptions shall not be made on the basis of scanty information, especially scanty radar information**
- Actions taken to avoid collision shall be carefully monitored until all risk of danger is past, particularly when such actions are based on scanty radar information
- Conclusions must not be drawn from scanty radar information that may have been used in an automatic radar plot

In which part of a narrow channel or fairway is a vessel to proceed?

- **As near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable**
- As near to the outer limit of the channel or fairway which lies on her port side as is safe and practicable
- In the middle

Is a power-driven vessel underway, of less than 50 metres in length, obliged to exhibit a second masthead light?

- **No, but may do so by Rule 23: "Power-driven Vessels Underway"**
- No, but may be required to do so in compliance with Rule 27 "Vessels Not Under Command or Restricted in their Ability to Manoeuvre"
- Yes
- Yes, but only in the vicinity of a narrow channel or fairway

Under which of the following circumstances would you consider yourself an overtaking vessel?

- **When coming up with another vessel from a direction of more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights**
- When approaching from behind another vessel, such that with reference to the vessel she is overtaking, she would see the sternlight at the same time as the appropriate sidelight become visible
- When coming up with another vessel from a direction of more than 60 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that by day, she would see the masts in a line, or nearly in a line

What additional identification signal may be sounded by a pilot vessel "engaged in pilotage duty"?

- **Four short blasts**
- One prolonged blast
- One prolonged blast, followed by three short blasts
- Two short blasts

What additional whistle signal might a vessel aground sound in restricted visibility, to give warning of her presence to an approaching vessel?

- **The code letter "U"**
- The code letter "H"
- The code letter "I"
- The code letter "S"

What additional whistle signal might a vessel at anchor sound in restricted visibility, to give warning of her presence to an approaching vessel?

- **The code letter "R"**
- The code letter "A"
- The code letter "C"
- The code letter "T"
- Three short blasts

What additional whistle signal might this vessel sound to aid identification in conditions of restricted visibility?

- **Four short blasts**
- One prolonged blast
- The code letter "R"
- Two short blasts

What alternative arrangement to sidelights and sternlight may be exhibited by a sailing vessel of less than 20 metres in length?

- **A combined lantern**
- A blue light, where it can best be seen
- A conical shape, apex down, where it can best be seen
- Placing the lights close together near the top of the mast

What colour is a towing light?

- **Yellow**
- Blue (flashing)
- Orange
- White

What day signal must be displayed by a power-driven vessel engaged in towing, when the length of the tow exceeds 200 metres?

- **A diamond shape, where it can best be seen**
- A cone, apex downwards, in the direction of he tow
- A cylinder shape on the foremast
- Two black balls where they can best be seen

What day signal must be displayed by a vessel or object being towed when the tow length exceeds 200 metres, (other than an inconspicuous or partly submerged vessel or object or combination of such vessels or objects)?

- **A diamond shape, where it can best be seen**
- A black ball, at the aft end of the tow
- A black ball, at the fore end of the tow
- The International Code signal flag 'T' at the aft end of the tow

What do the two red lights on this vessel indicate?

- **That an obstruction exists on this side of the vessel**
- That an object is being towed alongside on that side of the vessel
- That no obstruction exists on this side of the vessel
- That the vessel is "Not Under Command"

What does Rule 8 "Action to Avoid Collision" recommend as potentially the most effective action to avoid a close-quarters situation?

- **Alteration of course alone, provided that it is made in good time, is substantial and does not result in another close-quarters situation**
- Alteration of course alone, provided that an alteration to course to port for a vessel forward of the beam is avoided
- Alteration of course and/or speed, provided that it is made in ample time, at an adequate distance and does not result in increasing the risk of collision with a vessel on the port side

What does a vessel under sail exhibit to indicate that it is also being propelled by machinery?

- **A conical shape, apex down, where it can best be seen**
- A conical shape, apex upwards, in the fore part of the vessel
- A cylindrical shape, where it can best be seen
- A diamond shape, at the mainmast head

What does this signal mean?
- Narrow channel
- Clear conditions
- Vessel in sight

- **"I intend to overtake you on your starboard side"**
- "I am dropping my anchor"
- "I am engaged in changing my pilot"
- "I intend to overtake you on your port side"

What does this signal mean?
- Narrow channel
- Clear conditions
- Vessel in sight

- **"I intend to overtake you on your port side"**
- "I am engaged in embarking / disembarking my pilot"
- "I am operating astern propulsion"
- "I intend to overtake you on your starboard side"

What does this signal mean?
- Narrow channel
- Clear conditions
- Vessel in sight

- **"I agree to your overtaking manoeuvre"**
- "I am operating astern propulsion"
- "I intend to overtake you on your port side"
- "I intend to overtake you on your starboard side"

What does this signal mean?
- Open water
- Clear conditions

- **"I am altering my course to starboard"**
- Fog signal of a power-driven vessel
- Fog signal of a vessel under 20 metres in length, at anchor
- "I am altering my course to port"

What does this signal mean?
- Open water
- Clear conditions

- **"I am altering my course to port"**
- Fog signal of a power-driven vessel
- Fog signal of a vessel under 20 metres in length, at anchor
- "I am altering my course to starboard"

What does this signal mean?
- Open water
- Clear conditions

- **"I am operating astern propulsion"**
- Fog signal of a vessel under 20 metres in length, at anchor
- "I am altering my course to port"
- "I am dropping my anchor"

What does this signal mean? (Select all applicable answers)
- Open water
- Clear conditions
- Vessel in sight

- **"I am in doubt that you are taking sufficient action to avoid collision"**
- **"I fail to understand your intentions"**
- "I agree to your overtaking manoeuvre"
- "I am operating astern propulsion"

What does this signal mean?(select all applicable answers)
- Single signal (not repeated at regular interval)
- Narrow channel
- View obscured by intervening obstruction

- **"Response to a vessel, obscured from view, which has indicated her presence by a sound signal"**
- **"Signal to indicate presence to a vessel out of sight"**
- "I am engaged in embarking / disembarking my pilot"
- "I am in doubt that you are taking sufficient action to avoid collision"

What is the arc of visibility for a masthead light, as described in Rule 21: Definitions?

- **225º, from right ahead to 22.5º abaft the beam on either side of the vessel**
- 180º, from right ahead to directly on the beam on either side of the vessel
- 200º, from right ahead to 10º abaft the beam on either side of the vessel
- 250º, from right ahead to 35º abaft the beam on either side of the vessel

What is the arc of visibility for a sidelight, as described in Rule 21: Definitions?

- **112.5º, from right ahead to 22.5º abaft the beam on the respective side of the vessel**
- 100º, from right ahead to 10º abaft the beam on the respective side of the vessel
- 120º, from right ahead to 20º abaft the beam on the respective side of the vessel
- 225º, from right ahead to 22.5º abaft the beam on either side of the vessel

What is the arc of visibility of the additional red and green lights exhibited at or near the top of the mast of a sailing vessel?

- **All round**
- 120º, from right astern to 60º on either side of the vessel
- 200º, from right ahead to 10º abaft the beam on either side of the vessel
- 225º, from right ahead to 22.5º abaft the beam on either side of the vessel

What is the definition of a fishing vessel engaged in trawling?

- **One which drags a dredge net or other apparatus used as a fishing appliance through the water**
- One where there is outlying gear extending more than 150 metres horizontally from the vessel
- One which displays green and white lights
- One which has been assessed and categorized as capable of operating purse-seine equipment by the appropriate Classification Society

What is the duration of a "prolonged" blast as defined in the COLREGs?

- **4-6 seconds**
- 10 seconds
- 15-20 seconds
- 3 seconds

What is the fog signal for a vessel aground, 150 metres in length?

- **At not more than 1 minute intervals: 3 strokes on bell forward, rapid ringing of the bell for about 5 seconds, 3 strokes on the bell, sounding gong aft for about 5 seconds**
- At not more than 1 minute intervals: 5 strokes on bell forward, rapid ringing of the bell for about 5 seconds, 5 strokes on the bell, sounding gong aft for about 10 seconds
- At not more than 2 minute intervals: 5 strokes on bell forward, rapid ringing of the bell for about 5 seconds, 5 strokes on the bell, sounding gong aft for about 5 seconds

What is the fog signal of a vessel "Constrained by her Draught"?

- **At not more than 2 minute intervals, one prolonged blast, followed by two short blasts**
- At not more than 1 minute intervals, two short blasts
- At not more than 2 minute intervals, one prolonged blast
- At not more than 2 minute intervals, one short blast

What is the fog signal of a vessel "Restricted in her Ability to Manoeuvre"?

- **At not more than 2 minute intervals, one prolonged blast, followed by two short blasts**
- At not more than 1 minute intervals, two short blasts
- At not more than 2 minute intervals, one prolonged blast
- At not more than 2 minute intervals, one short blast

What is the frequency of a "flashing light", as defined in Rule 21: Definitions?

- **120 flashes or more per minute**
- 100 flashes or more per minute
- 150 flashes or more per minute
- 60 flashes or more per minute

What is the required minimum range for a masthead light on a vessel of 50 metres or more in length?

- **6 miles**
- 10 miles
- 1000 metres
- 3 miles

What is the required minimum range for a sidelight on a vessel of 50 metres or more in length?

- **3 miles**
- 1000 metres
- 2 miles
- 6 miles

What is the "safe distance" around a vessel engaged in mine clearance operations as defined by the Rules?

- **1000 metres**
- 2000 metres
- 250 metres
- 500 metres
- 5000 metres

What length is the vessel making this signal?
- (Repeated - not more than 1 minute interval)
- In or near an area of restricted visibility

- **Less than 100 metres**
- Any length
- Less than 12 metres
- More than 100 metres

What lights are required on an inconspicuous, partly submerged vessel or object of less than 25 metres in breadth, (other than a dracone) being towed?

- **One all-round white light at or near the forward and one all-round white light at or near the aft end**
- One all-round blue light at or near the aft end
- One all-round yellow light at or near the forward and one all-round yellow light at or near the aft end
- Two all-round yellow lights at the extremities of breadth on the aft end

What term is applied to a partly submerged flexible container used for transporting liquid cargoes?

- **Dracone**
- Flexi-cone
- Seacone
- Subcone

What vessel makes this signal?

- **A vessel in distress**
- A vessel aground
- A vessel engaged in underwater operations, about to deploy divers
- A vessel requesting a pilot

What vessel makes this signal?

- **A vessel in distress**
- A vessel aground
- A vessel engaged in underwater operations, about to deploy divers
- A vessel requesting a pilot

What vessel makes this signal?
- (Repeated - not more than 2 minute interval)
- Open water
- In or near an area of restricted visibility

- **A pilot vessel engaged in pilotage duty, less than 100 metres in length, at anchor**
- A pilot vessel, engaged in pilotage duty, more than 100 metres in length, stopped and making no way through the water
- A sailing vessel, making way through the water
- A vessel engaged in underwater operations, stopped and making no way through the water

What vessel makes this signal?
- (Immediately after a sound signal comprising one prolonged blast plus two short blasts)
- In or near an area of restricted visibility

- **A vessel being towed, if manned**
- A vessel indicating that an earlier signal is to be disregarded
- A vessel operating astern propulsion
- The last vessel in a series of vessels in a single tow, if manned

What vessel makes this signal?
- (Repeated - at 1 minute intervals)

- **A vessel in distress**
- A vessel engaged in underwater operations, about to deploy divers
- A vessel requesting a pilot

What vessel makes this signal?
- (Repeated - not more than 1 minute interval)
- In or near an area of restricted visibility

- **A vessel at anchor, less than 100 metres in length**
- A vessel at anchor, less than 12 metres in length
- A vessel at anchor, more than 100 metres in length
- A vessel engaged in underwater operations, at anchor

What vessel makes this signal?
- (Repeated - not more than 1 minute interval)
- In or near an area of restricted visibility

- **A vessel at anchor, more than 100 metres in length**
- A vessel at anchor, more than 20 metres in length
- A vessel at anchor, more than 50 metres in length
- A vessel engaged in mine-clearance operations, at anchor

What vessel makes this signal?
- (Repeated - not more than 1 minute interval)
- In or near an area of restricted visibility

- **A vessel aground, less than 100 metres in length**
- A vessel aground, more than 100 metres in length
- A vessel at anchor, less than 100 metres in length
- A vessel at anchor, more than 100 metres in length

What vessel makes this signal?
- (Repeated - not more than 1 minute interval)
- In or near an area of restricted visibility

- **A vessel aground, more than 100 metres in length**
- A vessel aground, less than 100 metres in length
- A vessel at anchor, less than 100 metres in length
- A vessel at anchor, more than 100 metres in length

What vessel makes this signal?
- (Repeated - not more than 2 minute interval)
- In or near an area of restricted visibility

- **A power-driven vessel, making way through the water**
- A power-driven vessel, stopped and making no way through the water
- A vessel at anchor
- A vessel operating astern propulsion

What vessel makes this signal?
- (Repeated - not more than 2 minute interval)
- In or near an area of restricted visibility

- **A power-driven vessel, stopped and making no way through the water**
- A power-driven vessel, making way through the water
- A vessel at anchor
- A vessel operating astern propulsion

What vessel makes this signal?
- (Repeated - not more than 2 minute interval)
- Open water
- In or near an area of restricted visibility

- **A pilot vessel engaged in pilotage duty, making way through the water**
- A sailing vessel making way through the water
- A vessel engaged in underwater operations, stopped and making no way through the water
- A vessel in distress

What vessel makes this signal?
- (Repeated - not more than 2 minute interval)
- Open water
- In or near an area of restricted visibility

- **A pilot vessel, engaged in pilotage duty, stopped and making no way through the water**
- A sailing vessel, making way through the water
- A vessel engaged in underwater operations, stopped and making no way through the water
- A vessel in distress

What vessel makes this signal? (select all applicable answers)
- (Repeated - not more than 2 minute interval)
- In or near an area of restricted visibility

- **A sailing vessel**
- **A vessel constrained by her draught**
- **A vessel engaged in fishing**
- **A vessel engaged in towing or pushing another vessel**
- **A vessel not under command**
- **A vessel restricted in her ability to manoeuvre**

What whistle signal can a vessel proceeding along a narrow, inland channel and approaching a bend around a high bluff or headland use to indicate its presence to another vessel which may be out of sight around that bend?

- **One prolonged blast**
- One prolonged blast, followed by two short blasts
- Three prolonged blasts
- Three short blasts

What whistle signal would you use to indicate to another vessel in sight that you had put your engines astern?

- **Three short blasts**
- One prolonged blast
- One prolonged blast, followed by two short blasts
- Three prolonged blasts

When a power-driven vessel is stopped and making no way through the water, what would her fog signal be?

- **At not more than 2 minute intervals, two prolonged blasts, with about 2 seconds between them**
- At not more than 1 minute intervals, two short blasts, with about three seconds between them
- At not more than 2 minute intervals, one prolonged blast
- At not more than 2 minute intervals, one prolonged blast, followed by two short blasts

When a vessel is in doubt as to whether she is overtaking another vessel, what must she do according to the COLREGs?

- **She shall assume that this is the case and act accordingly**
- She shall assume that this is not the case and act accordingly
- She shall maintain her course and speed
- She shall sound one short blast and make a broad alteration of course to starboard

When in open waters and in sight of one another, what whistle signal should a power-driven vessel underway make to indicate that she is altering her course to port?

- **Two short blasts**
- One prolonged blast
- One prolonged blast, followed by two short blasts
- One short blast

When in open waters and in sight of one another, what whistle signal should a power-driven vessel underway make to indicate that she is altering her course to starboard?

- **One short blast**
- One prolonged blast
- One prolonged blast, followed by two short blasts
- Two short blasts

When two sailing vessels are approaching on another, so as to involve risk of collision and each has the wind on a different side, which one is obliged to keep out of the way?

- **The vessel which has the wind on the port side**
- The vessel which has the wind closest to directly astern
- The vessel which has the wind on the starboard side
- The vessel which is geographically to the north
- The vessel which is geographically to the north

When will a power-driven vessel engaged in pushing and a vessel being pushed be considered as simply a "power-driven vessel"?

- **When they are rigidly connected as a composite unit**
- When both the vessel engaged in pushing and the vessel being pushed have independent propulsion machinery
- When the vessel being pushed is shorter in length than the vessel pushing
- When they are not considered to be restricted in their ability to manoeuvre

Which Annex to the Rules lists the "Additional Signals for Fishing Vessels Fishing in Close Proximity"?

- **Annex II**
- Annex I
- Annex III
- Annex IV

Which Annex to the Rules lists the "Distress Signals"?

- **Annex IV**
- Annex I
- Annex II
- Annex III

Which Annex to the Rules lists the "Positioning and Technical Details of Lights and Shapes"?

- **Annex I**
- Annex II
- Annex III
- Annex IV

Which of the following is a statement extracted from the International Regulations for Preventing Collisions at Sea?

- **Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case**
- Any vessel, or the owner, master or crew thereof shall be exonerated from the consequences of any neglect to comply with these Rules or neglecting any precautions normally taken by seamen, or by any special circumstances applicable in the case
- Nothing in these Rules shall benefit the vessel, her owner, master or crew from any arrangement whereby commercial considerations can be brought to bear over precautions that might normally be applied by seamen, or by the special circumstances of a case

You are in open water in clear visibility and you meet a vessel on a reciprocal course. What action would you reasonably expect the other vessel to take?

- **To alter her course to starboard in compliance with Rule 14**
- No action
- To alter her course to port in compliance with Rule 15
- To make contact to discuss a suitable manoeuvre

0178 – Container Fires

Version 1.0 (3420)

A flammable liquid, which is also toxic, appears on a shipper's declaration as class 3 (sub risk class 6.1). What would be the main concern to fire fighters? (Select all applicable answers).

- **Fire-fighting procedures associated with IMDG class 3 goods**
- **The flammability of the substance**
- Fire-fighting procedures associated with IMDG class 6 goods
- Fire-fighting procedures associated with IMDG classes 3 and 6 goods
- The toxicity of the substance

By what criteria are explosives sub-classed, under the IMDG Code? (Select all applicable answers).

- **The degree of hazard**
- **The nature of hazard**
- The extent of risk whilst carrying in containers
- The extent of risk whilst carrying in package form

By what other names is Calcium Hypochlorite shipped? (Select all applicable answers).

- **B-K Powder**
- **Bleaching powder**
- **Calcium hypochloride**
- **Calcium oxychloride**
- **Chloride of lime**
- **Chlorinated lime**

Choose the combination of actions given in the options by which you are most likely to extinguish a reefer container electrical fire? (Select all applicable answers).

- **By turning off the electrical supply to the container.**
- **Using, preferably, a portable dry powder fire extinguisher.**
- Using a chemical fire extinguisher.
- Using a foam fire extinguisher.

For ships carrying poisonous substances, what does the IMDG Code provide? (Select all applicable answers).

- **Fire fighters should use self-contained breathing apparatus**
- **Fire fighters should wear protective clothing**
- Fire fighters should allow a fire to be extinguished on its own
- Fire fighters should spray burning containers with water
- Fire fighters should stay as far as possible from such substances

If water ingressed into a container of drums, containing chemical wastes, what could be a source of ignition? (Select all applicable answers).

- **Contact between the cargo and the drum wall**
- **Friction between the cargo and packaging**
- Heat
- Sparking

In general, what would extinguish corrosive substances on fire? (Select all applicable answers).

- **Any extinguishing medium**
- **Dry powder**
- **Foam**
- **Water**

In the EmS column of the IMDG Code, what is F-H?

- **Oxidising substances with explosive potential.**
- Explosive substances and articles.
- Non-water-reactive flammable liquids.
- Water-reactive substances.

In the IMDG Code, what class of cargoes are liable to spontaneous combustion?

- **Class 4.2**
- Class 3
- Class 5.1
- Class 5.2

In the IMDG EmS column, what does an underlined EmS code indicate? (Select all applicable answers).

- **A material for which additional advice is given in the emergency response procedures.**
- **A substance for which additional advice is given in the emergency response procedures.**
- **An article for which additional advice is given in the emergency response procedures.**
- A hazardous substance.
- Hazardous material.

In the IMDG EmS column, what does the first EmS code refer to?

- **The relevant fire schedule of a particular substance.**
- The properties of a particular substance.
- The relevant spillage schedule of a particular substance.
- The relevant sub-class of 'Explosives' of a particular substance.

Once a container catches fire, what could happen to the container above it? (Select all applicable answers).

- **It could collapse**
- **It would be affected by heat**
- **It would lose strength**
- **Its cargo could catch fire**

Under SOLAS, what is the requirement for cargo manifests, for ships carrying dangerous goods? (Select all applicable answers).

- **A detailed stowage plan may be carried**
- **The list must show the location of dangerous goods on board**
- **The manifest must show the location of dangerous goods on board**
- **There must be a manifest that sets forth the dangerous goods on board**
- **There must be a special list that sets forth the dangerous goods on board**
- Ships must carry a special list, manifest and a detailed stowage plan of the dangerous goods on board

Unless burning cargo has been positively identified, what must you do? (Select all applicable answers).

- **Maintain a safe distance from the container on fire.**
- **Use at least two fire-fighting teams.**
- Await master's orders.
- Cool adjacent containers.
- Fight the fire from as close as possible.

What are some of the major shipboard concerns when fighting container fires? (Select all applicable answers).

- **Communications.**
- **Deciding upon a safe distance from which to fight a fire.**
- **Not knowing whether there are stowaways in containers.**
- **The possibility of a sudden flare-up of under-deck fires.**

What are spontaneously combustible substances? (Select all applicable answers).

- **Substances with a potential to self heat**
- **Substances with a potential to self ignite**
- Substances liable to catching fire
- Substances liable to explosion
- Substances liable to heat when in contact with air

What are the additional fire safety measures for ships carrying dangerous goods? (Select all applicable answers).

- **Provisions for appropriate personal protective equipment**
- **Provisions for fire protection systems to protect the ship from added fire hazards**
- **Separation of dangerous goods from sources of ignition**
- Segregation of dangerous goods

What are the advantages of the more robust types of water mist lances? (Select all applicable answers).

- **They can penetrate steel containers.**
- **They can reach the heart of the fire.**
- They can replace fire hoses.
- They provide a form of protection to fire fighters.
- They supplement the use of fire hoses.

What are the characteristics of scrap metal? (Select all applicable answers).

- **It is liable to spontaneous heating**
- **It is likely to heat up due to vibration and friction**
- **It is likely to heat up when loaded wet**
- **It presents a low fire-risk except when mixed with swarf**
- It presents a high fire-risk

What are the concerns when using carbon dioxide gas, for extinguishing under-deck fires aboard mega carriers? (Select all applicable answers).

- **Carbon dioxide may be ineffective in extinguishing under-deck fires.**
- **Large amounts of carbon dioxide gas are required.**
- **The gas may leak via the hatch seals.**
- The supply of carbon dioxide gas is severely limited.

What are the disadvantages of using mobile high expansion foam generators when fighting container fires? (Select all applicable answers).

- **Foam cannot reach the heart of a container fire.**
- **Foam is not very efficient in cooling the boundaries of enclosed spaces on fire.**
- Foam cannot effectively flood large spaces.
- Foam generators must be attended to at all times.

What are the limitations of a lashing bridge, with respect to container fire-fighting operations?

- **It does not allow fire fighters access to both ends of twenty-foot containers.**
- It does not allow fire fighters access to both ends of forty-eight foot containers.
- It does not allow fire fighters access to both ends of forty-foot containers.
- It does not allow fire fighters access to both ends of reefer containers.

What dangers are associated with the carriage of empty tank containers? (Select all applicable answers).

- **An empty container could produce dangerous air-vapour mixtures**
- **They are not always empty and have been found to contain cargo residues**
- **When stowed close to a burning container there could be a high risk of explosion**
- They are not properly labeled
- They are often loaded in damaged condition

What general precautions should be exercised so as to minimise a fire risk on board ship? (Select all applicable answers).

- **Never carry out repairs in any compartment containing explosives.**
- **Never carry out welding in cargo spaces, unless in an emergency.**
- **Special care should be exercised when carrying out repairs in adjacent compartments.**
- Never carry out repairs in under-deck compartments.
- Never commence fire-fighting operations without master's orders

What information would you need to enter the IMDG dangerous goods list? (Select all applicable answers).

- **The proper shipping name of a particular substance.**
- **The UN number of a particular substance.**
- The trade name of a particular substance.
- The 'Not Otherwise Specified' (NOS) name of a particular substance.

What is a proper shipping name? (Select all applicable answers).

- **A Not Otherwise Specified name**
- **The technical name of a product listed in column 2 of the IMDG dangerous goods list**
- A name provided by a shipper
- An abbreviated trade name

What is the advantage of using the water screen over a conventional fire hose when fighting container fires? (Select all applicable answers).

- **It can deliver water in a vertical range, of up to twenty meters.**
- **It is made of more robust material than a fire hose.**
- It allows fire fighters to keep well clear of burning containers.
- It is guaranteed to extinguish container fires.

What is the fourth element in the fire tetrahedron? (Select all applicable answers).

- **Combustion.**
- **The sustaining chemical reaction.**
- A combination of fuel, heat and oxygen.
- Combustible material.

What is the greatest risk in the handling and transport of goods of class 1? (Select all applicable answers).

- **Explosions**
- **Fire from a source external to the goods**
- Hazards to personnel
- The cargo being carried in package form

What must a fire fighter, wearing self-contained breathing apparatus, ensure?

- **That the mask is fitted properly.**
- That any space to be entered does not contain carbon dioxide gas.
- That dangerous goods are not on fire.
- That there is a back-up team behind him.
- That there is a fire burning below decks.

What needs to be displayed on a fumigated container? (Select all applicable answers).

- **The date the fumigant was applied**
- **The IMDG fumigation warning sign**
- **The name of the fumigant**
- The name of the consignee
- The name of the shipper

What options could be considered when faced with a calcium hypochlorite container fire? (Select all applicable answers).

- **Attempt to jettison the container**
- **Fight the fire using water**
- **Flood the cargo hold with carbon dioxide gas**
- **Put into a port of refuge**
- **Use mobile fire-fighting equipment such as a piercing applicator**

What personal injuries can burning anhydrouns ammonia cause? (Select all applicable answers).

- **Loss of eyesight**
- **Severe burns, upon contact**
- **Severe damage to respiratory membranes**
- Numbness

What precautions are recommended when carrying Calcium Hypochlorite? (Select all applicable answers).

- **Containers should be stowed clear of living quarters**
- **Containers should be stowed on-deck**
- **Containers should be stowed on-deck, away from sources of heat**
- Containers should be stowed under-deck

What should you remember when fighting chemical fires? (Select all applicable answers).

- **Chemical fires are very difficult to control**
- **Heated chemicals leaking from a container could ignite cargoes in the containers below**
- **Liquid chemicals may turn to gas**
- **Many chemicals provide their own oxygen**
- **Solid chemicals may turn to liquid or gas**

When a corrosive substance decomposes to produce toxic gases, what remark will you find in the Dangerous Goods List?

- **"When involved in a fire, evolves toxic gases"**
- "When involved in a fire, a health hazard"
- "When involved in a fire, corrosive to mild steel"
- "When involved in a fire, decomposes at high temperature"

When fighting container fires what is the purpose of flooding empty tanks adjacent to or inside a cargo hold on fire, with seawater?

- **To cool down essential structures.**
- To cool down burning containers.
- To prevent the ship from being damaged.
- To prevent the spread of fire.

Where would you expect to find the 'Class' of a particular dangerous cargo? (Select all applicable answers).

- **In the Dangerous Goods Manifest**
- **In the IMDG Code.**
- **In the IMDG Dangerous Goods List**
- In the ship's stowage plan

Where would you find a list of certain substances forbidden for transport?

- **In chapter 3.3 of the IMDG code.**
- In chapter 6 of SOLAS 74.
- In the appendix to the ISM Code.
- In the dangerous cargo supplement.
- In the United Nations Convention on Law of the Sea.

Which class of dangerous goods is associated with the term, 'Flashpoint?

- **Flammable or inflammable liquids**
- Explosives
- Flammable Solids
- Oxidising substances

Which of the following options could be considered as a dangerous cargo chemical fire? (Select all applicable answers).

- **Burning magnesium.**
- **Burning potassium.**
- **Burning sodium.**
- Burning paper.
- Burning wood.

Which of the following options could be considered to be elements of a fire triangle? (Select all applicable answers).

- **Combustible material**
- **Fuel**
- **Heat**
- **Oxidising agent**
- **Oxygen**

Which of these cargoes can ignite spontaneously, leading to an explosion? (Select all applicable answers).

- **Direct reduced iron**
- **Steel swarf**
- **Zinc skimmings**
- Acetylene
- Paint in drums

Why are escaping gases from containerised cargoes, dangerous? (Select all applicable answers).

- **Because they give no warning of an outbreak of fire**
- **Because they may give rise to explosive mixtures with air**
- **Because, if conditions are favourable, they may cause an explosion**
- Because they pressurise the container walls

Why is it necessary to wear self-contained breathing apparatus prior to entering a cargo hold? (Select all applicable answers).

- **Because excessive carbon dioxide gas could be present.**
- **Because of the company's safety procedures.**
- **Because phosphine gas could be present.**
- **Because there could be methyl bromide gases present.**
- **Because undeclared containers may have been shipped in hazardous condition.**
- Because of the ISM Code requirements.

With reference to the fire triangle, when would a fire occur?

- **When heat, fuel and oxygen are present in the right combination.**
- When fuel is heated in a closed tank.
- When there is more heat and less oxygen.
- When there is more oxygen and less heat.

With reference to the text, what are oxidising substances sensitive to? (Select all applicable answers).

- **Chemical Reactions**
- **Decomposition under certain conditions.**
- **Impact.**
- **Rise in temperature.**
- **Spontaneous ignition.**

With respect to container fire-fighting operations on board, what is the concern regarding megaships? (Select all applicable answers).

- **As these ships become bigger, their crews become smaller.**
- **Fire-fighting operations are extremely labour-intensive.**
- The fixed innovation systems make fire-fighting operations easier.
- The larger crews also make fire-fighting operations easier.
- There are fixed innovation systems on board.

With respect to detection systems, for ships carrying dangerous goods, what do the SOLAS regulations state? (Select all applicable answers).

- **A sample extraction smoke detection system must prevent the leakage of toxic fumes into occupied areas**
- **Cargo spaces may be fitted with a sample extraction smoke detection system**
- **Cargo spaces must be fitted with a fixed fire detection and fire alarm system**
- **Ro-ro spaces must be fitted with a fixed fire detection and fire alarm system**
- Ro-ro spaces may be fitted with a sample extraction smoke detection system

With respect to sources of ignition, for ships carrying dangerous goods, what do SOLAS regulations state? (Select all applicable answers).

- **Electrical equipment and wiring may be fitted in enclosed cargo spaces if essential for operational purposes.**
- **Electrical equipment and wiring must be certified as being safe.**
- Cables may ordinarily penetrate decks and bulkheads.
- Cables within cargo spaces may be unprotected.
- Cables within cargo spaces may not be fire-proof.

0179 – Marine Fuel Properties II

Version 3.0 (1906)

A value for fuel density is required for a number of operational reasons. Select all of the operations for which a density value for the fuel is required from the options given.

- **Calculating the mass of fuel.**
- **Setting up purifiers.**
- **Vessel stability calculations.**
- Setting the temperature to achieve correct fuel injection viscosity.

A value for fuel viscosity is required for a number of operational reasons. Select all of the operations for which a viscosity value for the fuel is required from the options given.

- **Calculating storage temperatures to ensure transfer of fuel is possible.**
- Calculating a safe storage temperature.
- Calculating flow rates for purifiers.
- Calculating the quantity of the fuel onboard.

According to this module, what is the most likely cause of marine pollution by fuel oil?

- **Spillage/overflow during bunkering operations.**
- Disposal of leakage and drainage from onboard use.
- Disposal of waste product from treatment/cleaning operations.
- Spillage/overflow during fuel transfer operations.

Air pollution from the use of marine fuels, particularly in diesel engines, is an ongoing problem. Select, from the options given, all of those which contribute directly to air pollution.

- **Air pollution by exhaust emissions following combustion of the fuel in a diesel engine.**
- **Air pollution by exhaust emissions from an incinerator when burning sludge residues from the treatment of fuel oil.**
- **Air pollution by vapour and gas given off by the fuel in storage and during treatment.**

Although an energy equivalent indicator is not included within the ISO 8217 fuel standard tables it is still an important indication of a fuels quality. Which of the following range of energy levels is most representative of marine fuels?

- **39 to 42 MJ/kg**
- 22 to 25 MJ/kg
- 32 to 35 MJ/kg
- 49 to 52 MJ/kg

Although the stability of a fuel is important the ISO 8217 standard tables do not include any absolute stability indicators. Which of the following statements gives the best description of a stable fuel?

- **A stable fuel is one which remains in its original condition during storage and handling under varying conditions.**
- A stable fuel is one which can be mixed with other fuels.
- A stable fuel is one which contains no other elements other than hydrogen and carbon.
- A stable fuel is one with only one component.

Aluminium and silicon may be present in the fuel. What is the most likely source of these contaminants?

- **Refinery processes.**
- Crude oil source.
- Onboard treatment.
- Transport and storage.

Asphaltenes are complex, unsaturated hydrocarbon molecules contained in residual fuels. Select any of the likely problems from the following options which may result from fuels with high asphaltene content.

- **A poor combustion performance.**
- **An increase in sludge formation.**
- **Heavy fouling of separators.**
- An increase in abrasive wear.

During combustion of fuel oil in a diesel engine there are a number of pollutants that may be present in the exhaust emissions. From the options given select all of those which constitute air pollutants.

- **Carbon monoxide.**
- **NOx.**
- **SOx.**
- Water vapour.

Each residual fuel grade is designated by a code consisting of three letters and a number. An example of this is the grade RMG 380. What does the number signify?

- **Kinematic viscosity at 50° C**
- Density at 15° C
- Density at 50° C
- Kinematic viscosity at 150° C

From the following processes select all of those which are used in the refining of petroleum oil.

- **Atmospheric distillation.**
- **Catalytic cracking.**
- **Vacuum distillation.**
- **Visbreaking.**

From the options given select those which can be used to effectively remove water from fuel oil.

- **Centrifuge set up as a purifier.**
- **Regular draining of settling and service tanks.**
- Centrifuge set up as a clarifier.
- Passing the fuel through a mesh filter.

If a person swallows a quantity of fuel which of the following actions should be taken?

- **Seek urgent medical help.**
- Induce vomiting.
- Move the victim to an area where fresh air is present.
- Provide oxygen.

In oil refining the term 'cracking' is applied to a process based on which one of the following?

- **The breakdown of heavy hydrocarbons using either heat or heat and chemical means.**
- Product separation by atmospheric distillation.
- Product separation by vacuum distillation.
- Removal of salt and water from the crude.

Information regarding the health hazards associated with different materials, including marine fuels, should be obtained before handling and working with the materials. Where would be the most likely source of this information?

- **Material safety data sheets.**
- Bunker delivery note.
- Fuel sample container label.
- ISO 8217 standards tables.

Limits for Micro Carbon Residue, MCR, are included in both of the ISO 8217 fuel tables. Which of the following is the best description of the effect of a high value of MCR?

- **Heavy fouling of engine cylinder and exhaust system may occur in the event of poor combustion conditions.**
- Blockage of fuel oil filters may occur.
- Frequent cleaning of purifiers will be required.
- Seizure of injection system may occur.

Operating a diesel engine on fuel oil containing used lubricating oil may result in engine damage due to high levels of debris or remaining additives it may contain. Which of the following options is recognised as an indication of used lubricating oil being present in fuel?

- **Zinc, calcium and phosphorus levels all above limit values.**
- Copper, calcium and phosphorus levels above limit values.
- Zinc, calcium and lead levels above limit values.
- Zinc, copper and phosphorus present but levels below limit values.

Select from the options given all of those which apply to the four compounds methane, ethane, propane and butane.

- **They are all alkanes.**
- **They are all gases at atmospheric conditions.**
- **They are all hydrocarbons.**
- **They are all saturated.**

Select, from the options given, all of the possible effects of using fuel with a high sulphur content.

- **Cold corrosion may occur if correct temperatures are not maintained.**
- **Increased SOx emissions.**
- Hot corrosion may occur if sodium is also present.
- Ignition delay will be increased.

Soluble contaminants such as vanadium may be present in residual fuel. What can be done to remove such contaminants from the fuel prior to onboard use?
Select the ONE correct answer.

- **Fuel soluble contaminants cannot be removed.**
- Fuel soluble contaminants can be removed by centrifuging.
- Fuel soluble contaminants can be removed by filtration.
- Fuel soluble contaminants can be removed in the settling tank.

The ash content of a fuel is limited according to the ISO 8217 fuel tables. Which one of the following is the highest risk when operating an engine on fuel with a high ash content?

- **Risk of abrasive damage to injection equipment and cylinder components.**
- Risk of fuel system filter blockage.
- Risk of overloading purifiers.
- Risk of poor combustion.

The best way to remove residual fuel oil from the skin is to wash with clean distillate fuel. True or False?

- **False.**
- True.

The compatibility of one fuel with another cannot be measured directly and may lead to an unstable mixture. To avoid compatibility problems, which of the following options is recommended?

- **If mixing of different fuels cannot be avoided then mix new bunkers with the minimum quantity of old bunkers possible.**
- If mixing of different fuels cannot be avoided then mix all of the new bunkers with all of old bunkers remaining onboard.
- If mixing of different fuels cannot be avoided then mix new bunkers with the equal quantities of old bunkers.
- If mixing of fuels cannot be avoided then the mixture should be kept in storage for a long period of time to check for problems.

The contaminants normally found in marine fuel oil usually originate from which of the following? Select all of the likely sources from the options given.

- **Crude oil.**
- **Refining process.**
- **Transport and storage.**
- Fuel injection system.

The two tables in the current edition of the ISO Standard for marine fuels both contain which of the following indicators?

- **Sulphur limits according to MARPOL Annex VI.**
- CCAI limits for residual fuels.
- Conradson carbon residue limits.
- Energy equivalent values.

The various contaminants that may be present in marine fuel may originate from different sources. Drag and drop the contaminants listed to the most likely source.

- Crude oil. **Vanadium and Sulphur.**
- Refining processes. **Aluminium and silicon.**
- Storage of fuel. **Water and microbes.**
- Transport of fuel. **Rust and foreign material.**

Total sediment (potential) is a measure of which of the following?

- **Deposit of sediment due to storage of fuel over a period of time at elevated temperatures.**
- Deposit of sediment due to mixing with other fuels.
- Deposit of sediment due to poor combustion.
- Deposit of sediment due to suspended solids.

What is the ratio of hydrogen atoms to carbon atoms in the naphthene or cyclo-alkane series of hydrocarbons?

- **2 to 1**
- 1 to 2
- 1 to 4
- 4 to 1

When testing diesel engine fuel injectors care must be taken to avoid direct skin contact with the high pressure fuel spray. From the options given select the ONE most likely effect on health that may result from this particular hazard.

- **Blood poisoning due to penetration of the skin by fuel.**
- Cancer due to prolonged or repeated contact.
- Dermatitis due to prolonged or repeated contact.
- Scalding due to hot fuel.

Which of the following can be used to check that the fuel delivered exactly meets the specification of the fuel ordered?

- **Analysis by a specialist facility.**
- Bunker delivery note.
- ISO 8217 standard tables.
- Simple on board testing.

Which of the following is not a fossil fuel?

- **Wood.**
- Coal.
- Natural gas.
- Petroleum oil.

Which of the following is the best description of the alkane or paraffinic group of hydrocarbon compounds?

- **Saturated hydrocarbon molecules with a straight chain structure.**
- Saturated hydrocarbon molecules with a ring structure.
- Unsaturated hydrocarbon molecules with a ring structure.
- Unsaturated hydrocarbon molecules with a straight chain structure.

Which of the following non-hydrocarbon elements found in marine residual fuels is the most likely to originate from part of the refining process?

- **Aluminium and silicon.**
- Nickel and zinc.
- Sulphur and calcium.
- Vanadium and sodium.

Which of the following options is the best description of marine diesel oil?

- **Mainly distillate fuel with a small proportion of heavy fuel.**
- High grade distillate fuel.
- High grade residual fuel.
- Mainly residual fuel with a small proportion of distillate fuel.

Which of the following processes allows the chemical energy in fuel oil to be converted into mechanical work?

- **Combustion.**
- Blending.
- Heating.
- Refining.

Which of the following quality indicators is included in the ISO 8217 tables?

- **An ignition quality indicator for distillate fuels.**
- A combustion quality indicator for distillate fuels.
- A combustion quality indicator for residual fuels.
- An ignition quality indicator for residual fuels.

Which of the following statements is the most accurate with reference to the term 'combustion quality'?

- **The combustion quality of a fuel refers to how easily the fuel burns once it has been ignited.**
- Combustion quality and ignition quality of a fuel are the same thing.
- Combustion quality of a fuel depends on the pre-treatment before use.
- The combustion quality of a fuel refers to how easily the fuel can be ignited.

Which of the following temperature limits is included in the ISO 8217 standard because of safety reasons?

- **Flash point.**
- Cloud point.
- Pour point (Summer).
- Pour point (Winter).

Which of the following temperature ranges is the most likely to be employed for preheating the crude oil for the atmospheric distillation process.

- **350 – 400° C**
- 150 – 250° C
- 50 – 150° C
- 500 – 700° C

Which one of the following chemical elements is the most likely to increase the risk of high temperature corrosion when present with vanadium during combustion of fuel oil?

- **Sodium.**
- Calcium.
- Phosphorus.
- Sulphur.

0181 – Ship to Ship Transfer Operations

Version 3.0 (1293)

According to the requirements contained in, the MARPOL Annex I – STS Plan, – the national and local authorities should be notified prior to the commencement of an STS (ship to ship) transfer operation. How much notice is required for such an operation?

- **48 hours**
- 12 hours
- 6 hours
- 72 hours

In relation to STS (ship to ship) transfer operation, select the three methods by which personlle would be permitted to transfer between the ships? (Select all applicable answers)

- **By lightweight, insulated gangway fitted with rails, safety net and securing ropes**
- **By ship's crane suitable for this task, using a, certified personnel basket and an experienced crane operator.**
- **By workboat transfer**
- By pilot ladder via the fender
- By ship's derrick and certified personnel basket
- By the accommodation ladder

In relation to STS (ship to ship) transfer operations which two of the given options are considered as "organisers" as defined in the module?

- **STS agency**
- **STS service provider**
- The local agent and National Authority
- The person in overall advisory control

In relation to STS (ship to ship) transfer operations who should fill in the STS Check List no. 1?

- **Ship Owners/Operators.**
- The Master of each ship.
- The person in overall advisory control.
- The STS Service provider.

In relation to STS (ship to ship) transfer operations, when should the STS Check-List No. 5 be completed?

- **Before unmooring**
- After un-berthing is completed
- Before disconnecting the hoses
- Just as the ships are moving apart

In relation to STS (ship to ship) transfer operations, when should the STS check list No. 3 completed and confirmed?

- **Before run in and mooring commences.**
- Just before cargo transfer commences.
- When mooring is completed.
- When the STS operation commences.

In relation to STS (ship to ship) transfer operations, when should the STS check list No. 4 be completed?

- **When both ships are securely moored and before cargo transfer starts.**
- After first line is made fast.
- Before unmooring.
- When cargo transfer is completed.

In relation to STS (ship to ship) transfer operations, when should the STS check list no. 2 be completed and confirmed?

- **Before operations commences.**
- At the same time as check list no. 1 is confirmed completed.
- When operations are underway but before mooring commences.
- When the service provider is ready.

In relation to STS (ship to ship) transfer operations, which of the given options identifies a common feature that is required for both a dedicated ship and an established ship?

- **Both must have an experienced crew.**
- Both must be contracted for a period in a specific location.
- Both must be equipped to secure all STS equipment safely for ocean passages.
- Both must be equipped with dedicated hose handling gear.

Select the option which best describes the preferred procedure to adopt when unmooring and un-berthing following an STS (ship to ship) transfer operation carried out with both vessels under way.

- **Keep the wind and sea on the port side when unmooring, then bring the two ships to head into the wind in order to spread them apart**
- Keep sea and wind dead ahead throughout the entire operation
- Keep the wind and sea abreast on the port side throughout the entire operation
- Keep wind and sea on the starboard while unmooring, then bring the wind in from astern in order to split the two ships

Select the option which best describes the type of communication which is recommended between the two ships during STS (ship to ship) transfer operation?

- **Intrinsically safe VHF/UHF plus visual/voice contact .**
- Any portable and fixed VHF.
- Any portable or fixed UHF.
- Only UHF and visual/voice.

Select the three options that identify the intervals at which the fenders used in STS (ship to ship) transfer operation should normally be visually inspected.

- **After each transfer**
- **Annually**
- **Prior to each transfer**
- Every six month by shore-based service company
- Every six month by STS service provider
- Monthly

Select the three options which detail the factors that will influence a mooring plan for STS (ship to ship) transfer operations?

- **Expected difference in freeboard and displacement**
- **Size of each ship and the difference between their sizes**
- **The efficiency of mooring line leads available**
- Parallel body length of the ships
- The number of fenders to be used
- The quality and experience of the crews

Select the two options which best describe the patterns that are usually employed when positioning fenders on larger ships involved in STS (ship to ship) transfer operations?

- **4 primary fenders rigged in one string at equal distance and 1 secondary fender fore and aft.**
- **4 primary fenders rigged in pairs clear from manifold areas and 1 secondary fore and aft.**
- 1 primary fender at each end and 1 in between, and 1 secondary at the bow
- 1 primary fender at each end and 2 secondary fenders in between
- 4 primary fenders equally distributed along the parallel body, 1 secondary fender on deck ready for use where required

Select the two options which identify the publications which contain the requirements and recommendations to be followed in order to plan and execute an STS (ship to ship) transfer operation?

- **International Safety Guide for Oil Tankers and Terminals**
- **MARPOL Annex I, STS Transfer Guide (ICS/OCIMF)**
- International Seafarer's Guide for Oil Tanker Trade
- International Shipping Guide for Offshore Tanker Terminals

Select the two options which identify the test and maintenance requirements for cargo hoses carried on board for the use in STS (ship to ship) transfer operations.

- **Pressure test on board every 6 month and a, test record to be kept on board**
- **Regularly inspected for damages and deterioration and, a record of inspections to be kept on board**
- Annually inspected by Class
- Pressure tested and inspected annually by shore-based company and, test and maintenance record supplied to ship
- Pressure tested annually by service provider and a, test record given to ship

Select two options which identify who has a responsibility for carrying out a risk assessment prior to an STS (ship to ship) transfer operation?

- **Both ships Masters.**
- **The person in overall advisory control.**
- Only the Master of the ship at anchor.
- The Ship Owner or Operator.

Select, from the given options, the main reason indicated in the module as to why STS (ship to ship) transfer operations are carried out?

- **A shortage of suitable terminals in the region means an STS (ship to ship) transfer area will allow effective distribution of part cargoes to local reception facilities.**
- In order to reduce the risk as STS transfer operations are always safer than ship to terminal transfers.
- In order tos ave time as STS transfer operations always take less time than loading or discharging at a terminal
- In the event of congestion in ports it is quicker to carry out STS operations than to wait for a berth to be available.

Under which maritime law and regulation are STS (ship to ship) transfer operations covered?

- **MARPOL Annex I**
- IMO – IMDG Code
- ISPS Code
- SOLAS

What is a risk mitigation and contingency plan in relation to STS (ship to ship) transfer operation?

- **It is a product of a risk assessment.**
- It is a guide line for emergency planning.
- It is a product of all STS requirements.
- It is the basis for a risk assessment.

What should be the first action to be taken if a cargo leakage or spill is observed during an STS (ship to ship) transfer operation?

- **Report to the officer in charge in order to stop the transfer in a controlled manner.**
- Activate the emergency alarm system to notify all personnel.
- Activate the emergency shut-down system tp immediately stop the transfer.
- Inform the Master for his permission to stop the transfer.

What speed is recommended for two ships manoeuvring alongside each other while underway as part of an STS (ship to ship) transfer operation?

- **4 to 5 knots**
- 2 to 4 knots
- Dead slow, approximate 2 knots
- Slow speed – 6 to 7 knots

When an STS (ship to ship) transfer operation is being carried out at anchor, according to which of the publications given in the options is it required to maintain anchor watch?

- **STCW**
- COLREG
- ICS
- ISGOTT

When an STS (ship to ship) transfer operation is planned who is responsible for informing the local authority?

- **The master of each ship.**
- The person in overall advisory control.
- The service provider.
- The Ship Owner or Operator.

When berthing alongside an anchored ship, inpreparation for an STS (ship to ship) transfer operation, which ship will normally deploy the lines?

- **The manoeuvring ship**
- Always the anchored ship
- Both ships deploy equal number of lines
- The anchored ship deploys the first line, the manoeuvring ship deploys the remaining

When two ships are manoeuvring alongside underway, in preparation for an STS (ship to ship) transfer operation, from which direction is it recommended to keep wind and sea?

- **On the port bow of the constant heading ship**
- Dead ahead of both ships
- On the beam of the larger ship to give shelter to the smaller ship
- On the starboard side of the manoeuvring ship

When two ships are manoeuvring alongside underway, in preparation for an STS (ship to ship) transfer operation, what is the main reason for securing the forward head- and spring lines first?

- **Because the forward motion of the ships driving the water between them can push the bows apart**
- Because the first touch on the fenders will always push the ships' bows apart
- Because the interaction between the ships will push the smaller ship away
- To avoid misalignment of the two ship's manifolds

Whenever possible STS (ship to ship) transfer operations should be carried out in daylight conditions, If, however, it is decided to carry out such operations at night which one of the given options is required in order to make such operation acceptable?

- **Personnel involved must have suitable experience in night STS transfer operations and all parties must agree.**
- An additional STS Superintendent must be employed.
- Transfer must be carried out with both ships underway.
- Tug boat assistance must be made available.

Which contingency plan should be activated in the event of an oil spill during an STS (ship to ship) transfer operation within the Dominican Republic EEZ?

- **Shipboard Oil Pollution Emergency Plan (SOPEP)**
- Local Authority Contingency Plan
- OPA 90 Vessels Response Plan
- The ship owner's contingency plan

Which daylight signal is required to be hoisted during the STS (ship to ship) transfer operations?

- **Two black balls with a diamond shape in the middle.**
- A black cylinder.
- Two diamond shapes with a black ball in the middle.
- Two diamond shapes.

Which factor would have a vital influence on the unmooring operation following an STS (ship to ship) transfer operation with one ship at anchor.

- **A change of tide**
- An acceptable but strong wind from port side
- The difference in the two ship's displacement
- The hydrodynamic interaction caused by underwater hull shapes

Which fender type is most commonly used as primary fender for STS (ship to ship) transfer operations?

- **High pressure pneumatic**
- A combination between high pressure and foam filled
- Foam filled
- Low pressure pneumatic

Which five of the following conditions have to be considered when a location for STS transfer operation that allows for drifting or steaming during transfer is being planned?

- **Safe distance to offshore installations.**
- **Sufficient sea room.**
- **The availability of a safe zone free from small craft.**
- **The need to notify and obtain approval from coastal authority.**

- **The tidal current condition.**
- A safe anchorage and holding ground
- Ship repair yard available in the local area
- Tug-boat availability

Which four important factors need to be considered when selecting the correct fender type to be used for an STS (ship to ship) transfer operation?

- **Berthing speed/berthing velocity**
- **Sea and swell conditions**
- **Size of ships involved**
- **Weather conditions**
- Fender mooring types
- Sea water temperature
- What kind of cargo to be transferred

Which one of the actions given in the options should be taken if communication is lost during an STS (ship to ship) transfer operation?

- **Sound the emergency signal and suspend all operations if safe to do so. Transfer operations should not be resumed until communication is restored.**
- Continue the operation at a reduced pumping rate and establish voice communication at the manifold.
- Continue the operation at a reduced pumping rate until normal communication is re-established.
- Reduce pumping rate and inform the Master and the person in overall advisory control of the problem.

Which one of the actions given in the options should be taken if communication breakdown occurs during a closing up manoeuvre as part of an STS (ship to ship) transfer operation?

- **The manoeuvre should be aborted if appropriate and safe to do so and further actions to be indicated by whistle signals according to COLREG recommendations?**
- Adjust course and speed in relation to the other ship and try to make a parallel, soft berthing against the fenders.
- Continue the closing up operation by communication via load- speaker (Megaphone).
- Reduce the speed and make a new close up manoeuvre when communication is restored.

Which ships used for STS (ship to ship) transfer operations are required to have closed fairleads within 35 meters fore and aft of the centre of the starboard manifold?

- **All tankers over 160-.000 dwt**
- All tankers and LPG ships above 50-.000 dwt
- All tankers over 120-.000 dwt
- LPG ships larger than 85-.000 cub.mtrs.

Who is normally "the person in overall advisory control" of an STS (ship to ship) transfer operation?

- **An STS Superintendent or one of the ship's Masters.**
- The pilot , if a pilot is used.
- The Ship Owner or Operator.
- The STS service provider.

Who is responsible for the STS (ship to ship) transfer operation?

- **Each master is responsible for his ship.**
- The master on the manoeuvring ship.
- The service provider.
- The Superintendent in overall advisory control.

Who is responsible for the correct positioning of the fenders for an STS (ship to ship) transfer operation being carried out underway?

- **The person in overall advisory control**
- The Master of the constant heading ship
- The pilot (if pilot is involved)
- The STS service provider

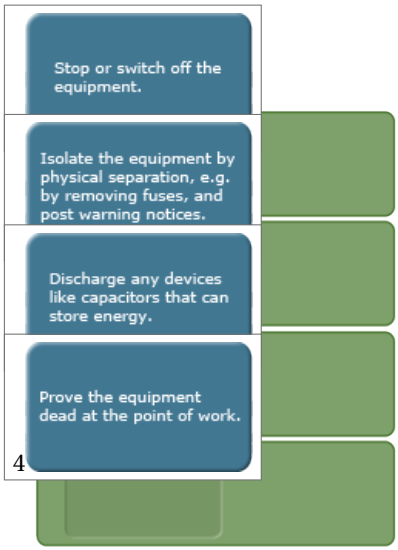
0183 – Electrical safety

Version 1.0 (1303)

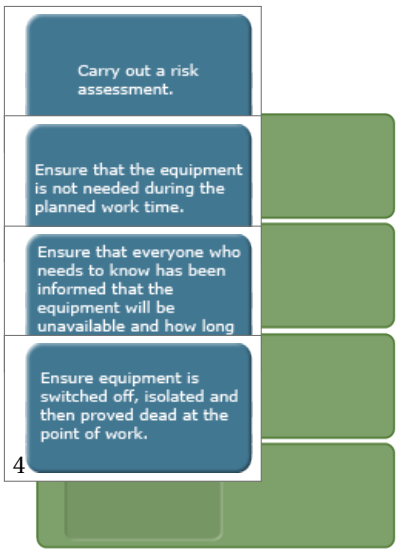
At voltages of less than 1,000 V, what is the generally accepted minimum value of insulation resistance for electrical equipment onboard ships?

- **One megohm. (1 MΩ)**
- One hundred kilohms. (100 kΩ)
- One hundred megohms. (100 MΩ)
- One thousand ohms. (1 kΩ)

Drag and Drop the following steps for working on electrical equipment into the order that you would carry them out.



Drag and drop the following steps for working on electrical equipment into the order that you would carry them out.



For a ship's electrical system, which one of the following statements best describes the meaning of the term 'earth'?

- **The steelwork of the hull, whose electrical potential can be regarded as zero.**
- An electrical connection only used in dry-dock.
- Any electrical conductor not on the live side of the supply.
- Any part of the electrical system insulated from live conductors.

If a fluorescent tube or energy saving light bulb breaks, in addition to ensuring good ventilation, which one of the following statements best describes how the pieces should be cleaned up?

- **Use rubber gloves, sticky tape or wet towels to clean the area, place all materials in a marked sealed container for disposal ashore.**
- Call in a specialist firm to clean up and dispose of the waste.
- Collect the pieces using rubber gloves, wet towels or sticky tape, grind all materials to less than 25 mm and then they can be thrown overboard once more than 25 miles from land.
- Sweep the pieces up with a brush and place all materials in a marked sealed container for disposal ashore.

If any part of a live electrical circuit is brought into accidental contact with earth then which one of the following statements should we assume to be correct?

- **It is an unsafe situation.**
- All circuits are protected against any damage occurring.
- Nothing will ever happen if the ship has an earthed neutral.
- Nothing will ever happen if the ship has an insulated neutral.

If unsure whether electrical equipment is live or not, the best course of action to take is to:

- **Treat it as though it were live until a competent person has proved it dead.**
- Assume it must be dead if it looks as though it has been disconnected.
- Connect an ammeter between any two terminals.
- Wear gloves and apply a short across the terminals to see if there is any sparking.

If work is to be carried out only on one part of a lighting circuit controlled by fuses or a miniature circuit breaker (MCB), which one of the following options is the best way to proceed?

- **Switch off all the other lights first and then withdraw the fuses or MCB for the whole circuit.**
- Leave the fuses or MCB in, but switch off the circuit to be worked on.
- Use a rubber mat and insulated tools and work carefully, knowing the system is live.
- With the system still live, carefully disconnect the part of the circuit to be worked on.

Once you are sure the power has been turned off, what is the next step in helping someone who has received an electric shock?

- **Check airway, breathing and circulation.**
- Get them out into fresh air.
- Raise the alarm and do nothing until someone with first aid training arrives.
- Start looking for burn injuries.

Outside of a hazardous area, which one of the following statements best describes the choice of portable electrical equipment when working in wet areas or galleys?

- **Portable electrical equipment supplied from a 110 Volt centre-tapped transformer should be used rather than mains voltage equipment.**
- Any portable equipment will be suitable, as this is a safe area.
- Only portable electrical equipment supplied at less than 12 Volts should be used.
- Portable electrical equipment should never be used in such an area.

The flow of unwanted leakage current through surface deposits on electrical cables is known as which one of the following?

- **Tracking.**
- Creeping.
- St. Elmo's fire.
- The corona effect.

The relationship between voltage, current and resistance in an electrical circuit can be described by which one of the following statements?

- **The higher the voltage, or the lower the resistance, the higher the current.**
- The higher the voltage, or the higher the resistance, the lower the current.
- The higher the voltage, or the lower the resistance, the lower the current.
- The lower the voltage, or the lower the resistance, the lower the current.

What should be your first action if you see someone receiving an electric shock?

- **Turn off the electrical supply or get the victim clear of any live electrical conductors.**
- Check the victim's breathing.
- Check the victim's pulse.
- Grab hold of the victim to prevent them falling.

Which THREE of the following are true reasons why electrical equipment needs to be designed and approved for use in a hazardous area?

- **Explosive gas can enter equipment.**
- **High energy sparks can ignite flammable gasses.**
- **Working temperatures can reach the ignition temperature of the gas.**
- Even very low voltage or current is dangerous in a hazardous area.

Which TWO of the following conditions should be met before working on or testing any electrical equipment in a hazardous area?

- **A permit to work system should be used.**
- **Only competent, suitably trained and experienced persons should carry out the work.**
- A gas free certificate must always be obtained.
- A zero flammable gas reading must be obtained.

Which one of the following current paths for an electric shock is considered the most dangerous to the human body?

- **From hand to hand, across the chest.**
- From both hands to the ground.
- From one foot across to the other foot.
- From one hand to the ground.

Which one of the following gasses is given off during the normal charging of vented lead-acid batteries?

- **Hydrogen.**
- Carbon Dioxide.
- Hydrogen Peroxide.
- Hydrogen Sulphide.

Which one of the following is the main purpose of a rubber mat placed on the floor next to electrical equipment?

- **To provide an insulating surface to reduce the risks from an electrical shock.**
- To be a conductive surface and reduce the risk of sparks from static electricity.
- To prevent damage to equipment during maintenance.
- To provide a heat resistant surface.

Which one of the following options correctly completes the following statement?
"Electrical equipment is said to be live . . .

- **. . . when it is supplied with or contains an electrical voltage or current".**
- . . . only when it is connected to an electrical supply".
- . . . only when it is taking current from the supply".
- . . . when it is in operation or still moving".

Which one of the following options correctly completes the following statement?
"Electrical equipment is said to be “dead” when . . .

- **. . . it is not connected to an electrical supply and any stored electrical energy has been discharged".**
- . . . it has been switched off and disconnected from an electrical supply".
- . . . it has been switched off by means of a double pole switch".
- . . . it is no longer in operation".

Which one of the following statements best describes a safer alternative to checking voltages on live electrical equipment during fault finding?

- **Make the circuit dead and carry out low resistance (continuity) tests.**
- Isolate the equipment, then dismantle it and find the fault.
- Keep the equipment live and check for loose connections using an insulated screwdriver.
- Operate contactors by hand using an insulated screwdriver.

Which one of the following statements best describes how much electrical resistance the human body has?

- **The resistance of the human body decreases with higher voltages, sweat or damp conditions.**
- The resistance of the human body increases with temperature and humidity.
- The resistance of the human body is fairly constant at around 2,000 Ohms for the average person.
- The resistance of the human body is fairly constant at around 5,000 Ohms for the average person.

Which one of the following statements best describes the testing of live electrical equipment?

- **Live testing should only be carried out by competent persons using approved equipment and PPE, and only when essential.**
- Live testing should never be carried out under any circumstances.
- Provided approved instruments are used, there is no problem with live testing.
- Provided insulated leads with finger guards are used, there is no problem with live testing.

Which one of the following statements best describes which voltage level can be considered as "safe?"

- **Below 50 Volts is generally considered safe as regards electric shock, but lower voltages still present some risks.**
- Below 110 Volts is considered completely safe, apart from in hazardous areas.
- Below 12 Volts is considered completely safe.
- Below 24 Volts is considered completely safe, apart from in hazardous areas.

Which one of the following statements best describes why meter test leads should have finger guards fitted?

- **To prevent accidental contact between the hands and conductors which might be live.**
- To prevent damage to insulated gloves from the sharp edges of test probes.
- To prevent injury to the hands from the sharp edges of test probes.
- To provide a better grip for holding the test leads.

Which one of the following statements correctly describes how batteries or their contents should be disposed of?

- **Batteries and their contents should always be treated as controlled, special waste.**
- Apart from Cadmium batteries, most other batteries do not present any special risk and can be treated as other, non-food, waste.
- Once drained of any liquids, batteries can be treated as other, non-food, waste.
- Once drained of any liquids, batteries can be treated in the same way as plastics.

Which one of the following statements defines what we mean by a 'hazardous area'?

- **An area where a flammable gas atmosphere is, or may be, expected to be present.**
- An area that should be kept off limits to passengers.
- An area where work or operations mean that it is dangerous to enter.
- Any compartment containing electrical equipment operating above 110 Volts.

Which one of the following statements most accurately defines the term 'portable electrical equipment?'

- **Electrical equipment that is connected to the supply by a cable through a plug and socket.**
- Electrical equipment that can be carried by two persons.
- Electrical equipment that can be lifted by one person.
- Electrical equipment that can operate on dual voltage supplies.

Which one of the following would be the minimum current level above which an electric shock could be considered potentially fatal for to the average person?

- **Twenty five milliamps. (25 mA)**
- One hundred milliamps. (100 mA)
- Thirteen amps. (13 A)
- Three amps. (3 A)

Which one of these is the main purpose of electrical cable insulation?

- **To prevent current from flowing anywhere, other than through the conductors.**
- To prevent heat loss to the area around the cable.
- To protect the conductors from mechanical damage.
- To protect the conductors in the event of a fire.

Which one of these is the most important effect of deposits of dirt, chemicals, or oil on electrical cables?

- **The cable insulation resistance will be reduced.**
- The cable insulation resistance will be increased.
- The cable sheath will become discoloured.
- There will not be any effect.

Which one of these statements best describes persons who should carry out electrical work onboard ship?

- **Only those who are trained, experienced and therefore competent.**
- Any person nominated by the Chief Engineer.
- Only the designated electrical and engineering officers.
- Only the designated electrical officer.

With regard to an insulation resistance test on electrical equipment, which one of the following statements best describes where the insulation resistance should be measured?

- **Between each conductor and earth, and between each of the conductors.**
- Between each cable sheath and earth.
- Between each conductor and the equipment frame or enclosure.
- Between the ends of each conductor, and between the ends of the cable sheath for each conductor.

With regard to an insulation resistance test on electrical equipment, which one of the following statements gives the correct sequence for connecting and disconnecting the test leads?

- **Connect the earth lead first, before the live lead. Remove the live lead first, and the earth lead last.**
- Connect the earth lead first, before the live lead. Remove the earth lead first, and the live lead last.
- Connect the live lead first, before the earth lead. Remove the earth lead first, and the live lead last.
- Connect the live lead first, before the earth lead. Remove the live lead first, and the earth lead last.

With regard to work on electrical equipment, which one of the following statements best describes how equipment should be proved dead?

- **By using an approved voltage testing device or voltmeter at the point of work.**
- By confirming that the local starter no longer causes the equipment to run.
- By using an approved multimeter at the point of work.
- By using an approved voltage testing device or voltmeter at the point of isolation.

With regard to work on electrical equipment, which one of the following statements best describes how equipment should be safely identified?

- **By observing at the point of work that it starts/stops when its control circuit is operated.**
- By asking the Chief Engineer.
- By checking the labels at the point of work and in the control room.
- By looking at the circuit diagrams onboard ship.

With regard to work on electrical equipment, which one of the following statements best describes how it can be ensured that equipment remains safely isolated?

- **By posting a caution notice anywhere that the equipment has been isolated, and locking off switches or removing fuses whenever possible.**
- By disconnecting wiring or cables at the point of work.
- By informing the engineer and deck officers on duty.
- By removing fuses.

With regard to work on electrical equipment, which one of the following statements best describes the term "isolated"?

- **Disconnected from any electrical supply, and physically separated from it by an adequate gap.**
- Switched off, with a warning notice posted.
- Switched off, with the start button locked off.
- Switched off.

With regard to work on electrical equipment, why is the use of multimeters not recommended for proving equipment dead?

- **It is too easy to have them on the wrong measurement scale, and in some cases this can also be dangerous.**
- Multimeters can give incorrect readings if the batteries are nearly used up.
- Multimeters tend to be less accurate.
- Only analogue instruments should be used, and multimeters tend to be digital devices.

With regard to work on electrical equipment, why should a special insulation resistance meter be used for checking insulation resistance, rather than a multimeter?

- **The insulation resistance meter applies a test voltage of 500 Volts which gives a more realistic test.**
- Multimeters can give incorrect readings if the batteries are nearly used up.
- Multimeters cannot usually measure resistance.
- Using multimeters can damage some equipment.

0186 – Vessel General Permit - VGP

Version 2.0 (3315)

Apart from the EPA who else is empowered to inspect ships to ensure that the Vessel General Permit (VGP) is complied with?

- **The US Coast Guard**
- Certain classification societies
- Pilots in the US
- Port authorities in the US

How many violations of the Vessel General Permit (VGP) is a ship guilty of if paint chips have been allowed to wash off the deck into permit waters for 8 consecutive days?

- **8**
- 1
- 16
- 2

How often is a report of non-compliances required to be submitted to the EPA on behalf of each ship using the Vessel General Permit (VGP)?

- **Every year**
- Every time a dry dock inspection is completed
- Every time a ship goes to dry dock
- Every time an inspection is completed

Part of the annual inspection required by the Vessel General Permit (VGP) is a check on underwater oil seals. If this cannot be done during the year what should be done?

- **The annual inspection report must state that the item has not been inspected and has been scheduled for inspection at dry dock**
- An inspection by EPA certified divers must be arranged
- The annual inspection report must state that the item is still considered satisfactory
- The item can be deferred until next dry dock

The Vessel General Permit (VGP) controls oil/sea interfaces like CPP and stern tube seals. Which of the following is also considered an oil/sea interface by the VGP?

- **Lubricated wire ropes and chains subject to immersion**
- Any lubricated machinery on open decks
- Any lubricated wire ropes and chains used externally on the ship
- Oil/ water mixtures on decks

The Vessel General Permit (VGP) recommends that the least toxic anodes are used for hull protection. Drag the anode materials into order of their toxicity.

- Least toxic **Magnesium**
- Most toxic **Zinc**
- Toxic **Aluminium**

To which of the following types of ships does the Vessel General Permit (VGP) apply special requirements?

- **Oil and petroleum tankers**
- Gas tankers
- Offshore Support Vessels
- Tugs

When must a Notice of Intent be submitted for a ship planning to enter waters controlled by the Vessel General Permit (VGP)?

- **At least 7 days before entering permit waters for the first time**
- At the 1st port after entering permit waters for the first time
- Every time before entering permit waters
- Within 7 days of entering permit waters for the first time

Which of the following best describes a specific requirement of the Vessel General Permit (VGP) for oily bilge water discharges from a ship of more than 400 gross tons?

- **Do not discharge while in waters federally protected for conservation purposes**
- Do not discharge while in permit waters
- Do not discharge within 12 miles of the US coast
- Do not discharge within 50 miles of the US coast

Which of the following best describes a type of general discharge regulated by the Vessel General Permit (VGP)?

- **Toxic and hazardous ship's stores**
- Chemical cargoes
- Oil cargoes
- Seawater cooling

Which of the following best describes a way of meeting the specific requirements of the Vessel General Permit (VGP) for antifouling coatings?

- **Comply with the International Antifouling Systems Convention**
- Antifouling coatings must be certified by the US Coast Guard (USCG)
- Antifouling coatings must be certified by the US Environmental Protection Agency (EPA)
- Antifouling coatings must be copper based instead of organotin

Which of the following best describes a way of meeting the specific requirements of the Vessel General Permit (VGP) for deck runoff on a general cargo ship?

- **Maintain the deck areas to prevent excess corrosion and leaks**
- Fit spill rails around the deck edges
- Keep a supply of oil dispersant available on deck
- Wash down the main deck area at least once a week

Which of the following best describes how the ship can prove a Notice of Intent (NOI) under the Vessel General Permit (VGP) scheme has been submitted at the correct time?

- **The status of NOI processing can be viewed via the EPA website**
- A VGP permit is e mailed to the ship's operators
- A VGP permit is sent to the ship's agent
- Proof of posting must be on board

Which of the following best describes one of the Vessel General Permit (VGP) requirements for grey water?

- **Discharge of grey water while in port must be minimised**
- Discharge of grey water while in permit waters is not allowed
- Discharge of grey water while in permit waters must be minimised
- Discharge of grey water while in port is not allowed

Which of the following best describes one of the specific requirements of the Vessel General Permit (VGP) for discharges from chain lockers?

- **Chain lockers must be cleaned thoroughly during dry docking**
- Anchor chains must be washed using environmentally sound detergents following any lowering into near-shore waters
- Chain lockers must be cleaned thoroughly once a month
- Chain lockers must be coated with rubber based paint

Which of the following best describes one of the specific requirements of the Vessel General Permit (VGP) for discharges of ballast?

- **The person-in-charge of the discharge must be trained on the application of ballast water and sediment management and treatment procedures.**
- All crew members must be trained on the application of ballast water and sediment management and treatment procedures.
- No discharges of ballast are allowed while in permit waters
- No discharges of ballast are allowed while in US ports

Which of the following best describes one of the specific requirements of the Vessel General Permit (VGP) for lifeboat engine wet exhausts?

- **Engines must be maintained in good order**
- Engines must be run only for drills and in case of emergency
- Engines must be run only in case of emergency
- Engines must not be run

Which of the following best describes the Vessel General Permit (VGP) requirements for condensate discharges from refrigeration and air conditioning plant?

- **Oils and toxic materials must be excluded from condensate discharges**
- Condensate must be retained on board
- Condensate must be stored and discharged outside permit waters
- Condensate must be treated with biocide before discharge

Which of the following best describes the Vessel General Permit (VGP) requirements for handling fouling organisms within ship's seawater piping?

- **Fouling organisms must be removed regularly and, if discharged overboard, be done outside permit waters**
- Fouling organisms must be prevented by continually using biocide
- Fouling organisms must not be disturbed
- Fouling organisms must not be removed using biocide while in permit waters

Which of the following best describes the action required by the ship to ensure compliance with the Vessel General Permit (VGP) when it returns to the US from voyages outside permit waters?

- **All types of scheduled inspections required by the VGP must be up to date**
- A new schedule of inspections required by the VGP must be drawn up
- All types of scheduled inspections not carried out while the ship was outside permit waters must be re-started
- Entries for any inspections not carried out while the ship was outside permit waters must be signed for in the VGP record book

Which of the following best describes the action required by the ship's watch ratings to keep the ship compliant with the Vessel General Permit (VGP)?

- **Regular visual monitoring of the water around the ship**
- Blocking up the deck scuppers
- Careful greasing of winches and windlasses
- Mopping up rain water on deck

Which of the following best describes the action to be taken if a violation of the Vessel General Permit (VGP) discharge conditions is discovered by a crew member?

- **An assessment must be carried out by the ship into the nature, cause, and potential options for eliminating the problem**
- The EPA must be requested to approve the preferred option for eliminating the problem
- The EPA must be requested to assess the nature, cause, and potential options for eliminating the problem
- The US Coast Guard must be requested to assess the nature, cause, and potential options for eliminating the problem

Which of the following best describes the method of submitting a Notice of Intent (NOI) under the Vessel General Permit (VGP) scheme?

- **It must be submitted electronically via the EPA's eNOI system**
- It must be delivered by hand to the local EPA office after the ship docks
- It must be given to the US Coast Guard if the ship is boarded
- It must be sent via e mail

Which of the following best describes the nature of the annual inspections required by the Vessel General Permit (VGP)?

- **A comprehensive inspection by qualified persons covering all affected areas of the vessel**
- A comprehensive inspection by qualified persons covering all safely accessible areas of the vessel
- An inspection by the chief officer covering all safely accessible areas of the vessel
- An inspection by the Designated Person Ashore (DPA) covering all areas of the vessel

Which of the following best describes the purpose of a Notice of Intent (NOI) under the Vessel General Permit (VGP) scheme?

- **It notifies the EPA what types of discharges a ship intends to make while in US waters**
- It is a way of charging permit fees
- It makes it easy for the EPA to inspect a ship
- It notifies the EPA which ships are in US waters at any time

Which of the following best describes the requirements of the Vessel General Permit (VGP) for cleaning of hull underwater areas?

- **In-water cleaning must minimise discharges of fouling organisms and antifouling coatings**
- In-water cleaning is allowed only if conducted by EPA approved contractors
- In-water cleaning is not allowed
- In-water cleaning is not allowed within the first year after painting unless there is a significant visible hull fouling

Which of the following best describes the requirements of the Vessel General Permit (VGP) for sewage discharge?

- **It is regulated, but outside the VGP**
- It is not allowed to be mixed with grey water
- It is not regulated by the VGP if mixed with grey water
- The discharge must contain no metals

Which of the following best describes the specific requirements of the Vessel General Permit (VGP) for non-oily machinery waste water discharges into permit waters?

- **It must be free from toxic or bioaccumulative additives**
- It must be passed through an oily water separator
- It must be used to dilute oily machinery waste water
- It must not be discharged within 1 mile of the coast

Which of the following best describes why the Vessel General Permit (VGP) regulates seawater cooling discharges?

- **Seawater coolant is hot and contains dissolved materials from the system**
- Seawater coolant flows can cause seabed mud to be disturbed
- Seawater coolant is hot and allows alien species to grow on the outside of the hull
- Seawater coolant is not regulated by the VGP

Which of the following discharge sources does the Vessel General Permit (VGP) regulate?

- **Discharges incidental to the normal operation of a vessel engaged in transportation**
- Discharges following damage to a vessel engaged in transportation
- Discharges from permanently moored vessels
- Discharges incidental to the normal operation of a vessel engaged in transportation or oil exploration

Which of the following is a legal option for ships that cannot comply with the discharge standards in the Vessel General Permit (VGP)?

- **The operator can apply for an individual permit for the ship**
- The operator can apply for an exemption certificate for the ship
- The ship can exceed the VGP standards for 30 days
- The ship can exceed the VGP standards so long as the discharge is recorded

Which of the following is a valid reason for submitting a Notice of Termination under the Vessel General Permit (VGP)?

- **The ship has permanently ceased operating in permit waters**
- The ship is bound for a port outside permit waters
- The ship is bound for Hawaii from the US mainland or vice versa
- The ship is planning on being outside permit waters for over 12 months

Which of the following is required as part of the annual inspections required by the Vessel General Permit (VGP)?

- **Checking all ballast tanks for sediment and flaking anodes**
- Checking a sample of ballast tanks for sediment and flaking anodes
- Checking all ballast pumps still meet original pumping rate
- Checking ballast pipework is not leaking

Which of the following is the maximum time normally allowed for corrective action if a violation of the Vessel General Permit (VGP) is found to have resulted in a need for replacement parts?

- **3 months**
- 1 month
- 2 weeks
- 3 weeks

Which of the following is the most accurate for ships subject to the Vessel General Permit (VGP) that are over 400 gross tons and leave permit waters at least once per week?

- **Boiler blowdown is permitted for safety purposes**
- Boiler blowdown is permitted no more than once per day
- Boiler blowdown is permitted without restriction
- No boiler blowdown is permitted

Which of the following was the main reason for the Vessel General Permit (VGP) being introduced in 2008?

- **The exemption for ships in the US Clean Water Act was ruled illegal**
- Pollution from ships was found to have increased rapidly
- The US Clean Water Act came into force
- The US National Pollutant Discharge Elimination System was created

Which of the following would be considered as a violation of the Vessel General Permit (VGP)?

- **Failure to meet any requirement of the permit**
- Any shipboard procedure that was designed to minimise pollutant discharges
- Failure to be inspected by the EPA at least once per year
- Failure to be inspected by the US Coast Guard at least once per year

Which one of the following best describes the Vessel General Permit (VGP) requirements for discharge of swimming pool water from a cruise ship?

- **It must not be discharged in protected/conservation areas**
- It can be discharged in permit waters if it contains chlorine
- It can be discharged in permit waters if the ship is anchored
- It must not be discharged in permit waters

Which one of the following best describes the areas that the Vessel General Permit (VGP) applies to?

- **US internal waters and the sea up to 3 nautical miles from the coast of the US.**
- Internal waters and the sea up to 3 nautical miles from the US and Canada.
- The sea up to 3 miles from the coast of the US and Canada.
- US and Canadian internal waters and their territorial seas out to 3 nautical miles from land.

Which one of the following best describes the ballast records required by the Vessel General Permit (VGP) to be kept?

- **Comprehensive records, including tank capacities, filling and emptying, ballast treatment and sediment discharges**
- Copies of ballast water reporting forms sent to the National Ballast Information Clearinghouse only
- Details of ballast to be discharged into permit waters only
- Details of ballast to be discharged into permit waters or to a US shore facility only

Which one of the following best describes the discharges controlled by the Vessel General Permit (VGP)?

- **5 general types of discharge and 27 specific types of discharge**
- 27 specific types of discharge
- 32 specific types of discharge
- 5 general types of discharge

Which one of the following best describes the form of record keeping required by the Vessel General Permit (VGP)?

- **Written records must be kept, in any form that suits the operator**
- Computer records must be kept, in a format to suit the EPA
- Records must be uploaded to the EPA website, or otherwise sent to the EPA
- Written records must be kept at a place within the US

Which one of the following best describes the requirements of the Vessel General Permit (VGP) for liquids collected in elevator shaft pits?

- **In permit waters it must be discharged overboard through an oily water separator**
- Discharges in permit waters are allowed
- Discharges in permit waters are allowed if the ship is sailing at more than 6 knots
- No discharges are allowed in permit waters

Which one of the following best describes the ships that are allowed to use the Vessel General Permit (VGP)?

- **Commercial ships over 79 feet long and non-recreational vessels**
- All commercial ships
- Commercial ships over 79 feet long
- Non-recreational vessels that carry more than $8m^3$ of ballast

Which one of the following best describes the ships that must submit a Notice of Intent (NOI) in order to be allowed to use the Vessel General Permit (VGP)?

- **Ships over 300 GT or those that carry more than 8 m^3 of ballast**
- All commercial ships
- Ships over 300 GT
- Ships that carry more than $8m^3$ of ballast

Which one of the following best describes the specific requirements of the Vessel General Permit (VGP) for fire main discharges while in conservation areas?

- **Only allowed in case of emergency or for anchor wash downs**
- Only allowed for anchor wash downs
- Only allowed for drills, in case of emergency or for anchor wash downs
- Only allowed in case of emergency

Which one of the following best describes the weekly inspections required by the Vessel General Permit (VGP)?

- **A visual inspection of safely accessible deck and cargo areas and areas where chemicals, oils, dry cargo or other materials are stored, mixed, and used**
- A visual inspection of all deck and cargo areas and areas where chemicals, oils, dry cargo or other materials are stored, mixed, and used
- A visual inspection of all deck and engine room areas
- An inspection of safely accessible deck and cargo areas including testing of water on deck or in bilges for oil

Which one of the following is required as part of the dry dock inspections required by the Vessel General Permit (VGP)?

- **Cathodic protection anodes be cleaned and/or replaced to reduce flaking**
- Anchor chains must be ranged for inspection
- Oil lubricated shaft seals must be opened up for inspection
- The antifouling paint must be renewed

While a ship is regulated by the Vessel General Permit (VGP) how often would you expect its crew to monitor the ballast water treatment plant for correct operation?

- **At least once a month**
- At least once a week
- At least once a year
- Every 3 months

0187 – Marine environmental awareness

Version 2.0 (6199)

A food chain has five levels and 1,000,000 kg of phytoplankton. The food chain consists of phytoplankton – small zooplankton – large zooplankton – small fish – predator fish.
How much of the top predator will be produced according to the 10% rule?

- **100 kg**
- 1 kg
- 1,000 kg
- 10 kg

A food chain has four levels and 1,000,000 kg of phytoplankton. The food chain consists of phytoplankton –large zooplankton – small fish - predator fish
How much of the top predator will be produced according to the 10% rule?

- **1,000 kg**
- 1 kg
- 10 kg
- 100 kg

Approximately how long does it take for a plastic bottle to break down in the marine environment?

- **450 years**
- 150 years
- 20 years
- 200 years

How big are the 'pools of garbage' in the Pacific Ocean estimated to be?

- **The size of the continent Europe**
- The size of New York
- The size of Singapore
- The size of the USA

Over the past 50 years, the phytoplankton population has declined by 50% because the sea water temperature has increased. Is this a problem and why?

- **Yes, it is a problem because, without phytoplankton, the world food supply will be in danger**
- No, it is not a problem because phytoplankton is not a crucial link in marine food webs
- No, it is not a problem because there is too much phytoplankton in the sea anyway
- Yes, it is a problem because phytoplankton produce lots of the carbon dioxide that people breathe

Select from the given options the statement which is most correct in relation to the marine environment.

- **Food chains are connected in more complex food webs**
- Food chains are connected by bacteria
- Food chains have no relation with each other
- Multiple food chains together form a food pyramid

Select the option that most accurately completes the following statement. 'A typical food chain for salt marshes and tidal flats would be . . .

- **. . . phytoplankton – zooplankton – worms – fish'**
- . . . phytoplankton – zooplankton – worms – dugongs'
- . . . phytoplankton – fish – birds'
- . . . zooplankton – phytoplankton – worms – fish'

Select the option that most accurately completes the following statement. 'Animals sometimes eat solid waste (and even feed it to their young), because:

- **. . . solid waste may look like their normal food'**
- . . . animals are not smart enough'
- . . . solid waste always contains lots of nutrients'
- . . . solid waste is colorful and is therefore always attractive to animals'

Select the option that most accurately completes the following statement. 'Bioaccumulation can be defined as . . .

- . . . the process by which chemicals accumulate as sediment on the sea floor'
- . . . the process by which chemicals are broken down by organisms'
- . . . the process by which chemicals have toxic effects in animals'
- . . . the process by which chemicals tend to reach higher concentrations high in the food chain'

Select the option that most accurately completes the following statement. 'Climate change is defined as . . .

- **. . . any change of climate over time which may be due to natural causes or as a result of human activity'**
- . . . a change of climate over time caused only by human activity'
- . . . any change in temperatures worldwide'
- . . . any significant warming of the earth'

Select the option that most accurately completes the following statement. 'Dumping plastic at sea is . . .

- **. . . never allowed'**
- . . . only allowed when sailing faster than 10 knots'
- . . . only allowed after breaking it up into small pieces'
- . . . only allowed in the open seas'

Select the option that most accurately completes the following statement. 'In relation to climate change, the scientific community agrees . . .

- **. . . that the climate is changing and that humans play a role in this'**
- . . . that all of the glaciers will disappear this century'
- . . . that sea levels will rise at least 0.5 meter this century'
- . . . that the temperature on earth will rise more than 4.0 degrees Celsius'

Select the option that most accurately completes the following statement. 'Once animals get entangled in solid waste it is problematic because . . .

- **. . . they may drown, suffocate, get injured or have trouble finding food'**
- . . . they can no longer digest food and will starve to death'
- . . . they cannot find mates because they are no longer attractive'
- . . . they starve to death because plastics remain in their stomachs'

Select the option that most accurately completes the following statement. 'Once microplastics are ingested by small animals at low levels in the food chain . . .

- **. . . the chemicals contained in the plastic enters their body tissues and are passed on to organisms higher in the food chain'**
- . . . other animals do not eat them anymore because they taste like plastic'
- . . . the small animals excrete the plastics rapidly'
- . . . they become instantly toxic'

Select the option that most accurately completes the following statement. 'Research shows that the main reason for dumping solid waste from ships is . . .

- **. . . simply habit'**
- . . . that the rules are not clear'
- . . . the cost of collection'
- . . . the lack of storage space'

Select the option that most accurately completes the following statement. 'The main problem with microplastics being present in the marine environment is that . . .

- **. . . they may be eaten by small animals and enter the food chain'**
- . . . they are too small to be seen'
- . . . they sink to the sea floor'
- . . . they stick together and to larger plastic items'

Select the option that most accurately completes the following statement. 'The natural greenhouse effect is a good thing because . . .

- **. . . without it the average world temperatures would be -18 degrees Celsius'**
- . . . it cools down the earth'
- . . . it produces the oxygen that humans need to exist'
- . . . it protects the ozone layer'

Select the option that most accurately completes the following statement. The People P in sustainable development relates to:

- **. . . acceptable working conditions for employees and societal acceptance'**
- . . . care for the natural world'
- . . . simply making a commercial profit'
- . . . the number of people employed by an environmentally friendly company'

Select the option that most accurately completes the following statement. 'A typical food chain in a kelp forest would consist of . . .

- **. . . kelp – sea urchins – sea otters'**
- . . . kelp – turtles – sharks'
- . . . sea urchins – kelp – sea otters'
- . . . sea urchins – turtles – humans'

Select the option that most accurately completes the following statement. 'Antifouling paint containing tributyltin has been prohibited by IMO because . . .

- **. . . it caused toxic effects in marine organisms'**
- . . . it kept the hull too smooth'
- . . . it killed invasive species'
- . . . it was not very effective as antifouling paint'

Select the option that most accurately completes the following statement. 'Bacteria are an important element in the marine environment primarily because . . .

- **. . . they break down the waste produced in the food chain'**
- . . . they are eaten by zooplankton'
- . . . they eat phytoplankton'
- . . . they kill organisms and provide food for other animals'

Select the option that most accurately completes the following statement. 'Emissions of SOx and small particles (PM) from ships into the atmosphere are relatively high because . . .

- **. . . heavy fuel oil contains high concentrations of sulphur'**
- . . . leaking refrigerant gases contain high amounts of sulphur'
- . . . ships engines are not very efficient'
- . . . SOx evaporates from cargo spaces'

Select the option that most accurately completes the following statement. 'In coastal seas a typical food chain normally has . . .

- **. . . 4 levels'**
- . . . 3 levels'
- . . . 5 levels'
- . . . 6 levels'

Select the option that most accurately completes the following statement. 'In open oceans a typical food chain normally has . . .

- **. . . 6 levels'**
- . . . 3 levels'
- . . . 4 levels'
- . . . 5 levels'

Select the option that most accurately completes the following statement. 'MARPOL first entered into force in . . .

- **. . . 1978'**
- . . . 1973'
- . . . 1980'
- . . . 1983'

Select the option that most accurately completes the following statement. 'Mangroves are important as they provide . . . .

- **. . . nursery grounds for fish and natural flood barriers'**
- . . . a home for turtles and natural flood barriers'
- . . . a wood supply for humans and shelter grounds for dolphins'
- . . . nursery grounds for fish and shelter grounds for dolphins'

Select the option that most accurately completes the following statement. 'Oceans are important for humans because . . .

- **. . . they provide fish and regulate our climate'**
- . . . they provide fish and 80% of the oil we use'
- . . . they provide fish and enhance the green house effect'
- . . . they regulate the climate and enhance the greenhouse effect'

Select the option that most accurately completes the following statement. 'Plankton in the open ocean are usually small because . . .

- **. . . open oceans are nutrient-poor'**
- . . . open oceans are deep'
- . . . the small size allows them to capture sunlight energy more efficiently'
- . . . the small size allows them to hide from predators more easily'

Select the option that most accurately completes the following statement. 'Rocky shore inhabitants are generally tough because . . .

- **. . . they are used to rough circumstances, like the waves'**
- . . . the rocky shores are crowded with a lot of other animals'
- . . . the rocky shores are often polluted areas'
- . . . the rocky shores are slippery'

Select the option that most accurately completes the following statement. 'Sustainable development has three aspects, often represented by the three P's, which are . . .

- **People, Planet and Profit'**
- Participation, Pride and Profit'
- People, Planet and Pride'
- Promotion, Profit and Pride'

Select the option that most accurately completes the following statement. 'The Planet P in sustainable development relates to . . .

- **. . . care for the natural world'**
- . . . acceptable working conditions for employees and societal acceptance'
- . . . making a business profit'
- . . . preventing invasive species from spreading around the planet'

Select the option that most accurately completes the following statement. 'The Profit P in sustainable development relates to . . .

- **. . . making a business profit while recognizing sustainable demands'**
- . . . acceptable working conditions for employees and societal acceptance'
- . . . care for the natural world'
- . . . simply making a commercial profit regardless of the consequences'

Select the option that most accurately completes the following statement. 'The coral triangle is located . . .

- **. . . in South-East Asia'**
- . . . at the North Pole'
- . . . in California'
- . . . in Europe'

Select the option that most accurately completes the following statement. 'The main ecological impacts of oil on the marine environment are . . .

- **. . . death of seabirds, fouling of coastal areas'**
- . . . ingestion by fish, fouling of coastal areas'
- . . . ingestion by mammals, death of seabirds'
- . . . ingestion by mammals, fouling of rocky shores'

What percentage of the global man made CO2 emissions was international shipping estimated to be responsible for in 2007?

- **2,7 %**
- 1,0 %
- 12,0 %
- 5,0 %

Which IMO convention contains the main environmental regulations for shipping?

- **MARPOL**
- COLREG
- SOLAS
- STCW

Which group of oil products given in the options is most likely to require a clean-up response in the event of a spill into the sea?

- **Crude oil, heavy diesel and lubricating oils**
- Crude oil, gasoline and kerosene
- Crude oil, heavy diesel and kerosene
- Gasoline, light diesel oil and kerosene

Which of the combinations given in the options are the main components of diesel engine exhaust gas emissions that may cause health problems for humans?

- **SOx and PM**
- CO2 and SOx
- CO2 and PM
- PM and CFC's

Which of the given options are the components of ships' diesel engine exhaust gas emissions that are the main contributors to acidification and acid rain formation?

- **SOx and NOx**
- CFC's
- CO2 and PM
- SOx and PM

Which of the statements given in the options is NOT true in relation to oil pollution of the sea?

- **Economic effects of oil pollution are always connected to the size of the spill**
- Accidents with oil tankers damage the image of shipping
- Oil pollution of beaches lead to economic losses in the tourist industry
- Shellfish ingest oil and this may give them an oily taste

Which of the statements given in the options is NOT true in relation to sewage discharges from ships?

- **Discharge of sewage often leads to problems in the open ocean**
- Discharge of sewage may lead to problems in enclosed areas
- Sewage can cause problems when it contains cleaning chemicals
- When discharging sewage nutrients are added to the sea giving algae extra food

Which one of the IMO conventions given in the options is most concerned with the difference between open oceans and coastal seas?

- **Ballast water convention**
- MARPOL, Annex I - oil
- MARPOL, Annex V – solid waste
- Ship Recycling convention

Which one of the IMO conventions given in the options is most concerned with the difference between open oceans and coastal seas?

- **MARPOL, Annex IV – sewage**
- Anti fouling convention
- MARPOL, Annex V – solid waste
- Ship Recycling convention

Which one of the given options best describes the process of photosynthesis?

- **Plants use sunlight to produce sugars and oxygen**
- Animals and plants use sunlight to produce sugars
- Animals use sunlight and oxygen to produce sugars
- Plants use sunlight and oxygen to produce sugars

Which one of the given options best represents the main constituents that make up the exhaust gas emissions from ships' diesel engines?

- **CO2 , SOx, NOx and PM**
- CO2 , NH4, NO and, PM
- CO2 , NH4, NOx and CFC's
- CO2 , SOx, NOx and CFC's

Which one of the given options is considered a chronic toxic effect of chemicals rather than an acute effect?

- **Problems with reproduction**
- Coma
- Death
- Vomiting

Which one of the given options is the estimated operational discharge of oil into the sea in 2002?

- **24% of oil entering the sea**
- 16% of oil entering the sea
- 32% of oil entering the sea
- 8% of oil entering the sea

Which one of the given options is the estimated quantity of oil entering into the sea as a result of shipping accidents in 2002?

- **8% of oil entering the sea**
- 16% of oil entering the sea
- 24% of oil entering the sea
- 32% of oil entering the sea

Which one of the given options is the most accurate definition of the term 'invasive species' in relation to the marine environment?

- **Alien species that become a pest in a new environment**
- Alien species found in open seas
- Alien species that eat other species
- Alien species that invade coastal areas

Which one of the marine ecosystems given in the options is normally only found in tropical areas?

- **coral reefs and mangroves**
- coral reefs and seagrass beds
- mangroves and kelp forests
- mangroves and seagrass beds

Which one of the marine environments given in the options is most vulnerable to oil pollution?

- **Mangrove forest**
- Deep sea
- Open ocean
- Rocky shore

Which one of the statements given in the options is INCORRECT in relation to open oceans?

- **Open oceans have a richer sea life than coastal seas**
- Open oceans are deeper than coastal seas
- Open oceans have less nutrients than coastal seas
- Open oceans have longer food chains than coastal seas

Which one of the statements given in the options is NOT correct in relation to seafarers, shipping and the environment?

- **Seafarers have no influence on how things work onboard; they should just follow orders**
- Seafarers are an important part of the shipping industry
- Seafarers are essential to make regulation and procedures work
- Seafarers can make a difference in the environmental performance of shipping

Which one of the statements given in the options is NOT true in relation to invasive marine species?

- **Invasive toxic algae kill shellfish like mussels**
- Invasive species cause risks to human health
- Invasive species have caused the collapse of fisheries
- Whole ecosystems are changed by the invasion of species

Which one of the statements given in the options is NOT true in relation to litter dumped into the sea?

- **Litter may cause damage to underwater constructions**
- Fishermen lose fishing time when they have to remove litter from their nets
- Large amounts of litter make beaches unattractive and tourists stay away
- Litter may cause damage to boats when ropes get entanglement around propellers

Which one of the statements given in the options is NOT true in relation to plastics dumped into the sea?

- **When animals eat plastics they always pass them out with body waste**
- Animals may eat plastic because it looks like their normal food
- Plastic cannot be digested and may clog the intestines
- When animals eat plastics they have a false sense of fullness so they stop eating and may die

Which one of the statements given in the options is NOT true in relation to shipping and the environment?

- **Disasters with oil tankers are the only important environmental impact from shipping**
- Environmental impacts from ships include emissions into the atmosphere and dumping of waste and oil into the sea
- Most of the harmful emissions from ship come from the daily operational release of substances
- Over 90% of world trade is carried by sea

Which one of the statements given in the options is NOT true in relation to the transfer of marine life through ballast water operations?

- **Faster ships make it harder for marine species to travel**
- At least 10.000 different species travel in 'ships ballast tanks around the world
- Every year, 10 to 15 billion tons of ballast water is carried along by ships
- Most species do not survive inside ballast tanks

Which one of the statements given in the options is NOT true?

- **Marine mammals use sound only for communication**
- Background noise from ships makes it harder for animals to hear other sounds
- Marine mammals, fish and even coral reefs make sounds
- Ships' noise, military sonar and underwater construction are examples of human made sounds under water

Why are coral reefs considered to be vulnerable?

- **Because they are slow growing animals and can be easily damaged**
- Because they are often situated in polluted regions
- Because they are often situated in shallow waters in coastal areas
- Because they are slow growing plants and can be easily damaged

food chain has three levels and 1,000,000 kg of phytoplankton. The food chain consists of phytoplankton –large zooplankton – predator fish.
How much of the top predator will be produced according to the 10% rule?

- **10,000 kg**
- 1.000 kg
- 10 kg
- 100 kg

0190 – H2S - Hydrogen Sulphide Awareness

Version 3.0 (1772)

Can H_2S be dissolved into water?

- **Yes**
- No

How does the density of H_2S compare to that of air?

- **H_2S is heavier than air**
- Air is heavier than H_2S
- H_2S is the same density as air

How many PPM of H_2S is in a sample with 2% H_2S gas?

- **20,000**
- 0.002
- 2,000
- 50

ISGOTT recommends that workers wear a personal alarm when they may be exposed to H_2S. At what level does ISGOTT recommend that the alarm be set?

- **TLV-TWA**
- TLV-A
- TLV-C
- TLV-STEL

ISGOTT refers to TLVs when talking about H_2S exposure. What does 'TLV' stand for in this case?

- **Threshold Limit Value**
- Time Limited Value
- Time Lower Value
- Total Level Value

In which one of the following places on a ship is H_2S most likely to be found?

- **Sewage tanks**
- Around diesel engine cylinder heads
- Drinking water tanks
- The galley

Over what time are the H_2S levels averaged when calculating ISGOTT's TLV-TWA?

- **8 hours**
- 1 hour
- 15 minutes
- 24 hours

What does ISGOTT recommend that you do when the H_2S level is expected to rise rapidly, such as when removing cargo line blanks?

- **Wear SCBA**
- Run a fan nearby to blow the gas away
- Take H_2S readings more often
- Undo the bolts at arms length

What is the ISGOTT Time Weighted Average exposure limit (TLV-TWA) quoted for H_2S in this module?

- **5 ppm**
- 10 ppm
- 15 ppm
- 20 ppm

What is the allowable exposure limit for H_2S?

- **There is no single allowable limit. Different regulators set different limits**
- 10 %
- 2 ppm
- 30 ppm

Which of the following best describes how a gas detection tube is used to find out if H_2S is present in a space?

- **The amount of colour change in the tube gives an H_2S reading**
- The amount of electric current passing through the tube gives an H_2S reading
- The glass tube will become cloudy if H_2S is present
- The tube turns blue if H_2S is present

Which of the following best describes the effect on your skin caused by exposure to H_2S over long periods?

- **Various skin disorders**
- No effect, provided a cotton boilersuit is worn
- Skin turns blue
- Skin turns yellow

Which of the following best describes the possible effect of drinking alcohol before being exposed to H_2S?

- **The effects of H_2S are worse**
- H_2S reacts with the alcohol on your breath causing mouth burns
- It has no effect
- The effects of H_2S are less

Which of the following is the best action if you begin to show symptoms of H_2S exposure?

- **Get fresh air, leave the work area and call for help.**
- Ignore the symptoms and continue working.
- Keep working and call for help.
- Take a 15-minute break and resume work.

Which one of the following best describes a major safety hazard of H_2S?:

- **It is explosive**
- It burns with a yellow flame.
- It causes corrosion
- It loses its rotten egg smell

Which one of the following best describes the effect of exposure to a low level of H_2S on your eyes?

- **Minor eye irritation**
- Change in eye colour
- Permanent eye damage
- Reduced vision due to the gas cloud

Which one of the following best describes the factors that decide how much H_2S exposure affects your body?

- **Frequency, intensity and duration of exposure as well as individual factors**
- Frequency and intensity of exposure
- Frequency and intensity of exposure as well as individual factors
- Intensity and duration of exposure

Which one of the following best describes why we often use a time weighted average when measuring H_2S exposure?

- **It measures two of the most important factors which determine the harm caused by H_2S**
- Concentrations of H_2S are constantly changing and readings need to be averaged
- It allows you to work in high levels of H_2S so long as you take the next day off
- It is easier to measure than the instant value of H_2S

Which one of the following best describes why you cannot only use a time weighted average reading when measuring exposure to H_2S?

- **High gas levels can cause harm even if exposure time is short**
- Time weighted averages are measured in % but exposure limits are set in PPM
- Time weighted averages are not accurate
- You have to stop work to calculate the time weighted average exposure

Which one of the following describes how H_2S would normally enter your body while onboard a ship?

- **Absorption and inhalation**
- Ingestion and absorption
- Injection and absorption
- Injection and ingestion

Which one of the following describes how an electric meter can be used to safely measure H_2S, a flammable gas?

- **The meter is certified for use in explosive atmospheres**
- The meter is kept in the accommodation and a sample is taken via a rubber tube
- The meter is powered by a hand crank
- The meter is powered by a piezo-electric cell

Which one of the following describes the main source of the H_2S found onboard ships?

- **Natural organic decay**
- Anti-corrosion paint
- Atmospheric pollution
- Rusting of steel

Which one of the following describes the most likely way H_2S can be brought onboard a ship?

- **Bunkering residual fuel oil**
- Bunkering lube oil
- Loading fresh water
- Taking on harbour water as ballast

Which one of the following does ISGOTT recommend you do if your personal H_2S alarm sounds while you are at the bottom of a cargo tank?

- **Use an emergency air supply while you move to a safe area**
- Climb the vertical ladder to the tank entrance
- Start to calculate your time weighted average exposure since you started work
- Test the alarm

Which one of the following is a symptom of H_2S exposure?

- **Nausea or sickness**
- Dilated pupils
- Loss of hearing
- Shivering

Which one of the following is the best way to approach a workplace where H_2S is expected to be, according to this module?

- **Approach while using a detector which takes continuous H_2S measurements.**
- Do not enter the workplace
- Keep clear by testing for H_2S using a sample tube on a long stick
- Wear an EEBD set

Which one of the following is the best way to approach a workplace where H_2S is suspected to be, if continuous monitoring is not available?

- **Wear a positive pressure SCBA until H_2S levels have been shown to be safe**
- Do not approach the workplace
- Hold your breath until H_2S levels have been shown to be safe
- Keep clear by testing for H_2S using a long rubber sample tube

Which one of the following may protect you from H_2S, while you give artificial respiration on a ship to someone who has been exposed to H_2S?

- **A pocket face mask**
- A defibrillator
- A mechanical resuscitation pump
- Using gloves

Which one of the following statements about H_2S is true?

- **H_2S may still remain in an oil cargo tank after gas freeing**
- An oil cargo tank with an oxygen level of 20% will not contain H_2S
- Crude oil washing will remove all H_2S from an oil cargo tank
- Water washing will remove all H_2S from an oil cargo tank

Which one of these statements correctly describes the colour of H_2S?

- **Colourless**
- Blue
- Green
- Yellow

Which one of these statements correctly describes the smell of H_2S?

- **It smells like rotten-eggs**
- It has no smell
- It smells like lemons
- It smells like wet grass

Which one of these statements correctly describes what happens when we try to burn H_2S?

- **H_2S burns in air with a blue flame**
- H_2S burns in air with an orange flame
- H_2S will burn in pure oxygen but not in air
- H_2S will not burn

Which option best describes what happens if you breathe so much H_2S into your lungs that your body is unable to break it down?

- **Your lungs will stop working and you can suffocate and die**
- You will develop skin cancer
- You will die of pneumonia
- Your throat will become very red and sore

Why is it dangerous to rely on your sense of smell to detect the presence of H_2S?

- **H_2S can cause you to lose your sense of smell**
- H_2S does not have a recognisable smell
- H_2S has no smell
- H_2S only has a smell when the concentration is above safe working limits

You start to smell H_2S while working. Which one of the following would you expect to be the first symptom you notice if H_2S levels continue to rise?

- **Eyes start to water**
- Hands turn red
- Skin turns red
- Trouble breathing

Your company procedures for H_2S hazards give a ppm level for TLV-STEL. What does 'STEL' stand for?

- **Short Term Exposure Limit**
- Short Term Entry Level
- Sulphur Test Expiry Limit
- Sulphur Time Exposure Life

0191 – Maritime Labour Convention 2006, an Introduction

Version 3.0 (4605)

According to the MLC 2006 what will be the minimum age for seafarers to undertake normal night working on-board ship?

- **18**
- 16
- 17
- 21

Certain ship types are exempted from compliance with the requirements of the MLC 2006. From the options given select the type which is among this exempt group.

- **Vessels engaged in fishing**
- Commercial yachts
- Offshore support vessels
- Oil tankers

For how long is a Maritime Labour Certificate, issued for compliance with the MLC 2006, normally valid?

- **5 years**
- 1 year
- 2 years
- 3 years

How does the MLC 2006 define a seafarer onboard a ship to which the Convention applies?

- **Any person doing any work on the ship.**
- Only persons holding an STCW qualification.
- Only persons signed on ships articles.
- Only persons who hold seamen's identity documents.

How should the non-mandatory Part B of the Code of the MLC 2006 be treated by ratifying Member States?

- **It must be fully considered and reasons given for non-compliance with it.**
- It can be ignored completely as it is not mandatory.
- It must be covered by substantial equivalence.
- It must still be fully implemented.

If a shipowner fails to repatriate a seafarer, or pay the cost of the repatriation, following satisfactory completion of a contract, who, according to the MLC 2006, is responsible for making the arrangements and covering the cost for the repatriation?

- **Competent authority of the Flag State**
- One of the seafarers' welfare organisations
- The seafarer
- The ship's Master

In order for a Maritime Labour Certificate, issued for compliance with the MLC 2006, to remain valid it is necessary that an intermediate inspection is carried out on behalf of the Competent Authority. When should this inspection be carried out?

- **Between the 2nd and 3rd anniversary of issue of the certificate.**
- 6 months after issue of the certificate.
- After the 3rd anniversary of issue of the certificate.
- Between the 1st and 2nd anniversary of issue of the certificate.

In which element of the MLC 2006 would you find a purpose statement?

- **In each Regulation**
- In each Article
- In Part A of the Code
- In Part B of the Code

Seafarer recruitment and placement services operating under the jurisdiction of a ratifying Member State should be approved and licensed by that State. Which of the statements given in the options is the most accurate in relation to these services?

- **Once licensed the service should be monitored and inspected regularly to ensure compliance with the requirements of the convention.**
- Once licensed the service should be self regulated.
- Once licensed the service should be spot checked to ensure compliance with the requirements of the Convention.
- Once licensed the service should only be inspected regularly in the event of an alleged or suspected non-compliance.

Select the option which best completes the following statement. A recognised Organisation is one which is appointed by . . . . . to carry out the inspection and certification requirements of the MLC 2006.

- **A Member State or a Competent Authority.**
- A Classification Society.
- A Shipowner.
- The ILO.

The ILO is split into a number of geographical regions. How many of these regions are there?

- **5**
- 10
- 12
- 2

The MLC 2006 contains a number of Articles. Which of the given options best describes the content of the Articles?

- **A set of broad statements of principles and obligations.**
- A specific list of requirements to be met.
- Part A and Part B of the MLC 2006 Code.
- The specific national laws and regulations that must be introduced by member states.

The MLC 2006 contains a requirement that a copy of the Convention must be carried onboard all ships flying the flag of Member States. Who should have access to this document onboard the ship?

- **All seafarers.**
- All of the ship's Officers
- Only the ship's Master and Chief Officer.
- Only the ship's Master.

The MLC 2006 requires that Member States provide medical and dental care facilities available ashore. Under what conditions should seafarers have access to these facilities?

- **As required when the necessary care cannot be provided onboard.**
- Only in emergency situations.
- Only when the seafarer agrees to pay for any costs incurred.
- Only when the ships hospital is already fully occupied.

The MLC 2006 requires that either hours of work or hours of rest are regulated and a notice displayed showing the detail of this. When hours of rest are regulated what should the minimum hours of rest be in a seven day period?

- **77**
- 40
- 65
- 84

The MLC 2006 requires that either hours of work or hours of rest are regulated and that a notice is clearly displayed showing the detail of this. When hours of rest are regulated what should the minimum hours of rest be in a 24 hour period?

- **10**
- 12
- 14
- 8

The MLC 2006 requires that either hours of work or hours of rest are regulated and that a notice is clearly displayed showing the detail of this. When hours of work are regulated what should the maximum hours of work be in a seven day period?

- **72**
- 40
- 48
- 60

To ensure flexibility for Member States in ratifying the MLC 2006 it is possible for them to show substantial equivalence of their existing laws to meet the requirements of these laws. Which parts of the Convention can be covered by substantial equivalence?

- **Part A of the Code**
- Part B of the Code
- The appendices
- The Articles.

Under the MLC 2006 what is the validity for medical certificates for seafarers aged 18 years or older?

- **2 years**
- 1 year
- 4 years
- 5 years

Under which of the following circumstances might an Interim Maritime Labour Certificate be issued?

- **When a ship is newly built and delivered.**
- Following a successful intermediate inspection.
- Following corrective action for a non-compliance under MLC 2006
- When the DMLC is incomplete.

What does the abbreviation DMLC stand for?

- **Declaration of Maritime Labour Compliance.**
- Detailed Maritime Labour Certificate.
- Dispensation from Maritime Labour Certificate.
- Document of the Maritime Labour Convention.

What is seen as the key advantage to the shipowner resulting from the introduction of the MLC 2006?

- **It should give them a level playing field in which to operate.**
- It should allow them to build cheaper ships.
- It should allow them to reduce crew numbers onboard their ships.
- It should reduce the number of inspections onboard their ships.

What is the main responsibility of a Master on a ship flying the flag of a Member State that has ratified the MLC 2006?

- **Ensure ongoing compliance with the Convention.**
- Carry out checks on the recruitment and placement service supplying seafarers to the ship.
- Ensure that the DMLC part 2 is completed.
- Hear all complaints raised by individual seafarers.

What is the maximum period of validity for an Interim Maritime Labour Certificate issued under the MLC 2006?

- **6 months**
- 12 months
- 3 months
- 36 months

What is the minimum age specified for a person to work as a cook onboard ship according to the MLC 2006?

- **18 years of age**
- 16 years of age
- 21 years of age
- 25 years of age

What is the minimum leave rate that seafarers should be entitled to under the MLC 2006?

- **2.5 days per month**
- 1.5 days per month
- 5 days per month
- 6 days per month

When Port State Control Officers carry out ship inspections according to the requirements of the MLC 2006 they must do so with the regard to the 'no more favourable treatment' clause. Which ships is this clause meant to prevent from having more favourable treatment?

- **Foreign flag ships flying the flag of a non-ratifying State.**
- Foreign flag ships flying the flag of a ratifying Member State.
- Ships built or delivered before the entry into force date of the Convention.
- Ships flying the flag of the Member State visiting a home port.

When deciding on safe manning levels for vessels flying their flag the primary concern of a competent authority is always the safety of the vessel. Which factor should always be taken into account when setting the safe manning level?

- **Fatigue risk due to vessel operational requirements.**
- Overall wage bill for the shipowner.
- Size of the mess room.
- The capacity of the galley.

Which body has the final responsibility for implementation and enforcement of the MLC 2006 on ships flying the flag of a ratifying Member State?

- **The Member State**
- Port State Control
- The ILO
- The IMO

Which sizes of ship does the MLC 2006 apply to?

- **All sizes**
- Only ships above 10,000 grt
- Only ships above 200 grt
- Only ships above 500 grt

Who would normally complete part 2 of the DMLC?

- **The ship Owner.**
- A Recognised Organisation.
- The Competent Authority.
- The ship Master.

0192 – MLC 2006 Onboard Responsibilities

Version 2.0 (1647)

A renewal inspection must be carried out prior to expiry of a Maritime Labour Certificate. Which of the given options is responsible for carrying out the renewal inspection?

- **Flag State.**
- Port State control.
- The International Labour Organisation.
- The ship's Master.

From the options given select the one which identifies the types of vessel for which certification is mandatory under the MLC 2006.

- **Only commercial vessels over 500 grt trading internationally or between foreign ports.**
- All vessels trading internationally or between foreign ports.
- Only commercial vessels over 1500 grt trading internationally or between foreign ports.
- Only commercial vessels over 200 grt trading internationally or between foreign ports.

How are existing ships affected by the accommodation requirements of the Maritime Labour Convention 2006.

- **They only need to comply with the requirements if substantial modifications, alterations or change of use has occurred.**
- Existing ships are subject to the accommodation requirements of the Convention only if a change of flag occurs.
- Existing ships are totally exempt from any of the accommodation requirements of the Convention.
- There is no legislation for the accommodation requirements for existing ships.

How many areas of inspection are identified as mandatory for the purpose of MLC 2006 certification?

- **14**
- 10
- 20
- 6

Onboard medical care for seafarers is covered by Regulation 4.1 of the Convention. The regulation stipulates the conditions under which a medically qualified doctor must be carried onboard. Which of the given options would require such provision?

- **All vessels with over 100 persons onboard on voyages exceeding 3 days.**
- All passenger ships on international voyages.
- All vessels on voyages exceeding 14 days.
- Only vessels carrying more than 100 passengers.

Once the MLC 2006 enters into force, which of the given options will have the practical responsibility for maintaining ongoing compliance onboard a vessel?

- **Ship's Master and Senior Officers.**
- Flag State authorities.
- Port State authorities.
- Shipowner/operator.

Regulation 3.1 of the Convention deals with accommodation and recreational facilities. To which vessels do the accommodation requirements directly apply?

- **All vessels built or delivered after entry into force of the Convention.**
- All new and existing vessels.
- All ships over 10,000 grt.
- All vessels of ratifying Member States.

Regulation 3.2 of the Convention deals with food and catering requirements. Which of the following statements is one of the requirements of this regulation?

- **The Master or an authorised officer must carry out regular inspection of all onboard catering facilities.**
- All officers and crew must be provided with exactly the same food and drinking water.
- Anyone onboard may take on the role of ship's cook as long as they hold a cook's qualification.
- Shipowners may make a reasonable charge for providing food and drinking water.

Regulation 4.1 of the Convention deals with onboard medical care requirements. During a detailed inspection of a vessel by port State control which of the options best describes the evidence that the Master would be expected to provide to demonstrate compliance with the requirements for the medical chest?

- **Access to the medical chest and full records of contents and inspections, including arrangements for replenishment.**
- A list of contents of the medical chest and purchase orders for medicines.
- A list of the contents of the medical chest.
- Details of the qualifications of the person in charge of medical care onboard should be sufficient.

Regulation 5.1.5 of the Convention deals with the requirements for arrangements to deal with seafarers onboard complaints. Which of the statements given in the options is true?

- **Every seafarer has the right to take his complaint directly to the Master.**
- Complaints can only be dealt with if a seafarer is accompanied by a union representative.
- Every complaint by a seafarer must be dealt with directly by the Master.
- Seafarers may be removed from a vessel if they make repeated complaints.

Select from the given options those vessels which will be subject to port State control inspections under the MLC 2006.

- **Foreign flag vessels.**
- Only vessels flying the flag of a ratifying State.
- Only vessels over 500 grt.
- Vessels flying the flag of the Member State for that port.

Suggested evidence sources to show compliance with the requirements of the Convention are given in the ILO guidelines for flag and port state control inspectors. Which of the given options is true with regard to evidence of compliance?

- **Evidence from other sources may be used for MLC 2006 inspection purposes if it is relevant.**
- All evidence used for the purpose of demonstrating compliance with MLC 2006 must be generated solely for that purpose.
- Only written evidence is acceptable for the purpose of demonstrating compliance with MLC 2006.
- Visual inspection of all areas is a requirement of any port State control inspection of a vessel.

The Maritime Labour Certificate can only remain valid for the full period on the condition that an interim inspection is carried out by flag State. During which part of the validity period must this be completed if only one interim inspection is carried out?

- **Between second and third anniversary of issue.**
- At any time before the expiry date.
- Between first and second anniversary of issue.
- Between third and fourth anniversary of issue.

The arrangements for payment of seafarers' wages are dealt with under Regulation 2.2 of the Convention. What arrangements, if any, are required regarding allotments under this regulation?

- **Seafarers must be able to allot all or part of their wages on a regular basis.**
- Seafarers can only allot their full wage each month.
- Shipowners may charge seafarers up to 10% of their wages to cover the cost of allotment arrangements.
- There is no requirement to have arrangements in place to deal with allotments; it is just recommended.

The majority of the evidence required to show that the seafarers working onboard are suitable is likely to be detailed in existing or suitably modified crew lists. What is the Masters responsibility regarding the details entered on the crew list?

- **The Master should inspect seafarers documents and carry out reasonable checks for authenticity if there is any doubt.**
- If the seafarer has been recruited through an approved service the Master can assume they will have carried out sufficient checks.
- Provided all of the documentation is in date there is no need for any further checks by the Master.
- The Master can assume that the shipping company has carried out adequate checks.

The requirements for manning levels onboard are dealt with by Regulation 2.7 of the Convention. What should be the main consideration for flag States when setting the manning level for a particular vessel?

- **Fatigue risk and safe working.**
- Crew costs.
- Size and capacity of the galley and mess rooms.
- Total number of cabins onboard the vessel.

The start date of the period of validity of a Maritime Labour Certiificate issued following a renewal inspection depends on the relative timing of the inspection date and the expiry date of the original certificate. When is the most beneficial period for the renewal inspection to be carried out?

- **From 3 months before up to the expiry date.**
- From 1 month before to 1 month after the expiry date.
- From 12 months before up to the expiry date.
- From 6 months before up to the expiry date.

There are certain conditions regarding the suitability of a person to serve as a ship's cook and the Master must provide evidence that the requirements are being met. Select from the options the one which is correct?

- **A person appointed as ship's cook must be at least 18 years of age.**
- A person appointed as ship's cook must be at least 16 years of age.
- A person appointed as ship's cook must be at least 21 years of age.
- A person appointed as ship's cook must hold a professional chef's certificate.

There have been a number of publications produced to try to ensure an acceptable level of consistency in monitoring compliance with the MLC 2006. Which publication is recommended for detailed inspections of foreign flag vessels?

- **The guidelines for port State control officers carrying out inspections under the MLC 2006 (ILO).**
- Ratification of the Maritime Labour Convention 2006.
- The guidelines for flag State inspections under the Maritime Labour Convention 2006 (ILO).
- The Maritime Labour Convention: Action plan 2006 - 2011.

To which of the following groups of vessels will the MLC 2006 apply?

- **Commercial vessels on international trade.**
- Commercial vessels exclusively navigating in inland waterways.
- Fishing vessels.
- Warships.

Under the requirements of the MLC 2006 what is the main role of the competent authority or the recognised organisation appointed by them?

- **To monitor and enforce compliance.**
- To adapt and implement national legislation.
- To ensure that each Member State is in compliance.
- To generate policies and procedures to ensure onboard compliance.

Under what circumstances would the ship's Master expect to receive written notification of any deficiencies discovered during a port State control inspection?

- **Only if the deficiencies are considered significant.**
- For every minor deficiency found during the inspection.
- Only for deficiencies arising from seafarers complaints.
- Only if the deficiency results in a health or safety hazard.

What is the minimum age for a seafarer to work onboard a ship under Regulation 1.1 of the MLC 2006?

- **16**
- 18
- 20
- 21

What should the ship's Master expect to happen if a port State control inspector discovers some minor deficiencies onboard during a detailed inspection under the MLC 2006?

- **The Master should receive a verbal notification and asked to agree a deadline for rectification.**
- The Master should not expect any action to be taken in the event of minor deficiencies being discovered.
- The Master should receive a written notification a copy of which will also be sent to the flag State authority.
- The vessel is likely to be detained until the deficiency is rectified.

When would a ship's Master expect the ship to be detained following an MLC 2006 port State control inspection?

- **Whenever a deficiency is found which presents a safety or security hazard to crew members.**
- Whenever any deficiency cannot be immediately rectified by ship's staff.
- Whenever any deficiency is found relating to one of the 14 specified areas during the inspection.
- Whenever more than one minor deficiency is discovered onboard.

Which of the given options is a requirement before a Maritime Labour Certificate can be issued?

- **Flag State inspection and a completed DMLC.**
- Expiry of an Interim Maritime labour Certificate.
- Onboard inspection by the ILO.
- Port state control inspection and a completed DMLC.

Which of the given options is a requirement of Regulation 2.1 of the Convention which deals with seafarers' employment agreements?

- **All seafarers must be given a signed copy of their employment agreements.**
- Only seafarers over the age of 18 years are required to have employment agreements.
- Seafarers must sign their employment agreement immediately upon joining the ship.
- The Master must retain the original and all copies of the seafarers employment agreements for inspection by port State control inspectors.

Which of the given options is likely to be necessary to show compliance with Regulation 2.2 which deals with the payment of seafarers' wages when the payroll is handled directly by the shipping company office ashore?

- **Seafarer interviews and records of claims for overtime and other additional work.**
- A written statement from the company's finance department.
- A written statement from the Master.
- Copies of the seafarers' employment agreements.

Which of the given options is one of the main aims of the MLC 2006?

- **Good working and living conditions for seafarers.**
- Enhanced safety of the vessel due to better construction standards.
- Improved training and qualifications of seafarers.
- Increased onboard safety of seafarers.

Which of the given options is the main requirement for ratification before entry into force of the MLC 2006?

- **30 Member States and 33% of world grt.**
- 12 Member States and 25% of world grt.
- 25 Member States and 12% of world grt.
- 33 Member States and 30% of world grt.

Which of the given options is the required minimum number of hours of rest for a seafarer in any 24 hour period according to Regulation 2.3 of the Convention?

- **10**
- 12
- 14
- 8

Which of the given options should be accepted as prima facie evidence by port State control inspectors when inspecting vessels for compliance with the MLC 2006?

- **A valid Maritime Labour Certificate and DMLC.**
- A signed statement from the Master that the vessel complies with the convention.
- No complaints from the seafarers onboard the vessels.
- No obvious deficiencies sighted when boarding the vessel.

Which of the given options would be suitable evidence to show that a vessel is meeting the requirements of Regulation 2.3 of the Convention which deals with hours of work and hours of rest?

- **A detailed matrix of seafarers work activities.**
- A record of the weekly and monthly aggregate of hours worked by each seafarer.
- A written statement from the heads of each department onboard.
- An absence of overtime claims.

Which of the given options would likely be accepted as primary evidence to present to port State control inspectors to show that the conditions contained in seafarers' employment agreements are being met?

- **Written statements signed by the Master and the individual seafarers.**
- Absence of any recorded complaints in the onboard complaints records.
- Tick boxes on the crew list against each seafarers name.
- Verbal statement from the Master.

Which of the options given should result in a port State control inspector deciding that a detailed inspection under MLC 2006 was required onboard a vessel?

- **An expired Maritime Labour Certificate.**
- A valid Maritime Labour Certificate with a completed DMLC attached.
- An unfounded complaint from a seafarer.
- No obvious signs of deficiencies.

Which of the statements given in the options is the most accurate in relation to port State control inspections under the MLC 2006?

- **All foreign flag vessels may be inspected by port State control under the 'no more favourable treatment' clause**
- Full port State control inspections are carried out in the event of a seafarer complaint.
- Full port State control inspections are only carried out if the vessels Maritime Labour Certificate has expired.
- Only vessels flying the flag of a ratifying Member State are subject to port State control inspections.

Which of the statements given in the options is true in relation to Regulation 1.2 of the MLC 2006 which deals with medical certification?

- **All seafarers must hold a valid medical certificate issued by an approved medical practitioner.**
- A medical certificate issued by any medical practitioner from the seafarers home country is acceptable.
- A seafarer must hold a valid medical certificate issued by the flag State of the vessel that they are working on.
- Only seafarers over the age of 40 years require a medical certificate.

Which of the statements given in the options is true in relation to Regulation 1.3 of the MLC 2006 which deals with seafarers qualifications?

- **All seafarers must be trained or qualified to carry out their duties onboard ship.**
- All seafarers must hold an STCW qualification.
- Only those seafarers with watchkeeping or cargo handling duties need to have any qualifications.
- Young seafarers can be given full training onboard once they have joined a vessel.

Which of the statements given in the options is true in relation to the hours of work for seafarers under the MLC 2006, Regulation 2.3?

- **No seafarer should normally work more than 72 hours in any 7 day period.**
- After working for 14 hours a seafarer must be given a minimum of 6 hours rest.
- No seafarer should ever work more than 72 hours in any 7 day period .
- The hours of work requirement is fully met if a seafarer never works more than 14 hours in a 24 hour period.

Which one of the given options is most likely to be the minimum evidence necessary to satisfy port State control inspectors that the requirements for dealing with onboard complaints, as detailed in Regulation 5.1.5, are being met?

- **Records of complaints and confidential interviews with seafarers.**
- A copy of the onboard complaints procedure.
- A signed list showing that each seafarer has been given a copy of the procedures.
- A written statement from the Master detailing any complaints dealt with during previous voyage.

Which one of the given options would be the most suitable primary evidence for the Master to provide to a port State control inspector to show that a vessel was in compliance with the onboard health and safety and accident prevention requirements detailed in Regulation 4.3 of the Convention?

- **Records from the onboard safety management system (ISM).**
- The minutes from a safety committee meeting.
- The shipping company's health and safety policy.
- The vessels ISM certificate.

Which one of the given options would part 1 of the DMLC be expected to contain?

- **Details of any exemptions and substantial equivalences for the Member State.**
- A record of inspections.
- Reference to the shipping companies procedures for compliance with the MLC 2006.
- The Maritime Labour Certificate for the vessel.

0193 – MLC 2006 Shipowner's Responsibilities

Version 2.0 (1284)

A Declaration of Maritime Labour Compliance is in two main parts. Who is responsible for completing part 1 of the declaration?

- **The Competent Authority.**
- A port State control officer.
- The International Labour Organisation.
- The Shipowner.

A renewal inspection of a ship is to be carried out prior to the expiry of the vessels existing Maritime Labour Certificate. If the inspection is carried out 2 months before expiry of the original certificate when will the new certificate expire?

- **On the fifth anniversary of the expiry date of the original certificate.**
- On the date that the Declaration of Maritime labour Compliance expires.
- On the fifth anniversary of the date of the renewal inspection.
- On the fifth anniversary of the date that the previous intermediate survey was carried out.

An Interim Maritime Labour Certificate can only be issued under certain circumstances and if certain conditions are met. Which of the given options is one of the conditions of issue of an IMLC?

- **The vessel's Master must be familiar with the requirements of MLC, 2006.**
- The Shipowner must state their intention to draw up suitable policies and procedures to ensure compliance.
- The vessel must have previously held an Interim Maritime Labour Certificate.
- The vessel must hold a valid DMLC.

Complete the following statement by selecting the most accurate of the given options. Shipowners can expect that each Declaration of Maritime Labour Compliance issued to them by . . .

- **. . . an individual Member State will be very similar.**
- . . . an individual Member State will be identical.
- . . . an individual Member State will be totally different.
- . . . all Member States will be identical.

Complete the following statement from the options given. In meeting the requirements of the MLC, 2006, relating to medical certification of seafarers, the Shipowner . . .

- **. . . must include arrangements for revalidating medical certificates which may expire during a voyage.**
- . . . can only accept medical certificates issued by an approved medical practitioner from the seafarer's own country.
- . . . need only have checks in place to ensure each seafarer's medical certification is valid at the start of their employment.
- . . . should accept medical certificates issued by any medical practitioner.

Complete the following statement with regard to the requirements for onboard medical care under the MLC, 2006. To meet the requirements of the Convention regulation dealing with medical care . . .

- **. . . all ships carrying more than 100 persons on international voyages of more than 3 days must carry a qualified medical doctor.**
- . . . all ships carrying more than 100 persons must carry a qualified medical doctor.
- . . . all ships carrying passengers must carry a qualified medical doctor.
- . . . all ships with more than 10 crewmembers onboard must carry a fully qualified doctor.

How many main parts are there in a Declaration of Maritime Labour Compliance?

- **2**
- 14
- 3
- 4

If, during an MLC, 2006, renewal inspection, carried out two months before the expiry date of the certificate, a number of deficiencies and non-conformances are found but the Shipowner cannot agree a rectification action plan with the inspector, what are the likely consequences?

- **The inspector would refuse to renew the Maritime Labour Certificate and may withdraw the existing certificate.**
- The inspector would issue an Interim Maritime Labour Certificate until a plan could be agreed.
- The vessel would be allowed to sail and keep the existing certificate as it had not expired.
- The vessel would be immediately removed from the registry.

In addition to a signed copy of their employment agreement each seafarer is also entitled to be provided with other documentation under the requirements of the MLC, 2006. Which one of the given options should a seafarer also expect to receive?

- **Documented record of his work onboard ship.**
- A letter to hand to the Master of the seafarer's next ship detailing any complaints the seafarer had raised during the current employment.
- An invoice covering the cost of food provided during employment onboard ship.
- An invoice detailing the charges for medical care provided onboard ship.

In order to meet the MLC, 2006, requirements covering food and catering certain evidence of compliance must be provided to inspectors. Which of the given statements is the most accurate in relation to Shipowners requirements for the documented evidence that should be maintained onboard their ships?

- **The documents maintained onboard should include reference to menu planning and nutritional quality of the meals.**
- A record of the checks carried out to verify the age and qualifications of the cook will be sufficient evidence in the majority of cases.
- If the galley is clean and none of the crew has had any complaints about the food there is no need to maintain documented evidence.
- Only a record of inspections of the provision stores and catering facilities will be required.

It is a requirement of MLC, 2006 that either hours of work or hours of rest must be regulated. Where hours of rest are regulated a minimum rest of 10 hours in any 24 hours must be allocated which may be split into a maximum of two periods no more than 14 hours apart. What is the minimum duration for at least one of the rest periods?

- **6 hours.**
- 5 hours.
- 7 hours.
- 8 hours.

MLC, 2006 requires that each vessel has an onboard complaints procedure. Which of the given statements is accurate with regard to these requirements?

- **Seafarers should have the right to complain directly to the ship's Master if they wish.**
- Complaints must be presented to the ship's Master by an appointed representative of the seafarers.
- Seafarers must address their complaints directly to the ship's Master at all times.
- Seafarers must only take their complaints to their immediate superior who will refer them to a higher level if they feel it appropriate.

Seafarers' employment agreements are covered by Regulation 2.1 of MLC, 2006. Which of the given options is a requirement of this regulation?

- **All seafarers must be issued with a copy of their employment agreement.**
- As long as a specimen of the seafarers' employment agreement is displayed onboard the ship there is no need to issue copies to individual seafarers.
- Seafarers do not need to be issued with an employment agreement if it is based on a collective bargaining agreement.
- Seafarers only need to be issued with a copy of their agreement if they request it.

Shipowners often use placement and recruitment services to supply crews for their vessels. What requirement does the MLC, 2006 have in relation to such services which are based in a ratifying Member State?

- **The service must be licensed or certified.**
- Only public services are allowed to operate in a ratifying Member State.
- Shipowners can only employ seafarers supplied by such a service.
- The service can only provide seafarers who are nationals of that Member State.

Ships' accommodation requirements under MLC, 2006 are detailed in Regulation 3.1 of the Convention. Which of the statements given in the options is the most accurate in respect of application of these requirements?

- **The full accommodation requirements apply to all ships of ratifying Member States constructed after the Convention enters into force unless specifically exempted.**
- The full accommodation requirements apply to all ships flying flags of ratifying Member States.
- The full accommodation requirements apply to all ships once the Convention enters into force.
- The full accommodation requirements apply to all ships over 3000 gross tonnage regardless of the flag they fly.

The ILO guidelines for flag State and port State inspections under MLC, 2006, detail the actions that inspectors may and must take in the event of deficiencies, non-conformances and non-compliance being found onboard. Where would a Shipowner find details of any additional penalties or fines that may be applied in such circumstances?

- **Contained in the national laws of the Member State.**
- In the appendices of the Maritime Labour Convention, 2006.
- In the text of the Maritime Labour Convention, 2006.
- On the vessels Maritime Labour Certificate.

The MLC 2006 requires that Shipowners ensure that certain arrangements are in place for the payment of seafarers' wages. Which one of the given options is a mandatory requirement of the regulation dealing with payment of wages?

- **Each seafarer must be issued with an account of wages due, on a regular basis, onboard the ship.**
- An account of all wages paid during employment onboard must be sent to the seafarer's home after they have left the vessel.
- Arrangements must be in place to allow seafarers to allot up to a maximum of 10% of their wages each month.
- Wages must be paid weekly.

The MLC, 2006 regulation dealing with the minimum age of seafarers also includes other requirements relating to young seafarers. Which one of the given options will the Shipowner need to have in place to meet these requirements?

- **A policy identifying suitable or restricted work activities for young seafarers.**
- A policy prohibiting training activities for young seafarers at night.
- A policy prohibiting young seafarers from working at night under any circumstances.
- A policy which allows young seafarers to work at any time under supervision.

The MLC, 2006, requires that seafarers are paid regularly for the services. What is the recommended maximum interval for the payments to be made?

- **Monthly.**
- At the end of each voyage.
- Every 3 months.
- Weekly.

The appointed competent authority will normally complete part 1 of the Declaration of Maritime Labour Compliance on behalf of a Member State. Which of the given options is the most accurate with regard to part 1 of the declarations issued by an individual Member State?

- **There may be some variations in the details within the document to reflect any exemptions granted to a shipowner for a particular ship by the Member State.**
- All of the DMLCs issued to all shipowners will be identical apart from the vessel's name, number and tonnage as they only contain details of the national laws and regulations of that Member State.
- All of the DMLCs issued to an individual Shipowner by a particular Member State will be identical.
- Every part 1 of a DMLC issued by a Member State will be different as it must be written to accommodate the measures entered into part 2 by the Shipowner.

The manning levels required onboard ship are covered by Regulation 2.7 of the MLC, 2006. Which one of the options identifies the manning level specified in this regulation?

- **Sufficient personnel for safe, efficient secure operation.**
- Sufficient personnel so that no overtime work is required.
- Sufficient personnel to ensure two persons on a bridge watch at sea.
- The number of personnel stated on the Safe Manning Certificate.

The reference periods used in the regulation relating to hours of work and hours of rest are 24 hours and 7 days. How is the period of 24 hours apparently being interpreted by the majority of flag States?

- **Any consecutive period of 24 hours.**
- 08.00 to 08.00 hours.
- Midnight to midnight.
- Noon to noon.

The requirements for persons responsible for cooking and food preparation onboard ships are detailed in the Convention under the regulation dealing with food and catering. Which of the given statements best describes the requirement?

- **The person must be properly trained and qualified.**
- The person responsible for cooking is not required to be trained or qualified at all if there are less than ten crewmembers onboard ship.
- The person responsible for cooking must always be a qualified ships cook.
- The person responsible for cooking must hold an STCW cook's qualification.

What are the age requirements for seafarers undertaking regular night work onboard ship according to Regulation 1.1 of the MLC, 2006?

- **Seafarers must be at least 18 years old.**
- Seafarers must be at least 16 years old.
- Seafarers must be at least 21 years old.
- There is no minimum age as long as seafarers are qualified for the work.

What is the likely consequence for a vessel which has an invalid Maritime labour Certificate that is boarded by a port State control officer in a foreign port?

- **The vessel will be subject to a full detailed inspection.**
- The port State control officer will carry out a renewal inspection in order to issue a new certificate.
- The vessel will be detained indefinitely.
- The vessel will not be allowed to complete cargo operations in that port.

What s the maximum period of validity of a Maritime Labour Certificate issued under MLC, 2006?

- **5 years.**
- 2 years.
- 3 years.
- 6 months.

What would a Shipowner expect the consequences of a serious deficiency being uncovered during an MLC, 2006, inspection where the deficiency was not considered to present a health or safety risk?

- **The vessel would be allowed to sail once a rectification action plan had been agreed.**
- The Maritime Labour Certificate would be withdrawn.
- The vessel would be detained until the deficiency was rectified.
- There would be no action taken as there was no health or safety risk associated with the deficiency.

When considering the role of the port State control authorities in relation to the Maritime Labour Convention, 2006, which one of the statements given in the options is true?

- **PSC is responsible for inspecting all foreign vessels trading from a port under its jurisdiction.**
- PSC is only responsible for inspecting foreign vessels flying the flag of a non-ratifying Member State trading from a port under its jurisdiction.
- PSC is only responsible for inspecting foreign vessels flying the flag of any ratifying Member State trading from a port under its jurisdiction.
- PSC is responsible for inspecting all vessels trading from a port under its jurisdiction.

When determining the level of manning onboard ship the Shipowner must obviously refer to the minimum levels shown on the vessels safe manning certificate or document. What other factor does the MLC, 2006, require be taken into account in relation to manning levels?

- **The fatigue risk due to operational requirements.**
- How many ships the Shipowner operates and the number of seafarers employed.
- The availability of suitable seafarers.
- The overall cost of crewing the vessel.

When drawing up the policies addressing the requirements of the MLC, 2006, regulation dealing with medical care which of the given options should be included by the Shipowner.

- **The arrangements for provision of medical and dental care in port.**
- Details of the cost to the seafarer of medical care onboard and ashore when in port.
- Details of the medical conditions which are excluded from a seafarer's entitlement to medical care whilst in employment.
- The maximum number of treatments that a seafarer is entitled to whilst employed aboard ship.

Where existing policies and procedures exist primarily for other purposes but which also meet the requirements of part of the MLC, 2006 what action is required by the Shipowner with respect to their responsibilities under the Convention?

- **Identify the evidence sources in part 2 of the DMLC.**
- Duplicate the evidence records so that there is a separate record specifically for MLC, 2006.
- Rename the policies and procedures to reflect the role they have in meeting MLC, 2006 compliance.
- Rewrite the policies and procedures specifically for MLC, 2006.

Where would a Shipowner find details of any ship specific requirements that a competent authority may have under the MLC, 2006?

- **In part 1 of the Declaration of Maritime Labour Compliance.**
- In part 2 of the Declaration of Maritime Labour Compliance.
- On the Maritime Labour Certificate issued to a vessel.
- On the vessels certificate of registry document.

Where would a record of all MLC, 2006, inspections be found?

- **On or appended to the vessel's DMLC.**
- In the vessels official log book.
- On or appended to the vessel's Maritime Labour Certificate.
- On the vessels certificate of registry.

Where would details of exemptions granted by a competent authority, under MLC, 2006, be recorded?

- **In part 1 of the Declaration of Maritime Labour Compliance.**
- In the ships official log book.
- On a vessels Maritime Labour Certificate.
- On the Safe Manning Certificate.

Which date is considered as the construction date of a new vessel according to MLC, 2006?

- **The date the keel is laid.**
- The date the order is placed for the vessel.
- The date the vessel enters into service.
- The date the vessel is launched.

Which element of the Maritime Labour Convention, 2006, is considered as guidance rather than being mandatory?

- **Part B of the Code.**
- Part A of the Code.
- The articles
- The regulations.

Which of the following statements is true in relation to a Declaration of Maritime labour Compliance issued under the MLC, 2006?

- **A Shipowner will be issued with a separate DMLC for each vessel that they operate which is covered by the Convention.**
- A Shipowner will be issued with a single DMLC to cover all of the vessels registered with a particular Member State.
- A Shipowner will be issued with a single DMLC to cover all the vessels they operate which are covered by the Convention.
- A Shipowner will only require a DMLC for vessels over 3000 gross tonnage.

Which of the following statements is true when considering MLC, 2006, compliance?

- **All of the requirements of MLC, 2006 must be complied with.**
- Only the 14 areas identified for inspection and certification requirements need to be complied with.
- Port State control and flag State inspections are restricted to the 14 areas identified for inspection and certification.
- Ships flying the flag of non-ratifying states do not need to comply with any of the requirements of MLC, 2006.

Which of the given options best describes one of the main aims of the Maritime Labour Convention, 2006?

- **It will ensure a level playing field for the Shipowner.**
- It will increase the number of complaints from seafarers.
- It will increase the number of seafarers working onboard ships.
- It will reduce the number of inspections carried out onboard ships.

Which of the given options is the gross tonnage limit for which certification is a mandatory requirement under MLC, 2006?

- **Vessels over 500 gross tonnage.**
- Vessels over 200 gross tonnage.
- Vessels over 3000 gross tonnage.
- Vessels over 5000 gross tonnage.

Which of the given options would be a reasonable example of non-compliance in relation to the requirements of MLC, 2006?

- **An invalid Maritime Labour Certificate.**
- A faulty light fitting in the ship's galley.
- A lack of books in the ship's library.
- Record of accommodation inspections incomplete.

Which of the given statements is the most accurate in relation to an Interim Maritime Labour Certificate, IMLC, issued as part of the MLC, 2006 certification requirements?

- **An IMLC has a validity of 6 months maximum.**
- An IMLC can be renewed when it expires if a Shipowner has applied for a Declaration of Maritime Labour Compliance for a vessel.
- An IMLC can only be issued to a newly delivered ship.
- An IMLC is issued to a vessel if the vessel fails to meet the initial inspection requirements.

Which of the statements given in the options best describes the Maritime Labour Convention, 2006?

- **It is the result of a review and consolidation of existing labour standards.**
- It is a collection of existing labour standards.
- It is a completely new set of labour standards.
- It is a new revision of STCW 95.

Which of the statements given in the options is the most accurate with respect to recreational facilities specified in the MLC, 2006?

- **The mandatory part of the Code only requires that appropriate recreational facilities should be provided onboard by the Shipowner.**
- Shipowners must ensure that all of the recreational facilities detailed in part B of the Code are provided onboard their ships.
- Specific details of the type and range of recreational facilities which must be provided onboard ship are given in the mandatory part A Code.
- There is no requirement for the Shipowner to provide seafarers' recreational facilities onboard a ship.

Who has the responsibility for drawing up suitable policies and procedures to ensure that the requirements of the national laws of the flag State and therefore the requirements of the Maritime Labour Convention, 2006, are being met?

- **The Shipowner.**
- A Recognized Organisation.
- The Competent Authority.
- The Member State.

With reference to the requirements for recreational facilities onboard ship under the MLC, 2006, which of the following statements is the most accurate?

- **Shipowners must have programmes to maintain and continuously improve the recreational facilities onboard ship.**
- Seafarer's welfare organisations are responsible for providing and maintaining recreational facilities onboard ship.
- The Shipowner has no duty to provide any recreational facilities under the requirements of the Convention.
- The Shipowner must provide recreational facilities onboard ship but the crew must cover the cost of maintenance and improvement.

With regard to qualifications and training covered by Regulation 1.3 of the Maritime Labour Convention, 2006, which of the options is accurate.

- **All seafarers' must be appropriately qualified or trained to carry out their duties onboard ship.**
- Only seafarers identified on the Safe Manning Certificate need to be qualified or trained.
- Only STCW approved qualifications and training is recognised under MLC, 2006.
- Seafarers under the age of 18 years do not require any previous training or qualifications to work onboard a ship as they are only considered as trainees.

0194 – Lifestyle assessment and advice

Version 1.0 (1048)

According to the World Health Organisation approximately what percentage of the population move too little during their normal daily activities and therefore need to exercise?

- **60%**
- 20%
- 40%
- 80%

According to the World Health Organisation what percentage of deaths can be linked to so called lifestyle diseases?

- **50%**
- 20%
- 30%
- 40%

According to the World Health Organisation's recommendations, which of the statements given in the options is correct?

- **An adult man may consume two standard glasses of alcohol and a woman may consume one glass of alcohol per day.**
- All adults may consume two standard glasses of alcohol per day.
- An adult man may consume one standard glass of alcohol and a woman may consume two glasses of alcohol per day.
- An adult man may consume two standard glasses of alcohol and a woman may consume one glass of alcohol per week.

Alcohol is normally passed out of the body through the liver. How long does it take approximately for the liver to deal with one standard glass of alcohol?

- **1.5 hours**
- 2.5 hours
- 2.5 hours
- 5 hours

How long after quitting smoking does it take the body to clear itself of nicotine?

- **Approximately one week**
- Approximately one day
- Approximately one month
- Approximately one year

Select from the given options the level of activity which is rated as having a metabolic equivalent of one by the World Health Authority?

- **Sitting relaxing**
- Intense physical activity
- Moderately intense physical activity
- Sleeping

Select from the options given the one which correctly complete the following statement. Smoking tobacco . . .

- **. . . increases the risk of various forms of cancer.**
- . . . increases the appetite for healthy food.
- . . . reduces the effects of alcohol.
- . . . reduces the risk of cardiovascular diseases.

Select from the options given the one which is the most accurate regarding fat content of some common foods?

- **Fish generally has a high unsaturated fat content.**
- Approximately 50% of the fat in most food is unsaturated fat.
- Chocolate and potato crisps generally have a high unsaturated fat content.
- Fruit and vegetables generally have a high unsaturated fat content.

Select from the options given the statement which is accurate in relation to the effect of fats on the cholesterol level in the human body?

- **Unsaturated fats help to reduce cholesterol levels.**
- Fats have no effect on the bodies cholesterol levels.
- Saturated fats help to reduce cholesterol levels.
- There is no difference between saturated and unsaturated fats as far as cholesterol levels are concerned.

Select the option that correctly completes the following statement regarding lifestyle. Your lifestyle is . . .

- **. . . learned and automatic but can be altered by your own conscious choices.**
- . . . can only be altered if you become ill and receive medication.
- . . . only influenced by your culture and the groups you belong to.
- . . . predetermined and cannot be altered by individual choices.

Smoking is generally regarded as bad for your health with approximately 40 poisonous substances contained in tobacco smoke? Which substance contained in the smoke is particularly responsible for so called smokers cough?

- **Tar**
- Ash particles
- Carbon monoxide
- Nicotine

The Body mass index, BMI, is used to indicate the ratio of body fat. Obesity is a condition where an individual has far too much body fat for their size. What is the World Health Organisations identified BMI level to indicate obesity?

- **BMI of 30 or higher**
- BMI of 19 to 25
- BMI of 25
- BMI of 40 or higher

The energy we require for our daily activities comes from the food we eat. Which of the statements given in the options is correct?

- **We should eat sufficient to provide the energy we need and to maintain a healthy weight**
- The more we eat the more energy we have.
- We should always eat a heavy meal before undertaking any strenuous activity so that we have plenty of energy.
- We should eat a lot of fats as they provide more energy than other foods.

What does MET stand for in relation to human physical activity?

- **It is the metabolic equivalent**
- It is maximum exercise time
- It is the manual exercise test.
- It is the maximum exertion time

What does the World Health Organisation physical activity standard recommend as the minimum required frequency and duration for intense exercise to be beneficial to an individual?

- **20 minutes activity 3 days a week**
- 20 minutes activity every day
- 30 minutes activity 5 days a week
- 30 minutes activity every day

What does the World Health Organisation recommend as the responsible daily consumption of alcohol for women?

- **One standard glass per day**
- Half a standard glass per day
- One standard glass every day
- Two standard glasses per day

What does the World Health Organisation recommend regarding frequency of alcohol consumption if it is not to be considered habitual?

- **Consume no alcohol on at least two days each week.**
- Consume alcohol every day but never more than one standard glass.
- Men can drink every day but women should only drink five days a week.
- Only consume alcohol once a week.

What is the main effect on the body of the carbon monoxide contained in tobacco smoke?

- **Reduced oxygen in the blood leading to low energy levels**
- Clogging of the cilia in the lungs resulting in smokers cough
- It is addictive resulting in a craving to smoke.
- It results in headaches.

Which of the given options is considered as moderately intense physical activity for an adult who is under 55 years of age according to the world health organisation?

- **Walking 5km per hour**
- Cycling 10km per hour
- Cycling 5km per hour
- Walking 16km per hour

Which of the given options is part of the recommended standard for healthy physical activity for an adult, according to the World Health Organization?

- **For adults this is at least 30 minutes of moderately intense physical activity at least 5 days per week.**
- For adults this is at least 40 minutes of moderately intense physical activity at least 3 days per week.
- For adults this is at least 60 minutes of intense physical activity at least 5 days per week.
- For adults this is at least 60 minutes of intense physical activity at least 5 days per week.

Which of the given options is true according to the World Health Organisation with regard to the food we eat?

- **The majority of the food we eat contains more than enough salt to meet our requirements and no further salt should be added.**
- Eating too much fruit, vegetables and bread will make you feel full and you will have insufficient energy to do normal daily activities.
- Energy rich foods should form the majority of our diet in order to maintain a healthy weight.
- We should all eat meat every day to ensure we have sufficient vitamins and fibre in our diets.

Which of the groups of aspects would form part of a healthy lifestyle according to the World Health Organisation?

- **Avoiding smoking, having moderate alcohol consumption and avoiding excess stress.**
- Eating plenty of nutritious food, taking enough exercise and relaxing by watching a lot of TV.
- Sleeping for as long as possible, eating only fruit and vegetables and only smoking occasionally when having a drink of alcohol.
- Taking enough exercise, eating plenty of fats and consuming large quantities of alcohol.

Which one of the given options is an accurate statement in relation to the recommendations for healthy eating?

- **You should regularly eat fish.**
- Candy is better than fruit.
- Totally avoid salt and sugar.
- You may never eat French fries of potato chips.

0195 – Physical exercise and lifestyle coaching

Version 3.0 (1022)

According to the World Health Organisation how should daily physical exercise recommended activity times be met?

- **The time can be broken down into convenient periods during the day.**
- The activity should be carried out for no more than 5 minutes at any one time.
- The time should be continuous
- The time should be split into two equal parts

All physical exercise is classed as sport. Is this statement true or false?

- **False**
- True

Carbohydrates and fats are both considered as fuels, and therefore energy sources for the human body. What is the approximate ratio between the energy available from equal amounts of carbohydrates and fats?

- **Fats provide approximately 37 times more energy than an equivalent amount by weight of carbohydrates.**
- Carbohydrates provide approximately 37 times more energy than an equivalent amount by weight of fats.
- Equal amounts by volume of carbohydrates and fats provide equal amounts of energy.
- Equal amounts by weight of carbohydrates and fats provide equal amounts of energy.

During physical exercise activity, as the body works harder, breathing will quicken and the heart rate will increase. What else will occur to allow more oxygen to be passed through the body to the working muscles?

- **The blood vessels will expand to increase the blood flow**
- The arteries will contract to make the oxygen rich blood flow more quickly
- The blood supply will shut off to the liver and kidneys to allow increased blood flow to the muscles
- The heart will enlarge to pump more blood

How does the human body deal with the heat energy generated during physical exercise?

- **It produces sweat which helps get rid of the heat energy.**
- It causes the body to shiver.
- It increases urine production in the body which, when passed, gets rid of some of the heat.
- It stores it in the body until the individual can take a cold shower.

How many energy systems are normally available to the human body?

- **3**
- 2
- 4
- 5

In addition to speed, stamina, flexibility and co-ordination which one of the given options is the other aspect considered as a measure of physical performance capacity?

- **Power**
- Gender
- Height
- Weight

In terms of physical training what does the Principle of Decreasing Output predict will happen?

- **The better trained that an individual is the more difficult it will be to further improve performance.**
- The greater the effort put into training by an individual the less the output will be.
- The more food an individual consumes the less output energy is available.
- The more training an individual does the less they can achieve.

One of the principles of physical training is the Principle of Reversibility. What does this principle predict will occur following a period of a few weeks of interruption to training?

- **Training should be started again slowly but built up rapidly as the body 'remembers' the necessary adjustments**
- Training can be picked up at the level that was being done prior to the interruption
- Training must be started again from basics and built up only very slowly
- Training needs to be done only once per week until the individual has returned to the fitness level they were at prior to the interruption.

One of the principles of physical training is the Principle of Specificity? Select, from the options given the one which best describes the main theme of this principle.

- **You should mainly train the muscle groups that are required for the activity you intend to take part in.**
- Cycling is the best training activity to do to prepare for a 5 kilometre run.
- You should mainly train the muscle groups that you do not normally use in your regular physical activities to prevent them becoming too weak.
- You should train all of the muscles groups in your body equally.

Select the most accurate statement, in relation to the human body's energy systems, from the options given . . .

- **The phosphate system is used for very short, explosive activities and the oxygen system is used for long, steady exertion.**
- The lactic acid system is used for intense activity and the oxygen system is used for very short, explosive activities.
- The phosphate system is used for long, steady exertion and the oxygen system is used for very short, explosive activities.

Select the option which correctly completes the following statement. By doing sufficient physical activity you can decrease the chance of developing . . .

- **Type 2 diabetes, headache or proteins**
- Cardiovascular disease, the flu or sweaty feet
- Type 1 diabetes

Select the option which correctly completes the following statement. By inhaling, oxygen goes . . .

- **. . . first to your lungs and is then transported to your muscles through the blood.**
- . . . directly into your blood and then to your muscles.
- . . . first to your muscles and then into your blood.

Select the option which correctly completes the following statement. The heart is . . .

- **. . . a muscle that pumps blood through your body.**
- . . . comprised of protein and carbohydrates.

Select the option which correctly completes the following statement. There are three types of fuel in your body . . .

- **. . . proteins, carbohydrates and fats.**
- . . . muscles, tendons and joints.
- . . . the phosphate, lactic acid and oxygen systems.

Select the option which correctly completes the following statement. When you engage in physical activity . . .

- **. . . heat is released which makes you sweat.**
- . . . lactic acid gets into your lungs.

Select the option which correctly completes the following statement. You can globally calculate your maximum heart rate as follows:

- **220 – (your age)**
- 180 – (your age)
- 200 – (your age) + 10

Some forms of physical exercise contribute more to weight loss than others. Which one of the physical activities given in the options is most likely to result in weight loss?

- **Bike riding**
- Horse grooming
- Power lifting
- Sprint running

The Lactic acid system is one of the energy systems of the human body? What is the main fuel source used in this system?

- **Carbohydrates**
- Fats
- Meat
- Protein

The Principle of Overload is one of the five principles of physical training. What does this principle predict will happen if training overload occurs?

- **The muscles will hurt and may need to be rested for a few days.**
- If an individual tries to lift too heavy a load they will risk breaking a bone.
- Individuals will rapidly lose weight due to the additional training load.
- Lactic acid will build up in the muscles.

What does the World Health Organisation physical activity standard recommend as the minimum required frequency and duration for intense exercise to be beneficial to an individual?

- **20 minutes activity 3 days a week**
- 20 minutes activity every day
- 30 minutes activity 5 days a week
- 30 minutes activity every day

Which of the given options is considered as one of the three main energy systems of the human body?

- **Lactic acid system**
- Amino acid system
- Arterial system
- Carbon dioxide system

Which of the given options is one of the five main principles of physical training?

- **Super-compensation**
- Added-compensation
- Extra-compensation
- Over-compensation

Which of the health conditions given in the options is regular physical exercise most likely to reduce the risk of an individual developing?

- **Type 2 diabetes**
- Headaches
- Liver disease
- Stress fractures

0196 – Volatile Organic Compound

Version 1.0 (3560)

As a crude oil tanker cargo tank fills with liquid cargo what will occur in the ullage space of the cargo tank?

- **Saturated and unsaturated vapour mix together and become the Total Vapour Pressure.**
- Saturated vapour from the cargo pushes the inert gas out through the vent system.
- The space above the liquid becomes saturated.
- The vapour above the liquid becomes neutral when mixed with the inert gas.

As part of the VOC control measures certain operational records must be maintained. Select, from the given options, all of those that should be recorded when any VOC is released from a crude oil tanker.

- **Location of vessel at the time of release.**
- **Person in charge of the VOC release.**
- **Target operating pressure for the voyage.**
- **Time and pressure within the system after the releasing the VOC.**
- **Time and pressure within the system before releasing the VOC.**

During loading of a crude oil cargo, the vapour displaced by the cargo is released to the atmosphere via the mast riser. Which of the given options best describes the most likely composition of the released vapour?

- **All compounds in VOC including inert gas.**
- Only Methane, Ethane and inert gas.
- Only NMVOC and inert gas.
- Only the inert gas and Methane.

Record keeping is necessary in order to document compliance with the requirements of VOC emission control. Is this statement true or false?

- **True**
- False

Select from the given options all of those that are considered to be 'best practices' to be adopted to reduce VOC emissions during a loaded voyage on a crude oil tanker.

- **Set the target operating pressure for the cargo tanks as high as safely possible.**
- If crude oil washing reduce the cycle time as much as possible.
- If filling inert gas to prevent under pressure, use as little inert gas as possible.
- When venting to reduce excessive tank pressure, vent as little as possible.

Select from the given options all of those that are considered to be 'best practices' to be adopted to reduce VOC emissions during cargo loading on a crude oil tanker.

- **Communicate with the terminal before loading starts.**
- **Keep the cargo tank pressures as high as safely possible.**
- **Plan the loading procedure to reduce cargo agitation in the oil tanks.**
- **Reduce the loading rate.**
- **Use the vapour returns lines when shore facilities are available.**
- Always leave the mast riser valve open.

Select, from the given options, all of those that are considered to be 'best practices to be adopted to reduce VOC emission during cargo discharge and COW operations on a crude oil tanker.

- **All COW operations should comply with the requirements and procedures set out in the class approved COW manual.**
- **As far as possible a closed cycle method should be used for COW operations.**
- **Continuously monitor the pressure of the cargo tanks during discharge to avoid excessive supply of inert gas.**
- **Excessively gassy cargoes should not be used for crude oil washing.**
- **Use a defined volume of crude oil for the COW from a specified cargo tank instead of fresh crude oil.**
- The duration of any COW operations should be extended as much as possible.

Select, from the given options, the one which most accurately completes the following statement. The temperature of the gas phase in a crude oil cargo tank will change more rapidly . . .

- **. . . during day and night cycle.**
- . . . during discharging of the cargo tank.
- . . . during loading of the cargo tank
- . . . during transportation.

Select, from the options given, all of those which are criteria that influence the VOC release from the cargo of a crude oil tanker.

- **The pressure setting or control of the vapour phase within the cargo tank.**
- **The size or volume of the vapour phase within the cargo tank.**
- **The temperature of the liquid and gas phases of the crude oil tank.**
- **The volatility or vapour pressure of the crude oil.**
- The atmospheric pressure at the location where VOC is produced.
- The place of production of the VOC.

The Gothenburg Protocol has set emission limits for which of the pollutants?

- **Sulphur, NOx, VOCs, and ammonia.**
- Carbondioxide and carbon monoxide released from ships.
- Inert gas.
- Nitrogen and carbon dioxide released from ships.

The temperature of the liquid phase in a crude oil cargo tank does not vary much except:

- **When heating or cooling of the cargo is being carried out.**
- When proceeding from a cold region to a warm region.
- When proceeding from a warmer region to a colder region.
- When the ship is stationary in warm climate.

Vapours that are released directly into the air during cargo operations onboard crude oil tankers are considered hazardous. Which of the given options is the most likely to be adversely affected by the release of these vapours?

- **Coastal Plants and land animals**
- **Human health and the atmosphere**
- **Marine environment**
- The ships paintwork

What is the main purpose of P/V breaker fitted to the cargo system of a crude oil tanker?

- **It provides secondary protection to the tank structure from over and under pressure.**
- It provides a means of venting the gases and vapours from the tank as cargo is loaded.
- It provides oxygen content readings of the cargo tanks.
- It provides primary protection to the tank structure from over pressure.

What is the main purpose of the P/V valve fitted to a cargo tank on a crude oil tanker?

- **It provides primary protection to the tank structure from over and under pressure.**
- It provides a means of venting the gases and vapours from the tank as cargo is loaded.
- It provides an indication of the pressure in the cargo tanks.
- It provides secondary protection to the tank structure from over pressure.

What is the main purpose of the VOC management plan?

- **To ensure that the operation of a tanker, to which regulation 15.6 of the revised Annex VI applies, prevents or minimizes VOC emission to the extent possible.**
- It is a requirement of port State control
- It is a requirement of the Kyoto protocol
- It totally prevents VOC emission into the atmosphere.

What is the main purpose of the vapour emission control pipe line fitted onboard a crude oil tanker?

- **Direct the cargo vapours from cargo tank to the engine room.**
- **Direct the cargo vapours from cargo tank to vapour absorption system.**
- **Direct the vapours from the cargo tanks to the shore system.**
- Direct the cargo vapours from cargo tank to burn in the boilers.

What is the major source of volatile organic compound from ships?

- **The majority of VOC comes from cargo tanks of oil tankers.**
- The majority of VOC comes from incomplete combustion in ships' engines.
- The majority of VOC is released from ships' air conditioning compressors.
- The majority of VOC is released from ships' boilers.

What is the procedure for the maintenance of the P/V valves fitted to the cargo system of crude oil tankers?

- **The maintenance should be carried out in accordance with the directions in the vessel's planned maintenance system and the maker's manual.**
- The maintenance should be carried out during cargo discharge.
- The maintenance should be carried out during each loaded passage.
- The maintenance should not be carried out until the valves start leaking.

What is the purpose of VOC emission control from ships?

- **To prevent or minimize the release of VOC as much as possible during loading, discharging and transit.**
- Only to prevent or minimize release of VOC as much as possible during cargo transfer.
- Only to prevent or minimize release of VOC as much as possible during production.
- Only to prevent or minimize the release of VOC as much as possible during loading at offshore terminals.

What is the purpose of the vapour recovery systems installed on-board crude oil tankers?

- **To control the VOC emissions to the atmosphere, when the plant is in operation.**
- To extract the hydrocarbon gases from the cargo to be used to run the ship's engines.
- To liquefy the hydrocarbon gases emitted from the cargo and then use the liquid to run the ships compressors.
- To maintain the cargo specifications of the loaded cargo.

What is the requirement, if any, with regard to the experience of the person intended to act as a designated person in relation to VOC control?

- **They should have at least one year experience on crude oil tankers where their duties included all cargo handling operations relevant to VOC management and COW operations.**
- There is no such requirement for the designated person to have any experience of crude oil tankers.
- They should have at least 6 months experience on oil tankers with some involvement in COW operations.
- They should have at least 9 months experience on oil tankers with some involvement in COW operations.

What target pressure, according to this training module, should be maintained in the ullage spaces of tanks during loading to minimise VOC release to the atmosphere?

- **As high as safely possible, at about 70% of the pressure release setting of the P/V valves.**
- 100 % of the P/V valve setting pressure.
- 90 % of the P/V valve setting pressure.
- As low as possible and no more than 50% of the pressure release setting of the P/V valves.

What was the main goal of the Kyoto Protocol?

- **To lower overall emissions from six greenhouse gases - carbon dioxide, methane, nitrous oxide, sulphur hexafluoride, HFCs, and PFCs.**
- Arrange to keep a record of each nations green house gas emissions
- To arrange developed nations to pay underdeveloped nations to control emissions
- To arrange the developed nations to pay for the maintenance of climate change.

When considering crude oil tanker operations what is meant by the term volatile organic compounds?

- **They are organic chemicals that easily vaporize at normal conditions and enter in to the atmosphere.**
- They are organic chemicals that vaporize and enter in to the atmosphere only after a chemical reaction.
- They are organic chemicals that vaporize and enter in to the atmosphere only when agitated.
- They are organic chemicals that vaporize and enter in to the atmosphere only when heated.

Which of the given options best describes the main factors that will influence the effect on health of human of exposure to VOC emissions onboard a crude oil tanker?

- **The concentration of the VOCs and the level and length of time of exposure.**
- Only the concentration of NMVOCs in the VOCs.
- Only the level and length of time of exposure.
- Only the weight of the person and the length of time of exposure.

Which types of ship are required to carry and follow an approved VOC management plan onboard?

- **Every oil tanker carrying crude oil regardless of whether existing or new.**
- Every Cargo ship 500 GT or more
- Every Chemical tanker regardless of existing or new.
- Every new oil tanker built after 10 October 2010.

Why are the emission control requirements for crude oil tankers limited to non- methane volatile organic compounds, NMVOCs, and do not include methane?

- **As it is not deemed cost-efficient to recover methane.**
- As methane constitutes such a small part of the VOC.
- As methane will be removed during the production process.
- As methane will be transported by dedicated ships.

0197 – MLC 2006, Basic Introduction

Version 3.0 (1589)

According to the MLC 2006 what will be the minimum age for seafarers to undertake normal night working on-board ship?

- **18**
- 16
- 17
- 21

Before entry into force of the MLC 2006 at least 30 Member States must have ratified the convention. Which one of the given options is the minimum share of world gross tonnage that is also required before entry into force can happen?

- **at least 33%**
- at least 15%
- at least 40%
- at least 50%

During an MLC 2006 inspection seafarers may be interviewed by either a flag State or port State control inspector. Which one of the given statements about these interviews is true?

- **The interview must be treated as private and confidential.**
- Only seafarers selected by the Master may be interviewed.
- The interview must be recorded so it can be replayed at a later date if required.
- The Master must be present throughout such an interview.

For how long is a Maritime Labour Certificate, issued for compliance with the MLC 2006, normally valid?

- **5 years**
- 1 year
- 2 years
- 3 years

If a shipowner fails to repatriate a seafarer, or pay the cost of the repatriation, following satisfactory completion of a contract, who, according to the MLC 2006, is responsible for making the arrangements and covering the cost for the repatriation?

- **Competent authority of the Flag State**
- One of the seafarers' welfare organisations
- The seafarer
- The ship's Master

In order for a Maritime Labour Certificate, issued for compliance with the MLC 2006, to remain valid it is necessary that an intermediate inspection is carried out on behalf of the Competent Authority. When should this inspection be carried out?

- **Between the 2nd and 3rd anniversary of issue of the certificate.**
- 6 months after issue of the certificate.
- After the 3rd anniversary of issue of the certificate.
- Between the 1st and 2nd anniversary of issue of the certificate.

Seafarer recruitment and placement services operating under the jurisdiction of a ratifying Member State should be approved and licensed by that State. Which one of the statements given in the options is the most accurate in relation to these services?

- **Once licensed the service should be monitored and inspected regularly to ensure compliance with the requirements of the convention.**
- Once licensed the service should be self regulated.
- Once licensed the service should be spot checked to ensure compliance with the requirements of the Convention.
- Once licensed the service should only be inspected regularly in the event of an alleged or suspected non-compliance.

Select the option which agrees with how the MLC 2006 defines a seafarer onboard a ship to which the Convention applies.

- **Any person doing any work on the ship.**
- Only persons holding an STCW qualification.
- Only persons signed on ships articles.
- Only persons who hold seamen's identity documents.

The ILO is split into a number of geographical regions. How many of these regions are there?

- **5**
- 10
- 12
- 2

The MLC 2006 contains a number of Articles. Which one of the given options best describes the content of the Articles?

- **A description of how the MLC 2006 is meant to work and the basic rights of seafarers.**
- A specific list of requirements to be met.
- Part A and Part B of the MLC 2006 Code.
- The specific national laws and regulations that must be introduced by member states.

The MLC 2006 contains a requirement that a copy of the Convention must be carried onboard all ships flying the flag of Member States. Which one of the options best describes who should have access to this document onboard the ship?

- **All seafarers.**
- All of the ship's Officers
- Only the ship's Master and Chief Officer.
- Only the ship's Master.

The MLC 2006 requires that either seafarers' hours of work or hours of rest are regulated. When hours of rest are regulated what should the minimum number of hours of rest be in a 24 hour period?

- **10**
- 12
- 14
- 8

The MLC 2006 requires that either seafarers' hours of work or hours of rest are regulated. When hours of rest are regulated what should the minimum number of hours of rest be in a seven day period?

- **77**
- 40
- 65
- 84

The MLC 2006 requires that either seafarers' hours of work or hours of rest are regulated. When hours of work are regulated what should the maximum number of hours of work be in a seven day period?

- **72**
- 40
- 48
- 60

The onboard complaints procedures required under the MLC 2006 allows seafarers to make complaints relating to compliance with the convention. Which one of the options best identifies the level at which a seafarer can make their complaint to under these procedures?

- **At whichever level is required to resolve the issue.**
- Only to the onshore authorities.
- Only to the ship's Master.
- Only to their direct line manager.

There are four types of vessel that are exempt from the requirements of the MLC 2006. From the options given select the one which is one of the four exempt types.

- **Vessels engaged in fishing**
- Commercial yachts
- Offshore support vessels
- Oil tankers

What is seen as the main advantage to a shipowner that should result from the introduction of the MLC 2006?

- **It should give them a level playing field in which to operate.**
- It should allow them to build cheaper ships.
- It should allow them to reduce crew numbers onboard their ships.
- It should reduce the number of inspections onboard their ships.

What is the approximate number of existing ILO maritime labour standards (conventions and recommendations) that the MLC 2006 will replace?

- **70**
- 10
- 30
- 50

What is the main responsibility of a Master on a ship flying the flag of a Member State that has ratified the MLC 2006?

- **Ensure ongoing compliance with the Convention.**
- Carry out checks on the recruitment and placement service supplying seafarers to the ship.
- Ensure that the DMLC part 2 is completed.
- Hear all complaints raised by individual seafarers.

What is the maximum period of validity for an Interim Maritime Labour Certificate issued under the MLC 2006?

- **6 months**
- 12 months
- 3 months
- 36 months

What is the minimum age specified for a person to work as a cook onboard ship according to the MLC 2006?

- **18 years of age**
- 16 years of age
- 21 years of age
- 25 years of age

What is the minimum leave rate that seafarers should be entitled to under the MLC 2006?

- **2 ½ days per month**
- 1 ½ days per month
- 5 days per month
- 6 days per month

What is the requirement under the MLC 2006 regarding the person issuing a seafarer's medical certificate?

- **They must be approved and properly qualified.**
- They must be the seafarer's own doctor.
- They must be the shipping company's own doctor.
- They must be the ship's doctor.

When Port State Control Officers carry out ship inspections according to the requirements of the MLC 2006 they must do so with the regard to the 'no more favourable treatment' clause. Which one of the options identifies the ships that this clause is meant to prevent from having more favourable treatment?

- **Ships flying the flag of a non-ratifying State.**
- Foreign flag ships flying the flag of a ratifying Member State.
- Ships built or delivered before the entry into force date of the Convention.
- Ships flying the flag of the Member State visiting a home port.

Which body has the final responsibility for implementation and enforcement of the MLC 2006 on ships flying the flag of a ratifying Member State?

- **The Member State**
- Port State Control
- The ILO
- The IMO

Which one of the given options do the letters ILO stand for in connection with the MLC, 2006?

- **International Labour Organisation**
- International Labour Office
- International Law Organisation
- International Leaders Organisation

Which one of the given options of sizes of ship does the Maritime Labour Convention, 2006 apply to?

- **All sizes of ship**
- Only ships above 10,000 gross tonnage
- Only ships above 200 gross tonnage
- Only ships above 500 gross tonnage

Which one of the given statements is true according to the MLC 2006 regulation dealing with medical care of seafarers?

- **Free medical care must be provided onboard and in port if required.**
- Medical care only has to be provided in emergency situations.
- Medical care only has to be provided onboard ship.
- Medical care only has to be provided when the seafarer agrees to pay for it.

With respect to the MLC 2006 what does the abbreviation DMLC stand for?

- **Declaration of Maritime Labour Compliance.**
- Detailed Maritime Labour Certificate.
- Dispensation from Maritime Labour Certificate.
- Document of the Maritime Labour Convention.

0211 – ECDIS system and chart types

Version 1.0 (3134)

Approximately how many countries are producing electronic charts?

- **Between 40 and 50**
- Between 10 and 40
- Between 50 and 100
- More than 100

Can an ECS chart plotter be approved as an ECDIS?

- **No, as it has unofficial charts on it**
- Yes, an ECS is always approved as an ECDIS
- Yes, but only when connected to a navigation signal
- Yes, but only when in raster mode

How can a quick check on the quality of the navigation signal to ECDIS be obtained?

- **By viewing the radar overlay**
- By asking the US Coastguard how the system is operating
- By checking the LAN to ascertain if the signal is still running into the ECDIS
- By setting up software to observe the navigation signal

How many files of chart information does a raster chart have?

- **One**
- Five
- Ten
- Two

IMO made an important decision for the production of electronic charts. Which of the given options best describes their decision?

- **They asked countries to produce their own charts for their territorial waters**
- They allowed countries to go anywhere in the world to make charts without seeking permission
- They asked the American Hydrographer to complete the task for the world
- They decided to have little influence in the activity

Official electronic charts have various names; select the correct list from the options given.

- **Vector, S57, official ENC**
- Digital, raster, C-Map, ENC
- ENC, Transas, vector, BSB
- HCRF, SENC, raster, official

Select the option which correctly completes the following statement.
'An ECDIS compliant electronic chart is best described as . . .

- **. . . one constructed under the S57 standard.'**
- . . . a digital chart constructed under S52 standard.'
- . . . a raster chart made under S61 standard.'
- . . . any electronic chart.'

Select the option which correctly completes the following statement.
'An ECDIS position filter protects the ECDIS from . . .

- **. . . showing a fluctuating position of the vessel.'**
- . . . airborne dust in sandy areas of the world.'
- . . . incorrectly set up system software on the ECDIS.'
- . . . operating under conditions that are extreme e.g. cyclones.'

Select the option which correctly completes the following statement.
'An ECDIS will alarm if . . .

- **. . . the chart datum and navigation signal datum mismatch.'**
- . . . the chart datum is not WGS 84.'
- . . . the navigation signal datum is not WGS 84.'
- . . . there is a discrepancy between WGS 84 and ETRS 89 datum.'

Select the option which correctly completes the following statement.
'Digitally produced charts are suited to computer use because . . .

- **. . . they are produced as a series of files that give flexibility.'**
- . . . the operator can change the file to suit their needs.'
- . . . they are easily made by a cartographer.'
- . . . they can never be changed by the manufacturer.'

Select the option which correctly completes the following statement.
'The SCAMIN attribute will influence the chart display because . . .

- **. . . objects will be removed as the operator zooms.'**
- . . . the chart cannot be used in the official mode with SCAMIN.'
- . . . the colours of the display will be wrong when the SCAMIN is set wrongly.'
- . . . the SCAMIN affects the saved files.'

Select the option which correctly completes the following statement.
'The presentation library on the ECDIS is important because . . .

- **. . . if it is not the latest version the ECDIS will not show all detail**
- . . . it allows the ECDIS to decide on the scale setting of the chart
- . . . it influences the colours on the display
- . . . it is where the files for maintenance of the system software are stored

Select the option which correctly completes the following statement.
'When loading updates on the ECDIS it is important to delete old files because . . .

- **. . . the ECDIS may display old versions not the update one.'**
- . . . it stops the ECDIS computer from slow running.'
- . . . the auto delete function on the ECDIS may not work properly.'
- . . . the old files can be confusing to the operator when selecting charts.'

Select the option which correctly completes the following statement.
'When using ECDIS, out of position navigation buoys will be noticed because . . .

- **. . . the radar return and buoy symbol do not coincide**
- . . . a local port authority would send an automatic signal to the ECDIS
- . . . there will be a flashing alarm on the ECDIS showing which one is out
- . . . they are coloured orange on the ECDIS display

Select the option which correctly completes the following statement.
'When the ECDIS is in DR mode . . .

- **. . . the course and speed through the water are used for navigation.'**
- . . . it is in normal operational mode.'
- . . . it stops working not showing the position on the screen.'
- . . . the bridge team can rely on this trusted method of navigation.'

Select the option which describes the correct scale to view chart detail on a vector chart.

- **At the same scale as the compilation scale**
- Any scale, as the scale setting makes no difference to the detail seen
- At the scale the operator decides to select
- Using the scale window to see what the objects are

Select the option which is normally one of the main feeds into the ECDIS system

- **The navigation signal**
- GMDSS communication system from the radio equipment
- Ship's stability information
- The ship's internal telephone system via the LAN

Select the option which most accurately describes the following statement.
'When raster charts are made, new data is always used to make them.'

- **The statement is totally incorrect**
- The statement is correct for the UKHO only
- The statement is partially correct for all manufacturers
- The statement is totally correct

Under the ISM code what are ECDIS Operators required to have?

- **Proper equipment and generic training for ECDIS**
- Computer skills that can be adapted to other equipment
- Equipment specific training only
- Some awareness of the basic functionality of ECDIS

What does the acronym HDOP stand for?

- **Horizontal dilution of precision**
- High datum of precision
- High dilution of precision
- Horizontal depth of precision

What does the acronym RNC stand for in relation to ECDIS?

- **Raster Navigational Chart**
- Raster Nautical chart
- Regional Navigation Committee
- Retailer of Navigational Charts

What does the acronym SENC stand for in relation to ECDIS?

- **System Electronic Navigational Chart**
- Software Electronic Navigational Chart
- Survey Electronic Navigational Chart
- Swedish Electronic Navigational Chart

What is WEND in relation to electronic navigation?

- **A world system of supply and updating of charts**
- An IMO working group looking at the manufacture of charts
- Part of the ECDIS software which changes object attributes
- The organisation where charts are purchased

What is the height datum for water depth on an ENC chart usually taken as?

- **Lowest astronomical tide**
- Low water Newlyn
- Mean high water springs
- Mean higher high water

What is the main advantage to the Mariner of a vector chart?

- **The stored data base has intelligence**
- It is cheap to purchase as it is only a computer file
- Simple to use with an easily understood structure on the ECDIS
- The chart can be copied and passed to other vessels on a disk

What is the main disadvantage to the Mariner of a raster chart?

- **They have little flexibility in their use**
- They are very expensive to buy costing the company lots of money
- They cannot be used on an ECDIS display
- They have a series of files so are very flexible to use

What should a chart, with usage code 1, be used for?

- **Over viewing an ocean when planning**
- A coastal passage
- Berthing and un-berthing
- General navigation

What should a chart, with usage code 6, be used for?

- **Berthing and un-berthing**
- A coastal passage
- Crossing an ocean
- General navigation

What would a large magenta question mark on the ECDIS display indicate?

- **That the presentation library is out of date**
- That there is a detected problem with the ECDIS software
- That there is a wrong action by the operator
- That there is no chart loaded in the ECDIS

What would a small magenta question mark on the ECDIS display indicate?

- **That the ECDIS is uncertain as to some object's attribute**
- That there is a wrong action by the operator
- That there is a detected problem with the ECDIS software
- That there is no chart loaded in the ECDIS

What would be contained in the 'Exchange set' of files in relation to ECDIS?

- **The vector chart file and other information files**
- Data that will drive the chart display colours
- Files showing the standard protocol for ECDIS
- Information that the ECDIS needs to assist the integration of the navigation signal

When can a raster chart be used on the ECDIS display?

- **When there is no vector chart available**
- At any time as they are more recognizable than vector charts
- Never because they are not official charts
- Only in harbour areas as they show more detail

When does the ENC data set become an SENC?

- **When the ECDIS converts it**
- When the chart displays on the ECDIS
- When the hydrographic office finish making the data set
- When the value added retailer obtains the data set

Which of the given options best describes a data set in relation to ECDIS and electronic charts?

- **It is the compiled information making up the electronic chart**
- It is a set of data that allows the ECDIS to communicate with the bridge equipment
- It is the electronic raster chart delivered from the chart agent
- It is the information necessary to make the ECDIS boot up

Which of the given options best describes how ECDIS fits in with the chart carriage requirements as identified in SOLAS?

- **An approved ECDIS is acceptable under the chart carriage requirements**
- ECDIS is not allowed
- Electronic chart systems (ECS) comply with SOLAS and can be used as an ECDIS
- Paper charts have to be always carried with the ECDIS

Which of the given options best describes the effects on an ECDIS if the navigation signal has errors on it?

- **The indicated position will be wrong**
- It will go into DR mode
- There will be an alarm
- There will be no effect

Which of the given options best describes the function of RENC?

- **A centre for collating and distributing charts**
- A centre where chart anomalies can be reported
- A place that is tasked to develop standard ECDIS software
- A world wide body for chart manufacture

Which of the given options best describes the function of data protection on the chart?

- **Anti piracy and controls authenticity**
- Allows the chart files to be passed through customs without issues arising
- Makes sure the updates are correct
- The bridge team hacking into the chart for their laptops

Which of the given options best describes the validity of the following statement?
'After the first of July 2012, new passenger ships will have to be fitted with ECDIS.'

- **Correct, but only if they are above 500 GRT**
- Correct, but only if the vessel crosses oceans
- Incorrect, it will be left to each Flag state to decide
- Incorrect, it will depend on the shipping company

Which of the given options best describes the validity of the following statement?
'An approved ECDIS can replace paper charts on a vessel.'

- **Only with approval from the flag state**
- Only when it used as a 'dual fuel' system
- Only when official charts are loaded
- Only when official charts are loaded

Which of the given options best describes the validity of the following statement?
'In the future existing ships will have to be fitted with ECDIS.'

- **Correct, in accordance with a rolling time program depending on ship type**
- Correct, but only if they are going to be kept in service for many more years
- Incorrect, it will be left to each Flag state to decide
- Incorrect, it will depend on the shipping company

Which of the given options best describes the validity of the following statement?
'Ships approved for ECDIS are robust enough to sail the world with one system.'

- **Incorrect, there should be a backup system fitted**
- Correct, but it should be used only when crossing oceans
- Correct, but only if the system is on the bridge of the vessel
- Correct, when there are vector charts on the ECDIS

Which of the given options best describes the validity of the following statement?
'The 1996 amendment to the ECDIS standards allowed the use of raster charts on an ECDIS.'

- **Correct, but only when an official chart is not available**
- Correct, but only when an official chart has not been obtained for use
- Correct, but only when the ECDIS is older than 1996
- Incorrect, raster charts can never be used

Which of the given options describes the normal types of input to an ECDIS?

- **Any useful input from bridge and engine room equipment**
- Only an input from a Satellite navigator
- Only an input from a wind speed and direction indicator
- Only an input from engine room equipment

Which of the given options is correct in relation to the following statement?
'When using ECDIS, a general chart, with usage code 2, can be zoomed in on for harbour viewing.'

- **Correct, it can be but should not be**
- Correct and it should be done as it saves money on charts
- Correct but the chart could only be zoomed in on once
- Incorrect as the chart cannot be zoomed in on

Which of the given options is the meaning of the letters GB and the number 5 in the chart cell identifier GB50344?

- **A chart made by the UKHO for harbour usage.**
- A British chart, number 5 in the series.
- The chart is from the UKHO, the 5 means it is an updated version
- The GB means from Great Britain, the 5 means it is for ocean use

Which of the given options is the most accurate in relation to the following statement?
'Vector charts are no more accurate than paper charts.'

- **True since vector charts are usually digitised versions of the paper chart**
- This is untrue as vector charts are always more accurate
- True since the digitising tool is very complicated to use and is prone to large errors
- True since the file in the vector chart stores only rough approximate values

Which of the given options is the typical scale of a harbour chart?

- **1:4000**
- 1:1,000.000
- 1:24,000
- 1:30,000,000

Which of the given options would be the most appropriate action to take if the datum of the raster chart is not WGS84 and GPS is being used?

- **Apply the stated offsets on the chart**
- Change the datum in the GPS receiver
- Do nothing as the GPS will give the correct charted position
- Tell the Master

Which of the options is an important task given for RENCs?

- **They are responsible for ensuring the chart data is protected**
- They have to colour code the products for distribution
- They have to oversee and be in charge of the chart makers
- They set a price on the product for onward despatch to AVRs

Which of the requirements given in the options is necessary when loading charts on a vessel's ECDIS?

- **A license or permit to allow the charts to install.**
- To always undertake the operation in port in case the computer crashes.
- To have the Master undertake the operation.
- To have verbal permission from the chart supplier.

Which of the standards given in the options must a vector chart be manufactured under in order for it to be considered an official ENC?

- **S57**
- S100
- S52
- S61

Which one of the authorities, given in the options, sets the hydrographic standards for electronic charts?

- **IHO**
- IMO
- NOAA
- UKHO

Which one of the authorities, given in the options, sets the performance standards for ECDIS?

- **IMO**
- IHO
- NOAA
- UKHO

Which one of the given options can be considered as summarising the disadvantages of an ECDIS to a navigator?

- **The navigator can be misled by the computer display believing it to be correct**
- It is complicated and older persons tend to be less able to use computers
- More work is involved looking after it than for paper charts
- The navigator will become board as the ECDIS will do all of the navigation automatically

Which one of the given options can be considered as the main advantage to the bridge team of having an ECDIS?

- **Various types of information can be displayed to the bridge team**
- It can use raster charts that are recognisable
- It saves time for the 2nd Officer
- The team can sit back and totally trust the ECDIS

Which one of the options is considered as the main disadvantage to the Mariner of a vector chart?

- **The electronic display can make the Mariner over reliant**
- It has no flexibility compared to other chart types
- It is very costly to update the charts and time consuming
- The Mariner gets lost in the menu system when trying to install charts

Which type of electronic chart is considered to have intelligence when used on ECDIS?

- **A digital vector chart**
- A digital raster chart
- Non official charts
- None as only humans have intelligence

Why is it important that an approved method is used for the conversion of ENC to SENC?

- **Because the conversion process allows the chart electronic files to be converted properly.**
- Because the process is always completed by the retailer and they know how to do it.
- Because the raster chart would not be seen if you did not
- It is not important as the ECDIS is intelligent

Why is it important to have the same datum for chart and navigation signal when using ECDIS?

- **Because navigation positions can be plotted directly on the chart**
- Because the attributes on the symbols will not be understood by the ECDIS
- Because the operator will not be able to change the plotted positions
- It is not important as ECDIS can cope with any datum

Why is it that some ECDIS devices show symbols that look different for the same object?

- **The ECDIS manufacturer interpreted the S57 standard differently**
- The differences come from the raster chart data set
- The speed of the ECDIS computer varies so the shapes can be different
- They should all display the same shape and looking symbols

Why will a VAR sometimes convert an ENC file to a SENC file?

- **To save time on the vessel**
- It is a necessary part of the data protection process
- The ENC cannot be loaded directly onto the ECDIS on board the vessel
- The VAR does this to check the contents of the file

0212 – ECDIS display features

Version 1.0 (2038)

Select the correct meaning of the abbreviation ZOC in relation to ECDIS:

- **Zone of Confidence**
- Zero on Carbon
- Zone of Communications
- Zone of Confusion

Select the option that correctly completes the following statement in relation to ECDIS. 'A precautionary area is shown on the ENC as . . .

- **. . . a magenta coloured i in a box inside a diagonal hatched area.'**
- . . . a magenta coloured i in a box.'
- . . . a magenta filled box with red borders.'
- . . . words saying 'Precautionary Area – take care' in the area.'

Select the option which best describes a cautionary area on the ECDIS?

- **Orange cross hatching on the chart display**
- Magenta cross hatching on the chart display
- Red cross hatching on the chart display
- Vertical grey bars on the chart display

Select the option which best describes how ECDIS informs the operator of an approaching chart at different scale to the one being used.

- **There will be a thick continuous black line visible**
- A visual alarm will activate
- An audible alarm will sound
- The chart detail will not be visible

Select the option which best describes how Temporary and Preliminary notices are shown on the vector chart in relation to ECDIS:

- **The ECDIS may not display them – a navigator's note could be set up to show them**
- Ts & Ps are always automatically loaded and visible alongside the area of concern being
- Ts & Ps are always shown on the chart in blue text being loaded manually
- Ts & Ps are shown in a magenta colour near the area of the notice

Select the option which best describes the meaning of a magenta coloured 'i' in a box seen on a charted sea area in relation to ECDIS.

- **There is some information appertaining to the sea area**
- It is an indication that a chart correction has been applied
- The ECDIS is trying to say something to the operator about the route
- There is an object missing from the chart

Select the option which best describes the meaning of a magenta coloured 'i' in a box with a line to an object in relation to ECDIS.

- **There is more information available on the object than is shown on the display.**
- It is a recently added object to the chart data set.
- The object has been moved with a chart correction.
- The position of the object is approximate.

Select the option which best describes the meaning of the text 'bn 10' on an ECDIS display.

- **Navigation beacon number 10**
- Attribute set at level 10
- Depth figure of 10 metres
- VHF channel 10 to be used

Select the option which best describes the meaning of the text 'by 12' on an ECDIS display.

- **Navigation buoy number 12**
- Attribute set at level 12
- Depth figure of 12 metres
- VHF channel 12 to be used

Select the option which best describes why vertical grey lines can sometimes be seen all over the ECDIS display?

- **ECDIS is set at the wrong scale for the vector chart loaded**
- The chart scale is the same as the compilation scale
- The lines show where shallow water begins
- There is not enough detail to sow soundings correctly

Select the option which correctly completes the following statement in relation to ECDIS.
'A ZOC A1 indicates the depth accuracy on the vector chart to be . . .

- **. . . 0.5m + 1% of depth value.'**
- . . . 1m + 2% depth.'
- . . . 2m + 5% depth figure.'
- . . . 5m + 10% depth.'

Select the option which correctly completes the following statement in relation to ECDIS.
'A navigation buoy cancelled with a chart update will . . .

- **. . . have an orange diagonal line through the symbol.'**
- . . . be removed from the chart display altogether.'
- . . . have a magenta box around it.'
- . . . have an orange box around it.'

Select the option which correctly completes the following statement in relation to ECDIS.
'An isolated danger buoy is represented on an ENC chart with . . .

- **. . . two red spheres slightly inclined.'**
- . . . a single black sphere.'
- . . . a single red sphere.'
- . . . two upright red spheres.'

Select the option which correctly completes the following statement in relation to ECDIS.
'Isolated danger symbols are not seen on a vector chart when they should be because . . .

- **. . . the safety depth contour is set with too small a value.'**
- . . . the draught of the vessel is not great enough to hit it.'
- . . . the object is not a threat to the vessel.'
- . . . there will be some other symbol there that gives the same visual information.'

Select the option which correctly completes the following statement in relation to ECDIS.
'Updates to official ENCs cover . . .

- **. . . all relevant chart corrections.'**
- . . . all Temporary and Preliminary notices.'
- . . . changes to navigation buoys only.'
- . . . changes to the chart in navigable areas.'

Select the option which correctly completes the following statement in relation to ECDIS. 'A grey question mark alongside an object indicates that . . .

- **. . . the position of the object is doubtful.'**
- . . . the ENC data set does not know what the object is.'
- . . . there is an attribute missing from the object.'
- . . . there is more information available on the object.'

Select the option which correctly completes the following statement in relation to ECDIS. 'An isolated danger beacon is represented on an ENC chart with . . .

- **. . . two upright red spheres**
- . . . a single black sphere
- . . . a single red sphere
- . . . two red spheres slightly inclined

Select the option which correctly completes the following statement.
'ECDIS alarms . . .

- **. . . should never be ignored but must be investigated.'**
- . . . are a nuisance and only happen when a sensor is failing to the ECDIS.'
- . . . are the responsibility of the Master alone.'
- . . . only need to be acknowledged.'

Select the option which correctly completes the following statement.
'ECDIS display various colours and shades on vector chart depth areas because . . .

- **. . . they are easily seen by the operator.'**
- . . . it lets ECDIS know they are there.'
- . . . it stops the ship grounding on them.'
- . . . they set off alarms on the bridge.'

Select the statement that best describes why a button on ECDIS shows with a red highlight.

- **The ECDIS is in an alarm state**
- It is a warning to the operator that they are using a raster chart instead of a vector chart
- It is an indication that the vector chart has loaded successfully
- It shows an incoming email

Select the statement which best answer this question. 'Do some ECDIS devices show tidal stream values as an information overlay?'

- **'Yes – if they have dedicated software loaded on the ECDIS computer.'**
- 'No – this feature can be seen on standalone bridge computers only.'
- 'No – this feature is not possible with ECDIS.'
- 'Yes – this is a standard feature embedded in all ECDIS software.'

Select the statement which best answers this question. 'Do ECDIS depth soundings show height of tide on them?'

- **'Most do not but in the future most will.'**
- 'Always as the ECDIS can calculate tide heights automatically.'
- 'No - because the tide changes.'
- 'Yes -they show chart datum with the maximum tide rise added on.'

Select the statement which best describes a buoy symbol which is NOT properly defined in the ENC data set.

- **A spherical buoy grey in colour**
- Always green for a port lateral mark, red for a starboard lateral mark
- Always red for a port lateral mark, green for a starboard lateral mark
- ECDIS will always define the correct characteristic of the buoy

Select the statement which best describes how an operator establishes the nature of an isolated danger on an ENC.

- **Interrogate it and ask for a pick report**
- Check the chart symbols catalogue to ascertain the meaning
- Check the surrounding area for any clues on the sea floor
- Check with a senior officer

Select the statement which best describes the anomalies that are sometimes seen on ECDIS.

- **They are features of the ECDIS that sometimes seem illogical**
- They are caused by a mismatch in the navigation signal datum
- They are ECDIS features dictated to manufacturers by Port State
- They are errors caused by the vector chart having bad information

Select the statement which best describes the use of docking mode on ECDIS.

- **'ECDIS used with the vessel close to berths.'**
- 'The ECDIS switched to stand by awaiting updates.'
- 'ECDIS used only when the vessel enters a dry dock.'
- 'The ECDIS set to stand by with the vessel alongside.'

Select the statement which best describes the use of the text 'bn' on an ENC display.

- **Stops confusing beacon numbers and depth sounding figures**
- Helps the operator understand sounding figures
- They are adjectives and are used to describe land objects
- 'bn' is a code for the object in the data set.

Select the statement which best describes the use of the text 'by' on an ENC display.

- **'Stops confusing buoy numbers and depth sounding figures.'**
- 'by' is a code for the object in the data set.'
- 'by' means close to an object.'
- 'They are adjectives and are used to describe sounding figures.'

Select the statement which best describes why a Port State Control Officer is interested in ECDIS when they undertake vessel inspections.

- **Because ECDIS has to comply with IMO requirements**
- There is a section on their check sheet that has to be ticked
- They are required to register the ECDIS Type for the IMO
- They enjoy playing with the new features

What happens on the ECDIS display when the selected safety contour value does NOT exist in the chart data set?

- **The next highest value will be shown highlighted as a bold line**
- The depth areas will be coloured dark blue
- The ECDIS will not display a contour value at all
- The next lowest will be highlighted as a bold line

Which of the following statements is the most accurate in relation to the safety depth contour setting on an ECDIS?

- **'The safety depth contour is best set at a value to reflect the vessel's draught.'**
- 'The safety depth contour value has no effect on ECDIS operation.'
- 'The safety depth contour value is best set at a high value.'
- 'The safety depth contour value is best set at a low value.'

Which of the given features is indicated by a grey vertical rectangle on an ENC in relation to ECDIS?

- **An undefined beacon**
- A house on land
- A symbol associated with oilfields
- The absence of chart information

Which of the given options best describes an ECDIS being in 'browse' mode?

- **A mode allowing the operator to look ahead on the chart**
- A mode allowing the operator, not sure of their position, to fix using radar
- A mode for casual operation of the system
- A mode putting ECDIS into a stand by condition to investigate files

Which of the statements best describes how the raster chart gives advice on data quality in relation to ECDIS?

- **There are notes on the chart that can be read.**
- Data quality comes from text that describes the sea floor type.
- Data quality comes from the help menu on the stand alone computer.
- The chart has a ZOC function where the data quality can be assessed.

Which of the statements best describes what happens to the sounding figures on the display when the safety depth figure is set on an ECDIS?

- **'Depth values below the depth are highlighted in bold.'**
- 'The display looks the same as it did before and is unchanged.'
- 'The whole sea area below that depth turns dark blue.'
- 'There will be alarms activated to let the operator know the depth under 'keel.

Which option best describes the reason behind this statement?
'There should be regular ECDIS monitor colour tests.'

- **To ensure the monitor shows correct colours**
- It says that in the bridge Procedures guide
- Monitor test complies with IMO's and IHO's requirements
- The manufacturer's warranty indicates to do this

Which option best describes what lies below an isolated danger symbol on an ENC?

- **There could be any seafloor obstruction**
- There is a rock outcrop not seen by the survey team
- There is a wreck from a sunken vessel
- There is probably a man made subsea structure associated with oil or gas

Which statement best describes how the vector chart gives advice on data quality in relation to ECDIS?

- **'The chart has a ZOC function.'**
- 'It does not - the operator has to go to the help menu on the stand alone computer.'
- 'The chart has text on it that describes the sea floor type.'
- 'There are source data diagrams on the chart that can be read.'

Which statement best describes orange cross hatching as might be seen on an ECDIS sea area?

- **'Orange cross hatching is an indication of a Cautionary area.'**
- 'It is an indication of a port approach area.'
- 'It means that a chart correction has been added somewhere in the area.'
- 'The ECDIS sees an area with dredged sounding values and shows it in orange hatching.'

Which statement best describes the action needed if ECDIS cannot automatically display Temporary and Preliminary notices?

- **'Annotate the ECDIS using the notes function.'**
- 'Ask a VAR to encode them into SENC format.'
- 'Ignore them as they will be made into a Notice to Mariners soon.'
- 'Inform the Master.'

Which statement best describes the assistance the operator gets from the predictive vector when monitoring in a coastal area in relation to ECDIS?

- **'To protect the vessel from grounding.'**
- 'Assess anti collision manoeuvres.'
- 'Assess the relative approach of target vessels.'
- 'See how the vessel can be turned.'

Which statement best describes the meaning of the Paris Memorandum?

- **An understanding for Port State Control Officers in Europe**
- A government paper in France concerning hydrography
- A reminder that ECDIS should be kept up to date.
- An agreement between worldwide nations for ECDIS development

Which statement best describes what happens to land in North up relative motion mode on ECDIS?

- **'Land on screen moves relative to the ship symbol.'**
- 'Land is stopped with the vessel symbol moving.'
- 'The land is always in the middle of the screen.'
- 'This is a feature only seen on radar.'

Which statement best describes what happens to the land in North up true motion mode on ECDIS?

- **'Land on the screen does not move until it resets.'**
- 'Land is moving with the vessel symbol stopped.'
- 'The land is always in the middle of the screen.'
- 'This is a feature only seen on radar.'

Which statement best describes what happens to the ship symbol in North up relative motion mode on ECDIS?

- **'Vessel symbol is stopped on the screen.'**
- 'This is a feature only seen on radar.'
- 'Vessel symbol is always in the middle of the screen.'
- 'Vessel symbol moves across the screen.'

Which statement best describes what happens to the ship symbol in North up true motion mode on ECDIS?

- **'Vessel symbol moves across the screen.'**
- 'This is a feature only seen on radar.'
- 'Vessel is stopped on the screen land moves relative.'
- 'Vessel symbol shows the heading fixed vertically on the screen.'

Which statement best describes why it is important to remove cancelled chart cells from the ECDIS?

- **'To stop them being loaded onto the ECDIS display.'**
- 'It is part of Flag State requirements.'
- 'New charts will not load if old versions are there.'
- 'So that the computer hard drive stays uncluttered.'

Which statement best describes why the ECDIS has settings for ambient light conditions?

- **'Observation of the screen is necessary under all light conditions at sea.'**
- 'The screen will automatically set light levels day on night.'
- 'The sensitivity of the operator's eyes is a factor allowing changes.'
- 'To allow the bridge team time to adjust their eyes at night.'

0213 – Passage planning with ECDIS

Version 2.0 (1705)

Apart from charted information select from the list which two features are permanently displayed on the chart display of the Seagull ECDIS Simulator and a real ECDIS.

- **Chart scale**
- **North direction indicator**
- chart number
- chart permit
- EBL
- route name
- VRM

Select a statement which best describes where the consistent common reference point (CCRP) is set on the Seagull ECDIS Simulator.

- **'The CCRP is set amidships.'**
- 'The CCRP is set at the conning position.'
- 'The CCRP is set near the bow.'
- 'The CCRP is set near the stern.'

Select a statement which best describes where the consistent common reference point (CCRP) should be set on ECDIS.

- **'The CCRP should be set at the conning position.'**
- 'The CCRP should be set amidships.'
- 'The CCRP should be set near the bow.'
- 'The CCRP should be set near the stern.'

Select a statement which best describes why isolated danger symbols inside the safety depth contour are always visible on the Seagull ECDIS Simulator.

- **'The object underneath has no depth attribute.'**
- 'The object underneath has a depth less than the safety depth contour.'
- 'The object underneath has a depth more than the safety depth contour.'
- 'The object underneath is not seen to be dangerous.'

Select an option which best describes the anomaly seen on the Seagull ECDIS Simulator in position 52° 51.2'N, 003° 14.1'E.

- **'There is an area boundary line without an associated area.'**
- 'There is a line indicating the next chart to use in the wrong place.'
- 'There is a pipeline with the wrong symbol in this area.'
- 'There is a wrong colour for the depth of water.'

Select an option which best describes the state of the tide as seen on the Seagull ECDIS Simulator.

- **'Probably close to or at high water.'**
- 'Probably at half tide.'
- 'Probably close to or at low water.'
- 'The tide height cannot be assessed on an ECDIS.'

Select an option which best describes what might be giving the radar return on the Seagull ECDIS Simulator in position 52° 51.6'N, 003° 16.2'E.

- **'It is a vessel called the Sea Venture.'**
- 'It is a false echo.'
- 'It is a fishing boat with nets deployed.'
- 'It is probably a buoy adrift.'

Select from the given options one answer to this question. 'How many hours is the ECDIS required to have recorded detailed voyage data?'

- **12 hours.**
- 24 hours.
- 6 hours.
- For the whole voyage.

Select from the given options which statement best answers this question. 'When does a particular object have an isolated danger symbol displayed on it?'

- **'When an object in safe water has less depth than the safety contour.'**
- 'When an object in safe water has more depth than the safety contour.'
- 'When an object in safe water is considered safe to pass over.'
- 'When there is an isolated danger buoy in safe water.'

Select from the given options which statement best explains what an isolated danger symbol represents.

- **'There is an underwater obstruction.'**
- 'There is a navigation mark in that position.'
- 'There is a new wreck located in that position.'
- 'There is a surface obstruction.'

Select the ONE option which correctly answers this question. 'With an ECDIS in track control, will the vessel's position always be correct if the ship symbol is on the route?'

- **No - because there may be a position error on the navigation signal.**
- No - because at some waypoints the ECDIS will show an over shoot.
- Yes - because that is the meaning of track control.
- Yes - because the bridge Officers are skilled at steering the vessel.

Select the best option which answers the following question. 'When passage planning on ECDIS which two functions should always be set?

- **Safety depth contour and safety depth value.**
- Course and speed of the vessel.
- Safety depth and UKC at a waypoint number.
- Speed of the vessel and ETA to destination.

Select the correct option for the default setting of the safety depth contour on ECDIS.

- **30m**
- 0 m
- 20 m
- 50 m

Select the given option which correctly completes the following statement.
'The datum of the sounding figures shown on ECDIS . . .

- **. . . is normally lowest astronomical tide.'**
- . . . is normally mean high water springs.'
- . . . is normally not known by the chart maker.'
- . . . is normally referenced to the local country.'

Select the given option which correctly describes the meaning of the text found on the Seagull ECDIS Simulator in position 52° 51.4' 003° 16.1'?

- **A 15m depth sounding figure.**
- A buoy number
- A chart number.
- A leg distance

Select the one option which correctly completes the following statement.
'An event mark on ECDIS is best described as . . .

- **. . . a time and date related happening.'**
- . . . a date in the future.'
- . . . when the crew mail arrives.'
- . . . when there is a company visit to the vessel.'

Select the option which best answers the following question. 'What should be set as the cross track limit on a vessel with 25 metres beam and length 100 metres entering Newtown harbour on the Seagull ECDIS Simulator in reasonable weather?'

- **'Set a cross track limit of 70 metres about quarter of the channel.'**
- 'Set a cross track limit of 140 metres about half of the channel.'
- 'Set a cross track limit of 20 metres about the vessel's beam.'
- 'Set a cross track limit of 278 metres about the full channel.'

Select the option which best answers this question. 'Why does an object on ECDIS always display an isolated danger symbol?'

- **'It displays an isolated danger symbol when the object has no depth attribute.'**
- 'It displays an isolated danger symbol when the object has a depth attribute greater than the depth.'
- 'It displays an isolated danger symbol when the object lies between the 20 and 30metre contours.'
- 'It displays an isolated danger symbol when there is an isolated danger buoy located on the object.'

Select the option which best answers this question? 'When will an alarm sound on an ECDIS operating in track control?'

- **When approaching a wheel over position.**
- When the route is half completed.
- When the vessels speed is too low.
- When the vessel's speed is too high.

Select the option which best describes AIS fusion on an ECDIS?

- **Bringing together a displayed radar return and AIS target.**
- The discrepancy between radar and AIS displayed positions.
- When a target's AIS signal has a large error on it.
- When two adjacent AIS targets come together.

Select the option which best describes how sleeping AIS targets on the ECDIS are shown.

- **Blue triangles.**
- Green circles.
- Orange circles.
- Orange triangles.

Select the option which best describes the motion of the vessel symbol when ECDIS is in true motion mode?

- **The symbol moves across the screen in the direction of COG and SOG**
- The symbol moves across the screen from corner to corner.
- The symbol stays on the display in a position chosen by the operator.
- The symbol will always be in the centre of the display.

Select the option which best describes the motion of the vessel symbol when ECDIS is set in relative motion mode.

- **The symbol stays on the display in a position chosen by the operator.**
- Nothing happens to the symbol as this is a radar feature.
- The symbol moves across the screen.
- The symbol will always be in the centre of the display.

Select the option which best describes the use of the predictive vector on an ECDIS.

- **If set to the correct length it will help protect against grounding.**
- It is a feature that comes from ARPA with acquired targets.
- It is there to assist in ports when turning is critical.
- The predictive vector is only a prediction and not trustworthy.

Select the option which best describes the validity of this statement. 'When sailing with ECDIS voyage data has to be hand written to maintain a record.'

- **False - ECDIS will record voyage data in an electronic log book.**
- False - because the VDR records all voyage data that can be used.
- False – because the bridge computer does this.
- True - because computer recorded data files can become corrupt

Select the option which best describes this statement? 'With an ECDIS in track control the ship symbol will always be on the route line.'

- **Correct - because the system is set to follow the planned track.**
- Correct - because the bridge Officer will steer the vessel along the route.
- Not correct - as route mode is when the passage is planned.
- Not correct - because the vessel is not capable of following a route.

Select the option which best describes what happens to ECDIS functionality when a voyage play back is running.

- **The ECDIS stops working in real time.**
- It will be saving all the data to make a back up copy.
- Nothing changes as the ECDIS still fixes position.
- There will be two charts running together if the GPS is still connected.

Select the option which correctly completes the following statement in relation to ECDIS.
'Orange text on a vector or raster chart display will be . . .

- **. . . navigator notes.'**
- . . . place names on the vector chart.'
- . . . system information from the ECDIS software.'
- . . . system information from the operating system on the computer.'

Select the option which correctly completes the following statement.
'A bold red, dashed line on ECDIS represents.'

- **. . . the route in monitor mode.'**
- . . . a fairway limit set up approaching a port.'
- . . . an information link via the pick report.'
- . . . the edge of the traffic scheme.'

Select the option which correctly completes the following statement.
'Insert mode when route planning with ECDIS means . . .

- **. . . waypoints are being added to a route list.'**
- . . . chart corrections are being downloaded.'
- . . . the route is ready for validation.'
- . . . waypoints are being edited.'

Select the option which correctly completes the following statement.
'On ECDIS spot depth soundings with less value than the safety depth contour are shown . . .

- **. . . as bold figures.'**
- . . . in a greatly enlarged fashion.'
- . . . in bright red.'
- . . . the same as any other depth figures.'

Select the option which correctly completes the following statement.
'Setting the turn radius on ECDIS is important because . . .

- **. . . the turn catenary on the display is set from it.'**
- . . . a Port State Control Officer would expect to see it.'
- . . . the ECDIS procedures indicate it should be done.'
- . . . the route could not be validated without it.'

Select the option which correctly completes the following statement.
'The conning position, GPS aerial position and radar antenna position are adjusted into one position on ECDIS which is called the . . .

- **. . . consistent common reference point.'**
- . . . common closest reference point.'
- . . . common position for navigation.'
- . . . the CCPR.'

Select the option which correctly completes the following statement.
'The green highlighted area seen on the Seagull ECDIS Simulator in position 52° 52.4'N, 003° 19.3'E indicates . . .

- **. . . the exposed land between high and low water.'**
- . . . a radar return from the land close by.'
- . . . an AIS symbol for a vessel close inshore.'
- . . . land that is always covered by the tide.'

Select the option which correctly completes the following statement.
'The own vessel past track on ECDIS is shown as . . .

- **. . . a solid black line.'**
- . . . a broken orange line.'
- . . . a broken red line.'
- . . . a solid red line.'

Select the option which correctly completes the following statement.
'The submarine cable found on the Seagull ECDIS Simulator in position 52° 50.2'N, 003° 16.4'E is probably . . .

- **. . . a communications cable.'**
- . . . a cable guiding a ferry.'
- . . . a cable supporting a shark net.'
- . . . a power cable.'

Select the option which correctly completes the following statement.
'The value of cross track limit when automatically validating a route on ECDIS is important because . . .

- **. . . ECDIS sets up a cross track either side of the route to look for dangers.'**
- . . . if the alarm sounds it annoys the watch keeper.'
- . . . it is the value of the margin of safety on the passage plan.'
- . . . the wheel over position is set up from the value of cross track.'

Select the option which correctly completes the following statement.
'When validating a route using ECDIS it is important to have the correct chart at the correct scale because . . .

- **. . . if not at the correct chart scale, dangerous objects could be missed.'**
- . . . ECDIS cannot automatically load charts at the correct scale.'
- . . . ECDIS has no intelligence and has to be helped.'
- . . . if there are raster charts on the ECDIS they would be used.'

Select the option which correctly completes the following statement.
'Yellow text on a vector or raster electronic chart display will be.

- **. . . manufacturer notes.'**
- . . . down loaded chart corrections.'
- . . . information in the pick report.'
- . . . system information from the ECDIS software.'

Select the option which correctly completes the following statement.
'When the ECDIS display is set in different orientations there should always be a visible indication of . . .

- **. . . true north.'**
- . . . compass north.'
- . . . grid north.'
- . . . gyro north.'

Select the option which correctly describes the meaning of the letter found on the Seagull ECDIS Simulator in position 52° 51.9'N, 003° 22.6'E?

- **It is an S indicating sand on the sea floor.**
- It is an S indicating a sounding figure.
- It is an S indicating shells on the sea floor.
- It is an S indicating stones on the sea floor.

Select the option which correctly describes the meaning of the line found on the Seagull ECDIS Simulator in position 52° 55.2'N, 003° 16.4'E?

- **It is an indication of a land contour.**
- It is an indication of a depth contour.
- It is an indication of a field boundary.
- It is an indication of a road.

Select the option which correctly describes the meaning of the magenta highlighted area found on the Seagull ECDIS Simulator in position 52° 50.7'N, 003° 18.3'E?

- **It is an indication of a traffic scheme round about.**
- It is a symbol indicating a cautionary area.
- It is a symbol indicating a highlighted depth area.
- It is a symbol indicating a precautionary area.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 43.5'N, 003°20.7'E?

- **Simplified symbol for port hand lateral buoy.**
- Simplified symbol for port hand lateral beacon.
- Simplified symbol for starboard hand lateral beacon.
- Simplified symbol for starboard hand lateral buoy.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 50.1'N, 003° 14.5'E?

- **A general tree**
- A coniferous tree
- A deciduous tree
- A mangrove swamp

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 50.3'N, 003° 22.6'E?

- **It is a symbol indicating a safe water mark.**
- It is a symbol indicating a lateral mark.
- It is a symbol indicating an emergency wreck buoy.
- It is a symbol indicating an isolated danger.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 50.4'N, 003° 19.2'E?

- **It is an indication of a ZOC level A2.**
- It is an indication of a ZOC level A1.
- It is an indication of a ZOC level D.
- It is an indication of a ZOC level U.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 50.7'N, 003° 15.7'E?

- **It is a symbol indicating a lighthouse.**
- It is a symbol indicating a building on the land.
- It is a symbol indicating a buoy.
- It is a symbol indicating a silo.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 50.9'N, 003° 20.6'E?

- **It is a simplified symbol indicating the presence of a south cardinal buoy.**
- It is a simplified symbol indicating the presence of a south cardinal beacon.
- It is a traditional symbol indicating the presence of a south cardinal beacon.
- It is a traditional symbol indicating the presence of a south cardinal buoy.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 51.3'N, 003° 14.9'E?

- **It is a symbol indicating anchoring prohibited.**
- It is a symbol indicating a lost anchor.
- It is a symbol indicating an anchorage area.
- It is a symbol indicating anchoring allowed.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 52.4'N, 003° 13.1'E?

- **It is a symbol indicating a dangerous wreck.**
- It is a symbol indicating a dangerous rock outcrop.
- It is a symbol indicating a safe wreck.
- It is a symbol indicating a subsea manifold.

Select the option which correctly describes the meaning of the symbol found on the Seagull ECDIS Simulator in position 52° 53.5'N, 003° 16.7'E?

- **It is a symbol indicating a conspicuous church.**
- It is a symbol indicating a conspicuous minaret.
- It is a symbol indicating a regional post office.
- It is a symbol indicating the port Customs House.

Select the option which correctly describes the meaning of the wavy line found on the Seagull ECDIS Simulator in position 52° 50.2'N, 003° 16.4'E?

- **A submarine cable**
- A cautionary boundary
- A pipeline
- A port limit

Select the option which correctly describes the symbol found on the Seagull ECDIS Simulator in position 52° 52.0'N, 003°19.4'E?

- **Isolated danger symbol**
- A no entry symbol
- A wreck mark
- An information note

Select the option which correctly identifies the light sectors on Seagull Island light?

- **White sector 212°/014° - red sector 014°/212°**
- Red sector 212°/014° - white sector 014°/212°
- White sector 014°/212° - red sector 212°/014°
- White sector 104°/212° - red sector 212°/104°

Select the option which correctly identifies the value of the channel width between the inner pair of lateral marks at the entrance to Newtown harbour on the Seagull ECDIS Simulator. One nautical mile = 1852metres.

- **Cross channel width is approximately 278 metres.**
- Cross channel width is approximately 2130 metres.
- Cross channel width is approximately 2334 metres.
- Cross channel width is approximately 482 metres.

Select the option which correctly indicates the width of the channel between the inner pair of lateral marks at the entrance to Newtown harbour on the Seagull ECDIS Simulator?

- **Cross channel width is approximately 0.15 of a mile.**
- Cross channel width is approximately 0.26 of a mile.
- Cross channel width is approximately 1.15 miles.
- Cross channel width is approximately 1.26 miles.

Select the statement which best answers this question. 'When route validating how does the ECDIS inform the planner of a possible problem?'

- **'It makes a report showing the findings.'**
- 'By highlighting the problem text in bold.'
- 'It does not because it corrects any problems found.'
- 'There will be an alarm sound both visual and audible.'

Select the statement which best describes the affect on the display when the ECDIS safety depth value is set.

- **'Depth values shallower than the depth are highlighted in bold.'**
- 'Depth values deeper than the depth are highlighted in bold.'
- 'The whole sea area below that depth turns dark blue.'
- 'There will be alarms activated to let the operator know the depth under keel.'

Select the statement which best describes the answer to the following question.
'What is the meaning of NUP/TM in relation to ECDIS?

- **North up true motion.**
- Navigation under precision for true motion.
- North up total motion.
- North up transverse Mercator.

Select the statement which best describes what the Consistent Common Reference Point is when established on an ECDIS.

- **'It is a position on the vessel common to all sensors.'**
- 'It is a position that the GPS aerial is in.'
- 'It is a position that the radar scanner is in.'
- 'It is a term associated the ECDIS time constant.'

Select the statement which best describes why a 'man over board button' is always visible on the ECDIS display.

- **To ensure an accurate position of the MOB.**
- So that a record can be kept of the MOB position.
- To allow track control to return to the position of the MOB.
- To have the MOB position sent to the Coastguard.

Select the statement which best describes why on the Seagull ECDIS Simulator or a real ECDIS the North up TM button is highlighted in yellow.

- **It is an ECDIS indication.**
- It is a warning to the operator.
- It was a mistake and should be grey.
- The ECDIS shows this as an alarm.

Select the two options which correctly complete the following statement.
'On passage the difference between course steered and course over ground is . . .

- **. . . the effect of gyro error.'**
- **. . . the environmental effects on the vessel.'**
- . . . the difference between heading and course steered.'
- . . . the effects of track control on ECDIS.'

Select two of the options which correctly complete the following statement.
'On passage the difference between speed through the water and speed over ground is . . .

- **. . . the effect of log error.'**
- **. . . the environmental effects on the vessel.'**
- . . . the difference between ground log and GPS.'
- . . . the effects of track control on ECDIS.'

Select which option best answers this question. 'Is there one disadvantage with passage planning on an ECDIS?'

- **Yes the view of the chart is not as big as the paper chart.**
- No there is not because all the appraisal information the planner needs is on the ECDIS.
- No there is not because it saves the bridge team lots of time.
- Yes there is because the process is totally different to paper charts.

Select which statement best answers the following question. 'Can ECDIS be used to assist in the assessment of the manoeuvre to avoid collision with another vessel?'

- **'Yes, but not totally, the Collision Regulations take precedent.'**
- 'No because it cannot give target information on CPA and TCPA.'
- 'No, it should not be used at all; the IMO Collision Regulations should be followed.'
- 'Yes, action can be taken just by watching the track of the other vessel.'

The illustration shows a vessel on passage with ARPA radar target information over laid on ECDIS, vector length is 12 minutes. Select from the options which statement best answers the following question. 'Is there any risk of collision with the target?'

- **'No - risk of collision does not exist.'**
- 'No - risk of collision does not exist as the target will pass ahead.'
- 'Not sure because ECDIS cannot be used for collision assessment.'
- 'Yes - there is risk of collision.'

The illustration shows a vessel on passage with ARPA radar target information over laid on ECDIS, vector length is 12 minutes. Select from the options which statement best answers the following question. 'Is there any risk of collision with the target?'

- **'Yes - there is risk of collision as the vectors meet.'**
- 'No - there is no risk of collision as the target vector does not point at own ship.'
- 'No – risk of collision does not exist as the target will pass astern.'
- 'Not sure because ECDIS should not be used for collision assessment.'

The illustration shows a vessel on passage with ARPA radar target information over laid on ECDIS, vector length is 12 minutes. Select the option which best completes the following statement . . . 'The action that should be taken is to . . .

- **. . . correctly apply the Regulations for Preventing Collisions.'**
- . . . .check the CPA and TCPA if close alter course to port.'
- . . . assess the situation from ECDIS then manoeuvre accordingly."
- . . . stand on with caution as the target is the give way vessel.'

When appraising Newtown harbour for berthing on the Seagull ECDIS Simulator it is essential to know the berth heading and berth length. Select which of the following statements correctly identifies the heading and length.

- **'Heading is 002°/182° and length approximately 167metres.'**
- 'Heading is 002°/182° and length approximately 567metres.'
- 'Heading is 002°/272° and length approximately 167metres.'
- 'Heading is 092°/182° and length approximately 267metres.'

When appraising Southern Island for berthing on the Seagull ECDIS Simulator it is essential to know the berth heading and berth length. Select which of the following statements correctly identifies the berth heading and length.

- **'Heading is 094°/274° and length approximately 185metres.'**
- 'Heading is 094°/184° and length approximately 285metres.'
- 'Heading is 184°/274° and length approximately 185metres.'
- 'Heading is 184°/274° and length approximately 585metres.'

Which of the given options best describes the GPS position information that should be shown on the ECDIS display?

- **Primary or secondary sensor used and if a differential signal.**
- The error likely on the primary and secondary GPS signal.
- VDOP figure on the primary receiver.
- Whether in DR mode and the problem with the GPS receiver.

Which of the options best describes the information seen on the ECDIS route monitor page?

- **Basic route information as the vessel proceeds.**
- All the voyage data being recorded
- Details of the whole route planned.
- Navigational information from own vessel and AIS targets.

Which option best describes the correctness of the following statement?
'ECDIS can be relied upon totally to always display the vessel's correct position.'

- **Not correct - the position should be routinely checked with other means.**
- As GPS feeds most ECDIS devices this is a correct statement.
- Correct as the system will always show the right position using GPS
- Not really correct - once a week the 2nd Officer should make a check.

Which statement best describes docking mode on an ECDIS?

- **'ECDIS showing a large scale chart for berthing.'**
- 'As the vessel is approaching the berth the ECDIS voyage recording has been stopped.'
- 'For use when tugs are connected to the vessel so the pilot can see them.'
- 'When the vessel is going into a dry dock – it shows the dock outline.'

Which statement best describes how a dangerous AIS target is seen on the ECDIS display?

- **It shows with a red triangle and vector.**
- It shows as a blue triangle.
- It shows as a red triangle.
- It shows with a blue triangle and vector.

Which statement best describes the difference between route and track when passage making?

- **'The route is planned; the track is where the vessel has been.'**
- 'The route dictates the heading; the track is all the legs on the route.'
- 'The route is where the vessel is now, the track is up ahead.'
- 'There is no difference between the two.'

Which statement best describes why a previously validated route would have to be re-validated on an ECDIS?

- **'Revalidation is necessary when there have been chart updates.'**
- 'It would not need re-validating because once validated that is sufficient.'
- 'Revalidation is necessary when it has been put out of monitor mode.'
- 'Revalidation is necessary when it has been removed from the display.'

Which statement best describes why the value of the turn radius is important in route creation on an ECDIS?

- **'Because at waypoints the wheel over position is set from it.'**
- 'Because the value should be on the passage plan.'
- 'It is not important as the Master always changes course his way.'
- 'Turn radius allows ETAs to be worked from it.'

Which statement correctly describes the process of ECDIS automatic route validation?

- **'It is the ECDIS checking the route for safety.'**
- 'It is the ECDIS computer checking files are correct.'
- 'It is the only way to make sure the route is safe.'
- 'This is when the ECDIS checks that the route is not a copy.'

0215 – Voyage planning, Pre-departure

Version 2.0 (446)

A Voyage Plan is considered essential for the safe navigation of the ship by many authorities and IMO provide guidelines to assist vessels with this task. Which of the statements given in the options is the most accurate in relation to the requirement to construct a Voyage Plan?

- **All vessels must construct a Voyage Plan.**
- Fishing vessels are exempt from having to construct a voyage plan.
- It is a voluntary procedure for ships to construct a voyage plan but strongly recommended by IMO and most Flag States.
- Only Merchant vessels are required to construct voyage plans; pleasure craft do not need to plan a voyage.

At the Appraisal stage when Planning a Voyage and collecting suitable data, which, if any, specific document is more important than others?

- **All documents containing significant data contribute information to an effective voyage plan so no one document is any more important than any other.**
- Routeing charts are the most important contributor to an effective Voyage Plan.
- The large scale charts are the most important contributor to an effective Voyage Plan.
- The ship's manoeuvring particulars are the most important contributor to an effective Voyage Plan.

In addition to close and continuous monitoring of the ships progress and position which one of the given options best describes the primary objectives of a Voyage Plan?

- **Safe and effective navigation of the ship, the protection of the environment and the safety of crew and passengers.**
- Protection of the environment, ensuring the shortest route and the safety of the crew and passengers.
- Safe and effective navigation of the ship, minimising running costs and ensuring the shortest route.
- The protection of the environment, the safety of the crew and passengers and ensuring the shortest route.

What are the four stages of Voyage Planning as specified by IMO guidelines?

- **Appraisal, Planning, Execution and Monitoring.**
- Appraisal, Planning, Monitoring and Contingencies.
- Fact gathering, Planning, Monitoring and Passage run.
- Planning, Passage run, Execution and Monitoring.

What is the Master's responsibility regarding the production of Voyage Plan?

- **The Master must ensure the Bridge team is suitably qualified, competent and well rested prior to the start of the voyage.**
- The Master must personally research and plan the Voyage Plan and then discuss the plan with the Bridge Team.
- The Master should ensure the Voyage Plan is safe and effective from leaving the departure port to arrival at the destination port.
- The Master's only role in Voyage Planning is to ensure the Navigation Officer has produced a plan which is safe for the ship, crew and environment.

When Planning a passage into an anchorage, what type of information, if any, should be marked on the approach line indicated on the chart?

- **"Distance to run" with suggested speeds at various points along the approach line.**
- Suggested final anchorage position.
- The approach line should indicate any influencing currents and tidal heights.
- There is no need to mark anything other than just the approach course to make good.

When Voyage Planning there may be a temptation to use a Voyage plan from a previous voyage. Which of the statements given in the options is most accurate?

- **A previous voyage plan could be for a different time of the year and will therefore not necessarily be suitable.**
- A previous voyage plan should never be used in any circumstance.
- A previous voyage plan will always be acceptable and avoid lengthy research and development of another Voyage plan.
- A previously prepared voyage plan can be used successfully by other ships.

When Voyage Planning, what reference should be examined to determine any seasonal restrictions and geographical areas which a vessel may not enter due to the draught at which the vessel is loaded?

- **The International Load Line Zones, Areas and Seasonal Periods Map.**
- The International Load Line Certificate on board the vessel.
- The marks on the side of the vessel.
- The Weather Routeing map.

When Voyage Planning, would the fact that the vessel had two fully operational ARPA radars allow the "Safe Speed" to be faster than for a vessel with only one ARPA?

- **It may be sometimes slower or sometimes faster, dependant on the individual circumstances.**
- It should always be possible to proceed faster with two operational ARPA radars, than with only one.
- Never
- Two ARPA radars would provide more information and indications of danger and normally make the vessel progress slower than with only one radar and limited information.

When executing a Voyage Plan, what information should been known by the Officer of the Watch (OOW) at the start of his/her watch?

- **All alterations of course and all of the navigational hazards, navigational marks and the lights likely to be encountered during their watch.**
- Only navigational hazards and the lights and navigational marks likely to be encountered during their watch.
- Only the main alterations of course and navigational hazards likely to be encountered during their watch.
- Only the main navigational hazards likely to be encountered during their watch.

When involved in Planning a Voyage, is every vessel obliged to comply with the International Regulations for the Prevention of Collision at Sea?

- **Yes, all vessels must comply with the Collision Regulations.**
- No, small boats are exempted from complying with the Collision Regulations.
- No, vessels engaged in dredging are exempted from complying with the Collision Regulations.
- Yes, all vessels except military navy vessels must comply with the Collision Regulations.

When planning a Voyage is it necessary to mark estimated times of arrival (ETA) at different places on the voyage?

- **Yes, always**
- No, never
- Occasionally, but only when significant
- Sometimes

When planning a voyage, can all the relevant information be found on the ship, or is there some important data required from external sources?

- **There is important sources outside the vessel.**
- All relevant data is found on board.
- It depends on destination to which the voyage is planned.
- It depends on the type of ship whether outside sources are required.

Will the type of cargo on board affect the selection of route to follow when planning a Voyage?

- **Yes, the cargo may impose restriction when the vessel enters environmental sensitive areas.**
- No, all environmental sensitive areas must be avoided irrespective of the cargo onboard.
- No, the route is selected for the safety of the navigation and will not be affected by the cargo onboard.
- No, the type of cargo will only affect the ports to which the vessel will trade to load or discharge cargo.

With regard to Voyage Planning, who is responsible for radical deviations from the planned route?

- **The Master**
- Any of those mentioned
- The designated Navigation Officer
- The Officer of the Watch

0216 – Voyage planning, The sea passage

Version 1.0 (340)

In the execution stage of a Voyage Planning, can the use of Parallel Index lines remove the need to take regular fixes?

- **Never**
- Always
- Depends on the location of the vessel
- Sometimes

When Voyage Planning and marking the "wheel over" position on the chart, what factors could possibly change the size of the turning circle and need to be considered?

- **Engine Revolutions, direction and speed of any current and depth of water.**
- Direction and speed of any current and depth of water.
- Speed of ship, direction of any current.
- Speed of vessel, direction and speed of any current and depth of water.

When Voyage Planning is it possible to be totally reliant on the vessels position as shown on the ECDIS?

- **No, there should never be total reliance on one method of fixing the ship's position.**
- . It can only be totally relied upon when the HDOP of the GPS is small.
- Yes, because the ECDIS cross references visual bearings and electronic fixes automatically.
- Yes, the modern ECDIS is totally reliable and can be depended upon.

When Voyage Planning why is it important to consider strategic Weather Routeing advice?

- **Weather Routeing can reduce structural stress and damage and reduce fuel consumption.**
- The significance of weather routeing is normally only important for passenger ships to ensure a smooth passage for their passengers.
- Weather routeing is not normally important to consider, unless on a long trans-ocean voyage.
- Weather routeing only requires the Officer of the Watch to carefully monitor the local weather forecasts.

When Voyage Planning, is it ever possible to prepare for reduced visibility and "blind" pilotage?

- **Sometimes, since reduced visibility can be seasonal in some places and an increased likelihood of its occurrence can be predicted.**
- No, reduced visibility is a meteorological occurrence and cannot be predicted.
- Only on coastal voyages since reduced visibility can only be forecasted locally and there is no long distance prediction.
- Only on short voyages since reduced visibility can be predicted within a few days and never long term.

When Voyage Planning, is there any advantage to have a waypoint, or wheel-over position, when a significant headland, (or visible navigational mark), is abeam?

- **Yes, it makes it easier to know that the vessel has arrived at that position.**
- No, a waypoint should be positioned to suit the overall plan.
- No, with the use of GPS there is no advantage.
- Yes, it will always be significant to any changes in the direction of the current and the allowance to the course to steer

When Voyage Planning, what is the main consideration when selecting a Great Circle route between two locations on the voyage?

- **The main consideration is to reduce the overall steaming distance between departure and destination ports.**
- The main consideration is to avoid environmentally sensitive areas.
- The main consideration is to keep the vessel out of areas where ice may be expected.
- The main consideration is to keep the vessel within limiting latitudes and better weather.

When Voyage Planning, what other information, if any, is provided by standard Routeing Charts in addition to suggested routes through the area covered by the chart?

- **Traffic separation schemes and prevailing weather patterns are provided.**
- The density of traffic the vessel will experience on the suggested route is provided.
- There is only the suggested routes to steer through different areas, no other information is provided.
- Weather particulars the vessel will experience are provided.

When Voyage Planning, where is information found about how to navigate in the vicinity of a Traffic Separation Scheme?

- **International Regulations for the Prevention of Collision at Sea**
- List of Radio Signals Volume 5
- Notice to Mariners
- Routeing Chart for that area

When executing a Voyage Plan and the vessel has been forced off the planned route by crossing traffic, should the Officer of the Watch (OOW) bring the vessel back to the planned route as quickly as possible?

- **No, the vessel should be brought back to the planned course in convenient manner, without radical alterations.**
- No, the ship should proceed on a parallel course to the planned course providing it is within the safety margins.
- No, the vessel should always be brought back to the planned track at the next waypoint.
- Yes, the ship should be returned to the planned course as quickly as possible.

When executing a Voyage Plan, how often should the position of the ship be obtained and marked on the chart?

- **It depends on the location and how close the ship is to danger.**
- Every half hour.
- Every Hour.
- When the Officer of the Watch has the opportunity and there are good visual navigational marks.

When involved with Voyage Planning a Bridge Note Book is used to record information about the voyage plan. Is the Bridge Note Book a mandatory document which follows an approved standard format?

- **No, the Bridge Note Book is a voluntary method to record voyage planning data notes effectively.**
- No, a Bridge Note Book is not mandatory, but if used must follow IMO guidelines and be a well defined and structured book to record relevant information.
- No, all relevant information must be kept together for easy use and should be noted on the chart.
- Yes, IMO stipulates that when Voyage Planning, all relevant information must be recorded into a Bridge Note Book and be available to the OOW.

When planning a voyage is it mandatory for all vessels to comply with a Traffic Separation Scheme (TSS)?

- **Yes, All vessels must comply with the TSS when navigating in that area.**
- No, Coastal traffic may ignore the TSS.
- No, only deep draught vessels are compelled to comply with a TSS.
- No, TSS are voluntary schemes to aid the traffic in that area and are not mandatory

When planning a voyage what dangers to a ship must be considered when trading in areas where ice may be expected?

- **Collision with an iceberg, loss of stability due to ice build up and structural damage.**
- Loss of stability due to ballast water and fuel freezing in the tanks is the main danger.
- Structural damage to the ship is the only danger to be considered.
- Water and hydraulics pipes freezing and stopping machinery operation is the main danger.

Which one of the given options best describes the way in which information should be entered onto the chart when using large scale charts during Voyage Planning?

- **The track, or course to make good is entered on the chart after the shallow patches are hatched and danger limit lines and safety margins are drawn on the chart.**
- It depends on the region covered by the chart, but the course to steer is always the first line drawn on the chart.
- The course to steer is drawn on the chart first and then other relevant information highlighted around that course.
- The shallow patches are hatched and then course to make good is entered on the chart in the middle of the channel or area. The safety limits are then entered from the course to make good.

Which one of the statements given in the options is true in relation to route selection during Voyage planning?

- **It should be the safest for the ship and the environment, which is not always the quickest.**
- The quickest route should be selected in order to save costs for the ship owner.
- The route to select when planning a voyage should prioritize the voyage time and potential bunkers consumed.
- The shortest distance is always the best route to select when planning a voyage

Which type of chart is used to determine the track of a Great Circle Route?

- **Gnomic Chart**
- Large scale coastal charts
- Mercator Chart
- Raster Chart

Will the study of ocean currents and consequent adjustment of courses to take them into account during the planning of a voyage, have any affect on the overall time at sea of the vessel when the plan is executed?

- **Almost definitely**
- No, never
- Possibly
- Unlikely

0217 – Voyage planning, Pilotage and berthing

Version 2.0 (255)

What is the quickest means of knowing whether a vessel is being affected by currents, when coasting?

- **Use of parallel indexing.**
- By the Doppler log.
- By visual bearings of conspicuous objects.
- Use of the echo sounder.

What should be done if a deviation creates considerable time delays?

- **All subsequent waypoint ETAs should be adjusted accordingly.**
- The closest VTS should be informed.
- The vessel should be brought back onto its intended track.
- The vessel's speed should be increased.

When Voyage Planning, should the primary method of navigation always be the GPS?

- **No, it depends on the location and circumstances.**
- No, visual positions should always be the primary method to fix the vessel's position.
- Yes, because GPS provides positions in the middle of the ocean.
- Yes, because the GPS is accurate within a few metres.

When Voyage Planning, why should the plan suggest alternative methods to fix the vessels position?

- **Alternative methods to fix the vessels position are needed because no single method is 100% reliable.**
- Alternative methods to fix the ship's position are taken because the OOW needs to practice these various competencies.
- Alternative methods to fix the vessel's position are needed because sometimes there are no visual marks identified.
- Alternative methods to fix the vessel's position are not needed because GPS is always accurate.

When approaching port, why should a voyage plan have to identify more than one possible anchorage area?

- **Because in an emergency a vessel should have an alternative anchorage option.**
- Because it is mandatory, under international regulations.
- So that an appropriate anchorage may be chosen, for better shore communications.
- So that the Master may choose an anchorage with the least number of ships.

When conducting a Voyage Plan and using GPS, ECDIS and other modern navigational aids, is it still necessary to determine the error of the magnetic compass?

- **Yes, always**
- No, never
- Yes, but only in special circumstances
- Yes, very occasionally

When executing a Voyage Plan, what would be the significance to the OOW when the predicted time at a way point is significantly different to the actual time arriving at that position?

- **The OOW should treat it as a warning something may be wrong.**
- It could be ignored as the Voyage Plan would have been constructed a long time before it was executed.
- It indicates that the Voyage Plan is incorrect.
- The OOW would consider altering the time at the next waypoint to correct the time difference.

When passage planning, what should be prepared in advance for coming alongside?

- **A berthing plan.**
- A copy of the ship's manoeuvering characteristics.
- A docking plan.
- Composition of the bridge team.

When passage planning, where should VTS reporting locations be made?

- **On charts and in the Bridge notebook.**
- In the Bridge notebook.
- On the bridge console.
- On the VHF.

Why does a radar picture not always follow the charted coastline?

- **Because of poor radar reflections from some coastal areas.**
- Because by the time the radar receives a signal, the vessel has moved along its track.
- Because of interference from radars on other ships.
- Because of poor surveys of coastlines in some areas of the world.

Why should buoys not be used, with regard to position fixing?

- **Because they could be adrift.**
- Because navigational charts warn against this practice.
- Because they are only meant to mark fairways and dangers.
- Because they are too small to obtain a fix.

Why should the echo sounder be frequently used in shallow waters?

- **To provide a useful position line when crossing a particular depth contour.**
- Because it is an international requirement.
- To cross-check the position of underwater obstructions with the chart in use.
- To warn the navigator of possible squat.

Why would a voyage plan possibly need adjustment when a pilot arrives on board?

- **Because he/she will have the latest local knowledge, for which the voyage plan may need to be adjusted**
- Because the pilot may not be satisfied with the voyage plan.
- Because this is standard practice with most pilots.
- Because voyage plans do not cover the pilotage phase of a voyage, in detail.

0218 – Oil Record Book, Part I

Version 3.0 (2686)

Bunkering of heavy fuel oil started at 9pm on May 19th and took 5½ hours. Based upon the recommendations in this training program which of the following is the correct way to record these times in the oil record book against operation Code H 26.2?

- **Start 19-MAY-2011 21:00 Stop 20-MAY-2011 02:30**
- Start 19-MAY-2011 9 pm for 5½ hours
- Start 21:00 Stop 02:30
- Start 9pm Stop 2:30am

Just before the engine room goes unmanned for the night you find that the oily bilge pump, which supplies the oily water separator, will not start. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **F 19 17:45**
  20 Not known
  21 Not known – OWS pump not starting
- F 19 17:45
  20 Tomorrow
  21 OWS pump not starting
- F 19 17:45
  21 Not known – OWS pump not starting
- F 19 17:45

The Incinerator Sludge Service Tank has been heated for 4 days in preparation for sludge burning. During this time 0.2 m^3 of water has evaporated. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 12.4 0.2 m^3 water evaporated from Incinerator Sludge Service Tank, 1.7 m^3 retained**
  Between 9-MAY-2011 – 13-MAY-2011
- C 12.3 0.2 m^3 removed from Incinerator Sludge Service Tank
  4 days
- C 12.3 0.2 m^3 removed from Incinerator Sludge Service Tank
  9-MAY-2011 08:20 – 13-MAY-2011 08:40 (96.3 hours)
- C 12.3 0.2 m^3 water evaporated from Incinerator Sludge Service Tank

The contents of the Incinerator Sludge Service Tank have settled so that you can drain some water to the bilges. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 12.4 0.2 m^3 water drained from Incinerator Sludge Service Tank – 3.1m^3 retained**
- C 11.4 0.2 m^3 water drained from Incinerator Sludge Service Tank – 3.1m^3 retained
- C 12.2 0.2 m^3 water drained from Incinerator Sludge Service Tank – 3.1m^3 retained
  to Port Forward ER Bilge Well – about 0.5 m^3 retained
- C 12.3 Water drained from Incinerator Sludge Service Tank – 3.1 m^3 retained

The contents of the Main Sludge Tank have been pumped to a road tanker while in the port of Miami. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 12.1 12 m^3 sludge from Main Sludge Tank – 0.3 m^3 retained to Acme Waste road tanker during port stay @ Miami**
- C 12.1 12,000 Litres sludge from Main Sludge Tank – 0.3 m^3 retained to road tanker Miami
- C 12.1 13.3 m^3 sludge from Main Sludge Tank and Purifier Drains Tank to Acme Waste road tanker during port stay @ Miami
- C 12.2 12 m^3 sludge from Main Sludge Tank – 0.3 m^3 retained to Acme Waste road tanker

The incinerator has just been shut down after 8 hours burning waste cardboard and sludge from the Incinerator Sludge Service Tank. The incinerator sludge burning capacity is noted on the IOPP certificate as 480 kW (approx. 50 litres/hour). Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 12.3 0.3 m^3 sludge from Incinerator Sludge Service Tank, 1.7 m^3 retained Burned in Incinerator for 8 hours**
- C 12.1 0.5 m^3 sludge from Incinerator Sludge Service Tank Burned in Incinerator for 8 hours
- C 12.1 0.5 m^3 sludge from Incinerator Sludge Service Tank – 3.5 m^3 retained
- C 12.4 0.3 m^3 sludge from Incinerator Sludge Service Tank Burned in Incinerator for 8 hours

When you take the oil record book to the master for his signature he points out that last week's sludge discharge ashore has not been recorded. Which one of the following is the best way to correct this mistake?

- **A code I entry must be made explaining that the following is a missed entry, underneath write the late entry**
- Destroy the sludge receipt and pretend it never happened
- The missed entry must be recorded in the next blank line
- The missed entry must be written on paper and stapled to the book in the correct date position

Which of the following is the correct format to be used for writing dates in an oil record book entry?

- **25-JUN-2011**
- 25/06/11
- 6.25.11
- JUNE- 25-2011

Which one of the following best describes how the ship's positions in oil record book entries can easily be found to be false, even ones from a month ago?

- **The positions in the oil record book do not match those in the deck log book**
- The position of the ship is recorded in the oily water separator control system and can be downloaded later
- The positions are written with East/West co-ordinates before North/South ones
- The positions of ships are now tracked by satellite

Which one of the following best describes how you can enter the quantity of water pumped overboard from the engine room bilge wells through the oily water separator (OWS) into the oil record book when you have no sounding tables for the bilge wells?

- **You need to estimate the quantity pumped from the OWS capacity and the time it was used for**
- The quantity does not need to be entered into the oil record book because it has not been in a calibrated tank
- You can estimate the quantity by looking into the bilge well before and after pumping
- You cannot pump water directly from bilge wells without putting it in a tank to measure the quantity first

Which one of the following best describes the action to take if you realise the entry you have just made in the oil record is partially incorrect?

- **The incorrect part should be struck through with a single line in such a way that the incorrect entry is still legible. The incorrect entry should be signed and dated, with the new corrected entry following**
- The incorrect part should be erased and replaced with the new correct entry
- The incorrect part should be struck through repeatedly so that it cannot be read. The incorrect entry should be signed and dated, with the new corrected entry following
- The whole entry should be struck through with the new corrected entry underneath

Which one of the following best describes the likely reaction of a Port State Control Officer (PSCO) if unusually repetitive entries are found in the oil record book?

- **The PSCO will be suspicious that the entries are false**
- The PSCO will think that the crew speak limited English
- The PSCO will think the crew are committed to complying with MARPOL
- The PSCO will think the ship is run to tight routines

Which one of the following best describes the number of signatures you would expect to find on one page of an oil record book when it is completed?

- **One for every entry plus the master's signature**
- One
- One for every entry made
- One for every line in the book

Which one of the following best describes what should be done with an oil record book when it has been filled up?

- **It must be kept onboard for 3 years from the date of closure**
- It must be kept ashore by the company for 3 years from the date of closure
- It must be kept onboard for 5 years from the date of opening
- It must be kept onboard until the next oil record book is filled up

Which one of the following best describes what should be written under each oil record book entry by the officer in charge of the operation?

- **His signature, name, rank and the date**
- His signature and the date
- His signature, name and the date
- His signature, rank and the date

Which one of the following best describes what you should do if your old oil record book is full and the only available new one is found to contain out of date MARPOL operation codes?

- **Use the new book after replacing the out of date codes with a copy of the current ones**
- Do not make any oil record book entries until an up to date book has been sent
- Make oil record book entries in the engine room log until an up to date book has been sent
- Use the new book and use the out of date codes for entries in it

Which one of the following best describes where you would expect to find a receipt for sludge that was discharged ashore?

- **With the oil record book**
- In the Annex VI fuel record book, with the bunker receipts
- In the chief engineer's cabin
- In the master's office

Which one of the following best describes why MARPOL requires ships to maintain an oil record book?

- **To provide the authorities with a record of the handling of oil residues on the ship**
- To give port State control officers something to find fault with
- To provide the authorities with a record of the handling of liquid wastes on the ship
- To warn ships' crews when the sludge tanks are nearly full

Which one of the following best describes why the total quantity of sludge, measured by the weekly soundings of sludge tanks, usually goes up every week even if the oily water separator has not been used?

- **Oil residues from save-alls and other oily drains gradually build up in the sludge tanks**
- Bilge water transferred from the bilge wells into the bilge holding tank adds to the quantity of sludge measured each week
- The volume of sludge increases when the tank is heated
- Water tends to drain into the sludge tanks

Which one of the following describes the best way of finding out if your ship has been fitted with an alarm and automatic stop on the oily water separator which complies with MARPOL Annex I requirements?

- **Check the IOPP certificate**
- Ask the chief mate
- Check the oil record book
- Check the oily water separator maintenance manual

Which one of the following describes the best way of making sure that the quantity of sludge discharged to reception facilities matches the reduction in sludge retained on board?

- **Check the quantity of sludge on board immediately before and after discharge ashore and make sure the reception facility receipt matches the change**
- Check the bilge holding tank levels have not changed during the sludge discharge
- Check the quantity of sludge on board at the next weekly soundings and make sure the reception facility receipt matches the change
- Make sure you get a detailed receipt for the sludge from the reception facility

Which one of the following describes the best way to find out which tanks have been designated for sludge and which for holding bilge water on your ship, as far as oil record book entries are concerned?

- **Check the IOPP certificate**
- Check the bilge and waste oil system drawings
- Check the capacity plan
- Check the tank plan

Which one of the following is recommended as the first thing to do when starting a new oil record book?

- **Check that it contains the current operation codes**
- Check that it has a blue cover
- Check that it has a green cover
- Write the ship's name on it

Which one of the following is the maximum fine that could be imposed in Germany for an incorrect entry in the ship's oil record book according to this program?

- **25,000 Euros**
- 2,000 Euros
- 200 Euros
- 5,000 Euros

Which one of the following is the maximum penalty that could be imposed in the United States for a deliberately false entry in the ship's oil record book according to this program?

- **$250,000 and 5 years in jail for each violation**
- $50,000
- $50,000 and 1 year in jail for each violation
- 1 year in jail

Which one of the following operation codes would be the best to use to record the fact that the oily water separator overboard discharge valve has been sealed while the ship passes through the Mediterranean MARPOL Special Area?

- **Code I**
- Code D
- Code F
- Code G

While loading heavy fuel oil at Yokohama the bunker barge flexible hose bursts leading to fuel entering the water. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **G 22 14:45**
  23 Yokohama Deep Water Terminal Berth # 2
  24 Approx 0.3 m^3 HFO
  25 Bunker barge flexible hose burst
- G 22 14:45
  23 35° 40' N, 139° 69' W
  24 Approx 0.3 m^3 HFO
  25 Bunker barge flexible hose burst

- G 22 14:45
  23 35° 40' N, 139° 69' W
  24 Approx 0.3 m^3 HFO

- G 22 14:45
  23 Yokohama Deep Water Terminal Berth # 2
  25 Bunker barge flexible hose burst

You have just finished bunkering diesel fuel. Based upon the recommendations in this training program which of the following is the correct operation code to use in the oil record book?

- **Code H**
- Code D
- Code G
- Code I

You have just finished bunkering diesel with a specific gravity of 0.9. 20 m^3 was loaded to the Diesel Oil Settling Tank and 100 m^3 to the Diesel Oil DB Tank. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **108 Tonnes of ISO 8217 DMB 0.9% Sulphur bunkered to tanks:**
  90 T added to Diesel Oil DB Tank – now contains 167 T
  18 T added to Diesel Oil Settling Tank – now contains 32 T
- 108 Tonnes of ISO 8217 DMB 0.9% Sulphur bunkered to tanks:
  100 m^3 added to Diesel Oil DB Tank – now contains 167 m^3
  20 m^3 added to Diesel Oil Settling Tank – now contains 32 m^3
- 120 m^3 of ISO 8217 DMB 0.9% Sulphur bunkered to tanks:
  100 m^3 added to Diesel Oil DB Tank
  20 m^3 added to Diesel Oil Settling Tank
- 120 m^3 of ISO 8217 DMB 0.9% Sulphur bunkered to tanks:
  100 m^3 added to Diesel Oil DB Tank – now contains 167 m^3
  20 m^3 added to Diesel Oil Settling Tank – now contains 32 m^3

You have just finished pumping clean water from the bottom of the Primary Bilge Tank into the Main Bilge Holding Tank. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **D 13 1.1 m^3 bilge water from Primary Bilge Tank, 1.2 m^3 retained**
  14 Start 08:10, Stop 08:35
  15.3 To Main Bilge Holding Tank, 15.3 m^3 retained
- D 13 1.1 m^3 bilge water from Primary Bilge Tank
  14 Start 08:10, Stop 08:35
  15.3 To Main Bilge Holding Tank, 15.3 m^3 retained
- D 13 1.1 m^3 bilge water from Primary Bilge Tank
  15.3 To Main Bilge Holding Tank, 15.3 m^3 retained
- No entry is needed

You have just finished pumping the sump oil from No 2 generator to the Main Sludge Tank. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 11.1 Main Sludge Tank**
  11.2 15.3 m^3
  11.3 5.6 m^3
  11.4 1.1 m^3 collected from No 2 Generator sump tank

- C 11.1 Main Sludge Tank
  11.4 1.1 m^3 collected from No 2 Generator sump tank
- C 11.1 Main Sludge Tank
  11.4 1.1 m^3 collected from No 2 Generator sump tank 0 m^3 retained
- C 11.1 Main Sludge Tank 5.6 m^3 retained
  11.4 1.1 m^3 collected from No 2 Generator sump tank

You have just finished transferring bilge water from the Main Bilge Holding Tank in the engine room into the starboard cargo slop tank of the crude oil tanker you are sailing on. Based upon the recommendations in this training program which of the following is the correct way to record this in the Oil Record Book Part I (machinery space operations)?

- **Make an entry using operation Code D**
- Make an entry using operation Code D and another using Code J
- Make an entry using operation Code E
- Make an entry using operation Code J

You have just finished transferring oil residue from the Bilge Holding Tank to the Incinerator Sludge Service Tank ready for evaporation of the water. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **Record the increase in sludge tank level as a manual collection (Code C 11) followed by an entry showing a non-automatic transfer of bilge water (Code D)**
- Record the operation as a non-automatic disposal of bilge water (Code D) only
- Record the operation as a sludge disposal (Code C 12) and ignore the Bilge Holding Tank change
- Record the operation as a sludge disposal (Code C 12) followed by an entry showing a non-automatic transfer of bilge water (Code D)

You have just finished transferring the sludge from the Main Sludge Tank in the engine room into the starboard cargo slop tank of the crude oil tanker you are sailing on. Based upon the recommendations in this training program which of the following is the correct way to record this in the Oil Record Book Part I (machinery space operations)?

- **Make an entry using operation Code C 12.4**
- Make an entry using operation Code C 12.1
- Make an entry using operation Code C 12.2
- Make an entry using operation Code C 12.4 and another using Code J

You have just set the automatic bilge pumping system to discharge overboard through the oily water separator for the next few days, while the ship crosses the Atlantic. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **E 16 Pump start 08:00 at 49° 30' N, 8° 27' W from ER bilge wells**
- D 13 from ER bilge wells
  14 Pump start 08:00
  15.1 49° 30' N, 8° 27' W
- E 16 Pump start 08:00 from ER bilge wells
- E 17 Pump start 08:00 at 49° 30' N, 8° 27' W

You have used the 6 m^3/hr oily bilge pump to pump the aft ER bilge well to the Bilge Holding Tank. The Bilge Holding Tank is listed in section 3.3 of the IOPP certificate. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **2.3 m^3 bilge water from aft ER well**
  Start 09:10, Stop 09:40
  To Bilge Holding Tank, 3.5 m^3 retained
- 2.3 m^3 bilge water from aft ER well
  Start 09:10, Stop 09:20
  To Bilge Holding Tank, 3.5 m^3 retained
- 2.3 m^3 bilge water from aft ER well
  Start 09:10, Stop 09:40
  To Bilge Holding Tank
- from aft ER well
  Start 09:10, Stop 09:40
  To Bilge Holding Tank, 2.3 m^3 transferred

Your Company has just taken over a ship from other owners. Which one of the following is the best way to decide which tanks MARPOL requires to be sounded weekly and recorded in the oil record book?

- **Sound the tanks listed on the IOPP certificate as sludge tanks**
- Sound all the tanks listed on the tank plan as being for waste oil
- Sound the same tanks as the previous operator's crew did
- Sound the tanks listed on the IOPP certificate as sludge tanks or bilge holding tanks

Your Company's policy is to sound all tanks daily and to record the results for the bilge tanks in the oil record book every week. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **Make an entry using operation Code I which includes the capacity and amount retained in each tank**
- Make an entry using operation Code C 11 which includes the amount retained in each tank
- Make an entry using operation Code C 11 which includes the capacity and amount retained in each tank
- Make an entry using operation Code I which includes the amount retained in each tank

Your ship is allowed to ballast No 2 Centre Diesel Oil Tank and did so after the fuel was used on the last voyage. The ship has now arrived in port and needs to deballast to shore reception facilities before bunkering for the next voyage. Based upon the recommendations in this training program which of the following is the correct operation code to use in the oil record book?

- **Code B**
- Code C
- Code D
- Code I

Your ship needs to de-bunker the remaining fuel from No 27 Bunker Tank so that it can be steamed out and surveyed. Based upon the recommendations in this training program which of the following is the correct operation code to use in the oil record book?

- **Code I**
- Code C
- Code G
- Code H

0219 – Oil Record Book, Part II

Version 2.0 (1411)

A Port State Control Officer (PSCO) wishes to remove your oil record book ashore as evidence in a possible pollution case. What should the master do about this?

- **Make copies of all the completed pages and get the PSCO to certify they are true copies**
- Make copies of all the completed pages and mark them with the ship's stamp
- Phone the Designated Person Ashore
- Refuse to let the PSCO take the oil record book

A small 'product tanker' and a 'tanker engaged in specific trades' are both discharging dirty ballast from their cargo oil tanks to a reception facility ashore. Which one of the following statements about their oil record book entries is true?

- **The tanker engaged in specific trades must have the entry signed by a shore official**
- Both ships must record the operation under code H
- The product tanker must record whether the ODME was used during the discharge
- The tanker engaged in specific trades must record the operation under code L

After Crude Oil Washing a cargo tank how is the washing time recorded in the oil record book?

- **The start and stop times are recorded as separate items**
- Only the washing time is recorded
- The start and stop times are recorded as a single item
- The start time and washing time are recorded

After which one of the following would you need to make a Code N entry in the oil record book part II?

- **An oil sheen is seen on the sea while discharging segregated ballast overboard**
- An oil sheen is seen on the sea while discharging engine room bilges overboard
- Oil is found on the surface of ballast water in a segregated ballast tank
- The engineer doing the annual ODME service reports that it requires a new sampling pump

Ballasting of No1 Centre cargo oil tank started at 9pm on May 19th and took 5½ hours. Based upon the recommendations in this training program which of the following is the correct way to record these times in the oil record book against operation Code E 19.2?

- **Start 19-MAY-2011 21:00 Stop 20-MAY-2011 02:30**
- Start 19-MAY-2011 9 pm for 5½ hours
- Start 21:00 Stop 02:30
- Start 9pm Stop 2:30am

During discharge of water from the slop tank your ship's ODME alarm operates because no ship position input is being received. This is traced to a faulty GPS and the backup GPS is connected within 30 minutes instead. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **M 70 0915 (1015 UTC)**
  71 0945 (1045 UTC)
  72 Primary GPS position output failed – switched to back up
- M 70 0915
  71 0945
  72 Position off
- M 70 0915 (1015 UTC)
  71 0945 (1045 UTC)
  72 Unknown
- O ODME Primary GPS position output failed
  switched to back up
  Position missing for 30 minutes from 0915

How are amounts of cargo loaded to be recorded in the oil record book part II?

- **In cubic metres, corrected to volume at 15 degrees Celsius**
- In cubic metres
- In metric tonnes, corrected to 15 degrees Celsius
- In US barrels

The flow rate sensor on the ODME has failed. Which of the following is the correct operation code for recording this?

- **Code M**
- Code J
- Code N
- Code O

When you take the oil record book to the master for his signature he points out that last week's discharge of slops to shore has not been recorded. Which one of the following is the best way to correct this mistake?

- **A code O entry must be made explaining that the following is a missed entry, underneath write the late entry**
- Destroy the receipt for the slops discharge and pretend it never happened
- The missed entry must be recorded in the next blank line
- The missed entry must be written on paper and stapled to the book in the correct date position

Which of the following best describes what should be recorded in the oil record book after Crude Oil Washing (COW) a cargo tank as item 17?

- **An explanation of why COW manual requirements were not met**
- Item 17 is not related to COW
- Remarks on how well the COW went
- Remarks on why COW was not carried out

Which of the following best explains why the washing line pressure is recorded in the oil record book after Crude Oil Washing (COW) a cargo tank?

- **To show that the COW manual requirements were met**
- To show that the cargo pump is working properly
- To show that the correct number of cleaning machines were used
- To show that the washing line did not leak

Which of the following is the correct format to be used for writing dates in an oil record book entry?

- **25-OCT-2011**
- 10.25.11
- 25/10/11
- OCTOBER 25 2011

Which of the following statements about a Clean Ballast Tank (CBT) tanker is true?

- **The Clean Ballast Tanks remain connected to the cargo oil system**
- Clean Ballast Tanks are completely segregated from the cargo oil system
- Clean Ballast Tanks are only found on ships built since 2001
- Oil is sometimes carried in Clean Ballast Tanks

Which one of the following best describes how the ship's positions in oil record book part II entries can easily be found to be false, even after some time has passed?

- **The positions in the oil record book part II do not match those in the bridge log book**
- The position of the ship is recorded in the bilge water separator control system and can be downloaded later
- The positions are written with East/West co-ordinates before North/South ones
- The positions of ships are now tracked by satellite

Which one of the following best describes the action to take if you realise the entry you have just made in the oil record is partially incorrect?

- **The incorrect part should be struck through with a single line in such a way that the incorrect entry is still legible. The incorrect entry should be signed and dated, with the new corrected entry following**
- The incorrect part should be erased and replaced with the new correct entry
- The incorrect part should be struck through repeatedly so that it cannot be read. The incorrect entry should be signed and dated, with the new corrected entry following
- The whole entry should be struck through with the new corrected entry underneath

Which one of the following best describes the number of signatures you would expect to find on one page of an oil record book when it is completed?

- **One for every entry plus the master's signature**
- One
- One for every entry made
- One for every line in the book

Which one of the following best describes what should be done with an oil record book when it has been filled up?

- **It must be kept onboard for 3 years from the date of closure**
- It must be kept ashore by the company for 3 years from the date of closure
- It must be kept onboard for 5 years from the date of opening
- It must be kept onboard until the next oil record book is filled up

Which one of the following best describes what should be written under each oil record book entry by the officer in charge of the operation?

- **His signature, name, rank and the date**
- His signature and the date
- His signature, name and the date
- His signature, rank and the date

Which one of the following best describes what you should do if your old oil record book is full and the only available new one is found to contain out of date MARPOL operation codes?

- **Use the new book after replacing the out of date codes with a copy of the current ones**
- Do not make any oil record book entries until an up to date book has been sent
- Make oil record book entries in the bridge log book until an up to date book has been sent
- Use the new book and use the out of date codes for entries in it

Which one of the following best describes where you would expect to find a receipt for cargo slops that were discharged ashore?

- **With the oil record book part II**
- In the master's cabin
- In the master's office
- Pinned to the cargo control room notice board

Which one of the following best describes why MARPOL requires ships to maintain an oil record book part II?

- **To provide the authorities with a record of the handling of cargo oil and residues on the ship**
- To give port State control officers something to find fault with
- To provide the authorities with a record of the handling of liquid wastes on the ship
- To warn ships' crews when the slop tank is nearly full

Which one of the following describes the best way of making sure that the quantity of slops discharged to reception facilities matches the reduction in slops retained on board?

- **Check the quantity of slops on board immediately before and after discharge ashore and make sure the reception facility receipt matches the change**
- Check the levels in the cargo tanks next to the slop tank have not changed during the discharge of slops
- Check the quantity of slops on board at the next weekly soundings and make sure the reception facility receipt matches the change
- Make sure you get a detailed receipt for the slops from the reception facility

Which one of the following describes the best way to find out which tanks have been designated as slop tanks on your ship, as far as oil record book entries are concerned?

- **Check the IOPP certificate**
- Check the capacity plan
- Check the cargo oil system drawings
- Check the tank plan

Which one of the following is recommended as the first thing to do when starting a new oil record book?

- **Check that it contains the current operation codes**
- Check that it has a blue cover
- Check that it has a green cover
- Write the ship's name on it

Which one of the following is the maximum fine that could be imposed in Germany for an incorrect entry in the ship's oil record book according to this program?

- **25,000 Euros**
- 2,000 Euros
- 200 Euros
- 5,000 Euros

Which one of the following is the maximum penalty that could be imposed in the United States for a deliberately false entry in the ship's oil record book according to this program?

- **$250,000 and 5 years in jail for each violation**
- $50,000
- $50,000 and 1 year in jail for each violation
- 1 year in jail

Which one of the following operation codes would be the best to use to record the fact that the overboard discharge valve from the slop tank has been sealed while the ship passes through the Mediterranean MARPOL Special Area?

- **Code O**
- Code I
- Code J
- Code M

You have just finished Crude Oil Washing the forward half of No 2 Centre while discharging at the SupaOil berth in Singapore.

Which of the following is the correct operation code?

- **Code D**
- Code C
- Code G
- Code J

You have just finished Crude Oil Washing the forward half of No 2 Centre while discharging at the SupaOil berth in Singapore. Based upon the recommendations in this training program which of the following is the correct way to start this entry in the oil record book?

- **D 9 SupaOil terminal Singapore**
  10 2C Forward section
- D 9 1º 14'N, 103º 55'E
  10 2C Forward section
- D 9 1º 14'N, 103º 55'E
  10,11 3 forward machines
- D 9 SupaOil terminal Singapore
  10,11 3 forward machines

You have just finished discharging all of the crude oil in 3P and all but 2,000 m^3 of 3S ashore to the Staten Oil terminal in New York. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **C 6 Staten Oil terminal, New York**
  7 3P&S
  8 3P – Yes 3S – 2,000 m^3 retained
- C 6 New York
  7 3P&S
  8 3P – Yes 3S – 2,000 m^3 retained
- C 6 Staten Oil terminal, New York
  7 3 wings
  8 2,000 m^3
- C 6 Staten Oil terminal, New York
  7 3P&S
  8 2,000 m^3 retained

You have just finished loading sea water ballast into No 2 Port and Stbd oil cargo tanks while at sea on passage. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **E 18 Start 49º 45'N 13º 10'W, Stop 50º 55'N 13º 02'W**
  19.1 2P&S
  19.2 Start: 0300 Stop: 0445
  19.3 1,180m^3 to each
- E 18 Gibraltar – Le Havre
  19.1 2P&S
  19.2 Start: 0300 (0200UTC) Stop: 0445 (0345UTC)
  19.3 1,180m^3 to 2P 1,180m^3 to 2S
- E 18 On passage Gibraltar – Le Havre
  19.1 2P&S
  19.2 Start: 0300 Stop: 0445
  19.3 1,180m^3 to each
- E 18 Start 49º 45'N 13º 10'W,
  19.1 Stop 50º 55'N 13º 02'W
  19.2 Start: 0300 Stop: 0445
  19.3 1,180m^3 each to 2P&S

You have just transferred all the remaining cargo in 1C to top up 5C, adding 125 m^3 to the 450m^3 already in 5C. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **B 4.1 From 1C**
  4.2 To 5C 125m^3 5C = 575m^3
  5 Yes
- B 4 From 1C to 5C 125m^3 5C = 575m^3
  5 Yes
- B 4 From 1C to 5C 125m^3
  5 Yes 5C = 575m^3
- J 55 From 1C to 5C
  56 125m^3
  57.2 125m^3

You have to make an oil record book entry relating to the loading of a cargo of 350 m^3 of diesel fuel. Which of the following is the correct operation code to use in the oil record book?

- **Code A**
- Code C
- Code N
- Code P

You see that a previous oil record book entry shows that dirty ballast was discharged overboard (item 35) but also to the slop tank (item 39). Which of the following best explains this?

- **The oily residues were transferred to the slop tank after the clean water was discharged to the sea**
- Item 35 should have read 'N/A' as dirty ballast would have to be pumped to the slop tank, not into the sea
- Item 39 should have read 'N/A'
- Items 35 and 39 should have read 'N/A' as dirty ballast would have to be pumped to reception facilities ashore

Your ship has discharged its slops to shore reception facilities. Based upon the recommendations in this training program which of the following is the correct way to record this in the oil record book?

- **J 55 Slop Tks P & S**
  56 Trans P=37.4m^3 S=10m^3 Ret P=1.2m^3 S=3.6m^3
  57.1 Southern Tank Cleaners, New Orleans 47.4 m^3
- C 6 Southern Tank Cleaners, New Orleans
  7 Slop Tks P & S
  8 No - P=1.2m^3 S=3.6m^3
- J 55 Slop Tks P & S
  56 Trans P=37.4m^3 S=10m^3 Ret P=1.2m^3 S=3.6m^3
  57.4 to Southern Tank Cleaners, New Orleans 47.4 m^3
- O Discharged Slop Tks to Southern Tank Cleaners, New Orleans
  P=37.4m^3 S=10m^3

Your ship has discharged water from the slop tank into the sea. Which of the following best describes why it is a good idea to record the ship's times and UTC times in the oil record book for this operation?

- **The oil record book times should match those stored in the ODME and they are normally set to UTC**
- All times in oil record books should be recorded as local and UTC times
- All times in oil record books should be recorded as UTC times
- Pollution spotter planes record video with UTC time stamps

Your ship has discharged water from the slop tank into the sea. Which of the following best describes why there are 2 items for recording discharge flow rates in the oil record book for this one operation?

- **It is expected that the discharge rate will be slowed as the oil interface reaches the outlet pipe**
- It is expected that the discharge rate will be faster as the oil interface reaches the outlet pipe
- One is the maximum discharge rate and the other the average flow rate
- Only one of the items is recorded, depending on whether bulk flow is used or not

Your ship is a 'tanker engaged in specific trades'. Which of the following describes the best way of finding out where your ship is allowed to trade?

- **Check the IOPP certificate supplement**
- Ask the chief engineer
- Check the Class certificate
- Check the oil record book

Your ship's oil cargo tanks have been cleaned by washing with seawater and a chemical cleaning additive before drydock. Which of the following best describes how the use of chemical additives should be recorded in the oil record book?

- **It should be recorded as part of the method of cleaning (item 30)**
- It is not recorded
- It should be recorded as a separate Code O operation
- It should be recorded as part of the remarks (item 17)

0220 – Galley Operations 1, Workplace and Equipment Safety

Version 3.0 (4011)

(Not counting a vessel that carries special extinguishers for fires involving cooking fat and oil), which two classes of fire would the extinguishers normally found in a galley be capable of controlling?

- **Flammable liquids and electrical fires.**
- Electrical equipment and combustible metals.
- Flammable liquids and flammable gases.
- Ordinary combustibles and flammable gases.

A walk-in refrigerator has been out of service for an extended period of time and needs cleaning. When standing near the open door, people feel a little dizzy and there is a suspicion that there may be a leak from the cooling system, requiring a repair. What are you going to do?

- **Notify the ship's Safety Officer, who will perform a Risk Assessment and advise on the safety precautions necessary.**
- Go inside and check but ensure that you get someone else to go inside the space with you.
- Hold your breath as you enter, locate the gas leak and stick some tape over the hole.
- Nothing special, it's only a refrigerator and everything will be OK.

An electrical food processing unit in the galley has multiple attachments. Can they be used for any task?

- **No, attachments should be used for the designed purpose only.**
- Sometimes - it depends on whether they have the same colour coding.
- Yes, but only if they fit the same socket on the equipment.
- Yes, if it looks like they might do the job.

As best practice, how often should the tip and cog wheel of a tin opener be cleaned?

- **After every time it has been used.**
- Once a week.
- Once aday.
- The tip and cog of a tin opener are automatically cleaned the next time it is used.

For an ordinary cargo ship at sea, which of the following policies would generally apply to the watchman and his safety rounds during the night?

- **The watchman should include the galley in his night-time safety rounds.**
- Provided the cook reports to the bridge that all is safe at the end of the day, there is no need to include the galley in the night-time safety rounds.
- The galley is specifically excluded from a watchman's night-time safety rounds.
- The galley should be locked shut and the watchman will not include it in his night-time safety rounds.

How might you safely and practically prevent spillage of hot liquid from a pan on a galley range in bad weather?

- **Fill the pan to a lower level.**
- Dry cook the food; do not use any water or fat.
- Place absorbent cloths around the pan to catch any spillage.
- Use a tight-fitting pan lid.

How should a knife be washed and dried?

- **With the cutting edge down and away from the hand.**
- Knives should only be wiped with a thick cloth, never washed and dried.
- With the blade in the hand and handle facing away from you.
- With the cutting edge up and towards the hand.

How should knives be laid down on a work surface?

- **With the cutting edge flat and not facing upwards.**
- Knives should never be laid down on a work surface.
- On top of a cloth laid on the work surface.
- With the cutting edge facing upwards.

How should pans with long handles be arranged on a galley range during cooking?

- **The handles should be turned inwards, so they are not caught by people passing by.**
- If a pan with a long handle is to be used, it should be the only pan on the range at that time.
- Pans with long handles should only be used at the back of a range.
- The handles should be turned outwards, so that the pans can easily be lifted.

How should wooden cutting boards be arranged immediately after they have been washed?

- **They should be stacked in such a way that air can flow over all surfaces.**
- They should be placed in the top of a warm oven for five minutes.
- They should be stacked flat, one on top of the other.
- They should be stacked vertically, packed close together.

How should you pick up a knife from a bench?

- **By the handle, with the point of the blade facing away from you.**
- By sliding it to the edge of the workbench and picking it up by whichever part is overhanging the edge.
- By the handle, with the point of the blade facing towards you.
- It doesn't matter really and depends how it was left by the last person who used it.

If a permanently-mounted electrical mixer breaks down, what should the catering staff do about it?

- **They should report the defect through the proper channels, isolate the equipment from the power supply and place a safety notice on the unit.**
- The catering staff are fully responsible at all times for any repairs necessary to galley equipment and should therefore make it serviceable again at the earliest opportunity.
- They should immediately order a replacement unit for delivery in the next port.
- They should try to adapt another piece of equipment to perform mixing tasks.

If so equipped, is it always necessary to fit guards to electrical equipment when in use?

- **Yes, they are there for the purposes of safety.**
- No, I only need to fit them when other people are working close by.
- No, if I know what I am doing, they are unnecessary and get in the way.
- Sometimes, but only if I think things could get dangerous.

If the plate of a microwave oven is overloaded with food, which of the following will result?

- **The cooking will be uneven and there will be 'cold-spots'.**
- The food in the middle will be burned.
- The food on the top will be burned.
- The pile of food will 'explode' as the middle starts to boil.

In general, what speed setting should I use on galley electrical equipment?

- **That required for the task and as recommended in the operating instructions.**
- I should always start with the fastest and reduce the speed if necessary.
- The fastest possible, in order to save time.
- The slowest.

In what common cleaning material is Phenol (carbolic acid) found in significant quantities?

- **Disinfectant.**
- Dishwasher additive.
- Floor polish.
- Liquid soap.

In what particular part of a knife are food and grease particles most likely to become trapped?

- **The groove where the blade is set into the handle.**
- Food and grease will never become trapped on a properly sharpened knife.
- On and around the point of the knife.
- On the handle.

Manual meat saws are divided into two types. The bow saw is one type – what is the other?

- **Tenon saw.**
- Tempering saw.
- Tendon saw.
- Tension saw.

The large walk-in freezer on your ship has a door secured by a padlock. Which of the following is a good policy to follow when working inside?

- **Carry the padlock with you, so that nobody thinks it has been forgotten and re-secures the door with you still inside.**
- Leave a notice in the galley that you are down below in the freezer.
- Only work in the freezer where you can keep watch on the door.
- Put the padlock on the deck, close to the door, so that it doesn't get lost.

Trussing needles should be stowed in a labelled box after they have been used and cleaned. This is to prevent injury and for what other purpose?

- **To prevent contamination that might occur if they are left around close to food or food waste.**
- To comply with US Food and Drug Administration guidelines.
- To prevent them being mistaken for canvas sewing needles.
- To stop them going rusty.

What colour is generally used on tools and equipment associated with the preparation of raw fish?

- **Blue.**
- Green.
- White.
- Yellow.

What commonly happens to glassware when hot, such as after cleaning in a dishwasher?

- **It becomes brittle and will easily crack.**
- Food particles will stick to it.
- It becomes soft and may go out of shape.
- The surfaces become cloudy and scratched.

What is a fire blanket used for?

- **To smother a liquid fires in a pan, such as burning cooking oil.**
- To hang around an area in which an open flame is burning, such as from a barbeque grill, to safely isolate it from other food preparation areas.
- To protect a cook from a dish that is "spitting" hot oil, by wrapping it around his upper body.
- To wrap around a casualty who has been burned.

What is a palette knife normally used for?

- **Spreading, filling and other baking related tasks.**
- General purpose cutting where a wide flat cut is required.
- Removing fish from the bone.
- Working with sliced cooked meat.

What is the main reason why a knife must be cleaned after sharpening with a "butcher's steel"?

- **To remove grease and small iron particles that may have stuck to the blade.**
- Because the manufacturer's instructions require the knife to be cleaned.
- Because the steel will have become hot during sharpening and cleaning it with water will also cool it down.
- To seal the newly sharpened edge.

What is the major safety issue associated with using a blunt knife?

- **Extra pressure is required to achieve a cut and this increases the chance that it may slip.**
- More time is required to achieve a cut.
- The cut will not be neat and clean and particles of bone and sinew may be drawn into the flesh.
- The extra effort required can put undue strain on the muscles in the cook's forearm.

What is the safest policy in respect of washing knives?

- **Be responsible for washing your own knives; never leave them in the sink.**
- Always leave them in the sink for the steward to wash.
- Do not wash knives, only wipe them.
- Leave them on a work surface; it will be obvious which ones need washing.

What kind of marking might be found on the deck directly beneath a storing hatch?

- **An area marked with yellow and black diagonal stripes.**
- A blue circle, containing the words “Danger Above”
- A red square containing the word “Caution” in black letters.
- There is not usually any special marking on the deck under a storing hatch.

What risk exists if food with a high water content is cooked in a deep fat fryer?

- **Violent bubbling and spitting may occur as the food is added to the hot oil.**
- The food may be scalded and discoloured when removed after cooking.
- The food may break apart when added to the hot oil.
- The water will cause the hot oil to solidify.

What should be on the deck immediately outside the door to a walk-in refrigerator or freezer space?

- **Non-slip matting.**
- A pressure sensor.
- An area marked with yellow paint.
- Plastic sheeting.

What should the fixed blades of slicing machines be cleaned with?

- **A sanitizer spray.**
- A dry cloth.
- A synthetic fibre brush.
- Nothing - the blades of slicing machines are self-cleaning and need no further attention.

What should you rest meat on when cutting it with a cleaver or chopper?

- **A thick, heavy-duty cutting block.**
- A plastic sheet, laid over a work bench.
- A thin, red, plastic cutting board.
- The deck.

What small electrical safety device could be fitted to a coffee maker, to ensure that it does not burn dry during the night?

- **A timer-switch.**
- A flashing light.
- A transformer.
- An audio alarm.

When cleaning and servicing a slicing machine, which of the following is a critical safety requirement?

- **That the machine is switched off and isolated before work starts.**
- That I am wearing rubber gloves.
- That I have anti-static (natural fibre) cleaning brushes available.
- There are no special or critical safety requirements. This is a routine task.

When filling pans in a galley at sea, what must always be considered?

- **The prevailing weather conditions and movement of the ship.**
- The amount of fresh water available.
- The time required for cooking the food in the pan.
- The type of metal used in the pan.

When not in use, where should a knife be kept?

- **In a drawer, rack, block, roll or box.**
- In a sink or dish-washer.
- In sight, on a work surface.
- In the cook's cabin.

When the ship is to leave port and bad weather is expected, which of the following precautions apply?

- **All loose equipment must be safely stowed, benches cleared and doors secured shut.**
- All doors must be closed and the lights switched off.
- All electrical equipment must be switched off, sinks drained and unused food disposed of.
- The cook must be on stand-by in the galley.

When using a large steamer unit, what is it essential to do before opening the lid or door after cooking?

- **Pressure must be safely released.**
- All free water must be drained from the unit.
- The unit must be switched off and isolated.
- Warn everyone in the vicinity that you are about to open the steamer.

Which of the following actions is considered essential before the oil is drained from a deep fat fryer?

- **It must be allowed to cool first.**
- A layer of hot water must be added to loosen any solid residues on the bottom of the fryer.
- Everyone in the vicinity of the fryer must be warned that the draining operation is about to take place.
- The oil must be heated up to at least 70o Celsius, to ensure it will flow easily during draining.

Which of the following is most likely to have an Equipment Operating Notice next to it?

- **A meat slicing machine.**
- A coffee maker.
- A ready-use refrigerator.
- A rice cooker.

Which of the following precautions should be taken when using a chemical oven cleaner?

- **The area should be well-ventilated.**
- The flooring around the oven should be flooded continuously with fresh water.
- The oven should be warmed until the surfaces are between 60°-70° Celsius.
- The surfaces should be pre-treated with bleach.

Which part of a knife is referred to as its "toe"?

- **The tip or point.**
- The inner end of the handle.
- The thickest part of the blade.
- The unsharpened side of the blade.

Why can all glassware not be cleaned in a dishwasher?

- **Because some glassware is not dishwasher proof.**
- Because some cleaning programmes depend on there being a mixture of glassware, crockery and cutlery in the dishwasher at the same time.
- Because the racks in dishwashers are not designed to handle only glassware.
- Because the salt additive is extremely abrasive on glass that is low in silicate content.

Why do exhaust ducts above galley ranges require regular inspection and maintenance?

- **Because the build-up of fat deposits is a fire hazard and it restricts the extraction of polluted air.**
- Because food will not cook properly on the galley range if the vents are unclean.
- Because the emissions from a galley exhaust duct are controlled under MARPOL Annex VI.
- Because the fatty deposits create a poisonous atmosphere around the galley range.

Why must all open wounds, once treated, then be covered with a suitable protective dressing, such as a waterproof band-aid or special sleeve or glove?

- **To further protect the casualty against secondary infection and to protect others against blood-borne bacterial infection.**
- So that everyone else knows the casualty has had an accident.
- So that the casualty can resume working as soon as possible.
- To protect the wound against hot water.

Why should a wet cloth never be used for handling a hot pan?

- **A wet cloth will conduct heat quickly and may scald the hands of the user.**
- A wet cloth may drip into other pans on the range.
- A wet cloth may make the pan handles slippery.
- A wet cloth will generate steam and restrict the visibility of the user.

Why should you NOT remove items, such as scissors or tape from the galley first aid kit for everyday purposes?

- **Because you may cause delay when they are needed for a genuine accident.**
- Because there will be trouble when an inspection is carried out.
- Because they are of types dedicated to first aid and will be useless for other purposes.
- Because they may be costly to replace in some ports.

You accidentally cut the end of your finger whilst preparing food in the galley. Should you report it?

- **Yes.**
- It depends on what the ship's Safety Officer says.
- No.
- Only if there is a problem with it afterwards.

You and the cook are both preparing food at the same bench. The cook is peeling vegetables with a small hand knife. You want a cloth which is on the other side to where the cook is working. Which of the following would be considered an UNSAFE act?

- **Stretching across the bench in front of the cook to get the cloth.**
- Asking the cook to change places with you.
- Asking the cook to stop peeling the vegetables and pass the cloth to you.
- Walking round behind the cook to get the cloth.

You are newly signed on as a 2nd cook. What colour coding system should you follow for the tools and equipment used in food preparation?

- **The one that is defined in the ship's Safety Management System.**
- I will not follow any particular system.
- The one defined by the US Food and Drug Administration.
- The one that you used at College when you did your training.

You have some plastic containers, containing pre-prepared meals, sealed with a cellophane film lid. They are to be heated up in your microwave oven. What precautions must be followed to achieve this safely?

- **The lid must first be pierced.**
- The contents must be transferred to fully open dishes before cooking in a microwave oven.
- They must be inserted into the oven upside-down.
- They must only be placed around the edge of the cooking plate, never in the middle.

Your ship has received stores, some of which have been packed in dry ice. What is the significant risk if the packages are left in a storeroom at ambient temperature for any length of time?

- **Dry ice is carbon dioxide in its solid state. If significant quantities of gas are released as it melts, then oxygen depletion in the storage space is a possibility.**
- Dry ice is carbon monoxide in a frozen condition. If it is allowed to melt, the liquid it releases is corrosive and can damage human tissue it comes into contact with.
- Dry ice is ice that has been created in a controlled, dry environment. If it is placed in a warm atmosphere, it becomes brittle and can shatter, throwing sharp needles of glass several metres.
- Dry ice is super-cooled nitrogen. If it melts, then the atmosphere can become positively charged and an increased explosion risk will develop.

0221 – Galley Operations 2, Hygiene

Version 2.0 (1641)

A Garbage Management Plan will indicate the locations and intended use of stations onboard for the storage of garbage. Which other piece of information must the Plan specify?

- **Their capacity.**
- The maximum size of garbage item permitted.
- The type of seal required on any containers.
- Their colour coding and labelling requirements.

A person designated in a Garbage Management Plan is responsible for which of the following?

- **Implementing the procedures described in the Plan.**
- Arranging garbage collection contracts in foreign ports.
- Maintaining the records in the Garbage Record Book.
- Operating the garbage handling equipment.

According to the table displayed in this program, how long will a plastic bottle take to degrade in the ocean?

- **450 years.**
- 1000 years.
- 250 years.
- 450 days.

Annex V of MARPOL 73/78 (as amended) requires placards to be displayed. What should these placards indicate?

- **Garbage disposal requirements.**
- How plastic should be disposed of at sea.
- The daily permitted garbage limit for every individual on board.
- Who the designated person is for completing entries in the Garbage Record Book.

Apart from ships of their own flag, which ships might a country's national food safety regulations apply to?

- **Foreign ships trading in that country's waters.**
- All ships of 400 gross registered tons and over.
- Those ships which have been in drydock since their last visit to the port in question.
- Those ships which have previously been detained.

Besides listing the requirements in respect of garbage handling by vessels, what else does Annex V of MARPOL 73/78 (as amended) prescribe?

- **The provision of facilities for the reception of garbage from ships at ports and terminals.**
- The certification requirements for Responsible Persons under shipboard Garbage Management Plans.
- The regulations controlling the disposal of oily waste by vessels.
- The regulations governing the operation of incinerators in ports and harbours.

Certain regions of the seas are subject to particular controls under MARPOL 73/78 (as amended). What are they called?

- **Special Areas.**
- Controlled Regions.
- Designated Sea Areas.
- MARPOL Areas.

For how long must a Garbage Record Book be kept after the date of the last entry?

- **2 years.**
- 5 years.
- 6 months.
- Once completed, it may be disposed of.

For which of the certificates listed below, if applicable, will a ship's galley and storerooms be inspected?

- **The de-ratting certificate.**
- The cargo ship safety certificate
- The loadline certificate.
- The safe manning certificate.

How can minor stains on storeroom shelves used for foodstuffs best be removed?

- **By scrubbing with a suitable antibacterial cleaner.**
- By gentle washing with warm fresh water.
- By painting over the stains.
- By soaking in bleach.

How do individual countries apply food safety regulations to ships?

- **Through national legislation.**
- Countries do not act independently with health and safety regulations. They apply them through international agreements.
- Food safety regulations only apply to places ashore.
- Through the IMO.

How may a flying beetle be recognised?

- **By its horny wing cases, which cover the wings when not in use.**
- By its bright green eyes.
- By its large rear legs.
- By the sour smell that it gives off.

If a ship lands garbage at a port, what should the Master obtain?

- **A receipt specifying the amount of garbage landed.**
- A certificate of compliance with port sanitary regulations.
- A customs declaration for the estimated amount of garbage landed.
- An invoice for the measured amount of garbage landed.

If gloves are to be worn for food preparation, what type should they be?

- **Powder-free, disposable latex or vinyl gloves.**
- Colour-coded rubber gloves.
- Heavy-duty leather gloves.
- Lightweight cotton gloves.

If spraying to control cockroaches, which of the following areas should you concentrate on?

- **Entry points, such as conduits and pipes.**
- Around doors and windows.
- Exhaust ducts above galley ranges.
- In and over machinery, near the motors in particular.

If your ring will not come off your finger, what alternative precaution may be taken to protect the food with which you are working?

- **I can wear powder-free disposable latex or vinyl gloves.**
- I can ensure the food, once prepared, is heated up to a minimum of 70'C to kill any bacteria present.
- I can use wooden or plastic utensils to work with the food.
- I can wrap cling-film around my hands.

In what language should the Garbage Management Plan be written?

- **The working language of the crew.**
- English, Spanish, Chinese and Russian.
- It doesn't matter what language is used, provided the ship has a certified Plan.
- Versions should be available for all languages spoken on board, plus English.

On which of the following occasions is it most important that you wash your hands from a hygiene point of view?

- **After blowing my nose.**
- After finishing the service of a meal.
- Before collecting vegetables from the storeroom.
- Before operating the incinerator.

There are generally three elements to a Garbage Management Plan. Besides 'Recycling' and 'Disposal', which of the following is the third element?

- **Reduction at Source.**
- Green packaging.
- Incineration.
- Return to Sender.

What English term is used to describe the process by which garbage breaks down and dissolves in the sea?

- **Degrade.**
- Deflate.
- Delaminate.
- Destabilise.

What danger does a studded finger ring present when preparing food?

- **The clasps can hold bacteria and the stones may even become loose and drop into the food.**
- A gold ring may introduce a metallic taste into the food.
- Certain precious stones used in rings are poisonous.
- The metal in the ring may produce an electric shock if it accidentally contacts with the metal casing of a microwave oven.

What is the larger cousin of the common house fly?

- **The blow fly.**
- The blue fly.
- The fruit fly.
- The sand fly.

What is the main risk that may arise if you wear your catering department uniform in the dayroom or ashore?

- **The uniform clothing may pick up germs that then get transferred to the galley when you return to work.**
- You might get in trouble with the authorities.
- You will get asked for food when you are off duty.
- You will transfer germs if you accidentally brush up against another person.

What is the minimum time for which hands should be washed, as recommended by the US CDC?

- **20 seconds**
- 10 seconds
- 45 seconds
- 5 seconds

What kind of footwear is acceptable for working in the galley and storerooms?

- **Heavy-duty, with a non-slip sole and reinforced toecap, covering the foot fully.**
- Comfortable, casual, soft leather and low cut.
- Easily removed, well-ventilated, flip-flop or sandal style.
- Smooth-soled, open-backed clogs.

What kind of lamp is generally used in the electronic insect device?

- **Ultra-violet.**
- Infra-red.
- It doesn't matter what kind of lamp is used.
- Night-vision.

What material should galley work clothing be made of?

- **Cotton.**
- Nylon.
- Plastic fibres.
- PVC

What should be located next to the wash basin in the galley?

- **A soap-dispenser and hand-dryer or disposable towels.**
- A mirror.
- A rack for a razor and toothbrush.
- A warning sign.

What virus do the faeces of the common house fly carry?

- **The Salmonellae virus.**
- Clostridium botulinum.
- E-coli 0157
- Staphylococcal enterotoxin.

When can you smoke in the galley?

- **Smoking is prohibited in the galley at all times.**
- At any time, provided no food preparation is underway.
- Between 1900 and 0600 hours only.
- Once food service is finished.

Which Annex of MARPOL 73/78 (as amended) governs the handling of waste aboard ships?

- **Annex V**
- Annex IV
- Annex VI
- Annex VII

Which of the following is a description of a compactor?

- **A machine designed to compress garbage into manageable shapes for subsequent disposal.**
- A machine designed to grind down waste material to a size less of than 25mm and thereby pass it through a mash with openings of that diameter.
- A processing which will cut and shred plastic, then compress and melt it into solid shapes for subsequent disposal.
- An apparatus designed for the combustion of waste materials through high-temperature thermal treatment and by which waste will be reduced to ash.

Which of the following is a positive value of a properly maintained Garbage Record Book?

- **It may be used to demonstrate compliance by the ship in the event of a suspected pollution incident.**
- It may be used to demonstrate compliance with a QA system in the event of an oil pollution incident.
- It may be used to verify the validity of the certificates of the catering staff.
- It provides reference to operating instructions for the garbage incinerator.

Which of the following is the Master required to do under flag State regulations?

- **To inspect the galley, storerooms and dining areas at regular and frequent intervals.**
- To confirm that all Catering personnel are in possession of MLC2006 Hygiene Compliance certificates.
- To inspect all catering facilities at least once every three months.
- To verify, by observation and within 48 hours, that a newly signed-on cook can produce meals of an acceptable quality and quantity for his ship's crew.

Which of the following statements correctly applies to a container of used cooking oil?

- **Disposal into the sea is prohibited.**
- It can be disposed of into the sea if mixed with other forms of garbage.
- It can be disposed of into the sea if the vessel is more than 12 miles from the nearest land.
- It can be disposed of overboard without restriction.

Which organisation operates the Vessel Sanitation Program?

- **The US Centers for Disease Control and Prevention.**
- The Food and Drug Administration.
- The Norwegian maritime Directorate.
- The UK Maritime and Coastguard Agency.

Which ships are required by MARPOL 73/78 (as amended) to operate a Garbage Management Plan?

- **All ships of 100 grt and above and all ships certified to carry 15 persons or more.**
- All ships of 500 grt and above and all ships certified to carry 12 persons or more.
- All vessels certified to trade in coastal or international waters.
- All vessels where catering operations are carried out on board.

Who must countersign each page of the Garbage Record Book?

- **The Master.**
- The Senior Catering Officer.
- The Ship Safety Officer.
- The Superintendent.

Why is plastic considered to present the greatest danger to marine life?

- **Because of the time it takes to degrade.**
- Because it becomes poisonous if left in salt water over six months.
- Because it floats.
- Because it is difficult to digest.

Why should trousers worn in the galley be loose fitting around the lower legs?

- **To prevent or lessen injury if hot liquids are accidentally spilled.**
- Because that is the fashion style adopted by all seafaring cooks.
- Because ventilating the lower legs will lessen the chance of cramp.
- Because ventilating the lower legs will lessen the chance of heat rash.

Why should you wear a hat or hair net when preparing food?

- **To prevent odours being absorbed and loose hair or dry skin falling as I work directly over food.**
- Because it is required by the STCW Code.
- So that other crew members can easily identify me.
- To stop hair getting into my eyes.

You are a steward. The cook is working in the galley, preparing food. Can you use an insect aerosol spray at the other side of the galley while he is working?

- **No, the spray might be carried across and contaminate the food.**
- Yes, but only if the extractor fan is switched on.
- Yes, killing any insects is the priority.
- Yes, provided I ask him first.

You are the Steward. You are suffering from a cut on the third finger of your right hand, caused by a staple which was sticking out from a carton of tins in the storeroom. The wound has not closed over and continues to bleed slightly. What should you do?

- **I should report it, have it treated, then covered with a waterproof band-aid and a glove.**
- I will carry on and work as best I can. I will try to keep my injured finger clear of any food.
- I will continue to work, but will wash my hands more frequently than usual.
- The important thing is the dangerous carton. I will get something to knock the staple flat.

0222 – Galley Operations 3, Food handling

Version 1.0 (2049)

According to the Marine Guidance Note quoted in this training module, deep freeze units should be maintained at or below which of the following temperatures?

- **-18°C (0°F)**
- -32°C (-26°F)
- 0°C (32°F)
- 18°C (64°F)

According to the standard quoted in this training module (UK MCA MGN 61) what is the minimum temperature to which those foods suitable for re-use must be re-heated?

- **75°C (167°.F)**
- 100°C (212°F)
- 50°C (122°F)
- 63°C (145°F)

As quoted in this training module, at what minimum temperature should 'hot' food be kept?

- **63°C (145°F)**
- 100°C (212°F)
- 50°C (122°F)
- 72°C (162°F)

Besides contaminated food, what else may be a common and direct source of viral food poisoning on board ship, as described in this module?

- **The water supply**
- The cutlery and glassware
- The ovens
- The refrigerators and freezer stores

How can wastage best be prevented with prepared cold dishes?

- **Part of the dish can be kept back in the refrigerator or coldroom until needed to replace food that has been consumed**
- Immediately freeze half of the dish when prepared and defrost it again in the microwave immediately before service
- Keep the dish covered during service, to keep it cool
- Only make a limited amount and accept there may not be enough to serve everybody

How might you best determine if a joint of meat is cooked right through?

- **By using a meat thermometer**
- By allowing between 1 and 2 hours after the outside of the joint is seen to start changing colour
- By cutting the joint in half and tasting a piece from the middle
- By ensuring that the oven is set at a minimum of 180°C 356°F

In recent years, what type of food poisoning has been particularly associated with outbreaks on cruise ships?

- **Viral**
- Bacterial
- Mould related
- That caused by the improper use of chemicals

In which one of the following can the Escherichia coli (E-Coli) bacteria commonly be found?

- **Contaminated water**
- Cooked meats
- Pasteurised milk
- Processed vegetables

Is it permitted to mix together batches of raw and cooked minced beef in the same dish?

- **No, raw meat and cooked meat should never be mixed together**
- Yes, but only if the mixture is to be heated immediately
- Yes, provided more than half is the cooked portion
- Yes, this is a good way to prevent wastage

Is it safe to eat food that has reached its 'use by' date?

- **No**
- Yes, provided it looks alright
- Yes, provided it smells alright
- Yes, so long as it is thoroughly cooked before serving

Prior to cooking, which one of the following is the usual technique employed, to ensure raw beans are safe to eat?

- **They should be thoroughly soaked**
- They should be dry roasted for at least 3 minutes
- They should be salted and left for at least one hour
- They should be stored in the refrigerator overnight (or for a minimum of 12 hours)

The 'Staphylococcus Aureus' bacterium is commonly associated with cream-filled cakes and pies and other dairy products. Improperly chilled foods are also often a source. By which one of the following additional routes may the infection also be spread?

- **Bacterial infections originating in the noses and throats of food handlers or cuts on their skin**
- Contaminated water
- Cooked meat, which has been allowed to cool and is later re-used in another meal
- Overheated milk

To prevent the growth of bacteria in food, we must ensure that food is maintained outside a temperature range known as the 'danger zone'. What are the generally accepted limits of that zone, as quoted in this training module?

- **5°C to 63°C (41°F to 145°F)**
- 0°C to 100°C (32°F to 212°F)
- 10°C to 75°C (50°F to 170°F)
- 5°C to 75°C (41°F to 170°F)

What are 'mycotoxins'?

- **Poisonous substances, often resulting in liver or kidney disease, produced by some moulds under certain conditions**
- Highly toxic reef-living fish, of which the barracuda is an example
- Poisonous plants of the tomato family
- This is the name given to the form of viral food poisoning which accounts for approximately one third of the cases recorded in Western Europe

What is the maximum period that both raw and cooked perishable foods should be left at room temperature?

- **2 hours**
- 24 hours
- 30 minutes
- 6 hours

What is 'Ciguatera'?

- **The name given to food poisoning which results from eating contaminated fish**
- A Spanish herb commonly used in fish dishes
- A West-Indian cooking technique involving heated stones
- The flowery green leaves on the tops of carrots

What kind of arrangement should exist for handling 'incoming' dirty plates and cutlery?

- **They should be kept completely separate from 'outgoing' food and clean plates and cutlery**
- All dirty plates and cutlery should be left on the table until the end of the meal
- They can be brought into the galley and placed on the work benches, alongside the outgoing food and plates
- They should be given a quick wipe and re-used

What reason can the growth of mycotoxins usually be traced back to?

- **Foods that have been improperly stored, especially in warm or humid conditions**
- A contaminated water source
- Crowded conditions, such as may be found on cruise liners
- Soil that has been packed around the roots of vegetables

What should be done with ice cream that has defrosted?

- **It must be used immediately or thrown away**
- It should be carefully mixed with frozen ice cream
- It should be re-frozen immediately
- It should be set to one side for use in coffee

What should happen to any cooked rice left after a meal?

- **It should be thrown away**
- It should be frozen for future use
- It should be refrigerated and can be re-heated later
- It should be washed and used in a salad dish the following day

What should you do if you discover an egg with a cracked shell in the tray you are using to prepare omelettes?

- **Throw it away**
- Break it into a separate dish and see what it smells like
- Put it to one side to hard-boil for use in a salad dish
- Use it, as it will not matter in a mix of several eggs

What should you do with a batch of whipped cream you have prepared for use on a fruit desert to be served in around two hours' time?

- **It must be placed in a refrigerator until required**
- It must be covered and left on the work bench until required
- It must be spooned over the fruit immediately and the dishes set to one side ready for serving when required
- It should be monitored until needed and, if it appears to be thickening, a few spoons of milk should be added and stirred in

When chilling food, why is it important to achieve rapid but controlled cooling of the surface?

- **Because this is where the bulk of bacterial contamination occurs**
- Because discoloration will occur if cooling is too slow
- Because freeze contamination will occur if the process takes too long
- Because that is the part in contact with the packaging

When defrosting frozen meat in cold water, it must be in a leak-proof package or plastic bag. If the bag leaks, bacteria from the air or surrounding environment could be introduced into the food. What is the other risk described in this training module?

- **The meat tissue may absorb water, resulting in a watery product**
- The meat will discolour
- The meat will give off an offensive smell
- The packaging will start to disintegrate and small bits of plastic may stick to the meat during cooking

When food has been cooked in a microwave oven, why is it important to allow the required 'standing time'?

- **Because it completes the cooking process**
- So that any food which has risen during cooking returns to its proper size and shape
- So that any oil or fat drains away
- So that the stewards do not burn their fingers during service

When marinating foods, where should they be placed?

- **In a refrigerator or coldroom**
- In the oven, as marinating should take place during cooking
- It doesn't matter where marinating takes place
- On a work bench

When planning your food preparation schedule, generally speaking what is the intention after food is cooked?

- **That it will be served immediately**
- That it should be served within 3 hours
- That it will be served within 24 hours
- That it will be served within 6 hours

When taking a carton of fresh tomatoes out of the cold store to use in a salad, you notice that some of the fruits are bruised, broken or discoloured. What will you do with them?

- **Dispose of them immediately**
- Put them back on the shelf and maybe use them in something else later
- Use them for a soup instead
- Use them in the salad anyway and hope that nobody notices

Which of the bacteria listed below is commonly associated with eggs?

- **Salmonella**
- E-coli
- Listeria
- Shellacter

Which of the following best describes the short-term risk that would be encountered if a frozen chicken was just left on the bench to defrost?

- **There would be an increased chance of bacterial infection**
- The flesh would start to decompose
- The outer parts of the bird would defrost, but the inner parts would remain frozen
- The skin would discolour

Which one of the following bacteria is commonly associated with soil, often found clinging to raw vegetables?

- **Clostridium**
- Listeria
- Salmonella
- Staphyloccus Aureus

Which one of the following can safely be eaten without risk of a stomach disorder?

- **The leaves of celery**
- Apple seeds
- Raw kidney beans
- The leaves of tomato plants

Which one of the following cheeses has a deliberate surface mould, created during its manufacture?

- **Brie**
- Danish Blue
- Gorgonzola
- Jarlsberg

Which type of pan is a risk in terms of food poisoning if left in prolonged contact with foodstuffs?

- **Copper**
- Aluminium
- Steel
- Teflon-coated

Which viral food poisoning is alternatively known as the 'Cruise Ship Illness'?

- **Norovirus**
- Hepatitis-A
- Rotavirus
- Salmonella

Why can the Ciguatera toxins not be neutralised simply by cooking the fish to a specified temperature?

- **Because these toxins are particularly heat resistant**
- Because the toxins are contained in the bones of the fish and are released into the flesh by cooking
- Because the toxins inhabit the skin of the fish and they are transferred to the cooking liquor as it is cooked
- Because they will return as the fish cools to eating temperature

0223 – Galley operations 4, Nutrition

Version 1.0 (539)

Select the ONE option which accurately completes the following statement. 'A balanced diet is . . . '

- **. . . a meal that contains all the necessary nutrients in the recommended proportions.**
- . . . a meal that contains a balance between breakfast, lunch an dinner.
- . . . a meal that contains equal amounts of carbohydrates and proteins.
- . . . a meal that contains equal amounts of fats and carbohydrates.

Select the ONE option which accurately completes the following statement. 'Breads, cereals, rice and pasta, provide complex carbohydrates . . . '

- **. . . which are an important source of energy.**
- . . . which are essential for the development of the organs.
- . . . which are important to the development of the muscles.
- . . . which are necessary for weight gain.

Select the ONE option which accurately completes the following statement. 'Calcium helps to ensure growth and strengthening of the . . . '

- **. . . bones.**
- . . . brain.
- . . . hair.
- . . . heart.

Select the ONE option which accurately completes the following statement. 'Compared to many other dairy products cottage cheese is . . . '

- **. . . low in calcium and low in fat.**
- . . . high in calcium and high in fat.
- . . . high in calcium and low in fat.
- . . . low in calcium and high in fat.

Select the ONE option which accurately completes the following statement. 'Egg yolk has a relatively . . . '

- **. . . high cholesterol content.**
- . . . high salt content.
- . . . low protein content.
- . . . low vitamin content.

Select the ONE option which accurately completes the following statement. 'In situations of higher physical activity . . . '

- **. . . the calorie intake must be higher.**
- . . . the calorie intake must be lower.
- . . . the calorie intake must be twice as much as the usual intake.
- . . . the calorie intake must not change.

Select the ONE option which accurately completes the following statement. 'Meat, poultry, fish, dry beans, eggs and nuts are a main source of . . . '

- **. . . proteins.**
- . . . carbohydrates.
- . . . minerals.
- . . . salts.

Select the ONE option which accurately completes the following statement. 'Nuts and seeds can be . . . '

- **. . . high in fats and therefore must be eaten in moderation.**
- . . . high in carbohydrates and therefore must be eaten in moderation.'
- . . . high in proteins and therefore must be eaten in large quantities.'
- . . . low in fats and can therefore be eaten in large quantities.'

Select the ONE option which accurately completes the following statement. 'To keep the intake of sodium to an acceptable level, you must . . . '

- **. . . limit the consumption of salt.**
- . . . have lean meats.
- . . . have only one banana a day.
- . . . limit the amount of ketchup used in cooking.

Select the ONE option which accurately completes the following statement. 'While stocking a vessel with stores and food provisions it is . . . '

- **. . . important for the chief cook to consider the route or run of the vessel.**
- . . . important for the chief cook to discuss the stores list with the crew.
- . . . not important for the chief cook to consider the route or run of the vessel.
- . . . not important for the chief cook to keep the crews preferences in mind.

Select the ONE option which accurately completes the following statement. ' . . . of fruits and vegetables per day are recommended for the average adult.'

- **5 portions**
- 2 portions
- 3 portions
- 6 portions

Select the THREE options that are true in relation to the diet and well-being of the crew.

- **A balanced diet must provide sufficient calories to a crew member.**
- **It is important to be aware that high risk food stuffs must be avoided or consumed sparingly.**
- **It is important to consider the preparation method.**
- It is important to provide as much food as possible to the crew members.

Select the THREE options which accurately complete the following statement. 'Potassium helps . . . '

- **. . . reduce blood pressure.**
- **. . . reduce the risk of developing kidney stones.**
- **. . . remove elements that bind calcium.**
- . . . reduce cholesterol in the blood.

State whether the following statement is true or false 'Fried chicken is as healthy as chicken stew.'

- **False**
- True

State whether the following statement is true or false 'It does not matter how much fat is added while cooking vegetables as they are always a healthy food option.'

- **False**
- True

State whether the following statement is true or false 'Punches, squashes and most fruit drinks are as healthy as having fresh fruit.'

- **False**
- True

State whether the following statement is true or false.
'A long haul cargo vessel will usually undertake a major storing operation only once a year.'

- **False**
- True

State whether the following statement is true or false.
'The calorie requirement of the crew is directly proportional to the physical activity they are expected to carry out.'

- **True**
- False

State whether the following statement is true or false.
'It is better to eat fats contained in fish and nuts as compared to those contained in meat and fried foods.'

- **True**
- False

State whether the following statement is true or false. 'Calories derived from fatty acids should be less than 10% of the total calorie intake.'

- **True**
- False

State whether the following statement is true or false. 'It is important to try and provide low fat milk and milk products each day.'

- **True**
- False

State whether the following statement is true or false. 'Pre-packed foods, snacks and bottled sauces are a good and healthy food option.'

- **False**
- True

Which ONE of these is not one of the five food group?

- **Vitamins**
- Breads, cereals, rice and pasta
- Fruits
- Vegetables

Which THREE of the factors given in the options are the most likely to affect the purchase of stores and food supplies for a ship?

- **Company's policies**
- **Duration of the voyage**
- **Route of the vessel**
- Currency available

0224 – Nutrition and well-being

Version 1.0 (1005)

As suggested by the food guide pyramid, how many servings of fruits must you have each day? Select ONE appropriate option.

- **2 to 4 servings**
- 1 to 3 servings
- 5 to 7servings
- 6 to 8 servings

In accordance with MLC 2006, which THREE of the following will be inspected regularly with respect to food and nutritional well-being on board a vessel?

- **All spaces and equipment used for the storage and handling of food and drinking water**
- **Galley and other equipment for the preparation and services of meals**
- **Supplies of food and drinking water**
- Methods of cooking adopted by the chief cook

Select THREE options that correctly describe the importance of nutrition.

- **It provides the body with energy**
- **It stimulates the body**
- **It supports the repair of muscles**
- It helps the body build muscles

Select TWO correct options to identify the food groups that form the second tier from the top of the food guide pyramid.

- **Dairy products and meats**
- **Fruits and vegetables**
- Bread and pastas
- Fats and sugars

Select TWO options that correctly define the food guide pyramid.

- **It is an information source that helps you choose a healthy diet.**
- **It is an outline of what to eat each day.**
- It is a diet plan that helps you lose weight.
- It is an outline chart that provides some easy recipes.

Select TWO options that correctly describe the ill effects of over consumption of fatty foods.

- **Lower concentration and fatigue**
- **Obesity**
- Backache
- Constipation

Select the TWO most appropriate options, from those given, that correctly complete the following statement.
'Proteins help . . . '

- **. . . to build and repair the muscles in your body.'**
- **. . . to facilitate chemical reactions in the body.'**
- . . . to build immunity.'
- . . . to improve absorption of nutrients in the body.'

State whether the following statement is true or false.
'Cottage cheese has far less fat than cheddar cheese.'

- **True**
- False

State whether the following statement is true or false.
'Cottage cheese is a rich source of calcium.'

- **False**
- True

State whether the following statement is true or false.
'Egg white is rich in cholesterol'

- **False**
- True

State whether the following statement is true or false.
'The food guide pyramid suggests that you have 2 to 3 servings of milk and other dairy products each day as these foods contain calcium that keeps the bones strong.'

- **True**
- False

State whether the following statement is true or false. 'A seafarer requires a well balanced diet as his job onboard a vessel involves physical activity and mental preparedness.'

- **True**
- False

State whether the following statement is true or false. 'According to MLC 2006, there shall be no regard for religious requirements and cultural practices that pertain to food.'

- **False**
- True

State whether the following statement is true or false?
'Green leafy vegetables are rich in vitamins C and E.'

- **True**
- False

Which ONE of the following foods does not contains added sugars?

- **Fresh fruit juices**
- Jams
- Jellies
- Soft drinks

Which ONE of the following options are fruits and fruit juices considered to be rich sources of?

- **Vitamins A and C**
- Folate
- Minerals
- Vitamin E

Which ONE of the following options correctly defines a balanced diet?

- **It is a diet that contains all the food groups in the recommended proportions.**
- It is a diet that contains fats and carbohydrates in equal proportions.
- It is a diet that contains fats and proteins in equal proportions.

Which ONE of the following options correctly defines nutrition?

- **Absorption of nutrients in the body.**
- Providing the body with carbohydrates
- Providing the body with fats

Which ONE of these food groups is considered to be a rich source of protein, vitamin B and zinc?

- **Meat, poultry, eggs and nuts**
- Breads, cereals and pastas
- Fruits
- Vegetables

Which ONE of these is NOT a nutrient class?

- **Amino acids**
- Carbohydrates
- Minerals
- Proteins

Which ONE of these is a healthier option?

- **Wholegrain flour**
- Multipurpose flour
- Rice flour
- Self-raising flour

Which ONE of these is at the tip of the food guide pyramid?

- **Oils, fats and sugars**
- Fruits
- Milk and milk products
- Vegetables

Which ONE of these two has more sugar?

- **One large cup of Chocolate shake with ice cream**
- One large cup of chocolate milk

Which THREE of these foods are rich sources of calcium?

- **Cheese**
- **Milk**
- **Yoghurt**
- Soya curd

Which TWO of the following are considered healthier cereal options with respect to whole grain and fibre?

- **Muesli**
- **Wheat bran flakes**
- Cornflakes
- Rice flakes

Which TWO of the following fruit types are most likely to be the healthiest options?

- **Fruit juices**
- **Whole fruits**
- Canned fruits
- Stewed fruits in thick syrup

Which TWO of these bread options are likely to be the least healthy?

- **Sweet fruity bread**
- **White bread**
- Pita bread
- Whole meal bread

0230 – High Voltage, Fundamentals

Version 1.0 (1913)

A 440 Volt AC source supplies 2 loads in parallel. The left load has an impedance of 44 Ohms and the right of 11 Ohms. Which of the following is the current taken from the AC supply?

- **50 Amps**
- 44 Amps
- 8 Amps
- It cannot be calculated from the information given

A man accidently receives an electric shock resulting in a current of 2 mA flowing through his hands and chest. Which of the following best describes the likely effect on him?

- **He will feel tingling or mild pain**
- He will be killed
- He will feel nothing
- He will feel severe pain from burns

An AC circuit delivers an active power of 30 kW when supplied at 400 Volts and 125 Amps. Which of the following is the reactive power?

- **40 kVAr**
- 30 kW
- 4 kVAr
- It cannot be calculated from the information given

An AC electrical circuit operates at 110 Volts. Which one of the following is the circuit's voltage rating according to this module?

- **Low Voltage**
- Extra Low Voltage
- High Voltage
- Medium Voltage

In analysing AC electrical circuits it is common practice to use impedances. Which one of the following best describes what an 'impedance' is?

- **It expresses the practical effect of resistance, capacitance and inductance for a single or group of components**
- It expresses the practical effect of inductance for a single or group of components
- It expresses the practical effect of reactance in a circuit
- It expresses the theoretical effect of reactance in a circuit

PVC insulated cable is very popular ashore. Which one of the following best describes the reason why it is less frequently used on ships?

- **It gives off poisonous halogen gas if on fire**
- Ambient temperatures on ships trading in the tropics are too high
- It is expensive
- It is only used on high voltage cables

Sometimes it is possible to see electrical conductors, such as busbars, which have no insulation covering parts of them. Which one of the following best describes why these uninsulated live parts do not normally cause a short circuit?

- **The surrounding air has been used as the insulation**
- Each conductor is surrounded by a magnetic field which contains the electricity
- High speed overload trips are fitted to disconnect the supply
- Insulation is fitted but it is so thin that it is not noticeable

The basic strategy to stay alive while working with high voltage systems may be summarised as D.I.E. What does D.I.E. stand for?

- **Disconnect, Isolate, Earth**
- Direct Insulation to Earth
- Direction, Inertia, Evaluation
- Disconnect, Isolate, Evaluate

The battery in the circuit shown has an output of 1.5 Volts. The resistors are 32.75 Ohms each. What voltage is dropped across the right hand resistor?

- **1.5 Volts**
- 0.05 Volts
- 0.75 Volts
- It cannot be calculated from the information given

The battery in the circuit shown has an output of 12 Volts. If the resistance of the resistor is 3 Ohms what current should be flowing in the circuit?

- **4 Amps**
- 36 Amperes
- 4 Farads
- It cannot be calculated from the information given

The battery in the circuit shown has an output of 12 Volts. If the resistance of the upper resistor is 3 Ohms what current should be flowing in the circuit?

- **It cannot be calculated from the information given**
- 36 Amps
- 4 Amperes
- 4 Amps

The battery in the circuit shown has an output of 12 Volts. The resistors are 3 Ohms each. What current should be flowing through the battery connections?

- **12 Amps**
- 0.75 Amps
- 4 Amps
- It cannot be calculated from the information given

The battery in the circuit shown has an output of 24 Volts. If the resistance of the upper resistor is 3 Ohms and the lower resistor is 5 Ohms what current should be flowing in the circuit?

- **3 Amps**
- 4 Amps
- 6 Amperes
- It cannot be calculated from the information given

The battery in the circuit shown has an output of 6 Volts. If the resistance of the upper resistor is 3 Ohms and the current flowing in the circuit is 1 Amp then what voltage is dropped across the upper resistor according to Kirchhoff's Law?

- **3 Volts**
- 1 Volt
- 2 Volts
- It cannot be calculated from the information given

The battery in the circuit shown has an output of 6 Volts. The resistance of the upper resistor is 3 Ohms and the lower resistor is 9 Ohms. What voltage is dropped across the upper resistor according to Kirchhoff's Voltage Law?

- **1.5 Volts**
- 1 Volt
- 2 Volts
- It cannot be calculated from the information given

Two electrical conductors are circular in cross section and are otherwise the same except that conductor 1 has twice the diameter of conductor 2. If the resistance of conductor 1 is 2 Ohms what is the resistance of conductor 2?

- **8 Ohms**
- 1 Ohms
- 2 Ohms
- 4 Ohms

Two electrical conductors are the same except that conductor 1 is four times as long as conductor 2. If the resistance of conductor 1 is 2 Ohms what is the resistance of conductor 2?

- **0.5 Ohms**
- 0.2 Ohms
- 2 Ohms
- 8 Ohms

Which of the following allows the heat being produced in a conductor carrying a current to be calculated?

- **$I^2.R$**
- I.R
- $I^2.Z$
- V/R

Which of the following best describes how a capacitor works?

- **It stores energy like a fast charging battery**
- It converts electricity to heat
- It converts electricity to higher voltages
- It converts electricity to magnetism

Which of the following best describes how the apparent power of an AC electrical circuit with a single coil as a load can be calculated?

- **P = V.I**
- $P = I^2.R$
- $P = I^2.X$
- $P = V.I \cos \Phi$

Which of the following best describes one of the ways that might cause an electrical conductor to suffer voltage overload?

- **Accidental connection to the wrong supply**
- An earth fault
- Fuse failure
- Insulation too thin

Which of the following best describes one reason why people receiving electric shocks from high voltage (HV) systems are more likely to die than those receiving shocks from low voltage (LV) systems?

- **The safe exposure time to current falls rapidly as the voltage increases**
- They are less well trained than people working on LV
- They are likely to fall further from HV equipment
- They are more likely to have sweaty hands than people working on LV

Which of the following best describes the effect of over voltage when applied to an electrical conductor?

- **It will cause insulation breakdown**
- It will be indicated by a colour change in the insulation
- It will cause build up of static charge
- It will cause thermosetting insulation to go hard

Which of the following best describes the meaning of the symbol Ω when found in an electrical calculation?

- **The quantity is expressed in Ohms**
- It is the impedance of the component
- It is the resistivity of the conductor material
- The quantity is expressed in Coulombs

Which of the following best describes what we mean by 'power factor' in an AC electrical system?

- **It is the proportion of apparent power which is delivered as active power**
- It is ratio of reactive power to active power
- It is the proportion of apparent power which is converted to reactive power
- It is the ratio of power drawn from the generator to its maximum capacity

Which of the following best describes why traditionally ships' electrical systems have operated with an insulated neutral?

- **So as to minimise the risk of essential equipment tripping**
- So as to minimise the cost of the electrical network
- So as to minimise the risk of electrocution
- So as to minimise the risk of generator damage

Which of the following is the level at which electricity becomes high voltage, according to the International Electrotechnical Commission standard for ships?

- **AC electricity above 1,000 Volts**
- AC electricity above 11,000 Volts
- AC electricity above 50,000 Volts
- DC electricity above 1,000 Volts

Which of the following options best explains how we ensure that electricity flows from the power supply to the load without it 'leaking' away?

- **The conductor is surrounded by insulation to prevent leakage**
- The conductor is magnetised to contain the electricity
- The conductor is surrounded by wire mesh to prevent leakage
- The conductor material is polarised during cable manufacture

Which one of the following best describes reactive power in an AC electrical circuit?

- **Reactive power is easily converted back to electricity**
- Reactive power reacts against the force of gravity
- Reactive power reacts against the rotation of a rotor
- The more reactive power in a circuit the more efficient it is

Which one of the following best describes the meaning of 'voltage stress' when applied to ships' electrical cables?

- **It is the voltage per millimetre of insulation**
- It is the damage done to insulation following a voltage being applied across it.
- It is the voltage drop caused by a right angle bend with a radius of 5 times the cable diameter
- It is the voltage drop per metre of cable length

Which one of the following best describes the primary method of protecting against overloading of an electrical conductor?

- **The conductors must be adequately rated for maximum load current under the worst environmental conditions expected**
- The best quality current overload trip must be fitted
- The best quality voltage overload trip must be fitted
- The conductors must be placed as far apart as possible

Which one of the following best describes what the difference is between the thermosetting and the thermoplastic insulations used on ships electrical equipment?

- **Thermoplastic insulations soften as they get warmer and thermosetting insulations do not**
- Thermoplastic insulations are grey and thermosetting insulations are black
- Thermoplastic insulations are rigid and thermosetting insulations are not
- Thermosetting insulations soften as they get warmer and thermoplastic insulations do not

Which one of the following best describes why a length of '10 Amp' cable bought from a shop ashore may not be suitable for a circuit carrying 10 Amps on a ship?

- **Temperatures on ships are often higher than ashore so that the conductor may not be cooled as effectively as assumed by the seller**
- A 10 Amp overload trip on a ship operates at a different current to those used ashore
- Cables on ships are not allowed to carry more than 60% of their rated current
- Cables on ships often have to be bent around tight corners

Which one of the following best describes why arc and blast are more of a problem with high voltage (HV) electrical systems?

- **The amount of energy available in a HV circuit tends to be much higher**
- Arcs cannot form in air at less than 800 Volts
- Blasts only happen when faults occur in a confined space like an HV cabinet
- HV circuit breakers are big and slow to operate

Which one of the following best describes why most circuits on ships are supplied by AC electricity rather than DC?

- **The voltage level of AC is easily changed**
- 100 Volts of AC is safer than 100 Volts of DC
- DC generators are not available for ships
- The waveform of AC makes electric motors run more smoothly

Which one of the following describes the best way of protecting the conductors deep inside electrical machines from the effects of excessive temperature?

- **Thermistors can be built into them to sense the temperature**
- Bi-metallic thermal cut outs can be fitted
- Clean ventilation filters on a regular basis
- The current overload trip can be used

Which one of the following describes the best way to protect an electrical cable from mechanical damage where it passes through an area on deck where cargo is loaded and unloaded?

- **Run the cable through steel conduit**
- Run the cable through plastic pipe to keep the water out
- Use cable with a steel centre core
- Use multicore cable

Which one of the following formulas agrees with Ohms Law?

- **R = V/I**
- I = V.R
- R = V.I
- R = ρL/A

Which one of the following is the most common reason for a rapid rise in the current flowing through an electrical circuit, to a level many times its rated value?

- **A short circuit**
- Bearing failure
- Excessive ambient temperature
- Weakened conductors

Which one of the following options best describes a significant problem with thermosetting types of insulation on ships?

- **Thermosetting insulation is not flexible and can become damaged by impacts**
- Thermosetting insulation is black and hides burn marks
- Thermosetting insulation is too expensive
- Thermosetting insulation is too flexible and needs excessive support

Which one of the following options best describes why connecting cables are usually fitted with flexible insulation?

- **Flexible insulation is necessary so connections can be made**
- Flexible insulation is cheaper than hard insulation
- Flexible insulation is necessary so that cables can be pushed through conduit
- Flexible insulation is thinner and lighter than hard insulation

Which one of the following would be a typical power factor for a 440 Volt galley stove?

- **1**
- 0.5
- 0.8
- 2

0231 – High Voltage, Ship's electrical networks

Version 1.0 (994)

Earth leakage circuit breakers are commonly used ashore to protect people from earth faults. Which of the following best describes why earth leakage circuit breakers have not been commonly used on ships' electrical networks in the past?

- **Continuity of supply is preferred**
- Earth leakage circuit breakers are expensive
- Earth leakage circuit breakers are not reliable
- Earth leakage circuit breakers do not work on insulated neutral networks

Even on a large passenger ship which generates electricity at 6,000 Volts lighting is normally supplied at voltages of 220 Volts or less. Which one of the following best describes why lighting is not supplied at a higher voltage?

- **Electrical loads for lighting are low enough to allow lower voltages to be used without excessive cable losses**
- High voltage lights have an annoying flicker
- Light fittings are not available for electrical supplies over 220 Volts
- Lighting switches are operated by people with no electrical training

High voltage ships tend to use earthed neutral electrical networks. Which one of the following best describes how excessive fault currents are prevented in the event of an earth fault on these ships?

- **An impedance is placed in the neutral earthing line**
- An impedance is placed in each output cable from the generators
- Earth leakage circuit breakers are used on every circuit
- The risk of earth faults is reduced by extra planned maintenance

If the earth lamps on your ship looked as shown which of the following would you assume was the most likely problem?

- **An earth fault on phase 'S'**
- A faulty lamp on phase 'S'
- An earth fault on phase 'R'
- An earth fault on phase 'T'

On shore it is normal practice to connect all exposed metalwork and the neutral point of the supply to earth. Which of the following best describes what normally happens on shore when an earth fault occurs?

- **A large current flows causing the overcurrent protection to disconnect the supply**
- A large current flows causing an earth alarm to sound
- A small current flows causing an earth alarm to sound
- A very small current flows until the overcurrent protection disconnects the supply

On the main switchboard cabinet for each generator there are normally instruments fitted. Which one of the following best describes the purpose of the 4 position switch often found beneath the generator voltmeter?

- **It is to check the balance of voltage between the generator's phases**
- It is to allow one meter to read the voltage of 2 different generators
- It is to allow one meter to read the voltage of 4 different generators
- It is to turn off the meter as it is not rated for continuous use

The kilowatt load on the ship's only running generator is steady but the power factor meter shows a fall from 0.9 to 0.75. Which one of the following best describes the risk that this change in power factor poses for the ship's electrical network?

- **The increase in apparent power may cause an overcurrent trip to operate**
- The decrease in reactive power could cause supply cables to get hotter
- The fuel consumption of the generator will be increased and the fuel could run out before arrival in port
- The galley stove might burn out

When a conductor is moved through a magnetic field an EMF is produced. Which one of the following is a way to increase the value of EMF produced?

- **Move the conductor more quickly**
- Change the conductor for one with a lower resistance
- Change the conductor for one with a lower resistivity
- Move the conductor in circles

When talking about electrical power generation on ships we often refer to the 'prime mover'. Which one of the following best describes a 'prime mover'?

- **The device where the energy is first converted to mechanical energy**
- The pump that supplies fuel during the starting of a generator after a total blackout
- The rotor of the exciter for the main generator
- The rotor of the main generator

Which of the following best describes what the result would be if earth faults occurred on 2 different phases of a ship's 3 phase distribution system at the same time?

- **A short circuit**
- A long circuit
- Two earth fault alarms sound
- Two earth faults cannot occur at the same time

Which of the following best describes why earth faults are a common problem on ships' electrical networks?

- **Due to the steel structure, the vibration and the wet conditions**
- Due to poor crew training
- Due to poor maintenance procedures
- They are not common as ships are not connected to earth except when in port

Which of the following best describes why the risk of serious electric shock is higher onboard ships than it is ashore?

- **The large amount of metal can provide a low resistance path to earth**
- A ship's crew are not as well trained as people ashore
- Ships tend to use an insulated neutral distribution system
- Voltages used on ships are higher than those used ashore

Which one of the following best describes the fundamental difference in the typical construction of a single phase generator and a 3 phase electrical generator?

- **A three phase generator has 2 more pairs of stator coils spaced 120 degrees apart than a single phase generator**
- A three phase generator has 2 more pairs of rotor coils than a single phase generator
- A three phase generator has 2 more rotors than a single phase generator
- A three phase generator has a commutator and a single phase generator does not

Which one of the following best describes the general principle used when a designer is deciding the positions of automatic disconnection devices within a ship's electrical network?

- **The higher parts of the network 'tree' must be protected against a fault on the lower 'branches'**
- Every cable must be protected at the point where power is supplied to it
- Every load must be protected at the point where power is supplied to it
- The lower 'branches' of the network 'tree' must be protected against a fault on the higher parts

Which one of the following best describes the main consideration when a designer selects the voltage level for a particular area of a ship's electrical network?

- **The maximum voltage level that the equipment can withstand**
- The frequency of that area of the network
- The maximum voltage level available from a ship's generator
- The minimum voltage level available from a ship's generator

Which one of the following best describes the main purpose of fitting reverse power protection to an electrical generator's circuit breaker?

- **To prevent the generator driving the prime mover**
- To prevent circulating currents between 2 or more generators
- To prevent the generator being run backwards
- To prevent the generator circuit breaker being closed if the generator output cables are incorrectly connected

Which one of the following best describes the main purpose of fitting under voltage protection to an electrical generator's circuit breaker?

- **It prevents the connection of a 'dead' generator to a 'live' switchboard**
- It corrects the output voltage of a generator after a heavy load has started
- It prevents overcurrents being taken from a generator due to low voltage
- It prevents the synchronisation of 2 generators with different output voltages

Which one of the following best describes the main reason why ships with high voltage electrical networks do not usually have a floating neutral system?

- **It tends to produce voltage stresses on the system cables**
- High voltage equipment is not designed for floating neutral systems
- High voltage networks on ships just copy shore practice
- It is essential to use earth leakage circuit breakers on high voltage supplies

Which one of the following best describes the meaning of 'discrimination' when applied to electrical distribution systems on ships?

- **Protection devices are to 'discriminate' between healthy circuits and circuits with faults, shutting off only the faulty circuits**
- Colour coding is used to 'discriminate' between cables made with PVC and those of other materials
- Colour coding is used to 'discriminate' between cables with flame retardancy and those without
- Protection devices are to 'discriminate' between high voltage circuits and low voltage circuits, shutting off only the low voltage circuits

Which one of the following best describes the need for cooling of 3 phase electrical transformers?

- **They need cooling due to their internal iron and copper losses**
- They need cooling due to the magnetic interaction between the 3 coils
- They need cooling due to their moving parts
- They only need cooling when the ambient temperatures go above 50°C

Which one of the following best describes the primary function of a ship's electrical distribution system?

- **To ensure that at any given point in the system the voltage and frequency are of almost rated value**
- To ensure that at any given point in the system there is always a voltage present
- To ensure that at any point in the system there is always some electricity available
- To ensure that the voltage, current and frequency are always the same at any point in the system

Which one of the following best describes why few merchant ships have been fitted with nuclear electrical power generation despite the increasing price of oil fuels?

- **The generation plant is quite big and very costly**
- Nuclear powered ships find it hard to meet SOLAS requirements
- Nuclear ships do not contain high value materials and fetch a low price when scrapped
- Nuclear ships need regular supplies of fuel which is hard to obtain in some countries

Which one of the following best describes why it is preferable to use electromagnets when constructing an electrical generator, rather than permanent magnets?

- **We can control the amount of flux that electromagnets produce**
- The electromagnets are bigger and therefore more robust
- The electromagnets are easier to shape
- We need conductors around the magnets to carry the generator output

Which one of the following best describes why many electrical transformers are not fitted with overcurrent protection on the secondary circuit output?

- **The output from the secondary circuit is fixed in relation to the input to the transformer so it is protected by the overcurrent trip on the supply**
- No overcurrent protection on the secondary circuit output is used when the output is taken from more than one voltage tapping
- The output from the secondary circuit is limited by the impedance of the coil
- The output from the secondary circuit is limited by the magnetic inertia of the transformer core

Which one of the following best describes why many modern ships are fitted with a computerised Power Management System (PMS)?

- **It can reduce crew workload and allow more precise control of the network**
- A PMS has been a requirement of SOLAS since 1995 for ships over 1,600 GT
- A PMS must be fitted where ships have 3 or more generators
- SOLAS requires ships with electrical main propulsion have a PMS fitted

Which one of the following best describes why most ships have 3 or more generators which can provide electrical power?

- **It provides greater reliability of supply and allows more efficient operation**
- SOLAS requires at least 3 generators to be provided
- Three generators allows better balance of the ship and reduced permanent list
- Three smaller generators are usually cheaper than 2 large ones

Which one of the following best describes why physical forces can be created from the interaction of 2 magnets?

- **The magnets move to make the lines of magnetic flux as short as possible**
- The magnets move because of an electrical charge stored within them
- The magnets move if the lines of magnetic flux are forced to cross
- The magnets move to even out the flux density

Which one of the following best describes why the frequencies of 3 phase electrical generators being connected together must be 'in phase'?

- **So that the output voltage peaks coincide**
- So that the current peaks coincide
- So that the firing of No. 1 cylinders on each driving diesel engine coincide
- So that the frequency peaks coincide

Which one of the following best describes why the overall system power factor is important when operating electrical generators on a ship?

- **It determines the total current that the generators must produce for a given active power**
- It determines the fuel consumption of generators
- It determines the kilowatt load on the generators
- The overall system power factor cannot safely exceed 0.95

Which one of the following best describes why the overall system power factors tends to change while operating electrical generators on a ship?

- **It is the sum of the power factors of all the individual loads and these may vary**
- It decreases as the generator load approaches 100% of capacity
- It increases as the generator load approaches 100% of capacity
- It varies if the ship hits bad weather and pitches heavily

Which one of the following best describes why we do not normally run generators that are supplying the ship's main switchboard at their maximum continuous output where their fuel efficiency is usually best?

- **There needs to be enough reserve capacity in the generating system to deal with unexpected load increases**
- Fuel efficiency of generators is unimportant compared to that of the main engine
- Starting and stopping an extra generator to supply short term increases in load uses more fuel than running it all the time
- The generators on ships are rarely able to run at maximum continuous output without breaking down

Which one of the following describes when the under voltage trip in the starter panel of an electric motor is most likely to operate?

- **During an electrical black out**
- After a large electrical load is started nearby in the network
- As the motor is started
- If the motor becomes seized

Which one of the following describes where a 'bus tie' might be found on a ship with a high electrical power demand?

- **In the main switchboard near the centre of the busbars**
- In any distribution board which is subject to high vibration
- In any starter panel which is subject to high vibration
- In the main switchboard near the ends of the busbars

Which one of the following options best describes why controlling the electrical network on a ship can be more challenging than controlling a larger network ashore?

- **Ships' electrical networks are subject to proportionately larger variations in load demand**
- Ships' electrical networks are more complex than a network ashore
- Ships' electrical networks are subject to the movement of the ship
- Ships' electrical networks use more different voltages than a typical network ashore

Which one of the following options best describes why each feed normally leaves the section boards in a ship's electrical network via some form of isolation?

- **To allow for maintenance on the circuit**
- To allow lights to be switched off remotely
- To allow motors to be stopped remotely
- To protect against slow overloads

0232 – High Voltage, Equipment and testing

Version 1.1 (1901)

After maintenance of high voltage equipment has been completed the person carrying out the work is often asked to sign the Permit to Work before the permit is cancelled. Which one of the following best describes what the person is signing for?

- **That all tools, test equipment and any additional earths have been removed**
- So that the signatures can be compared before and after permit issue
- To confirm that he knows the permit is cancelled
- To record the time at which he surrendered the permit

If isolation is not thorough, which of the following is the most dangerous stage in making a piece of high voltage electrical equipment safe to work on, according to this module?

- **Earthing**
- Deading
- Disconnection
- Reconnection

Insulation readings for high voltage equipment are sometimes used to calculate a polarisation index.

Which one of the following is the value of polarisation index which is normally considered to be the lowest acceptable before investigation of the condition of the insulation is recommended?

- **2**
- 1
- 4
- 5

Maintenance of high voltage equipment has been completed and the Permit to Work cancelled. Which one of the following best describes what needs to be done to put the equipment back into service?

- **A switching plan, approved by the authorised person, must be carried out**
- The competent person will remove the main circuit earth and turn the supply back on before informing the watchkeeper
- The competent person will turn the supply back on before informing the watchkeeper
- The engine room watchkeeper is told that the permit is cancelled and the equipment can be used again

On low voltage electrical systems insulation is normally tested using voltages above the normal rated supply voltage of the equipment. Which one of the following best describes why high voltage insulation is normally tested using a voltage below the normal rated supply voltage?

- **Testing at voltages above the rated voltage will eventually lead to damage to the insulation material**
- Insulation testers cannot produce high enough voltages from batteries
- Testing at voltages above the rated voltage can generate high static charges
- Testing at voltages above the rated voltage is too dangerous

Some insulation test meters for high voltage equipment are fitted with a 3rd lead, often called a 'guard'. Which one of the following best describes the reason for using the guard lead?

- **The guard lead cuts down leakage currents which can give a falsely low insulation reading**
- The guard lead cuts down absorption currents which can give a falsely high insulation reading
- The guard lead cuts down the risk of electrocution from the meter
- The guard lead is used to check for a good earth connection

The electrician has been issued with a Permit to Work on the high voltage thruster starter. Which one of the following best describes why he may also require a Sanction to Test Permit?

- **The circuit cannot be energised for testing while a Permit to Work is in force**
- A Sanction to Test is needed when testing high voltage circuits which run outside the main engine room
- High voltage insulation testers are very expensive and can only be issued by the stores to someone holding a Sanction to Test Permit
- No testing of any kind is allowed on high voltage systems without a Sanction to Test Permit

When measuring the insulation resistance of high voltage equipment a 'polarisation index' test may be used. Which one of the following best describes what a 'polarisation index' is?

- **It is the ratio of insulation reading obtained after 10 minutes of testing divided by the reading obtained after 1 minute of testing**
- It is the angle between the central axis of a rotor and the indicated North of a compass needle while a test voltage is applied to the rotor windings
- It is the ratio of insulation reading obtained after 1 minute of testing divided by the reading obtained after 10 minutes of testing
- It is the strength of polarising lens that is necessary to see the magnetic field of the equipment under test

Which of the following best describes one of the most important reasons for using high voltage electricity on a ship?

- **High voltages may be necessary to limit fault currents to manageable levels**
- High voltages may increase the power of main electric propulsion motors
- High voltages may reduce earth currents to manageable levels
- High voltages may remove the need for the use of D.C. in high power systems

Which of the following is the voltage above which electricity is considered to be high voltage according to this module?

- **1,000 Volts A.C.**
- 1,000 Volts D.C.
- 250 Volts A.C. or D.C.
- 3,000 Volts A.C.

Which one of the following best describes a method which can be used to continuously monitor high voltage electrical machinery for insulation condition?

- **A partial discharge monitor**
- A power factor meter
- A vibration monitor
- Earth lights

Which one of the following best describes the action to take if the polarisation index calculated from insulation readings on a piece of high voltage equipment has risen by 35% in 6 months?

- **The change should be investigated unless maintenance is known to be responsible for the rise**
- Pat yourself on the back for a good job done
- Send an e mail to the company telling them the new insulation meter is better than the old one
- Take the readings again, there must be a mistake

Which one of the following best describes the function of the insulating hooks often found near high voltage electrical switchboards?

- **They are used to remove an unconscious person from areas near high voltage electricity**
- They are used to dissipate high voltage electricity
- They are used to form a barrier between workers and nearby high voltage sources
- They are used to test high voltage conductors to see if they are dead

Which one of the following best describes the main advantage in using high voltage electricity through a 'step up' system from low voltage generators?

- **Most of the electrical system will operate at familiar voltages**
- A high voltage fault can not affect the essential low voltage consumers
- Low voltage generators are smaller than high voltage ones
- Most of the system is made up of low voltage components which are cheap

Which one of the following best describes the main reason why the azimuth thrusters used for propulsion on some ships are normally supplied with high voltage electricity?

- **High voltage allows the azimuth thrusters to be kept small, so reducing hull resistance**
- Azimuth thrusters are only fitted on passenger ships where high voltage is available
- High voltage allows the azimuth thrusters to be larger, so a larger propeller can be fitted
- High voltages remove the need for transformers to be installed on the ship

Which one of the following best describes the meaning of the term 'disconnected' in relation to high voltage electrical systems?

- **The connection of power supply to the equipment has been broken**
- The equipment is safe to work on
- The power supply cables to the equipment have been removed from the contactor terminals and taped up
- There is an open circuit in at least one supply conductor to a piece of equipment

Which one of the following best describes the meaning of the term 'tracking' when applied to high voltage electrical systems?

- **It is a noticeable current that flows through dirt on the surface of insulation**
- It is a current that flows within insulation layers
- It is the method of securing busbars within a switchboard
- It is the method of separating busbars within a switchboard

Which one of the following best describes the possible effect of supplying a solid insulator with a voltage above that for which it was designed?

- **Its insulating ability may be permanently reduced**
- Its insulating ability can be temporarily improved
- Its insulating ability will not be affected provided that the voltage is no more than 150% of the rated voltage
- Its insulating ability will only be temporarily reduced

Which one of the following best describes the possible effect of the air between conductors becoming ionised?

- **Current arcs can continue to flow after the minimum safe separation is regained**
- Cable insulation can become cracked as the ionisation effects its make up
- Current arcs can form if the conductors move towards each other
- "Ball" arcs can appear

Which one of the following best describes the process of drawing up and implementing a switching plan when connecting or disconnecting high voltage electrical equipment?

- **Anyone can draw up the switching plan but the authorised person must approve it**
- The authorised person must draw up the switching plan
- The authorised person must draw up the switching plan but the chief engineer must approve the plan
- The competent person must approve the switching plan

Which one of the following best describes the reason for 'dissipating' charge from high voltage electrical systems?

- **Stored electrical charge must be drained away from equipment after it has been 'isolated' to make it 'discharged'**
- A charge must be applied to equipment in order to test its function
- A charge must be applied to equipment in order to test its insulation resistance
- A charge must be transferred to the equipment using transformer action between coils

Which one of the following best describes the role of fresh air within high voltage switchboards?

- **Fresh air can be used as an insulator between suitably spaced busbars**
- Fresh air is often dried before being supplied to high voltage switchboards
- Fresh air is often removed from high voltage switchboards to create a vacuum
- Fresh air is used to cool high voltage circuit breaker contacts during faults

Which one of the following best describes what someone issued with a Limitation of Access Permit is allowed to do in relation to high voltage electrical systems?

- **They are allowed to enter an area where access to high voltage equipment is normally restricted**
- They are allowed to climb on top of high voltage equipment which is more than 2 metres high
- They are allowed to enter and operate equipment within a restricted high voltage switchboard room
- They are allowed to enter engine rooms containing high voltage propulsion motors

Which one of the following best describes what will happen if a low impedance connection is accidently created across a charged capacitor within a high voltage electrical network?

- **Large amounts of power can be quickly discharged**
- A large amount of A.C. power can be stored
- A small amount of A.C. power can be discharged
- A small leakage current will flow

Which one of the following best describes what you should do after you have got together the protective, test or other equipment which is to be used during work on a high voltage electrical system?

- **It must be carefully examined for visual defects before use**
- It must all have a test certificate which was issued less than 12 months before
- Personal protective equipment must have its insulation resistance checked in at least 2 places per item
- Test equipment must have its batteries checked

Which one of the following best describes why extra care should be taken when preparing a high voltage transformer circuit to be worked on after it has been fully isolated?

- **The transformer contains a lot of inductance and may still hold a deadly charge**
- High voltage transformers are normally oil filled and may leak toxic oil
- The transformer contains a lot of capacitance and may still hold a deadly charge
- The transformer may be supplied from high voltage and low voltage sides

Which one of the following best describes why high voltage (HV) cables tend to store electrical charge?

- **HV systems are made up of two or more parallel conductors, separated by an insulator with a voltage between them, like a capacitor**
- HV cables are made up of many coils of conductor with a voltage between them like a capacitor
- The insulation on HV cables is harder than for low voltage and holds a static charge
- The insulation on HV cables is softer than for low voltage and creates static charge

Which one of the following best describes why it is important for someone to read the detailed risk assessment for the job before they start work on high voltage equipment?

- **So they are aware of the dangers that have been identified and the safety measures in place**
- So they can see exactly what they need to do to complete the job efficiently
- So they can see if a Limitation of Access Permit will be required
- So they can see who should issue the Electrical Permit to Work

Which one of the following best describes why it is necessary to record insulation readings taken from high voltage equipment after a fixed period of test time?

- **There are 3 currents created when testing, all of which vary over time**
- It is advisable to ensure any undetected charge has decayed before taking a reading
- The electronics within the meter need to warm up before an accurate reading is produced
- The equipment gets warm while the test current is present, so changing the reading

Which one of the following best describes why it is normal practice to place the high voltage (Hi-pot) tester back in its test unit after a circuit has been tested as being dead?

- **To make sure it is still working correctly**
- To make sure the test unit batteries are still working
- To make sure the test unit reading agrees with the circuit reading
- To protect the ceramic end of the tester

Which one of the following best describes why someone on a high voltage ship may want to use a 'multilock'?

- **Multilocks are used where more than one person wishes to ensure that a circuit breaker cannot be operated without their permission**
- Multilocks are used so all padlocks on circuit breakers can be opened by the same key
- Multilocks are used where more than one size of circuit breaker is required to be locked using a standard size lock
- Multilocks have long wires so that more than one switch can be locked off with a single lock

Which one of the following best describes why the earth connections applied to high voltage circuits while maintenance is being carried out are of much larger diameter than those typically used in low voltage lighting circuits?

- **The prospective earth fault currents on high voltage are much larger and the connection must be large enough not to fuse**
- Earth faults on high voltage need larger diameter earth connections with low resistance to allow the current to flow freely
- Owners of high voltage systems can afford the more expensive, thicker copper earth connections
- The earth connections must be the same diameter as the current carrying conductors being earthed in both cases

Which one of the following best describes why, when taking insulation readings from high voltage cables, the reading is poor to start with but usually gets better quite quickly?

- **Initially there is a capacitance charging current which shows as a low resistance**
- The cable insulation gets warm while the test current is present, so changing the reading
- The electronics within the meter need to warm up before an accurate reading is produced
- The test current takes a while to stabilise as the meter batteries build up a high voltage output

Which one of the following describes the best way for the person assigned to carry out work on high voltage equipment to ensure that the equipment will remain dead until their Permit to Work is withdrawn?

- **Keep the lock keys to all the isolating breakers under their control**
- Disconnect the output cables from all the isolating breakers
- Keep the lock keys to all the isolating breakers in the control room
- Take the control fuses out of all the isolating breakers

Which one of the following describes the best way for the person assigned to carry out work on high voltage equipment to ensure that the switching plan has been properly carried out?

- **They should go with the person doing the switching to ensure that the plan has been completed**
- They should check the signature of the person doing the switching is on the Permit to Work
- They should check the signature of the person doing the switching is on the Sanction to Test
- They should count the number of keys in the electrician's key safe

Which one of the following is normally the first person to touch the conductors of high voltage equipment after they have been made safe to work on?

- **The person issuing the Permit to Work**
- The Competent Person
- The person issuing the Limitation of Access Permit
- The senior engineer on watch

Which one of the following is the main reason why you may be sent to stay with the motorman while he paints the high voltage switchboard room?

- **To ensure the motorman does not touch any of the local controls on the switchboards**
- To ensure that any warning signs are put back exactly where they were before painting
- To ensure that no paint is allowed to enter the switchboard vents
- To ensure that the switchboards are properly covered against paint splashes

With regard to the management of high voltage electrical systems, which one of the following best describes the difference between an 'Authorised Person' and a 'Senior Authorised Person', as stated in this module?

- **An Authorised Person may have control over certain parts of the network with overall network control being held by a Senior Authorised Person**
- An Authorised Person can become a Senior Authorised Person after 18 months sea service
- Only a certified Chief Engineer can be a Senior Authorised Person
- Senior Authorised Persons need to reach the standards set out in STCW for electrical officers and Authorised Persons can be engineers

You are carrying out an internal examination of a high voltage switchboard. Which one of the following best describes the faults that might be detected using your sense of smell?

- **Excessive arcing, overheating, burning and leakage from some types of circuit breaker**
- Insulation breakdown, excessive voltage
- Loose connections, wrong type of terminations used
- Tracking

You have been appointed the Competent Person for checking a high voltage distribution cable. While accompanying the authorised person you see that the high voltage (Hi-pot) tester shows a voltage is present when the cable is tested to see if it is dead.
Which one of the following best describes the action you should take?

- **Immediately move away from the live parts, making sure no other persons will be at risk of electrocution**
- Change the batteries in the Hi-pot tester
- Get the authorised person to test the Hi-pot tester in its test unit
- Keep the Hi-pot tester in place until the voltage dissipates, or for 2 minutes, whichever is shorter

You have been appointed the Competent Person for checking a high voltage motor of a cargo pump. While accompanying the Authorised Person you see that the high voltage (Hi-pot) tester shows a voltage is present when the motor is being tested to see if it is dead. The voltage falls away to zero within 2 seconds. Which one of the following best describes what you should do before signing the Permit to Work to say that you are satisfied with the safety arrangements?

- **You should ask for the motor to be retested before you sign the permit , even though the reading indicates the probable successful dissipation of inductive charge**
- You must definitely not sign the permit as the motor is not properly isolated
- You should check the batteries in the Hi-pot tester test unit
- You should start the work carefully but not sign the permit or any industrial injury claim will fail

You have been issued with a Sanction to Test Permit to allow you to test the insulation of a 3 phase high voltage motor. In which order should you carry out the necessary steps?

Drag the steps into their correct order in the procedure

- 1 **Connect meter leads to 2 earth points and check for effectiveness**
- 2 **Move 1 meter lead to a phase conductor**
- 3 **Remove all circuit earths**
- 4 **Test insulation resistance**
- 5 **Connect all circuit earths**

0233 – High Voltage, Management on Ships

Version 1.1 (2775)

A joint between one supply cable conductor and its terminal on a 6.6 kiloVolt electric motor has a resistance of 2 milli-Ohms. If a short circuit current of 20,000 Amps flows through the joint how much heat is generated at the joint?

- **800 kW**
- 40 kW
- 400 kW
- 80 kW

In relation to electrical equipment, which one of the following best describes the relationship between current levels and overheating in a conductor?

- **Current causes a heating effect of I^2R Watts**
- Current causes a heating effect of VI Watts
- Current levels do not affect heating
- High currents can cause high magnetic fields and hysteresis heating

In relation to high voltage electrical equipment, which one of the following best describes how equipment designers try to deal with the effects of blast following short circuits?

- **Most blasts occur at joints and these are normally sited within strong enclosures**
- Conductors are spaced further apart than necessary to minimise the chance of short circuits
- Most blasts occur within cables and these are not tightly enclosed so the pressure can escape
- Quick response short circuit protection can prevent blasts

In relation to high voltage electrical equipment, which one of the following best describes how the maximum safe exposure time to electricity changes with relation to its voltage level?

- **The safe exposure time falls rapidly from a low threshold voltage**
- The safe exposure time remains the same before rising above a threshold voltage
- The safe exposure time remains the same regardless of voltage
- The safe exposure time rises gradually with voltage

In relation to high voltage electrical equipment, which one of the following best describes the main danger from a conductor which becomes overheated?

- **It can start a fire**
- The insulation around the conductor can become discoloured
- The resistance of the conductor can rise leading to power reduction at the load
- The smell can be unpleasant for passengers

In relation to high voltage electrical equipment, which one of the following best describes the main way to ensure workers do not suffer high voltage electric shock while working out of sight of the circuit main earth?

- **An additional earth within sight of the work area should be used as well**
- A voltage tester fitted with an alarm should be left connected between one phase and earth
- An insulation tester should be left connected between one phase and earth
- The incoming cables to the equipment being worked on should be disconnected

In relation to high voltage electrical equipment, which one of the following best describes the meaning of 'dead'?

- **At or about zero voltage and disconnected from all sources of electrical energy**
- Disconnected from all power supplies
- Isolated from its power supply
- Not capable of movement

In relation to high voltage electrical equipment, which one of the following best describes the meaning of 'disconnected'?

- **The power supply to the equipment has been broken**
- The equipment contains no electrical energy
- The equipment is dead
- The power to the equipment is isolated from all power supplies

In relation to high voltage electrical equipment, which one of the following best describes the purpose of a 'Limitation of Access' Permit?

- **It allows designated persons to enter a restricted area**
- It allows designated persons to be kept out of an area where work is going on
- It limits the number of persons who can be in a designated area at any one time
- It restricts access to an area between specified times

In relation to high voltage electrical equipment, which one of the following best describes the purpose of a 'danger notice'?

- **It calls attention to the danger of approach or interference with the apparatus to which it is attached**
- It calls attention to the danger of approaching high voltage equipment when rubber matting has been removed
- It is a warning against interference with the apparatus to which it is attached
- It is a warning that the equipment to which it is attached is live

In relation to high voltage electrical equipment, which one of the following best describes the reason for getting a Sanction to Test Permit issued?

- **It is required when testing operations require removal of an earth**
- It is required before an insulation tester can be withdrawn from the stores
- It is required before the power supply can be reconnected after a permit to work is cancelled
- It is required when testing operations require the use of test equipment with an output of 500 Volts or higher

In relation to high voltage electrical equipment, which one of the following best describes the reason why arcs are dangerous to people who can see them?

- **They give off strong ultra violet light**
- They can go on for a long time
- They can set off fire detectors
- They can throw off a lot of heat

In relation to high voltage electrical equipment, which one of the following best describes when fuses may be used to protect circuits rather than circuit breakers?

- **Fuses are used in circuits where overcurrent is the only protection required**
- Fuses are not used in high voltage circuits
- Fuses are used everywhere except where power supplies leave a switchboard
- Fuses are used in circuits where there is no space for circuit breakers

In relation to high voltage electrical networks, which one of the following best describes the most likely place to find a circuit main earth?

- **At a high voltage circuit breaker or isolator**
- At every large motor
- At intervals of no more than 40m throughout the network
- Where the switchboard is bolted down

It is common on ships which generate at high voltage to supply the low voltage services via 2 transformers which can be interconnected on the low voltage side via a bus-tie circuit breaker. Which of the following best describes why it is not normal practice to operate with the low voltage bus-tie closed?

- **It would allow both transformers to supply a fault, at currents too high to break**
- It is impossible to synchronise the 2 transformer outputs
- The loads on the 2 transformers cannot be balanced
- The loads on the main generators cannot be balanced

Most high voltage systems on ships will have their neutral permanently connected to earth, unlike low voltage systems. Which of the following best describes how earth fault currents are limited on neutral earthed systems?

- **A high resistance earth is used to keep earth currents down**
- All circuit breakers are set to trip as soon as any earth leakage is detected
- All circuit breakers rated at over 1,000 Amps are set to trip as soon as any earth leakage is detected
- All circuit breakers rated at over 1,000 Volts are set to trip as soon as any earth leakage is detected

Most high voltage systems on ships will have their neutral permanently connected to earth, unlike low voltage systems. Which of the following best describes the reason for this?

- **It minimises voltage stresses**
- It maximises conductor efficiency
- It maximises safety
- It minimises earth fault currents

On some high voltage systems circuit breakers may operate automatically to re-route power supplies. Which of the following best describes when this might be programmed to happen?

- **When a slow acting fault is detected by the control system**
- When a circuit breaker is due for service
- When a load requires more current
- When a short circuit fault is detected by the control system

**The STCW Code has recommendations for the training of engineering personnel having management responsibilities for the operation and safety of electrical power plant above 1,000 Volts.
Which of the following is the commonly used name for persons trained to this level?**

- **Authorised Persons**
- Chief Electricians
- Competent Persons
- HV Aware Persons

The high rupture capacity (HRC) fuses used in low voltage systems look similar to high voltage (HV) fuses. Which one of the following best describes the main difference between them?

- **HV fuses are longer than HRC fuses of the same capacity**
- HV fuses are normally square and HRC fuses are round
- HV fuses are thicker than HRC fuses of the same capacity
- HV fuses have a calibration point

The processes for issuing a high voltage Permit to Work and a Sanction to Test Permit are similar, as are the form layouts normally. Which of the following best describes the relationship between the two permits?

- **The Sanction to Test allows equipment to be energised where the Permit to Work does not**
- The Permit to Work allows the equipment to be started up where the Sanction to Test does not
- The Permit to Work allows the supply to be reconnected to equipment where the Sanction to Test does not
- The Sanction to Test allows the supply to be reconnected to equipment where the Permit to Work does not

Two identical generators are supplying a high voltage switchboard at 3,300 Volts, each with a synchronous impedance of 0.5 Ohms, when a solid, symmetrical three phase, short circuit occurs at point A.

What is the prospective fault current through circuit breaker B1?

- **13,200 Amps**
- 26,400 Amps
- 3,300 Amps
- 6,600 Amps

Which of the following best describes how the condition of the inert gas in a circuit breaker is monitored after it is put into service?

- **A pressure alarm is normally fitted to continuously monitor the gas**
- A connection is provided to allow a pressure gauge to be fitted for regular checking
- A visual pressure indicator is normally fitted which must be checked every day
- Inert gas breakers must be serviced at least once a year by specialists

Which of the following best describes the best action to take after a high voltage circuit breaker has opened under fault conditions and the current broken is suspected to have been in excess of the breaker's continuous rated current capacity?

- **The breaker should not be operated until it has been checked by specialists**
- The breaker should be wound into the test position and test closed before being put back into service
- The breaker should have its vacuum or gas pressure checked
- The resistance through the breaker contacts should be proved to be less than 1 Ohm before being put back into service

Which of the following best describes the normal way to ensure that high voltage equipment is ready to be reconnected to the supply following maintenance?

- **The Competent Person will sign the permit to say that it is ready for reconnecting**
- The Authorised Person will check the equipment over for problems before reconnecting
- The Chief Engineer will check the equipment over for problems before reconnecting
- The Competent Person will tell the engine room watchkeeper that it is ready for reconnecting

Which of the following best describes the reason why care must be taken when handling or overhauling gas filled circuit breakers?

- **Arcing breaks down the quenching gas into various toxic gases**
- The quenching gas is hard to buy
- The quenching gas is toxic
- The surface of the main contacts is very easy to damage

Which of the following best describes why more than one Competent Person may be needed to put a high voltage switching plan into effect?

- **Physically remote breakers may need to be operated promptly to minimise danger**
- One is needed to check that the other has operated the correct breaker
- Switching breakers can take the strength of 2 men
- The switching plan may be difficult to hold while operating breakers

Which of the following best describes why the prospective fault current levels should not be reached in a real world short circuit situation?

- **The build up of current is limited by the impedance of the conductors which are inductive**
- The build up of current is limited by the impedance of the conductors which are capacitive
- The build up of current is limited by the impedance of the conductors which are resistive
- The heat build up in the conductors increases circuit resistance

Which of the following describes how isolated high voltage electrical equipment can be changed from a 'charged' to a 'discharged' state?

- **Any charge has to be dissipated**
- The power supply must be disconnected
- The power supply must be earthed
- The tension of all springs in the equipment must be released

Which of the following describes one of the main reasons for having a switching plan drawn up and approved on high voltage ships?

- **To ensure no essential equipment is accidently disconnected**
- To allow the correct caution and danger notices to be removed
- To ensure that network circuit diagram is up to date
- To ensure the bridge know what machinery is available

Which of the following is the name of the gas most commonly used in the inert gas filled circuit breakers used on ships' high voltage systems?

- **Sulphur hexafluoride (SF_6)**
- Carbon tetrachloride (CCl_4)
- Halon ($CBrF_3$)
- Sulphur dioxide (SO_2)

Which one of the following best describes one of the main criteria to be used when an Authorised Person decides whether to issue someone with a Limitation of Access Permit to enter a high voltage switchboard room?

- **Whether they are HV aware**
- Whether they are officers or ratings
- Whether they have been on the ship before
- Whether they have ever been inside the room before

Which one of the following best describes the basic principle for deciding where protective devices should be positioned within an electrical network?

- **Anywhere where they will prevent damage or harm while maintaining electrical supply to as much of the network as possible**
- Near the input to every load in the network
- On the high voltage side of every point where the circuit voltage changes
- On the supply side of every point where the circuit voltage changes

Which one of the following best describes the main criteria to be used when an Authorised Person chooses a Competent Person to carry out maintenance on, or switching of, high voltage equipment?

- **They must have sufficient technical knowledge or experience to enable them to avoid danger**
- They must be certified as a STCW III/2 second engineer
- They must be certified as a STCW III/6 electrical engineer
- They must be certified by the company

Which one of the following best describes the reason for using vacuum circuit breakers in high voltage (HV) networks instead of the air circuit breakers normally used for low voltage (LV) networks?

- **We can break the HV arc more quickly if there is no air to ionise**
- The special material used in HV breaker contacts can corrode in air
- We can break the LV arc more easily using air movement to blow away the arc
- We can break the LV arc more easily using the resistance of the air between the contacts

Which one of the following best describes the resistance reading of the human body when tested at different voltages?

- **An increase in test voltage causes a reduction in the body's resistance**
- An reduction in test voltage causes a reduction in the body's resistance
- The body's resistance is more or less constant at different test voltages
- The body's resistance is steady until test voltages rise above 1,000V when it starts to fall

Which one of the following best describes what is defined as 'High Voltage' with reference to ships' electrical systems?

- **AC electricity above 1000 Volts**
- AC electricity above 3 kiloVolts
- AC electricity above 3000 Volts
- DC electricity above 1000 Volts

Which one of the following best describes who decides whether a person is a 'Competent Person' for the purposes of high voltage electrical work, according to this module?

- **The Authorised Person**
- The Head of the personnel department
- The Master
- The Technical Superintendent

Which one of the following best describes who should hold the keys for the circuit breaker locks while maintenance is carried out on high voltage equipment if 4 people are to carry out the work?

- **The person given overall responsibility for carrying out the work should keep all the keys**
- The Authorised Person should keep all the keys
- The engine room watchkeeper should keep all the keys
- Whoever locks the breaker should keep the key for it

Which one of the following best describes who should sign to accept the high voltage Permit to Work from the Authorised Person if 3 people are to carry out the work?

- **One person must be made responsible for overseeing the work and sign the permit**
- All 3 should sign the permit
- All officers should sign the permit, ratings do not need to sign
- The permit does not normally need to be signed by the workers

0234 – Electric Propulsion, Fundamentals

Version 1.0 (1358)

A single phase AC generator produces a 50 Hz output. How long does it take for one complete cycle of the voltage?

- **20 milliseconds**
- 1 second
- 100 milliseconds
- 50 milliseconds

An electrical network consists of 3 loads with different power factors. Which of the following best describes how the overall network power factor can be calculated?

- **The three powers would need to be added together using vector addition**
- The apparent powers of all 3 loads can be added together and divided by the total power
- The arithmetic mean of the three power factors can be used
- The three powers would need to be added together using simple addition

Here are the terminal connections on an electric induction motor. Which of the following best describes the arrangement?

- **3 phase - Delta connected**
- 3 phase - Star connected
- 3 phase - Star connected with earthed neutral
- 3 phase - Star connected with insulated neutral

Here are the terminal connections on an electrical machine. Which of the following best describes the arrangement?

- **3 phase - Star connected with neutral**
- 3 phase - Delta connected with neutral
- 3 phase - Delta connected with split winding
- 4 phase - Star connected

In relation to an electrical motor: which of the following best describes what a 'star point' is?

- **It is the point at which one of the ends of all 3 windings are shorted together**
- It is a point provided for test equipment
- It is the point at which the motor is unable to turn
- It is the point where the end of one winding is connected to the start of the next

In relation to electrical systems: which of the following best describes what the definition of 'power' is?

- **Power is the rate of energy conversion**
- Power is the amount of electricity available at any point in the system
- Power is the amount of energy converted
- Power is the rate of generator fuel being burnt

Study of an electrical circuit results in the calculation of current phasors for resistive, inductive and capacitive components of the total current as shown. If the circuit voltage is 200 V what will be the value of reactive power (Q)?

- **Q = 2,000 VAr**
- Q = 12,000 VAr
- Q = 5,000 VAr
- Q = 8,000 VAr

The power factor of a large electric motor changes from 0.8 to 0.7 while its kW load remains constant. Which of the following best describes the effect of this on the electrical network?

- **The current taken from the network will increase**
- The current taken from the network will decrease
- The network power factor will be automatically increased to compensate for the change
- The network will not be affected as the active power has not changed

What is the minimum number of stator coils required to make a 3 phase electrical generator?

- **3**
- 1
- 2 pairs
- 4

When a conductor is moved through a magnetic field an EMF is produced. Which one of the following is a way to increase the value of EMF produced?

- **Move the conductor more quickly**
- Change the conductor for one with a higher resistivity
- Change the conductor for one with a lower resistance
- Move the conductor in circles

When considering an electromagnet which one of the following factors effects its relative permeability?

- **The material of the core**
- The current applied
- The diameter of the coil
- The voltage applied

When considering electrical systems: which of the following best describes why we may use 'Fleming's left hand rule'?

- **To determine the relative directions of motion, current and field in a motor**
- To count the number of vibrations per cycle in a machine
- To determine the best way to twist conductors in a multicore cable
- To determine the phase angle of the load

When used to construct a 'power triangle' for an electrical circuit which of the following best describes what is represented by a phasor's direction?

- **Its phase position with respect to some reference, usually the system voltage**
- The angle of the generator rotor with respect to a reference point
- The value of the quantity being represented
- Whether the quantity being measured is active or reactive

Which of the following best describes one of the major advantages of using 3 phase electricity on a ship rather than single phase?

- **Reduced cable costs**
- Reduced maintenance
- Reduced number of conductors
- Reduced number of transformers

Which of the following best describes one of the major problems with using only single phase electricity for a ship's electrical system?

- **The torque variations cause excessive torsional vibrations in single phase machines**
- Single phase machines require higher maintenance
- Single phase networks require fewer conductors with larger diameters
- Single phase networks require more conductors

Which of the following best describes the construction of a simple electrical capacitor?

- **Two conducting plates separated by a gap of a dielectric substance**
- Loops of conductor separated by an insulating substance
- Two conducting plates separated by a semi conductor
- Two insulating plates separated by a conducting substance

Which of the following best describes the relationship between the current and voltage in a circuit with pure capacitance which is supplied with single phase AC?

- **The peak current will lead that of the voltage by 90°**
- The current and voltage will reach their maximum at the same time
- The peak current will lag that of the voltage by 90°
- The voltage cycle does not affect the times at which the current reaches its maximum

Which of the following best describes the relationship between the current and voltage in a circuit with pure inductance which is supplied with single phase AC?

- **The current will reach its maximum 90° after the voltage reaches its maximum**
- The current and voltage will reach their maximum at the same time
- The current will reach its maximum 90° before the voltage reaches its maximum
- The voltage cycle does not affect the times at which the current reaches its maximum

Which of the following best describes what will happen in this inductive electrical circuit when the switch is closed?

- **A current will flow, rising quickly before levelling off**
- A current will flow, rising instantly then remain at a steady value
- A current will flow, rising quickly until the supply is switched off or reversed
- No current will flow

Which of the following best describes what will happen to the current in this capacitive electrical circuit when the switch is closed?

- **A current will flow clockwise but will fall to zero after a time**
- A current will flow anticlockwise
- A current will flow anticlockwise but will fall to zero after a time
- No current will flow

Which of the following correctly describes what is happening at the instant represented by this phasor diagram?

- **The value of current is zero**
- The current leads the voltage
- The value of current is falling
- The value of voltage is falling

Which of the following is the best description of a 'current' in an electrical circuit?

- **A flow of electrons**
- A flow of neutrons
- A flow of protons
- The energy that produces a voltage

Which one of the following are the units used when measuring capacitance in an electrical component?

- **Farads**
- Coulombs
- Teslas
- Teslas per metre^2

Which one of the following best describes why it is preferable to use electromagnets when constructing an electrical generator, rather than permanent magnets?

- **We can control the amount of flux that electromagnets produce**
- The electromagnets are bigger and therefore more robust
- The electromagnets are easier to shape
- We need conductors around the magnets to carry the generator output

Which one of the following best describes why physical forces can be created from the interaction of 2 magnets?

- **The magnets move to make the lines of magnetic flux as short as possible**
- The magnets move because of an electrical charge stored within them
- The magnets move if the lines of magnetic flux are forced to cross
- The magnets move to even out the flux density

Which one of the following can be assumed to have an unchanging value in an electrical circuit, whether it is an AC or DC circuit?

- **Resistance**
- Capacitance
- Impedance
- Inductance

Which one of the following describes a way of increasing the Capacitance value of a capacitor?

- **Increase the area of the conducting plates**
- Increase the voltage applied to the conducting plates
- Reduce the area of the conducting plates
- Reduce the voltage applied to the conducting plates

Which one of the following describes how the inductance of an electrical coil can be increased?

- **Wind the conductor around a core that can be magnetised**
- Increase the voltage applied to the inductor
- Use smaller diameter conductor
- Wind the conductor around a long plastic core

Which one of the following describes the way of calculating the amount of electrical energy that can be stored in a capacitor?

- **Energy stored is = ½ CV^2 Joules**
- Energy stored is = ½ CV^2 Watts
- Energy stored is = ½ LI^2 Joules
- Energy stored is = ½ LI^2 Watts

Which one of the following is an example of an electrical load with a power factor of approximately one?

- **A galley hotplate**
- A fluorescent light
- An induction motor while running
- An induction motor while starting

Which one of the following is the name given to the quantity used to combine the effects of resistance, capacitance and inductance when calculating currents in an electrical circuit?

- **Impedance**
- Dielectric
- Insulation
- Reluctance

Which one of the following would increase the capacitive reactance, X , of an electrical circuit containing a single capacitor?

- **A decrease in supply frequency**
- An increase in circuit current
- An increase in circuit voltage
- An increase in the capacitor's capacitance

Which one of the following would increase the inductive reactance, X , of an electrical circuit containing a single coil?

- **An increase in supply frequency**
- A decrease in the coil's inductance
- An increase in circuit current
- An increase in circuit voltage

With reference to an AC electrical circuit: which of the following best describes the meaning of 'in phase'?

- **The voltage and current rise and fall together**
- The circuit contains a resistor in series with an inductor
- The voltage and current have the same value
- The voltage starts to rise as the current starts to fall

With regard to a delta connected electrical machine: which one of the following statements is true?

- **The line voltage is the same as the phase voltage**
- The line current is smaller than the phase current
- The line current is the same as the phase current
- The line voltage is larger than the phase voltage

With regard to a star connected electrical machine: which one of the following statements is true?

- **The phase current is the same as the line current**
- The phase current is smaller than the line current
- The phase voltage is larger the line voltage
- The phase voltage is the same as the line voltage

0235 – Electric Propulsion, Electrical machines

Version 1.0 (963)

A 3 phase synchronous AC motor is supplied at 60 Hz. If the supply frequency falls by 3% what will be the effect on the motor speed?

- **It will also fall by 3%**
- It will be unchanged
- It will fall by 9%
- It will rise by 3%

A DC motor is driving a load at constant torque. Which one of the following best describes how best to reduce the speed of the motor?

- **Increase the magnetic field density**
- Decrease the field coil current
- Increase the armature current
- Switch in the starting resistors

If an electric induction motor is supplied at 50 Hz and contains 1 pair of poles what is its synchronous speed?

- **50 revs/s (3,000 RPM)**
- 25 revs/s (1,500 RPM)
- 30 revs/s (1,800 RPM)
- 60 revs/s (3,600 RPM)

Some larger electric induction motors have the number of starting attempts limited during a set time period. Which one of the following best describes why this is done?

- **Too many high starting currents in a short time may overheat the motor and associated control equipment**
- Too many high motor starting currents in a short time may overload the generator
- Too many high starting currents in a short time may cause damage to the motor drive coupling and shaft
- Too many high starting currents in a short time mean that there is a problem with the equipment the motor is driving

The most common AC motor has a cage type rotor. Which one of the following best describes this type of rotor?

- **It has a set of parallel conductors fixed around an iron core which are shorted together at each end**
- It has a set of conductors wound around a ring of iron cores
- It has a set of parallel conductors fixed around a copper core which are not connected at the ends
- It is made up of 3 conductor 'cages' fitted one inside the others

Which of the following best describes how 3 phase electricity can be produced from a diesel generator?

- **We need at least 3 coils, so there are 3 outputs**
- We need at least 6 coils, so there are 12 terminals
- We need at least 6 coils, so there are 6 outputs
- We need only 1 coil, and the output can be split using thyristors

Which of the following best describes how a solenoid works?

- **A current in the coil produces a magnetic field which moves an iron rod**
- A current in the central rod overcomes the permanent magnetic field around it
- A current in the coil induces an opposing current in an iron rod
- A current in the coil produces a magnetic field which moves a copper rod

Which of the following best describes how 'brushless rectification' is carried out to supply the field coils of a typical synchronous AC generator?

- **The output of an AC exciter is converted to DC by rectifiers mounted on the shaft**
- A DC exciter is used with a commutator instead of slip rings
- A DC output is taken from the rotor of a DC exciter
- An external DC supply is fed to the rotor through slip rings

Which of the following best describes the basic operation of the voltage control system of a typical large modern synchronous AC generator?

- **The voltage regulator changes the main field strength by controlling the output of an exciter**
- The voltage regulator changes the driving engine speed by controlling its governor
- The voltage regulator changes the main field strength by means of a sliding permanent magnet
- The voltage regulator controls the back EMF through a set of brushes and slip rings

Which of the following best describes the convention for the magnetic field produced when a conductor carries an electrical current flowing directly away from you?

- **The flux runs clockwise around the conductor**
- The flux runs anticlockwise around the conductor
- The flux runs in the opposite direction to the current
- The flux runs in the same direction as the current

Which of the following best describes the electrical output when a coil is rotated at a steady RPM in a fixed magnetic field?

- **The output is a sine wave**
- The output is a square wave
- The output is a steady DC voltage
- The output is a steady DC voltage which changes direction

Which of the following best describes the electrical output when a magnet is rotated at a steady RPM inside a fixed coil?

- **The output is a sine wave**
- The output is a square wave
- The output is a steady DC voltage
- The output is a steady DC voltage which alternates in direction

Which of the following best describes the main difference in the construction between a 3 phase and a single phase AC generator

- **The 3 phase generator has 3 pairs of stator coils and the single phase generator only 1 pair**
- The 3 phase generator has no commutator and the single phase generator does
- The single phase generator has 3 pairs of rotor coils and the 3 phase generator only 1 pair
- The single phase generator has no commutator and the 3 phase generator does

Which of the following best describes the meaning of 'synchronous' when applied to electrical generators?

- **The stator output frequency varies in direct proportion to the speed of the rotor**
- The stator output frequency is always the same as the main switchboard frequency
- The stator output frequency is fixed
- The stator output voltage varies in direct proportion to the speed of the rotor

Which of the following best describes the term 'slip' when applied to an asynchronous electric motor?

- **It is the difference between the rotor speed and the synchronous speed**
- It is the difference between speed of the load and the speed of the rotor
- It is the difference between speed of the load and the speed of the rotor during start up acceleration
- It is the difference between the rotor speed and the synchronous speed only when measured during start up acceleration

Which of the following best describes what happens to an electric induction motor if more torque is needed to turn the pump it is driving?

- **The rotor will slow down and the supply current will increase**
- The rotor will slow down but the supply current will stay the same
- The rotor will speed up, which reduces the torque needed
- The supply current will increase but the rotor will stay at the same speed

Which of the following best describes when the back EMF produced in a DC machine will be higher than the terminal voltage?

- **When the machine is run as a generator**
- When a supply is first connected to the machine
- When the machine is run as a motor
- When the machine is run as a synchronous compensator

Which of the following best describes why single phase AC generators are more likely to suffer from mechanical failure than 3 phase machines?

- **Single phase generators tend to have larger torsional vibrations**
- Single phase generators tend to be lower quality
- Single phase generators tend to be smaller
- Single phase generators tend to be used for larger capacities

Which of the following best describes why the rotor of an asynchronous electric motor can never rotate at synchronous speed?

- **At synchronous speed the rotor would feel no change in stator flux and so would produce no current to make its own field**
- The number of pole pairs in an asynchronous motor is different in the rotor and stator so their synchronous speeds are always different
- The rotor can reach synchronous speed but will lag behind the stator position
- The rotor construction is not strong enough to stand running at synchronous speed

Which of the following is a type of DC motor which can give accurate positional control?

- **Stepper motor**
- Sensor motor
- Staged motor
- Synchronous motor

Which of the following is the level of starting current typically taken by an electric induction motor when first started?

- **6 - 8 times rated full load**
- 0.6 - 0.8 times rated full load
- 1.25 times rated full load
- 2 - 3 times rated full load

Which of the following is the most accurate description of the magnetic field of a synchronous AC generator?

- **The field is produced from an electromagnetic pole arrangement supplied with a DC current**
- The field is produced from a molybdenum permanent magnet pole arrangement
- The field is produced from an electromagnet pole arrangement with a fixed strength
- The field is produced from an electromagnet pole arrangement supplied with an AC current

Which one of the following best describes how a synchronous AC motor can be used to reduce a lagging power factor on a network?

- **By increasing the strength of its rotor field**
- By arranging it for dynamic compensation
- By reversing its direction
- By running it above synchronous speed

Which one of the following best describes how the magnetic field is created in the cage rotor of an electric induction motor?

- **Current is produced in the rotor conductors by transformer action. This creates an electromagnetic field**
- A magnetic field is induced in the rotor cage of opposite polarity to that in the nearest stator winding
- Current is passed to the rotor conductors by brushes. This creates an electromagnetic field
- The cage of the rotor is made of highly magnetic materials

Which one of the following best describes the affect on a conductor passing from left to right through a magnetic field if its direction of travel is reversed?

- **The voltage produced in it will be reversed**
- The voltage produced in it will be increased
- The voltage produced in it will be reduced
- The voltage produced in it will stay the same

Which one of the following best describes the affect on a conductor when passed through a magnetic field at double the previous speed?

- **The voltage produced in it will be doubled**
- The voltage produced in it will be halved
- The voltage produced in it will be quadrupled
- The voltage produced in it will stay the same

Which one of the following best describes the affect on a conductor when passing a magnetic field if the strength of the field is doubled?

- **The voltage produced in it will be doubled**
- The voltage produced in it will be halved
- The voltage produced in it will be quadrupled
- The voltage produced in it will stay the same

Which one of the following best describes the difference between the 2 basic types of asynchronous AC electrical machines?

- **One has a wound rotor and slip rings, the other has a cage rotor without slip rings**
- One has a wound rotor and a commutator, the other has a cage rotor without a commutator
- One has a wound rotor and the other has a cage rotor. Neither has slip rings
- One has a wound rotor without slip rings, the other has a cage rotor and slip rings

Which one of the following best describes the effect of a current carrying conductor when a magnetic field is moved across it?

- **The conductor will try to move**
- The current flow in the conductor will increase
- The current flow in the conductor will reduce
- There will be no effect

Which one of the following best describes the main criteria when designing an electrical generator?

- **The conductors and magnetic field must move in relation to each other**
- The conductors and magnetic field must move at the same speed
- The conductors must be stationary and the magnetic field must rotate
- The magnetic field must be stationary and the conductors must rotate

Which one of the following best describes the reason why the commutator of a DC machine has so many individual segments?

- **Each armature conductor on the rotor must have its polarity reversed twice per revolution**
- Each armature conductor on the rotor must have its polarity reversed four times per revolution
- Each coil on the rotor must have its polarity reversed twice per revolution
- The number of commutator segments is determined by the need to convert the AC supply frequency to DC while the motor turns

Which one of the following best describes the usual meaning of 'EMF' when talking about electrical machines?

- **Electromotive force**
- Electro mechanical force
- Electromagnetic field
- Electromagnetic force

Which one of the following best describes what happens when 2 magnets are placed next to each other with their North poles alongside each other?

- **Their fields will be distorted and the flux density between them will increase**
- Their fields will be distorted and the flux density between them will decrease
- Their magnetic fields are in the same direction, so there will be no change in flux density
- They will jump together

Which one of the following best describes what happens when the brushes of a DC motor reach a gap in the commutator?

- **The conductor not being supplied is moving parallel to the magnetic field and so would produce little force**
- The change in current in the conductor not being supplied generates a boost to the rotation
- The conductor not being supplied is moving at right angles to the field and so would produce little force
- The conductor not being supplied would be subject to maximum current so time for cooling is created

Which one of the following best describes what is between the segments of a DC machine's commutator?

- **Insulation**
- A groove cut in the solid commutator
- A wear down sensor
- Glue

Which one of the following best describes why DC motors are normally fitted with brushes and commutators?

- **The armature is mounted on the rotor and must be supplied with electricity**
- The frame is mounted on the rotor and must be supplied with electricity
- The rotor is prone to dirt build up and must be cleaned
- The stator must be supplied via the commutator

Which one of the following best describes why an electric induction motor takes a large current when starting?

- **The low resistance of the rotor conductors and the high rate of change in the magnetic field passing over them causes the large starting current**
- The load needs to be accelerated, which takes more current
- The low resistance of the stator conductors causes the large starting current
- The slow rate of change in the magnetic field within the rotor causes the large starting current

Which one of the following describes the forms of energy used on ships in the process of producing electrical energy?

- **Chemical energy, mechanical energy then electrical energy**
- Electrostatic energy then electrical energy
- Mechanical energy then electrical energy
- Mechanical energy, chemical energy then electrical energy

Which one of the following is a correct statement in relation to the speed of a DC motor?

- **The speed is proportional to the back EMF**
- The speed is fixed by the number of pole pairs
- The speed is inversely proportional to the back EMF
- The speed is proportional to the flux density

Which one of the following is a way of increasing the torque output of a DC motor without losing speed?

- **Increase the field density**
- Decrease the armature current
- Decrease the field density
- Increase the armature current

Which one of the following is an application where a DC motor is likely to be used?

- **A servo motor**
- A pump room fan motor
- A sea water pump motor
- An air compressor motor

Which one of the following statements about synchronous and asynchronous AC electrical machines is correct?

- **The asynchronous machine has no separately controlled field supply**
- A synchronous machine has no separately controlled field supply
- The speed of the synchronous machine cannot be controlled
- The speed of the synchronous machine is dependent on its ‘slip’

0236 – Electric Propulsion, Power electronics

- **30W**
- 250W
- 25kW
- 3W

A diode has 2 electrical terminals. Which of the following are the correct names for these terminals?

- **Anode and cathode.**
- Anode and electrode.
- Anyline and catadyne.
- Electrode and cathode.

An inverter circuit is adjusted to supply a motor with a 3 phase waveform which has a cycle time of 50 milliseconds. What is the effective frequency of this supply?

- **20 Hz**
- 10 Hz
- 50 Hz
- 60 Hz

An inverter circuit is used to supply a fan motor with a 3 phase waveform. What is the main reason for inverting the supply to a motor like this?

- **The adjustable switching rate can be used to control the speed of the motor.**
- It is hard to find DC motors with enough power.
- The adjustable switching rate can be used to control the voltage to the motor.
- The adjustable switching rate can be used to improve the power factor.

If a single phase AC supply is to be full wave rectified to DC from a simple transformer, what is the minimum number of diodes that will be required?

- **4**
- 1
- 2
- 3

If a single phase AC supply is to be full wave rectified using diodes and thyristors to a DC voltage that can be controlled in value, what is the minimum number of thyristors that will be required?

- **2**
- 1
- 4
- 6

If a three phase AC supply is full wave rectified to DC how many ripples (peaks) would you expect to see on the DC waveform for each complete cycle of the supply?

- **6**
- 12
- 3
- None

If a three phase AC supply is to be full wave rectified to DC what is the minimum number of thyristors (or equivalent) that will be required?

- **6**
- 12
- 3
- 9

The inverter bridge shown is being used to convert DC to supply a motor with a 3 phase AC. If we want a current to flow from V1 to V2 which of the thyristors must be switched on?

- **Thyristors 3 and 4.**
- Thyristors 1 and 2.
- Thyristors 1 and 4.
- Thyristors 3 and 6.

The inverter bridge shown is being used to supply an electric motor. Which of the following best describes this inverter's basic principle of operation?

- **Inversion is achieved by establishing a controlled switching sequence.**
- Inversion is achieved by amplifying variations in supply.
- Inversion is achieved by converting AC to DC.
- Inversion is achieved by reversing the polarity of each wave.

When talking about electrical motor control systems the PWM control method is often mentioned. What does PWM stand for?

- **Pulse Width Modulation.**
- Power Wave Modulation.
- Proton Wave Modification.
- Pulse Wave Modulation.

Which of the following are the names given to the terminals of a bipolar transistor?

- **Collector, emitter and base.**
- Anode, cathode and base.
- Collector and emitter.
- Collector, emitter and gate.

Which of the following best describes one of the main benefits of transmitting digital control signals rather analogue ones?

- **Digital signals are less likely to be changed by external interference.**
- Digital signals are more accurate than analogue signals.
- Higher currents can be used with digital signals.
- Higher voltages can be used with digital signals.

Which of the following best describes the average voltage applied across the load if the bipolar transistor in this circuit is sent trigger pulses as the supply voltage goes positive and is switched off after 90 degrees (electrical) of the supply cycle?

- **¼ of the supply voltage.**
- The supply voltage.
- ½ of the supply voltage.
- $^1/_8$ of the supply voltage.

Which of the following best describes the main advantage of PWM drives using transistors over thyristor based inverter drives to control electric motor speeds?

- **The PWM output waveform is closer to a sine wave.**
- Motors with PWM drives can be made to go faster.
- Motors with PWM drives can be made to go slower.
- The PWM output waveform is closer to a square wave.

Which of the following best describes the main difference between a diode and a thyristor?

- **Thyristors have an extra terminal called the gate.**
- Diodes are made of silicon and thyristors made of germanium.
- Diodes have an extra terminal called the gate.
- Thyristors have an extra terminal called the cathode.

Which of the following best describes the main weakness of field effect transistors when considering them for control of the main propulsion motors?

- **They tend to fail when working at high voltages.**
- They are good for a limited number of switching operations before failure.
- They have high heat losses.
- They tend to fail when working in extreme electrical environments.

Which of the following best describes the meaning of 'self commutating' when talking about power electronic devices?

- **The device will stop conducting when the triggering supply is turned off.**
- The device does not need a triggering supply to conduct.
- The device has no brushes.
- The device will continue to conduct when the triggering supply is turned off.

Which of the following best describes the method usually used to ensure that water leaks from water cooled electronic devices do not cause short circuits?

- **The conductivity of the water is kept low and constantly monitored.**
- A water leak detection system is installed in the bottom of every cabinet.
- The conductivity of the water is kept low by adding chemicals to the water.
- The electronic devices are sealed in plastic.

Which of the following best describes the possible result of uncontrolled triggering of a transistor in the inverter circuit of a speed controller?

- **It could cause a short circuit.**
- It could cause the motor to reverse.
- It could cause the motor to slow down.
- It could cause the motor to speed up.

Which of the following best describes the properties of analogue signals when talking about control and monitoring functions?

- **They can have an infinite number of values.**
- Analysis by a computer is impossible.
- They can easily analysed by a computer.
- They have a finite number of values set by the sampling rate.

Which of the following best describes the values transmitted by a digital sensor to a monitoring system?

- **It normally uses only 2 values.**
- It normally uses 10 values.
- It normally uses 256 values.
- It normally uses an infinite number of values.

Which of the following best describes what happens when a diode becomes 'reverse biased'?

- **Conduction will stop.**
- Conduction will start.
- The diode output will be reversed.
- The diode will start to conduct if a gate pulse is received.

Which of the following best describes what happens when the gate pulse to a thyristor is turned off?

- **The thyristor will continue to conduct as long as it forward biased.**
- The gate of a thyristor is fed a steady signal not a pulse.
- The thyristor will continue to conduct as long as it reverse biased.
- The thyristor will stop conducting as soon as the pulse stops.

Which of the following best describes what may happen if a diode is not efficiently cooled?

- **Differential rates of expansion may damage the bond between the P-N junction and the external metal terminals.**
- The hysteresis losses will be increased.
- The inductance of the diode will increase leading to the power factor worsening.
- The resistance of the diode will increase.

Which of the following best describes why an inductance is often fitted to the DC link between a rectifier circuit and an inverter circuit?

- **It helps to smooth the supply of current to the inverter.**
- It is used to improve the power factor.
- It is used to tune out harmonics in the supply.
- It reverses the polarity of the DC power.

Which of the following best describes why diodes in power circuits need to be cooled while operating?

- **There is a forward voltage drop which causes heat to be given off.**
- Heat is caused while the diode is stopping current flow.
- High voltages cause high heat generation.
- The control circuits for the diodes operate at high currents.

Which of the following is one of the newer types of semiconductor device for power electronic uses?

- **Insulated gate bipolar transistor.**
- Conductive gate bipolar transistor.
- Insulated base polar transformer.
- Insulated glove bipedal motion.

Which of the following is the name given to the piece of metal often found attached to a thyristor?

- **A heat sink.**
- A radiator.
- The backing strip.
- The emitter.

Which one of the following best describes a potential cause of damage to power electronic devices while they are being handled?

- **Static discharge from wearing man-made fabrics.**
- Conductor etching due to fingerprints.
- Excessive heat from hands.
- Excessive vibration when stored onboard.

Which one of the following best describes how a non conducting material such as silicon can be made into a semiconductor?

- **The silicon is doped with another material to free up electrons.**
- The silicon core is surrounded by a conducting material.
- The silicon has a core of conducting material inserted into it.
- The silicon is subjected to radiation to free up electrons.

Which one of the following best describes the atomic structure of a material that is an electrical conductor?

- **The material will have incomplete outer valence rings.**
- The material will have completely filled outer valence rings.
- The material will have few valence rings.
- The material will have no valence rings.

Which one of the following best describes the method commonly used to minimise the number of signal cables that need to be run on a ship using digital monitoring and control signals?

- **The signals are carried along a common conductor known as a data bus.**
- The signals are converted to optical format.
- The signals are modulated on a carrier wave.
- The signals are multiplied into a carrier wave.

Which one of the following best describes the method usually used to reduce the risk of uncontrolled triggering of thyristors?

- **An R-C filter is used.**
- An N-P filter is used.
- Analogue signals are used.
- Thyristors do not suffer from uncontrolled triggering.

Which one of the following best describes what happens when a diode is connected in a circuit with an AC supply as shown?

- **The resistor will receive a half wave rectified DC voltage.**
- The diode will prevent a current from flowing in the circuit.
- The resistor will receive a full wave rectified DC voltage.
- The resistor will receive an AC voltage of half the supply voltage.

Which one of the following best describes what is happening at the atomic level when a voltage is applied to an electrical conductor?

- **An electrical current flows due to the movement of free electrons from atom to atom.**
- An electrical current flows due to the movement of bosons from atom to atom.
- An electrical current flows due to the movement of neutrons from atom to atom.
- An electrical current flows due to the movement of protons from atom to atom.

Which one of the following is the major advantage of using semiconductor devices as switches in motor control circuits?

- **A low voltage, low power circuit can control a supply with higher voltage and current.**
- Semiconductors are cheaper to buy than mechanical switches.
- Semiconductors are quieter than mechanical switches.
- The semiconductor and motor can be controlled by the same voltage supply.

Which one of the following is the other name by which silicon controlled rectifiers are often known?

- **Thyristors.**
- Bridge rectifiers.
- Thermadors.
- Thermistors.

Which one of the following statements about the construction of a transistor is true?

- **It can be constructed as a P-N-P or as a N-P-N device.**
- It can be constructed as a P-N or as a N-P device.
- It can be constructed as a P-N-P or as a P-N-P-N device.
- It is always constructed as a P-N-P-N device.

0237 – Electric Propulsion, Systems for ships

Version 1.0 (891)

A cycloconverter is supplied from the main switchboard at 60 Hz. What is the maximum frequency that can be output to the propulsion electric motor?

- **20 Hz**
- 120 – 180 Hz
- 30 Hz
- 60 Hz

A synchroconverter has a DC link between its 2 bridge arrangements. Which of the following best describes the nature of the DC link?

- **It has a large reactance and a very low resistance**
- It has a large reactance and a very high resistance
- It has no reactance and a very high resistance
- It has no reactance and a very low resistance

An electric propulsion system suffers from harmonics. Which one of the following harmonic frequencies will have the largest value?

- **Three times the fundamental frequency**
- Eleven times the fundamental frequency
- Seven times the fundamental frequency
- Thirteen times the fundamental frequency

On a ship with electric propulsion which is slow steaming, you may find that 2 generators are connected to the main switchboard even when one could supply all the load comfortably. Which of the following is the most likely reason for this?

- **The power management system has been programmed to use a minimum of 2 generators to provide redundancy in the event of an unexpected fault**
- All ships with power management systems use a minimum of 2 generators, whether they have electric propulsion or not
- The power management system has been programmed to use a minimum of 2 generators to provide waste heat
- The power management system has been programmed to use a minimum of 2 generators to reduce harmonic vibrations

On a ship with electric propulsion you may find that many more generators are connected to the main switchboard than are necessary to meet the current load. Which of the following is the most likely reason for this?

- **The power management system is set to manoeuvring mode**
- The power management system is broken
- The propulsion motors are in regenerative braking mode
- The propulsion motors are not working properly

PWM technology is sometimes used to control propulsion electric motors. Which of the following best describes what PWM stands for?

- **Pulse Width Modulation**
- Power Width Modulation
- Propulsion Waveform Modulation
- Pulse Width Management

Regenerative braking can be applied to propulsion electric motors supplied by synchroconverters. Which of the following best describes what happens when this is used?

- **The propeller drives the machine as a generator and the roles of the control bridges are reversed, feeding back power to the switchboard**
- The control bridges are used to apply a braking torque to the machine by altering the timing of inverter firing
- The control bridges are used to apply a braking torque to the machine by reversing voltage polarity
- The propeller drives the machine as a generator. The control bridges are bypassed, so power is delivered straight to the switchboard

Synchroconverters for electric propulsion are made with 2 bridge circuits. What is the main function of the 2nd bridge (the machine bridge)?

- **To create AC with controllable frequency from DC**
- To create AC with controllable voltage and frequency from AC
- To create AC with controllable voltage and frequency from DC
- To create AC with controllable voltage from DC

Which of the following best describes a common way of preventing low order harmonics in an electric propulsion systems?

- **Use half motors and phase shifting transformers to increase the pulse number**
- Enclose all propulsion control gear in an earthed metal box
- Use a variable pulse number control system
- Use twisted pair cables

Which of the following best describes how a cycloconverter can produce a motor supply waveform that has positive and negative voltages?

- **The controlling thyristors are arranged to make one positive and one negative bridge for each phase**
- The gate pulse of the controlling thyristors can be changed to allow conduction in the reverse direction
- The motor supply voltage changes, but does not reverse. Negative voltages are not needed.
- The supply diodes are arranged to make one positive and one negative bridge for each phase

Which of the following best describes how the current output of a PWM controller suitable for electric propulsion is varied?

- **The width of the pulses that allow conduction is varied**
- The height of the pulses that allow conduction is varied
- The interval between pulses that allow conduction is varied
- The number of the pulses that allow conduction per second is varied

Which of the following best describes how the output frequency of a PWM controller suitable for electric propulsion is varied?

- **The interval between pulses that allow conduction is varied**
- The DC voltage level is varied
- The height of the pulses that allow conduction is varied
- The width of the pulses that allow conduction is varied

Which of the following best describes how the output from a cycloconverter bridge to the U phase of a motor can be maintained at a positive voltage when the polarity of the R phase supplying the cycloconverter has gone negative?

- **The S and T phase supplies can be used to maintain the cycloconverter output**
- It cannot. The U phase output will be the same polarity as the R phase supply
- The U phase output is produced from a steady DC and so can be controlled at any value or polarity
- The U phase output is supposed to reverse at the same time as the R phase supply

Which of the following best describes how the risk of short circuit from the water used to cool electrical components in an electric propulsion system is minimised?

- **The water has very low conductivity and is monitored by an conductivity alarm**
- Cooling pipework is regularly pressure tested
- Fresh water is used for cooling, not sea water
- Water is not used for cooling of electrical components

Which of the following best describes one of the advantages of using a cycloconverter for electric propulsion over use of a synchroconverter?

- **It produces excellent torque particularly at low speeds**
- It allows more effective regenerative braking
- The propulsion motor can be run at higher speeds
- The propulsion motor can be smaller

Which of the following best describes the basic functions of a closed loop control system, as used in electric propulsion systems?

- **The actual output speed is compared against the desired speed and any error is sent back to the controller to correct the output**
- The actual speed is compared against the desired speed and any error is indicated and used for an alarm
- The desired speed is input to the controller which sets the output speed automatically
- The number of revolutions is counted over a 5 minute period and any error is sent back to the bridge, so the output can be changed

Which of the following best describes the basic layout of a PWM controller suitable for electric propulsion?

- **A diode bridge supplies a DC link before a transistor bridge converts the power back to a variable voltage and frequency output**
- A thrystor bridge supplies a DC link before a thyristor bridge converts the power back to a variable voltage and frequency output
- A transistor bridge converts the supply directly to a variable voltage and frequency output
- A transistor bridge supplies a AC link before a diode bridge converts the power back to a variable voltage output

Which of the following best describes the basic operation of a cycloconverter, as used in an electric propulsion system?

- **It changes a fixed frequency AC supply directly to a controllable frequency AC motor supply**
- It changes a DC supply to a controllable frequency AC motor supply
- It changes a fixed frequency AC supply to a controllable frequency AC motor supply via a DC link
- It changes an AC supply directly to a controllable voltage DC motor supply

Which of the following best describes the function of a 'snubber' when fitted across thyristors in a control system for an electric propulsion system?

- **It diverts voltage spikes from the thyristor gate, preventing uncontrolled turn on**
- It diverts voltage spikes to the thyristor gate where they can do no harm
- It smoothes the step change in output current from the thyristor, improving power factor
- It smoothes the step change in output voltage from the thyristor, reducing harmonics

Which of the following best describes the likely action that will be taken by an electric propulsion system if a potential generator overload is detected?

- **The propulsion power will be limited**
- At least one propulsion motor will trip
- The propulsion system will do nothing and wait for the power management system to start another generator
- The propulsion system will go into regenerative braking mode

Which of the following best describes the main advantage of using PWM to control a propulsion electric motor, rather than a synchroconverter?

- **It allows the use of a three phase induction motor**
- It allows the use of a three phase 60Hz supply
- It allows the use of a three phase synchronous motor
- It allows the use of a three phase synchronous motor with no field excitation

Which of the following best describes the meaning of the 'spinning reserve' on a ship with electric propulsion?

- **It is the unused capacity of all the generators currently connected to the main switchboard**
- It is the unused capacity of all the generators that can be connected to the main or emergency switchboards
- It is the unused capacity of all the generators that can be connected to the main switchboard
- It is the unused capacity of the propulsion electric motors (PEMs)

Which of the following best describes the output from the network bridge in a synchroconverter used for electric propulsion?

- **A variable voltage DC supply**
- A steady 3 phase AC supply
- A steady 3 phase DC supply
- A variable single phase AC supply

Which of the following best describes the reason for programming operational modes into the power management system (PMS) on a ship with electric propulsion?

- **They allow the PMS to better deal with expected load changes and power restrictions**
- They allow the PMS to run as few generators as possible for any given switchboard load
- They force the PMS to run a fixed number of generators according to the mode set
- They prevent the PMS from restricting power to the propulsion motors while at sea

Which of the following best describes why DC motors are now rarely used for propulsion electric motors?

- **DC motors are costly and need more maintenance than AC motors**
- DC generators big enough for modern propulsion motors are not available
- DC propulsion motors create severe harmonic distortions in the electrical network
- Rectifiers big enough for modern propulsion motors are not available to convert AC to DC

Which of the following best describes why it is normal to control the voltage supplied to an AC propulsion motor as well as its frequency?

- **If the voltage is not controlled the motor may produce dangerously high torques**
- Harmonics can be reduced by constantly changing the voltage
- If the voltage is not controlled the motor may over speed
- The voltage can be controlled to improve the power factor

Which of the following correctly describes the factors which set the speed of a synchronous AC motor?

- **The speed equals the supply frequency divided by the number of pole pairs in the motor**
- The speed equals the square of the supply frequency divided by the number of pole pairs in the motor
- The speed equals the supply frequency divided by the number of poles in the motor
- The speed is directly proportional to the supply voltage

Which of the following correctly describes the factors which set the torque available from a synchronous AC motor?

- **The motor torque is proportional to the applied voltage squared**
- The motor torque is inversely proportional to the applied voltage
- The motor torque is proportional to the applied voltage
- The motor torque is proportional to the field current

Which one of the following best describes the output from a typical electric propulsion synchroconverter?

- **It will be AC with variable frequency and voltage**
- It will be AC with variable frequency only
- It will be AC with variable voltage only
- The output will be DC

Which one of the following best describes the problem that occurs because of the shape of the output waveform from a synchroconverter machine bridge?

- **Harmonic interference**
- Chordal frequencies
- Synchronous impedance
- Vibrating pulses

Which one of the following best describes what a 'firing pulse' does in a large electric propulsion control system?

- **It turns a thyristor on**
- It is the command sent from the bridge to start the motor turning
- It transmits the shaft position back to the propulsion control system
- It turns a thyristor off

Which one of the following best describes why a synchroconverter needs to operate in 'pulse' mode to rotate a motor at low speeds?

- **The DC link voltage must be turned off to make the inverter thyristors commutate**
- The AC output voltage must be turned off to make the rectifier thyristors commutate
- The AC output voltage must be turned on and off to make the thyristors conduct
- The DC voltage must be turned off to make the thyristors conduct

Which one of the following best describes why electric propulsion is becoming popular for LNG carriers?

- **The associated generating capacity can be used for cargo pumping or reliquefaction**
- Electric propulsion is more efficient than geared propulsion when used with steam turbines
- LNG boil off cannot be burned in direct drive diesel engines
- The ability to fit azimuth propulsors aids manoeuvrability

Which one of the following best describes why many propulsion electric motors (PEMs) are supplied via step down transformers?

- **It is efficient to generate high powers at high voltages but most PEM converters cannot operate above 3,000 Volts**
- PEM control technology has been taken from ashore where supply voltages are lower than on ships
- Step down transformers are needed to increase the currents available to the PEMs
- The controller systems for most PEMs cannot operate above 48 Volts

Which one of the following is a typical constraint that must be designed into the control system for an electric propulsion system?

- **Limiting torque on the drive train and couplings**
- Axial shaft position
- Over speed
- Propeller immersion

Which one of the following is an advantage of installing electric propulsion, particularly on a large passenger ship?

- **Economy at reduced power operation**
- Lower capital cost
- Reduced number of diesel engines installed
- Simpler propulsion system

Which one of the following is claimed as an advantage of fitting azimuth podded propulsors (Azipods) to a ship with electric propulsion?

- **Stern thrusters and a conventional rudder are not required**
- Cycloconverters can be used which are simpler
- Maintenance is easier
- There are no shaft seals to leak

Which one of the following is normally used for the switching devices in a PWM controller for electric propulsion?

- **Transistors**
- Diodes
- Thermistors
- Thyristors

Which one of the following motor types would you expect to find used with a synchroconverter for speed control?

- **A synchronous AC machine**
- An asynchronous AC machine
- An asynchronous DC machine
- An induction motor

Which one of the following statements about propulsion electric motor excitation is most accurate?

- **Synchronous motors always have external excitation**
- Asynchronous motors always have external excitation
- External excitation for asynchronous motors is controlled by an AVR
- Synchronous motors occasionally have external excitation

0240 – Bunker Fraud

Version 2.0 (1204)

A bunker quality claim may include several issues depending on the extent of loss and associated cost. Which of the following statements describes the issues that may be included in a quality claim? Tick all that you think apply.

- **Cost of de-bunkering and re-supplying ship with fuel oil of the correct grade**
- **Cost of pursuing the claim including legal costs, survey costs, additional analyses**
- **Cost of replacing damaged machinery, parts and labour**
- **Indemnity against the consequences of receiving fuel oil that does not comply with the requirement of MARPOL annex 6**
- **Loss of time**

Apart from bunker fraud relating to fuel quality, there is also bunker fraud that relates to fuel quantity. Please identify from the following list the five issues that are bunker QUANTITY fraud.

- **Adding air to the fuel, the so-called "cappuccino effect"**
- **Adding water**
- **Deliberately declaring incorrect densities and temperatures of the delivered fuel or fuels**
- **Falsified tank tables for the bunker barge supply tanks**
- **Shortening of measuring tapes used by the barge/supplier**
- Detection of off-specification fuel oil during pre-test

Normally, as a local requirement in a Singapore bunkering Port, 3 samples should be taken on board ship or by the buyer. Which three of the following correctly describe these samples?

- **One MARPOL sample for the vessel's retention**
- **One retained sample for the vessel**
- **One sample for the vessel, if the vessel is on a fuel quality testing programme.**
- One sample for Flag State Inspection
- One sample for Port State Control
- One SOLAS-sample for the vessel's retention

The Chief Engineer is in charge of the whole process of bunkering and his position demands certain responsibilities. Pick four of the following selections below which are the responsibility of the Chief Engineer during the whole bunkering process.

- **As the one in charge, the Chief Engineer delegates tasks to others**
- **The Chief Engineer directly communicates with the barge master and everyone involved in the bunkering**
- **The Chief Engineer makes a risk assessment of the tasks required for bunkering**
- **The Chief Engineer takes control of the whole bunkering process**
- The Chief Engineer personally monitors the pumping system during the bunker operation
- The Chief Engineer signs all maintenance checklists prior to the bunkering

Which five of the following are bunkering practices which help prevent bunker QUALITY fraud?

- **On the Bunker delivery note record the details of the sample seals**
- **Take continuous drip samples throughout bunkering**
- **Take multiple samples**
- **Take samples at the vessel's bunker receiving manifold**
- **Use tamper-proof seals and labels**
- Check bunker fuel density and temperature prior to bunkering
- Use valid bunker tank conversion tables

Which four of the following statements correctly describe ways of knowing if the integrity of bunker sample bottles has been maintained?

- **Both the supplier and receiver representatives' signatures are visible on the sample labels**
- **Details of the seal are recorded on the bunker delivery note**
- **Sample containers are sealed with uniquely numbered tampered-proof seals**
- **Sample labels can be checked against the uniquely numbered seals**
- Sample bottles are stamped "approved samples"

Which of the following statements describes why we need to check the general terms and conditions of the bunker contract? Tick all that you think apply.

- **It gives the applicable standard for the sampling method**
- **It names the jurisdiction for any legal questions that may arise**
- **It says how claims should be handled**
- **It spells out distribution of liabilities**
- **It states the delivery procedures and the subsequent standards applied.**

Which one of the following is the correct name for the main document which forms the basis for indicating grievances encountered in bunkering to settle disputes and claims relating to quality and quantity of the bunker oil delivered?

- **Letter of Protest**
- Letter of Claims
- Letter of Disputes
- Letter of Indemnity

Which one of the following statements is the name given to the additional level of evidence collection covering a particular bunker fraud problem, and might include, the taking of tank samples, retention of parts apparently damaged as a result of using the fuel oil, ships general arrangement plan, statements from the ship's engineers and, perhaps, additional surveys attended by representatives of the parties involved.

- **Non-routine evidence collection**
- Document evidence collection
- Proof of liability evidence collection
- Routine evidence collection

Which one of the following statements is the name given to the level of evidence collection consisting of the documents and notes that normally accompany a bunkering operation – for example a chief engineer's notebook, the loading plan, ullage reports and the routine sampling and analysis of fuel oil.

- **Routine evidence collection**
- Document evidence collection
- Non-routine evidence collection
- Proof of liability evidence collection

Which one of the following terms best describes a deliberate act intended to deceive a buyer about the true nature of supplied bunker quality and/or quantity, including provision of incorrect information or other irregularities?

- **Bunker fraud**
- Breach of contract
- Bunker misinformation
- Deceit

Which three of the following are bunkering practices which help prevent bunker QUANTITY fraud?

- **Check bunker fuel density and temperature prior to bunkering**
- **Measure all tanks in the supply barge including dedicated tanks for fuel on board**
- **Use own measurement equipment when measuring tanks**
- Determine sulphur content of the bunker fuel
- Use tamper-proof seals and labels

Which three of the following statements describe the purpose of a Bunker Delivery Note?

- **A statutory document**
- **An agreement validating the samples indicated in it when signed by both parties.**
- **An anti-fraud tool**
- A document originating from the buyer or the receiving vessel

Which two of the following documents should the Chief Engineer ask the barge officer for before starting to load bunker oil?

- **Certified and valid sounding tables for the bunker barge**
- **Official loading papers**
- Bunker receipt
- Completed sounding report of the barge

Which two of the following statements describe practices used to determine off-specification fuel oil or bunker oil?

- **Independent laboratory testing provided by shore-based fuel testing services program**
- **Shipboard fuel analysis**
- Access to Port State Control database of fuel suppliers
- Bunker Delivery Note

0241 – Development of Heavy Weather

Version 1.0 (4443)

'When there is moving water on the deck from breaking waves the force it generates will be insufficient to knock a person over'? Select the best option that comments on this statement.

- **The statement is incorrect as the force may be sufficient to knock a person over**
- The statement is correct as the person's weight will hold him/her upright
- The statement is correct when the person wears boots
- The statement is incorrect as people will not be on deck with moving water present

Indicate from the given options the definition of phase wave period in relation to sea waves.

- **The time for two successive crests to pass a fixed point**
- The time it takes a crest to pass the vessel
- The time it takes for the wave to travel one mile
- The way that the wave moves relative to the vessel

Select an option that best answers the following question. In low equatorial latitudes which pressure system is associated with extremely strong wind?

- **A tropical revolving storm**
- Anabatic and katabatic winds
- The Azores High pressure system
- The weather associated with the doldrums

Select from the following options why very large waves build up in the sea.

- **It is because all required environmental factors are present**
- It is only because of wind against tide effect
- It is only because there is a large fetch
- It is only because there is a strong wind

Select from the given options the best definition of a vessel slamming.

- **The bow re-entering the water when the vessel is pitching**
- The bow being affected by the orbital motion in a wave
- The sound of unhooked doors opening and closing due to violent motion in heavy weather
- The sound that occurs when the vessel has water breaking on deck

Select from the given options the statement that best describes how much water force there can be on a vessel compared to wind force.

- **Water force can be considerably more than wind force on some vessels**
- Water force and wind force are usually about the same
- Water force is usually considerably less than wind force on most vessels
- Water force is usually fairly small on a vessel and can be ignored

Select from the given options the statement that best describes water force on a vessel.

- **The force is made up from form drag and skin friction**
- The force comes only from air drag as the vessel moves through the water
- The force comes only from friction found on the hull surface
- The force is not of great concern with a vessel in a harbour

Select from the given options the statement that best describes wind force on a vessel.

- **The force is made up from form drag and skin friction**
- The force comes only from friction found on surfaces
- The force comes only from hull drag
- The force is not of great concern in an ocean

Select from the possible options one definition of heavy weather given in this module.

- **When the wave height exceeds 4 metres**
- When the vessel is in a high pressure weather system
- When the vessel starts to roll
- When the wave height exceeds 2 metres

Select the option that completes the following statement. 'If the wind speed doubles . . .

- **. . . the force on the vessel increases by an approximate factor of four'.**
- . . . the force on the vessel increases by an approximate factor of five'.
- . . . the force on the vessel increases by only a small amount'.
- . . . the force on the vessel will also double'.

Select the option which correctly completes the following statement. 'Wind waves are classed as gravity waves because . . .

- **. . . gravity is the restoring force when wind stops'.**
- . . . gravity causes vertical motion of particles in the wave'.
- . . . their wave height is bigger than 4 metres'.
- . . . they suffer the effects of gravity making them bigger than the wind would indicate'.

Select the option which correctly completes the following statement. 'A Beaufort wind force 10 will have a velocity of . . .

- **. . . 48 – 55 knots'.**
- . . . 22 – 27 knots'.
- . . . 34 – 40 knots'.
- . . . greater than 64 knots'.

Select the option which correctly completes the following statement. 'A Beaufort wind force 10 will have associated deep water waves of average height . . .

- **. . . 9 metres'.**
- . . . 2 – 4 metres'.
- . . . 5.5 metres'.
- . . . greater than 14 metres'.

Select the option which correctly completes the following statement. 'The Beaufort scale is used to . . .

- **. . . indicate wind strength'.**
- . . . describe the types of cloud cover'.
- . . . give some indication of swell height'.
- . . . measure the pressure value in a depression'.

Select the option which correctly completes the following statement. 'The group wave period of a deep water wave explains why . . .

- **. . . there are bigger and smaller waves in the same wave train'.**
- . . . there are only swell waves visible'.
- . . . there are only wind waves visible'.
- . . . there is a long time between crests'.

Select the option which correctly completes the following statement. 'The strongest winds in a temperate latitude depression often occur . . .

- **. . . behind the cold front'.**
- . . . in advance of the cold front'.
- . . . in advance of the warm front'.
- . . . in the warm sector'.

Select the option which correctly completes the following statement. 'Very large waves can be destructive because they have . . .

- **. . . lots of energy'.**
- . . . long wave length'.
- . . . short wave length'.
- . . . steep sides'.

When manoeuvring a vessel in a shallow water tidal area with tugs where should the tugs be positioned, select the best option?

- **In positions to control the water forces**
- Centre leads fore and aft
- In positions to suit the berth
- Where ever the Pilot chooses

Which of the following options best completes the following statement? 'A seiche is . . .

- **. . . the oscillation of water in a harbour'.**
- . . . a type of wind blowing in Antarctic areas'.
- . . . a wave seen in the tropics from earthquake activity'.
- . . . the oscillation of water in an ocean'.

Which of the following options best completes the following statement? 'With waves ahead a vessel with a very fine bow will . . .

- **. . . cut through the waves relatively easily'.**
- . . . be badly affected in heavy weather'.
- . . . be very dry on deck forward'.
- . . . pitch heavily on meeting the waves'.

Which of the following statements best explains why some vessels cope with heavy weather better than others?

- **Heavy weather effects can be relative to the vessel**
- It is totally dependent on the hull shape
- Some vessels do not react to wind as much as others
- Some vessels have better reserve stability than others

Which of the given options best completes the following statement in relation to sea waves? 'The height of a wave is the distance from . . .

- **. . . the crest to the trough vertically'.**
- . . . one crest to the next crest'.
- . . . one trough to the next trough'.
- . . . the crest to the midpoint of the wave vertically'.

Which of the given options best completes the following statement? 'A high pressure centre will . . .

- **. . . temporarily depress the sea surface'.**
- . . . affect the tide raising forces'.
- . . . have no affect on the sea surface'.
- . . . temporarily raise the sea surface'.

Which of the given options best completes the following statement? 'A low pressure centre will . . .

- **. . . temporarily raise the sea surface'.**
- . . . affect the tide raising forces'.
- . . . have no affect on the sea surface'.
- . . . temporarily depress the sea surface'.

Which of the given options best completes the following statement? 'Water breaking over the bow of a large vessel with aft accommodation . . .

- **. . . may be causing damage not seen immediately by the crew'.**
- . . . is acceptable in heavy weather as the vessel will be at high speed'.
- . . . looks impressive from the aft bridge'.
- . . . will not impact on the vessel'.

Which of the given options best describes a Tsunami sea wave?

- **A long wavelength wave caused by an earthquake**
- A seismic wave in the earth
- A very short wavelength wave caused by an earthquake
- A wave found in the South Pacific only

Which of the given options best describes a cross swell in relation to sea state.

- **Two swells from different directions meeting at a large angle**
- A swell found off the island of Malta
- Two swell meeting from the same direction
- Two swells meeting from opposite directions

Which of the given options best describes a katabatic wind?

- **A wind that blows from the land towards the sea**
- A wind found only in USA
- A wind localised to desert areas
- A wind that blows towards land from the sea

Which of the given options best describes a tidal storm surge?

- **Raising or lowering of the predicted tide height**
- A sudden increase in the severity of the weather
- Lowering of the predicted tide height
- Raising of the predicted tide height

Which of the given options best describes an anabatic wind?

- **A wind that blows towards land from the sea**
- A wind found only in Indonesia
- A wind localised to arctic areas
- A wind that blows from the land over the sea

Which of the given options best describes parametric rolling on a vessel?

- **Very heavy rolling when ahead or astern waves are encountered**
- It is a type of rolling that only tankers and bulk carriers suffer from
- Very heavy rolling in a beam sea on a container vessel
- Very heavy rolling on a container vessel because of the deck cargo

Which of the given options best describes the direction of wind circulation in a tropical storm in the northern hemisphere?

- **Anti clockwise when viewed from above**
- Anticlockwise then clockwise when crossing the equator
- Clockwise then anticlockwise when crossing the equator
- Clockwise when viewed from above

Which of the given options best describes the direction of wind circulation in a tropical storm in the southern hemisphere?

- **Clockwise when viewed from above**
- Anti clockwise when viewed from above
- Anticlockwise then clockwise when crossing the equator
- Clockwise then anticlockwise when crossing the equator

Which of the given options best describes the location of the world's Tropical Revolving Storms

- **In lower latitudes across all oceans**
- In lower latitudes close to the equator
- In temperate latitudes only
- In the western Pacific only

Which of the given options best describes the motion of water particles in a shallow water wave?

- **They move horizontally only**
- They do not move at all
- They move both vertically and horizontally
- They move vertically only

Which of the given options best describes the significant wave height in relation to sea waves?

- **The mean height of the third largest waves**
- The height of a wave measured from crest to trough
- The mean height of all waves
- The wave that causes most damage

Which of the given options best describes the speed of swell waves?

- **Long wavelength swell travels faster than short wavelength**
- Short wavelength swell travels faster than long wavelength
- They are very slow compared to wind waves
- They travel faster in shallow water

Which of the given options best describes what happens to a wave entering shallow water.

- **It feels the bottom and it's characteristics change**
- It immediately starts to break
- Nothing happens to the wave but the bottom gets eroded
- The wave speeds up causing the sea to flatten

Which of the given options best describes when a large vessel will start rolling in waves?

- **When the encounter angle is near the beam and waves have sufficient force**
- As soon as any waves hit the vessel's hull
- If the transverse GM is large and there are waves present
- When there is a wind above Beaufort force 4

Which of the given options best describes why a vessel may be rolling or pitching when there is no wind?

- **The vessel is suffering swell waves**
- Cargo is moving on board the vessel
- The vessel's stability is suspect
- The wind is gusting up and down in speed

Which of the given options best describes wind against tide effect?

- **Wind blowing in an opposite direction to water flow**
- Where the wind causes a build up of water increasing the tide height
- Where the wind causes a drop of water height
- Wind blowing in the same direction as water flow

Which of the given options can be considered a definition of temperate latitudes?

- **Latitudes between 30 and 70 degrees north and south**
- Latitudes between 0 and 30 degrees close to the equator
- Latitudes between 30 and 70 degrees only in northern hemisphere
- Latitudes between 30 and 70 degrees only in southern hemisphere

Which of the given options can be considered correct for the time when a large vessel starts to feel the bottom and have increased drag?

- **When the depth becomes less than six times the draught**
- When the depth is less than ten times the draught
- When the depth of water is 1.2 times the draught
- When the under keel clearance is very big

Which of the given options can be considered correct to complete this statement? 'Breaking water on the deck of a vessel . . .

- **. . . can impose enormous loads on deck and deck machinery'.**
- . . . can be seen as acceptable with a vessel at high speed'.
- . . . is not a major issue as vessels are built to withstand it'.
- . . . looks impressive as the spray flies'.

Which of the given options can be considered correct to complete this statement? 'The bow of an aft accommodation vessel with wind on the beam when moving ahead . . .

- **. . . will probably seek the wind'.**
- . . . will be difficult to control'.
- . . . will not be affected by the wind'.
- . . . will probably pay off downwind'.

Which of the given options can be considered the best answer to a definition of the dangerous semi circle in a Tropical Revolving Storm (TRS)?

- **It is the semicircle where the wind is in the same direction as the TRS movement**
- It is the northern half of the TRS
- It is the semicircle where the wind is in the opposite direction to the TRS movement
- It is the southern half of the TRS

Which of the given options can be considered the most accurate for this statement? 'The position of the centre of water pressure will affect how the vessel handles when being berthed'.

- **A correct statement – as a berthing vessel usually has force from moving water on it**
- A correct statement – as a berthing vessel is under tug control at all times
- A correct statement – as a berthing vessel uses the centre of pressure to push on the berth fenders
- An incorrect statement – as a berthing vessel is not affected by moving water

Which of the given options correctly completes the following statement? 'On a large container vessel in heavy weather with waves ahead slowing down may . . .

- **. . . increase the likelihood of parametric rolling'.**
- . . . increase the likelihood of being broached
- . . . increase the likelihood of being pooped'.
- . . . not make any difference to the ETA next port'.

Which of the given options correctly completes the following statement? 'There is a risk of parametric rolling if . . .

- **. . . the natural roll period is twice the wave encounter period**
- . . . a vessel puts the waves on the beam
- . . . the natural roll period is the same as the wave encounter period
- . . . the natural roll period is half the wave encounter period

Which of the given options correctly describes a standing wave at sea?

- **One that moves in the same direction and speed as the vessel**
- One that has a steady wave height
- When the vessel runs down waves at a faster speed than the waves
- When the vessel runs down waves at a slower speed than the waves

Which of the given options correctly describes a standing wave in a harbour?

- **One that has vertical movement only**
- One that has both vertical and horizontal movement
- One that has horizontal movement only
- Sea waves entering a harbour

Which of the given options correctly describes a vessel being pooped?

- **When a wave breaks on the stern of a vessel**
- A condition that exists with the engine after a long sea passage
- When a vessel has water breaking over the bow
- When waves break over the vessel's sides

Which of the given options correctly describes a vessel broaching?

- **An uncontrolled manoeuvre putting the vessel beam on to waves**
- A controlled turn in heavy weather to heave to
- A manoeuvre executed by the Master with manual steering engaged
- Movement similar to parametric rolling that happens on container vessels

Which of the given options correctly describes what happens to a vessel's stability in heavy weather?

- **Passing waves cause increases and decreases in the vessel's transverse GM**
- Passing waves cause the vessel's longitudinal GM to change
- There is no change in the vessel's stability in waves
- Water on deck increases the vessel's GM values

Which of the given options correctly describes why fin stabilisers are sometimes not as effective in heavy weather as in calmer conditions?

- **If the vessel slows down lift from the fin is reduced**
- In heavy weather they turn more slowly to reduce inertial loads
- The varying depth of water in waves means they work differently to each other
- There is no difference in their effectiveness in heavy or calm conditions

Which of the given options correctly identifies the speed of a deep water sea waves of 150 metres length and period 10 seconds?

- **Approximately 30 knots in deep water**
- Approximately 10 knots in deep water
- Approximately 100 knots in deep water
- It cannot be calculated because it is a function of wave amplitude

Which of the given options is correct for a local name in Australia for a Tropical Revolving Storm?

- **Willy – willy**
- Bonga – wonga
- Hurricane
- Silly – billy

Which of the given options is correct for the following question? 'If the wind force on a vessel is calculated to be 50 tonnes how many 50 tonne tugs will be required with 25% redundancy to overcome the wind force?

- **A minimum of two tugs**
- A minimum of four tugs
- A minimum of three tugs
- One 50 tonne tug is sufficient

Which of the given options is correct in relation to the following statement? 'Fetch is the important factor in allowing the formation of fully developed waves'.

- **The statement is only part correct as water depth is also important to wave formation**
- The statement is false as developed waves will always form
- The statement is false as the wind strength determines the wave height
- The statement is fully correct as developed waves will always result

Which of the given options is correct in relation to the following statement? 'Storms causing heavy weather conditions occur seasonally'.

- **The statement is part correct – it depends upon location**
- The statement is correct as storms always happen in winter
- The statement is only correct for the North Atlantic
- The statement is only correct for the South Atlantic

Which of the given options is the most accurate for the following statement? 'Slamming is an acceptable part of being in big waves'

- **The statement is not correct as slamming can cause major damage**
- The statement is correct as slamming happens away from the crew accommodation
- The statement is correct as vessels are built to withstand slamming
- The statement is not correct as slamming is noisy keeping the crew awake at nights

Which of the given options is the most accurate in relation to a rogue wave?

- **There will often be a large trough before the wave front**
- It will be proceeded by a flat bit of sea
- The wave height is similar to surrounding waves
- There will be a flat bit of sea behind the wave

Which of the given options is the most accurate in relation to the following statement? 'A vessel when sailing deep sea may not notice a Tsunami wave passing'

- **A correct statement – because of the wave characteristics**
- A correct statement – because the wavelength is very short
- An incorrect statement – as it is a large wave and will always be seen
- An incorrect statement – as there will be violent motion on the vessel as it passes

Which of the given options is the most accurate in relation to the following statement? 'Synchronous rolling only occurs with beam seas'.

- **It is not correct as quartering seas can produce the same affect**
- It is correct as a beam sea is always of the correct characteristic to start it
- It is correct because a beam sea is needed to start rolling
- It is not correct as head waves can produce it

Which of the given options is the most accurate in relation to the following statement? 'Very large waves can be encountered in any sea area regardless of environmental conditions'.

- **Not correct - as environmental conditions have to be right**
- Correct – they can occur with or without wind blowing
- Correct – they will always be there because of the group velocity of waves
- Not correct – they are only seen in high latitudes

Which of the given options most accurately completes the following statement? 'A category 5 hurricane on the Saffir-Simpson scale is . . .

- **. . . a severe tropical storm'.**
- . . . a down grading code for a tropical cyclones to a depression'.
- . . . a moderate tropical cyclone'.
- . . . one of the lightest tropical storms'.

Which of the given options most accurately completes the following statement? 'The transverse GM of the vessel . . .

- **. . . has an influence on a vessel rolling in waves'.**
- . . . has an influence on a vessel pitching in waves'.
- . . . has no influence on a vessel pitching in waves'.
- . . . has no influence on a vessel rolling in waves'.

Which of the given options most accurately completes the following statement? 'Water on deck of a vessel . . .

- **. . . can increase top weight reducing the vessel's stability**
- . . . has no affect on the vessel's stability
- . . . tends to stay there till the vessel next rolls
- . . . will temporarily increase the vessels stability

Which one of the given options best describes swell waves found at sea?

- **Waves resulting from previous wind or wind in another area**
- Waves bigger than sea waves
- Waves caused by winds in the same area as the swell is sighted
- Waves from an American registered vessel's hull

Which one of the given options can be considered a definition of fetch?

- **The distance which the wind has been blowing over the sea surface**
- The distance over the land surface which the wind has been blowing
- The size of the depression causing the wind to blow
- The time the wind has been blowing

0242 – Heavy Weather Navigation

Version 2.0 (2235)

IMO Resolution A.950(23) requires coastal states to provide a Maritime Assistance Service. Which one of the given options best describes the reason for this?

- **To monitor a disabled vessel and be a point of contact**
- It is a requirement making the VTS operation assist the disabled vessel
- To help ship owners get through their legislation
- To lead a salvage operation on a disabled vessel

Select from the following options the reason why it is important to have adequate reserve stability on a vessel when approaching an area of heavy weather.

- **It is because heavy weather conditions vary the initial stability**
- It is because heavy weather conditions permanently change the initial stability
- It is because IMO regulations require this for heavy weather
- It is because strong wind causes large heeling moments

Select from the given options the best reason why, in heavy weather, greater under keel clearance should be allowed on a passage in coastal waters.

- **Vessel motion causes potentially large reductions in clearance values**
- As waves pass they reduce the depth of water so the vessel bodily sinks
- Squat values are greater as a vessel maintains higher speed
- The vessel's propeller needs more water to have efficient propulsion

Select from the given options the most accurate definition of bulk cargo liquefaction.

- **When a bulk cargo starts to behave like a liquid**
- It is a process used to discharge the product quickly using suction
- It is when the cargo gets wet from sweat due to poor ventilation
- When a bulk cargo changes to a liquid

Select from the given options the one that best describes how land can be used to best navigational advantage whilst on a coastal passage in heavy weather.

- **Land will afford some shelter if relatively close and upwind**
- Best navigational advantage is to stay away from it as it is dangerous
- Land will afford some shelter if relatively close and downwind
- Use it to take lots of visual fixes as it will be prominent

Select from the given options the one where information on heavy weather procedures would most likely be found on board a vessel.

- **In the vessel's safety management system**
- In the Admiralty List of Radio Signal
- In the vessel's official log book
- Printed on the chart where heavy weather is likely to be found

Select from the given options the statement that best describes a major consideration for ensuring that a vessel has adequate under keel clearance during a coastal passage in heavy weather.

- **That a negative storm surge will reduce the predicted water depth**
- That a negative storm surge will increase the predicted water depth
- The clearance should be the normal minimum as per Company instructions to load a full cargo
- There should always be a 50% increase in normal clearance in such conditions

Select from the given options the statement that best describes the location in which anchors should be secured.

- **In a location safe for the deck crew**
- At the pilot station when the pilot is clear
- In a location as directed by Company procedures
- In a location outside port limits

Select the option which correctly completes the following statement. 'If GMDSS equipment is logged into one Inmarsat satellite storm warnings broadcast from another satellite will . . .

- **. . . be received if the message is classed as distress or urgency'.**
- . . . be missed as the broadcast will not be relevant to the area the vessel is located'.
- . . . be missed as the equipment does not listen to the other satellite'.
- . . . be received but stored till the equipment is switched to that satellite'.

Select the option which correctly completes the following statement. 'On a tanker save all drains should . . .

- **. . . be open in heavy weather'.**
- . . . always be closed in heavy weather'.
- . . . be checked once a watch in heavy weather'.
- . . . be sealed with wooden plugs in heavy weather'.

Select the option which correctly completes the following statement. 'The World Wide Navigation Warning Service offers storm warnings in . . .

- **. . . all sea areas'.**
- . . . coastal areas only'.
- . . . high seas areas only'.
- . . . NAVAREAS 1 to 6 only'.

Select the option which correctly completes the following statement. 'To ensure all personnel are safe during periods of extreme vessel motion . . .

- **. . . rig safety lines and use personal harnesses in exposed places'.**
- . . . act normally and wear a safety harness at all times'.
- . . . do not go on the vessel's main deck under any circumstances'.
- . . . tell crewmen to restrict their movements on the vessel staying in their cabins'.

Select the option which correctly completes the following statement. 'When navigating around the shores of Australia weather information is received via . . .

- **. . . the SafetyNet system'.**
- . . . coast radio stations only'.
- . . . the Internet as there is no NAVTEX there'.
- . . . the NAVTEX receiver'.

Under SOLAS requirements which one of the given options best describes the action a vessel's Master's should take if they suspect a tropical storm is developing in the vessel's vicinity?

- **Report it to other vessels and the nearest competent authority**
- Record the weather at the end of the navigational watch as normal
- Record the weather particulars every hour and monitor the situation
- Record the weather particulars in the deck log book and inform the company

When a forecast of heavy weather is received who should be informed immediately?

- **The vessel's master and crew members**
- Other vessels in the area
- The Chief Engineer
- The designated person ashore

Which of the given options best describes the main cause of bulk cargoes movement in a cargo space which has water tight integrity on a vessel that is at sea in heavy weather?

- **They can slide under the force of gravity due to severe motion**
- Movement is caused by liquefaction due to water ingress
- Ship sweat due to lack of ventilation makes decks slippery for the cargo
- Torsional stresses in the hull change the shape of the holds

Which of the given options best describes the main precaution necessary with bulk cargoes on a vessel approaching heavy weather?

- **Ensure that the cargo will not shift with vessel motion**
- Check the cargo temperature
- Send crewmen into the holds to trim the cargo
- Shut down the cargo space ventilation system to stop cargo sweat

Which of the given options describes possible action that needs to be taken on board a vessel with a dragging anchor, in strong wind, to make it hold?

- **Pay more chain out**
- Make sure the bow stopper is on
- Shorten the chain
- Tighten up the windlass brake

Which of the given options is correct for the number of NAVAREAs or METAREAS in the world?

- **21**
- 16
- 18
- 24

Which of the given options is the most accurate in relation to the following statement? 'A vessel with structural problems has an automatic right to go to any safe haven'.

- **An incorrect statement as the requested country has to grant permission**
- A correct statement to maintain safety of life on board the vessel
- A correct statement to maintain safety of the vessel
- An incorrect statement as countries do not want damaged vessels in their areas

Which of the given options is the most accurate in relation to the following statement? 'If heavy weather conditions are expected the Master should have navigational contingency plans'.

- **This is a correct statement as there may be unforeseen happenings**
- This is a correct statement as SOLAS require this
- This is an incorrect statement as the vessel should never get into heavy weather conditions
- This is an incorrect statement as the vessel will always follow the original plan

Which of the given options is the most accurate in relation to the following statement? 'The galley on a vessel is just another living space in heavy weather'.

- **The statement is incorrect as it is more dangerous due to their being hot items in the galley**
- The statement is correct as it is a place to get warm on a cold, heavy weather day
- The statement is correct no matter what the weather conditions are
- The statement is incorrect as the galley is a place no crew member should enter

Which of the given options is the most accurate in relation to the following statement? 'When a vessel is weather routed all routing communications are from the service provider to the vessel only'.

- **This is an incorrect statement as the vessel also has to relay regular position information back to the service provider**
- This is a correct statement as the service is paid for and all inclusive to the vessel
- This is a correct statement as the vessel only has to take instructions
- This is an incorrect statement as the vessel is required to send weather observations

Which one of the following options best completes the following statement? 'A vessel lying to a single anchor in a strong wind will drag the anchor if . . .

- **. . . there is not enough chain out'.**
- . . . the anchor is snagged on the sea floor'.
- . . . the engine is turning dead slow ahead'.
- . . . the wind changes direction'.

Which one of the following options best completes the following statement? 'The two types of weather routing are . . .

- **. . . optimum weather routing and voyage optimisation'.**
- . . . maximum weather routing and voyage optimisation'.
- . . . minimum weather routing and voyage optimisation'.
- . . . optimum weather routing and voyage minimisation'.

Which one of the following options best completes the following statement? 'Voyage optimisation software uses . . .

- **. . . feedback from vessel sensors to enable setting of course and speed'.**
- . . . a UNIX based computer software system'.
- . . . feedback from the weather forecast to enable setting of course and speed'.
- . . . only a vessel's manoeuvring parameters to set course and speed'.

Which one of the following options best completes the following statement? 'Water ingress to void spaces on a vessel causes . . .

- **. . . a free surface effect reducing the vessel's GM'.**
- . . . a resulting trim by the vessel's head'.
- . . . the GM to increase making the vessel more stable'.
- . . . the vessel no problem as it is automatically pumped out'.

Which one of the following options best completes this statement? 'In storm force winds the Master must . . .

- **. . . report them to other vessels and the nearest competent authority if the conditions have not been forecast'.**
- . . . question the Coast Radio station as to why there was no warning'.
- . . . report them to other vessels and the port state where the vessel is registered'.
- . . . send a message to the nearest competent authority only'.

Which one of the following statements best explains the meaning of term weather routing?

- **This is when a vessel is guided away from bad weather**
- This is a route set on a passenger vessel in a sunny area
- This is a vessel's route when blown off course in strong wind
- This is when a vessel has a route laid into a bad weather area

Which one of the given options best completes the following statement? 'Cargo lashings should . . .

- **. . . be checked before and regularly during heavy weather'.**
- . . . be checked before heavy weather only'.
- . . . be checked on deck cargo only before heavy weather'.
- . . . be checked only on cargo in holds that cannot be seen from the bridge before heavy weather'.

Which one of the given options best completes the following statement? 'Weight distribution on a vessel in large waves is important because . . .

- **. . . vessel movement is influenced by weight distribution when meeting waves'.**
- . . . it influences how the vessel is able to be manoeuvred'.
- . . . stability is affected with large amounts of weight in the bow area'.
- . . . the motion of the vessel may cause weights in the middle of the vessel to move'.

Which one of the given options best describes how a vessel in ballast is most likely to react when anchored in good holding ground when there is a strong wind blowing?

- **The vessel can range from port to starboard across the cable direction**
- The vessel will be rolling heavily because of the anchor force
- The vessel will sit quietly downwind of the anchor
- The vessel will start to drag the anchor

Which one of the given options best describes how bow anchors on a vessel should be prepared for heavy weather conditions?

- **They should be hove tight and have extra lashings applied**
- The chains should be lashed together below the spurling pipe
- The windlass should be left in gear with the brake on
- They should be hove tight and have the windlass brake applied

Which one of the given options best describes one of the actions to be taken on the deck of a vessel before heavy weather?

- **All items of portable equipment should be properly stowed away**
- All items of portable equipment should be lashed in place on deck
- Heavy items of portable equipment are best left behind a deck house
- Only light portable items need to be removed as might float off the deck

Which one of the given options best describes the action to be taken by the bridge team if the main engine is totally disabled in heavy weather with a close by lee shore?

- **Call for tug assistance and drop both anchors**
- Call for tug assistance
- Drop the port anchor in the northern hemisphere
- Drop the starboard anchor in the southern hemisphere

Which one of the given options best describes the action to be taken by the bridge team if the main engine stops in heavy weather?

- **Start up any thrusters to maintain heading**
- Call the engine room to find out why it has stopped
- Get the crew forward quickly to take off anchor lashings
- Start rudder cycling to slow the vessel down

Which one of the given options best describes the action to be taken by the engine room staff if the main steering system fails in heavy weather?

- **Ensure the auxiliary system is available for immediate use**
- Call the bridge and ask them what has happened
- Call the Chief Engineer to sort out the problem while the watch engineer looks after the engine
- Send the Motorman to the steering flat to investigate

Which one of the given options best describes the function of a commercial weather routing company?

- **They are in business to route vessels away from bad weather**
- Their function is to supply vessels with completed passage plans
- They are in business to forecast the weather
- They function because of IMO requirements

Which one of the given options best describes the immediate action to be taken by the watch Engineer if the main engine stops in heavy weather?

- **Assess the reason for it stopping and restart it as soon as possible**
- Call the Chief Engineer to the engine room
- Restart the engine immediately as it is essential for control of the vessel
- Run a diagnostic check using the engine management system

Which one of the given options best describes the operation of making fast a salvage tug to a vessel in open water in heavy weather conditions?

- **It is a very high risk operation where safeguards need to be in place to protect personnel**
- It is a low risk operation simply requiring the tow line to be heaved on board
- It is a very high risk operation that should never be attempted
- It is very similar to making fast a tug in a harbour except the tug will be bigger

Which one of the given options best describes the risk factor of being on a vessel's exposed main deck in heavy weather?

- **It is a very dangerous place**
- It is safe because of the vessel's freeboard
- It is safe being well protected from the sea
- There is a possibility of flying spray

Which one of the given options best describes the vents on a vessel that should be closed before the onset of heavy weather?

- **Fore deck vents into chain lockers, stores and void spaces particularly**
- All deck accommodation vents into living spaces
- Engine room vents should have the flaps closed
- Tank vents should be plugged along the whole length of the vessel

Which one of the given options best describes vessel motion prediction software?

- **Software that predicts vessel movement for forecast weather conditions**
- Software that can be used to predict movement of the vessel if ballast is taken
- Software that is used for stability calculations
- Software that records vessel movement for cargo insurance claims in bad weather

Which one of the given options best describes what happens to a vessel with a very light bow area meeting waves head on?

- **The vessel's bow will very quickly rise over the wave**
- The vessel will be greatly slowed down when the wave hits
- The vessel will tend to plough into waves
- There will be a lot of water on deck forward

Which one of the given options best describes why water ingress forward can endanger a vessel?

- **The added weight reduces the vessel's freeboard**
- Because the added weight can cause the main engine to deliver less power
- Unwanted water adds to forward weight increases the likelihood of slamming
- Unwanted water on board affects rudder control hence steering capability

Which one of the given options best explains how to use a salvage tug to help a vessel in heavy weather when the vessel has a disabled steering system but an operational main engine?

- **Attach the tug to the after part of the vessel and use it to steer**
- Always attach a salvage tug to the bow of the vessel on the Smit bracket
- Attach the vessel to the bow on a very short lead to help steering
- Lash the tug alongside to act like a thruster and steer the vessel

Which one of the given options can be considered correct to complete the following statement? 'When avoiding a tropical revolving storm the safe semicircle has the wind . . .

- **. . . astern or near astern'.**
- . . . ahead or nearly ahead'.
- . . . on the port beam in the northern hemisphere'.
- . . . on the starboard beam in the southern hemisphere'.

Which one of the given options can be considered the best secondary action on a vessel with a structural problem in heavy weather?

- **Risk assess the damage before deciding on any further course of action**
- Attempt a repair to the damaged structure
- Request that the nearest coastguard arrange for the crew to be taken off the vessel
- Send out an urgency message on the GMDSS equipment

Which one of the given options correctly completes the following statement? 'Tugs working a vessel in heavy weather conditions . . .

- **. . . are not as effective as in normal conditions'.**
- . . . are just as effective as in normal conditions'.
- . . . only make fast on the lee side of the vessel'.
- . . . will never make fast for fear of parting the tow line'.

Which one of the given options correctly describes a meteorological related tidal storm surge?

- **A time when tides are higher or lower than predicted**
- A time with increased tidal currents because of the tide raising forces
- Only a time when tides are higher than predicted
- Only a time when tides are lower than predicted

Which one of the given options describes the main action that should be taken to control the effects of heavy weather on a vessel moving in a port area?

- **Risk assess considering environmental forces on the vessel**
- Double up the mooring lines
- Hand over the vessel to the harbour pilot as he knows best
- Take extra tugs if the company allow it

Which one of the given options explains what should happen with the vessel's mooring ropes before entering an area of heavy weather?

- **Stow them off deck in a locker or storeroom**
- Make sure they are well lashed on pallets on deck
- Stow them behind a deck house clear of possible deck water
- Turn them up on the bits with the ends lashed

0245 – Safe operation and maintenance of grinding wheels

Version 2.0 (7743)

A new cutting disc is taken from the stores. When subjected to a ring test it fails. What is the most likely reason for this?

- **It is reinforced**
- It is still dirty
- It was not stored flat
- The wrong type of hammer was used

A new grinding wheel is taken from the stores. It has a red line across the middle. What does this mean?

- **It has a maximum operating speed of 80 m/s**
- It is a cutting wheel
- It is part of the maker's logo
- Someone has scrapped it

A new grinding wheel is taken from the stores. When subjected to a ring test it fails. Which of these is the most likely reason for this?

- **It is too small**
- It has a red stripe
- It was held using wire
- It's edge is uneven

Abrasive wheel machines are often used for grinding operations. Which of these may cause excessive heat to be created when grinding?

- **Pushing wheel and work together too hard**
- Strong sunlight
- Too small a wheel
- Twisting the wheel

Abrasive wheel machines found onboard ship are sometimes fitted with a wire brush wheel. What is the problem with bristles coming off a brush wheel?

- **The bristles are sharp and can puncture skin**
- The bristles are hot and can burn through boilersuits
- The job will take longer
- They make a lot of noise

Can you see anything wrong with the abrasive wheel machine shown in the picture?

- **The guard is not original type**
- No, there is nothing wrong
- The dressing tool is missing
- The wheel is too small

Complete the following statement with the option that best describes the action to be taken when you are finished using a pedestal grinder. When you have finished using a pedestal grinding machine you should . . .

- **. . . keep other people away until it has stopped**
- . . . stop the wheel with your gloved hands
- . . . just walk away
- . . . stop the wheel with your bare hands

Complete the following statement with the option that best describes the correct action that should be taken after a grinding operation is completed. When you leave a work piece which is hot after grinding on a bench you should . . .

- **. . . post a warning that it is hot**
- . . . coat it with oil to stop rust
- . . . direct the vent towards it
- . . . go for a coffee before touching it

Complete the following statement with the option that best describes the likely result of banging a grinding wheel with the work piece. Banging the work piece against a grinding wheel may . . .

- **. . . cause the wheel to break**
- . . . clear clogging in the wheel
- . . . make the finish better
- . . . make the job go faster

Complete the following statement with the option that best describes the reason for jamming of an abrasive wheel machine. On a fixed abrasive wheel machine jamming may occur because . . .

- **. . . the gap between rest and wheel is too big**
- . . . the machine's fixing bolts are loose
- . . . the work is pressed too hard against the wheel
- . . . the work piece is too big

Complete the following statement with the option that best describes the reason why vibration on abrasive wheel equipment is considered to be dangerous.

Excessive vibration on abrasive wheel equipment is considered to be very dangerous because . . .

- **. . . it can lead to wheel breakage**
- . . . it makes it a bit more noisy
- . . . it makes your arms ache after a while
- . . . the bearings will have to be changed sometime

Complete the following statement with the option that best describes the reason why you should not press too hard when using a brush wheel on an abrasive wheel machine. When using a brush wheel on an abrasive wheel machine you should not press too hard because . . .

- **. . . you will wear the brush out quickly**
- . . . you may hurt your back
- . . . you may melt the steel
- . . . you will get a rough finish

Complete the following statement with the option that best describes the reason why you should not press too hard with the work piece against a grinding wheel. When using a grinding wheel on a bench machine you should not press too hard because . . .

- **. . . the extra heat may cause the wheel to break**
- . . . you are not paid enough to rush things
- . . . you may get sore hands
- . . . you may hurt your back

Grinding wheels are sometimes classified by the bonding method used in the manufacture of the wheels. Which class of grinding wheel is most easy to break?

- **Vitrified (V)**
- Resinoid (B)
- Rubber (R)
- Shellac (E)

Here are some of the stages necessary to replace a wheel on a pedestal grinding machine. Drag the stages on the right into the correct order.

- 1. **Make the machine safe to work on**
- 2. **Note position of parts**
- 3. **Remove old wheel**
- 4. **Move rest clear**
- 5. **Check wheel fits spindle or flange step**
- 6. **Check wheel rotates freely and is "true"**

How can you find out the RPM that a pedestal grinding machine is intended to run at?

- **Check the nameplate on the grinding machine**
- Check any motor nameplate – RPMs are all the same
- Check what RPM is written on the spare wheel in the store
- Check your ISM manual

How could the noise from grinding possibly lead to a crewmember being injured?

- **By drowning out a warning shout**
- By causing the ship to vibrate until it cracks
- By causing vibration in their teeth and head
- By driving them crazy

On a bench mounted abrasive wheel machine what should the maximum gap be between wheel and rest before it is adjusted?

- **3 mm**
- 1 mm
- 10 mm
- 5 mm

On a grinding wheel label, which of the given options is the code letter for a wheel made mainly of Aluminium Oxide?

- **A**
- C
- O
- Z

On a grinding wheel what does 'MOS' stand for?

- **Maximum Operating Speed**
- Maximum Operating Size
- More Operations Style
- Multi Operation Science

Select the option that correctly completes the following statement. A grinding wheel is said to be "true" when . . .

- **. . . it is perfectly round and the rest gap is constant**
- . . . it is not cracked
- . . . it sounds good during a ring test
- . . . the label on the side is present

Select the option which best defines the term 'abrasion' when used in the context of abrasive wheels.

- **Scoring or wearing away by friction**
- To be able to stick
- To be rough
- Using coolant

The gap between the grinding wheel and the rest on a bench machine is 4 mm but cannot be made smaller as the rest has no more adjustment. What should you do?

- **Change the wheel**
- Bend the rest bracket in a bit
- Carry on working and tell the engineers later
- Make the rest bracket slot longer

The visor on a bench mounted grinding machine is cracked. Which of the actions given in the options is the correct one in such a situation?

- **Secure it out of the way and wear goggles instead**
- Put tape over the crack to seal it
- Use the visor and wear safety glasses "just in case"
- Use the visor but keep the crack from blocking your view of the contact area

When changing a brush wheel on a bench grinding machine to a grinding wheel you notice that the securing flanges are smaller than those on the grinding wheel at the other end and only cover 1/4 of the diameter of the grinding wheel. What should you do?

- **Stop until you find larger flanges**
- Ignore the difference
- Make the securing nut extra tight so the wheel does not slip
- Mention it to your boss at dinner

Which of the following options gives the right wheel bond and grain for removing burrs after flame cutting mild steel?

- **Organic bond + medium grain**
- Inorganic bond + fine grain
- Inorganic bond + medium grain
- Organic bond + coarse grain

Which of the following options gives the right wheel hardness and porosity for removing burrs after flame cutting mild steel?

- **A medium hardness wheel with medium porosity**
- A hard wheel with medium porosity
- A medium hardness wheel with closed porosity
- A soft wheel with closed porosity

Which of the given options is the main reason to 'dress' a grinding wheel?

- **To remove clogged abrasive from the wheel.**
- To add sharp grains to the wheel.
- To allow the side of the wheel to be used.
- To make the wheel look better.

Which of the given options is the most appropriate when starting up a pedestal grinding machine?

- **Keep everybody clear**
- Look at the lights to see if they dim
- Press the work against the wheel so it speeds up slowly
- Put you ear near the wheel to listen for noises

Which one of the given options is the main reason why you should not grind wooden broom handles on grinding wheels?

- **The wheel will get clogged**
- A broom handle is too long
- The wood may catch fire
- The wood will be rough afterwards

Why do abrasive cutting wheels normally contain some reinforcement?

- **Because they are thin and easily broken**
- Because cutting is harder than just grinding
- Because of the special abrasive they use
- Because they spin faster

Why is a grinding wheel breaking up during a grinding operation considered bad?

- **The pieces are big and move fast enough to kill**
- Grinding wheels are expensive
- It will take an hour to fit a new wheel
- The damaged edge can lead to jamming

With regard to grinding very small work pieces, which of the given options is the recommended way to hold the work piece?

- **Clamped in locking pliers or a vice**
- Glued to a larger piece of scrap
- Held using pliers
- Held without using gloves

You go to use a bench machine and see that the electrical cable insulation has a few cuts in it so that you can see the blue wire. Which of the actions given in the options is the correct one in such a situation?

- **Mark it "Do not use" and report it to the electrician**
- Use the machine and make a note to tell the electrician later
- Use the machine but keep away from the cut cable
- Wrap the cuts with electricians tape before using it

You notice that the edge of a grinding wheel on a bench machine is quite rounded on the outer side. Which one of the given options is the best action to take?

- **Get the wheel dressed**
- Be careful to use only the inner part of the wheel where it is still flat
- Ignore it, you are only sharpening 1 scraper
- Use the side of the wheel

You see that the brush wheel on the abrasive wheel machine has a place where the bristles are bent lower than the rest. Which one of the given options is the best action to take?

- **Change the wheel**
- Cut the higher bristles so they are all the same height again
- Use the wheel but be extra careful
- Use the wheel, they will straighten out when spinning

0246 – Piracy and armed robbery II

Version 2.0 (3320)

Select one option which correctly identifies the action a vessel should take to arrive at points A or B of the IRTC in the Gulf of Aden?

- **The vessel should arrive at a time advised by the MSCHOA guidance**
- The vessel should arrive and wait at points A or B to set off at the correct time
- The vessel should consult with the Company Security Officer
- The vessel should get advice from the nearest war ship

Select the option which correctly completes the following statement.

'A pirate safe area is an area . . .

- **. . . where the crew can muster in safety inside the vessel'.**
- . . . in the engine room'.
- . . . on an outside upper deck adjacent to the bridge'.
- . . . where pirates can retire to'.

Select the option which correctly completes the following statement.

'Blast resistant film on the bridge windows will . . .

- **. . . stop personal injury caused by flying glass from broken windows'.**
- . . . help deter pirate attacks'.
- . . . stop pirates from seeing into the wheelhouse'.
- . . . stop the bridge windows from being broken with small arms fire'.

Select the option which correctly completes the following statement.

'If pyrotechnics are used to scare pirates . . .

- **. . . they should not be those designated for life saving purposes'.**
- . . . ideally orange smoke floats should be used as they are a sign of distress'.
- . . . the Chief Officer is the only one who should use them'.
- . . . they should be fired off in pairs one from each bridge wing'.

Select the option which correctly completes the following statement.

'In an area known to have a high risk of pirate attacks a vessel's master should . . .

- **. . . liaise with local coastguards or naval authorities for information'.**
- . . . liaise with the International Maritime Organisation (IMO)'.
- . . . maintain radio silence for the transit'.
- . . . slow down and proceed with caution keeping a good lookout'.

Select the option which correctly completes the following statement.

'In some piracy high risk areas . . .

- **. . . vessels are encouraged to report to a regional centre'.**
- . . . a cloud of secrecy is best to safeguard the vessel'.
- . . . vessels are encouraged to have radio silence'.
- . . . vessels are given private radios so they can hide communications'.

Select the option which correctly completes the following statement.

'The primary aim of the organisation 'Regional Cooperation Agreement on Combating Piracy and Armed Robbery against Ships (ReCAAP) is . . .

- **. . . to combat piracy in the sea areas of Asia'**
- . . . to be a worldwide piracy reporting centre'.
- . . . to give advice to shipping Companies with vessels in a high risk area'.
- . . . to record statistic for pirate attacks on behalf of NATO'.

Select the option which correctly completes the following statement.

'Under the vessel Safety Management System a shipping company is responsible for ensuring that . . .

- **. . . risk control measures are in place on a vessel'.**
- . . . razor wire has been supplied to the vessel for a voyage in the Indian Ocean'.
- . . . the Company has a copy of the risk control measures for the vessel'.
- . . . there are sufficient stores for the intended voyage'.

Select the option which correctly completes the following statement.

'Publication BMP4 requires that vessels in the Horn of Africa area initially report to . . .

- **. . . UKMTO'.**
- . . . EU NAVFOR'.
- . . . MSCHOA'.
- . . . NATO'.

Select the option which correctly completes the following statement.

'When an electric fence is deployed on a vessel to stop unwanted boarders there should . . .

- **. . . be warning signs in various languages to warn persons'.**
- . . . be no signs or indications that it is there so that it comes as a surprise to pirates'.
- . . . be some sort of earth on it to stop dangerous static electrical discharges'.
- . . . be warning signs in the English language only'.

Select the option which correctly completes the following statement.

'When navigating in an area with high piracy risk lookouts . . .

- **. . . need to know what to look for'.**
- . . . should be made to wear flak jackets'.
- . . . should be on the bridge wing not in the wheelhouse'.
- . . . should report only pirate craft'.

Select the option which correctly describes what NOT to do when manoeuvring a vessel under pirate attack.

- **Zigzag so much that the vessel slows down**
- Change the vessel's course
- Manoeuvre to expose the vessel's sides to the weather
- Use the rudder

Select the option which most accurately completes the following statement with respect to this module.

'Risk assessment is . . .

- **. . . part of the defensive strategy to combat piracy'.**
- . . . a lot of work by vessel's staff with nothing gained'.
- . . . all that is needed to combat pirate attacks'.
- . . . part of the extra, unnecessary work given by the ISM code'.

Select the option which most accurately completes the following statement.

'A pirate citadel is . . .

- **. . . a proper, secure space crew can stay in safety with pirates on board'.**
- . . . a safe haven for pirates'.
- . . . an area inside the vessel, normally the mess room'.
- . . . any space that can be locked up with crew inside and pirates outside'.

Select the option which most accurately completes the following statement.

'As a vessel transits an area with high risk of pirate attack, a Company Security Officer should . . .

- **. . . continue to gather information and pass relevant bits to the Master'.**
- . . . continue to gather information and record it in the company office'.
- . . . hand over the responsibility for the vessel's security to the Master'
- . . . rest easy as their job is done'.

Select the option which most accurately completes the following statement.

'Hijacking is the act of . . .

- **. . . forcibly detaining a vessel against the wishes of the Master'.**
- . . . armed robbery from a vessel on the high seas'.
- . . . boarding a vessel and stealing cargo within port limits'.
- . . . robbing from a vessel in a state's territorial waters'.

Select the option which most accurately completes the following statement.

'Pirates attempting to board a vessel will . . .

- **. . . look for the most vulnerable part of the vessel'.**
- . . . always attempt it during the hours of darkness'.
- . . . always attempt it during the hours of daylight'.
- . . . request that a ladder be deployed for them'.

Select the option which most accurately completes the following statement.

'Armed robbery is . . .

- **. . . taking money or goods from another's possession whilst armed'.**
- . . . an act of piracy on the high seas'.
- . . . forcibly boarding a vessel in a state's territorial waters'.
- . . . taking money or goods from another's possession'.

Select the option which most accurately completes the following statement.

'Physical defences should be . . .

- **. . . set up before entering a high piracy risk area'.**
- . . . arranged by the Company Security Officer'.
- . . . set up by third party contractors in a port'.
- . . . set up in the high risk area before a pirate attack'.

Select the option which most accurately completes the following statement.

'The main reason why centres have been set up by a number of international organisations is to . . .

- **. . . help reduce fraud and piracy'.**
- . . . safeguard the ship owners' assets'.
- . . . safeguard the vessel's crew'.
- . . . stop pirates making a lot of money'.

Which of the given options best describes why it is necessary to send information reports to the international centres set up to combat piracy?

- **Because their work is only effective when they have a lot of information on pirate activity**
- Because insurance companies need the information to set their premiums
- Because the centres record information on attacks for world statistics
- Because there are IMO directives to send information reports

Which one of the given options best describes NATOs simple attitude to piracy?

- **Deter – Disrupt - Protect**
- Deter – Distract - Prosper
- Develop – Destroy - Punish
- Disrupt – Deploy – Panic

Which one of the given options best describes a precaution that can be taken to protect a vessel's crew from personal injury during a pirate attack?

- **Make all crew aware of attack risks with training beforehand**
- Conduct a pre-transit risk assessment
- Tell them not to leave their cabins during a pirate attack
- Tell them not to wear high visibility jackets so they are not easily seen by pirates

Which one of the given options best describes how razor wire should be deployed on board a vessel to stop unwanted boarders?

- **Firmly secured outboard of the rails in coils**
- In a line along the main deck level
- In vertical strips hanging from the rails
- On stairways rising off the main deck only

Which one of the given options best describes how the Ship Security Alert System operates?

- **It is a covert method of sending an alarm from a vessel**
- It is a part of the GMDSS system for communications
- It is a system on board specially for the Indian Ocean/Gulf of Aden area
- It is an internal system to alert the crew of a pirate attack

Which one of the given options best describes operation 'ATLANTA'?

- **It is an operation being undertaken by EU NAVFOR**
- It is an operation being undertaken only by the US navy
- It is an operation happening in the western Atlantic ocean and Caribbean sea
- It is an operation involving European navies working together only in the Gulf of Aden

Which one of the given options best describes passage planning in an area with high piracy risk?

- **The plan should take into account any piracy risk control measures**
- Passage planning is not necessary as naval escort will guide vessels
- Passage planning will take a lot longer because of the risks
- The plan should be no different to usual

Which one of the given options best describes the meaning of the abbreviation MSCHOA?

- **Maritime Security Centre – Horn of Africa**
- Marine Simulation Centre – Horn of Africa
- Maritime Security Centre – Half of Africa
- Maritime Security Centre – Hormuz onto Aden

Which one of the given options best describes the worst overall risk associated with piracy and armed robbery?

- **Crew injured and the vessel hijacked**
- Cargo theft from broached containers
- Large ransoms being paid by the insurance companies
- The vessel not taken care of and therefore needing dry docking after the event

Which one of the given options best describes what should happen to external doors and windows before entering an area with high piracy risk?

- **They should be secured and locked to stop pirates gaining access to internal spaces**
- Ideally all doors should be welded up and all windows should have steel plates fitted
- They should be as normal to help crew escape from the accommodation in an emergency
- They should be secured to stop pirates opening them and gaining access but not locked

Which one of the given options best describes why it is important to stop pirates accessing the vessel's bridge?

- **Because the vessel's bridge is the control centre for the vessel**
- Because the vessel's bridge gives access to the accommodation
- Because the vessel's bridge has the cargo manifest on it
- Because the vessel's bridge has the communications equipment

Which one of the given options best describes why there should be distinctive signals on board a vessel attacked by pirates?

- **Crew actions under pirate attack are different to emergency actions**
- Signals should not let pirates know the vessel is aware of them
- So that crew and passengers are not panicked
- So that pirates are aware that the crew know they are attacking

Which one of the given options can be considered correct as to the attitude of the United States government to US vessel defence in a high piracy risk area?

- **They permit and require an armed defensive capability**
- They do not allow a US registered vessel to carry arms
- They leave it up to the shipping company to decide on armed guards
- They will supply US troops to ride on a vessel to protect it

Which one of the given options can be considered correct as to the presence of armed guards on board a merchant vessel?

- **There should be procedures set up to cover additional risk with them on board**
- The vessel will have to do nothing as they bring with them all that is necessary
- They are on board for the transit but there is little for them to do
- They will sign on the vessel as crew members and come under the Master for instructions

Which one of the given options can be considered most correct as to the attitude to armed guards on board non-US merchant vessels?

- **Governments and organisations allow their deployment in certain cases**
- They are not allowed on board by most government regulations
- They are unacceptably dangerous to the vessel's crew
- They do not really stop the vessel being highjacked

Which one of the given options can be considered the correct action to be taken by the vessel's master in relation to the crew before entering an area with a high risk of pirate attack?

- **Ensure all crew are sufficiently briefed and that drills have been carried out**
- Give the crew minimal information as it is not really their responsibility
- Tell the crew everything including the whereabouts of the keys to the safe
- Tell the crew nothing so that they are not alarmed at the prospect

Which one of the given options can be considered the correct answer to the geographical areas covered by publication BMP4?

- **Publication BMP4 covers only the Indian Ocean, Gulf of Aden and Arabian Sea**
- Publication BMP4 covers only the Gulf of Aden
- Publication BMP4 covers only the Indian Ocean and Caribbean areas
- Publication BMP4 covers only the Indian Ocean and Gulf of Aden

Which one of the given options correctly identifies a simple defensive approach to avoid a pirate attack?

- **Avoid areas known to be frequented by pirates**
- Always communicate with naval authorities
- Avoid every small boat seen at sea
- Ensure all crewmen know what to do in case of attack

Which one of the given options describes the simplest method that a vessel's crew can use to prevent pirates from hijacking their vessel in order to demand a ransom?

- **Stop them boarding the vessel**
- Always have a naval escort
- Avoid trading in pirate areas
- Complete a thorough risk assessment

Which one of the given options describes the two methods of risk assessment that can be used to help combat pirate attacks?

- **It can be completed on board the vessel or by using a commercial company**
- It can be completed on board the vessel or separately by the Company Security Officer
- One is the on board method the other is performed by the P&I Club
- One method is to ignore the risk and the second method take a chance in pirate areas

Which one of the given options is the best defence against pirate attack in an area with high piracy risk?

- **Have an armed vessel in attendance during the transit**
- Do not switch on the vessel's navigation lights at night so pirates do not see the vessel
- Have a merchant vessel transit alone at its maximum speed
- Mount dummies as lookouts around the vessel to scare pirates

Which one of the given options is the most accurate in relation to the following statement?

'Foam is a useful anti boarding deterrent'

- **Foam is slippery and makes climbing ladders or ropes more difficult**
- Foam is a sign that the vessel is prepared for pirates
- Foam is poisonous so pirates avoid it
- Foam, dispensed over board, can be part of the vessel's fire fighting stock

Which one of the given options is the most accurate in relation to the following statement?

'Vessels are safe from being boarded and robbed when using major shipping routes '.

- **The statement is incorrect as some major routes have had several boarding incidents**
- The statement is correct as there is always naval patrols in these areas
- The statement is correct as there is always VTS surveillance in these areas
- The statement is incorrect as a vessel is safe from being boarded with other vessels around it

Which one of the given options is the primary point of contact for a vessel in Asia wanting communications on piracy?

- **ReCAAP centre in Singapore**
- The Indian Maritime Administration
- The vessel's classification society
- UK Maritime Trade Organisation

Which one of the given options is the primary point of contact for a vessel in the Gulf of Aden wanting communications on piracy?

- **UK Maritime Trade Organisation**
- International Maritime Bureau – Piracy Reporting Centre
- Interpol via port sates administration
- Maritime Security Centre – Horn of Africa

0247 – Anchor Mooring Procedures

Version 1.0 (1939)

Once a vessel is brought up, what is the advantage of disengaging the bow chain-stopper and allowing the chain cable to be held by the brake?

- **In case of bad weather, more cable can be paid out easily to increase the Scope and improve anchor holding power.**
- Strong winds can release the bow chain-stopper.
- The bow stopper may get jammed by the cable link when strain increases in bad weather, requiring use of engine and windlass power to release it.
- The brake has more strength than the bow chain-stopper.

What are the two disadvantages of the Standing Moor over the Running Moor?

- **The Standing Moor is more dependent upon wind and current, which are less reliable than using the vessel's engine.**
- **The Standing Moor takes longer to execute than the Running Moor.**
- In the Standing Moor, the anchor cables are more prone to fouling.
- Pilots never execute the Standing Moor.

What are the two principal risks that could lead to injury and damage, whilst executing the Mediterranean Moor? (Select two answers)

- **A fouled propeller.**
- **Overstraining the aft ropes and winches.**
- Excessive strain on the forward windlass.
- Failure of the anchor cable.

What is the advantage of using 2 anchors to moor a vessel?

- **The vessel will not move over such a large area of the anchorage when swinging to the wind and tide.**
- In case of another vessel drifting down upon her, the vessel will hold firmly in her position.
- Officers and crew can rest without having to monitor the anchor and cable.
- The anchors will not drag.

What is the most common procedure used for anchoring in an emergency?

- **Letting go anchor from outside the hawse pipe.**
- Letting go anchor from just above the seabed.
- Using the Running Moor technique.
- Walking out anchor all the way.

What is the most significant risk of letting go anchor from just above the sea bed?

- **The anchor may take hold whilst the ship speed is too high, and damage to equipment or injury to personnel may result**
- The anchor chain may foul the anchor
- The anchor chain may foul the hull
- The anchor may not take hold properly

What risks are associated with letting go anchor from the hawse pipe? (Select all applicable answers)

- **Anchor damage if falling onto hard rock.**
- **The cable taking charge and running out fully.**
- Failure of the anchor cable.
- The windlass being overstrained and damaged.

When executing a running moor in calms or when wind and current are acting the same way, which anchor should be dropped first on a vessel with a right hand propeller?

- **The starboard anchor**
- The port anchor.

When executing the Mediterranean Moor with a light cross wind, from where should the first ropes be sent to mooring buoys and bollards?

- **From the lee quarter to the windward buoys.**
- From the lee quarter to the leeward buoys.
- From the vessel's centreline to the buoys directly astern.
- From the windward quarter to the windward buoys.

When executing the Mediterranean Moor, what should be the distance on the seabed between the first and second anchor?

- **One third the final length of cable to be paid out for each anchor.**
- Equal to the final length of the cable to be paid out for each anchor.
- Equal to the ships length.
- Half the final length of cable to be paid out for each anchor.

When executing the Mediterranean Moor, which anchor should be dropped first?

- **The offshore anchor.**
- The anchor on the vessel's shore-side.
- The port anchor.
- The starboard anchor.

When executing the Standing Moor in conditions where wind and current are running across each other, which anchor should be dropped first?

- **The anchor on the vessel's weather side.**
- The anchor on the vessel's leeward side.
- The port anchor
- The starboard anchor.

When unmooring from the Mediterranean Moor, which stern ropes should be cast off first?

- **Ropes leading to the leeward buoys.**
- Ropes leading to the windward buoys.

Which of the following is the main reason for letting go anchor from just above the sea bed?

- **To reduce the chance of the momentum of the anchor and cable becoming too high**
- To allow more time for the vessel to manoeuvre in the anchorage
- To increase the holding power of the anchor
- To speed up the anchoring process

Which one of the following options best describes what is meant by the scope of an anchor cable?

- **The length of the cable compared to the depth of water**
- The length of cable above the waterline
- The length of cable below the waterline
- The length of the cable compared to the length of the ship

Which one of the following options would be the best speed over the ground when dropping anchor using the windlass brakes?

- **0.5 knots astern.**
- 0 knots.
- 1 knot astern.
- 1.5 knots astern.

Which one of the following options would be the best speed over the ground when walking the anchor out all the way?

- **Less than 0.3 knots astern.**
- 0 knots.
- 0.5 knots ahead.
- 0.5 knots astern.

Which three of the following are the main disadvantages of walking out anchor all the way?

- **Disuse of brakes and operating skills.**
- **Straining the windlass.**
- **Trying to judge the cable lead visually, which may be difficult in the dark or due to bow overhang.**
- Straining the windlass brakes.
- Uncontrolled payout of anchor cable.

0250 – Conflict Management.

Version 3.0 (213)

A chief engineer, who has investigated a conflict between two engineroom ratings, finds that the fitter does not want the oiler to enter the workshop while he is busy working inside. Which one of the following types of conflict is occurring?

- **Interest conflict**
- Personal conflict
- Technical conflict
- Values conflict

Conflicts in meetings can be very disruptive, but they can also be very helpful. Which one of the following is NOT a good conflict handling technique for meetings that get 'out of control'?

- **Present your view and force for an agreement.**
- Agree that the person has a valid point and there may be some ways to make the situation work for both parties.
- Ask to speak with the person after the meeting or during the break.
- Document the subject and discuss it in the next meeting.

One of the techniques in avoiding and/or resolving conflicts is to accept that healthy disagreement can build better decisions. Which one of the following statements reflects this principle?

- **"Please try to explain what you mean so that I would be in a position to understand your point better and the reason why we disagree."**
- "I still cannot accept that we see this thing very differently."
- "I think I should stop talking to you. We have both better things to do than argue."
- "Please refrain from saying unnecessary things. I feel disappointed every time I hear irrelevant comments."

One of the techniques in avoiding and/or resolving conflicts is to discuss differences in values openly. Which one of the following statements does NOT reflect this principle?

- **"I do not accept your solution."**
- "I realize that I am disorganized at times. So you want me to be organized, right?"
- "I understand that we do the pre-departure preparations differently."
- "So it is important to you that I refrain from using foul languages."

People deal with the so-called "pure" conflict relating to facts at a certain level. Which one of the features given in the options is the level of conflict that is about facts?

- **Disagreement**
- Dialogue disappears
- Personalising
- Problem is growing

Select from the options the term which best describes the level of the following conflict.

'You have an unresolved conflict with the messman. You no longer have any contact with him and both of you still have a deep-rooted anger. There seems to be no way to come to terms with each other.'

- **Polarizing**
- Dialogue disappears
- Disagreement
- Personalising

Select the option that most accurately completes the following statement.

“A conflict is constructive when it . . .

- **. . . results in clarification of important problems and issues.”**
- . . . polarizes people and groups.”
- . . . reduces cooperation.”
- . . . takes away the attention from other important activities.”

Select the option that most accurately completes the following statement.

“A conflict is destructive when it . . .

- **. . . polarizes people and groups, and reduces cooperation."**
- . . . builds cooperation among people by learning more about each other."
- . . . causes effective communication."
- . . . results in clarification of important problems and issues."

Select the option that most accurately completes the following statement.

“The advantage of using I-language is that you communicate and express your emotions and intentions without . . .

- **. . . attacking the other person.”**
- . . . listening actively.”
- . . . understanding the other person’s point of view.”
- . . . asking questions.”

Select the option that most accurately completes the following statement.

“In order to arrive at a win-win solution to problems, it is best to . . .

- **. . . define the problem in such a way that it is mutually acceptable to both parties.”**
- . . . amplify the faults of the other party.”
- . . . personalize the problem and be defensive.”
- . . . accept the fact that somebody has to lose.”

Select the term from those given in the options which most accurately fills the gap in the following statement.

The . . . expresses what is going on inside us through obliging, cooperating and non-violent means. We tell about how we feel by avoiding to judge, blame or interpret the other person’s actions.

- **I- Language**
- Maritime English language
- We - Language
- You - Language

The Master is angry because the third officer was not able to prepare the lifesaving appliances for inspection. Pointing a left finger at the third officer the Master shouts "You are such a lazy officer!" Which one of the options best describes this form of language in the context of conflict?

- **You-Language**
- English language
- I- Language
- Maritime language

The chief mate has been very critical of the crew's performance after seeing him asleep while on lookout duties on several occasions. If you are the Chief Mate, which of the following statements will you say to AB in order to avoid destructive conflict?

- **"I get frustrated every time I see a crew member doing that."**
- "If you do that one more time, you will be dismissed."
- "You are unbelievably useless."
- "You are very irresponsible!"

There has been a complaint from the crew regarding 'alleged' unfair usage of the television. Which one of the following types of conflict is this considered to be?

- **Interest conflict**
- Personal conflict
- Technical conflict
- Values conflict

Two crewmen are heard to be arguing with each other about their religious beliefs. Which one of the following types of conflict is this?

- **Values conflict**
- Interest conflict
- Personal conflict
- Technical conflict

When you are attacked personally by another person, which one of the following actions should you take in order to resolve the conflict?

- **Try to understand the real problem behind the conflict.**
- Do not pay attention.
- Let the other party hook you into an emotional reaction.
- Start to get personal with the other party to get even.

Which one of the activities given in the options is least likely to lead to a conflict?

- **reading a book**
- interpersonal relationships
- meetings
- negotiations

Which one of the following is NOT a constructive technique for avoiding and resolving conflicts option?

- **Focus on the past.**
- Communicate honestly.
- Discuss your differences in values openly.
- Stick to the facts.

Which one of the following is NOT a good handling technique when dealing with interpersonal relationship issue?

- **Ensure that you are understood first before trying to understand the other person's point of view.**
- As you talk, ask for feedback. Do not 'attack' the other person with accusations.
- Be sure to respect his/her opinions.
- Try to listen to the other person with open minds.

Which one of the following is NOT a recognised type of conflict?

- **Complicated conflict**
- Interest conflict
- Technical conflict
- Values conflict

Which one of the following statements about conflicts is TRUE?

- **We cannot solve conflicts by ignoring them and hoping that they will go away**
- Conflicts are part of our daily lives. We cannot do anything about them.
- Conflicts onboard are only temporary. Therefore they can just be ignored as shipboard personnel are signing on and off in a matter of months.
- We cannot normally solve conflicts by using the "I-Language".

Which one of the following statements best describes I-Language?

- **Focusing on the 'wrong doing'**
- Asking leading question
- Blaming in a precise and unclear manner
- Focusing on the 'wrong doer'

Which one of the following statements best describes a technical conflict?

- **It is about a concrete target, means, method, structure, or work task.**
- It is about identity, self-esteem, loyalty, breaking one's trust, or rejection.
- It is due to limited resources such as money, work, space, materials and time.
- It is often about opposing views on ethical and moral standards.

Which one of the following statements best describes the disagreement level in conflicts?

- **It is easily dealt with through a discussion where both parties try to understand each other's point-of-view.**
- Both parties are blaming and misunderstanding each other.
- Communication lines between the involved parties create more animosity and less understanding.
- It is characterized by the strong desire of both parties to win over the other party.

Which one of the following statements refers to the 'You'-language?

- **Finger pointing and blaming language**
- Asking an open ended question
- Avoiding to judge the person
- Listening carefully

Which one of the following types of conflict applies to situations when people want to fight for their personal and cultural values?

- **Values conflict**
- Interest conflict
- Personal conflict
- Technical conflict

Which one of the following types of conflict is linked to identity, self-esteem, loyalty, or breaking one's trust and rejection?

- **Personal conflict**
- Interest conflict
- Technical conflict
- Values conflict

Which one of the following types of conflict is said to be happening when people disagree on what must be done or how to carry out certain tasks?

- **Technical conflict**
- Interest conflict
- Personal conflict
- Values conflict

Which one of the language types given in the options is characterized by blaming, focusing on the past, finger-pointing and asking leading questions, and, when used in a conversation, may make a conflict worse.

- **YOU- Language**
- English Language
- I- Language
- Maritime English

Which one of the reasons given in the options is considered the main reason why people tend to avoid conflicts?

- **Because of the emotional pressures that come along with it**
- Because conflicts are rare on board ships
- Because conflicts tend to go away as time passes by
- Because people are often too busy and so there is little time left to address conflicts

0251 – Culture.

Version 2.0 (320)

A 4th Engineer has seen that some of the entries made by the 1st Engineer in the Oil Record Book are incorrect. The 4th Engineer is aware that such incorrect entries may lead to heavy penalties if found during inspection by port authorities. However, the 4th Engineer is hesitant to raise the issue in case the 1st Engineer either sees it as questioning his capability or gets insulted. Which one of the following cultural orientations is the 4th Engineer showing?

- **High power distance**
- High uncertainty avoidance
- Individualism
- Low power distance

A newly embarked young officer from a high context communication culture wants to approach his superior from a low context communication culture to pass on some information. How should the young officer effectively deliver the message?

- **Be factual and direct in saying his message.**
- Avoid communicating with his superior
- He should ensure that he makes a lengthy introduction before getting to his point.
- Use indirect words to avoid confrontation.

A newly promoted 3rd Mate from a high-power distance culture joined the ship recently. Most of the shipboard officers and crew are from a low-power distance culture. Which one of the following actions can you take, as the Chief Officer, to help the 3rd Mate to adapt to the culture onboard?

- **Encourage him to speak up and become more assertive.**
- Act on his behalf and be his spokesperson during meetings.
- I will do nothing.
- Let him come up with his own ways to adjust.

According to Edgar Schein, culture is divided into three layers. Select, from the following, the one that contains these three layers.

- **Artifacts, norms and values, and basic assumptions**
- Artifacts, beliefs, and stereotypes
- Outer, middle, and inner
- Reactive, calculative, and proactive

According to Edgar Schein, which layer of culture refers to a group's idea of right or wrong and what is good or bad?

- **Norms and values**
- Artifacts
- Basic assumptions
- Perception

According to Edgar Schein, which one of the following is the term used to describe a group's unspoken and subconscious rules on how to act, think and behave?

- **Basic assumptions**
- Artifacts
- Norms and values
- Perception

Although it is not an official rule, it is widely accepted by all seafarers that smoking is only allowed in the day room. Select the option containing the word or phrase that correctly fills the gap in the following statement. According to Edgar Schein this is an example of . . . in the layers of culture.

- **basic assumptions**
- artifacts
- expressed values
- standard practice

As a 2nd Engineer, you observe that a number of maintenance jobs involving engine ratings are not completed as scheduled. Considering that most of the engine ratings are from a collectivist culture, how would you notify them of this concern?

- **Ask the concerned ratings individually in private about their reasons for not completing the scheduled jobs.**
- Call for a meeting and tell them directly that I am not happy with the maintenance backlog.
- Give them additional time to finish all pending jobs.
- Report it to the Master and let them decide the next actions.

Despite the fact that Bosun has several years of shipboard experience, he has been very open to accept ideas of other crew members. He believes that his ways of doing things may not always be the best ways at all times. Which one of the intercultural competencies given in the options does the Bosun show?

- **Adaptability**
- Coaching
- Intercultural Communication
- Knowledge

On your first day of working on board the ship, you were amazed to find out that no safety signs are posted in the workshop. In spite of this, all crew members are wearing the necessary personal protective equipment, PPE. Select the option containing the word or phrase that most accurately fills the gap in the following statement. This display of good safety behaviour is an example of . . . in the layers of culture identified by Edgar Schein.

- **basic assumptions**
- artifacts
- norms and values
- reactive level

Select the option containing the word or phrase that most accurately fills the gap in the following statement. 'People from . . . culture tend to think only of themselves as individuals; they are more focused on their own goals.'

- **an individualist**
- a collectivist
- a high uncertainty avoidance
- a maritime

Select, from the following, the phrase that most accurately completes the following statement. 'Emotional need for rules, agreements, and regulations to avoid unfamiliar situations is a characteristic of a culture with . . .

- **. . . high uncertainty avoidance.'**
- . . . high-context communication.'
- . . . low uncertainty avoidance.'
- . . . low-context communication.'

Select, from the following, the phrase that most accurately completes the following statement. 'When dealing with people from collectivist culture, you have to remember that it is only appropriate to deliver critical feedback . . .

- **. . . in private in order to avoid embarrassing the person.'**
- . . . during the monthly meeting.'
- . . . in the crew mess room.'
- . . . in the presence of other crew.'

Select, from the following, the word or phrase that most accurately completes the following statement. 'In a low-power distance culture, subordinates are expected to be consulted, however, in a high-power distance culture the subordinates are expected to be . . .

- **. . . told what to do.'**
- . . . careful in making decisions.'
- . . . independent.'
- . . . safety-conscious.'

The following signs about smoking were handed to you. Which of these are you going to use if the target group are more used to high-context communication?

- **Thank you for not smoking.**
- No smoking here.
- Smoking is not allowed here.
- Smoking is strictly prohibited.

There are seafarers from some cultures who will simply say "Yes" even if they do not really understand the instructions. They simply do it to avoid losing face but in reality, they really mean "maybe" or "I don't understand". For which one of the following communication styles is this behaviour typical?

- **High-context communication**
- Active listening
- Direct communication
- Low-context communication

When you go onboard a ship, you will get a first impression of the shipboard safety culture by observing items such as helmets, coveralls, safety shoes, goggles and safety signs. Select the option containing the word or phrase that correctly fills the gap in the following statement. According to Edgar Schein these items are considered as . . . in the layers of culture.

- **artifacts**
- basic assumptions
- expressed values
- safety equipment

Which of the following scenarios is an example of a poor personal attitude towards different cultures?

- **The other crew members make fun of the unique voice tone of a crew member from Eastern Europe.**
- The Captain allows his Muslim crew to do their regular prayers.
- The Chief Cook ensures that appropriate meals are prepared for all nationalities.
- The German Chief Engineer knows very well when to get firm and when to take it lightly when dealing with the Indian Motorman.

Which one of the following is typical in a low power distance culture?

- **Leader and subordinates work closely together and consult each other.**
- Privileges and status symbols for leaders are both expected and popular.
- Subordinates are hesitant to express their doubts and disagreements with leaders.
- Subordinates expect to be told what to do.

Which one of the following is typical of people from an individualistic culture?

- **Every one usually has a right to a personal opinion.**
- Leaders are expected to take all initiatives.
- People value harmony and loyalty to the company.
- Subordinates expect to be told what to do.

Which one of the following statements about collectivist culture is FALSE?

- **Everyone has a right to privacy.**
- Opinions are predetermined by group membership.
- People avoid direct confrontation.
- People avoid disagreeing with someone's opinion in the presence of other people.

Which one of the following statements is a feature to be expected in an individualistic culture?

- **Laws and rights are equal for all.**
- Direct confrontation is avoided.
- Individuals prefer to use high-context communication style.
- Relationship prevails over task.

Which one of the following statements is typical of a group orientation or collectivist culture?

- **Individuals focus on how to ensure good interpersonal relationships.**
- Each individual competes with other group members to be regarded as the strongest member.
- Individuals speak frankly about personal opinions, regardless of what other crew members will say.
- Individuals think only of personal interests and benefits.

Which one of the following terms is NOT a dimension of culture according to Hofstede?

- **Safety maturity**
- Collectivism versus individualism
- Power distance
- Uncertainty avoidance

While the Captain is presenting his ideas during a safety meeting, the Messman offers his opinion on the matter. The Captain acknowledges him and even encourages others to come out with their own ideas, even if they will be in conflict with the Captain's own ideas. Select, from the following, the phrase that correctly completes the following statement. 'This scenario is typical in cultures defined as . . .

- **. . . low power distance.'**
- . . . group orientation.'
- . . . high power distance.'
- . . . uncertainty avoidance.'

You are a junior officer from a culture with high context communication style while your Scandinavian Captain communicates in low context. Which of the following actions best describes how you should act in order to communicate effectively with the Captain?

- **Express my ideas and opinions in a straightforward manner.**
- Be very careful with my choice of words in conveying my message.
- Think twice before speaking and speak politely.
- Wait for his permission before I speak.

You are an officer on board a ship and you have a multi-cultural crew. In communicating with other shipboard personnel, which of the following practices should you do?

- **Use the communication style matching the culture of the person you are talking to.**
- Always use the low context communication style.
- Only use the high context communication style.
- Use high context communication style for ratings and low context for officers.

You are scheduled to meet an American executive next week. As part of your preparation you have read some material on culture and found out that most Americans prefer the low-context communication style. In order to ensure that the meeting will go smoothly, which of the following measures will be most appropriate?

- **I will be straightforward and concise with my statements during our discussion.**
- I will start by introducing myself. Before the end of the meeting, I will also ensure that I have gained a lot of knowledge about their personal life.
- I will take it as an informal meeting.
- In between discussions, I will tell jokes and funny stories in order to spice-up the meeting.

You are the Chief Engineer from a low-context communication culture. You are about to set a meeting with your crew members who come from a high-context communication culture. How will you address the crew in an effective way?

- **I will bear in mind that I need to be more patient in encouraging everyone to speak-up, and avoid embarrassing individual crew members in front of others.**
- I have to impose my authority. Straightforward communication style is most applicable in this case.
- I will be direct in my communication. I may hurt other people's feelings but at least I am honest.
- There's no need to go around the bush. I will say what I want to tell them. They will definitely respect what I will say.

You want to develop the intercultural competencies of your crew. In one of your toolbox meetings, a rating comes up with an idea which is exactly the same as yours. How should you handle this situation in order to encourage others to participate in the meeting?

- **Acknowledge the rating and give him the credit.**
- Ignore him since you have the same idea.
- Tell him that you have thought of the same idea earlier.
- Tell him to come up with a better idea.

0252 – Active Listening.

Version 2.0 (184)

Active listening consists of verbal and non-verbal skills that you should use in a conversation.

Which one of the following is a non-verbal response that indicates your interest to the speaker?

- **Nodding your head**
- Looking at the surroundings
- Saying “I see” or “hmm”
- Saying “Is that so?”

Active listening is a fundamental part of good communication.

Which one of the following statements best describes active listening?

- **Active Listening involves listening with the purpose of understanding.**
- Active listening forces people to talk.
- Active listening is applicable to officers only.
- Active listening is the same as hearing.

Active listening is a tool for building and maintaining pleasant interpersonal relationships.

Which one of the following is NOT a positive effect of listening actively?

- **Avoiding stress.**
- Connecting with other people.
- Maximizing positive responses
- Minimizing negative responses.

Active listening is a very useful communication tool for shipboard personnel in building interpersonal relationship.

To which one of the following activities can the tool NOT be applied?

- **Listening to music**
- Conflict handling
- Motivating people
- Problem solving

Active listening is one of the important tools in effective communication. Which one of the following practices does NOT normally result in effective communication?

- **Using the ‘You’ language**
- Avoiding distractions
- Maintaining an appropriate level of eye contact
- Showing relaxed, open body language

In communication, body language has the biggest effect on how the message is conveyed to the listener.

Which one of the following is a desirable body language?

- **Leaning slightly forward while seated facing the speaker.**
- Folding hands across your chest.
- Leaning backwards on the chair while talking.
- Yawning while talking to a person.

In relation to active listening, which one of the following is the main reason for paraphrasing?

- **To ensure that you understood what the speaker has said.**
- To convey that you don't understand what the speaker is talking about.
- To ensure agreement.
- To ensure that the speaker understands you.

In relation to active listening, which one of the following terms is used to identify short expressions such as "I see . . . ", "Hmm . . . ", "Ok . . . " that are used to show that you are listening to another person?

- **Manifestations**
- Affirmations
- Confirmations
- Interrogations

Select from the following options the phrase which most accurately completes the following statement.

"Active listening is different from hearing as it . . .

- **. . . involves listening and the use of other senses."**
- . . . always involves more than two people."
- . . . does not require any understanding in a conversation between two people."
- . . . requires an above average sense of hearing."

Select, from the options given, the phrase or word which most accurately completes the following statement.

"In relation to active listening, avoiding eye contact when listening to another person can be interpreted as . . .

- **. . . lack of interest."**
- . . . lack of awareness."
- . . . politeness."
- . . . shyness."

Select, from the options given, the phrase which most accurately completes the following statement.

"Asking an open ended question is one of the elements of active listening and can be used for the purpose of . . .

- **. . . encouraging the speaker to elaborate more and ensure understanding."**
- . . . establishing that the speaker is telling the truth."
- . . . leading the speaker to what you want to hear."
- . . . proving that what the speaker is saying is wrong."

Select, from the options given, the phrase which most accurately completes the following statement.

"One way of demonstrating good listening skills while listening to another person is to ask open-ended questions as they are generally . . .

- **. . . not answerable by either "yes" or "no"."**
- . . . answerable by "yes" only."
- . . . questions that can generate a correct answer."
- . . . questions that involve the word "open"."

Select, from the options given, the phrase which most accurately completes the following statement.

"Active listening is an effective communication tool which can be applied to many shipboard applications, but not to . . .

- **. . . writing safety reports."**
- . . . conflict management."
- . . . safety meeting."
- . . . toolbox meeting."

Select, from the options given, the phrase which most accurately completes the following statement.

"Asking open questions is a way to help . . .

- **. . . you draw out more information and focus attention on topics of interest."**
- . . . help you to find faults in the other person's statement."
- . . . others settle their conflicts."
- . . . you clarify certain points with the person you are talking to especially when you disagree with their point-of-view."

Select, from the options given, the phrase which most accurately completes the following statement.

"In a conversation, an indication that the other party is actively listening is when the speaker . . .

- **. . . feels that they are understood and taken seriously."**
- . . . feels unimportant."
- . . . is embarrassed."
- . . . responds with "yes" or "no"."

Select, from the options given, the phrase which most accurately completes the following statement.

"In order to use your body language in a constructive manner, you should . . .

- **. . . communicate interest by facing the speaker, keeping an open stance and paying attention."**
- . . . avoid facing the speaker and concentrate on your surroundings."
- . . . fold your arms over your chest."
- . . . lean back on your chair and glance in another direction."

Select, from the options given, the phrase which most accurately completes the following statement.

"In relation to active listening, paraphrasing means . . .

- **. . . expressing in your own words what you think the speaker has just said."**
- . . . asking the speaker to repeat what they have just said."
- . . . giving a different meaning to what the speaker has just said."
- . . . writing down what the speaker has just said."

Select, from the options given, the phrase which most accurately completes the following statement.

"In relation to active listening, summarizing is a useful way to . . .

- **. . . recap the segments of a conversation to make sure you . . . understood what the other person has just said."**
- . . . explain your idea on what the other person should do."
- . . . introduce other people to the subjects that should be discussed."
- . . . learn how to manage conflicts properly."

Select, from the options given, the phrase which most accurately completes the following statement.

"In relation to active listening, when you paraphrase, you try to . . .

- **. . . translate into your own words what the other person has said."**
- . . . avoid the use of negative words."
- . . . create a situation where the person you are talking to will take a defensive position."
- . . . repeat the exact words that the speaker has used."

Select, from the options given, the phrase which most accurately completes the following statement.

"When you listen actively, you are sending a signal to the person you are talking to that . . .

- **. . . they are important to you."**
- . . . problems can be solved if dealt with properly."
- . . . you are important to the speaker."
- . . . you have experience in solving problems."

Select, from the options given, the phrase which most accurately completes the following statement.

"When you use your body language in a constructive manner, you should . . .

- **. . . show interest by facing the person and keep an open stance."**
- . . . avoid facing the speaker and monitor the events in the surroundings."
- . . . fold your arms across your chest."
- . . . lean back on your chair and glance in a different direction."

Select, from the options given, the word which most accurately fills the gap in the following statement.

"Good listeners ask questions, however, questions that begin with the word . . . should be avoided because they may lead the speaker to become defensive".

- **. . . 'why' . . .**
- . . . 'how' . . .
- . . . 'what' . . .
- . . . 'when' . . .

The Chief Officer is talking to the AB and at the same time working on the cargo loading computer. As the Chief Officer, which one of the following should you AVOID in order to demonstrate good listening skills?

- **Telling the AB to stop talking as you need to focus on your work.**
- Asking open questions.
- Making appropriate level of eye contact.
- Stop working momentarily and face the AB.

The Second Engineer has a habit of interrupting you while you are talking and does not let you finish your statement before suggesting what to do. What should the Second Engineer do in order to be an active listener?

- **They should give you sufficient time to explain and understand your side before they respond.**
- They should ensure that they are well understood all the time.
- They should force you to listen to them so you will learn from their experience.
- They should give themselves more time to explain further so that you understand them.

Which of the descriptions given in the options best describes the type of eye contact that should be used to show that you are actively listening to another person?

- **Maintain reasonable eye contact, but avoid a challenging stare.**
- Avoid looking into the eyes of the speaker.
- Keep a constant focus on the eyes and nose of the speaker.
- Wink from time to time to avoid being sleepy.

Which one of the descriptions given in the options most accurately describes the meaning of the following statement?

"Do not get distracted by your own ideas, or by other events in your surroundings, while talking to another person."

- **Put down your work and look at the person you are talking to.**
- Continue your work while talking to the other person.
- Put down your work, and avoid arguments on what is going on inside the room.
- Put down your work, look around and observe what is going on inside the room.

Which one of the following actions is an example of the use of manifestations in active listening?

- **Nodding your head in agreement or shaking your head in disagreement.**
- Asking questions once in a while.
- Clarifying the impact of the other person's performance, conduct, attitude or behaviour.
- Describing what you have seen, heard, observed or experienced.

Which one of the following is likely to be associated with a person who is considered to be a poor listener?

- **Someone that interrupts you while you are talking.**
- Someone that encourages you to elaborate your thoughts further.
- Someone that makes you feel important.
- Someone that makes you feel understood and taken seriously.

Which one of the following options is a poor reason to use paraphrasing during a conversation?

- **To inform the speaker about your own ideas.**
- To clarify certain points.
- To maintain a shared level of understanding.
- To restate the speaker's ideas in your own words.

Why is it important to summarize the key points when you are engaged in a conversation?

- **It is a useful way to affirm your understanding of the speaker's ideas.**
- In order to explain your own point of view.
- So that the speaker and listener can agree to disagree.
- So that the speaker can agree with you.

0253 – Corrective Feedback.

Version 2.0 (141)

Based on the third step of the S.O.E.D. Feedback Model, which one of the following statements can be used in order to describe or clarify the impact of the other person's performance, conduct, attitude or behaviour?

- **"The effect of that action is . . . "**
- "I have been thinking about the way we . . . "
- "In the future I would like you to remember that . . . "
- "It could be better if . . . "

Generally, good feedback should highlight the different aspects of the person's performance.

Which one of the following can be considered a feature of ineffective feedback?

- **Highlighting only poor performances**
- Highlighting positive performance.
- Identifying areas for improvement.
- Suggestions for improvement.

In the fourth step of the S.O.E.D. Feedback Model, the person giving the feedback describes the changes they wish to see.

Which one of the following phrases can be used to state these changes?

- **"I would like you to . . . "**
- "Please remember to . . . "
- "What happens when you do this is that . . . "
- "You must . . . or else . . . "

Select, from the following options, the phrase which most accurately fills the gap in the following statement.

"In giving corrective feedback using the Sandwich Feedback Model, you state the . . . in the second layer or the beef."

- **area/s of improvement you wish to see from the person**
- effect of the actions of the concerned person
- general positive statement that you see from the concerned person
- positive things that you see in the concerned person

Select, from the options given, the combination that most accurately completes the following statement.

"The acronym SOED stands for . . .

- **. . . subject, observation, effect and demand."**
- . . . sample, obedience, education and demand."
- . . . social, opportunity, empowerment and development."
- . . . subject, observation, effect and disposition."

Select, from the options given, the phrase which most accurately completes the following statement.

"In the three-layer Sandwich Feedback Model, the second layer, referred to as 'Beef', is where you . . .

- **. . . state the changes that you would like to see."**
- . . . end your feedback with a general positive statement of the subject."
- . . . mention three positive things that the person has done."
- . . . state the effect of a negative performance."

Select, from the options given, the phrase which most accurately completes the following statement.

"It is important to communicate critical feedback because, if the problem is left unaddressed, . . .

- **. . . the situation will probably just get worse."**
- . . . it is important that someone is blamed for the problem."
- . . . someone else will have to deal with the problem."
- . . . the issues will be forgotten."

Select, from the options given, the phrase which most accurately completes the following statement.

"One of the main reasons why some officers find it difficult to give feedback is because they . . .

- **. . . feel awkward, especially when giving critical feedback."**
- . . . don't think it is part of their job."
- . . . think it is pointless if the error has already been made."
- . . . think only the Captain should do it."

Select, from the options given, the phrase which most accurately completes the following statement.

"One of the reasons for a feedback model is to determine that the performance feedback is delivered in a . . .

- **. . . concise and direct manner which is formulated to have the best possible impact."**
- . . . concise but humane manner."
- . . . concise but straightforward manner."
- . . . direct but flowery manner."

Select, from the options given, the phrase which most accurately completes the following statement.

"The three layers of the Sandwich Feedback Model in sequential order are the . . .

- **. . . top bun, the beef and the final bun."**
- . . . beef, the dressing and the final bun."
- . . . top bun, the final bun and the beef."
- . . . top bun, the sauce and the final bun."

Select, from the options given, the phrase which most accurately fills the gap the following statement.

"In third layer, or the final bun, of the Sandwich Feedback Model . . . should be included.

- **. . . a general positive statement about the receiver . . .**
- . . . the areas of improvement required from the receiver . . .
- . . . the effect of the actions of the receiver . . .
- . . . three positive things that you observe about the receiver . . .

Select, from the options given, the word which most accurately completes the following statement.

"Giving feedback is difficult especially when the type of feedback you have to deliver is . . .

- **. . . critical."**
- . . . appreciative."
- . . . motivational."
- . . . positive."

Select, from the options given, the word which most accurately completes the following statement.

"When moving from layer 1 to layer 2 in the Sandwich Feedback Model, you should avoid using the word . . .

- **. . . 'but'."**
- . . . 'and'."
- . . . 'maybe'."
- . . . 'yes'."

Select, from the options given, the word which most accurately fills the gap in the following statement.

"Describing what you expect from the person and the ways that they can change or improve their conduct refers to . . . in the S.O.E.D. Feedback Model."

- **Demand**
- Effect
- Objective
- Subject

Select, from the options given, the word which most accurately fills the gap in the following statement.

"When using the Sandwich Feedback Model, the word . . . should be used in moving from the first to second layer in order to avoid the receiver taking a defensive position."

- **. . . 'and' . . .**
- . . . 'however' . . .
- . . . 'otherwise' . . .
- . . . 'yes' . . .

The Chief Mate has to give some corrective feedback to one of the crew regarding the wearing of incorrect PPE.

Using the S.O.E.D. Feedback Model, which one of the following statements should the Chief Mate use in starting the corrective feedback?

- **"I want to talk to you about . . . "**
- "I am very disappointed . . . "
- "Please remember to . . . "
- "You seem not to impress me on . . . "

The Sandwich Feedback Model identifies that positive comments delivered in the first layer should motivate the receiver.Select, from the options given, the term which most accurately completes the following statement.

"The receiver should become very motivated and be ready to digest the . . . "

- **. . . beef."**
- . . . final bun."
- . . . top bun."
- . . . topping."

The following statements were mentioned by the Chief Engineer to the Fitter while giving corrective feedback.

Which one of the statements refers to the "Effect step" in the S.O.E.D. Feedback Model?

- **"The impact of such action is that it creates a safety hazard in the engine room."**
- "I noticed that when you leave your workplace, you often fail to keep your tools and clean up the workbench."
- "I want to talk to you about our cleanliness and safety practices in the engine room before you finish your work today."
- "I would like you to go back and put your tools in proper place. From now on, clean up the workbench before you leave."

The second step in the S.O.E.D. Feedback Model is to describe what you have observed, seen, heard, or experienced.

Which one of the following phrases can be used in this layer?

- **"I noticed that . . . "**
- "In the future I would like you to remember that . . . "
- "The effect of your action is that . . . "
- "There is something I would like to talk to you about"

To which one of the steps of the S.O.E.D. Feedback Model do the following statements correspond?

"Lately, I have noticed that you were not paying much attention to our safety policies. There were several instances when I saw that you were not wearing the correct PPE while working."

- **Observation**
- Demand
- Effect
- Subject

To which one of the steps of the S.O.E.D. Feedback Model does the following statement correspond?

"The next time you come across this situation I want you to check our procedures, or consult me if you are unsure."

- **Demand**
- Effect
- Observation
- Subject

Which of the following options is NOT one of the main reasons why officers find it difficult to give corrective feedback to their subordinates?

- **They believe that it is not their duty to give corrective feedback.**
- They don't know what to say and how to say it.
- They find the situation awkward in giving critical feedback.
- They hope that the problem will just take care of itself.

Which one of the following is the main reason why we should avoid using negative words when giving feedback?

- **It could result in the receiver taking a defensive position.**
- It could boost the morale of the receiver.
- It could make the receiver content.
- It could resolve any conflicts.

Which one of the following options is a poor reason for using feedback models in communicating performance feedback?

- **To have a guidance on focusing the negative aspect of the performance feedback.**
- To boost your confidence in saying the feedback.
- To communicate the feedback in an organized way.
- To have a guidance on what to say and how to express performance feedback.

Which one of the following reasons explains why it is NOT advisable to use the word "but" when moving from layer 1 to layer 2 of the Sandwich Feedback Model?

- **Most people only remember the words after the word 'but'**
- It creates a feeling of appreciation
- It provides a positive reinforcement to the person
- There is no significant effect on the feedback

Which one of the following reasons is most accurate as to why it is important to follow a certain structure in communicating feedback?

- **To boost your confidence in giving feedback.**
- To emphasize to the receiver their wrongdoings.
- To ensure that the receiver will be forced to follow your demand.
- To make the receiver lose face.

Which one of the following statements best describes the reason why people avoid giving feedback?

- **They hope that the problem will just take care of itself.**
- They think that it is always easy to ask for a replacement for the person.
- They think that it is not their primary task.
- They will just report the person to the office onshore.

Which one of the following statements corresponds to layer three of the Sandwich Feedback Model?

- **"In general I can see how you strive to maintain a good performance and it is very much appreciated."**
- "I would like you to do this job according to our standard procedure."
- "The way you manage your time is very inefficient."
- "What I want to see from you is improved consistency on how you perform your duties and responsibilities."

Which one of the following statements describes what should be used to start the first layer of feedback when using the Sandwich Feedback Model?

- **State three positive things about the person.**
- State your observation on any errors or failings.
- Tell the person how to change their behaviour.
- Tell the person not to react violently to the feedback.

Which one of the options is the name given to the tools that can help you communicate performance feedback in an organized way?

- **Feedback models**
- Criticism models
- Feedback analysis
- Feedback guidance systems

0254 – Meeting Management.

Version 2.0 (147)

A Chairman of a meeting needs to be aware of both positive and negative gestures of the participants in order to maintain a smoothly flowing discussion.

Which one of the following gestures would be considered to be negative?

- **Checking the watch frequently**
- Appropriate level of eye contact
- Open posture- relax arms and hands
- Slightly leaning towards the speaker

A Chief Engineer had to stop the main engine during navigation due to a cooling water leak. All of the engineers were called to investigate why it had happened and then to an emergency meeting to find a solution to the problem.

What type of meeting did the Chief Engineer facilitate?

- **Decision meeting**
- Idea development meeting
- Information meeting
- Safety meeting

A number of action items or tasks have been identified during a meeting.

Which one of the following options is the next most appropriate step to take?

- **Assign responsible person for each task and set deadlines.**
- Close the meeting right away.
- Include these actions in the agenda for the next meeting.
- Let the participants settle among themselves who will be assigned to do the tasks.

If the Information Phase of a meeting answers the question "What do we know?" which one of the following questions is addressed in the Suggestion Phase?

- **What can we do?**
- What do we expect?
- What do we want?
- What should we do?

In deciding whether or not to hold a meeting, which one of the following questions would NOT normally be considered necessary?

- **Should there be any minutes of the meeting recorded?**
- What is the relevance of the meeting to the participants?
- Who shall participate?
- Why do we need to meet?

In preparing for a meeting, which one of the following questions should be disregarded?

- **What should be the agenda for the follow-up meeting?**
- How are the subjects for discussion going to be introduced?
- What is the agenda of the meeting?
- What is the purpose of the meeting?

In which one of the following meeting types, a special form of which is brainstorming, are participants encouraged to share ideas which are helpful to group activities?

- **Idea development meeting**
- Consultation meeting
- Emergency meeting
- Information meeting

One of the participants in a safety meeting yawns from time to time and also leans backward and frequently checks their watch.

As the chairman of the meeting, how would you interpret this type of body language?

- **The participant shows negative signs that should be immediately addressed in order to sustain the smooth flow of the meeting.**
- The person is highly motivated during the meeting.
- The person is just quiet but in reality, he is attentive.
- The person shows positive signs that should be sustained.

Select, from the following, the phrase which most accurately completes the following statement.

"After making introductions, participants should next be reminded of the . . .

- **. . . purpose, expected output, and duration of the meeting."**
- . . . date of the next meeting."
- . . . minutes of the last meeting."
- . . . summary of discussions and duration of the meeting."

Select, from the following, the phrase which most accurately completes the following statement.

"Ensuring good meeting management is mainly the responsibility of . . .

- **. . . the meeting Chairman."**
- . . . both the Chairman and participants."
- . . . the participants."
- . . . the shipboard management."

Select, from the following, the phrase which most accurately completes the following statement.

"When following-up after a meeting the most important thing to ensure is that . . .

- **. . . the agreed decisions are being acted upon."**
- . . . everyone has read the minutes."
- . . . you remember the names of each participant so that you don't need to check the minutes."
- . . . your own ideas are implemented."

Select, from the following, the word which most accurately fills the gap in the following statement.

"Participants in a meeting who show open posture, relaxed arms and hands, and apply an appropriate level of eye contact are displaying . . . , non-verbal signs."

- **. . . positive . . .**
- . . . discouraging . . .
- . . . negative . . .
- . . . neutral . . .

Select, from those given in the options, the pair of terms which most accurately fill in the gap and complete the following statement.

"In the context of good meeting management, the S.M.A.R.T. tool stands for specific, measurable, attractive, . . . and . . .

- **realistic, time framed**
- realistic, time consuming
- relevant, time framed
- relevant, time limited

The Captain invites the senior officers to a brainstorming session to exchange ideas on how to improve the safety performance of the vessel.

Which one of the following types of meeting would this be classed as?

- **Idea development meeting**
- Emergency meeting
- Information meeting
- Instruction meeting

The Chief Mate gave a final briefing to the Bosun and ratings before they painted the main mast, emphasizing the risks involved and giving specific instructions on carrying out the job safely.

Which one of the following types of meeting is this an example of?

- **Instruction meeting**
- Emergency meeting
- Idea development meeting
- Information meeting

Which one of the following actions is the most appropriate to take if there are still disagreements at the end of a meeting?

- **Record the disagreements in the minutes.**
- Cancel any follow-up meetings.
- Continue with the discussion in private after the meeting.
- Ignore the disagreements.

Which one of the following actions would normally be carried out during a meeting rather than after a meeting?

- **Asking clarifying questions.**
- Evaluating the meeting.
- Making sure that decisions are implemented.
- Writing the minutes of the meeting.

Which one of the following best describes the action that should be taken before ending a meeting?

- **Summarize discussions and make sure that there is a common understanding of the problem and solutions.**
- Introduce the subjects that should be discussed.
- Make sure that participants are provided with necessary documents and information kits and that the agenda has been approved.
- Remind the participants about the meeting's purpose.

Which one of the following guidance statements regarding the writing of the minutes of the meeting is considered inappropriate?

- **Unclear issues should be recorded based entirely on the Secretary's interpretation.**
- Minutes should be concise and accurate.
- Stick to the facts and avoid biases.
- The minutes should be easily understood by those who were unable to attend the meeting.

Which one of the following is NOT normally a task for the Chairman of a meeting?

- **Preventing discussion of some agenda topics.**
- Allowing participants to express their opinions.
- Asking clarifying questions to keep the discussions flowing.
- Making sure that only one person speaks at a time.

Which one of the following is NOT normally included in a meeting agenda?

- **The action items.**
- The details of participants.
- The purpose.
- The subject.

Which one of the following is NOT normally included in the minutes of a meeting?

- **Agenda of the next meeting**
- Attendees and absentees
- Summary of discussions
- Time started and adjourned

Which one of the following is the last phase of the meeting and answers the question "What should we do?"?

- **Plan of action phase**
- Decision phase
- Discussion phase
- Information phase

Which one of the following is the least suitable type of characteristic for a chairman of a meeting to have?

- **To be highly emotional.**
- A sense of authority.
- To be impartial and factual.
- To be knowledgeable about the agenda.

Which one of the following meeting types aims to assess, and possibly revise, the basis for a decision before it is implemented?

- **Consultation meeting**
- Brainstorming meeting
- Decision meeting
- Emergency meeting

Which one of the following meeting types aims to follow up a decision by providing participants with information on how they can efficiently perform certain tasks?

- **Instruction meeting**
- Decision meeting
- Idea development meeting
- Investigation meeting

Which one of the following meeting types builds one-way communication and is held to inform participants about certain decisions.

- **Information meeting**
- Decision meeting
- Instruction meeting
- Safety meeting

Which one of the following meeting types is also called a 'problem-solving meeting' as it ensures that decisions are made on a sound basis?

- **Decision meeting**
- Emergency meeting
- Idea development meeting
- Instruction meeting

Which one of the following reasons best describes why it is important to define the objective and relevance of a meeting to all participants?

- **To ensure that participants feel that the meeting is worthwhile so avoiding any annoyance and frustration.**
- To avoid any interest from participants as this may result in a long meeting.
- To ensure that none of the participants need to ask any questions.
- To ensure the meeting finishes early.

Which one of the following statements best describes when an information meeting should be conducted?

- **When it is time to inform crew members about a decision that has been made and about its consequences.**
- When there is a need to assess and possibly revise the basis for a decision prior to its implementation.
- When there is a need to have some brainstorming.
- When there is a need to make a decision.

0255 – Question Techniques.

Version 2.0 (242)

Every second, approximately two million pieces of information are fed into your brain much of which must be screened out to prevent brain overload.

Which one of the modelling processes given in the options is the name for this screening out of information?

- **Deletion**
- Distortion
- Generalization
- Levelling

In a lost performative statement, the person states a value judgment without mentioning the performer or speaker, for example: "Boys shouldn't cry."

Which one of the following questions should you ask in order to challenge this statement?

- **"Who says boys shouldn't cry?"**
- "How do boys cry?"
- "What stops boys from crying?"
- "What would happen if boys stopped crying?"

In generalizations, a person makes one category represent a whole group, resulting in statements that leave no room for any exceptions, for example: "All Captains are aggressive."

To challenge this statement, which one of the following questions should be asked?

- **"All of them?"**
- "How aggressive are they?"
- "In what ways are Captains aggressive?"
- "What stops Captains from being aggressive?"

In which one of the modelling processes given in the options do you alter the meaning and structure of information as you simplify the descriptions of your experiences?

- **Distortion**
- Deletion
- Generalization
- Levelling

Necessity words describe a model of reality that believes in necessity. Such words define some governing rule the person operates from for example: "I should remember all these policies by heart."

Which one of the following questions should you ask in order to challenge this statement?

- **"What would happen if you didn't remember all these policies?"**
- "All of them?"
- "How specifically do you want to remember all these policies by heart?"
- "Really?"

Select from the options given the one phrase that correctly completes the following statement.

"Question technique intends to reconnect a speaker's language with the . . .

- **. . . experience out of which it came."**
- . . . experience of the other person."
- . . . language of the other person."
- . . . language used on board the ship."

Select from the options given the one word that correctly completes the following statement.

"A person's verbal description of an experience is often distorted, deleted and . . .

- **. . . generalized."**
- . . . clarified."
- . . . interpreted."
- . . . summarized."

Select the option which contains the two terms that correctly fill the gaps in the following statements.

" . . . lies in the unconscious parts of your mind and it is represented in your experiences. On the other hand, . . . is where you seek to present, articulate, express and clarify your experiences."

- **Deep structure, surface structure**
- Deep structure, communication
- Memory, language
- Surface structure, language

Select, from the options given, the one phrase or word that correctly completes the following statement.

"When a person thinks and asserts that they know the thoughts, motives and intentions of another person, they are engaged in . . .

- **. . . mind reading."**
- . . . comparative deletion."
- . . . generalization."
- . . . problem solving."

Select, from the options given, the one phrase that correctly completes the following statement.

"The aim of asking challenging questions is to . . .

- **. . . uncover the missing information that showed up in the surface structure of the speaker's words."**
- . . . elaborate more on his feelings and emotions."
- . . . make the speaker feel more comfortable."
- . . . talk more."

Select, from the options given, the one phrase that correctly fills the gap in the following statement.

"When a sentence lacks a . . . , it fails to specify the name, term, or phrase that it points to."

- **referential index**
- complex equivalence
- second adjective
- situation awareness

Select, from the options given, the one term that correctly fills the gap in the following statement.

"Challenging questions inquire about the incompleteness that shows up in the . . . and these enable the speaker to restore the material that was deleted, distorted, and generalized."

- **surface structure**
- deep structure
- experience
- manifestations

Select, from the options given, the one word that correctly completes the following statement.

"When moving from the deep structure in your unconscious mind to the surface structure that comes out of your conscious mind and mouth, you do all of these modelling processes except . . .

- **. . . levelling."**
- . . . deletion."
- . . . distortion."
- . . . generalization."

Select, from the options given, the one word that correctly fills the gap in the following statement.

"You generate a . . . statement whenever you use a part of an experience as being equivalent to the whole of its meaning. When you become aware of the external cue, you then assume the meaning of the whole experience.

- **complex equivalence**
- generalization
- mind reading
- operators of necessity

The statement, "He is much better off.", is an example of comparative deletion where the specific reference or standard by which the comparison was made is deleted.

In order to challenge this statement, which one of the following questions should you ask?

- **"Better off than whom?"**
- "How did you say that?"
- "Really?"
- "Why is he better?"

The statement, "I can't learn efficiently.", depicts operators of possibility and reveals the person's belief in what they can or cannot do in life.

In order to challenge this statement, which one of the following questions should you ask?

- **"What would happen if you did change that belief?"**
- "Are you sure?"
- "How did you say that?"
- "Who told you that?"

The statement, "The Fourth Engineer committed a mistake.", is an example of a statement with unspecified verbs, where the description is vague and the action is not specified enough to make a clear representation.

Which one of the responses given in the options should you use to challenge this statement?

- **"How exactly did he commit the mistake?"**
- "Are you sure?"
- "How often does he commit the mistake?"
- "When did he commit the mistake?"

The statement, "They did not come to the meeting.", is an example of one with a 'lack of referential index'.

Which one of the responses given in the options should you use to challenge this statement?

- **"Who specifically did not come to the meeting?"**
- "According to whom do you say such?"
- "When was the meeting?"
- "Why did they not come to the meeting?"

The statement, "You did not finish the maintenance job due yesterday; you are lazy.", is an example of complex equivalence, where the speaker uses a part of an experience as the equivalent of the whole of its meaning.

In order to challenge this statement, which one of the following questions should you ask?

- **"How specifically does my inability to finish the maintenance job due yesterday mean that I am lazy?"**
- "Are you sure that I am lazy?"
- "Really?"
- "Who said that I am lazy?"

The statement, "You make me bad!", illustrates a cause-effect statement.

In order to challenge this, which one of the following questions should you ask?

- **"How specifically do I cause you to feel bad?"**
- "Are you serious?"
- "Since when did I cause you to feel bad?"
- "Who said that I cause you to feel bad?"

The statement, "You make me mad!", is an over-used accusation which illustrates a cause-effect statement.

Which one of the responses given in the options should you use to challenge this statement?

- **"How specifically did I cause you to feel mad?"**
- "All of the time?"
- "Is that right?"
- "Who said that?"

Which one of the actions given in the options does a human brain NOT do to information it receives?

- **React to it**
- Delete it
- Distort it
- Generalize it

Which one of the following questions should you ask in order to challenge a mind reading statement?

- **"How specifically do you know how I think or feel that . . . ?"**
- "Really?"
- "What would happen if you did/didn't . . . ?"
- "Who said that?"

Which one of the following statements about question technique is NOT true?

- **It is possible to fully describe an experience.**
- In describing an experience, the speaker inevitably shortens the description and picks out the essentials.
- No verbal descriptions can fully describe an experience.
- People can only draw a map of reality but cannot express the whole scenario.

Which one of the following statements does NOT use any operators of possibility?

- **"I must go on shore leave tomorrow."**
- "I am not capable of doing the job."
- "I am not ready with the inspection."
- "I can't learn efficiently."

Which one of the options identifies the modelling process in which your brain compares new information with similar information previously learned therefore allowing you to learn quickly?

- **Generalization**
- Deletion
- Distortion
- Levelling

Which one of the question types given in the options should be used to inquire on the incompleteness that shows up in surface structures?

- **A challenging question**
- A closed question
- An investigative question
- An open question

Which one of the reasons given in the options best identifies why it is difficult to drag out the right information from another person during a conversation?

- **Because none of us gives a complete picture or description of the thought behind the words.**
- Because we are not speaking the same language.
- Because we intentionally don't tell the truth.
- Because we never listen properly.

Which one of the statements given in the options best describes what would happen if you attempted to completely describe your thoughts?

- **You would never be able to finish speaking.**
- You would always create a hostile atmosphere.
- You would be able to do it in an instant.
- You would easily be able to tell everything in your thoughts.

Which one of the terms given in the options can be defined as the "internal representation of the deep structure"?

- **Experiences**
- Actions
- Mistakes
- Reactions

0256 – Team Leadership.

Version 2.0 (150)

A newly promoted Captain finds that each one of the multi-cultural officers and crew onboard is different in terms of culture, experience, competence and personality.

Which one of the following styles should be used, in this situation, to build an effective onboard team?

- **Situational Leadership style**
- Authoritarian leadership style
- Consultative leadership style
- Democratic leadership style

During which phase of the UPI Team Leadership Model are questions, such as, "Who will do the job?", "When do we start/finish?", "What resources are required?", and "How do we do it?" answered?

- **Planning**
- Implementation
- Involvement
- Understanding

In creating the environment for the team to flourish and assemble, which one of the following principles is NOT correct?

- **The leader should use force to assemble team members.**
- The leader should create an environment that is pleasant for the team to assemble itself.
- The leadership style of the leader should have an important, positive influence on the behaviour of team members.
- The leadership style should match the team's culture.

Is the following statement correct in relation to work teams onboard ship?

"Being on board a ship does not necessarily mean being part of a team."

- **It is correct because not all shipboard organizations possess the characteristics of a team, such as common goals, interdependency and team consciousness.**
- It is correct because teams do not really exist on board the ship.
- The statement is incorrect because even the engine and deck departments are considered to be teams.
- The statement is incorrect because the shipboard organization is always considered to be a team.

Select, from the options given, the phrase that correctly completes the following statement.

"The three basic interpersonal needs, according to Schutz, are collectively known by the acronym CIA, which stands for control, . . .

- **. . . inclusion and appreciation."**
- . . . inclusion and allocation."
- . . . independence and appreciation."
- . . . integration and allocation."

Select, from the options given, the phrase that most accurately completes the following statement in relation to the four levels of readiness.

"Low follower readiness is shown by those who are . . .

- **. . . unable and unwilling."**
- . . . able and willing."
- . . . able but unwilling."
- . . . unable but willing."

Select, from the options given, the phrase that most accurately completes the following statement in relation to the four levels of readiness.

"Low to moderate follower readiness refers to followers who are . . .

- **. . . unable but willing."**
- . . . able and willing."
- . . . able but unwilling."
- . . . unable and unwilling."

Select, from the options given, the phrase that most accurately completes the following statement.

A good leader is capable of performing their duties by varying their role and their conduct. When carrying out risk assessments and dealing with critical situations they are acting as the . . .

- **. . . conflict manager."**
- . . . front figure."
- . . . informer."
- . . . negotiator."

Select, from the options given, the phrase that most accurately completes the following statement.

" The goal of the first phase of the UPI model, which is described as the think, know and feeling phase, is to . . .

- **. . . create a common understanding of the task that the team is confronted with."**
- . . . give orders."
- . . . implement action plan."
- . . . start planning how to implement the ideas of the team."

Select, from the options given, the phrase that most accurately completes the following statement.

"A work team is defined as a small collection of people who share all of the following characteristics EXCEPT . . .

- **. . . common nationality."**
- . . . a common goal."
- . . . a sense of shared purpose."
- . . . team consciousness."

Select, from the options given, the phrase that most accurately completes the following statement.

"Democratic leadership style is NOT considered lasting and ideal because . . .

- **. . . majority is not necessarily the best group to make competent decisions."**
- . . . it focuses on achieving the goal."
- . . . it is solely based on the principle of status power."
- . . . it prevents crew members taking the initiative."

Select, from the options given, the phrase that most accurately completes the following statement.

"During the third phase of the UPI model, the leader . . .

- **. . . gives orders to the team."**
- . . . creates a common understanding of the task that the team is confronted with."
- . . . generates options for the team."
- . . . starts planning how to implement the ideas of the team."

Select, from the options given, the phrase that most accurately completes the following statement.

"Empowered team members tend to . . .

- **. . . take initiative and be more productive."**
- . . . be less assertive and authoritative."
- . . . feel less power and authority."
- . . . lack safety awareness."

Select, from the options given, the phrase that most accurately completes the following statement.

"One of the outcomes of empowering team members, especially when they are involved in team activities such as the development of ideas, is that they will . . .

- **. . . be more committed to the team's goal because they feel they own the idea."**
- . . . just wait for the leader's decision."
- . . . only participate because they are forced to do so."
- . . . usually hesitate to cooperate because they are vulnerable to criticism."

Select, from the options given, the phrase that most accurately completes the following statement.

"The UPI Team Leadership Model is a way to understand how to develop powerful, creative and self-empowered teams. The abbreviation UPI stands for . . .

- **. . . understanding, planning and implementation."**
- . . . understanding, progressing and implementation."
- . . . uniformity, planning and implementation."
- . . . uniformity, planning and interpretation."

Select, from the options given, the phrase that most accurately completes the following statement.

"The readiness level of the crew, which is the basis for Situational leadership thinking, is defined as . . .

- **. . . the extents to which followers have the will, knowledge and ability to accomplish a specific task."**
- . . . subordinate's level of readiness in dealing with conflicts."
- . . . the crew's way of dealing with their leader."
- . . . the speed of subordinate's reaction to emergencies."

Select, from the options given, the phrase that most accurately completes the following statement.

"The second phase of the UPI model is when the leader and other responsible employees . . .

- **. . . start planning how to implement the ideas of the team."**
- . . . create a common understanding of the task that the team is confronted with."
- . . . give orders."
- . . . try to understand the situation."

Select, from the options given, the phrase that most accurately completes the following statement.

"Typically, an authoritarian leadership style is solely based on the principle of . . .

- **. . . status power."**
- . . . expert power."
- . . . information power."
- . . . personal power."

Select, from the options given, the word that most accurately completes the following statement.

"Team leaders seek to create the conditions that enable team members to reach common . . .

- **. . . goals."**
- . . . impressions."
- . . . power."
- . . . preferences."

Select, from the options given, the word that most accurately fills the gap in the following statement.

"In authoritarian leadership, other people's opinions and beliefs are seldom appreciated resulting in a misuse of power based on . . . and fear."

- **. . . pressure . . .**
- . . . authority . . .
- . . . feedback . . .
- . . . motivation . . .

The Chief Engineer observes that, although highly motivated, the Fourth Engineer lacks competence to do the assigned maintenance tasks.

Based on the principles of situational leadership, which one of the following should the Chief Engineer do?

- **Provide more guidance and coaching to the Fourth Engineer.**
- Observe further and wait until Fourth Engineer takes the initiative and asks for guidance.
- Report this to the Master and let them address the situation.
- Show lower assertiveness.

The Chief Mate observes that both the Second and Third Mates are confident and competent watch keepers.

Based on the principles of situational leadership, which one of the following actions should the Chief Mate take in this situation?

- **Provide minimal guidance and coaching to the Second and Third Mate.**
- Provide a high level of guidance and coaching to the Second and Third Mate.
- Show higher assertiveness with the Second and Third Mate.
- Use authoritarian leadership style with to the Second and Third Mate.

The Second Engineer ensures that new crew members are properly informed before they start any maintenance work by explaining the task, the risks involved, based on the latest risk assessment, and by answering their queries.

Which phase of the UPI Team Leadership Model is represented in this example?

- **Understanding**
- Follow-up
- Implementation
- Planning

The ratings and junior officers highly respect the Bosun because of his experience, age and personality and they often seek his advice on personal and family matters.

In the context of leadership, which one of the types of power does the Bosun have?

- **Personal power**
- Expert power
- Information power
- Status power

The ship's Electro-Technical Officer (ETO) has the respect of the officers, the crew and the Superintendent because of his excellent work performance.

In the context of leadership, which one of the types of power does the ETO have?

- **Expert power**
- Information power
- Personal power
- Status power

The three interpersonal needs which must be fulfilled in order to create good teams are known by the acronym CIA, according to Schutz.

Which one of the following terms is the interpersonal need represented by the letter 'A' in the acronym CIA?

- **Appreciation**
- Application
- Attitude
- Authority

Which one of the following is NOT a means by which both the team leader and team members can reach common goals?

- **By each making their own decisions.**
- By building on each other's strengths.
- By sharing information and knowledge.
- By working together.

Which one of the following statements best describes the term 'leadership'?

- **Getting the job done through others.**
- Directing team members on what to do.
- Making people happy and satisfied.
- Using resources economically.

Which one of the following statements best describes why it is important to have effective team leadership?

- **To ensure that the collective and complementary skills of each crew member can make up for individual limitations.**
- To ensure that each crew member works independently.
- To ensure that uncooperative crew members will be easily identified and removed from the team.
- To satisfy the management.

Which one of the following statements relating to team leadership is NOT correct?

- **A leader has the sole accountability to the team members.**
- A leader can create an environment that is pleasant for the team to assemble itself and flourish.
- The leadership style that is applied should match the culture of the team.
- The style of leadership has an important influence on the behaviour of the team members.

0257 – Stress Management

Version 3.0 (463)

A Chief Engineer begins to feel irritable, restless and unapproachable upon discovering that more than 5% of the planned maintenance jobs were not completed on time.

What behavioural symptom of stress is being shown by the Chief Engineer?

- **Unapproachable**
- Irritable
- Irritable and restless
- Restless

A newly promoted chief mate is to be in charge of the safety committee meeting for the first time. While waiting for the meeting to commence the chief mate starts getting nervous and restless with the hands becoming cold and sweaty.

For which one of the types of stress given in the options are these typical symptoms?

- **Acute stress**
- Chronic stress
- Positive stress
- Work stress

In the Four Step Model of managing stress, what should you do next after identifying the stress symptoms and acknowledging your stress?

- **Identify the stressors that provoked the stress.**
- Determine the applicable coping strategies.
- Determine your personality type.
- Take time to relax.

Of the two ABs, Peter works faster than Greg. While Peter is usually much more competitive and aggressive, Greg takes his assigned jobs one at time and works steadily.

Which one of the two ABs is LESS prone to stress?

- **Greg will probably be less prone to stress.**
- Both of them will probably be equally prone to stress.
- Neither of them is likely to be less prone to stress.
- Peter will probably be less prone to stress.

Of two ABs, Peter works faster than Greg. While Peter is often very competitive and aggressive, Greg takes his assigned jobs one at time and works steadily.

Which one of the two ABs is MORE prone to stress?

- **Peter will probably be more prone to stress.**
- Both of them will probably be equally prone to stress.
- Greg will probably be more prone to stress.
- Neither of them is likely to be more prone to stress.

Peter, the second officer, is losing interest in his work as there are always new demands and all of his time is spent on crucial tasks. Often, he feels behind schedule, finds it difficult to concentrate or sleep and has headaches.

For which one of the following types of stress are these typical symptoms?

- **Chronic stress**
- Acute stress
- Personal stress
- Work stress

Peter, the second officer, is losing interest in his work as there are always new demands and all of his time is spent on crucial tasks. Often, he feels behind schedule, finds it difficult to concentrate or sleep and has headaches.

In the above scenario what psychological symptom of stress does Peter show?

- **He has a hard time concentrating on his responsibilities.**
- He has a headache most of the time.
- He has trouble sleeping.
- He is losing interest in his work.

Peter, the second officer, is losing interest in his work as there are always new demands and all of his time is spent on crucial tasks. Often, he feels behind schedule, finds it difficult to concentrate or sleep and has headaches.

In the above scenario, what physical symptom of stress does Peter show?

- **He has a headache most of the time.**
- He has a hard time concentrating on his responsibilities.
- He has trouble sleeping.
- He is losing interest in his work.

Research shows that a high level of stress has a large impact on the safety awareness and capability of an individual to perform the tasks.

Which one of the ranges given in the options is recognised as the percentage of workplace accidents caused by stress?

- **60-80 %**
- 10-30 %
- 30-60 %
- 80-90 %

Select one phrase from those given in the options which correctly completes the following statement.

"A person experiences chronic stress when they are faced with a stressful situation and they . . .

- **. . . cannot cope with it."**
- . . . can cope with it."
- . . . cannot identify the stressors."
- . . . cannot identify the symptoms."

Select the one term from those given in the options which most accurately completes the following statement.

"All of the following are behavioural symptoms of stress EXCEPT for . . .

- **. . . lapse of memory."**
- . . . difficulty in making decisions."
- . . . lack of interest."
- . . . reduced performance."

Select the one term from those given in the options which most accurately completes the following statement.

"All of the following are physical symptoms of stress EXCEPT for . . .

- **. . . sadness."**
- . . . difficulty in breathing."
- . . . dizziness."
- . . . headaches."

Select the one term from those given in the options which most accurately completes the following statement.

"All of the following are psychological symptoms of stress EXCEPT for . . .

- **. . . reduced performance."**
- . . . irritability."
- . . . lapse of memory."
- . . . sadness."

Select the one term from those given in the options which most accurately completes the following statement.

"Negative thinking causes stress because it damages your confidence. All of the following tools are useful in combating negative thinking EXCEPT . . .

- **. . . prioritization."**
- . . . positive thinking and affirmation."
- . . . rational thinking."
- . . . thought awareness."

Select the one term from those given in the options which most accurately completes the following statement.

"The ability to manage stressors is referred to as . . .

- **. . . coping."**
- . . . problem solving."
- . . . resource management."
- . . . stress relieving."

The Second Mate, who is about to go on watch so cannot take any rest, is stressed and exhausted after more than 5 hours of cargo operations. For motivation, they repeatedly think of an inspirational line - "I can do it!".

What one of the following stress coping strategy is the Second Mate using?

- **Positive thinking and affirmation**
- Prioritization
- Rational thinking
- Thought awareness

Uncertainty is one of the causes of stress. Which one of the actions identified in the options should you do in order to combat uncertainty?

- **Ask for the information that you need.**
- Convince others to entertain you.
- Divert your focus to other concerns.
- Motivate yourself that there is nothing to worry about.

Which one of the factors given in the options is the most likely to make someone stressed.

- **Stressors**
- Coping
- Demands
- Goals

Which one of the following is an example of workplace stressor?

- **High job demands**
- Contentment
- Harassment
- Self-sacrifice

Which one of the following is the first step in the Four Step Model of managing stress?

- **Identifying your stress symptoms.**
- Acknowledging your stress.
- Finding your coping strategies.
- Identifying the stressors that provoked the stress.

Which one of the following is the fourth step in the Four Step Model of managing stress?

- **Finding your coping strategies.**
- Acknowledging your stress.
- Identifying the stressors that provoked the stress.
- Identifying your stress symptoms.

Which one of the following options is NOT a measure that you can take in order to reduce stress?

- **Spend some relaxation time every day.**
- Plan your time.
- Prioritize tasks.
- Say no if you think you cannot manage the task well because of time and resource constraints.

Which one of the following statements about acute stress is correct?

- **Acute stress is helpful in moving us towards the achievement of our goals.**
- Acute stress causes the release of a damaging amount of adrenalin.
- Acute stress happens to us when we feel that things are beyond our control.
- Acute stress is caused by frustration, anger, rejection, distraction, distrust, and depression.

Which one of the following statements about chronic stress is FALSE?

- **Chronic stress is considered a life energy force in the body and a key ingredient of optimal performance.**
- Chronic stress arises when you feel that the situation is beyond your control.
- Chronic stress is harmful and must be controlled.
- Chronic stress makes our body sensitive as it tries to combat threat until we run out of energy.

Which one of the statements given in the options about individual stressors is NOT correct?

- **People who are reflective and show no interest in social status are vulnerable to stress.**
- Ambitious and harassed people have a high risk of developing stress.
- People who have a tendency to put other people or cases before themselves are more vulnerable to stress.
- Perfectionist people are vulnerable to stress.

Which one of the statements given in the options about personalities and stress is NOT correct?

- **People with marked B-type behaviour are aggressive and competitive.**
- People who tend to expect too much from themselves even though such expectations are seemingly unrealistic and unattainable are more exposed to stress.
- People with marked A-type behaviour have the biggest risk of developing stress.
- People with marked B-type behaviour work steadily and take things one at a time.

Which one of the terms given in the options is used to identify the process by which you observe your thoughts for a while, especially when under stress, and become aware of what is going through your head?

- **Thought awareness**
- Positive thinking and affirmation
- Prioritization
- Rational thinking

Which one of the terms given in the options is used to identify the tool that is used to combat negative thinking by reviewing all negative thoughts and assessing if they are true?

- **Rational thinking**
- Positive thinking and affirmation
- Prioritization
- Thought awareness

Which one of these groups of stress symptoms are behavioural symptoms?

- **Difficulty in making decisions, being unapproachable and quick-tempered, lack of empathy**
- High blood pressure, headache, difficulty in breathing, dizziness
- Nervousness, lapse of memory, difficulty in concentrating, sadness
- Sweaty hands, eczema, nausea, lack of appetite

Which one of these groups of stress symptoms are psychological symptoms?

- **Nervousness, lapse of memory, difficulty in concentrating, sadness**
- Difficulty in making decisions, being unapproachable and quick-tempered, lack of empathy
- High blood pressure, headache, difficulty in breathing, dizziness
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0258 – SEEMP

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A review of electrical services on board can reveal the potential for unexpected efficiency gains. Which three of the following forms of modern technology might be employed to increase those savings?

- **More efficient thermal insulation**
- **Sensor-based lighting controls**
- **Solar panels**
- Industrial standard washing machines
- Integrated Navigation and Command Systems
- Lithium cell technology

According to the IMO Guidelines, what should a Ship Energy Efficiency Management Plan (SEEMP) preferably be linked to?

- **The company's broader corporate energy management policy**
- The company's strategic financial policy
- The ship's IOPP certificate
- The ship's onboard training procedures

Although the potential for savings in energy efficiency is high through the use of commercial weather routing as part of a Ship Energy Efficiency Management Plan (SEEMP), what, as indicated in this training module, may also perhaps happen for a given voyage?

- **Fuel consumption may actually increase**
- The data collected for later self-analysis may be inaccurate
- The lube oil consumption will increase
- The main engine temperatures will be erratic and unreliable

As described in this training module, how can cargo stowage involving reefer containers contribute towards energy efficiency management?

- **The careful and planned stowage of reefer containers may reduce the effect of heat transfer from compressor units. By concentrating the stowage of reefer containers, discharge will be more efficient and lead to energy savings in the cargo operation**
- By limiting the stowage of reefer containers to inside the ship's hull, the heating impact from direct sunlight will be minimised
- By limiting the stowage of reefer containers to the outside of a stack, maximum benefit will be secured from wind cooling

As described in this training module, how can cargo stowage on board a parcel chemical tanker contribute towards energy efficiency management?

- **Cargoes with similar heating requirements stowed in adjacent tanks will benefit from heat transfer across common bulkheads and can reduce the need to heat individual tanks**
- Cargoes with a higher Relative Density can be stowed to assist in achieving the optimum trim
- Cargoes with cooling requirements should be stowed in forward tanks to initially benefit from cold sea water temperatures
- The Company Energy Efficiency Management Plan will require a particular stowage to meet port rotation criteria

As explained in this training module and carried out as part of a Ship Energy Efficiency Management Plan (SEEMP), why is the documentation of energy efficiency measures beneficial?

- **For reference during the self-evaluation step in the SEEMP process**
- For reference during the planning step in the SEEMP process
- To demonstrate compliance with the IMO Guidelines
- To facilitate the application for an IEEC exemption

As explained in this training module, how might improved cargo handling contribute towards overall energy efficiency management?

- **Optimum cargo handling solutions matched to both ship and port requirements, with input from all appropriate parties, could be explored and implemented**
- Getting ships in and out of port as fast as possible will always provide the cheapest and therefore most energy efficient option
- If terminals and cargo berths focus on performing their side of cargo operations more effectively, then ships will not have to use so much fuel and energy
- Port Authorities should relax their regulatory controls on cargo operations, allowing ships to conduct them in more energy efficient ways

As part of each Ship Energy Efficiency Management Plan (SEEMP), the ship owner is required to do which one of the following?

- **Review current practices and energy usage onboard each ship with a view to determining any shortfalls or areas for improvement of energy efficiency**
- Apply for an annual IEEC
- Review personal energy consumption on board and restrict usage accordingly with a view to reducing the carbon footprint of the vessel
- Set targets for the reduction of environmental pollution resulting from cargo operations

As part of the monitoring step in a Ship Energy Efficiency Management Plan (SEEMP) process, how can any unnecessary additional administrative burden on ships' staff be avoided?

- **By monitoring being carried out as far as possible by shore staff, utilizing data obtained from existing required records such as the official and engineering log-books and oil record books**
- By only collecting data that does not involve any direct human activity
- By monitoring procedures being designed and written to take place at sea when ship's staff are less busy
- By only utilizing data that can be electronically transferred or remotely collected in off-duty hours

As part of the self- evaluation step of the Ship Energy Efficiency Management Plan (SEEMP) process, in which one of the following is it particularly recommended that time be invested, to improve the next step in the cycle?

- **The cause-and-effect of the ship's energy efficiency performance during the evaluated period**
- Ensuring that enough data has been collected and that it is from the right places to best guarantee an effective conclusion to the SEEMP cycle
- The effectiveness of instructions as determined during the implementation step of the SEEMP process
- Whether the tasks identified during the implementation step of the SEEMP process were allocated to the individuals best qualified for them

As part of the training carried out to ensure successful implementation of the company's energy efficiency management policy and as part of the crew's familiarization with recording and monitoring systems, which one of the following factors is the most important?

- **That there is full understanding of the instrument readouts**
- That the crew will know that the company are monitoring and recording their efficiency
- That the recording and monitoring system manufacturers have rigid quality control procedures in place
- That the ship's personnel will be able to download and modify data from critical recording devices

As part of the training carried out to ensure successful implementation of the company's energy efficiency management policy, the ship's crew must be able to recognize developing scenarios as they arise and know the appropriate action to take. What should they compare readings from monitoring and measuring equipment to when making these decisions?

- **Sea trial and manufacturer's data**
- Flag State performance records
- Industry Standard Energy Efficiency Performance (ISEEP) criteria
- Similar data secured from a ship of a different class and on a different trading route

As the amount of CO_2 emitted from a ship is directly related to the consumption of bunker fuel oil, what useful additional information will the Energy Efficiency Operational Indicator (EEOI) deliver to the ship's managers / operators?

- **It will indicate a ship's performance with regard to fuel efficiency**
- It will give a good indication of the quality of the ship's bunkers
- It will indicate the total operating costs of a ship over a defined period
- It will indicate whether the ship has been run at optimum speed over a defined period

Does sailing at less than optimum speed result in higher fuel consumption?

- **Yes**
- No
- Not generally, but it can occur in sub-zero sea temperatures
- Yes, but only in adverse weather

Does the Ship Energy Efficiency Management Plan (SEEMP) have to be a 'stand-alone' plan?

- **No, it may form part of the ship's Safety Management System**
- No, it depends on whether the plan is flag State approved or not
- No, it is up to the Master whether the SEEMP is integrated with other procedures on board
- Yes, each ship has to have an Energy Efficiency Management Plan and it must be independent of any other procedures

For new ships, the International Energy Efficiency Certificate (IECC) will state the required Energy Efficiency Design Index (EEDI) and which other of the following items of related information?

- **The attained EEDI**
- The applicable CO_2 conversion factor
- The EEOI
- The expiry date of the certificate

Goal-setting is voluntary as part of a Ship Energy Efficiency Management Plan (SEEMP), but which one of the following functions will it perform in support of the Plan?

- **It will drive incentive for energy reduction both at ship level and at corporate level**
- It will at least mean that the data collected has not been wasted
- It will demonstrate maximum compliance to an inspector or auditor
- It will guarantee that the SEEMP will be approved by the Flag State Authority

How are optimum ballast conditions primarily achieved?

- **Through good cargo planning**
- By always minimising the fuel, water and other consumables
- By calculating the cost of the fuel required to pump out ballast against the freight cost of the cargo to be loaded
- By pumping any ballast out at the earliest possible opportunity in a cargo operation

How are the tasks defined in the procedures for energy management within a Ship Energy Efficiency Management Plan (SEEMP) implemented on board?

- **They will be assigned to qualified personnel**
- It is expected that everyone on board will implement the tasks and therefore it is not necessary to identify anyone particularly responsible for them
- The Chief Engineer will be made wholly responsible for all the tasks on board his ship
- Volunteers will be sought on board to take responsibility for them

How can an Integrated Navigation and Command System achieve significant fuel savings as part of an energy efficiency management policy?

- **By reducing the distance sailed "off track" and delivering better course control and minimising losses due to rudder resistance**
- By automatic management of the engine RPM to achieve optimum speed
- By continuous incremental adjustment to the course steered to minimise water resistance
- By optimising fuel consumption through controlled interaction with the Engine Management System

How can the trim of a ship contribute towards a reduction in fuel consumption?

- **It can significantly influence the resistance to the movement of the ship through the water**
- A good trim will ensure better fuel flow to the engine system
- Trim has no significant impact on fuel consumption when considered over a complete voyage
- Trimming a ship so that it is “by the head” when at full speed at sea ensures there is less direct load on the propulsion system

How is the recognition by the IMO Guidelines for the Development of a Ship Energy Efficiency Management Plan (SEEMP) of an Environmental Management System established under ISO 14001 seen to be an advantage?

- **The monitoring of operational environmental efficiency as part of a SEEMP should be treated as an integral element of a broader company environmental management system, such as one established under ISO 14001**
- An Environmental Management System established under ISO 14001 will result in the automatic recognition of a SEEMP by the Flag State Authority and the granting of an IEEC
- ISO 14001 requires that a SEEMP is developed for every ship over 400 gross tonnage and it therefore demonstrates compliance with that standard
- The Guidelines for the Development of a Ship Energy Efficiency Management Plan (SEEMP) replaces existing procedures and the ISO 14001 Standard is close in outline and content and therefore easier to change

How might computer software be most effectively and proactively utilized by a shipping company in respect of energy efficiency management, as suggested by this module?

- **For the calculation of fuel consumption, leading to the establishment of an emissions "footprint" and the optimization of ship operations**
- To demonstrate to Flag State Authorities that maintenance procedures on board their vessels are all compliant with manufacturer's requirements
- To document the domestic electricity use of individual crewmembers on board selected vessels
- To list the approval dates of individual ships' certificates

How might the sourcing of fuel of improved quality, this is of higher calorific value, be of benefit as part of an energy efficiency management process, as explained in this module?

- **It may minimise the quantity of fuel required to provide a given power output**
- It may maximise the quantity of fuel required to provide a given power output
- It will cost the ship owner less money to renew his ships' International Energy Efficiency Certificates
- It will result in economic voyage planning

How should the energy efficiency of a ship be monitored?

- **Quantitatively**
- Both quantitatively and qualitatively
- Incrementally
- Qualitatively

If the Energy Efficiency Operational Indicator (EEOI) is used for the quantitative measurement of a ship's energy efficiency, which one of the following measurement tools may also be used?

- **Energy audits and consumption surveys**
- Archived historic maintenance records
- Personal tracking devices
- Port State Control inspection data

In order to implement a Ship Energy Efficiency Management Plan (SEEMP) onboard, which one of the following conditions must be complied with?

- **Procedures should be developed for energy management, defining the necessary tasks and assigning them to qualified personnel**
- A review of operating practices onboard each ship must be conducted by the Master and Chief Engineer, along with the ships Superintendent, to identify any areas which are failing to meet energy targets
- Procedures should be developed for energy reduction, defining the company targets and identifying those personnel responsible for achieving them
- The ship must have a valid IEEC

In order to maximise the efficiency of a ship's hull and its coating as part of an energy efficiency management policy, a ship's operator must consider a number of factors when planning dry-dock intervals. Which one of the following is an integral consideration, as described in this training module?

- **Ongoing assessment of ship performance**
- Average annual engine repair and maintenance costs
- Reduction in observed optimum speed to below charter party speed
- Renewal requirements for the vessel's IEEC

In respect of fuel efficient operations as part of a Ship Energy Efficiency Management Plan (SEEMP), which one of the following can careful planning and execution of a voyage lead to?

- **The identification of the optimum route and improved energy efficiency**
- Onward supply chain logistics being built around vessel schedules
- The identification of those responsible for previous, inefficient voyages
- The repeated application of the fastest route on all voyages

In respect of the development of a Ship Energy Efficiency Management Plan (SEEMP), which one of the following statements is correct?

- **Every ship has its own SEEMP**
- A SEEMP is developed on a selective basis for representative ships of identical class and equipment, trading on the same, or closely similar, trading routes
- Every Flag State Authority has a model SEEMP developed for it under license from the IMO
- Every shipping company has its own SEEMP

In terms of energy efficiency management, which one of the following is most likely to be a more efficient technique than continuously adjusting the vessel's speed through engine power?

- **Operation at constant shaft RPM**
- Adjustment of RPM based on historic data secured under the SEEMP
- Operation at minimum shaft RPM
- Periodic adjustment to achieve optimum shaft RPM

In the application of energy saving measures, which one of the following considerations does the IMO "Guidelines for the Development of a Ship Energy Efficiency Management Plan (SEEMP)" identify as paramount?

- **Safety**
- Cost saving
- Quality Assurance
- Speed

In the simplified Energy Efficiency Design Index (EEDI) formula, which one of the following does the unit "CO_2 emission" represent?

- **Emission from combustion of fuel, including propulsion and auxiliary engines and boilers**
- Emission from combustion of fuel in the main propulsion plant only
- Emission from combustion of fuel, including propulsion and auxiliary engines but excluding boilers and cargo handling machinery
- Projected emissions calculated at the design stage of a new-building project

Is there a legal requirement for a ship to keep a Ship Energy Efficiency Management Plan (SEEMP) onboard?

- **Yes, it is required by MARPOL Annex VI**
- No, provided one is held at the company's head office
- No, there is no requirement at all
- Yes, but only if the ship has been detained by Port State Control for environmental pollution

Select the three features from those listed below which may be aimed at the management of energy efficiency on new vessels.

- **Alternative low emission fuels**
- **Renewable energy sources**
- **Wind-assisted propulsion**
- Company strategic financial planning policy
- Conventional propulsion systems
- Traditional propeller arrangements

The Energy Efficiency Design Index (EEDI) has been designed to apply to certain ship types only. Which one of the following best describes them in respect of the carbon footprint they leave?

- **The most energy-intensive segments of the world fleet**
- All vessels over 400 gross tonnage
- Tankers only
- The most expensive vessels of the world fleet, measured in terms of building costs

The limits, conditions and ballast management arrangements are always to be observed when determining the optimum ballast conditions. Where are these to be found?

- **In the ship's Ballast Water Management Plan**
- In the ship's Cargo Operations Manual
- In the ship's Safety Management Plan
- Optimum ballast conditions are a matter of experience – no written guidance is available on such matters

The selection of the propeller is normally of course made at the design and construction stage of a ship's life and is beyond the control of the ship's staff, but what modern design features, as quoted in this training module, could increase propulsive efficiency power and hence reduce fuel consumption?

- **Fins and / or nozzles that better control the water inflow to propellers**
- Carbon-fibre propeller blades
- Contra-rotating propellers
- Paired propellers fitted at an oblique angle to the hull line

Under many charter parties, the speed of the vessel is determined by the charterer and not the operator. In practical terms and as described in this training module, how then might the owners/ managers and charterers help the ship to maximise energy efficiency?

- **By agreeing charter party terms that encourage the ship to operate at optimum speed**
- By adding a bonus clause which rewards energy efficiency
- By adding a clause which penalises ships which do not practice energy efficiency management
- By deleting any reference in the charter party to charter speed

What are the four component steps of creating and maintaining a Ship Energy Efficiency Management Plan (SEEMP)?

- **Planning / Implementation / Monitoring / Self-evaluation and improvement**
- Design / Implementation / Review / Evaluation
- Planning / Implementation / Monitoring / Recording
- Research / Design / Testing / Evaluation

What do the letters EEDI stand for in relation to the management of ship's energy?

- **Energy Efficiency Design Index**
- Energy Efficiency Data and Intelligence
- Energy Efficient Data Instrumentation
- European Energy Department Index

What do the letters IEEC stand for in relation to the management of ship's energy?

- **International Energy Efficiency Certificate**
- Indicative Energy Efficiency Calculation
- Inter-governmental Energy Efficiency Creditation
- International Energy Exemption Certificate

What do waste heat recovery systems capture for the purposes of electricity generation as part of energy efficiency management on board?

- **Potential heat losses from the exhaust gas**
- Cargo boil-off from LPG and LNG cargoes
- Excess engine heat via cooling water returns
- Thermal heat from the incinerator unit

What does the Energy Efficiency Operational Indicator (EEOI) deliver?

- **A quantitative indicator of a ship's or fleet's energy efficiency**
- A numerical measure of achievement which will indicate the success of a ship in reducing its energy targets
- A qualitative assessment of a ship's energy efficiency
- A quantitative index against which a ship's or fleet's energy consumption can be measured

What is the actual Energy Efficiency Design Index of a ship called?

- **Attained EEDI**
- Absolute EEDI
- Calculated EEDI
- Certified EEDI

What is the extent of the authority of a Port State Control Officer in respect of a vessel's energy efficiency measures?

- **He may only verify that there is an International Energy Efficiency Certificate on board and that it is valid**
- He has no powers at all in respect of energy efficiency measures on board a vessel he inspects
- He may place a "notation" against the vessel's SEEMP and require its re-submission within 3 months
- He may suspect the vessel's IEEC if he identifies a lack of, or inappropriate application of energy efficiency measures

What is the final step in the Ship Energy Efficiency Management Plan (SEEMP) continuous improvement cycle?

- **Self-evaluation and improvement**
- Accreditation
- Calculation and application
- Monitoring and recording

What is the intended outcome of a Ship Energy Efficiency Management Plan (SEEMP)?

- **To provide the framework against which a ship-owner or manager can develop a mechanism to improve the energy efficiency of a ship's operation**
- To have ship's crews trained to an approved standard in operational practices
- To reduce fuel prices
- To secure an International Energy management Certificate for the ship operator/owner

What is the legal requirement to have a Ship Energy Efficiency Management Plan (SEEMP)?

- **Each ship shall keep a ship specific Ship Energy Efficiency Management Plan on board**
- Each ship shall keep a company specific Ship Energy Efficiency Management Plan on board
- Every company shall keep a representative Ship Energy Efficiency Management Plan, applicable to selected vessels in their fleet, available for view in their office
- Every ship shall supply a Ship Energy Efficiency Management Plan to their Flag State Authority

What is the sequence of events in the development and application of a Ship Energy Efficiency Management Plan (SEEMP)?

- **The SEEMP establishes the mechanism, which then leads to improvements in energy efficiency**
- The SEEMP is approved by the IMO and then sent to the ship to implement
- The ship is inspected by the Flag State Authority and approved as suitable for inclusion in the Company's SEEMP
- The ship's crew carry out improvements in energy management, which are then consolidated in a SEEMP

What is the significant design feature of a "twist-flow" rudder, as described in this training module, in terms of energy efficiency management?

- **It reduces cavitation**
- It is made of carbon fibre material
- It is significantly cheaper to maintain in service
- It is specifically designed to work with an Integrated Navigation and Command System

What sort of data should be used in the self-evaluation step of a Ship Energy Efficiency Management Plan (SEEMP) process?

- **The data collected through the monitoring step of the SEEMP process**
- The data collected over the four weeks immediately prior to the self-analysis meeting
- The data collected over the previous 7 years
- The data collected since the last dry-dock or any major adjustment or alteration to the propulsion machinery

When considering potential energy efficiency measures in ship design, which one of the following is a typical thermal efficiency value achieved by a marine diesel engine?

- **50%**
- 20%
- 70%
- 80%

When should self-evaluation take place as part of a vessel's Ship Energy Efficiency Management Plan (SEEMP) cycle?

- **At defined intervals**
- At the end of the company's financial year
- Prior to entering dry-dock
- Whenever there is time

Which Annex of MARPOL 73/78 (as amended) contains the regulations relating to the prevention of air pollution from ships, including the requirements for a Ship Energy Efficiency Management Plan (SEEMP)?

- **Annex VI**
- Annex IV
- Annex V
- Annex VII

Which of the following best describes the requirement for goal setting within the Ship Energy Efficiency Management Plan process?

- **It is voluntary, although its use will serve as a good incentive for proper implementation**
- It is mandatory on all ships subject to the requirement to have a SEEMP
- It is mandatory, but only on ships greater than 10,000 grt
- It is voluntary, although it may be temporarily enforced by a Classification Society

Which of the following describes the key role of the ship's staff in the application and subsequent improvement of a Ship Energy Efficiency Management Plan (SEEMP)?

- **Implementation of any energy efficiency measures introduced as a result of the SEEMP and the on-going reporting of data and information for the purposes of future improvement measures**
- Ensuring that the Flag State Authority is updated on the success of any energy efficiency measures to facilitate the timely renewal of the IEEC according to the vessel's SEEMP
- Identification of the failure of any energy efficiency measures and communication of this information to the appropriate authorities
- Investigation and deletion of any energy efficiency saving measures felt by ship's staff to be inappropriate or too demanding on available resources

Which of those listed here will verify the Energy Efficiency Design Index and the associated technical file for a new ship?

- **The flag State administration**
- The IMO
- The port State administration of the place of building
- The shipyard

Which one of the following can a Ship Energy Efficiency Management Plan (SEEMP) form a part of?

- **The ship's Safety Management System**
- The company's Approved Operations List (AOL)
- The company's Security Management System
- The operator's ISO 14001 license

Which one of the following does the IMO "2012 Guidelines for the Development of a Ship Energy Efficiency Management Plan (SEEMP)" specifically recognize that many companies will already have in place?

- **An Environmental Management System (EMS) established under ISO 14001**
- An enhanced communication system with manufacturers of data recording and monitoring equipment
- An Environmental Safety Management Plan, created and applied in accordance with MARPOL 73/78
- An environmental training programme under STCW 2010

Which one of the following factors best facilitates economic voyage planning as part of an energy efficiency management policy?

- **The maximisation of weather routing**
- Running a ship at maximum speed whenever possible
- Sourcing good fuel
- Using a commercial computer program to perform all aspects of a ship's voyage planning

Which one of the following is a human resource measure that may be applied as part of a Ship Energy Efficiency Management Plan (SEEMP)?

- **Energy efficiency awareness initiatives**
- Centralised communication
- Efficiency compliance monitoring
- Energy maximisation training

Which one of the following is a statement that can accurately be applied in respect of improving the energy efficiency of a ship as part of the Ship Energy Efficiency Management Plan (SEEMP) process?

- **The specific measures for the ship should be identified in the first place**
- Proper monitoring equipment should be ordered by the company for fitting across the fleet at the earliest opportunity
- The company trading targets should be listed and circulated to all parties as the first step
- The ship's crew should be fully re-trained before any plan that involves changes is implemented

Which one of the following is an example of common modern energy saving technology that might be included in the design of a new vessel?

- **Solar panels**
- Biodegradable fuel filters
- Low tensile steel hull plating
- Two-bladed propellers

Which one of the following is currently the primary restriction on the use of emerging alternative fuels as a method to reduce CO_2 emissions?

- **The lack of an adequate fuel supply chain**
- Doubts over long-term availability of such fuels
- The product cost of such fuels
- Uncertainty over actual emission levels of such fuels

Which one of the following is responsible for developing the technical and operational measures related to Energy Efficiency Management, including the Ship Energy Efficiency Management Plan (SEEMP)?

- **The IMO Marine Environmental Protection Committee (MEPC)**
- The IMO Marine Safety Committee (MSC)
- The respective flag State Authority
- The World Maritime University

Which one of the following is responsible for the development of a Ship Energy Efficiency Management Plan (SEEMP)?

- **The company, with input from the ship's personnel**
- The company, based on instructions from the appropriate Flag State Authority
- The IMO Marine Environmental Protection Committee
- The ship's personnel, with input from the company

Which one of the following results, as outlined in this training module, might reasonably be expected if a company invests in developing communications with other stakeholders in their ships' operations?

- **Better coordination, resulting in improved energy efficiency**
- Greater awareness of the fleet's energy saving potential
- Longer overall operating life for the ships due to better operational management
- Lower communications costs by the use of selective time calling

Which one of the onboard record books listed below is most likely to be utilized as part of the monitoring step of a Ship Energy Efficiency Management Plan (SEEMP) process?

- **Engineering log books**
- Bridge VDR data
- Manufacturer's maintenance logs
- Working Hours records

0259 – Seafarer Appraisal

Version 1.0 (8885)

- **I hereby declare that I have completed all contents in this module and that it is my own work and effort.**

0260 – Behaviour Based Safety

Version 1.0 (2539)

Consider a situation where you are working in a team. Who is responsible for the situational awareness of the team?

- **Everyone involved in the job is responsible for the situational awareness relevant to their duties and responsibilities.**
- The Company or owner is responsible for situational awareness as they are financially in charge.
- The Master of the vessel is always responsible for all situational awareness, as they are in charge of the vessel.
- The officer in charge of the team is solely responsible for the situational awareness as he is the supervisor.

The concept of culture is often illustrated as an onion consisting of three layers. What does the innermost layer consist of?

- **It consists of the fundamental beliefs expressed through our actions, our behaviour and how we interact with others.**
- It consists of the artefacts of the culture, which is the physical evidence of a culture like signs, furniture, rituals, language, meetings etc.
- It consists of the expressed values of a culture like policies, and norms and values to be followed in the organisation.
- It consists of the procedures, written rules, campaign posters and reporting systems.

The process of observing safety conditions and behaviour (SCBO) is split into a number of sequential steps. Which of the given options fully identifies these steps?

- **Define and specify what to observe, prepare the observation, observe, intervene and provide corrective feedback and follow-up on actions.**
- Define and specify improvements, observe safety conditions and behaviour, provide corrective feedback and follow-up on actions.
- Identify the safety issues from last observation, discuss the improvements, observe present situation and intervene and provide corrective feedback.
- Observe, intervene and provide corrective feedback and follow-up on actions.

The risk management process can be divided into three levels; organisational level, risk management level, individual level. Which one of the given options best describes the level or levels typically overlooked and the mindset that should be adopted?

- **The individual level is typically overlooked. The mindset to adopt is that every individual is responsible for safety and should not only rely on procedures, but should make their own risk assessment of the present situation prior to any work task.**
- All three levels are typically overlooked. The mindset to adopt is that at least one of the levels should be addressed prior to any work task.
- The Organisational level is typically overlooked. The mindset to adopt is that the organisation is responsible for the safety and therefore all procedures should be followed at all times.
- The risk management level is typically overlooked. The mindset to adopt is that prior to every work task a formal risk assessment should be made by every individual and discussed in teams.

Which of the given options best describes the main difference between situational awareness and risk assessment?

- **The difference between situational awareness and risk assessment is that situational awareness is informal.**
- The difference between situational awareness and risk assessment is that risk assessment can only be used in certain situations.
- The difference between situational awareness and risk assessment is that risk assessment is informal.
- The difference between situational awareness and risk assessment is that situational awareness can only be used in certain situations.

Which of the given options is the first of the 7 principles of corrective feedback necessary to reach the desired result?

- **The first step to reach the desired result is to act quickly.**
- The first step to reach the desired result is to be clear and direct.
- The first step to reach the desired result is to set direction to improve behaviour.
- The first step to reach the desired result is to take ownership.

Which one of the actions in the options given is most likely to reveal why the feedback advice given to an employee was not followed by the employee?

- **Asking the employee questions and being enquiring in the approach.**
- Asking the employee's colleagues the reasons.
- Being appreciative and highlighting the positives in the outcome.
- Discussing the situation at a management meeting onboard.

Which one of the following statements best describes why it is important to observe and modify safety behaviour?

- **Good safety behaviour at a workplace creates a strong overall safety culture.**
- It is important because it is a requirement in the ISM code.
- It is important to ensure a good flow in the reporting system, so that shore based staff can take appropriate actions.
- So that unsafe actions can be reported.

Which one of the given options best describes the characteristics of an intervention culture?

- **It relies on peer-to-peer influence.**
- It relies on the captain's influence.
- It relies on the influence of superiors.
- It relies on top down influence.

Which one of the given options best describes the mindset that must be in place on the highest level of a safety maturity ladder?

- **Safety is a way of thinking, working and running the business and the safety aspect is part of every work related consideration.**
- Accidents are considered part of the job and the safety responsibility is placed in the safety department.
- It is important to involve the staff and to develop the individual's responsibility for the safety work.
- Safety is defined as compliance with rules and procedures. When accidents happen several actions are initiated to avoid recurrence.

Which one of the given options best describes the process that the crew must be able to manage as part of developing a good safety culture?

- **The process of observing, identifying and correcting unsafe behaviour, conditions and attitudes.**
- The process of identifying and note down unsafe behaviour, conditions and attitudes.
- The process of observing unsafe behaviour, conditions and attitudes.
- The process of reporting unsafe behaviour, conditions and attitudes.

Which one of the given options best describes the value of a formal and structured safety condition and behaviour observation?

- **It allows you to reflect, select and prioritise key areas to observe, to prepare and structure your observation and to write down important findings to be addressed.**
- It allows you to ask for assistance and back-up from the captain and the office staff and to check the SMS-system for updates.
- It allows you to prepare everybody on the ship in order for them to improve prior to the observation.
- It allows you to write down all the safety issues you believe should be addressed during the observation and to make plans for improvements.

Which one of the given options best describes who has responsibility for the safety onboard?

- **Everybody holds the responsibility for the safety onboard. Everybody should react upon poor and unsafe safety behaviour and conditions whether it involves a subordinate, a colleague or a superior.**
- Each person only has responsibility for their own safety. Being aware of their own safety behaviour is vital for overall safety on board.
- Only supervisors have responsibility for safety onboard. It is their responsibility to react to unsafe or poor safety behaviour and conditions by intervening, stopping or correcting the actions of subordinates.
- The captain holds the responsibility for safety on board ship. Anyone experiencing unsafe behaviour should therefore report it to the captain immediately.

Which one of the given options best describes why personal responsibility cannot be eliminated by procedures, equipment and generic risk assessments?

- **Because equipment, written generic procedures and risk assessments cannot capture all the relevant details of the current situation.**
- Because equipment and written generic procedures and risk assessments cannot protect you from the act of new and inexperienced crew on board.
- Because generic procedures and risk assessments are often made several years ago and therefore not updated in accordance with new equipment.
- Because written generic procedures and risk assessments are written only to comply with the ISM code and do not contain relevant details about the current situation.

Which one of the given options best explains why safety communication is important?

- **Ongoing verbal and visible safety communication is an effective way to develop a good safety culture and to communicate that safety is top priority of the company.**
- A company's safety culture benefits from regular newsletters about safety. Employees should then be updated on new updates in the SMS-system, which will influence their attitude towards safety.
- A company's safety culture is defined by the number of safety meetings on board, where safety reports are addressed and discussed.
- Monthly safety discussions are important for the attitude towards safety on board. Employees respond to formal meetings where safety is on the agenda.

Which one of the given options correctly completes the following statement? 'Situational awareness is . . .

- **. . . the responsibility of all crew members'**
- . . . only the responsibility of the cadet and able seaman'
- . . . only the responsibility of the officer in charge'
- . . . the responsibility of the company'

Which one of the given options describes the arrangement for feedback that is most likely to drive performance?

- **It should be informal, daily and constructive.**
- It should be given as a written plan where the employee can add their own ideas.
- It should be provided by the captain.
- It should be scheduled to take place every month at a meeting.

Which one of the given options has been identified in the module as one of the key elements of safety communication?

- **Feedback is an essential tool in safety communication and safety should be addressed daily and integrated into every operational activity.**
- Corrective feedback is an essential tool in safety communication and lists of reported safety issues should be put on notice boards to ensure improvements.
- The safety procedures are an essential tool in safety communication, and safety discussions should always take place in a formal meeting.
- Weekly observations are an essential tool in safety communication and safety should be addressed prior to all routine operations.

Which one of the given options identifies a factor that should be observed when observing safety behaviour?

- **Working conditions should also be observed, since poor working conditions can have a strong influence on people's behaviour.**
- Maintenance work should always be chosen when observing behaviour, as most accidents happen during this type of work.
- Mooring operations should always be observed as they involve many crewmembers at a time.
- The competence management system should also be addressed as the level of professional skills has a high impact on safety behaviour.

Which one of the given options identifies the five good safety behaviour patterns to strive for in creating a strong safety culture?

- **Proactiveness, assertiveness, involving, visibility, knowledge-based actions.**
- Assertiveness, integration, knowledge-based actions, involving, visibility.
- Assertiveness, knowledge-based actions, visibility, influence, intervention.
- Proactiveness, influence, appreciative, hands-on, knowledge-based actions.

Which one of the given options is an example of the simple ethics described in this module that should be followed when doing an observation?

- **Be well prepared, be enquiring in your approach, build on the relationship, provide immediate feedback.**
- Always have at least three persons doing the observation to avoid details being overlooked, don't get too friendly with the people you observe.
- Ensure that the captain is assisting you in the observation, write down both good and poor working conditions, provide feedback in writing as soon as possible after the observation.
- Prepare the observation by reviewing important procedures and ask the person who is being observed what they want you to observe.

Which one of the given options is the main reason why it is always important to follow up on safety behaviour modification feedback?

- **Because changing behaviour is like developing new habits - it requires continuous follow-up.**
- Because it is a requirement in the SMS-system.
- Because it is important for knowledge sharing between the crew.
- Because the captain needs to be informed about the result of the feedback.

Which one of the options below best describes what should be observed when it comes to safety condition and behaviour?

- **Incorrect use of tools and equipment, violations of procedures, incorrect body use, poor housekeeping.**
- If the crew member attends at least one safety meeting a year.
- If the crew member is familiar with the maintenance and emergency plans.
- If the crewmember knows the ISM code in detail and if procedures are followed no matter what.

Which one of the options given below best describes why it is important to track barriers to safe behaviour?

- **Because identification of barriers to safety makes it possible to start actions to remove those barriers and thereby make safe behaviour more likely and logical.**
- Because identification of barriers to safety provides an indication if procedures need to be changed.
- Because tracking barriers to safety gives management a chance to monitor the crew.
- Because tracking barriers to safety is part of the SMS.

Which one of the statements given in the options best describes what you need to do in preparation for giving corrective feedback?

- **In order to prepare for corrective feedback you need a clear idea of the desired result.**
- In order to prepare for corrective feedback you first need the Master's approval.
- In order to prepare for corrective feedback you first need to make a list of the items the employee has done wrong.
- In order to prepare for corrective feedback you need to confer with the Company's SMS Circular about corrective feedback.

0261 – Mentoring

Version 1.0 (3684)

Indicate the correct answer to the following question. 'Should the master take time out of his busy schedule to mentor his officers?'

- **Yes**
- No

Is the following statement true or false? A person of a minority nationality may feel alienated and lonely onboard a vessel.

- **True**
- False

Is the following statement true or false? 'A second Officer on board an operational tanker cannot possibly be both a candidate and mentor'.

- **False**
- True

Is the following statement true or false? 'As a result of mentoring, a junior officer can have performance that equates to a senior officer'.

- **True**
- False

Is the following statement true or false? 'For effective mentoring it is necessary to communicate in a common language on a multi-national crewed ship'.

- **True**
- False

Is the following statement true or false? 'On the job training serves as the best method of experiential knowledge transfer as it reinforces theoretical knowledge through practice'.

- **True**
- False

Is the following statement true or false? 'Senior officers should always monitor interpersonal relationships between crew members'.

- **False**
- True

Is the following statement true or false? 'The 10 minute challenge is a method of monitoring the performance of others'.

- **False**
- True

Is the following statement true or false? 'When the mentor recaps all they have taught, it is known as reverse mentoring.'

- **False**
- True

Select as many of the given options which best describe why some officers do NOT mentor?

- **Because they are frightened of someone taking their job**
- **Because they don't want others to be as knowledgeable as them**
- **Because they lack the confidence**
- **Because they lack the time and inclination**
- **Because they need the company's and master's instructions**
- Because they feel too superior

Select four of the given options which best explain the benefits of mentoring to members of a team.

- **Candidates are more adaptable within the team**
- **Candidates have increased confidence and ability**
- **Candidates quickly build experiential knowledge**
- **It results in a more united, stronger team**
- Candidates have quicker physical responses

Select four of the most appropriate options from the given list which describe barriers to mentoring.

- **Different languages spoken on a vessel**
- **Fatigue and fear to speak up**
- **Multinational crew of varying ethnic backgrounds**
- **Too rapid promotion**
- Risk management on a vessel
- Varying competencies found on a vessel

Select one appropriate option from those given which is NOT a step involved in mentoring?

- **Planning**
- De-briefing
- Execution
- Preparation

Select one of the given options that most accurately completes the following statement. 'In order to further experiential knowledge it is important that all accident investigations are . . .

- **. . . available in a relevant language'.**
- . . . kept confidential.'
- . . . printed in the English language only'.
- . . . written in a technical language'.

Select one option that most accurately completes the following statement. 'To be able to effectively perform their practical role, seafarers need . . .

- **. . . some form of mentoring on board a vessel'.**
- . . . a full week of on the job training before taking over'.
- . . . a high level of academic knowledge only'.
- . . . to have already sailed on the vessel'.

Select three of the given options that best describe the mentor-candidate relationship.

- **One that has a common language**
- **One that has acceptance of the process from both parties**
- **One that shows trust between both parties**
- One that always exists in a multicultural environment

Select three of the given options which are considered critical to effective mentoring.

- **Correct experiential knowledge**
- **Effective communication between candidate and mentor**
- **Sufficient time for the mentoring process**
- Formal teaching qualifications
- Mentor and candidate being of the same nationality
- The mentor senior in rank to the candidate

Select two of the given options which best explain the importance of mentoring in the maritime sector.

- **Promotion and career development is made easier because of experiential knowledge**
- **Transferred experiential knowledge allows informed decision making**
- Mentoring is part of any quality training plan mentioned in STCW
- The passing of experiential knowledge satisfies requirements under the ISM code

Select two of the given options which best explain the master's role as an effective mentor.

- **A master will encourage mentoring then mentor the mentors**
- **A master will ensure there is some time to mentor the officers**
- A master should not be involved with other peoples work as a mentor
- A master will reduce a person's work load if an effective mentor

Select two of the given options which best explain why reflection of life's experiences is important.

- **Reflection allows the buildup of experiential knowledge**
- **Reflection enables learning to be extracted from experiences**
- Reflection enables only negative experiences to be forgotten about
- Reflection enables only positive experiences to be enjoyed

Which one of the given options describes how mentoring can help the officer of the watch (OOW) to understand the correct avoiding action to take in compliance with the regulations for preventing collisions (coll regs)?

- **Mentoring helps the OOW better understand the practical application of the coll regs**
- Mentoring cannot help with practical knowledge of the coll regs
- Mentoring will always assist by telling the OOW what action to take in each case
- Mentoring will make the OOW comply with the coll regs

Which one of the given options describes the result of effective mentoring?

- **Mentoring helps in matching theoretical knowledge with practical application**
- Mentoring can effectively replace formal classroom training in all areas
- Mentoring can provide the theoretical principles of a task only
- Mentoring is mainly used to acquire theoretical underpinning knowledge

Which one of the given options is the most appropriate to define the term 'candidate'?

- **A person who receives experiential knowledge from another**
- A low ranking person on a vessel
- A person not able to work using their own initiative
- A person who needs physical assistance with a job in the work place

Which one of the given options is the most appropriate to define the term 'mentor'?

- **A person who is an experienced and trusted adviser**
- A person who has vast practical experience
- A person who likes to formally teach a group of people
- A senior officer on a vessel

Which one of the given options is the most appropriate to define the term 'personal reflection with respect to life's experiences'?

- **Thoughtful and honest consideration of a person's own experiences**
- A process of looking back to store bad memories of personal experiences
- A process of looking back to store good memories of personal experiences
- Critically reviewing the actions of another person in the work place

Which one of the given options is the most appropriate to explain the process of 'mentoring'?

- **Mentoring occurs when two persons exchange life's experiences**
- It is a classroom based method of knowledge transfers
- Mentoring is a personal activity undertaken by reflecting on life
- Mentoring occurs in a team when the leader monitors the activity

Which two of the given options are the most appropriate to define the term 'experiential knowledge'?

- **A person's own practical knowledge reflected upon**
- **Knowledge gained from professional on the job experiences**
- Knowledge from an opinion not based on life's experience
- Theoretical knowledge from classroom based training

Which two of the given options are the most appropriate to define the term 'mentoring'?

- **An informal means of knowledge transfer**
- **The transfer of experiential knowledge**
- A quick briefing session from a leader who is used to absolute authority
- Watching other personnel in the work place

0263 – Principles of assessment

Version 1.0 (2992)

An assessor suspects that there has been collaboration between candidates in an assessment that was intended to assess the individuals' knowledge or skill. Which one of the following actions is most likely to determine if collaboration has taken place?

- **Verify the knowledge of the suspected candidate by oral questioning**
- Make a final judgement, based on past performance of the candidate.
- Make all candidates take the assessment again.
- Set a new assessment for the suspected candidate.

Are there any advantages in using oral questioning when assessing a candidate on a one to one basis?

- **Yes, follow-up questions can be adjusted to the candidate's response and therefore allow assessment of different cognitive and affective skills.**
- No, there are no real advantages; the assessor would always be better using written questions to assess in a one to one situation.
- The only real advantage is that the assessor can see for themselves if the candidate knows anything related to the subject.
- Yes, this is the best method of assessment for the assessor to use in a one on one situation, no matter what skills are being assessed.

Bloom's taxonomy divides skills into three domains. Which one of the domains identified in the options deals mainly with knowledge and understanding?

- **Cognitive**
- Affective
- Elective
- Psychomotor

How can oral questioning be used to confirm a candidate's competence following an assessment by direct observation in which they have performed a practical task?

- **Answering oral questions allows the candidate to provide evidence of their knowledge, understanding and other skills required to confirm competence**
- Correctly answering oral questions is sufficient evidence of competence on its own, so it is the best method for an assessor to use if they didn't observe the candidate throughout the task.
- Oral questioning gives the candidate the opportunity to say what they did wrong during the observational assessment and that is sufficient for the assessor to judge them competent.
- Oral questions allow the candidate to verbally report on how they performed the task which is all that is required to demonstrate competence.

In relation to a programme of learning, select from the following the one which identifies when formative assessment would normally take place?

- **At various stages during the programme.**
- At the beginning and end of the programme.
- Only at the beginning of the programme.
- Only at the end of the programme.

In relation to assessing a candidate's language skills, which one of the following is best suited for assessment by oral questioning?

- **Speaking and listening skills.**
- Full competence in a language.
- Speaking and writing skills.
- Writing and listening skills.

In relation to assessment by direct observation, why is it important that the assessor clearly defines the required performance criteria?

- **The performance criteria define exactly what the candidate should be assessed against.**
- The performance criteria act as a guide for the assessor if he is unfamiliar with the assessment task.
- The performance criteria are important, as they are used solely to set a time limit on an assessment.
- The performance criteria will provide the candidate with a step by step guide on how to perform an assigned task.

In relation to assessment of learners, which one of the following is usually considered as the main purpose of using summative assessment?

- **To grade the learner after training is completed.**
- To grade the learner before training has started.
- To provide feedback to the assessor.
- To provide feedback to the learner.

In relation to assessment, what is meant by the term 'fairness'?

- **Fairness means that all candidates are given an equal opportunity to succeed.**
- Fairness is an alternative term which means that the assessment is reliable.
- Fairness means that all candidates will be given a second chance at the assessment.
- Fairness means that the assessment is fairly easy.

In relation to direct observation of a candidate, which one of the following is correct regarding any limitations of using this method of assessment?

- **Some candidates may not perform to their full potential if they feel uncomfortable being observed.**
- Assessing by direct observation is usually sufficient to fully establish whether or not a candidate is competent.
- The only limitation to using direct observational methods for assessment is that it is time consuming.
- There are no limitations in using direct observation as a method of assessment.

In relation to formative and summative assessment, which one of the following statements is true?

- **Formative and summative assessment can sometimes be combined for grading purposes.**
- Formative assessment can be used to give feedback whereas summative assessment cannot.
- Summative assessment can be used to give feedback to both the learner and the assessor, unlike formative assessment.
- The only purpose for using either summative or formative assessment is to grade learners.

In relation to the analyses of assessment results, which of the following can be considered necessary in order to ensure that any analyses is meaningful?

- **There should be sufficient data from different sources.**
- All of the data should come from a single assessment occasion.
- All of the data should come from an individual assessor.
- There should be no variance whatsoever between actual results and previous or expected results.

In relation to the assessment of seafarers, which of the following can be easily assessed using multiple choice type questions?

- **Knowledge and understanding.**
- Competence.
- Psychomotor skills.
- The higher cognitive skills and affective skills.

In relation to the assessment of seafarers, which of the following can be fully assessed using questions that require written descriptive answers?

- **Knowledge, understanding and other higher cognitive skills.**
- Affective skills.
- Competence.
- Psychomotor skills.

In relation to the assessment of seafarers, which one of the following can written assignments and projects be best used to provide?

- **Evidence to support entries in a work record book.**
- Evidence of affective skills ability
- Evidence of psychomotor skills ability
- Full evidence of competence in practical activities

It is the responsibility of an assessor to ensure that the candidate is provided with full information regarding the assessment. Which one of the following statements best describes the responsibility that the candidate has regarding this information?

- **The candidate should study it prior to the assessment so that they have time to clarify any doubts with the assessor.**
- The candidate only needs to use the information if there are any problems following the assessment.
- The candidate should study it at the beginning of the assessment so that the requirements are fresh in their mind.
- The candidate should take it to the assessment with them so that they can make reference to it during the assessment.

One reason why an assessor needs to produce a marking scheme when preparing an assessment is to indicate the mark allocation to the candidate. Which one of the following is another important reason for having a detailed marking scheme?

- **The assessor may not be the person who marks all of the assessments.**
- A detailed marking scheme is only required for quality assurance purposes.
- Flag State only accepts assessment decisions based on marking schemes that they have approved.
- Port State control will always require evidence of a marking scheme following any incident involving an individual who has been assessed onboard.

Oral questioning can be used by an assessor to fully assess a candidate's affective skills.' Is this statement accurate?

- **No, although oral questioning can be used to partly assess these skills, it is necessary to combine this method with observation of behaviour to fully assess affective skills.**
- No, oral questioning cannot usually be used to assess any part of a candidate's affective skills.
- Yes, this is the most common method of assessment used to fully assess affective skills.
- Yes, using oral questions is a much better method to use than any other method, particularly direct observation.

STCW is the main standard used for training of seafarers and contains a series of competence tables which should be the main source for defining competence requirements. Select from the following elements of the competence tables the one that is most likely to be used to determine learning objectives?

- **The knowledge, understanding and proficiency statements.**
- The competence evaluation criteria
- The competence statements
- The methods for demonstrating competence

Select from the following statements the one that is the most accurate in relation to the term 'assessment for learning'.

- **Assessment for learning is part of the learning process and provides instant feedback to the learner.**
- Assessment for learning is a process which is used solely for deciding the content of a programme of learning.
- Both pre assessment and summative assessment are usually used for assessment for learning.
- Only formative assessment can be used for the purpose of assessment for learning.

Select from the following statements the one which is the main reason why it is important to have a suitable marking scheme for an assessment?

- **It will provide evidence of the reliability and consistency of an assessment.**
- It allows all questions to be allocated equal marks, no matter how many parts there are to each question.
- It is required for quality assurance purposes.
- It makes it easy to count the total available mark.

Select from the following the most appropriate assessment method that can be used by an assessor to assess knowledge?

- **Direct oral or written convergent type questions.**
- Direct observation of the candidate performing a practical task.
- Direct oral or written divergent type questions.
- Direct written questions are the only appropriate method.

Select from the following the most appropriate assessment method that can be used by an assessor to assess understanding or comprehension?

- **Direct oral or written divergent type and follow-up questions.**
- Direct observation of the candidate performing a practical task.
- Direct oral or written convergent type questions.
- Direct written questions only.

Select from the following the one which is correct in relation to the importance of the timing of an assessment and the reason why it is important.

- **Assessments should only be scheduled for a time that allows both the assessor and the candidate to prepare properly.**
- Assessments should always be carried out in the morning, as the candidate will be less tired.
- Assessments should always take place in the afternoon so that both the assessor and candidate have the morning to prepare for them.
- Assessments should only be scheduled during normal working time, to avoid any additional costs.

Select from the options given the one that best defines a learning outcome in relation to a programme of learning.

- **It is a general statement to indicate what the learner should be able to do after completing the programme.**
- It is a general statement which describes what the assessment result should be at the end of the programme.
- It is a measurable, specific statement to indicate what the learner should be able to do after completing each section of the programme.
- It is a measurable, specific statement to indicate what the learner should be able to do after completing the programme.

Select from the options given the one that best describes the main characteristics of a learning objective.

- **It should be specific and measurable.**
- It should be general and measurable.
- It should be short and measurable.
- It should be specific and short.

Select from the options the phrase that most accurately completes the following statement. 'A programme of learning can be defined as . . .

- **. . . a series of activities designed to enable learners to gain further knowledge or skills or to improve existing knowledge or skills.'**
- . . . a course of study that is delivered through audio visual programmes at the learner's workplace.'
- . . . a series of lessons that is delivered in a classroom or lecture theatre involving more than one learner.'
- . . . one lesson from a series of computer based training programmes designed for self study.'

Select the statement which best describes how an assessor should act in relation to making a judgement of a candidate's competence based purely on the final product or end result.

- **Competence is about how the result is achieved as well as what result is achieved and therefore a judgement regarding competence should not be made based purely on the end result or final product.**
- In order to achieve an acceptable final product or end result a candidate must be competent and therefore the assessor can consider it is an effective test of competence.
- Providing a candidate can produce the correct end result on at least one occasion, the assessor can reasonably judge them to be competent.
- The only thing that really matters is the final product or end result, so if this is good then an assessor can judge a candidate as being competent.

Select, from the following statements, the one which is the main purpose of including pre-assessment in a programme of learning?

- **To establish a learner's existing skill or knowledge prior to a programme of learning.**
- To grade learners before they take on a new work role.
- To prepare learners for final, summative assessment.
- To reduce the number of learner's taking part in a programme of learning.

Select, from the following statements, the one which is the most accurate in relation to mark allocation in an assessment marking scheme?

- **An assessor should allocate marks to each question and part of a question based on their judgement as to the importance and difficulty of the required answer.**
- An assessor should allocate marks to each question and part of a question based on the anticipated time it will take to mark the answer.
- An assessor should allocate marks to each question and part of a question based on the complexity of the wording of the question.
- An assessor should allocate marks to each question and part of a question based on the expected length of the answer to each question.

Select, from the following, the phrase which correctly completes the following statement. 'If an assessor uses an inappropriate level of language during an assessment . . .

- **. . . some candidates may be disadvantaged so it is unacceptable.'**
- . . . it will probably not affect the performance of any of the candidates so it is acceptable.'
- . . . only the performance of the weakest candidates will be affected but it is still unacceptable.'
- . . . the performance of all candidates will be affected to the same degree so it is acceptable.'

Select, from the following, the statement which best describes the circumstances under which an onboard assessor would normally use formative assessment?

- **Only when the assessor is also the trainer.**
- An assessor would never use formative assessment.
- An assessor would normally use formative assessment as the main way of grading a learner.
- An assessor would use formative assessment before any training takes place to establish the learners existing knowledge and skills.

Select, from the phrases given in the options, the one that correctly completes the following statement. 'An assessment strategy is said to be reliable when . . .

- **. . . assessment decisions made by different assessors are consistent for different candidates over a period of time.'**
- . . . assessment results are the same for any candidate, no matter how many times they are assessed.'
- . . . each year the same percentage of candidates successfully complete the assessments and obtain the same grades.'
- . . . it requires that assessors have a draft marking scheme for each assessment.'

Select, from the phrases given in the options, the one that correctly completes the following statement. 'An assessment strategy is said to have transparency . . .

- **. . . when the exact requirements for each assessment are clearly communicated to all candidates.'**
- . . . when all assessors agree with the assessment methods to be used.'
- . . . when all candidates are fully informed of their results for all assessments.'
- . . . when the assessment results are made publically available to everyone.'

Select, from the phrases given in the options, the one that most accurately completes the following statement. 'An assessment strategy can be said to be valid if . . .

- **. . . the assessment tests all the learning objectives and therefore all of the intended outcomes are fully measured.'**
- . . . an acceptable number of candidates successfully complete each assessment test.'
- . . . at least some of the assessment tests are related directly to the defined learning objectives.'
- . . . the assessment tests the majority of the learning objectives and therefore a good sample of the outcomes is measured.'

Summative assessment is usually carried out either at the end of a programme of learning or as each section of the programme is completed. Which one of the following statements is the most accurate in relation to summative assessment?

- **Summative assessment can be used to provide feedback as to the effectiveness of a programme.**
- Summative assessment is another term for pre-assessment.
- Summative assessment is mainly used to provide feedback to the learner to assist in further learning.
- Summative assessment is only used for grading of learners.

When assessing a candidate's affective skills, which one of the following are the best methods for an assessor to use?

- **One to one or group discussion and direct observation.**
- Direct observation and written examination.
- One to one discussion and oral questions.
- Written projects and examinations.

When assessing a candidate's psychomotor skills, which one of the following is the most appropriate for an assessor to employ?

- **Direct observation of a work task.**
- Oral questioning and written tests.
- Oral questioning only.
- Written tests only.

Which of the following is the main reason why the duration of an assessment should be limited?

- **If an assessment lasts too long, candidates may not be able to perform to their full ability throughout.**
- Candidates have to get used to keeping to limited time allowances for their work role.
- The longer the assessment lasts, the more marking an assessor will have to do.
- With unlimited time, anyone will be able to pass an assessment so time should be limited.

Which of the following is the most appropriate method that an assessor can use to test a candidate's ability to apply knowledge or understanding?

- **By setting the candidate a problem to solve or a task to complete.**
- By oral questioning of the candidate.
- By setting a written examination.
- Only by direct observation.

Which of the following most accurately defines the term 'collaboration' in relation to assessment of candidates?

- **Work submitted for assessment to which individuals other than the candidate have knowingly contributed.**
- Claiming work submitted for assessment is their own work when another person has contributed to it.
- It is concealing prohibited material during an examination for reference purposes.
- It is when a candidate uses prohibited material during an examination.

Which of the following most accurately defines the term 'plagiarism' in relation to assessment of a candidate?

- **Falsely claiming work submitted for assessment is their own without the knowledge of the person it actually belongs to.**
- Claiming work submitted for assessment is their own work when another person has contributed to it.
- It is concealing prohibited material during an examination for reference purposes.
- It is when one candidate gives another candidate the correct answer during an examination.

Which of the following statements is the most accurate in relation to assessment by direct observation?

- **It can normally be used alone to assess a candidate's proficiency in a physical activity, but not always their competence.**
- It can normally be used alone to assess a candidate's proficiency in a physical activity and their knowledge and understanding.
- It can normally be used alone to fully assess both a candidate's proficiency in a physical activity and their competence.
- It cannot normally be used alone to assess a candidate's proficiency in a physical activity or their competence.

Which of the following statements is the most accurate in relation to candidates' actions in an assessment?

- **Collaboration is both expected and acceptable during assessment of group work.**
- A candidate using prohibited, concealed information during an assessment is guilty of collaboration.
- Collaboration and plagiarism are exactly the same and both are unacceptable.
- Collaboration should never be allowed during any assessment.

Which of the following statements is the most accurate in relation to using oral questioning in a group assessment situation?

- **The assessor will need to identify the less proactive group members and give them the opportunity to demonstrate their knowledge, usually by asking direct questions.**
- As long as some members of the group answer each of the questions asked by an assessor the assessor should award the whole group a pass mark.
- If individual candidates within a group do not voluntarily contribute answers the assessor should just fail them.
- In a group assessment situation the more proactive candidates always encourage the less proactive candidates to involve themselves.

Which of the following statements most accurately describes learning that is referred to as 'associative learning'?

- **It is when, through practice, individual skills are combined together with knowledge and understanding to demonstrate competence.**
- It is learning based on copying the actions of associates or colleagues.
- It is the process where an improvement in language skills is associated with general learning.
- It is when a candidate helps an associate or colleague to learn.

Which one of the following best describes a performance improvement plan that is drawn up following an assessment?

- **It is a plan agreed between the assessor and candidate following feedback to improve candidate performance in areas of weakness.**
- It is a plan drawn up by the assessor to improve candidate performance in areas of weakness.
- It is a plan drawn up by the assessor to improve the format of the assessment and the way it was carried out.
- It is a plan drawn up by the candidate to improve their performance in areas of weakness.

Which one of the following criteria can be used to determine the reliability of an assessment?

- **Consistent scores, irrespective of the assessor, timing of assessment or the candidate**
- An equal number of candidates meeting and failing to meet the assessment criteria.
- Every candidate achieves the same score for a given assessment.
- Every marker awards exactly the same score for a selected candidate.

Which one of the following criteria can be used to determine the validity of an assessment strategy?

- **It is judged to have validity if it fully tests the stated learning objectives and no other knowledge or skills.**
- It is judged to have validity if good candidates achieve a 100% score
- It is said to have validity if it fully tests the stated learning objectives and also other knowledge and skills.
- It is said to have validity if it is judged to be reliable.

Which one of the following is 'assessment by direct observation' the best method of assessment for an assessor to use?

- **Psychomotor skills**
- Cognitive skills
- Competence
- Knowledge

Which one of the following is the main limitation of using oral questioning for a group assessment?

- **It is difficult for the assessor to be sure that all candidates have all of the required knowledge.**
- It is difficult for the assessor to think of sufficient questions to ask.
- The assessor will not be able to hear the answer if everyone in the group is answering at the same time.
- The assessor will not be able to remember all of the individuals in the group.

Which one of the following is the use of 'open book' tests best suited for assessing?

- **Some of the higher order cognitive skills.**
- Affective skills
- Knowledge
- Psychomotor skills

Which one of the following statements is correct in relation to the value of self assessment by a candidate during a programme of learning?

- **Self assessment can motivate a candidate to improve performance, as it provides evidence of progress and identifies where improvement can be made.**
- Self assessment by the candidate can be used as an alternative to assessor led assessment and therefore reduces the required assessor input.
- Self assessment has little value in helping to improve performance for most candidates.
- When candidates self assess their own performance they only ever identify their strengths, and therefore it has little value.

Which one of the following statements is most accurate in relation to an assessor's preparation for assessment of a candidate by direct observation?

- **It is essential that an assessor should define the learning objectives and performance criteria as an initial part of their preparation for the assessment.**
- As long as the assessor has selected some task for the candidate to perform their preparation is complete.
- The assessor doesn't need to check that the assessment venue is properly prepared; this is usually the candidate's responsibility.
- The assessor should ensure that the candidate is aware of the venue and time for the assessment; any other information can be given when the assessment takes place.

Which one of the following statements is most accurate in relation to the use of oral questions for group assessment?

- **Some candidates may be more comfortable in a group situation and this may encourage them to contribute more.**
- It is a good method to adopt, as the weak candidates will probably gain all of the necessary knowledge from the strong candidates.
- It is easy for an assessor to identify a candidate who is not contributing and directly question them to show how weak they are.
- It is the best form of assessment, as someone in the group is sure to know the correct answer and so the group will pass the assessment

Which one of the following statements is the most accurate in relation to a person who is given a completely new work role?

- **There may be a need to give additional training to the person in order to improve their existing skills or teach them additional skills**
- If a person has been competent in their former work role then they will be competent in any new role.
- The only way to learn how to perform a new work role is 'on the job' so providing additional training is a waste of time and effort
- Training is expensive, so it should only be given when new technology is introduced in the work place.

Which one of the following statements is the most accurate in relation to assessment of competence?

- **Since competence usually involves a combination of skills, then more than one assessment method will normally be required to test whether or not a candidate is competent.**
- As long as a candidate can complete a task of work then they are competent regardless of how they do it.
- Competence can only be assessed by direct observation of a candidate carrying out a physical task.
- The best method to use for testing a candidate's competence is always a combination of written and oral tests.

Which one of the following statements is the most accurate in relation to assessment of the competence of a group of candidates?

- **Assessors should consider the best methods for assessing each individual candidate in order to ensure that assessments are effective and fair.**
- All candidates should be assessed by exactly the same means in order to ensure fairness in the assessment.
- Assessors are usually extremely busy and cannot concern themselves with individual candidate needs.
- If any of the candidate's cannot complete an assessment due to their individual characteristics then the assessor should just fail them.

Which one of the following statements is the most accurate in relation to the limitations of using oral questioning when assessing a candidate?

- **Oral questioning does not allow an assessor to make a proper judgement of the candidate's psychomotor skills.**
- Oral questioning does not allow an assessor to make a proper judgement of the candidate's knowledge and understanding.
- Oral questioning, combined with an assessment by direct observation, cannot usually be used to confirm a candidate's competence.
- There are no limitations to the use of oral questioning when assessing any of a candidate's proficiencies.

Which one of the following statements is the most accurate in relation to the potential for assessment to encourage learning?

- **All assessment can encourage learning when appropriate feedback is given.**
- Candidates who are unsuccessful in assessment are always put off any further learning activity.
- Only candidates that are successful are motivated by assessment and wish to learn more.
- Only summative assessment can provide motivation for candidates to learn more.

Which one of the given statements is the most accurate in relation to the learning objectives used in a programme of learning?

- **A learning objective is a specific statement which describes what the learner should be able to do, or will know, after completing a section within a programme of learning.**
- A learning objective is a general statement which describes what the learner should be able to do, or will know, after completing a programme of learning.
- A learning objective is a specific statement which describes what the learner should be able to do, or will know, after completing a programme of learning.
- A learning objective is a specific statement which describes what the teacher or trainer will do in order that the learner will learn something during a programme of learning.

Which one of the phrases given in the options correctly completes the following statement? 'Associative learning is improved by feedback from assessment which . . .

- **. . . encourages a learner to practice and improve component parts of a skill and therefore the skill overall.'**
- . . . compares a candidate's performance with the performance of others.'
- . . . is negative and concentrates only on the poor elements of a candidate's performance.'
- . . . is positive and concentrates only on the good elements of a candidate's performance.'

Which one of the phrases given in the options most accurately completes the following statement? 'Feedback to a candidate following assessment is important because . . .

- **. . . it should motivate the candidate by highlighting areas of strength as well as areas for improvement.'**
- . . . it is better to offer feedback than to draw up an improvement plan for the candidate.'
- . . . it is the only way to let the candidate know the result of the assessment.'
- . . . it is usually a requirement of the assessment strategy quality assurance process.'

Why is it important that assessors follow best practice when planning and carrying out an observational assessment?

- **It gives the candidate the best opportunity to demonstrate their proficiency.**
- A totally inexperienced assessor will always make the correct judgement if they follow best practice.
- Best practice means that there are fixed methods that work for every candidate in every situation.
- Following best practice means that nothing will ever go wrong during an assessment.

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Assessors should always use best practice when planning and carrying out assessments. Which one of the options explains why this is necessary?

- **It will produce consistently effective assessments and reliable assessment results.**
- It will always produce higher candidate grades.
- It will reduce the time taken for assessments and effectively minimise costs.
- It will result in consistently effective assessments and the highest pass rates.

Different types of oral question can be used to test different levels of cognitive skills. Which of the types given in the options is best suited to test understanding?

- **Divergent questions that require deductive reasoning.**
- Convergent questions, each with a small number of probable correct answers.
- Convergent questions, each with a unique correct answer.
- Divergent questions, each with a small number of possible correct answers.

How can learning objectives and performance criteria be used to ensure that assessment is relevant and that all of them are covered within the assessment?

- **By forming each of the learning objectives and performance criteria into tasks and/or questions.**
- By asking questions in the assessment on everything related to the main topic of the learning objectives and performance criteria.
- By having the same number of questions as there are learning objectives and performance criteria in the programme of learning.
- By writing a learning objective and performance criteria for each question in the assessment.

In relation to assessment of a candidate by direct observation, which of the following statements best describes the environment that the candidate should be required to perform the assessment task in?

- **Wherever possible the assessor should ensure that the condition of the environment is as close as possible to that in which the candidate would normally perform the work activity.**
- The assessor should ensure that the environment is as quiet as possible to avoid distractions for the candidate.
- The assessor should ensure that the environment is similar to the most extreme conditions that the candidate is likely to be exposed to in order to test the candidate fully.
- The assessor should ensure that the environment is warm and well ventilated so as to make the candidate as comfortable as possible.

In relation to assessment of a candidate by direct observation, which one of the following statements best describes how an assessor should, wherever possible, select a task for the candidate to perform?

- **The assessor should select a task which would be a normal work activity for the candidate.**
- The assessor should select a task which is a simulation of a normal work activity.
- The assessor should select a task which is based on an unusual work activity.
- The assessor should select a task which is not based on a normal work activity.

In relation to assessment of a candidate, which one of the following statements best describes what an assessor should consider when setting the performance criteria for an assessment by direct observation?

- **The expected level of proficiency in all of the required skills, and the full range of conditions that the task might be carried out under.**
- Only the expected level of proficiency that the candidate needs to perform the task to.
- The expected level of proficiency and a time limit for completing the task.
- The expected level of proficiency, but only under the conditions that exist at the time the task is being undertaken.

In relation to setting the required performance criteria for assessment of competence or proficiency in a practical skill which one of the following statements is the most accurate?

- **The performance criteria should relate directly to the practical skill and should have sufficient detail to fully meet the required standard defined in the competence or proficiency statement**
- Performance criteria are not necessary where only practical skills are being assessed as it is only the end result which is important.
- Performance criteria should only ever cover the main part of the task and never any preparation before the task is started or testing after the task has been completed.
- The performance criteria for assessment of a candidate performing a practical task must always cover the knowledge and understanding requirement as well as the practical aspects of the task

In relation to the time and venue chosen for assessment by direct observation of a candidate's proficiency in performing a practical task, which one of the following statements is correct?

- **Both the time and venue should be chosen to ensure that the assessment will be as fair and effective as possible.**
- Any venue that is available when the assessor and the candidate have some free time will be suitable for the assessment.
- Anytime that a suitable venue is available is a good time to carry out the assessment.
- As long as the time is suitable to both the candidate and the assessor and the venue is not being used by anyone else it will be alright to go ahead with the assessment.

In relation to the use of convergent questions during oral assessment, which one of the cognitive skill levels given in the options can be best tested using this question type?

- **Knowledge**
- Application
- Evaluation
- Understanding

In relation to the use of different oral question types, which one of the following statements is most accurate?

- **Divergent questions that require deductive reasoning can be suitable for testing a candidate's cognitive and affective skills**
- Divergent questions that require deductive reasoning are only suitable for testing a candidate's affective skills.
- Divergent questions that require deductive reasoning are only suitable for testing a candidate's cognitive skills.
- Divergent questions that require deductive reasoning can be suitable for testing a candidate's psychomotor and affective skills.

In relation to the use of divergent questions during oral assessment, which one of the statements given in the options is the most accurate?

- **This type of question is best used to test the higher cognitive skills as the candidate will need to evaluate and analyse the possible answers in order to decide on the most appropriate.**
- This type of question is best used to test knowledge as the candidate will need to recall the correct answer from memory.
- This type of question is best used to test the higher cognitive skills as the candidate will need to remember the correct answer.
- This type of question is best used to test the range of psychomotor skills associated with a particular competence.

It is a general principle that assessors should produce and keep to an agreed assessment plan when carrying out an assessment by direct observation. Which one of the actions given in the options should an assessor follow if circumstances develop which mean that the assessment plan cannot be followed?

- **Postpone or suspend the assessment unless the candidate agrees to any necessary changes.**
- Discard the assessment plan and carry on assessing using oral questioning only to see what the candidate might have done.
- Just pass the candidate if they have performed well up to that point in the planned assessment.
- Stop the assessment and make a final judgement based on the part of the assessment that has been completed.

STCW is the main standard used for the training and certification of seafarers. Which one of the following best describes the main structure of STCW?

- **It is based on a number of functional areas and 3 levels of responsibility.**
- It is based on all of the departments usually found onboard the ship.
- It is based on an officer's rank and their associated tasks.
- It is based solely on the levels of responsibility assigned to each rank onboard ship.

Select from the following options the one which contains the most important criteria when selecting a suitable practical task for testing a candidate when assessing by direct observation of performance.

- **The task selected should directly relate to the competence or proficiency that is being assessed and give the candidate a fair opportunity to demonstrate their proficiency.**
- The task selected should be one that requires the least resources and is the most cost effective.
- The task selected should be the easiest and quickest to complete of the available options.
- The task selected should be the one which allows the maximum number of proficiencies to be assessed at the one time and reduces the time that the assessor uses in preparation for assessment.

Select from the following statements the one which describes the circumstances under which an assessor would be most likely to use pre-assessment and formative assessment in a competency based learning programme.

- **When the assessor is also the trainer.**
- An assessor would never use these forms of assessment in a competency based programme of learning.
- Pre-assessment and formative assessment would always be used by an assessor along with summative assessment in any learning programme.
- The assessor would use these forms of assessment whenever they use written questions to test a candidate.

Select from the given options the main reason why a written assessment plan or brief should be drawn up as part of the preparation even for an assessment by direct observation?

- **To make it clear to the assessor and candidate exactly what is required from the candidate during the observed activity.**
- To ensure the candidate turns up at the correct place and time for the assessment.
- To help the candidate, as it will contain detailed information of how to complete the task being observed.
- To remind the assessor of the correct sequence that the candidate should follow in completing a task.

Select from the given options the main reason why it is important to limit the expected duration of an assessment during the planning stage?

- **If it lasts too long the candidate could become fatigued, which may affect their performance.**
- If it lasts too long it may result in the candidate exceeding normal working hours.
- If it lasts too long the assessor may forget details of the early part of the candidate's performance.
- If it lasts too long the assessor may get bored.

Select the option which contains a definition of a question which is classed as a truly convergent question.

- **It is a question with only one unique correct answer.**
- It is a question either two or three correct answers.
- It is a question that has no correct answer.
- It is a question with a number of correct answers.

Select the option which contains a definition of a question which is classed as a truly divergent question.

- **It is a question with an infinite number of correct answers.**
- It is a question that has no correct answer.
- It is a question with either two or three correct answers.
- It is a question with only one unique correct answer.

Select the option which correctly completes the following statement.

'A candidate's ability to follow a defined procedure when completing a practical task . . .

- **. . . will often be one of the performance requirements for an assessment by direct observation.**
- . . . never matters as long as the end result is acceptable.
- . . . will always be one of the performance requirements for an assessment by direct observation.
- . . . will never be one of the performance requirements for an assessment by direct observation.

Select the option which correctly completes the following statement.

'When an assessor is selecting the venue for an assessment to take place one of the main considerations is that it . . .

- **. . . provides a suitable environment for the assessment.'**
- . . . is away from the normal workplace.'
- . . . is the most convenient place for the assessor.'
- . . . is the most convenient place for the candidate.'

Select the statement which best describes how an assessor should act in relation to making a judgement of a candidate's competence based only on the final product or end result.

- **Competence is about how the result is achieved as well as what result is achieved and therefore a judgement regarding competence should not be made based only on the end result or final product.**
- In order to achieve an acceptable final product or end result a candidate must be competent and therefore the assessor can consider it is an effective test of competency.
- Providing a candidate can produce the correct end result on at least one occasion the assessor can reasonably judge them to be competent.
- The only thing that really matters is the final product or end result so if this is good then an assessor can judge a candidate as being competent.

When forming an assessment plan for an assessment by direct observation what would normally be the first thing an assessor should do?

- **Select a competence which has proficiencies that are suited to assessment by direct observation.**
- Ask the candidate which work activity they would prefer to have observed.
- Create some learning objectives and write some supplementary questions.
- Select a time and venue for the assessment.

Which of the following best describes an inductive question that may be used by an assessor during an oral assessment of a candidate?

- **It is a question which contains only sufficient information for the candidate to be able to arrive at a probable answer.**
- It is a question which contains all of the information for the candidate to be able to arrive at the correct answer.
- It is a question which contains full information but the candidate will only be able to give a probable answer
- It is a question which contains incomplete information but the candidate should still be able to give the correct answer.

Which of the following statements is the most accurate in relation to the use of oral questioning and listening techniques by an assessor?

- **By using oral questioning and listening techniques the assessor can make a judgement as to a candidate's proficiency in cognitive and affective skills.**
- By using oral questioning and listening techniques only, an assessor can usually make a judgement as to a candidate's competence in a practical task.
- By using oral questioning and listening techniques the assessor can make a judgement as to a candidate's proficiency in cognitive and psychomotor skills.
- By using oral questioning and listening techniques the assessor can make a judgement as to a candidate's proficiency in psychomotor and affective skills.

Which one of the combinations of skills given in the options is best suited to assessment by direct observation rather than by written or oral questioning techniques?

- **Psychomotor and affective skills.**
- Affective and cognitive skills.
- Psychomotor and cognitive skills.
- Psychomotor, affective and cognitive skills.

Which one of the following best describes the main limitation, if any, of using only observational methods when carrying out an assessment of a candidate's competence?

- **Direct observation of candidate performance does not necessarily test the associated knowledge and understanding required for competence.**
- Assessment by direct observation cannot be used to assess across the range that the candidate is required to perform a given work activity.
- Assessment by direct observation is not the best method for assessing psychomotor skills.
- There are no limitations in using only direct observational methods for assessing a candidate's competence.

Which one of the following best represents the elements that form part of a competence as required in a competency based learning strategy?

- **A combination of proficiency, knowledge and understanding.**
- A combination of practical skills and the ability to communicate.
- The ability to answer written and oral questions.
- The ability to pass written tests and to perform practical tasks.

Which one of the following contains the two methods of assessment most frequently used by an onboard assessor when assessing a candidate during a work activity onboard ship?

- **Direct observation and oral questioning.**
- Direct observation and a formal written test.
- Direct observation and written questions and answers.
- Oral questioning and written questions and answers.

Which one of the following statements about effective oral assessment to confirm a candidate's competence is the most accurate?

- **Divergent questions that require deductive thinking with follow-up questions allow both cognitive and affective skills to be assessed and are most suitable for confirming competence.**
- Convergent questions that require deductive thinking allow all skill types to be assessed and are therefore most suitable to confirm competence.
- Pre-selected convergent questions are suitable for assessing all cognitive skills and therefore most suitable for confirming competence.
- Pre-selected divergent questions that require deductive thinking are most suitable for fully confirming competence.

Which one of the following statements best describes the main advantages of assessing a candidate performing a normal work activity by using direct observation?

- **It is authentic and relevant, and an appropriate means of assessing psychomotor skills.**
- It allows the candidate to demonstrate the full range of knowledge and understanding associated with the work activity.
- It is the cheapest way to fully assess competence.
- It is the only way for an assessor to assess a candidate's competence in a work activity.

Which one of the following statements correctly describes competence in a work activity onboard ship?

- **Competence in a work activity usually involves a combination of different types of skills.**
- Assessment of any competence relating to onboard work activities will always require testing by visual, written and oral methods.
- Competence in a work activity usually only involves psychomotor skills.
- It is usually possible to assess competence in a work activity using a single method of assessment.

Which one of the following statements is the most accurate definition of a good learning objective?

- **It is a specific statement which describes what the learner should be able to do after completing a section within a programme of learning.**
- It is a general statement which describes the material that will be covered during a programme of learning.
- It is a generic statement relating to a functional role at work.
- It is a specific statement which describes what the trainer wishes to do in a specific part of a programme of learning.

Which one of the given options best describes the relationship between learning objectives or performance criteria and assessments in a competency-based programme of learning?

- **Assessments should be designed to provide evidence that one or more learning objectives or performance criteria have been met.**
- Learning objectives and performance criteria are only included as guidance for the candidate and do not need to be linked to the assessment.
- Learning objectives and performance criteria are only included as reminders to the assessor of what has to be assessed.
- Learning objectives and performance criteria are simply statements of intent and do not have to be directly assessed.

Which one of the given options can be considered as a key element of a good oral questioning technique to be used by an assessor following an assessment by direct observation?

- **Consider follow-up questions based on the possible responses that a candidate may give to initial oral questions in order to test higher cognitive and affective skills across the required range.**
- Consider oral questions that only test the candidate's ability to recall facts and figures relating to the work activity.
- Consider oral questions that relate to other learning objectives associated with different work activities.
- Prepare a set number of oral questions and keep to them no matter what responses the candidate may give.

Which one of the given options is an accurate definition of the term 'inductive reasoning'?

- **It is a process where a limited amount of information given to the candidate is used to infer a probable answer to an oral question.**
- It is a process where a candidate is given all of the information and facts and they use it to deduce the correct answer to an oral question.
- It is a process where a candidate is given all of the information and facts and they use it to infer the probable answer to an oral question.
- It is a process where a limited amount of information given to the candidate is used to deduce the correct answer to an oral question.

Which one of the given options is the main reason why the use of leading questions should be avoided by an assessor when testing candidates using oral questions?

- **Leading questions usually contain at least part of the answer that is required or an obvious clue to the answer and therefore they do not effectively test the candidate.**
- Leading questions are only suitable for confirming a candidate's competence and do not test their cognitive skills.
- Leading questions tend to unsettle candidates which may affect their performance.
- The language required for leading questions is too complicated and this will only confuse the candidate.

Which one of the options identifies the most important action that an assessor can take to ensure that a candidate is suitably relaxed for an assessment by direct observation?

- **Make sure that the candidate is fully informed of the requirements for the assessment.**
- Make sure that the assessment takes place during the candidate's normal work hours.
- Make sure that the candidate has unlimited time to complete the assessment.
- Make sure that the task chosen for the assessment is one that the candidate has done before.

Which one of the options should be considered as the main purpose of a post assessment debrief, following an assessment by direct observation?

- **To provide detailed feedback and, if required, to produce a performance improvement plan.**
- It is only to inform the candidate of the result of the assessment.
- To allow the assessor to reflect on the assessment process.
- To inform the candidate of the result of the assessment and to highlight what went wrong during the assessment.

Which one of the statements given in the options describes one of the main responsibilities of a candidate for assessment by direct observation?

- **They must familiarise themselves with all details of the assessment provided by the assessor before the assessment.**
- They must complete the assessment in the shortest time possible.
- They must ensure that the assessment decision is properly recorded in a secure record system.
- They must ensure the assessment venue is properly prepared for the assessment.

Which one of the statements given in the options is correct in relation to individuals who have responsibility for assessing seafarers onboard ship?

- **Assessors should not be influenced by external pressures.**
- Assessors should always consider how long the candidate has served in their current rank when assessing a candidate's competence.
- Assessors should consider the cost of an assessment as the main criterion when planning an assessment.
- Assessors should only be allocated a fixed period of time when planning assessments.

Which one of the statements given in the options is the best description of the term 'performance criteria' in relation to assessment in competency based programme of learning?

- **They are clear statements which describe the level and quality of performance that a candidate must meet in order to demonstrate competence.**
- They are clear statements which define the time that is allowed for an assessment task.
- They are general statements which describe the competence that is being assessed.
- They are general statements which describe the performance that a candidate must demonstrate.

Which one of the statements given in the options is the main reason for an assessor to use both inductive and deductive types of question during oral testing of a candidate?

- **Using a mixture of inductive and deductive questions allows the assessor to test the whole range of a candidate's cognitive skills as well as some affective skills.**
- Using a mixture of inductive and deductive questions allows the assessor to test the candidate's knowledge and understanding, although other methods are required to test higher cognitive and affective skills.
- Using a mixture of inductive and deductive questions allows the assessor to test the whole range of a candidate's cognitive skills although none of their affective skills.
- Using a mixture of inductive and deductive questions is less monotonous for both the candidate and the assessor.

With reference to an assessor including arrangements for providing feedback to the candidate in their assessment plan, which one of the following statements is correct?

- **Arrangements for feedback should be included as it is essential that the assessor informs the candidate about the standard of the performance and gives guidance as to any future actions required.**
- Arrangements for feedback should be included in the assessment plan just in case the assessor has to tell the candidate what they have done wrong.
- Arrangements for feedback should only be included in the assessment plan if the assessor is sure that there will be time to give any to the candidate.
- Arrangements for feedback should only be included in the assessment plan so that the assessor can praise the candidate if they have performed well to further motivate them.

0265 – Shore based assessment

Version 1.0 (1198)

Consider a multiple-choice test made up of 20 items each with four answer options with one option correct. What would be the expected percentage score for a candidate simply guessing the answers if each test item has equal scoring?

- **25%**
- 0%
- 5%
- 75%

Errors in multiple-choice test items can be minimised before they are used by careful wording of the stem and the options and by peer review. Which one of the following actions is likely to be the most effective in identifying poor test items after the test has been used?

- **Analysis of scores for individual test items to identify trends in scoring.**
- Just check the test items that good candidates get wrong.
- Look for individual items that all candidates got right or wrong; these will all be poor and will need to be rewritten.
- Run a spell check on the test items.

How can learning objectives be used to ensure that a written assessment is relevant and that all learning objectives are covered within the assessment?

- **By forming each of the learning objectives into tasks and/or questions.**
- By asking questions in the assessment on everything related to the main topic of the learning objectives.
- By having the same number of questions as there is learning objectives in the programme of learning.
- By writing a learning objective for each question in the assessment.

In addition to providing to some cognitive and affective skills which one of the following does completion of written assignments or projects by seafarers provide direct evidence of?

- **Report writing and research.**
- Competence in a practical skill.
- Psychomotor skills.
- The full range of language skills.

In relation to the development of multiple-choice test items, which one of the following best explains why it is important to consider the skill that is being tested?

- **Because not all skills can be effectively tested using multiple-choice tests.**
- Because the exact method for testing each skill will be defined in the standard so that the assessor knows when to use multiple-choice tests.
- Because the standard will contain sample test items to be used to test each skill.
- Because this is the only way to determine how many test items are required for the test.

One reason for considering the duration of an assessment is to allow candidates sufficient time to complete the test. Select from the given options the other main reason why it is important to limit the duration of a written assessment?

- **If the test takes too long some of the candidates could become fatigued which may affect their performance.**
- If the test takes too long candidates will require too many additional answer books.
- If the test takes too long it will be difficult to find a venue that is available for the time required.
- If the test takes too long the assessor will have too much marking to do.

STCW is the main standard used for the training and certification of seafarers. Which one of the following best describes the main structure of STCW?

- **It is based on a number of functional areas and 3 levels of responsibility.**
- It is based on shipboard ranks and their associated tasks.
- It is based on the departments usually found onboard ship.
- It is based solely on the levels of responsibility assigned to each rank onboard ship.

Select from the comments given in the options the one which is most accurate in relation to the following statement.

'When developing multiple-choice test items it is important to consider who is being tested.'

- **True, the test should be set at an appropriate content and language level for the intended target group.**
- False, it isn't important, as each target group is required to have the same knowledge for a given subject.
- False, it isn't important; all seafarers are required to have the same level of language skills.
- True, but it is only necessary to identify the nationality of the group being tested.

Select from the following the best way that an assessor can ensure that written assignments and projects are fit for purpose when they are to be used by a number of seafarers onboard different ships.

- **By offering a number of alternative tasks which each tests the skills to be assessed.**
- By allowing the candidate to select any task they wish.
- By only writing assignment and project task specifications for equipment which is common to all ships.
- By selecting a specific task to test the skills to be assessed.

Select the phrase from those given in the options which most accurately completes the following statement.

'The use of written assignments or projects for assessing seafarers onboard ship . . .

- **. . . is a relevant assessment method as they can be based on actual work activities.'**
- . . . has only limited use, as it is only suitable for testing knowledge.'
- . . . is not a suitable assessment method as each ship is different so it is not possible to prepare suitable assignments or projects.'
- . . . is not a suitable assessment method as they only provide evidence of the learner's report writing skills.'

Select the phrase which most accurately completes the following statement.

'When selecting the types of test item to use in an online multiple-choice test . . .

- **. . . a number of factors, including the target group and the skills being tested, need to be considered.'**
- . . . only simple multiple-choice questions with a single correct answer should be used.'
- . . . the only factor to consider is the availability of test item templates in the test editor.'
- . . . the selected scoring system is the main factor to consider.'

What would be the main purpose of having a peer review of multiple-choice test items carried by a person who is an experienced assessor but not a subject matter expert?

- **To check only the format, structure and language level of the test items.**
- There is no purpose if the person is not a subject matter expert.
- To check only the accuracy of the content.
- To check the accuracy of the content and also the structure of the questions.

When assigning a scoring scheme for a multiple-choice test, which one of the following is the most accurate?

- **Where test items have varying difficulty and importance, weighted scoring should be considered.**
- It doesn't matter if the test items have varying difficulty and importance, equal point scoring should always be used.
- Where test items have equal difficulty and importance, weighted scoring is the best scheme to use.
- Where test items have varying difficulty and importance, negative scoring is always the best scheme to use.

When developing multiple-choice test items why is it important to consider the purpose of the test? Select the most relevant reason from those given in the options.

- **The purpose of the test will help the assessor select the types of test items to use and define the type of feedback required.**
- If the test is to be used for formative assessment then the assessor will know that feedback is not necessary.
- If the test is to be used for summative assessment the assessor will know that feedback must be included.
- The purpose of the test will identify exactly how many test items to include.

Which one of the following best describes the elements that form part of a competence, as required in a competency-based learning strategy?

- **A combination of proficiency, knowledge and understanding.**
- A combination of practical skills and the ability to communicate.
- The ability to answer written and oral questions.
- The ability to pass written tests and to perform practical tasks.

Which one of the following correctly completes the following statement?

'When using predefined model answers for marking tests which require written descriptive answers, assessors should . . .

- **. . . apply some flexibility and be prepared to accept reasonable alternative answers.'**
- . . . limit the marks that can be awarded for answers which do not exactly match the model answer.'
- . . . never consider that questions may be misleading unless every candidate gives the same wrong answer.'
- . . . only ever accept the predefined model answer as being correct.'

Which one of the following definitions most accurately describes the main feature of surface learning?

- **Candidates memorise information without fully understanding it.**
- Candidates understand information for a short period of time.
- It is associated with long term memory.
- It results in a better learning outcome.

Which one of the following identifies the skills that are most likely to be effectively assessed by questions requiring written descriptive answers?

- **Cognitive and affective skills.**
- Affective and psychomotor skills.
- Cognitive and psychomotor skills.
- Most psychomotor skills including writing skills.

Which one of the following is a main requirement for ensuring a question requiring a written descriptive answer effectively assesses a candidate?

- **The question should be related directly to a defined learning objective.**
- The question should be written in as few words as possible.
- The question should contain direct clues to the required answer.
- The question should not be related to the candidate's normal work role.

Which one of the following is considered to be one of the limitations of using multiple-choice tests as an assessment tool?

- **This type of assessment cannot be used to fully assess competence.**
- It is not possible to test higher cognitive skills using this type of assessment.
- There is no way to prevent candidates from cheating with this type of test.
- This type of testing is not suitable for use online or for automated scoring.

Which one of the following is considered to be one of the main advantages of using a written assignment or a project as an assessment tool?

- **It is an appropriate means of testing the higher cognitive skills.**
- It is a good means of fully testing competence.
- It is easy to write effective questions that test all skill levels.
- It takes very little time to mark candidates' scripts.

Which one of the following is considered to be one of the main advantages of using multiple-choice tests as an assessment tool?

- **It is an efficient way of testing knowledge for a large number of objectives.**
- Candidates have no chance of correctly guessing answers.
- It is a good means of testing affective skills.
- It is very easy to write questions that test high level cognitive skills.

Which one of the following is considered to be one of the main advantages of using questions that require written descriptive answers as an assessment tool?

- **It is a suitable method of testing the higher cognitive skills.**
- It is a suitable method of fully testing competence.
- It is easy to write effective questions that test all skill levels.
- It takes very little time to mark candidates' scripts.

Which one of the following is correct in relation to testing affective skills using questions that require written descriptive answers?

- **Where questions are written to explore candidate's feelings and behaviour towards others they can be used to test affective skills.**
- Affective skills can only be tested by direct observation of a candidate in the workplace so questions that require written descriptive answers are not really suitable.
- Questions requiring written descriptive answers are only suitable for testing cognitive skills and cannot be used for testing affective skills.
- Questions requiring written descriptive answers that test the higher cognitive skills usually test the affective skills at the same time.

Which one of the following is the best way for an assessor to ensure that the requirements for a written assignment or project are properly communicated to a seafarer working onboard ship

- **By providing the seafarer with a comprehensive assignment or project brief, covering the full requirements.**
- A list of questions that need to be answered in the report will normally be sufficient.
- A report specification which identifies what should be included in the final report will normally be sufficient.
- By sending an email to the onboard training officer so that they can inform the seafarer.

Which one of the following is the best way that an assessor can ensure that written questions are properly focussed and therefore effective?

- **Relate them directly to the learning objectives and performance standards.**
- Keep the questions as short as possible.
- Review the questions a number of times
- Use simple language even when the subject is complex.

Which one of the following is the main method of assessment most frequently used by a shore-based assessor for onboard assessment of a seafarer?

- **Written questions.**
- Direct observation.
- Oral questioning.
- Third party evidence.

Which one of the following is the main source of the competence requirements for seafarers?

- **STCW**
- ISGOTT
- MARPOL
- SOLAS

Which one of the following publications defines the requirements for a person who is to assess seafarers' competence?

- **STCW**
- MARPOL
- MLC 2006
- SOLAS

Which one of the following scoring methods can be used to counter the effect of candidates guessing the answer to test items in a multiple-choice test?

- **Formula or negative scoring.**
- Equal point scoring.
- Raw scoring.
- Weighted scoring.

Which one of the following should be the main consideration for an assessor when they are using a weighted marking scheme?

- **The importance and difficulty of the question.**
- Ensuring the total marks allocated for a test adds up to 100.
- The expected length of the candidates answer.
- The length of the question.

Which one of the following skills are open book tests an appropriate assessment method for?

- **Higher cognitive skills.**
- Affective skills.
- Psychomotor skills.
- Recall skills (Knowledge).

Which one of the following statements best describes the main purpose of using written assignments or projects for seafarers onboard ship?

- **Written assignments and projects give seafarers an opportunity to gather evidence of their knowledge, understanding and skills when working onboard ship.**
- Completion of written assignments and projects by seafarers onboard ship reduces formal training requirements for seafarers.
- Completion of written assignments and projects by seafarers onboard ship removes the need for any other form of onboard assessment.
- Written assignments and projects allow seafarers to gather all the evidence they require for assessment of their competence in any work activity onboard ship.

Which one of the following statements correctly describes competence in a work activity onboard ship?

- **Competence in a work activity usually involves a combination of different types of skills.**
- Assessment of any competence relating to onboard work activities will always require testing by visual, written and oral methods.
- Competence in a work activity usually only involves psychomotor skills.
- It is usually possible to assess competence in a work activity using a single method of assessment.

Which one of the following statements is the most accurate in relation to designing multiple-choice tests?

- **In order to test deep learning, multiple-choice test items must be designed to test the higher level cognitive skills.**
- It doesn't matter how multiple-choice tests are designed, they can only ever test basic knowledge.
- Multiple-choice tests which are used for testing deep learning must have at least 6 options for each test item.
- The only way to test deep learning using multiple-choice tests is to have a minimum of 100 test items.

Which one of the following statements is the most accurate in relation to testing application skills?

- **Testing application skills can require the candidate to use ideas, concepts, and principles related to their knowledge and understanding.**
- Application skills are one of the psychomotor skills and always need to be assessed by practical tests.
- Application skills cannot be tested using questions requiring written descriptive answers.
- Application skills do not rely on candidates having either knowledge or understanding of the related topic.

Which one of the following statements is the most accurate in relation to testing comprehension or understanding?

- **Candidates can effectively demonstrate their understanding or comprehension of a topic by translating, summarising or interpreting information in a written form.**
- Recalling a piece of information is acceptable evidence of comprehension or understanding for most topics.
- Understanding and comprehension are easily assessed using simple direct questions which require a single sentence answer.
- Understanding or comprehension cannot be tested using questions that require written descriptive answers.

Which one of the following statements is the most accurate in relation to the use of written formative assessment for assessing seafarers onboard ship?

- **Formative assessment will normally only be used during training or instruction of seafarers onboard ship.**
- Formative assessment is never used when assessing seafarers onboard ship.
- Formative assessment is the main assessment tool used when selecting suitable candidates for promotion.
- Formative assessment will always be used when testing seafarers for competence onboard ship.

Which one of the following statements is the most accurate in relation to the use of written summative assessment for assessing seafarers onboard ship?

- **Summative assessment is one of the main assessment tools used when selecting suitable candidates for promotion.**
- Summative assessment is never used when assessing seafarers onboard ship.
- Summative assessment will only be used during training or instruction of seafarers onboard ship.
- Summative assessment will only be used for seafarers to identify training needs.

Which one of the following statements is the most accurate in relation to types of learning?

- **Deep learning can help to develop a learner's higher cognitive skills.**
- Deep learners usually forget information very quickly.
- Rote learning is a form of deep learning often used by learners in preparation for examinations.
- Surface learning can help to develop a learner's higher cognitive skills.

Which one of the following statements relating to the distracters that are used for a multiple-choice test item is the most accurate?

- **All distracters should be plausible and should not contain repeated information which could be included in the stem.**
- Only one of the distracters used in a test item needs to be plausible.
- The more distracters that are used the better, as it will be more difficult for a candidate to guess the correct option.
- Two of the best distracters to use are 'None of the above' and 'All of the above'.

Which one of the following statements relating to the use of formula or negative scoring in a multiple-choice test is correct?

- **It is normally used in order to counter the effect of candidates guessing the answers.**
- It is normally used to prevent candidates achieving a 100% score.
- This type of scoring is used to differentiate between easy and more difficult test items.
- This type of scoring system is never used in this type of test.

Which one of the following statements relating to the use of weighted scoring in a multiple-choice test is correct?

- **It is normally used in order to differentiate between easy and more difficult or important and less important items.**
- It is normally used to prevent candidates achieving a 100% score.
- This type of scoring is used to counter the effect of guessing.
- This type of scoring system is never used in this type of test.

Which one of the following would be the usual source of subjects for written assignments and projects that are to be used for the assessment of seafarers onboard ship?

- **The competence tables in STCW.**
- SOLAS
- The ISM Code.
- The MARPOL annexes.

Which one of the given options best describes the relationship between learning objectives and written assessments in a competency-based programme of learning?

- **Assessments should be designed to provide evidence that the knowledge linked to the learning objectives have been met.**
- Learning objectives are only included as guidance for the candidate and do not need to be linked to the assessment.
- Learning objectives are only included as reminders to the assessor of what has to be assessed.
- Learning objectives are simply statements of intent and do not have to be directly assessed.

Which one of the phrases given in the options correctly completes the following statement?

'Defined marking schemes are essential with written assessments to help ensure that . . .

- **. . . scoring is reliable and consistent.'**
- . . . all questions carry equal marks.'
- . . . candidates only attempt questions with the highest mark allocation.'
- . . . each part of a question carries equal marks.'

Which one of the reasons given in the options is the most accurate in relation to the following statement?

'When deciding on how a multiple-choice test is to be compiled it is important to consider the scoring system used in order to ensure the test is balanced.'

- **It is important since a balanced test must have test items for each topic and also for each weighting group if weighted scoring is used.**
- As long as the test contains test items covering each topic it is balanced so the scoring scheme is not important.
- It is not possible to have a balanced test unless it is made up from a discreet number of test items.
- The test will only be balanced if it is made up of test items with equal point scoring.

Which one of the statements given in the options describes one of the main responsibilities of a candidate for a written assessment?

- **The candidate must familiarise themselves with all details of the assessment provided by the assessor before and in the assessment.**
- The candidate must complete the assessment in the shortest time possible.
- The candidate must ensure that the assessment decision is properly recorded in a secure record system.
- The candidate must ensure the assessment venue is properly prepared for the assessment.

Which one of the statements given in the options is correct in relation to individuals who have responsibility for assessing seafarers onboard ship?

- **Assessors should not be influenced by external pressures.**
- Assessors should always consider how long the candidate has served in their current rank when assessing a candidate's competence.
- Assessors should consider the cost of an assessment as the main criterion when planning an assessment.
- Assessors should only be allocated a fixed period of time when planning assessments.

0271 – Leadership, Communication for maritime leaders

Version 2.0 (1789)

According to Mehrabian, which ONE of these elements provides the biggest impact in face-to-face communication?

- **Body language**
- Appearance
- Voice tone
- Words

As the chairman of a meeting, which ONE of these actions should NOT be taken?

- **Be selective in allowing participants to speak up.**
- Ensure that all participants are involved in the meeting.
- Explain the purpose of the meeting to all participants.
- Follow the approved agenda of the meeting.

As the duty officer on watch during manoeuvring, which ONE of the following actions should you take in order to secure a closed loop communication between the captain and helmsman?

- **Ensure that the captain's orders are clearly understood, repeated and executed correctly by the helmsman.**
- Check the ship's position.
- Listen closely to the captain's orders and repeat them as necessary.
- Relay the captain's helm order to the helmsman.

As the receiver of a message, which ONE of these actions would you take to ensure that a closed loop communication is attained?

- **Repeat the message in the way you understood it**
- Listen carefully to the message and execute it right
- Respond affirmatively
- Understand the message

Communication on board can be done using various means.

Which ONE of these is NOT applicable to onboard or internal communication?

- **Use of punctuation marks**
- Use facial expression
- Use of body language
- Use of hand signals

Here is a corrective feedback message forwarded by the master to the chief cook.

Based on the SOED feedback model, which ONE of these statements refers to the 'effect' part of the model?

- **This is unsafe and sets a bad example for the messman.**
- Hi chief, I would like to talk to you about safety improvement in the galley.
- I noticed that you have been wearing slippers while working inside the galley.
- I would like you to wear appropriate shoes always while you are inside the galley.

Select from these the ONE which is NOT an example of external communication.

- **The Captain contacts the Chief Mate through VHF at forward mooring station.**
- An officer makes a DSC call to another ship.
- The third mate reports the ship's position to Vessel Traffic Information System.
- The vessel reports ETA to port control.

The chief engineer asked the fourth engineer to give him the fuel ROB figures. The instruction was not clear to the fourth engineer, but he simply said "Yes sir!" in order to avoid looking foolish.

Which ONE of the types of communication barrier is the above an example of?

- **Fear**
- Competence level
- Lack of time and planning
- Noise

Which ONE of these actions is generally associated with good listening skills?

- **Maintaining an appropriate level of eye contact.**
- Answering your phone while in the middle of a conversation with another person.
- Leaning back in your chair while in a meeting.
- Talking to a person while working on the computer.

Which ONE of these best describes how communication should be onboard the ship?

- **It should be characterized by simplicity, directness, and usage of standard marine communication phrases.**
- It should be based on the language of the majority group.
- It should be characterized by lengthy exchange of messages through email and phone calls.
- There are no fundamental differences between communication with people ashore and communication between people onboard the ship.

Which ONE of these components of communication confirms to the sender that the message was correctly received and interpreted?

- **Verification**
- Context
- Question
- Sender

Which ONE of these groups of headings does the acronym TIAS stand for in relation to open communication?

- **Trust, Involvement, Appreciation, Safety**
- Training, Integration, Appreciation, Safety
- Training, Involvement, Authority, Safety
- Trust, Integration, Appreciation, Sociable

Which ONE of these is a closed question?

- **How many were injured in the accident?**
- How did the accident happen?
- What happened?
- Why did the accident happen?

Which ONE of these statements does NOT support the importance of effective communication on board the ship?

- **Communication cannot change seafarer's performance.**
- Communication builds good relationship among the crew.
- Knowledge sharing is one of the positive results of effective communication
- Poor communication may lead to accidents onboard.

Which ONE of these would be the best course of action when issues are raised in meetings that are not originally part of the meeting agenda.

- **Take note of this new issue and put it on the agenda for the next meeting.**
- Disregard any new issues that are not part of the original agenda.
- Extend the time until all newly raised issues are tackled to everyone's satisfaction.
- Tackle this new issue instead of the original agenda.

Which THREE of the following are recognized as components of communication?

- **Context**
- **Medium**
- **Message**
- Delivery

Which THREE of these items are best suited for use in internal communication?

- **Intercom**
- **VHF Radio**
- **Written messages**
- Satellite phone

Which THREE of these personal attitudes are barriers to effective communication?

- **Defensiveness**
- **Distorted perception**
- **Selective hearing**
- Trustworthiness

Which TWO of these are examples of voice distance communication?

- **Phone call to ship's agent**
- **Ship to ship VHF call**
- Hand signals
- Job instruction

Which TWO of these are practical uses of communication in the maritime setting?

- **Establish common expectations**
- **Facilitate effective performance**
- Establish work ethics
- Facilitate new jobs

Words stand alone in which ONE of the following forms of communication?

- **Written**
- Face-to-face
- Non-verbal
- Voice distance

You are about to give the bosun feedback on his poor performance.

Which ONE of these best describes how you should give the feedback?

- **Convey your feedback to the bosun face-to-face.**
- Call him in his cabin and discuss your feedback over the phone.
- Complete the appraisal form and send it to the shore office.
- Provide him with a sealed written performance report.

You are doing some paper work when the fitter knocks on your door and tells you that he has some concerns about his appraisal.

Which ONE of these actions should you avoid in order to show good active listening skills?

- **Continue working while listening to the fitter.**
- Have a relaxed, open body language
- Maintain an appropriate level of eye contact.
- Temporarily stop doing the paper work.

You are the 2nd Engineer and you notice that tools are scattered and improperly secured in the workshop. You expect the Fitter to secure these tools after every usage.

Which ONE of these best describes the most appropriate course of action that you can take?

- **Talk to him face-to-face**
- Disregard it
- Instruct the 3rd Engineer to make a report about this
- Instruct the Fitter to call you

You are the chief engineer and while trying to resolve a conflict among ratings, you ask them a number of questions. Which ONE of these is an open question?

- **Tell me, how did this conflict start?**
- Are you sure?
- Do you agree?
- How many of you are involved here?

You observed that the third mate has not been doing the maintenance jobs for lifesaving and firefighting equipment.

Following the TIAS principle, which ONE of these statements would it be best to use in approaching the issue with the third mate?

- **"I observed that you were not able to inspect all life saving and fire fighting equipment again. I would like to know your challenges so that I may be able to assist you."**
- "Give me three reasons why I should not send you home. You failed to check the life saving and firefighting equipment again."
- "I just don't understand why you failed to inspect all life saving and firefighting equipment again. You owe me an explanation."
- "You are irresponsible. Explain why you failed again to inspect all life saving and firefighting equipment."

'Face-to-face communication is the preferred method when . . .

Select the TWO phrases that could accurately complete the above statement.

- **. . . giving feedback.'**
- **. . . misunderstanding is likely to happen.'**
- . . . relaying engine orders to the duty engineer.'
- . . . the subject is rather casual.'

'In the Sandwich feedback model, you start by saying three positive things about the person. This is followed by the second layer called beef which should be preceded by the word . . . instead of 'but'.

Select the ONE word which most accurately completes the above statement.

- **and**
- however
- nevertheless
- yet

'The . . . could be physical, social, chronological or cultural dimension where a particular communication is based upon.'

Select the ONE word which most accurately completes the above statement.

- **context**
- listener
- message
- verification

'The key idea that a person wants to convey which is composed of verbal and non-verbal symbols.'

Which ONE of these components of communication is the above statement an example of?

- **Message**
- Context
- Medium
- Verification

'When composing e-mails, you should be extra cautious when capitalizing the letters because it . . . '

Complete the above statement by selecting the ONE, most appropriate phrase.

- **. . . is synonymous with shouting and/or emphasizing**
- . . . encourages trust and involvement
- . . . is accepted as a standard
- . . . is indicating your agreement

' . . . can be used during crane and mooring operations where the use of voice communication, with or without the aid of communication equipment, is not feasible.'

Complete the above statement by selecting the ONE, most appropriate word or phrase.

- **Hand signals**
- Email messages
- Shouting
- Texting

0272 – Leadership, Maritime conventions

Version 2.0 (1373)

An SMC is issued to a ship as . . .

Complete the above statement by selecting ONE, most appropriate phrase.

- **. . . proof that a safety management system is in place in accordance with the ISM.**
- . . . proof of compliance with MLC 2006.
- . . . proof that management strategies are being followed. .
- . . . proof that the vessel complies with the ISPS code.

Based on the IMO principles of safe manning, which THREE of the following principles must be observed in determining the safe manning of a ship?

- **Capability to ensure safe carriage of cargo during transit.**
- **Capability to maintain general surveillance of the ship.**
- **Capability to maintain the safety arrangements and cleanliness of all accessible spaces to minimize the risk of fire.**
- Capability to become totally independent of the shore-based office.

For which of these cases have the IMO and ILO jointly produced guidelines?

Select ONE correct answer.

- **A tanker caught fire after an oil spill.**
- A group of officers and ratings onboard a ship accused the master of violating the alcohol policy of the company.
- Delivered spare part to the ship does not meet the technical specification, according to the Chief Engineer.
- Ship cargo was damaged due to a super typhoon.

In order to minimize potential delays and costs for ships visiting certain ports, several memorandum of understanding (MoU) or regimes such as the Paris and Tokyo MoU were established to . . .

Complete the above statement by selecting ONE, most appropriate phrase.

- **. . . harmonize inspections.'**
- . . . shorten inspections.'
- . . . simplify inspections.'
- . . . waiver inspections.'

In which year was the STCW adopted by the IMO?

- **1978**
- 1984
- 1995
- 2010

The IMO adopted MARPOL in . . . and it has been amended by the protocols of 1978 and 1997 and kept updated with relevant amendments.

Complete the above statement with the ONE, most appropriate year.

- **1973**
- 1958
- 1964
- 1978

The International Labor Organization (ILO) is a tripartite agency of the United Nations. Tripartite means that . . .

Complete the above statement by selecting the ONE, most appropriate phrase.

- **. . . it brings together representatives of governments, employers, and workers to jointly shape labour policies.**
- . . . representatives from the employers, employees, and clients sit in the organisation's policy-making body.
- . . . resolutions follow a three-stage process before final enforcement.
- . . . the deck, engine and catering departments are equally represented.

The Occupational Health and Safety Act 1993 was enforced by . . .
Complete the above statement by selecting ONE, most appropriate country.

- **Australia**
- Denmark
- New Zealand
- Norway

The . . . is an agency under the United Nations (UN) that is responsible for drawing up and overseeing international labour standards.

Complete the above statement by selecting the ONE, most appropriate option.

- **ILO**
- IMO
- Seafarers Union
- UN

The first version of the SOLAS was adopted in 1914 in response to the . . . disaster.

Complete the above statement by selecting the ONE, most appropriate word or phrase.

- **Titanic**
- Exxon Valdez
- Herald of Free Enterprise
- Torrey Canyon

The guidelines on the mitigation of fatigue were developed by IMO. Which ONE of the following modules is NOT a part of these guidelines?

- **Fatigue and stress**
- Fatigue
- Fatigue and the rating
- Fatigue and the ship officers

Which MARPOL Annex is concerned with control of pollution by noxious liquid substances in bulk?

- **Annex II**
- Annex I
- Annex III
- Annex IV

Which ONE of the following conventions was not conceived through the IMO?

- **Maritime Labour Convention**
- MARPOL
- SOLAS
- STCW

Which ONE of the following is ILO's main concern in connection with the shipping industry?

- **Working conditions of seafarers**
- Environmental protection
- Ship safety
- Ship security

Which ONE of these conventions specifies the minimum standards required to ensure safe construction, equipment and operation of ships?

- **SOLAS**
- MARPOL
- MLC
- STCW

Which TWO of these are IMO's recommendations that deal with the human factor?

- **Framework for consideration of ergonomics and work environment**
- **The organisations strategy to address the human element.**
- Checklist for operations on board.
- Guidelines for the operational implementation of company goals

Which certificate does the flag state issue to a company as a proof of its compliance with the MLC?

- **Maritime Labour Certificate**
- Crew Satisfaction Certificate
- Document of Compliance
- Quality System Certificate

Which code established safety management objectives and required companies to set up a safety management system (SMS)?

- **ISM**
- ISPS
- MLC
- STCW

Which code reflects an amendment to SOLAS and deals with special measures to enhance maritime security for ships, ports and government agencies?

- **ISPS**
- COLREG
- ISM
- STCW

Which convention includes several regulations aimed at preventing and minimizing pollution at sea from ships?

- **MARPOL**
- ISM
- ISPS
- SOLAS

Which international maritime regulation establishes basic requirements for training, certification and watch-keeping for seafarers?

- **STCW**
- ISM
- MARPOL
- SOLAS

'Flag states are responsible for implementing provisions of maritime conventions. They can delegate these responsibilities to . . .

Complete the above statement by selecting ONE, most appropriate phrase.

- **. . . classification societies.'**
- . . . customs.'
- . . . port States.'
- . . . the United nations.'

'Recommendations are . . . '
Complete the above statement by selecting ONE, most appropriate phrase.

- **. . . non-mandatory guidance and specifications formulated by IMO to help authorities comply with international regulations.'**
- . . . mandatory standards set up by ILO through international conventions.'
- . . . mandatory standards set up by IMO through international conventions.'
- . . . non-mandatory guidance and specifications formulated by ILO to help authorities comply with international regulations.'

'The ILO develops international labour standards in the form of conventions and recommendations that set minimum standards on work-related issues. The following are examples of such labour standards except . . . '

Complete the above statement by selecting the ONE, most appropriate phrase.

- **. . . STCW training requirement.'**
- . . . basic labour rights.'
- . . . collective bargaining.'
- . . . equality of opportunity and treatment.'

'The IMO is responsible for . . . '

Complete the above statement by selecting the TWO, most appropriate phrases.

- **. . . encouraging the ratification of maritime conventions by as many flag States as possible.**
- **. . . the development of international conventions and recommendations concerning safety and security of shipping and prevention of marine pollution by ships.**
- . . . the development of local and company policies of companies across the world.
- . . . the implementation of state legislations related to shipping.

'The MLC sets out . . . '

Complete the above statement by selecting ONE, most appropriate option.

- **. . . seafarers' rights to decent work conditions and create conditions of fair competition for shipowners.'**
- . . . guidelines to company policies on marketing and chartering.'
- . . . labour laws and jurisdiction.'
- . . . pollution prevention guidelines.'

'The sub-committee of Standards of Training and Watchkeeping considers matters related to . . . '

Complete the above statement by selecting THREE, most appropriate options.

- **. . . principles of safe watchkeeping**
- **. . . promotion of awareness from lessons learnt**
- **. . . review, updating and revision of IMO model courses**
- . . . consultation with maritime companies regarding training requirements

0273 – Leadership, Shipboard Personnel Management and Training

Version 2.0 (1306)

A structured shipboard training programme helps . . .

Complete the statement by selecting TWO, most appropriate phrases.

- **. . . to develop officers' proficiency to a higher position.**
- **. . . to ensure crew competence as set by the company.**
- . . . to develop a friendly and harmonious environment onboard a vessel.
- . . . to ensure compliance with company policies and performance standards.

A third engineer is onboard a large new tanker. The vessel is well equipped with the latest technology and as such the third engineer is complacent due to the automation bias. What can the third engineer do to avoid the boredom that is beginning to set in? Select TWO most appropriate options.

- **Do some special projects to improve the ship's performance.**
- **Get involved in other tasks that are being done on board without relying only on those assigned by a senior.**
- Discuss the reasons for boredom with others in the peer group.
- Write an email to the shore office so that they are aware of the same.

Automation bias leads to reduced level of situational awareness.

State whether the statement is true or false.

- **True**
- False

Errors of commission occur when there is no response or action in situations where it is required.

State whether the statement is true or false.

- **False**
- True

How can the company help reinforce the learning process and implement a training programme onboard ship?

Select ONE most appropriate option.

- **The company must demonstrate its commitment to the programme by providing time and opportunity to the ship's staff.**
- The company must demonstrate its commitment by making all training compulsory.
- The company must demonstrate its commitment by making the staff complete the training programme while home on leave.
- The company must demonstrate its commitment by making the staff pay for the training programme.

In management without leadership there is no direction for the team.

State whether the statement is true or false.

- **True**
- False

Informal structures generally stem from . . .

Complete the statement with the ONE, most appropriate option.

- **. . . personal relationships.**
- . . . mentor relationships.
- . . . professional relationships.
- . . . workplace relationships.

Mentoring is more structured than coaching.

State whether the statement is true or false.

- **False**
- True

Not wearing the correct or complete PPE is an error of omission.

State whether the statement is true or false.

- **True**
- False

Select ONE, most appropriate option that defines SMART goals.

- **SMART goals are Specific, Measureable, Achievable, Realistic and Time-bound goals.**
- SMART goals are Short, Meaningful, Achievable, Realistic and Time-bound goals.
- SMART goals are Specific, Measureable, Accurate Rational and Time-bound goals.
- SMART goals are Specific, Measureable, Accurate, Realistic and Time-bound goals.

The Company Training Officer or CTO is responsible for organising the practical training programme at sea.

State whether the statement is true or false.

- **False**
- True

The deck officers and ratings report to the Chief Mate.

State whether the statement is true is false.

- **True**
- False

The fourth quadrant of the Johari window represents the area where your personal abilities are unknown to you and your team members.

State whether the statement is true or false.

- **True**
- False

The master is responsible for . . .

Complete the statement by selecting the TWO, most appropriate phrases.

- **. . . ensuring that all concerned are effectively carrying out the onboard training programme.**
- **. . . ensuring the continuity of the role if the shipboard training officer is relieved during a voyage.**
- . . . ensuring that no one misses any important session.
- . . . ensuring the shipboard training curriculum is up to date.

The organisational chart illustrated in the Company's Safety Management System (SMS) is a representation of . . .

Complete the statement by selecting the ONE, most appropriate option.

- **. . . a formal structure in the shipboard organisation**
- . . . ad-hoc committees
- . . . information structure in the shipboard organisation
- . . . roles and responsibilities of the ship's crew

The vessel is in port loading cargo. Company policy means that the chief mate cannot go ashore as he is responsible for loading the cargo. However, the master realises that the chief mate has had a busy few weeks. He therefore takes over cargo loading and sends the chief mate ashore. Which ONE of these is the most likely reason for the master's action?
Select the ONE, most appropriate option.

- **The master did this to build a better work relation with his chief mate.**
- The master did this as he was aware that the chief mate was about to sign off.
- The master did this because of respect and influence.
- The master did this on account of their cultural diversity to show respect for another culture.

Which ONE of the following is NOT a normal method of shipboard training?

- **Classroom lectures**
- Instructions and supervision from shipboard trainers
- Videos
- Workbooks

Which ONE of the following most appropriately defines an error chain?

- **It is a sequence of casual factors that leads to an accident, loss or damage.**
- It is a sequence of communication gaps that result out cultural differences.
- It is a sequence of events that results from one error.
- It is a sequence of many overpowering factors that leads to an accident, loss or damage.

Which ONE of these most appropriately defines fatigue?

- **Fatigue is a reduction in physical and/or mental capability as a result of exerting too much mental, emotional, or physical effort.**
- Fatigue is a mentally stressful situation arising from too much pressure from seniors.
- Fatigue is a state of boredom on account of reduced workload and tasks that do not carry any challenge.
- Fatigue is a state of mental illness where the person does not realize the importance of completing work on time.

Which THREE of the following factors are the most common reasons for fatigue?

- **Heavy weather**
- **Night work**
- **Sleep disturbance**
- Type of vessel

Which THREE of these are methods to assess shipboard training?

- **Direct observation of work activities**
- **Onboard assessment and examination**
- **Projects and assignments**
- Group discussions

Which TWO of the following are factors that are most likely to affect informal structures?

- **Culture**
- **Desire to achieve a common goal**
- Environment
- Rank

Which TWO of these are considered as management focus areas?

- **Directing**
- **Problem solving**
- Motivation
- Vision

'Active failures are . . . '

Complete the statement by selecting the ONE, most appropriate phrase.

- **. . . acts or conditions that result in accidents.'**
- . . . acts or conditions that people have no control over.'
- . . . acts or conditions that result in machinery failure.'
- . . . acts or conditions that result out of crew dispute.'

'After a near miss has been observed . . . '

Complete the statement by selecting TWO, most appropriate phrases.

- **. . . it should be reported, documented, and shared.**
- **. . . recommendations or preventive actions to avoid similar near miss events need to be presented.**
- . . . the master must call the port authorities for an inspection.
- . . . those involved must have a meeting.

'Automation complacency leads to . . . '

Complete the statement by selecting ONE, most appropriate phrase.

- **. . . boredom, as the crew member has not much to do.'**
- . . . efficiency, as the crew member has a lot of equipment to do his daily tasks.'
- . . . inefficiency, as the crew member may not be aware of how these sophisticated systems need to be used.'
- . . . interest in work, as the crew member has a lot of help.'

'Automation complacency occurs when . . . '

Complete the statement by selecting the TWO, most appropriate phrase.

- **. . . manual tasks compete with automated tasks.**
- **. . . there is multi-task load.**
- . . . manual tasks take preference over automated tasks.
- . . . the crew is not skilled.

'Designated leadership is . . . '

Complete the statement by selecting the ONE, most appropriate phrase.

- **. . . taking responsibility for the team and guiding the team towards achieving its goals**
- . . . making one person responsible for a team of people
- . . . one person being assigned a task
- . . . self initiative to complete a task

'Examples of latent conditions are . . . '

Complete the statement by selecting the ONE, most appropriate phrase.

- **. . . company management factors and social pressures that make up the shipboard culture and influence its procedures.**
- . . . conditions that impact the vessel due to shore activities.
- . . . daily tasks onboard a vessel that impacts the ship's environment.
- . . . work pressures that affect crew performance.

'In leadership with management . . . '

Complete the statement with the ONE, most appropriate option.

- **. . . there is a direction and the resources are managed to achieve that direction.**
- . . . there is direction and the team members are not expected to manage resources.
- . . . there is direction but the management has no idea of what to do.
- . . . there is no direction and the team members are expected to do their best.

'Mentoring is . . . '

Complete the statement by selecting ONE, most appropriate phrase.

- **. . . a long-term career relationship between the mentor and mentee.'**
- . . . a long-term personal relationship between the mentor and mentee.'
- . . . a short-term career relationship between the mentor and the mentee.'
- . . . a short-term personal relationship between the mentor and mentee.'

'SWOT analysis stands for . . . '

Complete the statement by selecting the ONE, most appropriate phrase.

- **Strengths, Weaknesses, Opportunities and Threats**
- Strengths, Workload, Options and Threats
- Stress, Weaknesses, Options and Threats
- Stress, Workload, Opportunities and Timeline

'The second quadrant of the Johari window represents . . . '

Complete the statement by selecting the ONE, most appropriate option.

- **. . . the blind area.'**
- . . . the hidden area.'
- . . . the open area.'
- . . . the unknown area.'

'When there is leadership without management . . . '

Complete the statement by selecting the ONE, most appropriate phrase.

- **. . . subordinates are left clueless about their roles**
- . . . productivity is monitored by no one
- . . . team members do what they think is right
- . . . the team does not have a sense of direction

0274 – Leadership, Task and workload management

Version 2.0 (2419)

A senior officer who provides too much input, direction and review of the delegated work to a junior officer or rating is doing . . . , which is the opposite of effective delegation.

- **micromanagement**
- assignment
- coordination
- leadership

A task is simply the described activity that should be carried out by an individual or a team, while a workload is the . . . that is being demanded from an individual for a particular period.

- **perceived amount of effort**
- amount of time
- number of pending tasks
- required resources

All of the following are external factors that can cause resource constraints EXCEPT . . .

- **company budget**
- commercial demand
- supplier delay
- trade route

All of the following are internal factors that can cause resource constraints EXCEPT . . .

- **trade route**
- company budget
- safe manning
- shipboard emergencies

All of the following are task and workload management tools EXCEPT . . .

- **replacement**
- coordination
- planning
- prioritization

As an officer, which of the following techniques should you NOT use to maintain an appropriate level of shipboard workload?

- **Elimination**
- Delegation
- Planning
- Prioritization

As tasks become . . . , the higher the workload becomes.

- **urgent**
- completed
- coordinated
- fewer

During cargo loading operation of a crude tanker, the Deck Watchkeeper continuously monitors the cargo level. He considers this task as . . . that he needs to prioritize to achieve the team's goal of safe operation and to avoid overflow. If indeed it overflows, it will trigger an urgent and important activity that is considered critical.

- **not urgent but important**
- not urgent and not important
- urgent and important
- urgent but not important

In which of the following situations will shipboard crew have a high possibility of experiencing time constraints?

- **Ship is engaged in short voyages**
- During long navigation
- Ship is at anchor waiting for berth
- Ship is drifting and awaiting voyage instructions

It is the perceived amount of effort that is being demanded from an individual for a particular period.

- **Workload**
- Maintenance
- Schedule
- Task

It refers to giving authority and responsibility to another person to carry out specific tasks. However, the person who gives the authority remains accountable for the outcome.

- **Delegation**
- Assignment
- Conflict handling
- Coordination

It refers to the activities, jobs or tasks that are part of the standard duties and responsibilities of shipboard personnel as stated in the Safety Management System (SMS).

- **Assignment**
- Conflict handling
- Coordination
- Delegation

It refers to the particular activity described and characterized to be done by an individual or team for a certain purpose. The issue on how much effort it requires from the person or team is not yet relevant here.

- **Task**
- Maintenance
- Schedule
- Workload

Planning outcomes are measured by . . .

- **comparing the actual result to the desired result**
- calculating profit margins
- conducting an employee satisfaction survey
- monitoring the usage of resources

Preparing the applicable work permit before starting the job ensures that all involved parties will become aware and no other activities that will use the same resources will be carried out. This is an example of . . .

- **coordination**
- consolidation
- merging
- prioritization

Seafarers with low workload have the tendency to become all of the following EXCEPT . . .

- **productive**
- bored
- complacent
- homesick

Shipboard workload can be assessed by considering the perceived difficulty level of the tasks, its urgency and importance and . . .

- **available time**
- results of previous inspections
- type of cargo
- vessel speed

Soliciting feedback is done during . . . to find the outcome and to determine what needs to be improved and what practices should be retained.

- **Debriefing**
- Briefing
- Performance appraisal
- Toolbox talk

The Chief Mate is assisting the port inspector with cargo documentation. He needs to read the draft for calculating the cargo quantity but it is more urgent and important that he provides the document to the inspector right away. Which of the following options should the Chief Mate do to address this time constraint?

- **Delegate the task to the Third Mate**
- Ask the Captain for advice
- Attend a training on time management
- Wait until the first task is completed

The Third Engineer was issued with five work orders by the Chief Engineer. All of these maintenance jobs are due within the week. Assuming that the Third Engineer has no other activities except for these five maintenance jobs, how should he/she prioritize the tasks?

- **Prioritize the work order with the most immediate due date.**
- Ask the Second Engineer how to prioritize these work orders.
- Prioritize the work order which can be completed in the shortest time.
- Prioritize the work order which is easiest to complete.

These activities in the Urgent/Important Matrix do not lead to the achievement of any goals. These are considered distractions and should be avoided as much as possible.

- **Not urgent and not important**
- Not urgent but important
- Urgent and important
- Urgent but not important

This tool is used for prioritizing tasks. It is represented as a matrix with four quadrants that vary in the level of urgency and importance of tasks.

- **Urgent/Important Matrix**
- Planned Maintenance System
- Task Selection Tool
- Work Plan Matrix

Time constraints are caused by factors that you may or may not have control on. Some examples of factors that you have control on are work habits, . . . and prioritization skills.

- **competence**
- commercial demands
- environmental conditions
- trade route

To minimize critical activities in quadrant 1 of the Urgent/Important Matrix, you should spend more time doing tasks that will achieve your personal and/or team goals in quadrant . . .

- **2**
- 3
- 4
- 5

Which of the following options correctly describes coordination?

- **It avoids overlapping of tasks.**
- It eliminates less prioritized tasks.
- It ensures completion of tasks as scheduled.
- It makes all shipboard crew motivated.

Which of the following options is a feedback mechanism to check if planning outcomes are achieved after a shipboard operation?

- **Debriefing**
- Briefing
- Safety meeting
- Toolbox talk

Which of the options below exemplify proper ways of addressing time and resource constraints? There are two correct answers.

- **Ensure that the planned maintenance system (PMS) is religiously followed.**
- **Request the necessary spares, provisions, or supplies according to inventory and projected future requirements**
- Put pressure on the subordinates to meet the deadlines.
- Send a complain to the company that the scheduled tasks were not accomplished due to limited time available.

While the ship is at port, an AB became sick and was immediately sent home for medication. The replacement was still being processed. While waiting for the replacement's arrival, the ship was short of one deck hand in cleaning the cargo hold. This lack of personnel is an example of . . .

- **resource constraint**
- accident
- safe manning
- time constraint

While urgent activities lead to the achievement of another person's goals, important activities have outcomes that lead to the achievement of . . .

- **personal and/or team goals**
- higher safety standards
- shipboard management's goals
- the Master's goals

You have no control on the following factors that can cause time constraints EXCEPT . . .

- **work habits**
- commercial demands
- environmental conditions
- safe manning standards

0275 – Leadership, Effective resource management

Version 1.0 (320)

A situation where additional crew cannot be assigned on board due to lack of available cabin falls under what resource constraint?

- **Physical**
- Dependencies on other tasks
- Inadequate resources
- Regulatory

A situation where additional crew cannot be assigned on board due to lack of available cabin falls under what resource constraint?

- **Physical**
- Dependencies on other tasks
- Inadequate resources
- Regulatory

A watchkeeping officer is expected to maintain high situation awareness while on watch. Which action of the watchkeeping officer can be considered as an indicator of declining situation awareness?

- **Finds it difficult to answer questions pertaining to the watch.**
- Always seeks factual information from the different navigational systems and through visual observations.
- Completely aware of the entire situation and can provide projections.
- Confidently provides accurate information pertaining to the watch when asked.

All maintenance jobs were suspended while the vessel was transiting Panama Canal. This scenario illustrates what method of resource levelling?

- **Delay**
- Reduce
- Replace
- Split

As a watchkeeper, how should you react upon hearing an alarm?

- **Acknowledge the alarm and investigate.**
- Acknowledge the alarm then reset.
- Put the alarm to silent mode then reset.
- Reset the alarm then wait if it goes on again.

As a watchkeeper, how should you react upon hearing an alarm?

- **Acknowledge the alarm and investigate.**
- Acknowledge the alarm then reset.
- Put the alarm to silent mode then reset.
- Reset the alarm then wait if it goes on again.

Assertiveness is shown using all of the following communication techniques EXCEPT . . .

- **shout to the person**
- avoid being apologetic
- clearly and honestly say what you need to say
- use supportive body gestures

High assertive leadership requires having the confidence to express feelings and opinions directly and honestly without violating other people's rights. This is most appropriate to use during . . .

- **emergencies**
- barbeque parties
- investigating accidents
- resolving conflicts

In coordination with Chief Engineer, you are assigned to monitor the fuel oil tank level during bunkering operation. The bunker tanks are fitted with automatic ullaging system. What appropriate action should you do in order eliminate the factor of complacency and over-reliance on the ullaging system to ensure safe bunkering operation?

- **Prior and during the bunkering operation, take manual sounding and compare the readings with the tank level gauge.**
- Always trust that the indicators will show accurate reading as what has been observed in the past.
- Take manual sounding only when you hear the high-level alarm.
- Take manual sounding when ordered by the Chief Engineer.

In coordination with the Chief Engineer, you are assigned to monitor the fuel oil tank level during bunkering operation. The bunker tanks are fitted with automatic ullaging system. What appropriate action should you do to eliminate the factor of complacency and over-reliance on the ullaging system to ensure a safe bunkering operation?

- **Prior and during the bunkering operation, take manual sounding and compare the readings with the tank level gauge.**
- Always trust that the indicators will show accurate reading as what has been observed in the past.
- Take manual sounding only when you hear the high-level alarm.
- Take manual sounding when ordered by the Chief Engineer.

In order to project potential events that may happen and maintain a high level of situational awareness, what question should you ask?

- **What if?**
- How often does it occur?
- When does it usually happen?
- Who are involved?

In this decision-making style, you tell the team about your intention and the problem, facilitate the discussion, involve everyone, and arrive at a consensus.

- **Collaborative**
- Group consultation
- Individualism
- Open autocratic

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Reduce?

- **The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.**
- During meal times, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- Securing of deck after transit may be arranged in such as way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Replace?

- **During mealtimes, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.**
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- Securing of deck after transit may be arranged in such as way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.
- The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Split?

- **Securing of deck after transit may be arranged in such a way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.**
- During mealtimes, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.

It is defined as having the confidence to act and speak out for oneself and express feelings and opinions directly and honestly without violating other people's rights.

- **Assertiveness**
- Safety consciousness
- Situational awareness
- Visibility

It refers to the person's ability to identify, process and comprehend the critical elements of information about what is happening around at that particular moment.

- **Situational awareness**
- Risk management
- Situation assessment
- Workload management

It refers to the person's ability to identify, process and comprehend the critical elements of information about what is happening around at that particular moment.

- **Situation awareness**
- Risk management
- Situation assessment
- Workload management

Low assertive leadership is suitable in situations where involvement and commitment of subordinates are necessary. This is most appropriate to use in which of the following situations?

- **toolbox talk**
- abandon ship
- fire
- manoeuvring operation

Lower assertive leadership is suitable in situations where involvement and commitment of subordinates are necessary. This is most appropriate to use in which of the following situations?

- **Toolbox talk**
- Abandon ship
- Fire
- Maneuvering operation

Messman suffered back pains upon lifting a sack of potatoes. The Master temporarily relieved him of his duties and ordered to take rest. The Master initially thought of delegating some of the Messman's duties to other ratings, but the latter have to perform also their respective tasks. This is an example of which resource constraint?

- **Inadequate resources**
- Dependencies on other tasks
- Physical
- Regulatory

Of the two ABs, Eric is better at working on heights, while Thomas has excellent skills in splicing. Two jobs should be completed within the day: some minor deck works and greasing of crane wires. If you are the Chief Mate, how will you allocate these works in order to finish both faster and safer?

- **Greasing of crane wires will be assigned to Eric, while minor deck works will be given to Tomas.**
- Assign the tasks to Bosun and other ratings instead.
- Eric and Thomas will do the jobs together, starting with the greasing of crane wires followed by minor deck works.
- Greasing of crane wires will be assigned to Thomas, while minor deck works will be given to the Eric.

Rescheduling the task on another date/time when the necessary resources are available depicts what method of resource levelling?

- **Delay**
- Reduce
- Replace
- Split

Second Mate was heavily involved in cargo operation, port inspection and unberthing. Vessel has just started sailing out, but he will not be able to take rest yet because of watchkeeping duties. What is the potential danger of having the Second Mate take the watch for another 4 hours?

- **His situational awareness will be reduced.**
- He will be motivated.
- His situational awareness will increase.
- Nothing will change.

Shipboard Resource Management Process is characterized by four main steps arranged in which order?

- **Definition – Allocation – Consolidation – Levelling**
- Definition – Allocation - Levelling – Consolidation
- Definition – Consolidation – Levelling – Allocation
- Definition – Levelling – Allocation – Consolidation

Shipboard resource management process is characterized by four main steps arranged in which order?

- **Definition – Allocation – Consolidation - Levelling**
- Definition – Allocation – Levelling - Consolidation
- Definition – Consolidation – Levelling - Allocation
- Definition – Levelling – Allocation - Consolidation

Standards on safe manning and rest hours are categorized under what type of resource constraints?

- **Regulatory constraints**
- Dependencies on other tasks
- Inadequate resources
- Physical constraints

The Chief Mate is assessing the resource requirements for chipping and painting the main deck. He asked the Bosun about the area size that could be covered by one tin of paint, without telling him of the intention. This decision-making style is . . .

- **Open autocratic**
- Group consultation
- Individual consultation
- Questioning technique

The Chief Mate is assessing the resource requirements for chipping and painting the main deck. Without telling the Bosun his intention, he asks him about the area size that can be covered by one tin of paint. This decision-making style is . . .

- **open autocratic**
- group consultation
- individual consultation
- questioning technique

The Messman suffered back pains upon lifting a sack of potatoes. The Master temporarily relieved him of his duties and ordered him to rest. The Master initially thought of delegating some of the Messman's duties to other ratings, but the latter have to perform their respective tasks too. This is an example of which resource constraint?

- **Inadequate resources**
- Dependencies on other tasks
- Physical
- Regulatory

The level of situation awareness of a watchkeeper can also be measured in terms of how he/she utilizes information from different equipment. Which of the following options shows lack of situation awareness?

- **Relies only on the information that the equipment provides in making decisions.**
- Believes that some alarms are merely preset by previous users and that these settings can be changed when desired.
- Constantly checks and compares the information obtained from the equipment to other available data.
- Utilizes his/her senses to verify and gather more information in additional to what the equipment provides.

The vessel is approaching the pilot station. Two crucial tasks should be carried out simultaneously - anchor preparation and rigging the pilot ladder. The Bosun, two OS, the AB and the Second Mate are the only available crew at that time. If you were the Master, to whom would you assign these two tasks?

- **The Bosun and the first OS will prepare the anchor while the AB, the second OS and the Second Mate will rig the pilot ladder.**
- The Bosun and the Oiler will rig the pilot ladder while the AB and the two OS will prepare the anchor.
- The Bosun and the Second Mate will prepare the anchor, while the two OS and the AB will rig the pilot ladder.
- The two OS will rig the pilot ladder while the Bosun and the AB will prepare the anchor.

The vessel will sail in a high piracy-risk area in the next few days. The shipboard management analyzed the situation and agreed that the deck crew will rig hoses around the ship, engine ratings and OS will post as bridge wing lookouts at night time, while the Chief Cook and Messman will also participate as additional lookouts. These are part of which step in the resource management process?

- **Allocation**
- Definition
- Levelling
- Work

These are company solutions based on past experiences, current trends and future projections that are geared for longer period of time.

- **Long-term solutions**
- Emergency solutions
- Medium-term solutions
- Short-term solutions

These are temporary operational arrangements or actions that address the immediate problem arising from resource constraints.

- **Short-term solutions**
- Emergency solutions
- Long-term solutions
- Medium-term solutions

These are temporary operational arrangements or actions that address the immediate problem arising from resource constraints.

- **Short-term solutions**
- Emergency solutions
- Long-term solutions
- Medium-term solutions

This is the human condition where a reduction in physical and/or mental capability becomes evident as a result of physical, mental or emotional exertion.

- **Fatigue**
- Assertive
- Awareness
- Confusion

This resource management principle aims to increase efficiency by resolving potential conflicts arising from over or under allocation of resources.

- **Levelling**
- Consolidation
- Prioritization
- Splitting

This type of resource is depleting and non-renewable.

- **time**
- machinery
- navigation equipment
- tools

This type of shipboard resource is depleting and nonrenewable.

- **time**
- machinery
- navigation equipment
- tools

To make a decision on how he will allocate the resources, the Captain shares his intention then asks the crew members for suggestions on the subject during a meeting. What decision-making style did the Captain use?

- **Group consultation**
- Collaborative
- Individual consultation
- Open autocratic

Vessel is approaching pilot station. Two crucial tasks should be carried out simultaneously: anchor preparation and rigging the pilot ladder. The Bosun, 2 OS, AB and Second Mate are the only available crew resource at that time. If you are the Master, to whom will you assign these two tasks?

- **Bosun and first OS will prepare the anchor while the AB, 2nd OS and Second Mate will rig the pilot ladder.**
- Bosun and Oiler will rig the pilot ladder while the AB and two OS will prepare the anchor.
- Bosun and Second Mate will prepare the anchor, while the two OS and AB will rig the pilot ladder.
- Two OS will rig the pilot ladder while Bosun and AB will prepare the anchor.

Vessel is scheduled to arrive in port on the next day. While performing the pre-arrival preparation and testing of winches, the ratings observed that one of the winches is dysfunctional. Which of the following options is a short-term solution?

- **Make an arrangement where another working winch can be used.**
- Ensure that a spare break lining is available at all times.
- Ensure that the operator is adequately trained and competent to do the job.
- Perform break test and other maintenance checkups according to PMS schedule.

Vessel is scheduled to arrive in port on the next day. While performing the pre-arrival preparation and testing of winches, the ratings observed that one of the winches is dysfunctional. Which of the following options is a short-term solution?

- **Make an arrangement where another working winch can be used.**
- Ensure that a spare break lining is available at all times.
- Ensure that the operator is adequately trained and competent to do the job.
- Perform break test and other maintenance checkups according to PMS schedule.

What principle should be borne in mind when keeping a watch using modern integrated equipment systems?

- **Garbage in-garbage out.**
- All information is accurate at all times.
- Equipment is functioning perfectly unless it sounds an alarm.
- Less talk, less mistake.

What should you do to build a more personal and authentic relationship with colleagues and subordinates?

- **Increase your assertiveness**
- Avoid being assertive
- Increase the power distance
- Lower your assertiveness

What should you do to set a good atmosphere for higher participation, involvement, and commitment of subordinates?

- **Lower your assertiveness**
- Avoid being assertive
- Increase the power distance
- Increase your assertiveness

When you are carefully observing a potential danger, your situational awareness is at . . .

- **Level 3**
- Level 1
- Level 2
- Level 4

When you are carefully observing potential danger, your situation awareness is at . . .

- **Level 3**
- Level 1
- Level 2
- Level 4

Which of the following actions and behavior indicates high situational awareness while watchkeeping?

- **Constantly monitoring the vessel or machinery performance both visually and through the control display panel**
- Believing that everything is in order unless an alarm is activated
- Having a relaxed and carefree attitude
- Relying only on the centralized display monitor of different equipment in the watch station

Which of the following actions and behaviour indicates high situational awareness while watchkeeping?

- **Constantly monitoring the vessel or machinery performance both visually and through the control display panel.**
- Believing that everything is in order unless an alarm is activated.
- Having a relaxed and easy-go-lucky attitude.
- Relying only on the centralized display monitor of different equipment in the watch station.

Which of the following is not an ideal action of an assertive watchkeeper?

- **Makes suggestions when permitted by senior officer.**
- Challenges the senior officers' action when he thinks something is not safe or wrong.
- Monitors orders given by person in command and verifies if they are executed properly.
- Provides corrective feedback to ratings when he/she observes one.

Which of the following options is NOT considered a shipboard resource?

- **pilot boat**
- chipping hammer
- fuel oil
- shipboard officers

Which of the following options is not considered a shipboard resource?

- **pilot boat**
- chipping hammer
- fuel oil
- shipboard officers

Which of these steps in the resource management process aims to increase efficiency by resolving potential conflicts arising from over or under allocation of resources?

- **Levelling**
- Allocation
- Consolidation
- Definition

With respect to the Cadet's Training Record Book, the Chief Mate decided to breakdown the tasks into smaller milestones distributed throughout the 12-month period. His involvement in coaching, monitoring and assessing the cadet's performance for each task will require lesser amount of time. This approach falls under which type of resource levelling?

- **split**
- distribute
- reduce
- replace

With respect to the engine cadet's Training Record Book, the Chief Engineer decided to breakdown the tasks into smaller milestones distributed throughout the twelve-month period. His involvement in coaching, monitoring and assessing the cadet's performance for each task will require lesser amount of time. This approach is under which type of resource levelling?

- **Split**
- Distribute
- Reduce
- Replace

You are a junior officer. You heard that an engine telegraph order "dead slow ahead" was requested. You observed that your experienced senior officer acknowledged the order but set the telegraph to "dead slow astern" instead. How will you react in this situation?

- **Tell the senior officer right away that it should be "dead slow ahead".**
- Ask someone to confirm your understanding.
- Never question the senior officer's actions.
- Wait and see until the officer realizes that he made a wrong action.

You are a junior officer. You heard that an engine telegraph order "dead slow ahead" was requested. You observed that your experienced senior officer acknowledged the order but set the telegraph to "dead slow astern" instead. How will you react in this situation?

- **Tell the senior officer right away that it should be dead slow ahead.**
- Ask someone to confirm your understanding.
- Never question the senior officer's actions.
- Wait and see until the officer realizes that he made a wrong action.

You just finished your risk assessment meeting with Bosun and the rest of the deck ratings for your ballast exchange plan. You coordinated with Chief Engineer and found out that Electro Technical Officer is supposed to repair two electrical junction boxes on deck near the ballast tank vents and both activities have overlapping schedule.
Which step in the resource management process is this action categorized?

- **Consolidation**
- Allocating
- Checking
- Levelling

You just finished your risk assessment meeting with the Bosun and the rest of the deck ratings for your ballast exchange plan. You coordinated with the Chief Engineer and found out that the Electro Technical Officer is supposed to repair two electrical junction boxes on deck near the ballast tank vents and both activities have overlapping schedule.
Which step in the resource management process is this action categorized?

- **Consolidation**
- Allocation
- Checking
- Levelling

0275.01 – Leadership, Effective resource management, Part 1

Version 2.0 (1459)

All maintenance jobs were suspended while the vessel was transiting Panama Canal. This scenario illustrates what method of resource levelling?

- **Delay**
- Reduce
- Replace
- Split

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Reduce?

- **The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.**
- During meal times, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- Securing of deck after transit may be arranged in such as way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Replace?

- **During mealtimes, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.**
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- Securing of deck after transit may be arranged in such as way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.
- The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.

In transiting Panama Canal, which of the following measures illustrates the resource levelling method, Split?

- **Securing of deck after transit may be arranged in such a way that the Second Mate and two ratings will be responsible for the aft station, while the Chief Mate and Bosun will be responsible for fwd station.**
- During mealtimes, it is possible to alternate the bridge duty of the Second Mate and the Third Mate, and the Chief Mate and the Bosun at the fwd station.
- Panama Canal transit is very crucial that other maintenance jobs should be suspended while the vessel is in transit. Also, it is necessary to delay ballast transfer to adjust the vessel's trim after crossing Panama canal.
- The vessel will be anchored upon reaching Gatun Lake and only one shipboard personnel will be retained in the fwd and aft stations. The rest of the crew can rest.

Rescheduling the task on another date/time when the necessary resources are available depicts what method of resource levelling?

- **Delay**
- Reduce
- Replace
- Split

Shipboard resource management process is characterized by four main steps arranged in which order?

- **Definition – Allocation – Consolidation - Levelling**
- Definition – Allocation – Levelling - Consolidation
- Definition – Consolidation – Levelling - Allocation
- Definition – Levelling – Allocation - Consolidation

The Chief Mate is assessing the resource requirements for chipping and painting the main deck. Without telling the Bosun his intention, he asks him about the area size that can be covered by one tin of paint. This decision-making style is . . .

- **open autocratic**
- group consultation
- individual consultation
- questioning technique

The vessel is approaching the pilot station. Two crucial tasks should be carried out simultaneously - anchor preparation and rigging the pilot ladder. The Bosun, two OS, the AB and the Second Mate are the only available crew at that time. If you were the Master, to whom would you assign these two tasks?

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The vessel will sail in a high piracy-risk area in the next few days. The shipboard management analyzed the situation and agreed that the deck crew will rig hoses around the ship, engine ratings and OS will post as bridge wing lookouts at night time, while the Chief Cook and Messman will also participate as additional lookouts. These are part of which step in the resource management process?

- **Allocation**
- Definition
- Levelling
- Work

This type of shipboard resource is depleting and nonrenewable.

- **time**
- machinery
- navigation equipment
- tools

To make a decision on how he will allocate the resources, the Captain shares his intention then asks the crew members for suggestions on the subject during a meeting. What decision-making style did the Captain use?

- **Group consultation**
- Collaborative
- Individual consultation
- Open autocratic

Which of the following options is NOT considered a shipboard resource?

- **pilot boat**
- chipping hammer
- fuel oil
- shipboard officers

Which of these steps in the resource management process aims to increase efficiency by resolving potential conflicts arising from over or under allocation of resources?

- **Levelling**
- Allocation
- Consolidation
- Definition

With respect to the engine cadet's Training Record Book, the Chief Engineer decided to breakdown the tasks into smaller milestones distributed throughout the twelve-month period. His involvement in coaching, monitoring and assessing the cadet's performance for each task will require lesser amount of time. This approach is under which type of resource levelling?

- **Split**
- Distribute
- Reduce
- Replace

You just finished your risk assessment meeting with the Bosun and the rest of the deck ratings for your ballast exchange plan. You coordinated with the Chief Engineer and found out that the Electro Technical Officer is supposed to repair two electrical junction boxes on deck near the ballast tank vents and both activities have overlapping schedule.
Which step in the resource management process is this action categorized?

- **Consolidation**
- Allocation
- Checking
- Levelling

0275.02 – Leadership, Effective resource management, Part 2

Version 2.0 (629)

A situation where additional crew cannot be assigned on board due to lack of available cabin falls under what resource constraint?

- **Physical**
- Dependencies on other tasks
- Inadequate resources
- Regulatory

As a watchkeeper, how should you react upon hearing an alarm?

- **Acknowledge the alarm and investigate.**
- Acknowledge the alarm then reset.
- Put the alarm to silent mode then reset.
- Reset the alarm then wait if it goes on again.

Assertiveness is shown using all of the following communication techniques EXCEPT . . .

- **shout to the person**
- avoid being apologetic
- clearly and honestly say what you need to say
- use supportive body gestures

In coordination with the Chief Engineer, you are assigned to monitor the fuel oil tank level during bunkering operation. The bunker tanks are fitted with automatic ullaging system. What appropriate action should you do to eliminate the factor of complacency and over-reliance on the ullaging system to ensure a safe bunkering operation?

- **Prior and during the bunkering operation, take manual sounding and compare the readings with the tank level gauge.**
- Always trust that the indicators will show accurate reading as what has been observed in the past.
- Take manual sounding only when you hear the high-level alarm.
- Take manual sounding when ordered by the Chief Engineer.

In order to project potential events that may happen and maintain a high level of situational awareness, what question should you ask?

- **What if?**
- How often does it occur?
- When does it usually happen?
- Who are involved?

It is defined as having the confidence to act and speak out for oneself and express feelings and opinions directly and honestly without violating other people's rights.

- **Assertiveness**
- Safety consciousness
- Situational awareness
- Visibility

It refers to the person's ability to identify, process and comprehend the critical elements of information about what is happening around at that particular moment.

- **Situational awareness**
- Risk management
- Situation assessment
- Workload management

Lower assertive leadership is suitable in situations where involvement and commitment of subordinates are necessary. This is most appropriate to use in which of the following situations?

- **Toolbox talk**
- Abandon ship
- Fire
- Maneuvering operation

Standards on safe manning and rest hours are categorized under what type of resource constraints?

- **Regulatory constraints**
- Dependencies on other tasks
- Inadequate resources
- Physical constraints

The Messman suffered back pains upon lifting a sack of potatoes. The Master temporarily relieved him of his duties and ordered him to rest. The Master initially thought of delegating some of the Messman's duties to other ratings, but the latter have to perform their respective tasks too. This is an example of which resource constraint?

- **Inadequate resources**
- Dependencies on other tasks
- Physical
- Regulatory

These are company solutions based on past experiences, current trends and future projections that are geared for longer period of time.

- **Long-term solutions**
- Emergency solutions
- Medium-term solutions
- Short-term solutions

These are temporary operational arrangements or actions that address the immediate problem arising from resource constraints.

- **Short-term solutions**
- Emergency solutions
- Long-term solutions
- Medium-term solutions

Vessel is scheduled to arrive in port on the next day. While performing the pre-arrival preparation and testing of winches, the ratings observed that one of the winches is dysfunctional. Which of the following options is a short-term solution?

- **Make an arrangement where another working winch can be used.**
- Ensure that a spare break lining is available at all times.
- Ensure that the operator is adequately trained and competent to do the job.
- Perform break test and other maintenance checkups according to PMS schedule.

What should you do to build a more personal and authentic relationship with colleagues and subordinates?

- **Increase your assertiveness**
- Avoid being assertive
- Increase the power distance
- Lower your assertiveness

What should you do to set a good atmosphere for higher participation, involvement, and commitment of subordinates?

- **Lower your assertiveness**
- Avoid being assertive
- Increase the power distance
- Increase your assertiveness

When you are carefully observing a potential danger, your situational awareness is at . . .

- **Level 3**
- Level 1
- Level 2
- Level 4

Which of the following actions and behavior indicates high situational awareness while watchkeeping?

- **Constantly monitoring the vessel or machinery performance both visually and through the control display panel**
- Believing that everything is in order unless an alarm is activated
- Having a relaxed and carefree attitude
- Relying only on the centralized display monitor of different equipment in the watch station

You are a junior officer. You heard that an engine telegraph order "dead slow ahead" was requested. You observed that your experienced senior officer acknowledged the order but set the telegraph to "dead slow astern" instead. How will you react in this situation?

- **Tell the senior officer right away that it should be "dead slow ahead".**
- Ask someone to confirm your understanding.
- Never question the senior officer's actions.
- Wait and see until the officer realizes that he made a wrong action.

0276 – Leadership, Decision making techniques

Version 2.0 (1052)

A decision made in simple situations and based solely on personal experience, common sense or intuition is called . . .

- **Intuitive**
- Analytic
- Complex
- Fast tracked

According to the STCW Manila Amendments, all of the following shipboard personnel are required to have the knowledge and ability to apply decision-making techniques EXCEPT . . .

- **ratings**
- deck officers
- Electro-Technical Officers
- engine officers

All of the following are conditions where the open and closed autocratic decision-making styles are applicable EXCEPT which option?

- **You are not fully competent on the subject**
- Team member are mandated to follow your decision
- You have expertise on the matter
- You have little time available

All of the following are decision-making tools you can use to define the problem and set the goal EXCEPT which option?

- **Advantages/Disadvantages Technique**
- Active listening
- Brainstorming
- Questioning Technique

All of the following are forms of decision-making EXCEPT which option?

- **Reading the SMS Manual**
- Going ahead or stopping
- Saying yes or no
- Turning left or right

In decision-making, the two options that are assessed to be the most workable are called . . .

- **alternatives**
- best option
- feasible options
- raw options

In the questioning techniques, what type of question should you use to encourage generous response?

- **Open**
- Closed
- Formal
- Probing

In this decision-making style, you have the absolute authority to make the decision based on what you currently know or aware of without consulting anybody.

- **Closed autocratic**
- Collaborative
- Defensive
- Open autocratic

It is defined as the process of selecting the course or courses of action based on a generated list of options conceived upon careful assessment of the situation.

- **Decision-making**
- Coaching
- Observation technique
- Resource management

SWOT Analysis is a decision-making tool that can be used for which step in the decision-making process?

- **Assessing the situation**
- Allocating resources
- Evaluating outcome effectiveness
- Generating options

Seafarers should possess good decision-making skills because . . .

- **wrong decisions may result into accidents, injuries, lost business, jeopardized career, etc.**
- crucial decisions are made based on intuition
- it is all right to commit errors at the beginning
- they are not responsible for the safety of the vessel, its cargo and crew

The Advantages/disadvantages and CS2 techniques are decision-making tools for . . .

- **Assessing the options**
- Assessing the situation
- Defining the problem and setting the goal
- Generating options

The CS2 technique is a tool for assessing options by comparing their relative strengths based on all of the following issues EXCEPT which option?

- **Compliance**
- Cost-effectiveness
- Safety
- Suitability

The Captain called for a meeting to discuss where to allocate the ship's recreational fund. He facilitated the meeting and allowed the officers and crew to give their suggestions. After the discussions, all voted and the Captain followed the decision of the majority. What decision-making style did the Captain use in this scenario?

- **Collaborative**
- Autocratic
- Consultative
- Dictatorship

The acronym SWOT stands for strengths, weaknesses, opportunities and . . .

- **threats**
- team
- time constraints
- training

The decision-making process starts with . . .

- **defining the problem and setting the goal**
- assessing the options
- assessing the situation
- selecting the course/s of action

The decision-making style of individual consultation is most suitable in which of the following situations?

- **Investigate an accident**
- Conduct a risk assessment
- Contain oil spill
- Handle an emergency situation such as pirate attack

This decision-making tool is used to make a collective assessment involving different individuals without fear of being criticized, opposed or taken for granted.

- **Brainstorming**
- Active listening
- Observation techniques
- Questioning techniques

This step in the decision-making process deals with creating the correct picture of the situation.

- **Step 2 - Assess the situation**
- Step 1 - Define the problem and setting the goal
- Step 3 - Generate the options
- Step 4 - Assess the options

This step in the decision-making process is carried out by comparing the results of the decision with the goal defined in the first step.

- **Step 6 - Evaluate outcome effectiveness**
- Step 3 - Generate the options
- Step 4 - Assess the options
- Step 5 - Select the course/s of action

This type of options remains after eliminating those that do not appear to be realistic, practical or appropriate to the situation.

- **Feasible options**
- Alternatives
- Decision
- Raw options

What decision-making style do you use when you disclose your intention and ask other individuals for specific information or opinion?

- **Individual consultation**
- Closed autocratic
- Collaborative
- Observation technique

When are consultative and collaborative decision-making styles applicable?

- **The team member's commitment to the decision is important.**
- The affected individuals are mandated to follow your decision.
- You are fully competent on the subject.
- You have little time available to make the decision.

Which of the following options is an appropriate non-verbal skill for active listening?

- **Avoid distractions**
- Continue what you are doing
- Cross your hands
- Show an intimidating stare

Without disclosing his intention, the Chief Mate asked the Chief Engineer about remaining onboard (ROB) quantities of fuel oil (FO) and diesel oil (DO) to compute the maximum loadable cargo. What decision-making style did the Chief Mate use?

- **Open autocratic**
- Collaborative
- Individual consultation
- Intuitive

0277 – Marine environmental awareness, Environmental challenges

Version 2.0 (5275)

According to the BWM Convention, Ballast Water Exchange will be phased out and replaced by Ballast Water Treatment; TWO reasons for this include:

- **Exchange can be dangerous for the ship's crew**
- **Treatment is more effective**
- Treatment is cheaper
- Treatment is less work for the ship's crew

According to the International Labour Organisation (ILO), working at a ship recycling facility where ´beaching´ takes place is one of the most dangerous jobs in the world. Which THREE statements below best explain why?

- **Because heavy machinery is often not available**
- **Because protective clothing is seldom provided**
- **Because workers lack proper training**
- Because at the same time ships are grounded, people are walking around on the beach

According to this module, which THREE negative effects for humans were caused by mercury poisoning in the Minamata Bay case study:

- **Brain damage**
- **Dementia**
- **Mentally disabled babies**
- Blindness

By 2020 the Global Sulphur Cap will be tightened to:

- **0.5%**
- 0.1%
- 1.0%
- 2.0%

List TWO pollutants from a ship that are NOT formed in the engine:

- **HCFCs**
- **TBT**
- PM
- SOx

ONE of the processes that remove chemicals permanently from the environment is called:

- **Biodegradation**
- Bioaccumulation
- Dissolution
- Evaporation

Of the four chemical categories X, Y, Z, and OS in MARPOL annex II which ONE category is the most dangerous?

- **X**
- OS
- Y
- Z

Select ONE correct option that correctly describes problems caused by carbon dioxide emissions.

- **Climate change**
- Acidification of freshwater lakes
- Health problems – smog
- Ozone depletion

Select THREE effects of ingestion of plastics.

- **Animals starve to death**
- **Animals stop eating because the stomach is full**
- **Sharp litter objects cause internal injuries**
- Plastics in their stomach cause them to be hungry all the time

Select THREE of the following statements that apply to bioaccumulation:

- **Chemicals are passed on in the food chain from prey to predator**
- **Some chemicals have bioaccumulation factors of one hundred thousand, meaning that the concentration in predators are one hundred thousand times higher than in the surrounding water**
- **The concentration of bio-accumulating substances increases higher up in the food chain**
- Because of bioaccumulation, low concentrations of chemicals are broken down quickly

Select THREE potential consequences of climate change for the marine environment:

- **Acidification of the oceans**
- **Coral bleaching**
- **Decline of phytoplankton communities**
- Change of plastic distribution patterns

Select TWO air emission problems that are caused by both sulphur oxides and nitrogen oxides.

- **Acidification**
- **Smog**
- Ozone depletion
- Particulate Matter

Select TWO options that complete the following statement: Chronic effects of HNS are often unknown . . .

- **. . . because many different processes occur over long time periods**
- **. . . because the effects over a long time are hard to measure**
- . . . because the environmental damage caused by chronic effects is always small
- . . . therefore chronic effects are not considered important

Select the ONE option that describes what special areas with stricter regulations under MARPOL annex VI are called:

- **Emission Control Areas**
- Emission Restriction Areas
- Marine Protected Areas
- Particularly Sensitive Sea Areas

Select the ONE option that most accurately completes the following statement: Climate change can cause coral bleaching. This means that . . .

- **Corals release the algae that live within the coral tissues**
- Acidification caused by climate change turns the corals white
- Corals need to be painted white to protect them from higher temperatures
- White corals grow better in higher temperature water

Select the ONE option that most accurately completes the following statement: Discharging sludge and waste oil is prohibited in all circumstances, but discharge of engine room bilge water . . .

- **Is allowed when the oil content is no higher than 15 ppm and the ship is proceeding en route**
- Is allowed everywhere in the marine environment
- Is allowed when a ship is proceeding on route further than 200 nm from the shore
- Is never allowed

Select the ONE option that most accurately completes the following statement: The theory of Ballast Water Exchange in mid-ocean is based on . . .

- **The difference in living circumstances between open ocean and coastal seas**
- The amount of clean water available in the open ocean
- The difference in depth between open ocean and coastal seas
- The difference is turbulence between open oceans and coastal seas

Select the TWO options that complete the following statement: Crude oil is a complicated mixture containing . . .

- **Hydrocarbons, aromatic hydrocarbons and several metals**
- **Hydrocarbons, sulphur and several metals**
- Hydrocarbons, plastics and chemicals
- Hydrocarbons, sewage and several metals

The ONE goal of the Ship Energy Efficiency Management Plan (SEEMP) is:

- **To reduce fuel consumption and decrease CO_2 emissions**
- To check the attitude of the ship's crew
- To label the energy performance of the ship
- To manage the sulphur emissions of the vessel

The ONE option that describes the toxicological principles of the categories in MARPOL annex II are:

- **Bioaccumulation, biodegradation and toxicity**
- Bioaccumulation, biodegradation and density
- Density, bioaccumulation and toxicity
- Toxicity, molecule size and density

The abbreviation HNS stands for:

- **Hazardous and Noxious Substances**
- Health, NOx and SOx
- Huge Numbers of Ships
- Hydrocarbons and Noxious Substances

The mission of the IMO includes promoting environmentally sound and sustainable shipping. Which ONE of the following conventions has had the largest contribution to a significant decrease in pollution from international shipping?

- **MARPOL**
- COLREG
- SOLAS
- STCW Code

What is the ONE most relevant difference between the open seas and coastal seas with respect to sewage regulations?

- **Open oceans are nutrient-poor, while coastal seas contain many nutrients**
- More whales live in the open oceans
- Open oceans are deeper than coastal seas
- The coastal seas are more turbulent than open oceans

When oil enters the sea, there are four main weathering processes that occur first, within a couple of days. These are best describes by which ONE option?

- **Spreading, evaporation, dispersion and emulsification**
- Spreading, evaporation, sedimentation and emulsification
- Spreading, oxidation, dispersion and dissolution
- Spreading, sedimentation, dissolution and emulsification

Which ONE group of chemicals are most likely to affect organisms living in water?

- **Dissolvers**
- Evaporators
- Floaters
- Sinkers

Which ONE invasive species is best described by the following statement: This species has nearly wiped out the anchovy population in the Black Sea and Caspian Sea, causing losses of around 500 million euro's per year and causing fishermen to stop fishing.

- **The comb jelly**
- Golden mussel
- Green sea fingers
- The blue mussel

Which ONE of the following combinations of emissions to air is especially due to the increased use of heavy fuel oil by ships?

- **PM and SOx**
- CO_2 and NOx
- NOx and SOx
- PM and CO_2

Which ONE of the following is NOT a potential consequence of climate change?

- **Depletion of the ozone layer**
- Extreme weather conditions
- Global temperature rises
- Sea level rise

Which ONE of the following methods contributes to the transport of large marine species to other sea areas by ships?

- **In ballast tanks - marine organisms are transported during their planktonic phase**
- By seafarers that keep aquaria on board
- In ballast tanks - they hide in the sediments
- In ballast tanks – often filters don't work

Which ONE of the following statements is NOT true?

- **Small particles (PM) are mainly a problem for humans because of their bad smell**
- Air pollution can be a threat to human health
- Air pollution types that have most impact on human health are smog and particulate matter (PM)
- Ozone is one of the main components of smog

Which ONE of the following statements is true:

- **In general, lighter oils are more likely to mix in water and are toxic to marine life**
- In general, lighter oils cause no harm to the environment
- In general, lighter oils do not mix in the water column
- In general, lighter oils remain on the water surface and may smother birds

Which ONE of the following statements regarding the Anti-Fouling Convention is true?

- **The Convention prohibits the use of harmful organic compounds containing tin in anti-fouling paints**
- The Convention prohibits the use of all toxic substances in anti-fouling paint
- The Convention prohibits the use of copper in anti-fouling paints
- The Convention prohibits the use of heavy metals in anti-fouling paints

Which ONE oil weathering process is best described by the following statement: 'The oil slick on the water surface takes up sea water'.

- **Emulsification**
- Dispersion
- Dissolution
- Spreading

Which ONE option describes the areas affected by harmful invasive species?

- **Fisheries, water supply systems and harbours**
- Fisheries
- Harbours
- Water supply systems

Which ONE option is treated as an exception to the discharge of marine litter in the marine environment since 2013?

- **Ground food waste**
- Metals
- Paper
- Very fine plastics

Which ONE part of the oil is most toxic to plankton?

- **Aromatic hydrocarbons**
- Atoms containing sulphur
- Cyclic alkanes
- Heavy hydrocarbons

Which ONE statement about the breaking down of plastics is true?

- **Plastics break down into ever smaller pieces**
- Plastics are broken down by fish
- Plastics are made of oil and some bacteria eat plastics
- Plastics dissolve in the water column when they break down

Which ONE statement about tributyltin, TBT, is FALSE?

- **TBT caused animals to lose their reproductive organs**
- TBT bioaccumulates and is highly toxic
- TBT is considered one of the most toxic substances ever deliberately released into the sea
- TBT was a very effective anti-fouling method

Which ONE statement applies to the pools of garbage?

- **Ocean currents drive plastic wastes together and the pools of garbage are like a plastic soup**
- Most waste in the pools of garbage floats on top of the water surface
- The pools of garbage are like solid islands one could walk on
- The pools of garbage are located in busy shipping lanes

Which THREE of the following conventions are designed to protect the marine environment?

- **MARPOL**
- **The Anti-Fouling Systems Convention**
- **The Ballast Water Management Convention**
- Maritime Labour Convention

Which THREE of the following statements apply to problems caused by microplastics?

- **Chemicals and contaminants can be transferred to tissues of marine animals via ingested microplastics**
- **Microplastics are so small, we hardly recognize them as litter**
- **Microplastics may transfer toxic chemicals to humans eating contaminated fish**
- Microplastics are so small, they do not cause any problems

Which THREE of the following statements are true?

- **Air emissions from ships contribute to both worldwide and local problems**
- **Seventy percent of all shipping traffic takes place close to shore, therefore air emissions from ships contribute to environmental issues on land**
- **SOx and NOx emitted at the middle of the ocean, far away from land, do not contribute to local problems on land such as acidification and human health issues**
- Only seventy percent of all shipping traffic takes place close to shore, therefore air emissions from ships do not contribute to environmental issues on land

Which THREE of the following statements are true?

- **CO_2 is the most important human-based greenhouse gas**
- **Scientist agree that since 1750, the overall effect of human activities on the climate has been a warming one**
- **Without the greenhouse effect, life on earth as we know it would not be possible**
- Humans do not play a role in climate change

Which THREE of the following statements are true?

- **In certain local port areas (like Long Beach, California), the percentage of SOx and NOx emitted by ships is much higher than the global average**
- **In certain local port areas, ships contribute more to SOx emissions than road transport**
- **The contribution of SOx is related to the type of fuel used on ships**
- The contribution of NOx is related to cargo residues

Which THREE of the following statements is true?

- **Carbon dioxide is formed when carbon in the fuel reacts with oxygen in the air**
- **Nitrogen oxides are formed when nitrogen and oxygen in the air react under high temperatures in the ships' engine**
- **Particulate matter are extremely small solid particles that are mainly carbon-based and are a by-product of the combustion process**
- Particulate matter does not contain sulphur

Which THREE processes determine the ecological effects of chemicals for the marine environment?

- **Bioaccumulation**
- **Biodegradation**
- **Toxicity**
- Biofluorescence

Which TWO of the following environmental problems are an outcome of emissions from a ship sailing in the middle of the open ocean?

- **Climate Change**
- **Ozone depletion**
- Corrosion of old buildings
- Health problems (smog)

Which TWO of the following statements are true?

- **In sea-going ships, species travel along in ballast water or attach themselves to ship hulls**
- **Shipping is considered to be the most important source of marine invasive species across the world**
- Ballast water is responsible for 90% of the introduction of marine invasive species worldwide
- Biofouling is responsible for 90% of the introduction of marine invasive species worldwide

Which TWO of the following statements are true?

- **Most oil pollution results from operational discharges**
- **Small discharges can have a big impact on the environment**
- Only large amounts of oil have negative effects on the environment
- The effects of operational discharges are well studied

Which TWO statements apply to the inventory of hazardous materials (IHM) on ships?

- **Ship owners will be responsible for maintaining and updating the IHM throughout the life of the ship**
- **This is a list of all materials present on board that may be hazardous to man and nature**
- An alternative to the IHM is to apply for a 'Green Passport' which is required for ships moving across borders for recycling
- Ship recycling facilities will be responsible for maintaining and updating the IHM throughout the life of the ship

Which of the TWO answers below describe ecological impacts of chemicals for marine life?

- **If a chemical negatively affects one species, this may also impact other species in the food web that are dependent on the affected species**
- **Substances such as caustic soda cause fish to die**
- Chemicals do not effect whales
- In the marine environment, chemical substances are always diluted by sea water which makes them harmless for marine life

Which of the TWO options correctly describes what happens when zooplankton is exposed to dissolved oil compounds in the laboratory?

- **The zooplankton dies**
- **The zooplankton stops moving**
- The zooplankton eats the oil
- The zooplankton form colonies

Which of the following TWO statements are true?

- **CO_2 emissions contribute to climate change, which is seen as one of the most serious problems facing the world today**
- **IMO estimate that, if no action is taken, by 2050 shipping CO_2 emissions will have tripled**
- CO_2 emissions of ships are relatively high compared to road transport
- CO_2 regulations for ships are stricter than for most other industries

0278 – Marine environmental awareness, Sustainable shipping

Version 2.0 (2776)

Describe THREE sustainable options for handling of solid waste:

- **Compacting**
- **Minimize plastic packaging**
- **Segregation**
- Discharge into the sea

It is possible to cut emissions by 'after treatment' of exhaust gases. NOx emissions can be reduced by using which ONE of the following:

- **Selective catalytic reduction (SCR)**
- Incinerator
- Sea water scrubber
- UV treatment system

Oceans provide food and regulate our climate. Select TWO statements that describe other important aspects of the oceans to humans.

- **The oceans are important as a 'blue highway' for world trade**
- **The oceans provide large quantities of oxygen**
- The oceans prevent earthquakes
- The oceans provide most of our drinking water

Regarding the profit P, choose the ONE option a sustainable company focuses on:

- **Short and long term profits**
- A positive reputation for the sector
- Care of the environment
- Short term profits

Select ONE method of waste management on board that is encouraged by MARPOL annex V:

- **Segregation**
- Compacting
- Discharge into the sea
- Incineration

Select ONE option that correctly describes how economic incentive schemes may lead to a better environmental performance of ships.

- **Making a better environmental performance financially attractive**
- Educating ships' crew
- Forcing ships to comply to regulations
- Increasing the fines for environmental crimes

Select ONE option that describes the main source of oxygen in the oceans.

- **Tiny algae**
- Green bacteria
- Green sea weeds
- Viruses

Select ONE option that describes the reasons why the diversity of deep sea life can be astounding.

- **Animals are fully adapted to the harsh living circumstances of the deep sea**
- Deep sea animals need to be able to make light to survive in the dark deep sea
- Deep sea life has evolved only recently
- Deep sea life is adapted to high levels of carbon dioxide

Select ONE option that is mandatory for all ships in MARPOL annex V:

- **Keeping a garbage record book**
- Carry a Garbage Management Plan
- Discharge of food waste into the sea
- Segregation of Garbage on board

Select ONE option why open oceans produce most phytoplankton (and most primary production).

- **Open oceans are big – more than 90% of the world ocean area**
- Food chains are long
- Food chains are very productive
- Lots of nutrients are available for the phytoplankton

Select THREE Particularly Sensitive Sea Areas, PSSA's from the list below.

- **Baltic Sea**
- **Canary Islands**
- **Great Barrier Reef**
- Antarctic

Select THREE appropriate actions where every individual onboard can contribute to minimizing the environmental impact of shipping.

- **Don't throw waste over the side, even if it is only a small amount and you are miles from land.**
- **Ensure proper maintenance of technical equipment.**
- **Speak up if you think environmental rules are not being followed.**
- Read books that make you aware of your rights and responsibilities onboard.

Select THREE appropriate actions where officers onboard can contribute to minimizing the environmental impact of shipping.

- **Be a good example for the crew.**
- **Give suggestions for improvement of procedures to shore staff and company management.**
- **Motivate the crew to follow procedures that protect the environment.**
- Always verify what your colleagues are doing.

Select THREE differences between open oceans and coastal seas.

- **Nutrient levels**
- **Salinity**
- **Vertical water movement or turbulence**
- Intensity of sunlight at sea surface

Select THREE different treatment methods used for killing organisms in ballast water

- **Chlorine**
- **Ozone**
- **UV light**
- Extreme temperatures

Select THREE economic benefits of sustainable shipping

- **Ensuring a good environmental performance can lead to a reduction of port dues (e.g. in the Green Award System)**
- **Ensuring a good environmental performance will attract some cargo owners**
- **Ensuring a good reputation will provide a license to operate and create new business opportunities**
- Using the cleaner fuel MDO instead of HFO is cheaper

Select THREE examples of possible new solutions for energy supply on ships:

- **Fuel cells**
- **Wave energy**
- **Wind energy**
- Water hydroid energy

Select THREE examples of voluntary best practices related to sustainable shipping

- **Cargo owner initiatives like the clean shipping index**
- **Maersk ship recycling who work with a safe and clean ship recycling facility**
- **The green award system that certifies ships that are extra clean and extra safe**
- The application of a ship energy efficiency management plan

Select THREE of the following technical installations and/or procedures that prevent discharges of oil to the sea.

- **Incinerator**
- **Informing the crew about effects of oil in the sea**
- **Keeping an oil record book**
- Installing a magic pipe

Select THREE of the given options that best describe the three aspects of sustainable development.

- **Economic prosperity**
- **Environmental quality**
- **Social equity**
- Scientific certainty

Select THREE options that apply to the Planet P, which is one of the three pillars of sustainability.

- **It refers to the natural resources of our Planet**
- **It refers to what is known as 'the natural capital'**
- **It stands for minimizing environmental impact of human activities**
- It stands for eliminating environmental impact of human activities

Select THREE options that best describe the advantages of using shore side electricity accurately.

- **Ships do not have to use auxiliary engines in port**
- **Shore side power reduces local air emissions in port areas**
- **Shore side power reduces the risk of health issues in port areas**
- Shore side power reduces the use of energy on the ship

Select THREE options that describe how proper personal behaviour leads to prevention of pollution:

- **Compliance with (international) regulations**
- **Correct use of technical innovations**
- **Follow company procedures designed to protect the environment**
- Try and save time and money disposing of waste whenever possible

Select THREE reasons why sustainable shipping is becoming increasingly important.

- **Every sector should play a role in acting more sustainably, so shipping should too**
- **Shipping activities have environmental impacts on a global scale**
- **Ships use fossil fuels as a power source, and fossil fuel resources are not endless**
- Shipping is not very important for world economy and has to compete with cleaner modes of transport like planes and trucks

Select THREE statements that best describe the living circumstances in the deep sea.

- **Cold and dark**
- **High pressures**
- **Low food availability**
- Photosynthesis only by bacteria

Select TWO Special Areas from the list below.

- **Arabian Seas**
- **Southern South African Waters**
- Malpelo island, Columbia
- Paracas National Reserve, Peru

Select TWO advantages that smaller open ocean phytoplankton cells have, compared to larger coastal plankton.

- **Because they are smaller, they are not heavy and this prevents sinking**
- **Small means optimal surface to volume ratio and therefore the best chance to take up nutrients**
- It is easier to hide for predators
- It is easier to stick together and form floating colonies

Select TWO consequences of the complexity of food webs.

- **If one species disappears, this may affect other species in the same food web**
- **It is extremely difficult to predict the consequences of changes in food webs**
- Species can disappear without having any effects on other species
- Species may eat the wrong prey

Select TWO examples of deep sea areas that are extremely productive for sea life.

- **Hydrothermal vents**
- **Underwater mountains or 'sea mounts'**
- Deep sea seagrass areas
- Mariana Trench

Select TWO options that apply to food chains in open oceans.

- **A typical food chain is phytoplankton – small zooplankton – copepods – bigger zooplankton – sardines – sharks**
- **Food chains generally have six levels**
- A typical food chain is phytoplankton – copepods – sardines – sharks
- Food chains generally have four levels

Select TWO options that best describe the balance between Planet, People and Profit (three P's).

- **All three P's should be considered**
- **Balancing the three P's can be done in different ways**
- All three P's should always get the same level of attention
- Environmental care (Planet P) is the most important P

Select TWO options that best describe the combination of methods to treat ballast water.

- **Combination of filtration and chemical methods**
- **Combination of filtration and mechanical methods**
- Combination of chemical and mechanical methods
- Combination of chemical treatment and after treatment

Select TWO options that best describe the consequences of the food chain 10% rule.

- **Animals higher in the food chain are lower in numbers**
- **Coastal seas with shorter food chains are more productive than open oceans**
- Fishermen are only allowed to catch 10% of the total fish present in the sea
- Open oceans with longer food chains are more productive than coastal seas

Select TWO options that describe IMO's focus on good reputation of the shipping industry.

- **Ensuring focus on quality agenda**
- **Raising the public image of shipping**
- Economic growth of the industry
- Fair and healthy competition

Select TWO statements that best describe how oceans affect the climate.

- **Oceans absorb large quantities of carbon dioxide from the air**
- **Oceans redistribute heat across the globe**
- Oceans release nutrients to the atmosphere
- The 'ocean conveyer belt' is driven by carbon dioxide concentration

Select the ONE correct option that defines why more fish biomass can be found in coastal areas than in open oceans.

- **Coastal areas have higher levels of primary production**
- Most coastal seas are protected
- Most fishing takes place in the open ocean
- The 10% rule does not apply to coastal areas

Select the ONE option that best describes why, in coastal waters, sinking of plankton to the sea floor (where it is darker) is not a big risk.

- **Coastal waters are very turbulent**
- They are big, but made of light materials
- They are eaten before they reach the sea floor
- They have special floating devices

Select the ONE option that most accurately completes the following statement: Reducing NOx emissions means . . .

- **Lowering the combustion temperature; this often leads to higher CO2 emissions**
- Lowering the combustion temperature; this often leads to lower CO2 emissions
- Using different fuel; this leads to higher NOx emissions
- Using different fuel; this leads to lower NOx emissions

Select the ONE option that most accurately completes the following statement: 'The differences in plankton size between open oceans and coastal seas have the following consequence for food chains . . .

- **Open ocean food chains are longer**
- Open ocean food chains are less complex
- Open ocean food chains are less efficient
- Open ocean food chains are shorter

Select the ONE option that states how the process where plants in the sea (mainly phytoplankton) turn carbon dioxide (CO2) and water in sugar and oxygen (O2) is called:

- **Photosynthesis**
- Bioluminescence
- Bioproduction
- Eutrophication

Select the ONE situation where use of shore side electricity is a particularly practical way to reduce local air emissions.

- **Ferries and RoRo carriers**
- Large container ships
- Small container ships
- VLCC's

State whether the following statement is true or false.
'The Black Sea is a PPSA'

- **False**
- True

The IMO has estimated that combined operational and technical measures have the potential to reduce ships CO2 emissions by as much as:

- **25-75%**
- 10-20%
- 15-50%
- 20-40%

The following food chain: bacteria – mussels – octopuses, is typical for:

- **Deep Sea vents**
- Kelp forests
- Rocky shores
- Tidal flats

The total area covered by sea is.

- **70% of the world surface**
- 50% of the world surface
- 60% of the world surface
- 80% of the world surface

Which ONE of the following methods can be used to reduce SOx emissions.

- **Sea water scrubber**
- Exhaust gas recirculation
- Selective catalytic reduction (SCR)
- Water in fuel technology

Which ONE option most accurately completes the following statement? 'Switching from HFO to distillate fuels such as MDO significantly reduces emissions of . . . '

- **SOx and PM**
- CO2 and PM
- CO2 and SOx
- SOx and NOx

Which ONE statement applies to life in the open ocean?

- **Even though biomass is lower than in coastal areas, some deep sea areas are filled with many life forms**
- Because biomass is higher than in coastal areas, deep sea life is more important
- Because biomass is higher than in coastal areas, deep sea life is very diverse
- Because biomass is lower than in coastal areas, deep sea life is less important

Which of the following TWO statements apply to the use of oily water separators?

- **After passing through this device, oily water with a maximum of 15 ppm oil may be discharged overboard**
- **They treat oily water from the bilge tank in such a way that oil is separated from water**
- After passing through this device, oily water with a maximum of 25 ppm oil may be discharged overboard
- Oil is separated from water by making use of gravity alone

Which of the following TWO statements apply to waste collection procedures in ports?

- **If waste collection in a port is lacking or insufficient, a ship's Master can complete a form and inform the IMO**
- **Ports play a critical role in the proper management of ship-generated waste**
- If waste collection in a port is lacking or insufficient, there is nothing that a ship's Master can do
- Waste that is collected in a port is never segregated into different waste streams

0280 – Engineering operations in cold environments

Version 1.0 (1822)

According to the IMO's Polar Classes, which ships can operate all year round in all ice covered waters?

- **Only those of Polar Class 1**
- Only those classed as Ice-breakers
- Only those of Polar Class 4 or better
- Only those of Polar Class 7

According to the IMO's Polar Classes, which ships can operate in Second-year Ice? Pick the best answer.

- **Only those of Polar Class 3 or better**
- Only those classed as Ice-breakers
- Only those of Polar Class 1
- Only those of Polar Class 7

According to this module, what is the most common way of protecting the firemain in extreme cold weather?

- **Drain the firemain and leave exposed hydrants just cracked open**
- Drain the firemain and then close all hydrants tightly
- Keep the firemain pressurised and allow water to flow through the anchor wash valves
- Keep the firemain pressurised with all valves closed

According to this module, what type of construction is becoming popular on modern Ice Class vessels? Pick the best answer

- **Simpler construction using heavier frames spaced further apart**
- Using light frames and longitudinal stiffeners
- Using longitudinal girders and transverse stiffeners
- Using more frames spaced closer together

According to this module, what type of major hull damage is most critical for vessels operating in the Arctic?

- **Frame buckling and tripping of frames against the side shell**
- Damage caused by excessive bending stress
- Indentation of bottom plating
- Indentation of side plating

According to this module, when the firemain it is drained for extreme cold weather, where should the position of all valves on the firemain be recorded? Pick the best answer.

- **On the bridge, in the engine room and at the emergency headquarters**
- In the engine room
- On the bridge and in the Chief Engineer's cabin
- On the emergency fire plan and on the bridge

First-year Ice can sometimes be recognised by its colour. Which one of these options is sometimes used to describe First-year Ice?

- **White ice**
- Black ice
- Blue ice
- Green ice

How effective is steam injection for clearing ice from sea suctions? Select the best answer.

- **Operational experience has shown that steam injection on its own cannot always clear ice**
- It has been shown to be effective at clearing ice in nearly all circumstances
- It has been shown to have no effect
- It is highly effective, but too costly to be used regularly

If a ship is not designed for operation in heavy ice, how should the propulsion machinery be designed? Pick the best answer.

- **To give the best efficiency in the areas it operates in for most of its service**
- To give the best efficiency in open water
- To give the best efficiency in thin ice
- To give the maximum power to weight ratio

If the vessel has to go astern in ice, what should be done with the rudder? Select the best answer.

- **It should be moved to mid-ships**
- It should be moved from side to side to keep it free of ice
- It should be put hard over to help turn the vessel and break the ice
- The steering gear motors should be stopped to avoid excessive force on the rudder

Should ships which are not Polar Class operate in polar waters? Pick the best answer.

- **Only if those waters can be confirmed as being ice free along the route**
- No, never
- Only if there is nothing worse than First-year Ice confirmed along the route
- Only in the summer months

Some propellers may be fitted inside a duct or nozzle. Select TWO of the following statements about this that you think are true.

- **This helps protect propellers when in ice**
- **This increases the available thrust**
- This can only be done with controllable pitch propellers
- This can only be done with fixed pitch propellers

What are the advantages of supplying a diesel engine with air through its own ducting on a vessel operating in very cold weather? Pick the best TWO answers.

- **It helps avoid making the machinery spaces too cold**
- **It makes it easier to pre-heat the air**
- It means a higher air pressure can be delivered to the engine
- The air supply can be regulated using a flap valve

What are the reasons for fitting an exhaust gas bypass for the main engine turbocharger on a vessel operating in very cold weather? Pick the best TWO answers.

- **It allows more heat to be recovered from the exhaust gas**
- **It helps prevent overpressure in the cylinders**
- It allows the engine to be operated without the turbocharger
- It protects the turbocharger from overpressure

What precaution should be taken for the deck seal of an inert gas system in extreme cold weather? Pick the best answer.

- **Make sure the deck seal heating coil is in use and working**
- Close the deck isolating valve
- Ensure a flow of warm inert gas through the seal at all times
- Isolate the deck seal

When operating in ice, what is the best draught to be at to avoid problems? Select the best answer.

- **The deepest draught practical**
- The draught that allows the propeller to break up ice on the surface
- The draught that allows the rudder to break up the ice sheet
- The shallowest draught practical

When operating in ice, which sea suctions should be used? Select the best answer.

- **Use low sea suctions**
- Keep all sea suctions open
- Sea suctions should be changed over each day
- Use high sea suctions

When sailing in ice, what is the best draught to use? Pick the best answer.

- **Operate at a deep draught, to help protect the rudder and propellers and keep sea suctions free of ice**
- Operate at shallow draught and trim by the stern to help the bulbous bow cut through the ice
- Operate at shallow draught but trim by the head to help the bulbous bow cut through the ice
- Operate at shallow draught to reduce the force required to move through the ice

Which Annexes of MARPOL designate North American sea areas as a Special Area? Select any you think apply.

- **Annex VI (Air Pollution)**
- Annex I (Oil)
- Annex IV (Sewage)
- Annex V (Garbage)

Which Annexes of MARPOL designate the Baltic as a Special Area? Select any you think apply.

- **Annex I (Oil)**
- **Annex IV (Sewage)**
- **Annex V (Garbage)**
- **Annex VI (Air Pollution)**

Which Annexes of MARPOL have special requirements for the Antarctic Area? Select any you think apply.

- **Annex I (Oil)**
- **Annex II (Noxious liquid substances)**
- **Annex V (Garbage)**
- Annex VI (Air Pollution)

Which THREE of these are advantages of a controllable pitch propeller for operations in ice?

- **Pitch can be reduced to avoid overloading the engine**
- **The direction of thrust can be changed more quickly and easily**
- **The propeller can continue to turn when the vessel is stopped**
- The efficiency at service speed is increased

Which TWO of the following are advantages of electrical propulsion for vessels operating in ice?

- **High torque at low speeds**
- **Improved redundancy**
- Lower fuel costs at maximum service speed
- Lower installation costs

Which TWO of these statements about brittle fracture are true?

- **Brittle fracture can be caused by shock loading, such as when a vessel is moving through ice**
- **Brittle fracture is more likely at low temperature and usually occurs suddenly and without warning**
- Brittle fracture occurs gradually
- Brittle fracture of the hull in ice usually occurs below the water-line

Which items of repair equipment do the IMO recommend that all icebreakers and ships of Polar Class 1 to 5 should carry? Select TWO answers

- **A portable submersible pump of 100 tonnes/hour capacity, with a set of hoses**
- **Portable gas welding and cutting equipment with a reserve of gas and consumables**
- A portable electric arc welding set and spare electrodes
- Spare I-section girders for temporary hull repairs

Which of these is a possible diesel engine problem caused by low air temperature? Pick the best TWO answers.

- **The fuel will not ignite in the cylinder**
- **The pressure in the cylinder may become excessive**
- More air needs to be supplied to the engine
- There will be excessive condensation at the charge air cooler

Which one of these has been found to be the most effective method of keeping sea suctions clear of ice?

- **Use of a sea bay and recirculation of the heated cooling water back to the suction**
- Compressed air and duplex filter arrangements
- Sailing at deep draught and keeping air or steam on the sea suctions
- Steam injection and duplex filter arrangements

Which one of these is an IMO recommendation for the lubrication of stern tube bearings used in polar regions?

- **That stern tube systems should not leak pollutants into the sea**
- That only fresh water lubricated stern tubes should be used
- That only oil lubricated stern tubes should be used
- That only sea water lubricated stern tubes should be used

Which one of these is the best way for most vessels to get through ice?

- **Keep moving and avoid heavy ice even if it means taking a longer route**
- Continually steer to port then starboard
- Move at high speed and use the momentum of the ship to break the ice
- Take the shortest possible route through the ice

Which part of a ship's hull experiences the highest loads when sailing through ice?

- **The bow**
- An area just forward of mid-ships
- The bottom plating
- The side shell plating

Why are frozen ballast and fuel tank vent pipes a danger? Select the best answer.

- **Because they can cause the tank to be over pressurised when filling or a vacuum to form when emptying and the tanks may rupture**
- It makes it harder to get an accurate sounding
- It makes it harder to pump fluid into the tanks
- It makes it harder to see when the tank is overflowing

Why do IMO guidelines say that ballast water to be discharged in the Antarctic should be exchanged outside of the region? Select the best answer.

- **Because of damage to the environment caused by invasive species and sediments.**
- Because of stability concerns
- So that the vessel remains low in the water at all times
- To avoid sea suctions becoming blocked

Why is it important to be able to tell Old Ice from Young Ice? Pick the best answer.

- **Both types can cause hull damage, but Old Ice is thicker and therefore the risk of hull damage increases**
- Young Ice is an early warning for Old Ice
- Young Ice is not thick enough to cause hull damage, whilst Old Ice can
- Young Ice is thicker and therefore the risk of hull damage increases

Why is it recommended that fresh water ballast is replaced with sea water before entering extremely cold weather? Select the best answer.

- **Because fresh water is more likely to freeze than sea water**
- Because it reduces corrosion in part filled tanks
- Because most cold weather regions have strict environmental controls
- Because sea water will give more weight for the same volume

Why should oils be stored at temperatures above their pour point? Select the best answer.

- **So that they can be pumped**
- So that they can be purified
- To prevent filters becoming blocked with wax
- To prevent vapours from being given off

0281 – Personal safety in cold environments

Version 1.0 (788)

After abandoning ship, you are being recovered from the sea into a rescue ship. Why is best to stay horizontal during your recovery? Select the ONE best answer.

- **To avoid your heart failing.**
- Being horizontal stops you getting too cold.
- It does not matter what position your body is in.
- It is easier for your rescuers.

Can any of the contents of personal survival kits be used if the crew are short of equipment? Select the ONE best answer.

- **No, they should only be used when abandoning ship.**
- Only the Master can authorise their use to make up shortages.
- They can be used, provided any items used are replaced at the next port.
- They can be used, provided items used are put back in the bag each day.

Can personal or group survival kits be used for training exercises? Select the ONE best answer.

- **No, they should never be used for training exercises.**
- Only the Master can authorise their use for training exercises.
- They can be used for training exercises, as long as any damaged items are replaced.
- They can be used for training exercises, under strict supervision.

What affect does cold weather have on fabrics used for personal protective equipment? Select the ONE correct answer.

- **Some fabrics may become less flexible.**
- Most fabrics become more flexible.
- Most fabrics bend more easily.
- There is no effect on most fabrics.

What affect does cold weather have on steel? Select the ONE correct answer.

- **It may shatter if hit suddenly.**
- It becomes harder and stronger.
- It is easier to weld and cut.
- There is no effect.

What are the early signs of 'hypothermia' (extreme cooling of the body)? Select the ONE best answer.

- **You shiver a lot.**
- You cannot breathe.
- You feel a little bit cold.
- You shiver a little bit.

What effect does "wind chill" have on you? Select the ONE best answer.

- **Your body loses heat more quickly and you get colder quicker.**
- It is uncomfortable, but there is no real effect.
- You feel colder, but there is no real effect on you.
- You feel warm, but it is colder than you realise.

What is "frostbite"? Select the ONE best answer.

- **Damage to the skin and muscles caused by extreme cold.**
- A slight loss of feeling, particularly in your fingers in cold weather.
- A sudden sensation of feeling cold when you go out into cold weather.
- It is when your hand sticks to cold surfaces.

What is the biggest cause of accidents onboard ships? Select the ONE best answer.

- **Slips, trips and falls.**
- Drinking alcohol.
- Heat exhaustion.
- Lack of sleep.

What should cold weather clothing be able to do? Select the ONE best answer.

- **It should keep out the wind and rain but allow sweat to evaporate.**
- It should allow heat to escape quickly.
- It should be loose fitting and allow air to circulate.
- It should not allow any moisture to pass through it.

When abandoning ship in cold water, what do the IMO say about drinking alcohol? Select the ONE best answer.

- **Do not drink alcohol, because it reduces your chances of survival.**
- A small amount of alcohol can be helpful to make you less afraid.
- A small amount of alcohol can be helpful to warm you up.
- Do not drink alcohol until you are in the liferaft.

When abandoning ship in cold water, which of these do the IMO recommend? Select the ONE best answer.

- **Put on as many layers of warm clothing as possible, including on your feet.**
- Do not put clothing on underneath an immersion suit.
- Do not switch on emergency beacons until you are in the water.
- Do not tighten your lifejacket straps until you are in the water.

When abandoning ship, when do the IMO say that you should take anti-seasickness tablets? Select the ONE correct answer.

- **Before leaving the ship, or immediately after boarding the survival craft.**
- Only after medical advice from the person in charge of the survival craft.
- Only after the first 24 hours.
- Only if you are feeling seasick.

Which ONE of these statements about the effects of sunlight is correct?

- **It can cause eye damage in cold regions.**
- It can only cause damage in tropical areas.
- It has less effect in cold regions.
- Snow and ice reduce the effects of sunlight.

Which of these is nearest to the normal human body temperature? Select the ONE correct answer.

- **36.3 °C (97.3°F)**
- 32.3 °C (90.14°F)
- 34.3 °C (93.7°F)
- 38.3 °C (100.9°F)

Which of these is the best way to stay warm in a liferaft? Select the ONE best answer.

- **Stay as dry as possible.**
- Being in a liferaft will keep you warm enough.
- Drink a small amount of alcohol.
- Try and lie in any pools of water inside the liferaft.

Which of these parts of the body is most at risk from frostbite? Select the ONE best answer.

- **The ears.**
- The arms.
- The legs.
- They are all equally at risk.

Which of these statements about "wind chill" are correct?
Select the THREE correct answers.

- **The wind causes moisture in the skin to evaporate and you lose heat faster.**
- **The wind removes the insulating layer of air around your body and you lose heat faster.**
- **You will suffer from frostbite much quicker than in still air.**
- It makes you feel colder, but there is no real effect.

Which parts of the body are MOST at risk from frostbite? Select the ONE best answer.

- **Fingers and toes**
- The arms
- The chest
- The legs

Why are clothes such as denim jeans a bad choice for working in cold weather? Select the ONE best answer.

- **They absorb water, which evaporates and takes heat away from the body.**
- They are too tight to work in and prevent free movement.
- They make you sweat more because the air can't circulate.
- They wear out too quickly and shrink when wet.

Why do you need to eat more high energy foods in cold weather? Select the ONE best answer.

- **Your body works faster to create heat, for instance by shivering.**
- Being overweight helps to fight the cold.
- You do not. Your clothes will keep you warm.
- Your body burns less energy wearing heavy clothes.

Why you should get into survival craft without jumping into cold water. Select the TWO correct options.

- **There is less chance of drowning.**
- **There will be less chance of hypothermia.**
- You will not need to wear an immersion suit.
- You will not need to wear or carry a lifejacket.

You have abandoned ship and are in cold water. What are the two best ways to keep warm? Select the TWO best answers.

- **Keep as still as possible if you are wearing a lifejacket.**
- **Keep your arms folded over your body.**
- Raise your arms and shout for help.
- Start swimming in circles to get warmer.

Your ship is in freezing weather. You try to use the fire main for washing down and there is no water. What should you do? Select the ONE correct answer.

- **Report this to the bridge and engine room.**
- Assume the fire main has been drained and do nothing.
- Open a few more valves and leave them open overnight.
- Try hitting the pipes to clear any blockages.

0285 – Enclosed space entry, Awareness

Version 3.0 (1093)

According to IMO guidelines, how much oxygen, by volume, is required for entry into an enclosed space?

- **21%.**
- 15%.
- 16%.
- 18%.

According to IMO guidelines, if the atmosphere in an enclosed space is suspect, then . . .

- **Entry should only be made for essential operation, further testing, safety of life or safety of the ship.**
- Entry can only be made provided there is no flammable gas
- Entry can only be made provided there is no toxic gas.
- No entry is allowed under any circumstances.

According to the IMO, which one of the following options best completes the following statement "an enclosed space is one which might have any of the following characteristics . . .

- **. . . limited openings for entry and exit, inadequate ventilation or is not designed for continuous worker occupancy"**
- . . . has only one opening and is bounded by steel bulkheads and deckheads".
- . . . is kept locked, with a warning sign posted".
- . . . is not open to the atmosphere or is used as a store ".

How can the presence of toxic gasses in an enclosed space be detected?

- **Selective tests have to be carried out with knowledge of which toxic products have previously been in the space.**
- Test for flammable gasses. If there are no flammable gasses and no bad smells, the space should be safe.
- Test for oxygen. Provided there is enough oxygen, and no bad smells, the space should be safe.
- Use a personal monitor, calibrated for carbon monoxide or hydrogen sulphide.

If the ventilation fails whilst you are working in an enclosed space, what should you do?

- **Leave the space immediately and ensure everyone else does the same**
- Carry on working as long as your gas alarm does not sound
- Ensure someone is regularly checking the atmosphere
- Report the ventilation failure, but carry on working if the atmosphere is still safe

If there is any doubt about pockets of flammable gas remaining in an enclosed space, what particular precaution should be taken?

- **Portable lights and other electrical equipment should be of a type approved for use in an explosive atmosphere.**
- No more than one person should enter the space at any time.
- Only 24 Volt hand lamps should be used.
- Personal gas monitors must be used by all members of the team.

If there is any doubt about pockets of gas or toxic cargo remains in an enclosed space, what precaution should be taken?

- **Breathing apparatus must be worn at all times.**
- An EEBD should be used.
- Personal gas monitors must be used.
- The time each person can spend in the space should be limited.

If there is any doubt about toxic vapours remaining in an enclosed space, what particular additional precaution should be taken?

- **Breathing apparatus and suitable protective clothing should be worn.**
- Personal gas monitors must be used by all members of the team.
- The time each person can spend in the space should be limited.
- Ventilation should be stopped to avoid spreading vapours.

If you are given a respirator to wear in an enclosed space, but don't like wearing it whilst working, what can you do?

- **If it is part of the agreed safety procedures you must continue to wear it.**
- If your personal monitor says that the atmosphere is okay then you can take it off.
- Take it off now and again for short periods.
- Take it off, and if there are no bad smells then it will be okay.

If you are working as part of a team in an enclosed space and one of your colleagues starts acting as though they were drunk what should happen?

- **All the team should leave the space immediately and seek medical attention.**
- The person who appears drunk should be given plenty of water.
- The person who appears drunk should be reported.
- The person who appears drunk should be told to leave the space.

If you are working in an enclosed space and the attendant instructs you to leave, what should you do?

- **Leave the space immediately, and ensure the rest of the team does the same.**
- Ask for an SCBA to be brought in so that you can continue working.
- Ask why work is being stopped and contact the officer of the watch.
- Check your personal gas monitor.

Once an emergency team has been prepared for an enclosed space rescue, which one of these options is their first priority in most situations?

- **To supply the casualty with clean air and then get them out of the space and into fresh air.**
- To find out the cause of the incident.
- To see if they can get the casualty sufficiently recovered to avoid having to lift them out.
- To treat all of the casualty's injuries at the scene of the incident.

Stagnant water, some sewage systems and some oil cargoes can all be the source for a gas which is both flammable and toxic. This gas is:

- **Hydrogen sulphide.**
- Carbon dioxide.
- Hydrogen chloride.
- Nitrogen.

Under what circumstances should someone acting as the attendant at the entrance to an enclosed space go into that space?

- **Only when relieved of that duty and as part of an organised rescue team.**
- If called for by the persons in that space.
- If the persons in the space get into any difficulty.
- If they can do so and remain in sight of the entrance.

Vehicle and diesel engine exhausts can be the source for toxic gas. This gas is:

- **Carbon monoxide.**
- Hydrogen chloride.
- Hydrogen sulphide.
- Nitrogen.

What is the main role of the attendant outside an enclosed space?

- **To remain at the entrance to the space, keep in contact with those in the space, and raise the alarm if there are any signs of danger.**
- To be available to immediately go in and get anyone out if they become ill.
- To close the entrance once the team have entered the space and open it up again once they are ready to leave.
- To pass tools or cleaning materials down into the space and help remove any cargo residues.

What is the maximum level of flammable gas allowed for safe entry into an enclosed space?

- **1% of the lower flammable limit.**
- 1% by volume.
- 10% of the lower flammable limit.
- 50% of the lower flammable limit

What percentage of the Occupational Exposure Limit (OEL) do the IMO say is safe before seafarers enter an enclosed space?

- **No more than 50%.**
- No more than 0.1%.
- No more than 100%.
- No more than 20%.

What should you do if you start feeling tired, or ill, feel your eyes watering, or start developing a headache in an enclosed space?

- **Leave the space immediately, report the matter and seek medical attention**
- Leave the space, have a rest and some fresh air and then return
- Mention it to a colleague, just in case.
- Try working on and see if you get better.

When rusting of ship's steelwork occurs, which one of the following is the main effect on the air inside a ballast tank or cargo tank?

- **The oxygen content of the air decreases.**
- Carbon dioxide is given off by the rusting process.
- The air develops a bad smell.
- The oxygen content of the air increases.

When should an Emergency Escape Breathing Device (EEBD) be used in an enclosed space?

- **Only as a means of escape when an unexpected hazard occurs.**
- It can be worn by a rescuer in an emergency.
- It can be worn for short periods of work when there is a lack of oxygen.
- It can be worn for short periods of work when toxic gasses are present

Which one of the following options best completes the following statement?
"Onboard ship, an enclosed space is any space which . . .

- **. . . has limited ventilation or could contain toxic or flammable vapours"**
- . . . has a danger sign at the entrance".
- . . . has four walls and a ceiling".
- . . . is not open to the atmosphere".

Which one of the following options correctly completes the following statement?
"Before entry into any space which has limited ventilation or might contain toxic or flammable gasses . . .

- **. . . shipboard procedures, including risk assessment by a competent person, should be followed".**
- . . . the door should be left open for at least an hour".
- . . . the ventilation fan should be run for at least two hours".
- . . . you should let someone know where you are going".

Which one of the following statements best describes how enclosed space rescues should be organised?

- **All rescues should be by a team following pre-arranged procedures.**
- As speed is vital, the person nearest the space should attempt a quick rescue where possible.
- If the casualties can be seen from the entrance to the space then the attendant should rescue them.
- Senior officers should act on their own initiative.

Which one of the following statements best describes the term 'Occupational Exposure Limit' (OEL)?

- **It is a guide to safe exposure to toxic products, based on the amount of a given substance and the time in contact it.**
- It is a guide to safe exposure to harmful ultra violet (UV) rays when working on deck.
- It is a guide to safe exposure to toxic products, based only on the amount of a given substance present.
- It is a guide to safe exposure to toxic products, based only on the time in contact with the substance.

Which one of the following statements best describes the use of a safety harness when going down into an enclosed space?

- **They should be worn whenever possible to save time in the event of an emergency.**
- They need not be worn, provided one is available onboard ship
- They need not be worn, provided they are available in the emergency headquarters.
- They should not be worn, as they are only for rescue and may get caught on obstructions.

Which one of the following statements best describes why a ship's chain locker should be classed as an enclosed space?

- **Because there is limited ventilation and rusting may use up the oxygen in the space.**
- Because access may be restricted.
- Because it may be below the waterline.
- Because it would be dangerous to be in there during anchoring operations.

Which one of the following statements best describes why a ship's sewage tank should be classed as an enclosed space?

- **Because it is likely to contain toxic or flammable gasses.**
- Because it is a covered space.
- Because it is inside the engine room.
- Because it might be quite small and difficult to move around in.

Which one of these statements best describes the effects some cargoes can have on the atmosphere in a cargo space?

- **They can use up or displace oxygen and give off toxic or flammable gasses.**
- No effect, because any harmful cargoes are sealed.
- The atmosphere can develop a bad smell.
- Very little effect, since the spaces are ventilated.

Who can take part in enclosed space entries, rescues or act as an attendant at the entrance to an enclosed space?

- **Only those who have had specific enclosed space training and have taken part in enclosed space rescue training drills.**
- Anyone holding a certificate of competency.
- Anyone who has been trained how to wear SCBA.
- Anyone who has had an onboard safety briefing for the work.

0286 – Crane operations, Heavy lift

Version 1.0 (1387)

How is the risk associated with adverse weather conditions controlled during a heavy lifting operation? Pick the one option you think is best.

- **Risk is controlled by obtaining forecasts and observing the weather.**
- Risk is controlled by completing an operation quickly.
- Risk is controlled by only using the crane during the hours of daylight.
- Risk is controlled by setting a 5 knot wind limit.

Select the one option which best completes the following statement. 'Briefing the team before a heavy lifting operation . . .

- **. . . makes for a cohesive team'.**
- . . . is always undertaken by the vessel's master'.
- . . . is often a waste of time, as they forget everything'.
- . . . takes too much time'.

Select the one option which best completes the following statement. 'Cargo lashings are essential to . . .

- **. . . resist inertial and gravity loads on stowed cargo'.**
- . . . assist with load stability when the heavy lift is airborne'.
- . . . keep various components of the cargo together whilst in transit'.
- . . . resist gravity loading only on a heavy lift cargo'.

Select the one option which best completes the following statement. 'Risk assessment for a heavy lifting operation . . .

- **. . . covers the vessel, cargo and personnel involved'.**
- . . . covers risks to the cargo only'.
- . . . covers risks to the personnel involved only'.
- . . . covers risks to the vessel only'.

Select the one option which best completes the following statement. 'To prepare the crane for a heavy lifting operation first . . .

- **. . . ensure there is a permit to work'.**
- . . . make sure there is power to the crane'.
- . . . remove all sea fastenings'.
- . . . test the slewing action'.

Select the one option which best completes the following statement. 'When proposing to work a heavy lift on a vessel in extremely low air temperatures . . .

- **. . . consideration has to be given to the possibility of brittle fracture'.**
- . . . consideration has to be given to slings reducing in length due to the cold'.
- . . . it is necessary to start up the crane much earlier'.
- . . . the operation should be stopped because of the dangers of slipping on ice'.

Select the one option which best completes the following statement. 'When taking a heavy lift a vessel has to be securely moored . . .

- **. . . to stop the suspended load moving relative to the quay'.**
- . . . to make it easier to take the lift'.
- . . . to reduce wear in the crane slewing bearing '.
- . . . to stop fender damage'.

Select the one option which best completes the following statement. 'When taking a heavy lift on a vessel, company procedures . . .

- **. . . should be followed at all times'.**
- . . . are a not always relevant as they cannot be referenced on the deck of a vessel'.
- . . . are unnecessary as personnel all know what to do'.
- . . . do not cover the lifting operation'.

Select the one option which best describes how health and safety issues on a heavy lift vessel are addressed DURING a heavy lifting operation.

- **Keep non essential personnel clear of the lifting operation.**
- Complete a full and detailed risk assessment.
- Inform all personnel that the lift is taking place.
- Leave the operation to dock workers and cargo officers.

Select the one option which best describes how the risk of crane malfunction is controlled on a heavy lift vessel.

- **Risk is controlled by correctly maintaining the crane.**
- It is controlled by having the technical superintendent organise crane maintenance schedules.
- It is controlled by using the crane well within its SWL rating.
- Risk is controlled by only using the crane in daylight hours.

Select the one option which best describes the meaning of the 'management of change' principle.

- **Changes to a plan are effectively managed on a ship.**
- As changes are made, shore managers will supervise them.
- People's attitudes at sea to safety have to change.
- Personnel at sea will be changed by management.

Select the one option which best describes the use of a 'toolbox' talk before or during an operation on a heavy lift vessel.

- **The team involved are briefed with the latest information.**
- It is a discussion concerning tools to be used.
- It is a social get together in the deck workshop.
- It is the management telling the team the lifting plan.

Select the one option which best describes those personnel required to maintain a specialised heavy lift crane.

- **Personnel who have undertaken crane maintenance training.**
- Any of the deck crew on a heavy lift vessel.
- Crane manufacturers' maintenance personnel only.
- Deck crew on a heavy lift vessel who have shadowed a senior person undertaking maintenance.

Select the one option which best describes when a 'stop the job' call can be made on board a heavy lift vessel.

- **A team member notices a safety issue.**
- It is lunch time on board the vessel.
- The operation is running beyond its allotted time.
- When an alarm bell is heard from a quayside warehouse.

Select the one option which best describes why a contingency plan is necessary on board a heavy lift vessel when working a heavy lift.

- **The plan copes with unwanted events.**
- Because the vessel's SMS requires it to be there.
- In case the original plan is flawed in some way.
- In case the warranty surveyor asks to see it.

Select the one option which best describes why wind effect is important to a heavy lift vessel when taking a heavy lift.

- **Because wind force on the load acts at the crane head.**
- Because the load will swing around a lot.
- Because wind noise makes communications more difficult.
- Because wind will cause a vessel to move away from the quay side.

Select the one option which best identifies items on a vessel requiring appraisal when planning for a heavy lift operation.

- **Items include crane capacity, stowage position and lifting equipment.**
- Items include crane capacity, main engine power and lifting equipment.
- Items include crane capacity, stowage position and communications equipment.
- Items include crane capacity, stowage position and type of lashings held on board.

Select the one option which best identifies why a heavy lifting operation on a vessel should be individually risk assessed.

- **Because a heavy lift is a unique event, unlike everyday crane operations.**
- Because generic crane risk assessments are poorly completed.
- Because the heavy lift or vessel might be damaged.
- Because the warranty surveyor demands this action.

Select the one option which best identifies why having adequate reserve stability is important on a heavy lift cargo vessel.

- **To reduce heel angles with a load on the hook.**
- To decrease the vessel's roll period at sea.
- To increase the vessel's roll period at sea.
- To minimise pitching affects with a vessel at sea.

Select the one option which best identifies why other non essential work should be stopped when a heavy lift cargo is being worked.

- **So that all personnel give priority to safety.**
- Because the whole vessel's crew are needed on deck.
- In case it is dropped which causes a vessel to incline to a large angle.
- To allow ship's personnel to watch the operation.

Select the one option which correctly completes the following statement. 'Any non tested loose gear on a heavy lift cargo vessel should . . .

- **. . . be isolated and not used for lifting operations'.**
- . . . be kept with the other loose cargo gear'.
- . . . be sent ashore for testing and certification'.
- . . . be sent ashore permanently'.

Select the one option which correctly completes the following statement. 'Heavy lift crane maintenance should . . .

- **. . . follow guidelines recommended by the manufacturer'.**
- . . . be completed every twelve months'.
- . . . be undertaken only by shore personnel following company guidelines'.
- . . . be undertaken only when the crane stops working'.

Select the one option which correctly completes the following statement. 'It is important to know the location of the centre of gravity of the load because . . .

- **. . . the crane head has to be positioned above it'.**
- . . . it will affect the stowage position on a vessel'.
- . . . the type of cargo bearers used are influenced by it'.
- . . . this is where the lashing should be attached'.

Select the one option which correctly completes the following statement. 'Loose gear on a heavy lift vessel is an expression given to . . .

- **. . . equipment connecting the load to the crane hook'.**
- . . . cargo bearers surplus to requirements'.
- . . . spare luffing and lifting wires'.
- . . . wires and blocks hanging from a crane structure'.

Select the one option which correctly completes the following statement. 'Loose gear on a heavy lift vessel should be tested . . .

- **. . . as thoroughly as the crane itself'.**
- . . . every month when the crane is tested'.
- . . . only by the supplier when new'.
- . . . to destruction to check the breaking load'.

Select the one option which correctly describes a particular hazard associated with a tandem crane operation.

- **The crane heads can clash.**
- The load may become distorted.
- There could be a shortage of qualified crane drivers.
- There may not be enough generating capacity on a vessel.

Select the one option which correctly describes how the risks to a heavy lift crane are controlled when it is operating offshore.

- **Risk is controlled by using offshore crane curves.**
- It is controlled by only using the crane when the wave height is less than 1 metre.
- It is controlled offshore by limiting the use of the crane.
- Risk is controlled by using the crane at some distance from an oil installation.

Select the one option which correctly describes the contents of a briefing on a heavy lift crane vessel before taking a heavy lift.

- **The lifting plan, duties of personnel and the anticipated time.**
- The lifting plan, any changes to the plan and the cargo shipper's details.
- The lifting plan, GM of the vessel and weather forecast.
- The lifting plan, weather forecast and vessel's draught.

Select the one option which correctly describes the main tests a new built heavy lift crane will be put through before use.

- **Factory acceptance then ship installation tests.**
- Factory acceptance then docks regulation tests.
- Rate of slewing and luffing.
- Twelve monthly tests for overload and limit switches.

Select the one option which correctly identifies a typical proof load applied to a heavy lift crane when it is load tested.

- **1.1 times the crane's SWL.**
- 2 times the crane's SWL.
- 6 times the crane's SWL.
- The crane's SWL.

Select the one option which correctly identifies how the chain register requires lifting equipment to be marked.

- **Lifting equipment is to have the SWL, an identifying number and testing authority.**
- Lifting equipment is to have the SWL and identifying number.
- Lifting equipment is to have the SWL, maker and testing authority.
- Lifting equipment is to have the SWL, vessel's name and testing authority.

Select the one option which correctly identifies the devices on a heavy lift crane that are required by a classification society to be tested daily when the crane is in use.

- **Audible alarms and holding capacity.**
- Control system failure.
- Fire and gas alarm response.
- Vessel blackout.

Select the one option which correctly identifies the devices on a heavy lift crane that are typically required to be tested every 12 months by a classification society.

- **Emergency stops and load limit devices.**
- Boom up protection system.
- Shut down for detected gas.
- Shut down for fire.

Select the one option which correctly identifies the function of a crane signaler.

- **A signaller directs the crane using a radio or hand signals.**
- A signaller is always positioned at the hatch coaming to observe the operation.
- A signaller is positioned close to the stowage position to give directions
- A signaller only directs the crane when the crane driver cannot see the load.

Select the one option which correctly identifies the markings seen on a crane, as required by a classification society.

- **The SWL and any limiting outreach values.**
- The item number and the testing authority.
- The name of the classification society that issues the crane certificate.
- The name of the crane manufacturer.

Select the one option which correctly identifies the use of a chain register on a heavy lift cargo vessel.

- **The chain register is a list of cranes and lifting equipment.**
- The chain register is a certificate certifying the strength of the anchor chain.
- The chain register is a list of chains used for lashing cargo.
- The chain register is only a list of lifting slings used on a vessel.

Select the one option which correctly identifies which persons are qualified to drive a heavy lift crane on board a vessel.

- **Persons that have been trained and properly assessed as competent.**
- Any of the crewmen with a bit of training are able to drive the crane.
- Crane drivers employed by stevedores.
- Only the chief officer is allowed to drive the crane.

Select the one option which correctly identifies why all personnel involved in a heavy lift operation should be aware of the content of the lifting plan.

- **So that a cohesive team is formed to support each other.**
- So that team members can relieve each other to do other work.
- So the lift supervisor has less to do when the lift is taking place.
- To cover the insurance requirements for the lift

Select the one option which correctly identifies why some space is needed around a heavy load when it is stowed on a vessel.

- **To be able to work the lift safely.**
- So that it does not touch the side of the vessel.
- So there is a fire break in case of fire in the hold.
- To allow other cargo to be packed around it.

Select the one option which correctly identifies why there is a limit on how short the legs in a multi leg lifting sling can be.

- **To minimise the horizontal force component in each leg when on a wide load.**
- Because the longer the sling the weaker it will be.
- So that the crane head does not have to be raised so far.
- To minimise the vertical force component on each leg when on a wide load.

Select the one option which is correct for the following statement. 'The supervisor of a lifting operation should not appoint themselves as the crane signaler'.

- **The statement is correct, as the supervisor needs to keep an overview of an operation.**
- The statement is correct, as the supervisor will be on the vessel's bridge not close to the crane.
- The statement is incorrect, as the supervisor is most experienced and should direct the crane.
- The statement is incorrect, as the supervisor knows best what to do.

Select two of the given options to best identify why effective communications are important during a team activity on a vessel.

- **Effective communications keep the team together.**
- **Effective communications produce better management.**
- Effective communications have to be practised because of the ISM code.
- Effective communications satisfy the warranty surveyor.

Which one of the following options correctly explains why a vessel has a very small roll period when rolling to large angles?

- **The vessel's GM is very large.**
- The vessel has waves on the beam.
- The vessel's GM is very small.
- The wind force is very strong.

Which one of the following options is correct for this statement? 'Unforeseen events are taken care of in a contingency plan'.

- **This statement is incorrect as contingency planning covers unwanted events.**
- The statement is correct as contingency planning looks at all possibilities.
- The statement is correct as this is what contingency planning is designed for.
- The statement is incorrect as contingency planning covers expected events.

Which one of the given options best completes the following statement? 'If a heavy lift breaks its lashings in bad weather . . .

- **. . . it is an extremely dangerous operation trying to re-fasten them'.**
- . . . just leave it till the weather dies down'.
- . . . the company will not be very happy about it'.
- . . . the vessel should return to port to get them re-attached'.

Which one of the given options best completes the following statement? 'Lashings applied to a heavy load should . . .

- **. . . follow the arrangements seen in the heavy lift plan'.**
- . . . be always more than necessary for safety'.
- . . . be those decided by the warranty surveyor'.
- . . . be what seems necessary to secure the cargo'.

Which one of the given options best completes the following statement? 'The best scenario with lashings on a heavy lift is that . . .

- **. . . they are always tight'.**
- . . . they make work for the deck crew'.
- . . . they need tightening every day'.
- . . . they totally fail in a rough sea'.

Which one of the given options best completes the following statement? 'The worst scenario with lashings on a heavy lift is that . . .

- **. . . they totally fail in a rough sea'.**
- . . . they are always tight'.
- . . . they have to be applied at all'.
- . . . they need tightening every day'

Which one of the given options correctly describes why slack tanks should be avoided on a vessel when taking a heavy lift?

- **To stop the vessel's effective GM being reduced.**
- Because a liquid moving across a tank can damage it.
- Because a liquid moving in a tank increases a vessel's GM.
- To stop water or oil overflowing through air vents.

Which person ashore first signs off the heavy lift plan when it is completed? Pick the one option you think is best.

- **The competent person ashore.**
- The P&I Club representative.
- The vessel master.
- The warranty surveyor.

0287 – Inspection of mooring lines

Version 2.0 (1031)

According to ISO guidelines, what type of material is indicated by a blue yarn in the strand of a synthetic fibre mooring rope? Select the ONE correct answer.

- **Polyester.**
- Polyamide (Nylon).
- Polyethylene.
- Polypropylene.

According to ISO guidelines, what type of material is indicated by a green yarn in the strand of a synthetic fibre mooring rope? Select the ONE correct answer.

- **Polyamide (Nylon).**
- Polyester.
- Polyethylene.
- Polypropylene.

According to ISO guidelines, what type of material is indicated by a red yarn in the strand of a synthetic fibre mooring rope? Select the ONE correct answer.

- **Polypropylene.**
- Polyamide (Nylon).
- Polyester.
- Polyethylene.

According to OCIMF guidelines, how often should a synthetic fibre rope tail be replaced? Select the ONE correct answer.

- **At least every 18 months.**
- At least every 24 months.
- At least every 36 months.
- At least every 48 months.

According to this module, at what percentage of the original minimum breaking load should a mooring rope be replaced? Select the ONE correct answer.

- **75%.**
- 33%.
- 50%.
- 60%.

How should a synthetic fibre rope tail be joined to a mooring wire? Select the ONE best answer.

- **With an approved shackle.**
- With a series of Bulldog clips.
- With an eye splice in both ropes.
- With an eye to eye, cow hitch.

Where would you find the Safe Working Load of mooring fittings on-board a ship? Select the ONE correct answer.

- **You will find the Safe Working Load of mooring fittings in the ship's towing and mooring plan.**
- You will find the Safe Working Load of mooring fittings in the ship's capacity plan.
- You will find the Safe Working Load of mooring fittings in the ship's fire and safety plan.
- You will find the Safe Working Load of mooring fittings in the ship's general arrangement plan.

Which ONE of the options best completes the following statement? 'A synthetic fibre tail attached to a mooring wire . . .

- **. . . makes the wire easy to handle and helps absorb sudden loads'.**
- . . . can be far too elastic, allowing the vessel to surge on the berth'.
- . . . introduces a weak link, because the fibre is not as strong as wire'.
- . . . is bad practice and not used on large vessels'.

Which ONE of the options best completes the following statement? 'A wire mooring rope that has been heaved around a tight corner under load . . .

- **. . . will get a permanent kink in it'.**
- . . . will be safe to use in the future'.
- . . . will not endanger the safety of the crew.
- . . . will not suffer any damage'.

Which ONE of the options best completes the following statement? 'Slipping a synthetic fibre mooring rope on a drum end . . .

- **. . . should be avoided, because it damages the rope'.**
- . . . should be done by experienced seamen only'.
- . . . should be done if excessive strain comes on the rope'.
- . . . should be done with only three turns on the drum end'.

Which ONE of the options best completes the following statement? 'Synthetic fibre mooring ropes . . .

- **. . . should be stowed in a safe area when not in use'.**
- . . . should be hung in the boiler room to dry when not in use'.
- . . . should be stored in direct contact with the deck'.
- . . . should never be washed with fresh water'.

Which ONE of the options best completes the following statement? 'When spooling a new mooring wire on a fixed drum . . .

- **. . . suspend the coil on a turntable or roller to stop kinks'.**
- . . . clamp one end, then drop the coil overboard to unravel it'.
- . . . place the coil on deck, just in front of the drum'.
- . . . start with the end of the wire found inside the coil'.

Which ONE of these options best completes the following statement? 'Inspection records for mooring ropes should be maintained on-board because . . .

- **. . . everyone on-board, and any inspectors, can see that the ropes have been inspected and are fit for purpose.**
- . . . the flag State require this to be done on all vessels'.
- . . . the ISM code requires it to be followed on all vessels'.
- . . . there is an OCIMF regulation stating this practice is to be followed on board all tankers'.

Which ONE of these options correctly completes the following statement? 'Six strand, 'wire laid' synthetic fibre mooring ropes . . .

- **. . . are particularly suited to use on self-tensioning winches'.**
- . . . are never used for mooring ropes'.
- . . . are not suited to use on fixed mooring drums and winches'.
- . . . stretch excessively when under tension'.

Which of the options correctly describe the easiest way to identify a mooring rope? Select the ONE correct answer.

- **From the tag, if fitted.**
- By its colour.
- By its position on the vessel.
- From the stored certificate.

Which of the options correctly describes 'actinic action' seen on a mooring rope? Select the ONE correct answer.

- **It is when UV radiation changes the chemical nature of the rope fibre.**
- It is a technical description for a mooring rope stretching.
- It is the action of the rope when it parts with excessive strain.
- It occurs when acid has attacked the fibres of a mooring rope.

Which of the options correctly describes the typical construction of a wire mooring rope? Select the ONE best answer.

- **6 x 36 or 6 x 41.**
- 6 x 24 or 6 x 36.
- Only ever 6 x 36 and a wire core.
- Only ever 6 x 41 and a fibre core.

Which one of these options best describes the desirable characteristics of a wire mooring rope? Select the ONE best answer.

- **The wire should be strong, easy to handle and with low stretch.**
- The wire should be strong, but stretch easily.
- The wire should be strong, with absolutely no stretch at all.
- The wire should have a small diameter to make for easy handling.

Which one of these options best describes why mooring ropes have chafing mats and sleeves fitted to them? Select the ONE best answer.

- **To protect the rope from being chafed.**
- It is mandatory to do so.
- To increase the strength of the rope.
- To protect the fairlead through which the ropes pass.

Which one of these options correctly describes the characteristics of polyester rope? Select the ONE correct answer.

- **Expensive to buy, heavy, good abrasion resistance.**
- Cheap to buy, heavy, good resistance to stretch.
- Light to handle, water resistant, stretches easily.
- Resistant to ultra violet, always three stranded, good abrasion resistance.

Which one of these options correctly describes why mooring ropes are part of a SIRE inspection? Select the ONE best answer.

- **To ensure that a vessel can be safely moored.**
- To check for poor quality ropes that need replacing.
- To comply with the ISM requirements.
- To save the vessel crew from having to inspect them.

Why are 'snap back' zones painted in the mooring areas of a ship? Select the ONE correct answer.

- **To warn crew not to stand in that zone when a rope passing through that lead is under strain.**
- Crew can seek protection in these areas if the wind picks up.
- It serves no purpose.
- To warn of a tripping hazard.

Why is it NOT recommended to 'short splice' a synthetic fibre mooring rope if it has parted? Select the ONE correct answer.

- **Because the short splice weakens the overall strength of the rope.**
- Because it creates an unwanted strong point.
- Because the short splice looks out of place on a modern ship.
- Because the short splice makes it difficult to work the rope on a drum end.

Why is it important to spool a wire mooring rope onto a traction drum evenly? Select the ONE best answer.

- **To stop the wire burying and being damaged when under load.**
- Because the friction on the traction drum will be greater.
- Because the wire looks nice when spooled on neatly.
- Because the wire takes up less space.

0288 – Green Passport (Inventory of Hazardous Materials)

Version 1.0 (1407)

A new piece of equipment is to be supplied to your ship.
Which one of the following will be the best source of information to help you decide whether the ship's Inventory of Hazardous Materials will need to be revised?

- **Ask for a Materials Declaration and a Supplier's Declarations of Conformity for the equipment**
- Ask your superintendent if the equipment contains hazardous materials
- Email the supplier and ask if the equipment contains hazardous materials
- Hazardous materials are no longer allowed to be supplied to ships, so no changes are needed

A ship has some material listed in Appendix II of the 'Hong Kong' Convention onboard.
Which one of the following is the most likely reason that the material might NOT be recorded in the Inventory of Hazardous Materials (IHM)?

- **The material is below IMO specified concentrations**
- Appendix II materials are banned and must not be recorded on the IHM
- Appendix II materials are recorded in Part 3 of the IHM, but only when the ship is sold
- It must be a mistake, all Appendix II materials should be recorded in the IHM

According to the 'Hong Kong' Convention, which materials must be listed in the Inventory of Hazardous Materials of an EXISTING ship which is in service? Select the most accurate answer.

- **Any listed in Appendix I of the Convention, and any from Appendix II where practical**
- Any materials considered hazardous by the flag State
- Only those listed in Appendix I of the Convention
- Only those listed in Appendix II of the Convention

According to the 'Hong Kong' Convention, which parts of an Inventory of Hazardous Materials should have been completed for an EXISTING ship which is in service? Select the most accurate answer.

- **Part 1**
- Part 2
- Parts 1 and 2
- Parts 2 and 3

Approximately how much of a ship is recycled by a typical ship breaking yard?

- **99%**
- 38%
- 50%
- 90%

Even before the 'Hong Kong' Convention becomes international law, your ship could be detained or fined if it does not have an Inventory of Hazardous Materials.

Which one of the following reasons most accurately explains why the above statement is true?

- **Your ship trades to a State that has signed the Convention, making it a requirement there**
- Your ship carries cargoes of hazardous material
- Your ship has hazardous material onboard
- Your ship is registered in Hong Kong

It has been decided that a ship is to be scrapped.
Which one of the following most accurately identifies what must be recorded in the Inventory of Hazardous Materials for operationally generated wastes?

- **Their location, type and quantities**
- The fact that there are some onboard
- Their location and quantities
- Their type and quantities

It has been decided that a ship is to be scrapped.
Which one of the following most accurately identifies what should be done before completing the 'operationally generated wastes' part of an Inventory of Hazardous Materials (IHM)?

- **It must be decided which wastes will be disposed of before the ship is sold for recycling**
- All the operationally generated wastes onboard must be recorded in Part 1 of the IHM
- All the operationally generated wastes onboard must be recorded in Part 3 of the IHM
- All the operationally generated wastes onboard must be sent ashore

Part 3 of the Inventory of Hazardous Materials is to be completed. Which one of the following should be recorded in the section for 'Regular consumable goods potentially containing Hazardous Materials'?

- **Computers in the ship's office**
- Paint in the deck store
- Pencils in the stationary locker
- Tinned food in the dry store

Select the option which correctly identifies the categories to be used when assessing information on the location of hazardous materials on an EXISTING ship which is in service?

- **Contained, Not contained, or Unknown**
- Contained or Not contained
- Contains hazardous or Unknown
- Highly hazardous, Moderately hazardous, or Not hazardous

Under the Hong Kong Convention, which ships need to have an International Certificate on Inventory of Hazardous Materials? Select the most accurate answer.

- **All ships over 500 GT that leave their flag State's waters**
- All ships over 400 GT that leave their flag State's waters
- All ships over 500 GT
- Any ship that requires a Safety Construction Certificate

Under the 'Hong Kong' Convention how will the accuracy of a ships Inventory of Hazardous Materials be checked? Select the most accurate answer.

- **A certifying authority will conduct a check of it before issuing a statutory certificate**
- Accuracy is guaranteed by the ship's ISM procedures
- The ship's flag State will endorse it
- The State where the ship was built will endorse it

What is the purpose of a 'Supplier's Declaration of Conformity' when completing an Inventory of Hazardous Materials? Select the best answer.

- **To provide assurance that the related Material Declaration conforms to 'Hong Kong' Convention requirements**
- It confirms that the supplier's products conform to US rule 33 CFR Part 151.05
- It confirms that the supplier's products contain no hazardous materials
- To provide assurance that the product contains no materials banned under the 'Hong Kong' Convention

When must an Inventory of Hazardous Materials on an EXISTING ship list Appendix II materials? Select the best answer.

- **When new equipment containing Appendix II materials is supplied**
- Before the first renewal survey for the International Certificate on Inventory of Hazardous Materials
- It must list all Appendix II materials known or suspected to be onboard
- It never needs to list Appendix II materials

When would 'Indicative' testing be justified while making an Inventory of Hazardous Materials on an EXISTING ship? Select the best answer.

- **To confirm that a hazardous material is present, as expected**
- To indicate that further testing of the material IS NOT required
- To indicate that further testing of the material IS required
- To indicate that the material IS hazardous

Which one of the following best describes how an Inventory of Hazardous Materials is most likely to change procedures at dry docking?

- **The dockyard must be advised that Materials Declarations and Supplier's Declarations of Conformity will be required for any new equipment they provide**
- An International Certificate on Inventory of Hazardous Materials survey will be required at the end of docking
- No antifouling paint can be applied
- No antifouling paint can be removed

Which one of the following correctly identifies the documentation that must be completed before the ship is sold to a ship breaker, under the 'Hong Kong' Convention?

- **An International Ready for Recycling Certificate must be issued**
- All the ship's statutory certificates must be cancelled by the certifying authority
- An export license must be obtained
- The International Certificate on Inventory of Hazardous Materials must be endorsed by the certifying authority

Which one of the following correctly identifies what a ship breaker must do with a ship's Inventory of Hazardous Materials, under the 'Hong Kong' Convention?

- **Make a ship recycling plan, for its Government to approve**
- Calculate how much the hazardous materials can be sold for
- Decide how much it will cost to make oil cargo tanks safe for hot work
- Nothing

Which one of the following correctly identifies what must be done to an oil tanker before it can be sold for scrap, under the 'Hong Kong' Convention?

- **The cargo tanks and pump room must be made 'safe for entry'**
- All the cargo residues must be disposed of ashore
- Fuel tanks, cargo tanks and pump room must be made 'safe for entry'
- Fuel tanks, cargo tanks and pump room must be made 'safe for hot work'

Which one of the following identifies the best way in which the presence of hazardous materials can be confirmed on an EXISTING ship which is in service without inspection or sampling?

- **Material Declarations may be available for recently supplied equipment**
- Suppliers Declarations of Conformity may be available for recently supplied equipment
- The chief engineer might know
- The stores department can confirm this

Which one of the following identifies the main use that an Inventory of Hazardous Materials is to the ship's crew?

- **It can be used to check for hazardous materials before starting maintenance**
- It can be used to calculate the ship's resale value
- It can be used to claim higher pay
- No particular use, it is just extra work

Which one of the following identifies when the listing of equipment as 'Potentially containing hazardous material' in the Inventory of Hazardous Materials can best be justified on an EXISTING ship?

- **The costs of sampling and testing the material far exceed the cost of disposal**
- The chief engineer decided it was not possible to shut the equipment down for testing
- The Supplier's Declaration of Conformity is not available
- You have not had time to check before the survey

Which one of the following identifies why it could be worth paying a little more for a ship which is documented as having been built with little or no hazardous materials?

- **Hazardous materials are increasingly expensive to dispose of**
- Hazardous materials are typically inefficient when used as coolants
- Hazardous materials cannot be recycled
- Many of the best insulation materials are not hazardous

Which one of the following is responsible for updating an Inventory of Hazardous Materials (IHM) after a ship's dry docking?

- **The ship's crew**
- The class society
- The dockyard
- The local Health and Safety Authorities

Which one of the following is the correct name for the document commonly known as a ship's 'Green Passport'?

- **Inventory of Hazardous Materials**
- Hazardous Materials Passport
- Inventory of Recyclable Materials
- Statement of Green Credentials

Which one of the following is the correct name of the certificate that a ship gets after passing a survey for compliance with the 'Hong Kong' Convention?

- **International Certificate on Inventory of Hazardous Materials**
- Certificate of Fitness for Scrapping
- International Certificate of Safe Materials
- International Ship Safety Certificate

Which one of the following is the main aim of the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes?

- **To reduce the dangers to workers and to reduce pollution in poorer countries**
- To allow the movement of hazardous wastes without customs duties
- To extend the IMDG Code and control the packaging used for the transport of hazardous waste
- To remove barriers to the movement of hazardous wastes between countries

Which one of the following is the maximum time of validity for an International Certificate on Inventory of Hazardous Materials?

- **5 years**
- 4 years
- The life of the ship
- Until the ship's next drydocking

Which one of the following is the proper title of the 'Hong Kong' Convention?

- **The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships**
- The Hong Kong Ship Scrapping Convention
- The International Convention for the Safe Transport of Hazardous Wastes
- The Prevention of Dumping Convention

Which one of the following most accurately identifies the additional parts of an Inventory of Hazardous Materials that must be completed after it is decided to scrap a ship?

- **Parts 2 and 3**
- Only Part 2
- Only Part 3
- Parts 1 and 2

Which one of the following most accurately identifies the materials that should have been listed in the Inventory of Hazardous Materials before delivery of a NEW ship, according to the 'Hong Kong' Convention?

- **Any materials listed in Appendix I or II of the Convention**
- Any materials considered hazardous in the State where the ship was built
- Any materials listed in Appendix I of the Convention
- Any materials listed in Appendix II of the Convention

Which parts of an Inventory of Hazardous Materials should have been completed before delivery of a NEW ship, according to the 'Hong Kong' Convention? Select the best answer.

- **Part 1 and perhaps Part 3**
- Part 1 and perhaps Part 2
- Part 2
- Parts 2 and 3

Which two of the following are the most likely to be useful when collecting information on the location of hazardous materials on an EXISTING ship which is in service?

- **The Insulation plan**
- **The International Anti-Fouling System Certificate**
- The Damage control plan
- The Safety Construction Certificate

Who would complete a 'Material Declaration' so that an Inventory of Hazardous Materials for a NEW ship can be completed? Select the best answer.

- **Suppliers of equipment to the ship**
- Class surveyors
- Flag State surveyors
- The shipbuilder

0297 – Recovery of persons from the water

Version 2.0 (996)

According to SOLAS, where should you be able to see a life-saving equipment and survival training manual? Select the ONE best answer

- **In each mess room or recreation room or in every cabin**
- In the emergency headquarters
- In the ship's library
- Only on the bridge

According to the IMO guidelines used to produce this module, what are the two greatest risks when recovering persons from the water? Select the TWO best answers

- **Injury to the casualties from contact with the ships side or by the recovery appliance itself**
- **The safety of those on the rescue ship whilst manoeuvring to carry out the rescue**
- Any delays to the recovery process if there is too much caution
- Ships not willing to assist in the rescue process

As a general rule, what treatment should be given to casualties who have been recovered after being in cold water for a long time? Select the ONE best answer

- **Remove their wet clothing, put them in a warm room and cover them with blankets**
- Get them to do vigorous exercise to get the circulation going again
- Get them to move their limbs around to quickly restore blood circulation
- Get them to stand up and move around

Can lifeboats on the falls be used to recover persons from the water? Select the ONE best answer

- **Yes, provided weather conditions are good and the lifeboat is not overloaded**
- No, casualties may be injured in the lifeboat
- No, this method would be too slow to be practical
- Yes, if they are on the falls this can be done in any conditions

Can stores cranes be used to recover persons from the water? Select the ONE best answer

- **Yes, provided they are not overloaded and are equipped with at least a rescue strop or a secure loop**
- No, they should never be used for that purpose
- No, this method would be too slow to be practical
- Yes, but only if they are specially designed for that purpose

How can the effects of the weather on the recovery operation best be assessed? Select the ONE best answer

- **Consider running by the target first, if time allows, or approach the target at slow speed to make the best judgement**
- Approach the target quickly and adjust your own course to maintain a steady bearing
- Approach the target quickly, but note its relative movement
- Keep downwind of the target in all circumstances

How often do the IMO recommend that drills for recovery of persons from the water should be held? Select the ONE best answer

- **As often as the ship's master thinks is needed to make sure that all crew know their duties**
- At every man overboard rescue drill
- Before the start of each voyage, if over 50% of the crew have changed
- Once every month

How should the ship approach survivors during a recovery operation? Select the ONE best answer

- **With the strongest element (wind, sea or swell) on your weather bow and the target on your lee bow**
- Keeping the target dead ahead
- With the target on your weather side
- With the wind and target on your weather bow

If it is impossible to reach the people or survival craft in the water using your own vessel, should you launch a lifeboat or rescue craft? Select the ONE best answer

- **Yes, but only if it is safe to do so**
- Only if you have a dedicated fast rescue craft
- Yes, because the safety of those in the water is more important than any risk to your own ship's crew
- Yes, but only if there are volunteers to operate it and they understand it will be very risky

If you are on a ship with high freeboard, it may be difficult to see survivors in the water from the bridge. What should you do? Select the ONE best answer

- **Post good lookouts with direct communications to the bridge while in the incident area**
- Do not take part in search and rescue operations with this type of ship
- Launch a rescue craft to patrol ahead of the ship
- Reduce speed when in the incident area

It is too dangerous to launch and recover lifeboats or rescue craft to pick up survivors. What can be done? Select the ONE best answer

- **Pass lines, or lifebuoys or inflated liferafts on lines, to the survivors**
- Ask for volunteers who are prepared to take the risks to man the lifeboats
- Do nothing, but inform the search and rescue authorities
- Make sure another rescue ship is on its way, then proceed on voyage

On a ship with high freeboard, should openings in the hull be used as a recovery point for persons in the water? Select the ONE best answer

- **Yes, provided there is no risk to the ship's stability and that persons can be recovered there**
- No, not if they are below the ship's freeboard deck
- Yes, but they must be certified as approved recovery points
- Yes, provided that they are on the lee side of the vessel

Should relevant IMO circulars be included in your plans for recovery of persons from the water? Select the ONE best answer

- **Yes, because although you should be familiar with them, they should also be available for reference**
- No, because the ship's master can direct operations
- No, because they will not be needed once a good plan has been made
- No, because well trained staff will never need them

Should your ship's particulars be included in your plans for recovery of persons from the water? Select the ONE best answer

- **Yes, because search and rescue co-ordinators will need these to decide if you are able to carry out the recovery operation**
- No, because any competent master will know enough information to readily pass it on to the authorities
- No, because search and rescue authorities can easily find these by other means
- No, because these are already available on the bridge

What are the two biggest risks when recovering casualties from cold water? Select the TWO best answers

- **Casualties are at risk from heart failure and need to be recovered horizontally**
- **Casualties will be unable to grip things and climb ladders**
- Casualties may be drowsy
- Casualties may be shivering uncontrollably

What is the best position for a person to be in when recovered from the water? Select the ONE best answer

- **Horizontal**
- It makes no difference
- Vertical, with the head down
- Vertical, with the head up

What is the biggest danger when manoeuvring your ship to recover persons from a survival craft? Select the ONE best answer

- **You could run over the persons or craft to be rescued**
- You may have to make several approaches to judge the conditions
- You might not be able to get alongside the craft to be rescued
- You might not be able to see the craft from the bridge

What is the greatest risk to persons after they have been recovered from cold water? Select the ONE best answer.

- **The sudden return of cold blood from the limbs can cause heart failure**
- They may panic at being on a foreign ship
- They may suffer from intense shivering
- They may suffer from poor blood circulation

When should risk assessment for recovery of persons from the water be done? Select the ONE best answer

- **When the plans for recovery operations are first done, and then again once the specific conditions at the scene of the recovery are known**
- Only when the plans for recovery operations are first done
- Only when the specific conditions at the scene of the recovery are known
- When the plans for recovery operations are first done, and then again at every ISM audit

Where is the best part of the ship to bring people onboard? Select the ONE best answer

- **Clear of the ship's propellers and usually within the ship's parallel mid-body section**
- Anywhere there is an access point and good lighting
- Under the lee of the bow
- Under the lee of the stern

Where would you find advice on preparing your ship for the transfer of survivors using a helicopter? Select the ONE best answer

- **The IAMSAR manual which should be available onboard**
- The IACS manual which should be available onboard
- The IMO manual which should be available onboard
- The SOLAS manual which should be available onboard

Which IMO rules require you to have a plan for the recovery of persons from the water on board? Select the ONE best answer

- **SOLAS and the ISM code**
- COLREGS
- IAMSAR and AMVER
- STCW

Which ONE of these statements best describes the plans for the recovery of persons from the water carried on your ship? Select the ONE correct answer

- **They are ship specific plans covering a range of possible recovery situations**
- They are company specific plans covering a range of possible recovery situations
- They are plans produced by senior technical staff in the office for your ship and should be followed in detail
- They are plans produced by the IMO or classification societies covering a range of possible recovery situations

Who is at risk during an operation for the recovery of persons from the water? Select the ONE best answer

- **Those being rescued and all those involved in the rescue**
- Nobody is at risk if the correct safety equipment is used
- Only those being rescued
- Only those being rescued and anyone in a small rescue craft

Who will need to see the ship's plan for the recovery of persons from the water? Select the ONE best answer

- **Everyone onboard if it applies to their duties, and any external audit body checking compliance with SOLAS and the ISM Code**
- All bridge and deck crew, and the chief engineer
- All deck and engineer officers
- The master and all the bridge team

0298 – Use of Liferafts

Version 1.0 (2327)

After boarding a liferaft, which one of these actions will best slow down the drift of the liferaft in strong winds? Select the ONE best answer.

- **Streaming the sea anchor.**
- Closing the liferaft entrances.
- Cutting the painter.
- Using the paddles.

During abandonment, what must you check first before lowering a liferaft embarkation ladder? Select the ONE best answer.

- **That it is secured to the deck.**
- That it can be brought back on board.
- That it is long enough.
- That the steps are evenly spaced.

How could a crewmember become familiar with the steps needed to launch a remote liferaft? Select the TWO correct answers.

- **By checking the illustrated liferaft instruction poster near the liferaft.**
- **By checking the launching instructions on the container.**
- By checking the nearest fire and safety plan.
- By referring to the life-saving training manual in the focsl'e head. It is a requirement to have a copy of the life-saving training manual forward.

In a liferaft, where is most of the survival equipment stored?

- **In a large bag inside the liferaft.**
- In a large bag floating in the water. The bag is connected to the liferaft by a rope.
- In a net, suspended from the canopy.
- Survival equipment is individually taped to the floor of the liferaft.

There is a knife, in a pocket, close to the entrance to the liferaft. What is it's main use? Select the ONE best answer.

- **To cut the painter, once all survivors are on board.**
- As a can opener, to open cans of drinking water.
- To cut away the boarding ladder, once everyone is on board.
- To cut away the rope for the sea anchor in a hurry.

Underneath a liferaft there are webbing straps spreading across the underside of the floor. What are they used for? Select the ONE correct answer.

- **To turn the liferaft the right way when it is upside down.**
- They are for survivors in the water to hold on to.
- They hold the gas cylinder in place.
- To connect a towing line from a lifeboat or rescue boat.

What can help you most when boarding a liferaft from the water? Select the ONE best answer.

- **The webbing ladder at the entrance.**
- Holding on to the upper buoyancy chamber.
- The lifeline.
- The righting strap.

What equipment in a liferaft is provided to help rescue survivors who are in the water? Select the ONE best answer.

- **The buoyant rescue quoit (or ring) and line.**
- First Aid Kit.
- Sea anchor.
- The paddles.

What is the first thing you must do once everyone is in a liferaft? Select the ONE correct answer.

- **Cut the painter.**
- Close the entrances.
- Issue anti-sea-sickness tablets.
- Remove your lifejacket.

What is the main purpose of the painter connected to the liferaft? Select the ONE correct answer.

- **To secure the liferaft to the vessel during launching.**
- To hold a davit-launched liferaft alongside during launching.
- To prevent the container falling into the sea, after launching a davit-launched liferaft.
- To secure a liferaft to the cradle.

What is the minimum number of people needed to transfer a large throw-over liferaft?
Select the ONE correct answer

- **Four**
- Five
- Three
- Two

What is the purpose of a liferaft hydrostatic release unit? Select the ONE correct answer.

- **It releases the liferaft from its stowage point when the vessel sinks.**
- It is used in heavy weather as an additional securing measure.
- It lets the liferaft inflate automatically as soon as it touches the water.
- It releases a davit-launched liferaft as soon as it is in the water.

What is the purpose of the liferaft "container line"?
Select the ONE correct answer.

- **It is a line which is secured on board and prevents the empty container falling into the sea.**
- It is a rope for securing the liferaft in the cradle.
- It is used to secure the liferaft to the vessel before throwing overboard.
- When pulled fully out of the liferaft, and then pulled hard, it inflates the liferaft.

What part of a liferaft is used to help a survivor in the water to get inside the liferaft? Select the ONE correct answer.

- **The boarding ramp.**
- The canopy
- The lifeline around the buoyancy chamber
- The righting strap.

When abandoning ship, when would you have to use liferafts instead of lifeboats? Select the ONE correct answer.

- **When it is not possible to launch the lifeboats due to the list of the ship.**
- When power to the lifeboat winches has been lost.
- When the liferafts are nearer to where your place of work is.
- When there is a fire in the engine room.

When might you need to launch a remote liferaft on the weather side?
Select the ONE correct answer.

- **When there is fire on the water or thick black smoke on the leeside.**
- When the embarkation ladder is stowed nearer to the weather side.
- When the remote liferaft is stowed nearer to the weather side.
- You must never launch a liferaft on the weather side.

Where would you find details of the types of liferaft types found onboard your ship? Select the ONE best answer.

- **The life-saving appliances training manual.**
- Details of all types of liferafts onboard are shown on the illustrated liferaft instruction poster.
- The fire and safety plan
- The muster list.

Which one of these options best describes when you might have to launch a liferaft without being given orders? Select the ONE best answer.

- **You are in immediate danger and there is no officer available to supervise launching**
- In any emergency, if you have had shore-based training in personal survival.
- The general emergency alarm has sounded.
- You are in immediate danger and there is an officer supervising the launching.

Which one of these options is most important when preparing to launch and board a liferaft? Select the ONE best answer.

- **Wearing extra clothing and a survival suit.**
- Collecting as much food as possible from the galley.
- Getting the EPIRB and SART from the bridge.
- Getting two UHF radios from the bridge.

Which type of liferaft does not need to be fitted with a hydrostatic release unit? Select the ONE correct answer.

- **An additional remotely located liferaft.**
- A davit-launchable liferaft.
- A throw-over liferaft.
- All liferafts must be fitted with a hydrostatic release unit.

Why is it most important to board a liferaft from the water as quickly as possible? Select the ONE best answer.

- **To stop your body temperature getting too low.**
- Because your lifejacket light battery will run down.
- In case the weather gets too rough.
- You need to board the liferaft before it drifts away from you.

Why is it necessary to close the liferaft entrances as soon as possible after boarding? Select the ONE best answer.

- **It will protect the survivors in the liferaft from the sea and the wind and stop them getting too cold.**
- It is always better to keep at least one entrance open at all times.
- It will help to stop the liferaft drifting too much in strong winds.
- It will protect the survivors from sunlight.

Why should you NOT bring a sharp knife with you when boarding a liferaft? Select ONE best answer.

- **It could puncture the liferaft.**
- Because the liferaft does not have a knife in the survival pack.
- Because the liferaft has a knife in the survival pack.
- You might get injured going down the ladder.

Why should you look over the side before launching a throw-over liferaft?
Select ONE best answer.

- **There could be debris or survivors in the water.**
- It is not necessary to check overboard in an emergency.
- To check that the boarding ladder is long enough.
- To check that the painter is long enough.

0299 – SIRE inspections

Version 2.0 (854)

During the closing meeting, following a SIRE inspection, the SIRE inspector raises an observation that is clearly incorrect and is not supported by regulation or industry standard. What should the Master do? Select the ONE best answer.

- **Challenge the SIRE inspector in a professional and business-like manner**
- Accept that the SIRE inspector is always right.
- Plead with the inspector to change his mind
- Tell the inspector he is wrong and bring the meeting to an end.

During vetting, why do SIRE subscribers check for repeat observations in previous SIRE inspection reports? Select the ONE correct answer.

- **Repeat observations may be an indication that the standard of management on a tanker does not meet their expectations.**
- So that they can carry out an onboard inspection to check that repeat observations have been dealt with.
- So that they can send an email to the ship warning them about repeat observations.
- To check that that there are no errors in the inspection reports.

How is the SIRE database kept up to date? Select the ONE correct answer.

- **The SIRE database is kept up to date by input of standardized vessel inspection reports by submitting members.**
- All tanker operators are required to update the SIRE database weekly.
- Each tanker Master regularly updates the SIRE database with cargo information.
- It is updated by OCIMF only when a tanker is involved in an accident.

How long does a typical SIRE inspection normally take? Select the ONE correct answer.

- **8-10 hours**
- 18 hours
- 24 hours
- Usually no more than 4 hours

How often would SIRE inspections be typically carried out on a five year old VLCC? Select the ONE best answer.

- **Every four to six months**
- Annually
- Every two years
- Monthly

The SIRE inspector refuses to accept the observation is incorrect, in spite of clear, unambiguous evidence. What action can the Master take? Select the TWO best answers.

- **Make remarks on the observation record before signing and stamping**
- **Record the details on the SIRE Inspection feedback form and return as soon as possible to the tanker operator**
- Accept the observation list but do not co-operate with the SIRE inspector
- There is no point taking action the contracting oil company will only accept the report of the SIRE inspector

What are the benefits of conducting a pre-SIRE inspection? Select the ONE best answer.

- **Any unknown deficiencies can be identified and dealt with before the SIRE inspection.**
- The contracting oil and gas company will include the pre-inspection result in the SIRE inspection report.
- The SIRE inspector will only need to perform a partial inspection.
- There are no benefits; the SIRE inspector will always be looking for things wrong with the ship.

What are the benefits of planning and organizing the inspection process? Select one best answer.

- **It will make the inspection process easy and straight forward for the inspector**
- The inspection will take less than 6 hours
- The inspector will not need to carry out a full inspection
- There is no benefit of planning and organizing the inspection. All inspections are different

What does a SIRE subscriber mean when they refer to a 'nominated third party tanker'? Select the ONE correct answer.

- **A tanker not operated by them but made available to carry specific cargoes.**
- A tanker operated by them but carrying their cargo.
- A tanker operated by them.
- Any type of oil tanker, which is not a gas tanker.

What is a Harmonized Vessel Particulars Questionnaire (HVPQ)? Select the ONE correct answer.

- **It is a questionnaire designed to provide comprehensive information about a vessel, the technical operator, commercial operator, and crewing agency.**
- It is a pre-hire questionnaire used by SIRE subscribers before hiring third party tankers.
- It is a standard questionnaire used by the SIRE inspector when inspecting a tanker.
- It is a table providing information on qualifications and experience of the officers onboard.

What is tanker vetting? Select the ONE correct answer.

- **The assessment of a tanker to ensure it meets the required expectation of onboard safety and pollution prevention management.**
- It is the combination of A) and B).
- It is the process where a tanker is inspected to ensure that all the certificates onboard are valid and that the vessel is seaworthy.
- It is the process where the tanker is inspected by a SIRE inspector to ensure that it meets the required standard of safety management.

What technique would give the Master and Chief Engineer the best overview when performing a pre-SIRE inspection? Select the ONE best answer.

- **A thorough inspection using the V.I.Q. as a basis for the inspection.**
- A random inspection of ships certificates, main deck and engine room.
- Delegating the pre-inspection to junior officers and crew
- Starting at the bow and working aft so that nothing is missed.

Which of the following possible answers to a question in the V.I.Q.is required to include a comment? Select the TWO correct answers

- **No (N)**
- **Not seen (NS).**
- Not applicable (NA)
- Yes (Y)

Which of the following would be the preferred way to present the certificates and documents to a SIRE inspector? Select the ONE best answer.

- **Arrange in one place with ready-reference index using V.I.Q. number.**
- It is enough to have photocopies or scanned digital copies available.
- Just arrange in one place.
- The Master should explain the filing system on board.

Why might a submitting member reject a nominated third-party tanker? Select the TWO best answers.

- **The nature and number of the observations on a SIRE report suggest that the way the tanker is managed does not meet their expectations.**
- **The tanker operator's response is inadequate.**
- The oil company has never hired the third party tanker before.
- The tanker is not owned by the oil company.

0300 – Rigging and Slinging

Version 5.0 (272)

At which angle of lift will a Linked Eyebolt be rated higher than a normal Collar Eyebolt?

- **At all angles above 15° angle.**
- At 45° and above.
- At 60° and above.
- Never, a Linked Eyebolt will always be weaker than a normal Collar Eyebolt.

At which of the following situations is the winch capacity at it's highest?

- **When the winch drum is at first layer.**
- When the winch drum is 36% full.
- When the winch drum is full.
- When the winch drum is half full.

At which of the following situations will the pulling speed of the wire be at its highest?

- **When the winch drum is full.**
- When the winch drum is 36% full.
- When the winch drum is at first layer.
- When the winch drum is half full.

How many falls will you need to lift a 3 ton load with winch cap. of 1.5 ton?

- **3 fall**
- 2 fall
- 4 fall
- 5 fall

How much luggage can be allowed inside a personnel basket during personnel transfer?

- **Only small amounts of luggage are allowed together with personnel, it is preferable to send luggage in a separate lift.**
- 20 kg per person to be transferred.
- As much as can be safely stowed inside the ropes.
- No luggage must be transferred together with personnel.

How much wear can be accepted on a single link on a chain before the chain must be discarded?

- **10%**
- 15%
- 2%
- None

How much will the SWL of a Hoist Ring have to be de-rated for a lift at 30° vertical angle ?

- **Nothing, a SWL of a hoist ring does not change at any angel of lift.**
- The SWL will be reduced by 25%.
- The SWL will be reduced by 50%.
- You must never do an angular lift with a Hoist Ring.

How often shall a personnel basket undergo thorough examination?

- **Minimum every 6 months.**
- At least once every year.
- Every month.
- Minimum every 3 months.

How often should lifting machinery/appliances, which is not used for personnel transfer, undergo thorough inspection?

- **Once every year.**
- Every month.
- Once every six months.
- Once every two years.

How often should lifting machinery/appliances, which is used for personnel transfer, undergo thorough inspection?

- **Once every six months.**
- Every month.
- Once every year.
- Weekly

How often should loose gear/tackle like slings, shackles etc. undergo thorough inspection?

- **Once every six months.**
- Every month.
- Once every year.
- Weekly

If all of the slings are of equal strength, which of the following configuration is the strongest?

- **A**
- B
- C
- D

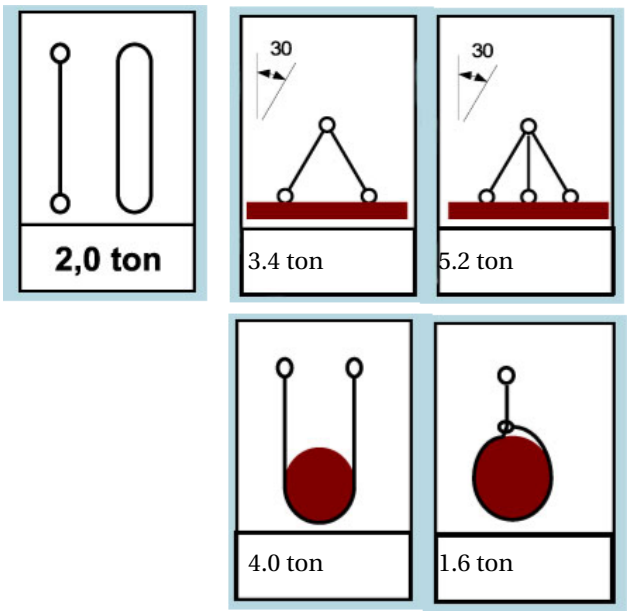
In which position should the ram be when you are refilling oil on the jack?

- **Fully retracted.**
- At 80% of full stroke.
- At full stroke.
- Half retracted.

Is use of drum slings an accepted method of lifting steel barrels?

- **Yes, but only for internal transfer.**
- No.
- Yes, but only for single drums in case of external transfer.
- Yes, no restrictions.

Place the correct SWL at the correct sling configuration. All the individual slings have the same SWL (2 ton).



What is the maximum angle you can apply on a side loading on a beam trolley?

- **You must never apply side loadings on a beam trolley.**
- 20°
- 45°
- 60°

What is the maximum angle you can have on an angular lift with a Dynamo type eyebolt?

- **You must never do an angular lift with this kind of eyebolt.**
- 15°
- 5°
- 60°

What is the maximum face-to-face angle (A) allowable for a standard pair of Horizontal Plate Clamps?

- **90°**
- 120°
- 45°
- 60°

What is the maximum side angle (B) allowable for a standard two pair set of Horizontal Plate Clamps?

- **20°**
- 10°
- 30°
- 45°

What is the maximum stroke you should use on a hydraulic jack?

- **80% of usable stroke.**
- 100% of usable stroke.
- 70% of usable stroke.
- 90% of usable stroke.

What is the minimum SWL required for the lifting beam in this scenario?
The blocks 4 fall are fitted with roller bearings and the total weight of the gear is 250 kg.

- **(6 ton X 1,29)+0,25=8 tons**
- (6 ton X 1,0)+0,25=6,3 tons
- (6 ton X 1,21)+0,25=7,3 tons
- (6 ton X 1,54)+0,25=9 tons

What is the minimum SWL required for the lifting beam in this scenario?
Use 8% friction force and 100 kg total weight of gear.

- **(2,2 ton X 1,08 X 1,84)+0,1=4,5 tons**
- (1,08 X 1,41) X (2,2 ton+0,1)+0,1=3,5 tons
- (2,2 ton X 1,08 X 1,73)+0,1=4,2 tons
- (2,2 ton X 1,84) / 1,08)+0,1=3,1 tons

What is the purpose of the stabilizer on a personnel basket?

- **To prevent the basket collapsing when it surges up and down on the boat deck.**
- It serves as a secondary safety sling in case the main sling fails.
- To keep the basket balanced (horizontal) during the transfer.
- To prevent the basket from capsizing if it falls into the water.

What kind of safety precautions must be taken if personnel unfamiliar with basket transfer are to be transferred by this method?

- **A person who is well familiar with such transfer should accompany them.**
- None in particular.
- They must be given strict instructions in how to behave.
- They must stand inside the ropes.

What kind of special requirements exists for the tag line?

- **It must be snag free and have no knots on it.**
- It must be minimum 2 meters long.
- It must be of a bright visible colour.
- It must not be more than 3 meter long.

What must you check before you are applying an angular load to a beam clamp?

- **That the beam clamp is designed for angular lifts and that correct de-rating is used.**
- That the lifting beam is designed for side loads.
- That the total lifting height does not exceed more than 75% of the distance between deck and beam clamp.
- That the weight of the load is not more than 2 tons.

Which advantage does the Bow Shackle have compared to a same size Dee Shackle of same material?

- **It is suitable for connection to multiple components, while the Dee Shackle is most suitable for connection of only 2 components.**
- It is lighter.
- It is smaller.
- It is stronger.

Which of the below statements are true?

- **You must never weld on grade 80 chains.**
- **You should always clean the slings before you inspect them for wear.**
- An easy and good method of shortening a chain sling is to tie a knot on it.
- You must never have more than 60° angle between the slings in a multi leg configuration.

Which of the following items must be checked before you use a chain hoist? (select all applicable answers)

- **That there are no knots on any of the chains.**
- **That there are safety catches fitted on the hooks.**
- **The SWL of the hoist and the weight of the load.**
- The type of slings to be used.

Which of the following requirements for the beam must be fulfilled in order to use a beam trolley on it?

- **It must be fitted with end stoppers.**
- **It must be levelled, so that the trolley will not run away by gravity.**
- It must be higher than 2 meter above the deck.
- It must have a SWL of two times the weight of the load.

Which of these 3 illustrations shows the correct way of slinging tubulars?

- **A**
- B
- C

Why do we have to be extra careful when lifting electrical cables?

- **They can be very sensitive and the slightest damage may make them unusable.**
- They are very expensive.
- They can be electrically charged and cause sparks if landed to hard.
- They can be extremely heavy.

Why should you use a saddle on the hydraulic jack?

- **It distributes the load evenly on the ram.**
- It increases the lifting capacity.
- It increases the max lifting height.
- It prevents the jack from retracting unintentionally.

0301 – Crane Operations

Version 2.0 (137)

A seal is found to be damaged on a panel, allowing water into the mechanism of a crane fitted on the bow. What should NOT be done. Select the ONE correct answer.

- **Leave the panel loose, so that it may be checked more frequently**
- Remove the water and dry the chamber
- Replace the seal and refit the panel
- Treat and re-coat any area of rusty steel

A ship is taking stores, using a provisions gantry crane and cargo net, lifting approximately ten loads, some palletised, from a barge alongside into the stores flat behind the accommodation. How should the loads be controlled and why? Select the ONE best answer.

- **Due time should be allowed to unload one net and clear the handling area before the next lift is brought on board. This is to avoid anyone working under a suspended load**
- The loads should be brought on board as quickly as possible, in order to save paying overtime for the delivery barge. It is up to the cooks and stewards to clear the loads as quickly as they can
- The loads should be reduced in size and number of boxes and packages to that which can be handled by the cook and stewards without any overlap or delay. The rate of loading should be such that they just have enough time to clear one load before the next arrives

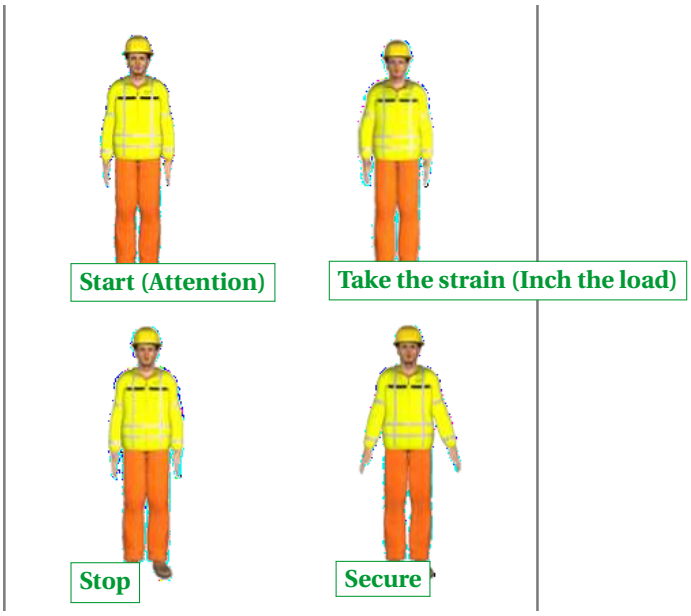
Can a crane be used to "drag" a heavy load closer and make the lift easier? Select the ONE correct answer.

- **No**
- Yes

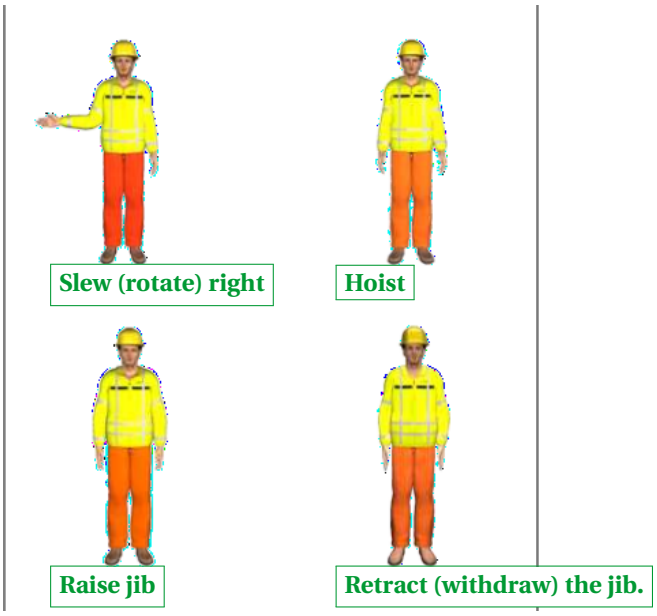
Crew are lifting hoses and reducers using the deck manifold cranes. What must happen if a pilot is to soon board by helicopter? Select the THREE correct answers.

- **All associated loose gear must be secured**
- **The crane must be housed**
- **The Responsible Officer must verify these facts and indicate he has done so by ticking the relevant box on the pre-helicopter operations checklist**
- Crew must continue lowering hoses into position, so that the ship is ready for work when it gets alongside

Drag and drop the labels onto the corresponding signals.



Drag and drop the labels onto the corresponding signals.



How might low temperature and ice beginning to form, affect the operation of the manifold crane? Select the TWO correct answers.

- **The control station, access ladder and work area can become slippery and dangerous**
- **The hydraulic oil temperature might initially be below the minimum operating temperature**
- The manifold crane cannot be used below 4oC
- They will not affect the operation at all

How might the ship rolling and pitching affect the safe operation of a crane? Select the TWO correct answers.

- **A heavy load may swing out of control**
- **Excessive dynamic stresses may be induced on all or part of the crane's structure and fittings**
- The hydraulic oil temperature might be, at least initially, above maximum operating temperature
- Use of the crane when the vessel is rolling more than 15° from the vertical is prohibited under the Merchant Shipping (Lifting Devices) Regulations 1975

How should an operator gain access to the control platform of a crane sited above the deck? Select the TWO correct answers.

- **Only ever by the proper means, such as the permanent ladder**
- **The crane should be stationary at the time**
- By any convenient method, such as jumping from an adjacent rail or structure
- The crane can be moving, but the jib should be pointing away from the operator as they gains access

How should crane controls be marked? Select the THREE best answers.

- **Permanently and legibly**
- **The function of each control should be clearly identified**
- **The operating direction of each control should be identified by arrows or similar means**
- The markings should all be applied using luminous green paint

Under what circumstances can a crane be used for the lifting and transfer of personnel? Select the TWO correct answers.

- **For rescue in an emergency situation**
- **Only when the crane is fitted with special equipment designed for that purpose, such as a basket hoist**
- No crane can ever be used to lift and transfer personnel under any circumstances
- There is no restriction in using a crane to lift and transfer personnel, provided the safe working load is not exceeded

Under what circumstances might it be necessary to temporarily isolate a power source or limit switch, during the operation of a crane? Select the ONE correct answer.

- **As part of an emergency procedure in the event of crane failure, in order to lower a jib to the stowage position**
- In order to change operators in the middle of a lift
- In order to operate the crane temporarily beyond its designed specification
- So that a replacement bulb can be fitted to the crane's floodlight

What criteria is used to add crew names to a list of "Approved Crane Operators"? Select the ONE best answer.

- **Crew are assessed by the Responsible Officer as competent in the operation of the crane and authorised to operate it unsupervised**
- The list is issued by the company safety department, that indicates those personnel who are allowed to operate the crane
- The list must also contain the names of crew who are approved service engineers authorised by the crane manufacturer
- The list shows those engineers and fitters on board that have attended and passed a training course run by the manufacturer

What does the abbreviation "SWL" stand for? Select the ONE correct answer.

- **Safe working load**
- Safe weight limitation
- Standard weight limit
- Standard working load

What does this approved crane signal mean? Select the ONE correct answer.

- **Vertical distance**
- Cease operation
- Travel (move) backwards
- Travel (move) forwards

What does this approved crane signal mean? Select the ONE correct answer.

- **Stop (interruption or end of movement)**
- Hoist
- Raise jib
- Slew right

What does this approved crane signal mean? Select the ONE correct answer.

- **Secure**
- Hoist
- Stop (interruption or end of movement)
- Travel away from me

What does this approved crane signal mean? Select the ONE correct answer.

- **Retract (withdraw) the jib**
- Extend (telescope) the jib
- Lower the jib
- Travel towards me

What does this approved crane signal mean? Select the ONE correct answer.

- **Start, or attention!**
- Danger, or emergency stop!
- Finish
- Secure

What does this approved crane signal mean? Select the ONE correct answer.

- **Danger, or emergency stop!**
- Hoist
- Lift the jib
- Start, or attention!

What does this approved crane signal mean? Select the ONE correct answer.

- **Lower the jib**
- Lower
- Move backwards (travel away from me)
- Move forwards (travel towards me)

What does this approved crane signal mean? Select the ONE correct answer.

- **Slew left**
- Move aft
- Move forward
- Slew right

What kind of treatment should be applied to foot-operated controls on a crane? Select the ONE correct answer.

- **Non-slip coating**
- Bare steel - no coating should be applied
- Black and yellow stripes
- Heavy duty, chip resistant paint

What permanent markings are required to be displayed on a crane? Select the THREE correct answers.

- **Any limiting conditions that apply to the device**
- **The identification number**
- **The safe working load**
- An arrow indicating the direction of operation

What should an operator know about the emergency stops for a crane? Select the TWO correct answers.

- **How to operate them**
- **Where they are located**
- How to disconnect or over-ride them for routine work
- It is not necessary for an operator to know anything about emergency stops. That is just for the engineers during overhaul

What should the Responsible Officer do prior to a lifting operation with an increased level of danger? Select the ONE correct answer.

- **Carry out an enhanced risk assessment**
- Assemble as many extra personnel as possible
- Clear the engine room of everyone not immediately involved in the operation
- Rig chain blocks

When should the emergency stop on a crane be tested? Select the THREE correct answers.

- **At any other time felt necessary or prudent**
- **At periodic intervals, as required under the Planned Maintenance system**
- **Prior to the commencement of lifting operations**
- During a lifting operation to ascertain that the load will stop

When the wire on a hoist drum is replaced, which of the following items should be given particular attention? Select the TWO correct answers.

- **The applicable limit switches should be adjusted and re-set**
- **The end of the wire should be securely fitted by the approved method**
- The drum face should be re-painted
- The speed of the hoist motor should be permanently adjusted, so that the wire coils satisfactorily

Where can a new crew member find instructions and guidelines for operating a crane? Select the TWO correct answers.

- **The Code of Safe Working Practises for Merchant Seamen**
- **The company Safety Management System manuals**
- The IMDG Code
- The master's Standing Orders
- The SOLAS Training Manual

Where should a crane signaller position themselves when directing a crane? Select the THREE best answers.

- **A safe location**
- **One from which they have a clear view of the path of travel of the load**
- **One that is in plain view of the crane operator, unless an effective system of radio communication is in use**
- Always forward of the crane operator

Which ONE of the following is NOT a movement limit switch found on a crane? Select the ONE correct answer.

- **On the hydraulic system to stop over pressure with a large weight**
- At or near the end of the rails of a travelling or gantry crane, to slow down the rate of approach and prevent hard impact with the end buffers
- On the chassis or frame, to prevent the jib being over-rotated or rotated in the wrong direction
- On the hoist mechanism, to prevent it from continuing to be operated once the hook is lifted up close to the head of the jib

Which of the following conditions can place excessive dynamic stresses on a crane structure or its fittings when it is in use? Select the TWO correct answers.

- **Heavy list or trim**
- **Heavy rolling and pitching**
- Heavy rain
- Ice formation on deck

Which of the following factors contribute to the safe and efficient operation of a shipboard service crane? Select the THREE correct options.

- **Having competent people operate it.**
- **Having good teamwork during a lifting operation**
- **Making sure that maintenance is properly carried out**
- Having an electric powered crane rather than a hydraulic one.

Which of the following is a safety principle that a crane operator should observe when other personnel are working in the handling area? Select the ONE correct answer.

- **Loads should not be lifted over any person or hatch with personnel working below and adequate warning should be given, so that personnel avoid passing under a load which is being lifted**
- Loads can only be lifted over other personnel if they are wearing safety helmets
- Loads lifted over other personnel should be raised to the maximum possible height, to give a greater chance to those below if a strop or wire breaks
- That the crane driver devotes all of their attention to monitoring the movements of personnel in the area

Which of the following items is NOT usually included in an operational briefing before using a crane? Select the ONE correct answer.

- **The planned maintenance schedule**
- The methods to be used in transferring the load
- The potential hazards that have been identified in the operation
- The purpose of using the crane

Which of the following might be included in the pre-operation checks on a hydraulic powered crane? Select the THREE correct answers.

- **A function test of all controls**
- **A function test of limit and emergency stop switches**
- **A visual inspection of hoses and crane hook**
- A detailed inspection of the rotation assembly

Which of the following roles would NOT be identified at the pre-operations briefing before using a crane? Select the ONE best answers.

- **The Port Safety Officer**
- The crane operator
- The signaller (flag-man), if required
- Those assisting with handling the load

Which of the following statements describe the characteristics of the warning device fitted to indicate movement of an engine room gantry crane? Select the TWO best answers.

- **A continuous flashing light, prominently mounted where it can best be seen, on the trolley frame, which starts automatically as the crane starts to move, but of a type that cannot be confused with other visual alarms in the vicinity**
- **A continuous two-tone claxon, which start automatically as the crane starts to move, but of a type that cannot be confused with any other audible alarm in the vicinity**
- A general automatic warning claxon that indicates a number of features to personnel in the engine-room, including, but not exclusively, that the crane is being used.
- An alarm light mounted in the alarm panel on the console in the engine control room, which is activated whenever the engine-room crane is used

Who is allowed to enter the handling area during a lifting operation? Select the ONE correct answer.

- **Authorised personnel only**
- Anybody
- Nobody at any time
- Ship's personnel only

0302 – Permit to work

Version 3.0 (2013)

According to this module, where should a work permit be compulsorily displayed? Select ONE correct answer.

- **At the work site.**
- Bridge
- Crew recreation room.
- Engine control room.

From the given list of tasks select the ONE that does NOT require a permit.

- **Navigation in fog.**
- Entry into enclosed spaces.
- Replacing a valve in a cargo oil tank.
- Welding on deck.

In which ONE document would you find all ship specific work permits listed?

- **Safety Management System**
- Chief Engineer's standing orders
- Document of Compliance
- Master's standing orders

What action should be taken, if, during the course of work one of the conditions, specified on the permit, cannot be complied with? Select the ONE correct answer.

- **Stop work, remove all personnel from site and inform issuer.**
- Continue work and inform issuer of the new changes to the work permit.
- Continue work and undertake new risk assessment.
- Stop work and inform issuer.

What permit would be used to undertake a task such as 'disconnection of section of a bunker pipeline for maintenance'? Select ONE CORRECT answer.

- **Permit involving pressurized fluid in pipeline.**
- Electrical isolation permit.
- Working at height permit.
- Working overside permit.

Who is permitted to issue a permit to work? Select ONE CORRECT answer from the given choices.

- **Master or Chief Engineer**
- Bosun
- Chief Officer or Second Engineer
- Electrician

Why should permits and associated documents be retained on board? Select the ONE CORRECT answer.

- **They are required during International Safety Managementinspections, audits and accident investigation.**
- The only reason for retaining them is for inspection by superintendent.
- To reuse the permit if same job comes up again.
- To use them to build a database for research purposes.

With reference to permit to work system, select the ONE CORRECT statement from below.

- **Permit to work is one of the control measures to reduce risks.**
- Permit to work is an easy way to eliminate or reduce hazards.
- Permit to work is required for all hazardous shipboard tasks.
- Permit to work is simply a permission to carry out a dangerous job

0303 – COLREGs, Lights, shapes and sound signals

Version 2.0 (815)

A vessel is proceeding along a narrow channel and hears the following signal "two long blasts and two short blasts on a whistle" from astern. How should the vessel respond?

- **The response should be one long, one short, one long and one short blast on the whistle, if in agreement to be overtaken.**
- The response is one long blast on the whistle if not in agreement to be overtaken.
- The response should be one long, one short, one long and one short blast on the whistle.
- The response should be two long blasts and two short blasts on the whistle if in agreement to overtaken on the port side.

A vessel is seen showing a black ball in the fore part of the vessel. What does this day time signal indicate?

- **It is a ship at anchor.**
- It is a cable-laying vessel.
- It is a ship that it is showing that it is "under way" but not making any way through the water.
- It is a vessel that has run aground.

A vessel is seen showing a rigid replica of international code flag A. What does this day time signal indicate?

- **The flag is a signal shown by certain vessels involved in a diving operation.**
- The boat has engine failure and is calling for help.
- The flag has no particular meaning under these Rules.
- The flag is a signal that the vessel has a pilot on board.

A vessel is seen showing a white light in the fore part of the vessel. What does this night time signal indicate?

- **It is a ship at anchor, under 50 metres in length.**
- It is a cable-laying vessel.
- It is a ship that it is showing that it is "under way" but not making any way through the water.
- It is a vessel that has run aground under 100 metres in length.

A vessel is seen showing three black balls in a triangle at the head of the fore mast of the vessel and another single black ball forward. What does this day time signal indicate?

- **A ship involved in mine clearance operation at anchor.**
- A ship involved in compass adjustment.
- A ship involved in underwater operation.
- A ship that is dredging where the dredging equipment extending more than 150 metres in the direction of the single black ball.

A vessel is seen showing three black balls. What does this day time signal indicate?

- **It is the signal for a ship that has run aground.**
- It is the ship at anchor.
- It is the signal for a ship that is taking provision.
- It is the signal of a ship involved in an underwater operation.

A vessel is seen showing three green lights in a triangle at the head of the fore mast of the vessel and another single white light forward. What does this night time signal indicate?

- **A ship involved in mine clearance operation at anchor.**
- A ship involved in compass adjustment.
- A ship involved in underwater operation.
- A ship that is dredging where the dredging equipment extending more than 150 metres in the direction of the single black ball.

A vessel is seen showing three lights in a vertical line; the top and bottom ones are red and the middle one is white in the fore part of the vessel. What does this night time signal indicate?

- **It is a vessel restricted in her ability to manoeuvre, underway but not making way.**
- It is a ship at anchor.
- It is a ship dredging.
- It is a ship engaged in a mine clearance operation.

A vessel is seen showing three shapes in a vertical line; the top and bottom ones are black balls and the middle one is a black diamond in the fore part of the vessel. What does this day time signal indicate?

- **It is a vessel restricted in her ability to manoeuvre.**
- It is a ship at anchor.
- It is a ship dredging.
- It is a ship engaged in a mine clearance operation.

A vessel is seen showing two black balls on one side; two black diamonds on the other side of the vessel and also three shapes in a vertical line, the upper and lower black balls and the middle one a diamond. What does this day time signal indicate?

- **It is a vessel engaged in dredging or underwater operations which restrict its ability to manoeuvre - showing the side where the obstruction exists and the side it is safe to pass.**
- It is a vessel engaged in mine clearance operations, which is restricting its ability to manoeuvre - showing the side where the obstruction exists.
- It is a vessel engaged in towing which is restricting its ability to manoeuvre - showing the side where the tow exists.
- It is power driven vessel not under command due to damaged steering gear which is restricting its ability to manoeuvre.

A vessel is seen showing two black balls on one side; two black diamonds on the other side of the vessel and also three shapes in a vertical line, the upper and lower black balls and the middle one a diamond. Which is the correct side to pass this vessel?

- **The side which displays the two diamond shapes is the side it is safe to pass.**
- It doesn't matter on which side a vessel passes the vessel as long as it is done at a slow safe speed.
- Other vessels should pass well clear of either side and avoid passing within 100 metres of the other vessel.
- The side which displays the two balls is the side it is safe to pass.

A vessel is seen showing two black cones, points together and another cone point upwards on one side of the vessel. What does this day time signal indicate?

- **It is a vessel engaged in fishing (other than trawling) where outlying gear extends more than 150 metres horizontally from the vessel.**
- It is a vessel engaged in fishing (other than trawling) where outlying gear extends more than 100 metres horizontally from the vessel.
- It is a vessel engaged in trawling where outlying gear extends less than 100 metres horizontally from the vessel.
- It is a vessel engaged in trawling where outlying gear extends more than 100 metres horizontally from the vessel.

A vessel is seen showing two black cones, points together. What does this day time signal indicate?

- **It is a vessel engaged in any type of fishing.**
- It is a vessel engaged in fishing, but not trawling.
- It is a vessel engaged in trawling only.
- It vessel engaged in underwater operations.

A vessel is seen showing two red lights in a vertical line and also a white light forward and another white light aft. What does this night time signal indicate?

- **It is the signal for a ship that has run aground.**
- It is the ship at anchor.
- It is the signal for a ship that is taking provision.
- It is the signal of a ship involved in an underwater operation.

A vessel is seen showing two red lights on one side; two green lights on the other side of the vessel and also three lights in a vertical line, the upper and lower red and the middle one white. What does this night time signal indicate?

- **It is a vessel engaged in dredging or underwater operations which is restricted in its ability to manoeuvre, showing the side where the obstruction exists and the side it is safe to pass – under way but not making way.**
- It is a vessel engaged in mine clearance operations, which is restricting its ability to manoeuvre, showing the side where the obstruction exists – under way but not making way.
- It is power driven vessel not under command due to damaged steering gear which is restricting its ability to manoeuvre - – under way but not making way.
- It is vessel engaged in towing which is restricting its ability to manoeuvre, showing the side where the tow exists – under way but not making way.

A vessel is seen showing two red lights on one side; two green lights on the other side of the vessel and also three lights in a vertical line, the upper and lower red and the middle one white. Which is the correct side to pass this vessel?.

- **The side which displays the two green lights is the side it is safe to pass.**
- It doesn't matter on which side a vessel passes the vessel as long as it is done at a slow safe speed.
- Other vessels should pass well clear of either side and avoid passing within 100 metres of the other vessel.
- The side which displays the two red lights is the side it is safe to pass.

Two ships are seen close to each other and both are showing a black diamond in the fore part of the vessel. What does this day time signal indicate? Select the ONE best answer.

- **One ship is towing the other and the length of the tow is more than 200 metres.**
- Both ships are engaged in towing another vessel or craft where the length of tow in both cases is under 200 metres.
- One ship is towing the other.
- The ships are trawling together as a single unit.

What fog signal should be sounded by a power driven vessel underway and making way through the water, when in reduced visibility?

- **One long blast on the whistle every two minutes.**
- Four short blasts every two minutes.
- One long and two short blasts on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

What fog signal should be sounded by a power driven vessel underway but stopped and making no way through the water, when in reduced visibility?

- **Two long blasts on the whistle every two minutes.**
- Four short blasts every two minutes.
- One long and two short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.

What is the clearance distance required by a ship when navigating in the vicinity of a vessel engaged in a mine clearance operation?

- **1000 metres**
- 1500 metres
- 2000 metres
- 500 metres

What is the fog signal for a 150 metre long power driven vessel when at anchor in reduced visibility?

- **Sound the bell for about 5 seconds and in the after end of the vessel immediately after the bell, a gong for about 5 seconds every minute.**
- Sound the bell for 5 seconds and after that, sound a gong for 5 seconds every minute.
- Sound the bell for about 10 seconds and after that sound the gong for 10 seconds in the aft part of the vessel every two minutes.
- Sound the bell for about 5 seconds and after that sound the gong for about 5 seconds in the aft part of the vessel every two minutes.

What is the fog signal for a power driven vessel engaged in pushing another vessel ahead, but not as a composite unit?

- **One long blast followed by two short blasts every two minutes.**
- One long blast every two minute.
- Two short blasts and one long every one minute.
- Two short blasts followed by one long blast every two minutes.

What is the fog signal for a power driven vessel engaged in towing another vessel?

- **One long blast followed by two short blasts every two minutes.**
- One long blast every two minute.
- Two short blasts and one long every one minute.
- Two short blasts followed by one long blast every two minutes.

What is the fog signal to be sounded by a 150 metre long power driven vessel when aground in reduced visibility?

- **Every minute a ringing of the bell for about 5 seconds and in addition three distinct strokes before and after the ringing of the bell and at the after end of the vessel immediately after the bell, a gong for about 5 seconds.**
- Sound the bell for about 10 seconds, plus three distinct strokes before and after the bell and in the after end of the vessel the gong for about 10 seconds every minute.
- Sound the bell for about 5 seconds, plus three distinct strokes before and after the bell and in the after end of the vessel the gong for about 5 seconds every 2 minutes.
- Sound three blasts on the whistle, namely one long blast followed by two short blasts every two minutes.

What is understood by the term "prolonged blast"?

- **A blast of from four to six seconds in duration.**
- A blast of about one second in duration.
- A blast of about three seconds in duration.
- A blast of about two seconds in duration.

What is understood by the term "short blast"?

- **A blast of about one second in duration.**
- A blast of about four seconds in duration.
- A blast of about three seconds in duration.
- A blast of about two seconds in duration.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when altering course to port?"

- **Two short blasts.**
- A minimum of five short blasts.
- One short blast.
- Three short blasts.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when altering course to starboard?

- **One short blast.**
- A minimum of five short blasts.
- Three short blasts.
- Two short blasts.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when going astern on the engines?"

- **Three short blasts on the whistle.**
- Five short and rapid blasts on the whistle.
- One short blast on the whistle.
- Two short blasts on the whistle.

Which fog signal should a power driven pilot vessel under way and making way through the water on pilotage duty, sound in reduced visibility?

- **One long blast on the whistle every two minutes and may sound an identity signal of four short blasts.**
- An identity signal of four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes and an identity signal of four short blasts.

Which fog signal should a power driven vessel constrained by her draught, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a sailing vessel sound in reduced visibility?

- **One long blast followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel engaged in fishing, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel engaged in towing or pushing another vessel, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel not under command, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

Which fog signal should a vessel restricted in her ability to manoeuvre, sound in reduced visibility?

- **One long followed by two short blasts on the whistle every two minutes.**
- Four short blasts on the whistle every two minutes.
- One long blast on the whistle every two minutes.
- Two long blasts on the whistle every two minutes.

You are Officer of the Watch and you are approaching a bend in a narrow channel, where you can not see round the bend. What action should you take?

- **You should give one long blast on the whistle.**
- You should give one short blast on the whistle.
- You should give three short blasts on the whistle.
- You should give two short blasts on the whistle.

You are officer of the watch and you are approaching a bend in a narrow channel, where you can not see round the bend and you sound one long blast on the whistle. What action would you expect an approaching vessel on the other side of the bend to answer to your signal?

- **It should answer with one long blast on the whistle.**
- It should answer with one long one short one long and one short blast on the whistle.
- It should answer with one short blast on the whistle.
- It should answer with two short blasts on the whistle.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What is the meaning of this signal? Select the ONE best answer.

- **It is a ship that is intending to pass you on your starboard side where you need to take some action to allow this manoeuvre.**
- It is a pilot vessel coming up from the stern.
- It is a ship that is intending to pass you on your port side.
- It is a ship that is intending to pass you on your starboard side.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What should you give in response to this signal, if you feel doubtful of the safety of their intended manoeuvre?

- **At least five short and rapid blasts.**
- One long blast.
- One long one short one long and one short blast.
- One short blast.

You are officer of the watch of a power driven vessel proceeding in a narrow channel and hear two long blasts followed by one short blast coming from astern of your vessel. What should you give in response to this signal? Select the ONE best answer.

- **If in agreement to being overtaken - one long, one short, one long and one short blast and take the appropriate action to allow passing.**
- If in agreement to being overtaken - one long, one short, one long and one short blast.
- One long blast.
- One short blast.

0304 – COLREGs, Conduct of vessels

Version 2.0 (539)

A bearing is taken of an approaching vessel and shortly afterwards another observation is made of the same vessel. The bearing is changing slowly, but under what circumstances (as defined in the Rules) could there still be a risk of a collision in this situation? Select the ONE best answer.

- **All of the suggested answers.**
- When approaching a very large vessel.
- When approaching a vessel at close range.
- When approaching a vessel engaged in towing.

A power driven vessel is sighted coming towards own ship from 45 degrees on the port side at a distance of two nautical miles and the bearing is still not changing. What is the first action you should take?

- **There is a close quarter situation developing and the other vessel is not keeping clear, therefore I would sound at least 5 short and rapid blasts to indicate my concern and monitor the situation.**
- There is a close quarter situation developing and the other vessel is the "keep clear" vessel, therefore I should maintain my speed and course.
- There is a close quarter situation developing and under the Rules both vessels should give way to the other.
- There is a close quarter situation developing, but I am the "give way" vessel and therefore I shall keep well clear of the other ship.

A power driven vessel is sighted coming towards your ship from 45 degrees on the port side, at a distance of five nautical miles and the bearing is not changing. What is the initial action you should take?

- **There is a risk of a close quarter situation, but the other ship is the "give way" ship, so I should maintain my course and speed, but monitor the situation.**
- There is a risk of a close quarter situation and therefore both vessels should give way to each other.
- There is a risk of a close quarter situation, but I am the "give way" vessel and therefore I shall keep well clear of the other ship.
- There is no risk of a close quarter situation and therefore I should maintain my speed and course.

A power driven vessel is sighted coming towards your ship from 45 degrees on the port side. The bearing is not changing and it is getting closer and appears to be taking no action. You have sounded five or more short and rapid blasts and it still appears to be taking no action. As "stand-on" vessel can you take action under the rules at this time?

- **Your vessel may take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to you that the vessel required to keep out of the way is not taking appropriate action.**
- You must take substantial action altering your course to the port and give two short blasts.
- You must take substantial action and give two short blasts and change your course to the starboard.
- You must take substantial action, and give one short blast and change your course to the starboard.

A vessel is observed approaching from the port side showing two black balls. A series of bearings indicate that a close quarter situation is likely to occur. As Officer of the Watch, indicate what action should be taken within the meaning of the Rules.

- **Take early and substantial action to keep well clear.**
- Give two short blasts on the whistle and alter course to port.
- It would be expected that the vessel will be slowing down and therefore no action should be necessary.
- Maintain your course and speed.

A vessel is observed approaching from the port side, showing two red lights and also a green side light. A series of bearings indicate that a close quarter situation is likely to occur. As Officer of the Watch, indicate what action should be taken within the meaning of the Rules.

- **Take early and substantial action to keep well clear.**
- Give two short blasts on the whistle and alter course to port.
- It would be expected that the vessel will be slowing down and therefore no action should be necessary.
- Maintain your course and speed.

Another vessel is observed dead ahead, steaming in the opposite direction. What action would the Officer of the Watch expect from the other vessel?

- **Expect the other vessel to alter its course to the starboard.**
- Expect the other vessel to alter its course to the port.
- Expect the other vessel to maintain their course and speed.
- Expect the other vessel to slow down.

Are there any particular rules for sailing vessels and vessels less than 20 metres in length, when they are proceeding in a narrow channel?

- **They should not impede the passage of a vessel that can only safely navigate within a narrow channel of fairway.**
- They should always keep to the port side in a narrow channel.
- They should always stop and allow larger vessels to pass in the narrow channel.
- They should not pass a vessel that can only safely navigate within a narrow channel of fairway.

As Officer of the Watch you sight another power driven vessel on the port side approaching your vessel on a collision course. Which ONE of the following answers is the correct action under the Rules if the other vessel appears to be taking no action to avoid a close quarter situation?

- **You give a signal to the approaching vessel consisting of at least five short and rapid blasts combined with a light signal and keep monitoring the situation.**
- You give a signal to the approaching vessel consisting of one short blast combined with a light signal and altering your course to the starboard.
- You give a signal to the approaching vessel consisting of two short blasts combined with a light signal and altering your course to the port.
- You slow down and let the other vessel pass ahead.

As Officer of the Watch you sight another power driven vessel on the port side approaching your vessel on a collision course. Which ONE of the following answers is the correct initial responsibility under the Rules?

- **Your vessel should maintain its speed and course.**
- Your vessel is obliged to keep out of the way of the other vessel therefore you alter your course to the port.
- Your vessel is obliged to keep out of the way of the other vessel therefore you alter your course to the starboard.
- Your vessel is obliged to keep out of the way of the other vessel therefore you should slow down and allow the other vessel to pass ahead.

As Officer of the Watch you sight another power driven vessel on the starboard side approaching your vessel on a collision course. Which ONE of the following answers is the correct initial action to be taken under the Rules?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed.
- Make an early and substantial alteration of course to starboard.
- Sound two short blasts on the whistle and alter course to port.

As Officer of the watch of a power driven vessel, what should you take into consideration when entering a narrow channel?

- **Proceed at a safe speed, keeping to the starboard side of the fairway.**
- Be prepared to reduce your speed and keep to the starboard side of the fairway.
- Proceed with great care and keep to the port side of the fairway.
- Reduce speed to a speed just enough to maintain steerage.

The visibility is about one mile and as Officer of the Watch you visually see a vessel dead ahead on a reciprocal course coming out of the fog towards you. What should be the action to take?

- **Give one short blast and make a substantial alteration of course to starboard.**
- Alter course to starboard and continue sounding the fog signal, but do not sound one short blast, as this is for clear visibility only.
- Maintain your course and speed.
- Stop your vessel and change the fog signal to two long blasts every two minutes until the vessel has passed.

Under what circumstances should a vessel consider itself to be an overtaking vessel within the meaning of the Rules? Select the ONE best answer.

- **A vessel is overtaking when coming up with another vessel from a direction more than 22.5 degrees abaft her beam or if in any doubt she shall assume that she is.**
- When you are faster than the other vessel.
- When you are overtaking the other vessel from a direction more than 20 degrees abaft her beam.
- When you are overtaking the other vessel from abaft her beam.

What manoeuvring signal would a power-driven vessel give to another vessel when they are in sight of each other, when altering course to starboard?

- **One short blast.**
- A minimum of five short blasts.
- Three short blasts.
- Two short blasts.

Which of the following statements is correct within the Rules?

- **You should join a traffic separation system at as small angle to the general direction of traffic flow as practicable.**
- It is important that all vessels in a traffic separation system keep the same speed.
- You are not allowed to fish in a traffic separation area.
- You can enter a traffic separation only at the end points.

Which vessels are allowed to use an inshore traffic lane of a Traffic Separation Scheme?

- **All of the suggested options.**
- A vessel shorter than 20 metres.
- Sailing vessels.
- Vessels engaged in fishing.

You are Officer of the Watch and see another vessel displaying two cones, points together and a cone point up on one side. This vessel is sighted dead ahead and the bearing is not altering. What actions should be taken?

- **Take early and substantial action to keep well clear of the other vessel.**
- Sound not less than five short and rapid blasts on the whistle.
- The other vessel should give way to your vessel.
- You give it right of way and alter your course to best keep to your track.

You are Officer of the Watch of a power driven vessel and your vessel is proceeding in fog, when you hear the fog signal of another power driven somewhere ahead of you. The signals seem to be getting closer, but nothing is detected on the radar. How should you proceed within the meaning of these rules?

- **Reduce the speed to lowest manoeuvring speed and if necessary take all the way off and proceed with extreme caution.**
- Slow down and proceed with caution.
- Slow down, keep a sharp lookout and change the fog signal to two long blasts.
- Stop the vessel and sound one long blast followed by two short blasts.

You are Officer of the Watch of a power driven vessel when your vessel encounters fog during your watch. What are the important initial actions that should be taken under these Rules?

- **All of the suggested answers.**
- Call the Master.
- Ring Standby on the engines and reduce the speed to a safe speed as soon as possible.
- Start sounding the appropriate fog signal.

You are Officer of the Watch on a power driven vessel in a Traffic Separation Scheme and sight a fishing vessel engaged in fishing inside the Traffic Separation Scheme? Who has the responsibility to keep clear?

- **A vessel engaged in fishing shall not impede the passage of any vessel following a traffic lane.**
- A fishing vessel is not allowed to fish within the Traffic separation Scheme and should be forced to leave the area.
- A power driven vessel should keep clear of a vessel engaged in fishing.
- Your vessel is in the Traffic Separation Scheme and therefore does not need to keep clear of other vessels.

You are Officer of the Watch on a vessel meeting another vessel in clear visibility on a collision course and you are the "give way" vessel. Under the Rules what action should you take to avoid a close quarter situation?

- **You should take early and substantial action to keep well clear.**
- You should avoid a collision by altering course to port and giving two short blasts, under these Rules.
- You should avoid a collision by sounding one short blast and altering course to starboard.
- You take whatever action is necessary to avoid a collision.

You are Officer of the Watch on a vessel when you observe another two vessels crossing from port. Both of these vessels are showing a black diamond and one is also showing three shapes, the highest and lowest are black balls and the middle one a black diamond. What would be the most suitable initial action to be taken by your vessel under these Rules, if the bearing of the nearest vessel is only changing slowly?

- **Slow down your vessel and allow the two vessels to pass ahead.**
- Maintain your course and speed as the vessels are crossing from port.
- Maintain your course and speed until you are close to the other vessel, in case it makes an alteration of course.
- Sound one short blast and make a series of small alterations of course to starboard.

You are Officer of the Watch on a vessel when you observe another two vessels crossing from port. Both of these vessels are showing a black diamond. What would be the correct initial action to be taken by your vessel under these Rules, if the bearing of the nearest vessel is only changing slowly?

- **Maintain your course and speed and continually monitor the situation.**
- Slow down your vessel and allow the two vessels to pass ahead.
- Sound one short blast and alter your course to the starboard.
- Sound two short blasts and alter course to port.

You are Officer of the Watch on a vessel when you observe another vessel under full sail showing a black cone apex downwards and on a constant bearing of about 50 degrees on your port bow. What would be the correct initial action to be taken by your vessel?

- **Maintain your course and speed.**
- Alter your course to the port and sound two short blasts on the whistle.
- Alter your course to the starboard and sound one short blast on the whistle.
- Reduce your speed until the other vessel has passed ahead.

You are Officer of the Watch proceeding in fog and hear one long blast every two minutes coming from ahead by a vessel approaching, which you have been tracking on your ARPA. You know that there is a risk of a close quarter situation. What would be the best action to take?

- **You should take avoiding action in goog time, avoiding an alteration to port.**
- You reduce your speed to a minimum manoeuvring speed, navigate with extreme caution and keep your engines ready to stop.
- You should alter your course 15 degrees to the starboard.
- You should make a series of small alterations to starboard.

You are Officer of the Watch proceeding on your voyage in bad visibility. You have plotted a target on the radar screen and see that it is a ship on a collision course. What are the recommended actions to be taken in this situation?

- **All of the suggested answers.**
- Avoid an alteration of course towards a vessel abeam or abaft the beam.
- Avoid an alteration to port for a vessel forward of the beam, other than for a vessel being overtaken.
- Take action in ample time and avoid a series of small alterations.

You are bridge Officer of the Watch and observe another vessel approaching from the port side across your course line and showing two black cones, points together and a cone point upwards on its starboard side. What would be the correct action to be taken by your vessel, if a close quarter situation is developing?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed for a crossing vessel from port.
- Sound one short blast and alter course to port.
- Sound two short blasts and alter course to starboard.

You are bridge Officer of the Watch and observe another vessel approaching from the port side across your course line, showing a red light over a white light and another white light on its starboard side and also a green side light. What would be the correct action to be taken by your vessel, if a close quarter situation is developing?

- **Take early and substantial action to keep well clear.**
- Maintain your course and speed for a crossing vessel from port.
- Sound one short blast and alter course to port.
- Sound two short blasts and alter course to starboard.

You are bridge Officer of the Watch in poor visibility and plot another vessel on the radar approaching your ship from the port side so as to involve a close quarter situation. Which of the following is the most suitable action required under these Rules?

- **Make an early and substantial alteration of course to starboard.**
- Maintain your course and speed as the “stand-on” vessel.
- Make a small reduction in speed to allow the other vessel to pass ahead.
- Make an early and substantial alteration of course to port.

You are on a power driven vessel approaching the vessel illustrated (showing one black ball), directly ahead on your course line. What is the correct action to be taken by your vessel?

- **Taking into account any navigational hazards, take early and substantial action to keep well clear.**
- Slow down and allow the other vessel to pass clear of your vessel.
- Taking into account any navigational hazards alter course to starboard.
- Taking into account any navigational hazards, you should adjust your course to avoid passing ahead of the other vessel.

You are on a power driven vessel approaching the vessel illustrated (showing two white lights), directly ahead on your course line. What is the correct action to be taken by your vessel?

- **Taking into account any navigational hazards, take early and substantial action to keep well clear.**
- Slow down and allow the other vessel to pass clear of your vessel.
- Taking into account any navigational hazards alter course to starboard.
- Taking into account any navigational hazards, you should adjust your course to avoid passing ahead of the other vessel.

You are on a vessel approaching a Traffic Separation Scheme and wish to enter a port within the inshore traffic zone on the side of the Scheme which is used by vessels coming in the opposite direction. What is the correct approach to safely gain entry into the destination port in compliance with the Rules?

- **Take the "inshore traffic zone" and keep out of the main lanes of the scheme.**
- Enter the scheme in the correct main lane and at the appropriate position alter hard over and cross the other lane on a course at right angles to the main traffic flow.
- Enter the scheme into the correct main lane and cross over the scheme at as small an angle as possible to the general direction of traffic flow.
- Take the correct main lane until the end of the scheme and then return down the other lane until arriving at the appropriate position to exit for the destination.

Your vessel is a deep draught vessel and is approaching the entrance to a narrow channel. How should you proceed? Select the ONE best answer.

- **Reduce to a safe speed, keep a sharp lookout and keep to the starboard side of the fairway.**
- Engines on standby ready to reduce your speed when required and stay to the starboard side of the fairway.
- Reduce the speed and keep in the centre of the channel for maximum depth.
- Reduce your speed and keep a sharp lookout.

Your vessel is approaching a bend in a narrow channel with an obstructed view ahead. How should you proceed?

- **All of the suggested answers.**
- Keep to the starboard side of the fairway.
- Proceed at a safe speed.
- Sound one long blast.

0306 – Incident investigation, Cause and effect

Version 2.0 (2214)

A Junior Engineer is injured during the replacement of a main engine cylinder liner. An investigator, looking at the period just before the accident, may look for two distinct types of actions (referred to in this e-learning module). What are they? Select the TWO correct answers.

- **Commission.**
- **Omission.**
- Prevention.
- Solution.

A stack of containers has collapsed onto the deck causing damage, but no injuries. What is the first thing the investigator should identify? Select the ONE best answer.

- **The transfer of energy.**
- How the greatest damage was caused.
- The sequence of events leading to the incident.
- Where the biggest impact has taken place.

As described in this training module, an accident is generally caused by contact with which TWO of the following? Select the TWO correct answers.

- **A source of energy.**
- **A substance above the threshold limit of the body concerned.**
- A body capable of generating an extremely high level of energy.
- A body within which all energy has been contained during an impact.

Immediate causes are generally divided into which TWO of the following categories? Select the TWO correct answers.

- **Substandard acts.**
- **Substandard conditions.**
- Job factors.
- Underlying factors.

In an incident investigation, complacency, incompetence, negligence, recklessness and lack of communication are all examples of what? Select the ONE correct answer.

- **Fallibility.**
- Bad luck.
- Non-compliance.
- Pre-existing management failure.

Inadequate compliance is normally identified by which TWO of the following circumstances? Select the TWO correct answers.

- **Evidence that there was inadequate knowledge or understanding of a standard.**
- **Evidence that there was knowledge of a declared standard but that it was deliberately ignored.**
- Evidence that the injured party had undergone a training course.
- The feeling that those involved in an incident have not listened to instructions.

Root causes are generally divided into which TWO of the following categories? Select the TWO correct answers.

- **Job factors.**
- **Personal factors.**
- Fundamental causes.
- Substandard conditions.

The maximum safe working load of a crane has been exceeded. When the operator attempted the lift, the crane mounting sheared and the crane toppled over. Which TWO of the following may be considered as evidence of "fallibility" in this situation? Select the TWO correct answers.

- **Incompetence.**
- **Negligence.**
- Poor design.
- Poor maintenance.

What are the three common reasons for lack of control? Select the THREE correct answers.

- **An inadequate system.**
- **Inadequate standards.**
- **Poor compliance with standards.**
- Lack of personnel.
- Poor commitment to training.
- Poor communication.

What is the name of the O.C.I.M.F. inspection system which includes verifying if an accident and incident investigation policy exists within a company? Select the ONE correct answer.

- **The SIRE Programme.**
- The International Risk Assessment Guide for Ship and Terminal Operators.
- The IPPA Initiative.
- The SOVA Scheme.

What is the purpose of investigating incidents and using the knowledge gained from the investigation? Select the ONE best answer.

- **A reduction, or even elimination of such incidents in the future.**
- A more efficient system will be developed for the purposes of allocating blame.
- A reduction in personal injuries, but an increase in incidents involving material damage.
- An increased likelihood of such incidents in the future.

What name do we give to the condition that starts the process of cause and effect and ultimately results in an incident? Select the ONE correct answer.

- **Lack of control.**
- Improper risk assessment.
- Intentional abuse.
- Lack of training.

Which ONE of these options best describes a "blame culture"? Select the ONE best answer.

- **One where people look no further than "human error".**
- One in which the level of responsibility of each person involved in an incident is allocated as a percentage.
- One where investigators seek to blame as many people as possible.
- One which seeks to avoid giving any mention of blame.

Which TWO of the following are examples of abnormal behaviour? Select the TWO correct answers.

- **Operating a crane without having any training or familiarization.**
- **Starting to overhaul a ballast valve, without first checking that it has been properly isolated and the appropriate warning notices put in place.**
- Carrying out a risk assessment and completing a "Permit to work aloft" form before repairing a floodlight on the front of the accommodation.
- Wearing footwear with steel toe caps during storing.

Which TWO of the following are examples of an incident due to transfer of energy by contact? Select the TWO correct answers.

- **A steward opens a locker door and is hit by falling crockery as the ship rolls.**
- **Damage to spare parts in storage when a work bench broke loose in the electrician's workshop during heavy weather.**
- Failure of an electrical control panel has resulted in the bilge pump not starting.
- Wear and tear on the shaft of a sledge hammer has resulted in the head working loose.

Which TWO of the following are examples of latent, or hidden, factors that may be found to contribute to an accident? Select the TWO correct answers.

- **Design.**
- **Faulty maintenance.**
- New procedures.
- Substandard practices.

Which TWO of the following are examples of possible "hidden" costs resulting from an accident or incident? Select the TWO correct answers.

- **Increased insurance premiums.**
- **Lost time.**
- Material damage.
- Pollution.

Which TWO of the following are examples of post-contact controls? Select the TWO correct answers.

- **Emergency evacuation plans.**
- **Search and Rescue operations.**
- Briefing of the ship's crew before the operation.
- Preparation of firefighting equipment for possible use.

Which TWO of the following are examples of substandard conditions? Select the TWO correct answers.

- **Crowded or restricted work area.**
- **Inadequate or defective warning systems.**
- Operating equipment at an improper or unsafe speed.
- Working under the influence of alcohol or drugs.

Which TWO of the following are examples of substandard practises? Select the TWO correct answers.

- **Failure to use personal protective equipment.**
- **Overriding or bypassing safety devices.**
- Exposure to excessive high or low temperatures.
- Missing guards or barriers.

Which maritime Code requires shipping companies to investigate incidents in order to increase safety? Select the ONE correct answer.

- **The ISM Code.**
- MARPOL.
- SOLAS.
- The ISPS Code.

Which of the following may be categorized as 'lack of knowledge'? Select the THREE correct answers.

- **Inadequate orientation or briefing.**
- **Inadequate updating or refresher training.**
- **Lack of experience.**
- Excessive frustration.
- Poor support.

Within this e-learning module, which of the following is the definition of a "near-miss"? Select the ONE best answer.

- **A near miss is an undesired circumstance with the potential to cause harm, injury, ill health, damage to equipment or the environment.**
- A near miss is an undesired event which did not cause harm to people.
- A near miss is an undesired event which does not need to be reported.
- A near miss is an undesired event which only caused minor injury to people or minor damage to machinery.

0307 – Incident investigation, Investigation techniques

Version 2.0 (796)

A Superintendent is going on board to conduct an investigation into an accident in which a crewmember was seriously injured during the stowage of an accommodation ladder. Which THREE of the following should they particularly try to interview? Select the THREE best answers.

- **Any eye witnesses**
- **Anyone who can contribute to the understanding of the events leading up to, or the conditions at the time of, the accident**
- **The person in charge of the operation at the time of the accident**
- A representative from the manufacturer of the accommodation ladder
- All crewmembers

As described in this training module, Which THREE of the following are common interview techniques? Select the THREE correct answers.

- **Conversational approach**
- **Free recall**
- **Question and answer**
- Numerical assessment
- Trick questions

As described in this training module, what is the definition of an "impartial" witness? Select the ONE correct answer.

- **One who has no personal interest and nothing to gain, or lose**
- One who has a problem with authority and feels threatened by the interview
- One with a personal interest in the outcome of the inquiry
- One with whom there is a language-based communication problem

Following an accident onboard, there is an enquiry from a lawyer on the ship's satellite telephone, whilst at anchor awaiting port entry. Who should the Duty Officer direct the call to? Select the ONE correct answer.

- **The Master**
- The ship's Agent
- The Ship's Safety Officer
- The Ship's Security Officer

What is the purpose of incident investigation? Select the ONE best answer.

- **To establish the facts and prevent a repeat of the incident. It is not to allocate blame.**
- Mainly to allocate blame, and also identify relevant contributory factors. It may suggest corrective action.
- The completion of the necessary forms, so that an auditor can see that a system for documenting incidents exists.
- To draw up corrective action, including disciplinary procedures where appropriate. Contributory factors may not necessarily be identified.

What is the purpose of investigating incidents and using the knowledge gained from the investigation? Select the ONE best answer.

- **A reduction, or even elimination of such incidents in the future**
- A more efficient system will be developed for the purposes of allocating blame
- A reduction in personal injuries, but an increase in incidents involving material damage
- An increased likelihood of such incidents in the future

Where should you look first for information about how to investigate an incident on board ship? Select the ONE best answer.

- **Your company's Safety Management System**
- Classification Society Rules
- Information from your flag State
- The internet

Which ONE of the following best describes how the narrative section of an incident investigation report should be written? Select the ONE best answer.

- **An orderly description of the events leading up to the incident, what happened during the incident itself, the consequences and the post-incident actions, written in a neutral way, without opinion or judgment.**
- A detailed account of the incident, reflecting, so far as possible, the investigator's understanding of the circumstances and including recommendations for the attention of management.
- A random description of the events preceding an incident and the corrective actions necessary to prevent a recurrence, written in a neutral way, but including the investigator's summary assessment.

Which ONE of these statements best describes what should be done with the recommendations from an incident investigation report? Select the ONE best answer.

- **Recommendations from incident reports should be circulated in terms that can be readily understood by everyone on board**
- Recommendations from incident reports should be written in legal terms so that lawyers cannot dispute them
- Recommendations from incident reports should only be circulated at a high level within the company
- Recommendations from incident reports should only be circulated on the vessel where the incident took place

Which THREE of the following suggest that the original entries in an engine-room log-book have been amended at a later date? Select the THREE correct answers.

- **An entry between others in a different handwriting style**
- **An entry between others using a different pen**
- **An entry made by someone who was not on watch at the time**
- An entry found to have been made by the Chief Engineer

Which TWO of the following will usually be an investigator's aims when planning a series of interviews as part of an investigation? Select the TWO correct answers.

- **To establish and confirm the sequence of events**
- **To understand how the witness was involved in the incident**
- To determine if the witness is to blame for the incident
- To determine if the witnesses have "arranged a story" between them

You are investigating a minor incident involving a fall on a lifeboat access platform. A Seaman who was working nearby at the time is being interviewed and is having difficulty remembering some of the details. Which TWO of the following techniques could be used to help him remember? Select the TWO correct answers.

- **Talk through the incident with the Seaman**
- **Work through a list of potentially relevant items, asking the Seaman to include or reject them**
- Ask the Seaman to say anything, even if he is not absolutely sure of the facts, so that you have some testimony from him for the investigation report
- Get tough with the Seaman and tell him how important it is that he remembers

You are investigating an incident in the engine room and planning to interview a number of crewmembers. Which TWO of the following should you do, before starting the interviews? Select the TWO correct answers.

- **Analyse the evidence already found**
- **Identify evidence still needed**
- Work out in advance who is lying and how to trap them in the interview
- Write questions which will find evidence to support your ideas about what happened

You are investigating an incident which involved a broken wire on a mooring winch and obtaining evidence with photographs. How would you try to photograph the wire? Select the ONE best answer.

- **In place, still on the winch drum**
- Ashore, after removal from the vessel
- In the forecastle store, after being removed from the winch drum
- Photographs are of little value as evidence

0310 – Working at height

Version 1.0 (2066)

How are personnel best protected from an exposed opening? Select the best option.

- **Fence the opening.**
- Do not let personnel work near the opening.
- Light the opening with a portable floodlight.
- Tell everyone about it.

How should personnel be protected from work being undertaken above their heads? Select the best option.

- **They should be kept clear of the dangerous area.**
- There should be warning signs in the area.
- They should be told about the work being done.
- They should only be allowed to pass through the area for access.

Identify one issue which makes working from a vertical ladder difficult on a vessel? Select the best answer.

- **Two hands are needed to safely hold onto the ladder.**
- It is painful standing on a rung.
- Only one person can work in that location.
- The bosun is not present to give advice.

Is it normal to be afraid of heights? Select one of the options.

- **Yes – it is a normal human response.**
- No – fear of heights only affects nervous people.
- No – it is very unusual.
- Yes – it is a way of avoiding strenuous work.

Is it safe to carry a bag of tools in your hand, up a vertical ladder? Select the best option.

- **No it is not –two hands should always be used to climb the ladder.**
- Yes it is - provided the ship is not rolling too badly
- Yes it is – provided that the carrying hand is wrapped around the ladder
- Yes it is – provided three points of contact are maintained.

Select one option to correctly complete the following statement. 'A dropped object increases its' . . .

- **. . . kinetic energy as it falls'.**
- . . . latent energy as it falls'.
- . . . potential energy as it falls'.
- . . . thermal energy as it falls'.

Select one option to correctly complete the following statement. 'When working at height, the safety helmet chin strap . . .

- **. . . should be used to hold the helmet in place'.**
- . . . should be always detached from the helmet'.
- . . . should be fastened over the top of the brim'.
- . . . should be kept loose'.

Select the one option which correctly completes the following statement. 'A portable ladder . . .

- **. . . should be secured before working from it'.**
- . . . should be about 10 degrees to the vertical'.
- . . . should be about 20 degrees to the vertical'.
- . . . should be about 5 degrees to the vertical'.

Select the one option which correctly completes the following statement. 'A portable ladder . . .

- **. . . should be set up at about 30 degrees to the vertical'.**
- . . . should be set up at about 10 degrees to the vertical'.
- . . . should be set up at about 20 degrees to the vertical'.
- . . . should be set up at about 5 degrees to the vertical'.

Should a powered winch be used to haul a person aloft? Select one option.

- **No - it is not designed for that purpose.**
- No – unless they cannot do it themselves.
- Yes, because a person may be too heavy to do it manually.
- Yes, because it saves a lot of effort.

What is the greatest personal risk when working from a stairway on a vessel? Select the best option.

- **A person may fall down the stairs or over the handrail.**
- Other persons are inconvenienced when trying to pass.
- The stairway would have to be out of bounds for others.
- There are none, as the stairways are safe.

What should you do with a bosun's chair before using it? Select the best option.

- **Personally check it to make sure it is safe to use.**
- Ask the bosun to check it.
- Tie it on the gantline with a bowline.
- Wet the rope to tighten up the splice.

When someone is working at height, who is at risk? Select the best option.

- **There are risks to people at height and anyone below them.**
- There are risks to all members of a vessel's crew.
- There are risks to people working below only.
- There are risks to the people doing the job only.

Which of the options correctly describes the minimum load that an anchor point should be able to hold?

- **1000kg.**
- 10000kg.
- 100kg.
- 10kg

Which of these describes a basic safety harness? Pick the best option.

- **A waist belt.**
- A full body belt.
- A hip belt.
- A shoulder belt.

Which of these is the correct name for a more advanced safety harness than a simple waist belt? Pick the best option.

- **A full body harness.**
- A full body belt.
- A torso harness.
- A waist harness.

Which one of the options best completes the following statement? 'A fall arrest anchor point . . .

- **. . . is an attachment point for a safety lanyard'.**
- . . . is an attachment point for tools'.
- . . . is located at the base of a guided fall arrest system'.
- . . . is where safety barriers are connected to'.

Which one of the options best completes the following statement? 'Before a basket transfer, personnel . . .

- **. . . should be fully briefed about the operation'.**
- . . . should be attached to the vessel with safety harnesses.
- . . . should be told nothing'.
- . . . should be told of the dangers'.

Which one of the options correctly completes the following statement? 'A fall arrest system . . .

- **. . . stops a worker falling further'.**
- . . . is a system only found on a drill rig'.
- . . . is only needed for inexperienced climbers'.
- . . . stops a worker being over, or close to, a drop'.

Which one of the options correctly completes the following statement? 'A fall restraint . . .

- **. . . stops a worker getting close to a dangerous edge'.**
- . . . is a poor type of fall protection system'.
- . . . is too restricting in the work place'.
- . . . will suspend a worker that has fallen'.

Which one of the options correctly completes the following statement? 'A full body harness . . .

- **. . . minimises body injury in a fall arrest'.**
- . . . is only necessary for falls greater than 10 metres'.
- . . . maximises body injury in a fall arrest'.
- . . . stops a person falling'.

Which one of the options correctly describes how a safety harness should be attached to an anchor point?

- **Using an approved lanyard.**
- It is directly shackled onto it.
- Using a piece of gantline.
- With a hook and a line.

Which one of these is an advantage of a full body safety harness compared to a waist belt safety harness?

- **It is more secure on the body of a person.**
- It fits underneath the clothing worn by a person.
- It is lighter to wear.
- It takes a shorter time to put on.

Which one of these is an advantage of a full body safety harness compared to a waist belt safety harness?

- **A person is suspended head up.**
- A person can more easily remove the harness.
- A person is not allowed to fall.
- A person is suspended horizontally.

Which one of these options best describes a personnel basket transfer?

- **It is a method of transferring people from a vessel's deck.**
- It describes when a cargo basket is lifted off a vessel's deck.
- It is a system used to transfer the vessel's manifest.
- It is a way of loading food stores to a drill rig.

Which one of these options best describes how a safety harness should be worn?

- **Following the maker's instructions.**
- As slack as possible, to allow free movement to work.
- As tight as possible, so that it feels restricting.
- Underneath a jacket when the weather is cold.

Which one of these options correctly describes a guided fall arrest system?

- **It is a safety wire running the length of a ladder.**
- It is a horizontal safety wire across a work area.
- It is a system of pulleys only used on drilling rigs.
- It is a system to let personnel practice a fall.

Why does a heavy object cause more damage than a light object if dropped from the same height? Select the best answer.

- **It takes more force to stop a heavier object.**
- Because heavier objects are bigger in size.
- Because heavier objects are smaller in size.
- There is less force on the heavy object.

Why is working at height on a moving vessel more risky than working at height ashore? Select the best option.

- **The moving causes extra loads.**
- Safety standards ashore are higher than those at sea.
- The vessel can be far from land and rescuers.
- There is no difference between the two work places.

0311 – Vessel inspection and CMID

Version 1.0 (1345)

Are there any consequences following a poor CMID inspection? Select the ONE best answer.

- **Yes, as the vessel may be unsafe.**
- No, as it is only one person's opinion.
- No, as remedial actions will be taken.
- Yes, as port state inspectors would be informed.

At the end of a CMID vessel inspection, what should the inspector do before they leave a vessel? Select the ONE best answer.

- **Give verbal feedback to the master and crew.**
- Instruct the master about the work needed.
- Issue a paper copy of the full report to the vessel.
- Telephone the company owning or managing the vessel to report on it.

At the start of a CMID vessel inspection, what will the inspector often check first? Select the ONE best answer.

- **Any observations from the last CMID inspection.**
- Any areas reported by the chief officer as being doubtful.
- The deck area on a vessel.
- The engine room and machinery space on a vessel.

Choose ONE option that best completes the following statement. 'The International Marine Contractors Association supports the interests of . . .

- **. . . offshore, marine and underwater engineering companies'.**
- . . . offshore, drilling and oil companies'.
- . . . oil and gas exploration and processing companies'.
- . . . organisations such as IMO and the MCA'.

How long should a typical CMID inspection last? Select the ONE best answer.

- **One working day.**
- Five working days.
- One hour.
- Six hours.

How many numbered sections are there in a full CMID report? Select the ONE correct answer.

- **18.**
- 15.
- 20.
- 22.

How often should an offshore vessel be inspected using CMID? Select the ONE correct answer.

- **Once a year.**
- Every second year.
- Every six months.
- Once every five years.

Select ONE answer to correctly complete the following statement. 'Lifting equipment on an offshore vessel . . .

- **. . . should be managed properly'.**
- . . . is not part of a CMID vessel inspection'.
- . . . is the responsibility of the vessel charterer'.
- . . . should be tested once a year by an external organisation'.

Select ONE answer to correctly complete the following statement. 'The CMID common inspection document . . .

- **. . . is used and accepted in the offshore sector'.**
- . . . is not used or accepted in the offshore sector'.
- . . . is only fit for use at a basic level'.
- . . . was developed by academics and hence is not always practical'.

Select ONE answer to correctly complete the following statement. 'The CMID inspection . . .

- **. . . should be seen by the crew as a helpful process'.**
- . . . is a waste of time as it does not check everything'.
- . . . should be completed as quickly as possible'.
- . . . should be seen as inconvenient but necessary'.

Select one answer to correctly complete the following statement. 'Before a CMID inspection on a well run vessel . . .

- **. . . there should not be many extra preparations'.**
- . . . help will be needed from shore workers to clean the vessel'.
- . . . the vessel will have to stop operations'.
- . . . there will be a lot of work to get the vessel to an acceptable standard'.

Select the ONE answer that best completes the following statement: 'Sometimes on a vessel, areas listed on the standard CMID reporting form . . .

- **. . . cannot be inspected as they do not exist'.**
- . . . are only found on diving vessels'.
- . . . do not reflect on a vessel's management'.
- . . . should not have to be inspected'.

Select the ONE answer that correctly completes the following statement. 'Before a CMID inspection, a vessel's owners or managers . . .

- **. . . need to arrange timings with the inspection company'.**
- . . . need to detail the items to be inspected with the inspection company'.
- . . . should employ crew specifically for the inspection period'.
- . . . should have a pre-inspection check by an independent company'.

Select the ONE best answer to correctly complete the following statement. 'High priority findings during a CMID inspection . . .

- **. . . need to be addressed immediately by the crew'**
- . . . have to be passed on to the company owning or managing a vessel'.
- . . . need to be addressed within three months of the inspection'.
- . . . will be passed on to the vessel charterer'.

What categories can be used for CMID inspection observations? Select the ONE correct answer.

- **High, medium and low priority.**
- A scale from one to five.
- Acceptable or unacceptable.
- Complies with, or does not comply with, inspection criteria.

What is contained in e-CMID? Select one answer.

- **e-CMID is the electronic version of an inspection report.**
- e-CMID is a database of previous OVID inspections.
- e-CMID is a more basic version of the full CMID.
- e-CMID is a set of instructions to inspectors

What is the biggest advantage of CMID inspection to a vessel's crew? Select the ONE best answer.

- **There is a reduction in the number of inspections.**
- The CMID inspectors are easily satisfied.
- The inspection itself is much easier to undertake.
- The vessel's crew are not involved with the inspection.

What is the biggest advantage of a CMID inspection, compared to relying only on the judgment of a single inspector? Select the ONE best answer.

- **CMID criteria mean there is less subjectivity in the process.**
- IMCA state a minimum time for the inspection.
- Inspectors do not need to understand anything about vessel operations.
- The CMID document covers all vessel types.

What is the main purpose of an offshore vessel inspection using CMID? Select the ONE best answer.

- **To confirm that a vessel is fit for purpose.**
- To check up on a vessel's crew.
- To investigate areas needing work done.
- To satisfy elements of the ISM code.

What should be done with the CMID inspection report when the inspection is finished? Select the ONE best answer.

- **It should be kept onboard as a living, working document.**
- It should be sent to the company.
- It should be stored away on the vessel's office book shelf.
- It should be stored away until the next inspection.

What should the crew do if there are known defects on a vessel during a CMID inspection? Select the ONE best answer.

- **Tell the inspector, mentioning possible corrective actions.**
- Hide them from the inspector so that there are no defects in the report.
- Highlight the defect and complain about the company or owner.
- Say nothing unless the inspector finds them.

What types of vessels are subject to CMID inspections? Select one answer.

- **Offshore vessels.**
- Floating drill rigs.
- Jack-up rigs and platforms.
- Offshore shuttle tankers.

Which one of these should be arranged with an inspection company before a CMID inspection? Select the ONE best answer.

- **Time, duration, documents and personnel available.**
- If the inspector needs a cabin on the vessel.
- The name of the inspector who will undertake the inspection.
- The type and level of detailed to be surveyed.

Which organisation produces the Common Marine Inspection Document used for vessel inspection? Select ONE answer.

- **The International Marine Contractors Association.**
- The International Consultative Organisation.
- The International Marine Safety Forum.
- The International Maritime Organisation.

Who should undertake a CMID pre-inspection check? Select the ONE best answer.

- **The management team on a vessel.**
- A specialised inspection company.
- Nobody. A pre-inspection check is not necessary.
- The company that will undertake the full inspection.

Why is a CMID inspection report important to a vessel master and crew? Select the ONE best answer.

- **The report is a guide to the ability of the vessel's master and crew.**
- Because crew will be dismissed if it is poor.
- Because the yearly bonus is related to it.
- The report is only a guide if the company owning the vessel is good.

Why should an offshore vessel inspection be based on the CMID? Select the ONE best answer.

- **To ensure that the inspection is the same for all similar vessels.**
- To ensure that the inspection can be done quickly.
- To ensure that the inspection is done every five years.
- To ensure that the inspection regime can be varied.

0316 – Anchor handling operation

Version 1.0 (677)

Choose ONE option that best completes the following statement. 'An essential part of a rig move meeting . . .

- **. . . is to discuss the safety of personnel and the ship'.**
- . . . is to complete it quickly before a vessel has to sail'.
- . . . is to focus on how to get the rig move over quickly'.
- . . . is to have a warranty surveyor present'.

Choose ONE option that best completes the following statement. 'Once cargo is loaded for an anchor handling operation . . .

- **. . . it can be lashed for the sea passage'.**
- . . . it remains on deck for the duration of the rig move'.
- . . . the dock workers can assist in lashing it'.
- . . . there would be no other preparations necessary'.

Choose the ONE option that correctly completes the following statement. 'When loading wire onto a winch in port for an anchor handling operation . . .

- **. . . the bridge monitors and controls the operation'.**
- . . . it is best to avoid the working deck to avoid conflict with dock workers'.
- . . . make sure the wire is loaded towards the stern part of the deck'.
- . . . the dock workers can be left to do it without supervision'.

How do the deck crew confirm that a pennant wire is firmly held in a deck stopper? Select the ONE best option.

- **With a visual observation.**
- With confirmation from the bridge.
- With confirmation from the crane driver.
- With information from the winch tension.

How is a rig anchor placed in position on the sea floor by an anchor handling vessel? Select the ONE correct answer.

- **It is lowered using a pennant wire.**
- It is allowed to free fall over the stern roller.
- It is lowered on the anchor chain.
- It is lowered using a crane.

How should an anchor pennant wire be held on deck before releasing the crane hook? Select the ONE best answer.

- **Firmly in the deck stopper.**
- On the anchor winch.
- Over a deck pin.
- With a lashing on the rail.

What equipment should always be checked on an anchor handler before a rig move? Select the ONE best answer.

- **Equipment specific to anchor handling operations.**
- Equipment such as radios, winches and capstans.
- Main engine and thrusters.
- Winches and stern roller rotation system.

What is the outcome of a rig move meeting? Select the ONE best answer.

- **The outcome is the preparation of a deck working plan.**
- The outcome is a report added to the CMID document.
- The outcome is general confusion amongst the crew.
- The outcome is minutes of the meeting which are filed onboard.

What should be done with a decked and disconnected anchor on an anchor handling vessel? Select the ONE correct answer.

- **It should be secured to the barrier.**
- It should be hauled up hard to the anchor winch.
- It should be left close to the stern roller.
- It should be returned to the rig.

What should happen to the deck towing pins on an anchor handler when a work or tow-wire is deployed over the stern? Select the ONE correct answer.

- **They should be raised, closed and locked off to trap the wire.**
- Only one should be raised.
- They should be kept in the stowed position.
- They should both be made ready for immediate use.

When anchor handling, which part of the vessel is particularly dangerous? Select the ONE best answer.

- **The open stern.**
- The main deck.
- The side deck.
- The winch control area.

When spooling on extra wire to a winch in port on an anchor handler, why is it important to have effective communication from the deck? Select the ONE best answer.

- **To ensure the operation is proceeding as planned.**
- To make it easier for the bridge team.
- To make sure the deck crew do what they are told.
- To stop riding turns on the winch drum.

Where should disconnected rig anchor chain be kept on an anchor handling vessel? Select the ONE best answer.

- **In the dedicated rig chain lockers on the vessel.**
- In a deck container.
- In a pile on deck.
- In the vessel's own chain locker.

Which ONE of the options correctly completes the following statement? 'A rig anchor chain that has been intentionally broken to clear a sub surface obstruction . . .

- **. . . will be joined to a wire insert already loaded on a vessel's winch'.**
- . . . is now useless for being used to anchor a rig'.
- . . . would have to be replaced with a new section of chain'.
- . . . would now be too short to anchor a rig effectively'.

Which ONE of the options correctly completes the following statement? 'Clamps are attached to a wire insert in a rig anchor chain system . . .

- **. . . to enable buoys to be attached'.**
- . . . to act as stoppers when the wire is on the deck of an anchor handler'.
- . . . to enable the wire to be attached to a sub-sea structure'.
- . . . to join the wire and chain'.

Which ONE of the options correctly completes the following statement? 'The deck towing pins on an anchor handling vessel are . . .

- **. . . stowed when decking an anchor'.**
- . . . are located near the towing winch'.
- . . . raised when decking an anchor'.
- . . . used to stopper off a wire'.

Which ONE of the options correctly completes the following statement? 'When lowering a rig anchor the deck crew on an anchor handling vessel . . .

- **. . . should clear the working deck'.**
- . . . should be stood down'.
- . . . should observe the progress from the stern area'.
- . . . should start clearing up the working deck'.

Which ONE option correctly completes the following statement? 'An anchor handling safe area for deck crew . . .

- **. . . should be used during certain parts of an anchor handling operation'.**
- . . . is an area for the faint hearted'.
- . . . is the accommodation of the vessel, clear of the deck'.
- . . . should be used at all times during an anchor handling operation'.

Which ONE option correctly completes the following statement? 'When working on deck during an anchor handling operation . . .

- **. . . full protective equipment should always be worn'.**
- . . . dress as advised by the vessel master'.
- . . . only wear a safety hat and eye protection if under the rig'.
- . . . wear clothing to suit the air temperature, hence stay comfortable'.

Which option best explains the objective of having a rig move meeting with the ship's crew? Select the ONE correct answer.

- **To explain and discuss the rig move plan with the crew.**
- To enable a master to instruct the deck crew.
- To satisfy company procedures.
- To satisfy insurance company requirements.

Why are the deck and side pins raised whilst heaving on a wire on an anchor handling vessel? Select the ONE correct answer.

- **They provide additional safety.**
- To be used as a stopper when the wire is required to be held.
- To enable the wire to be attached to them.
- To reduce the tension on the winch.

Why is a person with good communication skills required on the bridge when anchor handling? Select the ONE best answer.

- **To get successful outcomes of co-ordinated activity.**
- To be able to counter false accusations received on the radio.
- To be able to understand a crane driver on a radio.
- To ensure there is always something being said on the radios.

Why is a rig move meeting important on a vessel? Select the ONE best answer.

- **To ensure that all involved in the operation are familiar with the rig move plan.**
- To ensure that the leader of the team has minimal communications during the rig move.
- To impress the attending oil company representative with the efficiency of a vessel's management.
- To satisfy the company and ISM management procedures on the vessel.

Why is it common to insert a section of buoyed wire in a rig anchor chain system? Select the ONE best answer.

- **To clear a sub-sea structure or pipeline.**
- To make the anchor chain system stronger.
- To make the anchor system more efficient.
- To save using lots of anchor chain.

Why should a locking pin be placed in a Karm fork when a wire is held? Select the ONE correct answer.

- **So the wire is retained.**
- To limit the possible tension on the Karm fork.
- To stop other wires fouling the stopper.
- To stop the Karm fork dropping.

Why should the deck of an anchor handler be clear of surplus equipment before starting an anchor handling operation? Select the ONE best answer.

- **To allow the anchor handling operation to proceed.**
- To maintain a vessel's stability during the operation.
- To make space for items loaded later in the operation.
- To stop the deck crew walking into them.

With regard to a rig's anchor chain system. Why are sub-surface buoys attached to the wire insert, fitted to clear a sub-sea obstruction? Select the ONE correct answer.

- **To support the wire over the obstruction.**
- To act as a warning to surface craft.
- To carry an underwater acoustic system.
- To mark the position of the chain and wire join.

With regard to working from an anchor handling vessel, which ONE of the options correctly completes the following statement? 'Once a rig anchor is positioned on the sea floor . . .

- **. . . the work wire is disconnected and the pennant wire can be attached to a buoy'.**
- . . . the anchor chain becomes very tight'.
- . . . the operation is completed'.
- . . . the pennant is disconnected and the work wire can be attached to a buoy'.

You are in port on an anchor handler and preparing for anchor handling operations. Which ONE of the following statements best describes your preparations?

- **Preparations include checking the vessel's machinery and equipment, and loading and stowing the necessary equipment for anchor handling.**
- Preparations include making sure the working deck is clear so that anchor handling operations can proceed safely.
- Preparations include servicing all deck winches and capstans to ensure that they are all working correctly.
- Preparations include topping up bulk commodities from ashore, including drill water, cement and barytes.

0317 – Lifting and Slinging, Equipment and operation

Version 1.0 (1094)

According to the International Labour Organisation, what is the time period for re-testing of lifting equipment? Select the ONE correct answer.

- **At least every 5 years.**
- At least every 1 year.
- At least every 3 years.
- At least every 6 months.

Before a pre-slung tank is lifted from a vessel, what special safety check should be made? Select the ONE best answer.

- **That drain covers are in place and there are no leaks**
- Attach a tag line to it to stop it swinging.
- To check the WLL of the sling.
- To make sure the destination is correct.

Choose the ONE option that best completes the following statement.
'Pre-use checks of lifting equipment are intended . . .

- **. . . to find any faults with the equipment'.**
- . . . to be undertaken when lifting is completed'.
- . . . to replace all other checks on the equipment'.
- . . . to satisfy the requirements of the SMS onboard'.

Choose the ONE option that best completes the following statement.
'Pre-use checks of lifting equipment . . .

- **. . . are completed by an operator before a lift takes place'.**
- . . . are undertaken by a vessel's management'.
- . . . are undertaken by service personnel before a lifting operation'.
- . . . are undertaken once a week'.

How does a double wrap choker hitch assist when using wire slings for lifting a length of pipe? Select the ONE best answer.

- **It stops the pipe slipping out of the sling**
- Because it allows the load to rotate.
- Because it is fitted at the centre of gravity.
- It is firmly attached with a shackle to the pipe.

How should tubular items be slung? Select the ONE correct answer.

- **With two wraps of choked and secured wire slings.**
- With a single wrap of a choked and secured wire sling.
- With the tubulars in groups of five using two nylon slings.
- With the tubulars laid in groups of five using a chain sling.

If a load is lifted with wire rope on a single vertical hitch above its centre of gravity, what is likely to happen to it? Select the ONE best answer.

- **The load will rotate**
- The load will rise without rotation.
- The load will tilt.
- The sling will part.

What are multi leg bridle hitches used for?
Select the ONE best answer.

- **To lift objects with attached lugs**
- To lift drill pipe.
- To lift objects without attached lugs.
- To lift riser pipe.

What are the most common types of sling, used on an offshore vessel?
Select the ONE correct answer.

- **Wire, chain and nylon.**
- Nylon, wire and dyneema.
- Sisal, chain and nylon.
- Wire, chain and manila.

What criteria can be used to condemn a certified chain sling used for lifting?
Select the ONE correct answer.

- **If chain links are stretched or cracked.**
- If there are signs of broken wires.
- When it is easier to use a wire sling.
- When there are signs of surface rust.

What criteria can be used to condemn a certified nylon sling used for lifting?
Select the ONE correct answer.

- **If there are signs of cuts or melting damage.**
- If it has been in water.
- When it has been stored in a hot storeroom.
- When steel wire slings are available.

What is meant by the 'Safe Working Load', (SWL), of an item of lifting equipment?
Select the ONE correct answer.

- **The load up to which the item can be safely used.**
- A figure assigned to the item from the lifting plan.
- Half of the failure load for the item.
- The load at which the item would fail.

What is meant by the 'Working Load Limit', WLL, of an item of lifting equipment?
Select the ONE correct answer.

- **The breaking load divided by a safety factor.**
- A figure assigned to the item from the lifting plan.
- The load at which the item would fail.
- The minimum breaking load (MBL).

What is meant by the term 'loose gear' in relation to lifting equipment?
Select the ONE correct answer.

- **Anything connecting a load to a hook.**
- A portable chain hoist.
- Items not secured in a storeroom.
- The purchase wires on a crane.

What is the main purpose of a lifting equipment management programme on a vessel?
Select the ONE best answer.

- **To ensure that all items used for lifting are safe to use.**
- To forecast the costs to the vessel owner or manager.
- To make sure that the chain register is kept up to date.
- To make sure the crew are aware of lifting operations.

What is the purpose of a lifting equipment colour coding system?
Select the ONE best answer.

- **The colour indicates the test year.**
- The colour indicates the equipment supplier.
- To be able to say which oil company the equipment belongs to.
- To show where each item of equipment will be used.

What is the safest way to load loose ship's stores onboard?
Select the ONE best answer.

- **In a basket or on a pallet**
- Hand carried up the gangway.
- In a cargo net.
- Using nylon slings around items.

What should be inspected before lifting tubulars with a choked wire sling?
Select the ONE best answer.

- **The choked wire, because of the small bend diameter.**
- Nothing, as they are pre-slung.
- The eye of the sling for damage.
- The length of the tubulars.

What should happen if a safety item does not function correctly during a pre-use lifting equipment check? Select the ONE correct answer.

- **The equipment should not be used until it is repaired.**
- Hold a toolbox talk to decide on possible action.
- Tell the chief officer, but continue with the lift.
- Use the equipment, but be aware of the fault.

What should happen to condemned items of loose gear? Select the ONE correct answer.

- **They should be tagged and isolated for disposal.**
- They should be repaired on board and re-used.
- They should be sent ashore next time the vessel is in port.
- They should be thrown over the vessel's side to get rid of them.

What type of hitch are most containers and baskets lifted with? Select the ONE correct answer.

- **With a 4 legged wire bridle hitch**
- With a 2 legged chain bridle hitch.
- With a 2 legged wire bridle hitch.
- With a 4 legged chain bridle hitch.

What would a user look for during a pre-use lifting equipment check? Select the ONE best answer.

- **Correct function of all equipment, including stops and limits.**
- If the manufacturers service schedule has been followed.
- That the equipment is in a clean state.
- That the lifting equipment is fully certificated.

When a load is lifted with a double vertical hitch, what is likely to happen to it? Select the ONE correct answer.

- **The load will rise without rotation.**
- One sling will take all the weight.
- The load will be bent.
- The load will rotate.

When a sling is choked, how much should the maximum load of the sling be reduced to? Select the ONE correct answer.

- **At least 75% of the SWL.**
- At least 50% of the SWL.
- At least 90% of the SWL.
- No reduction.

When should a certified wire sling not be used for lifting?
Select the ONE correct answer.

- **If it cannot be identified with a tag or stamp.**
- If a nylon one can be used instead, as it is softer to the touch.
- When the sling needs to be strong.
- When there are tubulars to be lifted.

When tubulars are slung what guidance do the slingers have?
Select the ONE correct answer.

- **To use the same type of slings.**
- Never choke the slings on a load.
- Sling drill pipe in single lengths.
- To use a mixture of chain and wire slings.

Where would you find the criteria used to condemn a wire sling?
Select the ONE best answer.

- **In the manual or information supplied with the sling.**
- From a port state inspector.
- In the Code of Safe Working Practices.
- On the test certificate associated with the sling.

Which one of these criteria would be used to condemn a certified wire sling used for lifting?
Select the ONE correct answer.

- **The number of broken strands per wire.**
- Colour degradation on the wire.
- The age, if greater than one year old.
- When it is surplus to requirements.

0318 – Lifting and Slinging, Roles and responsibility

Version 1.0 (982)

Choose ONE option that best completes the following statement. 'With a lifting operation, the competent person onboard a vessel will . . .

- **. . . plan and supervise the entire lifting operation'.**
- . . . be the person directing the crane driver'.
- . . . stay away from the lifting operation'.
- . . . will be a shore person appointed to a vessel'.

Choose ONE option that best completes the following statement. 'With a lifting operation, the lift supervisor is . . .

- **. . . actively in charge of the lifting operation'.**
- . . . always the same as the competent person'.
- . . . not directly involved in the lifting operation'.
- . . . usually located on the vessel bridge'.

Choose the ONE option that best completes the following statement. 'With a lifting operation, the landing place for a load . . .

- **. . . should be prepared before the lift takes place'.**
- . . . has to have protection barriers above it'.
- . . . is decided by the crane driver'.
- . . . will always be up against other items'.

Is the following statement true or false? "A lifting equipment operator does not need to follow any instructions or signals from signaller as long as they can see the load?
Select the ONE correct answer.

- **False**
- True

Is the following statement true or false? "The load handler, also known as the slinger or rigger, does not need to notify the signaller when the load is ready for lifting: the signaller can give instruction to start the lifting operation even if they cannot see the load handler.

- **False**
- True

What does the instruction 'slew left' from the signaller during a lifting operation mean?
Select the ONE correct answer.

- **Move the load to the crane driver's left.**
- Move the load to the crane driver's right.
- Move the load to the signaller's left.
- Move the load to the signaller's right.

What does the instruction 'slew right' from the signaller during a lifting operation mean?
Select the ONE correct answer.

- **Move the load to the crane driver's right.**
- Move the load to the crane driver's left.
- Move the load to the signaller's left.
- Move the load to the signaller's right.

What does this manual signal mean when used to direct a crane driver?
Select the ONE correct answer.

- **Emergency stop.**
- Carry on as you were.
- Extend the jib.
- Load is landed.

What is one important action that a lifting equipment operator must always take during a lifting operation?
Select the ONE correct answer.

- **To follow the signaller's instructions.**
- To keep the vessel in an upright position.
- To limit power consumption on the lifting equipment.
- To only follow a signaller when the load cannot be seen.

What is the job of a signaller during a lifting operation?
Select the ONE correct answer.

- **To direct the lifting equipment operator.**
- To ensure the load moves as quickly as possible.
- To inform all the crew that a lift is being taken.
- To supervise the lifting operation.

What is the minimum number of persons that should be involved in a lifting operation?
Select the ONE correct answer.

- **Three**
- Five
- Four
- Two

What obligation does a vessel's owning or managing company have, with respect to lifting equipment?
Select the ONE correct answer.

- **To provide procedures and appoint a competent person.**
- None, as all the responsibility lies with the vessel crew.
- None, as all the responsibility lies with the vessel crew.
- To inform port state control when a lifting operation is to proceed.

What risks are associated with a lifting operation?
Select the ONE best answer.

- **Damage to the load, people and equipment.**
- Damage to equipment only.
- Damage to the load only.
- That it can take too long.

What should happen if a lift is not going as planned?
Select the ONE best answer.

- **There should be a toolbox talk to manage change.**
- Nothing will happen, as this is normal.
- The lift should be cancelled until another time.
- The supervisor will direct the best action whilst the load is moving.

What should happen if a load is seen to be in an unsafe condition during a lifting operation?
Select the ONE best answer.

- **Stop the job until the problem with the load is rectified.**
- Inform the crane driver to take it slowly.
- Just carry on with the lift, hoping it will be successful.
- Make a report as to how the unsafe condition occurred.

What should happen if the lifting equipment operator cannot see or hear the signaller during a lifting operation?
Select the ONE correct answer.

- **The lifting equipment operator should stop immediately.**
- The lifting equipment operator should lower the load.
- The operation should carry on until communication is re-established.
- The operation should proceed cautiously until a new radio is found.

What should happen to a crane, if its power supply fails during a lifting operation?
Select the ONE correct answer.

- **The load should remain where it is**
- The crane will slew to a safe place.
- The load will be lowered slowly.
- The load will drop suddenly.

Which ONE of the following is the responsibility of the lifting equipment operator?
Select the ONE correct answer.

- **To make pre-use checks on the equipment.**
- To create a lifting plan.
- To keep the vessel charterer informed of the lift status.
- To provide a power supply for the lifting equipment.

Which ONE of the options best completes the following statement? 'During routine lifting operations never . . .

- **. . . let people stand under the load'.**
- . . . change the signaller'.
- . . . allow other work to continue'.
- . . . worry about items falling off the load'.

Why is clear leadership important on a vessel undertaking lifting operations?
Select the ONE best answer.

- **To encourage safety in the lifting operation.**
- To ensure the lifting operation proceeds quickly.
- To ensure there is a good record of working activity.
- To fulfil the requirements of the vessel's SMS.

Why is it necessary to have a lifting plan on a vessel?
Select the ONE correct answer.

- **To achieve a safe and successful lift.**
- Because oil companies require one.
- It is not, as most lifts are routinely done safely.
- So that the lift can be completed without supervision.

0319 – Periodic maintenance and inspection of lifting equipment

Version 2.0 (1379)

Choose ONE option that best completes the following statement. 'An industry standard . . .

- **. . . is a level of quality found within a particular industry'.**
- . . . does not usually improve safety'.
- . . . is often too high to achieve on a vessel'.
- . . . is often too low to be of use on a vessel'.

Choose ONE option that best completes the following statement. 'An equipment standard can be used . . .

- **. . . as the criteria for inspection'.**
- . . . to calculate the insurance requirement for a vessel'.
- . . . to ensure that equipment is rust free'.
- . . . to remove all risk'.

Choose ONE option that best completes the following statement. 'A defect found whilst maintaining lifting equipment . . .

- **. . . should be reported to management to ensure maintenance is carried out in a safe manner.**
- . . . is reported to the manufacturer of the lifting equipment'.
- . . . should be repaired immediately'.
- . . . would not affect the maintenance programme.

Choose ONE option that best completes the following statement. 'Information on the periodic inspection of equipment . . .

- **. . . can be found in the equipment user manual'.**
- . . . is always very hard to find'.
- . . . is never enough to make regular inspections'.
- . . . is usually too complicated to follow'.

Choose ONE option that best completes the following statement. 'On a vessel the person inspecting lifting equipment . . .

- **. . . will not be the same as the one that maintains it'.**
- . . . is always from a shore based inspection company.
- . . . is often the same as the one that has routinely maintained it'.
- . . . will complete the task every month only'.

Choose ONE option that best completes the following statement. 'On a vessel, correct maintenance of equipment . . .

- **. . . guarantees its operation for its lifespan'.**
- . . . does not really extend the lifespan of equipment'.
- . . . is an un-necessary task that interferes with the operation of equipment'.
- . . . is often in excess of what is needed'.

Choose ONE option that best completes the following statement. 'On a vessel, the people responsible for maintenance and inspection of equipment . . .

- **. . . are usually the chief officer and chief engineer'.**
- . . . are always difficult to locate'.
- . . . are told by the office staff when to undertake the task'.
- . . . often do not take their responsibility seriously'.

Choose ONE option that best completes the following statement. 'Oil and gas activity is mostly found . . .

- **. . . in shallow sea areas'.**
- . . . in deep sea areas'.
- . . . in shore locations only'.
- . . . in water depths of more than 2000 metres'.

Choose the ONE option that best completes the following statement. 'The term 'loose gear' is used to describe . . .

- **. . . shackles and slings used for lifting'.**
- . . . all types of baggy work clothing'.
- . . . the various types of crane hooks found on a vessel'.
- . . . unsecured equipment in a locker'.

Choose the ONE option that best completes the following statement. 'When maintaining lifting equipment, for other people's safety always . . .

- **. . . have the required tools secured at the workplace'.**
- . . . complete the job during the hours of darkness.
- . . . verbally tell them you are undertaking the work'.
- . . . wear a full body safety harness'.

Choose the ONE option that best completes the following statement. 'When maintaining lifting equipment, for your own safety always . . .

- **. . . take safeguards to prevent a fall'.**
- . . . complete the job during daytime.
- . . . have a workmate with you'.
- . . . have the required tools in a bag'.

Choose the ONE option that best completes the following statement. 'When maintaining lifting equipment, for your own safety always . . .

- **. . . wear all the correct PPE'.**
- . . . have one hand for the job and one for yourself.
- . . . undertake the job in daylight hours'.
- . . . wear a hard hat'.

How can a vessel find out if there are any local or regional standards which must be followed? Select the ONE best answer.

- **Check with the oil company or charterer.**
- Check in a vessel's ISM code.
- Check in the notices to mariners.
- Check with the country's safety agency.

How can one standard be applied to all vessel types? Select the ONE best answer.

- **The standard will relate to items common to all vessel types.**
- It cannot, as all vessels are different and require individual standards.
- The standard is set at a very low level.
- The standard only relates to the quality of the materials used.

If third party lifting equipment is used on a vessel, what should the vessel's management be told about it? Select the ONE best answer.

- **That it is maintained and inspected to a recognised standard.**
- How often it will be maintained by the vessel's crew.
- Where the equipment was built, as this may affect its quality.
- Which manufacturer has produced the equipment.

What is a crew person's responsibility when they are maintaining lifting equipment? Select the ONE best answer.

- **To complete the operation in a safe manner.**
- To complete the operation as soon as possible.
- To replace as few components as possible.
- To save the company money by not having a manufacturer's representative present.

What should a crane be inspected for, before and after use? Select the ONE best answer.

- **Any visible damage, controls not working and functionality.**
- Any cosmetic maintenance required on it.
- To make sure it will take its safe working load.
- To see that it complies with a classification society's requirements.

When maintaining equipment in a high place, how are others below the work area best protected? Select the ONE best answer.

- **By having a barrier below the work area, to ensure that no people are underneath in case of any dropped objects.**
- By attaching everything that may drop to the work area.
- By only completing the work at night, when others are not around.
- By posting up notices that work is being undertaken at height.

Which one of these statements best describes the term 'lifting equipment'? Select the ONE best answer.

- **Any equipment used to lift a load or connect it to a hook.**
- Cranes and gantries but not shackles.
- Cranes used only for stores.
- Shackles and slings but not cranes.

Who should carry out an inspection of lifting equipment on a vessel? Select the ONE best answer.

- **An operator before they use the lifting equipment.**
- An engineer officer only.
- The manufacturer's representative only.
- The vessel managers only.

Who should maintain and inspect lifting equipment on a vessel? Select the ONE correct answer.

- **Those persons qualified for the task.**
- Any of the vessel's crew.
- Only manufacturers' representatives.
- Only the engineering staff onboard.

Why is a daily or periodic inspection of lifting equipment important for crew? Select the ONE best answer.

- **To ensure it is safe to operate.**
- To check that the equipment is in working order.
- To circulate oil in the hydraulic system.
- To comply with a company's requirements.

Why is it important to have standards for lifting gear inspection and maintenance? Select the ONE best answer.

- **To be able to maintain quality and meet or exceed minimum requirements.**
- So that the vessel crew have records to keep.
- To be able to maintain quality and follow maximum requirements.
- To be used for blame if an accident occurs.

Why must an offshore vessel know if there is a regional standard? Select the ONE best answer.

- **A vessel will have to comply with the standard when working in that region.**
- A vessel will have to change all of its procedures to suit it when working there.
- It does not need to know, because a vessel's own standard will be the same.
- It does not need to know, because regional standards can generally be ignored.

0320 – Personal safety for offshore vessels

Version 1.0 (1276)

Are you allowed to smoke marijuana when outside US waters?

- **No, any use of drugs on board is strictly prohibited.**
- Yes, only for medicinal purpose.
- Yes, you are allowed to smoke marijuana if there are at least 4 hours until you go on duty, and you have permission from the master.
- Yes, you are allowed to smoke marijuana outside U.S. water

How can you best protect yourself against injuries to hands, and fingers? Select the ONE best answer.

- **By using gloves when practical, and being observant of the risk**
- By avoiding operations and jobs that can cause such accidents
- By never touching anything with your hands and always using some kind of tool
- It depends on the work requirement.

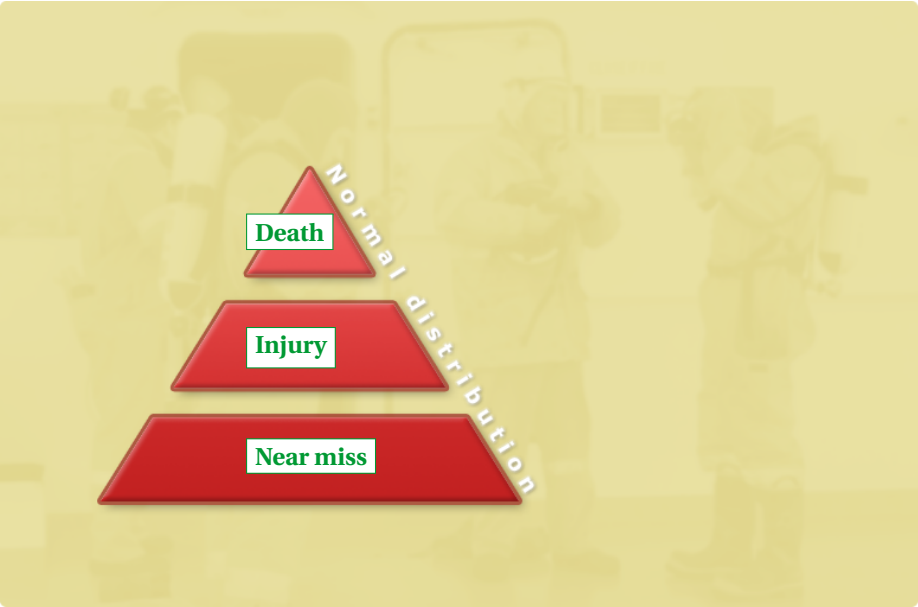
In general, which THREE of the following are the most common injuries on board?

- **Damage to feet or toes.**
- **Damage to hands or fingers**
- **Foreign body in eyes from welding, dust etc.**
- Burns from steam, hot oil or chemicals
- Damage to ears caused by loud noises
- Head injuries from falling objects
- Injuries of back or shoulders caused by poor working posture or wrong lifting techniques.
- Toxic injuries

In which type of jobs is there a risk of exposure to harmful substances? Select all answers that you think apply.

- **Cargo handling**
- **Cleaning**
- **Corrosion protection**
- **Painting**
- **Tank cleaning**
- **Working in the engine room**

Place the correct text right of the pyramid



What is the intention of the regular Safety Meeting?

- **The intention of this meeting is to inform and train the crew in safety matters, get the crew to take part in discussions and make their own suggestions for improvements.**
- The intention of this meeting is to discuss the activities ashore.
- The intention of this meeting is to gather the off duty crew so that they can get to know each other better, which will affect the safety on board.
- The intention of this meeting is to give planned work to the crew, and to ensure that checklists and permits have been filled in before the work starts.

Which FOUR accidents or injuries are the most common connected to maintenance work?

- **foreign bodies in eyes**
- **injuries from abrasive blasting**
- **injuries from falling**
- **injuries from using chemicals**
- falling overboard
- swallowing dust
- using the needle gun on a foot
- wet shoes

Which FOUR of the following give most help in preventing injuries? Select the best FOUR answers.

- **Following up incident reports**
- **Information and Training**
- **Maintaining equipment**
- **Wearing protective Equipment**

- Drug testing the crew
- Looking for oil spills
- Plenty of leisure time
- Seasickness protection

Which ONE of the following statements about hot work is most accurate?

- **Hot Work includes drilling and grinding and some forms of electrical work**
- Hot Work includes fitting blanks, valves and connecting pipe work
- Hot Work means only any work where flames are produced
- Hot Work means only welding and burning

Which ONE of the following statements is most accurate?

- **Safety Helmets need to be worn whenever working on Deck or in the Engine Room**
- Safety Helmets do not need to be worn in confined spaces
- Safety Helmets only need to be worn if there is anyone working above
- Safety Helmets only need to be worn in the Engine Room

Which ONE of the following statements is most accurate?
Select the ONE correct answer.

- **'Cold work involves jobs without any heat, or risk of sparks'.**
- 'Cold work is work done outside a vessel's accommodation'.
- 'Cold work means work done in a gas free environment'.
- 'Cold work means work in the vessel's freezer spaces only'.

Which types of work require a Permit to Work or the use of a checklist? Select all answers that you think apply.

- **Chipping down in a ballast tank**
- **Chipping on the bridge deck**
- **Lowering the anchor**
- **Painting on deck near the open stern**
- **Painting the mast with help of a bosun's chair**
- **Welding on the poop deck**

Why is it important that you become familiar with the vessel, the equipment and your own duties on board as soon as possible?

- **In an emergency situation, you need to know what to do and the position and use of different safety and fire fighting equipment.**
- In an emergency situation, it is important to know where to find the Safety Officer for reporting.
- It is important for you to know so that you can find your way around without someone guiding you all the time.
- It is important to know the port operational procedures for handling cargo.

Why is it that some jobs on a vessel require a permit to work or checklist, but others do not?
Select the ONE correct answer.

- **Because those jobs carry a particular risk.**
- Because the others have been forgotten about.
- Because there are not written procedures for them.
- Because those jobs are listed in the SMS.

With regard to Safe Lifting Technique, Which ONE of the following statements is most accurate?

- **The Load should be carried close to the body**
- An arched back reduces strain when lifting
- Safety Helmets may be removed when carrying heavy loads
- The Load should be carried away from the body

With regard to hearing protection, which ONE of the following statements is most accurate?

- **Ear Defenders should be worn whenever there is a noisy environment**
- Ear Defenders make it difficult to hear things in noisy environments
- Ear Defenders only need to be worn if there is a sign that says so
- Ear Defenders only need to be worn in the Engine Room

With regard to working at height, which ONE of the following statements is most accurate?

- **Working at height means any work where an unrestrained fall would cause injury**
- Working at height means any work with a possible fall of more than 1 metres
- Working at height means any work with a possible fall of more than 3 metres
- Working at height means working above deck

With regard to working in enclosed spaces, which ONE of the following statements is most accurate?

- **The person posted at the entrance is there to maintain communication and raise the alarm in the event of difficulty**
- The person posted at the entrance is there to give assistance if required
- The person posted at the entrance may assist with work in the space once the atmosphere has been checked
- The person posted at the entrance may carry out other duties provided they check the team in the space from time to time

0329 – Vessel inspection and OVID

Version 1.0 (1072)

Are there any consequences following a poor OVID inspection? Select the ONE best answer.

- **Yes, as it reflects badly on the vessel inspected.**
- No, as it is only one person's opinion.
- No, as remedial actions will be taken.
- Yes, as port state inspectors will be informed.

At the end of an OVID vessel inspection, what should the inspector do before they leave a vessel? Select the ONE best answer.

- **Give verbal feedback to the master and crew.**
- Give a paper copy of the full report to the vessel.
- Instruct the master as to the work that needs to be completed.
- Telephone the company owning or managing the vessel to report on it.

At the start of an OVID vessel inspection what will the inspector often ask? Select the ONE best answer.

- **If the OVPQ is up to date and completed.**
- If the OVIQ is completed.
- To see the deck area on a vessel.
- To see the engine room and machinery space.

Choose ONE option that best completes the following statement. 'The Oil Companies International Marine Forum . . .

- **. . . promotes the safe and environmentally friendly transport of oil'.**
- . . . is a US government organisation responsible for oil shipments'.
- . . . promotes the self interests of the major oil companies'.
- . . . represents and assists offshore engineering companies'.

How are OVIQ observations categorised in OVID? Select the ONE correct answer.

- **As 'Y', 'N', 'NS' or 'NA'.**
- As a high, medium or low priority.
- As either acceptable or unacceptable.
- On a scale from one to five.

How many numbered sections are there in the OVIQ document, which is part of the OVID system? Select the ONE correct answer.

- **15.**
- 18.
- 20.
- 22.

How often is an offshore vessel likely to be inspected using OVID? Select ONE answer.

- **Once a year.**
- Every second year.
- Once every five years.
- Six monthly.

Select ONE answer to correctly complete the following statement. 'The OVID inspection report has two weeks before it goes live . . .

- **. . . to allow the owner or managing company to add comments.'**
- . . . to allow OCIMF members to view it'.
- . . . to allow OCIMF to quality check it'.
- . . . to give the inspector time to see that it is correct'.

Select the ONE answer which completes the following statement correctly. 'High priority safety findings during an OVID inspection . . .

- **. . . need to be addressed immediately by onboard staff'.**
- . . . have to be passed on to the company owning or managing a vessel'.
- . . . need to be addressed within three months of the inspection'.
- . . . will be passed on to the vessel charterer'.

Select the ONE answer which correctly completes the following statement. 'A full OVID inspection report is . . .

- **. . . made available on the OCIMF web site'.**
- . . . available approximately six hours after the inspector leaves'.
- . . . available within one hour of the inspection ending'.
- . . . made available on the IMO's web site'.

Select the ONE answer which correctly completes the following statement. 'Before an OVID inspection, a vessel's owners or managers . . .

- **. . . need to arrange timings with the inspection company'.**
- . . . needs to arrange items to be inspected with the inspection company'.
- . . . should employ and man a vessel with more experienced crew'.
- . . . should have a pre-inspection check by an independent company'.

Select the ONE answer which correctly completes the following statement. 'Equipment used for lifting objects on an offshore vessel . . .

- **. . . should be managed properly'.**
- . . . is not part of an OVID vessel inspection'.
- . . . is the responsibility of the vessel charterer'.
- . . . should be tested at least once a year by an external organisation'.

Select the ONE answer which correctly completes the following statement. 'If a 'Y' is entered alongside an item on the OVID report . . .

- **. . . no further comment is needed'.**
- . . . a comment to support the judgement is made'.
- . . . the inspector has raised a question concerning the item'.
- . . . this item was not seen'.

Select the ONE answer which correctly completes the following statement. 'If an 'N' is entered alongside an item on the OVID report . . .

- **. . . the inspector has to add a comment'.**
- . . . the inspector raises a non conformance'.
- . . . the item should not have to be inspected'.
- . . . this item is not applicable'.

Select the ONE answer which correctly completes the following statement. 'Just before an OVID inspection on a well run vessel . . .

- **. . . there should be no need to have many extra preparations'.**
- . . . assistance will be needed from shore workers to clean the vessel'.
- . . . everyone will have to work hard to get the vessel to an acceptable standard'.
- . . . the vessel will have to down man and stop its operations'.

Select the ONE answer which correctly completes the following statement. 'The OVID inspection report . . .

- **. . . can be added to by an owner or managing company'.**
- . . . cannot be changed by anyone once it is posted on the web site'.
- . . . is a waste of time as it shows nothing'.
- . . . is too subjective as one inspector makes it'.

Select the ONE option that best describes the advantage to a vessel's charterer of the OVID inspection regime.

- **It assists their quality assurance with a chartered vessel.**
- The inspection costs are less than other inspection systems.
- The OVID inspection result is in the public domain.
- The vessel's crew are not aware of the inspection having taken place

Select the ONE option that best describes the advantage to a vessel's crew of the OVID inspection regime.

- The inspection itself is much easier to undertake.
- The OVID inspectors are easily satisfied.
- The vessel's crew are not involved with the inspection.
- There is a reduction in the number of inspections.

The offshore vessel inspection database (OVID) used for offshore vessel inspection reports is hosted by which organisation? Select the ONE correct answer.

- **The Oil Companies International Marine Forum.**
- The International Marine Safety Forum.
- The International Maritime Organisation.
- The Marine Contractors International Association.

What is the main purpose for having an offshore vessel inspection based on the OVID reporting system? Select the ONE best answer.

- **To confirm that a vessel is 'fit for purpose'.**
- To check up on a vessel's crew.
- To investigate areas needing work done.
- To satisfy elements of the ISM code.

What is the purpose of an offshore vessel inspection based on OVID? Select the ONE best answer.

- **To ensure that the inspection is common for all similar vessels.**
- To ensure that the inspection is limited in its duration.
- To ensure that the inspection is undertaken every five years.
- To ensure that the inspection regime is allowed to be very subjective.

What is the purpose of the Offshore Vessel Management and Self Assessment (OVMSA) tool? Select the ONE best answer.

- **It is a tool for an offshore company to assess, measure and improve their management system.**
- It is a tool to assess and inspect the quality of practical operations on an offshore vessel.
- It is a tool to show how to implement the Safety Management System on offshore vessels.
- It is a tool used to offer advice to the vessel crew on management issues.

What items should be arranged with an inspection company before an OVID inspection? Select the ONE best answer.

- **Time, duration, documents and personnel available.**
- The name of the inspector who will undertake the inspection.
- The type and level of detailed to be surveyed.
- To have the vessel alongside in a convenient port.

What should be done with the OVID inspection report once the inspection is over? Select the ONE best answer.

- **It should be used as a guidance document.**
- It should be printed and stored on the office book shelf.
- It should be sent to the company.
- It should be stored till the next inspection date.

What should happen if there are known defects on a vessel during an OVID inspection? Select the ONE best answer.

- **Tell the inspector, mentioning possible corrective actions.**
- Ensure that all defects are rectified before the inspection.
- Hide them from the inspector so they do not appear in the report.
- Highlight the defect complaining about the company at the same time.

What types of vessels are subject to OVID inspections? Select the ONE correct answer.

- **Any type of offshore oil and gas vessel.**
- Any commercial vessel.
- Any type of vessel.
- Shuttle tankers only.

Which document should to be completed before an OVID inspection? Select the ONE correct answer.

- **The OVPQ.**
- The FMEA.
- The OVID
- The OVIQ

Who should undertake an OVID pre-inspection check? Select the ONE best answer.

- **The management and crew on a vessel.**
- A specialised inspection company.
- No one, as a pre-inspection check is not necessary.
- The company that will undertake the full inspection.

Why is an OVID inspection report important to a vessel master and crew? Select the ONE best answer.

- **The report reflects the vessel's master and crew.**
- Because crew will be dismissed if it is poor.
- Because the yearly bonus is related to it.
- The report reflects the company owning the vessel.

0330 – GOMO awareness

Version 2.0 (3155)

According to the GOMO, what does "KATE" stand for? Select the ONE correct answer.

- **Knowledge, Ability, Training, Experience combine to give competence**
- Knowledge, Ability, Teamwork, Experience combine to give competence
- Knowledge, Age, Teamwork, Experience combine to give competence
- Knowledge, Age, Training, Experience combine to give competence

According to the GOMO, which THREE of the following MUST communicate with each other at the start of an outward voyage?

- **Area Co-ordinator**
- **Base Operator**
- **Vessel Master**
- Charterer

According to the GOMO, who would the vessel inform about any unusual operations during cargo operations at an installation? Select the ONE correct answer.

- **Offshore Installation Manager**
- Area co-ordinator
- Base operator
- Charterer

According to this module, what is a "near miss" incident?
Select the ONE best answer.

- **Any incident, accident or emergency which did not result in an injury**
- Any accident not causing death
- Any incident that could have caused damage to machinery
- Any incident which causes damage to machinery, but no injury to anyone

How is hydrogen sulphide usually created in wet bulk waste? Select the ONE correct answer.

- **From the activity of sulphate reducing bacteria on wastes that contain oil**
- By adding caustic soda to the tank
- From corrosion in the tank
- It is an additive used in some processes

How much deck space does the GOMO recommend should be kept free to receive initial backload? Select ONE correct answer.

- **Approximately 10%**
- Approximately 20%
- Approximately 30%
- No deck space should be wasted this way if there has been good planning

If the GOMO does not fully cover a particular operation, how can it still be of help? Select the ONE best answer.

- **There is a chapter within the GOMO giving references for further information**
- If an operation is not covered in the GOMO then it can be assumed to be safe in all circumstances.
- The GOMO says that you should refer all cases of doubt to the area co-ordinator
- The GOMO says that you should refer all cases of doubt to the Charterer

What are the TWO most frequent types of major accidents in the offshore industry? Use the information given in this module to select the TWO correct answers.

- **Hit by dropped objects**
- **Slips trips and falls**
- Electric shock
- Lifting and Handling

What does "GOMO" stand for? Select the ONE correct answer.

- **Guidelines for Offshore Marine Operations**
- Guidelines for Offshore Movement and Operations
- Guidelines for Onboard Marine Operations
- Guidelines for Operational Management Offshore

What does the GOMO recommend about "Cherry Picking" cargo? Select the ONE correct answer.

- **That the operation should be stopped and individually risk assessed**
- That it is a sign of bad planning and should be reported
- That it must be supervised by the OSV Master
- That this should never be done

What is the best way to avoid objects falling and causing injuries during lifting operations? Select the ONE best answer.

- **Ensure that the lift is checked to prevent the possibility of falling objects**
- Keep looking up at the load at all times
- Stand well clear
- Wear a hard hat and stay well clear of the intended path of the load

What is the biggest risk created when containers are stored on gravel? Select the ONE best answer

- **Gravel and stones get into fork lift pockets and can fall from height during lifting**
- Gravel and stones get into fork lift pockets, making it harder to get the forks in
- The doors jam when the container sinks into the ground
- The ground can be wet, which can cause corrosion

What is the main reason for following best practices in the GOMO? Select the ONE correct answer.

- **To reduce the number of accidents and pollution incidents**
- It is a legal requirement
- To cut costs
- To maximise profit

What is the purpose of reporting a "near miss" incident?
Select the ONE best answer.

- **To learn from what has happened and put procedures in place to prevent accidents happening**
- To comply with the law
- To find out who is to blame
- To gather statistics on types of accidents

What part of the body suffers the most injuries in the offshore industry? Use the information given in this module to select the ONE correct answer.

- **The limbs**
- The face
- The head
- The torso

What should an OSV do if asked to take slops when no analysis of the slops has been done? Select the ONE correct answer.

- **Refuse to take the slops**
- Accept verbal confirmation that no harmful substances are present
- Take the slops and then perform an analysis of them once they are onboard
- Take the slops and treat them as hazardous waste

What should you do if there is any difference between procedures in the GOMO and your own company's procedures, what should you do? Select the ONE best answer.

- **Follow your own company's procedures, but be aware of the GOMO guidance**
- Follow the guidance in the GOMO
- Follow your own company's procedures, and completely ignore any GOMO guidance
- Stop work immediately

Where do the rules for international marine legislation come from? Select the ONE best answer.

- **The IMO, a United Nations body, set regulations that are then made into laws by flag States**
- International Maritime Conventions are held to set new maritime laws
- International Maritime Law is set by the United Nations Assembly
- The International Court of Justice in The Hague

Which ONE of the following statements best describes the GOMO?

- **GOMO is a set of guidelines for safe offshore operations which can be used worldwide**
- GOMO is a set of guidelines which should only be used in Northern Europe
- GOMO is a set of rules upon which must be included in company marine offshore operations manuals in the European Economic Area
- GOMO sets out the regulations which must be followed in Northern Europe

Which ONE of these statements about hydrogen sulphide is most accurate?

- **It is toxic and flammable**
- It is flammable but safe to breathe in small amounts
- It is toxic but inert
- The main risk is that it causes corrosion

Which ONE of these types of work produces the most injuries in the offshore industry? Use the information given in this module to select the ONE correct answer.

- **Maintenance/construction work**
- Deck Operations
- Diving
- Drilling

Which of the following are recommended by the GOMO to reduce the risk of collision? Select the TWO statements that apply.

- **Establish a safety zone around the structure, unit or vessel**
- **Use a pre-entry check list before entering the safety zone**
- Do not enter the safety zone
- Never work on the weather side

0334 – Deck safety for supply vessels

Version 1.0 (1478)

Choose ONE option that best completes the following statement. 'A bulk hose laid across a supply vessel's deck . . .

- **. . . is a trip risk, when crew are working near to it'.**
- . . . is difficult to connect up to the vessel's deck connection'.
- . . . will damage itself, as it contacts other objects on the deck'.
- . . . will damage other cargo'.

Choose ONE option that best completes the following statement. 'Supply vessel movement . . .

- **. . . makes deck work more risky'.**
- . . . adds to the excitement of being at sea'.
- . . . can easily be allowed for'.
- . . . has no affect on deck working'.

Choose ONE option that best completes the following statement. 'The greatest risk with a supply vessel's dry bulk system is if . . .

- **. . . the crew do not realise there is pressure in the system'.**
- . . . cement gets damp and starts to set in a tank'.
- . . . cement gets damp and starts to set in the lines'.
- . . . the wrong grade of cement has been loaded'.

Choose ONE option that best completes the following statement. 'When a cargo item is suspended above the deck . . .

- **. . . never be underneath it'.**
- . . . approach it, ready to get hands to position it'.
- . . . approach it, to catch the tag line'.
- . . . never be on deck with it there'.

Choose ONE option that best completes the following statement. 'When a vessel sails out into bad weather, the anchors . . .

- **. . . should already be secured, or a safe heading and speed set to secure them'.**
- . . . can be left un-secured if the weather is expected to improve'.
- . . . have to be secured, even if the deck crew have to take some risks'.
- . . . will become loose and have the chains banging in the spurling pipes'.

Choose ONE option that correctly completes the following statement. 'When undertaking work at height . . .

- **. . . always wear a harness and fall protection'.**
- . . . complete the work as quickly as possible'.
- . . . never use tools'.
- . . . tie yourself on, once you are at the work place'.

Choose the ONE option that best completes the following statement. 'A natural risk taker . . .

- **. . . puts themselves in more danger'.**
- . . . is an efficient worker, because they get the job done quickly'.
- . . . is as safe as any other person'.
- . . . is representative of most people at sea'.

Choose the ONE option that best completes the following statement. 'A simple method of guiding crew away from deck obstructions . . .

- **. . . is to have dedicated walkways laid out on the decks'.**
- . . . is to have plans posted up in the mess rooms, showing the obstructions'.
- . . . is to keep crew off the decks'.
- . . . is to paint the deck with a bright colour'.

Choose the ONE option that best completes the following statement. 'Before removing the cap from the deck connection of a supply vessel's dry bulk system, always . . .

- **. . . confirm there is no pressure before removing the cap'.**
- . . . have the hose on deck ready to be connected'.
- . . . open the deck valve located on the connection'.
- . . . tell the installation crane driver that this is about to be done'.

Choose the ONE option that best completes the following statement. 'If there is a risk of water breaking on a vessel's deck . . .

- **. . . only go onto deck after risk assessment and permit to work'.**
- . . . only go onto deck after telling the bosun that you are going'.
- . . . only go onto deck between the waves breaking on it'.
- . . . only go onto deck when there is a lull in the weather'.

Choose the ONE option that best completes the following statement. 'When a crane driver is working the deck of a supply vessel, crewmen . . .

- **. . . should stay in sight of the crane driver'.**
- . . . should clear the deck totally'.
- . . . should stand close to where the load has to be lowered '.
- . . . should stand in the place where the load has to be lowered'.

Choose the ONE option that best completes the following statement. 'When working with a crane hook on the deck of a supply vessel, for safety . . .

- **. . . always have two crewmen'.**
- . . . always attach a tag line to it'.
- . . . always have three crewmen'.
- . . . only ever put one hose on the hook'.

Choose the ONE option that correctly completes the following statement. 'Without proper control of an operation . . .

- **. . . safety is reduced'.**
- . . . safety can often be reduced'.
- . . . safety may sometimes be reduced'.
- . . . safety might occasionally be reduced'.

For their safety, what should deck crew do as soon as a load is lowered onto the deck of a platform supply vessel? Select the ONE correct answer.

- **Confirm that the load will not be raised again.**
- Check the cargo container number.
- Check the contents of the cargo skip.
- Un-hook it immediately.

How are the risks associated with disconnecting a bulk hose controlled on a supply vessel? Select the ONE correct answer.

- **Drain the product back and replace the hose end cap, if available.**
- Disconnect the hose and keep the open end upright on the crane hook.
- Disconnect the hose and send it up as soon as possible.
- Drain the product using the deck bleed valve, then disconnect.

How is effective communication achieved? Select the ONE correct answer.

- **Using a correct, common language.**
- By leaving the spoken communications to the vessel's bridge.
- By only using radio communication between two people.
- Using a limited number of hand signals to avoid confusion.

How is it possible to let a vessel go without having men ashore? Select the ONE best answer.

- **By having the mooring lines on bights.**
- By cutting the mooring lines at the vessel's fairleads.
- By letting the moorings go and leaving them behind.
- By using a boat hook to lift the eye off a shore bollard.

How is the risk from an installation crane hook controlled when it is at deck level, ready to be hooked on? Select the ONE best answer.

- **Keep it in sight and in control at all times.**
- Always wear a pair of working gloves when handling it.
- Stay away from it at all times.
- Tell the crane driver to lift the hook off the deck.

How is the risk of slipping on a wet walkway reduced? Select the ONE best answer.

- **By having a non-slip coating.**
- By brushing the water off it.
- By not walking on a wet deck.
- By spreading salt on it.

How is the risk of slipping on an ice covered deck reduced? Select the ONE best answer.

- **By spreading salt on the ice and allowing it to melt.**
- By having a non-slip coating on the deck.
- By never walking on an ice covered deck.
- By spreading paper over the ice.

If a platform discharge falls on the deck of a supply vessel with crew present, what should they do? Select the ONE correct answer.

- **Clear the deck and report to the Bridge.**
- Accept that this sometimes happens, and stay on deck.
- Find out what the discharge is, and then act accordingly.
- Proceed directly to the deck shower.

On a platform supply vessel what should happen with dangerous cargo? Select the ONE correct answer.

- **It should be manifested and in a marked container.**
- It should always be stowed together with all other dangerous cargo.
- It should be discharged as soon as possible.
- It should not be accepted onboard.

On a platform supply vessel, how are the risks associated with obstructions on the mooring deck controlled? Select the ONE best answer.

- **By using dedicated walkways.**
- By walking very slowly.
- By wearing adequate footwear.
- By wearing full PPE.

On a platform supply vessel, how are the risks associated with the dry bulk system controlled? Select the ONE best answer.

- **By following operating procedures.**
- By being careful when opening lids and caps, in case there is pressure.
- By first opening the deck valve, then the tank valve.
- By only getting senior crewmen to operate it.

On a supply vessel working at an installation, why should deck cargo stay lashed for as long as possible? Select the ONE best answer.

- **To stop it falling over if there is a sudden vessel movement.**
- Because cargo should always be lashed at sea.
- Because it is company policy under the ISM code.
- Because it saves having to re-connect lashings so frequently.

On the deck of a supply boat working an installation, why is water on deck particularly dangerous? Select the ONE best answer.

- **A crush risk exists from floating containers.**
- Because the deck crew run the risk of getting wet.
- It causes the cargo to get wet.
- The water affects the crane driver's ability to stow cargo.

Select the ONE best answer relating to the following statement. 'When tying up a supply vessel , it is safe to jump onto the jetty, from the bulwark of the vessel'.

- **The statement is incorrect. This practice can be very dangerous.**
- The statement is correct, but only if the correct footwear is used.
- The statement is correct. It can save a lot of time.
- The statement is incorrect. This practice is only safe for experienced crew.

There is the possibility of water breaking on deck. When should crew go onto the deck? Select the ONE best answer.

- **Only when the bridge gives them permission.**
- As soon as a permit to work has been issued.
- As soon as the risk assessment is completed.
- When the bosun gives the word.

What are the safety risks associated with dropping an anchor? Select the ONE best answer.

- **There are risks associated with moving equipment and rust particles.**
- None, as it is such a routine, uncomplicated operation.
- The anchor cable may run too fast.
- The anchor may not be dropped in the correct place.

What can happen to a crane hook that brings the greatest risk to the crane on an installation? Select the ONE correct answer.

- **It can become foul and trapped on a vessel.**
- It can accidently open when a load is suspended.
- It can be damaged by the deck crew opening it incorrectly.
- It can be out of sight of the crane driver.

What is meant by a mooring rope 'snap back'? Select the ONE correct answer.

- **The reaction of the rope when it parts.**
- It is a type of stopper used on the mooring deck.
- It is the affect when a rope is surged on a drum end.
- The noise a rope makes when it is under lots of tension.

What is the greatest risk if a bulk hose is over-pressurised? Select the ONE correct answer.

- **Being injured if the hose or coupling were to part.**
- The deck valves will be damaged with the pressure.
- The hose will become plugged with product.
- The product will be damaged inside the hose.

What is the most dangerous risk associated with cargo work on a supply vessel? Select the ONE best answer.

- **A crush injury from an item of cargo.**
- Being burned on a hot surface.
- Falling over the side of a vessel.
- Tripping over cargo lashings.

What is the worst possible consequence of an over pressurised hydraulic windlass motor? Select the ONE best answer.

- **The casing may blow apart.**
- None, as the pressure will never be too great.
- The hydraulic oil will get hot.
- The windlass motor may stop working.

When heaving a synthetic fibre mooring rope on a drum end, how many turns should there be? Select the ONE correct answer.

- **No more than three.**
- No more than five.
- No more than four.
- No more than two.

When undertaking routine deck work, what are the basic requirements for a crewman to stay safe? Select the ONE best answer.

- **Being in a fit state, risk assessing and wearing PPE.**
- Following instructions from the bosun at all times.
- Let others undertake the more risky work.
- Only work on deck when conditions are perfect.

When using a windlass for mooring, what is the most important safety priority? Select the ONE correct answer.

- **To have a person at the windlass control at all times.**
- To have the windlass in a low gear at all times.
- To make sure that the hydraulic oil is warmed up.
- To only use one side of the windlass at a time.

When working cargo on deck, with the possibility of water breaking onto it, how can crew protect themselves? Select the ONE correct answer.

- **By remaining in safe areas, or the accommodation, as much as possible.**
- By only working deck cargo and avoiding hose work.
- By refusing to go on deck to work cargo.
- By sheltering amongst the cargo stow.

Which one of the given options best defines 'complacency'? Select the ONE best answer.

- **Complacency is when a person feels secure or smug in their ability to perform a task.**
- Complacency happens to a person when they have to undertake a task that is not a routine activity.
- Complacency is when a person deliberately decides not to follow procedures.
- Complacency only occurs with people who are not seafarers.

Why can a mooring rope being heaved on a windlass drum end be particularly dangerous? Select the ONE best answer.

- **Because the rope may snap back towards the person holding it.**
- Because the mooring deck is a dangerous place.
- Because the windlass may put excessive force on the rope.
- Because there is little control of the rope.

Why do people need to be regularly reminded about safety? Select the ONE best answer.

- **Because people can forget even basic safety principles.**
- Because companies have to do this under their SMS obligations.
- Because people forget what they see as unimportant.
- Because people generally have very poor memories.

Why does skills' training not change a person's attitude to risk? Select the ONE best answer.

- **Because skills' training does not always address changing behaviour.**
- Because once attitudes are formed, they cannot be changed.
- Because people tend to ignore training.
- Because skills' training will never include the risks associated with it.

Why should crew avoid walking on top of deck cargo on a supply vessel? Select the ONE best answer.

- **To reduce the risk of slips, trips and falls.**
- Because company procedures do not allow it.
- Because the cargo owner objects to this.
- To reduce the risk of damaging the cargo.

With regard to mooring operations, choose the ONE option that best completes the following statement. 'Never stand . . .

- **. . . in the whip lash zone of a rope under tension'.**
- . . . at the winch control when heaving on a rope'.
- . . . behind the person operating the winch'.
- . . . near the edge of the vessel in case of being hit by a heaving line'.

0335 – Deck safety for anchor handlers

Version 1.0 (1710)

Choose ONE option that best completes the following statement. 'A natural risk taker . . .

- **. . . puts themselves in more danger'.**
- . . . is an efficient worker, because they get the job done quickly'.
- . . . is as safe as any other person'.
- . . . is representative of most people at sea'.

Choose ONE option that best completes the following statement. 'An anchor handler moving in a seaway . . .

- **. . . makes deck work more risky'.**
- . . . adds to the excitement of being at sea'.
- . . . can easily be allowed for'.
- . . . has no affect on deck working'.

Choose ONE option that best completes the following statement. 'When a pennant wire is being disconnected . . .

- **. . . be careful, in case it wants to rotate dangerously'.**
- . . . have a crew person grip it, to stop it rotating'.
- . . . retreat to a safe place, as this is the most dangerous part of an anchor handling operation'.
- . . . the operation is without risk'.

Choose ONE option that best completes the following statement. 'When a pennant wire is being heaved . . .

- **. . . stay in a safe place'.**
- . . . it is the most dangerous part of an anchor handling operation'.
- . . . stay close, to inform the bridge of its position in the stopper'.
- . . . take the chance to have a clear up on deck'.

Choose ONE option that best completes the following statement. 'When a vessel sails out into bad weather, the anchors . . .

- **. . . should already be secured, or a safe heading and speed set to secure them'.**
- . . . can be left unsecured if the weather is expected to improve'.
- . . . have to be secured even if the deck crew have to take some risks'.
- . . . will become loose and have the chains banging in the spurling pipes'.

Choose ONE option that best completes the following statement. 'When towing on an anchor handler . . .

- **. . . stay well clear of the tow wire when on deck'.**
- . . . never go onto the vessel's after deck'.
- . . . only go onto the vessel's forward deck'.
- . . . the tow wire has no affect on deck working'.

Choose ONE option that correctly completes the following statement with respect to safety. 'Grapnels and 'J' hooks used for anchor handling . . .

- **. . . should be inspected in the same way as any other lifting equipment'.**
- . . . are extremely robust and do not need testing or inspection'.
- . . . last a very long time because they are made of cast iron'.
- . . . never carry much load, because all they do is snag wire or chain'.

Choose ONE option that correctly completes the following statement. 'A remotely operated winch control . . .

- **. . . offers more safety to the operator'.**
- . . . is not a good thing, it decreases safety'.
- . . . is not as safe as controlling the winch locally'.
- . . . is not found on anchor handling vessels'.

Choose ONE option that correctly completes the following statement. 'When undertaking work at height . . .

- **. . . always wear a harness and fall protection'.**
- . . . complete the work as quickly as possible'.
- . . . never use tools'.
- . . . tie yourself on, once you are at the work place'.

Choose the ONE option that best completes the following statement. 'A simple method of guiding crew away from deck obstructions . . .

- **. . . is to have dedicated walkways laid out on the decks'.**
- . . . is to have plans posted up in the mess rooms, showing the obstructions'.
- . . . is to keep crew off the decks'.
- . . . is to paint the deck with a bright colour'.

Choose the ONE option that best completes the following statement. 'If there is a risk of water breaking on a vessel's deck . . .

- **. . . only go onto deck after risk assessment and permit to work'.**
- . . . only go onto deck after telling the bosun that you are going'.
- . . . only go onto deck between the waves breaking on it'.
- . . . only go onto deck when there is a lull in the weather'.

Choose the ONE option that correctly completes the following statement. 'Without proper control of an operation . . .

- **. . . safety is reduced'.**
- . . . safety can be reduced'.
- . . . safety may be reduced'.
- . . . safety might be reduced'.

How is effective communication achieved? Select the ONE correct answer.

- **Using a correct, common language.**
- By leaving the spoken communications to the vessel's bridge.
- By only using radio communication between two people.
- Using a limited number of hand signals to avoid confusion.

How is it possible to let a vessel go without having men ashore? Select the ONE best answer.

- **By having the mooring lines on bights.**
- By cutting the mooring lines at the vessel's fairleads.
- By letting the moorings go and leaving them behind.
- By using a boat hook to lift the eye off a shore bollard.

How is the risk of slipping on a wet walkway reduced? Select the ONE best answer.

- **By having a non-slip coating.**
- By brushing the water off it.
- By not walking on a wet deck.
- By spreading salt on it.

How is the risk of slipping on an ice covered deck reduced? Select the ONE best answer.

- **By spreading salt on the ice and allowing it to melt.**
- By having a non-slip coating on the deck.
- By never walking on an ice covered deck.
- By spreading paper over the ice.

On an anchor handler when towing, why does bad weather make the after deck more dangerous? Select the ONE best answer.

- **Because the tow wire slides across the deck.**
- Because more tow wire is paid out.
- Because the deck is always wet.
- Because the vessel is forced to slow down.

On the deck of an anchor handler working an installation, why is water on deck particularly dangerous? Select the ONE best answer.

- **A crush risk exists from floating containers.**
- Because the deck crew run the risk of getting wet.
- It causes the cargo to get wet.
- The water affects the crane driver's ability to stow cargo.

Select the ONE best answer relating to the following statement. 'When tying up an anchor handler, it is safe to jump onto the jetty, from the bulwark of the vessel'.

- **The statement is incorrect. This practice can be very dangerous.**
- The statement is correct, but only if the correct footwear is used.
- The statement is correct. It can saves a lot of time.
- The statement is incorrect. This practice is only safe for experienced crew.

There is the possibility of water breaking on deck. When should crew go onto the deck? Select the ONE best answer.

- **Only when the bridge gives them permission.**
- As soon as a permit to work has been issued.
- As soon as the risk assessment is completed.
- When the bosun gives the word.

What are the risks when using burning gear to break an anchor chain? Select the ONE best answer.

- **Burns, eye damage and fire.**
- Burns, eye and hearing damage.
- That the chain cannot be broken.
- That there is insufficient gas to complete the operation.

What danger is posed by equipment, with wire connected, about to be lifted? Select the ONE best answer.

- **The wire may whip around when the equipment is lifted.**
- The equipment may be damaged.
- The wire may be very long.
- The wire may fall off before the cargo is ashore.

What extra safety measures should always be taken during darkness on a vessel whilst anchor handling? Select the ONE best answer.

- **Ensure that decks are well lit.**
- Shut the cabin curtains to stop light pollution.
- Wear extra clothing as it gets cold.
- Wear full PPE.

What is the best preparation for crew, to ensure their safety during an anchor handling operation? Select the ONE best answer.

- **Make sure they are well briefed and knowledgeable.**
- Check their certificates to confirm that they are qualified.
- Make sure they are all on deck when the operation starts.
- Tell them to follow the bosun's instructions at all times.

What is the greatest risk associated with connecting up a tow on an anchor handler? Select the ONE best answer.

- **Falling over the stern.**
- Falling over tools.
- Snagging the hand on a wire rag.
- Tripping on the work wire.

What precautions should the deck crew take with an awkward shaped load, such as a rig anchor, just landed on deck by a crane? Select the ONE best answer.

- **Make sure it is stable before approaching it.**
- Attach a lashing to it immediately.
- Immediately unhook it, the same as any other load.
- Unhook it, then move other cargo against it.

What safety check is essential when a wire has just been positioned in a deck stopper? Select the ONE correct answer.

- **That the wire is securely in place.**
- That the quick release system works.
- That the wire is in good condition.
- That the wire is not under tension.

What safety check is required on a wire or chain deck stopper before it is used? Select the ONE correct answer.

- **The insert is in good condition and of the correct size.**
- That the insert will allow the chain or wire to pass through it.
- That the rotating guide is well greased.
- The insert is well grooved from previous use.

What safety check is required on hand tools used during an anchor handling operation? Select the ONE best answer.

- **That they are approved for use.**
- That they are close at hand.
- That they are identifiable as deck tools.
- That they are made in Germany.

When anchor handling in marginal weather, why is the work more risky? Select the ONE best answer.

- **Because wires and chains have greater strain on them.**
- Because the rig jumps around a lot.
- Because the vessel's engines are likely to stop.
- Because the winches do not work as well as in good weather.

When buoy catching, how are the risks associated with the open stern totally controlled? Select the ONE best answer.

- **Wear a lifejacket and safety restraint.**
- Do not work close to the stern.
- Use a boat hook to extend your reach.
- Wear a lifejacket.

When undertaking routine deck work, what are the basic requirements for a crewman to stay safe? Select the ONE best answer.

- **Being in a fit state, risk assessing and wearing PPE.**
- Following instructions from the bosun at all times.
- Let others undertake the more risky work.
- Only work on deck when conditions are perfect.

When working cargo on deck, with the possibility of water breaking onto it, how can crew protect themselves? Select the ONE correct answer.

- **By remaining in safe areas or the accommodation as much as possible.**
- By only working deck cargo and avoiding hose work.
- By refusing to go on deck to work cargo.
- By sheltering amongst the cargo stow.

Which of the given options best defines 'complacency'? Select the ONE best answer.

- **Complacency is when a person feels secure or smug in their ability to perform a task.**
- Complacency happens to a person when they have to undertake a task that is not a routine activity.
- Complacency is when a person deliberately decides not to follow procedures.
- Complacency only occurs with people who are not seafarers.

Why does skills' training not change a person's attitude to risk? Select the ONE correct answer.

- **Because skills' training does not always address changing behaviour.**
- Because once attitudes are formed they cannot be changed.
- Because people tend to ignore training.
- Because skills' training will never include the risks associated with it.

Why does working cargo offshore pose a greater risk to crew, than working cargo onshore? Select the ONE correct answer.

- **When offshore they do the cargo work, but in port that work is done by dock workers.**
- Because the offshore cranes are of poorer quality compared to the onshore cranes.
- Because when offshore, the crane drivers are not as good as the ones onshore.
- When offshore there is less time to undertake the equipment transfer than onshore.

Why is it that people need to be regularly reminded about safety? Select the ONE best answer.

- **Because people can forget even basic safety principles.**
- Because companies have to do this under their SMS obligations.
- Because people forget what they see as unimportant.
- Because people generally have very poor memories.

Why is the size of the wire fitted to a tugger winch important for safety? Select the ONE correct answer.

- **Because the winch power should not be capable of snapping the wire.**
- Because if it is too big it is difficult to handle.
- Because the drum of the winch is usually quite small, the wire should be small.
- To allow the wire to go to the stern and back up the deck with some spare.

With regard to towing operations in following weather, choose the ONE option that best completes the following statement. 'Green water . . .

- **. . . may be breaking over the stern'.**
- . . . is not a threat on a well found vessel'.
- . . . on deck is fun, and not a big threat'.
- . . . will never be breaking over the stern'.

0336 – Back deck safety for seismic vessels

Version 1.0 (1233)

Before a small boat is launched, what should be agreed with the chase boat? Select the ONE correct answer.

- **Who should perform a man overboard recovery.**
- At what vessel speed the operation will be performed.
- How the boat will be launched from the vessel.
- How to tackle a rogue vessel located near the cables.

Choose ONE option that best completes the following statement. 'A natural risk taker . . .

- **. . . puts themselves in more danger'.**
- . . . is an efficient worker, because they get the job done quickly'.
- . . . is as safe as any other person'.
- . . . is representative of most people at sea'.

Choose ONE option that best completes the following statement. 'Bad weather causes noise . . .

- **. . . which interferes with effective voice communications'.**
- . . . which has no effect on the seismic operation'.
- . . . which makes the vessel impossible to steer'.
- . . . which speeds up the rate of data acquisition '.

Choose ONE option that best completes the following statement. 'To prevent falling in the water when the stern ramp doors are open, . . .

- **. . . crew should wear a body harness and restraint'.**
- . . . crew should approach the area with caution'.
- . . . crew should stay away from the top of the ramp'.
- . . . crew should wear a lifejacket'.

Choose ONE option that best completes the following statement. 'When a vessel sails out into bad weather, the anchors . . .

- **. . . should already be secured, or a safe heading and speed set to secure them'.**
- . . . can be left un-secured if the weather is expected to improve'.
- . . . have to be secured, even if the deck crew have to take some risks'.
- . . . will become loose and have the chains banging in the spurling pipes'.

Choose ONE option that best completes the following statement. 'When undertaking small boat work . . .

- **. . . a lifejacket and hard hat should be worn'.**
- . . . let everyone have a chance to drive the boat'.
- . . . the operation is without risk'.
- . . . the risks are the same either during the day or at night'.

Choose ONE option that correctly completes the following statement. 'When undertaking work at height . . .

- **. . . always wear a harness and fall protection'.**
- . . . complete the work as quickly as possible'.
- . . . never use tools'.
- . . . tie yourself on, once you are at the work place'.

Choose ONE option that for safety best completes the following statement. 'Home made tools . . .

- **. . . should not be used'.**
- . . . are really useful, as they are made just for the job'.
- . . . should be added to the equipment list'.
- . . . should be approved by the chief engineer before use'.

Choose the ONE option that best completes the following statement. 'A simple method of guiding crew away from deck obstructions . . .

- **. . . is to have dedicated walkways laid out on the decks'.**
- . . . is to have plans posted up in the mess rooms, showing the obstructions'.
- . . . is to keep crew off the decks'.
- . . . is to paint the deck with a bright colour'.

Choose the ONE option that best completes the following statement. 'If there is a risk of water breaking on a vessel's deck . . .

- **. . . only go onto deck after risk assessment and permit to work'.**
- . . . only go onto deck after telling the bosun that you are going'.
- . . . only go onto deck between the waves breaking on it'.
- . . . only go onto deck when there is a lull in the weather'.

Choose the ONE option that correctly completes the following statement. 'Without proper control of an operation . . .

- **. . . safety is reduced'.**
- . . . safety can be reduced'.
- . . . safety may be reduced'.
- . . . safety might be reduced'.

How is effective communication achieved? Select the ONE correct answer.

- **Using a correct, common language.**
- By leaving the spoken communications to the vessel's bridge.
- By only using radio communication between two people.
- Using a limited number of hand signals to avoid confusion.

How is it possible to let a vessel go without having men ashore? Select the ONE best answer.

- **By having the mooring lines on bights.**
- By cutting the mooring lines at the vessel's fairleads.
- By letting the moorings go and leaving them behind.
- By using a boat hook to lift the eye off a shore bollard.

How is the risk of slipping on a wet walkway reduced? Select the ONE best answer.

- **By having a non-slip coating.**
- By brushing the water off it.
- By not walking on a wet deck.
- By spreading salt on it.

How is the risk of slipping on an ice covered deck reduced? Select the ONE best answer.

- **By spreading salt on the ice and allowing it to melt.**
- By having a non-slip coating on the deck.
- By never walking on an ice covered deck.
- By spreading paper over the ice.

How many 'back deck' areas will a modern deep seismic vessel have? Select the ONE best answer.

- **Generally one, as the term relates to the seismic working areas.**
- At least four on a modern purpose built deep seismic vessel.
- Only the one, as the term relates to the deck where the seismic guns are stored.
- Only the one, as the term relates to the deck where the streamer cables are stored.

If there is a very bad weather forecast, what should be done with work schedules? Select the ONE correct answer.

- **Limit them to non-essential work only.**
- Carry on with the schedules, as work needs to be completed.
- Stop all work.
- Try and get the work done more quickly.

On a platform supply vessel, how are the risks associated with obstructions on the mooring deck controlled? Select the ONE best answer.

- **By using marked walkways.**
- By walking very slowly.
- By wearing adequate footwear.
- By wearing full PPE.

Select the ONE best answer relating to the following statement. 'When tying up a seismic vessel, it is safe to jump onto the jetty, from the bulwark of the vessel'.

- **The statement is incorrect. This practice can be very dangerous.**
- The statement is correct, but only if the correct footwear is used.
- The statement is correct. It can save a lot of time.
- The statement is incorrect. This practice is only safe for experienced crew.

There is the possibility of water breaking on deck. When should crew go onto the deck? Select the ONE best answer.

- **Only when the bridge gives them permission.**
- As soon as a permit to work has been issued.
- As soon as the risk assessment is completed.
- When the bosun gives the word.

What are the safety risks associated with dropping an anchor? Select the ONE best answer.

- **There are risks associated with moving equipment and rust particles.**
- None, as it is such a routine, uncomplicated operation.
- The anchor cable may run too fast.
- The anchor may not be dropped in the correct place.

What extra safety measure should be taken, on a seismic vessel during the hours of darkness? Select the ONE best answer.

- **Ensure that decks are well lit.**
- Shut the cabin curtains to stop light pollution.
- Wear extra clothing, as it gets cold.
- Wear full PPE.

What is meant by a mooring rope 'snap back'? Select the ONE correct answer.

- **The reaction of the rope when it parts.**
- It is a type of stopper used on the mooring deck.
- It is the affect when a rope is surged on a drum end.
- The noise a rope makes when it is under lots of tension.

What is the worst possible consequence of an over pressurised hydraulic windlass motor? Select the ONE best answer.

- **The casing may blow apart.**
- None, as the pressure will never be too great.
- The hydraulic oil will get hot.
- The windlass motor may stop working.

What should be followed by crew performing maintenance on seismic energy sources? Select the ONE best answer.

- **The safety procedures devised for the equipment.**
- The IAGC guide to gun maintenance.
- The IMCA guide to equipment maintenance.
- The maintenance that was completed by the last crew.

What should never be done when working over the side of a vessel? Select the ONE correct answer.

- **Over reaching, as it increases the chances of falling.**
- Using tools connected to a safety lanyard.
- Wearing a lifejacket, as it restricts movement.
- Wearing a safety helmet, because it can fall off.

When heaving a synthetic fibre mooring rope on a drum end, how many turns should there be? Select the ONE correct answer.

- **No more than three.**
- No more than five.
- No more than four.
- No more than two.

When undertaking routine deck work, what are the basic requirements for a crewman to stay safe? Select the ONE best answer.

- **Being in a fit state, risk assessing and wearing PPE.**
- Following instructions from the bosun at all times.
- Let others undertake the more risky work.
- Only work on deck when conditions are perfect.

When using a windlass for mooring, what is the most important safety priority? Select the ONE correct answer.

- **To have a person at the windlass control at all times.**
- To have the windlass in a low gear at all times.
- To make sure that the hydraulic oil is warmed up.
- To only use one side of the windlass at a time.

Which decks on a deep seismic vessel are classed as 'back deck'? Select the ONE correct answer.

- **Working decks where seismic equipment is found.**
- Any deck, including accommodation decks.
- The gun deck and mooring deck.
- The mooring deck of the vessel.

Which of the given options best defines complacency? Select the ONE correct answer.

- **Complacency is when a person feels secure or smug in their ability to perform a task.**
- Complacency happens to a person when they have to undertake a task that is not a routine activity.
- Complacency is when a person deliberately decides not to follow procedures.
- Complacency only occurs with people who are not seafarers.

Which one of the following best describes the precautions to be taken before maintenance work on a compressed air system? Select the ONE correct answer.

- **The system should be shut down and de-pressurised.**
- The relevant section should be isolated from high pressure
- The system should be maintained at working pressure
- The system should be put into stand-by mode.

Which one of these is the best way to stay safe in a small boat? Select the ONE best answer.

- **By following the coxswain's orders.**
- By facing the ahead direction.
- By facing the astern direction.
- By sitting on the boat floor.

Why can a mooring rope being heaved on a windlass drum end be particularly dangerous? Select the ONE best answer.

- **Because the rope may snap back towards the person holding it.**
- Because the mooring deck is a dangerous place.
- Because the windlass may exert excessive force on the rope.
- Because there is little control of the rope.

Why does skills' training not change a person's attitude to risk? Select the ONE correct answer.

- **Because skills' training does not always address changing behaviour.**
- Because once attitudes are formed they cannot be changed.
- Because people tend to ignore training.
- Because skills' training will never include the risks associated with it.

Why is it that people need to be regularly reminded about safety? Select the ONE best answer.

- **Because people can forget even basic safety principles.**
- Because companies have to do this under their SMS obligations.
- Because people forget what they see as unimportant.
- Because people generally have very poor memories.

Why is water on the back deck of a seismic vessel dangerous? Select the ONE best answer.

- **Loose items may float and cause a crush risk.**
- Because the seismic crew run the risk of getting wet.
- It causes the seismic equipment stowed on deck to get wet.
- The deck water creates noise on the seismic streamer.

With regard to mooring operations, choose the ONE option that best completes the following statement. 'Never stand . . .

- **. . . in the whip lash zone of a rope under tension'.**
- . . . behind the person operating the winch'.
- . . . on the top of cargo containers'.
- . . . on the vessel's bulwark cap'.

0337 – Fresh water management

Version 3.0 (3117)

According to this module, how often should potable water on a vessel be tested for E.coli bacteria? Select the ONE correct answer.

- **Every 3 months.**
- Every 6 months.
- Every month.
- Every week.

Approximately, how long does chlorine take to disinfect water? Select the ONE correct answer.

- **30 minutes.**
- 5 minutes.
- 6 hours.
- 60 minutes.

Choose the ONE option that best completes the following statement. 'A shore company supplying water to a vessel should' . . .

- **. . . ensure the water is free from pollutants and contaminants'.**
- . . . deliver the water free of charge because it is a basic need'.
- . . . ensure the water is delivered at a reasonable pressure'.
- . . . treat the water more thoroughly than they would for local people'.

Choose the ONE option that correctly completes the following statement. 'An upland water reservoir . . .

- **. . . is fed by rivers and streams draining surrounding land'.**
- . . . contains nothing but pure water'.
- . . . is filled with water from underground wells'.
- . . . is usually totally covered'.

Disinfectants should kill bacteria, viruses and micro-organisms when they are first added to potable water. What other property is desirable for a disinfectant? Select the ONE best answer.

- **It should give long term protection.**
- It should be easy to administer.
- It should protect pipework from corrosion.
- It should protect the water from chemicals.

How does sludge form in a fresh water tank? Select the ONE correct answer.

- **It is probably the result of coagulation within the water.**
- It is a result of adding chlorine to the water when it is loaded.
- It is a result of parasitic action within the water.
- It is caused by the action of the water with the tank coating.

How much time should there be between taking and testing a potable water sample? Select the ONE correct answer.

- **The shortest time possible.**
- As much time as is needed.
- Not more than a month.
- Not more than a week.

How should a vessel's potable water transfer hose be stored? Select the ONE correct answer.

- **With both ends capped.**
- Both ends should be left open to air it.
- It should be hung vertically to allow it to drain.
- It should be uncoiled and laid out to dry.

On a vessel, where should potable water samples be taken? Select the ONE best answer.

- **At multiple points around the vessel.**
- From one point.
- From the supply side of the fresh water pump
- From the water tank itself.

What are the effects of a person drinking water containing bacteria? Select the ONE correct answer.

- **If bacteria is harmful, sickness will result.**
- There will be a bad taste, so the person will not drink a lot.
- There will be no affect, as the body can cope with it.
- There will definitely be sickness and death.

What biological pollutants can be found in untreated fresh water? Select the ONE correct answer.

- **Bacteria, parasites and viruses.**
- Bacteria, parasites and chemicals.
- Parasites, chemicals and sediment.
- Sediment, viruses and bacteria.

What is meant by the expression 'Ultraviolet Germicidal Irradiation' when referring to water treatment? Select the ONE correct answer.

- **It is a process to disinfect water.**
- It is a process to make water clear.
- It is a process to produce nice tasting water.
- It is a process to sample the level of bacterial contamination.

What is required of a disinfectant used within potable water? Select the ONE best answer.

- **It should kill bacteria, viruses and micro-organisms.**
- It should make the water clear by removing sediment.
- It should make the water smell fresh.
- It should remove all parasites from the water.

What is the maximum level of chloramine in potable water as recommended by the USA Environmental Protection Agency? Select the ONE correct answer.

- **4 parts per million.**
- 14 parts per million.
- 40 parts per million.
- 46 parts per million.

What is the recommended treatment for a vessel's potable water system during dry-dock? Select the ONE correct answer.

- **It should be cleaned with chlorinated water at a strength of 50 ppm.**
- It should be cleaned with chlorinated water at a strength of 0.2 ppm.
- It should be cleaned with chlorinated water at a strength of 5 ppm.
- It should be cleaned with chlorinated water at a strength of 500 ppm.

What is the recommended treatment for a vessel's potable water tanks at intervals of not greater than five years? Select the ONE correct answer.

- **They should be pressure tested.**
- They should be inspected for bacterial build-up.
- They should be re-coated with epoxy paint.
- They should be scaled then re-coated.

What is the recommended treatment for cleaning a potable water transfer hose? Select the ONE correct answer.

- **By leaving it filled with a solution of 50 ppm free chlorine.**
- By applying a detergent cleaner inside and out.
- By heat-treating it with steam to remove bacteria.
- By pumping chlorinated water through it.

What items are typically found in an upland water reservoir? Select the ONE correct answer.

- **Fish, debris, mud and aggregate.**
- Chemicals from industry and human sewage.
- Items of domestic rubbish.
- None, as the water is intended for drinking.

What name is commonly given to chlorine when it is used to disinfect fresh water? Select the ONE correct answer.

- **Free chlorine.**
- Entrained chlorine.
- Fixed chlorine.
- Trapped chlorine.

What should potable water be routinely tested for on board a vessel? Select the ONE best answer.

- **The presence of chlorine, chloramine and bacteria.**
- The presence of calcium, chloramine and lead.
- The presence of chorine, iron and bacteria.
- The presence of fluoride, chloramine and pesticides.

What type of container should be used to hold a potable water test sample? Select the ONE correct answer.

- **One made of glass.**
- One made of metal.
- One made of plastic.
- One made of wood.

When filters are fitted to a potable water system, what maintenance do they require? Select the ONE best answer.

- **They should be removed and cleaned as procedures advise.**
- They should be cleaned daily.
- They should be replaced every day.
- They should not be disturbed.

When the cap from a deck potable water fill point is removed, what should happen to it? Select the ONE correct answer.

- **It should be kept in a safe place, free from contaminants.**
- It shoud be left close by, on a hatch top.
- It should be left on the bulwark, close to the filling point.
- It should be placed on deck, close to the filling point.

Which ONE of the following options correctly completes the following statement? 'Disinfected water passing through a vessel's pipework system . . .

- **. . . helps keep the internal parts free from bacteria'.**
- . . . corrodes the brass components in pumps'.
- . . . destroys the seals used in a pipework system'.
- . . . helps the water flow more efficiently'.

Which ONE of these options best completes the following statement? 'Filters used in a fresh water system . . .

- **. . . can be breeding grounds for bacteria'.**
- . . . are fitted to trap chemicals'.
- . . . must be five microns mesh size'.
- . . . should be changed once a year'.

Which ONE of these options correctly completes the following statement? 'Water made on board a vessel has chemicals added to it to . . .

- **. . . give it some flavour and make it less corrosive'.**
- . . . make it safe to consume'.
- . . . remove the colour found in seawater'.
- . . . remove the salt from the seawater used'.

Which ONE of these options describes the three conditions to be met for ultraviolet radiation to disinfect water? Select the ONE correct answer.

- **The emitted radiation is correct, the exposure time is long enough, and there are no particles shadowing.**
- The emitted radiation is correct, the water is at the correct temperature and there are no particles shadowing.
- The exposure time is long enough, there are no particles shadowing, and the water is at the correct pressure.
- There are no particles shadowing, the emitted radiation is correct and the water is clear.

Which of the following is a characteristic of water from a naturally occurring underground fresh water reservoir? Select the ONE correct answer.

- **It is generally free from fish, vegetable material and mud.**
- It can be heavily contaminated with fish, vegetable material and mud.
- It can contain quantities of naturally occurring chlorine.
- It is often not suitable for use as drinking water.

Which of these lasts longer as a disinfectant when used in fresh water? Select the ONE correct answer.

- **Chloramine.**
- Free chlorine.

Why does potable water need to be disinfected, even after it has been filtered? Select the ONE correct answer.

- **To remove harmful bacteria and viruses.**
- To comply with a water company's requirements.
- To make the water clear and nice to drink.
- To make the water smell like chlorine.

Why does the port authority often have water storage tanks? Select the ONE best answer.

- **To be able to quickly deliver relatively large quantities to a ship.**
- Because the water supplier has recommended this.
- Because the World Health Organisation has insisted on this.
- To purify water before it has been sent to ships.

Why is boiling fresh water not a method used at sea when disinfecting water? Select the ONE best answer.

- **It would take too much time and energy.**
- Because it is not allowed by flag State rules.
- It is not a proven way to disinfect water.
- There is not a container big enough to do this.

Why is chloramine used to treat fresh water? Select the ONE correct answer.

- **Because it gives long-term disinfectant protection to water.**
- Because it acts immediately, but then has no long lasting effects.
- Because it is completely safe to handle.
- Because it is the cheapest, readily available product.

Why is it important that a port has a Fresh Water Management Plan in place? Select the ONE best answer.

- **To maintain the quality of the water supplied to it.**
- Because the local water supplier will insist on it.
- Because they are audited under port state control.
- To be able to purify the poor quality water supplied to it.

Why should records of potable water treatment and storage be kept on a vessel? Select the ONE best answer.

- **To show that water is being managed properly on the vessel.**
- Because records show where the water was loaded or made.
- Because the company requires it to be done.
- To record water consumption on the vessel.

With regards to crew safety, what is the most important part of a vessel's Fresh Water Management Plan? Select the ONE best answer.

- **The procedures that ensure water is safe to drink.**
- Diagrams showing the fresh water system on a vessel.
- Information relating to tank ullages and specific gravities.
- Instructions on how to start the fresh water generator.

0338 – Fresh water system sanitation

Version 2.0 (880)

After maintenance and inspection of a potable water tank, how can the tank be sanitised quickly without filling it with water? Select the ONE correct answer.

- **The surfaces should be coated in a solution containing chlorine at 200 ppm.**
- The surfaces should be coated in a solution containing chlorine at 20 ppm.
- The surfaces should be coated in a solution containing chlorine at 50 ppm.
- The surfaces should be coated in a solution containing chlorine at 500 ppm.

How can a seafarer catch Legionnaires' disease on a vessel? Select the ONE correct answer.

- **By breathing infected aerosols.**
- By drinking contaminated water
- By getting contaminated water on the skin.
- By human contact.

How could a person become sick after drinking a vessel's potable water supply? Select the TWO best answers.

- **The vessel's disinfection plant is not operating correctly.**
- **The water has not been chemically treated correctly.**
- Their immune system was too low.
- They have drunk too much of it.

How is a virus different to a bacterium? Select the ONE best answer.

- **A virus is not a living organism.**
- A virus causes a worse illness than bacteria.
- A virus is larger than bacteria.
- A virus is made up of a string of bacteria.

How often should the water in a regularly used leisure pool, as found on a passenger vessel, be tested for contaminants and chlorine levels? Select the ONE correct answer.

- **It should be continuously tested.**
- Often, particularly when there are children using it.
- Once a day.
- Once every hour.

How severe can water borne illnesses be? Select the ONE best answer.

- **They can cause death or disability.**
- They are usually not that severe.
- They will only give people diarrhoea.
- They will usually stop people working.

How should a calorifier be sanitised? Select the ONE best answer.

- **By following the manufacturer's advice.**
- Drain the water out and flush it with detergent.
- Fill it with a solution containing chlorine at 500 ppm.
- Heat the water in it up to 100 degrees Celsius.

How should a vessel's air conditioning system be maintained, to keep it free from the Legionella bacteria? Select the ONE correct answer.

- **Clean, free from scale and sludge.**
- At a temperature above 25 degrees Celsius.
- Clean, but with some residual moisture in the system.
- The system should be left alone, as activity releases any bacteria present.

How should shower heads and taps be disinfected? Select the ONE correct answer.

- **By using a chlorinated solution with 50 ppm concentration.**
- By using a chlorinated solution with 200 ppm concentration.
- With a standard detergent or cleaning product.
- With tap water which is disinfected by the system used on board.

If a potable water tank has been entered for inspection, what should happen to it before it is put back into use? Select the ONE best answer.

- **It should be closed up and disinfected at 50 ppm chlorine before use.**
- It should be cleaned with a standard chemical detergent before filling.
- It should be closed up and disinfected at 5 ppm chlorine before use.
- It should be filled with fresh water, and then pumped out to clean the tank.

What are the consequences to a vessel operator of water borne illness on board one of their vessels? Select the ONE best answer.

- **Delays, costs and a possible damaged reputation.**
- None because the illness will be cleared up.
- The crew refuse to sail on the vessel.
- The loss of a quantity of potable water.

What contaminants are likely to be found in a leisure pool on a vessel? Select the ONE best answer.

- **Those originating from people.**
- Chemicals from grey water treatments.
- Molluscs that were brought with the water when it was bunkered.
- Pesticides that are washed off the land.

What does the World Health Organisation recommend should be sanitised on a regular basis? Select the ONE best answer.

- **Any object that a person contacts.**
- Shower heads and sink taps, each month.
- Swimming pools and spa bath waters, daily.
- Taps and toilet flush handles, on a weekly basis

What happens when a vessel arrives with severe sickness on board? Select the TWO best answers.

- **Sick people will immediately be taken to hospital on arrival.**
- **The port health authorities will attend upon arrival.**
- Nobody will want to go near it.
- The cargo operation will immediately start.

What is the most likely way that a contaminated aerosol causes infection in a person? Select the ONE best answer.

- **A person breathes in the aerosol.**
- The aerosol contaminates a person's food.
- The aerosol contaminates the vessel's water supply.
- Through an open cut in the skin.

What is the only type of water that should be supplied to a food preparation area? Select the ONE correct answer.

- **Potable water.**
- Grey water.
- Salt water.
- Untreated fresh water.

What is the recommended cleaning interval for a shower head? Select the ONE correct answer.

- **Every 3 – 6 months.**
- Every 4 weeks.
- Every week.
- Yearly.

What regular maintenance is needed for a potable water tank? Select the ONE correct answer.

- **It should be opened up, flushed and disinfected.**
- It should be re-filled with super chlorinated water and left for 12 days.
- Nothing, as it is always contains potable water.
- The water level should be maintained at least half-full.

What should a crew member ensure for water hygiene, when it is bunkered from ashore? Select the ONE best answer.

- **That the supply hose is clean and flushed through.**
- That the correct grade of water is supplied to the vessel.
- That the holding tank does not overflow.
- That the supply hose is well secured to stop it flailing around.

What should happen regularly to the surfaces of a whirlpool and spa pool? Select the TWO correct answers.

- **The surfaces should be physically cleaned.**
- **The surfaces should be super chlorinated.**
- The surfaces should be covered in a non-skid coating.
- The surfaces should be regularly painted.

What should happen to the water used in a leisure pool to keep it clean and free from debris? Select the ONE correct answer.

- **It should be circulated through filters.**
- It should be changed every week.
- It should be circulated through the hydrophore.
- It should be totally changed daily.

What sort of disinfection should be used to maintain the quality of leisure pool water? Select the ONE correct answer.

- **Some form of chemical protection.**
- A super strength chemical to protect the users.
- UV and silver ionisation combined.
- UV disinfection is adequate.

What type of water should always be used in a leisure facility on a vessel? Select the ONE correct answer.

- **Potable water.**
- Fresh water.
- Manufactured water.
- Salt water.

Which ONE of the following is NOT included in vessel sanitation records?

- **Treatments given to boiler feed water.**
- Actions taken to sanitise shower heads and taps.
- All areas where water borne infections may be found.
- Treatment given to potable water.

Which answer best completes the following statement? 'In a vessel's accommodation, bacteria can easily grow . . .

- **. . . because there is often moisture and an ideal temperature'.**
- . . . because the crew do not keep the accommodation very clean'.
- . . . because there is no direct sunlight to kill it'.
- . . . because there is still, dry air'.

Which answer best completes the following statement? 'When a fresh water generator is being used on a vessel . . .

- **. . . the automatic water testing system should be checked occasionally with a manual test'.**
- . . . it uses a lot of energy, so the water is very expensive compared to shore water'.
- . . . there is no need to treat the water made, as it is pure'.
- . . . there will never be a need to take fresh water from the shore'.

Why can a calorifier be a breeding ground for bacteria? Select the ONE correct answer.

- **Because there is a range of temperatures ideal for bacterial growth.**
- Because the water at the bottom can become stagnant.
- Because the water in a calorifier is very cold.
- Because the water in a calorifier is very hot.

Why do filters in a potable water system have to be opened up and cleaned regularly? Select the ONE best answer.

- **Because they can be a breeding ground for bacteria.**
- Because the vessel procedures demand it.
- To check that the filter is not damaged.
- To make sure they are not blocked.

Why is it important to have a planned maintenance system in operation for fresh water generators? Select the ONE best answer.

- **So that the water made is of excellent quality.**
- So that the company does not have to buy bottled water.
- To make sure that the generator lasts a long time.
- To stop the crew complaining about the taste of the tap water.

Why is it most important to keep air conditioning systems in a clean condition? Select the ONE best answer.

- **To stop the air flowing through it spreading bacteria.**
- To ensure that it does not get blocked up.
- To ensure that the cooling system works when a vessel is in the Tropics.
- To stop the air that is blown into the accommodation from smelling.

0340 – Introduction to bulk carriers

Version 1.0 (1086)

Approximately how long does a typical washing cycle on a fixed hold washing installation take? Select ONE correct answer.

- **2 hours**
- 1 hour
- 30 minutes
- 4 hours

Do accurate and complete written records provide supporting evidence of "due diligence" on the part of the ship under the Hague - Visby Rules?

- **Yes**
- No

How can you make sure that the safe working load limits of the mucking winch are met? Select ONE correct answer.

- **By limiting the size of the lifting bucket**
- By not over-filling the bucket or drum
- By only lifting low density cargoes
- By weighing each bucket before lifting

If a ship is floating in fresh water which ONE of these is the Dock Water Allowance?

- **The fresh water allowance for the ship's draft**
- 100 mm
- 25 mm
- Zero

Should a Draft Survey hydrometer be used for Load Line purposes?

- **No**
- Yes

Under a time charter who pays for the fuel used by the ship? Select ONE correct answer.

- **The charterer**
- The agent
- The shipowner
- The shipper

Under a voyage charter who pays for the fuel used by the ship? Select ONE correct answer.

- **The shipowner**
- The agent
- The charterer
- The shipper

Under the Common Structural Rules for Bulk Carriers, which notation is used for the most robust design? Select ONE correct answer.

- **BC-A**
- BC-B
- BC-C
- BC-D

What does a solid equilateral triangle mark on each side of a bulk carrier, near the deck line amidships, tell you? Select ONE correct answer.

- **That there are restrictions on the ship's ability to load higher density cargoes**
- That is the location of the centre of flotation
- The triangle mark makes it easier to locate the load line markings
- This is the deepest load line to which the ship may be loaded in fresh water

What does 'OBO' stand for in relation to bulk carriers? Select ONE correct answer.

- **Oil/Bulk/Ore carrier**
- Open Bulk/Ore carrier
- Optional Bulk Ore carrier
- Overseas Bulk Operator

What is the SOLAS upper limit for cargo densities for cargoes that are allowed to be loaded onto old bulk carriers? Select ONE correct answer.

- **1,780 kg/m^3**
- 0,780 kg/m^3
- 1,000 kg/m^3
- 1,100 kg/m^3

What is the approximate cargo-carrying capacity of a 'Capesize' bulk carrier? Select ONE correct answer.

- **100,000 – 190,000 dwt**
- 5,000 – 12,000 dwt
- 50,000 – 60,000 dwt
- 75,000 – 85,000 dwt

What is the approximate cargo-carrying capacity of a 'Handysize' bulk carrier? Select ONE correct answer.

- **25,000 – 40,000 dwt**
- 12,000 - 20,000 dwt
- 5,000 – 12,000 dwt
- 50,000 – 60,000 dwt

What is the approximate cargo-carrying capacity of a 'Panamax' bulk carrier? Select ONE correct answer.

- **40,000 – 80,000 dwt**
- 100,000 – 190,000 dwt
- 25,000 – 40,000 dwt
- 5,000 – 12,000 dwt

When a master signs a Bill of Lading who are they signing on behalf of? Select ONE correct answer.

- **The shipowner**
- The agent
- The charterer
- The shipper

When a psychrometer is used for measuring the relative humidity in the air in a hold should the hold ventilation be stopped?

- **Yes**
- No

Where can detailed information about cargo density restrictions be found? Select ONE correct answer.

- **In the ship's loading manual**
- In the Operational Limitations certificate
- On the general arrangement drawing
- On the midship section drawing

Which ONE of the following cargoes is a 'minor bulk'?

- **Cement**
- Coal
- Grain
- Iron ore

Which ONE of the following cargoes is described as a 'major bulk'?

- **Coal**
- Cement
- Salt
- Sulphur

Which ONE of these best describes a "Bill of Lading"?

- **A document that provides evidence that the cargo has been received onboard for shipment**
- A document issued by the shipowner to say that the cargo quantities are correct
- An invoice for the work of the stevedores in loading the ship
- The shippers' document that declares the type of cargo loaded and its characteristics in accordance with the IMSBC Code

Which ONE of these best describes a "Contract of Affreightment"?

- **It is a series of voyage charters over a long period**
- It is a contract to carry certain freight on a single trip
- It is a single voyage charter
- It is a time charter for a fixed period

Which ONE of these best describes a "charter party"?

- **A written agreement between a shipowner and a charterer for the hire of a vessel**
- A social event organised after successful completion of a charter
- A written agreement between a shipper and a charterer to carry a particular cargo.
- A written agreement between the cargo interests and the master to carefully carry a cargo

Which ONE of these best describes the colour of the triangle marking on some bulk carriers, required by SOLAS?

- **Of contrasting colour to the hull**
- Always black
- Always white
- The same as the hull colour

Which ONE of these best describes what giving "Notice of Readiness" means?

- **The charterer is notified in writing by the master that the ship is ready to load or to discharge**
- The harbourmaster is notified that the main engines are about to be started
- The port authority is notified that the ship is ready to sail and needs a pilot
- The shipper is notified in writing that the hatches are open and ready to receive cargo

Which ONE of these best describes what "mucking winches" are used for?

- **They are used during hold cleaning to lift buckets of cargo residues to deck level ready for disposal**
- For emergency closure of hatch covers
- For lifting cargo residues onto the jetty
- They are used for positioning hold washing machines

Which ONE of these best describes when "Laytime" starts?

- **When Notice of Readiness has been correctly issued and acknowledged**
- At the start of the 'laycan spread'
- On the laytime start date written in the charter party
- Within 24 hours of arrival at the port limits or anchorage

Which ONE of these best describes where or what the "laycan spread" is?

- **The time interval between the earliest layday date and the date of possible cancellation of the charter**
- The earliest date for cancellation of the charter
- The number of days after the Notice of Readiness is issued when the ship can load the cargo
- The port area where ships wait for berths

Which ONE of these best describes why accurate and complete written records are important during charter party disputes?

- **They provide evidence of facts in the event of any disputes under a charter party**
- They demonstrate to the shipowner's office that the ship was not at fault
- They help show that the ship was not at fault
- They help to divert blame from the ship to others in the event of a dispute

Which ONE of these best describes why "sump pumps" are used during hold cleaning operations?

- **They can pump water mixed with cargo residues, which would block the ship's fixed bilge pumping system**
- The pump provides a jet of water which helps keep the hold bilges free from cargo residues
- The pump provides a jet of water, to clean the cargo residues from the hold structures
- The ship's fixed bilge pumping system is not capable of pumping out small quantities of cleaning water

Which ONE of these best explains the difference between a 'geared' bulk carrier and an 'ungeared' bulk carrier?

- **A 'geared bulk' carrier has cranes and grabs for discharging cargoes: an 'ungeared' ship does not**
- A 'geared' bulk carrier carries her own cargo grabs for use by discharge port shore cranes: an 'ungeared' ship does not
- A 'geared' bulk carrier has a gearbox attached to the main engine: an 'ungeared' ship does not
- An 'ungeared' ship does not carry her own hold cleaning equipment; a 'geared' ship does.

Which ONE of these best explains why bulk carriers have a large water ballast capacity?

- **On un-laden passages the ballast is needed to keep the propeller immersed and to give sufficient forward draft**
- Lighter cargoes may not bring the ship down to her load line
- They need a large amount of water for hold cleaning
- They need large amounts of ballast to raise the centre of gravity when loaded

Which ONE of these explains how "sump pumps" are usually powered?

- **Water power**
- Compressed air
- Hydraulic power
- Low voltage electricity

Which ONE of these is a feature of topside tanks that is particularly helpful when carrying high density bulk cargoes, such as iron ore?

- **They can be filled with water ballast to raise the vertical centre of gravity of the ship**
- They are sloped helping to keep the cargo in place without the need for trimming
- They protect the cargo from collision damage
- They raise the centre of gravity of the cargo in the hold

Which ONE of these is a psychrometer used to measure?

- **The relative humidity of air in a hold, and its dew point**
- Cargo temperatures
- Moisture content of cargoes
- The concentration of fumigant left in the hold

Which ONE of these signs the Bill of Lading, so confirming the cargo condition and quantity?

- **The master, or an assigned agent onshore**
- The charterer
- The shipowner
- The shipper

Which ONE of these will usually use the Draft Survey hydrometer?

- **The draft surveyor**
- A Port State Control Officer
- The cargo shipper
- The ship's officers

Which TWO of these are ways to use a sump pump to drain the maximum water from a cargo hold?

- **Connect the vacuum hose to the pump**
- **Turn the hopper to face downwards**
- Turn the driving water on and off
- Use spades to fill the hopper

Which TWO of these can be used to sample the different layers of water density the ship is floating in?

- **A water sampling bottle with a small opening can be used to get an average sample while lowered to keel depth**
- **A water sampling bottle with removable plug can be used to get samples from different water depths**
- A bucket can be lowered to the deepest draft and then pulled up to the deck
- The surface water density provides a reasonable sample of the water density

Which TWO of these features of bulk carrier design particularly help with the discharge of cargoes?

- **Sloping hopper tank sides**
- **Wide hatch openings**
- Corrugated bulkheads
- Double bottom tanks

Which TWO of these must a master do upon receiving a copy of the charter party onboard?

- **Confirm receipt to all interested parties**
- **Ensure that he and all senior officers understand the key terms of the charter party**
- Keep the terms of the charter party secret
- Sign it and return the signed copy to the Agent

Which of these is the name of the principal rules governing the carriage of goods by sea? Select ONE correct answer.

- **The Hague - Visby Rules**
- The Carriage of Goods by Sea Regulations
- The London - Paris Rules
- The York - Antwerp Rules

Why are some bulk carriers designated by a place name, for example 'Dunkirkmax'? Select ONE correct answer.

- **Because their dimensions are at the limit for the places named**
- Because the cargoes they are allowed to carry only come from those places
- Because the ship's owners come from those places
- Because they are only allowed to load at those ports

Why is it important to monitor the water level in the hold during the fixed water jet hold washing cycle? Select ONE correct answer.

- **To make sure that the tank top is correctly washed by the water jets**
- To avoid damage to the bulkhead by the sloshing of water
- To avoid the bilge becoming blocked with debris
- To avoid the cleaning water tank emptying

0341 – Liquefying cargoes

Version 2.0 (1274)

A cargo of 60,000 tonnes of nickel ore is for shipment. Which ONE of the following is the number of samples that should be taken by the shipper, according to the IMSBC Code?

- **120**
- 10
- 50
- 6

Although the 'can test' is not a substitute for proper laboratory testing, which ONE of the following correctly identifies what a 'wet' result should tell the master or mate?

- **That the sample is likely to be above the TML and that there is a risk of cargo liquefaction in the remainder of the cargo presented to the ship**
- That the sample for the 'can test' was taken from too near the surface of the stockpile
- That the sample is likely to be above the TML but the rest of the cargo will probably be safe to load
- That the sample is wet but more 'can tests' should be carried out to find other dryer samples in the stockpiles so as to get an overall average moisture content which is below the TML

How many different test methods that can be used by the shipper for determining the Transportable Moisture Limit are identified in the IMSBC Code?

- **3**
- 1
- 2
- 4

If a shipper declares a cargo as IMSBC Code Group C but the master has suspicions that this might not be correct, which TWO of the following steps should the master take?

- **Contact the Competent Authority for the port**
- **Contact the ship manager**
- Check the IMDG Code
- Check the terms of the charter party

If no shipper's declaration has been received, should cargo loading be allowed to start?

- **No**
- Yes

If the moisture limits are exceeded in any cargo hold, once cargo loading has started, is the master allowed to suspend loading operations?

- **Yes**
- No

If the shipper does not provide satisfactory cargo information, or is not co-operative in safe loading, to which TWO of the following should the master report?

- **The Competent Authority for the port**
- **The ship's P & I Club**
- The ship's classification society
- The ship's Hull and Machinery insurers

In which ONE of these books can you find detailed guidance on cargoes which may liquefy?

- **IMSBC Code**
- FTP Code
- IMDG Code
- ISM Code

Is it acceptable to load cargo with a moisture content (MC) less than the Transportable Moisture Limit (TML) in some holds and cargo with a MC greater than the TML in other holds?

- **No, this is not correct**
- Yes, this is acceptable at any time
- Yes, this is correct, so long as the MC is below the flow moisture point
- Yes, this is correct, so long as the overall average MC is less than the TML

Is the following statement always true?

"When a can test shows no free fluid it means that the cargo is safe to load."

- **No**
- Yes

The cargo is not listed in the IMSBC Code but consists of fine particles and appears to be damp or wet.

Can it be safely loaded without testing for moisture content?

- **No**
- Yes

What is the IMSBC Code recommendation for the minimum number of times that the can is banged down when carrying out a 'can test'?

- **25 times**
- 15 times
- 5 times
- 50 times

Which ONE of the following IMSBC Code groupings should be used if 80% of a bulk coal cargo is made up of fine particles and the remaining 20% is larger lumps?

- **Group A and B**
- Group A only
- Group B only
- Group C only

Which ONE of the following actions is available to the master, before loading, to assess the moisture content of a cargo?

- **They can look at the cargo stockpile and, if it looks wet carry out 'can tests'**
- They can ask which moisture test method was used
- They can take a sample from the cargo stockpile and heat it in a pan
- They can take a sample from the cargo stockpile and look at it under bright light

Which ONE of the following best defines the term 'cargo liquefaction' in relation to solid bulk cargoes?

- **When the whole body of the cargo in a hold compacts during the voyage and the cargo starts to behave like a fluid.**
- When a Group B dry bulk cargo shifts due to excessive rolling angles
- When a Group C dry bulk cargo shifts due to excessive rolling angles
- When leaks of water through a hatch cover cause the cargo to become wet

Which ONE of the following best describes what happens to the moisture in a cargo when moisture migration takes place?

- **It moves down towards the bottom of the cargo stow**
- It dries out naturally
- It moves from side to side
- It moves up towards the surface of the cargo

Which ONE of the following cargo groups is subject to the IMSBC Code?

- **Solid bulk cargoes other than grain**
- Cargoes in intermediate bulk containers
- Crude oil
- Grain

Which ONE of the following correctly identifies how the master should be informed that the Competent Authority has approved the shipper's test procedures?

- **He is given a document by the shipper or the Competent Authority stating that the procedures have been approved**
- The charterer tells him
- The shipper gives him an email statement
- The shipper tells him

Which ONE of the following correctly identifies why loading should only be started after satisfactory declarations have been received from the shipper?

- **It is very difficult to unload wet cargo in the loading port once it has been loaded**
- If 'wet' cargo is loaded it will need to be mixed with some dry cargo to give an acceptable average moisture content in each hold
- The shipper is responsible for loading the ship safely
- The shipper is responsible for loading the ship within the charter party time limits

Which ONE of the following describes how the shipper should let the ship know the results of the moisture content tests of a dry bulk cargo?

- **They issue a signed certificate to the master**
- They inform the master by email
- They provide the information on completion of cargo loading.
- They tell the OOW on deck

Which ONE of the following identifies the cargo group(s) that a cargo that may liquefy belongs to, according to Appendix 1 of the IMSBC Code?

- **Group A**
- Group B
- Group B and C
- Group C

Which ONE of the following identifies the most important result of cargo liquefaction?

- **Cargo shift leading to loss of stability**
- Blockage of bilge suctions
- Difficulty in discharging the cargo
- Risk of cargo damage

Which ONE of the following is correct if a cargo stockpile is sampled, and the Moisture Content (MC) is found to be uniform at 95% of the Flow Moisture Point (FMP)?

- **It is not safe to load the cargo because the MC is more than 90% of the FMP**
- It is always safe to load the cargo as the MC is less than the FMP
- It will be safe to load the cargo, but only if it passes a 'can test'
- It will be safe to load the cargo, but only if the cargo looks dry

Which ONE of the following is responsible for certifying the Transportable Moisture Limit and the actual moisture content of a cargo, or part cargo?

- **The cargo shipper**
- The charterer
- The master
- The ship's agent

Which ONE of the following is responsible for issuing the cargo declaration if a cargo, offered for loading, is not shown in the IMSBC Code Appendices?

- **The Competent Authority**
- The ship manager
- The ship owner
- The shipper

Which ONE of the following is responsible for providing the Competent Authority with the cargo characteristics if a cargo, offered for loading, is not shown in the IMSBC Code Appendices?

- **The shipper**
- The master
- The port authority
- The ship owner

Which ONE of the following is responsible for warning a ship about the risk of 'wet base' cargoes?

- **The cargo shipper**
- The charterer
- The ship's agent
- The ship's master

Which ONE of the following is the maximum moisture content that is considered safe for the carriage of IMSBC Group A bulk cargoes?

- **100% of the Transportable Moisture Limit (TML)**
- 100% of the Flow Moisture Point (FMP)
- 75% of the Transportable Moisture Limit (TML)
- 90% of the Transportable Moisture Limit (TML)

Which ONE of the following is the maximum time that is allowed before loading that the declared moisture content can be measured by the shipper?

- **7 days**
- 10 days
- 2 weeks
- 5 days

Which ONE of the following is the most accurate if the shipper declares that the moisture content is different at various places in the stockpiles of an IMSBC Group A cargo?

- **Only cargo which has a moisture content equal to or less than the TML can be safely loaded**
- Cargo from the various stockpiles can be mixed: provided the average moisture content in each hold is equal to or less than the TML
- Cargo from the various stockpiles can be mixed: provided the average moisture content in the whole cargo is equal to or less than the TML
- If the average moisture content in the various stockpiles is equal to or less than the TML then loading can safely go ahead

Which ONE of the following is the most likely consequence of cargo shift due to cargo liquefaction?

- **The ship heels to an angle of loll where she has little residual stability and a free surface in the cargo spaces**
- Structural damage to the hatch covers
- The ship heels, but remains stable
- The ship trims by the stern

Which ONE of the following is the percentage of the Flow Moisture Point that is considered to be the Transportable Moisture Limit?

- **90%**
- 100%
- 50%
- 75%

Which ONE of the following most accurately describes the action that should be taken where the ship's master suspects that the various cargo stockpiles that are to be used for loading the ship may have varying moisture contents?

- **The master must insist that the shipper make a separate declaration for each stockpile or for the stockpile intended for each hold**
- Ensure that the cargo in each hold is mixed, so that the average moisture content is acceptable
- No action is required as long as the average moisture content over the whole ship will be OK
- The master should write a Note of Protest

Which ONE of the following most accurately describes when it is acceptable to load cargoes with a moisture content above the Transportable Moisture Limit (TML) on a normal bulk carrier?

- **It is never acceptable**
- Only if the cargo has been previously carried without problems
- Only if the forecast weather 'en route' is good
- Only if the moisture content is less than 5% above the TML

Which ONE of the following most accurately describes why cargo liquefaction causes loss of stability?

- **The liquefied cargo acts like a free surface, causing a virtual rise in the ship's centre of gravity**
- The centre of gravity of the cargo stow gets higher
- The centroid of the waterplane moves upwards
- The ship heels to small angle

Which ONE of the following most accurately identifies when loading of a bulk cargo should be allowed to start?

- **After satisfactory written cargo declarations have been received from the shipper**
- As soon as the hatches are open
- Only when the weather conditions are dry
- When the shipper tells the master to start

Which ONE of the following statements is the most accurate in relation to a sample of IMSBC Group A cargo taken from a stockpile which appears to be dry?

- **It might be safe to load but the shipper's declaration must be checked, to ensure the moisture content is below the declared TML**
- If it also passes a can test it is OK to load the cargo
- If it looks dry it is OK to load the cargo
- It might be safe to load but the shipper must be telephoned, to ensure the moisture content is OK before loading begins

Which ONE of the following terms is the acronym 'BCSN' used for in relation to the IMSBC Code?

- **Bulk Cargo Shipping Name**
- Bulk Cargo Shipment Number
- Bulk Cargo Statutory Name
- Bulk Carrier Ship Number

Which ONE of the options is the most accurate in relation to the following statement?

"If moisture migrates to the bottom part of the cargo stow, and 'wet base' conditions arise, it can lead to a dangerous cargo shift"

- **Yes, the whole body of the cargo can slide from side to side when the ship rolls with the water acting as low-friction lubricant**
- No, high moisture content only leads to the upper part of the cargo sloshing back and forth
- No, only liquefaction of the top surface of the cargo causes cargo shift
- Yes, the bottom of the cargo can move, leaving the higher parts unsupported

Which ONE of these is the easiest way for the master to find out which test method was used to find the Transportable Moisture Limit (TML)?

- **The shipper has to provide the master with test certificates for the test method used to find the TML**
- The master can contact the Competent Authority in the discharge port
- The master can contact the Competent Authority in the loading port
- The ship's agent can contact the shipper

Which TWO of the following dry bulk cargoes are most likely to liquefy?

- **Cargoes made up of fine particles**
- **Mixed fine particles and lumps**
- Cargo which is mainly medium to large lumps
- Large rocks

Which one of the following most accurately identifies when cargo liquefaction can occur?

- **At any stage while the cargo is onboard**
- After arrival at the discharge port
- Only during heavy weather
- Only near to the end of the voyage

0342 – Common bulk cargoes

Version 1.0 (1069)

According to the International Grain Code, in a modern bulk carrier design, do the ends of each hold have to be completely filled/trimmed? Select the ONE correct answer.

- **There is a dispensation which permits them to not be trimmed into the end spaces**
- It depends on the acceptability of the stability calculations
- The ends have to be fully trimmed at all times
- The ends of the holds must never be filled

Apart from rain, which ONE of these is the major cause of grain cargo wetness?

- **Ingress of sea water through leaky hatch covers**
- Cargo sweat
- Ice and snow
- Ship sweat

Does a 'partly-filled compartment' require to be over-stowed when a modern bulk carrier loads grain? Select the ONE correct answer.

- **Usually not, but the stability needs to be checked for acceptability**
- Always
- Never
- Only in the centre holds

For how long should hold ventilation be shut down before gas monitoring measurements are taken? Select the ONE correct answer.

- **4 hours**
- 1 hour
- 2 hours
- 3 hours

How can the amount of methane in the space over a coal cargo be measured? Select the ONE correct answer.

- **Using an explosimeter**
- By smelling the gas
- Litmus paper attached to a brass sounding rod
- Using a wet and dry bulb thermometer

How can the master decide what the maximum permissible grain heeling moment is for their ship? Select the ONE correct answer.

- **The maximum allowable grain heeling moment for the ship's displacement and KG**
- The maximum grain heeling moment found for any of the holds
- The maximum grain heeling moment permitted by the Load Line Convention
- The maximum value of calculated grain heeling moment for an angle of heel of 25º

How do the USA, Canada and Australia differ from the International Grain Code requirements in respect of stability calculations? Select the ONE correct answer.

- **They differ in their treatment of untrimmed ends of holds**
- They assume a final heel angle of the ship of 15º
- They differ in their treatment of down flooding angles
- They have lower requirements than the International Grain Code

How does the master know which fumigant has been used on a grain cargo? Select the ONE correct answer.

- **The charterer's fumigator will provide a hazardous chemical data sheet**
- The charterer will inform him
- The fumigator will speak to the master
- The shipper will inform him

How is the actual Grain Heeling Moment found for the whole ship? Select the ONE correct answer.

- **The Volumetric Heeling Moments for each hold are added together and corrected for the actual stowage factor**
- It is calculated from the statical stability curves for the ship
- It is found in the Grain Loading Manual
- The master should ask the shipper on completion of loading

How often should the hold bilges be sounded during a loaded voyage with an iron ore cargo? Select the ONE correct answer.

- **Every day**
- Every 12 hours
- Every watch
- Every week

How should non-cohesive cargoes, which have an angle of repose equal to, or less than, 30º (as defined in the IMSBC Code), be treated? Select the ONE correct answer.

- **As a grain cargo and the International Grain Code applied**
- As a cargo that may liquefy under the IMSBC Code
- As a Group B cargo under the IMSBC Code.
- As a standard Group C bulk cargo.

If infestation of a grain cargo is found at any inspection during the voyage or prior to arriving at the discharge berth what action should be taken? Select the ONE correct answer.

- **The master should inform his owners and arrange for expert advice on survey or fumigation**
- The master should get the crew to try to remove traces of the infestation
- The master should go alongside for discharge and hope that the infestation is not noticed
- The master should urgently arrange for fumigation on arrival

If self-heating has started which ONE action could cool the coal cargo?

- **Boundary cooling, if possible**
- Natural ventilation of the coal surface
- Open the hatches to let the breeze cool the cargo
- The hatches can be opened slightly and fire-fighting water sprayed into the coal

If self-heating is detected which TWO of these actions should be taken?

- **All ventilation should stop**
- **Hatch covers, coaming drains and hold accesses should be fully closed and sealed**
- Surface ventilation should continue
- Through-cargo ventilation should be started

If the angle of repose of most iron ore cargoes is about 35o, why they are required to be trimmed level? Select the ONE correct answer.

- **It spreads the heavy iron ore cargo weight evenly over the whole tank top**
- It is better for bulldozers when they are completing the stow
- It is easier to walk about on a level surface
- It makes the discharge easier

If there is any doubt about the accuracy of the cargo declaration what should the master do? Select the ONE correct answer.

- **Assume that the cargo might self-heat and act accordingly**
- Assume that the Declaration is correct
- Tell the Agent
- Tell the Owner

In addition to wetness, which ONE of these can make grain cargo self-heating more likely?

- **Stowage next to or above heated fuel tanks**
- Carbon monoxide
- Tropical weather
- Ventilation of the cargo

In the International Grain Code what does 'filled compartment - trimmed' mean? Select the ONE correct answer.

- **All of the spaces under the deck and hatches are filled as much as possible**
- The cargo is poured until it fills the hatch coamings
- The grain is trimmed to fill as much space as possible at the sides but not at the ends under the cross decks
- The grain is trimmed to give an even angle of repose

In the International Grain Code what does 'filled compartment – untrimmed' mean in the case of a modern bulk carrier? Select the ONE correct answer.

- **The hold is filled completely up to the topside tanks and hatch coamings but not the two fore and aft ends**
- The grain is trimmed to fill all of the available underdeck spaces
- The hold is filled but the cargo level does not reach the topside tanks
- The hold is partly filled

In the International Grain Code what does 'specially suitable compartment' mean? Select the ONE correct answer.

- **A hold that has longitudinal sloped topside tanks which are coincident with the hatch side girders**
- A hold that can be easily cleaned after the cargo is discharged
- A hold that is clean enough to receive a grain cargo
- A hold that is the correct size for the parcel of cargo to be loaded

In which ONE of these can the contents checklist for the loading plan be found?

- **The BLU Code**
- SOLAS Chapter XII
- The IMSBC Code Appendix 3
- The ship's loading manual

There will be certain times during grain loading when a ship will not have adequate stability to put to sea. What is the most important action to take? Select the ONE correct answer.

- **The master should include those times in the Ship-Shore Checklist and inform the terminal representative**
- The master should be prepared to go to sea at all times
- The master should notify the charterer
- The master should try to keep those periods to a minimum

To which IMSBC Code cargo Group(s) do hot and cold-moulded Direct Reduced Iron cargoes belong? Select the ONE correct answer.

- **Group B**
- Group A
- Group A and B
- Group C

Using temperature measuring ports or pipes, at how many levels should temperature monitoring be done? Select the ONE correct answer.

- **At 2 or 3 levels**
- At the lowest and mid-levels only
- Only at the lowest level
- Only at the surface

What TWO precautions should be taken to prepare for loading coal cargoes that are declared as giving off methane?

- **Instruments for measuring methane gas must be available onboard**
- **Ventilation systems should be checked and working**
- Coaming drain valves should be blocked
- Ventilation systems should be blanked off

What action should be taken if insects, insect larvae, rats or mice, are seen, either dead or alive, at the time of loading grain? Select the ONE correct answer.

- **The master should stop loading, inform his owner and arrange for a surveyor to attend**
- Bury the infestation as much as possible
- The master should have samples taken to show to the discharge terminal
- Try to remove the infestation as much as possible and continue loading

What are Volumetric Heeling Moments? Select the ONE correct answer.

- **Heeling moments due to a specified amount of shifting of grain in each hold for a range of hold loading levels**
- Heeling moments calculated using the volumes of each hold
- Heeling moments caused by shifting of water ballast during grain loading
- Heeling moments due to the volumes of water displaced during heeling

What are the TWO main issues when carrying grain cargoes?

- **Damage, contamination and infestation**
- **Reduction or loss of stability due to cargo shifting**
- Cargo liquefaction
- Explosion risk

What cargo would you expect to be shipped in a VLBC? Select the ONE correct answer.

- **Iron ore**
- Coal
- Containers
- Grain

What concentration of carbon monoxide does the IMSBC Code say indicates a coal cargo is self-heating?

- **50 ppm**
- 10 ppm
- 100 ppm
- 80 ppm

What does a methanometer measure? Select the ONE correct answer.

- **The percentage of methane in the air**
- The amount of air as a percentage of the methane LEL
- The amount of methane as a percentage of the methane LEL
- The likelihood of an explosion as a percentage

What does a solid equilateral triangle mark on each side of a bulk carrier's hull near the deck line amidships indicate? Select the ONE correct answer.

- **There are restrictions on the ship's ability to load higher density cargoes**
- That is the location of the centre of flotation
- The triangle mark makes it easier to locate the load line markings
- This is the deepest load line to which the ship may be loaded in fresh water

What does an 'explosimeter' measure? Select the ONE correct answer.

- **The amount of methane as a percentage of the methane LEL**
- The amount of air as a percentage of the methane LEL
- The likelihood of an explosion as a percentage
- The percentage of methane in the air

What does 'LEL' stand for? Select the ONE correct answer.

- **Lower Explosive Limit**
- Light Emitting Lamp
- Lower Energy Level
- Lower Environmental Limit

What does 'MHB' stand for in relation to bulk carriers and their cargoes? Select the ONE correct answer.

- **May be Hazardous in Bulk**
- Marine Hazard in Bulk
- Most Hatches loaded with Bulk
- Must Have Bulkheads

What extent of grain shifting is assumed when calculating Volumetric Heeling Moments? Select the ONE correct answer.

- **25º in every full hold and 15º in every part filled hold**
- 12º in every hold
- 25º in all holds
- 35º in every partly filled hold and 12º in full holds

What is a typical stowage factor for wheat or corn? Select the ONE correct answer.

- **1.36 m^3/t**
- 0.50 m^3/t
- 0.85 m^3/t
- 2.85 m^3/t

What is a 'Document of Authorisation'? Select the ONE correct answer.

- **A document issued by the ship's flag state certifying that the ship is capable of loading grain cargoes in accordance with the International Grain Code**
- A document issued by the ship's flag state certifying that the ship meets the requirements of the Load Line Convention.
- A document provided by the shipper to permit loading to commence
- A document required before discharge of grain cargo can proceed

What is likely to happen if a grain cargo is loaded which is wet? Select the ONE correct answer.

- **The cargo may self-heat which can encourage sprouting and/or mould**
- The bilges will need to pumped more frequently
- The cargo may give off toxic gases
- The cargo may shift during the voyage

What is the Grain Heeling Moment? Select the ONE correct answer.

- **It is the total heeling moment assumed to be caused by the shift of the grain cargo specified in the International Grain Code**
- It is the equal to the heeling moment when the cargo has shifted through 12º
- It is the heeling moment due to wind and wave action on the ship
- It is the moment in time when grain starts to shift causing a heel

What is the coal temperature above which the IMSBC Code says loading should not be allowed? Select the ONE correct answer.

- **55°C**
- 100°C
- 25°C
- 32°C

What is the main reason for greasing hatch cover wheels and steel-to-steel contact points before opening and closing of covers? Select the ONE correct answer.

- **To help prevent sparks which could ignite any explosive gases**
- To enable the hatch covers to operate efficiently
- To keep the noise down
- To prevent corrosion

What is the maximum acceptable level of methane in the space above a coal cargo according to the IMSBC Code? Select the ONE correct answer.

- **20% of the LEL**
- 5% of the LEL
- 70% of the LEL
- 90% of the LEL

What is the maximum permissible angle of heel after grain shifting has been calculated? Select the ONE correct answer.

- **12°**
- 10°
- 15°
- 5°

What is the minimum permissible GM fluid which must be achieved before considering grain stability? Select the ONE correct answer.

- **0.3 metres**
- 0.1 metres
- 0.55 metres
- 0.75 metres

What is the typical angle of repose of a grain cargo? Select the ONE correct answer.

- **20°**
- 10°
- 35°
- 5°

What is the typical stowage factor of lump ore? Select the ONE correct answer.

- **0.35 to 0.85 m^3/t**
- 0.80 to 1.5 m^3/t
- 1.00 to 1.2 m^3/t
- 1.3 to 2.0 m^3/t

What precautions should be taken when loading grain to avoid rain water affecting the cargo? Select the ONE correct answer.

- **Hatch covers should be closed on non-working hatches**
- Arrange tarpaulins to keep off the rain
- Never load in rainy conditions
- Ventilate the holds as much as possible between showers

What should the maximum variation in height be across the top of an iron ore cargo? Select the ONE correct answer.

- **2 metres or (Beam of ship/10) metres, whichever is the less**
- (Beam of ship/20) metres or 2.5 metres, whichever is the less
- 1 metre
- It should be absolutely flat and level

When should coal cargo temperatures be taken? Select the ONE correct answer.

- **Prior to loading and during loading**
- After the hatches are closed
- On completion of loading
- There is no requirement to take the cargo temperature

Where can the Master find detailed procedures for gas monitoring during the voyage? Select the ONE correct answer.

- **The IMSBC Code**
- IMO Circular A.(82) 743
- The ISM Code
- The shipper's declaration

Where can the Ship-Shore Safety Checklist for use by ships and grain loading terminals be found? Select the ONE correct answer.

- **The BLU Code.**
- SOLAS documentation
- The IMSBC Code Appendix 3
- The ship's Loading Manual

Where can the master find information on coal cargoes and their hazards? Select ONE correct answer.

- **The IMSBC Code**
- The International Grain Code
- The internet
- The ISM Code

Which IMSBC Code cargo group(s) does an iron ore cargo that may liquefy belong to? Select the ONE correct answer.

- **Group A**
- Group B
- Group C
- Groups A and C

Which ONE of these IMSBC Code cargo groups would "coal slurry" fall into?

- **Group A**
- Group A and B
- Group B
- Group C

Which ONE of these can give rise to an explosion hazard in a coal cargo?

- **Hydrogen**
- Carbon dioxide
- Nitrogen
- Propane

Which ONE of these contains details of cargoes that should be treated like grain, even though they are not grain?

- **The IMSBC Code, Appendix 3**
- MARPOL Annex V
- SOLAS Chapter XII
- The International Grain Code

Which ONE of these contains guidance on iron ore fines or concentrates which may liquefy?

- **The International Maritime Solid Bulk Cargo Code**
- IMO circular MEPC 157(19)
- International Grain Code
- MARPOL

Which ONE of these contains the permissible shear force and bending moment data for the ship?

- **The ship's loading manual**
- The shipper's declaration
- The ship's official log book
- They are on the ship's structural arrangement drawings

Which ONE of these describes why hold bilges should be regularly pumped during voyages loaded with Group C iron ore?

- **To limit the amount of free water left in the hold at the end of discharge**
- To minimise any possible free surface effect on ship stability
- To prevent liquefaction
- To prevent water damaging the lower part of the cargo

Which ONE of these gases typically causes an explosion risk with coal cargoes?

- **Methane**
- Nitrogen
- Oxygen
- Propane

Which ONE of these is a hazard associated with cold-moulded Direct Reduced Iron (DRI)?

- **Self-heating and hydrogen gas fires**
- Explosion
- Liquefaction
- Methane production

Which ONE of these is another name for 'alternate hold loading'?

- **Jump loading**
- Block loading
- Joint loading
- Non-adjacent loading

Which ONE of these is given off when coal self-heats?

- **Carbon monoxide**
- Carbon dioxide
- Methane
- Oxygen

Which ONE of these is given off when corrosive acids are formed in a coal cargo?

- **Hydrogen**
- Carbon monoxide
- Methane
- Oxygen

Which ONE of these is likely to cause structural corrosion in a hold containing coal? Select the ONE answer that is the most likely.

- **Sulphuric acid**
- Acetic acid
- Formic acid
- Nitric acid

Which ONE of these is necessary if cold moulded Direct Reduced Iron is carried?

- **The space above the cargo should be filled with inert nitrogen at all times**
- The cargo should be sprayed with water from time to time
- The holds must be sealed as air-tight as possible
- The space above the cargo should be kept surface ventilated at all times

Which ONE of these is the percentage of methane in air that creates an explosion risk?

- **5% to 16%**
- 0% to 5%
- 17% to 23%
- 25% to 30%

Which ONE of these is the solution to the problems caused by a very high maximum loading rate?

- **The loading and ballast discharge must be carefully planned, with the plan explained to the terminal**
- The loading can go ahead at a high rate and the water ballast discharged later
- The loading rate can be slowed down
- The master can stop the loading once per watch, so as to be able to discharge water ballast

Which ONE of these should the master do if self-heating is detected with smoke or heat?

- **Immediately contact the owners to get expert advice**
- Sail on to the discharge port to deal with the problem on arrival
- Sail to a port of refuge and then contact the owner
- Try to fight the fire immediately

Which TWO are reasons why hatch covers on holds that are not being worked should be closed?

- **It reduces the amount of air likely to blow through the cargo**
- **It reduces the possibility of rain or snow entering the coal cargo**
- It helps to avoid crew falling into open holds
- It shows the stevedores which holds are to be filled

Which TWO methods of monitoring the temperature of a coal cargo during loading are recommended?

- **Use a mercury thermometer probe**
- **Use an infrared thermometer**
- Ask the stevedores
- Use a carbon monoxide gas detector

Which TWO of these are grain cargoes?

- **Maize**
- **Wheat**
- Dolomite
- Gypsum

Which TWO of these are particular hazards of coal cargoes?

- **Explosion**
- **Toxic gases**
- Infestation
- Shifting

Which TWO of these are signs of self-heating in a coal cargo?

- **Cargo temperature measurements at any level in the stow reach 40°C**
- **The cargo temperature is found to be rising quickly**
- Hydrogen gas is detected
- Methane levels are found to be high

Which TWO of these help personnel reduce the risks from the dust made by iron ore cargoes?

- **Regular washing of outer clothing**
- **Wear the correct PPE, including a dust mask**
- Do not spend any time on the foredeck
- Wait for the wind to drop before carrying out work on deck

Which TWO of these increase the risk of suffocation in a hold containing coal?

- **Carbon monoxide replaces some or all of the breathable air**
- **The oxygen is depleted by chemical reactions in the coal**
- Nitrogen replaces some or all of the breathable air
- The coal gives off inert gases

Which TWO of these iron ores are IMSBC Code Group C cargoes?

- **Lump ore**
- **Run of Mine ore**
- Concentrates
- Fines

Which TWO of these might indicate self-heating of a coal cargo?

- **A carbon monoxide measurement of 50 ppm or more is measured**
- **A steady increase in daily carbon monoxide measurements over 3 days**
- Hydrogen gas is detected
- The hatch covers feel warm

Which TWO things are needed for self-heating to occur in a coal cargo?

- **Moisture**
- **Oxygen**
- Carbon monoxide
- Methane

Which intact stability requirements must a ship carrying grain comply with? Select the ONE best answer.

- **Both Load Line Convention and International Grain Code**
- MARPOL damage stability criteria
- The International Grain Code only
- The Load Line Convention only

Which of these best describes which ships are subject to the International Grain Code when carrying bulk grain? Select the ONE correct answer.

- **All ships regardless of size**
- Ships less than 75,000 dwt
- Ships over 100 metres in length
- Ships over 500 grt

Which of these cargoes groups are known as the 'major bulks'? Select the ONE correct answer.

- **Iron ore, coal and grain cargoes**
- All bulk cargoes carried in dry bulk carriers
- All mineral cargoes
- Grain cargoes and mineral concentrates

Which of these is the biggest hazard associated with iron ore cargoes? Select the ONE correct answer.

- **Over-stressing of the hull structures**
- Cargo sweat
- Explosion
- Loss of stability

Which of these precautions must be taken before entering a hold that contains iron ore? Select the ONE best answer.

- **All aspects of the enclosed space entry must be complied with according to the SMS**
- Hold lighting should be switched on
- The atmosphere should be checked for methane and carbon monoxide
- The bridge OOW should be told

Who should test the atmosphere of a fumigated space after ventilation at the discharge port in order to declare the space safe to enter? Select the ONE correct answer.

- **An expert chemist**
- The cargo receiver
- The chief mate
- The port agent

Why does most localised damage to ship structures take place during discharge? Select the ONE correct answer.

- **Grabs and bulldozers can cause impact damage**
- Lack of visibility due to dust
- The cargo has to be discharged as quickly as possible
- The hatch openings are relatively small

Why should hold access ways and adjacent compartments be sealed while carrying coal? Select the ONE correct answer.

- **To prevent explosive gases from moving into those spaces**
- To keep the dust down
- To prevent cargo theft
- To prevent ingress of water

Why should hold bilge pumping records be kept while carrying iron ore cargoes? Select the ONE correct answer.

- **There may be disputes about differences between loaded and discharged cargo weights**
- To enable stability calculations to be carried out prior to discharge
- To meet MARPOL requirements
- To prove that the hold bilge pumps continue to work well

Why should the condition of electrical systems in the holds and nearby access ways be checked before loading a coal cargo? Select the ONE correct answer.

- **To make sure that no electrical sparks can occur which could ignite explosive gases**
- As part of a planned maintenance system
- To avoid having to go into the hold to fix them during the voyage
- To save energy

Why should you never do through-ventilation into the body of a coal cargo? Select the ONE correct answer.

- **It will increase the likelihood of self-heating of the cargo**
- It might cause sweating of the cargo
- It might dry out the coal
- It will increase the likelihood of methane being emitted

0343 – Preparation for loading bulk carriers

Version 1.0 (936)

A bulk carrier is on a long term contract of affreightment and loading the same cargo each trip. What is the usual standard of hold preparation between cargoes? Select the ONE best answer.

- **Load on top**
- Grain clean
- Normal clean
- Shovel clean

According to this module, how much degreasing chemical would be required per hold on a 100,000 tonnes dwt bulk carrier? Select the ONE correct answer.

- **100 litres**
- 10 litres
- 25 litres
- 250 litres

After the bilges and strainers have been cleaned, why is burlap fitted over the bilge cover plates prior to loading? Select the ONE correct answer

- **To allow water to pass through the burlap but to stop as much of the cargo solids as possible from entering the bilge piping system**
- To keep the bilge cover plates clean and free from cargo solids
- To keep water out of the bilges
- To stop insect infestation accumulating in the bilges

How are cleaning chemicals usually applied during hold washing? Select the ONE correct answer

- **The chemicals are applied using a pressure spray**
- The chemicals are applied using a low-pressure hose
- The chemicals are applied using a mop from a mixing bucket
- The chemicals are hand brushed onto the surfaces to be treated

How are dry cargo residues brought up to the deck by the crew? Select the ONE correct answer.

- **Using a mucking winch to lift buckets of cargo residue**
- Cargo is loaded into a canvas or plastic bag and lifted to the deck manually
- Crew carry buckets up the access ladders
- Crew use a light line to lift the cargo residues manually

How are minor water leaks past the coaming seals, which might occur in heavy weather, stopped from falling onto cargo in the hold? Select the ONE correct answer.

- **The coaming drain channel catches the water and leads it safely to the drains and then to the deck.**
- High expansion foam is used to close all of the gaps around the sides of the hatch covers
- The cleats must be tightened up as much as possible
- The seal is weathertight, so no water can get inside

How does a Master know whether a cargo is Harmful to the Marine Environment (HME) or not? Select the ONE correct answer.

- **The shipper is required to say whether the cargo is HME in the Shipper's Declaration**
- The IMSBC Code can be checked
- The Master has to ask the Charterer
- The stevedore will tell the Master

How does the rubber seal on a hatch provide a weathertight seal? Select the ONE correct answer.

- **The compression bar fitted to the coaming or the cross joint presses into the rubber seal when the cover is closed.**
- Rubber seals are fixed to the coaming and to the hatch cover. When these are pressed together, they provide a weathertight seal
- The quick-acting cleats pull the hatch panels down onto the compression bar, forcing the seal to be compressed
- The seal absorbs water like a sponge stopping it passing through to the ship's hold.

How is the sea water for hold washing applied in the holds? Select the TWO best answers.

- **High-pressure water guns**
- **Hose and nozzle**
- Buckets and mops
- Portable pump

How is the sea water from the ship's system usually accelerated in a hold cleaning gun (water cannon)? Select the ONE best answer.

- **By compressed air from the ship's system**
- By a hand pump
- By a portable diesel driven pump
- By a pressure tank system

If a Chief Officer finds damage to the ship structure, infestation, smell or taint, that cannot be fixed onboard, what action should they take? Select the ONE correct answer.

- **They should, through the Master, immediately inform the owners and the charterers to get help**
- They should load on top as soon as possible to cover over the problems
- They should repeat the water wash
- They should ventilate the holds as soon as possible

In relation to the rubber joint in a hatch seal, what is 'permanent set'? Select the ONE correct answer.

- **This is when the rubber seal loses its flexibility and has a fixed groove in it.**
- This is the depth to which the compression bar should press into the seal rubber, according to the maker's instructions
- This is when the rubber seal is pushed sideways out of position and is permanently offset in the seal channel.
- This refers to the fixing of the seal height when it is first installed

In the bulk trades, how many standards of hold cleanliness are there? Select the ONE correct answer.

- **Five**
- Eight
- Three
- Two

On a bulk carrier, what is the highest standard of hold cleaning? Select the ONE correct answer.

- **Hospital clean**
- Grain clean
- Very clean
- Washed clean

On a bulk carrier, what standard of cleanliness would you expect if the next cargo were to be bulk fertilizer? Select the ONE correct answer.

- **Grain clean**
- Hospital clean
- Normal clean
- Shovel clean

Outside of MARPOL Annex V Special Areas, what is the general rule for discharging none HME cargo residues and wash water? Select the ONE correct answer.

- **Discharge is permitted, but not less than 12 miles from the nearest shore or ice shelf**
- Discharge is not permitted under any circumstances, except to a reception facility
- Discharge is permitted in seas and rivers as required
- Discharge may only be done at night

What are pre-wash chemicals used for? Select the ONE correct answer.

- **They are applied before loading and help to stop cargo sticking to the hold surfaces, making hold cleaning easier after discharge**
- The pre-wash chemical is applied before the cleaning chemicals to soften up dirt and reduce cleaning time
- They are mixed with the cleaning chemicals to help them work more effectively
- They dissolve rust scale and help clean the hold

What instructions on hold cleaning standard would be expected when loading a grain cargo after a previous grain cargo? Select the THREE correct answers.

- **Grain clean**
- **Hold sweep**
- **Hold washing**
- Normal hold clean

What is a 'bottom wash'? Select the ONE correct answer

- **A water wash of the tank top and lower areas of a hold**
- A method of cleaning the underside of the deck structures
- This is a method of cleaning the underside of the hatch covers
- This is carried out in dry dock to clean the outside of the hull

What is the best way of removing wash water mixed with lumpy cargo residues from the hold of a ship? Select the ONE best answer

- **Use a portable sump pump (mucking pump)**
- Use buckets to manually collect the water
- Use the bilge educator system
- Use the bilge pumping system

What is the definition of 'wash water' under MARPOL Annex V? Select the ONE correct answer.

- **Water used in washing the holds that contains some cargo residues and may contain small amounts of cleaning chemicals**
- Fresh water used to rinse the holds
- Water used in washing the holds that contains some cargo residues but no cleaning chemicals
- Water used in washing the holds that has been filtered to remove solids

What is the expected accuracy of a well-conducted pre-load draught survey? Select the ONE best answer.

- **+/- 0.5%**
- +/- 0.05%
- +/- 10%
- +/- 5%

What is the general guidance for when a hatch seal should be replaced? Select the ONE best answer.

- **When permanent set reaches half the design compression, replace the seal.**
- Replace the seal once reports of cargo damage have been received
- When permanent set reaches half the depth of the rubber seal it should be replaced.
- When permanent set reaches the design compression, replace the seal

What is the main purpose of the Chief Officer's inspection? Select the ONE best answer.

- **To make sure that the holds are ready to carry the intended cargo**
- To make sure that all cleaning materials have been removed from the holds
- To make sure that all hold sweepings have been removed from the holds
- To make sure that any damage to the ship structure has been recorded

What is the most common type of hatch cover on larger bulk carriers? Select the ONE correct answer.

- **Side rolling type**
- Hydraulic folding type
- Lift away pontoons
- Single pull type

What is the purpose of a water ingress detector?

- **The water ingress detector is fitted to detect ingress of water into the holds**
- The water ingress detector is fitted to detect high levels of water in the bulkhead stool spaces
- The water ingress detector is fitted to detect high levels of water in the hold bilges
- The water ingress detector is fitted to detect water ingress into the bilge piping system

What is 'fat cargo slip'? Select the ONE correct answer

- **A type of pre-wash chemical designed for treating holds before loading cargoes which stain, such as petcoke**
- A cleaning chemical specially designed for fatty cargoes
- A type of lubricant used on hatch cover wheels
- This occurs during cargo liquefaction

What special precaution is necessary when using portable ladders or staging in the holds of a ship which is at sea or at anchor? Select the ONE best answer.

- **Care must be taken to properly secure all portable ladders and staging against ship movements**
- If the weather is good, no special precaution should be needed
- The Chief Officer should be asked for permission to use the equipment
- The equipment is designed for use in ships and therefore no special precautions are needed

When might the requirement for holds to be 'dry' not be enforced? Select the ONE correct answer.

- **If the cargo to be loaded is already in a damp stockpile**
- If the Master reports that there will not be enough time to dry the holds
- If the moisture content of the cargo to be loaded is above the transportable moisture limit
- When the first part of the cargo to be loaded is expected to absorb the dampness in the hold

Where are the regulations covering the discharge of hold cargo residues and wash water? Select the ONE correct answer.

- **MARPOL Annex V**
- IMSBC Code
- SOLAS
- The Grain Code

Which TWO of these cargoes need the most hold cleaning and preparation? Select the TWO best answers.

- **Salt**
- **Sulphur**
- Alumina
- Coal

Which one of these hold cleaning standards would be expected when moving from a coal cargo to another coal cargo? Select the ONE best answer.

- **Load on top**
- Grain clean
- Hold wash
- Sweep clean

Which three of these instructions on hold cleaning standards would be expected when moving from a coal cargo to an iron ore cargo? Select the THREE correct answers.

- **Hold sweep**
- **No hold wash required**
- **Normal clean**
- Hold wash

Which three of these instructions on hold cleaning standards would be expected when moving from a petcoke cargo to an alumina cargo? Select the THREE correct answers.

- **Hold sweep**
- **Hold wash**
- **Hospital clean**
- No hold wash
- Normal clean

Which two of these instructions on hold cleaning standards would be expected when moving from a coal cargo to a grain cargo? Select the TWO correct answers.

- **Hold sweep**
- **Hold washing**
- No hold wash required
- Normal hold clean

Why are cleaning chemicals sometimes used during hold washing? Select the ONE correct answer

- **Cleaning chemicals are used to remove residues or staining from the previous cargo**
- Cleaning chemicals are used only to clean the bilge boxes
- Cleaning chemicals are used to clean the mops and buckets used in hold preparation
- Cleaning chemicals are used to remove loose scale and paint

Why are holds often rinse washed with fresh water after the sea water wash? Select the ONE best answer.

- **To remove any dried salts that remain on the hold structures after the seawater previously used has dried**
- To assist in dissolving any cargo residues
- To help remove the cleaning chemicals
- To reduce the level of salts in the bilge valves and pipes

Why are special precautions necessary prior to loading sulphur or salt cargoes? Select the ONE best answer.

- **Because the cargoes can cause corrosion of the ship's structure**
- Because they are easily damaged by moisture
- Because they can react with paint, causing damage to the cargo
- Because they can stick to the sides of the ship's holds

0344 – Loading bulk carriers

Version 1.0 (661)

A ship's draughts are: 7.2m forward; 8.6m aft; 7.8m midships. Is the ship hogged or sagged? Select the ONE correct answer.

- **Hogged**
- Sagged

For a ship arriving in ballast, why does a shore terminal need to supply information on the depth of water at the berth? Select the ONE best answer.

- **Because this information is required when deciding how much cargo the ship can load**
- So that the ship can calculate how to distribute cargo in order to complete loading on an even keel
- So that the ship can make adjustments for air draught
- To minimise the effects of squat

For bulk carriers that may be loaded differently from their design loading conditions as specified in the loading manual, which parameters must not be exceeded? Select the TWO best answers.

- **Maximum bending moments**
- **Maximum shear force values**
- Minimum bending moments
- Propeller immersion at the designed draught

For cargo trimming purposes, how are bulk cargoes categorised? Select the ONE best answer.

- **Cohesive and non-cohesive**
- By density
- Free-flowing and compact
- Hazardous and non-hazardous

If a bulk cargo is prone to liquefaction, what information in addition to the normal cargo declaration, must shippers provide to Masters of ships over 150m in length? Select the ONE correct answer.

- **The cargo density**
- A letter guaranteeing that the cargo will not liquefy
- The method of testing the cargo for moisture content
- The name of the Competent Authority at the load port

The quantity of cargo to be loaded, obtained by dividing the capacity of cargo holds by the Stowage Factor, exceeds the maximum summer deadweight of the ship. What does this mean? Select the ONE correct answer.

- **The ship will reach its summer draught before the cargo holds are full**
- Bending Moments will be exceeded in the still water condition
- Shear Forces will be exceeded in the still water condition
- The ship will reach its summer draught when cargo holds are full

The ship is to load a low density bulk cargo. How can the approximate quantity of cargo that can be carried in a particular hold be determined? Select the ONE correct answer.

- **By proportioning the cargo hold capacity to the ship's total cargo hold cubic capacity and applying that to the quantity of cargo to be loaded**
- By comparing the load density of the cargo hold with the cargo stowage factor

What action should be taken by the Master or Chief Officer if deviations from the agreed load plan are detected? Select the ONE best answer.

- **Adjust cargo or ballast operations, or both, to correct the deviations and inform the terminal**
- Contact the shipper and seek clarification
- Make appropriate entries in the log book
- Only ballast operations should be adjusted

What does a permanent equilateral triangle, marked amidships on both sides of a ship, indicate? Select the ONE best answer.

- **Certain cargo restrictions have been imposed upon the ship**
- The ship has been granted alternative tonnages
- The ship is certified for alternate hold loading
- The ship maintains a greater-than-minimum freeboard in all conditions of loading

What is the consequence of the Master of a time-chartered ship issuing Notice of Readiness (NOR) but the ship not being able to receive cargo due to the holds being dirty? Select the ONE correct answer.

- **The ship can be taken off the berth and placed 'off-hire'**
- The ship can withdraw the NOR
- The ship will be deemed unseaworthy
- The ship will be detained by port-state control

What method is used to accurately measure the quantity of cargo loaded? Select the ONE correct answer.

- **The draught survey**
- A weighbridge survey
- Tallying the number of grab loads
- The load line survey

What might happen if a ship does not request advance information about a terminal where cargo is to be loaded? Select the ONE best answer.

- **There could be a long delay between the time of berthing and commencement of cargo loading operations**
- The ship could run aground in the approaches to the berth
- The ship will be placed 'off-hire'
- The ship will not be allowed to enter the port

What should the Master do if advised by the shore terminal that cargo with a moisture content in excess of the Transportable Moisture Limit is to be loaded? Select the ONE best answer.

- **The Master would have to demonstrate to the flag-State adequate residual stability in case of cargo shifting and adequate structural integrity for additional stresses**
- The Master should never refuse to load if the ship is specially constructed for such cargoes
- The Master should never refuse to load under any circumstances
- The Master would have to start loading, but monitor and record the moisture content at regular intervals

What should the Master do if the shipper's cargo declaration has not been received? Select the ONE correct answer.

- **Await the shipper's declaration before commencing cargo loading operations**
- Commence loading cargo
- Inform the shipper that the cargo declaration must be provided before the ship departs from the load berth
- Seek approval to load cargo from the ship's classification society

When will a marine authority issue the Master with a Certificate of Readiness to Load? Select the ONE best answer.

- **After an inspection of cargo spaces has been satisfactorily carried out and pre-loading calculations checked**
- After the initial draught survey has been carried out
- After the Master has issued Notice of Readiness (NOR)
- When the terminal representative is satisfied that the ship is cargo-worthy

Where can information on trimming requirements for a bulk cargo with an angle of repose greater than 30 degrees be found? Select the ONE best answer.

- **In the I.M.S.B.C. Code**
- In SOLAS
- In the BLU manual
- In the ship's loading manual

Which ONE of the following is NOT acceptable during cargo loading operations? Select the ONE best answer.

- **Loading cargo high against one bulkhead and low against the opposite bulkhead in any one cargo hold**
- Distributing cargoes evenly
- Loading cargo high against both bulkheads in any one cargo hold
- Trimming cargoes to the boundaries of each cargo space

Which ONE of the following is NOT required to be shown in the loading plan? Select the ONE correct answer.

- **Outstanding items of class-related hull surveys**
- Permissible shear forces and bending moments
- The cargo loading rate
- The rate of de-ballasting

Which ONE of these options is NOT shown on a Certificate of Fitness to proceed to sea? Select the ONE correct answer.

- **Dock water density**
- Cargo precautions to be taken during the voyage
- The method(s) of cargo securing
- The quantity of cargo loaded

Which ONE of these stability criteria is necessary for a ship to depart from the load berth on an even keel? Select the ONE best answer.

- **The longitudinal centres of gravity and buoyancy must be equal**
- The longitudinal centre of gravity must be less than the longitudinal centre of buoyancy
- The longitudinal centre of gravity must be midships
- The longitudinal centre of gravity must be more than the longitudinal centre of buoyancy

Which THREE of the following must NOT be allowed whilst a draught survey is in progress? Select the THREE correct answers.

- **De-ballasting a tank aft of amidships**
- **De-ballasting a tank forward of amidships**
- **Receiving freshwater**
- Allowing persons to board the ship

Which THREE of the following must be monitored by the duty watchkeeper during cargo loading operations, as per the load plan? Select the THREE correct answers.

- **Tanks being ballasted or de-ballasted**
- **The amount of ballast pumped out from a particular tank by the end of the watch**
- **The ship's trim after each de-ballasting operation**
- Maintaining an even keel condition at all times

Which THREE of the following should not be permitted whilst taking draught survey readings and subsequent tank soundings? Select the THREE correct answers.

- **De-ballasting topside tanks**
- **Opening or closing hatch covers**
- **Transferring oil between tanks**
- Running auxiliary machinery

Which TWO precautions must be taken when commencing the loading of scrap metal or iron ore? Select the TWO best answers.

- **Loading should commence at a slow rate**
- **The conveyor (or grab) should be kept (or lowered) as close to the tank top as possible**
- Cargo hold ventilation must be in operation
- Cargo trimming must commence immediately

Who is responsible for issuing a ship with a Document of Authorisation for the carriage of grain? Select the ONE correct answer.

- **The ship's flag State**
- Surveyors representing charterers
- Surveyors representing the ship's P&I Club
- The shipper

Why is it important for the terminal to supply the Master in advance with the water density at the load berth? Select the ONE best answer.

- **For accurate stability calculations to be made before the loading plan is finalised**
- For accurate stability calculations to be made before the commencement of cargo loading operations
- For calculations to be made during the final draught survey
- For calculations to be made during the initial draught survey

Why must tidal information be included in the ship-shore safety checklist? Select the ONE best answer.

- **For the Master to know water depths at the berth and its approaches especially when un-berthing in an emergency**
- Because it is mandatory under SOLAS
- Because it is mandatory under the I.M.S.B.C. Code
- For moorings to be tended accordingly

With regard to bulk cargoes that may liquefy, what information must be provided by the shipper? Select the ONE best answer.

- **Moisture content and Transportable Moisture Limit**
- Certified Moisture Content only
- Certified results of 'Can Tests'
- Certified Transportable Moisture Limit only

With regard to draught surveys, which ONE of the following statement is correct? Select the ONE best answer.

- **Accurate draught readings must be taken forward, aft and amidships on both sides of the ship**
- Displacement corrections for heel need not be applied
- The observed draught readings are directly used in the calculations
- When the ship is trimmed beyond the limits specified in tank sounding tables, corrections to the soundings can be applied by extrapolation

With regard to the shipper's declaration for bulk cargoes that may liquefy, what additional information must the Competent Authority provide the Master with? Select the ONE correct answer.

- **Verification of the shipper's testing and sampling procedures**
- A guarantee that cargo will not shift during the voyage
- Labelled composite samples of the cargo to be shipped
- Verification of the angle of repose for fine-grained cargoes

0345 – Hull structural strength

Version 1.0 (732)

For a loaded, simply supported beam which ONE of the following statements about compressive and tensile forces is most accurate?

- **The load side is under compression and the opposite side is under tension**
- Both the load side and the opposite side are under compression
- Both the load side and the opposite side are under tension
- The load side is under tension and the opposite side is under compression

For a loaded, simply supported beam with uneven loading, which ONE of the following statements about the loading of the supports is most accurate?

- **If the loading is uneven, the support nearest to the load has the greatest force**
- Even if the loading is uneven, each support carries equal force
- Even if the loading is uneven, the reactions at each support will be zero.
- If the loading is uneven, the support furthest from the load has the greatest force

For a loaded, simply supported symmetrical beam which ONE of the following statements about the resulting bending stresses is most accurate?

- **If the beam is symmetrical, the bending stresses are highest at the outside surfaces and zero along the neutral axis**
- If the beam is symmetrical, the bending stresses are the same throughout any cross-section of the beam
- If the beam is symmetrical, the bending stresses are zero at the outside surfaces and highest along the neutral axis
- If the beam is symmetrical, the bending stresses are zero throughout any cross-section of the beam

For an evenly loaded, simply supported beam which ONE of the following statements about the bending stress is most accurate?

- **The bending stress is greatest half way along the length of the beam**
- The bending stress is greatest at each end
- The bending stress is zero half way along the length of the beam
- The ends of the beam cannot move

What does the abbreviation PMA stand for? Select the ONE correct answer.

- **Permanent Means of Access**
- Port marine Agencies
- Port marine Authority
- Post-Mooring Activities

What is Block Loading? Select the ONE best answer

- **When pairs of holds are fully loaded, with other empty holds between the loaded pairs**
- When alternate holds are loaded with high density cargo
- When the cargo is loaded in the form of blocks of material
- When the whole ship is loaded with one batch of cargo

What is a homogeneous loading pattern? Select the ONE correct answer

- **One in which the total amount of cargo is loaded more or less evenly along the length of the ship**
- A loading pattern in which pairs of holds are loaded, with empty holds between
- One in which cargo is only loaded in designated holds
- One in which low density cargo completely fills each alternate hold

What restrictions are usually placed on holds carrying water ballast? Select the ONE best answer

- **Holds must not be filled to between 20% and 90% capacity**
- Holds must be filled to approximately 60% at all times
- Holds must be filled to between 20% and 90% of capacity
- Holds should always be filled to 70% capacity

When is it safe to load the tank top with more than the maximum given in the Capacity Plan? Select the ONE correct answer

- **Never**
- If the cargo has a low stowage factor
- If the cargo is trimmed to give a level surface after loading is complete
- If the forecast weather conditions on voyage are good

When the distance between wave crests is equal to the length of the ship's hull, which ONE of the following statements is most accurate?

- **When the wave crests are fore and aft, they will make the ship tend to sag**
- The smallest bending moment caused by the wave motion will be when either the peak or the trough of the wave is at mid-ships
- There will be no overall effect on the hull
- When the wave crests are fore and aft, they will make the ship tend to hog

Which ONE of the following statements about bending of the hull is most accurate?

- **When the hull is hogging, the deck will be in tension whilst the keel will be in compression**
- When the hull is hogging, the deck will be in compression whilst the keel will be in tension
- When the hull is sagging, the deck and keel will be under tension
- When the hull is sagging, the deck will be under tension whilst the keel will be in compression

Which ONE of the following statements about corrosion is most accurate?

- **Corrosion can greatly reduce the strength of the ship**
- Corrosion of the hull is relatively unimportant, as there is an allowance for this at the design stage
- No permanent loss of hull strength occurs because of operation in heavy weather
- The strength of the hull always decreases rapidly in service

Which ONE of the following statements about forces on the hull is most accurate?

- **Shear forces on the hull will tend to be high if one hold is fully loaded and the next is empty**
- Shear forces on the hull will tend to be high if all holds are empty
- Shear forces on the hull will tend to be high if all holds are evenly loaded
- Shear forces on the hull will tend to be low if one hold is fully loaded and the next is empty

Which ONE of the following statements about forces on the ship's hull is most accurate?

- **Panting is the in and out movement of the shell plating due to the variation in water pressure on the shell plating.**
- Panting occurs when the vessel is lifted by the sea and then slams back down on the water
- Pounding damage is most likely at mid-ships
- Pounding is most likely to occur when the ship is fully loaded

Which ONE of the following statements about hull inspections is most accurate?

- **Ship's crew should regularly inspect the cargo holds, hatch covers and ballast tanks with a view to detecting damage and defects**
- Ship's crew are not allowed to inspect the cargo holds, hatch covers and ballast tanks
- Ship's crew have insufficient training to inspect the cargo holds, hatch covers and ballast tanks with a view to detecting damage and defects.
- There is no need to detect damage and defects between surveys

Which ONE of the following statements about hull strength is most accurate?

- **A ship regularly operating in heavy weather is subject to many stress reversals which lower the strength of the hull**
- A ship regularly operating in heavy weather will remain as strong as when new if properly maintained
- Fatigue cracking is never related to corrosion
- Small Fatigue cracks will normally be okay until the next dry-dock or survey

Which ONE of the following statements about sloshing of fluid in tanks is most accurate?

- **Excessive forces from fluids sloshing in tanks can be prevented by not having them between 20% and 90% full**
- Excessive forces from fluids sloshing in tanks can be prevented by having them between 20% and 90% full
- Excessive forces from fluids sloshing in tanks can be prevented by having them only half full
- Excessive forces from fluids sloshing in tanks can only be prevented by having them completely full

Which ONE of the following statements about the loads on the hull is most accurate?

- **Reducing the dynamic loads on the hull by slowing down or altering course in heavy weather is effective in both increasing safety and maintaining the strength of the vessel**
- Little can be done in heavy weather to reduce the dynamic loads on the hull
- Reducing the dynamic loads on the hull by slowing down or altering course in heavy weather should only be considered in an emergency for safety reasons
- Static loads on the hull can be reduced by slowing down or altering course

Which ONE of the following statements about the strength of the hull is most accurate?

- **Factors such as operation in rough weather and damage from cargo work can greatly reduce the strength of the hull**
- Incidental damage from cargo work has no effect on the strength of the hull
- The strength of the ship always decreases rapidly in service
- The strength of the ship never changes much after the building stage

Which ONE of these IMO publications requires that Bulk Carriers have a loading plan?

- **SOLAS**
- COLREG
- Load line
- MARPOL

Which of the following most accurately describes a disadvantage of High Tensile Steel used in ship structures? Select the ONE best answer

- **When corrosion occurs, the rate of loss of strength is more than for ordinary mild steel**
- It is heavier than mild steel
- It is more expensive
- It is more prone to corrosion

Which one of the following best describes why bulk carriers are more prone to corrosion than tankers or container ships? Select the ONE best answer.

- **Bulk carriers suffer much more mechanical damage to coatings due to grab damage and other impacts during loading and discharging**
- Bulk carriers are built of lower grades of steel
- Tankers and container ships have higher technology coating systems applied to their cargo carrying spaces
- They spend more time in port

Which one of the following statements best describes the Enhanced Survey Programme? Select the ONE best answer

- **More detailed inspections at each ship survey**
- Frequent inspections by a Port's terminal operators
- More frequent inspections of the ship's structures
- More inspections by Port State inspectors

With reference to crack formation in the hull structure, which ONE of the following statements is most accurate?

- **Cracks can occur due to fatigue, corrosion, or sharp changes of section**
- Cracks in hatch coamings are not a significant problem
- Cracks occur when the material can no longer withstand compressive forces
- Small cracks often occur and are not a significant problem

With reference to cracking of the hull structure, which ONE of the following statements is most accurate?

- **Cracks occur when the material can no longer withstand tensile forces**
- Corrosion does not affect whether or not cracks occur
- Cracks occur when the material can no longer withstand compressive forces
- Fatigue does not affect whether or not cracks occur

With reference to looking for defects in the hull structure around hatch openings, which ONE of the following statements is most accurate?

- **The hatch corners and hatch coaming brackets need close attention as they are prone to cracking**
- Hatch openings do not form part of the structure and therefore do not need close inspection
- The deck plating midway along the breadth of the hatch needs close attention as it is prone to cracking
- The deck plating midway along the length of the hatch needs close attention as it is prone to cracking

With reference to looking for defects in the hull structure, which ONE of the following statements is most accurate?

- **Indentations in the side and floor plating of Bulk carriers from cargo work can lead to corrosion and cracking**
- Double bottom tanks are not usually subject to corrosion
- Indentations in the side and floor plating of Bulk carriers from cargo work are usually of no great importance
- Large hatch openings on container and bulk vessels do not form part of the structure and therefore do not need close inspection

With reference to shear forces on a ship's hull, which ONE of the following statements is most accurate?

- **Shear forces cause further stress on the hull in addition to that caused by bending**
- Shear forces are not significant compared to the stress on the hull caused by bending
- Shear forces cancel out some of the stress on the hull caused by bending
- Shear stress always opposes the stress on the hull caused by bending

With reference to static and dynamic forces acting on the hull, which ONE of the following statements is most accurate?

- **High static forces, for instance due to poor cargo loading, mean that the vessel is less safe when operating in heavy weather**
- Dynamic loads on the vessel are insignificant compared to the weight of cargo and ballast
- Provided the maximum static loading for the hull has not been exceeded, the vessel should be safe in any weather conditions
- Static forces on the vessel are insignificant compared to sea forces

With reference to stiffening of side shell plating, which ONE of the following statements is most accurate?

- **The middle and ends of stiffeners and girders will be subject to the highest bending stress**
- Bending stress is equal along the entire length of stiffeners and girders
- The ends of stiffeners and girders will be subject to zero bending stress
- The middle and ends of stiffeners and girders will be subject to zero bending stress

With reference to the forces acting on the hull, which ONE of the following statements is most accurate?

- **Panting, pounding, and sloshing of liquids in tanks are examples of dynamic forces**
- Dynamic forces are always small compared to static forces
- Static forces tend to cancel out dynamic forces
- The weight of fuel, cargo, fresh water and ballast are examples of dynamic forces

With reference to the hull structure, which ONE of the following statements is most accurate?

- **Resistance to buckling depends on the length and cross section of the stiffener or girder**
- Buckling can occur in tension or compression
- Resistance to buckling depends only on the cross section of the stiffener or girder
- Resistance to buckling depends only on the length of the stiffener or girder

With reference to the hull structure, which ONE of the following statements is most accurate?

- **Fatigue is the reduction in strength of a material due to repeated cycles of stress**
- Fatigue cracking is most likely when the ship is new
- Fatigue is the reduction in strength of a material due to constant high load
- If protective coatings are maintained, fatigue will never affect a ship

With reference to the stress distribution along the length of a ship's hull subjected to bending, which ONE of the following statements is most accurate?

- **If the ship is evenly loaded, the bending stresses are at a maximum at mid-length**
- If the ship is evenly loaded, the bending stresses are at a maximum fore and aft
- If the ship is evenly loaded, the bending stresses are at a minimum at mid-length
- If the ship is evenly loaded, the bending stresses are equal along the length of the ship

With reference to the stress distribution of a ship's hull subjected to bending, which ONE of the following statements is most accurate?

- **The bending stresses will be maximum at the deck and bottom plating and reduce towards the neutral axis at approximately the mid-point of the side plating**
- The bending stresses will be equal at the deck, bottom plating and the neutral axis at approximately the mid-point of the side plating
- The bending stresses will be maximum at the deck plating and reduce to zero at the bottom plating
- The bending stresses will be zero at the deck and bottom plating and increase towards the neutral axis at approximately the mid-point of the side plating

With reference to water on the deck of a ship, which ONE of the following statements is most accurate?

- **Sea water coming over the deck will cause downward forces on the ship's structure and is most likely to affect the fore end of the vessel**
- Sea water coming over the deck causes additional buoyancy forces on the ship's structure
- Sea water coming over the deck has little overall effect on large vessels
- Sea water coming over the deck will cause downward forces on the ship's structure and is most likely to affect the aft end of the vessel

With regard to looking for defects in the hull structure, which ONE of the following statements is most accurate?

- **Connections between parts of the structure, or changes of section, are common areas to find defects**
- Connections between parts of the structure are so strong that they rarely fail
- Loss of protective coating above welds is fairly common and of no immediate concern
- Small cracks in shell plating above the waterline are of no concern

With regard to shear forces acting on the hull, which ONE of the following statements is most accurate?

- **Shear force is created when there are two parallel, opposite forces, like gravity and buoyancy**
- Shear force is created when there are two parallel forces in the same direction, like gravity and buoyancy
- Shear force refers to the horizontal force at the aft part of the vessel
- Shear force refers to the horizontal force at the forward part of the vessel

With respect to bending stresses on the ship's hull, which ONE of the following statements is most accurate?

- **Bending stresses will normally be highest at mid length and at the deck and bottom plating levels**
- Bending stresses will normally be highest at mid length and half way between the deck and bottom plating levels
- Bending stresses will normally be highest fore and aft and at the deck and bottom plating levels
- Bending stresses will normally be highest in way of the engine room and half way between the deck and bottom plating levels

With respect to flat bar cross sections of stiffeners used for the ship's hull, which ONE of the following statements is most accurate?

- **Flat bar is relatively poor at resisting buckling**
- Flat bar is difficult to apply a protective coating to
- Flat bar is relatively expensive to make
- Flat bar is relatively good at resisting buckling

With respect to framing types for the ship's hull, which ONE of the following statements is most accurate?

- **Combination framing uses both transverse and longitudinal framing**
- Combination framing is required on all new oil tankers
- Combination framing uses longitudinal frames and box girders
- Combination framing uses transverse frames and box girders

With respect to rolling causing stress on the ship's hull, which ONE of the following statements is most accurate?

- **Rolling mostly causes stresses in the transverse bulkheads**
- Rolling mostly causes stresses at the aft part of the ship
- Rolling mostly causes stresses at the forward part of the ship
- Rolling mostly causes stresses in the longitudinal stiffeners

With respect to the cross sections of stiffeners used for the ship's hull, which ONE of the following statements is most accurate?

- **T-profile performs better in most situations**
- Angle bar performs better in most situations
- Flat bar performs better in most situations
- It is difficult to apply protective coatings to Bulb profiles

With respect to the ship's hull, which ONE of the following statements is most accurate?

- **The main purpose of the longitudinal stiffeners is to resist bending**
- Longitudinal stiffeners are only used in longer vessels
- Longitudinal stiffeners are only used in shorter vessels
- The main purpose of the transverse frames is to support the deck structure

With respect to the ship's hull, which ONE of the following statements is most accurate?

- **Longitudinal framing tends to be used for longer vessels**
- Longitudinal and transverse framing should never be used on the same ship
- Longitudinal framing cannot resist bending stresses
- Transverse framing tends to be used for longer vessels

0346 – Bulk carriers, Personal safety and environmental protection

Version 1.0 (1900)

According to SOLAS, what is the minimum level of oxygen required before entry into an enclosed space? Select the ONE correct answer.

- **21%**
- 16%
- 18%
- 25%

According to the IMO, at least how many sets of Respiratory Protective Equipment (RPE), must be available when fumigation has been carried out? Select the ONE correct answer.

- **Four**
- Eight
- Six
- Two

Do the poisoning and suffocation risks of various cargoes only exist within the cargo spaces? Select the ONE correct answer.

- **No. gases and/or vapours can migrate to nearby compartments and spaces**
- Yes – unless there has been a structural failure, such as a crack
- Yes. Other compartments are usually not directly connected to the cargo holds
- Yes. The gases or vapours can only accumulate in the cargo holds where the cargo was stowed

How are portable water jet machines able to project a jet of water to a height of 30 metres or more? Select the ONE correct answer.

- **By using compressed air to assist the water jet**
- By using an electrically-powered pump
- By using the ship's fire main supply with a suitable venturi nozzle
- Portable water jet machines cannot project a water jet to this height

How are the risks of enclosed space entry on a bulk carrier controlled? Select the ONE correct answer.

- **The ship's SMS will require a Permit to Work system to be in place covering all aspects of enclosed space entry**
- The IMSBC Code provides guidance on enclosed space entry
- The Master will decide which compartments are safe to enter based on past experience
- The Shipper's Declaration will provide guidance on cargo toxicity

How is the safety of cleaning and preparation work in a hold controlled? Select the ONE best answer.

- **The Permit To Work System**
- The charterer's cleaning requirements are followed
- The OOW on the bridge maintains a listening watch on the radio communications
- The supervising officer keeps a watch from the deck

How may Port State Authorities check that a ship's ballast water exchange has been effective? Select the ONE correct answer.

- **Taking samples of ballast water and checking them for unwanted aquatic organisms.**
- By checking the Ballast Water Record Book
- By verifying the pumping times in the logbook
- Delegating an inspection to the ship's agent, who will provide a report

How often must enclosed space entry drills be carried out onboard? Select the ONE correct answer.

- **At intervals not exceeding two months**
- Annually
- At intervals not exceeding three months
- Every two weeks

How often must the Ship Safety Officer carry out a ship safety inspection? Select the ONE correct answer.

- **At least once every three months**
- At least once every month
- At least once every two weeks
- At least once every week

Is it possible for a Safety Management System (SMS) to be modified once it has been implemented onboard? Select the ONE correct answer.

- **Yes – it must be subject to continuous review and improvement**
- No – it cannot be changed once implemented
- No. It is a statutory requirement and cannot be changed
- Yes – but only in exceptional circumstances

Outside of MARPOL Annex V Special Areas, what is the general rule for discharging none HME cargo residues and wash water? Select the ONE correct answer.

- **Discharge is permitted, but not less than 12 miles from the nearest shore or ice shelf**
- Discharge is not permitted under any circumstances, except to a reception facility
- Discharge is permitted in seas and rivers as required
- Discharge may only be done at night

What PPE outfit would you expect to be used on deck when working a non-dusty cargo? Select the ONE best answer.

- **Boiler suit, protective gloves, safety helmet and safety boots**
- Boiler suit, safety hat
- Boiler suit, sports shoes, sunglasses
- Safety boots, protective gloves

What PPE would you expect to be used when on deck when working a dusty cargo? Select the ONE best answer.

- **Boiler suit, protective gloves, safety helmet, dust mask and safety boots**
- Boiler suit, safety hat, dust mask
- Boiler suit, sports shoes, sunglasses
- Safety boots, protective gloves, warm clothing

What actions should be taken if active fumigant material is seen in the cargo when the hatches are opened? Select the ONE correct answer.

- **The hatches should be closed and expert advice obtained**
- The fumigant material can be placed in the ship's garbage
- The fumigant material can be removed by the crew wearing P.P.E.
- The fumigant will no longer be active and can be discharged with the rest of the cargo

What is the definition of 'wash water' under MARPOL Annex V? Select the ONE correct answer.

- **Water used in washing the holds that contains some cargo residues and may contain small amounts of cleaning chemicals**
- Fresh water used to rinse the holds
- Water used in washing the holds that contains some cargo residues but no cleaning chemicals
- Water used in washing the holds that has been filtered to remove solids

What is the first step in any risk assessment? Select the ONE correct answer.

- **Identify the hazards in the activity being assessed**
- Appoint a risk consultant
- Estimate the consequences of a hazard occurring
- Estimate the likelihood of a hazard occurring

What is the main operational safety issue when using a mucking winch? Select the ONE best answer.

- **The lifting load limit must not be exceeded**
- Dust may be generated as the load is lifted
- Noisy operation
- The lifting speed is slow

What materials are sump pumps designed to handle safely? Select the ONE best answer.

- **Water with included solid particles and small lumps**
- Chemically contaminated water
- Oil and water emulsions
- Only water which is free of any cargo residues

What percentage of dust should a FFP1 standard dust mask filter out? Select the ONE correct answer.

- **80%**
- 100%
- 25%
- 50%

What system does the ISM Code require the ship owner or manager to implement onboard? Select the ONE correct answer.

- **A safety management system**
- A safe manning system
- A watchkeeping system
- A working hours system

What will a respirator provide protection against? Select the ONE correct answer.

- **Dusts and aerosols**
- Dusts and gases in the air
- Oxygen depletion
- Toxic gases in the air

When using a respirator, what is the recommended maximum working time in any one day? Select the ONE correct answer.

- **Three hours**
- Eight hours
- One hour
- Six hours

Where can information about the poisoning or suffocation risks of various bulk cargoes be found? Select the ONE correct answer.

- **The IMSBC Code**
- MARPOL
- SOLAS
- The Grain Code

Where can the regulations covering the discharge of hold cargo residues and wash water be found? Select the ONE correct answer.

- **MARPOL Annex V**
- SOLAS
- The Grain Code
- The IMSBC Code

Where can you find advice on the safe use of fumigants? Select the ONE best answer.

- **The IMO Recommendations on the Safe Use of Pesticides on Ships**
- The Charterer's instructions
- The Grain Code
- The Shipper's Declaration

Where can you find information about the PPE required for key onboard tasks? Select the ONE best answer.

- **The Permit To Work for the task**
- SOLAS rules
- The IMSBC Code
- The Master's Standing Orders

Where should details of the ballast water exchange be recorded? Select the ONE best answer.

- **In the Ballast Water Record Book**
- In a Ballast Water notebook
- In the Deck Logbook
- In the Engine Room Logbook

Which personnel are permitted to operate high pressure water guns? Select the ONE correct answer.

- **Only those crew members properly trained in their use can operate high pressure water wash guns**
- Any crew members on a bulk carrier can operate high pressure water wash guns
- Only the ship's officers can operate high pressure water wash guns
- Shore-based cleaning contractors are the only ones permitted to operate high pressure water wash guns

Why is a positive pressure respirator better than a mask type? Select the ONE correct answer.

- **The air pressure makes breathing easier and reduces face seal leakage**
- It is more comfortable to wear
- It stops the face mask steaming up
- The wearer can work for up to 8 hours in any one day

Why must goggles or a visor be worn when operating a light hand-held water jet? Select the ONE best answer.

- **To avoid the risk of flying debris getting into the operator's eyes**
- To help the operator to see where to clean
- To keep the operator warm in cold conditions
- To keep water spray out of the operator's eyes

0347 – Bulk carriers, Trim and stability

Version 1.0 (942)

According to the International Grain Code, how many partly filled holds should a bulk carrier have? Select the ONE correct answer.

- **As few holds as possible**
- All holds may be partly filled
- All holds must be filled, but may be untrimmed
- Not more than two holds

An angle of loll occurs due to cargo fluidisation. There is fuel in a deep tank and a free surface in fuel double bottom tanks. How could ship stability be improved? Select the ONE correct answer.

- **The fuel should be transferred from the deep tank to the double bottom tanks, starting with the tank on the low side**
- As much fuel as possible should be transferred from the double bottom tanks to the deep tank
- Fuel should only be transferred from the double bottom tank on the low side to the deep tank
- The fuel should be transferred from the deep tank to the double bottom tanks, starting with the tank on the high side

An angle of loll occurs due to cargo fluidisation. Which split double bottom tanks should be filled first in order to improve the ship's stability? Select the ONE correct answer.

- **The double bottom tanks on the low side of the ship should be filled first**
- Both sides of the double bottom tanks should be filled at the same time
- Neither: only deep tanks should be filled
- The double bottom tanks on the high side of the ship should be filled first

At what minimum angle must the maximum GZ occur for a bulk carrier? Select the ONE correct answer.

- **30 degrees**
- 25 degrees
- 40 degrees
- 90 degrees

At what point is the True Mean Draught always measured? Select the ONE correct answer

- **The Longitudinal Centre of Flotation**
- Half way between the forward and aft perpendicular
- The Forward Perpendicular
- The midships point

How are untrimmed hold ends dealt with by the authorities in the USA, Canada and Australia? Select the ONE correct answer.

- **They assume that the spaces are filled for the purposes of weight calculations, but that the grain is free to shift to 25°**
- They assume they are fully filled for the purposes of weight and therefore no grain shifting
- They assume they are partly filled for the purposes of the weight calculations and with an assumed grain shifting of 25°
- They assume they are partly filled for the purposes of weight calculations but with no grain shifting

How can a Water Ingress Detector indicate the rate of flooding? Select the ONE correct answer.

- **The time difference between the low and high level alarms being activated indicates the rate of flooding**
- The WID cannot help to find the rate of flooding
- The WID detects the water pressure at a low level in a flooded compartment
- The WID has multiple levels of detection and these give a timed read-out of flooding rates

How can the KG be determined for any loading condition of a bulk carrier? Select the ONE correct answer.

- **Using a table of moments**
- It can be found from the Maximum KG diagram
- It is calculated as part of the displacement calculations
- It is given in the ship's loading manual

How can the value of GZ be found from the value of KN at any value of heel, Theeta?

- **By subtracting a value of KG sin Theeta from the KN value**
- By adding a value of KG sin Theeta to the KN value
- By multiplying the KN value by sin Theeta
- By subtracting the value of KG at angle Theeta from the KN value

How do the USA, Canada and Australia differ from the International Grain Code requirements in respect of stability calculations? Select the ONE correct answer.

- **They differ in their treatment of untrimmed ends of holds for stability purposes**
- They assume a final heel angle of the ship of 15°
- They differ in their treatment of down flooding angles
- They have lower stability requirements than the International Grain Code

How is a Maximum KG diagram used? Select the ONE correct answer.

- **The actual KG and displacement are plotted on the diagram to make sure that the plotted point lies in the zone of acceptable stability**
- The actual GM and the displacement are plotted on the diagram to find KG
- The actual KG and draught are plotted on the diagram to make sure that the plotted point lies in the zone of acceptable stability
- The diagram is used to find the actual KG of the ship in the particular condition of loading

How is the actual Grain Heeling Moment found for the whole ship? Select the ONE correct answer.

- **The Volumetric Heeling Moments for each hold are added together and corrected for the actual stowage factor**
- It is calculated from the statical stability curves for the ship
- It is found in the Grain Loading Manual
- The Master should ask the Shipper on completion of loading

How is the reliability of a loading instrument checked? Select the ONE correct answer.

- **By carrying out regular checks against standard or test loading conditions**
- By checking that the computer is operating correctly
- By checking that the version being used is the most up to date version of the software
- By checking to see that the outputs are being printed in the right format

If an approved loading Instrument is provided onboard, does this make the loading manual redundant? Select the ONE best answer.

- **No. The loading manual should be consulted every time the loading instrument is used**
- No. The loading manual is there as a back-up if the loading instrument fails
- Yes. It is not required unless the loading instrument breaks down
- Yes. Modern computers are so reliable that the loading manual is not needed

In the International Grain Code, what does the term 'Filled compartment – trimmed' mean? Select the ONE correct answer.

- **All of the spaces under the deck and hatches are filled as much as possible**
- The cargo is poured until it fills the hatch coamings
- The grain is trimmed to fill as much space as possible at the sides, but not at the ends, under the cross decks
- The grain is trimmed to give an even angle of repose

In the International Grain Code, what does the term 'Filled compartment – untrimmed' mean? Select the ONE correct answer.

- **The hold is filled completely up to the topside tanks and hatch coamings, but not the two fore and aft ends**
- The grain is trimmed to fill all of the available underdeck spaces
- The hold is filled, but the cargo level does not each the topside tanks
- The hold is partly filled

In the International Grain Code, what does the term 'specially suitable compartment' mean? Select the ONE correct answer.

- **A hold that has longitudinal sloped topside tanks which are coincident with the hatch side girders**
- A hold that can be easily cleaned after the cargo is discharged
- A hold that is clean enough to receive a grain cargo
- A hold that is the correct size for the parcel of cargo to be loaded

Is it necessary to keep records of loading conditions that have been computed on a loading instrument? Select the ONE correct answer.

- **Yes, electronically stored and signed hard copies must be kept as part of the ship's Safety Management System**
- No, because it is very easy to reproduce any loading condition on the Loading Instrument, so records are not necessary
- No, only to keep electronic outputs on a memory stick
- No, the information can be printed and sent to interested parties and then deleted.

The ship is travelling at a constant speed and heading with respect to the waves. What can a slowing down of the ship's roll period indicate? Select the ONE correct answer.

- **Flooding, causing sloshing or a large free surface in a hold or other large compartment**
- Cargo has settled, causing the KG to reduce
- Cargo shift to one side
- Fuel has been shifted to double bottom tanks from deep tanks

What are Volumetric Heeling Moments? Select the ONE correct answer.

- **Heeling moments due to a specified amount of shifting of grain in each hold for a range of hold loading levels**
- Heeling moments calculated using the volumes of each hold
- Heeling moments caused by shifting of water ballast during grain loading
- Heeling moments due to the volume of water displaced during heeling

What does the term 'fluid GM' mean? Select the ONE correct answer.

- **The GM, including the effects of any free surfaces**
- The GM, corrected for dock water density
- The GM, ignoring the weight of any fixed objects
- The GM, with the ship floating in dock water

What extent of grain shifting is assumed when calculating Volumetric Heeling Moments? Select the ONE correct answer.

- **25° in every full hold and 15° in every part filled hold**
- 12° in all holds
- 25° in all holds
- 35° in every partly filled hold and 12° in full holds

What intact stability requirements must a ship carrying grain comply with? Select the ONE correct answer.

- **Both the Load Line Convention and the Grain Code(IGC)**
- MARPOL damage stability criteria
- The Grain Code (IGC) only
- The Load Line Convention only

What is KN? Select the ONE correct answer.

- **KN is the value of the righting lever, assuming KG is zero**
- KN is the height of the centre of buoyancy above the keel
- KN is the height of the transverse metacentre above the keel
- KN is the righting lever for the ship's loaded condition

What is the First Trim Correction also known as? Select the ONE correct answer.

- **The Layer Correction**
- The Free Surface Correction
- The Fresh Water Correction
- The Second Moment of Waterplane Area

What is the Fresh Water Allowance? Select the ONE correct answer.

- **It is the amount that the ship's draught changes when moving from salt water to fresh water and vice versa**
- It is the amount of fresh water allowed per day per crew member
- It is the correction to maximum deadweight to allow for ballast water being fresh water
- It is the correction to maximum deadweight to allow for the amount of fresh water that will be used before the next port

What is the Grain Heeling Moment? Select the ONE correct answer.

- **It is the heeling moment assumed to be caused by the shift of a grain cargo**
- It is equal to the heeling moment when the cargo has shifted through 12°
- It is the heeling moment due to wind and wave action on a ship carrying grain
- It is the moment in time when grain starts to shift, causing a heel

What is the best measure of stability at angles of heel greater than ten degrees? Select the ONE correct answer.

- **The Righting Moment, GZ**
- The displacement of the ship
- The metacentric height, GM
- The position of the centre of buoyancy

What is the effect of a very large GM? Select the ONE correct answer.

- **The ship will be very stiff, leading to rapid rolling in waves**
- The ship will be very tender, leading to a long rolling period in waves
- The ship will have a comfortable roll behaviour
- The ship will not roll very much

What is the main thing to avoid when a ship has developed an angle of loll? Select the ONE correct answer.

- **Filling tanks on the high side**
- Any quick action. Just wait until more fuel has been used.
- Filling tanks on the low side
- Shifting weights in the ship downwards

What is the maximum permissible angle of heel after grain shifting has been calculated? Select the ONE correct answer.

- **12°**
- 10°
- 15°
- 5°

What is the minimum GM (fluid) permitted by the Load Line Convention? Select the ONE correct answer.

- **0.15 metres**
- 0.015 metres
- 0.3 metres
- 0.51 metres

What is the name given to the difference between the draught at the midships point and the draught at the Longitudinal Centre of Flotation? Select the ONE correct answer.

- **The layer correction**
- The free surface correction
- The midships trim
- The second trim correction

What is the usual angle of repose for a grain cargo? Select the ONE correct answer.

- **20º**
- 12º
- 15º
- 35º

What is the usual measure of initial stability on a ship? Select the ONE correct answer.

- **The GM, or metacentric height**
- The KG, or vertical centre of gravity
- The metacentric height above the keel, KM
- The vertical centre of buoyancy, KB

What simple alternative is available for assessing the ship's stability without having to draw a GZ curve? Select the ONE correct answer.

- **The simplified stability information**
- Similar loading conditions in the loading manual
- The calculated GM
- The use of GM and angles of heel up to 900 to estimate the GZ

What size of ship does the International Grain Code apply to? Select the ONE correct answer.

- **All ships carrying grain, regardless of size**
- Ships over 150 metres in length carrying grain
- Ships over 500 g.r.t., carrying grain
- Ships over 5000 g.r.t., carrying grain

What will the Master particularly have to demonstrate to the national authorities prior to loading grain? Select the ONE best answer.

- **That on departure, and at all times during the proposed voyage, the ship will comply with the stability criteria in the International Grain Code**
- That all of the ship's certificates are up to date
- That the amount of cargo proposed to be loaded will not submerge the ship's load line
- That the ship has enough fuel for the proposed voyage

Where are the minimum standards for ship stability defined? Select the ONE correct answer.

- **International Load Line Convention**
- IMSBC Code
- MARPOL
- The Charterparty

Which ONE of the following is correct?

- **GM = KM - KG**
- GM = BM - KG
- GM = KG - KM
- GM = KM + KG

Why is it important to calculate the stability for the arrival condition before departure? Select the ONE correct answer.

- **This is likely to be a less stable condition after fuel and water have been used**
- This is a Classification Society requirement
- This is likely to be a more stable condition after fuel and water have been used
- This is required by the cargo receiver at the unloading port

0348 – Bulk carrier voyage

Version 1.0 (1091)

According to the IMSBC Code, what is the acceptable limit on methane concentration in the space above a coal cargo? Select the ONE correct answer.

- **20% of the Lower Explosive Limit, (LEL)**
- 5% of the Lower Explosive Limit, (LEL)
- 70% of the Lower Explosive Limit, (LEL)
- 90% of the Lower Explosive Limit, (LEL)

According to this module, what is expected to eventually replace ballast water exchange operations? Select the ONE correct answer.

- **Systems for ballast water treatment onboard during the ballast voyage**
- A return to discharging ballast water in the loading ports
- Complete discharge of ballast water prior to arrival at the loading port
- The provision of good reception facilities at loading ports

How can high acidity of the water in the bilges be detected? Select the ONE best answer.

- **A sample of water is taken from the sounding pipe and tested with litmus paper, or litmus paper is attached to the sounding rod**
- A sample of water is taken and tested onboard using a silver nitrate testing kit
- A sample of water is taken from the sounding pipe and sent ashore after arrival for testing
- After discharge, any water left in the bilges can be tested with litmus paper

How can the rate of hold flooding be quickly estimated using the Water Ingress Detector System? Select the ONE best answer

- **The time interval between activation of the low level alarm and the higher level alarm gives an indication of the rate that water is entering the hold**
- After the alarm has activated, a visual inspection of the hold will enable an estimate of the rate of flooding to be made
- The height of the detector above the tank-top indicates how deep the water is in the hold
- The time interval between the alarm activating and the space being fully flooded gives an indication of the rate that water is entering the hold

How is the Dew Point found? Select the ONE correct answer.

- **By using a digital hygrometer or by finding the wet and dry bulb thermometer temperatures**
- By measuring the air temperature and atmospheric pressure
- By testing the moisture content of a sample of air
- From long term weather forecasts

If a self-heating cargo is quickly increasing in temperature and smoke is seen at the hatches, who should the Master first inform? Select the ONE best answer.

- **The ship owner or operator, who should then contact the shippers and get expert advice and recommend immediate actions**
- The Charterer
- The ship's Classification society, who can offer advice on likely repairs
- The unloading port, so that preparations can be made for the ship's arrival

In what circumstances must the hatch covers be fully secured before leaving the berth? Select any answers that you think apply.

- **If a hatch closure certificate has been issued by an independent surveyor**
- **If the holds containing a grain cargo have been fumigated**
- Hatch covers must always be fully secured before leaving the berth
- If it is raining

In which ONE of the following circumstances would weather routeing services be most helpful? Select the ONE best answer.

- **Tropical areas in the cyclone season**
- In busy shipping lanes
- Routes in the Red Sea
- Tropical areas outside of the cyclone season

Is the Master required to follow weather routeing advice provided by the charterer? Select the ONE best answer.

- **No. The Master will use their own experience and appraise the present conditions**
- No. The Master can ignore the advice without having to provide any reasons
- Yes. The Charterparty requires the Master to use 'Utmost Despatch' and this requires the Master to follow their weather routeing advice
- Yes. The Master must follow the weather routeing advice provided by the charterer

Is ventilation always carried out during loaded voyages? Select the ONE best answer.

- **No. There are some cargoes and circumstances where ventilation can be harmful and should not be carried out**
- Yes. Air must be circulated over the cargo at all times, to prevent heating of the cargo
- Yes. Air must be circulated over the cargo at all times, to remove moisture from the cargo
- Yes. Continuous ventilation will remove any dangerous gases which might otherwise be present

Outside of MARPOL Annex V Special Areas, what is the general rule for discharging none HME cargo residues and wash water? Select the ONE correct answer.

- **Discharge is permitted, but not less than 12 miles from the nearest shore or ice shelf**
- Discharge is not permitted under any circumstances, except to a reception facility
- Discharge is permitted in seas and rivers as required
- Discharge may only be done at night

What action should be taken if bilge water levels are rising, or ballast tank levels are changing unexpectedly? Select the ONE best answer.

- **An immediate visual inspection should be carried out and the outcome reported to the owners**
- Charterers should be informed
- Class should be informed
- The agents at the next port should be requested to arrange a survey and repair contractor to attend the ship

What are the main types of test for weather tightness of hatch covers? Select any answers that you think are correct.

- **A chalk test**
- **A hose test**
- **An ultrasonic test**
- A drop test

What is the 'three degree' rule for cargoes that can absorb water? Select the ONE correct answer.

- **Consider ventilating the holds if the external air temperature is three degrees or more below the loaded temperature of the cargo**
- Consider ventilating the holds if the external air temperature is three degrees or more above the loaded temperature of the cargo
- Do not ventilate the holds if there is three degrees difference between the dry bulb and wet bulb temperatures of the air
- Never ventilate the hold if the external air temperature is three degrees or more below the loaded temperature of the cargo

What is the Relative Humidity at the dew point? Select the ONE correct answer.

- **100%**
- 0%
- 50%
- This varies with the temperature of the air

What is the biggest risk when using the flow-through ballast water exchange method? Select the ONE best answer

- **Over-pressuring a tank by pumping in faster than the outlets can discharge**
- Damage to the ship's coating systems by water from the tanks flowing over the decks
- Good weather is needed to carry out the operations safely
- Torsional forces on the hull due to non-symmetric ballasting

What is the definition of 'wash water' under MARPOL Annex V? Select the ONE correct answer.

- **Water used in washing the holds that contains some cargo residues and may contain small amounts of cleaning chemicals**
- Fresh water used to rinse the holds
- Water used in washing the holds that contains some cargo residues but no cleaning chemicals
- Water used in washing the holds that has been filtered to remove solids

What is the greatest risk when methane is given off by coal cargoes? Select the ONE best answer.

- **There is an explosion hazard in the hold and adjacent spaces**
- The cargo may be at risk of liquefaction
- The cargo may be damaged by the presence of the gas
- The ship's structures may be corroded

What is the preferred location for ballast water exchange? Select the ONE correct answer?

- **At least 200 miles from nearest shore and 200 metres water depth**
- At least 100 miles offshore and 200 metres water depth
- At least 12 miles offshore
- At least two days before arrival at the loading port

What is the purpose of Water Ingress Detector Systems fitted to bulk carriers? Select the ONE correct answer.

- **To give an early warning of water ingress to the holds and also the ballast, dry and void spaces forward of the collision bulkhead**
- They are fitted to the holds and ballast tanks of all bulk carriers to check the moisture content of the air in the hold
- Water ingress detector systems are fitted to the ballast tanks of all bulk carriers to remotely monitor the levels during filling
- Water ingress detector systems are fitted to the holds of all bulk carriers to monitor the level of water in the bilges

What percentage of ballast water needs to be exchanged during sequential ballast exchange? Select the ONE correct answer.

- **95%**
- 100%
- 25%
- 80%

When is the best time for the Chief Officer to carry out the final inspection of the holds prior to arrival at the loading port? Select the ONE correct answer.

- **Towards the end of the ballast voyage, but with time to fix any faults that are found**
- On departure from the unloading port
- On the day before arrival at the loading port
- The Chief Officer should inspect the holds on arrival, with the terminal representative

Where can guidance on ventilation of cargoes be found? Select the ONE correct answer

- **The IMSBC Code**
- MARPOL
- SOLAS
- The ship's loading manual

Where can the Master find advice on fumigation of cargoes? Select the ONE best answer.

- **The IMO Recommendations on the Safe Use of Pesticides on Ships**
- The Shipper's Declaration
- The Ships Sanitation Control Exemption Certificate (SSCEC)
- The ship's Loading Manual

Where can the regulations and guidelines relating to ballast water management be found? Select the ONE correct answer.

- **The Ballast Water Management Convention**
- MARPOL
- SOLAS
- The IMSBC Code

Where should details of the ballast water exchange be recorded? Select the ONE best answer.

- **In the Ballast Water Record Book**
- In a ballast water note book
- In the Deck Logbook
- In the Engine Room Logbook

Which ONE of these circumstances would cause 'cargo sweat' to occur in a hold? Select the ONE correct answer.

- **When a cold cargo surface meets warm moist air coming from outside the hold**
- When a cargo has a high moisture content and water leaches out
- When a cold cargo surface meets cold air during ventilation
- When cold external air flows over a warm cargo during ventilation

Which ONE of these circumstances would cause 'ship's sweat' to occur in a hold? Select the ONE correct answer.

- **When warm moist air in the space above the cargo meets the cold surfaces of the ship's steel structures**
- When cold air in the space above the cargo meets the cold surfaces of the ship's steel structures
- When cold air in the space above the cargo meets the warm surfaces of the ship's steel structures
- When the sea temperature is higher than the air temperature

Which ONE of these circumstances would cause sweat to occur in a hold? Select the ONE correct answer.

- **When warm moist air meets a cold surface**
- When cold dry air meets a warm surface
- When warm dry air meets a warm surface
- When warm moist air meets a warm surface

Which of the following amounts of ballast would give the 'heavy ballast' condition? Select the ONE correct answer.

- **55% of the loaded displacement**
- 25% of the loaded displacement
- 40% of the loaded displacement
- 95% of the loaded displacement

Which one of the following is a typical minimum forward draught? Select the ONE correct answer.

- **2.5% of the ship's length**
- 10% of the ship's length
- 5% of the ship's length
- 7.5% of the ship's length

Which one of the following is the main disadvantage of the sequential ballast water exchange method? Select the ONE correct answer.

- **Reasonably good weather is required for the whole of the operation**
- Damage to the ship's coating systems due to water flooding over the decks
- It is not as good as the flow-through method at removing organisms
- Over-pressuring the tanks

Whose direct responsibility is it to ensure that the hatch covers are properly secured prior to proceeding to sea? Select the ONE correct answer.

- **The Chief Officer**
- The Bosun
- The Master
- The Officer of the Watch

Why are daily soundings of hold bilges required during the ballast voyage? Select the ONE correct answer.

- **So that any signs of leakage into the hold can be found as soon as possible**
- To check that the ballast system is working correctly
- To confirm that the non-return valves are working correctly
- To confirm that the sounding pipes are clear

Why does the atmosphere above a coal cargo need to be monitored. Select the ONE best answer.

- **To monitor for explosive or toxic gases**
- This is a requirement of the charterers
- To determine the dew point in the hold
- To ensure that the relative humidity in the hold is at the correct level

Why is it important to keep good records of all ventilation during the voyage? Select the ONE correct answer

- **To be able to show that good care has been taken of the cargo during the voyage**
- To be allow the amount of power used to ventilate the cargo to be calculated
- To keep a log of the crew working hours with respect to ventilation of the cargo
- To record the running hours and wear on the ventilation system

Why is regular measurement of hold temperatures necessary when carrying cargoes such as coal or grain? Select the ONE best answer.

- **To monitor for self heating**
- So that the cooling ventilation can be planned
- To help with planning the unloading of the cargo
- To indicate if the hold bilges need pumping

With the exception of Australia, how many insects (dead or alive) in a hold will make it unfit to load a grain cargo? Select the ONE correct answer.

- **Three or more**
- Fifty or more
- One or more
- Ten or more

0349 – Bulk Carriers, Unloading

Version 1.0 (1082)

According to the BLU Manual, what is the definition of a 'terminal'? Select the ONE best answer.

- **A 'terminal' is any fixed jetty or quay, a fixed or mobile floating facility and any other facility capable of handling bulk cargoes**
- A 'terminal' is defined as applying only when there are fixed jetties or quays
- The word 'terminal' is applied only to floating jetties
- The word 'terminal' is applied only to specialized berths equipped with bulk unloading equipment

During the unloading of a bulk carrier, when should 'trimming' of the cargo be started? Select the ONE correct answer.

- **When about 85% of the cargo has been unloaded from a hold**
- After the unloading of the hold is completed
- As soon as unloading of all holds is declared complete
- When about 50% of the cargo has been unloaded from a hold

How can ship's officers show that they have been diligent in keeping the ship cargo-worthy? Select the ONE best answer.

- **Maintenance records for hatch covers and all hold piping systems**
- Crew certificates of qualification and medical records
- Log Book records of ventilation and bilge pumping
- Maintenance of Classification Society survey records

How is the heavy ballast condition normally achieved? Select the ONE correct answer.

- **The floodable hold is ballasted**
- Number One hold is filled
- Various holds are fully filled
- Various holds are partially filled

If stevedore damage is seen to be relatively minor, such as small indentations to plating, what repair action is necessary? Select the ONE correct answer.

- **No immediate action is required, but the stevedores should be recorded as being responsible and repairs done later**
- Immediate repair should be carried out
- In such cases, no stevedore damage report is necessary
- The Master will inform the stevedores that minor damage will be repaired at the next dry-docking, at no charge to the stevedore

Outside of MARPOL Annex V Special Areas, what is the general rule for discharging none HME cargo residues and wash water? Select the ONE correct answer.

- **Discharge is permitted, but not less than 12 miles from the nearest shore or ice shelf**
- Discharge is not permitted under any circumstances, except to a reception facility
- Discharge is permitted in seas and rivers as required
- Discharge may only be done at night

The BLU Manual is a supplement to which Code? Select the ONE correct answer.

- **The IMSBC Code**
- MARPOL
- The Grain Code
- The Green Code

The Master usually delegates the responsibility for producing the Unloading Plan. Who does the Master delegate this to? Select the ONE correct answer

- **The Chief Officer**
- The Second Officer
- The ship's shore-based marine superintendant
- The terminal representative

What action should the Master take if cargo damage is reported during the unloading? Select the ONE best answer

- **The shipowner's/operator's office must be informed immediately and the charterer and P&I Club informed**
- A meeting should be arranged with the terminal representative
- Nothing. The damage may not be noticed by the terminal operators
- The crew should dig out the damaged areas and set aside for disposal later

What action should the OOW take if they see large amounts of cargo leaking from the grabs during unloading? Select the ONE correct answer.

- **Protest to the terminal representative and give a written note of the complaint**
- Ask the terminal representative to have a look at the grabs at the earliest opportunity
- Get the crew to sweep up as much of the spillage so that it can be discharged to shore
- Nothing. This is the normal way that grabs work

What actions should the OOW take if they suspect that the Unloading Plan is not being followed to any significant extent? Select the ONE correct answer

- **The OOW must inform the Chief Officer immediately and, if necessary, suspend all cargo and ballasting operations**
- The OOW must contact the terminal representative and plan a meeting as soon as possible to resolve the issues
- The OOW should contact the ships agents to inform them of the problems
- The OOW should stop all ballasting operations

What can happen if a ballasted hold is de-ballasted without opening the Pressure/Vacuum (PV) valves in the hatch covers? Select the ONE best answer.

- **The hatch covers will collapse because of the suction under-pressure**
- Nothing, the de-ballasting will proceed as planned
- The ballast pumping will become very slow and eventually stop
- The hatch cover seals may be damaged

What guidance is there on the pre-arrival information exchange from ship to terminal? Select the ONE correct answer.

- **The BLU Code**
- The Green Code
- The MARPOL regulations
- There is no published guidance

What guidance is there on the pre-arrival information exchange from terminal to ship? Select the ONE correct answer.

- **The BLU Manual**
- The Green Code
- The MARPOL regulations
- There is no published guidance

What is gravity ballasting? Select the ONE correct answer.

- **When valves are opened to allow external water to flow into double bottom tanks and hopper tanks under gravity**
- When ballast water supplied by hose from the shore flows into the ship's tanks
- When external water is pumped into a ship's tanks using the ship's ballast pumping system
- When water ballast is first pumped into topside tanks and then allowed to drop by gravity into the double bottom tanks

What is recommended prior to opening the hatch covers if water-sensitive cargo is being carried? Select the ONE correct answer.

- **The crew should brush off any water or debris on the top of the hatch covers**
- An inspection of the cargo should be carried out prior to opening
- The hatch covers should be hose-tested
- The ship's mechanical ventilation system should be operated at maximum output for a few hours before opening

What is the best way for an OOW to monitor for stevedore damage? Select the ONE correct answer.

- **To be on deck as much as possible, observing the unloading operations**
- To ask the terminal representative to tell them if any damage occurs
- To sit in the ship's office listening for the sound of any damage occurring
- To wait until the end of discharge and then look for damage

What is the definition of 'wash water' under MARPOL Annex V? Select the ONE correct answer.

- **Water used in washing the holds that contains some cargo residues and may contain small amounts of cleaning chemicals**
- Fresh water used to rinse the holds
- Water used in washing the holds that contains some cargo residues but no cleaning chemicals
- Water used in washing the holds that has been filtered to remove solids

What is the expected accuracy of a well-conducted pre-load draught survey? Select the ONE best answer.

- **+/- 0.5%**
- +/- 0.05%
- +/- 10%
- +/- 5%

What is the most common ship-based cargo weight determination method for commercial purposes? Select the ONE correct answer.

- **A draught survey**
- Estimation by terminal representative based on discharge times
- Estimation by terminal representative of cargo weight unloaded
- Estimation of hold quantities by ship's officers

What is the purpose of the Ship Shore Safety Checklist? Select the ONE best answer.

- **To improve working relationships between ship and terminal, improving the safety of operations by better communications**
- The Checklist is required to monitor the progress of cargo unloading
- The purpose of the Ship Shore Checklist is to help avoid over-stressing of the ship's hull during unloading
- To protect the ship from cargo damage claims

What is the usual level of filling of a floodable hold? Select the ONE correct answer.

- **95%**
- 30%
- 50%
- 80%

What percentage of the loaded ballast draught is the light ballast draught? Select the ONE best answer.

- **45%**
- 25%
- 60%
- 80%

What records should the ship obtain if damage to the cargo is seen during unloading? Select all answers that you think apply.

- **Photographs of the affected areas**
- **Sketches of the affected areas**
- Nothing – an independent surveyor will look after all of the necessary records
- The name of the crane driver

What should be done before the hatch covers are opened if there is water or debris on top of the covers? Select the ONE correct answer.

- **The OOW should arrange for crew to sweep the water or debris off the hatch covers onto the deck**
- An independent surveyor should be employed to survey the hatch covers
- The hatch covers should be opened slowly to allow the water or debris to drop off onto the cargo below or the deck
- The terminal representative should be informed

What should the OOW do if they think that it will soon start to rain or snow and a water-sensitive cargo is being unloaded? Select the ONE correct answer.

- **Stop the unloading and close the hatch covers in good time to avoid water damage to the cargo**
- Ask the terminal representative for local advice on how long the rain is likely to last
- Continue cargo work on the basis that a small amount of water is unlikely to do much damage
- Partially close the hatches to reduce the amount of water falling into the holds and continue cargo work

What tools do the ship's officers have for producing the Unloading Plan? Select the ONE correct answer.

- **They will refer to the onboard Loading Manual and usually use the ship's Loading Instrument**
- They will make estimates based on hand calculations and complete the blank Unloading Plan form
- They will request the terminal to provide the Unloading Plan
- They will request their shore office to carry out the necessary calculations and provide a checked Unloading Plan

When is the Ship Shore Safety Checklist signed by the ship and terminal representatives? Select the ONE best answer.

- **As soon as the ship comes alongside and a safe access is available**
- Once all hatches are open and cargo work has been started
- The Checklist does not need to be signed
- Upon successful completion of discharge

Where are the regulations covering the discharge of hold cargo residues and wash water? Select the ONE correct answer.

- **MARPOL Annex V**
- IMSBC Code
- SOLAS
- The Grain Code

Where can details of the standard Ship Shore Safety Checklist be found? Select the ONE correct answer.

- **The BLU Code**
- Charterparty documentation
- The ship's logbook
- The ship's Loading Manual

Where does grab damage mostly occur during the final, trimming stage of unloading? Select the ONE correct answer.

- **The tank top.**
- The hatch coamings
- The hatch covers
- The topside tanks

Where does grab damage mostly occur during the first stage of unloading? Select the ONE correct answer.

- **The hatch coamings**
- The bilge covers
- The side frames and brackets
- The tank top

Where would you find the planned ballast sequence? Select the ONE correct answer.

- **In the Unloading Plan**
- In the Captains Standing Orders
- In the deck log book
- In the Ship-Shore Safety Checklist

Which of the following are 'continuous unloaders'? Select any answers that you think are correct.

- **Bucketwheel equipment**
- Cavaletto equipment
- Grab unloader gantry system
- Mobile luffing crane with grab

Which one of the following cargo types would you expect to be discharged by pneumatic elevator? Select the ONE correct answer

- **Grain**
- Coal
- Iron ore
- Iron ore fines

Which one of the following is an acceptable shore-based cargo measurement method? Select the ONE correct answer.

- **Electronic measurement of the weight of each grab load.**
- An estimate of the volume of stockpiles in the terminal
- Draught survey by ship's officers
- Estimation by terminal representative based on discharge times

Who is responsible for monitoring the Unloading Plan during unloading? Select the ONE correct answer.

- **The ship's officers, under the guidance of the Master, monitor the Unloading Plan**
- An independent surveyor is employed to monitor the Unloading Plan
- The cargo receiving representative is required to monitor the Unloading Plan
- The terminal representative is responsible for monitoring against the Unloading Plan

Who is responsible for producing the stevedore damage report? Select the ONE correct answer.

- **The Master**
- A Classification Society surveyor
- An independent surveyor
- The terminal representative

Who is responsible for real time monitoring of the Unloading Plan? Select the ONE correct answer.

- **The deck officer of the watch**
- The Chief Officer
- The Master
- The terminal representative

Why is the maximum permissible height from the waterline to the top of the hatch coamings or hatch covers included in the pre-arrival information? Select the ONE best answer.

- **So that the Master can check that the ship will safely fit under the unloaders at the highest tides predicted during the ship's stay**
- So that the Master can check that the hatch covers can be opened
- So that the Master can check the depth of water under the keel at high and low water
- So that the Master can check the steepness of the gangway

Why is wire cutting type damage to the compression bar on the coaming to be avoided? Select the ONE correct answer?

- **It can cause loss of the weathertight seal**
- It can damage the hatch cover running wheels
- It can lead to cracking of the coaming structure
- It can prevent the hatch covers from closing properly

Why might it be of benefit for the crew to join in trimming the holds during unloading? Select the ONE best answer

- **It will make it easier to clean the empty holds later**
- It will make sure the maximum amount of cargo is discharged
- It will save the cost of shore trimmers
- It will speed up the trimming activities

Why should pendulum swinging of the grab be avoided? Select the TWO best answers.

- **It can cause impact damage to the ship's structures**
- **It can cause puncturing damage to the ship's plating**
- It can lead to spillage of cargo from the grab
- It slows down the discharge rate

0360 – Crude oil tankers, Control of flow rates during loading

Version 1.0 (1855)

According to SOLAS, who should control closing and opening of the valves on the inert gas distribution line? Select the ONE correct answer.

- **The officer in charge of the cargo transfer**
- The Chief Engineer or the appointed deputy
- The duty rating when checking for cargo leaks on deck
- The Master, or the appointed deputy if ashore

According to the SIRE Vessel Inspection Questionnaire (VIQ), where should information regarding cargo and ballast loading rates and venting capacities be available? Select the ONE correct answer.

- **At the cargo control position**
- In the Chief Officer's standing orders
- Next to the manifold
- Noted on the Ship Shore Safety Checklist

According to this module, approximately how long should it take a 550mm diameter hydraulic actuated butterfly valve to close? Select the ONE correct answer.

- **55 seconds**
- 15 seconds
- 25 seconds
- 75 seconds

According to this module, in which of the following situations might the flow rate be greater than the maximum flow rate requested at the start loading? Select the ONE best answer.

- **When loading by gravity**
- When the crude oil has a high vapour pressure
- When the crude oil has not been stabilised
- When the terminal uses two pumps instead of one

According to this module, which of the following changes of ullage might be considered a suitable filling rate during loading? Select the ONE best answer.

- **5cm per minute**
- 1cm per minute
- 20cm per minute
- 50cm per minute

Before changing over to loading the next tank or batch of tanks during topping off, what precautions must you take with hydraulically actuated filling/suction valves? Select the ONE best answer.

- **Briefly open and close them to ensure they will open during change over**
- Disconnect the hydraulic connections and only open them using a portable hydraulic pump
- Put the loading facility on standby for emergency stopping, just in case the valves on the next batch do not open
- Stop loading when the batch of tanks is topped off, and resume when the valves on the next batch open and the valves on the previous batch close

During loading, there is a small but continuous change in the ullage of a filled tank not involved in the loading operation. What is the most likely cause? Select the ONE best answer.

- **The hydraulic suction valve for the tank has crept open very slightly**
- There has been a change of list
- There has been a change of trim
- There has been an increase in cargo temperature

During loading, which of the following must be monitored particularly closely while venting through a terminal's vapour return line? Select the TWO correct answers.

- **Cargo loading rate**
- **Cargo tank vapour pressure**
- Cargo vapour temperature increase
- Vapour growth in the cargo tank

During the topping off of the last few tanks of a batch, which of the following describes the action that could be taken to control the tank-filling rate without reducing the loading rate? Select the ONE best answer.

- **By partially opening and adjusting the position filling suction valve(s) on the next batch of cargo tanks to be loaded**
- Ensure the terminal loads by gravity only
- Ensuring the cargo filling suction valves on the remaining tanks are adjusted so that all reach the required ullage at the same time
- The loading rate must always be reduced to control the filling rate

How can the contents of a cargo tank that is not being loaded be checked to see whether or not the ullage is changing during loading? Select the TWO best answers.

- **Check that any ullage change can be explained by changes in trim, list or temperature**
- **Set the high and low level alarm close to either side of the ullage of the tank and monitor closely**
- Check that the loading rate for the tank is always shown as zero on the on-line loading instrument
- Check to see that the ullage remains constant; any change would indicate tank leakage

Which of the following actions will help minimise the risk of a pressure surge? Select the ONE correct answer.

- **Check that all in-line valves required to be open are fully open**
- Closely monitor the loading rate to ensure maximum rate is not exceeded
- Fit a pressure gauge to the manifold
- Slowly throttle two suction valves simultaneously during loading

Which of the following could be effects of loading at a higher rate than the maximum allowable? Select the TWO best answers.

- **Damage to cargo valve seals**
- **Eventual erosion of pipes leading to local pipe leakage**
- Damage to cargo tank coatings
- The average flow rate will be very difficult to calculate

Which of the following factors are used to determine the size and venting capacity of a cargo oil tank pressure-vacuum valve? Select the TWO best answers.

- **The design loading rate**
- **Vapour growth during loading**
- The height of the pressure-vacuum valve above the deck
- The horizontal distance between the inert gas inlet and the pressure/vacuum valve

Which of the following is the most likely source of leakage when loading starts? Select the ONE best answer.

- **Flanged manifold connection to a cargo loading arm**
- Flanged connection between two sections of deck cargo piping
- Flanged connection to a tee piece or bend on the deck cargo piping
- Flanged connection to the manifold valve

0380 – ISO 9001

Version 2.0 (2324)

According to this module, how do ISO define a 'standard'? Select the ONE correct answer.

- **A standard is a document that provides requirements, specifications, guidelines or characteristics that can be used consistently to ensure that materials, products, processes and services are fit for their purpose.**
- A standard is a minimum set of requirements that must be met in order to satisfy a customer, regulatory body or administration.
- A standard is any set of requirements or guidelines set down by a customer, regulatory body or administration.
- A standard is any set of requirements set down by a customer.

According to this module, what does ISO 9001 mean by the term 'Risk Management'? Select the ONE best answer.

- **It includes both avoiding risk, or deciding to retain some risk in order to pursue an opportunity**
- Risk Management in ISO 9001 is the same as Risk Assessment in the ISM Code.
- Risk Management means ensuring that all risks are properly insured.
- Risk Management means the avoidance of any risk.

Conformance to ISO 9001 can be proved by third party audit. What does this mean? Select the ONE correct option.

- **The organisations Quality management system can be audited by an independent, impartial third party, a certification body or 'registrar' who will conduct an assessment to verify conformity to ISO 9001 requirements.**
- ISO can act as a third party and carry out an audit to see if the organisations Quality management system meets the required standard.
- The organisations Quality management system can be audited by a customer.
- The organisations Quality management system can be audited by another organisation provided that organisation has ISO approval.

ISO 9001 makes reference to 'documented information'. Why is it important to define the documented information in a quality management system? Select the ONE best answer.

- **Because documented information in a quality management system needs to have controls covering distribution, access, use, storage and any changes made.**
- Because documented information can only be changed by senior management.
- Because documented information is fixed and cannot be changed.
- Because documented information must be made available for both internal and external auditors.

ISO 9001 requires a process for continuous improvement, often referred to as the 'PDCA' cycle. What does this mean? Select the ONE correct answer.

- **A Plan-Do-Check-Act cycle.**
- A Plan-Do-Check-Analyse.
- A Prepare-Do-Consult-Analyse cycle.
- A Product Delivery-Customer Analysis cycle.

ISO 9001 says that there should be measurable targets, in the form of objectives that are consistent with the company's quality policy. What properties should a quality objective have? Select ANY options that you think apply.

- **They should be communicated.**
- **They should be measurable.**
- **They should be monitored and updated.**
- **They should be relevant.**

ISO is derived from the Greek word 'isos'. What does isos mean? Select the ONE best answer.

- **Equal**
- Fair
- Justice
- Legal

In a Quality management system, which ONE of the following options is the best definition of a customer? Select the ONE best answer.

- **Customers can be anyone who receives goods or a service, whether they are inside an organisation or outside of it.**
- Customers are anyone outside of the organisation who receives goods or a service.
- Customers are anyone outside the organisation who receives goods or services, and the owners or shareholders of a company.
- Customers are anyone who pays for a service.

Select ONE option that best completes the following statement. 'Besides offering a number of benefits to the organization and the customer, ISO 9001 promotes International trade by . . .

- **. . . helping to prevent trade barriers and opening up global markets'.**
- . . . drafting contractual clauses'.
- . . . making standard sale and purchase agreements'.
- . . . providing a virtual meeting place for buyers and sellers'.

Should organisations be using a 'Quality Manual' following the 2015 revision to the ISO 9001 standard? Choose the ONE best option.

- **The use of a 'Quality Manual' is not specifically required, but documented information is required, and this might be grouped together and called a Quality Manual.**
- Quality Manuals are no longer allowed.
- Quality Manuals may be purchased from ISO or ISO approved organisations.
- Quality Manuals must be used and should always be available for auditors.

The Operations clause in ISO 9001 says that we should have procedures for non-conformances. According to this module, what is a 'procedure for non-conformance'? Select the ONE best answer.

- **It is what we should do if processes are not correct or procedures have not been followed.**
- It is an alternative way of doing things.
- It is the disciplinary procedure if anyone does not follow established procedures.
- It is the extra work carried out if there is something wrong with the product.

What is the full form of the abbreviation 'ISO'? Select the ONE correct answer.

- **International Organization for Standardization.**
- Inter-governmental Organization for Standardization.
- International Shipping Organization.
- International System Of Standardization.

What is the role of ISO in certifying organisations to ISO 9001? Select the ONE correct answer.

- **ISO is only responsible for producing standards; it has no role in certifying organisations.**
- ISO gives approvals for the other bodies which certify organisations to ISO 9001.
- ISO is only one of a number of organisations that can certify companies to ISO 9001.
- ISO is the only body which can certify organisations to ISO 9001.

Which ONE of these is NOT a Quality management principle in ISO 9001?

- **Lowest cost**
- Evidence based decision making
- Leadership
- Using a Process approach

Within ISO 9001, why does an organisation need to understand its own Context? Select the ONE best option.

- **The organisation needs to know its own regulatory obligations and what might be expected of them by third parties.**
- The organisation needs to establish a fair price for its goods and services.
- The organisation needs to know its own customers.
- The organisation needs to understand the market forces at work in the area it operates.

0385 – Occupational health and safety management system

Version 1.0 (1253)

All OHSAS organisations are required to have an occupational health and safety policy in place. Who drafts this policy? Select the ONE correct answer.

- **Senior management.**
- Auditor.
- Management representative.
- Workers.

All visitors should be informed about safety precautions to be followed during the course of the visit. What safety precautions should they be told? Select the THREE correct answers.

- **Actions to be taken in the event of an emergency.**
- **Entry in to areas permitted or not permitted during visit.**
- **Personal protective equipment to be worn.**
- Name and contact details of the management representative.

How is safety directly controlled in an OHSAS 18001 company? Select the TWO best answers.

- **By hazard identification.**
- **By risk control.**
- By appointing a management representative.
- By conducting frequent management review.

In the interests of safety, when should a task or work not be undertaken? Select the ONE correct answer.

- **If the risk assessed is unacceptable.**
- If the work is important it should be undertaken.
- If the work or task is low risk.
- If the work or task is medium risk.

KPI's are used to measure safety performance. What does the abbreviation K.P.I represent? Select the ONE correct answer.

- **Key Performance Indicator.**
- Key Performance Index.
- Key Personality Indicator.
- Key Persons Index.

OHSAS 18001 works on the principle of P.D.C.A. cycle. What does the letter P represent? Select the ONE correct answer.

- **Plan**
- Persevere
- Policy
- Protect

Select the ONE correct answer that describes the process of determining the frequency of conducting an audit.

- **Frequency of audit should be planned on the basis of the significance of risks.**
- A yearly audit should be carried out regardless of risk.
- Frequency of audit would depend on availability of auditors.
- Senior management would decide the frequency of audit depending on budget.

Training is necessary to ensure all personnel are competent in their health and safety duties. How is the effectiveness of the training assessed? Select the THREE best answers.

- **By conducting an end of course examination.**
- **By getting feedback from the trainees.**
- **By taking practical tests for skill based subjects.**
- The effectiveness can only be checked in an emergency.

What are the duties of a management representative in a company with an operational OHSAS 18001? Select the TWO best answers.

- **The health and safety management system is established, implemented and maintained in accordance with OHSAS 18001.**
- **The performance of the system is reported to the senior management on an ongoing basis.**
- Preparing planned maintenance schedules.
- Preparing work permit used prior to the start of a hazardous operation.

What does OHSAS stand for? Select the ONE correct answer.

- **Occupational Health and Safety Assessment Series.**
- Occupational Health and Safety Advisory Service.
- Occupational Health and Safety Assessment Specification.
- Occupational Health and Safety Assessment System.

What should be the MAIN priority to prevent emergencies from taking place? Select the ONE best answer.

- **Establish control measures that reduce the possibility of an emergency from occurring.**
- Maintain records of all safety training and drills carried out.
- Plan procedures to be prepared for an emergency should it occur.
- Review procedures using the lessons learnt from drills or actual emergencies.

What should generally be the first preferred option for risk reduction? Select the ONE correct answer.

- **Hazard elimination.**
- Administrative controls.
- Engineering controls.
- Use of personal protective equipment.

Which of the following documents, are found in an OHSAS company? Select the THREE correct answers.

- **Health and safety objectives.**
- **Health and safety targets.**
- **Occupational health and safety policy.**
- Age and physical fitness of employees.

Who defines the frequency and time of the management review? Select the ONE correct answer.

- **Senior management.**
- Auditors.
- Middle management.
- Workers.

'Near miss' is an incident that came very close to causing injury, ill health or fatality. Can a non-conformity be raised following a near miss incident?

- **Yes.**
- It would depend on auditors.
- It would depend on the health and safety policy in place.
- No.

0390 – Harassment and bullying

Version 1.0 (1750)

A Russian officer observes that the Filipino Rating takes too much time in finishing the job. The Russian tells him: "Don't waste time thinking, just eat some banana and get the job done". This is an example of . . .

- **Racial harassment**
- Corrective feedback
- Cyber bullying
- Sexual harassment

A certain behaviour which is acceptable to some people may be unacceptable to another individual or group. What common characteristic of harassment and bullying is described?

- **Differences in perception**
- Performance complaints
- Persistence
- Public humiliation

A person may lose his/her temper sometimes. This may be tolerated if the person apologises when calmed down. On the other hand, repeated sarcasm, belittling and abuse are considered harassment and bullying. What common characteristic of harassment and bullying is described here?

- **Persistence**
- Differences in perception
- Performance complaints
- Public humiliation

A senior officer initiates intimate advances to a female cadet in exchange for her faster promotion. The senior officer's behaviour is an example of . . .

- **Sexual harassment**
- Assault
- Racial harassment
- Sexual bullying

All of the following are among the nine characteristics protected by the laws of European Union and many countries EXCEPT which one?

- **Fashion preferences**
- Race
- Religion and beliefs
- Sexual orientation

All of the following are potential negative effects of harassment and bullying to the company EXCEPT which option?

- **reduced cost**
- crew resignations
- negative publicity
- reduced work performance

All of the following situations are examples of bullying EXCEPT which option?

- **The Bosun shouted at the OS who stepped inside the slip back zone while mooring operation is ongoing.**
- The AB who has a personal grudge against the OS took photos of the latter while drunk and posted these on Facebook.
- The Chief Officer ordered the Deck Cadet to deliver air supply using a bucket.
- The first timer Cadet is subjected to jokes as a form of entertainment to the experienced crew members.

Bullies do not . . . So during the direct confrontation, repeat what you have said, clearly and firmly, until it registers.

- **Listen**
- Ask for forgiveness
- Back-off
- Say too many words

Crew members who show depression, unhappiness, reluctance to communicate, or decrease in motivation and job performance are all . . .

- **Potential signs of harassment and bullying**
- Factors that cause harassment and bullying
- Fundamental problems on board ships
- Indicators of a good safety culture

How can the senior officer ensure that the complainant will feel at ease in reporting the harassment or bullying case?

- **The senior officer should ensure privacy of the complaint by treating it as a confidential matter.**
- The senior officer should ask the alleged bully to come forward and face the complainant right at that moment.
- The senior officer should ensure that he/she will definitely take side with the complainant.
- The senior officer should record his/her conversation with the complainant for future use.

In explaining the process, the senior officer should tell that he/she will ask the complainant a number of questions and write notes in order to . . .

- **get as many details as possible – not because he/she does not believe the complainant's story**
- Be able to present the answers to the alleged bully.
- Record the answers for the complainant's next performance appraisal.
- speed-up the investigation, and get the issue settled quickly

In order to avoid the perception of bullying, there should be no . . . and no picking on particular crew members when assigning jobs.

- **favouritism**
- compromise
- feedback
- name calling

In order to ensure that the responsible person will take time to hear your concern, how should you alert him/her? Choose the best answer from the statements below.

- **Captain, I have a serious concern to tell you. I prefer that we do it in private. Is this a good time?**
- Captain, I would like to ask if I can disembark on the next port. I am no longer happy working on this vessel.
- Captain, the Second Officer has been bullying me since I joined the ship. I cannot take it anymore.
- Captain, you should admonish the Second Officer. He is a bastard.

It is an offensive, embarrassing, intimidating or humiliating behaviour directed at a person or group because of his, her, or their skin colour.

- **Racial harassment**
- Conflict handling
- Cyber bullying
- Sexual harassment

Showing empathy to the other person is a way for the alleged bully to break the habit of bullying others. Which of the following is NOT a proper way of showing empathy?

- **If the person is sitting on kneeling, loom over him/her.**
- Do not stand too close to other people.
- Do not stare them down.
- Do not use your finger like a pointer to other's face.

The Chief Officer tells the first timer Cadet that everyone goes through the same initiation in their first vessel. If you are the Cadet, how should you reply to the Chief Officer?

- **I believe seafaring is a professional work. It did not come into my mind that I will be joining a secret society.**
- I have no time for that useless exercise.
- I wish I have the same guts to make through the initiation.
- That is a silly idea which is totally unacceptable to me.

The Second Engineer is a funny guy who always cracks jokes on other crew members. However, the Fitter finds the jokes not funny as his personal character has become the usual target. When the Fitter tells the Second Engineer that he should stop it, the latter reacts that it is just a joke that should not be taken seriously. If you are the Fitter, how should you reply to the Second Engineer?

- **A joke is only a joke if the person listening to it does not find it offensive.**
- I am asking you to treat people with respect.
- Really? I think you are just rude.
- That is totally unacceptable.

The Second Engineer is not aware that his subordinates perceive him as a bully because of the alleged unfair allocation of work among the engine personnel. They believe that in this respect, they are being bullied. What should the Second Engineer do to address the situation?

- **Ensure that there is a transparent system for equitable and fair allocation of work.**
- Explain to every crew member that he only wants to finish the job on time. For this reason, the more reliable and competent people are given more work.
- He does not need to be worried. The sentiment will die down naturally.
- Reprimand those who question him.

The female Third Officer feels embarrassed and humiliated every time the Second Officer touches her hand and looks at her with malice. The Second Officer's behaviour is an example of . . . ?

- **Sexual harassment**
- Cyber bullying
- Discrimination
- Racial harassment

The increase in heart rate and blood pressure and adrenaline rush are arousal signals that prepare the bully for his/her aggressive behaviour. What should the bully do in order to reverse these signals?

- **Take a deep breath and calm down**
- Divert the impending aggressive behaviour by hitting a punching bag.
- Just let the signals persist
- Take calming pills

The person being harassed or bullied should avoid doing all of the following actions EXCEPT which option?

- **Report the harassment or bullying to the Master or DPA.**
- Do nothing.
- Persuade yourself that this is part of the job.
- Quit or resign from the job.

The persons who might be harassing or bullying others should break this habit through behavioural changes. What are the four behavioural tendencies that they should be made aware of?

- **Triggers, positive thinking, arousal signals, empathy**
- Training, positive thinking, assertiveness, empathy
- Triggers, planning ahead, arousal signals, empathy
- Triggers, positive thinking, arousal signals, enticement

To get the facts from the AB who claims he was bullied, the Master asks him a series of questions and the AB responds with very few and basic answers. The Master would like to dig deeper into the allegations. How can the Master achieve this?

- **The Master should ask probing questions in relation to the previous answers to extract more information from the AB.**
- The Master should ask another resource person to bridge the gaps in the AB's statements.
- The Master should ask closed questions which are answerable by yes or no or specific facts.
- The Master should maintain an appropriate level of eye contact with the AB.

What is the positive impact to the person being harassed or bullied of seeking help and support from other crew members?

- **It prevents the person from feeling isolated.**
- It enables the person to confidently give the bully a taste of his own medicine.
- It makes both the person being bullied and the bully happy.
- It solves harassment and bullying problems.

When elaborating the details of the alleged harassment or bullying behaviour, you should AVOID . . .

- **making general remarks about the bully's personality**
- citing specific dates, occasions, or events when the harassment or bullying took place
- describing the bully's specific behaviour that you believe is harassment or bullying
- sticking to the facts

When you first realise you are being systematically picked on, start . . . where and when it occured, what happened, what was said by both parties, and how you felt.

- **recording**
- analysing
- memorising
- sharing

Which of the following actions should the senior officer do in order to handle the harassment and bullying complaint properly?

- **Know exactly what is the company procedure in handling harassment and bullying complaints**
- Dismiss the case instantly
- Forward the complaint to the alleged bully
- Record the arguments of the complainant, especially the emotional parts

Which of the following options is a form of discrimination which creates feelings of unease, humiliation, embarrassment, intimidation or discomfort to the person on the receiving end.

- **Harassment**
- Conflict
- Corrective feedback
- Favouritism

Which of the following situations can be classified as bullying?

- **The Chief Officer gives corrective feedback to the Third Officer who breached the safety procedures and highlights that he is not really surprised because the Third Officer comes from a carefree culture.**
- The Bosun gives short and sharp orders to the deck ratings in preparation for the inspection.
- The Chief Engineer calls the attention of the Fourth Engineer upon observing that the latter is not wearing the appropriate PPE.
- The Master reminds the Chief Cook about a previous commitment to adjust the weekly menu based on the collective feedback from the crew members.

Which of the following statements about bullying is FALSE?

- **Bullying always involves an abuse of power or position.**
- Bullying may be carried out through verbal or physical actions.
- Bullying may be done intentionally or not.
- Subordinates are also capable of bullying their superiors or colleagues.

Why is it important to be sensitive and patient when talking with the complainant or alleged bully?

- **Because they are expected to be too emotional and irrational.**
- Because they are always telling half-truths.
- Because they are emotionally strong and effective communicators.
- Because they are expected to be listening actively.

. . . is a form of harassment where a person or group is made to feel frightened or intimidated through verbal or physical actions that are unacceptable, hostile, or not funny.

- **Bullying**
- Racial harassment
- Sexual harassment
- Threatening

. . . involves the use of electronic media such as cellphones, tablets and computers to threaten, humiliate or embarrass a person or group.

- **Cyber bullying**
- Bullying
- Electronic harassment
- Technological terrorism

0391 – Fatigue management

Version 2.0 (1157)

A cabin with a suitable temperature and a good bunk of the right size, mattress, and bed linen are preconditions for good sleep.

- **True**
- False

A person sleeping for about 30-40 minutes is easily woken by ship rolls, noise in the passageway, or ringing phone. This description fits what stage of a normal sleep cycle?

- **Stage 2**
- Stage 1
- Stage 3
- Stage 4

All of the following are measures to develop a good sleeping habit, EXCEPT for which option?

- **Avoiding a hot shower before sleeping**
- Avoiding alcohol and caffeine prior to sleep
- Avoiding violent physical exertions just before going to sleep
- Getting up at the same time each working day and considering doing so on days off, too

All of the following are operational factors that contribute to sleep disruption, EXCEPT for which option?

- **Alcohol/coffee intake**
- Disruptions caused by drills
- Watch systems
- Working and rest hours

All of the following are techniques to maintain a high level of situational awareness during night watch on the bridge, EXCEPT for which option?

- **Taking short naps**
- Discouraging the use of chairs
- Drinking a cup of coffee or tea
- Eating some fruits

Coffee, tea, cola, and chocolate should be avoided at least three hours before going to sleep.

- **True**
- False

It is defined as the state of feeling tired, weary, or sleepy.

- **Fatigue**
- Sickness
- Sleepiness
- Stress

It is defined as the state of struggling to maintain wakefulness.

- **Sleepiness**
- Fatigue
- Overworked
- Stress

It is generally recommended that a person obtain, on average, . . . hours of sleep per day.

- **7–8**
- 5–6
- 6–7
- 8–9

It is the most effective strategy to fight fatigue.

- **Sleep**
- Drinking coffee
- Exercise
- Nap

It refers to the cycle in which a human body moves through various physical processes and states within a 24-hour period.

- **Circadian rhythm**
- Activities
- Body clock
- Daily routine

It regulates the body's circadian rhythm. It is perfectly synchronised with the usual pattern of daytime wakefulness and night-time sleep.

- **Biological clock**
- Heartbeat
- Nervous system
- Pulse

Poor judgement of distance, speed, time, etc., and having slow or no response to normal, abnormal, or emergency situations are . . . symptoms/indicators of fatigue.

- **mental**
- chronic
- emotional
- physical

The two periods of low alertness or low points/dips in each 24-hour period commonly occur between . . .

- **3–5 am and 3–5 pm**
- 12 mn–1am and 12 nn–1 pm
- 1–2 am and 1–2 pm
- 1–3 am and 1–3 pm

0395 – Gas carrier, Basic principles

Version 1.0 (3289)

According to SOLAS, what is the maximum allowable oxygen content of the inert gas produced by an inert gas generator? Select the ONE correct answer.

- **Less than 5% by volume.**
- Less than 3% by volume.
- Less than 8% by volume.
- There should be no oxygen present.

How can we increase the temperature at which a liquid boils? Select the ONE correct answer.

- **By increasing the pressure.**
- By decreasing the pressure.
- By moving it to high altitude.
- The boiling point is a fixed value.

In relation to cargo operations on gas carriers, what does the term 'gas freeing' mean? Select the ONE correct answer.

- **The removal of toxic and flammable gas from a tank with inert gas, followed by the introduction of fresh air.**
- The removal of cargo gas from a tank using fresh air, followed by the introduction of inert gas.
- The removal of oxygen from a tank with inert gas.
- The removal of toxic and flammable gas from a tank with inert gas.

In relation to cargo operations on gas carriers, what does the term 'gassing up' mean? Select the ONE correct answer.

- Replacing a gas free atmosphere in a tank with the vapour from the next cargo to allow cooling down and loading.
- Replacing an inert atmosphere in a tank with fresh air.
- Replacing an inert atmosphere in a tank with the vapour from the next cargo to allow cooling down and loading.
- Replacing the cargo vapour in a tank with inert gas.

Select the ONE statement that correctly completes the following statement.

'Cameras, phones, radios and torches used in or near the cargo areas onboard a gas carrier must be . . . . . .

- **. . . aproved as safe to use in a hazardous area'.**
- . . . kept switched off at all times'.
- . . . only operated on battery power'.
- . . . only used in aircraft mode'.

The IGC Code requires that the ESDS emergency shutdown valves close within 30 seconds. What is the reason for this? Select the ONE correct answer.

- **The valves need to close smoothly and not cause a pressure surge that would damage the pipelines.**
- It gives time for manual intervention to prevent closure in case of accidental tripping.
- It gives time for personnel to clear the area.
- That is as quickly as large valves can reasonably be expected to close.

What are the requirements of the IGC code for the pressure relief valves of cargo tanks? Select the ONE correct answer.

- **At least TWO pressure relief valves of equal capacity.**
- At least THREE pressure relief valves.
- ONE main relief valve and one smaller, auxiliary relief valve.
- ONE pressure relief valve, provided that the area for venting exceeds the area of the filling line.

What is the greatest problem when storing LPG at ambient temperature? Select the ONE correct answer.

- **The pressure in the tank will be about 8 bar, so this requires a very strong tank, which is both heavy and expensive.**
- The LPG can only be stored as a vapour at this temperature, so the amount that could be stored would not be economically viable.
- The pressure in the tank is too high to be contained.
- The pressure in the tank will continue to rise and must be vented.

What is the purpose of the insulation around the cargo tanks of gas carriers? Select the TWO correct answers.

- **To protect the structure of the ship from contact with cold cargoes, which could cause the steel to become brittle.**
- **To reduce the amount of heat transferred to the cargo and therefore reduce the amount of cargo that boils off.**
- To completely support the weight of the cargo tanks.
- To reduce the amount of heat transferred from the cargo and therefore reduce the amount of cargo that boils off.

Where would you find the regulations about how the cargo refrigeration plant on your ship is constructed? Select any answers that you think apply.

- **Classification Society Rules.**
- **Flag State Rules.**
- **The IMO's International Gas Code (IGC).**
- SOLAS.

Why is Nitrogen sometimes used for inerting the tanks of gas carriers, instead of inert gas produced by combustion? Select the ONE correct answer.

- **Because the products of combustion can spoil the purity of some cargoes.**
- Because Nitrogen can be produced more quickly.
- Because Nitrogen is cheaper to produce.
- Because the Nitrogen has a higher dew point.

Why is it important that the design and operation of the ship's gas detection system takes into account the types of cargo to be carried? Select ANY answers that you think apply.

- **Because gas detectors must be calibrated for the type of cargo carried.**
- **Because some gases are heavier than air and some lighter than air, so this affects where the gas detectors are located.**
- **Because the IGC code has different requirements for different cargoes.**
- Because an effective fixed system means that portable gas detection equipment is not needed.

0396 – Gas carrier, Cargo cooling systems, introduction

Version 1.0 (797)

Cooling or Liquefaction plants are one way of keeping the pressure inside cargo tanks within acceptable limits. What other methods are commonly used? Select the TWO correct answers.

- **Burning the gas in a gas combustion unit.**
- **Some cargoes like LNG can be used in the main propulsion plant.**
- Transferring some of the cargo to other tanks
- Venting the cargo

During the cargo liquefaction process, from what point is cargo taken? Select the ONE correct answer

- **It is taken as vapour, from the top of the cargo tank.**
- It is taken as liquid, from the bottom of the cargo tank.
- It is taken at the surface of the liquid, where its removal will have the greatest cooling effect.
- It is taken from just below the surface of the cargo, where its removal will have the greatest cooling effect.

If the working fluid is pure, then what determines the compressor discharge pressure in a vapour compression cycle? Select the ONE best answer.

- **The temperature in the condenser, which largely depends on the temperature of the coolant.**
- The inlet pressure of the compressor.
- The speed of the compressor.
- The volume of the condenser.

The pressures and temperatures at the compressor discharge in a liquefaction plant do not match those given for the cargo on the compressor data sheet. What could be the reasons for this? Select the TWO best answers

- **Non-condensable gases have collected in the condenser.**
- **The cargo is not pure and therefore not exactly the same as that used for the data sheet.**
- The compressor is old.
- The data sheet is wrong.

We are heating a liquid and it is boiling. How can we increase its temperature? Select the ONE correct answer.

- **Increase the pressure above the fluid.**
- Decrease the pressure above the fluid
- Increase the amount of heat added
- The temperature of the liquid will rise anyway, provided we keep the pressure constant

What do we mean by the 'critical point' of a fluid and why is it important? Select the ONE correct answer.

- **It is the temperature and pressure at which the liquid and vapour phases co-exist. There is no latent heat, so we cannot use the fluid in a refrigeration cycle**
- It is the point at which enough latent heat has been gained to turn a liquid into a vapour. It is the point at which we can start to compress the vapour.
- It is the point at which enough latent heat has been removed to turn a liquid into a solid. It must be avoided for good heat transfer.
- It is the point at which enough latent heat has been removed to turn a vapour into a liquid. It is the point at which the fluid exits the condenser.

What do we mean by the 'enthalpy' of a fluid? Select the ONE correct answer.

- **The total heat energy that it has, usually given per kg of fluid.**
- The amount of heat energy required to make it boil at atmospheric pressure.
- The amount of heat energy required to raise its pressure by one bar once it is boiling.
- The amount of heat energy required to raise its temperature by one degree at atmospheric pressure.

What is the most likely cause of a very sudden rise in the condenser pressure of an open cycle direct cargo cooling plant? Select the ONE best answer.

- **The condenser has become full of liquid cargo because its capacity has been exceeded.**
- The cargo contains fractions of another gas.
- The sea water temperature has risen .
- There are some impurities in the cargo.

What should be done with non-condensable gases in a vapour compression refrigeration cycle? Select the ONE correct answer.

- **They should be bled off from the top of the condenser, with due regard to any polluting or other effects.**
- Filter driers will absorb them and therefore these need to be regularly changed.
- Nothing needs to be done, they will circulate naturally.
- They should be bled off from the compressor discharge pipe, with due regard to any polluting or other effects.

What type of cargoes would need to use a cascade cooling plant? Select the ONE correct answer.

- **Those with a low boiling point**
- Those that are particularly flammable
- Those with a high boiling point
- Those with a high critical temperature

Why can Ethane not be condensed using a seawater cooled condenser? Select the ONE correct answer.

- **The condenser pressure would be too high.**
- Ethane cannot be turned into a liquid, except in a laboratory.
- Ethane reacts with seawater.
- The volume of seawater required to remove the required amount of heat would be too great.

Why do some impurities in the cargo cause a high pressure in the cargo tank? Select the ONE best answer.

- **Because they have a lower boiling point or do not condense.**
- Because they are denser.
- Because they form liquids which take up volume in the cargo tank.
- Because they react to form new compounds.

0403 – Container ships, Securing cargo awareness

Version 3.0 (1053)

How many degrees of freedom is a ship's motion defined by? Select the ONE correct answer.

- **6**
- 2
- 3
- 4

How should container stacks be secured? Select the ONE best answer.

- **In accordance with the Cargo Securing Manual**
- In accordance with the ship's shipboard management system
- In accordance with the terminal's cargo securing policy
- Sufficient enough to ensure the container stack doesn't become unstable

What do we mean by lashing force? Select the ONE best answer.

- **This is the tension applied to lashing rods to restrain the movements of containers in a seaway**
- This is compressive force acting down in the vertical direction on the corner-posts
- This is tensile force acting in the transverse direction on the corner-casting container
- This is the effect of forces acting on the pins of twist-locks and stacking cones

What produces dynamic force when a container ship is rolling in a seaway? Select the ONE correct answer.

- **Accelerations in the transverse direction**
- Acceleration due to gravity
- Accelerations in the longitudinal direction
- Accelerations in the vertical direction

What rolling phenomenon is experienced on ships with very fine and flared bows and wide transom sterns like those on container ships? Select the ONE correct answer.

- **Parametric rolling**
- Excessive Rolling
- Synchronous rolling
- Violent rolling

What type of force results from the weight of a container and its contents? Select the ONE correct answer.

- **Static force**
- Direct force
- Dynamic force
- Impulse force

What type of motion causes the greatest dynamic forces that act on a container? Select the ONE correct answer.

- **Rolling**
- Heaving
- Pitching
- Yawing

What type of ship's motion produces movement about the transverse axis? Select the ONE correct answer.

- **Pitch**
- Surge
- Sway
- Yaw

Which of the following are the main consequences of a collapsed stack of containers, some of which have fallen into the sea? Select the TWO best answers.

- **Navigational hazards to shipping**
- **Safety hazard to the ship's personnel**
- The ship will develop a faster rolling period
- The ships speed can be reduced significantly

Which of the following factors contribute to losses and damages to containers at sea? Select TWO best answers

- **Condition of securing components**
- **Inadequate lashing of the contents within a container**
- Age of the container; containers more than three years old will always be weaker than newer ones
- Type of container, twenty-foot general containers tend to be lost more easily

Which of the following ship motions are most likely to result in the loss of containers on large modern container ships? Select the ONE best answer.

- **Continued parametric rolling**
- Normal rolling and pitching in a three meter swell
- Pitching in heavy weather
- Rolling to a 15° angle of heel

Which of these might be a weak point in a stack of containers? Select the ONE best answer.

- **A container with worn corner-castings**
- A container secured with manual twist-locks
- A container stack more than three high
- An empty container located at the top of a stack

0404 – Container ships, Securing cargo arrangements

Version 3.0 (718)

During which of the following situations is a container stack of forty-foot containers always required to be secured with lashing rods? Select the ONE correct answer.

- **A container stack with four tiers.**
- A container stack with one tier.
- A container stack with two tiers.
- Containers on deck must always be secured with lashing rods regardless of tier height.

How are lashing rods tensioned when securing a stack of containers on deck? Select the ONE best answer.

- **They should be tensioned sufficiently to prevent any movement of the twist-locks.**
- It is good enough to just hold the twist-locks in place and no more.
- Lashings must always be tightened as much as possible. It is essential that a container stack is as rigid as possible.
- The lashing bars should always be loosely tensioned. It's the twist-locks that keep the stack secure.

How can the Cargo Securing Manual be helpful to deck officers who are new on board? Select the TWO best answers.

- **It can help them recognise the types of securing equipment onboard.**
- **It can help them to recognise the correct method of securing containers.**
- It can help them recognise how cargo is secured in the containers.
- It can help them recognise how to use the ballast system to increase the GM.

How is a container's securing components rated for strength according to the Code of Safe Practice for Stowage and Securing (The CSS Code)? Select the ONE correct answer.

- **According to Maximum Securing Load (MSL).**
- According to Calculated Strength (CS).
- According to Minimum Breaking Load (MBL).
- According to the Tensile Load (TL).

How is a semi-automatic twist-lock secured or locked in place when it engages a container corner-casting? Select the ONE best answer.

- **Once set to the locked position, it is automatically locked in place.**
- It has to be locked in place by hand. It only unlocks automatically.
- Once set to the locked position, it is locked in place by pulling a toggle or lever.
- The container being loaded has to be tipped slightly in order to properly engage.

How should lashing bars be tightened to ensure they are at the correct tension? Select the ONE correct answer.

- **Turnbuckles should be tightened hand-tight plus one quarter-turn with a spanner.**
- Lashing bars must always be slightly slack otherwise lashing bars may break in heavy weather.
- Turnbuckles must always be tightened hand-tight only and no tighter.
- Turnbuckles should be as tight as physically possible using the spanner provided.

On a container ship, where can information be found on lashing arrangements? Select the ONE best answer.

- **Cargo Securing Manual.**
- General Arrangement Plan.
- Ship Board Management System.
- Trim and stability booklet.

Under which of the following requirements is a Cargo Securing Manual required to be carried? Select the TWO correct answers.

- **Code of Safe Practice for Cargo Stowage and Securing (CSS Code).**
- **SOLAS: Chapter VI and chapter VII Carriage of Cargoes.**
- MARPOL Annex V Prevention of Pollution by Garbage from Ships.
- The International Convention on Load Lines.

Where might stevedores who are unfamiliar with the ship find cargo lashing patterns based on the ship's Cargo Securing Manual? Select the TWO best answers.

- **These are commonly posted on placards between container bays.**
- **These are commonly posted on placards on the lashing bridges (if fitted).**
- Stevedores don't need to check the Cargo Securing Manual as all container securing is always the same for cellular container ships.
- The Chief Officer will always issue the foreman with a copy of the Cargo Securing Manual.

Which of the following best describes the purpose of the Cargo Safe Access Plan (CSAP)? Select the ONE correct answer.

- **It is a plan that is used in conjunction with the Cargo Securing Manual to ensure safe working access and conditions during securing operations.**
- It is a plan that is used in conjunction with the Cargo Securing Manual to detail the order in which containers are secured.
- It is a plan that is used in conjunction with the Cargo Securing Manual which shows specifically the position of deck securing points and how to access them.
- It is a plan that is used in conjunction with the Cargo Securing Manual which specifically maps the route to be followed by stevedores when lashing containers.

Which of the following fittings can be used to help secure a container directly to the deck, hold or pedestal? Select the TWO correct answers.

- **A dovetail foundation.**
- **A raised socket.**
- A D-ring.
- A lashing plate.

Which part of a twist-lock resists lifting force? Select the ONE best answer.

- **The cones.**
- The collar.
- The housing.
- The intermediate plate.

0415 – Fuel Oil Bunkering, Effective Practice and Sampling

Version 1.0 (3074)

According to this module, in case of excessive sludge, if fuels have been mixed, how should the fuel oil separators be operated? Select the ONE correct answer.

- **In parallel, if possible, and at the lowest possible flow rate.**
- In parallel, if possible, and at the highest possible flow rate.
- In series, if possible, and at the highest possible flow rate.
- In series, if possible, and at the lowest possible flow rate.

According to this module, what should a bunker sample be collected in? Select the ONE correct answer.

- **A disposable 10-litre or 5-litre cubitainer that is screwed onto the sampler and threaded with a seal.**
- A clean, re-usable glass storage jar with a seal.
- A measuring jug for the easy distribution of the samples into other sealed plastic bottles.
- Four small plastic bottles with labels affixed to them.

According to this module, why is it not recommended to blend fuels on board? Select the ONE best answer.

- **Because it can result in an unstable blend that causes operational problems, such as large amounts of sludge.**
- Because it is easier to get them blended ashore.
- Because it makes it difficult to calculate an accurate density.
- Because they tend to separate very quickly if blended this way.

During bunkering, why should you check the temperature of the bunkers? Select the ONE best answer.

- **Because temperature affects density, which is used in the calculation of mass (weight) of bunkers delivered.**
- Because temperature affects density, which is used in the calculation of volume of bunkers delivered.
- Because temperature affects the pumping rate.
- To check that the bunker oil remains below its closed flashpoint.

How should a bunker sample be taken? Select the ONE correct answer.

- **By a continuous drip-method throughout the duration of the bunkering.**
- A series of samples should be taken throughout the bunkering operation.
- One sample should be taken at intervals not exceeding 15 minutes.
- The supplier should take and provide samples.

How should you fill the sample bottles from the cubitainer? Select the ONE correct answer.

- **All sample bottles should be filled 1/3 at a time, making several passes to fill up the bottles equally, capping and re-shaking the cubitainer before each filling.**
- Each sample bottle in turn should be completely filled, capping and re-shaking the cubitainer before each filling.
- Each sample bottle in turn should be completely filled, with no break between them.
- The contents of the cubitainer should be allowed to settle before each filling.

Sampling equipment should be fitted at the point where custody of the bunkers changes. Where is this? Select the ONE correct answer.

- **The ship's bunker manifold.**
- At the filling point for the bunker barge or terminal storage facility.
- The bunker barge or shore terminal manifold.
- The ship's storage tanks.

What drip rate should be used during bunker sampling? Select the ONE best answer.

- **The drip rate should be enough to give sufficient volume of fuel sample in the cubitainer by the end of the bunkering.**
- The drip rate should be enough to completely fill the cubitainer by the end of the bunkering.
- The drip rate should be enough to completely fill the cubitainer well before the end of the bunkering.
- The drip rate should be enough to give sufficient volume of fuel sample in the cubitainer well before the end of the bunkering.

What is the main reason that the Sulphur content of fuel must be known? Select the ONE best answer.

- **To ensure compliance with MARPOL and ECA regulations.**
- To ensure compliance with SOLAS regulations.
- To ensure the ignition qualities of the fuel.
- To limit engine wear.

What should be done with the MARPOL bunker sample? Select the ONE correct answer.

- **It should stay under the ship's control until the fuel oil is used up, and it should be kept for at least 12 months from the time of delivery.**
- It should be handed to an IMO representative.
- It should be sent to an approved laboratory for testing.
- It should be sent to head office and kept for 3 years.

What should you do if the supplier refuses the bunker sample taken by the ship? Select the ONE correct answer.

- **Record it in the ship's log with full details of the responsible person, the barge, the supplier's name, the time, date, and seal number of the sample.**
- Ask the supplier to provide a sample as a cross reference.
- Note this fact in the ship's log and offer to provide another sample from the bunker tanks.
- Nothing, this is the responsibility of the supplier.

When should you sign the Bunker Delivery Note? Select the ONE correct answer.

- **At the completion of bunkering.**
- Before starting the bunkering process.
- Once a commercial agreement to buy the bunkers has been completed.
- Once the bunker barge is alongside.

Which of the following could be a result of poor bunker management? Select all the answers you think are correct.

- **A big loss of money.**
- **Damage to the main engine.**
- **Failure to comply with rules for Sulphur emissions.**
- **Failure to comply with rules for the minimum flash point of main fuel oils.**

Why must the Sulphur content of fuel oils be known? Select the ONE best answer.

- **Because there is a maximum limit on the Sulphur content of all marine fuel oils and tighter limits in ECAs and elsewhere.**
- Because it affects the calorific value of the fuel.
- Because it affects the price of the fuel.
- Because there is a maximum limit on the Sulphur content of marine fuel oils in ECA's and in accordance with local regulations.

Why must the density of the bunker fuel be accurately known? Select the ONE correct answer.

- **To work out the quantity of bunkers received, since mass is volume multiplied by density.**
- To calculate the minimum storage temperature at which it can still be pumped.
- To calculate the temperature needed to atomise and burn it.
- To work out the quantity of bunkers received, since mass is volume divided by density.

Why should the bunker line pressure be checked continuously during bunkering? Select the ONE best answer.

- **Because it can vary when the barge switches tanks or if the fuel is not homogeneous and needs to be checked to ensure it stays below a safe maximum value.**
- Because it should be kept absolutely constant at a level agreed between the ship and barge representatives.
- To check that the fuel is not being mixed with any other fluid or air, which would explain any fluctuation of pressure.
- To check when bunkering is complete, so that the hose can be disconnected as soon as possible and not delay sailing.

0416 – Fuel oil bunkering, Understanding test results

Version 1.0 (2160)

According to this module, how can wax formation with distillate fuel oils best be avoided?
Select the ONE best answer.

- **Ensure the oil is kept at least ten degrees Celsius above its pour point.**
- Ensure the oil is kept at least ten degrees Celsius below its pour point.
- Ensure the oil is kept at least ten degrees Celsius below its wax point.
- Use the separators as clarifiers only.

According to this module, how long should Bunker Delivery Notes (BDN) be kept on-board?
Select the ONE correct answer.

- **Bunker delivery notes should be kept readily available on-board for a minimum of 3 years.**
- Bunker delivery notes can be sent to head office and stored there.
- Bunker delivery notes should be kept readily available on-board for a minimum of one year and until the fuel has been used up.
- There is no need to keep bunker delivery notes on-board, since the suppliers must keep a copy.

According to this module, how should Type 2 and 3 ECA compliant fuels be treated?
Select the ONE correct answer.

- **They should be treated as residual heavy fuel oil of corresponding grade, particularly with regard to heating.**
- They should be treated as distillate fuels.
- They should not be centrifuged, as this causes the constituents to separate.
- They should not be heated in storage.

According to this module, if distillate fuel onboard is found to have a high Acid number, what should you do?

- **Send a sample to a specialist testing agency to check for corrosion and/or chemical contamination.**
- Add an alkali to the fuel.
- Increase the rate of cylinder lubrication oil feed, and keep the main engine cooling water temperatures high.
- Water-wash the fuel using the separators.

According to this module, what special care is necessary when storing Type 2 and 3 ECA compliant fuels?
Select the ONE correct answer.

- **They should not be mixed with other residual fuels onboard, since there may be compatibility issues.**
- They should be regularly transferred between tanks, to prevent their component parts from separating out.
- They should not be heated in storage.
- They should not be heated to within twenty degrees of their closed flashpoint.

How long should the MARPOL Annex VI sample be kept?
Select the ONE correct answer.

- **Until the fuel has been substantially consumed, but in any case for a minimum of 12 months.**
- For three years.
- Only until the fuel has been consumed.
- Until the next Port State inspection of for at least 12 months.

If marine residual fuel is found to contain catalytic fines, how should the separators be operated?
Select the ONE correct answer.

- **In parallel, with minimum flow rate.**
- In parallel, with maximum flow rate.
- In series, with maximum flow rate.
- In series, with minimum flow rate.

In relation to distillate fuels, what is the significance of FAME?
Select the ONE correct answer.

- **Fatty Acid Methyl Esters increase the risk of microbial growth and filter blockage.**
- Fatty Acid Methyl Esters increase engine cylinder liner wear and deposits.
- Fatty Acid Methyl Esters reduce the ignition qualities of the fuel.
- Fuels with a high FAME content have a much lower calorific value.

The condition of distillate fuels can often be quickly assessed by a visual inspection of the sample, which should be bright and clear. At what temperature should this test be carried out?
Select the ONE correct answer.

- **Between 10 and 25 degrees Celsius.**
- The sample should be cooled to below 10 degrees Celsius.
- The sample should be warmed to at least 30 degrees.
- The sample should be warmed to within 10 degrees Celsius of its closed flash point.

What action should be taken if a significant amount of water is discovered in the ship's fuel oil?
Select the ONE best answer.

- **Keep the settling tank temperatures high and drain them frequently. Operate separators in parallel.**
- Increase bunker tank temperatures and try to evaporate as much water as possible.
- The water will turn to steam during combustion, so there may be some benefits.
- Transfer the fuel to other tanks, thereby mixing the fuel and lowering the percentage of water to a manageable amount.

What corrective action can be taken if the residual fuel bunkered has a low net specific energy?
Select the ONE correct answer.

- **None, it just means the fuel has a low energy value, so more will need to be burned for the same power.**
- It will need to be blended with distillate fuel onboard before it can be burned.
- The fuel oil temperature will need to be decreased to raise the viscosity for atomisation.
- The timing of the engine will need to be adjusted in order to burn the fuel.

What difficulties can occur when a fuel with a low Calculated Cetane Index (CCI) is used?
Select the ONE correct answer.

- **There may be starting difficulties at low temperatures, ignition delay and incomplete combustion.**
- Main engine cylinder pressures can be excessive.
- There may be excessive cylinder liner wear.
- There may be high exhaust temperatures at high loads.

What is the minimum allowable closed flash point of marine fuel oils used for propulsion?
Select the ONE correct answer.

- **60 degrees Celsius.**
- 50 degrees Celsius.
- 70 degrees Celsius.
- More than ten degrees below the auto-ignition temperature.

What is the 'CCAI' value of a fuel, and why is it significant?
Select the ONE correct answer.

- **It is the Calculated Carbon Aromaticity Index, and gives some indication of the ignition quality of the fuel.**
- It is the Calculated Carbon Aromaticity Index, and gives some indication of how volatile the fuel is.
- It is the Combustible Carbonic Acids Index, and gives some indication of the acidic deposits formed after combustion.
- It is the Concentration of Carbon Aromatics Index, and gives some indication of the deposits left after combustion.

When are deposits in the engine due to Vanadium most likely to be a problem?
Select the ONE best answer.

- **When the ratio of Vanadium to Sodium is 3:1 and the Vanadium content is greater than 150mg/kg.**
- At low operating temperatures.
- When there are equal amounts of Sodium and Vanadium in the fuel.
- When there is no Sodium in the fuel.

Why are catalytic fines so problematic when present in residual fuel oil?
Select the ONE correct answer.

- **They are small and difficult to remove, but very hard and cause rapid wear of engine parts.**
- They are an indication that the fuel has not been completely refined and may therefore give poor combustion.
- They are an indication that the fuel oil has been contaminated with other waste oils, such waste lubricating oil, which will give poor combustion.
- They cannot be burned and therefore lead to sticky deposits in the combustion chambers of the engine.

Why can burning distillate fuels present a problem onboard ship?
Select the ONE correct answer.

- **The viscosity of the fuel may fall below that recommended by manufacturers for successful injection.**
- Exhaust temperatures are higher when using distillate fuels.
- The fuel oil density is too low for effective separation.
- They are often contaminated with other products.

Why is the Kinematic viscosity of residual fuel oils important?
Select the ONE best answer.

- **We need sufficient heating capacity to be able to reduce the viscosity for pumping,**
- It directly affects the price of the fuel.
- It has the biggest influence on how we set up the separators and self-cleaning filters for fuel oil treatment.
- It is an indicator of fuel quality.

Why is the Micro Carbon Residue of a distillate fuel important?
Select the ONE correct answer.

- **High levels result in more carbonaceous deposits in the engine.**
- It is a measure of how abrasive the deposits left in the engine cylinder will be.
- It is a measure of how much energy is released when the fuel is completely burned.
- It is an indication of how likely it is that filters will be clogged, or need to be changed or back-flushed.

Why is the Sulphur content of marine fuel oils so important?
Select any answers that you think apply.

- **Sulphur oxides formed during combustion can lead to corrosion of engine parts.**
- **The amount of Sulphur in the fuel will determine the type and feed rate of main engine cylinder lubrication used.**
- **To protect the environment, the amount of Sulphur in fuel is limited by legislation, particularly in Emission Control Areas.**
- Sulphur does not burn.

Why is the pour point of a fuel important?
Select the ONE correct answer.

- **Below the pour point, the fuel will not flow. It will tend to solidify and be difficult to re-heat.**
- Above the pour point, the fuel will start to evaporate and pumps will gas up.
- Above the pour point, the fuel will start to stick to the surfaces of heat exchangers.
- If heated above the pour point, the fuel will start to separate out into its various fractions.

Why is water contamination of distillate fuels a particular problem?
Select the ONE best answer.

- **It can lead to microbial growth.**
- It changes the appearance of the fuel.
- Water cannot be easily removed in the separators.
- Water will not separate from the fuel in the settling tank.

0420 – Lockout Tagout

Version 2.0 (373)

Equipment has been isolated for work to be carried out. Why should the lock and tag at a point of isolation normally only be removed by the person who fitted it? Select the ONE best answer.

- **There is a risk that anyone else may not know enough about the isolation and the work done.**
- Anyone can remove the lock and tag; this is routinely done on many vessels.
- No one else will know the location of the lock and tag.
- Nobody else will have the key to the padlock used.

How can safe isolation be maintained when more than one person is working on the same system, for instance in dry-dock? Select the ONE best answer.

- **Each person should apply their own locks at the points of isolation.**
- One person should agree to take charge of the situation.
- There should be a verbal agreement to check with each other before restoring the equipment to service.
- They should leave the locking arrangements to the Master or Chief Engineer.

If the equipment you are working on can move downwards under gravity, what should you do to keep safe? Select the ONE best answer.

- **Make sure that it is securely locked into position or properly supported.**
- Always work from above it.
- Make sure that no-one can walk underneath it.
- Only do the job when in port, so that the ship is not rolling.

If the person that applied the lock and tag out to a system is not available, what should be done when work is completed? Select the ONE best answer.

- **Follow the on board procedure for such an event.**
- Anyone who has been involved in the work can remove them.
- Leave them in place till the original person is available.
- The Chief Engineer or Chief Mate will remove them.

If there is a powerful spring inside the equipment that you are working on, what are the two alternative ways to stay safe? Select the TWO correct answers.

- **Fit a locking or latching device to prevent the spring from being released.**
- **Make sure the spring tension is released before working on the equipment.**
- Just take care to keep your fingers out of the way from moving parts.
- Poke around with a screwdriver, and if nothing moves it is safe to work on the equipment.

Should equipment always be locked out and tagged out if it is being worked on? Select the ONE correct answer.

- **Yes, if it can be supplied with energy or fluids that could cause injury, fire, flood, explosion, or any similar dangers.**
- No, not if everyone has been told about the work and knows the situation.
- No, not unless it is electrical equipment, especially high voltage equipment.
- Yes, unless the work is of a very short duration.

Which ONE of these statements best describes the procedures known as 'Lockout Tagout'? Select the ONE best answer.

- **A set of precautions, including applying a lock and a tag, to make sure that equipment stays safely isolated when it is being worked on.**
- An isolation procedure only used when working on electrical equipment.
- The practice of issuing tags to authorized visitors, to control security at the gangway.
- The practice of locking off overboard discharge valves to prevent marine pollution.

Who should normally remove the lock and tag after completing work on equipment? Select the ONE correct answer.

- **The person who fitted the lock and tag.**
- Any suitably qualified person.
- The Master or Chief Engineer.
- The Master or Chief Officer.

You are going to work on equipment that has been isolated, but which was previously pressurised or full of fluid. Which TWO of these precautions should be followed? Select the TWO best answers.

- **Loosen the bolts at off flanges but leaving the nuts on. Then just break the joint at the flange.**
- **Use any drain and vent valves fitted, checking that they are clear.**
- Fully open a flange at the top of the equipment first.
- Give the isolating valves extra force with a wheel key.

You have been working on electrical, pneumatic or hydraulic equipment. What should be done about the controls for the equipment before re-energising it? Select the ONE correct answer.

- **The controls should be put into a safe position, usually the 'off' position.**
- Nothing needs to be checked, the equipment is not likely to start.
- The controls should be checked to make sure they are in the same position they were when the work started.
- The controls should be placed to the 'on' position.

0425 – Gas measurement, A safe atmosphere

Version 2.0 (1654)

If the oxygen content of an enclosed space is less than 20%, under what conditions should entry be made by personnel? Select the ONE correct answer.

- **In an emergency, only using suitable breathing equipment.**
- For a short period only using breathing equipment.
- Only in pairs using breathing equipment.
- Using breathing equipment.

Stagnant water, some sewage systems and some oil cargoes can all be the source for a gas which is both flammable and toxic. What is this gas? Select the ONE correct answer.

- **Hydrogen sulphide.**
- Carbon dioxide.
- Hydrogen chloride.
- Nitrogen.

The combustion control of an independent inert gas generator, supplying inert gas to a cargo tank system, is incorrectly set, giving incomplete combustion. What is the likely effect of this? Select the ONE correct answer.

- **Increased level of carbon monoxide in the inert gas.**
- Increased level of hydrogen sulphide gas in the inert gas.
- Increased level of NOx and SOx in the inert gas.
- Increased level of oxygen in the inert gas.

The oxygen content in an enclosed space falls below 21% during entry by personnel. What action should be taken? Select the ONE correct answer.

- **Stop working and leave the space immediately.**
- Complete the work as quickly as possible.
- Increase the ventilation and continue working.
- Stop the work and put on breathing apparatus before carrying on.

What are the TWO main reasons for measuring the oxygen content of an enclosed space onboard a ship? Select the TWO correct answers.

- **To check if the oxygen content is low enough to avoid flammable mixtures.**
- **To ensure the oxygen content is sufficient to allow safe entry.**
- To check the amount of hydrocarbon gas present.
- To check the amount of toxic gases present.

What is the effect on the oxygen content of the atmosphere of an enclosed space if hydrocarbon gases are present? Select the ONE correct answer.

- **Oxygen content is reduced.**
- No effect.
- Oxygen content is increased.
- The effect cannot be determined.

What is the most common source of hydrogen sulphide gas in enclosed spaces onboard a ship? Select the ONE correct answer.

- **Sour crude or petroleum products from sour crude in tanks.**
- Combustion of distillate fuel oil in the inert gas generator.
- Incomplete combustion of fuel oil in a boiler furnace.
- Low sulphur marine fuel oil being used in a diesel engine.

What is the most likely reason for the presence of high levels of oxides of Nitrogen and Sulphur (NOx and SOx) in the atmosphere of a cargo tank on an oil tanker? Select the ONE correct answer.

- **Inert gas from the combustion of fuel oil has been used to inert the tank.**
- A cargo of heavy fuel oil with a high sulphur content has been carried in the tank.
- Sour crude oil has been stored in the tank.
- The tank has been fully gas freed following discharge of cargo.

What is the normally accepted value for the lower explosive limit, LEL, for hydrocarbon gas in a cargo tank? Select the ONE best answer.

- **1% by volume.**
- 10% by volume.
- 21% by volume.
- 8% by volume.

What is the normally accepted value for the maximum oxygen content in an inerted cargo tank? Select the ONE correct answer.

- **8% by volume.**
- 1% by volume.
- 21% by volume.
- 5% by volume.

What is the normally accepted value for the upper explosive limit, UEL, for hydrocarbon gas in a cargo tank? Select the ONE best answer.

- **10% by volume.**
- 1% by volume.
- 21% by volume.
- 8% by volume.

What is the purpose of gas freeing a cargo tank? Select the ONE correct answer

- **To replace the contents with fresh air.**
- To remove the hydrocarbon gases only.
- To remove the inert gas only.
- To replace the contents with inert gas.

What is the time period that is normally used when stating threshold limit values for the Time Weighted Average for exposure, TLV – TWA, for toxic gases? Select the ONE correct answer.

- **8 hours.**
- 1 hour.
- 15 minutes.
- 24 hours.

What is the time period used when stating threshold limit values for the Short Term Exposure Limit, TLV – STEL, for toxic gases? Select the ONE correct answer.

- **15 minutes.**
- 1 hour.
- 5 minutes.
- 8 hours.

When rusting of ship's steelwork occurs, which ONE of the following is the main effect on the air inside a ballast tank or cargo tank?

- **The oxygen content of the air decreases.**
- Carbon dioxide is given off by the rusting process.
- The air develops a bad smell.
- The oxygen content of the air increases.

Which ONE of the following options best completes the following statement?
"Onboard ship, an enclosed space is any space which . . .

- **. . . has limited ventilation or could contain toxic or flammable vapours"**
- . . . has a danger sign at the entrance".
- . . . has four walls and a ceiling".
- . . . is not open to the atmosphere".

Which ONE of the following statements best describes the term 'Occupational Exposure Limit' (OEL)?

- **It is a guide to safe exposure to toxic products, based on the amount of a given substance and the time in contact it.**
- It is a guide to safe exposure to harmful ultra violet (UV) rays when working on deck.
- It is a guide to safe exposure to toxic products, based only on the amount of a given substance present.
- It is a guide to safe exposure to toxic products, based only on the time in contact with the substance.

Which ONE of these statements best describes the effects some cargoes can have on the atmosphere in a cargo space?

- **They can use up or displace oxygen and give off toxic or flammable gasses.**
- No effect, because any harmful cargoes are sealed.
- The atmosphere can develop a bad smell.
- Very little effect, since the spaces are ventilated.

Which of the following toxicity threshold limit exposure terms would normally have the lowest numerical value? Select the ONE correct answer.

- **TLV - TWA.**
- TLV - C.
- TLV - STEL.
- TLV.

Which term is used for the property of a gas that has a long term harmful effect on a living organism as a result of repeated or long term exposure to even low concentrations of the gas? Select the ONE correct answer.

- **Chronic toxicity.**
- Acute toxicity.
- Long toxicity.
- Low toxicity.

0426 – Gas measurement, Measuring instruments

Version 2.0 (894)

A sampling pump for an oxygen analyser has a capacity of 100ml per stroke. It is connected to a 10 metre sampling line with a capacity of 25 ml per metre. What is the minimum number of strokes of the pump needed to obtain a representative sample? Select the ONE correct answer.

- **4**
- 10
- 2
- 25

For common hazardous gases like hydrogen sulphide, what is the main advantage of using a personal gas monitor rather than a long term gas detector tube when working in a potentially hazardous space? Select the ONE correct answer.

- **The personal monitor will give an alarm if the exposure limits are exceeded.**
- The personal monitor can be used to check total exposure throughout the working period.
- The personal monitor can be used to obtain a time weighted average.
- The personal monitor can check for any hazardous gas.

How is a Time Weighted Average reading obtained from a diffusion type detector tube that has been used over a working period of eight hours? Select the ONE correct answer.

- **Divide the total reading on the tube by 8.**
- Half the reading on the tube to get a mid point reading.
- Multiply the total reading on the tube by 8.
- Take a direct total reading from the tube.

How is the correct volume of the sample controlled when using short term gas detector tubes with a manually operated hand pump? Select the ONE best answer.

- **By checking how many complete pump strokes are required for that particular tube.**
- By always operating the pump for the same period of time for each tube.
- By always using a fixed number of complete strokes of the pump for each tube.
- By always using a single complete stroke of the pump for each tube.

It is generally recommended that non metallic sampling lines and sensors are used when obtaining gas samples from cargo tanks. What additional precautions are required when taking samples during inerting of a cargo tank? Select the ONE best answer.

- **The inerting operation should be stopped for at least 30 minutes before introducing sampling equipment into the tank.**
- It is safe to take samples during inerting.
- Samples should never be taken from a partly inerted tank.
- The inerting operation should be stopped while samples are obtained.

Many of the oxygen analysers used onboard ship use paramagnetic type sensors. What is meant by the term 'Paramagnetic'? Select the ONE correct answer.

- **That a material is temporarily magnetized when exposed to a magnetic field.**
- That a material is non magnetic under any circumstances.
- That a material is permanently magnetized when exposed to a magnetic field.
- That a material is repelled by a magnetic field.

Metal oxide semiconductor type sensors are increasingly being used in personal gas monitors. What is the operating principle of this type of sensor? Select the ONE correct answer.

- **Change of electrical resistance as oxide adsorbs the target gas.**
- Absorption of specific wavelengths of light.
- Combustion of target gas on heated filament.
- Conduction of heat from oxide surface by target gas.

What TWO types of sensor are normally used in portable oxygen analysers onboard ship? Select the TWO correct answers.

- **Galvanic cell**
- **Paramagnetic**
- Catalytic filaments
- Infra red absorption

What are the TWO main reasons for measuring the oxygen content of an enclosed space onboard a ship? Select the TWO correct answers.

- **To check if the oxygen content is low enough to avoid flammable mixtures.**
- **To ensure the oxygen content is sufficient to allow safe entry.**
- To check the amount of hydrocarbon gas present.
- To check the amount of toxic gases present.

What are the minimum checks that should be carried out before a gas detecting instrument is used to check the contents of an enclosed space? Select all correct options.

- **Check battery condition**
- **Check calibration**
- **Check condition of sample pump and lines and any filters**
- **Check for general condition**

What facility is normally provided for making a zero adjustment to a paramagnetic oxygen analyser? Select the ONE correct answer.

- **Alter the relative position of the optical system.**
- Adjust the electrical circuit.
- Change the rating of the light source lamp.
- Reposition the meter needle to zero.

What is meant by the term 'bump test' in relation to gas detecting instruments? Select the ONE correct answer.

- **Checking the reading of the instrument against a known concentration of test gas.**
- Adjusting the span of the instrument.
- Adjusting the zero setting in fresh air.
- Knocking the instrument to see if the meter needle is sticking.

What is normally used to remove dirt and moisture from a gas sample prior to entry into a combustible gas indicator? Select the ONE correct answer.

- **A cotton filter.**
- A paper filter
- Cardboard
- Soda lime

What might indicate that a galvanic cell type oxygen measuring instrument requires a replacement sensor cell? Select the ONE correct answer.

- **Unstable readings during calibration**
- Meter indicates maximum reading during test
- Reading less than 8% oxygen when testing an inerted cargo tank atmosphere
- Reading of 20.9% when testing fresh air

What precaution should be taken before using an oxygen analyser that has automatic calibration? Select the ONE correct answer.

- **The analyser should be switched on in fresh air**
- No precautions are necessary
- The analyser should be switched on 30 minutes before use
- The analyser should be switched on with the sensor placed in the space to be tested

What reading will be obtained if an explosimeter, with a range of 0 – 100% LEL, is used to check the hydrocarbon gas content in an atmosphere containing a concentration of hydrocarbon gas which is three times the LEL. Select the ONE best answer.

- **The meter needle will move to full scale deflection and then fall close to zero.**
- The meter needle will move to full scale deflection and remain there.
- The meter reading will oscillate between full scale deflection and zero.
- The meter reading will stay at zero.

What should be used as the reference gas when carrying out the zero setting of a combustible or hydrocarbon gas detecting instrument? Select the ONE correct answer.

- **Fresh air**
- 2% butane in nitrogen
- Inert gas
- Pure oxygen

What simple test can be carried out to check for leaks in a bulb type aspirator pump for a gas detecting instrument? Select the ONE correct answer.

- **Nip the inlet line after squeezing the bulb to see if it stays deflated**
- Nip the outlet line after squeezing the bulb to see if it stays deflated
- Pressure test the pump with gas from a span test canister
- Submerge the pump in liquid and check for bubbles as it is operated

What test gas should be used to check the span during calibration of a combustible gas or hydrocarbon gas detector? Select the ONE best answer.

- **A known concentration of butane in nitrogen.**
- A known concentration of butane in air.
- A known concentration of butane in inert gas.
- Fresh air.

What would be the effect, if any, on the meter reading of a catalytic filament type combustible gas indicator if the oxygen content of the sample was less than 11%? Select the one best answer.

- **The reading would be unreliable.**
- The reading would be zero.
- The reading would show full scale deflection.
- There would be no effect and an accurate reading would be obtained.

When carrying out a span check of a combustible or hydrocarbon gas detector the sample gas should normally be taken from a gas bag rather than direct from the gas canister? Why should this procedure be followed? Select the ONE correct answer.

- **To avoid excess pressure or flow in the instrument.**
- Connection of the gas bag to the instrument is easier.
- To control the quantity of test gas used.
- To reduce the temperature of the test gas by expanding it into the bag.

When carrying out checks of the atmosphere in a cargo tank following gas freeing, at how many levels in the tank should samples be taken to ensure there are no variations in the tank atmosphere? Select the ONE correct answer.

- **At least three levels**
- At least two levels
- One level is sufficient
- Samples should be taken every metre depth in the tank.

Which TWO of the following gases are likely to result in poisoning of the sensor element in a catalytic filament type combustible gas detector? Select the TWO correct answers.

- **Carbon monoxide**
- **Hydrogen sulphide**
- Carbon dioxide
- Oxygen

Which TWO types of combustible/hydrocarbon gas detectors rely on resistance change with temperature for their operation? Select the TWO correct answers.

- **Catalytic filament type**
- **Non-catalytic heated filament type**
- Infra red absorption type
- Refractive index type

Which gas would be most likely to affect the reading from a refractive index type hydrocarbon gas measuring instrument? Select the ONE correct answer.

- **Carbon dioxide**
- Hydrogen sulphide
- Nitrogen
- Oxygen

Which material is recommended for the removal of carbon dioxide from a gas sample when measuring the hydrocarbon gas content of an inerted cargo tank atmosphere with a refractive index or infrared absorption type detector? Select the ONE correct answer.

- **Sodalime**
- Cotton wool
- Gauze
- Silica gel

Which type of hydrocarbon gas detecting instrument uses thermal conductivity of the gas as the means of measuring gas content in a sample? Select the ONE correct answer.

- **Non-catalytic heated filament type**
- Catalytic filament type
- Infra red absorption type
- Refractive index type

You are going to use a galvanic cell type oxygen analyser for checking LOW concentrations of oxygen. Which ONE of the following should be used for the span test?

- **A test gas with 2% oxygen and 98% nitrogen**
- A test gas with 2% butane and 98% nitrogen
- Fresh air
- Inert gas with 8% or less oxygen content

0481 – Separators, S-type separator

Version 1.0 (3574)

According to this module, after the separator is switched off, how long does it take the separator to come to a complete standstill? Select the ONE correct answer.

- **About 20 minutes**
- About 40 minutes
- About 5 minutes
- About one minute

During bowl discharge, how is movement of the sliding bowl bottom achieved? Select the ONE correct answer.

- **By pressure of operating water, directed by opening/closing mechanism**
- By pressure of compressed air, delivered to the separation system
- By pressure of oil from separator lubricating system
- Electromagnetic mechanism

During bowl discharge, what happens to the flow of uncleaned oil to the separator? Select the ONE correct answer.

- **The oil is diverted back to the return line**
- The capacity of the oil feed pump is decreased
- The oil feed pump is stopped
- The oil flow is maintained, but the heater is switched off

During normal operation, when does sludge discharge from the separator take place? Select the ONE correct answer.

- **At pre-set time intervals**
- Whenever there is a high solids content in the oil outlet
- Whenever there is a high water content at the oil inlet
- Whenever there is a high water content in the oil outlet

How fast does the bowl rotate during the separation process? Select the ONE correct answer.

- **9000 rpm**
- 2000 rpm
- 4000 rpm
- 6000 rpm

How should an emergency stop of the separator be carried out? Select the ONE correct answer.

- **By pressing the Emergency Stop button on the control cabinet**
- By pressing the DISCHARGE button and then the SEPARATION STOP button on control panel
- By pressing the SEPARATION button on the control panel
- By switching off the power supply

On the S-type separator display, what are the "+" or "-" buttons used for? Select the ONE correct answer.

- **To navigate through the menu and to change parameters**
- To acknowledge alarms
- To enter into the parameter list and accept a parameter value
- To initiate a bowl discharge

What happens to the oil separation process when the water drain valve is open? Select the ONE correct answer.

- **The oil separation process continues**
- The oil feed pump is stopped
- The oil separation process is interrupted
- The separator is stopped

What is the correct supply temperature of Heavy Fuel Oil for the S-type separator? Select the ONE correct answer.

- **95 to 98°C**
- 85 to 90°C
- 90 to 95°C
- 99 to 105°C

What is the correct way to stop the separator? Select the ONE correct answer.

- **Automatically, by pressing the SEPARATION STOP button**
- Automatically, if mode selection switch is put to the AUTO position
- Manually, by command from the power supply panel
- Manually, by pressing DISCHARGE button on control panel

What is the full name of the device used to securely close the bowl of an S-type separator? Select the ONE correct answer.

- **Centrilock ring**
- Centrishoot
- Lock ring
- Snap ring

What is the name of the device which pumps out the water accumulated in the bowl during the separation process? Select the ONE correct answer.

- **Paring tube**
- Centrifugal pump
- Paring disc
- Peristaltic pump

What is the 'Paring disc' used for in an S-type separator? Select the ONE correct answer.

- **Pumping cleaned oil out of the bowl**
- Pumping separated water out of the bowl
- Removing unprocessed oil from the bowl
- Supplying unprocessed oil to the separator bowl

What is 'Displacement water' used for? Select the ONE correct answer.

- **To reduce oil loss during discharge by displacing oil from the bowl**
- To fill the bowl with water
- To push water towards the disc stack
- To soften the sludge accumulated at the periphery of bowl

What types of oil can the S-type separator be used for? Select the ONE best answer.

- **All types of fuel and lubricating oils**
- Diesel Oil (DO)
- Heavy Fuel Oil (HFO)
- Lubricating Oil

When does the drain valve operate to remove separated water? Select the ONE best answer.

- **When the signal from water transducer exceeds a pre-set trigger value**
- When the signal from water transducer is high
- When the signal from water transducer is low
- When the signal from water transducer is removed

Where does dirty oil enters the bowl from? Select the ONE correct answer.

- **The top, through the inlet pipe**
- From the displacement water pipe
- From the side of disc stack
- From the sludge space

Where does sludge accumulate during the separation process? Select the ONE correct answer.

- **At the periphery of the bowl**
- At the centre of the bowl
- In the side of disc stack
- Under the top disc

Where does the separation process of solid particles and water mainly take place? Select the ONE best answer.

- **The spaces between the stack discs**
- The bowl sludge chamber
- The space between the top disc and the bowl cover
- The space near the centre of bowl

Which ONE of the following is used to drive the bowl of an S-type separator? Select the ONE correct answer.

- **A flat belt drive**
- A chain drive
- A gear drive
- A magnetic drive

Why is a clutch needed in the drive between the electric motor and the separator bowl? Select the ONE best answer.

- **To allow the motor to accelerate the bowl more gradually, and to allow free rotation of the bowl after the electric motor has been stopped**
- To allow for any misalignment in the drive
- To allow the separator speed to be varied
- To help slow the bowl down after it has been stopped

0483 – Introduction to two-stroke diesel engines

Version 1.0 (1665)

According to this module, what is the correct temperature for the air when it enters the cylinder? Select the ONE correct answer.

- **Between 40°C and 60°C**
- Between 10°C and 20°C
- Between 60°C and 80°C
- Between 80°C and 90°C

According to this module, what is the typical efficiency of a modern two stroke marine diesel engine? Select the ONE correct answer.

- **About 50%**
- About 40%
- About 80%
- About 90%

For the type of engine described in this module, what is a typical fuel injector opening pressure? Select the ONE correct answer.

- **Between 200 and 300 bar**
- Between 50 and 100 bar
- Between 100 and 180 bar
- Between 300 and 400 bar

For the type of engine described in this module, what is the maximum fuel pump pressure during fuel injection? Select the ONE correct answer.

- **Between 500 and 1500 bar**
- Between 100 and 200 bar
- Between 5 and 15 bar
- Between 50 and 100 bar

For the type of engine described in this module, what is the maximum pressure during the power stroke? Select the ONE correct answer.

- **100 to 140 bar**
- 180 to 220 bar
- 6 to 10 bar
- 60 to 100 bar

For the type of engine described in this module, what is the maximum temperature in the combustion space? Select the ONE correct answer.

- **Between 1500°C and 2000°C**
- Between 1000°C and 1200°C
- Between 1200°C and 1400°C
- Between 500°C and 900°C

For the type of engine described in this module, what is the pressure of the air at the end of the compression stroke? Select the ONE correct answer.

- **70 to 100 bar**
- 10 to 30 bar
- 3 to 5 bar
- 30 to 50 bar

For the type of engine described in this module, what is the temperature of the air at the end of the compression stroke? Select the ONE correct answer.

- **Between 400°C and 600°C**
- Between 100°C and 200°C
- Between 200°C and 300°C
- Between 600°C and 800°C

For the type of engine described in this module, which two engine parts are connected by the connecting rod? Select the ONE correct answer.

- **The crankshaft and the crosshead**
- The crankshaft and the piston
- The entablature and the A-frames
- The piston and the crosshead

How is the quantity of fuel injected controlled? Select the ONE correct answer.

- **By turning the plunger**
- By changing the length of the push rod
- By turning the barrel
- By turning the cam

In practice, how many kilograms of air are needed in order to burn 1 kg of a typical liquid marine fuel? Select the ONE correct answer.

- **30 to 45 kg of air for 1kg of fuel**
- 1 kg of air for 1kg of fuel
- 10 to 20 kg of air for 1kg of fuel
- 5 to 10 kg of air for 1kg of fuel

THEORETICALLY, how many kilograms of air are needed in order to burn 1 kg of a typical liquid marine fuel? Select the ONE correct answer.

- **15 kg of air for 1 kg of fuel**
- 1 kg of air for 1 kg of fuel
- 10 kg of air for 1 kg of fuel
- 20 kg of air for 1 kg of fuel

To provide a good scavenging process, what should the scavenge air pressure be? Select the ONE correct answer.

- **Higher than the pressure in the cylinder when the inlet ports are uncovered**
- Equal to the pressure in the cylinder
- Higher than the compression pressure
- Lower than the pressure in the cylinder when the inlet ports are uncovered

What are the start and end points for the intake and compression stroke? Select the ONE correct answer.

- **From Bottom Dead Centre to the next Top Dead Centre**
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Top Dead Centre
- From Top Dead Centre to the next Bottom Dead Centre

What are the start and end points for the power and exhaust stroke? Select the ONE correct answer.

- **From Top Dead Centre to the next Bottom Dead Centre**
- From Bottom Dead Centre to the next Top Dead Centre
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Top Dead Centre

What is the first place that the air from the air cooler goes to? Select the ONE correct answer.

- **The scavenge air receiver**
- The crankcase
- The cylinder liner inlet ports
- The turbocharger air side

What is the fuel delivery pressure before the fuel pump plungers starts to move up? Select the ONE correct answer.

- **Between 2.5 and 10 bar**
- Between 1 and 2 bar
- Between 100 and 200 bar
- Between 20 and 100 bar

What is the main purpose of the stuffing box? Select the ONE correct answer.

- **To prevent leakage of scavenge air into the crankcase**
- To form a seal around the connecting rod
- To keep the piston rod central
- To prevent exhaust gas from leaking into the crankcase

What is the main reason for cooling the air after the turbocharger? Select the ONE correct answer.

- **To increase the density of the air**
- To decrease the fuel quantity
- To prevent early ignition of the fuel
- To recover waste heat

What is the name of the engine part which directly supports the cylinder blocks? Select the ONE correct answer.

- **The A-frames**
- The bedplate
- The cylinder heads
- The liners

What is the name of the engine part which the piston has direct contact with? Select the ONE correct answer.

- **The liner**
- The crankshaft
- The cylinder block
- The cylinder head

What is the name of the fuel injection pump part which transfers the cam movement to the pump plunger? Select the ONE correct answer.

- **The push rod**
- The barrel
- The fuel rack
- The spring

What is the name of the fuel injection pump part which transfers the cam movement to the pump push rod? Select the ONE correct answer.

- **The roller**
- The fuel rack
- The plunger
- The spring

What is the purpose of the auxiliary blower? Select the ONE correct answer

- **To provide scavenge air to the engine at start up and low loads**
- To maintain a positive pressure in the crankcase
- To provide air to auxiliary equipment
- To provide more air at high engine powers

What is the typical maximum speed of a two stroke diesel engine? Select the ONE correct answer.

- **80 to 200 rpm**
- 200 to 400 rpm
- 400 to 1000 rpm
- 60 to 80 rpm

What is the typical pressure of the scavenge air entering the cylinder? Select the ONE correct answer.

- **Between 2 and 5 bar**
- Between 0.2 and 0.5 bar
- Between 10 and 15 bar
- Between 5 and 10 bar

What is the typical residual pressure in the fuel injection pipe after injection has finished? Select the ONE correct answer.

- **Between 50 and 70 bar**
- Between 10 and 20 bar
- Between 100 and 200 bar
- Zero

What is the typical temperature of the exhaust gases at the end of the power stroke? Select the ONE correct answer.

- **Between 300 °C and 500 °C**
- Between 100 °C and 200 °C
- Between 200 °C and 300 °C
- Between 500 °C and 700 °C

When does the exhaust process start? Select the ONE correct answer.

- **Shortly before Bottom Dead Centre**
- Shortly after Bottom Dead Centre
- Shortly after Top Dead Centre
- Shortly before Top Dead Centre

When does the fuel injection process start? Select the ONE correct answer.

- **Just before Top Dead Centre**
- At mid-stroke
- Just after Top Dead Centre
- Just before Bottom Dead Centre

0484 – Diesel engine diagnostics

Version 1.0 (2282)

A decreased pH of the main lubricating oil has been observed. What are the TWO most likely causes? Select the TWO correct answers.

- **Faulty stuffing box.**
- **High sulphur content in the fuel.**
- High TBN of the cylinder oil.
- Low sulphur content in the fuel.

A low dispersancy of the main lubricating oil has been observed during a spot test. What is the most likely cause? Select the ONE correct answer.

- **Ageing of the oil.**
- Ineffective oil filtering.
- Low oil level in the sump.
- Low viscosity of the oil.

An increased cylinder box temperature near one scavenge box has been observed. What are the most likely causes? Select the TWO correct answers.

- **Too much cylinder oil.**
- **Worn piston rings**
- Dirty air cooler
- Not enough cylinder oil.

An increased temperature of ONE main bearing has been observed. What are the TWO most likely causes? Select the TWO correct answers.

- **Cylinder overload.**
- **Seizure in the main bearing.**
- Engine overload.
- High temperature of the lubricating oil.

High contamination of the main lubricating oil has been observed during the spot test. What is the most likely cause? Select the ONE correct answer.

- **Ineffective oil filtering.**
- High sulphur content in the fuel.
- Ineffective oil cooling.
- Low viscosity of the oil.

The viscosity of the lubricating oil has been found to be low. Which TWO of the following could have combined to cause this? Select the TWO correct answers.

- **Worn nozzles of the fuel injectors.**
- **Worn piston rings.**
- High sulphur content in the fuel.
- Low TBN of the cylinder oil.

The Mean Indicated Pressure, M.I.P. of ONE cylinder has an increasingly negative deviation as shown. What are the TWO most likely causes? Select the TWO correct answers.

- **Late combustion caused by the wrong injection timing.**
- **Low compression pressure.**
- Dirty air filter.
- Early combustion caused by the wrong injection timing.

The air cooler temperature difference has an increasingly negative deviation as shown. What is the most likely cause? Select the ONE correct answer.

- **Dirty water side of the air cooler.**
- Dirty fresh water side of the central cooler.
- Low pressure of the fresh cooling water.
- Low temperature of the fresh cooling water.

The air pressure drop at the air cooler has an increasingly positive deviation as shown. What is the most likely cause? Select the ONE correct answer.

- **Dirty air cooler.**
- Dirty air filter.
- Dirty turbocharger.
- Faulty stuffing box.

The air pressure drop at the air filter has an increasingly positive deviation as shown. What is the most likely cause? Select the ONE correct answer.

- **Dirty air filter.**
- Dirty air cooler.
- Dirty turbocharger.
- Low ambient air pressure.

The combustion pressure in ONE cylinder changes as shown on the graph. What is the most likely cause? Select the ONE correct answer.

- **A leaking exhaust valve.**
- Early combustion caused by the wrong injection timing.
- Late combustion caused by the wrong injection timing.
- Low scavenge air pressure.

The combustion pressure in ONE cylinder changes as shown on the graph. What is the most likely cause? Select the ONE correct answer.

- **A decreased quantity of the fuel injected.**
- A leaking exhaust valve.
- Late combustion caused by the wrong injection timing.
- Low scavenge air pressure.

The cooling water temperature difference at ONE cylinder has an increasingly positive deviation as shown. What are the TWO most likely causes? Select the TWO correct answers.

- **Crack in the cylinder head.**
- **Crack in the cylinder liner.**
- Crack in the turbocharger.
- High temperature of the cooling water.

The exhaust gas temperature deviation of a single cylinder shows the following trend. What are the possible causes? Select any answers that you think apply.

- **A leaking exhaust valve.**
- Early combustion caused by the wrong injection timing.
- Low inlet pressure of the oil cooling the pistons.
- Low scavenge air pressure.

The maximum combustion pressure of ONE cylinder has an increasingly negative deviation as shown. What is the most likely cause? Select the ONE correct answer.

- **Late combustion caused by the wrong injection timing.**
- **Low compression pressure.**
- Dirty air filter.
- Early combustion caused by the wrong injection timing.

The scavenge air pressure has an increasingly negative deviation as shown. What are the TWO most likely causes? Select the TWO correct answers.

- **Dirty air filter.**
- **Dirty exhaust duct after turbocharger.**
- Faulty stuffing box.
- Worn piston rings.

The turbocharger speed has an increasingly negative deviation as shown. What are the TWO most likely causes? Select the TWO correct answers.

- **Corroded turbocharger nozzle ring.**
- **Too large a clearance between the turbine blades and the shroud.**
- Dirty exhaust duct after turbocharger.
- Dirty turbocharger.

The turbocharger speed has an increasingly positive deviation as shown. What are the TWO most likely causes? Select the TWO correct answers.

- **A dirty air fiter or air cooler**
- **Dirty turbocharger.**
- Corroded turbocharger nozzle ring.
- Too large a clearance between the turbine blades and the shroud.

The water content in the lubricating oil has been found to be high. What are the TWO most likely causes? Select the TWO correct answers.

- **Crack in the cylinder liner.**
- **Faulty operation of the lubricating oil separator.**
- High water level in the compensation tank.
- Low level of the lubricating oil in the sump.

There is dark smoke coming from the funnel. What are the possible causes? Select any answers that you think are correct.

- **Engine overload.**
- **Fault in the turbocharging system.**
- Fault in the fresh water cooling system.
- Low viscosity of the lubricating oil.

What are the possible causes of an increasing positive deviation of the average exhaust temperature? Select any answers that you think apply.

- **Low inlet pressure of the cylinder cooling water.**
- **Low inlet pressure of the oil cooling the pistons.**
- **Not enough combustion air.**
- Low inlet temperature of the oil cooling the pistons.

Which THREE of the following are typical symptoms of problems with combustion in the engine? Select the THREE best answers.

- **Increasing deviation of the exhaust gas temperature after a single cylinder.**
- **Increasing negative deviation of the maximum combustion pressure in a single cylinder.**
- **Increasing positive deviation of the average exhaust gas temperature**
- Increasing negative deviation of the scavenge air pressure.

Which TWO of these parameters are most commonly used for the evaluation of diesel engine combustion? Select the TWO best answers.

- **Exhaust temperature after each cylinder.**
- **Maximum pressure in each cylinder.**
- Scavenge air pressure
- Turbocharger rpm.

Which TWO of these parameters are most commonly used for the evaluation of engine lubrication? Select the TWO best answers.

- **Main bearing temperatures.**
- **Water content in the lubrication oil.**
- Cylinder head temperatures.
- Mean Indicated Pressure in each cylinder.

Which TWO of these parameters are most commonly used for the evaluation of engine performance? Select the TWO best answers.

- **Engine rpm.**
- **Load index.**
- Ship speed.
- Turbocharger rpm.

Which TWO of these parameters are most commonly used for the evaluation of turbocharging? Select the TWO best answers.

- **Scavenge air pressure**
- **Turbocharger rpm.**
- Exhaust temperature after each cylinder.
- Maximum pressure in each cylinder.

Which TWO of these statements about diesel engine diagnostic parameters are true? Select the TWO correct answers.

- **They help to avoid the most serious engine faults.**
- **They help to optimize the maintenance process.**
- Their value does not depend on the engine technical state.
- They can only be measured with a special electronic equipment.

Which engine parameter should be used as an independent variable when creating a model diagram for turbocharger speed? Select the ONE best answer.

- **Scavenge air pressure.**
- Engine power.
- Engine rpm.
- Exhaust gas temperature before the turbocharger.

Which of these factors might affect the measured parameters of a diesel engine? Select any answers that you think are correct.

- **Fuel quality.**
- **Technical state of the engine.**
- **The draught of the ship.**
- **Weather conditions.**

Which parameter should be used as the independent variable when creating the model diagram for cooling water temperature difference at cylinder? Select the ONE best answer.

- **Engine power.**
- Cooling water inlet temperature.
- Engine rpm
- Exhaust gas temperature after cylinder.

Why is the first step in engine diagnosis usually to use a model diagram, followed by plotting the deviation on a time-based deviation chart? Select the TWO best answers.

- **Because measurements need to be compared to their values at a similar load by using a model diagram.**
- **Because there may be some short term effects like weather, or tide, so we need to look at the trend of deviations over a long enough time period.**
- Because individual measurements are inaccurate.
- Because over time, the performance cannot be expected to be anything like it was when the ship was new.

0485 – Introduction to four-stroke diesel engines

Version 1.0 (1365)

According to this module, what is correct the temperature for the air when it enters the cylinder? Select the ONE correct answer.

- **Between 40°C and 60°C**
- Between 10°C and 20°C
- Between 60°C and 80°C
- Between 80°C and 90°C

According to this module, what is the typical efficiency of a modern four stroke marine diesel engine? Select the ONE correct answer.

- **About 50%**
- About 40%
- About 80%
- About 90%

For the type of engine described in this module, what is a typical fuel injector opening pressure? Select the ONE correct answer.

- **Between 200 and 300 bar**
- Between 50 and 100 bar
- Between 100 and 180 bar
- Between 300 and 400 bar

For the type of engine described in this module, what is the maximum fuel pump pressure during fuel injection? Select the ONE correct answer.

- **Between 500 and 1500 bar**
- Between 100 and 200 bar
- Between 5 and 15 bar
- Between 50 and 100 bar

For the type of engine described in this module, what is the maximum pressure during the power stroke? Select the ONE correct answer.

- **100 to 150 bar**
- 180 to 220 bar
- 6 to 10 bar
- 60 to 100 bar

For the type of engine described in this module, what is the maximum temperature in the combustion space? Select the ONE correct answer.

- **Between 1500°C and 2000°C**
- Between 1000°C and 1200°C
- Between 1200°C and 1400°C
- Between 500°C and 900°C

For the type of engine described in this module, what is the pressure of the air at the end of the compression stroke? Select the ONE correct answer.

- **70 to 100 bar**
- 10 to 30 bar
- 150 to 250 bar
- 30 to 50 bar

For the type of engine described in this module, what is the temperature of the air at the end of the compression stroke? Select the ONE correct answer.

- **Between 400°C and 600°C**
- Between 100°C and 200°C
- Between 200°C and 300°C
- Between 600°C and 800°C

For the type of engine described in this module, which two engine parts are connected by the connecting rod? Select the ONE correct answer.

- **The crankshaft and the piston**
- The crankshaft and the camshaft
- The cylinder head and the cylinder block
- The cylinder head and the cylinder cover

How is the quantity of fuel injected controlled? Select the ONE correct answer.

- **By turning the plunger**
- By changing the length of the push rod
- By turning the barrel
- By turning the cam

In practice, how many kilograms of air are needed in order to burn 1 kg of a typical liquid marine fuel? Select the ONE correct answer.

- **30 to 45 kg of air for 1kg of fuel**
- 1 kg of air for 1kg of fuel
- 10 to 20 kg of air for 1kg of fuel
- 5 to 10 kg of air for 1kg of fuel

THEORETICALLY, how many kilograms of the air are needed in order to burn 1 kg of a typical liquid marine fuel? Select the ONE correct answer.

- **15 kg of air for 1 kg of fuel**
- 1 kg of air for 1 kg of fuel
- 10 kg of air for 1 kg of fuel
- 20 kg of air for 1 kg of fuel

To provide a good scavenging process, what should the scavenge air pressure be? Select the ONE correct answer.

- **Higher than the pressure in the cylinder when the inlet valves open**
- Equal to the pressure in the cylinder
- Higher than the compression pressure
- Lower than the pressure in the cylinder when the inlet valves open

What are the start and end points for the compression stroke? Select the ONE correct answer.

- **From one Bottom Dead Centre to the next Top Dead Centre**
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Top Dead Centre

What are the start and end points for the exhaust stroke? Select the ONE correct answer.

- **From one Bottom Dead Centre to the next Top Dead Centre**
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Bottom Dead Centre
- From one Top Dead Centre to the next Top Dead Centre

What are the start and end points for the intake stroke? Select the ONE correct answer.

- **From one Top Dead Centre to the next Bottom Dead Centre**
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Bottom Dead Centre to the next Top Dead Centre
- From one Top Dead Centre to the next Top Dead Centre

What are the start and end points for the power stroke? Select the ONE correct answer.

- **From one Top Dead Centre to the next Bottom Dead Centre**
- From one Bottom Dead Centre to the next Bottom Dead Centre
- From one Bottom Dead Centre to the next Top Dead Centre
- From one Top Dead Centre to the next Top Dead Centre

What is the first place that the air from the air cooler goes to? Select the ONE correct answer.

- **The charge air receiver.**
- The crankcase
- The inlet valves
- The turbocharger air side

What is the fuel delivery pressure before the fuel pump plungers starts to move up? Select the ONE correct answer.

- **Between 2.5 and 10 bar**
- Between 1 and 2 bar
- Between 100 and 200 bar
- Between 20 and 100 bar

What is the main reason for cooling the air after the turbocharger? Select the ONE correct answer.

- **To increase the density of the air**
- To decrease the fuel quantity
- To prevent early ignition of the fuel
- To recover waste heat

What is the name of the engine part which the piston has direct contact with? Select the ONE correct answer.

- **The liner**
- The crankshaft
- The cylinder block
- The cylinder head

What is the name of the fuel injection pump part which transfers the cam movement to the pump plunger? Select the ONE correct answer.

- **The pushrod**
- The barrel
- The fuel rack
- The spring

What is the name of the fuel injection pump part which transfers the cam movement to the pump push rod? Select the ONE correct answer.

- **The roller**
- The fuel rack
- The plunger
- The spring

What is the typical maximum speed of a four stroke high speed diesel engine? Select the ONE correct answer.

- **2000 to 4000 rpm**
- 200 to 400 rpm
- 400 to 2000 rpm
- 80 to 200 rpm

What is the typical maximum speed of a four stroke medium speed diesel engine? Select the ONE correct answer.

- **400 to 4000 rpm**
- 200 to 400 rpm
- 60 to 80 rpm
- 80 to 200 rpm

What is the typical pressure of the scavenge air entering the cylinder? Select the ONE correct answer.

- **Between 2 and 5 bar**
- Between 0.2 and 0.5 bar
- Between 10 and 15 bar
- Between 5 and 10 bar

What is the typical residual pressure in the fuel injection pipe after injection has finished? Select the ONE correct answer.

- **Between 50 and 70 bar**
- Between 10 and 20 bar
- Between 100 and 200 bar
- Zero

What is the typical temperature of the exhaust gases at the end of the power stroke? Select the ONE correct answer.

- **Between 500 °C and 700 °C**
- Between 100 °C and 200 °C
- Between 200 °C and 300 °C
- Between 300 °C and 500 °C

When does the exhaust process start? Select the ONE correct answer.

- **Shortly before Bottom Dead Centre**
- Shortly after Bottom Dead Centre
- Shortly after Top Dead Centre
- Shortly before Top Dead Centre

When does the fuel injection process start? Select the ONE correct answer.

- **Just before Top Dead Centre**
- At mid-stroke
- Just after Top Dead Centre
- Just before Bottom Dead Centre

0500 – Pilot ladders

Version 2.0 (1209)

After a pilot ladder is finished with and before stowing it, what should the crew always check? Select the ONE best answer.

- **That it is not damaged and ready for use.**
- That it is still as long as it was initially.
- When it will next be used.
- Where the bridge want it to be stowed.

Before a pilot ladder is deployed, apart from the condition of the ladder, what should crewmembers also check? Select the ONE correct answer.

- **The other safety equipment required to be at hand.**
- How the master wants the ladder to be lashed.
- The name of the pilot boat bringing the pilot.
- The other safety equipment required to be at hand.

Before a pilot ladder is used, what should crewmembers do with it? Select the ONE correct answer.

- **Check it to make sure it is safe to use.**
- Add a tripping line to the bottom of it.
- Check it to make sure it is in a dry condition.
- Wash all the steps likely to be used.

Before a pilot ladder is used, what should crewmembers do with it? Select the ONE correct answer.

- **Check it to make sure it is safe to use.**
- Add a tripping line to the bottom of it.
- Check it to make sure it is in a dry condition.
- Wash all the steps likely to be used.

Can the old chock be used on a replacement step in a pilot ladder? Select the ONE best answer.

- **No, because the chock is now too small.**
- No, because the chock will also be damaged.
- Yes, with a normal lashing.
- Yes, with an extra thick lashing.

Even though all ships are different, what is the one thing in common with regard to their pilot ladders? Select the ONE best answer.

- **They should all be of an IMO approved design.**
- The deployment position should always be the same.
- They should all be made by the same manufacturer.
- They should all have the 'wheel mark' on them.

How does a pilot board a vessel where the freeboard is greater than nine metres? Select the ONE best answer.

- **Using the pilot ladder in combination with the accommodation ladder.**
- Using a longer pilot ladder.
- Using the accommodation ladder lowered to the sea surface.
- Using the pilot ladder as normal.

How much of the pilot ladder should be deployed? Select the ONE correct answer.

- **Enough to reach the deck of the pilot boat.**
- Enough to allow for vessel rolling.
- Enough to be one metre above the deck of the pilot boat.
- Enough to reach the water surface.

How should manropes be deployed on a pilot ladder? Select the ONE correct answer.

- **Outside the ladder.**
- Attached to the ladder's side ropes.
- Inside the ladder.
- Outside the ladder, attached to the bottom step.

What is the final check that a pilot ladder safe to use? Select the ONE correct answer.

- **When it has been checked by a responsible person.**
- As soon as it is deployed to right level.
- When it is checked by the crew that deployed it.
- When it is lashed at the top.

What other equipment should be standing by at the top of a deployed pilot ladder? Select the ONE best answer.

- **Lifebuoy and light, manropes and a heaving line for their bag.**
- Lifebuoy and light with a full body safety harness for the pilot.
- Manropes and a heaving line for their bag.
- Over shoes for the pilot's footwear and a heaving line for their bag.

When a combination ladder is rigged, how far above the water should the lower platform of the accommodations ladder be?
Select the ONE correct answer.

- **At least 5 metres above.**
- As far as instructed by the pilot boat.
- At least 2 metre above.
- Just above the water surface.

When a pilot gets to the top of a ladder, who should he see? Select the ONE best answer.

- **Members of the crew ready to greet him.**
- Members of the bridge team ready to greet him.
- No particular people, maybe crew members working.
- The master of the vessel.

When a pilot is approaching a vessel to board it, what will increase their confidence? Select the ONE best answer.

- **If they see a correctly rigged ladder.**
- If the ladder has white painted steps.
- If the vessel is stopped in the water.
- If there are manropes deployed with the ladder.

When a pilot is leaving a vessel, what responsibility does the escorting officer have towards them? Select the ONE best answer.

- **To ensure they make a safe return to the pilot boat.**
- To give them anything that they may request.
- To make sure the ladder is retrieved and stowed away.
- To make sure they leave the accommodation safely.

When a pilot is on a ladder, why is their life in the hands of the crew who rigged the ladder? Select the ONE best answer.

- **Because if it was done incorrectly, the pilot's life is at risk.**
- Because the pilot has a lifeline to the deck where they are.
- Because they will inform the bridge if they were to fall off the ladder.
- Because they will rescue him if he falls in the water.

When a responsible person checks the rigging of a pilot ladder, whom do they inform? Select the ONE best answer.

- **The bridge, if it is rigged properly.**
- Always the bridge.
- The deck crew who deployed it.
- The pilot boat.

Where is the ideal place to stow a pilot ladder when not in use? Select the ONE best answer.

- **In a dry locker away from contaminants.**
- In the paint store at the side of the accommodation.
- Lashed up against a side rail.
- On deck close to its deployed position.

Why does a pilot ladder stowed outside need a cover? Select the ONE best answer.

- **To stop the sun's rays from damaging the side ropes.**
- To stop the ladder being washed overboard.
- To stop the ladder getting wet.
- To stop the sun's rays from damaging the steps.

Why should the pilot boat crew lift the tail of the ladder onto the deck of the boat? Select the ONE best answer.

- **To stop the boat from damaging it.**
- To help the pilot get onto the ladder.
- To lift it out of the water.
- To stop the boat's propeller from fouling it.

Why should the steps of a pilot ladder not be painted? Select the ONE best answer.

- **Because it makes them slippery.**
- Because it is not worth the time.
- Because the paint affects the wood.
- Because the regulations say you cannot.

Why should there be a lashing on an accommodation ladder deployed on a vessel at sea? Select the ONE correct answer.

- **To hold it against the vessel's side.**
- In case the main support wire fails.
- To stop it being lowered further.
- To tie the pilot ladder to when rigged in combination.

0501 – Gangway security

Version 2.0 (744)

Before the ISPS code was introduced, how did port directors manage security? Select the ONE best answer.

- **In whatever way that they thought best.**
- A lot better than they do now.
- Using guidance contained in the Code of Safe Working Practices.
- Very poorly.

Depending upon the security level on the vessel, what other additional security checks might be necessary? Select the TWO best answers.

- **A full body search, possibly using a scanner.**
- **A search of bags and boxes brought onboard.**
- Asking to see a visitor's reference from their employer.
- Confirming a visitor's nationality.

If a visitor cannot produce photo identification what should the security watch do? Select the ONE correct answer.

- **Ask them to go and get some.**
- Call the dock security gate to seek clarification.
- Just let them onboard.
- Let them onboard if they are on the visitor list.

If an unexpected visitor, such as the padre is found to be genuine, what should the gangway watch do? Select the ONE best answer.

- **They should get permission for the visitor to stay onboard.**
- They should ask the duty officer to come to the gangway.
- They should ask the visitor to leave.
- They should ask the visitor to stay on the quay until an officer arrives.

If there is a very high level of security set on the vessel and in the port, what may have to be done? Select the ONE best answer.

- **The gangway raised and cargo stopped.**
- A pilot ladder used to board visitors from the waterside.
- Only visitors from the company allowed onboard.
- Only visitors from the port authority allowed onboard.

If you suspect an unwanted visitor has a weapon on their person what should you do? Select the ONE best answer.

- **Treat them with great caution and alert the duty officer.**
- Challenge them to hand over the weapon.
- Ignore the fact that they have a weapon then seek shelter.
- Try to bungle them down the gangway.

Is the following statement true or false? 'In a port with a good security system in operation, a ship does not need bother about their own security'.

- **False.**
- True.

What characteristics should a good gangway security watch person have? Select as many answers that you think are correct.

- **Be assertive.**
- **Be welcoming to all.**
- **Follow gangway procedures.**
- **Know their duties.**

What information should be shown on the notice, usually placed at the bottom of the gangway? Select the ONE best answer.

- **Relevant security and safety information.**
- The name of the master and the email address for the vessel.
- The nature of the cargo onboard.
- The number of crew onboard.

What is meant by ship security? Select the ONE best answer.

- **Being able to live and work onboard a vessel without danger or threat from others.**
- Always being dressed properly and wearing the correct PPE.
- Having an armed response unit onboard in piracy areas.
- Making sure that the cabin door is kept locked at all times in port and at sea.

What should be provided to the gangway security watch, to assist them decide if a visitor is legitimate? Select the ONE best answer.

- **There should be a list of visitors.**
- The visitor should be known by the security watch.
- There should be photographs of expected visitors.
- They should have a telephone to call the agent.

What should the gangway security watch do if an unwanted visitor stays onboard and heads for the accommodation? Select the ONE correct answer.

- **Call the duty officer for assistance.**
- Chase after them and try and stop them.
- Just let them proceed, as someone else will find them.
- Stop them at the gangway by grabbing them.

What should the gangway security watch do if there is any doubt about a visitor? Select the ONE correct answer.

- **Contact the duty officer by radio or telephone.**
- Contact the agent for further clarification.
- Escort the visitor to the duty officer.
- Let the visitor stay onboard the vessel.

What should visitors to a vessel show to the security watch, to verify their identity? Select the ONE correct answer.

- **Photographic identity of some sort.**
- A letter from the agent.
- A letter from the ship manager.
- A valid job requisition.

What two characteristic should a good gangway security watch show to a visitor? Select the ONE correct answer.

- **A welcoming attitude and firmness.**
- A dismissive attitude and just sign them in without question.
- A welcoming attitude and a subordinate behaviour.
- An aggressive attitude and behaviour that obstructs a visitor.

When a vessel is tied up in a port with good security procedures, how should the vessel's gangway security watch be seen? Select the ONE correct answer.

- **As the last line of defence for people coming on board.**
- As a waste of time because the port will deal with everything.
- As only necessary during the hours of darkness.
- It can be set but then the crewmembers deployed on other work.

When a visitor is signed in at the top of a gangway, what should the security watch make sure that the visitor knows? Select the ONE correct answer.

- **The security level and some basic safety information.**
- How they will assist if there is a fire onboard the vessel during their stay.
- How to log onto the ship's Wi-Fi network.
- Where the crew mess room is located.

Where are details of visitors and the numbers of the passes issued to them kept? Select the ONE correct answer.

- **In the visitor logbook at the security checkpoint.**
- In the bridge deck logbook.
- In the ship's office computer.
- Nowhere as the pass contains the details required.

Which answer best completes the following statement? 'If a vessel has more than one gangway deployed . . .

- **. . . each one will need a security watch on it'.**
- . . . it is difficult to maintain any security onboard'.
- . . . others should be stowed leaving one with a security watch'.
- . . . they will normally be alongside each other requiring one security watch'.

Which answer best completes the following statement? 'The gangway security watch . . .

- **. . . represents the master'.**
- . . . is ineffective at stopping unwanted visitors'.
- . . . is only there because of the ISPS code'.
- . . . should follow what the company security officer tells them when they visit'.

Which genuine visitors might be missing off a list of visitors supplied to a gangway security check? Select as many answers that you think are correct.

- **Coastguards.**
- **Port officials.**
- **Port state inspectors.**
- **Surveyors.**

0600 – CEMS - Crew Endurance Management System

Version 1.1 (0)

CEMS is a program totally focused on managing crew fatigue.

- **False**
- True

CEMS is a voluntary, non-regulatory program developed by the U.S. Coast Guard to provide tools to assist in Crew Endurance Management.

- **True**
- False

Caffeine and lack of sleep are not risk factors in crew endurance.

- **False**
- True

Dehydration does not affect your levels of fatigue, mood, alertness, and stamina.

- **False**
- True

Four to five hours of sleep allows you to complete the required 5-6 cycles of uninterrupted sleep needed to restore your energy and maintain your performance levels.

- **False**
- True

If you don't successfully manage the Red Zone, your performance is dramatically reduced.

- **True**
- False

Irregular sleep periods and poor quality sleep can cause depletion of your energy stores and reduce levels of alertness.

- **True**
- False

Light Management consists of staring at a light bulb of a specific intensity for a certain amount of time.

- **False**
- True

Light Management means exposing yourself to measured amounts of artificial light of specific intensity and color.

- **True**
- False

Once a CEMS program is in place, there is no further, ongoing support.

- **False**
- True

Poor diet choices can reduce alertness and performance in the short-term, and reduce health and well-being in the long-term.

- **True**
- False

Sleep debt can cause the Red Zone to expand and eventually cover the entire day.

- **True**
- False

Sleep inertia is the name for the time period during which you feel overly sleepy and sluggish right after awakening.

- **True**
- False

The Crew Endurance Working Group is a real benefit to you, your fellow crew, and your company because they are pro-active in implementing Crew Endurance Management practices.

- **True**
- False

The first few REM cycles are the most beneficial stages of sleep, as they are the shortest.

- **False**
- True

The five critical steps in implementing a Crew Endurance Management Program ensure processes are continually revisited and refined.

- **True**
- False

The three phases in implementing a Crew Endurance Management Program are Program Development, Program Deployment, and Program Assessment.

- **True**
- False

When you are standing night watch, one Light Management technique is to ensure you have bright light exposure from sunset to 3 hours prior to going to sleep.

- **True**
- False

You are naturally oriented by your body clock to work during daylight hours.

- **True**
- False

Your body metabolizes caffeine quickly, so you will sleep fine with a cup of coffee right before bed.

- **False**
- True

0601 – Vapor Control Systems

Version 1.1 (0)

After a dock automatic shutdown, it is OK to resume transfer operations:

- **When the reason for the shutdown is known and corrected and the information has been passed on to the dispatcher.**
- When dock operations appear to be back to normal.
- When the dispatcher gives the OK.

As the cargo tank space fills with liquid, the vapors:

- **Vent out of the tank by a vent stack or into the dock collection system.**
- Build up in the tank.
- Collect in the barge's collection system.
- Vent out through the vent stack into the air.

Barge vent piping is used to direct the tank vapors into a common manifold on the barge deck.

- **True**
- False

Carbon bed absorption is one method that may be used to reclaim carbon vapors once they are collected at the dock facility.

- **True**
- False

High initial loading rates and improper gauging techniques can generate . . . that can cause an explosion if there are vapors in the surrounding area in the explosive range.

- **Static electricity**
- Excessive pressure
- Incorrect ullage readings

If for any reason, the transfer operation goes out of control, creating a serious emergency, what should you do first?

- **Stop the transfer**
- Contact the dispatcher
- Run for help

In the US, vapor control regulations are under the jurisdiction of:

- **USCG**
- EPA
- IMO
- SOCP

One reason for over-pressurization of the vapor lines is:

- **Blocked with condensation**
- Filled with air
- Leaking

Over-pressurization occurs when:

- **The vapors leave the cargo tank more slowly than the liquid enters.**
- The vapor pressure in the cargo tanks or dock facilities builds to excessive levels due to a slow loading rate.
- The vapors leave the cargo tank more quickly than the liquid enters.

Vapors that vent directly into the air pose hazards to:

- **Personnel and the environment**
- Personnel
- The environment

What can happen if the cargo and vapor lines are misconnected?

- **The tank may be top-loaded with liquid vapor from the vapor collection piping.**
- The transfer will load slower.
- You will have more time to perform your other tankerman duties.

What is a function of the vapor vent stack?

- **Allows air to enter the tank and fill the space left after the cargo leaves the compartment.**
- Helps prevent mistakes in connecting the cargo line to the vapor header.
- Helps prevent the build-up and uncontrolled release of static electricity.
- Relieve vapor pressure at the same rate as the maximum loading rate.

What is the primary reason for adding vapor control systems to barges?

- **Reduce harmful emissions.**
- Improve barge design.
- Increase barge efficiency.
- Meet regulatory requirements.

When the liquid level in the cargo tank rises, the vapors are forced into the barge's collection piping. This is referred to as:

- **Positive pressure**
- Inverse pressure
- Negative pressure

Which of these factors affect the flow of vapor through your barge's collection piping?

- **Loading rate**
- **Pipe diameter**
- **Vapor density**

Which of these statements about normal cargo transfer operations is true?

- **Stop the cargo flow when the liquid approaches the bottom edge of the prescribed top-off in the last tank.**
- Disconnect the hoses and power cable after disconnecting the bonding cable.
- Observe the cargo liquid level by opening the tank hatch.
- Slow the cargo flow when the cargo level reaches the red color on the gauge stick.

Which of these steps are NOT completed as part of a normal transfer operation?

- **Maintain communication with the boat.**
- Ensure the barge and dock vapor header valves are open.
- Monitor the transfer rate.
- Slow the cargo flow.

Which of these tasks do you need to do when completing your final barge preparation and inspection for a cargo transfer?

- **Connect the dock's power supply for the high level alarms.**
- **Ensure that the barges vapor system is properly alarmed.**
- **Inspect all sight glasses for condensation or damage.**
- **Inspect and test the barge PV to determine that it operates freely.**

Which of these tasks is NOT a transfer preparation task?

- **Observe the cargo liquid level.**
- Assist with tank inspection.
- Complete transfer connections.
- Discuss upcoming transfer.

0602 – Benzene Cargo Safety

Version 2.0 (278)

A physician must clear you to work around benzene and wear a respirator. Is this statement true or false?

- **True**
- False

Benzene has an odor threshold of . . . parts per million, which makes it difficult to smell in small concentrations. Select the ONE answer that correctly completes the statement.

- **5**
- 2
- 3
- 4

Dave has an 8-hour shift to work. The average value of his exposure cannot exceed . . . parts per million. Select the ONE answer that correctly completes the statement.

- **1.0**
- 0.5
- 1.5
- 2.0

How are benzene regulated areas marked? Select the ONE correct answer.

- **A white, non-skid safety stripe around the edge of the barge.**
- A large printed sign, facing outboard.
- Decals on the bow and stern cargo tanks.
- Warning cones.

How are gangways, walkways and ladders marked to protect dockside third parties from entering benzene regulated areas? Select the ONE correct answer.

- **Warning cones.**
- A large printed sign, facing outboard.
- A white, non-skid safety stripe around the edge of the barge.
- Decals on the bow and stern cargo tanks.

Sorbent material may be used to contain a spill as long as you do not need to enter the spill area. Is this statement true or false?

- **True**
- False

What PPE do you need to wear during a benzene cargo transfer operation where there is a potential for exposure? Select the ONE correct answer.

- **A full face respirator and rubber boots, gloves and rain gear.**
- Rubber boots, safety goggles and a full-face respirator.
- Splash goggles and rubber boots, gloves and rain gear.
- Splash goggles, rubber boots, gloves and rain gear, and a full-face respirator.

What is the acceptable working limit for 15-minute short-term exposure? Select the ONE correct answer.

- **5 parts per million**
- 1 part per million
- 1.5 parts per million
- 2 parts per million

What is the first action that should be taken if a liquid leak is detected at the vapor hose connection during cargo loading? Select the ONE correct answer.

- **Stop the loading.**
- Call for assistance.
- Evacuate the area.
- Inspect the leak.

Which ONE of these engineering controls does NOT protect you from exposure to benzene?

- **Drip pans**
- High level alarms
- Sight glasses
- Vapor vent stack

Which ONE of these is NOT a symptom of short-term exposure to benzene?

- **Convulsions**
- Headache
- Irritability
- Nausea

Which ONE of these statements about benzene spill response is true?

- **Remain outside of the spill area.**
- Contain all sizes of spills.
- Stay downwind of the spill.
- Wear rubber gloves and safety goggles only.

Which ONE of these statements does NOT describe a preliminary action to be taken in a benzene spill situation?

- **Clean-up the spill.**
- Begin emergency communications.
- Evacuate the spill area.
- Secure the cargo transfer.

Which of these applies to a benzene mixture cargo? Select the ONE most accurate answer.

- **It will contain 0.5% up to 100% benzene liquid by volume.**
- It will contain 0.5% benzene liquid by volume.
- It will contain 0.5% to 50% benzene liquid by volume.
- It will contain 1% and 100% benzene liquid by volume.

Which of these is a property of benzene? Select the ONE correct answer.

- **It has a sweet odor**
- It is lighter than air
- It is non-flammable
- It is soluble in water

Which of these is the most common source of benzene leaks during cargo discharge? Select the ONE best answer.

- **Dock hoses and cargo pump.**
- Drip pans.
- Pump seal.
- Wing tanks and rake voids.

Which of these should be checked for leaks in preparation for a benzene cargo? Select the ONE correct answer.

- **Sight glasses.**
- Cargo pump.
- Pressure vacuum valve.
- Pump mechanical seal.

'Before returning to work after a benzene spill, you'll need to ensure that . . . Select the ONE answer that correctly completes the statement.

- **. . . benzene levels in the work area do not exceed the permissible exposure limits or that you are protected with the appropriate PPE.'**
- . . . air monitoring was used to assess the exposure risk.'
- . . . hazardous material contractors were contacted.'

0604 – Eye Injury Prevention

Version 2.0 (1284)

Blinking the eye may provide relief for which type of injury?

- **Specks of debris**
- Chemical burn
- Heat
- Impact

For which type of injury is it advisable to apply a cold compress to the eye?

- **Impact**
- Chemical burn
- Foreign object
- Heat

For which type(s) of injury should the eye be flushed with water?

- **Heat and chemical burns**
- Chemical burns and foreign objects
- Impact
- Puncture

Goggles can be used to protect against which type of eye injuries?

- **Impact, chemical burns, heat, and foreign objects**
- Foreign objects and chemical burns only
- Heat and impact only
- Heat, impact, and chemical burns only

How do your eyelashes help to prevent eye injury?

- **They act as a barrier by sweeping foreign particles away from the eyes.**
- They help spread tears over the surface of the eyes.
- They keep the eyes moist.
- They produce the watery portion of tears.

If you do not have the proper eye protection for a job, you should:

- **Contact your supervisor**
- Complain to your crewmates
- Go back to your rack and not tell anyone
- Work carefully without eye protection

Rich in antibodies, they help prevent infection. They are:

- **Tears**
- Eyelashes
- Eyelids
- Mucous glands

Safety glasses are designed to protect against which type of eye injuries?

- **Impact and heat**
- Chemical burns and foreign objects
- Heat and chemical burns
- Impact and chemical burns

True or False: For chemical splashes or burns, you should flush the eyes with water for at least 15 minutes.

- **True**
- False

True or False: If an object hits your eye, you should get medical help immediately.

- **True**
- False

True or False: If you wear safety goggles, you never need a face shield.

- **False**
- True

True or False: It's safe to wear protective eye wear that is scratched and cracked.

- **False**
- True

True or False: Many eye injury hazards can be avoided by using safety glasses or goggles that protect you from the side as well as the front.

- **True**
- False

True or False: One common reason for eye injures is not using protective eyewear.

- **True**
- False

True or False: Safety glasses can provide adequate eye protection against hazardous splashes.

- **False**
- True

True or False: Under a welding mask you should wear eye protection, such as goggles or safety glasses.

- **True**
- False

True or False: When an object penetrates your eye you should seek medical attention immediately.

- **True**
- False

What are the four major types of eye injuries?

- **Heat, impact, chemical burns, and foreign objects**
- Chemical burns, dust, dryness, and heat
- Foreign objects, punctures, heat, and infections
- Impact, foreign objects, heat, and bright light

What percentage of on-the-job eye injuries can be prevented or minimized if eye protection is worn?

- **90%**
- 60%
- 70%
- 80%

What type of injury can be made worse by keeping the eyes closed?

- **Chemical burn**
- Foreign object
- Heat
- Impact

What type of injury most often results from flying or falling objects hitting the eye?

- **Impact**
- Burns
- Collision
- Foreign objects

Which of these is NOT a requirement for an eye protection device?

- **Disposable**
- Easily cleaned and disinfected
- Fits snugly and doesn't overly interfere with the movements of the wearer
- Reasonably comfortable when worn under the designated conditions

Which of these is the responsibility of the supervisor in preventing eye injuries?

- **Conducting a hazard assessment**
- Inspecting the eye protection device for damage prior to use
- Regularly disinfecting the eye protection device
- Wearing the eye protection device, as directed

Which of these statements is true?

- **Safety goggles and safety glasses should fit snugly.**
- All eye protection devices are the same.
- Contact lenses are a proven protective device against mild hazards.
- Eye protection devices should be cleaned and disinfected once a year.

Which type of eye injury is NOT one of the four main types?

- **Eye strain**
- Chemical burn
- Heat
- Impact

Your eyeball is set into a socket surrounded by a bony ridge, which is known as the:

- **Orbit**
- Arc
- Iris
- Lacrimal gland

0605 – Workplace Ergonomics and Back Care

Version 1.0 (1267)

Building strong and flexible muscles through exercise helps prevent back pain and injury.

- **True**
- False

Falls and slips are examples of . . . , which can cause severe back pain.

- **Impact traumas**
- Intense traumas
- Repetitive traumas
- Sudden traumas

How can you avoid straining your back while carrying a load?

- **Use a hand truck or dolly.**
- Carry the load on your shoulder.
- Gently twist your back when changing directions.
- Walk quickly, taking large steps.

How does diet impact back health?

- **Poor food choices can cause weight gain and being overweight puts extra strain on your back.**
- A good diet causes back muscles to tighten and knot.
- A poor diet builds flexible back muscles.
- Poor food choices help to minimize the pain of back strain and injury.

How does eating healthily help to keep your back healthy?

- **A healthy diet helps you keep your weight down, which reduces the stress on your back.**
- A low caloric diet improves bone mass, making the spine stronger.
- A poor diet builds more muscle.
- A poor diet provides you with more energy for safer lifting.

If a doctor prescribes pain medication to relieve your back pain, you should:

- **Advise your supervisor that you are on prescribed pain medication.**
- Increase your prescribed dosage if your pain increases.
- Operate machinery.
- Return to your job without notifying supervisor.

One way to reduce back strain and fatigue if you have to stand for an extended period of time is to:

- **Periodically walk around.**
- Complain to your supervisor.
- Drink coffee.
- Lean against a bulkhead.

Placing a pillow under your knees can help to alleviate the pain, if lying on your back gives you low back pain.

- **True**
- False

To support your spine when you are sitting:

- **Sit straight with your back supported.**
- Adjust your screen so it is above eye level, if you are in front of a computer.
- Get up and move around every few hours.
- Sit on a stool.

What causes repetitive trauma injuries?

- **Repeating an unhealthy position, movement, or lifting technique.**
- Lifting a heavy, awkward load without assistance.
- Tripping or falling.
- Wearing incorrect footwear.

What is ergonomics?

- **An applied science that focuses on improving the fit between your body and an activity**
- A methodology for improving posture to minimize injury
- A model designed to improve workstation design
- A practical approach to working safely that emphasizes physical fitness

Which of the following is NOT a proper lifting technique?

- **Carrying the load away from your body.**
- Carrying the load waist high.
- Pivoting your feet when turning.
- Planning the lift before starting.

Which of the following is not indicative of a good work area?

- **You can work long periods standing if you lean over your desk.**
- Comfortable temperature.
- Lighting is directly on your work area.
- Work materials you use are within easy reach.

Which of the following statements about ergonomic risk factors is correct?

- **A longer period of exposure to a risk factor increases the risk of injury.**
- Poor health has no impact on ergonomic injury.
- Rest time is not necessary when performing repetitive tasks or assuming awkward postures.
- The potential for injury is reduced when risk factors are combined for a short period of time.

Which of these is NOT a common cause of back pain?

- **Regular exercise**
- Aging and disease
- Improper lifting
- Tension and stress

Which of these is NOT a risk factor associated with ergonomic injury?

- **Proper posture**
- Exposure time
- Inadequate work environment
- Repetition

Which of these is NOT a risk factor for ergonomic injury?

- **Sleep deprivation**
- Improper technique
- Incorrect use of equipment
- Repetitive motion

Which of these is a good lifting technique?

- **Planning the lift before you get started.**
- Lifting quickly.
- Stretching to lift objects to high places.
- Using your back.

Which of these statements about exercise and back health is incorrect?

- **Exercise should be avoided whenever you experience back pain.**
- Exercise along with proper diet is important for good health.
- Exercise builds strong and flexible muscles.
- Exercise should be done regularly for it to be the most beneficial.

Which of these strategies might you try to relieve fatigue if you have to stand in one place for a period of time?

- **Periodically walk around.**
- Eat a nutritious snack.
- Lean against a bulkhead.
- Lie down every half hour to relax your neck.

Which of these strategies will help you maintain good posture when seated?

- **Adjust the height of your chair so that your knees are even with your hips.**
- Cross your legs
- Lean forward with your elbows on the desk.
- Tilt your chair backward.

While enlisting a helper to lift a load, it's a good idea to:

- **Keep the load at the same level as you move smoothly to carry the load.**
- Find a person who is taller than you.
- Keep your knees straight and your back bent, and lift the load together.
- Lower the load at the same time, keeping your knees straight.

You are doing a routine job with a repetitive motion and your hand begins to hurt. What do you do?

- **Stop working and re-evaluate your risk factors with your supervisor.**
- Change hands.
- Continue working – it's good to just get the job done.
- Stop work and leave it for someone else.

. . . can cause the back muscles to tighten and knot, which can lead to muscle spasms and back pain.

- **Tension and stress**
- Aging
- Lack of exercise
- Poor posture

. . . can cause your muscles to lose their strength and ability to function.

- **Aging and disease**
- Exercise and a healthy diet
- Improper lifting
- Poor posture

0606 – Hearing Conservation

Version 1.0 (1555)

Choose one statement that can contribute to your hearing loss?

- **Entering the engine room for only a few minutes without hearing protection.**
- Being careful about wearing the correct hearing protection.
- Having your hearing checked regularly.
- Replacing your earplugs when they are worn out.

Earplugs work by:

- **Blocking your ear canal**
- Covering your outer ear
- Electronically filtering noises
- Protecting your ears from dirt

Foam earplugs should be discarded:

- **When they become hard or dirty**
- After each use
- When the begin sticking in your ear canal
- When they fall apart

How often do you need to wear your hearing protection device to ensure you are adequately protected?

- **Continuously**
- At least 50% of the time
- Most of the time
- Never

How often should you replace the cuffs of your earmuffs?

- **Whenever they become hard or cracked**
- Every 3 months
- Every 6-12 months
- Never

If you are exposed to high levels of noise you may suffer hearing loss and will be prone to:

- **Accidents and injuries**
- Forgetfulness
- Poor vision
- Working too fast

If you have to raise your voice to be heard by someone . . . away, the noise level is probably hazardous.

- **A couple of feet**
- A dozen or more feet
- Ten or more meters
- Twenty yards

One advantage of earmuffs over earplugs is that they are:

- **Designed so that one size fits most people**
- Convenient for use in confined spaces
- Easy to carry and store
- More comfortable in hot, humid work areas for long-term use

TWA is the average level of noise over the course of an eight-hour work shift. TWA stands for:

- **Time Weighted Average**
- Test Work Average
- Theoretical Weekday Average
- Typical Work Average

The higher noice reduction rate a hearing protection device is given, the . . . protection it offers against hearing loss.

- **Greater**
- Less

There are three basic types of hearing protectors. Which one if the following is NOT a recognized form of ear protection?

- **Cotton balls**
- Earmuffs
- Earplugs
- Semi-insert earplugs

True or false: Hearing loss occurs when the cochlea hair cells are damaged.

- **True**
- False

True or false: In most cases, any amount of noise-induced hearing loss is permanent – it cannot be cured.

- **True**
- False

True or false: Foam earplugs are expandable to conform to the shape of your outer ear.

- **False**
- True

True or false: If you have to raise your voice to be heard by someone a couple of feet away, then the noise level is probably too high.

- **True**
- False

True or false: Pre-moulded earplugs are made from silicone, rubber, or plastic.

- **True**
- False

True or false: Removing your hearing protection device even for a short period of time can substantially reduce your hearing protection.

- **True**
- False

True or false: The notation for the decibel measurement of sound is represented by the dB symbol.

- **True**
- False

True or false: When you're working on a job that has multiple hazards, you should prevent injuries by wearing protective equipment for all hazards.

- **True**
- False

What is the decibel dB Action level at which you need hearing protection?

- **85 dB over an eight-hour period**
- 85 dB over an six-hour period
- 90 dB over an eight-hour period
- 95 dB over an ten-hour period

Where does hearing damage occur?

- **Cochlea hair cells**
- Brain
- Inner ear
- Outer ear

Which of these statements about fitting earmuffs is true?

- **Push your hair back so that it does not interfere with the seal of the cuffs to your ears.**
- Adjust the headband so it keeps the earmuffs loosely in place.
- Drill holes in them if they don't feel comfortable.
- Ensure the cuffs fit snugly over only part of your ear.

Which of these statements about hearing loss is true?

- **It can occur gradually, with no immediately noticeable change.**
- In most situations, it is reversible with proper care.
- It is one of the least common industry health hazards.
- It is only caused by unprotected exposure to brief, sudden bursts of high noise levels.

Which of these statements about hearing protection devices is true?

- **Noise Reduction Ratings identify the noise reducing capability of a hearing protection device.**
- Earplugs are more effective than earmuffs.
- More hearing protection is always better.
- One style of protection will work for everyone.

Which of these statements is false?

- **Noise is one of the least common work-related health hazards.**
- Hearing testing measures the sharpness of your hearing.
- Monitoring consists of testing for and measuring continuous, irregular, or spontaneous noise.
- Training and education about hearing conservation can help you manage your hearing health and safety.

Which of these statements is false?

- **Suitable hearing protectors must reduce your exposure to an eight-hour TWA of 105 dB.**
- A certified technician conducts hearing tests and an audiologist or physician reviews the results.
- Hearing tests are repeated, to check for changes in your hearing over time.
- It is your responsibility to care for and use your hearing protection device properly.

Which statement describes an advantage of earplugs over earmuffs?

- **Small and easy to carry**
- Covers your entire ear
- Easily seen at a distance to assist in monitoring usage
- Waterproof

0608 – Drug and Alcohol Policy and Testing

Version 3.0 (1721)

A condition where a person's faculties are diminished so that his or her ability to see, hear, walk, talk, and judge distances is below the normal level, as set by governmental health agencies, is known as . . .

- **Impairment**
- Drunkenness
- Euphoria
- Intoxication

A ship operator is interested in implementing a drug-free ship program. Which of the following are elements that could help ensure a successful program?(Select all applicable options)

- **Clearly written company drug policy**
- **Drug testing**
- **Training**

Crewmembers who test positive for drug or alcohol may face what consequences? (Select all applicable options)

- **Employment termination**
- **Removal from duties which affect ship operations**
- **Suspension of license or mariner's document**

In terms of its effects on the body, alcohol is considered a . . .

- **Depressant**
- Hallucinogen
- Painkiller
- Stimulant

Maritime drug and alcohol testing is:

- **Required for all vessels**
- Required only after a serious maritime accident
- Required only for vessels carrying hazardous cargo
- Voluntary for all vessels

Of the substances that have the potential to affect human performance, . . . is by far the most frequently used and abused.

- **Alcohol**
- Cocaine
- Marijuana
- Methamphetamine

True or False: If you suspect another crewmember of using alcohol or drugs, the first thing you should do is inform your supervisor.

- **True**
- False

True or False: Illegal Drug use while ashore will likely put you at risk when onboard.

- **True**
- False

True or False: In today's maritime industry, alcohol and drug abuse continues to be a serious problem.

- **True**
- False

What is a common sense policy when taking a prescription medicine that may impair job performance?

- **Inform your supervisor**
- Don't tell anybody
- Drink lots of caffeine
- Take the medication only on a full stomach

Which of the following is NOT a potential long-term health effect of stimulant abuse?

- **Blindness**
- Headaches
- Irregular heartbeat
- Stroke

Which of the following is a narcotic?

- **Heroin**
- Alcohol
- LSD
- Methamphetamine

Which of the following is a possible consequence of inhalant abuse?

- **Brain damage**
- Blindness
- Fatigue
- Soft-tissue infections

Which of these is NOT a possible symptom or effect of alcohol intoxication?

- **Increased concentration**
- Decreased reaction time
- Impairment of balance, speech, vision, and hearing
- Loss of consciousness

Which of these is NOT a tell-tale sign of stimulant abuse?

- **Fatigue**
- Dilated pupils
- Excessive activity, talkativeness, or irritability
- Long periods without eating or sleeping

Which of these is NOT a typical sign of alcohol intoxication?

- **Attentiveness**
- Slurred speech
- Staggering
- “Glossy” eyes

Which of these is a tell-tale sign of hallucinogen abuse?

- **Extreme changes in mood and behaviour**
- Bloodshot eyes
- Loss of muscle coordination
- Rash, spots, or sores around the nose and mouth

Which type of drug causes altered perceptions and senses of sight, hearing, and touch?

- **Hallucinogens**
- Alcohol
- Depressants
- Narcotics

“Random testing” means crewmembers are subject to drug or alcohol testing when?

- **At any time**
- Only when they are off duty
- Only when they are on duty
- Only when they are working on shore duty

0612 – Respiratory Protection

Version 1.0 (1270)

A correctly worn respirator can safeguard your health and save your life.

- **True**
- False

A respirator leak test verifies the integrity of the mask, not the fit of the mask to your face. A leak test should be performed how often?

- **Before each use**
- Once a month
- Once a week
- Only before the first use

An air-purifying respirator will protect you from Dust?

- **True**
- False

Can you wear glasses while correctly wearing a full-face respirator?

- **No**
- Yes

Can you wear glasses with a half-face respirator?

- **Yes**
- No

If you're not sure which type of respirator is required for a job, you should:

- **Refer to the MSDS and your supervisor.**
- Ask somebody else on the same job.
- Use one hand to cover your mouth and continue working.
- Use the respirator with the biggest cartridge or canister.

Respiration is achieved through the mouth, nose, trachea, lungs, and . . .

- **Diaphragm**
- Appendix
- Eardrums
- Liver

The MSDS contains information that will guide you in selecting the proper respiratory protection equipment. What does MSDS stand for?

- **Material Safety Data Sheet**
- Manifest Safety Destiny Sheet
- Material Danger Sheet
- Material Safety Demerit Sheet

The primary function of the respiratory system is to provide the blood with . . .

- **Oxygen**
- Carbon dioxide
- Nitrogen
- None of the options are correct

The purpose of a respirator is to . . .

- **Protect against inhaling dangerous substances such as vapors, fumes, dust, and gases**
- None of the options are correct
- Protect against high temperatures
- Protect against low temperatures

Training must be provided to a crewmember before using a respirator.

- **True**
- False

True or False: A chemical's MSDS sheet will provide respiratory-protection information.

- **True**
- False

True or False: Respirator use protects you from hazards that can cause lung damage, respiratory diseases, cancer, poisoning, and asphyxiation.

- **True**
- False

True or false: A respirator that uses cartridges or canisters to filter out harmful airborne contaminants is considered an air-purifying type respirator.

- **True**
- False

True or false: Respirator cartridges are color coded to help identify the contaminant they're intended for.

- **True**
- False

True or false: Respirator use is not for everyone. Depending on your health history, you may not be allowed to use a respirator.

- **True**
- False

True or false: SCBA respirators use air tanks as a method of protecting against harmful airborne contaminants. Therefore, these respirators are considered air-supplying-type respirators.

- **True**
- False

True or false: When working in a confined space, which involves using a respirator, the best type of respirator to use is an air-supplying respirator.

- **True**
- False

True or false: When working in an oxygen-deficient atmosphere, the best type of respirator to use is an air-purifying respirator.

- **False**
- True

True or false: You may not be able to use a respirator safely if you have breathing problems or a heart condition.

- **True**
- False

True or false: You should only use the respirator that has been assigned to you.

- **True**
- False

True or false: You should use the hottest possible water to clean a respirator.

- **False**
- True

What are the two main types of respirators?

- **Air-purifying and air-supplying**
- Air-inducing and oxygen-generating
- None of these are correct
- Oxygen-replicating and nitrogen oxide-supplying

What is the proper way to store respirators?

- **At room temperature and away from direct sunlight**
- In a cleaning solvent
- In a toolbox
- In the cleaning locker next to chemicals

When does your supervisor provide you with a respirator?

- **After you have been properly trained and qualified**
- None of the statements are correct
- Only after one year with the company
- When you are ordered to enter a hazardous area with airborne contaminants

When inspecting a respirator, you should look for problems like:

- **Deteriorated rubber or plastic parts**
- Fresh canisters or cartridges
- Fully charged oxygen tanks
- Tight connections

When is respirator fit-testing required?

- **Prior to initial use**
- Following each use
- Within your first 30 days of employment
- You never need to fit-test a respirator

When using an air-purifying respirator a "Black" color coded cartridge is intended to be used for vapors of organic solvents.

- **True**
- False

Why can't you use a respirator over a beard?

- **It can cause the respirator to leak.**
- It will cause a skin irritation.
- It will clog the respirator.
- The beard will interfere with your breathing.

Why is respiratory protection necessary?

- **To protect you from airborne hazards**
- It is mandated by all governing bodies
- To protect others from becoming infected by your disease
- To protect the user from ocean spray

2500 – Self-contained breathing apparatus

Version 3.0 (2)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4010 – Maris ECDIS

Version 4.0 (1553)

Another important task of the Application Manager is to?

- **-Password protect the setup of ECDIS.**
- -Share protect the setup of ECDIS.
- -User protect the setup of ECDIS.
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Application Manager may be opened by double clicking the square icon located?

- **-Somewhere on the screen.**
- -On the option menu.
- -On the taskbar of ECDIS.
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By connecting the ships AIS transponder you may not only receive and plot AIS targets on the chart, but you may also, which statement is correct?

- **-The AIS is able to send and receive dedicated messages via the ECDIS**
- -The ARPA is sending out AIS message
- -The ECDIS sends/receives AIS messages
- -The GPS connected to ECDIS allows sending messages.

ECDIS 900- As a minimum each ECDIS is connected with?

- **-Gyro and GPS.**
- -Gyro, GPS and AIS.
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ECDIS 900- Logbook items to be stored in intervals. What alarms are stored in ECDIS logbook?

- **-Status of main alarms as waypoint, XTE, grounding, CPA and position offset.**
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- -Any RADAR
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- **-Contents of the Charts, ECDIS user folder and ECDIS sub folders.**
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How can you view a route by screen jump in Load / Unload ?

- **- Localize button**
- - Click on the import button
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How to order Ocean Current for use in voyage optimization module?

- **-From MDS application.**
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- -It is sent automatically every day.
- -Send email to Chartservice.

How to set default value of turn radius and speed in ECDIS900?

- **-Select settings button, select route tab and enter default value for turn radius and speed.**
- -Right click on the route leg and enter default value for turn radius and speed.
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- **-Select load/unload route, select converted folders and select route.**
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Is it possible to change color of the safety zone in front of the vessel?

- **-Yes.**
- -No.
- -Only in night display.
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Logbook recordings. Where to go to check/adjust logbook recordings?

- **-ECDIS logbook key in toolbar and select settings.**
- -Chart handling key in toolbar and select settings.
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Navtex software connected to ECDIS900. What is required in a navtex message in order for the ECDIS900 to automatically plot the position on the chart?

- **-Must contain geographical coordinates.**
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Objects on the ENC charts are stored in a database as identified objects and can be switched on and off. This allows?

- **-Manipulation of the chart display.**
- -Editing of charts.
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Official vector charts produced by National Hydrographical Offices are called?

- **-ENC.**
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On the Application Manager window, you will find two buttons on top. They are?

- **-Reboot & Shutdown buttons.**
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One of the basic principles of ECDIS is to prevent access to the PC operational system by use of?

- **-Application Manager.**
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One way to point at the position you want to centre the zoom in and out by pushing function key?

- **-F1 and F2.**
- -F3 and F4.
- -F5 and F6.
- -F9 and F 10.

Other sensors besides GPS and Gyro are connected to Ethernet based sensor collectors and?

- **-Serial Port.**
- -LPT1 Port.
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- -USB Slot.

Scaled icon option can be switched off by which method?

- **-By deactivating 'Scaled icon' under Display tab.**
- -By activating 'Display' under the Display tab.
- -By activating 'Scaled icon' under the Display tab.
- -By clicking on 'Default' under the Shape tab.

Sensors are picked up locally on the ECDIS by the application called?

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- -Sensor Converter.
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The Dotted line vector on the ship symbol shows?

- **-True Heading.**
- -Gyro Heading.
- -Magnetic Heading.
- -True Heading and Magnetic Heading.

The ENC charts are delivered in which format?

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The Editors panel gives access to?

- **-Editing tools to add User data on the electronic chart.**
- -Drawing tools to draw images on the electronic chart.
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- -Editing tools to edit data on the electronic chart.

The GPS set position contains datum and offset values for which datum?

- **-WGS 84.**
- -NAD 83.
- -WGS 66.
- -WGS 72.

The own ship symbol can be displayed in how many different ways?

- **-2.**
- -1.
- -3.
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The own ship symbol has 2 independent vectors. They are?

- **-Solid line vector and Dotted line vector.**
- -Solid line vector and Broken line vector.
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The position of the panels can be adjusted directly in the console area by?

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- -Selecting different panels in the Option menu.

The raster charts are to be considered as?

- **-Pictures transferred to right shape or projection size, and placed in a geographical grid to fit into the display of the ECDIS system.**
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The scaled icon where own ship is drawn in on the screen can be set in?

- **-Shape tab under Ship Properties.**
- -AIS tab under Ship properties.
- -Display tab under Ship Properties.
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The vector chart is more like a?

- **-Cad drawing of the same chart.**
- - Cad drawing of a different chart.
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The wind indicator displays?

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To test the configuration, press the Apply button followed by the OK button. What will happen after that?

- **-Panels will appear outside the console area as floating windows.**
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Voyage plan. How do you access voyage plan in ECDIS900?

- **-Right click on the route and select voyage plan in drop down menu.**
- -Double left click in console area and select voyage plan from drop down menu.
- -Open sensor monitor and select voyage plan.
- -Select voyage plan button in tool bar.

WOP-Wheel over point. In order for a vessel to follow the route without deviation, how is the WOP calculated?

- **-From planned turning radius and ships speed.**
- -Automatic.
- -From AIS and CPA.
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What are the two types of communication used between ECDIS units?

- **-Share and Synchronization.**
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What do append function means in ECDIS?

- **- Add extra waypoint in the end of the route**
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What does ENC charts abbreviation means?

- **-Electronic Navigation Chart.**
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What does the scale indicator displays?

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What is the name of the ECDIS VRM / EBL which is not centered on the ship?

- **-Point to point range and bearing**
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When Radar is connected to the ECDIS, radar image is shared as well as communication included are route, user data to radar, cursor and?

- **-Targets transmission to the ECDIS.**
- -Alarm Transmission to the ECDIS.
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Where can the content of the console be configured?

- **-From the Option menu.**
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Where do you find the functions for install charts from a CD?

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Which optional sensor gives out Navigational warning and messages to ECDIS workstations?

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Which parameter may trig anti grounding alarms by the ENC?

- **- Safety height, safety contour and safety depth**
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Which sensor on the Monitoring Station can be selected as the Master or as the common reference point for all ECDIS?

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You may copy the complete chart database to other ECDIS workstations by?

- **-Copy database in Chart handling.**
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4011 – Consilium ECDIS

Version 2.0 (671)

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You may copy the complete chart database to other ECDIS workstations by?

- **-Copy database in Chart handling.**
- -Share database in Chart handling.
- -Transfer database in Chart handling.
- -Transmit database in Chart handling. Wrong

4012 – Kelvin Hughes ECDIS

Version 1.0 (1663)

Is the following statement true or false? 'All optional features are available with the MantaDigital product for twelve months after purchase'.

- **False.**
- True

Is the following statement true or false? 'It is not possible to have a route composed of both rhumb line and great circle legs'.

- **False**
- True

Select one of the given options to best describe why the safety depth figure cannot be seen in the 'chart settings' menu.

- **The chart soundings are switched off.**
- The chart soundings are higher than the safety depth figure.
- The chart soundings are lower than the safety depth figure.
- There is no safety depth figure set.

Select one of the given options to best explain the function of the following icon.

- **It is for configuring chart settings.**
- It is for activating the docking display.
- It is for own-ship vessel predictor and past track.
- It is for seeing own-ship sensor data

Select one of the given options to best explain the function of the following icon.

- **It is for information on the loaded route.**
- It is for modifying route waypoints.
- It is for own-ship vessel predictor and look ahead.
- It is for passage planning.

Select one of the given options to best explain the function of the following icon.

- **It is for managing target data.**
- It is for changing own-ship vessel predictor from true to relative.
- It is for changing the length of the own-ship vector.
- It is for inserting lines of position.

Select one of the given options to best explain the function of the following icon.

- **It is for management and creating of user maps.**
- It is for configuration of chart settings.
- It is for importing chart update images.
- It is for inserting text and graphics.

Select one of the given options to best explain the function of the following icon.

- **It is for own-ship settings.**
- It is for radar and AIS target manipulation.
- It is for setting parameters for passage planning.
- It is for target settings.

Select one of the given options to best explain the function of the following icon.

- **It is for manual position fixing including line of position.**
- It is for creating user lines.
- It is for setting position for the docking display.
- It is for the navigation settings for own-ship.

Select one of the given options to best explain the function of the following icon.

- **It is for the management of mariner's notes and events.**
- It is for entering a manual fix position
- It is for the management of logbook recordings.
- It is for the management of text files.

Select one of the given options to best explain the function of the following icon.

- **It shows or hides the menu panel.**
- It is for entering a man overboard position.
- It is for moving to the next page of a menu.
- It shows or hides the 'other' symbols.

Select one of the given options to best explain the function of the following icon.

- **It is for displaying information on the man overboard position.**
- It allows recording of the man overboard position.
- It is for modifying a man overboard position.
- It is for sending the man overboard information to the GMDSS equipment.

Select one of the given options to best explain the function of the following icon.

- **It gives access to navigation tools.**
- It is for entering a manual fix position
- It is for the addition of automatic chart updates.
- It is for the addition of manual chart updates.

Select one of the given options to explain the easiest and quickest way to obtain a relevant MantaDigital handbook.

- **Obtain an electronic copy from the MantaDigital computer.**
- Ask the chart agent in the next port.
- Check on the ship's office computer as there will be one there.
- Contact Kelvin Hughes who will send one to the vessel.

Select one of the given options to explain the reason for the system alarm button on the control panel to be flashing red.

- **There is an unacknowledged system error.**
- An AIS target has infringed set limits.
- Some sensor data is corrupted.
- The displayed chart is at an incorrect scale.

Select one of the given options to explain what happens with the alarm window when an alarm has been acknowledged.

- **Active alarms scroll in the alarm window.**
- Active, acknowledged alarms are transferred to the 'alarm list' in a sub menu.
- The alarm button shows the alarm but turns a green colour.
- The alarm is removed from the window and cleared from the system.

Select one of the given options to explain where parameters are set for the CCRP.

- **In the set up utility using the 'own-ship' tab.**
- In ECDIS mode clicking on the 'navigation' icon.
- In ECDIS mode clicking 'own-ship' icon.
- In the set up utility using the 'firmware' tab.

Select one of the given options which describe the compound route function on the MantaDigital ECDIS.

- **The function allows changes in one route to be transferred to another.**
- The function allows a route to be constructed bit by bit at different times then joined together.
- The function allows the ECDIS to send a route to the auto-pilot to track steer.
- The function allows two or more routes to be displayed and monitored at once.

Select one option that correctly explains why the illustrated text from a drop down menu is shown in a black font.

- **The sensor is not available for use.**
- The sensor is available for use.
- The sensor is ground locked not sea locked.
- The sensor is not fitted to the vessel.

Select one option that correctly explains why the illustrated text from a drop down menu is shown in a white font.

- **The sensor is available for use.**
- The sensor is not available for use.
- The sensor is not fitted to the vessel.
- The sensor readings are outside set limits.

Select one option which correctly completes the following statement. 'A 'custom' display is . . .

- **. . . a chart display modified and saved for future use.'**
- . . . a chart display temporarily modified for use.'
- . . . ECDIS with a non approved ENC chart display'.
- . . . one that should never be used as it is not IMO approved'.

Select one option which correctly completes the following statement. 'In head-up or course-up mode the heading marker will reset itself . . .

- **. . . every one degree.'**
- . . . every five degrees.'
- . . . every ten degrees'.
- . . . on command from the operator'.

Select one option which correctly completes the following statement. 'The MantaDigital ECDIS when switched on from the standby screen . . .

- **. . . will automatically display a chart for the vessel's position.'**
- . . . will display the world chart and have to be zoomed in to the vessel's position'.
- . . . will need a chart to be selected and displayed for the vessel's position.'
- . . . will show the vessel's position on a small scale chart of the area it is in'.

Select one option which correctly completes the following statement. 'The MantaDigital computer is switched off . . .

- **. . . with a single push on the power button.'**
- . . . once a week at sea to clear temporary files'.
- . . . to clear irritating alarms'.
- . . . with a three second push on the power button.'

Select one option which correctly completes the following statement. 'The four digit product code . . .

- **. . . is used to release optional features.'**
- . . . is a code associated with the e-Token'.
- . . . is needed to order new charts.'
- . . . is the log-in code needed to start the ECDIS'.

Select one option which correctly completes the following statement. 'The route safety check . . .

- **. . . is also a manoeuvring limit and pilotage function check.'**
- . . . also checks for inserted mariners notes'.
- . . . covers dangerous objects in the chart dataset only.'
- . . . is only effective on rhumb line routes, not great circles'.

Select one option which correctly completes the following statement. 'When ticked, the 'Transfer' column in the alarm list . . .

- **. . . transfers the alarm to an external system.'**
- . . . transfers the alarm to a later time'.
- . . . transfers the alarm to the alarm log.'
- . . . transfers the cause of the alarm to the VDR'.

Select one option which correctly completes the following statement. 'With dangerous AIS targets the ECDIS . . .

- **. . . flashes their symbol and vector until acknowledged.'**
- . . . flashes their radar return as well as sounding an alarm.'
- . . . reverts to relative motion and relative vector'.
- . . . shows them the same as any other AIS targets'.

Select two of the given options which correctly identify why there is no tick seen in the route icon box.

- **The loaded route is not approved for use.**
- **There is no route loaded.**
- The loaded route has not been safety checked.
- This is normally how the icon looks with an approved loaded route.

Which one of the given options best describes the function activated when the word 'check' is clicked in the following illustration?

- **Navigation sensors one and two are displayed on the chart.**
- The position system datum is checked to see if it is the same as the chart datum.
- There is a validity check carried out on the primary position sensor.
- There is a validity check carried out on the secondary position sensor.

Which one of the given options best describes the meaning of the green dot seen to the right of the chart scale in the following illustration?

- **The chart scale is the same as the compilation scale but set by the operator.**
- The chart scale is sufficient but a bit on the small size because of the small dot.
- The ECDIS will automatically set the scale to match the compilation scale.
- The ECDIS will automatically set the scale to match the vessel position.

Which one of the given options best describes the meaning of the orange dot seen to the right of the chart scale in the following illustration?

- **The chart scale is outside two scales of the compilation scale.**
- The chart scale is outside three scales of the compilation scale.
- The dots give warning that 1:20.000.000 is far too big a scale.
- This is an alarm as the chart scale is insufficient and should be changed.

Which one of the given options best describes the meaning of the yellow dot seen to the right of the chart scale in the following illustration?

- **The chart scale is within two scales of the compilation scale.**
- The dot gives warning that 1:20.000.000 is far too small a scale.
- The ECDIS has automatically set the scale for a world chart.
- This is an alarm as the chart scale is insufficient and should be changed.

Which one of the given options best describes the meaning of 'true motion best' with reference to modes of display?

- **True motion with maximum look ahead.**
- True motion is preferred to relative motion.
- True motion with minimum look ahead.
- True motion with the best chart detail.

Which one of the given options best describes the use of the function available on the left mouse button seen in the following illustration?

- **When clicked a menu allows a position to be entered for the chart to be centered on.**
- When clicked a menu allows a position to be entered for the next waypoint on a route.
- When clicked a menu allows a position to be entered for the vessel to be centered on.
- When clicked a menu allows a position to be entered for viewing the next waypoint along a route.

Which one of the given options best describes why some of the fonts in the drop down sensor box are black?

- **The sensor is configured but not available.**
- The sensor has malfunctioned and is not available.
- The sensor is not fitted to the vessel.
- The sensor is outside set limits but available for use.

Which one of the given options best describes why the 'set chart' button has a magenta coloured background?

- **A 'base' chart display has been selected.**
- A 'standard' chart display has been selected.
- The ECDIS is being updated with chart corrections.
- There are manual updates to be seen on the loaded chart.

Which one of the given options best explains the function of the floating cursor.

- **The cursor button functions are displayed alongside the cursor position.**
- The cursor button functions are displayed alongside the operation panel.
- The cursor button functions are displayed at the bottom right hand part of the screen.
- The cursor button functions are displayed at the top right hand side of the screen.

Which one of the given options best explains the significance of the absence of a tick alongside 'Mariner Events – info' seen in the following illustration?

- **The note is not displayed on the chart.**
- The note is being edited on the secondary ECDIS.
- The note is displayed on the chart.
- The note is to be erased from memory on next ECDIS restart.

Which one of the given options correctly describes the significance of the two black dots seen on the line in the following illustration?

- **They are the start and end points for the turn.**
- They are time intervals on the heading line.
- They are undersea manifolds on a pipeline.
- They mark the position of manually updated objects.

Which one of the given options correctly describes the use of a 'critical point' on a route?

- **It is an alerting point on a route with an operator alarm.**
- It is a point in the alarm log where a system alarm sounded.
- It is a point where manual chart updates have been added.
- It is a waypoint on a track control route.

Which one of the given options correctly explains where information on the e-Token number can be found?

- **On the 'chart licence' tab in chart maintenance.**
- On the front of the MantaDigital computer.
- On the 'install log' tab of chart maintenance.
- On the 'system set-up' page.

Which one of the given options correctly explains why two routes are visible in the following illustration, the upper one being a red colour the lower one an orange colour?

- **The red, alternative route is being monitored instead of the initial orange route.**
- The orange route is one that is being planned whilst the red one is being monitored.
- The orange, alternative route is being monitored.
- The red route is part of the orange compound route.

Which one of the given options is correct in relation to the following statement? 'A monitored route is potentially dangerous if modified and should not be used'.

- **The statement is correct unless it is safety checked.**
- The statement is correct as the ECDIS software would not now allow it to be modified.
- The statement is incorrect as a competent navigator would always plan a safe route.
- The statement is incorrect as the ECDIS software would automatically detect any danger.

Which two of the given options correctly describe parameters used to calculate the path predictor?

- **It is calculated from course and speed over the ground.**
- **It is calculated from rate of turn and water flow.**
- It is calculated from course and speed through the water'.
- It is calculated from rate of turn and wind force.

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A route name is displayed in a red font does this mean the route is invalid? Select one option to correctly answer this question.

- **No – route names always have red fonts.**
- No it means it has not been safety checked.
- Yes – the route should not be used.
- Yes – there must be safety issues associated with it.

Click and drag either true or false to answers the following statements related to own-ship and target vectors on the ECDIS.

- It is only possible to select the speed over ground vector. **FALSE**
- The selection of target ship's vector type influences the choice of own ship vector in order to prevent misunderstanding. **TRUE**

Click and drag either true or false to answers the following statements related to own-ship and target vectors on the ECDIS.

- A target ship's vector is shown as an arrow. **FALSE**
- It is not possible to individually set the radar target vector and the target's AIS vector. **TRUE**

Click and drag the statements related to displays on the ECDIS into either the true or false box.

- It is possible to choose chart objects one by one in a 'Custom' display. **TRUE**
- The 'Base' display includes sufficient information for navigation purposes because it is defined in IMO performance standard. **FALSE**

Click and drag the statements related to displays on the ECDIS into either the true or false box.

- ECDIS has 3 types of display object sets. **FALSE**
- It is possible to choose chart objects one by one in a 'Custom' display. **TRUE**

Is the following statement true or false? 'A route waypoint can be assigned as an anchor position for ETA purposes'.

- **True**
- False

Is the following statement true or false? 'It is possible to export or import files of 'NavLine and No-Go Areas'.

- **True**
- False

Is the following statement true or false? 'It is possible to have only official ENC charts displayed on the ECDIS screen'.

- **True – if the operator chooses this.**
- False – all types of charts are needed to give coverage.

Is the following statement true or false? 'It is possible to synchronise files between connected primary and backup ECDIS computers using the ECDIS software synchronise command'.

- **True – if the operator chooses this.**
- False –files have to be loaded individually on each computer.

Is the following statement true or false? 'Latitude and longitude values can be used to adjust a waypoint position'.

- **True**
- False

Is the following statement true or false? 'The 'Guard Frame' or danger detection system is always active on the ECDIS in monitor mode.

- **False**
- True

Is the following statement true or false? 'When a Doppler log is fitted to a vessel the 'speed through the water' button on the ECDIS will always indicate water speed'.

- **False.**
- True.

Is the following statement true or false? 'When route monitoring is being executed and the operator edits the active route, it is possible to store the modified route in the ECDIS'.

- **False**
- True

Is the following statement true or false? 'When route monitoring is being executed it is not possible to edit the attributes of the route'.

- **True**
- False

Is the following statement true or false? 'When route monitoring is being executed it is not possible to numerically edit all of the waypoints'.

- **True**
- False

Is the following statement true or false? 'When route monitoring is being executed it is possible to edit all waypoint information'.

- **False**
- True

Select one of the given options to correctly complete the following statement. 'A 'Custom' display . . .

- **. . . allows the operator to choose objects to be displayed'.**
- . . . allows the operator to manipulate the detail of charted objects'.
- . . . is always a dangerous type of display as objects are missing from the chart'.
- . . . is one that is only used in a pilotage area'.

Select one of the given options which best describes the reason to view the 'Self Test' results.

- **The results show the status of the whole system.**
- The self test results are recorded as part of the SMS.
- The self test results show how accurate the navigation signal is.
- Tokyo Keiki engineers want to regularly see the self test results.

Select one of the given options which best describes where information can be found on the version of the IHO 'Presentation Library' used by the ECDIS

- **Press the 'Maintenance' button and view the 'Version' tab.**
- Consult the user handbook where the information is stored.
- Go to the standby screen of the ECDIS and view the software version.
- Press the 'Chart' button and then press the version tab for information.

Select one of the given options which best describes why a sensor signal message shows in a red font in the 'Maintenance' window.

- **That message has an error and is rejected by the ECDIS.**
- That message is in normal condition as they are always red.
- That message shows the data to be in error but within set limits.
- The message has an error but is still used by the ECDIS.

Select one of the given options which best describes why there are vertical lines displayed on the sea areas of the chart as seen in the following illustration.

- **The chart is set over the compilation scale.**
- The chart permit has expired.
- The lines indicate the chart boundary.
- There is an unofficial chart loaded.

Select one of the given options which correctly completes the following statement. 'A fixed, off centre display on the ECDIS . . .

- **. . . is only possible in relative motion'.**
- . . . is only possible in true motion'.
- . . . is only possible on a radar set'.
- . . . is only possible with a custom display'.

Select one of the given options which correctly completes the following statement. 'Setting a realistic underkeel clearance in the alarm setting menu . . .

- **. . . allows the ECDIS to warn if shallow water is encountered'.**
- . . . allows the ECDIS to warn if deep water is encountered'.
- . . . is only possible when inside a pilotage area'.
- . . . stops the ECDIS alarm'.

Select one of the given options which correctly completes the following statement. 'To obtain information on the chart cell being displayed . . .

- **. . . press the information icon then press again with the cursor on the chart'.**
- . . . check the chart list on the planning computer'.
- . . . press the information icon on the toolbar'.
- . . . press the information icon then open up the 'chart' menu'.

Select one of the given options which correctly describes the easiest place to find the value to the safety contour.

- **On the top right hand side of the screen.**
- In the 'chart' tab.
- In the 'own-ship' tab.
- On the bottom right hand side of the screen.

Select one of the given options which describe what happens to the ECDIS if the mains AC power fails.

- **After 60 seconds the ECDIS shuts down automatically if DC power is available.**
- After ECDIS has shutdown, if AC power is restored, the ECDIS will re-start automatically.
- All functionality is immediately maintained by DC power and the ECDIS continues to work as normal
- The LCD monitor continues working although AC power is lost.

Select one of the given options which explains the meaning of the GPS sensor source name being in a red font

- **The sensor data is invalid.**
- The sensor data is corrupted.
- The sensor data is doubtful.
- The sensor data is normal.

Select one of the given options which explains where the 'Chart Object' button is located.

- **On the tool bar (other buttons).**
- On the operation panel.
- On the setting menu.
- On the status bar.

Select one of the given options which is FALSE regarding the safety contour.

- **The safety contour setting is always same as safety depth value.**
- It is used for a route check by the guard frame
- It is used for danger detection by the route check functionality.
- The value of the safety contour depends on chart data.

Select one of the options to correctly identify the chart object seen in the following illustration.

- **It is a manually inserted update.**
- It is a symbol for a flare tower.
- It is a symbol for a tree in a mangrove swamp.
- It is an automatically inserted update.

Select one of the options to explain the correct stages of loading C-Map Jeppesen charts onto the ECDIS computer.

- **Install the chart cells from the supplied disc then the chart license.**
- Both sets of files load together from the 'D' and 'E' media drives.
- First install the chart cell license then the cell files.
- It does not matter which files are loaded first.

Select one of the options to explain the correct stages of loading official ENCs onto the ECDIS computer.

- **First install the chart cell permits then the cell files.**
- Both sets of files load together from the 'D' and 'E' media drives.
- First install the chart cell files then the cell permits.
- It does not matter which files are loaded first.

Select one of the options to explain why the background colour of the safety contour button at the top right hand side of the screen is black.

- **It indicates there is an ARCS chart loaded.**
- It indicates that no charts are installed.
- It indicates there is a C-Map chart loaded.
- It indicates there is an official ENC chart loaded.

Select one of the options to explain why the background colour of the safety contour button at the top right hand side of the screen is green.

- **It indicates there is a C-Map chart loaded.**
- It indicates that no charts are installed.
- It indicates there is an ARCS chart loaded.
- It indicates there is an official ENC chart loaded.

Select one option that best explains why there is a large, black cross between waypoints 1 and 2 in the following illustration.

- **A waypoint is to be inserted.**
- It is a mariner note indication.
- The leg has been selected to set limits.
- The leg is to be deleted.

Select one option that best explains why waypoint 2 in the following illustration has a large black cross over the top of it.

- **It is attached to the cursor.**
- It has been selected to set limits.
- It is the next waypoint on the route.
- It is to be deleted.

Select one option that correctly describes how wheel over points are represented on the route legs.

- **As a small orange circle or line.**
- As a red cross.
- As a red line and circle.
- As text.

Select one option that correctly describes the colour of a route being modified on the chart screen.

- **Red.**
- Blue.
- Orange.
- Yellow.

Select one option that correctly describes the colour of a route being monitored.

- **Orange.**
- Blue.
- Red.
- Yellow.

Select one option that correctly describes the use of a route 'Limit' check.

- **Course turns are checked to see if they are possible.**
- A route is checked for dangerous objects in its path.
- A route is checked to see if any depth limits are infringed.
- The route log file is checked for space remaining.

Select one option that correctly describes the 'Guard Frame' function on the ECDIS.

- **It is a danger detection system.**
- It is a function used to protect against computer virus.
- It protects the ECDIS from loading old versions of the chart files.
- It protects the power button from being accidently pressed.

Select one option that correctly identifies approximately how long a route is made invisible when the 'Route Display Off' button is clicked.

- **10 seconds.**
- 1 second.
- 15 seconds.
- 5 seconds.

Select one option that correctly identifies how a mark is identified when the 'Edit' button is pressed in the 'NavLine and No-Go Area line' menu.

- **A red dot appears on the mark.**
- A black cross appears on the mark.
- An orange dot appears on the mark.
- The mark flashes on the chart screen.

Select one option that correctly identifies how long the 24 hour log is stored on the system.

- **Ten days.**
- Indefinitely.
- One day.
- Three months.

Select one option that correctly identifies how long the 3 month log is stored on in active memory on the system.

- **Three months.**
- 12 months.
- Indefinitely.
- Thirty days.

Select one option that correctly identifies the black dot seen in the following chart illustration.

- **A text object will be added in this position.**
- It is a simplified buoy object.
- It shows the position of a seabed obstruction.
- This is the cursor symbol when off the screen border.

Select one option that correctly identifies the duration of the ECDIS log recordings.

- **24 hours and 3 months.**
- 12 hours and 1 month.
- 12 hours and 3 months.
- 24 hours and 1 month.

Select one option that correctly identifies the function of the 'Show History' button found in the 'Navigation Record' menu.

- **The function enables history files to be seen.**
- The current own-ship track can be seen.
- The function enables the contents of history files to be seen.
- The function shows the alarm log history.

Select one option that correctly identifies the meaning of the outlined shape with cross hatched infill shown in the following chart illustration.

- **This is a shape representing a No-Go area.**
- This is a chart feature showing a cautionary area.
- This is a shape representing a precautionary area.
- This is a shape representing an anchorage area.

Select one option that correctly identifies the reason for the arrow and red circle surrounding the pilot ground symbol seen in the following illustration.

- **They identify the specified mark.**
- The arrow and red circle mean a manual chart update.
- The circle indicates an area of uncertainty.
- They identify the inserted symbol as originating from a Notice to Mariners.

Select one option that correctly identifies the reason why the letter 'R' appears alongside the data on the ECDIS screen as seen in this illustration.

- **There is a 24 hour playback file running.**
- The radar is connected to the ECIDS.
- There is a 3 month playback file running.
- There is a radar range and bearing been used to position the vessel.

Select one option that correctly identifies the synthetically generated details available when the 'NavLine and No-Go Area Line' button is pressed.

- **Marks, remarks, text, lines and areas.**
- Marks, line and areas.
- Marks, remarks, symbols, text, lines and areas.
- Marks, remarks, updates, text, lines and areas.

Select one option that correctly identifies the 'Modify Metric' function available from the toolbar icon.

- **It is a function allowing changes to be made to NavLines and No-Go Areas.**
- It is a function that changes units of distance.
- It is a function that modifies route waypoints.
- It is a function which deletes NavLines and No-Go Areas.

Select one option that correctly identifies why the no-go line has a red dot at the vertex as shown in the following chart illustration.

- **The line is in edit mode.**
- It is a port hand buoy used to drop the line ends on.
- This is an area created between two lines.
- This is the normal appearance of two joined lines.

Select one option which correctly describes a limitation of a backup ECDIS connected to the primary ECDIS when the primary ECDIS fails.

- **The backup ECDIS does not take over route monitoring or track control from the primary ECDIS.**
- The backup ECDIS has no limitation as it takes over all functions that the primary ECDIS was undertaking.
- The backup ECDIS is usually located in an awkward position making it difficult to navigate with.
- The backup ECDIS takes a long time to start up as it is usually powered off when the primary ECDIS is running.

Select one option which correctly describes the meaning of the abbreviation CCI in relation to a route plan.

- **Course change indication.**
- Compass course impossible.
- Compass course indication.
- Course control issue.

Select one option which correctly describes the meaning of the abbreviation ECCI in relation to a route plan.

- **Early course change indication.**
- Early compass course indication.
- ECDIS control condition issue.
- Estimated course change indication.

Select one option which correctly describes the number of maximum tracked AIS targets possible with the ECDIS.

- **400**
- 300
- 350
- 450

Select one option which correctly explains when the 'Integrity Monitoring' warning would be raised.

- **When the ECDIS cannot compare primary and secondary sensor.**
- When the ECDIS has lost a primary sensor.
- When the ECDIS has lost a secondary sensor.
- When the primary sensor has exceeded limits

Select the one option which best completes the following statement. 'A backup ECDIS is . . .

- **. . . required to maintain continuity of electronic navigation'.**
- . . . always required on a ship with ECDIS'.
- . . . never connected to the primary ECDIS'.
- . . . required to maintain continuity of navigation'.

Select the one option which correctly completes the following statement. 'A blue frame around position sensor information indicates . . .

- **. . . it is the secondary sensor information'.**
- . . . it contains doubtful data.'
- . . . it is good quality data'.
- . . . it is the primary sensor information'.

Select the one option which correctly completes the following statement. 'A navigation signal mode letter shown as 'D' indicates . . .

- **. . . differential position information'.**
- . . . a dead reckoning position'.
- . . . a degraded level of position information'.
- . . . a doubtful position.'

Select the one option which correctly completes the following statement. 'A sleeping AIS target is . . .

- **. . . activated using the icon on the tool bar'.**
- . . . a vessel at anchor'.
- . . . activated from the 'targets' settings menu'.
- . . . is not dangerous to own-ship as there is no vector'.

Select the one option which correctly completes the following statement. 'A temporary waypoint . . .

- **. . . is set up on a route leg for a vessel to join the route'.**
- . . . is a waypoint assigned to the pilot pick up point'.
- . . . is a waypoint used for an anchor position'.
- . . . is set up for a man overboard incident'.

Select the one option which correctly completes the following statement. 'If log records are required to be permanently kept . . .

- **. . . they need to be recorded into the archive'.**
- . . . they need to be saved onto the ECDIS computer'.
- . . . set the ECDIS software to enable this'.
- . . . the ECDIS does this automatically'.

Select the one option which correctly completes the following statement. 'If target vectors are ground based . . .

- **. . . own-ship's ground vector will always be displayed'.**
- . . . it increases risk of collision'.
- . . . own-ship is able to have any type of vector selected'.
- . . . own-ship's vector will not be displayed'.

Select the one option which correctly completes the following statement. 'If the own-ship guard frame touches a no-go line . . .

- **. . . an alarm will sound'.**
- . . . an ECDIS indication is given'.
- . . . nothing happens'.
- . . . the line is removed from the chart screen'.

Select the one option which correctly completes the following statement. 'Inserted remarks are interrogated for information . . .

- **. . . by using the toolbar information icon'.**
- . . . by double clicking the remark'.
- . . . by opening the 'Route Monitoring and TC' menu'.
- . . . by opening up the help menu on the operation panel'.

Select the one option which correctly completes the following statement. 'Once the power switch on the front panel is on, the ECDIS is started . . .

- **. . . with a three second push of the power button on the operation panel.'**
- . . . automatically after the computer does the self test'.
- . . . by inserting a start-up disc in the media drive'.
- . . . with a one second push of the power button on the operation panel.'

Select the one option which correctly completes the following statement. 'The Tokyo Keiki ECDIS will automatically load . . .

- **. . . an official ENC if one is available'.**
- . . . a C-Map chart in preference to all other types of chart'.
- . . . an ARCS chart unless the operator decides otherwise'.
- . . . no chart until the operator selects one'.

Select the one option which correctly completes the following statement. 'To be able to safely monitor a planned route . . .

- **. . . safety check it and press the 'Start' button in the 'Route Monitoring and TC' menu'.**
- . . . check with the master to see that it has been correctly planned'.
- . . . click the start button then the 'Route Monitoring and TC' menu'.
- . . . it should have a limit check first'.

Select the one option which correctly completes the following statement. 'When placed on a route being monitored, the cursor . . .

- **. . . shows the ETA along the route to the cursor position.'**
- . . . opens up the route waypoint table'.
- . . . shows the cursor position only in latitude and longitude'.
- . . . shows the direct line ETA from the own-ship to the cursor position'.

Select the one option which correctly completes the following statement. 'When placed on the chart away from a route being monitored, the cursor . . .

- **. . . shows the direct line ETA from the own-ship to the cursor position.'**
- . . . shows information on the chart cell in that position'.
- . . . shows the cursor position only in latitude and longitude'.
- . . . shows the ETA along the route to the cursor position'.

Select the one option which correctly completes the following statement. 'When target association is active . . .

- **. . . there is a circle around the target on screen'.**
- . . . the own-ship is better protected from collision risk'.
- . . . the target shows the AIS symbol only'.
- . . . the target shows the radar return only'.

Select the one option which correctly describes the meaning of the expression 'Free Motion' when related to an ECDIS mode.

- **In free motion the own-ship symbol leaves the chart screen.**
- In free motion the own-ship symbol is always on the chart screen.
- In free motion the own-ship symbol is stopped on the chart screen.
- In free motion the target symbols are not visible.

Select the one option which correctly describes the meaning of the 'Range Filter' associated with target vessels.

- **The range filter allows limits for sleeping AIS targets to be set.**
- The range filter allows limits for all active AIS targets to be set.
- The range filter allows limits for displayed radar targets to be set.
- The range filter allows limits for setting target vectors.

Select the one option which correctly describes what happens when the 'Alarm' button, located at the top of the screen, is pressed.

- **The active alarms, warnings and indications can be viewed.**
- The active alarms are cancelled with audible alarm stopped.
- The alarm log is cleared into the archive memory.
- The audible alarm is switched off.

Select the one option which correctly explains the meaning of target association.

- **Radar and AIS targets are merged.**
- Radar and AIS targets are shown by themselves on the screen.
- Radar track targets are given a greater priority over AIS targets.
- Radar track targets are given a lesser priority over AIS targets.

Select the one option which correctly explains the meaning of the following image.

- **A lost AIS target.**
- A dangerous AIS target.
- A sleeping AIS target.
- Merged radar and AIS target.

Select the one option which correctly explains the meaning of the following image.

- **A lost but dangerous target.**
- A safe AIS target.
- A sleeping target.
- Merged radar and AIS target.

Select the one option which correctly explains the significance of the letters 'A' and 'W' seen in the following illustration taken from the alarm list window.

- **'A' stands for alarm, 'W' for warning.**
- 'A' stand for active, 'W' for warning.
- 'A' stands for alarm, 'W' for waypoint.
- 'A' stands for archived, 'W' for waiting.

Select the one option which correctly identifies the following icon seen on the toolbar at the top of the screen.

- **It is the information icon.**
- It is an icon that delivers alarm information.
- It is an icon to enable chart selection.
- It is the cautionary area icon.

Select the one option which correctly identifies the following icon seen on the toolbar at the top of the screen.

- **It is the chart scroll icon.**
- It is an icon to enable increase or decrease the chart scale.
- It is an icon to increase or decrease the screen brilliance.
- It is the icon that allows a table to be scrolled.

Select the one option which correctly identifies the following icons seen on the toolbar at the top of the screen.

- **They are the zoom in and out icons.**
- They are the icons to increase and decrease display contrast.
- They are the icons to increase or decrease the chart scale.
- They are the icons to select C-Map or an ENC.

Select the one option which correctly identifies the function of the following icon seen on the 'Object Info' window when pressed.

- **It highlights the selected chart object.**
- It is a cautionary indication for a missing chart object.
- It is a hand icon.
- When pressed it removes the selected chart object from the 'Custom' display.

Select the one option which correctly identifies the presence of the yellow colour seen in the following window on the chart screen.

- **Yellow indicates a warning.**
- Yellow indicates a measurement.
- Yellow indicates an alarm.
- Yellow indicates an alert.

Select two of the given options which best describe the reasons for a 'Fatal Error' report.

- **It is a report sent to the company to order a visit from a support engineer.**
- **It is a report sent to Tokyo Keiki if an ECDIS fails to start.**
- It is a report comprising of operator errors sent to the company.
- It is a report that is stored with the navigational records.

Select two of the given options which best describe why the over scale 'prison bar' warning, as seen on the following illustration, should not be ignored.

- **The chart in this scale is enlarged and not suitable for navigation.**
- **The 'prison bars' always show when the display scale is larger than the chart compilation scale.**
- The 'prison bars' appear and disappear but the chart is useable at all times.
- The 'prison bars' show a dangerous sea area.

Select two options which correctly explains the meaning of a 'Sensor Timeout' alarm.

- **This occurs when a sensor cannot be detected by the ECDIS.**
- **This occurs when the sensor signal to the ECDIS is abnormal.**
- This occurs when the sensor has been de-selected by the operator.
- This occurs when two connected navigation sensors show different positions.

When the ECDIS is running what position should the 24 volt switch be in? Select one of the given options.

- **The 24 volt switch should be in the 'on' position.**
- The 24 volt switch should be in the 'off' position.
- The 24 volt switch should be in the 'stand-by' position.
- The position of the 24 volt switch is irrelevant as it is always connected.

Which of the given options describes where the four ECDIS colour table settings can be found?

- **On the icon at the top left side of the screen.**
- In the ECDIS settings menu.
- On the icon at the bottom of the screen.
- On the right hand side of the operation panel.

Which one of the given options best describes how to remove radar and AIS targets from the screen?

- **Use the TT and AIS buttons on the operation panel.**
- Click on the screen display over the target.
- Use the 'own-ship' settings tab and targets on/off.
- Use the 'targets' settings tab and targets on/off.

Which one of the given options best describes the information given in the man overboard window?

- **Co-ordinates of and range and bearing to the position.**
- Initially the water track of the man overboard.
- Range and bearing only.
- The time of the man overboard incident.

Which one of the given options best describes the values a prediction line is based on?

- **Vessel ground speed and rate of turn.**
- Vessel ground speed and acceleration.
- Vessel water speed and acceleration.
- Vessels water speed and rate of turn.

Which one of the given options best describes where the tool bar is found on the ECDIS screen?

- **At the top of the screen.**
- At the bottom of the screen.
- At the left side of the screen.
- At the right side of the screen.

Which one of the given options best describes why set and drift should be entered in the man overboard window?

- **To improve the estimated position of the person.**
- To enable the vessel to return to the ground based position of the man overboard incident.
- To ground track the ECDIS.
- To ground track the radar.

Which one of the given options best describes why the 'position' button is in a red font as shown in the following illustration?

- **There is an offset applied to the receiver position.**
- The GPS data is invalid.
- The GPS data is not being used to position the vessel.
- The incoming GPS signal has a different datum other than WGS84.

Which three of the given options are correct with reference to system data?

- **System data in a green font means the data is good.**
- **System data in a red font means the data is invalid.**
- **System data in a yellow font means the data is doubtful.**
- System data in a green font means the data is 100% reliable.

With an executed route when might a 'Course Difference' alarm sound? Select one option.

- **When the route leg and gyro heading differ by a set amount.**
- When the gyro compass and magnetic compass differ by ten degrees.
- When the vessel is approaching a waypoint.
- When the vessel is sufficiently off track.

4014 – JRC ECDIS

Version 6.0 (287)

Click and drag one of the given options to correctly complete the following statement:

- To be able to see an event mark, the chart

  . . .
  setting must be within the range entered in
  the event mark 'location/attributes'. **scale**

Drag and drop one of the given options to correctly complete the following statement:

- With a loaded route the next waypoint is shown in a

  . . . colour. **blue**

Is the following statement true or false? 'The JRC ECDIS can be set to alarm if an AIS target infringes set limits'.

- **True**
- False

Is the following statement true or false? 'An event mark will always be seen on the chart display'.

- **False**
- True

Is the following statement true or false? 'Manual chart updates can only be installed with the ECDIS in 'Chart manager' mode'.

- **False**
- True

Is the following statement true or false? 'The ECDIS automatic route safety check is sufficient to see all dangers with the planned route'.

- **False**
- True

Is the following statement true or false? 'The size of the text fonts describing a lighthouse can be changed'

- **True**
- False

Is the following statement true or false? 'The volume of the alarm cannot be altered.'

- **False.**
- True.

Is the following statement true or false? 'There is only one button to stop the ECDIS alarm'.

- **False**
- True

Select an option that correctly completes the following statement. 'If there is no hardware key board fitted . . .

- **. . . the onscreen key board will be used'.**
- . . . it is not possible to enter text'.
- . . . one has to be obtained and connected to the ECDIS'.
- . . . only enter numbers using the up and down range button'.

Select an option that correctly explains the function of the 'HOME' button on the operation panel.

- **Pressing the button returns the ship symbol to the chart screen.**
- Pressing the button centres the VRM and EBL.
- Pressing the button locates the chart in the home port.
- Pressing the button returns the cursor to the centre of the display.

Select from the given options the correct meaning of the symbol in the illustration.

- **A manually inserted light vessel.**
- A manually deleted light vessel.
- An automatically deleted light vessel.
- An automatically inserted light vessel.

Select one of the given options that correctly complete the following statement. 'For full danger protection a vessel . . .

- **. . . should have both the danger detection vector and sector active'.**
- . . . should have both the danger detection vector and sector displayed'.
- . . . should have the danger detection sector displayed'.
- . . . should have the danger detection vector displayed'.

Select one of the given options that correctly describe how the depth values are shown when they have a value below the safety depth setting.

- **Values are shown in a bold font.**
- Values are not shown at all.
- Values are shown in a red coloured font.
- Values are shown in the same way as any depth value.

Select one of the given options that correctly explain the use of the 'auto sail' settings when the vessel is under track control (TCS).

- **'Auto sail' fine tunes the steering control settings for the conditions.**
- Clicking 'auto sail' activates the TCS function.
- Clicking 'auto sail' disables the TCS function on the ECDIS.
- 'Auto sail' fine tunes how the engine will be set on each leg of the route.

Select one of the given options that correctly explain what the red circle surrounding the vessel symbol represents on the following illustration.

- **The red circle represents the anchor swinging circle.**
- The red circle causes an automatic target detection alarm.
- The red circle represents a Racon in the area.
- The red circle represents the danger detection sector as a full sector.

Select one of the given options that correctly explain why the red anchor swinging circle in not centred on the vessel symbol but offset a small distance.

- **The circle is based on the position of the anchor not the CCRP.**
- The circle has been placed in an incorrect position on the vessel.
- The circle is based on an earlier position and the vessel has dragged anchor.
- The circle is based on the CCRP not the position of the anchor.

Select one of the given options that correctly explains the 'associate' function on the target menu

- **It allows the merging of AIS and radar targets**
- It associates a radar target with a VHF radio transmission.
- It attaches a radar target to a specified route for it.
- It distinguishes between a moving and static radar target.

Select one of the given options that would activate the danger detection function on the JRC ECDIS.

- **Ensure that there is a tick against 'vector' in the alarm setting menu.**
- Ensure that the ship vector is set to a long time period.
- Ensure that there is a tick against 'display vector area' in the alarm setting menu.
- Ensure there is a tick against 'detect danger' in the alarm setting menu.

Select one of the given options to correctly complete the following statement. 'On the 901B and 701B models the normal setting on the drive organisation tab for the 'D' and 'E' drives is to . . .

- **. . . select dual drive'.**
- . . . select 'D' as priority over 'E' drive'.
- . . . select 'D' drive only'.
- . . . select 'E' drive only'.

Select one of the given options to correctly complete the following statement. 'When a pre-planned route is loaded it . . .

- **. . . is automatically in monitor mode'.**
- . . . has to be set into monitor mode before use'.
- . . . may not have been safety checked previously.'
- . . . will always be safe.'

Select one of the given options to correctly describe how to set the ECDIS to be able to see the Admiralty Information Overlay on the display.

- **Select 'other' as the display type.**
- Select 'AIO' as the display type.
- Select 'base' as the display type.
- Select 'standard' as the display type.

Select one of the given options to correctly explain where the man overboard button is located.

- **On the menu and the operation panel.**
- On the menu panel only.
- On the operation panel and the drop down menu.
- On the operation panel only.

Select one of the given options to correctly explain why an event mark may not be seen on the chart display.

- **It is invisible at the selected chart scale.**
- Because the font colour is incorrect.
- The chart is not set at the compilation scale.
- The time of the event is outside the four hour time limit.

Select one of the given options to correctly identify the function of the following icon seen on the chart information part of the menu panel.

- **When clicked it opens the event mark menu.**
- When clicked it opens the log menu.
- When clicked it opens the man overboard menu.
- When clicked it opens the no-go area menu.

Select one of the given options to correctly identify the function of the red 'pizza' shape in the illustration.

- **It is the danger detection sector.**
- It is a danger area on the chart.
- It shows the extent of the forward scanning depth sounder.
- It shows the vessel is entering a red sector of a navigation light.

Select one of the given options to correctly identify the two methods of ETA calculation used on the JRC ECDIS.

- **Using a set value or the actual speed of the vessel.**
- Using a set value or the average speed over the last four hours.
- Using the actual water speed of the vessel or the passage plan set value.
- Using the average speed or the vessel's maximum speed.

Select one of the given options to explain the use of the 'auto detect' button on the ECDIS chart portfolio menu.

- **When clicked the ECDIS looks for new charts or updates.**
- When clicked the ECDIS connects to the internet to download new charts.
- When clicked the ECDIS connects with the internet to download updates.
- When clicked the ECDIS looks for out of date charts on the 'D' drive.

Select one of the given options which best describes a 'normal' route on the JRC ECDIS.

- **It is one that is steered by the bridge team.**
- It is one following all normal planning conventions.
- It is one that is created as a rhumb line.
- It is one use for track control steering.

Select one option that correctly explains the meaning of the word 'hand' when seen on the menu panel.

- **The route is 'normal' being manually steered.**
- The autopilot has been disengaged, the wheel being used instead.
- The cursor has changed to a hand symbol to grab information.
- The route is a 'TCS' route.

Select one option to correctly explain why the over scale 'prison bars' warning can be seen on one half of a chart display only.

- **There are two chart cells loaded with different compilation scales.**
- Only half is used so that the operator can see detail on the other half.
- The warning is only displayed over land.
- There are both an ARCS and a vector chart loaded together.

Select one option which best explains the significance of the letter 'D' alongside EBL1 in the illustration.

- **The EBL is offset.**
- The EBL is positioned on Racon 'D'.
- The EBL is temporarily disabled.
- The EBL origin is not visible on the chart display.

Select one option which correctly completes the following statement. 'The ECDIS alarm is acknowledged . . .

- **. . . using either the button on the menu or on the operation panel'.**
- . . . automatically after a ten second timeout'.
- . . . using a button located on the chart screen'.
- . . . when the operator changes the item causing the alarm'.

Select one option which correctly completes the following statement. 'The offset VRM and EBL can be returned to the CCRP . . .

- **. . . by clicking on the letter 'D' in the 'tools' menu'.**
- . . . by clicking on the ship symbol'.
- . . . by clicking on the VRM and EBL intersection point'.
- . . . by clicking on 'reset' in the 'tools' menu'.

Select one option which correctly completes the following statement. 'User created areas . . .

- **. . . can be set up to highlight and detect dangers'.**
- . . . are not useful as they do not cause an alarm'.
- . . . create an alarm when radar targets are in them'.
- . . . only cause an alarm when the ship symbol is on them'.

Select one option which correctly explains why the alarm button located on the menu panel is in two parts.

- **To show both immediate and active alarms.**
- So that multi alarms can be acknowledged as fast as possible.
- To enable two alarms to be read by the operator at the same time.
- To have some redundancy when one alarm acknowledge button fails.

Select one option which correctly explains why the illustrated alert has been activated.

- **A chart object normally seen in a 'standard' display is removed.**
- Some synthetic graphics have been added to the chart display.
- The chart display has the passage plan active on it.
- The 'other' type of chart display has been selected.

Select one option which explains the use of the 'rotation' function with the ECDIS set in true motion, north up.

- **The function allows the north direction to be set on the screen.**
- The function changes the radar rotation speed seen on the ECDIS display.
- The function is the 'course up' reset value.
- The function when activated allows the ship's head to vary with the gyro.

Select the one option that best describes the 'priority' function seen on the AIS target menu of the JRC ECDIS.

- **When over 300 targets are visible, dangerous AIS targets are shown in preference to others.**
- AIS targets ahead of the vessel are given priority.
- AIS targets that are close to the vessel are given priority.
- The ECDIS takes the AIS position as priority over the radar position.

Select the one option that correctly completes the following statement. 'An AIS target will be left sleeping if . . .

- **. . . it does not infringe the set limits for AIS targets'.**
- . . . it is a class B AIS device transmitting'.
- . . . it is the give way vessel under the collision regulations'.
- . . . it outside a five mile radius'.

Select the one option that correctly explains the function of the 'PANEL' button seen on the operation panel.

- **The button dims the operation panel in five steps.**
- The button brings the menu panel on and off the screen display.
- The button dims the display screen in five steps.
- The button tests the panel illumination.

Select the one option that correctly explains the function of the 'VIDEO' button on the operation panel.

- **It adjusts the brilliance of displayed radar echoes.**
- It adjusts the display screen brightness.
- It adjusts the speed that the log video files playback.
- It overlays radar return from connected radar.

Select the one option which best completes the following statement. 'When importing routes from a USB memory stick . . .

- **. . . always make sure the stick is virus free'.**
- . . . always delete them from the stick after importing'.
- . . . always use a JRC approved memory stick'.
- . . . always use a USB 3.0 stick'.

Select the one option which best describes how the second EBL is distinguished from the first EBL.

- **The second EBL is of a dash-dot construction.**
- The second EBL is a dash construction.
- The second EBL is coloured green.
- The second EBL is coloured yellow.

Select the one option which best describes when the display is 'course-up' how the ship's head automatically resets towards the top of the screen.

- **A numerical value is input that once exceeded causes a reset.**
- The display has a default value that once exceeded allows a reset.
- The 'course up' display will reset every ten degrees of deviation.
- There is a button on the 'chart info' menu to manually reset the display.

Select the one option which best explains the function of the 'view sync' seen in the waypoint table.

- **When clicked, the chart screen locates at the waypoint position.**
- When clicked, all waypoints are set with the same cross track limit.
- When clicked, all waypoints are set with the same speed value.
- When clicked, the view is set from waypoint to waypoint sequentially.

Select the one option which best explains why a leg of the loaded route shows as an enlarged red line.

- **The safety check has identified a problem with it.**
- It is a route leg where the safety check cannot be made.
- It is the final leg of the passage plan.
- It is the next leg of the passage plan.

Select the one option which best explains why a section of a depth contour shows as an enlarged black line.

- **The safety check has detected a route conflict with it.**
- It is a section of contour with a doubtful value.
- It is the one hundred metre contour.
- The safety check identifies that it is in an incorrect position.

Select the one option which best explains why a waypoint detail has a red coloured font in the waypoint table.

- **The safety check has identified a problem with it.**
- It is a route leg where the safety check cannot be made.
- It is the final leg of the passage plan.
- It is the next leg of the passage plan.

Select the one option which correctly completes the following statement. 'Clicking on the 'alarm list' button . . .

- **. . . gives access to the alarm log and information or warnings'.**
- . . . gives access to the alarm log and system status only'.
- . . . gives access to the alarm log only'.
- . . . gives access to the current alarm only'.

Select the one option which correctly describes the normal appearance of the cursor.

- **A vertical cross shape (+).**
- A box shape.
- A circular shape.
- A diagonal cross shape (x).

Select the one option which correctly explains the use of the icon with the red exclamation mark seen in the illustration.

- **Clicking on the icon returns the display to 'standard'.**
- Clicking on the icon returns the display to 'other'.
- Clicking this icon activates the route.
- Clicking this icon temporarily removes synthetic graphics.

Select the one option which correctly explains why the position information, seen on the top right hand side of the menu panel, has a blue background.

- **A position offset has been applied to the GPS position.**
- The GPS input has failed to the ECDIS.
- The position fix is at the incorrect datum for the chart in use.
- There is a discrepancy between the two connected GPS receivers.

Select the option that correctly completes the following statement. 'The JRC web site should be . . .

- **. . . consulted regularly for ECDIS software updates'.**
- . . . checked regularly for chart updates'.
- . . . checked regularly for new editions of charts'.
- . . . consulted to download the public key'.

Select the option that correctly completes the following statement. 'The chart portfolio window gives access to . . .

- **. . . delete old versions of permits and load new ones'.**
- . . . delete chart updates once applied to the chart'.
- . . . view chart cells loaded and rename regular use charts'.
- . . . view the stored passage plan files'.

Select two of the given options that each correctly complete the following statement. 'The 'safety check' button . . .

- **. . . checks a planned route for safety'.**
- **. . . checks the route turning parameters of a TCS route'.**
- . . . checks the ECDIS for sensor integrity'.
- . . . checks the operation panel button illumination'.

Select two options that correctly complete the following statement. 'The EBL can . . .

- **. . . be set as 'true' or 'relative'.**
- **. . . be set on the CCRP or offset on the chart display'.**
- . . . be set in degrees or radians'.
- . . . be set in kilometres or nautical miles'.

Select two options that correctly complete the following statement. 'The VRM can . . .

- **. . . be set in kilometres or nautical miles'.**
- **. . . be set on the CCRP or offset on the chart display'.**
- . . . be set as 'true' or 'relative'.
- . . . be set in nautical or statute miles'.

The danger detection box is not visible surrounding the ship symbol in the illustration; does this mean danger detection is not active? Select one of the given options that correctly answer the question.

- **The danger detection may be still active.**
- The danger detection has temporarily been disabled.
- The danger detection is active because the vector is visible.
- The danger detection is not active.

When a route safety check is completed with an ECDIS set in 'base' display how does the check highlight a dangerous but invisible object? Select one of the given options.

- **A black triangle is placed in the position of the dangerous object.**
- It changes the display to 'standard' to display all objects.
- It does not highlight the dangerous object as it is not detected.
- It makes the dangerous object appear on the chart display.

Why do some items in the 'chart settings' menu have the letter 'O' against them? Select the one correct option from those given.

- **This indicates they are always seen on the 'other' display.**
- This indicates they are always seen on an 'open' display.
- This indicates they are an overlaid item.
- This indicates they are optional items.

4015 – SAM Electronics ECDIS CHARTPILOT 93x0/1100

Version 1.0 (2737)

'On a CHARTPILOT menu, sometimes when a value has been changed, the . . . button has to be pressed to confirm the change'. Select the ONE correct answer for the missing word.

- **'Apply'.**
- 'Change'.
- 'Enter value'.
- 'Insert'.

How can an operator gain information on the loaded chart cell? Select the ONE best answer.

- **Use a pick report and the cell history information.**
- Check in the chart cell catalogue.
- Consult a senior officer.
- Look for the cell number on the source data diagram.

How can an operator using the CHARTPILOT see the range of IHO chart symbols? Select the ONE best answer.

- **Use the 'Legend' tab on a pick report.**
- Connect to the IHO website.
- Use a small chart scale.
- Use the 'Symbols' tab on a pick report.

How is the variable range ring hooked onto the trackball for adjustment? Select the ONE correct answer.

- **By pressing on the range readout with the 'DO' button.**
- By double clicking on the screen adjacent to the range ring.
- By pressing on the range readout with the 'INFO' button.
- By pressing on the range ring seen on the screen.

In this illustration from the CHARTPILOT, why do some buttons have a little triangle on them? Select the ONE correct answer.

- **The triangle shows that there are subordinate menus available.**
- That pressing the button will toggle the function on the chart screen.
- That there is information available on a help menu.
- That this button can be pinned to the menu bar.

On a CHARTPILOT keyboard, what can help you remember what the special function keys do? Select the ONE best answer.

- **Attach a template to the keyboard.**
- Do not use, them, then they can be forgotten about.
- Post a list up on the bridge noticeboard.
- Write the function on each key.

On the CHARTPILOT ECDIS, what happens when the 'DO' button on the mouse is pressed? Select the ONE correct answer.

- **It enables a selected function.**
- Information on the objected pressed is given.
- Sub menus open giving more possibilities.
- The ECDIS changes into 'Planning' mode.

On the CHARTPILOT ECDIS, what happens when the 'INFO' button on the mouse is pressed? Select the ONE correct answer.

- **Information on the chart objects is given, in the chart area.**
- It enables a selected function.
- Sub menus open giving more possibilities.
- The ECDIS goes into standby.

On the CHARTPILOT ECDIS, what happens when the 'MORE' button on the mouse is pressed? Select the ONE correct answer.

- **Sub menus open giving more possibilities.**
- Information on the objected pressed is given.
- It enables a selected function.
- The ECDIS changes into 'Conning' mode.

On the CHARTPILOT screen, what shape is the available true motion screen area? Select the ONE correct answer.

- **Circular.**
- Polygon.
- Rectangular.
- Square.

Select the ONE answer that correctly completes the following statement. 'On the CHARTPILOT tracks are modified . . .

- **. . . using commands in the 'Tracks' 'Edit' menu'.**
- . . . by entering values from the keyboard'.
- . . . by using a digitising system on a connected CHARTPILOT'.
- . . . only after the CHARTPILOT has been put into planning mode'.

Select the ONE answer that correctly completes the following statement. 'The 'Calculator' function on the CHARTPILOT . . .

- **. . . allows navigational calculations to be made, including sailings'.**
- . . . allows the total distance on a track to be estimated'.
- . . . is a simple mathematical calculator'.
- . . . offers some estimate of the time needed to load charts from a base disc'.

Select the ONE answer that correctly completes the following statement. 'With the CHARTPILOT in planning mode . . .

- **. . . any chart can be browsed'.**
- . . . only the chart on which the vessel is positioned can be seen'.
- . . . only viewing small scale charts is possible'.
- . . . passage planning is the only possible activity'.

Select the ONE best answer to complete the following statement. 'When monitoring a track, an approaching waypoint alarm will sound . . .

- **. . . between one and six minutes before the vessel reaches the wheel over point'.**
- . . . always five minutes before the waypoint is reached'.
- . . . if the forward speed is too great compared to the plan'.
- . . . just before the actual waypoint position is reached'.

What colour is a 'System Track' when loaded and activated on the CHARTPILOT? Select the ONE correct answer.

- **Red.**
- Blue.
- Green.
- Orange.

What could be the reason that the anchor watch function will not activate? Select the ONE correct answer.

- **The vessel is not located within the assigned anchorage area.**
- The function is not available on the CHARTPILOT version.
- The vessel is in the anchorage area, but moving at above 2 knots.
- The vessel is on the wrong heading for the water flow.

What do the three full stops indicate on this CHARTPILOT button? Select the ONE correct answer.

- **That there is a subordinate menu available.**
- That the 'INFO' button on the mouse must be pressed.
- That the 'Presentation' menu is not now available.
- That there is no more information available.

What does the CHARTPILOT do after a track has passed a geometry and safety check? Select the ONE correct answer.

- **In a dialogue, it informs the operator with any relevant errors and warnings concerning the track with violation points.**
- Automatically store the track and close the track edit page.
- Set the newly created track as the 'System Track', to enable monitoring to take place.
- Sound an alarm to say the track is safe to use.

What does the following orange star shape indicate, when seen on an ENC loaded on the CHARTPILOT? Select the ONE correct answer.

- **There is a chart update at its centre.**
- The ENC cell is out of date.
- The navigation mark at its centre has been used for navigation.
- There is an object detected by the 'look ahead' detection system.

What does the letter 'V' indicate, when seen on the chart screen? Select the ONE correct answer.

- **The CHARTPILOT is in playback mode.**
- The CHARTPILOT has been paused, so monitoring has been suspended.
- The CHARTPILOT is running as a virtual machine.
- The ENC cell has not been updated for over a month.

What does 'Text Mode' allow the operator to do when passage planning? Select the ONE best answer.

- **It allows the input and adjustment of track parameters.**
- It allows the operator to change the track line style.
- It allows the operator to use text to enter a track.
- It replaces the mouse entry with the keyboard.

What function does the 'Compose' button have with open text mode on the CHARTPILOT? Select the ONE correct answer.

- **From a waypoint file, it enables a list of waypoints to be added to a track.**
- It adds text files to a track already saved.
- It allows the existing track to have its parameters adjusted.
- It allows the operator to manually insert more waypoints.

What function has the 'Check' button, when used on a created track? Select the ONE best answer.

- **Within the track limit, it verifies this track for geometry and safety with charted and user chart objects.**
- At a distance equal to twice the track limit, it verifies this track for objects in the way.
- It verifies a track for cross track distance values.
- Within the track limit, it verifies a track for turn sizes.

What happens to an alarm when the parameters causing the alarm are removed? Select the ONE correct answer.

- **The alarm is removed from the 'Alarm List'.**
- The alarm remains in the 'Alarm List' but the buzzer stops.
- The alarm remains in the 'Alarm List'.
- The operator is notified on the 'Alarm List'.

What happens to the chart display when 'Safety' is selected in the 'Contours Labels' menu? Select the ONE correct answer.

- **Only depth values on the safety contour value are shown.**
- All contour values are shown.
- All contour values up to 100 metres are shown.
- Only contour values above the safety contour are shown.

What happens when the vessel symbol is set into 'Auto scale'? Select the ONE correct answer.

- **The outline of the symbol is the correct size for the chart scale.**
- The outline of the symbol is at half the correct scale.
- The outline of the symbol is circular, not representing the vessel's shape.
- The outline of the symbol is maintained at the same size.

What has to be done to ensure that text labels are seen on screen? Select the ONE correct answer.

- **Make sure that text labels are switched on in the 'Chart' menu.**
- Ensure that the screen brightness is high enough.
- Have the CHARTPILOT in moving or centre display true motion.
- Make sure that items requiring text labels are set up.

What has to be done to save a created track? Select the ONE correct answer.

- **In 'Track/Edit', press the 'Store' button, allocating a catalogue and file number and name.**
- Insert a flash drive into the computer.
- Open up the 'Chart Maintenance' menu and allocate a file name using the 'Store' function.
- Press the save button and the CHARTPILOT automatically creates a folder name.

What is a 'Parameter Point' seen on a track? Select the ONE correct answer.

- **It is a waypoint without a course change but with changes in speeds, and limits for XTD and course as well as rudder economy. It lies on the straight part of a track line.**
- It is a point where the turn radius needs changing.
- It is a temporary waypoint on a contingency track that the 'TRACKPILOT' will steer for, if the vessel were to be off track by more than the cross track distance.
- It is the same as a track waypoint where there will be course and speed changes that the 'TRACKPILOT' will follow.

What is an ECDIS 'passport' used for? Select the ONE correct answer.

- **It is system information sent to a chart provider.**
- It is a certificate that customs want to see on arrival.
- It is an approval certificate for chart cells issued by IHO.
- It is an ECDIS approval certificate issued by IMO.

What is meant by 'Course Limit', as seen in the track 'Text Mode' window? Select the ONE correct answer.

- **During ECDIS 'Track monitoring', it is the difference between the actual course and the course defined by the pre-planned track leg; if the limit is exceeded, an alarm will sound.**
- It is the maximum allowable course change based on predicted speed.
- It is the maximum allowable course change based rate of turn, and used when the ECDIS is in 'track monitoring' mode.
- It is the number of legs making up a track and part of the CHARTPILOT's 'track monitoring' ability.

What is meant by 'Track Limit', as seen in the track 'Text Mode' window? Select the ONE correct answer.

- **During ECDIS 'Track monitoring', it is the deviation of the vessel's position from the activated route, the 'System Track'. If the limit is exceeded an alarm will sound.**
- During ECDIS 'Track monitoring', it is the distance to the next waypoint for an alarm to sound. If the limit is exceeded an alarm will sound.
- During ECDIS 'Track monitoring', it is the limit on track leg numbers. If the limit is exceeded an alarm will sound.
- During ECDIS 'Track monitoring', it is the maximum radius of turn at the waypoint. If the limit is exceeded an alarm will sound.

What is one function of the 'Table of Contents' found in 'Chart Maintenance' on a CHARTPILOT? Select the ONE correct answer.

- **It gathers a list of chart cells on the selected storage media.**
- It shows all of the functions available to an operator.
- It shows the contents of the electronic manual.
- It shows the menu structure on the CHARTPILOT.

What is recommended for the setting of the alarm buzzer when the CHARTPILOT is being used to monitor the track? Select the ONE correct answer.

- **It should be left on if the CHARTPILOT is unattended by an observer.**
- It can be switched off, as important objects will cause an alarm.
- It cannot be switched off, it will always be on.
- There is no particular recommendation.

What is revealed when the 'Tree' button, found on the bottom menu bar, is pressed? Select the ONE correct answer.

- **The hierarchical menu structure of the CHARTPILOT software.**
- A menu tree covering the operating system of the computer.
- The chart screen shows the chart in the vessels current position.
- The chart showing the ENC symbols.

What is the maximum time that can be set by an operator for the path 'prediction'? Select the ONE correct answer.

- **180 seconds.**
- 120 seconds.
- 180 minutes.
- 240 seconds.

What is the maximum waypoint approach alarm time that can be set by an operator? Select the ONE correct answer.

- **6 minutes.**
- 1 minute.
- 10 minutes.
- 3 minutes.

What is the purpose of the red box on this button? Select the ONE correct answer.

- **The named function is switched on.**
- The grid is not available.
- The named function is switched off.
- There is an alarm on the grid displayed.

What is the purpose of the small grey rectangle as seen on this button from the CHARTPILOT? Select the ONE correct answer.

- **It shows that there are other options available.**
- It shows that there are no other options available.
- It shows that this button can be moved onto the chart screen.
- It shows that WGS 84 is the correct datum.

What is the purpose of the 'Tree' button seen on the bottom bar? Select the ONE correct answer.

- **It opens a page to see the hierarchical menu of the ECDIS.**
- It opens up the CHARTPILOT software at administrator level.
- When pressed, it opens up the chart display.
- When pressed, it opens up the conning display.

What is the quickest way to return the vessel symbol to chart screen? Select the ONE correct answer.

- **Press the standard display button on the bottom bar (Basic Operation bar).**
- Enter the menu and switch on the vessel symbol.
- Press the standard display button on the top bar.
- Use the 'Tree' menu and select the display mode.

What is the significance of the red dot seen in this illustration from the CHARTPILOT alarm list? Select the ONE correct answer.

- **It is alongside an alarm message.**
- It is alongside a security message.
- It is alongside a warning message.
- It is alongside an indication message.

What is the significance of the yellow triangle seen in this illustration from the CHARTPILOT alarm list? Select the ONE correct answer.

- **It is alongside a warning message.**
- It is alongside a safety message.
- It is alongside an alarm message.
- It is alongside an indication message.

What is the 'Planned Speed' seen in the track 'Text Mode' used for? Select the ONE correct answer.

- **It is the speed that is planned for the track segment; it is in use for all time schedule calculations.**
- It is an estimate of the voyage average speed.
- It is the minimum allowable speed on a particular track leg and the speed in use for ETA calculations.
- It is the speed used for automatic track steering.

What is the 'Profile Speed' seen in the track 'Text Mode' used for? Select the ONE correct answer.

- **It is the speed used for automatic speed control on each leg of the track; it needs the optional 'SPEEDPILOT' equipment on board.**
- It is a measure of how fast the CHARTPILOT processor is working.
- It is the maximum allowable speed on a particular track leg.
- It is the speed used for ETA calculations.

What might happen if the CHARTPILOT ECDIS is not powered down properly? Select the ONE correct answer.

- **The CHARTPILOT program may have to be re-installed.**
- Nothing, as the software will just run up again.
- The Linux host will end up corrupt.
- The Windows host will end up corrupt.

What must the operator do to ensure that a 'User Chart Object' causes an alarm? Select the ONE correct answer.

- **Activate the appropriate object class in the 'Alarm Setting' menu.**
- Make sure the 'User Chart Object' is ahead of the vessel.
- Only use a line as the user object.
- Only use an area as the user object.

What use is made of the illustrated squares seen on the CHARTPILOT? Select the ONE correct answer.

- **The squares are used to set the screen brilliance and contrast.**
- The squares indicate a dangerous object, outside the safety contour.
- They appear on the screen when the ambient light level is too high.
- They indicate the position of the cursor when the computer hangs.

What use is made of the 'Min. Speed' setting seen in the illustration from the alarm settings menu? Select the ONE correct answer.

- **Alarms will not sound at this speed, or speeds lower than this speed.**
- Alarms will not sound at this speed, or speeds higher than this speed.
- This is the minimum speed for steerage to be maintained.
- Two knots allows the ahead sector width to be set.

What useful speed information does the 'Docking' display give the bridge team when a vessel is being docked? Select the ONE correct answer.

- **Forward speed, and bow and stern lateral speed.**
- Average speed over the last track leg.
- Speed to reach the desired ETA.
- Water track speed and shaft RPM.

When carrying out chart maintenance on connected CHARTPILOTs, what is recommended? Select the ONE correct answer.

- **That all the work is carried out on one CHARTPILOT, and all others are available in the network.**
- That all connected CHARTPILOTs are closed down.
- That all connected CHARTPILOTS are put into standby.
- That each CHARTPILOT has the same done independently to it.

When conning a vessel, what particularly useful information does the 'Conning' information strip have on it, compared to the 'Track/Control' information? Select the ONE correct answer.

- **Rate of turn information.**
- Change of heading required.
- Radius of turn information.
- Time to the next waypoint.

When does the cursor on the CHARTPILOT ECDIS look like the one illustrated? Select the ONE correct answer.

- **The cursor is in a text box for an entry.**
- The cursor is inside the chart area.
- The cursor is outside the chart area.
- The ECDIS computer is busy.

When does the cursor on the CHARTPILOT ECDIS look like the one illustrated? Select the ONE correct answer.

- **The cursor is outside the chart, on a menu.**
- The cursor is on the chart area.
- The ECDIS computer is hanging.
- The ECDIS computer is idle.

When does the cursor on the CHARTPILOT ECDIS look like the one illustrated? Select the ONE correct answer.

- **The cursor is inside the chart area.**
- The cursor is in text box for entry.
- The cursor is outside the chart area.
- The ECDIS computer is busy.

When is the path prediction function useful to the bridge team? Select the ONE correct answer.

- **During a pilotage, ship manoeuvre operation.**
- On a deep-sea passage when steady steaming.
- When passage planning.
- Whilst a vessel is at anchor.

When used on a newly created track, the 'Check' function finds impossible geometry on the track. What will happen? Select the ONE correct answer.

- **It will inform the operator and not perform a safety check.**
- It will automatically adjust the track parameters.
- It will automatically adjust the vessel turning parameters to suit the track.
- It will inform the operator and perform a safety check.

Where are track files stored on the CHARTPILOT computer? Select the ONE best answer.

- **In folders called 'Catalogs'.**
- In folders called 'magazines'.
- On a connected CHARTPILOT.
- On a floppy drive.

Where on the various CHARTPILOT installations, would an operator go to find the vessel's pilot card? Select the TWO correct answers.

- **Into the 'Tree' menu of the 'Chart' mode.**
- Into the 'Presentation' menu.
- Into the 'Track' menu.
- Into the 'Tree' menu the 'Conning' mode.

Which ONE answer correctly completes the following statement? 'When ordering new C-MAP-ENC chart cells . . .

- **. . . always give the provider the generated USER Key File for C-MAP-ENC'.**
- . . . always give the provider the 'IHO' key number'.
- . . . always let them know the old permit and licence numbers'.
- . . . just tell them what chart cells are required'.

Which ONE answer correctly completes the following statement? 'When ordering new IHO-ENC chart cells, . . .

- **. . . always give the provider the 'User Permit' number'.**
- . . . always give the provider the 'IHO' key number'.
- . . . always let them know the old permit and licence numbers'.
- . . . just tell them what chart cells are required'.

Which ONE of the options correctly completes the following statement? 'If the CHARTPILOT is running, but the screen brilliance is set too low to see detail . . .

- **. . . press all three buttons on the mouse to set maximum brightness'.**
- . . . press the 'DO' and 'INFO' buttons on the mouse at the same time.
- . . . switch off the computer using a mains power switch, then restart it'.
- . . . use the mouse to try and guess where the brightness menu is located'.

Which ONE of these answers correctly completes the following statement? 'Before pressing the power button to start a CHARTPILOT ECDIS . . .

- **. . . make sure it is not already running'.**
- . . . check to see that there is enough generated power'.
- . . . check with the vessel master'.
- . . . make sure there are charts loaded'.

Which option correctly describes the use of the 'Transfer' button found in chart maintenance? Select the ONE correct answer.

- **It allows files to be sent across a network to another database of the system, e.g. other CHARTPILOTs.**
- It converts waypoints from the WGS 84 datum to the European datum.
- It sends voyage logbook recordings to inserted media.
- It starts the loading of digital overlays from inserted media.

Which part of a vessel has to leave the defined anchorage are to cause an 'Out of Area' alarm to sound? Select the ONE correct answer.

- **The whole vessel.**
- The anchor buoy position.
- The CCRP.
- The system reference point.

Which part of a vessel has to leave the defined anchorage area to cause an 'Exiting Anchor Watch Area' alarm to sound? Select the ONE correct answer.

- **The vessel's contour is violating the border of the anchor watch area.**
- The anchor position is violating the border of the anchor watch area . . .
- The CCRP is violating the border of the anchor watch area.
- The whole vessel is violating the border of the anchor watch area.

Whilst ECDIS is monitoring a track, what happens when the 'INFO' button is pressed on the 'TO WPT' – to waypoint? Select the ONE correct answer.

- **A safety check for charted objects is performed from current ship's position.**
- The ETA to the waypoint is given.
- The vessel automatically adjusts course to directly steer for the waypoint.
- The waypoint parameters are shown to the operator.

Why do some ranges seen in the illustration, have the black diamond shape alongside them? Select the ONE correct answer.

- **For those ranges, vector chart data is available and will match with the ideal purpose of the chart loaded.**
- At these ranges, a base display only is available.
- No chart cells are available covering these ranges.
- They are ranges that do not have acceptable detail for the chart loaded.

Why does this button have a red box around it? Select the ONE correct answer.

- **This was the last button pressed on this menu.**
- The button cannot now be pressed because of a technical reason.
- There is an alarm on the coordinate system.
- This is illumination for night use.

Why does this button not have a 3D profile? Select the ONE correct answer.

- **The button has been pressed.**
- The button is available for use.
- The presentation is now showing on the chart screen.
- This is an information only button.

Why is one track on the CHARTPILOT defined as a 'System' track? Select the ONE best answer.

- **Because this track will be monitored by the CHARTPILOT during ECDIS monitoring.**
- Because it is loaded on the CHARTPILOT system and visible on the screen.
- Because they are used by the autopilot when track steering.
- Because they follow, VTS and IALA traffic routes.

Why might a CHARTPILOT ECDIS look like it is not running when it is? Select the ONE best answer.

- **The screen brilliance is set too low for the light conditions.**
- The CHARTPILOT computer fans cannot be heard running.
- The power light is dimly illuminated.
- The screen brilliance is set too high for the light conditions.

Why might a log recording not play? Select the ONE best answer.

- **There is a system track already loaded.**
- The CHARTPILOT is in true motion mode.
- The log file is corrupt.
- There is not the hard drive space required by the CHARTPILOT.

Will the CHARTPILOT alarm if a dangerous 'User Chart Object' is seen in the 'look ahead' detection feature? Select the ONE correct answer.

- **Yes.**
- No, as IMO do not require it.
- Only if it is a line.
- Only if it is an area.

With a true motion display, if the 'Set Center' function does not correctly locate the vessel symbol, what is the probable cause? Select the ONE correct answer.

- **The cursor was outside the set screen limits.**
- The CHARTPILOT is at too big a range.
- The CHARTPILOT is probably in planning mode.
- The operator has not selected the correct function.

You are inserting a new track waypoint using the cursor, but find that you have to move the cursor off the screen. What has to be done to resume insertion? Select the ONE correct answer.

- **Press the 'Set/Continue' button again.**
- Press on the last inserted waypoint to pick up the track again.
- Press on the screen in the next waypoint position.
- Press the 'text Mode' button to enable the track to be activated again.

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How can the screen brilliance be increased on an ECDISPILOT when it is running, but the screen is too dark to see it? Select the ONE correct answer.

- **Roll the cursor to the top left side of the screen and repeatedly press the 'DO' key.**
- Roll the cursor to the top left side of the screen and repeatedly press the 'MORE' key.
- Roll the cursor to the top right side of the screen and repeatedly press the 'DO' key.
- Roll the cursor to the top right side of the screen and repeatedly press the 'MORE' key.

How can the wheel over point be set? Select the ONE correct answer.

- **As a point or a line.**
- As a line only.
- As a point only.
- As a text message.

How can user symbols, areas and lines be defined on the ECDISPILOT? Select the ONE best answer.

- **Global, user defined or track related.**
- Global, track related and attached to a particular chart cell.
- They cannot be defined, as they are available to all once created.
- User defined group and available to anyone.

How does an operator find out which version of the ECDISPILOT is loaded on the MFD? Select the ONE best answer.

- **By pressing the 'About' button on the menu.**
- By contacting SAM Electronics.
- By looking in the ECDIS handbook.
- By pressing the 'Version' button on the menu.

How does an operator maximise the chart screen? Select the ONE correct answer.

- **By using the arrow button located on the menu bar.**
- By using the small arrow on the navigation data strip.
- With a double press on the chart screen.
- y using the square symbol located on the menu bar.

How is a manual fix entered onto the chart after two LOPs have been entered? Select the ONE correct answer.

- **Press the 'Drop Pos Fix' button.**
- By entering a user text, object.
- By entering an onscreen text message.
- Press the 'Create Pos Fix' button.

How is an offset EBL centred back on the vessel symbol? Select the TWO correct answers.

- **Switch it off then on again.**
- **Use the on-screen menu with 'Center EBL1' or 'Center EBL2'.**
- Click and drag it to the symbol on a zoomed in display.
- Press the 'Fixed' button.

How is it possible for an operator to check for chart updates on the ECDISPILOT? Select the TWO correct answers.

- **In the 'Chart' frame of the 'Navigation Side bar', left side of Chart type ID. An alarm symbol displays a chart expiry warning or alarm.**
- **With activated, highlighted updates in the chart area.**
- By checking with the chart records on board the vessel where the ECDIS is located.
- By contacting a local RENC, or the chart supplier.

How many electronic bearing lines are there on the ECDISPILOT?

- **Two.**
- One blue one and one brown one.
- One blue.
- One.

How many parallel index lines are available on the ECDISPILOT? Select the ONE best answer.

- **4.**
- 1.
- 2.
- 3.

How often are voyage records stored on the hard drive of an ECDISPILOT? Select the ONE correct answer.

- **Every 10 seconds for a period of 90 days.**
- Every 12 hours for a period of 3 months.
- Every 20 seconds for a period of 100 days.
- Every 24 hours for a period of 3 months.

If the LED on the lower right side of the monitor shows a fixed red light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **The PC is switched off or has a technical problem.**
- The monitor has failed.
- The system is in standby mode.
- There is a dangerous target.

If the LED on the lower right side of the monitor starts to flash a green light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **The display is switched off, or the PC connection lost.**
- The ECDISPILOT is being used for chart maintenance.
- The navigation signal from the sensor unit has failed.
- The system is operating normally.

If the LED on the lower right side of the monitor starts to flash a red light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **It is at too high a temperature.**
- The ECDISPILOT PC has failed.
- The signal from the sensor unit has failed.
- The system is shutting down.

In the illustration, what is missing from the cross shaped cursor when it is located on the chart screen of the ECDISPILOT? Select the ONE best answer.

- **The text box with position and range and bearing.**
- A small number, indicating which MFD is being used.
- Nothing, as this is what the cursor looks like on the chart screen.
- The text box with position.

Is it possible to have more than one route open at the same time on the ECDISPILOT? Select the ONE best answer.

- **Yes.**
- Depends on the mode.
- No.
- Only when planning.

Is the predictive function based on water or ground track? Select the ONE best answer.

- **Ground track.**
- Ground track, when the sea floor is relatively close for the Doppler log.
- Water track, when the vessel speed is below three knots.
- Water track.

Is the safety detection sector active at all times, or just when seen on the chart screen? Select the ONE best answer.

- **At all times.**
- At all times when there is an 'Active' route.
- Just when seen on the chart screen.
- Only when seen and if there is an 'Active' route.

On an ECDISPILOT, what is used to define a 'Navigating' target as opposed to an 'anchored' or 'moored' target? Select the ONE correct answer.

- **A speed greater than three knots.**
- A speed greater than five knots.
- One that is in a recognisable fairway.
- The definition set into the other vessel's AIS transmission.

On an MFD, the graphical user interface for the ECDISPILPOT is based on Windows 7. Is it possible for a user to load other software using the Windows operating system? Select the ONE correct answer.

- **No it is not possible to load user software.**
- No it is not possible until the system is out of warranty.
- Yes it is, as the ECDISPILOT is loaded on a standard PC.
- Yes it is, as Word is installed to allow a passage planning route list to be made.

On the ECDISPILOT version 1.0.15.xx, what does this symbol indicate? Select the ONE correct answer.

- **A priority one alert.**
- A warning classed as number one in the alarm list.
- An alarm on MFD number one.
- Number one buoy.

On the ECDISPILOT version 1.0.15.xx, what does this symbol indicate? Select the ONE correct answer.

- **A priority two alert.**
- A warning classed as number two in the alarm list.
- An alarm on MFD number two.
- Number two 'special' mark.

On the ECDISPILOT version 1.1.xx.xx what would this symbol indicate seen on the alarm strip? Select the ONE best answer.

- **A 'Warning'.**
- A system 'Alarm'.
- An AIS 'Alarm'.
- An ECDIS 'Alarm'.

On the ECDISPILOT version 1.1.xx.xx, what would this symbol indicate seen on the alarm strip? Select the ONE best answer.

- **An 'Emergency alarm' or an 'Alarm'.**
- The acoustic horn has failed.
- This is a system alarm.
- This is an ECDIS alarm.

On the ECDISPILOT, how does an operator know that the chart is over scaled? Select the ONE correct answer.

- **A warning is given on the 'Chart Status Line'.**
- The chart screen has horizontal lines on it.
- The indicated range has a red highlight around it.
- The range cannot be changed any further down.

On the ECDISPILOT, how many man overboard positions are possible? Select the ONE correct answer.

- **5**
- 1
- 2
- 6

On the ECDISPILOT, if a route geometry check is not passed, what cannot be done? Select the ONE best answer.

- **A chart and user object check.**
- Another geometry check.
- The route cannot be saved.
- The route cannot be used.

On the ECDISPILOT, if the 'Display Off' button has been pressed on the menu bar, how is the display returned? Select the ONE correct answer.

- **By moving the pointing device.**
- The chart screen has horizontal lines on it.
- The indicated range has a red highlight around it.
- The range cannot be changed any further down.

On the ECDISPILOT, what choices are there in the route 'Controller' for following a route? Select the TWO correct answers.

- **DP for vessels with 'Dynamic Positioning'.**
- **'TP' or 'Manual'.**
- 'Manual' or 'Operator'.
- 'Steering' or 'Auto'.

On the ECDISPILOT, why does a dropped fix sometimes not appear at the crossing point of two LOPs? Select the ONE correct answer.

- **Because the ECDISPILOT knows that visual bearings are great circles.**
- Because the bearing values are round up to the nearest whole number for plotting.
- Because the ECDISPILOT knows that visual bearings are rhumb lines.
- Because there is always a small error in bearings.

On the SAM integrated bridge system (NACOS), what does the abbreviation MFD stand for? Select the ONE correct answer.

- **Multi-function display.**
- Many fixed displays.
- Maximum forward detection.
- Multi file display.

On the SAM integrated bridge system (NACOS), what is the most useful principle of the MFDs found within it? Select the ONE best answer.

- **They can show any application within the NACOS system.**
- They can be dispersed around a vessel.
- They can be used by service technicians when they board.
- They can show both ECDIS and radar displays.

Some ECDISPILOTs are fitted with a 'Panel PC', what is meant by this? Select the ONE best answer.

- **This is an ECDISPILOT Basic, where the PC is integrated into the monitor.**
- The PC can be more easily accessed for maintenance.
- The PC has a solid-state hard drive not commonly found on an ECDIS.
- This is a system where the PC is more like a laptop, being thinner.

What are the reference positions available on an ECDISPILOT? Select the ONE correct answer.

- **'CCRP', 'Radar' and 'Track'.**
- 'CCRP', 'Radar' and 'Route'.
- 'Pivot point', 'CCRP' and 'Track'.
- 'Track', 'CCRP' and 'AIS'.

What can a scroll wheel on a mouse or trackball be used for, when fitted to a SAM MFD? Select the ONE correct answer.

- **It scrolls a list or zooms the chart.**
- A long push of it restarts the PC running the MFD.
- It can be set up similar to a special function key on the keyboard.
- It is used to obtain a 'Pick' report on a charted object.

What can be done by an operator to maintain the hardware of an MFD in the NACOS system? Select the ONE correct answer.

- **Check for the security of cables and remove dust.**
- Complete earth leakage tests on all equipment.
- Remove the hard drive from the cabinet and make sure it is free running.
- Replace the fluid in the motion damping unit.

What can be done by an operator to maintain the software of an MFD in the NACOS system? Select the ONE correct answer.

- **Nothing, as this should only be done by SAM approved personnel.**
- Check that all the necessary files are in their correct folders.
- Re-install the software for the ECDISPILOT from the backup.
- Remove temporary folders from the hard drive.

What can happen when the 'DO' key on the pointing device of an ECDISPILOT is given a double push? Select the ONE correct answer.

- **A folder is opened, or a function is activated.**
- It can be set up similar to a special function key on the keyboard.
- The ECDISPILOT is put into standby mode.
- The value of the object in a cell is advanced by one unit.

What do the special function keys 'F11' and 'F12' do when pressed on a NACOS MFD? Select the TWO correct answers.

- **Stop the ECDIS acoustic alarm.**
- **Stop the ECDIS visual alarm, the flashing symbols.**
- Accept a newly acquired radar target.
- Set the ECDIS display into a 'Standard' mode.

What does a yellow triangle indicate when seen on ECDISPILOT version 1.0.15.xx? Select the ONE correct answer.

- **It indicates that there is a warning message alongside it.**
- It indicates that there is an update buoy in this position.
- It is an alarm that requires immediate operator action.
- It says that there are three warning messages unacknowledged.

What does the small arrow alongside the speed through the water indicate, as seen on this illustration from the navigation side bar? Select the ONE correct answer.

- **It shows the resultant of longitudinal and transverse water speed.**
- That a water track log is sending the speed.
- That the log sensor signal is received within limits.
- That there is a sub-menu available allowing the speed to be changed to 'ground'.

What does the small symbol seen alongside 'helgoland' indicate? Select the ONE best answer.

- **That the route is open.**
- That the route has geometry errors.
- That the route has not been validated.
- That the route is closed.

What does the small triangle seen on this ECDISPILOT menu represent? Select the ONE correct answer.

- **It is a button to access a sub-ordinate menu.**
- If pressed, it closes the display menu.
- It notifies the operator that the mode is 'Browse'.
- It scrolls the chart down as the mode is 'Browse'.

What does the thick red line seen on this illustration from route indicate? Select the ONE correct answer.

- **The route has not passed the geometry check.**
- It is because the heading change is greater than 90°.
- It is because the route is laid over a buoy.
- It is the normal way a turn is shown at a waypoint.

What happens on an ECDISPILOT when the vessel symbol seen on the chart screen is pressed? Select the ONE best answer.

- **The vessel symbol centres on the chart screen.**
- Chart browse mode is selected.
- The vessel symbol changes to an outline shape.
- The vessel symbol moves to the bottom of the chart screen.

What happens to an ECDISPILOT display mode when the 'S' button is pressed on the menu bar? Select the THREE correct answers.

- **It activates the safety contour alarm, and it selects the standard display category.**
- **It reverts to north up, true motion and 6-mile range.**
- **The 'own–ship'-symbol is centred on the chart display.**
- It reverts to course up, true motion.

What happens to an ECDISPILOT when the 'S' button is pressed on the menu bar? Select the ONE best answer.

- **The chart display turns to the well-known pre-defined, part IMO setting.**
- The display brightness is increased to maximum.
- The ECDIS goes into standby mode.
- The system loads an IHO S57 chart type.

What is a 'Customised' display as seen on the ECDISPILOT? Select the ONE best answer.

- **A display personalised by an operator.**
- A display created by SAM Electronics.
- An IMO 'Standard' display.
- One created for a harbour pilot.

What is contained in the 'Chart Status Line' of an ECDISPILOT? Select the ONE correct answer.

- **The type of chart, the level of detail, the depth units and the scale status of the displayed chart.**
- The level of detail, mode of display and the type of displayed chart.
- The man over board button, the home button and the type of displayed chart.
- The type of chart, the associated permit number and the scale status of the displayed chart'.

What is indicated by a chart cell seen on the 'Maintenance' screen with a blue outline? Select the ONE best answer.

- **The cell is one supplied by C-Map.**
- The cell has an incorrect compilation scale.
- The cell is corrupt.
- The chart cell is an IHO ENC cell.

What is indicated by a chart cell seen on the 'Maintenance' screen with a green outline? Select the ONE best answer.

- **That the cell is okay to use.**
- The chart cell has an error on it.
- The chart cell is a C-Map one.
- The ECDISPILOT considers that this cell should be used on the planned route.

What is indicated by a chart cell seen on the 'Maintenance' screen with a red outline? Select the ONE best answer.

- **An error occurred when the cell was being checked by the ECDISPLIOT.**
- The chart cell has been checked and is ready for use.
- The chart cell has no content as it is corrupted.
- This is the normal way that the ECDISPILOT lets an operator see loaded chart cells.

What is meant by an 'Active' route on the ECDISPILOT? Select the ONE correct answer.

- **It is a route that will be monitored by the ECDISPILOT.**
- It is a route that is being edited by an operator.
- It is one that crosses into very busy sea areas.
- It is one that needs more edits to be completed by an operator.

What is the easiest way to move the chart around on the ECDISPILOT screen? Select the ONE best answer.

- **Press and hold the 'DO' key while moving the pointing device.**
- Change scale then reposition the chart centre.
- Load a different chart for another location.
- Use the 'MORE' key and opened menu to set the chart centre.

What is the greatest time setting available on the predictive function? Select the ONE correct answer.

- **180 seconds.**
- 120 seconds.
- 240 seconds.
- 60 seconds.

What is the name given to the folders where route files are stored on the ECDISPILOT? Select the ONE best answer.

- **'Catalogues'.**
- 'Compendiums'.
- 'Folders'.
- 'Magazines'.

What is the red sector seen in this illustration from a chart screen? Select the ONE correct answer.

- **The guard zone.**
- A danger area.
- A safety zone.
- Then safety detection sector.

What is the significance of the little red box seen on the guard zone? Select the ONE correct answer.

- **It allows the guard zone to be adjusted.**
- It allows the guard zone to be closed down.
- It identifies that this is number one zone.
- It is an acquired target within the zone.

What is the slider shown in the illustration used for? Select the ONE correct answer.

- **It is used to change the chart display scale.**
- It adjusts the volume of the acoustic horn.
- It is the screen brightness adjuster.
- It is used to adjust the chart compilation scale.

What is the use of the small orange box seen on the route leg in this illustration? Select the ONE best answer.

- **It informs the operator about the vessel's schedule.**
- It is a charted object showing a 'Caution' area.
- It is a manual chart update.
- It is a 'User Area' set in by an operator.

What is the 'Escalation' time as seen on the alarms list? Select the ONE correct answer.

- **It is the time before a warning becomes an alarm.**
- It is a limit on how many alarms will be raised at the same time.
- It is how rapidly a dangerous target would cause a close quarter situation.
- It is the time it takes to respond to an alarm.

What is the 'S' used for as shown on the menu bar of the ECDISPILOT? Select the ONE correct answer.

- **When pressed it changes the ECDIS display to an IMO defined 'Standard' one.**
- It is an indication that the display mode is set to 'Standard'.
- It is an indication that the ECDISPILOT is in standby mode.
- When pressed it returns the display to the 'Super Home' screen.

What might happen if AN ECDISPILOT is not powered down properly? Select the ONE correct answer.

- **The ECDISPILOT program will not re-start, needing a qualified service person to attend.**
- Nothing, as the software will just run up again.
- The Linux host will end up corrupt and SAM Electronics will have to send a qualified service person.
- The Windows host will end up corrupt and Windows will have to be re-installed.

What would a yellow dot, as seen alongside a sensor, indicate? Select the ONE correct answer.

- **It indicates that the sensor information has been manually set.**
- It indicates that the sensor is normal.
- It shows that a sensor is in 'Auto' mode.
- It shows that the sensor is outside the normal operational limits.

What would happen to the small white arrow seen alongside the speed through the water, on a vessel with only sternway? Select the ONE correct answer.

- **It would point downwards.**
- It would not change.
- It would point upwards.
- It would turn a red colour.

When an LOP is being set on the chart screen, what function does the 'Adjust' button have? Select the ONE best answer.

- **It opens a menu to allow exact coordinates, distance and bearing of the charted reference object to be entered.**
- It adjusts the bearing of the LOP in one degree steps from the reference object selected.
- It adjusts the coordinates of the observed object.
- It adjusts the origin of the LOP, placing it on the reference object selected.

When an operator sets safety depth values, what should they consider? Select the ONE best answer.

- **The vessel's draught and the zone of confidence value along the route.**
- The vessel's hydrodynamic pivot point position.
- They should set large values to stop unnecessary alarms.
- They should set small values to stop unnecessary alarms.

When does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE best answer.

- **Because an onscreen function is attached to it.**
- Because it is being moved.
- Because it is being used as a tape measure.
- Because it is on the chart screen.

When does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE correct answer.

- **The cursor is outside the chart, on a menu.**
- The cursor is on the chart area.
- The ECDIS computer is hanging.
- The ECDIS computer is idle.

Where is the danger detection sector referenced to on an ECDISPILOT? Select the ONE best answer.

- **The 'Track' reference position near the bow.**
- The vessel's centre point.
- The vessel's pivot point.
- The 'CCRP' located at the conning position.

Which ONE answer correctly completes the following statement? 'A long range AIS interrogation on an ECDISPILOT . . .

- **. . . is not automatically answered if the setting is in 'Manual'.**
- . . . is automatically answered if the setting is in 'Manual'.
- . . . is automatically answered if the station is within VHF range.
- . . . is not possible, as the system does not support it'.

Which ONE answer correctly completes the following statement? 'A man overboard position on an ECDISPILOT . . .

- **. . . updates with the estimated drift'.**
- . . . stays in the initial ground position'.
- . . . stays in the initial water position'.
- . . . updates according to a vessel's course and speed.

Which ONE answer correctly completes the following statement? 'A virtual button on the ECDISPILOT . . .

- **. . . is recognised because there is shading around it'.**
- . . . must always be pressed using the 'MORE' key on the pointing device.
- . . . is recognised because it always has a letter on it'.
- . . . is recognised because it is yellow in colour'.

Which ONE answer correctly completes the following statement? 'A virtual button on the ECDISPILOT . . .

- **. . . is pressed using the 'DO' button on the pointing device'.**
- . . . can be pressed using the 'MORE' or 'DO' button on the pointing device.
- . . . is recognised because it always has a number on it'.
- . . . is there as a prompt only to remind the operator to access a menu'.

Which ONE answer correctly completes the following statement? 'An ECDISPILOT shows charted objects that have been updated . . .

- **. . . by surrounding them with an orange star shape, lines and a U-pattern.**
- . . . by adding a small symbol alongside the charted object.
- . . . by adding an update icon on the 'Chart Register' bar.
- . . . by surrounding them with a red star shape.

Which ONE answer correctly completes the following statement? 'User symbols created on an MFD . . .

- **. . . are available throughout the vessel's network'.**
- . . . are only available on one other MFD'.
- . . . are only available on the MFD that they were created on'.
- . . . only have 'danger areas' spread throughout the network'.

Which TWO answers correctly completes the following statement? 'The ECDISPILOT can indicate an ETA using the planned speed or . . .

- **. . . calculate the ETA using the actual speed on vessels with a conning screen'.**
- **. . . calculate the speed to make a required ETA'.**
- . . . be set so that it does not do this'.
- . . . calculate the ETA using an average speed'.

Which answer describes a use for 'Chart 1' with respect to the setup of an ECDISPILOT display? Select the ONE best answer.

- **There is a grey scale and colour test pattern for the monitor.**
- The chart shows how to set up display types.
- There is a help menu embedded within 'Chart 1'.
- There is guidance on the number of pixels available on the fitted monitor.

Which of the mouse or trackball keys are active on a SAM MFD? Select the ONE correct answer.

- **The left side 'DO' key and the right side 'MORE' key.**
- The left side 'DO' key, the centre 'INFO' key and the right side 'MORE' key.
- The right side 'DO' key and the left side 'MORE' key.
- The right side 'DO' key, the centre 'INFO' key and the left side 'MORE' key.

Which of the special function keys on a fitted keyboard are assigned for ECDISPILOT use? Select the TWO correct answers.

- **The 'F11' key.**
- **The 'F12' key.**
- The 'F10'.
- The 'F8' key.

Why does the ECDISPILOT have a setting to reduce the output power of the vessel's AIS transponder? Select the ONE correct answer.

- **To reduce the risks of sparks being generated.**
- To restrict the range of the AIS in a piracy area.
- To stop it interfering with GPS reception.
- To stop 'modal' affect when the transmitter is close.

Why does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE correct answer.

- **Because an on-screen function can be grabbed.**
- Because it is on the SCAMIN lines.
- The ECDIS computer is hanging.
- The ECDIS computer is idle.

Why is there a yellow dot alongside the time shown on an ECDISPILOT? Select the ONE correct answer.

- **The time information is coming from the internal clock of the PC.**
- It is an indication that the displayed time is LMT.
- It is an indication that the displayed time is UTC.
- There is a discrepancy between the internal clock of the PC and the navigation sensor.

Why might An ECDISPILOT look like it is not running when it really is running? Select the ONE best answer.

- **The screen brilliance is set too low for the light conditions.**
- The ECDISPILOT computer fans cannot be heard running.
- The power light is dimly illuminated.
- The screen brilliance is set too high for the light conditions.

Why might the arrow-button for maximizing the chart area not work when pressed? Select ONE correct answer.

- **The MFD is involved in surveillances around the vessel, with target tracking or with TRACKPILOT in operation.**
- The MFD is being used with an ARCS chart displayed.
- The version of the ECDISPILOT does not have this functionality fitted.
- There are too many tracked targets for the MFD to cope with as the screen display would become too cluttered.

Why would a red diamond, as seen alongside a sensor, indicate? Select the ONE correct answer.

- **It indicates that the sensor information has failed.**
- It indicates that the sensor is normal.
- It shows that a differential GPS position is being received.
- It shows that the sensor is being used by another MFD.

Why would the alarm 'Deviation from Route' sound? Select the ONE best answer.

- **A vessel is outside the set cross track limits.**
- A vessel is not on the planned time schedule.
- A vessel's heading is outside the set limits.
- A vessel's speed has dropped below the set value.

Why would the 'Chart Status Line' show the following: 'Base+'? Select the ONE correct answer.

- **There is a 'Customised' display category selected.**
- There is a duplicated screen display.
- There is a 'Base' display at an incorrect range.
- There is a 'Base' display with zone of confidence symbols added.

Why would two red alert triangles be seen at the same time, side by side, on the alarm strip? Select the ONE correct answer.

- **The alert is coming from another connected MFD.**
- Because the alarm 'Escalation' time is exceeded.
- Because there is a fire alarm sounding on the vessel.
- There are two alarms active at the moment.

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How can an operator easily find out which version of the ECDISPILOT is loaded on the MFD? Select the ONE best answer.

- **By pressing the 'About' button on the menu.**
- By contacting SAM Electronics.
- By looking in the ECDIS handbook.
- By pressing the 'Version' button on the menu.

How can the screen brilliance be increased on an ECDISPILOT when it is running, but the screen is too dark to see it? Select the ONE correct answer.

- **Press and hold all three of the buttons on the pointing device.**
- Press and hold the 'DO' and 'MORE' keys.
- Roll the cursor to the top left side of the screen and repeatedly press the 'MORE' key.
- Roll the cursor to the top right side of the screen and repeatedly press the 'DO' key.

How can the wheel-over position before a waypoint be set? Select the ONE correct answer.

- **As a point or a line.**
- As a line only
- As a point only.
- As a text message.

How can user symbols, areas and lines be defined on the ECDISPILOT? Select the ONE best answer.

- **Global, user defined or track related.**
- Global, track related and attached to a particular chart cell.
- They cannot be defined, as they are available to all once created.
- User defined group and available to anyone.

How can 'ECDIS Chart 1' be used to help the setup of an ECDISPILOT display? Select the ONE best answer.

- **There is a grey scale and colour test pattern for the monitor.**
- The chart shows how to set up display types.
- There is a help menu embedded within 'Chart 1'.
- There is guidance on the number of pixels available on the fitted monitor.

How does an operator maximise the chart screen? Select the ONE correct answer.

- **By using the arrow button located on the menu bar to remove the data strip.**
- By using the small arrow on the navigation data strip.
- By using the square symbol located on the menu bar.
- With a double press on the chart screen.

How is a manual fix entered onto the chart after two LOPs have been entered? Select the ONE correct answer.

- **Press the 'Drop Pos Fix' button.**
- By entering a user text, object.
- By entering an onscreen text message.
- Press the 'Create Pos Fix' button.

How is an offset EBL centred back on the vessel symbol? Select the TWO correct answers.

- **Switch it off then on again.**
- **Use the on-screen menu with 'Center EBL1' or 'Center EBL2'.**
- Click and drag it to the symbol on a zoomed in display.
- Press the 'Fixed' button.

How is it possible for an operator to check for ENC cell updates on the ECDISPILOT? Select the TWO correct answers.

- **With activated, highlighted updates in the chart area.**
- By checking with the chart records on board the vessel where the ECDIS is located.
- By contacting a local RENC, or the chart supplier.
- Using a pick report and 'Chart Information'.

How many Electronic Bearing Lines (EBL's) are there on the ECDISPILOT? Select the ONE correct answer.

- **Two, one blue and one brown**
- One only, black in colour.
- One only, blue in colour.
- One red one and one brown one.

How many parallel index lines are available on the ECDISPILOT? Select the ONE best answer.

- **Four.**
- One.
- Three.
- Two.

How often are voyage records stored on the hard drive of an ECDISPILOT? Select the ONE correct answer.

- **Every 10 seconds for a period of 90 days.**
- Every 12 hours for a period of 3 months.
- Every 20 seconds for a period of 100 days.
- Every 24 hours for a period of 3 months.

If the LED on the lower right side of the monitor shows a fixed red light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **The PC is switched off or has a technical problem.**
- The monitor has failed.
- The system is in standby mode.
- There is a dangerous target.

If the LED on the lower right side of the monitor starts to flash a green light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **The display is switched off, or the PC connection lost.**
- The ECDISPILOT is being used for chart maintenance.
- The navigation signal from the sensor unit has failed.
- The system is operating normally.

If the LED on the lower right side of the monitor starts to flash a red light, what has happened to the ECDISPILOT? Select the ONE correct answer.

- **The ECDISPILOT's processor is too hot.**
- The ECDISPILOT PC has failed.
- The signal from a sensor unit has failed.
- The system is shutting down.

In the illustration, what is missing from the cross shaped cursor when it is located on the chart screen of the ECDISPILOT? Select the ONE best answer.

- **The data box with position and range and bearing.**
- A small number, indicating which MFD is being used.
- Nothing, this is what the cursor looks like on the chart screen.
- The text box with position.

Is it possible to have more than one route open at the same time on the ECDISPILOT? Select the ONE best answer.

- **Yes.**
- Depends on the mode.
- No.
- Only when planning.

Is the predictive function based on water or ground track? Select the ONE best answer.

- **Ground track.**
- Ground track, when the sea floor is relatively close for the Doppler log.
- Water track, when the vessel speed is below three knots.
- Water track.

On an ECDISPILOT, what is used to define a 'Navigating' AIS target as opposed to an 'anchored' or 'moored' target? Select the ONE correct answer.

- **A speed greater than three knots.**
- A speed greater than five knots.
- One that is in a recognisable fairway.
- The definition set into the other vessel's AIS transmission.

On an MFD, the graphical user interface for the ECDISPILOT is based on Windows 7. Is it possible for a user to load other software using the Windows operating system? Select the ONE correct answer.

- **No, it is not possible to load user software.**
- No, it is not possible until the system is out of warranty.
- Yes, because the ECDISPILOT is loaded on a standard PC.
- Yes, Microsoft Word is installed to allow a passage planning route list to be made.

On the ECDISPILOT, how does an operator know that the chart is over scaled? Select the TWO correct answers.

- **A warning is given on the 'Chart Status Line'.**
- **The jail 'prison bar' pattern is visible.**
- The chart screen has horizontal lines on it.
- The range cannot be changed any further down.

On the ECDISPILOT, how many 'man overboard' positions are possible? Select the ONE correct answer.

- **Five.**
- One.
- Six.
- Two.

On the ECDISPILOT, if a route geometry check is not passed, what cannot be done? Select the ONE best answer.

- **A chart and user object check.**
- Another geometry check.
- The route cannot be saved.
- The route cannot be used.

On the ECDISPILOT, if the 'Display Off' button has been pressed on the menu bar, how is the display returned? Select the ONE correct answer.

- **By moving the pointing device.**
- By pressing the power button on the monitor.
- By pressing the 'DO' key on the pointing device.
- By re-starting the MFD.

On the ECDISPILOT, what choices are there in the route 'Controller' for following a route? Select the TWO correct answers.

- **DP for vessels with 'Dynamic Positioning'.**
- **'TP' or 'Manual'.**
- 'Manual' or 'Operator'.
- 'Steering' or 'Auto'.

On the Wärtsilä SAM integrated bridge system (NACOS), what does the abbreviation 'MFD' stand for? Select the ONE correct answer.

- **Multi-function display.**
- Many fixed displays.
- Maximum forward detection.
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On the Wärtsilä SAM integrated bridge system (NACOS), what is the most useful principle of the MFDs used? Select the ONE best answer.

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- They can be dispersed around a vessel.
- They can be used by service technicians when they board.
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- **This is an ECDISPILOT Basic, where the PC is integrated into the monitor.**
- The PC can be more easily accessed for maintenance.
- The PC has a solid-state hard drive, not commonly found on an ECDIS.
- This is a system where the PC is more like a laptop, being thinner.

What are the reference positions available on an ECDISPILOT with connected radar? Select the ONE correct answer.

- **'CCRP', 'Radar' and 'Track'.**
- 'CCRP', 'Radar' and 'Route'.
- 'Pivot point', 'CCRP' and 'Track'.
- 'Track', 'CCRP' and 'AIS'.

What can be done by an operator to maintain the hardware of an MFD in the NACOS system? Select the ONE correct answer.

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- Complete earth leakage tests on all equipment.
- Remove the hard drive from the cabinet and make sure it is free running.
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- Remove temporary folders from the hard drive.

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- It can be set up as a special function key on the keyboard.
- The ECDISPILOT is put into standby mode.
- The value of the object in a cell is advanced by one unit.

What does the small arrow alongside the speed through the water indicate, as seen on this illustration from the navigation side bar? Select the ONE correct answer.

- **It shows the resultant direction of longitudinal and transverse water speed.**
- That a water track log is sending the speed.
- That the log sensor signal is received within limits.
- That there is a sub-menu available allowing the speed to be changed to 'ground'.

What does the small symbol seen alongside 'helgoland' indicate? Select the ONE best answer.

- **That the route is open.**
- That the route has geometry errors.
- That the route has not been validated.
- That the route is closed.

What does the small triangle seen on this ECDISPILOT menu represent? Select the ONE correct answer.

- **It is a button to access a subordinate menu.**
- If pressed, it closes the display menu.
- It notifies the operator that the mode is 'Browse'.
- It scrolls the chart down as the mode is 'Browse'.

What does the thick red line seen on this illustration from route indicate? Select the ONE correct answer.

- **The route has not passed the geometry check.**
- It is because the heading change is greater than 90°.
- It is because the route is laid over a buoy.
- It is the normal way a turn is shown at a waypoint.

What does this symbol indicate when seen on the alert strip? Select the ONE correct answer.

- **An alarm.**
- A caution.
- A warning classed as number one in the alert list.
- An alarm on MFD number one.

What does this symbol indicate when seen on the alert strip? Select the ONE correct answer.

- **A warning.**
- A warning classed as number two in the alarm list.
- An alarm on another MFD.
- An alarm.

What does this symbol indicate when seen on the alert strip? Select the ONE correct answer.

- **It indicates that there is a caution message alongside it.**
- It indicates that there is an update buoy in this position.
- It is an alarm that requires delayed operator action.
- It is an alarm that requires immediate operator action.

What happens on an ECDISPILOT when the vessel symbol seen on the chart screen is pressed? Select the ONE best answer.

- **The vessel symbol centres on the chart screen.**
- Chart browse mode is selected.
- The vessel symbol changes to an outline shape.
- The vessel symbol moves to the bottom of the chart screen.

What happens to an ECDISPILOT display mode when the 'S' button is pressed on the menu bar? Select the THREE correct answers.

- **It activates the safety contour alarm, and it selects the standard display category.**
- **It reverts to north up, true motion and 6-mile range.**
- **The 'own –ship'-symbol is centred on the chart display.**
- It reverts to course up, true motion.

What happens to an ECDISPILOT when the 'S' button is pressed on the menu bar? Select the ONE best answer.

- **The ECDISPILOT reverts to the pre-defined 'standard' setting.**
- The display brightness is increased to maximum.
- The ECDIS goes into standby mode.
- The system loads an IHO S57 chart type.

What is a 'Customised' display as seen on the ECDISPILOT? Select the ONE best answer.

- **A display personalised by an operator.**
- A display created by SAM Electronics.
- An IMO 'Standard' display.
- One created for a harbour pilot.

What is contained in the 'Chart Status Line' of an ECDISPILOT? Select the ONE correct answer.

- **The type of chart, the level of detail, the depth units and the scale status of the displayed chart.**
- The level of detail, mode of display and the type of displayed chart.
- The man over board button, the home button and the type of displayed chart.
- The type of chart, the associated permit number and the scale status of the displayed chart'.

What is indicated by a chart cell seen on the 'Maintenance' screen with a blue outline? Select the ONE best answer.

- **The cell is one supplied by C-Map.**
- The cell has an incorrect compilation scale.
- The cell is corrupt.
- The chart cell is an IHO ENC cell.

What is indicated by a chart cell seen on the 'Maintenance' screen with a green outline? Select the ONE best answer.

- **That the cell is an IHO approved cell ready for use.**
- The chart cell has an error on it.
- The chart cell is a C-Map one.
- The ECDISPILOT considers that this cell should be used on the planned route.

What is indicated by a chart cell seen on the 'Maintenance' screen with a red outline? Select the ONE best answer.

- **An error occurred when the cell was being checked by the ECDISPILOT.**
- The chart cell has been checked and is ready for use.
- The chart cell has no content as it is corrupted.
- This is the normal way that the ECDISPILOT lets an operator see loaded chart cells.

What is meant by an 'Active' route on the ECDISPILOT? Select the ONE correct answer.

- **It is a route that will be monitored by the ECDISPILOT.**
- It is a route that is being edited by an operator.
- It is one that crosses into very busy sea areas.
- It is one that needs more edits to be completed by an operator.

What is the easiest way to move the chart around on the ECDISPILOT screen? Select the ONE best answer.

- **Press and hold the 'MORE' key while moving the pointing device.**
- Change scale then reposition the chart centre.
- Load a different chart for another location.
- Use the 'DO' key and opened menu to set the chart centre.

What is the greatest time setting available on the predictive function? Select the ONE correct answer.

- **180 seconds.**
- 120 seconds.
- 240 seconds.
- 60 seconds.

What is the name given to the folders where route files are stored on the ECDISPILOT? Select the ONE best answer.

- **'Catalogues'.**
- 'Compendiums'.
- 'Folders'.
- 'Magazines'.

What is the red sector seen in this illustration from a chart screen? Select the ONE correct answer.

- **The ECDIS guard zone.**
- A danger area.
- A safety zone.
- Then safety detection sector.

What is the significance of a sensor set in 'Auto', as seen in the illustration? Select the ONE best answer.

- **If the sensor were to fail, another is automatically selected.**
- If 'Gyro 1' were to be available, the system would use it.
- That the sensor has been manually selected.
- That there is another sensor apart from 'Gyro 1' that should be used.

What is the significance of the small red box seen on this guard zone? Select the ONE correct answer.

- **It allows the guard zone to be adjusted.**
- It allows the guard zone to be closed down.
- It identifies the number one zone.
- It is an acquired target within the zone.

What is the slider shown in the illustration used for? Select the ONE correct answer.

- **It is used to change the chart display scale.**
- It adjusts the volume of the acoustic horn.
- It is the screen brightness adjuster.
- It is used to adjust the chart compilation scale.

What is the use of the small orange box seen on the route leg in this illustration? Select the ONE best answer.

- **It informs the operator about the vessel's schedule.**
- It is a charted object showing a 'Caution' area.
- It is a manual chart update.
- It is a 'User Area' set in by an operator.

What is the 'Escalation' time as seen on the alarms list? Select the ONE correct answer.

- **It is the time before a warning becomes an alarm.**
- It is a limit on how many alarms will be raised at the same time.
- It is how quickly a dangerous target would cause a close quarter situation.
- It is the time taken to respond to an alarm.

What is the 'S' used for, as shown on the menu bar of the ECDISPILOT? Select the ONE correct answer.

- **When pressed, it changes the ECDIS display to an IMO defined 'Standard' one.**
- It is an indication that the display mode is set to 'Standard'.
- It is an indication that the ECDISPILOT is in standby mode.
- When pressed it returns the display to the 'Super Home' screen.

What might happen if an ECDISPILOT is not powered down properly? Select the ONE correct answer.

- **The ECDISPILOT program will not start, because of a damaged hard drive.**
- Nothing, as the software will be able to run up the system correctly next time.
- The Linux host will end up corrupt and SAM Electronics will have to send a qualified service person.
- The Windows host will end up corrupt, and Windows will have to be re-installed.

What would a yellow dot seen alongside a sensor value indicate? Select the ONE correct answer.

- **It indicates that the sensor information has been manually set.**
- It indicates that the sensor is normal.
- It shows that a sensor is in 'Auto' mode.
- It shows that the sensor is outside the normal operational limits.

What would happen to the small white arrow seen alongside the speed through the water, on a vessel which had only sternway? Select the ONE correct answer.

- **It would point downwards.**
- It would not change.
- It would point upwards.
- It would turn a red colour.

When an LOP is being set on the chart screen, what function does the 'Adjust' button have? Select the ONE best answer.

- **It opens a menu to allow exact coordinates, distance and bearing of the charted reference object to be entered.**
- It adjusts the bearing of the LOP in one degree steps from the reference object selected.
- It adjusts the coordinates of the observed object.
- It adjusts the origin of the LOP, placing it on the reference object selected.

When an operator sets safety depth values, what should they consider? Select the ONE best answer.

- **The vessel's draught and the zone of confidence value.**
- The vessel's hydrodynamic pivot point position.
- They should set large values to stop unnecessary alarms.
- They should set small values to stop unnecessary alarms.

When does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE best answer.

- **Because an onscreen function is attached to it.**
- Because it is being moved.
- Because it is being used as a tape measure.
- Because it is on the chart screen.

When does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE correct answer.

- **The cursor is off the chart screen.**
- The cursor is on the chart area.
- The ECDIS computer is hanging.
- The ECDIS computer is idle.

When fitted to a SAM MFD, what can a scroll wheel on a standard mouse or desktop trackball be used for? Select the ONE correct answer.

- **It scrolls a list or zooms the chart.**
- A long push of it restarts the PC running the MFD.
- It can be set up as a special function key on the keyboard.
- It is used to obtain a 'Pick' report on a charted object.

When is the safety detection sector active? Select the ONE best answer.

- **At all times.**
- At all times when there is an 'Active' route.
- Just when seen on the chart screen.
- Only when seen and if there is an 'Active' route.

Where is the danger detection sector referenced to on an ECDISPILOT? Select the ONE best answer.

- **The 'Track' reference position near the bow.**
- The vessel's centre point.
- The vessel's pivot point.
- The 'CCRP' located at the conning position.

Which ONE answer correctly completes the following statement? 'A long range AIS interrogation on an ECDISPILOT . . .

- **. . . is not automatically answered if the setting is in 'Manual'.**
- . . . is automatically answered if the setting is in 'Manual'.
- . . . is automatically answered if the station is within VHF range.
- . . . is not possible, as the system does not support it'.

Which ONE answer correctly completes the following statement? 'A man overboard position on an ECDISPILOT . . .

- **. . . updates with the estimated drift'.**
- . . . stays in the initial ground position'.
- . . . stays in the initial water position'.
- . . . updates according to a vessel's course and speed.

Which ONE answer correctly completes the following statement? 'A virtual button on the ECDISPILOT . . .

- **. . . is recognised because it stands out a little bit'.**
- . . . must always be pressed using the 'MORE' key on the pointing device.
- . . . is recognised because it always has a letter on it'.
- . . . is recognised because it is yellow in colour'.

Which ONE answer correctly completes the following statement? 'A virtual button on the ECDISPILOT . . .

- **. . . is pressed using the 'DO' key on the pointing device'.**
- . . . can be pressed using the 'MORE' or 'DO' key on the pointing device.
- . . . is recognised because it always has a number on it'.
- . . . is there as a prompt only to remind the operator to access a menu'.

Which ONE answer correctly completes the following statement? 'An ECDISPILOT shows charted objects that have been updated using inserted media . . .

- **. . . by surrounding them with an orange star shape, lines and a U-pattern.**
- . . . by adding a small symbol alongside the charted object.
- . . . by adding an update icon on the 'Chart Register' bar.
- . . . by surrounding them with a red star shape.

Which ONE answer correctly completes the following statement? 'User symbols created on an MFD . . .

- **. . . are available to all MFDs on a vessel.**
- . . . are only available on one other MFD'.
- . . . are only available on the MFD that they were created on'.
- . . . only have 'danger areas' spread throughout the network'.

Which TWO actions are achieved by the special function keys 'F11' and 'F12' on a NACOS MFD? Select the TWO correct answers.

- **The ECDIS acoustic alarm is stopped.**
- **The ECDIS visual alarm, (flashing symbols) is stopped.**
- A newly acquired radar target is accepted.
- The ECDIS display is set into a 'Standard' mode.

Which TWO answers correctly completes the following statement? 'The ECDISPILOT can indicate an ETA using the planned speed or . . .

- **. . . calculate the ETA using the actual speed on vessels with a conning screen'.**
- **. . . calculate the speed to make a required ETA'.**
- . . . be set so that it does not do this'.
- . . . calculate the ETA using an average speed'.

Which keys on a standard mouse or desktop trackball are active on a Wärtsilä SAM MFD? Select the ONE correct answer.

- **The left side 'DO' key and the right side 'MORE' key.**
- The left side 'DO' key, the centre 'INFO' key and the right side 'MORE' key.
- The right side 'DO' key and the left side 'MORE' key.
- The right side 'DO' key, the centre 'INFO' key and the left side 'MORE' key.

Which of the special function keys on a fitted keyboard are assigned for ECDISPILOT use? Select the TWO correct answers.

- **The 'F11' key.**
- **The 'F12' key.**
- The 'F10' key.
- The 'F8' key.

Why does the cursor on the ECDISPILOT look like the one illustrated? Select the ONE correct answer.

- **Because an on-screen function can be grabbed.**
- Because it is on the SCAMIN lines.
- The ECDIS computer is hanging.
- The ECDIS computer is idle.

Why does the time shown on this ECDISPILOT come from an internal clock? Select the ONE correct answer.

- **The external synchronisation with time from a connected GPS sensor has failed.**
- It is an indication that the displayed time is LMT.
- It is an indication that the displayed time is UTC.
- There is a discrepancy between the internal clock of the PC and the navigation sensor.

Why might an ECDISPILOT look like it is not running when it really is still running? Select the ONE best answer.

- **The monitor is hibernating.**
- The ECDISPILOT computer fans cannot be heard running.
- The power light is dimly illuminated.
- The screen brilliance is set too high for the light conditions.

Why would a red diamond be seen alongside a sensor value? Select the ONE correct answer.

- **It indicates that the sensor information has failed.**
- It indicates that the sensor is normal.
- It shows that a differential GPS position is being received.
- It shows that the sensor is being used by another MFD.

Why would the alarm 'Deviation from Route' sound? Select the ONE best answer.

- **A vessel is outside the set cross track limits.**
- A vessel is not on the planned time schedule.
- A vessel's heading is outside the set limits.
- A vessel's speed has dropped below the set value.

Why would the 'Chart Status Line' show 'Base+'? Select the ONE correct answer.

- **There is a 'Customised' display category selected.**
- There is a duplicated screen display.
- There is a 'Base' display at an incorrect range.
- There is a 'Base' display with zone of confidence symbols added.

Why would two triangles be seen at the same time, on the alert strip, one with a cross through it? Select the ONE correct answer.

- **The alert, from another MFD, cannot be acknowledged on this MFD.**
- Because the alarm 'Escalation' time is exceeded.
- Because there is a fire alarm sounding on the vessel.
- There are two alarms active at the moment.

4019 – Tokyo Keiki ECDIS, EC 8000A, 8100, 8500A, 8600

Version 1.0 (793)

Click and drag each text box, to correctly complete the following sentence describing the guard frame function.

'When the guard frame functionality detects a dangerous charted object, the danger is surrounded with a Red _____ box and a caution is surrounded with a Yellow _____ box'.

Click and drag each text box, to correctly complete the following sentences describing how to display NAVTEX information on ECDIS.

'Choose Navigational INFO from the top menu whilst in monitor mode. Then press the On _____ button for NAVTEX. When you want to see detail information of a NAVTEX message, move the cursor onto the message in the message column and press the ENT _____ button on the operation panel'.

Click and drag each text box, to correctly complete the following sentences describing how to display radar target trails on the ENC

'Choose Settings _____ from the top menu whilst in monitor mode and choose the Trails _____ tab. Choose the On _____ button to show echo trails on the display'.

Click and drag each text box, to correctly complete the following sentences describing how to display the Admiralty Information Overlay.

'Choose Chart _____ from the top menu whilst in monitor mode. Choose the Presentation ____ tab, then the On _____ button for the AIO layer to be displayed'.

Click and drag each text box, to correctly complete the following sentences describing how to set a danger bearing line on the chart display.

'Choose **Navline and No-Go Area/Line** menu in the monitor mode. Choose the **Other Object** tab and then **Danger Bearing** choose **Add New** to create a new Danger Bearing or **Edit** to change an already existing Danger Bearing. Move a cursor onto the navigation mark, then press ENT button to specify it as the reference object, or input the geographical co-ordinates of the reference target into the 'Clearing Line' table. Specify the **NLT/NMT** information and bearings. If needed, input supplemental information, then press the **Save** button to settle them'.

Click and drag each text box, to correctly complete the following sentences describing how to upgrade the ECDIS software.

'Connect a USB mouse to the ECDIS, then insert external media containing the new version of the ECDIS software in an appropriate system port or drive. Choose **Maintenance** from the top menu in the update mode. Press the **Version** tab to check the current version of ECDIS software. Press the **System** tab, and then press the **Upgrade Software** at the bottom of the menu. Follow the dialog window or instructions so as to complete the software upgrade'.

How is a radar trail displayed on the ECDIS? Select the ONE correct option.

- **With a dark to light grey shade.**
- Blue.
- Green.
- With a varying magenta shade.

How often are screen shots sent to the VDR? Select the ONE correct option.

- **Every 15 seconds.**
- Every 15 minutes.
- Every 60 minutes.
- Every 60 seconds.

If the radar trail 'Threshold' value is set with a very high value, what is the effect on the displayed radar trails? Select the ONE correct option.

- **Only strong radar targets will show trails.**
- A larger number of weaker radar targets will be seen.
- Radar targets beyond a six-mile range are removed.
- The length of the trail is increased in size.

Is the following statement true or false? 'First install a permit for cell number GB800001 and then AIO data to display AIO objects'.

- **True.**
- False.

Is the following statement true or false? 'When the acoustic horn on the ECDIS sounds and it is an 'Alarm', the horn will sound three short beeps'.

- **True.**
- False.

Is the following statement true or false? 'When the acoustic horn sounds the 'Alert' message on the ECDIS complies with the IMO 'Bridge Alert Management System' recommendation.

- **True.**
- False.

Is the following statement true or false? 'When undertaking an 'Add Chart' operation, it is NOT possible to see an error detected and the number of the relevant cell, as a result of the decryption process'.

- **False.**
- True.

What does the following icon mean when seen on the 'ALERTS' list? Select the ONE correct option.

- **The acoustic alarm has been silenced.**
- The acoustic alarm has been disabled.
- The alarm has been acknowledged.
- The alarm is no longer active.

What does the following icon mean when seen on the 'ALERTS' list? Select the ONE correct option.

- **A warning has been acknowledged.**
- A warning has just come in.
- An alarm has been acknowledged.
- There is a caution just received.

What type of alert is this?

- **Unacknowledged alarms**
- Acknowledged alarms
- Acknowledged cautions
- Acknowledged warnings
- Unacknowledged warnings

What type of alert is this?

- **Unacknowledged warnings**
- Acknowledged alarms
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- Acknowledged warnings
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- Acknowledged alarms
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- Unacknowledged warnings

What type of alert is this?

- **Acknowledged warnings**
- Acknowledged alarms
- Acknowledged cautions
- Unacknowledged alarms
- Unacknowledged warnings

When a warning is generated by the ECDIS alert system, how many times does the acoustic horn sound? Select the ONE correct option.

- **Twice.**
- Continuously.
- Intermittently.
- Three times.

Where is the 'ALERTS' button located on the screen display? Select the ONE correct answer.

- **On the bottom of the menu.**
- On the chart area.
- On the status bar.
- On the toolbar.

Which of the following symbols, when pressed with the ECDIS in monitor mode, will show the chart, full screen? Select the ONE correct answer.

- **Symbol A**
- Symbol B
- Symbol C
- Symbol D

Why is the position shown in the illustration from the ECDIS menu coloured orange? Select the ONE correct option.

- **Because it is invalid data.**
- Because it is received from GPS 1.
- Because the GPS position is not being received.
- Because there is a manual setting.

With the ECDIS in monitor mode, which of the definitions of the 'Active Chart Display' is correct? Select the ONE best answer.

- **A chart including the own-ship position.**
- A chart at the centre of the chart screen.
- A chart seen in the menu window.
- A chart that is loaded with a permit.

With the ECDIS in plan mode and a chart loaded that covers a different area to that where the vessel is located, which of the definitions of the 'Active Chart Display' is correct? Select the ONE correct answer.

- **A chart at the centre of the chart screen.**
- A chart including the own-ship position.
- A chart that is loaded without a licence.
- A small scale chart showing the ocean area.

4050 – Vetting, Background and history

Version 1.0 (224)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4051 – Vetting, Preparation for a vetting inspection

Version 1.0 (352)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4052 – Vetting, The vetting inspection

Version 1.0 (474)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4053 – Vetting, Follow-up on the vetting inspection

Version 1.0 (283)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4054 – Vetting, Basic assessment

Version 1.0 (118)

How many pre-vetting meetings with the crew are required prior to an inspection?

- **As many meetings as deemed necessary.**
- None.
- One.

If the inspector asks you a question, and you don't know the answer, what is the best way to reply?

- **I am not sure, but let me investigate and get back to you shortly.**
- I just arrived yesterday, and I don't know enough about this ship yet.
- I think it is better if you ask the captain.

Is a post inspection meeting with the crew required?

- **No it is not required; however it is suggested in order to inform the crew about the inspection result.**
- No, there is no point in wasting time on this.
- Yes.

Is the entire crew required to participate in the inspection?

- **Only as necessary.**
- No.
- Yes.

Is the inspector entitled to walk around on the vessel unaccompanied?

- **No.**
- No, however if the crew is busy he can occasionally do part of the inspection on his own.
- Yes.

Must a vetting inspector identify himself at the gangway?

- **Yes, all persons must show ID at the gangway.**
- No, his attendance has been approved by the agent and Master in advance.
- Yes, but only if the gangway watch requires it.

Must a vetting inspector wear appropriate personal protection equipment at all times?

- **Yes, he must set a good example with respect to personal safety procedures whilst on board the vessel and in the terminal.**
- No, as he is an accredited vetting inspector he is exempted from this.
- Yes, but only if the Master requires him to do so.

What is the overall objective of a vetting inspection?

- **To provide a factual record of the vessel's condition and standard of operation at the time of the inspection and, in turn, allow an assessment of the risk that use of the vessel might pose.**
- To provide a record of the vessel's condition and standard of operation at the time of the inspection and, in turn, allow other companies to see the report.
- To provide a record of the vessel's condition and standard of operation at the time of the inspection to the vessel's owner.

When does a vetting inspection commence?

- **From the moment the inspector takes his first step onto the gangway.**
- As soon as the vetting inspector informs the Master.
- Immediately after the opening meeting.

Which accident was the first to draw public attention to the danger of transporting oil at sea?

- **Torrey Canyon.**
- Erika.
- Exxon Valdez.

Why would it be a good idea to re-check the engine room compartment shortly before the vetting inspection?

- **To ensure that it is clean and tidy.**
- To ensure that all safety equipment is in place and has not been removed.
- To ensure that no work is ongoing.

4055 – Vetting, Advanced assessment

Version 1.0 (94)

Are cargo operations required during an inspection?

- **No, but cargo operations are preferred during an inspection since the value of the inspection as a whole would be questionable without it.**
- No.
- Yes.

Are onboard officers required to hold valid original Flag State endorsements?

- **Preferably yes. Alternatively evidence that an endorsement application has been issued must be onboard.**
- No.
- Yes. Valid Flag State endorsement must always follow the individual officer.

Does the VIQ indicate which edition each required publication must have?

- **Yes.**
- No, it is only required to have the publications on board, there is no requirement for specific editions.
- Yes, but in general this should be disregarded as the various publication are frequently renewed.

Handing out relevant VIQ chapters to the responsible officers is a good idea. But why?

- **Because it enables them to check all the questions and complete a good pre-vetting.**
- Because the VIQ chapters are prepared especially for pre-vetting inspections and contains the tick-boxes for exactly this purpose.
- Because the VIQ guidelines include directions for pre-vetting inspections.

How long does an inspection usually take?

- **6-8 hours.**
- 2-4 hours.
- 8-12 hours.

In addition to the vetting inspection and the owners' comments, the charterer will also:

- **Screen any other available information about the owner and his operation before a decision is made regarding "acceptance".**
- Contact the owners to discuss the observations before a decision is made regarding "acceptance".
- Screen the latest Port State Control report and then issue an "acceptance".

In case a CDI inspection is being planned the Inspector will:

- **Require a CDI VPQ.**
- Accept a SIRE VPQ in the absence of a CDI VPQ.
- Just need a VPQ irrespective of which format.

In which VIQ Chapter do you find questions about UKC (Under Keel Clearance)?

- **Chapter 4.**
- Chapter 5.
- Chapter 6.

In which VIQ Chapter is CAS (Condition Assessment Scheme) located?

- **Chapter 2.**
- Chapter 3.
- Chapter 4.

Is an inspection valid if conducted during night time operation only?

- **Yes, but the value of the inspection as a whole is questionable, hence it is not common practice.**
- No.
- Yes, this is common practice.

Is the Inspector always correct when he notes observations?

- **No, but if the Master disagrees the issue may be raised during the closing meeting.**
- No, but he is entitled to rely on his own subjective opinion.
- Yes.

Is the inspector obliged to respect rest hours on board during an inspection?

- **Yes.**
- No.
- Yes in general, but he may ask individual crew members to deviate if needed.

Is the vetting inspector obliged to leave a note with observations on board before he leaves?

- **No, and if he doesn't the Master should make a list of the observations during the closing meeting.**
- No, and it doesn't matter because the final report will be submitted to the Company later.
- Yes, always.

The closing meeting provides the Master with:

- **An opportunity to challenge a doubtful observation**
- An opportunity to discuss the final report and have all incorrect observations deleted immediately.
- An opportunity to have those observations deleted, which are not applicable.

What are Best Practices issued by Oil Majors?

- **Additional requirements highlighting areas of concern.**
- A company specific guideline for correct loading procedures.
- The best method of completing a vetting inspection.

What is Chapter 11 in the VIQ about?

- **Engine and steering compartments**
- Cargo and ballast systems
- General appearance and condition

What is INTERTANKO?

- **The International Association of Independent Tanker Owners.**
- The International Association of Indian Tanker Managers.
- The International Association of Individual Terminal Operators.

What is the HVPQ?

- **A software program that ship owners use to compile ship particulars data.**
- A list of vetting questions used by the inspector.
- The inspection scheme used by OCIMF.

What is the maximum validity of a Class Status Report before an observation regarding lack of an updated Class status report is raised during an inspection?

- **4 months.**
- 2 months.
- 3 months.

What is the overall purpose of a vetting inspection?

- **To ensure that your vessel is operated and maintained in a safe manner adhering to all relevant rules and regulations.**
- To assure the Charterers that your vessel is safe to charter.
- To provide the inspection company with a report for their filing system.

What is the purpose of having a pre-vetting meeting with the majority of the crew prior to an inspection?

- **To brief the crew about the role of involved individuals during the inspection**
- There is no need for a pre-vetting meeting with the crew
- To tell the crew that they must ensure that no reasons for observations are found by the inspector.

Which of the following pre-vetting checks should be completed in the office?

- **Officers matrix, technical inspection reports, open audit reports and near-miss reports.**
- Environmental reports, Port State reports and Document of Compliance.
- Integrated Contingency Plans, tank inspection reports and Class Status reports.

Which of the following reports should at least be reviewed during vetting preparation?

- **Latest Port State Control report, ISM & ISPS reports, Technical inspection report and vetting report**
- Latest internal reports only.
- Only the most recent vetting inspection report with unclosed observations.

Who should attend the closing meeting?

- **The vetting inspector, the Captain and any other senior officer he may decide upon.**
- The vetting inspector and the Captain.
- The vetting inspector and the entire crew.

Who should attend the opening meeting?

- **The vetting inspector, the Captain and any other senior officer he may decide upon.**
- The vetting inspector and the Captain.
- The vetting inspector and the entire crew.

4060 – Social Media awareness

Version 1.0 (305)

A post on social media can be copied and spread around the world in seconds?

- **TRUE**
- FALSE

Can journalists find information about you on your personal social media accounts?

- **Yes, if your privacy settings are not set to maximum**
- No, as long as you only share information with friends
- No, they can't track things you delete
- Yes, but only if they get special access

Can you lose your job from just one simple social media post?

- **Yes, because a social media post could be misinterpreted and cause serious damage to a company's reputation**
- If you aren't breaking any rules, they can't do anything
- No you cannot
- Yes you could be fired if you post something the company doesn't like

Do journalists use social media as a source for their stories?

- **Yes, many journalists search social media channels for information to use in their news stories**
- It is illegal for journalists to use social media for their stories
- It is illegal for journalists to use social media for their stories
- No, journalists do not use social media

Do things on the Internet get deleted after a certain amount of time?

- **No, nothing will ever disappear on the internet**
- If your accounts have been inactive for 5 years they will be deleted
- Yes, nothing is kept on the internet forever
- Your posts and photos are owned by you so even if they don't get deleted no one will be able to re-use them anyway

If an incident happens on board your vessel you should:

- **Keep calm and wait until your superiors instruct you**
- Send social media posts about the incident to your friends and family
- Take photos and post them on social media
- Tweet about the incident so that the media hear the story from your side before anyone else's

If the company you work for has a social media policy you should:

- **Make sure you read and understand it and follow it every time you post**
- Don't follow it word for word. This is simply a guide
- Don't take it too seriously . Social media should not be considered a priority
- Quickly look through it so you get a general idea of the policy

If there is a critical situation that has something to do with your company, what should you do?

- **Don't post anything at all**
- Post a photo and share only with family and close friends
- Send an anonymous tip/news scoop to journalists
- Send the photos to a friend privately and ask him/her to upload these

If you are contacted by a journalists to discuss company affairs or a recent incident you should:

- **Tell them that you are not the correct person to speak to but you can take their contact details and get the appropriate person to call**
- Be honest, tell them what you know and what you have seen
- Get angry
- Hang up on them

If you are in a critical situation and you need to give a message to family or friends, what should you do?

- **Give them a phone call**
- Contact the agent
- Post a photo of the situation and tag them
- Post a public status message

If you post information that is not accurately marked with a time and a place a potential consequence could be that . . .

- **Journalists will assume what they want to get a good story**
- Journalists are allowed to fill in the time and place themselves
- Researchers will have restricted access to your posts
- Your name and social media accounts will rank higher in search engines, e.g. google

Once you have uploaded an image to social media channels, can you take it back and delete it forever?

- **No, you can't delete a photo once it has been posted**
- If you contact the social media sites, then they can put a stop to it
- It can be deleted, but it might take 24 hours to delete from other internet sites
- Yes. Once you have deleted a photo it entirely deletes it from the internet

What should you do if the press asks you questions about you company's actions?

- **Follow company rules and guidelines on how to deal with the press and public**
- Go ahead, everyone is free to talk
- Speak only when they insist
- Tell the press some of the things you know, but don't provide the whole picture

What you post on social media can possibly damage your company's reputation.

- **True, because others might misunderstand the posts and get a different story from them**
- False, because photos cannot be interpreted in many ways
- False, because posts can be hidden and deleted permanently
- True, but only if my posts and photos are manipulated

You should check regularly that your privacy settings are set to the maximum?

- **TRUE**
- FALSE

4070.01 – Transas ECDIS, User Interface

Version 1.0 (4064)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.02 – Transas ECDIS, Basic features

Version 1.0 (485)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.03 – Transas ECDIS, Chart Work

Version 1.0 (259)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.04 – Transas ECDIS, Sensors

Version 1.0 (72)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.05 – Transas ECDIS, Route planning

Version 1.0 (177)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.06 – Transas ECDIS, Route monitoring

Version 1.0 (69)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.07 – Transas ECDIS, Chart maintenance

Version 1.0 (342)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.08 – Transas ECDIS, Backup arrangements

Version 1.0 (72)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.09 – Transas ECDIS, Navigational tools

Version 1.0 (177)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4070.10 – Transas ECDIS, Assessment

Version 1.0 (237)

**DGPS 1 is used as primary position source. No secondary position source is selected.
If the GPS signal breaks down, which secondary position source will be available?**

- **DR**
- DPGS 2
- EP
- ER

DGPS 1 is used as primary position source. Which of the following statement is not true?

- **If DGPS 1 breaks down, DGPS 2 will automatically become active after a specified period for fallback-sensor activation**
- If DGPS 1 breaks down, the selected secondary position source DGPS 2 become active
- If DGPS 1 breaks down, the selected secondary position source DR become active
- If DGPS 1 breaks down, the selected secondary position source EP become active

**Determine the bearing from "bn Aquatic Park Entrance light 1"
(Position: 37° 48.638' N 122° 25.418' W)
to "Alcatraz Light" (Position: 37° 49.574' N 122° 25.328' W) (see Task 3.3).**

- **Bearing: 004.3 degree**
- Bearing: 094.5 degree
- Bearing: 184.5 degree
- Bearing: 274.5 degree

Does the "Safety Frame" function trigger an alert if the "safety frame" is not displayed on the chart area ([Show safety frame] check box is deactivated)?

- **Yes**
- No
- Only if navigational danger occur within the safety frame
- Only if the safety contour occur within the safety frame

Following situation: All "Area alerts" are activated. The [Show safety frame] checkbox is deactivated and the "Safety frame" is not displayed in the chart area. Which of the following statements is true?

- **Crossing the safety contour will trigger an alarm**
- All "Area alerts" will trigger an alarm
- Entering a "Caution area" will trigger an alarm
- No alarm will be triggered because the [Show safety frame] checkbox is deactivated

Four Depth Shade Mode is activated. Which of the following statements regarding the "Light Blue" shade is true?

- **Depth between the deep- and safety contour**
- Depth between the safety- and the shallow contour
- No hydrodynamic effects on the own ship
- Risk of Grounding

Four Depth Shade Mode is activated. Which of the following statements regarding the "Medium Blue" shade is not true?

- **Non-navigable water**
- Entry Should be avoided
- Not safe for navigation
- Risk of grounding

How can the "Man over Board" function be activated in the ECDIS?

- **With the [MOB] icon in the chart panel bar**
- By right-clicking on the own ship and selecting [Man-over-board]
- In the [Nav Tools] task with the [Man over board] fuction
- In the [Tasks] menu with the [SAR] tab

How can the DR position be corrected if the "Dead Reckoning“ mode is selected as primary position sources?

- **Set a new DR position by cursor**
- DR position will be corrected automatically by Radar
- Not at all
- Set a new DR offset by cursor

How is an active waypoint displayed on the chart during route editing mode?

- **Blue diamond**
- Blue circle
- Red diamond
- Red diamond

How is an active waypoint displayed on the chart during route monitoring?

- **Red diamond**
- Blue diamond
- Blue diamond
- Red circle

How is the Safety contour displayed in the chart area?

- **As a thick, grey coloured contour in any display mode**
- As a thick, grey coloured contour in Standard display mode only
- As a thick, red coloured contour in any display mode if dangers are detected
- As a thin red coloured line

How many miles is the route "San Francisco Bay" (see Task 5.1) shorter than route "Approach San francisco" (see Task 5.5)? Note: Distances are rounded to full cables (0.1nm).

- **10.2 nm**
- 11.8 nm
- 20.2 nm
- 9.8 nm

How many nautical miles are saved by great circle compared to rhumb line distance between the following positions (see Task 5.10)? Start: 37° 38.9' N 122° 59.0'W End: 34° 55.2' N 140° 24.6' E - Note: All distances are rounded to full miles.

- **233NM**
- 213NM
- 223NM
- 243NM

How will the navigational hazard "Golden Gate Bridge" be displayed (echo) on a 3 cm radar (see Task 2.20)?

- **Radar Echo: Dash/Dash/Dot (Morse code for "G")**
- Radar Echo: Dash/Dot (Morse code for "N")
- Radar Echo: Dot/Dash/Dash/Dot (Morse code for "W")
- Radar Echo: Dot/Dot/Dash (Morse code for "U")

Referring to the "Four Fathom Bank Lighted Bell Buoy" (see Task 2.16), which of the following statement is true ?

- **Nominal range: 4 nm**
- Nominal range value: 10.9 nm
- Nominal range value: 18.2 nm
- Nominal range: 4 mm

Ship is in position 37°49.0N 122°27.1W. The watch officer has set the Safety Contour from 9m to 13m (see Task 2.11). What is the difference between the set and displayed Safety Contour in meters in this area (chart in use US5CA13M)?

- **+ 5,2m**
- + 13,0m
- + 4.8m
- 0m

The plotted line of position (LOP) and the ship's track are recorded in the Log Book feature.
Which of the following statement is not true?

- **The LOP entry will be automatically deleted after 30 day**
- An automatic Fix using the Dead Reckoning will only be displayed in the chart area if the "Own ship track" feature is activated
- LOPs cannot be changed in the Log Book
- The LOP entry cannot be deleted from the Log Book

The purpose of the "Safety frame" is to alert for dangers within its limits. When does the "Safety frame" function always trigger an alert, if all "Antigrounding alarms" remain activated?

- **Buoy**
- Anchor Prohibited Area
- ARPA target with CPA of 0nm
- Deep Contour

The selected primary position source is "Dead Reckoning“ mode. How can the DR position be corrected?

- **Enter a new DR position via keyboard**
- DR position will be corrected automatically by Radar
- Enter a new DR offset via keyboard
- Not at all

The ship is approaching the "Golden Gate Bridge" from open sea. The master orders to have the ECDIS set to the new navigational situation. Which setting has not to be checked or reevaluated accordingly?

- **[Ships Settings]**
- "Safety frame" settings
- "Safety parameters" settings
- [Basic Areas] alerts

The ship is moored parrallel to the pier. The compass heading and the pier course differ by 5 degrees. Data quality of the selected chart is A1. Which actions need to be done?

- **Set "Gyro error" via Master station**
- It is not necessary to set a "Gyro error"
- Order new chart with data quality A3
- Select a chart with higher data quality

**What is the distance from "bn Aquatic Park Entrance light 1"
(Position: 37° 48.638’ N 122° 25.418’ W)
to "Alcatraz Light" (Position: 37° 49.574’ N 122° 25.328’ W) (see Task 3.3)?**

- **Distance: 0.94 nm**
- Distance: 0.90 nm
- Distance: 0.91 nm
- Distance: 0.98 nm

Which additional method for manual position fixing is implemented in the ECDIS as navigational features?

- **Lines of Position (LOP)**
- Dead Reckoning
- Query/ Pick Report
- User Object Reference

Which alternative method for manual position fixing is implemented in the ECDIS as navigational features?

- **EBL & VRM**
- Artificial Target Plotting
- Circles of Position (COPs)
- User Object Reference

Which display category is available (see Task 2.9)?

- **All**
- All Other
- Customised
- Standard Plus

Which of the following alarms cannot be set individually while monitoring a route?

- **[Out of XTD]**
- [Off leg course]
- [Out of schedule]
- [WPT approach]

Which of the following items/information is not displayed in the Base Display Mode?

- **Floating aids to navigation**
- Coastline
- Safety contour
- Scale, orientation and display mode

Which of the following sensor information is not displayed on the default screen?

- **Air Temperature**
- Course
- Position
- Speed

Which of the following statement is not true regarding a ship, that is moored parrallel to the pier, the difference between compass heading and the pier course is 25 degrees and the data quality of the selected chart is A3?

- **The Gyro has to be brooken**
- No need to set a gyro error
- Select a chart with higher data quality if available
- The chart displays data with low quality

Which of the following statement regarding Positioning Sources is true?

- **Estimated Position (EP) considers additional manually set current and drift data**
- Dead Reckoning (DR) needs ARPA data to constantly correct the ships position
- Dead Reckoning (DR) should be used at any time to verify the current position
- Estimated Position (EP) does not consider additional manually set current and drift data

Which of the following statement regarding route handling is not true?

- **The [Route Monitoring] tab/ "Route Editor" panel are used to start/ stop route monitoring**
- A route cannot be edited via the [Route Monitoring] tab
- Route editing and start route monitoring can be done in the via "Route Editor" panel
- Route monitoring can be started via the [Route Monitoring] tab

Which of the following statement regarding the activation of route monitoring is not true?

- **It is impossible to modify a monitored route**
- A monitored route is displayed in red by default
- Route monitoring can be activated in the [Route Monitoring] tab
- Route monitoring can be activated via the "Route Editor" panel

Which of the following statements referring to Task 3.4 and Task 3.5 is true?

- **The planned route is 1.07 nm shorter than the distance from buoy 2 to buoy 8**
- The distance from waypoint 2 to waypoint 3 is as same as buoy 4 to buoy 8
- The length of "San Francisco North Channel" from buoy 2 to buoy 8 is 3 nm
- Total distance of the planned route is 2.12 nm

Which of the following statements regarding MOB function is true?

- **The start time can be changed**
- After MOB function has been deactivated, the MOB symbol/marker turns white
- The end position can be changed
- The Start position can be changed only

Which of the following statements regarding Task 3.2 and Task 3.3 is true?

- **The difference between both distance is 3.92 nm**
- The distance from "Alcatraz Light" to "Aquatic Park Entrance Light 1" is 1.84 nm
- The distance from "Point Diablo" to buoy no. 7 is 3.92 nm shorter than "Alcatraz Light" to "Aquatic Park Entrance Light 1"
- The distance from buoy no. 7 to "Point Diablo" is 3.92 nm

Which of the following statements regarding alarms is true?

- **Pushing the [Alarm] button on the keyboard will acknowledge the most recently generated alarm**
- A Flashing alarm disappears from the "Alarm" window if the reason does not longer exist
- Alarms and warnings are displayed in bold red fonts
- Audible signal will be active active as long as the alarm exists

Which of the following statements regarding alerts and alerts handling is not true?

- **Pushing the [Alarm] button on the keyboard will acknowledge all existings alerts**
- Audible signal will be active as long as the alarm is unacknowledged
- Unacknowledged alarms are displayed flashing
- Warnings do trigger an audible signal

Which of the following statements regarding chart updating is true?

- **A deleted manual correction can be restored**
- In the Navi-Sailor, soundings cannot be corrected manually. Other ECDIS can
- It is impossible to install a selection of particular updates, provided by the chart supplier
- The "Navi-Planner" must be started via the windows start menu link "Navi-Planner"

Which of the following statements regarding how the "Man over Board" function can be activated in the ECDIS is true?

- **Via the [MOB] key on the keyboard**
- In the [Nav Tools] task with the [Man over board] fuction
- In the [Tasks] menu with the [SAR] tab
- Only with the [MOB] icon in the chart panel bar

Which of the following statements regarding route loading and route handling is true?

- **If an geometric error exists, a saved route cannot be loaded for monitoring**
- A route check can be done via "Route Editor" task and/or [Route Monitoring] task
- A route with an existing geometric error cannot be saved
- An unchecked route cannot be saved

Which of the following statements regarding the dongle is not true?

- **Once the USB stick was plugged, the ECDIS runs without the dongle**
- The dongle is necessary to update charts
- The dongle works as a key to unlock the workstation
- The permit codes on the dongle works as a key to unlock the software

Which of the following statements regarding the security lock is true?

- **The permit codes on the dongle works as a key to unlock the workstation and the software**
- Once authorised, the ECDIS works without a dongle
- The dongle can be used for all TRANSAS ECDIS within the fleet
- The system configuration file works as a key to unlock the workstation and the software

Which of the following statements regarding warnings is not true?

- **Pushing the [Alarm/Warning] button will acknowledge the most recently generated warning**
- Audible signal lasts 2 seconds only
- Warnings are displayed in orange colour
- Warnings do trigger an audible signal

Which of the following statements, referring to the "Four Fathom Bank Lighted Bell Buoy" is true (see Task 2.16)?

- **The bouy is the red lateral bouy of "Bonita Channel"**
- The bouy indicates safe water to starbord on "Bonita Channel"
- The bouy is the green lateral bouy of "Bonita Channel"
- The bouy is the red lateral bouy of "San Francisco Main Ship Channel"

Which of the following statements, referring to the "Four Fathom Bank Lighted Bell Buoy" is true (see Task 2.16)?

- **The bouy is the green lateral bouy of "San Francisco Main Ship Channel"**
- The bouy is the a red bouy for both channel
- The bouy is the green lateral bouy of "Bonita Channel"
- The buoy indicates an isolated danger to starbord on "San Francisco Main Channel"

Which of the following statements, referring to the "Four Fathom Bank Lighted Bell Buoy" is true (see Task 2.16)?

- **Colour: Red/Green/Red**
- Colour: Green
- Colour: Green/Red/Green
- Colour: Red

Which of the following value cannot be set during trial manoeuvre?

- **Set COG: 10°**
- Delay: 0.4 nm
- Radius: 0.5 nm
- Set CTW: 250°

Which parameters can be changed after the MOB function has been activated?

- **Start time**
- Bearing
- Distance
- End time

Which position sources consider additional manually set current and drift data?

- **EP**
- DGPS 1
- DR
- LOG 1

4076.01 – Sperry ECDIS, User Interface

Version 1.0 (816)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.02 – Sperry ECDIS, Basic Features

Version 1.0 (454)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.03 – Sperry ECDIS, Sensors

Version 1.0 (540)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.04 – Sperry ECDIS, ChartWork

Version 1.0 (679)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.05 – Sperry ECDIS, Route Planning

Version 1.0 (452)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.06 – Sperry ECDIS, Route Monitoring

Version 1.0 (174)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.07 – Sperry ECDIS, Alarms

Version 1.0 (114)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.08 – Sperry ECDIS, Chart Maintenance

Version 1.0 (240)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.09 – Sperry ECDIS, Backup Arrangements

Version 1.0 (62)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.10 – Sperry ECDIS, Display of additional Info

Version 1.0 (221)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4076.11 – Sperry ECDIS, Assessment

Version 1.0 (233)

Assume that GPS1 and GPS2 are available to the ECDIS. What happens if GPS1 fails to transmit data?

- **Position sensor remains active but position is dead reckoned**
- Position sensor automatically transfers to GPS2
- Position sensor switches only if automatic sensor selection is activated
- Position sensor transfers according to the set sensor fall back chain

Four Depth Shade Mode is activated with the correct safety settings. Which of the following statements regarding the "Light Blue" shade is true?

- **Safe for navigation**
- Accessible for small vessels only
- Entry should be avoided
- Non-navigable water

How are the waypoints are displayed on the chart during route monitoring mode?

- **Red circle**
- Blue circle
- Orange diamond
- Purple circle

How can be assessed if "Dead Reckoning Mode" is active?

- **The status of the position sensor is indicated as "DR" in the lower toolbar**
- Active sensors can only be assessed in the "Nav Device Selection" mode
- No ownship vector is displayed
- The ownship symbol is coloured yellow

How can the "Man over Board" function in the ECDIS be activated instantly?

- **With the [MOB] button in the toolbar**
- By right-clicking on the own ship and selecting [Man-over-board]
- In the [Nav Tools] task with the [Man over board] function
- In the [Ship] menu with the [Gurad Zone] submenu

How can the MOB function be deactivated?

- **Click on [Cancel Oscar] in the "Man Overboard" Menu**
- Acknowledge the "Man Overboard" alarm in the alarm status indicator
- Click on the MOB button again
- Right-click on the MOB marker and select [Cancel Oscar]

How is an unacknowledged alarm distinguished from an acknowledged alarms?

- **All three alternatives**
- Acknowledged alarms are muted
- Acknowledged alarms show a steady background
- Unacknowledged alarms trigger a buzzer

How is the Safety Depth setting related to the chart presentation?

- **The Safety Depth value determines which spot soundings are highlighted by bold black numbers**
- The Safety Depth indicates water, where no hydrodynamic effects act on the ship's hull
- The Safety Depth is a bold black contour line, seperating navigable and non navigable water
- The Safety Depth is a feature only available in four shade mode

How is the Safety contour displayed in the chart area?

- **As a thick, grey coloured contour in any display mode**
- As a thick, grey coloured contour in Standard display mode only
- Depending on the display mode as a thick red or green coloured line
- Depending on the Safety Depth as a red or green cloured line

How many miles is the route "San Francisco Bay" (see Task 5.1) shorter than route "Approach San Francisco" (see Task 5.5)?
Note: Distances are rounded to full cables (0.1nm).

- **10.2 nm**
- 11.8 nm
- 9.2 nm
- 9.8 nm

How many nautical miles are saved by great circle compared to rhumb line distance between the following positions (see Task 5.10)?
Start: 37° 38.9' N 122° 59.0'W, End: 34° 55.2' N 140° 24.6' E. What is the total distance and time to go?
Note: Distances are rounded to full nautical miles and time is rounded to full minutes.

- **233NM**
- 213NM
- 223NM
- 243NM

How position fixes, obtained by LOPs, are recorded?

- **The position fixes by LOPs are recorded directly in the data log**
- Only as position marks on the chart with time label
- Only positions obtained by GPS may be recorded in the data log
- Positions obtained by LOPs are only recorded if LOP is the active position sensor

How will the "Racon of the Golden Gate Bridge" be indicated on the 10cm band Radar? (see Task 2.21)

- **No pattern is displayed**
- The pattern "DASH DASH DOT" is originating from the respective echo
- The pattern "DASH DOT DASH" is originating from the respective echo
- The pattern "DOT DOT DASH" is originating from the respective echo

How will the "Racon of the Golden Gate Bridge" be indicated on the 3cm band Radar? (see Task 2.21)

- **The pattern "DASH DASH DOT" is originating from the respective echo**
- No pattern is displayed
- The pattern "DASH DOT DASH" is originating from the respective echo
- The pattern "DOT DOT DASH" is originating from the respective echo

Ship is in position 37°49.0N 122°27.1W. The watch officer has set the Safety Contour from 9m to 13m (see Task 2.11). What is the difference between the set and displayed Safety Contour in metres in this area (chart in use US5CA13M)?

- **+ 5,2m**
- + 5,0m
- + 6,2m
- 0m

The XTD is crucial for the route check. Which width does it represent?

- **Width to each side of the intended track line**
- Width of the entire XTD corridor
- Width of the ship's beam line
- Width to each side to track line including added ship's width

The purpose of the "Look- Ahead" is to alert for dangers within its limits.
When does the "Look- Ahead" function always trigger an alarm?

- **Land area**
- Anchor Prohibited Area
- ARPA target with CPA of 0nm
- Deep Contour

The ship is approaching the "Golden Gate Bridge" from open sea. The Master orders, to set the ECDIS to the new navigational situation. Which settings have to be checked or re-evaluated accordingly?

- **All three alternatives**
- "Chart Depths/ Heights" settings
- "Look- Ahead" settings
- Display Mode

To determine between routes in editing and monitoring mode, different colour schemes are used. How is the track displayed on the chart during route editing mode?

- **Orange dashes**
- Black dashes
- Blue dashes
- Red dashes

To determine between routes in editing and monitoring mode, different colour schemes are used. How is the track displayed on the chart during route monitoring mode?

- **Red dashes**
- Blue dashes
- Orange dashes
- Purple dashes

Under which circumstances can a route be loaded for monitoring?

- **The ownship's attitude must meet certain criteria in relation to the route**
- No other route may be loaded for editing
- Routes can be monitored at any time
- The route may have geometric errors

What happens if a recently lost GPS sensor is functional again and DR was set as active position sensor?

- **It has to be set back to the recently lost GPS sensor manually**
- A window requiring confirmation of available sensor pops up automatically
- GPS will be selected automatically
- The most accurate sensor is always selected as set up in the sensor hierarchy

What is not recorded in the data log?

- **Manual chart update**
- Alarms
- GPS position
- Position derived by LOP

What is the bearing from "Point Diablo" to "buoy no.7"? (See Task 3.2)

- **242 degrees**
- 052 degrees
- 062 degrees
- 224 degrees

What is the light characteristic of Alcatraz Light?

- **Fl.5s**
- Gp Fl.(2)
- Iso
- Occ.

What should be done, before entering the Deep Water Way, referred in task 2.18?

- **Call "San Francisco Traffic" before entering the Deep Water Way**
- 3 cylinders in a vertical row have to be hoisted as a daylightsignal
- No entry is permitted when carrying dangerous goods
- Speed restriction of 12.0kt shall be met, bevor entering the Deep Water Way

What will happen if the dongle is damaged or detached while the ECDIS is running?

- **The System remains available for the next 7 days in a degraded state**
- An extra Super-PIN must be entered to bypass the dongle
- The ECDIS reboots in demo mode
- The ECDIS will shut down

Where can you get detailed information abot the status of all connected sensors and all their subfunctions?

- **"Sensor Status Details" window**
- Alarm Status Indicator
- Colour code indicating the sensor's health status
- Prompt Indicator

Which alternative methods of manual position fixing are implemented in the ECDIS as navigational features?

- **Lines of Position (LOP)**
- All three alternatives
- Query/ Pick Report
- User Object Reference

Which category of "zone of confidence" is allocated to the chart US5CA13M?

- **U**
- A1
- A2
- B

Which considerations for voyage planning may be necessary regarding the quality of data (accuracy) of charts?

- **All three alternatives**
- Adjusting the minimum required clearance to dangers
- Adjusting the safety settings
- Adjusting the XTD

Which display category is not available (see Task 2.7)?

- **Standard Plus**
- Base
- Other
- Standard

Which external program is used to install chart updates?

- **Chart Handler**
- Chart Installer
- Chart Manager
- Navi Planner

Which external program is used to install permits?

- **Chart Installer**
- Chart Handler
- Chart Manager
- Navi Planner

Which information is not included in the "Conning Information display (CID)"?

- **Central Alarm Window**
- Bearing to Wheelover
- Docking
- Steering Mode & Route Info

Which of following statements regarding route loading/ route handling is true?

- **A route check can be done via "Edit Route" and/or "Monitor Route"**
- A route check must be triggered manually to validate its execution
- A route with an existing geometric error cannot be saved
- If an geometric error exists, a saved route can be loaded for monitoring

Which of the following items / information are displayed in the Base Display Mode?

- **Safety contour**
- Buoy names
- Cables, pipelines
- Floating aids to navigation

Which of the following items / information is not an element of the Base Display Mode?

- **Buoy**
- All three alternatives
- Depth Shades
- Safety contour

Which of the following sensor information is not displayed on the default screen (see Task 1.1)?

- **Air Temperature**
- Course
- Position
- Speed

Which of the following sensor information is displayed on the default screen (see Task 1.1)?

- **Speed**
- Air Temperature
- All three alternatives
- Engine RPM

Which of the following statements, referring to the "Four Fathom Bank" buoy is true (see Task 2.17)?

- **The buoy is the green lateral buoy of "San Francisco Main Ship Channel"**
- The buoy indicates an isolated danger to starboard on "San Francisco Main Channel"
- The buoy indicates safe water margin on starboard on "Bonita Channel"
- The buoy is the red lateral buoy of "San Francisco Main Ship Channel"

Which parameter does the swing radius depend on using the anchor watch feature?

- **All three alternatives**
- Chain Scope
- Ship Length
- Water Depth

Which requirements have to be met, to obtain a position by LOPs?

- **The position can be obtained by a single reference object only by bearings**
- Range and Bearing must be known to obtain a position fix
- The position can be obtained by a single LOP only by bearing
- The position must at least refer to 3 objects of reference

Which statement is true regarding the "Look- Ahead"?

- **The Look-Ahead is always activated**
- The Look-Ahead can be only activated manually
- The Look-Ahead is active only if "Show Safety Region" is activated
- The Look-Ahead is automatically activated if route is monitored

Which statement is true regarding the MOB function?

- **MOB datum is adjusted automatically for set and drift**
- MOB datum cannot be adjusted manually
- MOB events cannot be triggered by a single operator button
- Opening the Man Overboard window automatically triggers a MOB event

Which statement is true regarding the Safety Depth?

- **The Safety Depth value determines which spot soundings are highlighted by bold black numbers**
- All three alternatives
- The Safety Depth is a bold black contour line, seperating navigable and non navigable water
- The Safety Depth is a feature only available in four shade mode

Which statement is true regarding the available help features?

- **The iHelp function is interactive and provides context related help**
- The "Vision Master Help" feature provides brief information as a tool tip
- The iHelp function is a reproduction of the entire system's manual
- There is no tool tip feature available

Which statement is true regarding the dongle?

- **On start-up, the system compares permit codes from dongle with product identifier**
- An extra Super-PIN can be entered to bypass the dongle
- The ECDIS powers up without external sensors if dongle is not connected
- The ECDIS will not power up if the dongle is missing

Which statement regarding monitored routes is true?

- **A monitored route can be edited, but must be reloaded**
- A route cannot be edited during route monitoring
- A route track will not change colour when activated for monitoring
- No route can be planned during route monitoring

Which vertical datum does the clearance of the "Golden Gate Bridge" refer to? (see Task 2.12)

- **Mean high water**
- Highest astronomical tide
- Neap high water
- Spring high water

4101 – Tank Cleaning 1, Introduction and Safety

Version 2.0 (1338)

Following basic stages will lead to a potential electrostatic hazard.

- **Charge separation, charge accumulation and electrostatic discharge**
- Charge separation and electrostatic discharge
- Charge separation, charge neutralization, electrostatic discharge
- Two different materials rubbing against each other

ISGOTT is published by ICS, OCIMF and IAPH. Why is it wise to have this publication on board a tanker?

- **It deals with industrial best practices on all aspects of the safe carriage and handling of oils on board tankers and at terminals**
- It deals with environmental issues in relation to tankers
- It is required by IMO
- It is required by international law

Is it allowed to recirculate the tank cleaning water after carrying a cargo of petroleum?

- **Yes - if the tanks are inerted, it is allowed to recirculate the tank cleaning water**
- No - recirculation the tank cleaning water will increase the risk of an electrostatic discharge, therefore it is not allowed under any circumstances
- No - recirculation will prevent the tank from becoming clean
- Yes - it is always allowed to recirculate the tank cleaning water

The common denominator for almost all incidents and accidents during tank cleaning is found to be:

- **Human factors**
- Bad weather
- Malfunctioning or insufficient equipment
- Wrong or insufficient cargo information

The fire hazard is controlled by “removing” the oxygen side in the fire-triangle when cleaning an inerted tank. How do we control the fire hazard if the tank is not inerted?

- **By "removing" both the fuel and the sources of ignition**
- By "removing" both the fuel, oxygen, and the sources of ignition
- By "removing" the fuel
- By "removing" the sources of ignition

What is the flashpoint?

- **The lowest temperature where a liquid will vapourize to form an ignitable mixture in air**
- The minimum energy required to ignite an explosive mixture
- The minimum gas percentage required for a gas/air mixture to ignite
- The temperature at which a substance will spontaneously ignite without an external heat source

What will happen with the flammable range if the oxygen content is lowered?

- **The range will become more narrow**
- Nothing will happen
- The flammable range is not oxygen dependent
- The range will become wider

When gasfreeing on board a tanker without inertgas, when can you start to gasfree at deck level instead of using the high velocity vents or the dedicated ventilation pipes?

- **When the gas concentration is less than 30 percent of the lower flammable limit**
- When the gas concentration is less than 10 percent of the lower flammable limit
- When the gas concentration is less than 20 percent of the lower flammable limit
- When the gas concentration is less than 40 percent of the lower flammable limit

When is a tank considered to be fully inerted in accordance with SOLAS?

- **at max 8% oxygen**
- At max 11% oxygen
- at max 15% oxygen
- at max 5% oxygen

When is it safe to clean a tank using steam?

- **When it has either been inerted or cleaned and gas freed**
- After a line and bottom flush where the residues have been transferred to a slop tank
- Before ventilation
- It is never safe to clean a tank using steam

When oil is flowing through a steel pipe both will be electrostatically charged in such a way that:

- **One will be negatively charged, the other will be positively charged**
- Both will be negatively charged
- Both will be positively charged
- None of them will be charged

Which event or operation could lead to a charge separation within a tank?
(More than one alternative may be correct)

- **Liquid flowing through pipes**
- **Steaming**
- **Tank cleaning**
- **Water and/or air in product**

Which of the following gases has the highest relative density?

- **Hydro-Carbon gases**
- Air
- Methane
- Nitrogen

While cleaning an non-inerted tank you are testing the atmosphere with an explosimeter. You measure 25% of the Lower Flammable Limit. Will you stop the cleaning and ventilate?

- **No, I will not stop and ventilate since the maximum limit is 35% of the Lower Flammable Limit**
- No, I will not stop and ventilate since there are no limitations
- Yes, I will stop and ventilate since the maximum limit is 10% of the Lower Flammable Limit
- Yes, I will stop and ventilate since there should be no flammable gases present

You have decided to add tank cleaning chemicals to the tank cleaning water. You are washing in a non inert atmosphere. What will you do?

- **Ensure that tank cleaning chemicals are only used if the temperature of the wash water is below 60 degrees Celsius**
- Ensure that the oxygen content of the tank is below 11%
- It is not allowed to use tank cleaning chemicals on board a non-inerted ship
- Start cleaning with the chemical using hot water

4102 – Tank Cleaning 2, Principles of Tank Cleaning

Version 1.0 (1104)

Chemical reaction is supported by

- **Heat**
- Low dew point
- The presence of hydrocarbons
- The presence of oxygen

If you want to prepare the tank by ventilation only the vapour pressure must normally be more than

- **5 kpa**
- 0,5 Bar
- 5 mB
- 500 mB

The most common reason for tank cleaning is :

- **Prepare cargo tanks for carriage of the next cargo**
- Facilitate gas freeing - tank entry
- Prevent build-up of cargo residues
- To protect the environment

The tank washing principle wet-in-wet means

- **Initial tank washing shall start as soon as possible to avoid cargo residues to dry up.**
- Before cleaning with chemicals you must flush the cargo tank bottom
- Cleaning of cargo lines must be done simultaneously with washing of cargo tanks
- It is not possible to ventilate the tanks due to a high dew-point

The viscosity of a liquid depends on the

- **Temperature of the liquid**
- Acid value of the liquid
- Flash point of the liquid
- Vapour pressure of the liquid

What is a wall wash test?

- **A test to determine the presence of contamination on the bulkheads**
- A test to determine that the coating is fully cured
- A test to determine the passive layer in Stainless steel tanks
- A test to examine the thickness of epoxy coating

What is an inhibitor

- **A chemical added to products which can polymerize**
- A chemical used to remove smell
- A chemical with freezing point below -25°C
- A cleaning chemical for cleaning water sensitive cargoes

What is the most common reason for performing crude oil wash

- **To prevent cargo sludge building up**
- To avoid contamination of the next cargo
- To comply with the Volatile Organic Compound Management Plan
- To make it easier to ventilate the tank

What is the pour point

- **The lowest temperature at which a liquid will flow**
- The lowest temeprature where it is possible to remove the product by ventilation
- The lowest temperature at which the product can be ignited
- The lowest temperature where the product can give off vapour

What will be the normal pressure when cleaning with dual nozzle tank cleaning machines?

- **8-10 bar**
- 10-12 bar
- 4-6 bar
- 6-8 bar

You are discharging Fuel Oil with a Specific Gravity of 0.85 direct to the sea. What will you expect to happen?

- **The oil will float**
- The oil will be diluted by the sea water
- The oil will evaporate
- The oil will sink

You are going to wash cargo tanks after unloading Heavy Fuel Oil with a pour point of 43°C. What should be the initial washing water temperature?

- **60°C**
- 35°C
- 45°C
- 80°C

4103 – Tank Cleaning 3, Planning of Tank Cleaning Operations

Version 1.0 (887)

Before using tank cleaning chemicals for cleaning the tanks it is important to check :

- **Coating resistance list**
- Air humidity
- Salinity of washing water
- Tank cleaning pump capacity

Epoxy coating is a

- **Organic coating**
- Aluminium powder coating
- Inorganic coating
- Rubber coating

The duration of the tank washing depends on :
(More than one alternative may be correct)

- **The type and condition of the coating**
- **Viscosity of the product unloaded**
- **Washing water temperature**

What is White Water Standard ?

- **A cleanliness Standard where the tanks are cleaned so they are visually clean, dry and odour free**
- A cleaning chemical especially for Stainless Steel Tanks
- A tank cleaning method used to clean tanks in order to carry drinking water
- Sea Water treated with chemicals to remove chlorides

What is a detergent?

- **A synthetic cleaning agent resembling soap**
- A chemical that reduces the PH value of the washing water
- A chemical that removes water hardness in sea water
- A cleaning chemical that removes smell after fish oil

What is meant by the phrase "tank history"

- **A list of the last three cargoes carried in the tank**
- A description of the last three tank cleaning operations
- Number of times the tank has been recoated
- The tank coating age

When alkalis and vegetable oil are mixed they form

- **Soap**
- Alcohol
- Butter
- Organic Acids

When washing after a flammable cargo in a non inerted cargo tank, warm water wash should be preceded by a cold water wash. Why?

- **To control the sources of ignition**
- In order to protect the coating
- To avoid excessive vapour pressure
- To avoid low pressure in the cargo tank

When washing with warm water, the washing temperature is normally between

- **30-60°C**
- 10-20°C
- 20-30°C
- 60-80°C

Why is salt water the preferred washing liquid?

- **It is readily available**
- It evaporates easily
- Salt water has a very good cleaning effect
- The components in salt water acts like a cleaning agent

Why should hand spraying with chemical be avoided

- **For safety reasons**
- It is not a good cleaning method
- It is prohibited by ISGOTT and Tanker safety Guide Chemicals
- The amount of cleaning chemicals to be used is too high

Why should washing in stainless steel tanks with salt water be avoided?

- **The presence of chloride ions in salt water may cause corrosion**
- Salt water discolours stainless steel
- Stainless steel tanks can be washed with salt water without any problems
- The relative density of salt water is too high

4121 – Ex Basic, Organization and rules

Version 1.0 (663)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4122 – Ex Basic, Ex theory and area classification

Version 1.0 (594)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4123 – Ex Basic, Ex equipment

Version 1.0 (292)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4124 – Ex Basic, Protection concepts

Version 1.0 (1435)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4125 – Ex Basic, Protection concepts Exi

Version 1.0 (1401)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4126 – Ex Basic, Ex protection in practice

Version 1.0 (654)

- **I hereby declare that I have completed all the contents in this module and that it is my own work and effort.**

4127 – Ex Basic, Assessment

Version 1.0 (149)

How big AC nominal tension requires Protective Earthing (PE)?

- **Over 50 V**
- 50 V
- Under 50 V

How many earthing wires may be connected to one screw?

- **1**
- 1 to 2
- 1 to 3

How often must you repeat the course on using electrical skills in the explosive areas?

- **Every 5 years**
- Every 3 years
- Every 7 years

In which Ex-zone is it allowed to use Exe motors?

- **1 and 2**
- 0
- 0 and 1

In which Ex-zone is it allowed to use the Exn motors?

- **2**
- 0
- 1

In which zone can we use this equipment: ATEX II 2 G?

- **Zone 1**
- Zone
- Zone 2

In which zone is it allowed to use Exib equipment?

- **1**
- 0
- 2

In which zone(s) must we use Ex certified electrical equipment?

- **0 and 1**
- 0
- 0, 1 and 2

To test the earthing resistance with MegaOhm-meter, is it necessary to disconnect the earthing wire?

- **Yes, both ends**
- No
- Yes, the end on testing side

What does the housing IP-grade show?

- **Protection against water and dust**
- Protection against gas
- Protection against power surge

What does the letter "e" mean in Ex-marking?

- **Ignition-proof equipment**
- Explosion-proof equipment
- Intrinsically safe equipment

What does the letter "i" mean in Ex-marking?

- **Intrinsically safe equipment**
- Explosion-proof equipmen
- Ignition-proof equipment

What does this marking mean: Eex ib IIB T5?

- **Intrinsically safe equipment, approved for zones 1 and 2.**
- Explosion-proof equipment, approved for zones 1 and 2.
- Ignition-proof equipment, approved for zones 1 and 2.

What is considered the normal ambient temperature for electrical equipment?

- **-20C to +40C**
- -10C to +40C
- -20C to +50C

What is the maximum allowed resistance between connection point and main earthing in the intrinsically safe circuits?

- **1,0 Ohm**
- 0,1 Ohm
- 10 Ohm

What is the maximum allowed voltage for an Exe motor ?

- **11 kV**
- 1,1 kV
- 110 kV

What is the minimum allowed wire cross-section for cables used for portable devices?

- **1,0 Square millimeters**
- 0,5 Square millimeters
- 0,75 Square millimeters
- 1,5 Square millimeters

What is the minimum required distance between intrinsically safe end connections and non-intrinsically safe connections?

- **50 mm**
- 30 mm
- 500 mm

What is the minimum required distance between the intrinsically safe end connections and earthed metal?

- **3 mm**
- 10 mm
- 5 mm

When is it possible to use electrical equipment at non-standard environmental temperatures?

- **If the marked temperature range for the given equipment matches the environment temperatures**
- If the user has tested the temperature range first
- It is not possible

Where does the FSE directive apply?

- **Onshore**
- Offshore
- Onshore and offshore

Where does the FUSEX directive apply?

- **Onshore and offshore**
- Offshore
- Onshore

Which Ex-equipment group is meant for using in the coal mines?

- **I**
- I and II
- II

Which Ex-equipment is allowed to use in zone 0?

- **Ex ia**
- Ex e
- Ex ib

Which Ex-protection class does this equipment use: equipment cabinet containing contactors and fuses, with direct cable entry?

- **Ex de**
- Ex qe
- Exd

Which colour marks the intrinsically safe circuit cables?

- **Light blue**
- Black
- Red
- Yellow

Which equipment is not allowed to use in zone 1?

- **Exn equipment**
- Exi equipment
- Zone 0 equipment

Which equipment requires Exd glands?

- **Ex d**
- Ex e
- Ex i

Which explosion-hazard zone do these areas belong to: tanks, pipes, vicinity of safety valves?

- **0**
- 1
- 2

Which gas group indicates the rapid increase in pressure?

- **IIC**
- IIA
- IIB

Which gas group indicates the slow increase in pressure?

- **IIA**
- IIB
- IIC

Which glands is it allowed to use with Exe equipment?

- **Exe glands**
- Exd glands
- Regular industrial glands

Which is the marking for oil-filled devices?

- **Ex o**
- Ex d
- Ex i
- Ex p

Which is the marking for over-pressure devices?

- **Ex p**
- Ex e
- Ex i
- Ex o

Which power-supply system is not allowed to use in the explosion-hazard areas?

- **TN-C**
- IT
- TN-S
- TT

Who is allowed to repair the Ex-equipment?

- **Certified workshop**
- An electrician who has taken the Ex-course
- End user

Who may not repair serious defects and do flame-path repairs?

- **End user**
- Certified workshop
- Manufacturer

With which equipment is it not allowed to use Exe glands?

- **Exd with direct cable entry**
- Exd with indirect cable entry
- Exe

With which gas group is it allowed to use the class IIB equipment?

- **IIA and IIB gases**
- All gases
- IIA gases

With which gas group is it allowed to use the class IIC equipment?

- **All gases**
- IIA and IIB gases
- IIA gases

4150 – Norwegian maritime rules and regulations

Version 1.0 (1498)

EEA citizens who are holders of certificates that will be recognised by the Norwegian Maritime Authority in accordance with the regulations will be issued a Certificate of Receipt of Application (CRA) once they have submitted a complete application for endorsement. For how long can the applicant work on board with a CRA?

- **Until the application has been processed and decided.**
- For a maximum of six months.
- For a maximum of three months.

How can the crew know whether or not exemptions have been made from Norwegian regulations on board?

- **The exemptions are noted in collective bargaining agreements.**
- Every ship will have its own booklet containing exemptions from regulations.
- It is not possible to make exemptions from Norwegian regulations.

If the Master is absent, who makes decisions that cannot be postponed?

- **The senior mate present.**
- The Chief Engineer.

What is The Excerpts from the Norwegian Passenger and Cargo Ship Legislation?

- **A collection of rules and regulations that are relevant for Norwegian flagged vessels.**
- Exeptions in rules and regulations that are not relevant for Norwegian flagged vessels.
- Norwegian rules and regulations that are relevant for all international vessels.

Where can you find a comprehensive list of publications and documents that must be carried on board?

- **In the Excerpts from the Norwegian Passenger and Cargo Ship Legislation.**
- At lovdata.no
- In the Ship Labour Act.

Where can you find information on who to contact in case of an accident?

- **On the Norwegian Maritime Authority's home page.**
- In the Excerpts from the Norwegian Passenger and Cargo Ship Legislation.
- In the Ship Safety and Security Act.

Who is responsible for ensuring safety at sea?

- **The state under whose flag the ship is registered.**
- The Ministry of Trade, Industry and Fisheries.
- The Norwegian Petroleum Safety Authority.

Who is responsible for ensuring that Norwegian laws are complied with on board?

- **The Master.**
- The Company.
- The Company's Designated Person.
- The Superintendent.

Who is responsible for supervising NIS flagged vessels?

- **The Classification Societies.**
- Norwegian Police.
- The NMA.
- The Norwegian Coast Guard.

Who shall ensure that loading and discharge are carried out and the voyage performed with due dispatch?

- **The Master.**
- The Chief Officer.
- The NMA.

7001 – TOTS 1A-Ship Handling and Characteristics

Version 1.0 (32)

A "Williamson Turn", can be used in a Man Overboard situation, what is the first action when executing such a turn?.

- **Put the wheel hard over towards the man in the water.**
- Alter course 60 degrees to starboard.
- Reduce engine speed to Dead Slow Ahead.
- Stop engines.

At which speed could the ship's bow (or stern) thruster be mechanically damaged?

- **At a ship's speed of more than five knots.**
- At a ship's speed of more than two knots.
- When the vessel is running astern at a speed of less than two knots.
- When the vessel is stopped.

At which speed is the bow (or stern) thruster ineffective?

- **At a ship's speed of more than five knots.**
- At a ship's speed of more than two knots.
- When the vessel is running astern at a speed of less than two knots.
- When the vessel is stopped.

From which direction does the wind have the greatest effect on the vessel's manoeuvring?

- **On the beam, with the vessel in ballast condition.**
- From right ahead.
- From right astern.
- On the quarter, with the vessel in loaded condition.

How long can the main engine be run on critical revolutions?

- **Long enough for the transfer from one speed setting to another.**
- As long as required.
- As long as the Chief Engineer permits.
- It should not be run on critical revolutions at all.

How should you secure the anchor cable when your vessel is anchored?

- **By using the cable stopper and windlass brake, with slight slack on the cable between the stopper and the brake.**
- By using lashing wires and the brake.
- By using the anchor windlass brake and leaving the motor in gear.
- By using the cable stopper.

If you are sailing in shallow water with 20% UKC (Under Keel Clearance), how will your ship's stopping distance be affected, relative to deep water?

- **Longer than normal.**
- About 1.5 ships longer.
- Shorter than normal.
- The same as for deep waters.

If your vessel is turning to starboard in shallow water with 20% UKC (Under Keel Clearance), what will be the diameter of the turn when compared with deep water?

- **The turn will be wider than a normal deep water turn.**
- The turn will be same as a deep water turn.
- The turn will be tighter than a normal deep water turn.
- The turn will be wider on the port side and tighter on the starboard side than a normal deep water turn.

What action should be avoided during the approach to the anchorage?

- **Crossing in front of an anchored vessel at short distance.**
- All three answers.
- Passing behind an anchored vessel at short distance and slow speed.
- Passing behind another vessel that is manoeuvring towards anchorage.

What actions should be taken immediately by the Officer of the Watch if the vessel starts dragging anchor?

- **Call the Master, inform the engine room that the main engine is required**
- Give one hours notice to the engine room that the main engine is required
- Inform the Port Authority or Coast Guard.
- Inform the vessel behind you that your vessel is dragging anchor.

What could reduce the effect of squat on your vessel?

- **Decreasing the speed of the vessel.**
- Increasing the ship's speed.
- Maintaining your ship's present speed and course.
- Trim the vessel by stern as much as possible.

What data can be found on a ship's Manoeuvring Poster?

- **Turning circles in the ballast and loaded conditions.**
- All three answers.
- Blind sectors for RADARs.
- Emergency contact numbers for CSO.

What equipment must be tested before anchoring?

- **The steering gear, anchor windlass, and main engine ahead and astern.**
- Main and emergency fire pumps.
- The anchor lights and deck lighting.
- The fire alarm system.

What information can you find from the ship's turning circles posted on the Manoeuvring Posters?

- **Information about the "wheel-over point".**
- All three answers.
- Information about the stopping distances in zig-zag manoeuvres.
- Information on the stopping distance while the vessel carries out a full turn in ballast condition.

What is the "pivot point"?

- **It is the imaginary point around which the vessel turns.**
- It is the point where the forward light mast is posted.
- It is the point where the forward tugboats should be secured for towing.
- It is the point where the ship's centre manifolds are fitted.

What is the correct way of anchoring in deep water?

- **Approach the anchorage at slow speed, walk out the anchor to above the water line, stop the vessel, and then walk out the chain by the windlass to the required amount when ready.**
- Approach the anchorage at slow speed, pay out the anchor for one shackle and drop the anchor by the brake.
- Approach the anchorage at slow speed, stop the vessel and drop the anchor from the hawe pipe.
- Approach the anchorage at slow speed, stop the vessel, pay out the anchor for one shackle and drop out the anchor by the brake.

What is the formula to calculate squat in open sea conditions?

- **The speed in knots squared, multiplied by the block coefficient.**
- One and a half times the speed in knots squared, plus the block coefficient.
- The draught in metres squared, multiplied by the block coefficient.
- The draught in metres, times the speed in knots, times the block coefficient.

What means of position-fixing should you use during the approach to anchorage?

- **All available means.**
- GPS, because it is more accurate.
- Radar only, due to the increased workload on the bridge team.
- Visual methods, such as buoys.

What means should be used when securing the anchor for sea?

- **It should be secured by the windlass brake, cable stopper and lashing wire.**
- In good weather it can be secured by the windlass brake only.
- It should be secured by engaging the gipsy and securing the brake.
- It should be secured by the windlass brake, ready for immediate use.

What should be done as a first response to the "Man Overboard Emergency"?

- **Raise the Emergency Alarm, drop the MOB buoy on the side of the man overboard, place wheel hard over on the side of man overboard, place engines on standby.**
- Inform the DPA.
- Inform the Master and await his arrival to the bridge.
- Mark the event on the GPS set.

What should be undertaken before you switch from sea to manoeuvring speed?

- **Give one-hour notice to the engine room before requesting them to slow down from Sea to Manoeuvring Speed.**
- Check the length of the remaining voyage before commencing to manoeuvring.
- Check the ship's ETA.
- Inform the Master and the Pilot.

What should you watch for if you are overtaking a similar size vessel in a narrow channel?

- **The bow of your vessel could be sucked towards the other vessel and the stern pushed away as you pass.**
- Make sure you maintain course because the bow of your vessel could be pushed away from the other vessel.
- That the other vessel is not decreasing its speed.
- That you keep your vessel at full sea speed during overtaking.

What would be the correct method of approach to a busy anchorage area?

- **Slow down well in advance, approach in a controlled manner and identify a suitable anchoring position.**
- Proceed at dead slow ahead until the anchorage area is clear.
- Proceed at full manouvering speed and look for a suitable anchorage.
- Stop a few nautical miles before the designated anchorage area and wait for the anchorage area to clear-up.

Where is the best place to anchor in a crowded anchorage?

- **At the edge of the anchorage.**
- In the centre of the crowded anchorage.
- In the deepest part of the crowded anchorage.
- In the fairway entrance.

Which communications links should be tested before anchoring?

- **Between the bridge and the forecastle, engine room, and steering gear**
- All GMDSS equipment.
- Between the bridge wings and the wheelhouse.
- Between the ship and the port authority or traffic information service.

Which movements of the vessel could increase the ship's draft and reduce the Under keel Clearance?

- **Squat, pitching in a seaway, heeling due to rapid alterations of course**
- Changes in the vessel's passage plan.
- Uncharted areas of seabed.
- Unmarked wrecks or seabed features.

Why is it important to have the length of the ship's forward and aft vertical blind sectors listed on the Manoeuvring Posters?

- **To know the length of the blind area under the ship's forecastle or behind the stern that is not directly visible from the wheel-house.**
- It enables the Officer of the Watch to estimate speed through the water.
- It is a class requirement for them to be listed on the Manoeuvring Posters.
- This information is required when taking compass bearings with the bridge wing gyro repeaters.

Why should the maximum air draft be carefully calculated?

- **So that the air clearance is known when passing under bridges or electrical cables over channels or rivers.**
- Because the Officer of the Watch requires this information when using the sextant.
- Because this information is always required by port operators.
- This information is required for the GMDSS Log book.

You are on a ULCC of 300,000 tonne DWT in calm conditions and you are supposed to anchor in 90 metres depth. How much chain should you use?

- **About eleven shackles.**
- About eight shackles.
- About five shackles.
- About three shackles.

You are sailing on a small coaster, approaching the berth port side at the speed of one knot, and have moderate wind from the starboard side and no tug. How could you assist yourself in the situation described?

- **Let go starboard anchor and pay out the cable controlling the bow as you are approaching the berth.**
- Give the rudder hard a starboard.
- Increase the speed and drop port anchor one shackle in the water.
- Let go port anchor and gradually release the chain as you are approaching the berth.

You meet another vessel port to port in a narrow channel. What interaction could you expect of both ships?

- **The bows of both ships will be pushed away from each other and the aft parts of the vessels could make contact.**
- Only the ship with the deepest draft will be affected.
- Only the ship with the lightest draft will be affected.
- The bows of both ships will be sucked together and the forward parts of the vessels could make contact.

Your vessel has a conventional right-hand propeller. How would you position your rudder before going full power astern?

- **Amidships.**
- Hard a port.
- Hard a starboard.
- Ten degrees to port.

7002 – TOTS 1A-Pilotage

Version 1.0 (6)

If the Captain is unable to contact the Pilot Authority directly, he may pass a message via the . . .

- **local agent.**
- local coastguard.
- local ship handler.
- Port State Control.

If you do not agree with the Pilot's intended actions, would you . . .

- **immediately discuss the action with the Pilot and Master.**
- consider asking him later after the passage.
- inform the VTS (Vessel Traffic Services).
- not interfere since he is the Pilot.

Is a Canal Helmsman compulsory?

- **It depends on the local Pilotage Authorities' regulations and the conditions of pilotage.**
- No, never.
- Only in poor visibility.
- Yes, always.

The Pilot requires a seaman to stand-by forward for the whole pilotage. Do you . . .

- **organise your available manpower accordingly and post a seaman forward?**
- explain to the Pilot that it is not in the vessel's SMS (Safety Management System) and so you are unable to do so?
- refuse on the basis of fatigue?
- simply refuse as you do not have the manpower?

The Wheel-house Poster contains information regarding . . .

- **specific manoeuvring information / characteristics of the vessel.**
- boarding arrangements for pilots.
- similar information to the Pilot Card.
- the owners / managers standing orders.

The agreed language is English, however the Canal Helmsman repeats orders in a different language. What action would you take?

- **Immediately remind the Pilot and the Helmsman that the agreed language is English and to comply accordingly.**
- As long as the vessel is on course and the pilot and helmsman understand each other it does not matter
- Let the Pilot and the Helmsman communicate how they wish.
- Make a complaint to the Pilotage Authority.

The presence of a pilot on the ship does not relieve the master or officer in charge of the navigational watch from their duties and obligations for the safe conduct of the ship.

- **True**
- False

The vessel's freeboard is 9.5 metres. How would you disembark the Pilot?

- **Using a pilot ladder combined with an accommodation ladder.**
- Being just inside the regulations, using a pilot ladder only would be satisfactory.
- Obtain the services of a helicopter.
- Using the accommodation ladder only.

The weather is extremely bad with a poor forecast, so disembarking the pilot would be very dangerous. What actions would you take?

- **Proceed with the pilot to the next port or suitable rendezvous position.**
- Anchor and wait for an improvement in the weather.
- Check with the charter party for a deviation clause.
- Return to the port.

The wind and sea are on the port bow. Which side would you rig the pilot ladder?

- **The starboard side.**
- Ask the pilot which side.
- Ask the VTS (Vessel Traffic Services) which side.
- The port side.

The wind and sea are on the starboard bow, however, the swell is on the port beam. What would your initial action be regarding pilot disembarkation?

- **To fully discuss the circumstances with the Pilot.**
- Ask the pilot boat.
- Ask VTS (Vessel Traffic Services).
- Rig the pilot ladder on the port side.

What information should be passed to the Pilot when he arrives on the bridge?

- **The Master/Pilot Exchange information card.**
- ISPS procedures.
- The location of life-saving appliances.
- The VHF channels being monitored.

What is a "Pilot Card"?

- **It's a detailed description of the vessel, the main engine characteristics and operational status, and the limitations of various equipment.**
- It contains details of the Pilot's name and the maximum size of vessel he can pilot.
- It's confirmation of the Pilot's attendance on board.
- It's the card the Master signs to enable the Pilot to receive his fee.

What is described by the "Air draft"?

- **The height of the highest point of the vessel above the waterline.**
- The ship's draft in different terms.
- The wind force at pilot boarding position.
- The wind force on board a vessel when approaching port.

What is your understanding of mixed moorings?

- **Natural or man-made fibre ropes that are used in conjunction with wire ropes.**
- A combination wire.
- Different diameters of wire ropes.
- Some mooring lines that are secured on a winch while others are secured on the bitts.

What would your actions be if the Canal Helmsman requested a short rest due to fatigue?

- **Upon obtaining the Pilot's agreement, the vessel's own Helmsman would take over under close supervision.**
- He should not be allowed to rest under any circumstances.
- Let the Pilot steer the vessel.
- The Officer of the Watch would take over.

When a canal helmsman is being used, the ship's helmsman should . . .

- **remain on the bridge in case he is required.**
- get some rest while he can.
- stand-by in the messroom.
- take the opportunity to clean the wheel-house.

When a helm order is given by the Pilot, the Helmsman will . . .

- **repeat and execute the order.**
- ask the Captain.
- execute the order.
- wait for the Officer of the Watch to confirm.

When discussing the UKC with the Pilot, what is your understanding of the term "controlling depth"?

- **The minimum depth of water within the limits of a channel.**
- An allowance for squat.
- The maximum draft the vessel can transit at.
- The sounding on a chart close to where you want to transit.

When is the final "pilotage passage plan" agreed with the Pilot?

- **As soon as he arrives on the bridge.**
- By VHF prior to his arrival.
- By VTS (Vessel Traffic Services).
- The local agent will keep him informed.

When making verbal contact with the pilot station, the ship will be identified by name and . . .

- **its call sign.**
- its IMO number.
- its nationality (flag).
- its telex number.

When the vessel is under pilotage, where would you find information regarding the bridge team compliment and respective responsibilities?

- **The company's SMS (Safety Management System).**
- The Admiralty Sailing Directions.
- The Master's standing orders.
- The Pilot Card.

When would you use a combination ladder for pilot boarding?

- **When the vessel's freeboard exceeds 9 metres.**
- Only when the pilot requests it.
- When the accommodation ladder faces forward.
- When the pilot ladder is damaged.

Where would you find your company instructions regarding UKC?

- **In the Company's SMS (Safety Management System).**
- In the approved Trim & Stability book.
- In the SOLAS training manual.
- On the Pilot Card.

Which volume of the Admiralty List of Radio Signals gives contact information for marine pilots?

- **Volume 6.**
- None. Information is only from Guide to Port Entry.
- The Mariner's Handbook.
- Volume 2.

Why does a berthing pilot need to know the propeller's direction of turn (left or right-handed)?

- **The direction of turn of the propeller affects which way the bow may move when operating astern propulsion.**
- It affects the speed of the vessel.
- It effects the response time of the main engine.
- The allowance for squat may be affected.

Why is it important to monitor the vessel's progress in pilotage waters?

- **To ensure that the vessel is proceeding safely as per agreed passage plan.**
- It is not important but it's advisable.
- It is only important in reduced visibility.
- So as to ensure that the ETA at the berth is maintained.

Would you expect a Canal Helmsman to only take helm orders from the Pilot?

- **No, he is under the Master's direct authority also.**
- It may depend upon the local Pilotage Authorities' regulations.
- It's not important as the Master would probably not interfere.
- Yes, the Pilot is in charge.

You observe that the Canal Helmsman is not following the Pilot's instructions. What action would you take?

- **Immediately alert the Pilot to the circumstances.**
- Alert the Pilotage Authority .
- Check that the course the Helmsman has taken is not dangerous.
- Take no action as he is an experienced helmsman for those waters.

Your gyro compass has failed. Would you . . .

- **inform the Pilot?**
- keep quiet in case it affects the pilot's concentration.
- switch on the radar?
- tell the agent?

7003 – TOTS 1A-Mooring Equipment and Operations

Version 2.0 (12)

At what height should the eyes of the emergency towing wires (fire wires) be positioned?

- **They should be run out and maintained at the waterline or about two metres above it.**
- They should be kept at least two meters above the minimum ballast draught level.
- They should be kept at least two metres above the maximum loaded draught level.
- They should be kept secured not less than two metres above the waterline.

During a vessel's stay on a SPM berth, it is required to continuously monitor the ship's mooring.

- **True**
- False

During ship-to-ship operations, vessels use fenders. What is the purpose of the "primary fenders"?

- **These are large fenders, which float between the ships, and are made to absorb the impact of two ships approaching during berthing and later when sailing together.**
- These fenders are fixed to the bow of the attending Tender Vessel during the transfer of the transfer hose.
- These small fenders (Manila Fender ball) are used by the crew during an emergency.
- These small fenders (tyre type) are used from the deck during an emergency.

During ship-to-ship operations, vessels use fenders. What is the purpose of the "secondary fenders"?

- **These small fenders which are suspended near the ship's bow or stern protect the vessel from ship-to-ship contact.**
- These large fenders are used on the "mother vessel" for separating the two ships.
- These small fenders (Manila Fender ball) are used by the crew during an emergency.
- These small fenders (tyre type) are used from the deck during an emergency.

How could you define "squat"?

- **The increase in the ship's draught of the point of your ship that is closest to the bottom.**
- The decrease of the draught of your ship.
- The decrease of the speed and draft of your ship in confined waters.
- The decrease of the speed of your ship in confined waters.

How often should the mooring winch brake on tankers be tested?

- **At least every 12 months or after each change of brake band lining.**
- At each repair or dry-docking.
- At fixed 12-month intervals.
- At fixed 24-month intervals.

How should you secure the anchor chain while your vessel is being anchored?

- **By using the chain stopper and windlass brake, with slight slack on the chain between the chain stopper and the brake.**
- By using lashing wires and the brake.
- By using the anchor windlass brake and leaving the motor in gear.
- By using the chain stopper.

If you are sailing in shallow water with 20% UKC (Under Keel Clearance), what will be your ship's stopping distance?

- **Longer than normal.**
- About 1.5 ships longer.
- Shorter than normal.
- The same as for deep waters.

If you are using one tugboat to assist the vessel for turning bow to port in a narrow channel (with the ship's main engine also being used), where should you position the tug and how should you use it?

- **Make the tug fast on port shoulder and use it for pulling.**
- Make the tug fast aft on starboard shoulder and use it for pulling.
- Make the tug fast forward through centre fairlead and start pulling.
- Make the tug fast on starboard aft shoulder and use it for pushing.

If your vessel is turning in shallow water with 20% UKC (Under Keel Clearance), what will be the diameter of the turn when compared with deep water?

- **The turn will be wider than a normal deep water turn.**
- The turn will be same as a deep water turn.
- The turn will be tighter than a normal deep water turn.
- The turn will be wider on the port side and slighter on the starboard side than a normal deep water turn.

Is it acceptable to travel at the speed of four knots in the approach to the berth?

- **It's too fast.**
- It is up to the Pilot to decide.
- It's normal.
- It's sufficient for manoeuvring.

What could minimise the effect of squat on your vessel?

- **Decreasing the speed of the vessel as much as safe navigation permits.**
- Increasing the ship's speed if you are not already on full ahead.
- Maintaining your ship's present speed and course.
- Trim the vessel by stern as much as possible.

What important design feature should be met for construction of the forward ETA (Emergency Towing Arrangement) system?

- **The forward ETA should be capable of being deployed in harbour condition in not more than one hour.**
- The forward ETA should be able to be used for normal berthing of a vessel alongside the terminal.
- The forward ETA should be capable of being deployed in not more than one hour by two crew members.
- The forward ETA should be fitted on the centre line of the vessel.

What important design feature should be met for the construction of the aft ETA (Emergency Towing Arrangement) system?

- **The aft ETA should be pre-rigged and capable of being deployed in a controlled manner in controlled conditions in not more than 15 minutes by one person.**
- The aft ETA should be able to be remote-controlled from the stern during deployment.
- The aft ETA should have a SWL of at least 200 tonnes.
- The aft ETA should have a towing cable of at least 100 metres in length.

What information should be available on board for each mooring rope or mooring wire?

- **A certificate of quality stating the construction, the material and MBL/SWL, the position of the rope on board and the age (time of use of particular rope).**
- The MBL of mooring ropes or wires.
- The SWL of mooring ropes or wires.
- The type of construction material of the mooring ropes and wires in order to avoid using mixed mooring in the same direction.

What is an advantage of split-type mooring drums?

- **All three answers.**
- The mooring wires are not "crushed" under load.
- The use of the split drum uses the winch braking capacity to its maximum.
- There is a smaller chance that the mooring rope will be "jammed" on the storage part of reel.

What is it important to avoid during berthing operations?

- **The mooring rope fouling around the turning propeller.**
- The forward and aft winches working at the same time.
- The mooring stations remaining without communication by hand-held walkie-talkies with the wheel-house.
- The mooring winches set on the correct speed and load setting.

What is it important to discuss during the pre-berthing briefing between the Master and the ship's officers?

- **The specific berthing operation for the forthcoming berthing, the use of tugs and the position of the mooring ropes/wires to be used.**
- How many crew members should be called for berthing operations.
- The size (pulling power) of the tugs to be used during berthing.
- Which officer will be on the forecastle and which on the stern during mooring operations.

What is the "Bollard Pull" of the tugboat?

- **All three answers.**
- It is the indicator of a tug's power to push or pull.
- It is the main criteria of a tug and displays the pulling force expressed in tonnes.
- It is the measure of a tug's thrust at zero speed through water with full engine RPM.

What is the CPP?

- **All three answers.**
- The CPP is a special type of propeller that can be rotated around its long axis to change its pitch.
- The CPP is the propeller that has a variable pitch in order to obtain the optimum efficiency in relation to a ship's load so that fuel can be saved.
- The CPP is when the pitch of the ship's propeller is controlled by the hydraulic system working through the propeller shaft.

What is the MBL of mooring ropes or wires?

- **Minimum Breaking Load.**
- Maximum Breaking Load.
- Minimum Braking Length
- Minimum Braking Load.

What is the correct procedure for carrying out the winch brake test?

- **The winch brake lever is tightened by the torque wrench and the torque force is increased/adjusted until the required rendering force of the winch brake is reached.**
- The MBL of the mooring rope/wire is calculated at 80% along with the required hydraulic jack force based on test equipment design characteristics.
- The MBL of the mooring ropes/wires is calculated at 60% along with the required hydraulic jack force based on test equipment design characteristics.
- The winch brake lever is tightened by hand and tested with the hydraulic brake until the required brake force is achieved.

What is the definition of "make fast"?

- **To secure the rope.**
- To act fast.
- To coil the rope around a storage drum.
- To release fast

What is the first task of the Officer in Charge of the mooring station upon arrival at the station?

- **To check the communication with the wheel-house / the Master.**
- To check that all of his crew are available at the mooring station.
- To check that all of his crew are stand-by.
- To check that all of the mooring equipment is ready for mooring.

What is the most dangerous part of mooring operations?

- **The crew staying in rope "snap-back" and "rope-tight" areas.**
- Anchoring in rivers.
- Berthing of a vessel in high tidal ports.
- Berthing with the use of a Pilot and tugs in restricted visibility conditions.

What is the principal purpose of the emergency towing wires (fire wires) on tankers?

- **To be used if there is a fire on board or at the terminal and the ship needs to be pulled off the terminal.**
- To be used for normal unberthing operation of the vessel.
- To be used for shifting the vessel alongside the jetty.
- To be used to extend the towing rope of the tugboat during unberthing.

What is the recommended angle to be maintained between two ships after unberthing from ship-to-ship operations?

- **5 degrees.**
- 2 degrees.
- 20 degrees.
- 45 degrees.

What is the recommended procedure for securing synthetic rope on a pair of bitts?

- **To make an initial full turn around the outboard bitt and then secure the rope by using a figure eight on both bitts.**
- To make the rope fast in a figure eight on both bitts.
- To make the rope fast on the inner bitt only.
- To make the rope fast on the outboard bitt only.

What is the recommended procedure if your vessel is equipped with "self-stowing automatic tension" winches and your vessel is moored alongside the loading jetty with fixed loading arms?

- **All mooring ropes should be left secured by the manual mooring brakes and all winches should be left out of gear.**
- All winches should be kept in "auto-tension" mode.
- Head and stern line winches should be kept in "auto-tension" mode and breast and spring ropes should be kept manually secured by brakes.
- Spring line winches should be kept in "auto-tension" mode and all other ropes should be kept secured by manual brakes.

What moorings should be avoided if leading in the same direction?

- **Mixed mooring ropes of various quality and strength.**
- Mooring wires not fitted with a non-friction link between the wire and tail.
- Mooring wires not fitted with mooring tails.
- Soft mooring ropes of various ages.

What safe practices should be enforced during mooring operations?

- **All three answers.**
- The familiarisation of new personnel with mooring equipment (e.g. winches, brakes, SWL of bitts).
- The mooring station should always be manned with a sufficient number of trained and skilled Officers in Charge and crew.
- The use of Personal Protective Equipment.

What ships are required to be fitted with the aft ETA (Emergency Towing Arrangement) system?

- **Tankers over 20,000 tonnes DWT.**
- All ships over 20,000 tonnes displacement.
- All ships over 20,000 tonnes DWT.
- Tankers over 20,000 tonnes displacement.

What should be done when a vessel is approaching the Pilot Station with the Pilot on board, and the Chief Engineer advises the bridge that engine trouble could be expected during mooring operations?

- **The Master should inform the Pilot, abort mooring operations and drop anchor in a safe location, if possible.**
- The Master should be advised so he can decide if mooring operations should continue.
- The Master should call the Company DPA for advice and instructions.
- The Master should do nothing and continue with berthing as planned.

What should be the type and size (SWL) of the rope stoppers used for securing synthetic mooring ropes during mooring operations?

- **Double tail stoppers with a rope SWL of 25% of the mooring rope SWL.**
- A single tail stopper with 25% SWL of the mooring rope.
- A single tail stopper with 50% SWL of the mooring rope.
- Double tail stoppers with a rope SWL of 50% of the mooring rope SWL.

What should you watch for if you are overtaking a similar size vessel in a narrow channel?

- **The bow of your vessel could be sucked towards the other vessel and the stern pushed away as you pass.**
- That the other vessel is not decreasing its speed.
- That you keep your vessel at full sea speed during overtaking.
- There is nothing particular you need to watch for.

What signals should be displayed by ships engaged in ship-to- ship operations (while berthed together and underway)?

- **Signals for vessels restricted in their ability to manoeuvre.**
- No special signals are required during daylight.
- Signals for vessels in tow.
- Signals for vessels not under command.

What signals should be displayed by ships engaged in ship-to- ship operations while approaching one another?

- **Signals for vessels restricted in their ability to manoeuvre.**
- No special signals are required during daylight.
- Signals for vessels in tow.
- Signals for vessels not under command.

When nylon mooring tails are used in combination with mooring wire, what SWL should they have?

- **37.5% more SWL than mooring wire.**
- 10% more SWL than mooring wire.
- 25% more SWL than mooring wire.
- 50% more SWL than mooring wire.

When should the emergency towing wires (fire wires) be deployed?

- **When the vessel is berthed at the terminals and tugs are available.**
- Only on terminals where the vessel is secured alongside the jetty.
- Whenever the vessel is anchored or berthed at the terminals.
- Whenever the vessel is not in navigation.

When synthetic (other than nylon) mooring tails are used in combination with mooring wire, what SWL should they have?

- **25% more SWL than mooring wire.**
- 10% more SWL than mooring wire.
- 37.5% more SWL than mooring wire.
- 50 % more SWL than mooring wire

When will a vessel need the assistance of Tugs?

- **When it reduces speed, is no longer under total self-control and does not have sufficient sea room to manoeuvre safely without external assistance.**
- Only when it starts the mooring operation (i.e. sending mooring ropes ashore).
- Only when it starts the unberthing operation (i.e. letting go of the mooring ropes).
- Only when the Pilot recommends it to the Master.

Who has the overriding authority for decision-making on berthing over the tide?

- **The Master.**
- The Charterers.
- The Company DPA.
- The ship's Commercial Operator.

Who is responsible for selecting the number of tugs to be used during mooring operations?

- **The Master.**
- The Pilot.
- The Port Authorities.
- The vessel agents in port.

Why are the mooring equipment records important for the safe operation of a vessel?

- **Because they contain all of the information on the construction of the mooring ropes and/or wires and their records of maintenance.**
- Because they are a requirement of the Company Planned Maintenance System.
- Because they are a requirement of the ISM Code, Port State Control and USCG.
- Because they are required by Oil Major Vetting Inspectors.

Why are the non-friction type of mooring tail shackles important to use between mooring wires and tails?

- **Because they are the safest connection between the mooring wire and the soft (i.e. nylon) mooring tail and they extend the lifetime of the mooring eyes on the wires and tails.**
- Because they are made of anti-sparking material and prevent sparking when the mooring wire is paid out or pulled in on board.
- Because they are required by Oil Major Vetting Inspectors.
- They are a requirement of the PSC and USCG for large crude carriers and ULCC ships.

Why is the Master-to-Pilot exchange of information procedure important for the safety of the berthing or un-berthing operation?

- **Because it is confirmation that the plan for berthing between the Pilot and Master has been discussed and agreed.**
- Because it contains the berthing plan by the Master.
- Because it contains the berthing plan by the Pilot.
- Because it contains the Pilot Card.

Why should the mooring rope (or wire) not be kept on winch brakes with an additional engaged winch gear?

- **It could cause damage to the winch, drum shaft and wire.**
- It could be broken more easily since the brake could not slip out.
- It is not a sign of good seamanship on board.
- It is not permitted by the terminal and OCIMF mooring equipment guidelines.

You are approaching the berth and only one tug is available. Where should you position it and how should it be used?

- **Make it fast on forward shoulder and use it for pulling and pushing.**
- Make it fast aft through the centre lid and use it for pulling.
- Make it fast forward through the centre lid and use it for pulling.
- Make it fast on the ship's amidship side and use if for pushing and pulling when required.

You are on a 300,000 tonne DWT ULCC and the terminal is asking if your vessel is provided with SPM mooring according to OCIMF requirements. What is the requirement for fittings that should be complied with?

- **Two tongue-type chain stoppers of 200 tons SWL for a 76mm chain and separate closed fairleads of at least 450 x 600 mm in size.**
- One Smith-type bracket of 150 tonnes SWL for a 76mm chain and one closed fairlead of at least 450 x 600 mm in size.
- One tongue-type chain stopper of 200 tonnes SWL for a 76mm chain and the centre line positioned closed fairlead of at least 450 x 600 mm in size.
- Two tongue-type chain stoppers of 200 tonnes SWL for a 76mm chain and one centrally positioned closed fairlead of at least 450 x 600 mm in size.

You are on a ULCC of 300,000 tonne DWT and you are supposed to anchor in 90 metres depth. How much chain should you pay out?

- **Eleven shackles.**
- Eight shackles.
- Five shackles.
- Three shackles.

You are sailing on a small coaster, approaching the berth port side at the speed of one knot, and have moderate wind from the starboard side and no tug. How could you assist yourself in the situation described?

- **Let go starboard anchor and pay out the cable controlling the bow as you are approaching the berth.**
- Give the rudder hard a starboard.
- Increase the speed and drop port anchor one shackle in the water.
- Let go port anchor and gradually release the chain as you are approaching the berth.

You are turning your vessel in a narrow channel by using two tugs of the same bollard pull (same size tugs). The wind is strong and of variable force. Where you should position the tugs and how should you use them?

- **Make them fast to the vessel and use them for pushing.**
- Do not make them fast to the vessel and use one tug for pushing and other for pulling.
- Make them fast to the vessel and use them for pulling.
- Use them for pushing but not made fast to the vessel.

You meet another vessel port to port in a narrow channel. What interaction could you expect of both ships?

- **The bows of both ships will be pushed away from each other and aft parts could make contact.**
- Both ships will proceed on their planned courses and tracks.
- The bows of both ships will be sucked together and the ships could make contact with forward parts.
- This does not apply to tankers.

Your ULCC-sized vessel is anchoring in deep water anchorage. Who should be in charge of anchoring operations on the forecastle?

- **An experienced licensed ship officer.**
- A licensed ship officer.
- An experienced bosun.
- Any experienced crew member appointed by the Master for this operation.

Your mooring winches are fitted with a brake-holding capacity of 60 tonnes. What size mooring ropes should you request?

- **100 tonne MBL.**
- 60 tonne MBL.
- 72 tonne MBL.
- 80 tonne MBL.

Your vessel has a conventional right-hand propeller. How would you position your rudder before going full power astern?

- **Amidship.**
- Hard a port.
- Hard a starboard.
- Ten degrees to port.

Your vessel is berthing starboard side on the terminal jetty. How could you use two tugs for berthing?

- **Make the tugs fast on the ship's flat sides and use them for pushing or pulling when required.**
- Make one tug fast forward through the centre lead for pulling, while the other is positioned on the aft port quarter for pushing.
- Make the tugs fast forward and aft through the centre lead.
- Use the tugs to push the flat side of the vessel but not fast.

Your vessel is unmooring from the terminal and you have to use two tugs. How should you position and use them?

- **Make fast forward and aft in the centre line and pull the vessel from the jetty.**
- Make fast fwd tug on shoulder and keep aft tug on stand-by.
- Make fast on ship's flat side and use the tugs for pulling the berth.
- Use both tugs for pushing only until the vessel's ropes are cleared.

7004 – TOTS 1A-Passage Planning and Navigation

Version 1.0 (3)

During appraisal what sources should the Navigating Officer use to gain information?

- **From every available source.**
- From official sources approved by Company regulations.
- From official sources approved by the relevant Hydrographic Office.
- From sources agreed by the Master.

If a vessel is fitted with an unapproved electronic chart system, how can it be used in Passage Planning?

- **It should not be used for Passage Planning.**
- Any electronic chart system can be used with confidence, even if it has not been officially approved.
- It can be used fully, but positions must be plotted on a paper chart.
- It can be used in the appraisal, but this must then be checked by another method.

If, during the Appraisal it is realised that one or more up to date charts at the best possible scale are not available on board, what action should be taken?

- **The necessary charts should be requisitioned.**
- Cancelled charts can be used with caution.
- The next best scale charts should be used.
- The route should be changed to use only acceptable charts.

What actions must the vessel be able to safely take at the abort point?

- **Turn around, stop in safe water, or anchor.**
- Pick up the Pilot, providing a lee if necessary.
- Reduce speed to minimum steerage way.
- Test the main engine ahead and astern.

What are the four stages of Passage Planning?

- **Appraisal, planning, execution, monitoring.**
- Appraisal, preparation, approval, execution.
- Pilotage outwards, sea passage, anchoring, pilotage inwards.
- Planning on paper, preparing the route on the charts, discussion with the Master, correction of any errors.

What information should you record in the Bell Book and the Deck Log Book about the navigation of the vessel?

- **Sufficient information to enable the reconstruction of the ship's movement if required at a later date.**
- Only the information that is instructed by the Master to be recorded.
- Only the manoeuvring changes and status of the main engine should be recorded.
- Only the times that are important for the charterers' commercial operations (like EOSP, Pilot on board, all fast etc should be recorded).

What is achieved by carefully selecting "no-go areas" on the nautical charts required for the ship's passage?

- **It indicates all areas where the vessel cannot go with present draft.**
- It indicates all areas where the vessel cannot go on deepest draft.
- It indicates all areas where the vessel is advised not to go on deepest draft.
- It indicates all areas where the vessel should not go when loaded.

What is achieved by the "appraisal" phase of passage planning?

- **Appraisal is the part of passage planning where all pertinent information is gathered and the firm foundation for the plan is built.**
- Appraisal is the part of passage plan where all the information on required charts is collected, and it is confirmed that all required charts are available on board.
- Appraisal is the part of the passage plan that needs to approved by office.
- Appraisal is the part of the passage plan where the Master decides which route the vessel will take between two ports.

What is the "tidal window"?

- **It is the times between which the tide has achieved a required height, and it is safe for the ship to transit an area.**
- It is the maximum time permitted for the vessel to reach the berth from safe anchorage in a tidal area.
- It is the time required for the vessel to reach the berth from safe anchorage in a tidal area.
- It is the time when the vessel can go alongside and start discharging cargo to reach the required safe draft.

What is the meaning of the term "abort point"?

- **It is the latest position where the vessel may abandon the planned passage and either return or stop safely.**
- It is any position on the passage where the vessel can abandon the planned passage.
- It is not a phrase which should be used within the plan.
- It is the safe area, specified in the plan and marked on the chart, where the vessel can wait if the plan is abandoned.

What is the meaning of the term "margin of safety"?

- **It is a line parallel to the track, within which the vessel will always be in safe water.**
- It is another phrase meaning "no-go areas".
- It is not a phrase which should be used within the plan.
- It is the distance from the nearest danger.

What is the meaning of the term "parallel index"?

- **A radar-based constant update of a ship's cross track error.**
- A navigational instrument used for plotting bearings on a chart.
- Information on radar about the ship's position in relation to a fixed object ashore.
- Information on radar about the ship's position in relation to a ship's longitudinal track error .

What is the primary reason for the ship's passage plan to be discussed between the navigating officers and the Master before the beginning of a voyage?

- **In order to fully inform all navigation officers about the particular voyage, its specific issues and concerns.**
- In order to inform the Chief Officer and the 3rd Officer about the particular voyage and its specific issues and concerns.
- In order to inform the officers and crew about the destination port.
- In order to satisfy the Company's Navigational Procedures requirement.

What type of markers can be used on nautical charts?

- **Soft 2B pencil and permanent magenta ink.**
- Ball-point pens.
- HB pencils only.
- White correction fluid.

What would you use a gnomonic chart for during Passage Planning?

- **To plot Great Circle courses and waypoints.**
- To plot Rhumb Line courses and waypoints.
- To show sections of the route for which suitable passage charts were not available.
- To show the complete passage conveniently on one chart.

When is the UKC (Under Keel Clearance) calculation required to be completed?

- **The UKC should be calculated and applied for all ships loading and discharging from ports, irrespective of any local regulations and requirements.**
- The UKC should be calculated for all tankers while arriving and departing from US ports.
- The UKC should be calculated for loaded tankers while sailing to US ports.
- The UKC should be calculated for tankers while sailing to US ports.

Why should the passage plan be updated on receipt of Navarea Warnings or T&P notices during the voyage?

- **Because they may contain important and urgent navigational information and warnings that were not available on board at the beginning of the voyage.**
- Because they may contain important commercial information regarding the availability of berths, tugs, pilots etc.
- Because they may contain instructions from owners or charterers regarding the voyage.
- Because they may contain urgent weather warnings.

Why would you use risk assessment techniques during Passage Planning?

- **To reduce, and where possible eliminate risks from navigational hazards.**
- There is no requirement for risk assessment to be used.
- To comply with Flag State or Company regulations.
- To reduce, and where possible eliminate risks to the Bridge Team personnel during the passage.

7005 – TOTS 1A-Navigating in Restricted Visibility

Version 1.0 (7)

At what intervals should a power-driven vessel under way and making way through the water, sound one prolonged blast in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

At what intervals should a power-driven vessel underway but stopped and making no way through the water sound two prolonged blasts in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

At what intervals should a vessel not under command, or a vessel restricted in her ability to manoeuvre, a vessel constrained by her draft, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel, sound three blasts in succession when in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

If the vessel is fitted with two whistles that are spaced apart more than 100 metres, how should the whistles be arranged?

- **They should be arranged so that they are not sounded simultaneously.**
- They should be arranged so that only one can be used at a time.
- They should be arranged so that the forward whistle always sounds first and then the aft whistle sounds with a 10 second delay.
- They should be arranged so that they are sounded simultaneously.

In which publications can specific guidance and instructions on navigation in restricted visibility be found?

- **In the Bridge Procedures Guide, in COLREGS and in the Mariners' Handbook.**
- In the Bridge Procedures Guide, in Bridge Team Management and in the Passage Planning Guide.
- In the Deck Log Book and in the Company Navigation Manual.
- In the Master's Night Order Book.

What action should you take when you hear the fog signal of another vessel apparently forward of the beam, or when you cannot avoid a close quarters situation with another vessel forward of the beam?

- **Reduce speed to the minimum and if necessary take all way off and in any event navigate with extreme caution.**
- Identify the other vessel by AIS, and then establish contact with the vessel to agree the intended course of action to avoid the danger.
- Identify the other vessel by VHF using the ships call sign, and then establish the other vessels intended course of action to avoid the danger.
- Sound five short blasts on the ships whistle to indicate doubt about the others intention.

What actions required by the Collision Regulations should the Officer of the Watch comply with during periods of restricted visibility?

- **All three answers.**
- To have the ship's engines ready for immediate manoeuvres.
- To proceed at a safe speed.
- To sound the fog signals.

What additional sound signal may a pilot vessel sound when engaged in pilotage duties?

- **Four short blasts.**
- Four long blasts.
- Three short blasts.
- Two short blasts followed two long blasts.

What additional sound signal may a vessel over 100 meters in length make at anchor in an area of restricted visibility?

- **Three blasts in succession, one short one prolonged and one short.**
- Five short blasts in succession.
- Three blasts in succession, one prolonged and two short.
- Three blasts in succession, two short and one prolonged.

What is the meaning of the term "restricted visibility"?

- **Any condition in which visibility is restricted by fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.**
- Where visibility is restricted to within five nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.
- Where visibility is restricted to within three nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.
- Where visibility is restricted to within two nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.

What sound signal should a power-driven vessel making way through the water sound during periods of restricted visibility?

- **One prolonged blast**
- one prolonged blast followed by two short blasts
- Two prolonged blast.
- Two prolonged blasts followed by one short blast.

What sound signal should a power-driven vessel underway but stopped and making no way through the water sound during periods of restricted visibility?

- **Two prolonged blasts.**
- Four prolonged blasts.
- One prolonged blast.
- Two prolonged blasts followed by two short blasts.

What sound signal should a towed vessel sound during periods of restricted visibility?

- **One prolonged blast followed by two short blasts.**
- Three prolonged blasts.
- Two prolonged blasts followed by two short blasts.
- Two prolonged blasts.

What standard sound signal must a vessel over 100 meters in length make at anchor in an area of restricted visibility?

- **Sounding the anchor bell forward immediately followed by sounding the gong aft.**
- Sounding the anchor bell aft immediately followed by sounding the gong forward.
- Sounding the gong aft followed immediately by sounding the anchor bell forward.
- Sounding the gong forward immediately followed by sounding the anchor bell aft.

When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite what sound signal should they sound when in or near an area of restricted visibility when underway and making way.

- **One prolonged blast.**
- Two prolonged blasts followed by two short blasts.
- Two prolonged blasts.
- Two short blasts

When is Rule 19 of COLREGS applicable, and to which ships?

- **This rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.**
- This rule applies only to power-driven vessels navigating in an area of restricted visibility.
- This rule applies to all power-driven vessels navigating in an area of restricted visibility.
- This rule applies to all vessels sailing in an area of restricted visibility.

When restricted visibility is encountered or is expected, who should be informed?

- **The Master.**
- The Company Operator / DPA.
- The Helmsman.
- The Lookout.

Where is the ship-specific information about restricted visibility declared?

- **In the Master's Bridge Standing Orders.**
- In the Company's Navigation Standing Orders.
- In the Deck Log Book.
- In the Master's Night Order Book.

7006 – TOTS 1A-Heavy Weather Navigation

Version 1.0 (7)

If the vessel is in port, what is the best option if a tropical cyclone is passing within 80 nautical miles or so?

- **The best option is to sail out to sea in time to avoid the worst of the storm.**
- The best option is to stay alongside the well fendered berth that is on the lee-side of the wind.
- The best option is to stay at anchorage and make the vessel as heavy as possible by adding as much full ballast as possible.
- The best option is to stay in sheltered waters, and, if possible, on special moorings.

In relation to navigating in ice conditions, the term beset means that the ship has found itself in a position where it is surrounded by ice and is unable to move. True or False?

- **True.**
- False.

The vessel is in port and is due to sail in a few hours, however severe weather is forecast that may endanger the vessel. What should the Master do?

- **Based on risk assessment, the Master should make the decision on whether the vessel is in a "Go" or "No Go" situation and therefore determine if the ship should sail or remain in port.**
- The Master should call the company's operator for advice and further instruction.
- The Master should evaluate the weather conditions and try to solve the problem without causing a commercial delay for the vessel.
- The Master should make the decision based on available weather reports/forecasts and then wait for the company's advice.

What action should you take if you find yourself within the dangerous semi-circle of a tropical storm in the Southern hemisphere and the wind is backing?

- **You should put the wind on the port bow and proceed at full available speed, altering the course to port as the wind backs.**
- You should bring the wind on to the port quarter and proceed with all available speed turning to starboard as the wind changes direction.
- You should put the wind on the port quarter and proceed as fast as possible in a southerly direction.
- You should put the wind on the starboard quarter and proceed as fast as possible in a northerly direction.

What are the characteristics of "trade winds" in the Southern hemisphere?

- **They have a south-easterly direction and a steady average wind speed of force 4 to 5**
- They have a north-easterly direction and a steady average wind speed of force 4 to 5
- They have a north-easterly direction and mainly a variable wind speed of above force 6.
- They have a south-easterly direction and mainly a variable wind speed of above force 6.

What are the main types of "sea ice" that ships encounter?

- **Drift and fast ice.**
- Glacier ice.
- Growlers.
- Icebergs and packed ice.

What are the signs of ice in the vicinity of the vessel?

- **All three answers.**
- Abrupt moderation of the sea and swell.
- Ice blink.
- Sighting of small fragments of ice.

What can cause vessel icing on ships in freezing conditions?

- **All three answers.**
- Fog.
- Rain or wet snow.
- Sea spray or water breaking over the ship.

What impact could occur at a high speed while sailing through the ice?

- **The ship's hull could be damaged by collision with floes.**
- A vessel navigating at a high speed through the ice could have problems with icing of the sea-boxes (water suctions).
- A vessel navigating at a high speed through the ice could not keep the track through the passage opened by the icebreaker.
- The ship could "run aground" on thick ice and not be able to back up if required.

What impact could vessel icing have on the ship?

- **Vessel icing may impair the stability and safety of the ship.**
- Vessel icing may impair the manoeuvring of the vessel.
- Vessel icing may impair the security of the vessel.
- Vessel icing may impair the visibility from the bridge

What is considered "new ice"?

- **Frazil ice, Slush, Sluga and Grease ice with a thickness of up to 2.5 cm**
- First-year ice that has a thickness of between 30 to 70 cm.
- First-year ice that has less than 30 cm of thickness.
- Young ice that has less than 10 cm of thickness.

What is it important to know in order to decide the best course of action if the storm is suspected to be in the vessel's vicinity?

- **It is important to know the bearing of the centre of the storm and the path of the storm.**
- It is important to know the difference of pressure between the centre of the storm and the position of the vessel to calculate the strength of wind.
- It is important to know where the nearest port of refuge is.
- It is important to monitor the barometric pressure and wind strength and direction.

What is of vital importance when encountering a tropical cyclone in the vessel's vicinity?

- **It is imperative to avoid passing at 80 Nm or closer to the centre of the storm.**
- It is imperative to carry out an emergency drill in order to prepare for a worse case scenario.
- It is imperative to sail in the opposite direction of the storms track
- It is imperative to send out a SECURITE message to all ships in the vicinity warning them of the approcahing eye of the storm.

What is the "dangerous semicircle" of a tropical storm in the Southern hemisphere?

- **The dangerous semicircle is the section of the tropical cyclone that is on the port side of the cyclone path and its eye.**
- The dangerous semicircle is the section of the tropical cyclone that is 250 Nm fwd of the eye of the storm directly in her path.
- The dangerous semicircle is the section of the tropical cyclone that is 80 Nm fwd of the eye of the storm directly in her path.
- The dangerous semicircle is the section of the tropical cyclone that is on the starboard side of the cyclone path and its eye.

What is the "navigable semicircle" of a tropical storm in the Northern hemisphere?

- **The navigable semicircle is the section of the tropical cyclone that is on the port side of the cyclone path and its eye**
- The navigable semicircle is the section of the tropical cyclone that is 250 Nm aft of the eye of the storm directly behind her path
- The navigable semicircle is the section of the tropical cyclone that is at least 80 Nm aft of the eye of the storm behind on her path
- The navigable semicircle is the section of the tropical cyclone that is on the starboard side of the cyclone path and its eye

What is the Master required to do by SOLAS when his ship encounters dangerous ice?

- **He is required to communicate the fact to all ships in the vicinity by all means available**
- He is required to communicate the fact to his DPA
- He is required to exhibit lights and shapes declaring that his vessel is restricted in her ability to manoeuvre.
- He is required to stop his ship and communicate with the nearest coastal state.

What is the difference between hurricanes and typhoons?

- **A hurricane develops in the Atlantic or East Pacific while Typhoons develop in the West Pacific.**
- A hurricane develops in the western Pacific and a typhoon in the central and eastern Pacific.
- A hurricane is a storm with winds of over 64 knots and a typhoon is between 48 and 56 knots.
- A hurricane is a tropical storm and a typhoon is a storm of the mid-latitudes (above 40 degrees).

What is the first basic ship-handling rule in ice concentrations?

- **Keep moving, even if very slowly.**
- Try to avoid icing of the ship's superstructure.
- Try to keep the vessel sailing only after the icebreakers.
- Try to keep the vessel sailing only in convoys through the ice.

What is the first principle for successful passage through the ice?

- **To maintain freedom of manoeuvre.**
- To maintain radio watch in order to receive all ice reports and information.
- To sail through the ice during the day only.
- To stay out of the ice as much as possible.

What is the general freezing point of seawater?

- **About - 2º C**
- About - 4º C
- About - 5 º C
- About 0º C

What is the indication of an approaching depression?

- **A falling barometer.**
- A rising barometer.
- The wind is rising and the temperature is falling.
- The wind is subsiding and the temperature is rising.

What is the meaning of the term "glacier ice"?

- **Ice originating from a glacier, either on land or floating on the sea as icebergs, bergy bits, growlers, or ice islands.**
- Any ice originating from fresh water melted from glaciers.
- Any ice originating from fresh water.
- Any ice originating from large icebergs made of fresh water.

What is the meaning of the term "growler"?

- **A rounded piece of glacier ice, often transparent but appearing green or black in colour, extending less than 1 metre above the sea surface and normally occupying an area of about 20 m2.**
- A smaller piece of iceberg, extending about 5 metres above the sea surface, normally occupying an area of about 100 m2.
- A special type of iceberg that is only found in the Antarctic that is about 5 metres in height and green or black in colour.
- An iceberg that grows from packed sea ice and extends less than 1 metre above sea level.

What is the meaning of the term "ramming"?

- **Attempting to break the ice by repeatedly driving the ship as far forward as possible, backing the ship out and repeating the process.**
- The noise caused by icebreakers breaking the ice.
- The unpleasant noise caused by the propeller grinding through the ice when the vessel is running ahead in the ballast condition.
- The unpleasant noise caused by the propeller grinding through the ice when the vessel is running astern.

What is the meaning of the term "sea ice"?

- **Any form of ice found at sea which originated from the freezing of seawater.**
- Any form of ice that is floating on the sea.
- Any form of ice that is found on the open sea irrespective of the origin of the ice.
- Any form of ice that is free floating away from the ice.

What is the meaning of the term the "wind is backing"?

- **It's an anticlockwise change in the direction of the wind (i.e. from northwest to west to southwest).**
- It's a change of direction in the wind from astern to ahead.
- It's a change of direction in the wind from ahead to astern.
- It's a clockwise change in the direction of the wind (i.e. from northeast to east to southeast).

What is the meaning of the term the "wind is veering"?

- **It's a clockwise change in the direction of the wind (i.e. from northeast to east to southeast).**
- It's a change of direction in the wind from astern to ahead.
- It's a change of direction in the wind from ahead to astern.
- It's an anticlockwise change in the direction of the wind (i.e. from northwest to west to southwest).

What is the most serious danger to a "beset" vessel (i.e. a vessel that has become surrounded by ice and is unable to move)?

- **Pressure from the ice that can crush the hull or nip off the bottom.**
- Danger of icebergs capsizing the vessel.
- Danger of running aground as it's carried by the moving drift ice.
- Danger of the ice causing damage to the rudder and the propeller.

What precautions should be carried out on deck before encountering heavy weather?

- **All loose objects and equipment should be secured.**
- All crew should be given heavy weather clothing, including raincoats and boots.
- All crew should be instructed not to leave their cabins.
- All equipment should be greased.

What problems can occur on board from icing from fresh water?

- **Ice can accumulate on the rigging, and radar, and radio equipment failures can occur due to ice on the aerials.**
- Fresh water supplies will stop functioning
- Heating systems will trip out on overload
- Ice can prevent the opening of all accommodation openings.

What should be the most important task when navigating in the vicinity of strong tropical storms (e.g. cyclones, hurricanes, typhoons)?

- **To stay as far as possible from the path forward of the tropical storm.**
- To frequently monitor the ship's barometer.
- To inform the crew to secure all loose objects and equipment to avoid them being lost or damaged.
- To monitor carefully all tropical storm area weather reports.

What type of ice formation is most dangerous for navigation?

- **Icebergs.**
- New ice.
- Old ice.
- Sea ice.

What warnings and instructions should be given before the vessel encounters heavy weather?

- **All three answers.**
- Instructions to rig safety lines where necessary.
- Warning and instructions should be issued to the crew on deck access.
- Warning and instructions should be issued to the engine room crew.

Where can the Master or Officer of the Watch find details regarding ice reporting schemes and the radio stations transmitting them?

- **In the Admiralty List of Radio Signals, Volume 3.**
- In the Bridge Procedures Guide.
- In the Company SMS manuals.
- In the Mariner's Handbook.

Why should the Master regulate the ship's speed in heavy weather?

- **All three answers.**
- To avoid the possibility of damage to the cargo.
- To avoid the possibility of damage to the ship.
- To prevent needless expenditure of fuel.

You are in the vicinity of a tropical storm in the Northern hemisphere and the wind is remaining in a steady direction. If the vessel is in the navigable semi-circle, you should continue the voyage in the planned direction. True or False?

- **False.**
- True.

7007 – TOTS 1A-Emergencies, Drills and Fire Fighting

Version 1.0 (22)

A few drops of oil spill into the water but no traces are visible anymore. Is it still necessary to report the incident?

- **The incident shall be reported regardless of the size of the incident.**
- It's advisable to ignore the incident to prevent legal issues arising, especially if there are few witnesses.
- Since no traces of oil are visible there is no need to report it.
- Wait and watch to see what the reaction is, then decide accordingly.

Can cargo be restarted immediately once a leaking or burst pipe has been repaired?

- **No. It's important to ensure that all the hazards from the released oil are eliminated.**
- No, cargo cannot be started until a full incident investigation is completed.
- Yes, cargo can be restarted since the problem has been rectified.
- Yes. Working of cargo is based on the commercial pressures on the ship.

Can the SMPEP and SOPEP be combined into one manual?

- **Yes. Since the requirements are similar, they can be combined into one plan.**
- No. The SMPEP is a dry cargo ship contingency plan while the SOPEP is for tankers.
- No. The SMPEP is for a particular cargo while the SOPEP is a general plan.
- No. They address different regulations so cannot be combined.

Does the paint locker require a fixed fire fighting installation?

- **Yes, a fixed system is mandatory.**
- A portable installation is acceptable if within easy reach.
- It is only important to be able to quickly access some type of extinguishing system.
- The regulations require a portable extinguisher in the locker.

How is the SMPEP different from a SOPEP?

- **It is the contingency plan for ships carrying Noxious Liquid Substances.**
- Different Flag Administrations use different terminology for the same plan.
- It is the same plan, only called by different names based on the individual company.
- SMPEP deals with Oil Pollution only.

How many foam applicators must a fixed foam extinguishing system have on deck?

- **A minimum of 4.**
- 10 in total.
- A minimum of 2.
- At least 4 on each side.

In the event of a collision what are the duties to the other vessel?

- **Render assistance without jeopardising the safety of your own personnel and vessel.**
- If oil has been spilt the Master should take down details and try to avoid becoming involved in a pollution incident.
- Issue a notice of arrest to the other ship on behalf of the Company
- Since it is a collision incident , the Master should take details before continuing with the voyage.

Is high expansion foam suitable to fight a fire on deck?

- **It is unsuitable for use in open areas in outside locations as it is easily dispersed in light winds.**
- It depends on the expansion ratio of the foam.
- It is better to use water jets to clear the deck of the combustible substance.
- It is suitable since it gives a larger coverage area on deck.

Is it mandatory to have a fire blanket in the galley?

- **Yes, it is mandatory as stated by SOLAS.**
- It depends how committed the ship manager is to safety.
- It is up to the chief cook to decide.
- The Port State Control insists on having one in the galley.

Is uncontrolled venting an allowed practice while loading cargo?

- **No. Venting of tanks should be only through mast risers in a controlled fashion.**
- Yes. If the wind speed is good there is no problem with uncontrolled venting.
- Yes. It is not a problem since so many precautions have already been taken.
- Yes. Uncontrolled venting is unavoidable due to the vapour pressure while loading.

The ship suffers a collision. The bridge waste paper basket contains some small pieces of paper with scribbled notes on. What should the Master do?

- **Collect these pieces of paper and keep them as a record, as they may contain relevant information.**
- It might be an idea to falsify the scribbled notes after the collision so you will not be held responsible.
- There is already so much evidence available that it's not necessary to keep these small pieces of paper.
- Treat them as waste and ignore them as they have already been thrown away.

What actions should be initiated by the damage control team after grounding?

- **All the answers are correct.**
- They should check the watertight integrity of the compartments in contact.
- They should sound all compartments, sound around the ship and check the nature of the seabed.
- They should visually inspect the ship and check the draft/trim/list.

What are SOPEP / VRP / SMPEP / PCSOPEP?

- **Oil Spill Contingency plans based on different regulations.**
- Different plans as required by different nations of the world.
- Manuals required by the ISO 9001 standards.
- Manuals segregated by the company for ease of use.

What are the IMO guidelines concerning the types or reports that should be sent in the event of a probable spill or actual spill?

- **An initial report, an additional report and a supplementary report should be sent.**
- A supplementary report should be sent.
- An additional report should be sent.
- An initial report should be sent.

What are the different types of foam extinguishing media that may be found on board?

- **High expansion, medium expansion and/or low expansion foam**
- High Expansion Foam only.
- Low Expansion Foam only.
- Medium Expansion Foam only.

What details of the main and emergency fire pumps are described on the Fire Control Plan?

- **The capacity of the pumps is clearly indicated.**
- Only the location of the fire pumps are marked.
- The identity of the makers and model number is included.
- There is no such requirement for details.

What is the first action that must be taken in the case of a hose burst, pipe work fracture or cargo overflow?

- **All operations must be immediately stopped and the alarm sounded.**
- Quickly repair the leakage and keep operations going to prevent the ship from being delayed.
- Sound the alarm, inform the Master and wait for his instructions.
- Stop all operations and wait for the Coast Guard and terminal people to arrive.

What is the first reporting requirement in the event of a toxic liquid release at anchor?

- **The incident should be reported to the nearest Coastal State in detail without delay.**
- If the effect is not visible on the surface of the water there is no need to report the incident.
- The agent should be informed immediately and he will take necessary action.
- The ship's staff must first check with the ship's managers before taking action.

What is the mariner's role in collecting evidence during and after an incident?

- **All the answers are correct.**
- Ensure that log books, rough note books and all other relevant documentation is collected and retained for the vessel's lawyers.
- Examine the scene of the incident, taking photographs and using sketches where appropriate.
- It is good practice to record facts straight after the incident while it is still fresh.

What is the minimum frequency (for the majority of vessels) for an abandon ship drill and a fire drill for the crew, according to SOLAS Chapter lll?

- **There shall be at least one abandon ship drill and one fire drill every month.**
- The frequency of the drill is left to the discretion of the Master.
- There shall be at least one abandon ship drill and one fire drill every three months.
- There shall be at least one drill every week while the ship is at sea.

What is the minimum required service duration of an EEBD?

- **10 minutes.**
- 12 minutes.
- 15 minutes.
- 20 minutes.

What is the order of priority when dealing with contingencies?

- **Safety of life, the environment and then property.**
- Safety and protection of the environment.
- Safety of property and cargo.
- Safety of the ship and the environment.

What is the purpose of a Safety Meeting on board?

- **To review safety matters and improve the Safety Management System.**
- To discuss operational problems and the maintenance programme.
- To ensure the paperwork, as required by ISM, is completed.
- To receive and review crew complaints with regard to crew matters.

What is the requirement for galleys that have deep-fat cooking equipment?

- **All the answers are correct.**
- There must be an automatic shut off of electrical power when the fire extinguishing system is activated.
- There should be clear instructions for crew concerning manual operation of the system.
- They shall be fitted with an automatic or manual fire extinguishing system.

What is the requirement for manually operated call points on a fixed fire detection and alarm system?

- **They shall be capable of immediate operation at all times.**
- In case the automatic system fails then the manual call point is to be used.
- Manual operation can only be overridden by the automatic system from the bridge.
- No manually operated call points are required since the system is fixed.

What kind of fire alarm detectors are used on a ship?

- **They can be smoke detectors, heat detectors, flame sensors or any combination of the three.**
- A smoke detection system.
- Flame sensors.
- Heat detectors.

What kind of fixed fire-fighting installations may be fitted for the cargo pump room?

- **It should be fitted with a system suitable for machinery spaces of category A.**
- Pump rooms must be fitted with a gas fire extinguishing system only.
- Pump rooms should be fitted with a fixed high expansion foam extinguishing system only.
- The type of system is based on the type of cargo the ship is carrying.

What kind of fixed fire-fighting installations may be fitted for the machinery spaces?

- **Any of the types mentioned.**
- A fixed high pressure water spraying system.
- Fixed Gas Fire Extinguishing system.
- Fixed High Expansion Foam system.

What kind of galley equipment must be fitted with an automatic or manual fire fighting system?

- **Deep fat cooking equipment.**
- Electrical equipment.
- Galley ovens
- Galley stove hotplates

What must be covered in the familiarisation of the fixed fire-fighting system on the cargo deck area?

- **All the answers are correct.**
- The fire-fighting medium storage area.
- The location and operation of the fixed fire extinguishing system.
- The remote and local activation points and procedures.

What should the Officer on Watch do in the event of a 'Man Overboard' ?

- **Raise the alarm and turn the ship towards the side of the man overboard.**
- Call the Chief officer.
- Inform The Master and wait for his arrival.
- Turn the ship away from the side of the man overboard.

What should you do if the ship spills oil into the water in a port and you are contacted by the Media?

- **Be courteous, state facts and suggest the reporter contacts the company, and do not commit yourself on any issues you are not sure about.**
- Blame the other parties and deny any wrongdoing by you or the ship.
- Don't be afraid to be direct and explain to them the difficulties of life on board and working with companies.
- If they become aggressive, treat aggression with aggression and ask the crew to physically remove them.

What size of tanker must be fitted with a fixed deck fire-extinguishing system?

- **Tankers of 20,000 tonnes deadweight and upwards.**
- All tankers must be fitted with a fixed system.
- Tankers of 5,000 tonnes deadweight and upwards.
- Tankers of 50,000 tonnes deadweight and upwards.

What would be the immediate action taken in the event of a bunker spill?

- **Stop all transfer operations, inform, sound the alarm and initiate Emergency response.**
- Stop operations and work out a solution to cover up in collusion with the barge.
- Stop operations, check if anyone has noticed and then take any necessary action.
- Stop operations, investigate the occurrence and inform based on the quantity of the spill.

What would be the required action if a steering gear failure takes place in a busy shipping lane?

- **All the answers are correct.**
- Alert the VTS station and check if it's possible to anchor.
- SECURITE call to other ships in the vicinity so they can take evasive action.
- Switch to emergency steering and reduce speed.

When should a new crew member undergo on-board safety familiarization?

- **As soon as practicable upon joining the vessel.**
- No later than one month after joining the ship.
- No later than two months after joining the ship.
- The training schedule is subject to the time available between operations.

When would you use the Emergency Response Service of the Class Society?

- **For damage stability and longitudinal strength calculations after a casualty.**
- It is useful for any emergency situation on board where Class is affected.
- They are called in for support if the ship is involved in a pollution incident.
- They are contacted if the ship has an accidental release of cargo in USA waters.

Where are the operating instructions for the emergency fire pump located?

- **In a clear, visible place near the emergency fire pump.**
- Adjacent to the international ship/shore connection for use by the shore fire services.
- On the bulkhead next to the ship's fire plan.
- On the ship's bridge and the Engine Control Room.

Where shall the isolating valve on the fire main be fitted?

- **It shall be fitted in an easily accessible and tenable position outside the machinery space.**
- It can be fitted anywhere as long as it serves the purpose of isolating the section of fire main for machinery space.
- It shall be fitted anywhere on open deck spaces; the ship's main deck being the most appropriate.
- It shall be fitted in the accommodation as the crew are always available in that area.

Which of the following elements should generally be included in a fire drill?

- **All of these answers**
- Checking of the fireman's outfits, personal rescue equipment and communications equipment.
- Checking the operation of watertight doors, fire doors, fire dampers and main inlets and outlets of ventilation systems in the drill area.
- The starting of a fire pump, using at least two jets of water

Which party is to be notified first in the event of a probable spill or actual spill?

- **The Coastal State Authority**
- The Company
- The local agents
- The P&I Club

Who approves the ship's Fire Control Plan?

- **The Flag Administration or Recognised Organisation.**
- The Port State Control.
- The ship's Protection and Indemnity Club.
- The ship's Technical Managers.

Who is the person identified in the Vessel Response Plan who acts on behalf of the Owner for oil spill response in USA?

- **A Qualified Individual.**
- A designated person ashore.
- An authrorized person.
- The owner's representative.

Why are Vessel Management Meetings held?

- **To promote effective team-working, planning, clarity on tasks, execution and monitoring of jobs.**
- The Management level staff is required by the regulations to hold these meetings.
- To comply with the Company Auditors who periodically check to make sure the meetings have been carried out.
- To provide an opportunity for the vessel management team to discuss the issues raised at the HSE meetings.

Why are environmental issues discussed during HSE or environmental team meetings?

- **To raise awareness among the crew on issues that affect the environment and promote pollution prevention.**
- Because there are so many rigid regulations to comply with.
- For financial reasons as the commercial viability of ship may be affected.
- To discuss tactics in the eventuality of an incident involving environmental activists such as Greenpeace.

Why are proper records required to be maintained for routine tests, checks and maintenance of fire-fighting equipment?

- **For ease of reference and to allow proper and effective planning of testing and maintenance. Also plans and records must be available for inspection when required by the Administration.**
- Because the company auditors can ask for all records of inspections.
- Because the Port State Control is likely to detain the ship if records are unavailable.
- Because the quality system requires records to be maintained to provide evidence.

Why are the Accident, Incident and Near-Miss Reporting system used?

- **To analyse and learn from the errors and to work towards improvement.**
- It is a requirement once The Master has signed off from the ship.
- To ensure the Company Policies are followed rigidly.
- To ensure the internal ISM and Auditor's paperwork requirements are met.

Why do dry powder installations require periodical checking?

- **Solidification of the compound can occur on board over a period of time.**
- Dry powder evaporates and disappears over time.
- Other extinguishing media are more stable.
- The powder tends to change colour and loose effectiveness.

Why do foam compounds after an initial 3 year period have to be tested annually?

- **All the answers.**
- Because deterioration of the foam can reduce its properties.
- Because it is a Flag requirement.
- Because the foam compounds are likely to deteriorate over time.

Why do tankers require isolation valves to be fitted in the fire main at the poop front and on the tank deck at intervals of no more than 40 metres?

- **To preserve the integrity of the fire main system in case of fire or explosion.**
- Because additional isolation valves help to provide quick access.
- Because deck fire fighting becomes easier when different sections of fire main can be isolated.
- Because it makes it easier to renew damaged sections of the fire main while the fire pump is running.

Why is Risk Assessment carried out before carrying out operations?

- **To improve safety and minimise or reduce the possibility of incidents and accidents.**
- Because The Port State Control Officer checks the Risk Assessment records.
- It is a requirement of the ISO 9001:2000 Quality standard.
- To hold senior staff accountable and responsible in the event of an accident happening.

Why is a fire control plan stored in a prominently marked weathertight enclosure outside the deckhouse?

- **It is to assist shore side fire-fighting personnel in an emergency.**
- It is kept outside in case of an accommodation fire on board.
- It is kept there because the Port State Control Inspector will look for it.
- It is stored so that crew can access the information during a drill on deck.

Why is it important to follow a drill with a debriefing session?

- **To review and discuss the lessons learnt and continually improve.**
- It is a requirement of the Company Safety Policy.
- To advise the crew about their faults and assign blame to individuals.
- To motivate crew and make good use of any remaining time left over from the drill.

You have a small bunker spill in port. All notifications are done and the cleanup is completed. What do you do concerning the record collection?

- **Obtain calculations of relative quantities on board, in the pipelines and in the barge.**
- Forget the records and hope the estimated spilt quantity will be relatively accurate.
- Leave matters to the local agent and feel relieved that the emergency is over.
- Leave the record collection to the company who are not on site.

7008 – TOTS 1A-Ship and Shore Interface

Version 1.0 (3)

At what stage will the agreed communication methods be confirmed?

- **During completion of the Ship / Shore Safety Check List.**
- At the Master's discretion.
- Prior to arrival at the terminal.
- The agent will inform the Master.

Crude Oil Washing (COW) is to be carried out. What will be the terminal's requirements relating to oxygen?

- **The oxygen content in the cargo tank must not exceed 8%.**
- The oxygen content in the cargo tank must not exceed 11%.
- The oxygen content in the tank will depend upon the COW manual.
- The oxygen content is not important as the tank is over rich.

Does the Ship / Shore Safety Check List address Crude Oil Washing?

- **Yes, it does.**
- No, it doesn't.
- Only regarding cargo tank atmosphere.
- Only regarding inert gas.

For a safe and efficient cargo transfer, the ship and terminal must . . .

- **maintain good communication and cooperation throughout the operation.**
- check the moorings regularly.
- ensure that safe access to the vessel is maintained at all times.
- keep a good record of events.

How would the Chief Officer normally express the maximum loading rate to the terminal?

- **In cubic metres per hour (m3/hr) or barrels per hour (brls/hr).**
- Always in long tonnes per hour.
- Always in metric tonnes per hour.
- He would ask the Loading Master which units he prefers.

ISGOTT recommends that the transmission power setting of VHF and UHF equipment is set at . . .

- **One watt or less during cargo operations.**
- five watts or less
- Ten watts or less
- The recommendation is only for AIS and not VHF/UHF equipment.

Mooring winches must be used on self-tensioning.

- **False**
- True

Portable VHF/UHF equipment must be . . .

- **intrinsically safe and of an approved type.**
- always worn in a case.
- capable of transmitting at 25 watts power.
- not have an aerial.

Referring to the Ship / Shore Safety Check List, what does the letter "R" indicate?

- **That the item has to be rechecked at agreed intervals.**
- That the item has been reinstated to the check list at the terminal's agreement.
- That the item is to be removed from the check list.
- That there is restricted access.

The hardarm (chickson) operating envelope is only important at low water.

- **False**
- True

The inert gas fails. What is your initial response?

- **To immediately inform the terminal and cease discharging.**
- To advise the terminal of the circumstances and pass along frequent updates as appropriate.
- To just reduce the discharge rate. There is no need to inform the terminal.
- To keep discharging and say nothing.

The loading rate is only about 60% of what was agreed at the pre-cargo transfer meeting. What action would the Master take?

- **He would inform the terminal / Loading Master and issue a Note of Protest.**
- He would Inform the terminal.
- He would protest verbally to terminal and make a note in the logbook.
- He would recalculate the ETD and inform the owners accordingly.

The terminal advises a low flash point. What is a "flash point"?

- **The lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture.**
- The auto-ignition temperature.
- The loading temperature of the cargo.
- The minimum temperature the cargo can be discharged at.

What cargo information is required to enable accurate quantity calculations?

- **The API (S.G.) and load temperature.**
- American Petroleum Institute tables.
- The density.
- U.S. barrels per long tonne.

What does the code letter "P" indicate when completing the Ship / Shore Safety Check List? (ISGOTT)

- **That the particular operation must not commence until permission has been granted by the proper authority.**
- Petroleum.
- That permission has been granted to continue with the operation.
- That permission is not required.

What is "topping up"?

- **The introduction of inert gas into a tank that is already in the inert condition.**
- Filling a cargo tank.
- Topping off a bunker tank.
- Topping up the water in the deck seal.

What is a "berthing plan"?

- **It's the plan provided by the terminal giving relevant information relating to the position and lines required to ensure the vessel's safety throughout the cargo transfer.**
- It's the list of moorings available on board.
- It's the plan indicating the crew and visitors' accommodation layout and sleeping arrangements.
- It's the plan the Navigating Officer makes which forms part of the passage plan.

What is the "Terminal Operating Manual"?

- **It is a working document that includes procedures, practices and drawings relevant to the specific terminal.**
- A general booklet applicable to most tanker terminals.
- An advertising brochure.
- It forms part of the Code of Safe Working Practice.

What is the "odour threshold"?

- **It is the lowest concentration of vapour in the air that can be detected by smell.**
- It is the highest concentration of vapour in the air that can be detected by smell.
- It is the permissible odour when handling cargos with high H2S content.
- It is the smell you get from spiked crude.

What is the "topping off rate"?

- **The rate at which the cargo tank is finally completed to the required ullage.**
- A steady rate to ensure safe loading.
- The rate at which it is safe to ullage cargo tanks.
- The rate requested by the Chief Engineer to commence bunkering.

What is the advantage to all parties if the vessel is advised of reducer size prior to arrival?

- **The vessel can be fully prepared thus saving time and contributing to efficient cargo transfer.**
- It is one more thing off the Ship / Shore Safety Check List.
- It means that a pre-cargo transfer meeting will not have to be conducted.
- To make sure the vessel complies with ISGOTT regarding reducers.

What restrictions on personnel movement would you make if the vessel was to load a cargo with a high content of H2S?

- **Only those crew members actually involved in the cargo operation would be allowed on deck and each would have to do the appropriate PPE.**
- No particular restrictions as it is only sour crude.
- To allow just two crew members on deck at any one time.
- To ensure that all personnel are wearing dust masks and are walking slowly.

When would emergency signals and procedures be discussed?

- **At the pre-cargo transfer meeting / safety meeting.**
- Prior to arrival when the Chief Officer has a meeting with the Master and the Chief Engineer.
- The emergency signal is already posted on the muster list.
- When the Pilot boards.

Which document gives the necessary information for the safe handling of the cargo?

- **The MSDS (Material Safety Data Sheet).**
- SOPEP.
- The code of safe working practices.
- The MARPOL consolidated edition.

Who completes the Ship/Shore Safety Check List?

- **Responsible personnel from the vessel and the terminal.**
- The Loading Master and Chief Engineer.
- The ship's agent with the terminal representative.
- The ship's duty Officer

Why are pre-cargo transfer meetings conducted?

- **In order to discuss all aspects of the proposed cargo transfer and, in particular, any safety issues or particular circumstances which may affect a safe and efficient cargo operation.**
- In order to agree on loading and or discharging rates.
- In order to complete the Ship / Shore Safety Check List.
- In order to discuss firefighting equipment related to the proposed cargo transfer.

Why is it important to compare cargo figures between ship and shore at regular intervals?

- **To identify any discrepancies.**
- It assists in determining the vessel experience factor.
- It is good to check the remote ullaging facility.
- To ensure the cargo transfer rate is being maintained.

Why is it important to have the vessel properly manned during cargo transfer?

- **So that there are sufficient personnel available to conduct cargo transfer operations safely and deal with any emergency that may arise.**
- The ship may have to take on stores.
- The vessel may have to bunker.
- To try and engage in maintenance operations during cargo transfer.

Why is the item "emergency towing off pennants" coded with the letter "R"?

- **Because the emergency towing off pennants need tending as the vessel's freeboard changes to ensure a tug boat can always reach the eye of the pennant.**
- Because only the emergency towing off wire needs tending, the fire wire does not.
- Because the deck crew need to be available to rig the pennant if required.
- Because the pennant is liable to part under strain.

Why must you advise of the cargo plan / loading sequence to the terminal prior to arrival?

- **To ensure that the cargo is available and, in particular, that the terminal can supply in the sequence required.**
- It is not really necessary to advise it as it will all change anyway.
- The Loading Master may not agree.
- To comply with the charter party.

Why will the terminal advise a maximum discharge pressure at the manifold?

- **Because the terminal will know its own safety limits thus ensuring a safe cargo transfer.**
- Because the discharge pressure will assist in complying with the charter party.
- Because the shore tanks are far away.
- Because the shore tanks may be almost full.

7009 – TOTS 1A-Ship to Ship Transfer

Version 1.0 (9)

If two ships are carrying out an STS transfer while underway, what rudder and engine movements should the former manoeuvring ship undertake?

- **None. Her engine should be stopped and rudder amidships.**
- Her rudder and engine should be used as requested by the constant heading ship.
- Her rudder and engine should conform as closely as possible to the constant heading ship.
- The Master of the manoeuvring ship must make such rudder and engine movements as he thinks necessary to maintain the agreed heading and speed.

If two ships are carrying out an STS transfer while underway, who is responsible for keeping a navigational watch?

- **Both ships must maintain a full navigational watch.**
- Either ship, as agreed by the two Masters.
- Neither. Both ships must maintain a full deck watch.
- The ship which is using her engine must maintain a full navigational watch.

The "weather threshold" is the point at which the mooring line loads become unacceptable. What is the effect on the weather threshold if the displacement of the discharging ship is increased?

- **The weather threshold is increased in line with the displacement of the discharging ship.**
- Increased displacement has no effect on the weather threshold.
- The effect of increased displacement depends on many more parameters, and cannot be accurately predicted.
- The weather threshold is reduced due to the extra wind resistance.

The "weather threshold" is the point at which the mooring line loads become unacceptable. What is the effect on the weather threshold of increasing the number of head and stern lines deployed?

- **Increasing the number of moorings has little or no effect on the weather threshold.**
- Increasing the number of moorings increases the weather threshold.
- The effect of increased head and stern lines depends on many more parameters, and cannot be accurately predicted.
- The weather threshold is reduced due to the extra rigidity of the mooring system.

The Berthing Coefficient is used when selecting the fenders to be used in an STS operation. What is the formula used to obtain the Berthing Coefficient?

- **(2 x displacement ship A x displacement ship B) / (displacement ship A + displacement ship B)**
- (2 x displacement ship A + displacement ship B) / (displacement ship A x displacement ship B)
- (2 x displacement ship A x displacement ship B) / (velocity ship A - velocity ship B)
- (2 x velocity ship A x velocity ship B) / (displacement ship A + displacement ship B)

To what extent can STCW or ILO 180 working hour's requirements be relaxed during STS operations?

- **They cannot be relaxed. STCW or ILO 180 requirements remain in force.**
- STCW working hour's requirements do not apply to STS operations.
- Working hour's requirements can be suspended for personnel actually involved in STS operations.
- Working hour's requirements can be suspended for up to 48 hours during STS operations.

Under what circumstances should helicopter operations be permitted?

- **When there is no cargo or bunker transfer, or ballasting into cargo tanks.**
- In an emergency only.
- When permitted by local regulations.
- When wind speed is less than ten knots.

What action must be taken if STS hoses are found to have been critically damaged while taking them aboard?

- **They should be withdrawn from use for further inspection or repair.**
- The Master should Note Protest in case a leak develops from the hose.
- The pumping rate should be reduced throughout the transfer.
- The pumping rate should be reduced until the hoses are proved not to be leaking.

What action must be taken if a significant leak occurs at the manifold of the receiving ship?

- **The STS transfer operation must be suspended until the cause of the leak is eliminated.**
- The discharging ship should be requested to reduce the pumping rate.
- The Master of the discharging ship should Note Protest due to the condition of the hose.
- The Master of the receiving ship should Note Protest due to the condition of the hose.

What action must be taken if excessive cargo vapours are detected around the manifold of the discharging ship?

- **The STS transfer operation must be suspended until the problem is resolved.**
- Pressure should be reduced on the manifold.
- Tests should be undertaken to verify the level of vapour concentration.
- The Master should Note Protest due to the condition of the hose.

What factors would influence the agreed STS cargo transfer rate?

- **Hose characteristics, design of the receiving ship's pipework and vents, and weather conditions which may cause movement in the hoses.**
- The instructions from the vessel's technical managers or local authorities.
- The temperature and density of the cargo.
- Total volume of cargo parcel, and time available within the STS window.

What information is required to be broadcast in an STS Navigational Warning?

- **The name and flag of the ships involved, location of operations, heading of ships, type of operations, starting time and expected duration and the request for wide berth.**
- The name and call sign, and the position and nature of the navigational hazard.
- The name and call sign, the nature of the emergency, the position of the ship, and the assistance required.
- The name, flag and call sign of the ships involved, the type of cargo being transferred, and the request for wide berth.

What is the effect on mooring loads of variations in the wave period, or period of encounter?

- **The load on the moorings increases as the wave period becomes longer.**
- The length of the wave period does not affect the load on the moorings.
- The mooring load increases as the wave period becomes shorter.

What is the position and function of secondary fenders?

- **Positioned at the bow and stern, they give protection against accidental contact during berthing operations.**
- As the smaller fenders, they are positioned along the ship's side to give additional protection.
- Positioned on the constant heading ship, they give protection after the initial contact with the primary fenders.
- Positioned on the side of the constant heading ship away from STS operations, they give protection to tugs and workboats etc.

What is the position and function of the primary fenders?

- **Positioned along the parallel body of the ship, they give maximum mutual protection.**
- As the largest fender, it is usually positioned to give protection during berthing.
- Positioned on the aft of the ship, it is the first fender that the manoeuvring ship makes contact with.
- The primary fenders are located at the forward end of the ship, and are used during berthing.

What is the recommended length for the flexible hose string normally used in an STS transfer operation?

- **Twice the maximum difference in manifold heights.**
- Approximately 50 metres.
- The total of the distances between the two ships' rails and their manifolds, plus 30%.
- Twice the horizontal difference in the manifold positions.

What language should be used for communications between two ships?

- **The language will be mutually agreed between the two Masters.**
- English must be used.
- Masters should use phrases from the International Code of Signals.
- The language of the local Port State should be used.

What measures can be taken to reduce abrasion damage to the rubber skin of the fender?

- **They should be fitted with rubber sleeves and tyres.**
- Only new fenders in good condition should be used for STS operations.
- The closing speed should be kept to a minimum during berthing operations.
- The two ships should be kept slightly apart with rudder and engine movements.

What navigational lights and shapes must be shown by the smaller of the two ships involved in an STS operation?

- **The lights and shapes for a vessel restricted in her ability to manoeuvre.**
- Normal lights for a vessel underway.
- The lights and shapes for a vessel under tow.
- The smaller vessel should not show any navigational lights or shapes.

What smoking regulations should be in force during an STS cargo transfer?

- **Those similar to a shore terminal.**
- It's at the Master's discretion.
- Strictly no smoking.
- Those of the ship's normal at-sea regulations.

What speed should the constant heading ship maintain during the STS berthing procedure?

- **Her minimum steerage way (usually about five knots).**
- Approximately one knot.
- Half ahead.
- Sufficient to allow for wind and current.

When mooring wires are used for STS operations, rope tails should be employed. What characteristics should the tails have?

- **At least 11 metres long, and at least 125% of the wire's breaking strength.**
- Less than 11 metres long, and the same breaking strength as the wire.
- The breaking strength must be at least 11% higher than the wire.
- They must float, and be of suitable size for the ship's mooring equipment.

When should Check List 1 normally be completed?

- **When the STS operation is being planned.**
- 24 hours before starting the STS operation.
- At a convenient time before or during the STS operation.
- Seven days before the STS operation.

When should Check List 2 be completed?

- **Before commencement of the STS operation.**
- 24 hours before starting the STS operation.
- As soon as the STS operation starts.
- At a convenient time before or during the STS operation.

When should Check List 3 be completed?

- **Before commencement of the approach and mooring operations.**
- As soon as Check List 2 has been completed, and the other ship has been advised.
- At a convenient time before or during the STS operation.
- Before commencement of cargo transfer operations.

When should Check List 4 be completed?

- **Before commencement of cargo transfer operations.**
- Before commencement of STS operations.
- Before completion of cargo transfer operations.
- Before completion of STS operations.

When should Check List 5 be completed?

- **Before commencement of unmooring operations.**
- After completion of unmooring operations.
- Before commencement of cargo transfer operations.
- Before commencement of STS operations.

When should a Navigational Warning be broadcast concerning an STS operation?

- **Before commencement of operations, and then regularly according to local requirements.**
- After completion of STS operations.
- At least 24 hours before the commencement of transfer operations.
- If an emergency occurs and immediate assistance is required.

When should the Ship / Shore Safety Checklist be completed?

- **It should be fully completed before starting cargo operations.**
- It need not be completed as it is only for use with cargo operations at shore terminals.
- It should be completed during the planning stage.
- It should be fully completed before finishing cargo operations.

Which STS check lists must be completed by each ship after completion of cargo transfer operations, and before unmooring?

- **Check List 5.**
- All Check Lists 3 to 5.
- Check Lists 3 to 5, depending on the circumstances.
- No check lists required after completion of cargo transfer.

Which STS check lists must be completed by each ship before commencement of cargo transfer operations?

- **Check Lists 2, 3 and 4.**
- All Check Lists 1 to 5.
- Check Lists 1 and 2.
- Check Lists 1 to 5, as required by the circumstances.

Which ship should complete the Ship / Shore Safety Checklist?

- **Both ships.**
- Neither ship.
- The discharging ship.
- The receiving ship.

Which two parties would normally exchange Check List 1?

- **The operator / charterer of each ship and the STS organiser.**
- The Master of each ship and his own operator / charterer.
- The Master of each ship and the organiser.
- The Masters of the two ships.

Who can abort the operation if an emergency arises during an STS berthing procedure?

- **The Master of either ship, or the STS Superintendent.**
- Only the STS Superintendent.
- The Master of the constant heading ship.
- The Master of the manoeuvring ship.

Who should advise the Master that Check List 1 has been satisfactorily completed?

- **The operator / charterer of each ship.**
- The coastguard.
- The Master of the other ship.
- The Port State Authority.

Who should initiate changes of heading during the STS berthing procedure?

- **Changes of course may be requested by the manoeuvring ship, and followed by the constant heading ship.**
- Each ship must be manoeuvred as her Master sees fit.
- No changes of heading should be undertaken once the berthing manoeuvre has started.
- The constant heading ship will make any changes her Master considers necessary. The manoeuvring ship will then conform to these.

Who should normally complete and sign Check List 3?

- **The Master or the Chief Officer.**
- A Company representative.
- The Chief Engineer.
- The STS Superintendent.

Why is it recommended that the diameter of fenders used in an STS operation should be less than half the freeboard of the smallest ship?

- **In order to prevent the fender from accidentally boarding the ship during rough weather.**
- In order to ensure that the distance between the two ships is not excessive.
- In order to reduce the drag of the fenders on the manoeuvring ship.
- So that the fenders can be more easily handled by rigging crews.

7010 – TOTS 1A-Helicopter Operations

Version 1.0 (10)

A single-engine helicopter is not suitable for . . .

- **hovering operations.**
- carrying personnel.
- landing on deck.
- night flying.

According to recommendations what should you bear in mind concerning muster lists and helicopter drills?

- **You should follow Appendix D of the "Guide to Helicopter/Ship Operations" and its suggested action plan.**
- You should bear in mind the capacity of the foam tank.
- You should consider the number of crew on board.
- You should consider the type of ship.

After a helicopter accident, what actions would the crew members who were not involved in helicopter operations take?

- **They would go to their emergency stations.**
- They maintain their current positions onboard the ship.
- They would assist with any casualties.
- They would report to the Officer in Charge on deck.

After a helicopter accident, why would the Master alter course?

- **To keep the fire and smoke clear of the deck.**
- Due to the swell.
- To bring the ship's head to the wind.
- To stem the tide.

After a helicopter crash, what is a water spray used for?

- **To cover for helicopter crew, passengers and rescue party.**
- To control the fire caused by igniting fuel.
- To cool the surrounding area.
- To ensure the crash area is clear of debris.

Communication with the bridge is by . . .

- **walkie-talkie.**
- flares.
- hand signals.
- torch.

How is normal communication conducted between ship and helicopter?

- **By VHF.**
- By signal lamp.
- With flags.
- With flares.

If a fire breaks out during helicopter operations, the Master/Officer of the Watch will immediately . . .

- **sound the appropriate fire alarm.**
- inform the owners/managers.
- make an announcement on the loud speaker
- request information from the Officer in Charge of the deck party

In landing and winching area diagrams, what does the letter "D" stand for?

- **The overall length of the helicopter with the rotors turning.**
- Danger zone.
- Designated helicopter area.
- Positioning of deck party personnel.

In which direction should the fire and foam nozzles be pointing?

- **Away from the helicopter operating area.**
- Directly at the operating area.
- It does not matter.
- Towards the accommodation.

Once a helicopter fire has been extinguished, what actions should be taken by the Officer in Charge and the Chief Engineer?

- **To investigate the damage and check the reignition risk.**
- To inform the owners/managers that the fire has been extinguished.
- To muster all personnel and check for any injuries.
- To stand by for instructions from the bridge.

The Captain decides if the helicopter is to be secured.

- **True**
- False

What is the maximum recommended winching height?

- **12m**
- 6m
- It's at the Pilot's discretion.
- Twice the height of the nearest obstruction.

What is the most likely use of the axe and wire cutters after a helicopter accident?

- **To cut into the cabin structure and rescue the air crew and passengers.**
- To cut the fuel lines, thus stopping the engines.
- To cut the winch cable.
- To cut the wreckage free of the ship's aerials and awnings.

What is the purpose of the Guide to Helicopter/Ship Operations?

- **To encourage safe and efficient helicopter operations.**
- Because it's part of SOLAS.
- For helicopter recognition.
- To assist with fire-fighting training.

When should the aerials be lowered or secured?

- **Before landing or winching operations commence.**
- At the helicopter pilot's request.
- It does not matter.
- Never, it is required for communication.

When should the excessive cargo tank (IG) pressure be released?

- **Not more than 30 minutes before helicopter operations commence.**
- It does not matter.
- Only when the helicopter is landing.
- When the ambient temperature is above 20C.

When should the fire pumps be started?

- **Before commencing helicopter operations.**
- If fire breaks out.
- Only if the helicopter Pilot requests it
- When the helicopter has just landed, or when winching operations have commenced.

When should the ship's rescue boat be ready for immediate lowering?

- **Prior to landing or winching operations.**
- Only if the weather is good enough.
- Only when the helicopter is landing.
- Only when the helicopter is winching.

When should the ship's side rails be lowered?

- **Prior to landing operations.**
- At the helicopter pilot's request.
- If the helicopter's landing gear consists of skids, not wheels.
- Only during the hours of daylight.

When should you consider using a lifeboat as an emergency operating area?

- **When it is not safe to use any part of the ship due to permanent obstructions.**
- At the helicopter pilot's request.
- Due to circumstances when fire- fighting equipment is not available.
- When the wind is less than force 4.

When using the poop deck as a winching area, the winch cable must be hooked onto the ship's rail.

- **False**
- True

Where can you view a suggested action plan for a helicopter accident?

- **In Appendix D of the "Guide to Helicopter/Ship Operations".**
- In the International Safety Management Code (ISM code).
- In the muster list on the bridge.
- In the SOLAS training manual.

Where on the helicopter is the most likely cause of ignition after an accident?

- **In the engine area.**
- Around the cabin structure.
- In the fuel tanks.
- In the winch motor.

Why are helicopter operations only carried out on chemical/parcel tankers in exceptional circumstances?

- **Because their special construction relating to vertical pipes, ventilator pipes and deck tanks means no suitable area is available.**
- Because helicopter pilots require a special endorsement to operate on this type of vessel.
- Because hull insurance is invalid if helicopter operations are conducted
- Because the nature of the cargo does not allow helicopter operations.

Why are landing or winching areas not placed near the bow?

- **Because of the increased air flow turbulence.**
- Because fire-fighting equipment is not readily available.
- Because the rescue boat is located aft.
- Because they are too far from the accommodation.

Why do some vessels have only a winching area?

- **Because they are unable to comply with safe landing area requirements.**
- Because of SOLAS regulations.
- Because the ship does not have sufficient fire fighting equipment onboard to land a helicopter on deck.
- Because they are unable to illuminate a large landing area.

Why may the control of a helicopter be difficult when using the poop deck as a winching area?

- **Because of the turbulent airflow above the accommodation and the funnel.**
- Because of blind sectors.
- Because of excessive down- draught.
- Because of the ship rolling.

Why must the hook handler be equipped with rubber gloves and rubber-soled shoes?

- **To avoid shocks from static charge.**
- So that the helicopter pilot can identify the hook handler.
- To protect him from the helicopter down-draught.
- To protect the hook handler against the wind and sea.

Why should an alternative winching area be located near the side of the ship?

- **So that most of the manoeuvring zone extends over the ship's side.**
- So that if the helicopter crashes, it will fall into the sea.
- To make the foam hydrants more accessible.
- To minimise the effect of air turbulence.

Why should rope messengers be available during landing operations?

- **In case the helicopter needs to be secured.**
- To assist with passing stores between the helicopter and the deck party.
- To latch down any loose equipment affected by the helicopter's down- draught.
- To rope off the landing area.

Why would the poop deck be used as a winching area?

- **Because there is no other suitable area available on board.**
- Because it's the safest area on the ship.
- Because mooring winches are available.
- Because of its proximity to the accommodation for easy access.

Why, if possible, should two emergency operating areas, one on each side of the ship, be selected?

- **To allow for wind direction if the ship is unable to manoeuvre.**
- It's safer for the deck party.
- One may have obstructions greater than 3 metres in height.
- To allow a greater chance of success.

7011 – TOTS 1A-Pollution Prevention and Environmental

Version 1.0 (6)

A ship demonstrates compliance with MARPOL Annex IV by . . .

- **having a valid International Sewage Pollution Prevention Certificate.**
- having a sewage holding tank volume of 100 litres per person on board.
- having a stand-by sewage treatment plant.
- having a surveyor inspect the sewage system every six years.

After having performed a ballast operation the master must . . .

- **complete the ballast water management log.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

An environmental policy in line with ISO14001 is a commitment from . . .

- **company top management.**
- class society.
- flag administration.
- the IMO.

Discharge of sewage is prohibited when . . .

- **the vessel is 24 nautical miles from the nearest land and travelling at 3 knots.**
- the sewage is comminuted and disinfected and the vessel is 5 nautical miles from the nearest land.
- the ship is 24 nautical miles from land and has a functioning certified sewage treatment plant.
- the vessel is 24 nautical miles from the nearest land and travelling at 7 knots.

False entries in the Oil Record Book can lead to criminal penalties for the master and chief engineer?

- **True.**
- False.

How long must the receipts be held with the Garbage Record Book for?

- **Two years.**
- Five years.
- One year.
- Six months.

In the event of an oil pollution incident MARPOL Annex I requires that an oil tanker of 5,000 tonnes deadweight shall have prompt access to . . .

- **computerised shore-based damage stability and residual structural strength programs.**
- an empty oil tanker of equal deadweight.
- computerised models of the ship's cargo tank system.
- the nearest port.

It is illegal to discharge bilge effluent overboard when the oil content is . . .

- **150ppm.**
- 0.15ppm.
- 1.5ppm.
- 15ppm.

Operational malpractice of the oil filtering equipment can lead to criminal penalties for the vessel master and chief engineer?

- **True.**
- False.

Prior to exchanging ballast the master must . . .

- **perform a risk assessment.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

The oil record book shall be completed on each occasion that ballast is discharged from segregated ballast tanks.

- **False.**
- True.

The oil record book shall be completed on each occasion that the cargo tanks are cleansed.

- **True.**
- False.

The shipboard oil pollution emergency plan shall contain details of a list of authorities to be contacted in the event of an incident.

- **True.**
- False.

To reduce cargo tank vapour release which of the following should be checked?

- **That the tightness of cargo tank hatch dogs are checked after every cargo operation.**
- That header valves are unseated.
- That the cargo pump deepwell bleed valve is open.
- That the hatch gasket is checked every year.

Uncontrolled release of volatile organic compounds (VOCs) are an environmental problem because . . .

- **they contribute to low-level ozone pollution in populated areas.**
- they contaminate the sea.
- they react with CO2.
- they react with hydrogen creating acid rain.

What can happen if an emulsion is taken up by the OWS?

- **It can lead to a false reading on the oil content monitor.**
- It can cause damage to the pump bearings.
- It can completely block the system piping.
- nothing.

What should the Master refer to In the event of an accidental discharge of oil?

- **The shipboard oil pollution emergency plan.**
- The ballast water management plan.
- The cargo tank plan.
- The ship environmental management system.

When first using a cleansing agent in the machinery space, the crew should . . .

- **confirm that it won't form an emulsion.**
- confirm that it will form an emulsion.
- use it liberally.
- use it sparingly.

When is it good practice to check the pressure-vacuum valves?

- **Prior to cargo transfer.**
- After cargo transfer operations.
- Before the next dry-docking.
- During cargo transfer operations.

When is the Master to sign the Garbage Record Book?

- **After each completed page.**
- After each discharge operation.
- After the incinerator has been used.
- At the end of the day.

Where on board might ozone depleting substances be found?

- **In refrigeration systems.**
- In compressed air systems.
- In electronic equipment.
- In sewage treatment plants.

Which of the following disposals would be permitted?

- **Food waste from a vessel outside a special area and 12 miles from the nearest land.**
- Food waste from a vessel inside a special area and 10 miles from the nearest land.
- Food waste from a vessel outside a special area and 10 miles from the nearest land.
- Paper products from a vessel inside a special area and 15 miles from the nearest land.

Which of the following is NOT a method for detecting leaks of refrigerant gases?

- **Torch.**
- Gas detector.
- Halide torch.
- Soapy water bubble test.

Which of the following is NOT an option to comply with MARPOL Annex IV?

- **An approved comminuting plant.**
- A certified sewage treatment plant.
- An approved comminuting and disinfection plant.
- Holding tanks with an approved volume.

Which of the following substances is ozone-depleting?

- **Halons.**
- Ammonia.
- CO2.
- Neon.

Which of the following would increase the fuel consumption on board the ship?

- **Leaving steam system leaks unattended.**
- Maintaining steam traps.
- Reducing speed in heavy weather.
- Turning off lights in empty compartments.

Which of the following would reduce energy consumption?

- **Maintaining an unfouled hull.**
- Cooling accommodation spaces to 19°C.
- Heating accommodation spaces to 25°C.
- Running at normal sea speed

7012 – TOTS 1A-General Tanker Section

Version 1.0 (11)

A Permit-to-Work on electrical circuits is important in order to prevent what?

- **Electrical shock.**
- All of these answers.
- Non-scheduled planned maintenance tasks being carried out.
- The use of non-intrinsically safe equipment on deck.

A Permit-to-Work should specify the period of validity. According to the Code of Safe Working Practices, this period should not exceed how many hours?

- **24 hours.**
- 12 hours.
- Eight hours.
- Six hours.

A Risk Assessment can be a useful tool when preparing a Permit-to-Work on electrical circuits.

- **True**
- False

A single work permit is issued . . .

- **only for the work to be undertaken.**
- for a variety of tasks.
- to keep paperwork up-to-date.
- to satisfy auditors.

According to the Code of Safe Working Practices, a Permit-to- Work does not in itself make the job safe but contributes to measures for safe working.

- **True**
- False

At what gas level must tank washing be stopped?

- **The maximum 35% LFL at any level.**
- More than 10%LFL at any level.
- The maximum 35% LFL at the bottom level.
- The maximum 35% LFL at the top level.

At what levels should the tank atmosphere gas level be measured prior to and during tank washing?

- **At various levels.**
- At the bottom level.
- At the middle level.
- At the top level.

Before cargo or ballast operations commence, the Master (or his representative) and the terminal representative should do what?

- **All three of the above**
- Agree in writing on the action to be taken in the event of an emergency during cargo or ballast handling operations
- Agree in writing on the transfer procedures, including the maximum loading or unloading rates
- Complete and sign the Ship/Shore Safety Checklist.

Bonding links on portable equipment must not be disconnected until . . .

- **after the equipment is no longer in use.**
- after cargo operations are started.
- after requesting permission from shore.
- after testing for gas levels.

By what means are ships naturally "earthed"?

- **Via the sea.**
- Via the loading arms.
- Via the main switchboard.
- Via the mooring lines.

During tanker operations what happens if the material has a low relaxation time?

- **Little or no static electricity accumulates.**
- Decanting takes longer.
- Sediments quickly settle at the bottom.
- Stratification into various densities occurs much faster.

During tanker operations, what is the relaxation time of the material mainly related to?

- **Conductivity.**
- The flashpoint.
- The pour point.
- Viscosity.

During ullaging, dipping, gauging or sampling of static accumulator oils, how long should all metallic equipment be banned from use?

- **30 minutes after completion of loading.**
- 10 minutes after completion of loading.
- 20 minutes after completion of loading.
- One hour after completion of loading.

H2S is normally associated with crude oils, particularly after loading. What other products can contain H2S?

- **All of these answers.**
- Bitumen.
- Bunkers.
- Gas oil.

If hot work is to be undertaken above the tank deck (higher than 500mm), then cargo and slop tanks must either be cleaned and gas freed, OR emptied purged and inerted, OR filled with water, within a minimum radius around the working area. What is this min

- **30 metres.**
- 10 metres.
- 20 metres.
- 40 metres.

If hot work is to be undertaken on the tank deck or at a height of less than 500mm above the tank deck, the precautions taken should be the same as for hot work within the tank.

- **True**
- False

If petroleum is considered to be volatile, what will the flashpoint temperature be?

- **Below 60 degrees Celsius.**
- 38 degrees Celsius and below.
- Above 40 degrees Celsius.
- Above 50 degrees Celsius.

If the conditions under which a Hot Work Permit has been issued change, what should be done immediately?

- **Stop the hot work.**
- Amend the work permit conditions.
- Extend the validity of the permit.
- Prepare additional firefighting equipment.

In what range does the TLV/TWA for H2S lie (Threshold Limit Value for the Time Weighted Average 8 hour day)?

- **5-15PPM.**
- 15-25PPM.
- 25-50PPM.
- More than 50PPM.

Liquids are considered to be non-conductors when they have conductivities less than 50 pS/m. Such liquids are often referred to as static accumulators. Which of the following liquids is NOT a static accumulator?

- **Alcohol.**
- Jet fuel.
- Kerosene.
- Xylene.

Loading cargo "over the top" delivers charged liquid to a tank. What danger can result from this operation?

- **The production of a charged mist and an increase in petroleum gas concentration in the tank.**
- The production of a gas mist.
- The production of an increase in petroleum gas concentration in the tank.
- The production of excess tank pressure.

No hot work should be carried out in the vicinity of bunker tanks unless which of the following is carried out?

- **The tank has been cleaned to hot work standard.**
- The tank contains fuel with a flashpoint above 60 degrees Celsius.
- The tank contains fuel with a flashpoint below 60 degrees Celsius.
- The tank is greater than 90% full.

On a Permit-to-Work for pipeline/pressure vessels, which of the following questions would NOT likely be included as part of the checks?

- **Is the lifeboat/rescue boat ready for use?**
- Has the part of the system to be worked upon been isolated?
- Has the pressure been bled off the section to be worked on?
- Have all personnel been equipped with the correct personal protective equipment?

The Code of Safe Working Practices requires that any faults relating to cargo handling equipment, testing instruments and alarm systems must be . . .

- **reported immediately.**
- repaired immediately.
- repaired in dry dock.
- reported when cargo operations are completed.

The International Shore Fire Connection is used for what purpose?

- **To enable the water fire main onboard to be connected to an external water supply from shore.**
- To enable carbon dioxide to be used from shore to fight engine room fires.
- To enable the foam line onboard to be connected to an external foam supply from shore.
- To enable the ship to assist in extinguishing a fire ashore.

The MSDS for a cargo usually contains H2S content in PPM by weight. The H2S content in the vapour space is going to be . . .

- **greater.**
- less.
- possibly all of these answers.
- the same.

The tank bottom and lines should be flushed with water prior to tank cleaning in a non-inert atmosphere. How should this be done?

- **Using main cargo lines.**
- Using a hose by a team inside the tank.
- Using a hose from the deck.
- Using tank washing machines.

What do the terms "threshold limit value" and "permissible exposure limit" refer to?

- **Toxicity.**
- Contact limit.
- Immersion limit.
- Suppression value.

What equipment is used for testing gas levels in a non-inert atmosphere?

- **An Explosiemeter.**
- A gascope.
- A tankscope.
- An oxygen analyser.

What gas level is to be obtained in a tank before tank washing can be started?

- **Less than 10% LFL (Lower Flammable Limit).**
- Less than 10% by volume.
- Less than 100% LFL (Lower Flammable Limit).
- Less than 25% by volume.

What happens to a Permit-to-Work upon completion of work?

- **It is cancelled by the responsible officer.**
- It is cancelled by the person carrying out the work.
- It is left in place in case further work has to be carried out.
- It is transferred to another location where similar work is to be carried out.

What is the correct definition of a "spark"?

- **It is an almost instantaneous discharge between two conductors where almost all of the energy is converted into heat.**
- It is a diffuse discharge from a highly charged non-conductive object to a single blunt conductor.
- It is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy.
- It is a rapid high energy discharge from a sheet of material of high dielectric strength with the two surfaces highly charged but of opposite polarity.

What is the correct method of closing off unused cargo and bunker connections?

- **All unused connections must be closed and blanked. Blank flanges are fitted and are to be fully bolted.**
- All unused connections must be closed and blanked. Blank flanges are fitted and have every second bolt secured by a nut.
- Unused connections may be left unblanked so long as two valve separation exists between the connection and the line in use.
- Unused connections on the shore-side of the vessel must be closed and fully blanked. The offshore-side may be left unblanked.

What is the flammable range of a hydrocarbon gas and air mixture?

- **Between the Lower and Upper Flammable Limits.**
- Above the Upper Flammable Limit.
- From 0% hydrocarbon to the Lower Flammable Limit.
- From the Lower Flammable Limit to 100% hydrocarbon content.

What is the flashpoint temperature of petroleum that is considered to be non-volatile?

- **60 degrees Celsius or above.**
- 38 degrees Celsius and below.
- Above 50 degrees Celsius.
- Below 60 degrees Celsius.

What is the name for the lowest temperature where a liquid gives off sufficient gas to form a flammable gas mixture near the surface?

- **The flashpoint.**
- Flammability temperature.
- Gas emission level.
- Thermal mass.

What is the name of a sudden increase in the pressure of the liquid in a pipeline, caused by a sudden change in flow velocity?

- **A pressure surge.**
- Back pressure.
- Backflow.
- Flashback.

What is the name of an atmosphere made incapable of burning by deliberate reduction of the hydrocarbon content to below Lower Flammable Limit?

- **Too lean.**
- Gas free.
- Inerted.
- Non-inert.

What is the name of an oil that is able to retain a significant electrostatic charge and has a conductivity of less than 50 pS/m?

- **Static accumulator oil.**
- Electrostatic charged oil.
- Static insulating oil.
- Static non-accumulator oil.

What is the name of the practice during tanker cargo operations where all cargo tank apertures, such as ullage pipes and sighting ports, are securely closed?

- **Closed operations.**
- Hermetically sealed operation.
- Non-intrusive procedure.
- Sealed load and discharge.

What is the name of the process where petroleum liquid is ignited through heating rather than using a naked flame?

- **Auto-ignition.**
- Heat regeneration.
- Spontaneous combustion.
- Thermal combustion.

What is the percentage of oxygen content reduced to in a tank if an inert condition has been achieved during tanker operations?

- **8%**
- 0%
- 21%
- 5%

What is the process where a type of gas has been introduced into the tank atmosphere and the oxygen content is reduced, thus making it incapable of burning?

- **Inerted.**
- Gas free.
- Non-inert.
- Too lean.

What is the recommended minimum breaking load (MBL) and length for emergency towing-off pennants (fire wires) for a ship of between 100,000 and 300,000 dwt?

- **MBL = 100 tonnes, L = 60 metres.**
- MBL = 100 tonnes, L = 45 metres
- MBL = 55 tonnes, L = 45 metres.
- MBL = 55 tonnes, L = 60 metres.

What is the word that describes petroleum when it's inhaled or ingested and causes adverse physiological symptoms?

- **Toxic.**
- Convulsive.
- Corrosive.
- Irritant.

What percentage of oxygen should be inside the tank when tank washing in an inert atmosphere?

- **The oxygen level should not exceed 8%.**
- The oxygen level should not exceed 5%.
- The oxygen level should not fall below 5%.
- The oxygen level should not fall below 8%.

What should happen to deck scuppers during periods of heavy rain when engaged in cargo or bunker handling?

- **One or two scuppers may be opened temporarily to allow rain water to escape but must be closed as soon as possible and in the event of a deck oil spill.**
- All deck scuppers must be kept fully closed at all times and never opened until cargo and bunker operations are completed.
- Only deck scuppers on the outboard side of the vessel are to be opened to release the rain water. When the rain stops, the scuppers can then be replugged.
- Open all deck scuppers to release the rain water. When the rain stops, the scuppers can then be replugged.

What should personnel on board barges or other craft obey when moored alongside?

- **All of these answers.**
- Prohibition of naked lights.
- Restrictions on the use of cooking appliances.
- Smoking restrictions.

What should the ship's Master do if an unauthorised craft comes alongside and attempts to secure to the vessel when at a terminal?

- **Report the craft to the terminal and if necessary stop cargo operations.**
- All of these answers.
- Sound the emergency alarm.
- Turn fire hoses onto the small craft.

What washing mediums must never be used?

- **Recirculated water.**
- Chemicals.
- Heated water plus chemical additive.
- Heated water.

What word describes an atmosphere that is above the flammable range?

- **Over rich.**
- Explosive.
- Flammable.
- Non-inert.

What word describes the unrestricted fall of liquid into a tank during tanker operations?

- **Free fall.**
- Impingement.
- Load on top.
- Overflow.

What would be the minimum recommended firefighting equipment to have available when alongside a terminal?

- **Fire hoses with spray/jet nozzles run out forward, aft and adjacent to the manifold in use. Also a dry chemical extinguisher near the manifold.**
- Fire hoses with spray/jet nozzles forward and aft of the manifold and a fire blanket adjacent to the manifold in use.
- Fire hoses with spray/jet nozzles or fire extinguishers adjacent to the manifold in use.
- One fire hose and one dry chemical extinguisher near the manifold in use.

When H2S is known to be present, self-contained breathing apparatus must be used in which of the following activities?

- **All of these activities.**
- Mopping up spills.
- Open gauging and sampling.
- Removing blanks for connecting the cargo loading arm.

When are items on the ship/shore safety check list checked when a vessel is at a terminal?

- **Before cargo or ballast operations commence and then at regular intervals until hoses are disconnected.**
- During cargo operations.
- Immediately upon berthing and before starting cargo operations.
- Upon commencing cargo operations.

When can controlled hot work be carried out within a hazardous area on a tanker?

- **When the ship is in ballast condition.**
- When the ship is carrying out cargo operations.
- When the ship is carrying out gas freeing operations.
- When the ship is carrying out inerting operations.

When carrying out hot work with electric welding equipment, which one of the following statements is correct?

- **The ship's structure should not be used as an earthing point.**
- The cable route to the working area should be the shortest and most direct route.
- The Direct Current output source should be in excess of 70 Volts.
- The person carrying out the welding operation can work alone so long as a work permit has been issued.

When must ventilation be stopped while carrying out an enclosed space entry?

- **Ten minutes before the pre-entry atmosphere test.**
- Never.
- Only when 20.7 % oxygen is achieved.
- Only when tests show no hazardous gas or oxygen deficiency.

When should tank washing in an inert atmosphere be stopped?

- **When the tank atmosphere pressure is no longer positive.**
- When the tank atmosphere pressure exceeds 250mmWC.
- When the tank atmosphere pressure falls below 1000mmWC.
- When the tank atmosphere pressure falls below 250mmWC.

When venting is being carried out during the loading of a cargo containing H2S, when should cargo operations be stopped?

- **When there is no wind to disperse the cargo vapours.**
- When the wind force is greater than 10m/s.
- When the wind is variable in direction.
- When there is a combination of wind and rain.

Where does the Code of Safe Working Practices recommend guidance on general precautions be taken from?

- **All of these answers.**
- Safety in Oil Tankers.
- The Company's Safety Management System.
- The International Safety Guide for Oil Tankers and Terminals.

Where hot work is planned to be carried out, what should the Lower Flammable Limit (LFL) reading of the atmosphere be?

- **Less than 1% LFL.**
- 1% LFL.
- Between 1% and 2% LFL.
- Greater than 1% LFL.

Where should the ship's emergency fire control plans be located?

- **In a prominently marked weather-tight enclosure outside the accommodation block.**
- At the bottom of the gangway.
- In a prominently marked wallet contained within the cargo control room.
- With the Deck Officer of the Watch at all times.

Where the concentration of H2S in cargo is high, what term is often used to describe the oil?

- **Sour.**
- Acidic.
- Salty.
- Sweet.

Which of the following could be considered a "designated space" where hot work is able to be carried out safely?

- **The engine room workshop.**
- The engine room.
- The forecastle deck.
- The poop deck.

Which of the following examples must NOT be electrically bonded?

- **The ship/shore insulating flange.**
- All of these answers.
- Manual ullaging and sampling equipment.
- Portable tank cleaning machines.

Which of the following is NOT considered to be an enclosed space?

- **The engine room.**
- A domestic fresh water tank.
- A duct keel.
- The pumproom.

Which of the following questions would NOT be part of a Permit-to-Work check list for allowing small craft to come alongside?

- **Has the ship been inerted or gas freed?**
- Are all deck and cargo openings closed?
- Has the Master of the small craft been advised of all appropriate safety regulations (e.g. emergency signals, smoking regulations etc)?
- Is adequate illumination available?

Why is there a limit to the maximum throughput of tank washing machines when washing in a non-inert atmosphere?

- **To reduce the risk of introducing electrostatically charged mist.**
- It's a MARPOL regulation.
- To prevent water build up in cargo tanks with resulting free surface.
- To reduce the risk of inflammable gas evolution.

7013 – TOTS 1A-The Flammability Diagram

Version 1.0 (13)

A gas turbine with afterburner will produce similar inert gas to an independent inert gas generator. What level of oxygen might typically be delivered?

- **1.5-2.5% by volume.**
- 5% by volume.
- 8-10% by volume.
- Approximately 12% by volume.

After inerting a cargo tank, the oxygen content was found to be 7.5%, and the hydrocarbon gas content 2%. Using the Flammability Diagram, would you consider the tank to be safely inerted before gas freeing?

- **Yes, the atmosphere is below the critical dilution line.**
- No, inerting must be continued.
- No, the tank atmosphere is in the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

Define the "Lower Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration below which there is not sufficient gas to support and propagate combustion.**
- It is the gas concentration above which there is not sufficient air to support and propagate combustion.
- It is the gas concentration above which there is too much gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Define the "Upper Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration above which there is not sufficient air to support and propagate combustion.**
- It is the gas concentration above which there is too much oxygen to support and propagate combustion.
- It is the gas concentration below which there is not sufficient gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Explain how very cold weather conditions can cause over-pressurisation of cargo tanks?

- **Ice could form on the vents, or the P/V or high velocity vent valves could freeze.**
- Cold weather may cause malfunctioning of cargo monitoring equipment, including pressure monitors.
- Low temperatures will increase the viscosity of the cargo, and may make it unpumpable.
- The increased density of the cargo could lead to overloading.

Explain why the atmosphere in a cargo tank must be sampled at different depths?

- **Due to the high relative density of hydrocarbon gas, it tends to layer, and collect in pockets.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

If cargo tank openings are opened frequently for operations such as sampling and dipping, inert gas quality may be affected. How so?

- **Air could enter the tanks through the openings.**
- Inert gas would escape, lowering the tank pressure.
- Moisture from the atmosphere could enter the tank.
- The hydrocarbon content of the tank atmosphere would be increased.

In practical terms, what figures can be used for the Lower and Upper Flammability Limits of oil tanker cargoes?

- **1% and 10% of gas.**
- 19% and 21% of oxygen.
- 5% and 21% of oxygen.
- 5% and 50% of gas.

Inert gas quality in slop tanks may be degraded when tank washings from non-inerted tanks are added: Why?

- **Air would be trapped in the washings, particularly if an eductor was in use.**
- Adding washings would increase the pressure in the tank, forcing inert gas out via the vent line.
- The higher temperature of the slops caused by adding washings would result in increased evolution of hydrocarbon vapours.
- The turbulence in the slops caused by adding washings would result in increased evolution of hydrocarbon vapours.

It is generally accepted that an atmosphere with less than 11% oxygen cannot burn. Why is the figure of 8% usually specified as the maximum oxygen content of a cargo tank?

- **To allow a safety margin.**
- Because gas detection equipment is not accurate enough.
- Because hydrocarbon gas will be produced during cargo operations.
- To allow for operator error using the equipment.

Pressure in the cargo tanks may be reduced due to low night time temperatures. How could this affect inert gas quality?

- **Air could be drawn into the system.**
- Components of the inert gas may break down at low pressure.
- The inert gas may condense into the cargo at low temperatures.
- The rate of evolution of cargo vapours could be increased.

The characteristics of the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the relative density of butane?

- **2.00**
- 1.00
- 2.55
- 5.50

The characteristics of the vapours given off by crude oil can be represented by those of propane. What is the relative density of propane?

- **1.55**
- 0.55
- 1.00
- 5.50

The characteristics of the vapours given off by natural gasoline's can be represented by those of pentane. What is the relative density of pentane?

- **2.50**
- 0.55
- 1.00
- 5.50

The flammable limits for the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the Lower Flammable Limit of butane?

- **1.9% by volume in air.**
- 19% by volume in air.
- 5.9% by volume in air.
- 9% by volume in air.

The flammable limits for the vapours given off by crude oil can be represented by those of propane. What is the Lower Flammable Limit of propane?

- **2.2% by volume in air.**
- 10% by volume in air.
- 22% by volume in air.
- 5.2% by volume in air.

The flammable limits for the vapours given off by natural gasoline's can be represented by those of pentane. What is the Lower Flammable Limit of pentane?

- **1.5% by volume in air.**
- 21% by volume in air.
- 4.5% by volume in air.
- 7.8% by volume in air.

What action must be taken if the inert gas plant fails during cargo discharge operations?

- **You should prevent air from entering the tanks, and stop discharge from inerted tanks.**
- You should carefully allow fresh air into the tanks to prevent under-pressurisation.
- You should continue discharge, monitoring oxygen and gas concentration in the cargo tanks.
- You should reduce the discharge rate until the plant is repaired.

What are the main components of inert gas delivered from a flue gas plant?

- **Nitrogen 83%, Carbon Dioxide 12-14%, Oxygen 2-4%**
- Carbon Dioxide 63%, Nitrogen 22-24%, Oxygen 10-12%
- Carbon Dioxide 83%, Nitrogen 12-14%, Oxygen 2-4%
- Nitrogen 60%, Carbon Dioxide 30%, Oxygen 10%

What inert gas monitoring instrumentation must be fitted in the cargo control room of a tanker?

- **Instruments to continuously indicate and permanently record the oxygen content and deck main pressure.**
- A device to permanently record the times of stopping and starting the inert gas plant.
- Instrumentation showing all operating parameters of the inert gas plant.
- Instrumentation showing the cargo tank temperature and pressure.

What is meant by the "Dilution" method of inerting a cargo tank?

- **The incoming inert gas is mixed with the tank atmosphere to form a homogenous mixture throughout the tank.**
- The inert gas is introduced slowly at the bottom of the tank so that a stable horizontal interface is formed.
- The inert gas is introduced slowly at the top of the tank so that a stable horizontal interface is formed.
- The inert gas is mixed with fresh air, and introduced rapidly into the tank.

What is meant by the "Displacement" method of inerting cargo tanks?

- **The inert gas is introduced slowly at the top of the tank so that a stable horizontal interface is formed.**
- The incoming inert gas is mixed with the tank atmosphere to form a homogenous mixture throughout the tank.
- The inert gas is introduced slowly at the bottom of the tank so that a stable horizontal interface is formed.
- The inert gas is mixed with fresh air, and introduced rapidly into the tank.

What is pyrophoric iron sulphide, and what hazard is associated with it?

- **It's a compound formed by hydrogen sulphide and iron oxide which becomes incandescently hot in the presence of oxygen.**
- It's a compound found in inert gas, which increases corrosion in ballast and cargo tanks.
- It's a compound which causes the "rotten eggs" smell of some crude oil, and it is toxic.
- It's a crystalline substance found in inert gas which can build up in vents, filters etc., and can cause a blockage.

What is the current deadweight, above which tankers must be fitted with an inert gas system?

- **20000 tonnes.**
- 15000 tonnes.
- 2000 tonnes.
- 50000 tonnes.

What is the maximum oxygen content allowed by SOLAS in the output from an inert gas plant?

- **Not more than 5%.**
- Not more than 1%.
- Not more than 21%.
- Not more than 8%.

What is the maximum oxygen content which should be maintained in a cargo tank during operations?

- **Not more than 8%, except when gas freeing.**
- Not more than 1%, except when gas freeing.
- Not more than 18%, except when loading.
- Not more than 21%, except when loading.

What is the minimum recommended pressure for cargo tanks connected with an inert gas main?

- **100 mm water gauge.**
- 0.25 bars.
- 500 mm water gauge
- One bar.

What is the operating principle of a dry type deck seal in an inert gas system?

- **A dump valve is fitted, which opens, filling the lower chamber with water to form a seal when the inert gas supply stops.**
- A large diameter non-return valve is fitted, which will close when the inert gas supply stops.
- The chamber contains a number of baffles which prevent solid materials from entering the cargo tanks, and help regulate the pressure of the inert gas main line.
- The chamber is drained when the inert gas plant is prepared for operation, but can be filled with water when the plant is not in use, forming a seal.

What is the relative density of hydrocarbon gas at concentrations near the Lower Flammable Limit?

- **Approximately 1.0, the same as air.**
- Approximately 0.7, lighter than air.
- Approximately 1.5, heavier than air.
- Approximately 2.0, much heavier than air.

What oxygen content would normally be expected from an independent inert gas generator?

- **1.5-2.5% by volume.**
- 5% by volume.
- 8-10% by volume.
- Approximately 12% by volume.

When a tanker is carrying sour crude, it is possible for pyrophoric iron sulphide to form. What special precautions must be taken following a failure of the inert gas plant?

- **The inert gas plant must be repaired, or an alternative supply arranged, before resuming discharge.**
- Cargo tanks must be thoroughly cleaned before loading the next cargo.
- Cargo tanks must be water washed after discharge to remove iron sulphide deposits.
- Continue discharge, monitoring oxygen and hydrogen sulphide concentration in the cargo tanks.

When using the "Dilution" method of inerting a cargo tank, inert gas is introduced rapidly into the tank. Why?

- **Because the incoming inert gas must have enough velocity to reach the bottom of the tank and mix thoroughly with the tank atmosphere.**
- Because inert gas is slightly less dense than hydrocarbon vapour, and a stable horizontal interface can be formed.
- Because the incoming inert gas must have enough velocity to reach the bottom of the tank and form a stable interface.
- Because the inert gas is hotter than the tank atmosphere, and it is necessary to mix thoroughly with the tank atmosphere.

When using the "Dilution" method of inerting, it is essential that only a small number of cargo tanks are open. Why?

- **Because the inert gas must be introduced at a high enough rate to mix thoroughly with the tank atmosphere.**
- Because it is difficult to monitor the hydrocarbon levels properly if a large number of tanks are open.
- Because the incoming inert gas must form a stable interface with the tank atmosphere.
- Because the inert gas plant works more efficiently when the back pressure is higher.

When using the "Displacement" method of inerting cargo tanks, inert gas is introduced slowly at the top of the tank. Why?

- **Because inert gas is slightly less dense than hydrocarbon vapour, and a stable horizontal interface can be formed.**
- Because inert gas is denser than hydrocarbon vapour, and a stable horizontal interface can be formed.
- Because the incoming inert gas will mix with the tank atmosphere throughout the tank.
- Because the inert gas is hotter than the tank atmosphere, and it is necessary to avoid violent mixing.

When using the "Displacement" method of inerting, it is common practice to inert several tanks at the same time. Why?

- **Because the inert gas must be introduced at a slow enough rate to enable the formation of a stable horizontal interface.**
- Because it is easier to compare the hydrocarbon levels in the tanks if a large number are open.
- Because the inert gas must be introduced at a high enough rate to mix thoroughly with the tank atmosphere.
- Because the inert gas plant works more efficiently when the back pressure is lower.

Why is an oxygen content of not more than 8% considered to be the safe limit in a cargo tank atmosphere?

- **Because the atmosphere is safely outside the Flammable Range.**
- Because 8% hydrocarbon gas is above the Upper Flammable Limit.
- Because 8% hydrocarbon gas is below the Lower Flammable Limit.
- Because the tank atmosphere was outside the Flammable Range at the time the readings were taken.

Why is it essential that the deck and engine departments cooperate in the maintenance and operation of the inert gas plant?

- **To ensure that the cargo tanks are in the required atmospheric condition at all times, and that there is no risk of flammable vapours entering the engine room.**
- To comply fully with the requirements of the planned maintenance system.
- To comply with working hours regulations, and keep overtime under control.
- To ensure the minimum time is spent working cargo in port.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, what would be the effect of diluting with fresh air?

- **The hydrocarbon concentration would be reduced, and the oxygen level increased, taking the atmosphere into the Flammable Range.**
- The hydrocarbon concentration would be increased, taking the atmosphere into the Flammable Range.
- The hydrocarbon concentration would be reduced, and the oxygen level increased, resulting in a safe tank atmosphere.
- The oxygen level would be increased, resulting in a safe tank atmosphere.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, why would it be necessary to dilute the atmosphere to point H with inert gas before gas freeing with fresh air?

- **To reduce the hydrocarbon level below the critical dilution line.**
- To help prevent corrosion in the tank.
- To increase the hydrocarbon level above the Upper Flammable Limit.
- To reduce the oxygen level to below the critical dilution line.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, would you consider it to be safe to purge with inert gas with an oxygen content of 8%?

- **Yes, the atmosphere would remain outside the Flammable Range.**
- No, the tank atmosphere is in the Flammable Range.
- No, the tank must first be gas freed with fresh air.
- Yes, the tank atmosphere would remain too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point A?

- **The tank contains fresh air, with no hydrocarbon gas, and cannot support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point B?

- **The tank atmosphere is too rich to support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere is too lean to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point C?

- **The hydrocarbon gas concentration is at its Lower Flammable Limit.**
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point D?

- **The hydrocarbon gas concentration is at its Upper Flammable Limit.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too lean to support combustion.

7014 – TOTS 1A-Fixed and Portable Gas Detec. Systems

Version 1.0 (3)

A fixed gas detection system for toxic gas normally uses electro-chemical detectors. What level of accuracy can be expected?

- **Approximately 5 ppm.**
- Approximately 10 % by volume.
- Approximately 30 % by volume.
- Approximately 50 ppm.

After washing and gas-freeing from a toxic cargo, what type of gas detection equipment should you use before entering a cargo tank?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

Cotton filters are sometimes used with hydrocarbon gas detectors. What is their purpose?

- **To remove water from the sample.**
- To prevent chemical contamination of the sample.
- To remove dust from the sample.
- To stop pressure surges in the sample line.

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

How will leaks in the sample pipework affect the operation of a fixed gas detection system?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

How will leaks in the sampling line affect the operation of portable gas detection equipment?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

If an Explosimeter is calibrated using span gas containing 1% butane in air, what reading may be expected?

- **Approximately 50% Lower Flammable Limit.**
- Approximately 1% Lower Flammable Limit.
- Approximately 10% Lower Flammable Limit.
- Approximately 100% Lower Flammable Limit.

Multi-toxic gas detectors normally use electro-chemical detectors. What level of accuracy can be expected?

- **Approximately 1 - 5 ppm . . .**
- Approximately 10% by volume.
- Approximately 100 - 500 ppm.
- Approximately 30% by volume.

Span gas is generally used for calibration of a fixed gas detection system. What is the composition of this gas?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure nitrogen.

What effect will inert gas have on the operation of an infrared type gas detector?

- **None. The instrument is not affected by the absence of oxygen.**
- It will Increase the reading by increasing the temperature of the chamber.
- It will reduce the reading by contaminating the sensor.
- Small particles may obscure the mirrors.

What instrument should you use to test for very low levels of oxygen before loading a sensitive cargo?

- **Chemical indicator tubes.**
- A paramagnetic oxygen analyser.
- An Explosimeter.
- An infrared gas detector.

What instruments must be provided to comply with the statutory requirements for ships fitted with an inert gas plant?

- **An oxygen analyser and an instrument to measure the percentage of hydrocarbon gas in an inerted atmosphere.**
- An Explosimeter and an oxygen analyser.
- At least two instruments to measure the percentage of hydrocarbon gas in an inerted atmosphere.
- At least two instruments to measure the percentage of oxygen in an inerted atmosphere.

What is a "Pellistor"?

- **It's a poison resistant catalytic filament.**
- It's a balancing resistor between the sensing element and a fixed compensator.
- It's a filter which prevents the spread of flame.
- It's a protective device which removes contaminants.

What is a "Sintered Filter", and what is its purpose?

- **It's a flashback arrestor used to prevent flame from escaping from the combustion chamber.**
- It's a cotton wad used to prevent water from entering the combustion chamber.
- It's a fine metal filter used to prevent contamination of the sensor.
- It's an electronic device used to smooth the signal from the sensor.

What is the composition of the span gas generally used for calibration of an Explosimeter?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure carbon dioxide.

What is the composition of the span gas generally used to zero a portable oxygen analyser?

- **High purity nitrogen or carbon dioxide.**
- Approximately 2% propane in air.
- Approximately 20% propane in air.
- Fresh air.

What is the gas concentration alarm level on a fixed gas detection system?

- **30% of Lower Flammable Limit.**
- 10% of flammable gas.
- 3% of Lower Flammable Limit.
- 30% of flammable gas.

What is the principle by which a thermal conductivity meter works?

- **To show a change in resistance of an electrically heated resistance element, due to heat loss to the gas sample by conduction.**
- To show a change in temperature of an electrically heated resistance element, due to heat loss to the gas sample by conduction.
- To show a change in the conductivity of a heated gas sample relative to that of air.
- To show change in the density of a heated gas sample relative to that of air.

What is the principle by which an infrared gas detector operates?

- **It senses the level of absorption of infrared radiation in the sample.**
- It compares the electrical resistance of the sample with fresh air.
- It compares the relative density of a sample which has been heated by an infrared source.
- It measures the resistance of a sample which has been heated by an infrared source.

What is the principle by which chemical absorption indicators operate?

- **By passing the sample through a proprietary chemical, the gas concentration is indicated by its changing colour.**
- To compare the chemical absorption rate of the sample with fresh air.
- To sense the proportion of a sample which is absorbed by a catalyst.
- To show whether a toxic gas will be absorbed by the human body.

What level of accuracy can be expected when using toxic gas detection tubes?

- **Approximately 1 ppm, depending on the type of tube in use.**
- Approximately 10%, depending on the type of tube in use.
- Approximately 100 ppm, depending on the type of tube in use.
- Approximately 30%, depending on the type of tube in use.

What operational tests should be made routinely before using a gascope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self- testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using a tankscope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using an Explosimeter?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas, and zero using atmospheric air.

What operational tests should be made routinely before using an oxygen analyser?

- **Check battery level, reference point with atmospheric air (21%), and zero using nitrogen or carbon dioxide.**
- Check battery level, reference point with span gas, and zero using nitrogen or carbon dioxide.
- Check the battery level, and self- test the alarm functions.
- Check the reference point with span gas and zero using atmospheric air.

What property of oxygen is utilised in a paramagnetic oxygen analyser?

- **Oxygen is strongly attracted to a magnetic pole.**
- Oxygen molecules will break up in a magnetic field.
- The electrical resistance of oxygen changes in a magnetic field.
- The strength of a magnetic field is reduced in the presence of oxygen.

What type of sensor is fitted in a tankscope?

- **A non-catalytic heated filament.**
- A catalytic filament resistor.
- A magnetic induction sensor.
- A PT100 temperature sensor.

What type of sensor is fitted in an Explosimeter?

- **A catalytic filament resistor.**
- A gas diffusion filter.
- A magnetic resonance chamber.
- A non-catalytic heated filament.

What will be the effect of hydrogen sulphide on the reading shown on a Combustible Gas Indicator (Explosimeter)?

- **It will reduce the reading by contaminating the sensor.**
- It will give a zero reading by destroying the sensor.
- It will have no effect because it is non-flammable.
- It will increase the reading by raising the sensitivity of the sensor.

What would you use a personal oxygen meter for?

- **For monitoring atmospheric conditions during an enclosed space entry.**
- For checking the atmosphere of a ballast tank before entry.
- For testing an enclosed space before starting hot work.
- For testing the atmosphere of an open space before starting hot work.

When changing high purity cargo grades, what instrument should be used to ensure that tanks are free of the previous product?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

When starting to test a sample using a Combustible Gas Indicator (Explosimeter), the needle immediately flicks to maximum deflection, then returns to zero. What is indicated?

- **That the flammable gas concentration is above the Lower Flammable Limit.**
- That the battery is incorrectly fitted.
- That the instrument was not warmed up, but is now working correctly.
- That the sensor element has been contaminated by toxic gas.

Why is it necessary to use a water-retaining filter in the sample line of an oxygen analyser?

- **Because water vapour in the sample will cause damage to the measuring cell.**
- Because water in the sample will corrode the internal parts of the instrument.
- Because water reacts with the measuring cell, giving a false reading.
- Because water will build up in the sample line, reducing the flow.

Why must chemical absorption tubes only be used with pumps from the same manufacturer?

- **Because the pump and tube are matched to ensure that the correct volume of sample gas is tested.**
- Because the tubes will not fit correctly into another manufacturer's pump.
- Because using different tubes breaks the terms of the contract with the manufacturer.
- Because using the wrong tubes may lead to contamination of the pump.

Why must the span gas pressure be carefully controlled when calibrating a fixed gas detection system?

- **Excess pressure may damage the catalytic sensing element.**
- Excess pressure may cause errors in the readings.
- Excess pressure may cause leaks in the system.
- If the pressure is too high, it could cause an explosion inside the equipment.

Why must the tank atmosphere be sampled at a variety of positions when testing before entry?

- **To allow for the density of the gas and to ensure that any collected in dead spots is detected.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

7015 – TOTS 1A-Enclosed Spaces and Tank Atmosphere . . .

Version 1.0 (6)

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

For what purpose would you use a personal gas detector?

- **To monitor atmospheric conditions during an enclosed space entry.**
- To check the atmosphere of an enclosed space before entry.
- To test an enclosed space before starting hot work.
- To test the atmosphere of an open space before starting hot work.

For what purpose would you use a personal oxygen meter?

- **To monitor atmospheric conditions during an enclosed space entry.**
- Testing an enclosed space before starting hot work.
- Testing the atmosphere of an open space before starting hot work.
- To check the atmosphere of an enclosed space before entry.

How long should ventilation be continued after the atmosphere of an enclosed space has been found suitable for entry?

- **Continuously while the space is occupied, and even during temporary breaks.**
- For at least one hour after successful testing.
- It can be stopped when testing shows that the space is suitable for entry.
- It should be continued after testing until the space is actually entered.

How many different locations within a tank should be sampled to test the atmosphere before entering a tank?

- **At various depths, and through as many openings as possible.**
- As close as possible to the centre of the tank.
- At least three locations.
- In one position, at a minimum of three levels.

How many people should be allowed to enter an enclosed space to carry out routine work?

- **Only the number actually required to carry out the work.**
- Only two people.
- The number required to carry out the work, plus one extra watchman.
- The number required to carry out the work, plus the person in charge and one extra watchman.

What action must be taken if the presence of a toxic gas is suspected for which no specific testing equipment is available?

- **Treat the space as unsafe and follow the procedures according to COSWP 17.11.**
- Only allow entry after further ventilation.
- Test the space for a similar gas using any available equipment, following the procedures according to COSWP 17.6
- Test using an Explosimeter, and allow any necessary conversion factors.

What action must be taken when an enclosed space entry permit expires?

- **The space must be vacated and closed.**
- Additional crew should attend for assistance.
- The work must be completed as soon as possible.
- Work can continue until it has been completed.

What action should be taken if personnel working in an enclosed space get into difficulties?

- **The general alarm should be sounded and the rescue team should enter using breathing apparatus under the direction of the Person In Charge**
- All personnel should be instructed to leave the space immediately.
- The first person on the scene should immediately enter the space to give assistance.
- The Watchman should immediately enter the space to give assistance.

What actions must be taken to isolate a tank before entry?

- **Blank off the pipelines to the tank, close and secure all valves and inform the Officer of the Watch.**
- Request the Officer of the Watch to stop all cargo and ballast pumps.
- Stop all cargo and ballast operations.
- Stop all ventilation and inform the Officer of the Watch.

What equipment is required to be kept at the entrance of an enclosed space during entry?

- **A resuscitator, breathing apparatus, a lifeline and a harness, a torch or a lamp.**
- A resuscitator or escape breathing set.
- Enough escape breathing sets for all personnel in the space.
- Spare tools and parts for the work being undertaken.

What gas detection equipment must be used to test the atmosphere of a cargo tank before entry is permitted?

- **An oxygen meter, a combustible gas indicator and a toxic gas detector if necessary.**
- A combustible gas indicator and a toxic gas detector if necessary.
- An oxygen meter and a tankscope.
- An oxygen meter and a toxic gas detector if necessary.

What indication will be given by an Explosimeter in case of oxygen deficiency?

- **No indication.**
- Approximately 20% of the Lower Flammable Limit
- Full scale deflection.
- The indication is erratic and jumpy.

What is the maximum permitted level of toxic gas for an enclosed space entry?

- **The Occupational Exposure Limit, which varies according to the particular gas.**
- 1% of the Lower Flammable Limit.
- 10% of the Lower Flammable Limit.
- It varies according to the particular gas, but 1% is an acceptable level.

What potential hazards would be caused in a machinery space by the release of carbon dioxide for fire extinguishing purposes?

- **A lack of oxygen.**
- Carbon dioxide which is toxic to breathe.
- The carbon dioxide gas would be very cold, and could cause severe cold damage to the skin.
- Vapour, which may reduce visibility.

Who is required to standby at the entrance of an enclosed space during entry?

- **At least one watchman, equipped to communicate with the personnel inside and with the Officer of the Watch.**
- Nobody, provided the personnel inside are equipped to communicate with the Officer of the Watch.
- The Officer of the Watch and one watchman, equipped to communicate with the personnel inside.
- The Officer of the Watch, or whoever signed the Entry Permit.

Why could an unprotected steel space, which has been closed, produce a hazardous atmosphere?

- **Because oxygen in the atmosphere may be depleted by reacting with the steel (i.e. corrosion).**
- Because hydrocarbons could be liberated from rust scale in the tank.
- Because the level of nitrogen in the atmosphere would be increased by corrosion.
- Because toxic gases may be given off by the process of corrosion.

Why could an unventilated ballast tank fitted with Cathodic Protection contain a hazardous atmosphere?

- **Because hydrogen is produced by the Cathodic Protection.**
- Because hydrocarbons could be produced by the Cathodic Protection.
- Because the Cathodic Protection could produce an oxygen enriched atmosphere.
- Because the Cathodic Protection would increase the level of nitrogen in the atmosphere.

Why must the enclosed space atmosphere be sampled at a variety of positions?

- **To allow for the density of the gas and to check for any gas pockets which may have accumulated in blind spots.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

7016 – TOTS 1A-Familiarisation prior Cargo Operations

Version 1.0 (6)

A Ship to Shore Safety Checklist can be found in which publication?

- **ISGOTT.**
- ISPS code.
- SOPEP.
- The Mariner's Handbook.

A cargo tank wash plate improves intact stability.

- **True**
- False

A combination carrier loaded with crude oil has two tanks, each only 60% full. What affect will this have on the vessel's stability?

- **There will be a virtual loss of GM.**
- It will increase in permissible bending moments.
- It will increase the rolling period.
- The vessel will trim by the head.

A vessel is to maintain light ballast condition, however if logged and permissible bending moments have been exceeded, you would . . .

- **reduce ballast in Fore & After Peaks, maintaining propeller immersion and minimum draught forward?**
- ballast amidship cargo tanks?
- do nothing, it is only a short voyage?
- move bunker fuel aft?

A warning may be posted in the cargo control room (CCR) when some ballast or cargo tanks are slack due to . . .

- **free surface effect and loss of intact stability.**
- an inability to maintain the cargo transfer rate.
- damage on a double hull vessel.
- the vessel developing a list.

As part of the Safety Management System (SMS), new personnel joining the tanker must be . . .

- **familiarised with their duties and responsibilities.**
- given a minimum of 24 hours to recover after travel.
- informed of their salary.
- introduced to the Captain.

At commencement of loading, the generation of static is controlled by . . .

- **limiting the flow rate at the tank inlet.**
- earthing the manifold.
- monitoring the cargo temperature.
- only having one loading arm connected.

Before loading, you should ensure that . . .

- **the offshore manifold is closed and blanked.**
- the manifold is blanked but the valve is open.
- the pressure gauge is fitted.
- the temperature gauge is fitted between the valve and the blank.

Cargo tank temperature and pressure gauges and reliquefaction plant instruments should be calibrated . . .

- **preferably at intervals not exceeding 30 months.**
- at intervals not exceeding 12 months.
- at the Master's discretion.
- whenever a false reading is suspected.

Cargo transfer operational information will normally be available in . . .

- **the cargo control room (CCR).**
- the Chief Engineer's office.
- the SOLAS training manual located in the officers' and crew lounge.
- the wheel-house.

Damage due to sloshing is a direct result of the tank . . .

- **being partially filled (slack).**
- being filled more than 98%.
- having the tank pressure too high.
- None of these answers.

Different grades on a product tanker are normally segregated by . . .

- **Two valve separation.**
- A single valve.
- A spade blank.
- It does not matter as different densities will keep them separated.

During cargo operations, VHF/UHF radios power output is . . .

- **one watt or less.**
- as per the GMDSS logbook.
- It does not matter because they are intrinsically safe.
- ten watts.

During cargo transfer, what action would you take if the Inert Gas System fails?

- **Cease all discharge operations and advise the terminal.**
- Advise the terminal and continue discharging.
- Increase the rate of discharge.
- Review the Ship / Shore Safety Check List.

Emergency shutdown valves in liquid piping should fully close under all service conditions within . . .

- **30 seconds.**
- 5 seconds.
- 60 seconds.
- 90 seconds.

Flammability and toxicity of proposed cargo can be obtained from . . .

- **Material Safety Data Sheets (MSDS).**
- ISGOTT.
- The Approved Trim and Stability Manual.
- The SOPEP file.

How do you ensure that pre-cargo transfer stability and stress calculations are correct?

- **Only by using an approved loading instrument which has been regularly tested with an approved test programme.**
- By looking at similar cargoes in the cargo record book.
- By trusting your personal previous experience.
- This should not be necessary.

How does "free surface effect" (sloshing) affect the ship's stability?

- **It reduces the metacentric height.**
- It causes an angle of loll.
- It has no effect.
- It increases the metacentric height.

How does the Chief Officer ensure the approved stress and stability computer is giving accurate results?

- **By running an approved test programme on the computer.**
- By comparing results to previous similar conditions.
- By comparing the results to his own homemade programme.
- It never goes wrong, and so does not need checking.

How long must logbooks and other cargo records be kept on board?

- **As detailed in the owners/managers Safety & Quality Management System.**
- At the Master's discretion.
- For 12 months after discharge.
- For the life of the vessel.

If hydrocarbon gases are found in double hull ballast tanks, what action should be taken?

- **Immediately utilise the emergency inerting procedure.**
- Close the tank down.
- Read ISGOTT.
- Vent to atmosphere.

If the shore fails to deliver cargo at the requested rate, what action would the Master take?

- **He would issue a letter of protest.**
- He would inform the Class.
- He would inform the local P&I Club.
- He would take no action.

If unable to ground GMDSS equipment during cargo operations, what action may be taken?

- **To switch the equipment off.**
- To ignore any rules as you cannot comply.
- To make a logbook entry.
- To turn the volume down.

In the loading limit formula, at which temperature is the pR value expressed?

- **At reference temperature.**
- At 45 degrees Centigrade.
- At ambient temperature.
- At sea temperature.

In which documentation would you expect to find duties and responsibilities relating to the cargo system?

- **In the Safety Management System (SMS).**
- In the Builders' Cargo Operations Book.
- In the Certificate of Fitness.
- In the SOLAS training manual.

In which language would you expect the operational and technical manuals to be written?

- **The official working language of the vessel.**
- English.
- It depends upon the nationality of the manufacturer.
- Several languages.

Night orders must be signed by the appropriate personnel . . .

- **every night when new orders are issued.**
- It does not matter.
- once per cargo grade only.
- once per port only.

Offshore manifold gauges need not be fitted?

- **False**
- True

On a combination carrier, where would the audible bilge alarm sound from a void space forward of the collision bulkhead?

- **The navigation bridge.**
- No alarm. It would just checked daily by soundings.
- On deck close to an accommodation entrance.
- The CCR (cargo control room).

On what occasions will a pre-cargo transfer meeting be minuted?

- **On each and every occasion.**
- When the Captain attends.
- When there is something special to say.
- Whenever the port stay is estimated to be greater than 24 hours.

Permissible bending moments are greater in which condition?

- **In harbour condition.**
- It depends upon the amount of segregated ballast.
- It depends upon the mean draught.
- There is no difference between sea and harbour condition.

Prior to cargo transfer, where would you expect to see a reference to valves and gauges?

- **In an appropriate Ship / Shore Safety Check List.**
- In the Safety Management System (SMS).
- In the statement of facts.
- In the Vessel Inspection Questionnaire.

Sequential and flow through methods form part of . . .

- **the ballast exchange plan.**
- the ballast water analysis.
- the composition of a sounding book.
- the plan to clean cargo tanks.

Shore cargo hard arms are commonly known as . . .

- **chicksans.**
- cargo lines.
- hose strings.
- manifold lines.

The Ship / Shore Safety Check List states that temporarily removed scupper plugs must be . . .

- **constantly monitored.**
- clearly labelled as to location.
- only removed by the person responsible for cargo operations.
- stowed in a safe place.

The Ship Inspection Report Programme (SIRE) states that the loading programme must be checked by means of an approved test programme at least . . .

- **Quarterly.**
- Annually.
- As directed by the Master.
- At the Chief Officer's discretion.

The Vessels Inspection Questionnaire (VIQ) states the maximum interval between the calibration of temperature and pressure gauges to be . . .

- **30 months.**
- 12 months.
- 24 months.
- at the Master's discretion.

The approved loading computer programme is malfunctioning, you would . . .

- **consult the Approved Trim and Stability Manual?**
- consult a general arrangement and a capacity plan?
- look for a similar cargo in the cargo record book?
- use the Chief Officer's personal programme?

The location of technical manuals relating to the cargo system . . .

- **would depend upon who is responsible as per the Plan Maintenance System.**
- are always with the Chief Engineer.
- are in the common library on the bridge.
- would depend upon where the Captain wants to put them.

The port / cargo operations log will form the basis of . . .

- **the statement of facts.**
- compliance with SOLAS.
- crew overtime.
- ISPS implementation.

The principle purpose of a pre-cargo transfer meeting is to . . .

- **ensure that all involved personnel are aware of the proposed cargo operations, services needed and safety requirements.**
- arrange shore leave.
- discuss the Ship / Shore Safety Check List.
- inform the technical officers when cargo pumps will be required.

Under normal operations, a cargo tank should not be filled higher than . . .

- **the level at which the overfill alarm is set.**
- 95%
- as per appropriate SOLAS regulation.
- as per the Approved Trim and Stability Book.

Vessels with large width tanks are more susceptible to . . .

- **a loss of intact stability due to free surface effect.**
- an inability to maintain a minimum draught forward.
- developing an angle of loll.
- having problems with propeller immersion.

What alternative is available when an approved test programme is not available to check the loading computer?

- **A condition can be taken from the Approved Trim and Stability Book.**
- A previous cargo condition can be inputted and results compared.
- The lightship condition can be inputted to see how it looks.
- The trimming scale on a capacity plan can be looked at.

What competent body would normally approve a loading instrument / programme?

- **The Classification Society.**
- SOLAS.
- The P&I Club.
- The ship's Master.

What does code "R" indicate in the Ship / Shore Safety Check List?

- **Re-checking at intervals not to exceed an agreed time.**
- Checks to be repeated only upon the instructions of the Chief Officer.
- Regular checks not required.
- Restricted access.

What is a schematic drawing?

- **A drawing in the form of a diagram and not to scale.**
- An approximate drawing.
- An exact technical drawing.
- The general arrangement plan.

What is meant by the operating envelope when referring to loading arms?

- **It is the maximum amount the vessel can safely range up and down the berth and generally move without damaging the loading arms.**
- It is the maximum cargo pressure the loading arm can withstand.
- It is the range of cargo temperatures the loading arm can safely withstand.
- It is the vertical reach of the loading arm.

What is the name of the diagram that is superimposed on the cargo control console?

- **The mimic diagram.**
- The cargo lines.
- The pumping plan.
- The tank plan.

What material should manifold reducers be made of?

- **Steel.**
- Cast iron.
- GRP (glass reinforced plastic).
- None of these answers.

What medium is used to purge cargo tanks that have previously carried crude oil?

- **Inert gas.**
- A mixture of oxygen and water.
- Oxygen.
- Water.

What must be monitored before entering the pumproom during cargo operations?

- **The hydrocarbon gases and the oxygen content.**
- The amount of bilge water.
- The mechanical ventilation.
- The rescue harness.

What precaution must be taken when removing a blank flange?

- **Ensuring that the section between the blank flange and last valve does not contain oil under pressure.**
- Ensuring that the cock on the pressure gauge is closed.
- Ensuring that the correct spanner size is used to remove the bolts.
- Ensuring that the pressure gauge is fitted correctly.

When calculating the UKC, what is the "controlling depth"?

- **It's the least depth of water on the vessel's proposed track.**
- It's a figure required to calculate squat.
- It's the depth of water allowing for height of tide.
- It's the draught of the vessel when proceeding in a narrow channel.

When can cellular mobile phones be used to assist with cargo transfer communications?

- **Never.**
- When GMDSS equipment is switched off.
- When portable VHF/UHF's are not available.
- When the terminal agrees.

When discharging segregated ballast, the overboard discharge must be monitored . . .

- **for contaminated ballast.**
- Monitoring is not required.
- to help calculate the rate of discharge.
- to observe when de-ballasting has been completed.

When must the Chief Officer's standing orders be followed?

- **Throughout all cargo transfer, ballasting operations and tank cleaning.**
- Only during ship to ship transfer operations.
- Only in port.
- When the Captain is ashore.

When should P/V valves be checked for free movement?

- **Prior to the commencement of each cargo operation as required by the Ship / Shore Safety Check List.**
- As per the ISM code.
- As per the ISPS code.
- As per the Plan Maintenance System.

When starting a centrifugal cargo pump, the discharge valve should be initially . . .

- **Closed.**
- It does not matter.
- Open.
- Slightly open.

When would you exceed the maximum pressure ratings of the cargo system?

- **Never. All pressure ratings must be adhered to.**
- When circumstances dictated a departure from safe operating procedures.
- When you think it is safe, according to your own personal experience exceeding pressures on other occasions.
- Whenever instructed to do so by the head office.

When would you need to check the intact stability at intermediate stages of cargo transfer?

- **On each and every occasion.**
- If the Master directed you to do so.
- Seldom as it's not really required.
- When the vessel developed a list.

When would you use the Chief Officer's personally developed loading / stress monitoring computer programme?

- **Never.**
- All the time, if the results appear correct.
- Occasionally, if the approved one is non-operational.
- Once a month to check the results of the approved system.

Whenever an AIS (Automatic Identification System) has to be kept on during cargo operations, what precautions must be taken?

- **If the unit has facility, reduce the power to one watt, otherwise isolate the aerial, and give the AIS a dummy load.**
- It must be guided by SOLAS.
- It must never be kept running during cargo transfer.
- You must Inform the terminal and keep it running normally.

Where are temperature and pressure gauges normally calibrated?

- **In a competent shore-based establishment.**
- Always at the head office.
- In an Engineer's workshop.
- With the local ship handler.

Where on the cargo manifold are pressure gauges fitted?

- **Out board of the manifold valves.**
- Fitted to the offshore manifold.
- Inboard of the manifold valves.
- On the manifold flange.

Where would a plan indicating the location of cargo measurement instruments, sensors, pressure gauges, etc normally be found?

- **In the CCR (cargo control room).**
- At the manifold.
- In the Chief Engineer's office.
- In the reliquefaction plant.

Where would you expect to locate a mimic diagram of the cargo system?

- **In the cargo control room (CCR) operational panel.**
- Adjacent to the bunkering manifold.
- At the manifold.
- In the Chief Engineer's office.

Where would you find approved information regarding the vessel structural strength?

- **The Approved Trim and Stability Book.**
- The Certificate of Class.
- The deck logbook.
- The OCIMF Ship Inspection Report Programme.

Where would you find information relating to intact stability and stress limitations?

- **The Approved Trim and Stability Book.**
- MARPOL publications.
- The Safety Management System.
- The SOPEP Manual.

Where would you normally expect to find a full set of cargo operation instructions?

- **The cargo control room (CCR).**
- In the wheel-house.
- The Master's cabin.
- The SOLAS training manual in the crew's lounge.

Who has overall responsibility for the maintenance of the cargo system?

- **It will depend upon the owner's/manager's work instructions as detailed in the Safety Management System (SMS).**
- The Chief Engineer.
- The Chief Officer.
- The Electrical Officer.

Why are main radio transmitter aerials grounded during cargo operations?

- **To prevent a potential ignition source.**
- Because it is written in the Safety Management System (SMS).
- Because they are not needed in port.
- Because they are not required for communications in port

Why is it important that the Plan Maintenance System (PMS) have specific instructions relating to valves and cocks for the pressure gauges?

- **So as to secure the whole safety of the cargo transfer and to ensure that the cargo system is not over pressurised.**
- So that the person responsible can be informed.
- So that the terminal knows.
- To satisfy the ISM code.

Why is it important to keep the cargo manifold area clear of obstructions?

- **So as not to interfere with the automatic release of a cargo hardarm.**
- For easy access in case of a pollution incident.
- Simply to keep the space tidy and ship shape.
- So ship and shore staff can work easily.

Why is it important to pre-plan all aspects of cargo operations?

- **To ensure a smooth, pollution free and safe cargo transfer.**
- It's a Company requirement.
- To calculate the time required.
- To satisfy the ISM code.

Why is it important to regularly check offshore manifold gauges during a cargo transfer?

- **To check for manifold valve leakage.**
- To check for pressure in the cargo system.
- To comply with the MSDS (Material Safety Data Sheet).
- To comply with the terminal requirements.

Why must a cargo tank atmosphere be monitored during crude oil washing?

- **To ensure that the oxygen level remains at 8% or less by volume.**
- Overall, to ensure sludge control.
- To ensure that the cleaning gun cycle is completed.
- To ensure that the washing process is adequate.

Why should you monitor void spaces, cofferdams and empty ballast tanks during a loaded passage?

- **To confirm that such spaces remain gas free.**
- To check for water content.
- To confirm that welded seams are not cracked.
- To practise the use of portable instruments.

7017 – TOTS 1A-Cargo Information, Precautions . . .

Version 1.0 (4)

According to the IBC code where should in use protective equipment be stowed?

- **In special lockers in easy accessible places.**
- At the forecastle head.
- At the top of the pumproom.
- In the accommodation.

In which publication would you find the MFAG table number?

- **The IMDG Code Supplement.**
- ISGOTT.
- The Ship Captain's Medical Guide.
- USCG CFR 33.

The charter party includes information on STS transfer. What is meant by STS transfer?

- **Ship To Ship.**
- Safe Tanker System.
- Suitable Tank System.
- Suitable Tanker Stability.

The vessel is on time charter. The master is concerned about bunker ROB's. Who should he address his concerns to?

- **The time charterer who is responsible for supplying bunkers.**
- He should inform the brokers who will deal with it.
- He should inform the local agent at the next port.
- The vessel is on bareboat charter and sublet so he must discuss it with the bareboat charterer.

What clause relating to unsuitable cargoes is incorporated by the charter party to protect the vessel's owner?

- **Cargoes Excluded (Exclusion clause).**
- Limitation of Liability clause.
- The Jason clause.
- The Paramount clause.

What describes the lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture near the surface of the liquid?

- **The Flash Point.**
- The auto Ignition Point.
- The load temperature.
- The threshold Limit Value.

What is Odour Threshold?

- **The lowest concentration of vapour in the air that can be detected by smell.**
- It is the smell you get from spiked crude.
- The highest concentration of vapour in the air that can be detected by smell.
- The permissible odour when handling cargos with a high H2S content.

What is a charter party?

- **A charter party is the contract between the owner of a vessel and the charterer for the use of a vessel.**
- A group of ship brokers.
- A non-binding agreement between any two parties.
- Renting a vessel for an event.

What is deadfreight?

- **The sum payable to the owner when the charterer has failed to load a full and complete cargo.**
- The amount of cargo not shipped on the vessel.
- The amount of cargo which is described as FOB (Free on Board).
- When the charterer is bankrupt it's the freight claimed by the owner.

What is essential for an oxygen/air resuscitator to operate efficiently?

- **The casualty's airway must be clear.**
- The casualty must be unconscious.
- The casualty must not have facial hair.
- The casualty must weigh over 90 KG.

What is the "Cargo Retain Clause"?

- **It addresses the charterer's right to claim for cargo not discharged.**
- It addresses the owner's right to claim for cargo not discharged.
- It deals with cleanage after bulk discharge.
- It deals with the amount of cargo retained as samples.

What is the "pumping clause"?

- **It's the maximum time to discharge the entire cargo or a minimum average pressure at the discharge manifold.**
- It's the maximum number of cargo pumps that can be operated at one time.
- It's the maximum time the cargo pumps can be running.
- None of these answers are correct.

What is the important feature of the goggles on a tanker carrying Noxious Liquid Substances?

- **They must be tight fitting and splash proof.**
- Nothing in particular, chipping goggles are adequate
- They must be lightweight.
- They must have 2 headbands for extra safety.

What is the important feature of the lifeline and belt that form part of the safety equipment on a tanker carrying Noxious Liquid Substances?

- **The line and belt must be fireproof and resistant to the cargo being carried.**
- The equipment must be inspected at least annually.
- The length of the lifeline must be at least double the tankers moulded depth.
- They must always be orange in colour.

What is the inspection criteria for breathing apparatus on a tanker carrying Noxious Liquid Substances?

- **A monthly inspection by a responsible officer and annual inspection and testing at an approved facility.**
- A weekly inspection by the 3rd officer.
- It should be inspected by a class surveyor annually.
- The Chief Engineer inspects all the apparatus monthly.

What would you do if you looked down a cargo tank and saw a colleague apparently unconscious?

- **Raise the alarm and wait for assistance.**
- Don breathing apparatus and enter the tank.
- Immediately enter the tank to give first aid.
- Look at the permit to see who is down the tank.

Where would you find information on the cargo to be loaded relating to health hazards?

- **The MSDS (Material Safety data Sheets).**
- SOLAS consolidated edition.
- The Ship Captain's Medical Guide.
- The World Health organisation.

Why is the first 0 to 4 minutes critical for the casualty?

- **After this time permanent brain damage may occur even if the casualty survives.**
- It gives the rescuers some time to get organised.
- The casualty will always die if oxygen is given after this time frame.
- The casualty will always survive if oxygen is given within this time frame.

Why would a vessel be in possession of an NLS (Noxious Liquid Substances) certificate?

- **It's required for any vessel carrying a MARPOL Annex II cargo that does not have a Certificate of Fitness for the Carriage of Chemicals.**
- The certificate is not mandatory and would usually be kept by the owner.
- The certificate is required on a petroleum product tanker.
- The certificate is required when carrying liquid petroleum gas.

7020 – TOTS 1B-Engine Room Operations - manoeuvring

Version 1.0 (5)

As per SOLAS requirements, where should the quick close fuel valves' release cabinets be located?

- **Outside the engine room.**
- Inside the engine room.
- Near the engine control room.
- On the navigation bridge.

Does SOLAS legislation exist stating how many steering gear hydraulic pump units are directly connected to the emergency switchboard?

- **Yes, according to SOLAS legislation there should be a minimum of one pump.**
- According to SOLAS legislation all the pumps should be connected.
- No such requirement exists.
- SOLAS do not have the power to legislate on this requirement.

During manoeuvring what condition would you expect to consistently apply to the starting air system?

- **The starting air pressure should be kept at the maximum possible level at all times.**
- At least two air compressors should be running at all times.
- Only one air compressor should be run at all times and the second one kept on stand-by.
- The bridge should be informed of the status of the starting air capacity on a regular basis.

How many generators would you expect to have running during manoeuvring?

- **Sufficient to allow for continuous supply should one alternator trip.**
- All available generators.
- One running and one on standby.
- Sufficient for 100% normal seagoing load.

If one unit of the main engine is out of order due to a fracture in the liner, what action should be taken with respect to the scavenge air supply to the unit before manoeuvring?

- **The scavenge air should be isolated from the unit.**
- No action needs to be taken with respect to the scavenge air.
- The exhaust valve should be adjusted to remain open during the full cycle to prevent build up of air pressure.
- The scavenge air should be increased to improve cylinder cooling.

If one unit of the main engine is out of order, what action should be taken concerning the cylinder liner lubrication?

- **Increase the lubrication.**
- Decrease the lubrication.
- No action is required.
- Shut off the lubrication.

If, during manoeuvring, you find that a fire had broken out in the purifier room what action should you take after raising the alarm and advising the bridge of the situation?

- **Try to extinguish the fire using portable fire fighting equipment.**
- Call a vessel in the vicinity for help.
- Call the designated person ashore for help.
- Evacuate all non-required personnel from the engine room.

In the event of a main engine scavenge fire, which of the following options is the best course of action to take immediately?

- **Reduce main engine rpm to a minimum.**
- Speed up the engine in an effort to blow the fire out.
- Start the auxiliary blowers.
- Stop the engine and keep it in one position until the fire is out.

In what configuration should the starting air system bottles be set prior to arrival manoeuvring?

- **All air bottles should be fully charged with one online and one on stand-by.**
- All air bottles should be charged and fully open.
- All air compressors should be switched to MANUAL mode.
- All air-operated deck equipment should be isolated.

Is there a requirement to increase Main Engine Cylinder Lubrication during manoeuvring?

- **No, unless special circumstances apply.**
- No, as per MARPOL requirements.
- Yes, as manoeuvring is considered a special circumstance.
- Yes, as per the lubrication oil company requirements.

What action should be carried out FIRST if a fire has broken out in the engine room during manoeuvring?

- **Ensure the fire alarm is raised.**
- Call the bridge and ask for assistance.
- Call the Emergency Party.
- Stop the main engine by pushing the emergency stop button.

What action should be carried out if an electrical black-out occurs in the engine room during manoeuvring?

- **After the emergency diesel generator starts, run only minimum machinery for safe manoeuvring.**
- Give assistance to the Electrician and try to bring back the electrical power as soon as possible.
- Inform the bridge of the situation and try to restart the diesel generators.
- Wait for the emergency diesel generator to start and evacuate the engine room immediately.

What action should be carried out if the CO2 alarm is raised in the engine room?

- **All personnel should be evacuated from the engine room immediately.**
- All running equipment should be stopped first and then all personnel should leave the engine room.
- It should be re-checked to make sure it is not a false alarm.
- The Master should be informed immediately of the alarm.

What action should be carried out when one fuel pump becomes defective during manoeuvring?

- **The fuel pump should be hung up, the inlet valve shut and the bridge informed of the problem.**
- The main engine should be stopped immediately and the bridge informed of the problem.
- The main engine's revolution should be decreased to avoid overloading other cylinders.
- The main engine's revolution should be increased to recover the loss of power on one cylinder.

What action would you take if the main engine lubricating oil pressure dropped suddenly during manoeuvring ?

- **Request permission from the bridge to stop the main engine as soon as possible and warn them that it may stop automatically.**
- The main engine revolutions should be increased to compensate.
- The main engine should be kept running while waiting for the stand-by oil pump to start.
- The main engine's revolutions should be reduced.

What action would you take if, during manoeuvring operations the main engine oil mist detector alarm was activated, the engine automatically slows down and the alarm condition clears?

- **Advise the Master and prepare to stop the engine as soon as it is practical for the manoeuvring operations.**
- Advise the Master and continue with slow speed operation, until the completion of manoeuvring.
- Advise the Master and stop the engine immediately to inspect the crankcase.
- Advise the Master of the alarm activation and continue with the normal engine operations.

What activities related to arrival manoeuvring does the International Safety Management Code (ISM) require to be logged in the Engine Log Book?

- **All pre-arrival checks and tests carried out.**
- All function tests carried out on communications (Bridge / Engine Room).
- All function tests carried out on the main propulsion system.
- All tests carried out as per arrival check lists.

What affects the distribution of electrical load between electric generators in parallel?

- **Changing individual prime mover revolutions.**
- Adjusting individual exitation current.
- Adjusting individual generator voltage.
- Removal of consumer load demand.

What condition should stand-by equipment be in prior to arrival / departure manoeuvring?

- **All stand-by equipment should be operable / available and in standby condition.**
- All stand-by equipment should be running but not delivering its intended service.
- All stand-by equipment should be switched OFF.
- All stand-by equipment should be switched ON and running.

What does "Power Management System" mean?

- **It is the electric generators automatic load sharing and distribution system.**
- It is the automatic shut-down of all "non-essential consumers".
- It is the automatic shut-down of the cargo pumps.
- It is the automatic start of the steam boilers.

What is one of the first actions that should be carried out during manoeuvring if one or more of the main engine's turbochargers is defective?

- **Check that all auxiliary electric-powered blowers are switched on and operating correctly.**
- The main engine should be stopped and shore assistance should be called.
- The main engine's revolution should be increased for more power.
- The main engine's revolutions should be reduced.

What is the first action you should take if the main engine oil mist detector alarm sounds?

- **Slow down the main engine immediately and inform the bridge of the problem.**
- Call the Technical Superintendent for help immediately.
- Check if the alarm is genuine before initiating any action.
- Inform the bridge and re-check the main engine's bearing temperatures.

What is the main purpose of the "Arrival / Departure Checklist"?

- **To confirm that all systems necessary for manoeuvring are in working order.**
- To be used as a teaching aid to junior ranks.
- To ensure personnel changes on board are documented.
- To provide Company data required by the office.

What is the maximum allowed sulphur content in "Low Sulphur Fuel"?

- **1.5%**
- 1%
- 1.75%
- 2%

What is the normally accepted method for testing automatic start-up of stand-by equipment?

- **By manually stopping the in service unit.**
- By checking that the standby unit switch is correctly set.
- By doing a black-out simulation.
- By running both units simultaneously.

What is the purpose of a steering gear "Creep Test"?

- **To check the condition of the steering gear hydraulic system seals and valves.**
- To check if the steering gear autopilot is in good order.
- To check if the steering gear non-follow up control system is in good order.
- To check that the steering gear does not fall back from the hard-over position.

What is the usual acceptable time for the steering rudder to swing from "hard to port" to "hard to starboard"?

- **28 seconds.**
- 18 seconds.
- 38 seconds.
- 48 seconds.

What minimum starting air pressure is normally required to start a main engine?

- **12 bars.**
- 10 bars.
- 15 bars.
- 17 bars.

What practical preparations can be carried out, before manoeuvring, if the stern tube is found to be defective and oil leakage is observed (apart from dealing with environmental issues)?

- **The stern tube oil gravity tank can be lowered until oil leakage stops.**
- Leakage may be ignored if only a minor quantity of oil is present.
- Stop all operations and call for shore assistance to stop overboard leakage.
- The stern tube oil gravity tank can be raised until oil leakage stops.

What preparations should be carried out first before manoeuvring if the main engine's electronic governor is found to be defective?

- **Disconnect the governor from the fuel rack and regulate the rack manually.**
- Contact the maker and ask for assistance.
- Stop manoeuvring until the governor has been repaired.
- The governor should be blocked into the middle position and then manoeuvring can proceed.

What should be done if the remote control of the steering system from the bridge becomes defective during manoeuvring?

- **Change over to local control and engage one person for steering.**
- Call tugboats for assistance and proceed with manoeuvring.
- Stop all operations immediately.
- Stop manoeuvring until the remote control has been repaired.

When, during manoeuvring, would you expect to changeover from heavy oil to diesel oil on a vessel where the two fuels are utilised?

- **Prior to manoeuvring.**
- At a suitable time during manoeuvring.
- It is optional dependent on whether overhaul work will be carried out on the engine.
- Just before shutting down allowing time to flush through the fuel system.

Which of the following tests would NOT normally be carried out as part of a pre-arrival standby routine?

- **The testing of the fuel system quick-closing valves.**
- Testing communications between the bridge and engine room.
- The testing of main engine starting ahead and astern.
- The testing of the steering gear.

While on Main Engine stand-by during a river passage towards a port, you get a high exhaust temperature alarm on one main engine cylinder. What would you do?

- **Call the bridge and advise the Master of the problem.**
- Acknowledge the alarm and watch to see if the temperature increases.
- Acknowledge the alarm until you get into port.
- Stop the main engine.

Why is water not suitable for extinguishing main engine scavenge fires?

- **It will flash off and the fire may develop into a hydrogen fire.**
- It will cause the liner, piston rod and scraper box parts to rust.
- It will cool the space too quickly and cause the liner to crack.
- It will leak into the crankcase space and contaminate the lubricating oil.

Would you expect the Control Room to be manned during manoeuvring in an UMS classed vessel?

- **Yes.**
- No manning is required due to UMS conditions.
- No manning is required even if no UMS conditions apply.
- Yes, as required by the Master.

Would you expect the Engine Room itself to have personnel engaged in watchkeeping duties during manoeuvring?

- **Yes.**
- No, due to Company requirements.
- No, due to safety requirements.
- Yes, as required by the Master.

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An intermittent failure to start astern at the first try is . . .

- **not very significant if the engine starts on the second or subsequent attempts.**
- not very significant provided the engine starts astern on local control.
- significant and must be immediately rectified, and pilots and shore management notified.
- very significant and must be investigated at the next port.

How is it ensured that important operations are carried out according to set procedures?

- **By using the appropriate critical operations check list.**
- By following the Chief Engineer's standing orders.
- By using the Company operating manual.
- By using the manufacturer's operating / instruction manuals.

If ,during the sea passage, oil is observed leaking from the sterntube inboard seal header tank, what should you do?

- **Fit an oil tight cap to the header tank and monitor the inboard seal for leakage until repairs can be carried out.**
- Add thicker oil to the sterntube system in order to minimise the oil loss until repairs can be carried out.
- Collect the leaking oil and return it to the header tank to minimise the oil loss until repairs can be carried out.
- Reduce the header tank level to reduce the leakage by maintaining minimum positive pressure on the seals.

If a fuel pump is identified as defective on one unit and spares are unavailable, the main engine . . .

- **can continue to be run at reduced power with the fuel pump hung up and isolated.**
- can continue to be operated at lower load or reduced RPM once the pump has been removed from the engine.
- can continue to be run at full power until the next port of call.
- can continue to be run, but all other fuel pumps should be adjusted to balance exhaust temperatures.

If a significant limitation of engine power due to cooling problems is experienced before stand by . . .

- **you need to resolve the problem as emergency power may be needed unexpectedly.**
- you can wait until you have finished manoeuvring as manoeuvring speed is less than full sea speed.
- you may ignore the problem until you have time to carry out repairs.
- you should stop immediately as any such limitation could result in engine damage.

If the local engine side control is used in order to overcome problems with starting astern . . .

- **only the low lub oil trip and mechanical overspeed trip continue to work in the normal way.**
- all interlock and protection devices are over ridden.
- all interlocks and protection devices continue to work in the normal way.
- all interlocks and protection devices with the exception of those connected with reversing, operate in the normal way.

If, during the sea passage, oil is observed leaking from the sterntube outboard seals, what should you do?

- **Reduce the header tank level to maintain minimum positive pressure on the seals, and monitor the water level in the sterntube oil to ensure any ingress is controlled.**
- Add thicker oil to the sterntube system in order to minimise the oil loss until repairs can be carried out, and reduce the engine speed to prevent overheating.
- Raise the sterntube oil pressure to ensure there is no possibility of water entering the sterntube and causing damage to the sterntube bearing.
- Reduce the header tank level to below sea level to ensure that there is no possibility of pollution, and monitor the sterntube temperature for any overheating.

In case of any malfunction or breakdown of critical equipment during navigation what is the first action you should take?

- **Inform the bridge to alert them.**
- Inform the Chief Engineer and 2nd Engineer.
- Man the engine room until normal manned operation commences and then inform Engine Staff.
- Raise the general alarm.

In the event of a low lub oil pressure alarm during standby you should . . .

- **keep the engine running and try to solve the problem, but advise the bridge of the situation.**
- keep the engine running and try to solve the problem without troubling the bridge.
- stop the engine and engage the turning gear.
- stop the engine because you must prevent damage to the engine at all cost.

In the event of full astern emergency being requested on a slow speed engine on sea passage you should . . .

- **immediately bring the speed control to stop, wait for the engine rotation to slow, then put starting air to turn the engine in the astern direction.**
- immediately bring the speed control to stop and put starting air to turn the engine in the astern direction.
- immediately bring the speed control to stop, wait for ahead rotation to stop, then put starting air to turn the engine in the astern direction.
- stop the engines and open the air bottles in preparation for starting astern.

In the event of telegraph / bridge control being unexpectedly rung to "stop engines" on sea passage you should . . .

- **stop the engine and call the bridge to ask the reason why the engine has been stopped.**
- slow down the engine and phone the bridge to ask what is happening.
- stop the engine, call the bridge and inform them you have opened the indicator cocks and engaged the turning gear so the engine will require 10 minutes notice before restart.
- stop the engine, engage the turning gear and open the indicator cocks.

In the event of the crankcase explosion door opening under pressure on sea passage you should . . .

- **slow the engine and call the bridge to request agreement to stop.**
- keep the engine running and call the bridge to request agreement to stop the engine.
- slow the engine and increase lubricating oil pressure.
- stop the engine because you must prevent damage to the engine at all cost.

It is normal good practice to operate the engine in standby mode with . . .

- **one air bottle on line and the other topped up with its supply valve shut.**
- both air bottles on line and the main air compressor start and stop on automatic control.
- both air bottles on line and the main air compressor start and stop on manual control.
- both bottles on line with one compressor on auto and the other on manual start/stop.

It is normally expected that starting on local engine side control uses up . . .

- **less starting air than bridge control.**
- more starting air than bridge control but only on astern movements.
- more starting air than control room control.
- the same amount of starting air as bridge control.

On a Steam turbine propulsion, set astern manoeuvring power is . . .

- **less than 40% of ahead power.**
- 90% of ahead power.
- equal to ahead power.
- greater than ahead power.

On a slow speed direct drive engine, the safety device which prevents fuel and starting air being simultaneously applied to the engine is . . .

- **the fuel air interlock.**
- the camshaft wrong-position indicator.
- the turning gear interlock.
- the wrong way alarm and cut out.

On a slow speed main engine installation, where damage to a connecting rod bottom end bearing necessitates the removal of the bearing shells, the main engine may still be operated at reduced power . . .

- **avoiding any speed range that may induce severe torsional vibration in the crankshaft once the damaged unit has been blanked off and the connecting rod removed.**
- and speed, taking care not to exceed the maximum exhaust temperature on any unit once the damaged unit has been blanked off and connecting rod removed.
- once the fuel to the damaged unit is shut off and the connecting rod secured clear of the crankshaft.
- with the scavenge ports blanked off, the fuel and cooling water to the damaged unit shut off and the connecting rod secured clear of the crankshaft.

On a slow speed main engine installation, where the fuel pump on one unit has to be hung up because of damage to the fuel cam, how should the main engine be operated?

- **At reduced power and speed ensuring the engine is not run within barred speed ranges specified by the manufacturer.**
- At normal power provided the exhaust temperature of the other units does not exceed the maximum permitted temperature.
- At reduced power indicated in the engine operating manual, even though this may result in surging of the turbocharger.
- At the highest possible power once the other fuel pumps have been adjusted to balance the exhaust temperatures.

On a slow speed main engine, direct drive installation, a turbo-charger failure . . .

- **requires that the auxiliary blowers be used to operate the engine at reduced power.**
- does not affect full power operation provided the auxiliary blowers are fully operational.
- does not affect full power operation provided the engine can be kept running.
- puts the propulsion plant out of use as the engine cannot operate with natural aspiration.

On a slow speed main engine, direct drive installation, where a crankcase oil mist has been detected, the likely cause of the oil mist is due to . . .

- **a bearing overheating.**
- a blockage of the crankcase vent pipe.
- a crankcase door not being closed properly allowing air to enter the crankcase.
- running the engine with the lub oil pressure too high.

On a slow speed main engine, direct drive installation, where a crankcase oil mist has been detected, the main engine should be . . .

- **run at slow speed for 10 minutes, then stopped, the turning gear engaged and the engine turned over for 20 minutes before opening the crankcase for inspection.**
- immediately stopped to prevent any further oil mist, and the turning gear engaged while the crankcase space is inspected for hot spots.
- operated as normal with lub oil pressure increased to cool down hot spots, and the OMD to be monitored for 24 hours for alarms before returning to normal procedures.
- run at normal revs while the oil mist detector is checked.

On a slow speed main engine, direct drive installation, where both the Bridge Control system and the engine control room system are defective, the main machinery should be operated from the local (engine side) control . . .

- **by an engineer, and only after operating procedures have been implemented and communications with the bridge tested satisfactorily.**
- by a deck department crew member after establishing direct communication to the bridge for manoeuvring instructions.
- only by the Chief Engineer who will have communications directly to the Master on the bridge, for manoeuvring instructions.
- while the engine is being manoeuvred and unmanned operation can be resumed when the vessel is on passage.

On a slow speed main engine, direct drive installation, with more than 1 turbo charger, a turbine failure . . .

- **requires that the rotor be removed and blanking plates fitted.**
- does not affect full power operation provided the engine can be kept running.
- does not require that the rotor be removed and blanking plates fitted.
- means the engine cannot be run on reduced power.

On a slow speed main engine, the engine can be brought back into service after extinguishing a scavenge fire . . .

- **only after the cause of the fire is determined and the scavenge space cleaned out and inspected for damage.**
- as soon as the bridge confirms that the engine may be restarted.
- once the engine has been turned over on the turning gear to ensure that it turns freely.
- only after the engine unit(s) in way of the fire have been fully overhauled.

On a vessel where the bridge control of the steering gear has failed during the voyage, the vessel should not continue on voyage until the steering gear . . .

- **has been changed to local control with the operating position manned at all times, tested and communications with the bridge established.**
- creep test has been carried out.
- has been fully overhauled and the defect fully identified and rectified.
- is secured in amidships position, with tug assistance for the duration of the voyage to a port of refuge.

On a vessel where the bridge control of the steering gear is found to be defective, how should the steering gear be operated?

- **Directly from the steering gear space local control position, only after local communications with the bridge have been set in place and tested.**
- By the non-follow up steering control.
- From the engine room by a crew member with communications via the engine control room.
- From the steering gear compartment with communications from the bridge via the engine control room and OOW.

On a vessel with 2 generators, while preparation is being made to depart on a 3 day voyage, one generator breaks down but cannot be repaired due to lack of a spare part. What should you do?

- **Cancel the departure and do not sail until the generator is back in service or a backup package generator is fitted.**
- Advise the office of the problem and agree to sail as the spare part will be arranged for the next port.
- Sail as planned and have the generator repaired in the next port as the voyage is very short.
- Sail as planned as the owner's representative is insisting that the vessel must not be delayed.

On restoration of electrical power following a short duration electrical power failure in open water, the main engine may be re-started . . .

- **only after checking that all the ancillary equipment has restarted and that all pressures and temperatures are normal.**
- directly on power being restored.
- only after another generator has been put on line to ensure there is no further failure of the electrical power supply.
- only after the turning gear has been engaged and the engine turned over.

On restoration of electrical power following an electrical power failure during a confined water transit, the main engine must be re-started . . .

- **only after checking with the bridge that it is OK to do so.**
- as soon as the engineer confirms that the steering gear is running and the engine is ready to start.
- as soon as the engineer is satisfied that the power supply is stable and that the engine is ready to start.
- directly power is restored in order to minimise any delay to the vessel.

Stand by coolant supply pumps should auto start in the event of . . .

- **pump failure alarm and low pressure alarm.**
- high coolant temperature alarm.
- low pressure alarm only.
- pump failure alarm only.

The safeguard against starting a slow speed direct drive engine with the fuel cam in the incorrect position is . . .

- **a wrong way alarm and interlock.**
- a fuel cut off interlock only.
- a wrong way alarm only.
- not fitted as it will automatically correct itself.

Unavailability of stand-by coolant supply pumps is . . .

- **significant and must be rectified as soon as possible.**
- not very significant and UMS operation of the engine room may be continued.
- not very significant as one pump is enough for cooling.
- significant and must be rectified before the engine is started.

What is the number of consecutive ahead and astern starts a slow speed direct drive engine should be capable of?

- **More than 8 on each air bottle.**
- At least 5 on each air bottle.
- Less than 8 but more than 5.
- Less than 8.

7022 – TOTS 1B-Watch keeping in the Engine Room

Version 1.0 (4)

Crankcase Lub oil consumption of a slow speed main engine should be monitored to confirm that it is in line with the makers specific consumption . . .

- **to verify that oil is not being lost due to leakage or worn scraper boxes.**
- so that the need for regular topping up of the crankcase is not required.
- to ensure that leakage does not contaminate the bilge spaces.
- to ensure that there is no air entrainment in the system oil due to the crankcase level running low.

During cargo operations, on a vessel with steam turbine driven cargo pumps, the condenser salinity alarm sounds intermittently but resets automatically. What action should you take?

- **Test the chloride level in the boiler to check if it is rising.**
- Check the sea water pressure to the condenser to check that it is not too high.
- Ignore the alarm as it is obviously an intermittent fault.
- Slow down the turbine to reduce the load on the condenser.

During night-time under-unmanned operations, which procedure for engine entry should be followed?

- **The bridge must be informed of the entry.**
- Entry is allowed only if two people enter together.
- Entry is allowed only with the Chief Engineer's permission.
- Entry is prohibited.

Duty engineers are NOT allowed to create their own personal standing orders for watchkeeping ratings.

- **True.**
- False.

How are new crew members instructed in their watch keeping duties?

- **By means of the Engine Room familiarization check list.**
- During the first week of working alongside the engine staff, their duties will become apparent.
- Via the Chief Engineer's standing orders.
- Via the hand over process of the departing crew members.

How are procedural requirements to the watch keepers of the Engine Room conveyed?

- **By means of the Chief Engineer's Standing Instructions.**
- By walking around the E/R indicating areas of concern / maintenance.
- Verbally, while handing over the watch.
- Via the Engine log book and night watch book.

How are the conditions of the cooling water, boiler water and lube oil maintained?

- **By regular testing and treatment.**
- By maintaining the correct liquid levels.
- By shore side service technicians on a quarterly basis.
- By top up of the water level and purification of lub oils.

How do the watch keepers ensure various equipment is maintained regularly?

- **A work list is produced from the planned maintenance system & the running hours sheet is followed.**
- Equipment to be maintained will be identified on a daily basis by the Duty officer.
- The Classification society issues a listing for each vessel of machinery to be overhauled / maintained, and time frames for doing such work.
- Through morning meetings and delegation of the 2nd engineer.

How do the watch keepers ensure various equipment is working correctly?

- **By regular performance monitoring of each piece of equipment.**
- By completing planned maintenance in accordance to maintenance schedules.
- If equipment is defective, the alarm panel will indicate that accordingly.
- Via daily log book readings (pressure / temp / etc).

How do you ensure the various equipment is maintained regularly?

- **By following a work list produced from the planned maintenance system.**
- The Classification Society issues a listing of machinery to be overhauled/maintained, and time frames for doing such work.
- The equipment to be maintained will be identified on a daily basis by the Duty officer.
- Through morning meetings and delegation of the 2nd engineer.

How do you ensure the various indicating instruments are reading correctly?

- **By calibrating the equipment for their accuracy.**
- By double checking, via portable equipment.
- If they are not reading correctly, an alarm will sound and the machinery will not operate.
- They are changed during planned maintenance.

How does the Chief Engineer convey any specific overnight instructions to the engine room watchkeepers.

- **By means of the Chief Engineer's Night Order Book.**
- The officer of the watch will pass on any instructions to the Duty officer prior to the handover of watch duties.
- Verbally, prior to the engine room going unmanned.
- Via the Bridge, who will inform the Engine room at 8pm.

How frequently should the various auto control devices be tested for correct operation?

- **At regular intervals as laid down in the vessel's planned maintenance system.**
- All auto control devices are checked during dry docking, and certification is issued.
- Auto control devices operate automatically, and therefore do not require intervention.
- When stand-by equipment is changed over, the auto control devices indicate any errors, via an alarm.

How is it ensured that important operations are carried out according to set procedures?

- **By using the appropriate critical operations check list.**
- By following the Chief Engineer's standing orders.
- By using the Company operating manual.
- By using the manufacturer's operating / instruction manual.

How many times per day is a regular equipment check to be carried out when the vessel is underway?

- **Once per watch.**
- Each hour.
- Twice a day.
- When ordered by the Chief Engineer.

If a Rating is part of an engine watch, who is he directly responsible to?

- **To the Duty Engineer.**
- To the Chief Engineer.
- To the Master.
- To the Second Engineer.

If an alarm sounds and you are on duty during a UMS period what are the first actions you should take?

- **Cancel the alarm and investigate the cause.**
- Advise the Chief Engineer that it is necessary to return to full watchkeeping routine.
- Call the Bridge to inform them of an alarm state.
- Cancel the alarm and reduce the speed of the Main Engine.

If the vessel has unmanned machinery space, what precautions should you take before entering, when on UMS mode?

- **Inform the bridge watch keeper**
- Inform the 2nd Engineer.
- Inform the Chief Engineer.
- Inform the master.

In normal circumstances, which parameters on steam boilers are the most important?

- **The water level and steam pressure.**
- The water level and the steam temperature.
- The water pressure and the steam temperature.
- The water temperature and the steam pressure.

In unmanned machinery space, before the Engine room is put to unmanned, how will you ensure watch keepers check all the important items?

- **The UMS check list.**
- By reading the chief engineer's standing orders.
- By signing the engine room log book.
- Phoning the bridge before leaving the Engine room.

Information about engine problems for watch hand-over is required in written form by whom?

- **No one. Such a requirement does not exist.**
- By MARPOL.
- By SOLAS.
- By the ISM (International Safety Management) Code.

What is a dead man alarm?

- **A switch which should be operated at regular intervals which, if not activated, will give an alarm on the bridge.**
- A direct link alarm to the office in the event of an emergency.
- An Alarm that is activated when an injured crew member is discovered.
- Terminology for an alarm which is malfunctioning or out of order.

What is done to ensure watchkeepers check all important items before switching the engine room to unattended condition.

- **The watchkeeper should complete a UMS check list.**
- The watchkeeper should phone the bridge to confirm all is set before leaving the Engine room.
- The watchkeeper should read the Chief Engineer's Standing Instructions.
- The watchkeeper should sign the engine room log book.

What is the basic rule for good watchkeeping?

- **Regular visual checks.**
- To have a good engine rating as a watchkeeper.
- To have good lighting in the engine room.
- To keep the engine room clean at all times.

What is the diesel oil service tank sludge cock used for?

- **To drain any water out of the tank.**
- For draining the tank for cleaning.
- To draw off a sample for analysis.
- To draw off clean oil for cleaning equipment.

What is the purpose of the Engineers Call system and when should this be activated?

- **In case of any emergency situation requiring the support of all Engine room staff.**
- During Fuel Oil and Lub Oil Bunker operations when extra assistance is required.
- The Engineers call is used in the mornings to ensure all engine staff are present in the E/R.
- When the duty officer is changing rotation, all engine officers are to be present for handover.

What is the usual required preheating temperature for 380 cSt (centistokes) of heavy fuel (HFO)?

- **130-140˚C.**
- 100-110˚C.
- 110-120˚C.
- 140-150˚C.

What should be done by the watchkeeping engineer if the relieving engineer is not ready for watch hand-over?

- **He should continue with the watch and inform the Chief Engineer.**
- He should continue with the watch.
- He should leave the engine room and inform the Chief Engineer.
- He should leave the engine room and inform the Master.

What should be done if any problem to the propulsion power or steering gear occurs during the watch?

- **Immediately inform the Officer of the Watch on the bridge.**
- Call the Chief Engineer immediately.
- Call the Master immediately.
- Immediately call for help.

What should be done regarding the engine watch in restricted visibility?

- **The engine watch should be kept as per manoeuvring condition.**
- Close contact should be kept with the bridge.
- Restricted visibility has no influence on the engine watch.
- The Chief Engineer should be called for the engine watch.

What temperature should HFO be maintained at when passing through the purifier?

- **At a temperature calculated according to the density of the fuel being purified.**
- It should be maintained as close to 99 degree C as possible.
- It should be maintained at 101 to 103 degree C.
- It should be maintained at 90 to 95 degree C.

What temperature should lubricating oil be maintained at when passing through the purifier?

- **It should be maintained at 90 to 93 degree C.**
- At a temperature calculated according to the viscosity of the oil being purified.
- It should be maintained at 60 to 65 degree C.
- It should be maintained at 75 to 80 degree C.

What will happen if the "dead man alarm" is not reset on time?

- **The general alarm will sound.**
- The Chief Engineer will be notified of the alarm.
- The engineer call alarm will sound.
- The Master will be notified of the alarm.

What would rapid fluctuation of the main engine cooling water pressure indicate?

- **The possibility of a cracked cylinder liner or cylinder cover.**
- A problem with the cooling water circulating pump impeller.
- Problems with the cooling water temperature control valves.
- Speed fluctuation of the cooling water pump due to a motor problem.

When is the Engine Order Book filled in?

- **Whenever the Chief Engineer has additional orders.**
- From time to time.
- Only in port.
- When ordered by the Master.

When the engine watch is changing over from manned to unmanned (UMS) mode, which of the following equipment must be switched on?

- **The Engineer call alarm.**
- Another diesel generator.
- The second steering gear pump.
- The stand -by cooling pump.

When the engine watch is changing over from manned to unmanned (UMS) mode, who should be informed?

- **The bridge.**
- All crew.
- The Chief Engineer.
- The Master.

Which of the following statements is correct?

- **The watchkeeper must call the Bridge every 20 minutes to advise them if no "dead man" alarm is fitted?**
- Ear defenders must not be used in the ER as it is essential to hear all noises.
- It is expected that the maintenance of equipment will be carried out when on UMS.
- When an alarm sounds, no watchkeeper must enter the engine room under UMS until they have advised the Bridge.

Who is the Duty Engineer directly responsible to?

- **To the Chief Engineer.**
- To the Company.
- To the Master.
- To the Superintendent.

Who should be informed FIRST about unusual fuel consumption?

- **The Chief Engineer.**
- The Charterers.
- The owner.
- The Technical Superintendent.

With a vessel on extended tailshaft survey who (excluding the Chief Engineer) is required to see the tailshaft lubricating oil analysis results?

- **The Class.**
- The Flag State Inspection.
- The Port State Control.
- The Vetting Inspection.

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Version 1.0 (15)

A heat exchanger where two liquids flow in opposite directions is called . . .

- **counter-flow.**
- horizontal flow.
- opposite flow.
- parallel flow.

A high vacuum reading on the suction side of a gear pump . . .

- **is indicative of partially opened suction valves or blocked strainers.**
- is indicative of damaged gears.
- means that the discharge valve is shut.
- shows normal condition.

A lub oil purifier fault alarm, if not attended to immediately . . .

- **may result in good lub oil discharging to the sludge tank.**
- may result in a breakdown.
- may result in an overload condition.
- may result in contamination of the oil.

A ship's air conditioning system is normally designed to give . . .

- **six to eight changes of air per hour.**
- ten to twelve changes of air per hour.
- twelve to sixteen changes of air per hour.
- two to three changes of air per hour.

An electrical driven ballast pump will not start because one of the main fuses has blown. Which of the following would you check, prior to trying to restart the pump motor?

- **The motor and pump rotation.**
- The control system fuses.
- The electrical power supply.
- The insulation of the motor windings.

An overcharge or excess refrigerant in the system will be seen as . . .

- **high condenser pressure.**
- frost formation.
- high cooling effect.
- high lube oil pressure.

Any Oily Water Separation operation has to be recorded and noted in the Oily Record Book.

- **True**
- False

As the temperature of air is reduced, its capacity to carry water vapour . . .

- **is reduced.**
- is marginally increased.
- is rapidly increased.
- remains the same.

Cavitation is the result of . . .

- **the formation and collapse of air bubbles.**
- a too high discharge pressure on the pump.
- impurities in the water impacting on the impeller.
- overloading of the impeller.

During discharge, problems with the inert gas plant result in the production of clean inert gas being reduced by 50%. What do you do?

- **Advise the Chief Officer so that he can reduce the discharge rate in line with the reduction of the gas supply.**
- Adjust the boiler controls to increase the fuel and reduce the air in an attempt to reduce the oxygen content of the gas.
- Allow the discharge to continue as the oxygen content in the tanks may not rise significantly during the remaining discharge period.
- Shut down the discharge until the inert gas system problem is resolved.

During normal operation the main engine speed reduces and the engine stops. The engine can only be restarted at local emergency control station. The fuel lever is difficult to move. What do you suspect is the problem?

- **The pneumatic stop solenoid is defective.**
- A fuel pump has seized.
- There is a defective air start distributor.
- There is low control air pressure.

For any maintenance work, the sewage system should be considered as an enclosed space.

- **True**
- False

For the most efficient and practical separation of oil from water, what is required?

- **As long as possible dwell time in the separation tank.**
- A three stage separation tank.
- A two stage separation tank.
- A very high temperature in the separation tank.

How is the rudder travel stopped in the event of total telemotor failure?

- **It reaches the physical stop-limits of the steering gear.**
- No other means are provided.
- The emergency diesel alternator will start and supply power.
- The rudder cannot be stopped.

If the auxiliary boiler feed water hotwell was found to be heavily contaminated with oil, what initial action would you take?

- **Overflow the hotwell until the water is clear then top scum the boiler.**
- Drain the hotwell and blow down the boiler, then refill with treated fresh water.
- Drain the hotwell and refill with fresh water to the correct level.
- Overflow the hotwell until clear water is seen in the sightglass.

If the safety valves fitted on an air reservoir can be isolated from the reservoir, then . . .

- **the reservoir must be fitted with a fusible plug.**
- it can be used legally without any added safety measures.
- the compressor must be fitted with a fusible plug.
- the safety valve must be fitted with a fusible plug.

In a SW cooling water system where the piping is made of yorcalbro the sacrificial anodes should be made of . . .

- **soft iron.**
- aluminium.
- cast iron.
- zinc.

In a boiler, the system is set so that when the boiler is switched on, the following sequence of operation takes place . . .

- **(1) heat up and circulate fuel; (2) purge the combustion chamber; (3) turn on the fuel and ignite the flame.**
- (1) heat and circulate the fuel(2) turn on the fuel and ignite the flame (3) purge the combustion chamber.
- (1) purge the combustion chamber; (2) heat and circulate the fuel; (3) turn on the fuel and ignite the flame.
- (1) purge the combustion chamber; (2) turn on the fuel and ignite the flame (3) heat and circulate the fuel.

In a cargo system fitted with a vertical submerged pump, the pumped fluid contained in the vertical column will fall down into the tank when the pump is stopped after discharge. The normal remedy for this is by . . .

- **using the purge connection fitted to clear the discharge pipe.**
- stripping.
- the use of manually operated valves.
- the use of non-return valves.

In a typical tubular heat exchanger, the expansion of the tubes are accommodated by . . .

- **a floating header.**
- choosing the right material.
- leaving one end of the cooler free to move axially.
- using pre-expanded tubes.

In order to test a weld repair to a lubricating oil storage tank the tank is to be tested with compressed air to check for leakage. What is the maximum pressure that should be used?

- **0.2 Bar**
- 0.02 Bar
- 1.0 Bar
- 2.5 Bar

In ships other than those engaged in short voyages, the steering gear should be thoroughly checked and tested . . .

- **within 12 hours before departure.**
- on standby.
- when one hour's notice is given.
- within 24 hours before departure.

In the event that the sludge generated on board exceeds the incinerator capacity the excess sludge should be . . .

- **transferred to a shore facility.**
- discharged only when vessel is 12 nautical miles or more from land.
- discharged via OWS before entering port.
- transferred to fuel oil bunker tank.

It is recommended that the Oil Content Monitor of an Oily Water Separator is calibrated at least . . .

- **annually.**
- once every five years.
- once every ten years.
- once weekly.

Moisture in a refrigeration system could lead to . . .

- **ice formation and a drop in pressure on the evaporator side.**
- excessive discharge pressure.
- the dryer getting clogged.
- the loss of refrigerant.

On a slow speed main engine diesel the main engine turbocharger revolutions per minute are higher than normal at a fixed load, with reduced engine performance. What is the cause of this speed increase?

- **Reduction in the gas passage of the nozzle ring.**
- Burnt out exhaust valve seats or valves not closing.
- Poor fuel injection causing after burning in the cylinders.
- The turbine blades require cleaning.

On a tanker with 3 centrifugal cargo pumps, but with discharge reduced to 2 pumps due to shore restrictions you note that the idle pump is running backwards. What would you suspect as being the main problem?

- **The idle cargo pump non-return valve is not closing properly.**
- The non-return valve on one of the running pumps is defective and causing it to drive the idle pump.
- The Pumpman has not closed the idle pump suction and discharge valves.
- The Pumpman has set the valve incorrectly allowing the running pumps to drive the idle pump.

On a tanker, when on voyage with a heated cargo, the hotwell is found to be overheating due to steam returning from the cargo heating coils. What should you do?

- **Check the steam traps drains to see if the traps are functioning correctly.**
- Check that the Cargo Officer has not opened up the heating coil valves too far.
- Check the cargo heating log to see if the cargo tank temperatures are too high.
- Check the heating coil steam trap drain filters to see if they are clean.

On a vessel with 2 auxiliary boilers where only one boiler is required on a voyage of approximately one month, is any action required to preserve the internals of the out of use boiler?

- **The boiler should be put into wet lay-up - i.e. filled completely with water dosed with a raised level of oxygen scavenging chemical such as hydrazine.**
- No action is required as the voyage time is sufficiently short.
- The boiler should be fired periodically to keep it warm but not enough to raise steam.
- The boiler should be put into dry lay-up - i.e. drained and opened up with ventilation provided to keep the internals dry.

On completion of a generator engine overhaul you find that the bottom end bolts have been over-tightened beyond their elastic limit. What should you do?

- **Refuse to run the engine until you obtain and fit new bolts and have them tightened to the correct torque.**
- Leave the bolts as they are and run the engine, as checking of the bottom end shows that it is free and the bearing is not binding on the pin.
- Run the engine, as you know that the bolts are tight and will not work loose.
- Slacken back the bolts and have them re-tightened to the correct torque before you run the engine.

Plate type heat exchangers can effectively deal with . . .

- **low pressure fluids.**
- high pressure fluids.
- large volumes.
- vapour and gases.

Sewage discharge regulation can be found in the . . .

- **MARPOL regulations.**
- Health & Safety regulations.
- ISGOTT regulations.
- SOLAS regulations.

The cause of an abnormally high condenser pressure within a refrigeration system would be . . .

- **incondensable gases in the refrigerant.**
- a defective thermal expansion valve.
- a low cooling water flow through the condenser.
- an above normal level of refrigerant in the condenser.

The closed flashpoints of fuel for general use should not be less than . . .

- **60 degrees Celsius.**
- 65 degrees Celsius.
- 70 degrees Celsius.
- 75 degrees Celsius.

The control of temperature in heat exchangers when the cooling medium is seawater is normally achieved by . . .

- **by-passing a proportion or all of the hot fluid flow.**
- by-passing or limiting (controlling) the seawater flow.
- recirculating seawater discharge back to suction.
- throttling the flow of hot liquid.

The easing gear fitted to boiler safety valves is there to . . .

- **allow the valves to be opened to release pressure in the event of a boiler problem.**
- allow testing of the valve travel during safety checks.
- allow testing to make sure that the valves are not stuck shut.
- allow the valve to be eased off the seat slightly when adjusting the relief pressure.

The exhaust temperature deviation between units should be maintained within . . .

- **10%**
- 0%
- 40%
- 60%

The incinerator loading door is interlocked . . .

- **to stop the fan and the burner when opened.**
- to activate a shutdown on high temperature.
- to prevent opening when the inside temperatures are very high.
- to prevent operation in port.

The inert gas blowers should automatically shut down in the event of . . .

- **high temperature.**
- a high motor load.
- deck seal failure.
- high pressure.

The maximum permissible oxygen content of the inert gas going into the cargo tank should be . . .

- **5% by volume.**
- 6% by volume.
- 8% by volume.
- 9% by volume.

The oil content of bilge water is normally reduced from 100 to 15 ppm and below by the use of . . .

- **coalescent filters.**
- centrifugal separation.
- chemical treatment.
- steam heating.

The water side of an intercooler is protected against overpressure by . . .

- **bursting the disc on the cylinder jacket.**
- pressure testing it with the required pressure.
- the fusible plug on the cylinder jacket.
- the relief valve on the outlet side.

There is a relief valve in a refrigerant compressor lubricating oil system which is set to about . . .

- **2.5 bar above the crankcase pressure.**
- 10% above the crankcase pressure.
- 110% of normal working pressure.
- 3.5 bar above the crankcase pressure.

To prevent damage of the compressor, the refrigerant return line . . .

- **should not contain any liquid.**
- should be a mixture of gas and liquid.
- should be condensed to a liquid state.
- should not go back to the compressor.

What do you believe is most important to ensure trouble free function of pneumatic controls?

- **The supply of clean dry air.**
- Checking operation of the solenoid valves.
- Maintaining the correct air pressure.
- Regular checking / testing of the controls.

What do you predict would be the cause, if all the auxiliary diesel engines / generators, when running in parallel operation, started shedding load and hunting?

- **It could be caused by a reduction in the flow of fuel through the main filters.**
- It could be caused by a defective droop setting on one of the automatic voltage regulators.
- It could be caused by a faulty governor speed adjustment motor on one auxiliary engine.
- It could be caused by a loss of oil from the governor control circuit.

What is considered to be the main value of using ammonia as a refrigerant gas.

- **It doesn't deplete the ozone layer.**
- It is non-corrosive and non-irritant.
- It is non-toxic and non-flammable.
- It is readily available world wide.

What is meant by hunting in a control system?

- **The system being controlled does not settle on the set point.**
- The system being controlled does not reach the set point and remains below the set point.
- The system being controlled overshoots the set point and remains above the set point.
- The system being controlled reaches the set point but then falls back and remains offset.

What is the cause if, on a two stage main air compressor the discharge pressure is low and the 1st stage discharge temperature is higher than normal?

- **The 2nd stage suction valve is defective.**
- The 1st stage discharge valve is defective.
- The 2nd stage discharge valve is defective.
- The 2nd stage piston rings are damaged.

What is the discharge regulation if the sewage is not comminuted or disinfected?

- **The ship should be on voyage and more than 12 nautical miles from the nearest land.**
- The ship should be more than 12 nautical miles from the nearest land.
- The ship should be on voyage and four nautical miles from the nearest land.
- The ship should not be in special areas and should be proceeding at not less than three knots.

What is the first action you should take when there is no water in the boiler gauge glasses?

- **Shut the fuel off from the boiler.**
- Blow down the gauge glass locally and check the water level.
- Increase the feed water supply so that the boiler doesn't run dry.
- Reduce the load on the boiler.

What is the most common cause of the lowering in performance of heat exchangers?

- **Fouling of the heat transfer surface.**
- An air lock.
- Low water flow.
- The division plate joint failure.

What would cause a medium speed main engine exhaust gas temperature to be higher and the scavenge air pressure to be lower than normal, at a fixed load?

- **A partly blocked air cooler.**
- Incorrect fuel pump timing.
- Incorrect valve timing.
- Poor fuel injection.

What would cause the reduction in viscosity and flash point in the lubricating oil of an auxiliary engine operated on marine gas oil?

- **Fuel leakage into the crankcase.**
- A build up of combustion products in the lubricating oil.
- The lubricating oil is at an operating temperature that is too high.
- The wrong grade of lubricant being added to the crankcase.

What would you expect the main source of inert gas to be in a VLCC?

- **The main and auxiliary boiler.**
- A gas turbine plant with an after burner.
- An independent inert gas generator.
- The main engine exhaust.

What would you expect to cause smoky exhaust?

- **Sticky fuel injector.**
- A sticking air valve.
- A sticking fuel pump.
- Unsuitable cylinder lubricating oil.

When carrying out a generator engine overhaul, why is it important to check the bottom end bearings for ovality?

- **Because if the ovality of the bottom end bore is over the allowable tolerance, the new bearing shell may bind on the bottom end pin and cause overheating.**
- Because the ovality of the rod bore may allow the bearing to knock and cause early failure.
- Because the ovality of the rod bore may cause misalignment of the but serrations and overloading of the bottom end bolts.
- Because the ovality of the rod bore will allow increased clearance of the bearing and loss of oil pressure.

When charging the refrigeration system with more refrigerant gas . . .

- **the main liquid valve should be closed and the gas introduced before the regulating valve.**
- always inform the Chief Engineer.
- it is essential to weigh the refrigerant cylinder before and after the operation.
- you may choose whether to introduce the gas in the suction or discharge side.

When cooling a piston by system oil rather than water, the return temperature of the oil must be lower than it would be if using water because . . .

- **it avoids carbon deposits on metal surfaces.**
- it reduces the danger of fire.
- it reduces the danger of leaks.
- the specific heat capacity of water is less than oil.

When do you expect to have to change from low to high SW suction?

- **When the ship is entering an area where there will be restricted under keel clearance.**
- When the ship is entering a canal.
- When the ship is in heavy weather.
- When the ship loads a full cargo and is at maximum draught.

When the bunker fuel needs more purification, you should normally . . .

- **operate two purifiers in parallel.**
- change the gravity disc.
- increase the purifier throughput.
- operate two purifiers in series.

When using a cargo stripping eductor . . .

- **always open the cargo line after creating enough vacuum.**
- always open the cargo line first, and then start to build up a vacuum using seawater.
- it is always recommended to use remote operated valves.
- you do not have to wait for the vacuum to build up.

Where the main steering gear comprises two or more identical power units, then . . .

- **there must be two identical control systems, both controlled from the bridge.**
- both should be powered from the emergency supply.
- they should be interlocked to prevent simultaneous operation.
- two steering wheels are required.

Which of the following is most likely to make the engine run unevenly?

- **Sticky fuel pump control racks or linkage.**
- The fuel system not being primed.
- Unsuitable fuel.
- Water in the fuel.

Which of the following proportions make up good quality inert gas, from an inert gas generator plant?

- **nitrogen 77% by volume, carbon dioxide 13%, oxygen 4%, no water content.**
- nitrogen 75% by volume, carbon dioxide 13%, oxygen 12%, some water content.
- nitrogen 80% by volume, carbon dioxide 13%, oxygen 6%, some water and sulphur content.
- nitrogen 90% by volume, carbon dioxide 1.8%, oxygen 8%, no water content.

Which of the following refrigerants do not contain CFC?

- **R134a.**
- R11.
- R12.
- R22.

Which of the following safety components would you expect to find in the air starting system of a main engine?

- **All three devices.**
- Flame traps.
- Non-return automatic valves.
- Relief valves.

Why is it critical that all cargo instrumentation is maintained in good order at all times?

- **For safe operations, safety of personnel, the vessel and the environment as applicable to each type of vessel.**
- Because all items are covered under the vessel's planned maintenance and should be completed in accordance to the overhaul schedules.
- To ensure that the correct quantities of cargo are being loaded / discharged.
- To prevent defects being raised by Port State Control or Vetting inspectors.

Why is it necessary to preheat the engine before attempting to start it?

- **To reduce thermal stresses.**
- To ensure that it starts first time.
- To reduce start failure.
- To reduce the temperature gradient.

You would expect a deck seal to be . . .

- **of wet or dry types.**
- gland packings.
- mechanical type seals.
- normal non-return valves.

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A pump motor is found to be running at abnormally high temperatures. Which of the following would be the most likely cause of this condition?

- **A fouled or damaged impeller.**
- A loose terminal connection.
- Low winding insulation.
- The thermal overload is set too low.

A three phase DOL (Direct on line) motor is rotating in the wrong direction. How can this be corrected?

- **Interchange two of the three phases at the motor OR starter.**
- Fit a reversing contactor.
- Install a larger cable.
- Stop and restart the motor.

An insulation resistance (megger) test on a motor located on the open deck of a vessel provides an apparent low reading (1MΩ). Which course of action should be taken?

- **Check the maintenance history to identify any deterioration in insulation properties since previous tests.**
- Check the running current of the motor under full load conditions.
- Immediately remove the motor from service.
- Replace the motor with a spare.

Despite its vessel safety function, which of the following shutdowns must be fitted on emergency generator engines?

- **Over speed.**
- High cooling water temperature.
- High oil mist concentration.
- Low lub oil pressure.

For Direct on Line (DOL) Starting, what would you expect the starting current to be?

- **6 - 10 x normal current.**
- 0.5 - 1 x normal current.
- 10 - 20 x normal current.
- 2 - 3 x normal current.

For Star-Delta Starting, what would you expect the starting current to be?

- **2 - 3 x normal current.**
- 0.5 - 1 x normal current.
- 10 - 20 x normal current.
- 6 - 10 x normal current.

Governor droop is required for the prime mover of a synchronous generator in order to . . .

- **ensure stable kW load sharing.**
- ensure stable kVAR load sharing.
- maintain constant frequency.
- maintain constant voltage.

How is it possible to ensure that various alarms and cut outs are working?

- **By a regular simulation test to confirm their function.**
- The Alarm Panel would indicate an error if not operating correctly.
- They are checked during the daily rounds of the Duty Officer.
- They are tested and certified during dry docking.

How many methods of starting are required for an emergency generator?

- **Two methods.**
- Four methods.
- One method.
- Three methods.

How often would you expect the operation of standby equipment to be checked? How would you do this?

- **Weekly, by switching over to alternate standby machinery.**
- Daily, by switching over to alternate standby machinery.
- Monthly, by switching over to alternate standby machinery.
- Quarterly, by switching over to alternate standby machinery.

How often would you routinely take insulation resistances (Megger) readings on electrical equipment?

- **Every 3 months.**
- Every 2 weeks.
- Monthly.
- Weekly.

How should you check to see if an item of electrical equipment is overloaded?

- **By checking the ammeter to ensure it is not drawing excess current.**
- By conducting Megger tests to confirm that the insulation value is satisfactory.
- Monitor the heat generated, and also vibration and noise levels.
- Overlaod trips are fitted so that, if the equipment does not trip, then it is operating correctly.

How would you determine the value of a resistor?

- **By colour coding bands.**
- By its shape.
- By its size.
- By the colour of the resistor.

How would you ensure that all Electrical equipment has a satisfactory insulation value?

- **By regular megger testing.**
- By ensuring the equipment is dry and free from moisture.
- By operating and observing the operational performance and running amperage.
- By routine maintenance.

If a circuit breaker is fitted with a shunt release, the circuit breaker will trip . . .

- **when voltage is applied to the shunt release.**
- when the voltage is removed.
- when there is a faulty contact.
- when there is a wire break in the control circuit.

If you suspected that an electric motor had burnt out what would you check to confirm your diagnosis?

- **You would check motor insulation,the terminals and the starter box for any single phasing.**
- You would check the main switch board for earth fault indications.
- You would check the motor casing for discolouration or the smell of smoke.
- You would check the shaft and bearings for wear.

In a battery system, the most effective method of determining a cell's capacity to maintain its charge is by measuring the cell voltage . . .

- **under fully loaded condition.**
- when off load.
- when removed from the circuit.
- while on boost charge.

In addition to maintaining the output voltage of a generator at a pre-set level, what is the function of the Automatic Voltage Regulator (AVR)?

- **To ensure equal loading of kVAR between generators operating in parallel.**
- To control the system power factor.
- To ensure equal loading of kW between generators operating in parallel.
- To ensure that generators operating in parallel always run at matching speeds.

In preparing for maintenance on any critical equipment what should be considered first?

- **Has a risk assessment been prepared?**
- Are the spare parts onboard that are required for the work?
- Is permission from the port authorities required?
- Is there sufficient man power to complete the job?

In which compartments would you expect to find intrinsically safe electrical apparatus?

- **The battery compartment.**
- The engine control room.
- The engine room.
- The wheel-house.

On loss of main electrical power, how should the emergency generator start?

- **Automatically.**
- By pressing the local start button.
- By pressing the remote start button.
- Manually.

On starting a ship's auxiliary generator, the voltage does not rise to the required level while the correct system frequency is steady. What is the most probable cause of this fault? condition?

- **Excitation failure.**
- Governor failure.
- Low insulation.
- Reverse power.

The emergency generator automatic starting system must be capable of a minimum of . . .

- **three starts.**
- five starts.
- one start.
- ten starts.

The main reason for using an insulated neutral supply system is to . . .

- **prevent tripping of essential circuits when an earth fault occurs.**
- avoid the need for earth fault detection equipment.
- give automatic tripping of faulty circuits.
- prevent system over voltages.

The power supply at a ship's main switchboard has an unsteady frequency. What is the most likely cause of this condition?

- **A faulty generating engine governor.**
- Alternator winding over-temperature.
- An Automatic Voltage Regulator fault.
- The loss of excitation.

The starting of a large induction motor may result in high surge currents that can cause . . .

- **a voltage dip on the supply.**
- a long run up time.
- an increase in supply frequency.
- starting of the standby generator.

To enable two or more generators to be connected to a common busbar, which of the following electrical parameters must be matched?

- **The voltage and frequency.**
- The current.
- The power.
- The reactive power.

What action should you take if the insulation on the earth meter shows low readings?

- **Isolate each piece of equipment in turn to identify which one has low insulation.**
- Megger test all related equipment to identify the one which has low insulation.
- Observe the operation of all equipment for abnormalities.
- Switch all the equipment on, then turn off one by one to identify which has low insulation.

What condition would result in the operation of a generator's Reverse Power Relay?

- **When the generator engine is running at low speed.**
- When the generator engine is running at high speed.
- When the generator is in overload condition.
- When there is a high temperature of the alternator windings.

What does a PT100 Sensor measure?

- **Temperature.**
- Level.
- Pressure.
- Voltage.

What does the abbreviation UPS stand for?

- **Uninterruptible Power Supply.**
- Under-voltage Pre Set.
- Universal Power Supply.
- Unprotected Power System.

What element of electrical power is credited with being responsible for damaging the human body.

- **Amps.**
- Frequency.
- Volts.
- Watts.

What is the correct initial action that should be carried out prior to the inspection of a ship's auxiliary generator?

- **Obtain a Permit-to-Work.**
- Isolate/Lock off the generator circuit breaker at the main switchboard and display isolation notices.
- Prove the generator terminals are dead and earth/short circuit using approved equipment.
- Remove the alternator covers.

What is the fail safe condition for digital input switches connected to a ship's central alarm and monitoring system?

- **It can be open or closed depending on the application.**
- It does not matter.
- It is normally closed.
- It is normally open.

What is the most appropriate method of simulating a particular temperature output from a PT100 probe?

- **Connect a known resistance to the sensor cabling.**
- Apply a 4-20mA signal to the sensor cabling.
- Apply a mV signal to the sensor cabling.
- Short circuit the sensor cabling.

What is the most important aspect of portable or fixed electrical equipment in a hazardous area?

- **The equipment should be intrinsically safe.**
- The equipment is class approved.
- The equipment is safely secured to prevent damage.
- The equipment should be correctly earthed.

What is the purpose of a preferential trip?

- **When a generator goes into a predetermined overload condition, certain non-essential power is tripped to reduce load and prevent tripping of the generator main breaker.**
- It is the process of shutting down machinery when entering dry dock, prior to switching to shore power.
- When a generator is overloaded, the preferential trip automatically starts the standby alternator / engine to share the load / demand, and prevent damage and blackout.
- When two generators / alternators are operating and the load does not dictate the requirement for 2 alternators, one will automatically stop.

What is the purpose of a reverse power trip?

- **It trips the main breaker of a generator that is losing power, to prevent it being driven by another generator(s) on the same bus bar.**
- If a motor's electrical terminals are incorrectly connected, it trips the motor breaker and activates an alarm.
- In an emergency, it senses a dropping load demand and trips an alarm.
- It trips the main breaker of a generator to prevent it driving another generator.

What is the purpose of a shaft earthing device?

- **To ensure that there is no voltage differential between the shafting, propeller and the hull.**
- It is associated with the operation of the vessel's cathodic protection system.
- It prevents static electricity within the crankcase of the main engine and therefore reduces the possibility of a crankcase explosion.
- To provide one single point for all earths to be dissipated into the stern of the vessel.

What should the maximum setting of the thermal overload of a three phase induction motor be, in relation to the full load current rating?

- **A set percentage above the full load current.**
- A set percentage below the full load current.
- At the full load current.
- Twice the full load current.

What system is the most important in order to prevent the collapse of a computerised control system in the event of a power failure.

- **The UPS system must be maintained in good order.**
- Back up of files / documents must be completed.
- There should be an annual service and health check of the computer systems. This should be carried out by the manufacturer's technician.
- Up-to-date anti virus software must be installed.

What type of protective device would normally be installed at a vessel's main switchboard to serve the incoming cables from a 1.2 MW generator?

- **An Air Circuit Breaker.**
- A Miniature Circuit Breaker.
- An Isolator.
- FUSES.

What would happen in a Star/Delta Starter if the Star and Delta contactors were closed at the same time?

- **A short circuit would occur.**
- An overload would occur.
- The motor would rotate in the wrong direction.
- The motor would run at 50% normal speed.

When an earth fault occurs on an insulated neutral system, it must be located and cleared because . . .

- **a second earth fault causes tripping.**
- the faulty circuit is not working correctly.
- the system frequency is low.
- the system voltage is low.

When checking that the casing of a motor starter is adequately earthed, which resistance value would indicate a good earth connection?

- **Almost zero.**
- 1 MΩ.
- Greater than 100Ω.
- Greater than 200 MΩ.

When checking the insulation resistance between a motor winding and earth, which value would indicate a motor in good condition?

- **Greater than 200 MΩ.**
- 1 MΩ.
- 100 Ω.
- Zero.

When using an extension cable reel, which of the following statements is correct?

- **Do not exceed the reeled or unreeled rating.**
- Leave as much as possible coiled up on the reel.
- Only uncoil what you need.
- Uncoil it fully every time.

Where might you expect to find static electricity on a tanker?

- **In an electrical charge produced during tank cleaning operations.**
- In an alternator, as a differential electrical voltage between two generators during synchronisation.
- In any motor, as it is the electricity that remains as a charge within a motor when it has stopped running.
- In the battery compartment when they are being charged.

Which instrument would be used to determine if an earth fault exists on a three phase 440V heating element?

- **A megger.**
- A voltmeter.
- An ammeter.
- An ohmmeter.

Which of the following equipment should not be tested using a Megger insulation tester?

- **A fire detection system.**
- A heater element.
- A supply cable to a hydraulic pump.
- Electric motor windings.

Which of the following should be entered into a vessel's maintenance system following work carried out on an electric motor? (1) Whether work scheduled or unplanned. (2) Details of the work done. (3) Details of parts used while carrying out the work.

- **(1), (2) and (3).**
- (1) and (2).
- (1) and (3).
- (2) and (3).

Which of the following would provide the most reliable assessment of the insulation resistance of a motor winding?

- **A megger tester.**
- A digital multimeter on resistance setting.
- An analogue AVO meter on resistance setting.
- Earth lamps/monitoring unit on the main switchboard.

Which test instrument could be used to test the condition of a diode?

- **An ohmmeter.**
- A megger.
- A voltmeter.
- An ammeter.

Which test instrument would be used to determine the running current of a three phase motor?

- **An ammeter.**
- A megger.
- A voltmeter.
- An ohmmeter.

Why should equipment be earthed?

- **To discharge electrical current to earth in the event of a short circuit.**
- To enhance machinery performance and extend the life cycle of equipment.
- To prevent an earth alarm indication on the main switch board.
- To prevent damage to the vessels computer systems.

You are about to megger test a 440V motor. What is the minimum test voltage to be used?

- **500 Volts.**
- 110 Volts.
- 220 Volts.
- 400 Volts.

7025 – TOTS 1B-Maintenance and Repair

Version 1.0 (6)

After the overhaul of a main bearing and after crankcase inspection, what needs to be done first before manoeuvring with the engine?

- **Turn the engine on turning gear with the lub oil pump running.**
- Blow through engine using Local / Manual control
- Blow through the engine from Control room to test controls.
- Turn the engine on turning gear.

An increase in main engine crankshaft deflection readings may be an indication of . . .

- **excessive wear of main bearing shells.**
- excessive torsional stress.
- excessive wear of thrust bearings.
- excessive weight of piston and connecting rod.

Crankshaft deflections are normally taken when the engine is . . .

- **hot.**
- at any temperature.
- cold.
- either hot or cold.

During routine rounds of the main engine, you note that the pipe leading to one of the air start valves is hot. Do you?

- **Contact the bridge and request permission to stop immediately.**
- Advise the watch keepers to monitor it and report if the temperature increases.
- Apply portable ventilators to cool the pipe down.
- Attach a portable temperature gauge to record the temperature.

Excessive crankshaft deflections are taken as an indication of . . .

- **an engine misalignment or worn main bearings.**
- a bent crank shaft.
- only an engine misalignment.
- only worn main bearings.

For accurate readings the deflection gauge should be . . .

- **at the same temperature as the engine.**
- at a higher temperature than the engine.
- at a lower temperature than the engine.
- at room temperature.

How often would you normally expect to carry out water washing of the exhaust gas economiser?

- **Once a month.**
- Once a year.
- Once every 6 months.
- Once every two years.

How will you ensure the fire fighting and life saving equipment in the Engine room are in readiness?

- **By regular inspection, testing and maintenance.**
- By conducting drills every 3 months.
- By servicing and Certification annually.
- By servicing and Certification during dry docking.

If preventive maintenance work on critical equipment is overdue, what is the accepted course of action to proceed with this work?

- **Carry out a Risk Assessment and discuss maintenance options before making decision.**
- Carry it out before completing other items whose overhaul is due at the same time.
- Carry on and schedule the work at the next port of call.
- This has to be carried out immediately, even if it involves stopping the ship.

In a large 2 stroke diesel engine, which bearing half takes the highest loading?

- **Crosshead bearing lower half.**
- Connecting rod bearing lower half.
- Main bearing upper half.
- The crosshead bearing upper half.

In a large slow speed diesel engine, which of the following would you expect to be directly cooled by Sea Water?

- **Scavenging air.**
- Cylinder heads.
- Fuel valves.
- Turbocharger.

In addition to a deck mounted emergency stop button what type of safety device would you expect to find on a grinder to protect the operator?

- **A glass shield.**
- A high voltage trip.
- A reverse current trip.
- An over current trip.

In the event of a main engine scavenge fire, which of the following options is the best course of action to take immediately?

- **Reduce main engine rpm to a minimum.**
- Speed up the engine in an effort to blow the fire out.
- Start the auxiliary blowers.
- Stop the engine and keep it in one position until the fire is out.

In which direction is the engine turned when taking crankshaft deflections?

- **Ahead.**
- Any direction.
- Astern.
- Both ahead and astern.

Main engine cylinder liners normally have 'directional' air inlet ports in order to?

- **Induce swirling of the air inside the cylinder.**
- Cool the piston crown.
- Minimize air turbulence.
- Push hot gases away from the liner wall.

On a diesel engine, if the tappet clearances are greater than the manufacturer's recommend setting, the cylinder head valves will . . .

- **open late and close early.**
- open early and close early.
- open early and close late.
- open late and close late.

Prior to putting an Oily Water Separator into operation, should you?

- **Fill it completely with clean sea water.**
- First drain it completely.
- Flush the oily contents through with sea water prior to switching on the oil content monitor.
- Partially fill it and pressurize it with air.

The purpose of fine mesh screens on bunker tank vent heads is to?

- **Prevent flames from passing into the tank.**
- Prevent accidental overflow from the tank.
- Prevent dirt from getting into the fuel tank.
- Prevent the fumes from the tank affecting personnel working close by.

What action is needed if the fuel has a high water content?

- **Frequent draining of settling and service tanks.**
- Advance the fuel timing.
- Maintain a higher fuel oil temperature.
- Maintain a lower fuel oil temperature.

What checks would you carry out before starting an air compressor after an overhaul?

- **Check for debris from the overhaul, freedom of movement of the bottom end bearing and turn the compressor by hand.**
- Check for freedom of movement of the bottom end bearing.
- Check pressure in the starting air bottles.
- Turn the compressor by hand.

What could be the most common reason for the hydraulic pump in a hydraulic pack to trip on high oil temperature?

- **A fouled oil cooler.**
- A broken discharge valve.
- A broken pump.
- A broken suction valve.

What do you particularly check for, in the crank case, after an engine liner replacement?

- **Water ingress.**
- Abnormal noise.
- Lub oil ingress.
- Piston Clearance

What does the presence of copper in the crank case lub oil sample indicate?

- **Bearing wear.**
- Liner wear.
- Piston wear.
- Scraper box wear

What is a piston profile gauge used for?

- **To check for burn down of the piston crown.**
- To measure the piston clearance in the liner.
- To measure the piston groove wear.
- To measure the piston rod wear.

What is the main reason for honing the liners of a medium speed engine during routine overhaul?

- **To remove the glazing of the liner and re-establish a surface which retains the lubricating oil, thus reducing. liner wear**
- To reduce any ovality or barrelling of the liner.
- To remove scratches from the liner wall.
- To remove the wear ridge at the top of the liner.

What is the maximum allowable sulphur content of marine heavy fuel oil as per MARPOL Annex VI?

- **4.5%**
- 1.5%
- 2%
- 3%

What is the most common cause of turbocharger failure?

- **Debris from the main engine.**
- Bad fuel quality.
- Bearing failure.
- Lubrication failure.

What is the most common source of damage found on centrifugal pump impellers?

- **Cavitation damage.**
- Corrosion damage.
- Fatigue cracks.
- Impact damage.

What is the most important issue which must be borne in mind while inspecting critical equipment?

- **Reduction of possible risk to the vessel**
- Failure Mode and Effects Analysis (FMEA).
- It must not be carried out at sea.
- Repair methodology.

What is the possible consequence of prolonged oil carry over from a starting air compressor?

- **It could cause an air line explosion.**
- It could cause a high starting air temperature.
- It could cause lifting of the air compressor relief valve.
- It could cause low starting air pressure.

What is the recommended DC voltage output from a welding machine?

- **Less than 70V.**
- Less than 110V.
- Less than 240V.
- Less than 440V.

What is the requirement for the space where oxygen and acetylene bottles are stored on a ship?

- **They should be stored in separate well ventilated compartments on the deck.**
- They should be stored in a well ventilated compartment in the accommodation.
- They should be stored in an enclosed space.
- They should be stored in separated well ventilated compartments in the engine room workshop.

What main harmful effects does the presence of excessive quantities of metals like sodium and vanadium in the fuel have?

- **High temperature corrosion of exhaust valves.**
- Ineffective cylinder lubrication.
- Low calorific value.
- Piston ring wear.

What material are welding gloves normally made of?

- **Leather.**
- Cotton.
- Rubber.
- Wool.

What measurements are taken during the overhaul of a unit of the Main engine?

- **Liner calibration, piston groove measurements and piston rod measurements.**
- Liner calibration.
- Piston groove measurements.
- Piston rod measurements.

What measurements would you take during the overhaul of a reciprocating pump?

- **Plunger, slide valves and bottom end bearing measurements.**
- Bottom end bearing measurements.
- Plunger measurements.
- Slide valve measurements.

What needs checking if there are high levels of chromium in the crank case lub oil?

- **The piston rings.**
- The crosshead bearings.
- The cylinder liner.
- The main bearings.

What needs to be checked on the liner before putting back the piston during the replacement of the main engine liner?

- **The flow of cylinder lub oil.**
- The flow of camshaft lub oil.
- The flow of jacket cooling water.
- The flow of system lub oil.

What position of the piston are exhaust valve clearances adjusted?

- **During the compression stroke.**
- During the exhaust stroke.
- During the intake stroke.
- During the power stroke.

What type of fire detection would you find in the engine room workshop?

- **A smoke detector.**
- A flame detector.
- A thermal detector.
- Solely a Manual Push / Break Glass system.

What type of pump would you expect to find in a hydraulic power pack?

- **A positive displacement pump.**
- A multi stage centrifugal pump.
- A single stage centrifugal pump.
- A sliding shoe pump.

What type of pump would you use for a high volume, low pressure application?

- **Centrifugal.**
- Lobe.
- Reciprocating.
- Screw.

Where does the concept of critical equipment come from?

- **The ISM code.**
- MARPOL.
- SOLAS.
- The ISPS code.

Where should the dial gauge be placed while taking crankshaft deflections?

- **As far as possible from the axis of the crank pin.**
- Anywhere on the crank web but as indicated in the Planned Maintenance System.
- As close as possible to the axis of the crank pin.
- The centre of the crank web.

Where would you expect to find damping wire used in a turbocharger?

- **The gas side.**
- Both gas and air sides
- The air side.
- The bearings.

Where would you find the correct value for clearances of the main engine main bearings?

- **In the main engine instruction manual.**
- In the main engine lubricating oil piping diagram.
- In the main engine spare parts manual.
- In the Official log book

Where would you look for, and expect to find, the position of various engine room double bottom tanks?

- **In the general Arrangement plan.**
- In the Engine Room Piping Schematic drawings.
- In the stability booklet.
- In the tank calibration tables.

Which clearance would have the maximum effect on the pumping efficiency of a centrifugal pump?

- **Impeller and wear ring clearances.**
- Bearing clearances.
- Coupling clearances.
- Shaft seal clearances.

Which of the following is the first and most basic step you take when you start work on a unit of the Main engine?

- **To engage the turning gear.**
- To remove the fuel pipes.
- To remove the jacket water pipes.
- To remove the starting air pipes.

Which of the following parts of the bilge discharge system need calibration?

- **The 15 ppm monitor.**
- Bilge water Inlet flow meter.
- The opening angle of overboard discharge solenoid valve.
- The recirculation solenoid valve position.

Which of the following tools are required for overhauling a turbocharger?

- **A rotor locking tool, a bearing extracting tool and a spanner for the bearing nut.**
- A bearing extracting tool.
- A rotor locking tool.
- A spanner for the bearing nut.

Which of the following tools would be required for replacement of a main engine cylinder liner?

- **A liner lifting tool, a piston lifting tool and a cylinder head lifting tool.**
- A cylinder head lifting tool.
- A liner lifting tool.
- A piston lifting tool.

Which property of fuel oil is used to select the gravity disc of the fuel oil purifier?

- **The density.**
- The sulphur content.
- The viscosity.
- The water content.

Why is an air compressor started in the unloaded condition?

- **To avoid high starting torque.**
- To avoid condensate carry over.
- To avoid damage to the suction and discharge valves.
- To avoid high discharge pressure.

Why is it important to ensure Boiler Sliding Feet are maintained and kept correctly lubricated.

- **To ensure the Boiler is allowed to expand freely when it changes temperature.**
- It is not important as the provision of the sliding feet were to allow easy location of the boiler when it was being fitted.
- To allow limited movement of the boiler as the ship rolls and pitches in a seaway.
- To allow the boiler to move as the steam pipes expand under high loading.

Why is the TBN value of the crankcase lubricating oil of a Medium/High speed Diesel Engine so important?

- **The TBN is necessary to combat any acidic combustion products that enter the crankcase.**
- The TBN is necessary to maintain the lubrication of the liner in the combustion area.
- The TBN is necessary to maintain the viscosity of the oil at high temperatures.
- The TBN is necessary to reduce any corrosion caused by any water entering the crankcase.

7026 – TOTS 1B-Control of operations - Care of personnel

Version 1.0 (7)

A Permit-to-Work should normally NOT exceed . . .

- **one working day.**
- one hour.
- one week.
- three days.

A Port State Control Inspector advises a ship in good time on the date of a planned inspection.

- **False**
- True

A chemical from an unlabelled package should never be used unless its identity has been positively identified.

- **True**
- False

A ship is subject to Port State Control inspection in a particular region covered by MOU . . .

- **once in six months.**
- annually.
- every three months.
- every twelve months.

According to the ISPS Code, how often should security drills be conducted?

- **Every three months.**
- Every two months.
- Every two weeks.
- Monthly.

Any failure of the oil filtering equipment shall be noted in the Oil Record Book Part 1.

- **True**
- False

As required by the ISM Code, what certificate must a company have for it to be allowed to operate ships?

- **Document of Compliance.**
- Certificate of Class.
- International Ship Security Certificate.
- Safety Management Certificate.

Bunker procedures for ships are controlled under procedures that are incorporated in the Company's Safety Management System.

- **True**
- False

Bunkering operations should be planned and executed in accordance with . . .

- **MARPOL regulations.**
- ISM Code.
- the Code of Safe Working Practices.
- the ship's safety record book.

Compressed air should never be used for cleaning work spaces.

- **True**
- False

During a vetting inspection, the oily water separator is checked for . . .

- **All of these answers.**
- alterations in the piping system.
- correct operation.
- testing of the oil sensing alarm.

During topping of operations during bunkering, what hazard is most likely to occur?

- **Air lock in the tank causing oil mist carry over from the vents.**
- Crew fatigue.
- The ship's stability might be affected.
- The temperature of bunkers might increase.

Elements of Risk include:

- **the likelihood that a hazard might occur, and the consequence of the hazardous event.**
- the completion of a job in the slowest manner possible.
- the timely completion of a given task in as little time as possible.
- the unlikelihood that a hazard might occur, and the consequence of the hazardous event not occurring.

Engine room bilge oily water pumping and disposal arrangements are checked during a vetting inspection.

- **True**
- False

For how long should the Oil Record Book be preserved for?

- **Three years after the last entry.**
- Five years after the last entry.
- One year after the last entry.
- Two years after the first entry.

For which voltage should a General Electrical Work Permit be issued?

- **Under 1000 volts.**
- Under 120 volts.
- Under 230 volts.
- Under 70 volts.

How many years is a ship's IOPP Certificate normally valid for?

- **Five years.**
- Four years.
- One year.
- Two and a half years.

How often and where should the quantity of oil residues collected as a result of machinery space operations be recorded?

- **Weekly, in the ORB Part I.**
- After every operation, in the ORB Part I.
- At end of voyage, in ORB Part I.
- Daily, in the ORB Part II.

How often must a passenger lift be inspected by a competent person?

- **Every six months.**
- Every five years.
- Every two years.
- Yearly.

How often must a ship's lifting equipment be tested by a competent person?

- **Every five years.**
- Every six months.
- Every two years.
- Yearly.

How quickly should the main steering gear be capable of putting the rudder over from 35 degrees on one side to 30 degrees on the other side?

- **Not more than 28 seconds.**
- Not more than 30 seconds.
- Not more than 35 seconds.
- Not more than one minute.

Hydraulic jacks should be applied . . .

- **to safe jacking points on equipment.**
- after visual inspection of the equipment to be raised.
- to all suitable equipment for lifting purposes.
- to no equipment on board.

If the Vetting Inspector notes that the planned maintenance system being followed is not up to date and not class approved, he will . . .

- **raise an observation.**
- issue a Non-Conformance to the owners.
- request the Chief Engineer to provide a written statement.
- suspend the inspection of the engine room.

If the conditions under which a Hot Work Permit is issued should change . . .

- **hot work should be stopped immediately.**
- hot work should be continued with caution.
- hot work should be stopped for a few minutes and then started again.
- hot work would not have any impact with the change in conditions.

If work is to be performed on pressurised systems such as filters, what should be done to ensure that pressure in the system is off?

- **Open the drain cocks and air vents.**
- Place a rag over the cover when opening.
- Shut off any heating system.
- Wait for 24 hours to allow the pressure to drop.

It is a good practice to transfer E/R bilges into the cargo area slop tanks using a flexible hose.

- **False**
- True

Many of the bunkering operations overflows are due to . . .

- **human error.**
- technical failures.
- temperature of the bunkers.
- weather conditions.

On completion of the vetting inspection, the inspector passes the final report onto which database?

- **SIRE.**
- IMO.
- MSC.
- WIRE.

On completion of the vetting inspection, the inspectors initial report is . . .

- **discussed "on site" and the ship staff requested to comment.**
- handed over to port authority.
- handed over to the agent.
- kept confidential from the ship staff.

Prior to bunkering operations, the responsible officers should . . .

- **agree in writing the handling procedures and action to take in the event of an emergency.**
- delegate the responsibility of this operation to shore authorities.
- signal visually to agree the handling procedures and action to take in the event of an emergency.
- verbally agree the handling procedures and action to take in the event of an emergency.

Rest hour regulations can be found in . . .

- **ILO 180.**
- ILO 200.
- ILO 205.
- ILO 680.

Risk Assessment is essential prior to moving around heavy weights in the machinery space.

- **True**
- False

Risk Assessments conducted on board should be . . .

- **recorded and retained within the SMS documentation of the ship.**
- destroyed on completion of the task.
- handed over to the ship owner every month.
- put up on a notice board in the mess room.

Risk Assessments should be structured in order to identify . . .

- **potential hazards.**
- a simplified method of conducting work.
- short cuts to be taken to complete a job.
- the time frame required to complete a task.

Risk management should include Risk Assessments.

- **True**
- False

Something that is a source of potential harm or damage, or a situation with potential for harm or damage is known as . . .

- **a hazard.**
- a perception.
- an act of God.
- invincible.

The Code of Safe Working Practices is applicable and addressed to . . .

- **all ranks.**
- Junior officers.
- Ratings.
- Senior officers.

The Oil Record Book Part 1 is required for every oil tanker over . . .

- **150 Gross Tonnage.**
- 250 Gross Tonnage.
- 450 Gross Tonnage.
- 500 Gross Tonnage.

The Vetting Inspector has the authority to operate ship equipment on his own during the inspection.

- **False**
- True

The appointment of the Master as "Safety Officer" is . . .

- **generally not advisable.**
- ideal.
- not allowed on tankers.
- prohibited on all ships.

The flow chart for guidance on Hot Work can be found in which publication?

- **ISGOTT.**
- IAMSAR.
- MARPOL.
- SOLAS.

The main aim of Risk Assessment is to . . .

- **minimise accidents and ill health on board a ship.**
- increase profitability of the Company.
- man a ship with the minimum number of crew.
- minimise time wastage to complete a task.

The presence of a Superintendent onboard the ship during a vetting inspection . . .

- **does not relieve the ship staff of their duties during the inspection.**
- allows the ship staff to relax and seek assistance from him.
- allows the ship to keep any defects confidential.
- permits the Inspector to deal with the Superintendent directly without assistance from the ship staff.

The purpose of a Port State Control inspection is to . . .

- **ensure a ship's compliance with international regulations and conventions.**
- check the ship's service history.
- confirm that the ship has passed vetting inspections.
- make a list of additional suggestions that would benefit the ship's performance.

There are certain overriding factors that might subject a ship to re-inspection by Port State Control even though a ship was recently inspected.

- **True**
- False

What action should be taken if a Risk Assessment shows that an activity carries a very low risk?

- **Proceed with the task.**
- Do not continue with the task.
- Eliminate the risk totally.
- Redo the Risk Assessment.

What are the legal requirements for test intervals of lifting equipment?

- **Every five years.**
- Every 10 years.
- Every 15 years.
- Every six months.

What could happen if steam is admitted to a section of the steam pipe while the drains on the pipe are still closed?

- **Water hammer.**
- Excess pressure.
- Loss of feed water.
- Nothing, as this is normal procedure.

What data is necessary to correctly quantify the bunker quantity on completion of operations?

- **Specific gravity of the bunkers.**
- The chemical properties of the bunkers.
- The colour of the bunkers.
- The viscosity of the bunkers.

What is the maximum angle of list that the emergency generator and its prime mover can be capable of operating from?

- **22.5 degrees.**
- 10 degrees.
- 15 degrees.
- 6 degrees.

What is the maximum duration of validity of an Interim Safety Management Certificate?

- **Six months.**
- Five years.
- Three months.
- Two and a half years.

What is the meaning of a chemical's MSDS?

- **Material Safety Data Sheet.**
- Manual on Safety of Dangerous Substances.
- Maritime Substances Danger Statistics.
- Minimum Solubility Data Sheet.

What is the minimum capacity required for fire pumps?

- **25 m3/hr.**
- 10 m3/hr.
- 100 m3/hr.
- 50 m3/hr.

What is the minimum height requirement for guard rails fitted to all exposed parts of freeboard and superstructure decks?

- **1 metre.**
- 0.75 metres.
- 1.2 metres.
- 1.5 metres.

What is the minimum number of hours required for emergency electrical power to be available for lighting machinery spaces and machinery control position during periods of blackout in a cargo ship?

- **18 hours.**
- 24 hours.
- 36 hours.
- 8 hours.

What is the minimum sea service that a seafarer must have to qualify for the position of "Safety Representative"?

- **Two years.**
- No minimum requirement.
- One year.
- Six months.

What plans should be available on board the ship fitted with derricks?

- **The Rigging Plan.**
- The Distribution Plan.
- The Isolation Plan.
- The Movement Plan.

What publication contains information on the equipment for the control of oil discharge from machinery space bilges?

- **MARPOL.**
- ILO Conventions.
- ISPS Code.
- SOLAS.

What publication provides information on hazards related to dangerous substances carried on board?

- **IMDG Code.**
- ISPS Code.
- MARPOL.
- SOLAS.

What section of the Code of Safe Working Practices deals with various working practices common to all ships?

- **Section 3.**
- Section 1.
- Section 5.
- Section 6.

What tool/system on board provides the minimum basis for safe bunkering operations as agreed through the mutual exchange of critical information?

- **The Bunker Safety Check List.**
- The Bunker Loading Master.
- The Chief Engineer.
- The Chief Officer.

What toxic hazards are to be considered when handling fuel oil bunkering?

- **The presence of H2S.**
- The presence of Halon gases.
- The presence of helium gas.
- The presence of VCM.

When completing a work permit, at least . . . should be made.

- **two copies.**
- five copies.
- one copy.
- six copies.

When is a Declaration of Security NOT required?

- **In every port visited.**
- When the port has no Security Plan.
- When the ship is operating at a higher security level than the port.
- When there has been a security incident.

When is an oily water separator required to be fitted with an automatic stopping device?

- **If it is to be operated in a Special Area.**
- Always.
- Never.
- Only if effluent is 15 ppm or less.

Where can a description of the oily water separator be found?

- **In the IOPP Certificate Supplement.**
- In the Oil Record Book Part I.
- In the Oil Record Book Part II.
- In the SOPEP.

Where can information on machinery space chemicals be found?

- **MSDS.**
- IMTS.
- MSTS.
- RSPCA.

Where will the engineer's alarm sound if operated from the engine control room?

- **The engineer's accommodation.**
- The bridge.
- The Chief Engineer's office.
- The officer's smoke room.

Where would you find the record of construction and pollution prevention equipment for oil tankers?

- **In the supplement to the IOPP Certificate.**
- In the cargo record book.
- In the IOPP Certificate.
- In the Safety Construction Certificate.

Which among these bodies of water have been declared as SECA? 1) The Antarctic Sea 2) The Baltic Sea 3) The North Sea 4) Persian Gulf.

- **2) and 3).**
- 1) only.
- All of these answers.
- None of these answers.

Which among these operations is NOT required to be recorded in the Oil Record Book Part I?

- **None of these answers.**
- Ballasting of oil fuel tanks.
- Cleaning of oil fuel tanks.
- Disposal of oily residues.

Which annex of MARPOL Publication deals with regulations for the prevention of air pollution by ships?

- **Annex VI.**
- Annex I.
- Annex III.
- Annex VII.

Which chapter of the Code of Safe Working Practices describes the Risk Assessment procedures?

- **Chapter 1.**
- Chapter 11.
- Chapter 16.
- Chapter 8.

Which chapter of the Code of Safe Working Practices describes the safety system of work with reference to unmanned machinery spaces?

- **Chapter 15.**
- Chapter 2.
- Chapter 3.
- Chapter 7.

Which chapter of the Code of Safe Working Practices describes working procedures for high pressure/hydraulic/pneumatic equipment?

- **Chapter 20.**
- Chapter 14.
- Chapter 4.
- Chapter 8.

Which chapter of the Code of Safe Working Practices provides advice on maintenance of various machinery on board the ship?

- **Chapter 22.**
- Chapter 11.
- Chapter 14.
- Chapter 8.

Which chapter of the Code of Safe Working Practices provides information of lifting plants on board a ship?

- **Chapter 21.**
- Chapter 14.
- Chapter 4.
- Chapter 9.

Which component of inert gas is NOT toxic?

- **Nitrogen.**
- Carbon monoxide.
- Nitrogen dioxide.
- None.

Which document on board a ship provides vessel details to a vetting inspector during inspection?

- **VPQ.**
- OCIMF.
- PSC Report.
- SIRE.

Which of the following is NOT to be recorded in the Oil Record Book Part 1?

- **Cargo slop tank residues.**
- Accidental discharge of bilge tank residue.
- Bunkering of fuel or bulk lubricating oil.
- The condition of the oil filtering equipment.

Which of the following would give a Port State Control Inspector clear grounds to detain a ship?

- **If the Master or crew member is not familiar with the essential shipboard operational procedures that relate to the safety or security of the ship and pollution prevention.**
- If the last Port State Control inspection was conducted more than three months ago.
- If the ship is registered under a flag of convenience.
- If two senior officers have gone ashore in port.

Which publication highlights the parameters a Vetting Inspector will use during his inspection?

- **VIQ.**
- MARPOL.
- The ISM Code.
- VPQ.

Which publication would assist the ship staff to prepare for a vetting inspection?

- **VIQ.**
- SOLAS.
- The ISM Code.
- VPQ.

Who is responsible for keeping the vessel's particular questionnaire updated?

- **The ship's staff.**
- The ship owner.
- The shore-based office staff.
- The vetting inspector.

Working on electrical equipment in hazardous areas of the ship requires

- **a Permit-to-Work on an electrical circuit.**
- permission from the Duty Officer.
- permission from the owners.
- the Chief Engineer to be on stand-by while the work is being done.

Would the Vetting Inspector request to sight detailed bunker transfer instructions?

- **Always.**
- Never.
- Not applicable.
- Sometimes.

7027 – TOTS 1B-Management of the Engineering Department

Version 1.0 (33)

A Class Surveyor has imposed a Condition of Class on a machinery item. What does this mean?

- **It needs to be repaired and resurveyed in a given time.**
- It has been removed from the required survey items.
- It has to be resurveyed at the next external audit.
- It needs to be repaired in a given time.

A Garbage Record Book must be kept. When plastic has been incinerated what should you do with the ash?

- **Land the ash to reception facilities.**
- Dispose of it into the sea 25 miles from the nearest land.
- Mix it with plastic garbage and dispose to shore.
- Mix it with the other ash and dispose of it into the sea.

A failure of the bridge control of the main engine on a vessel with Certified Unattended Machinery Space results in having to revert to manned watchkeeping. If repairs cannot be done in a reasonable time which one of the following Ship's Certificates could this affect?

- **The Minimum Safe Manning Document.**
- The Cargo Ship Safety Equipment Certificate.
- The Certificate of Class.
- The Certificate of Fitness.

A risk assessment should be carried out and a Hot Work Permit issued before any hot work is carried out . . .

- **outside the Engine Room workshop.**
- at night.
- during bunker operations.
- during manoeuvring.

After taking over a vessel as Chief Engineer, you notice a Fitter trying to connect a hose to the oily water separator overboard discharge line. What do you do?

- **Stop this job and advise the Master and Managers of this onboard practice.**
- Advise him that this is not permitted and place the hose back in the store.
- Allow him to continue unaided.
- Instruct someone to assist him to avoid injury.

An Industry Vetting Inspector finds a deficiency in the Engine Room. What would the Chief Engineer expect the Vetting Inspector to do?

- **To discuss the deficiency with him, but take no responsibility for any corrective actions discussed.**
- Just to record the deficiency.
- To assist him to rectify the deficiency.
- To discuss the deficiency with him and agree on how to correct it.

An auxiliary engine has reached the running hours for major overhaul, but is due a survey in 2 months. Which of the following would be the correct course of action?

- **Overhaul it, and bring forward the survey to suit.**
- Overhaul it, and do the survey later.
- Postpone the overhaul until the survey date.
- Seek advice from the Class surveyor.

An internal audit of the vessel's Safety Management System must be carried out . . .

- **every 12 months.**
- every 18 months.
- every 24 months.
- every 6 months.

Before entering an empty Fresh Water tank for inspection and cleaning, you should . . .

- **ventilate and test the atmosphere with a calibrated oxygen indicator.**
- open and ventilate the tank at least 1 hour before entry.
- put on a Breathing Apparatus set and enter the tank.
- ventilate and test the atmosphere with a calibrated combustible gas detector.

During a Port State Inspection a discrepancy is found in the Engine Room. The item is recorded, but the Ship is not detained and sails without the discrepancy being cleared. What action should the Inspector take?

- **The Inspector should inform the Port State authority at the vessel's next port.**
- The Inspector does not need to take any action as the discrepancy is not a "detainable" item.
- The Inspector should contact the Company.
- The Inspector should inform the Class Society.

Having emptied, cleaned and ventilated a bunker tank for hot work, how would you ensure that the atmosphere in the tank is not combustible?

- **Test the atmosphere with an explosimeter.**
- Leave the tank lids open for 24 hours.
- Take a sample of the atmosphere in the tank and test it with an open flame.
- Test the atmosphere with a portable oxygen meter.

If a new motor is being fitted to a pump room fan how important is the direction of rotation of the motor?

- **It is very important as the fan must act as an extraction fan.**
- It is not important as the fan direction is switchable.
- It is not important as the fan is used for both supply and extraction.
- It is very important as the fan must act as a supply fan.

In order to prevent inadvertent direct overboard discharge of bilge water the main Engine Room bilge system should be . . .

- **secured.**
- blanked.
- checked regularly by a watchkeeper.
- displaying a warning notice.

Lifting and loose gear which is not part of the ship's equipment should be examined by a competent person at least once every . . .

- **12 months.**
- 24 months.
- 3 years.
- 5 years.

Oil Pollution Regulations require that any transfer of oil onboard a vessel should be recorded in . . .

- **the Oil Record Book.**
- the Bridge log book.
- the Engine Room log book.
- the Master's Official log book.

On a tanker, which of the following best describes a segregated Ballast System?

- **All ballast tanks, pipes and valves are independent of those used for cargo.**
- All ballast is discharged via an oil discharge monitor (O.D.M.E).
- Ballast and cargo tanks are separated by void spaces.
- Ballast is only taken after cargo has been flushed from the line.

On an indicator card, taken to calculate the power of a single acting 2-sroke engine, where is the atmospheric line on the card?

- **At the bottom.**
- At about mid-point of the diagram.
- At the side.
- At the top.

One result of poorly maintained fire dampers could be . . .

- **feeding the fire with oxygen.**
- a build up of toxic fumes.
- faster cooling of the fire.
- the fixed foam system being ineffective.

Paints and thinners used in the Engine Room for cosmetic maintenance should always be . . .

- **returned to the paint locker after every use.**
- kept in a closed container to prevent contamination.
- kept in a cool place.
- stored in the Engine Room away from the Deck paint supplies.

Ships receive a Quarterly Listing. What does this contain?

- **The status of Surveys from Class.**
- A list of spares the ship has used over 3 months.
- Budget reports for the last 3 months.
- The status of Surveys by Port State Control.

The Chief Engineer's Standing Orders must contain which of the following?

- **Duties of the watchkeeping Engineers and Ratings.**
- Duties of the Engineers in an emergency.
- Duty times of the watchkeeping Engineers and Ratings.
- Operation of the Dead Man Alarm.

The Emergency Generator is required to have clear operating instructions. Who must be familiar with the operation?

- **The ship's crew.**
- Deck Ratings.
- The Chief Engineer.
- The Training superintendent.

The Planned Maintenance System should be inspected by which of the following to confirm that it has been implemented according to Company procedures?

- **An Auditor.**
- A Class Surveyor.
- A Port State Control inspector.
- An authorised shore technician.

The Steering Gear has 2 electric motors. How is the power normally supplied?

- **Independently from the Main switchboard.**
- Both are connected via the Emergency Switchboard.
- Both are supplied from the Main Switchboard on a common power supply.
- One is supplied from the shaft generator.

The US Pollution Prevention Regulations (33 CFR) require that any oil spillage in US waters should be reported immediately to . . .

- **the Agents and local Harbour Authority.**
- the local Oil Spill Resource Centre.
- the US Coast Guard.
- the vessel's Owners.

The difference between the distance the vessel travels through the water and the theoretical distance it should have travelled through the water is commonly known as . . .

- **the propeller slip.**
- the engine distance.
- the engine slip.
- the propeller distance.

The exhaust gas temperature entering the turbo-blower in relation to the main exhaust manifold temperature should be . . .

- **higher.**
- any of these.
- lower.
- the same.

Weekly testing of the Engine Room CO2 alarm should confirm that . . .

- **the Engine Room ventilation supply and exhaust fans stop automatically.**
- the Engine Room mechanical ventilation dampers close automatically.
- the Engine Room ventilation supply fans stop automatically.
- the fuel quick closing valves shut automatically.

What do the letters UMS indicate in a vessel's Class notation?

- **The vessel may be run with Unattended Machinery Spaces.**
- It defines certain Machinery Spaces on the vessel which are Unused and do not require any maintenance.
- It defines those Spaces on the vessel which must remain Unused for Machinery.
- It is a classification for Underwater Survey Maintenance.

What instrument would be used to measure the area of a power card taken from the main engine?

- **A planimeter.**
- A micrometer.
- A power meter.
- A vernier calliper.

What is the normal survey interval for machinery items?

- **Once every 5 years.**
- Once every 2.5 years.
- Once every 3 years.
- Once every 4 years.

What would the Auditor report if he found that a laid down procedure was not being correctly carried out?

- **A non-conformity.**
- A non -appliance.
- A non- adherence.
- A non-correctness.

When an Exhaust Gas Boiler is surveyed who is responsible for setting the safety valves?

- **The Chief Engineer.**
- The Class Surveyor.
- The Company Superintendent.
- The Shore Technician.

When briefing your team prior to bunkering, what would you describe as the most critical part of the operation in terms of risk of spillage?

- **When topping off the bunker tanks.**
- When blowing the lines through with air.
- When fuel first starts coming onboard.
- When the hoses are being disconnected.

When bunkering, you note that you have some leakage from the connection manifold flange, but the Master is pressing you to depart as soon as possible to make the tide. What do you do?

- **Stop bunkering and repair the leak?**
- Notify the local authority of a potential oil spill?
- Reduce the pumping pressure to minimize the leakage?
- Spread absorbent material on the deck to soak up any spillage?

When calculating bunkers for a voyage, the amount of extra bunkers required to provide a safe margin is calculated according to . . .

- **the Company's procedures.**
- the Chief Engineer's own experience.
- the direction given by the Charterer.
- the Master's instructions.

When main engine cooling water is used to heat a fresh water generator, the chemical treatment used in the cooling system must be . . .

- **non-toxic in case it leaks.**
- coloured to make it easier to detect a leakage.
- given a salt content to make it easier to detect a leakage.
- self-sealing in the event of a leak.

Where would you look to find your prime responsibilities in the event of a fire?

- **The Company Management manuals.**
- The Emergency Stations (Muster List) poster.
- The Safety Equipment Certificate.
- The Shipboard Training Manual.

Which of the following would contribute to an increased risk of accidents?

- **Unsafe working conditions.**
- Good housekeeping in the Engine Room.
- Proper risk assessments.
- Regular inspections.

Why are self-closing cocks fitted to Engine Room double bottom tank sounding pipes?

- **To prevent the contents of the tank being forced into the Engine Room in the event of hull leakage.**
- To prevent oil fumes escaping from the tank into the engine room.
- To prevent the tank being flooded in the event of the Engine Room flooding.
- To protect the contents of the tank being contaminated by pollutants from the engine room.

With reference to the Safety Management Certificate, what is a document of compliance?

- **It is a document that the company is issued with by the flag state.**
- It is a certificate that a ship receives after being audited.
- It is a document that certifies that the ship complies with all the regulations.
- It is a document that the ship receives when it has no non - conformities.

7028 – TOTS 1B-Emergencies, Drills and Fire Fighting

Version 1.0 (30)

A few drops of oil spill into the water but no traces are visible anymore. Is it still necessary to report the incident?

- **The incident shall be reported regardless of the size of the incident.**
- It's advisable to ignore the incident to prevent legal issues arising, especially if there are few witnesses.
- Since no traces of oil are visible there is no need to report it.
- Wait and watch to see what the reaction is, and then decide accordingly.

Can the SMPEP and SOPEP be combined into one manual?

- **Yes. Since the requirements are similar, they can be combined into one plan.**
- No. The SMPEP is a dry cargo ship contingency plan while the SOPEP is for tankers.
- No. The SMPEP is for a particular cargo while the SOPEP is a general plan.
- No. They address different regulations so cannot be combined.

Does the paint locker require a fixed fire fighting installation?

- **Yes, a fixed system is mandatory.**
- A portable installation is acceptable if within easy reach.
- It is only important to be able to quickly access some type of extinguishing system.
- The regulations require a portable extinguisher in the locker.

How can the fire stations and lockers be identified on board?

- **Because they are marked with internationally recognised IMO signs.**
- By following the crew reporting to their stations when the emergency alarm sounds.
- It is the duty of the ship's Safety Officer to show the crew where the specific locations are.
- One can check the general arrangement plan and search for the lockers/stations when necessary.

How is it possible to ensure thermal stresses do not affect the compressor when the BA set air bottles are filled?

- **All the answers are correct.**
- By maintaining adequate lubrication at all times.
- By making sure overheating does not take place during operation.
- By making sure the compressor is located in a well ventilated place.

How is the SMPEP different from a SOPEP?

- **It is the contingency plan for ships carrying Noxious Liquid Substances.**
- Different Flag Administrations use different terminology for the same plan.
- It is the same plan, only called by different names based on the individual company.
- SMPEP deals with Oil Pollution only.

How many foam applicators must a fixed foam extinguishing system have on deck?

- **A minimum of 4.**
- 10 in total.
- A minimum of 2.
- At least 4 on each side.

If a man falls overboard, what type of message should be sent to alert the ships in the vicinity?

- **PAN - PAN**
- MAYDAY.
- NONE.
- SECURITE.

In the event of a collision what are the duties to the other vessel?

- **Render assistance without jeopardising the safety of your own personnel and vessel.**
- If oil has been spilt the Master should take down details and try to avoid becoming involved in a pollution incident.
- Issue a notice of arrest to the other ship on behalf of the Company
- Since it is a collision incident, the Master should take details before continuing with the voyage.

Is high expansion foam suitable to fight a fire on deck?

- **It is unsuitable for use in open areas in outside locations as it is easily dispersed in light winds.**
- It depends on the expansion ratio of the foam.
- It is better to use water jets to clear the deck of the combustible substance.
- It is suitable since it gives a larger coverage area on deck.

Is it mandatory to have a fire blanket in the galley?

- **Yes, it is mandatory as stated by SOLAS.**
- It depends how committed the ship manager is to safety.
- It is up to the chief cook to decide.
- The Port State Control insists on having one in the galley.

Is uncontrolled venting an allowed practice while loading cargo?

- **No. Venting of tanks should be only through mast risers in a controlled fashion.**
- Yes. If the wind speed is good there is no problem with uncontrolled venting.
- Yes. It is not a problem since so many precautions have already been taken.
- Yes. Uncontrolled venting is unavoidable due to the vapour pressure while loading.

The ship suffers a collision. The bridge waste paper basket contains some small pieces of paper with scribbled notes on. What should the Master do?

- **Collect these pieces of paper and keep them as a record, as they may contain relevant information.**
- It might be an idea to falsify the scribbled notes after the collision so you will not be held responsible.
- There is already so much evidence available that it's not necessary to keep these small pieces of paper.
- Treat them as waste and ignore them as they have already been thrown away.

What actions should be initiated by the damage control team after a grounding?

- **All the answers are correct.**
- They should check the watertight integrity of the compartments in contact.
- They should sound all compartments, sound around the ship and check the nature of the seabed.
- They should visually inspect the ship and check the draft/trim/list.

What are SOPEP / VRP / SMPEP / PCSOPEP?

- **Oil Spill Contingency plans based on different regulations.**
- Different plans as required by different nations of the world.
- Manuals required by the ISO 9001 standards.
- Manuals segregated by the company for ease of use.

What are the different types of foam extinguishing media that may be found on board?

- **High expansion, medium expansion and/or low expansion foam**
- High Expansion Foam only.
- Low Expansion Foam only.
- Medium Expansion Foam only.

What details of the main and emergency fire pumps are described on the Fire Control Plan?

- **The capacity of the pumps are clearly indicated.**
- Only the location of the fire pumps are marked.
- The identity of the makers and model number is included.
- There is no such requirement for details.

What information is available on the ship's Fire Control Plan?

- **The firefighting arrangements and location of all firefighting equipment on board.**
- The location of all portable fire-fighting equipment on board.
- The location of all Safety Equipment (LSA & FFA) on board.
- The location of the Fixed fire fighting system fitted on board.

What is the first action that must be taken in the case of a hose burst, pipe work fracture or cargo overflow?

- **All operations must be immediately stopped and the alarm sounded.**
- Quickly repair the leakage and keep operations going to prevent the ship from being delayed.
- Sound the alarm, inform the Master and wait for his instructions.
- Stop all operations and wait for the Coast Guard and terminal people to arrive.

What is the first reporting requirement in the event of a toxic liquid release at anchor?

- **The incident should be reported to the nearest Coastal State in detail without delay.**
- If the effect is not visible on the surface of the water there is no need to report the incident.
- The agent should be informed immediately and he will take necessary action.
- The ship's staff must first check with the ship's managers before taking action.

What is the importance of a running hours log on the BA set air bottle filling compressor?

- **To ensure periodical maintenance is carried out as required by the manufacturer.**
- It ensures the crew on board are doing their job by looking at the running hours log.
- It indicates when a new compressor needs to be ordered.
- The running hour's log is maintained as the company periodically requests to see it.

What is the importance of operating instructions posted at local positions for fire-fighting equipment?

- **Ease of operation when it's necessary to use the equipment in an emergency.**
- Instructions are provided because training standards are not uniform globally.
- It helps with the problem of language barriers between different nationalities.
- It is instructed by the Master as part of normal standard procedures.

What is the minimum frequency (for the majority of vessels) for an abandon ship drill and a fire drill for the crew, according to SOLAS Chapter lll?

- **There shall be at least one abandon ship drill and one fire drill every month.**
- The frequency of the drill is left to the discretion of the Master.
- There shall be at least one abandon ship drill and one fire drill every three months.
- There shall be at least one drill every week while the ship is at sea.

What is the minimum required service duration of an EEBD?

- **10 minutes.**
- 12 minutes.
- 15 minutes.
- 20 minutes.

What is the order of priority when dealing with contingencies?

- **Safety of life, the environment and then property.**
- Safety and protection of the environment.
- Safety of property and cargo.
- Safety of the ship and the environment.

What is the purpose of a Safety Meeting on board?

- **To review safety matters and improve the Safety Management System.**
- To discuss operational problems and the maintenance programme.
- To ensure the paperwork, as required by ISM, is completed.
- To receive and review crew complaints with regard to crew matters.

What is the requirement for galleys that have deep-fat cooking equipment?

- **All the answers are correct.**
- There must be an automatic shut off of electrical power when the fire extinguishing system is activated.
- There should be clear instructions for crew concerning manual operation of the system.
- They shall be fitted with an automatic or manual fire extinguishing system.

What is the requirement for manually operated call points on a fixed fire detection and alarm system?

- **They shall be capable of immediate operation at all times.**
- In case the automatic system fails then the manual call point is to be used.
- Manual operation can only be overidden by the automatic system from the bridge.
- No manually operated call points are required since the system is fixed.

What kind of fire alarm detectors are used on a ship?

- **They can be smoke detectors, heat detectors, flame sensors or any combination of the three.**
- A smoke detection system.
- Flame sensors.
- Heat detectors.

What kind of fixed fire-fighting installations may be fitted for the cargo pump room?

- **It should be fitted with a system suitable for machinery spaces of category A.**
- Pump rooms must be fitted with a gas fire extinguishing system only.
- Pump rooms should be fitted with a fixed high expansion foam extinguishing system only.
- The type of system is based on the type of cargo the ship is carrying.

What kind of fixed fire-fighting installations may be fitted for the machinery spaces?

- **Any of the types mentioned.**
- A fixed high pressure water spraying system.
- Fixed Gas Fire Extinguishing system.
- Fixed High Expansion Foam system.

What kind of galley equipment must be fitted with an automatic or manual fire fighting system?

- **Deep fat cooking equipment.**
- Electrical equipment.
- Galley ovens
- Galley stove hotplates

What must be covered in the familiarisation of the fixed fire-fighting system on the cargo deck area?

- **All the answers are correct.**
- The fire-fighting medium storage area.
- The location and operation of the fixed fire extinguishing system.
- The remote and local activation points and procedures.

What should the Officer on Watch do in the event of a 'Man Overboard'?

- **Raise the alarm and turn the ship towards the side of the man overboard.**
- Call the Chief officer.
- Inform The Master and wait for his arrival.
- Turn the ship away from the side of the man overboard.

What should you do if the ship spills oil into the water in a port and you are contacted by the Media?

- **Be courteous, state facts and suggest the reporter contacts the company, and do not commit yourself on any issues you are not sure about.**
- Blame the other parties and deny any wrongdoing by you or the ship.
- Don't be afraid to be direct and explain to them the difficulties of life on board and working with companies.
- If they become aggressive, treat aggression with aggression and ask the crew to physically remove them.

What size of tanker must be fitted with a fixed deck fire-extinguishing system?

- **Tankers of 20,000 tonnes deadweight and upwards.**
- All tankers must be fitted with a fixed system.
- Tankers of 5,000 tonnes deadweight and upwards.
- Tankers of 50,000 tonnes deadweight and upwards.

What would be the immediate action taken in the event of a bunker spill?

- **Stop all transfer operations, inform, sound the alarm and initiate Emergency response.**
- Stop operations and work out a solution to cover up in collusion with the barge.
- Stop operations, check if anyone has noticed and then take any necessary action.
- Stop operations, investigate the occurrence and inform based on the quantity of the spill.

What would be the required action if a steering gear failure takes place in a busy shipping lane?

- **All the answers are correct.**
- Alert the VTS station and check if it's possible to anchor.
- SECURITE call to other ships in the vicinity so they can take evasive action.
- Switch to emergency steering and reduce speed.

When should a new crew member undergo on-board safety familiarization?

- **As soon as practicable upon joining the vessel.**
- No later than one month after joining the ship.
- No later than two months after joining the ship.
- The training schedule is subject to the time available between operations.

When would you use the Emergency Response Service of the Class Society?

- **For damage stability and longitudinal strength calculations after a casualty.**
- It is useful for any emergency situation on board where Class is affected.
- They are called in for support if the ship is involved in a pollution incident.
- They are contacted if the ship has an accidental release of cargo in USA waters.

Where are the operating instructions for the emergency fire pump located?

- **In a clear, visible place near the emergency fire pump.**
- Adjacent to the international ship/shore connection for use by the shore fire services.
- On the bulkhead next to the ship's fire plan.
- On the ship's bridge and the Engine Control Room.

Where can the fixed fire fighting installation be located?

- **All three answers are correct.**
- It can be located in a room behind the fwd collision bulkhead, used for no other purpose.
- It can be located inside protected space.
- It can be located no more than one deck below open deck and directly accessible from open deck.

Where can you find the action to be taken and procedures to follow in an emergency?

- **In the Shipboard Oil Pollution Emergency Plan.**
- In SOLAS
- In the International Safety Guide for Oil Tankers and Terminals.
- In the Shipboard Safety Manual.

Where is the fire alarm control panel located?

- **On the ship's navigating bridge.**
- At the general muster point.
- In the Emergency headquarters.
- In the Engine Control Room.

Where shall the isolating valve on the fire main be fitted?

- **It shall be fitted in an easily accessible and tenable position outside the machinery space.**
- It can be fitted anywhere as long as it serves the purpose of isolating the section of fire main for machinery space.
- It shall be fitted anywhere on open deck spaces; the ship's main deck being the most appropriate.
- It shall be fitted in the accommodation as the crew are always available in that area.

Where should the main control station of the deck fixed foam fire fighting system be installed?

- **Outside the cargo area, adjacent to the accommodation, readily accessible and operable in event of fire.**
- Inside the accommodation so that it is accessible and operable in event of fire.
- Inside the cargo pump room since all cargo operations are controlled from there.
- On one of the superstructure decks in an easily accessible location for use in event of fire.

Where would you find details of the fire dampers, fire flaps and ventilation on board the ship?

- **In the Fire Control Plan.**
- In the ship's Airconditioning Plan.
- In the ship's Emergency Action Plan.
- In the ship's General Arrangement Plan.

Which of the following elements should generally be included in a fire drill?

- **All of these answers**
- Checking of the fireman's outfits, personal rescue equipment and communications equipment.
- Checking the operation of watertight doors, fire doors, fire dampers and main inlets and outlets of ventilation systems in the drill area.
- The starting of a fire pump, using at least two jets of water

Who is the person identified in the Vessel Response Plan who acts on behalf of the Owner for oil spill response in USA?

- **A Qualified Individual.**
- A designated person ashore.
- An authorised person.
- The owner's representative.

Why are Vessel Management Meetings held?

- **To promote effective team-working, planning, clarity on tasks, execution and monitoring of jobs.**
- The Management level staff are required by the regulations to hold these meetings.
- To comply with the Company Auditors who periodically check to make sure the meetings have been carried out.
- To provide an opportunity for the vessel management team to discuss the issues raised at the HSE meetings.

Why are environmental issues discussed during HSE or environmental team meetings?

- **To raise awareness among the crew on issues that affects the environment and promotes pollution prevention.**
- Because there are so many rigid regulations to comply with.
- For financial reasons as the commercial viability of ship may be affected.
- To discuss tactics in the eventuality of an incident involving environmental activists such as Greenpeace.

Why are proper records required to be maintained for routine tests, checks and maintenance of fire-fighting equipment?

- **For ease of reference and to allow proper and effective planning of testing and maintenance. Also plans and records must be available for inspection when required by the Administration.**
- Because the company auditors can ask for all records of inspections.
- Because the Port State Control is likely to detain the ship if records are unavailable.
- Because the quality system requires records to be maintained to provide evidence.

Why do dry powder installations require periodical checking?

- **Solidification of the compound can occur on board over a period of time.**
- Dry powder evaporates and disappears over time.
- Other extinguishing media are more stable.
- The powder tends to change colour and loose effectiveness.

Why do foam compounds after an initial 3 year period have to be tested annually?

- **All the answers.**
- Because deterioration of the foam can reduce its properties.
- Because it is a Flag requirement.
- Because the foam compounds are likely to deteriorate over time.

Why do tankers require isolation valves to be fitted in the fire main at the poop front and on the tank deck at intervals of no more than 40 metres?

- **To preserve the integrity of the fire main system in case of fire or explosion.**
- Because additional isolation valves help to provide quick access.
- Because deck fire fighting becomes easier when different sections of fire main can be isolated.
- Because it makes it easier to renew damaged sections of the fire main while the fire pump is running.

Why does the ship require an International Shore Connection?

- **To provide a standardised connection between the ship's fire main and shore systems which may have different couplings.**
- It is a requirement of tanker terminals and a practice in the 'wet' trade.
- It is called by that name since the ship trades in International waters.
- It is used only in the drydock to keep the fire main pressurised since the ship's pumps are off.

Why is a fire control plan stored in a prominently marked weathertight enclosure outside the deckhouse?

- **It is to assist shore side fire-fighting personnel in an emergency.**
- It is kept outside in case of an accommodation fire on board.
- It is kept there because the Port State Control Inspector will look for it.
- It is stored so that crew can access the information during a drill on deck.

Why is it important to follow a drill with a debriefing session?

- **To review and discuss the lessons learnt and continually improve.**
- It is a requirement of the Company Safety Policy.
- To advise the crew about their faults and assign blame to individuals.
- To motivate crew and make good use of any remaining time left over from the drill.

You have a small bunker spill in port. All notifications are done and the cleanup is completed. What do you do concerning the record collection?

- **Obtain calculations of relative quantities on board, in the pipelines and in the barge.**
- Forget the records and hope the estimated spilt quantity will be relatively accurate.
- Leave matters to the local agent and feel relieved that the emergency is over.
- Leave the record collection to the company who are not on site.

7029 – TOTS 1B-Pollution Prevention and Environmental

Version 1.0 (6)

A ship demonstrates compliance with MARPOL Annex IV by . . .

- **having a valid International Sewage Pollution Prevention Certificate.**
- having a sewage holding tank volume of 100 litres per person on board.
- having a stand-by sewage treatment plant.
- having a surveyor inspect the sewage system every six years.

After having performed a ballast operation the master must . . .

- **complete the ballast water management log.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

Discharge of sewage is prohibited when . . .

- **the vessel is 24 nautical miles from the nearest land and travelling at 3 knots.**
- the sewage is comminuted and disinfected and the vessel is 5 nautical miles from the nearest land.
- the ship is 24 nautical miles from land and has a functioning certified sewage treatment plant.
- the vessel is 24 nautical miles from the nearest land and travelling at 7 knots.

During bunker transfer operations deck scuppers should be . . .

- **plugged to prevent spilled oil flowing overboard.**
- attended by crew to collect any spilled oil that approaches.
- only fitted to the side of the vessel that bunkers are being loaded to.
- Plugged when required

During cargo or bunker operations, oil response equipment is to be . . .

- **ready for immediate use.**
- in its store room.
- in the cargo pump room.
- used elsewhere in the ship.

False entries in the Oil Record Book can lead to criminal penalties for the master and chief engineer?

- **True.**
- False.

How long must the receipts be held with the Garbage Record Book for?

- **Two years.**
- Five years.
- One year.
- Six months.

How long should the oil record book be kept for after it has been completed?

- **Three years.**
- One year.
- Six months.
- Two years.

How many categories of garbage are there in a Garbage Record Book?

- **2**
- 10
- 12

In the event of an oil pollution incident MARPOL Annex I requires that an oil tanker of 5,000 tonnes deadweight shall have prompt access to . . .

- **computerised shore-based damage stability and residual structural strength programs.**
- an empty oil tanker of equal deadweight.
- computerised models of the ship's cargo tank system.
- the nearest port.

It is illegal to discharge oily mixtures from the ship when the vessel is stationary and outside of a special area.

- **True.**
- False.

It is legal to discharge oil outside a special area when the oily mixture originates from the cargo pump room bilges on an oil tanker.

- **False.**
- True.

No discharge into the sea shall contain chemicals or other substances in sufficient concentration so as to cause damage to the marine environment.

- **True.**
- False.

Operational malpractice of the oil filtering equipment can lead to criminal penalties for the vessel master and chief engineer?

- **True.**
- False.

Prior to exchanging ballast the master must . . .

- **perform a risk assessment.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

The cargo manifold working platform surface should be made of . . .

- **open mesh grating.**
- chequered plate.
- perforated sheet steel.
- wooden laminate.

The oil record book makes references to factors affecting quantity of oil. The accuracy of measuring quantities are affected by . . .

- **All three answers**
- Accuracy of tank measurement devices.
- Temperature variations.
- Tendency of oil to cling to the inside tank walls.

The oil record book part 2 requires which of the following to be recorded after cleaning cargo tanks?

- **All three answers.**
- The duration of cleaning.
- The method of cleaning.
- The ship's position or port.

The oil spill removal equipment store must be . . .

- **prominently marked and easily accessible.**
- a storeroom close to the pump room.
- a storeroom in the forward area of the ship.
- just off the main engine room.

The portable pump equipment within the oil spill response kit must be . . .

- **All three answers**
- fitted with sufficient hoses.
- safe for use in hazardous areas
- Self priming

The shipboard oil pollution emergency plan shall contain details of a list of authorities to be contacted in the event of an incident.

- **True.**
- False.

The spill response station inventory should include . . .

- **personal protective equipment.**
- a supply of water.
- brushes.
- metal shovels.

To prevent inadvertent leaks, bunker manifolds not in use should . . .

- **have blank flanges fitted with gaskets and tightened with all bolts in place.**
- have blank flanges fitted and tightened with all bolts in place.
- have blank flanges fitted and tightened with some bolts in place.
- have blank flanges fitted with gaskets and tightened with some bolts in place.

To reduce cargo tank vapour release which of the following should be checked?

- **That the tightness of cargo tank hatch dogs are checked after every cargo operation.**
- That header valves are unseated.
- That the cargo pump deepwell bleed valve is open.
- That the hatch gasket is checked every year.

Uncontrolled release of volatile organic compounds (VOCs) are an environmental problem because . . .

- **they contribute to low-level ozone pollution in populated areas.**
- they contaminate the sea.
- they react with CO2.
- they react with hydrogen creating acid rain.

What can happen if an emulsion is taken up by the OWS?

- **It can lead to a false reading on the oil content monitor.**
- It can cause damage to the pump bearings.
- It can completely block the system piping.
- nothing.

What should the Master refer to In the event of an accidental discharge of oil?

- **The shipboard oil pollution emergency plan.**
- The ballast water management plan.
- The cargo tank plan.
- The ship environmental management system.

When first using a cleansing agent in the machinery space, the crew should . . .

- **confirm that it won't form an emulsion.**
- confirm that it will form an emulsion.
- use it liberally.
- use it sparingly.

When is it good practice to check the pressure-vacuum valves?

- **Prior to cargo transfer.**
- After cargo transfer operations.
- Before the next dry-docking.
- During cargo transfer operations.

When is the Master to sign the Garbage Record Book?

- **After each completed page.**
- After each discharge operation.
- After the incinerator has been used.
- At the end of the day.

Where on board might ozone depleting substances be found?

- **In refrigeration systems.**
- In compressed air systems.
- In electronic equipment.
- In sewage treatment plants.

Which of the following is NOT an option to comply with MARPOL Annex IV?

- **An approved comminuting plant.**
- A certified sewage treatment plant.
- An approved comminuting and disinfection plant.
- Holding tanks with an approved volume.

Which of the following is not part of the spill response equipment inventory?

- **The pressure washer.**
- Emulsifiers.
- Non sparking scoops, shovels and buckets.
- Sorbents.

Which of the following must be recorded in the oil record book part 1?

- **All three answers**
- Bunkering of fuel oil.
- Collection and disposal of oil residues.
- Condition of oil filtering equipment.

Which of the following should be checked prior to cargo operations?

- **All three answers**
- That pipeline drains are blanked correctly.
- That save-all drains in way of tank vents are plugged.
- That spill tank drain plugs are properly secured.

Which of the following substances is ozone-depleting?

- **Halons.**
- Ammonia.
- CO2.
- Neon.

Which of the following would increase the fuel consumption on board the ship?

- **Leaving steam system leaks unattended.**
- Maintaining steam traps.
- Reducing speed in heavy weather.
- Turning off lights in empty compartments.

Which of the following would reduce energy consumption?

- **Maintaining an unfouled hull.**
- Cooling accommodation spaces to 19°C.
- Heating accommodation spaces to 25°C.
- Runnning at normal sea speed

7030 – TOTS 1B-General Tanker Section

Version 1.0 (2)

A Permit-to-Work on electrical circuits is important in order to prevent what?

- **Electrical shock.**
- All of these answers.
- Non-scheduled planned maintenance tasks being carried out.
- The use of non-intrinsically safe equipment on deck.

A Permit-to-Work should specify the period of validity. According to the Code of Safe Working Practices, this period should not exceed how many hours?

- **24 hours.**
- 12 hours.
- Eight hours.
- Six hours.

A Risk Assessment can be a useful tool when preparing a Permit-to-Work on electrical circuits.

- **True**
- False

A company's Safety Management System (SMS) includes the following functional requirements: safety & environmental policy, procedures to ensure the safe operation of the ship, levels of authority amongst ship and shore personnel, procedures to prepare for

- **Procedures for reporting accidents and non-conformities.**
- All of these answers.
- Procedures for helicopter operations.
- Procedures for training ship's staff.

A single work permit is issued . . .

- **only for the work to be undertaken.**
- for a variety of tasks.
- to keep paperwork up-to-date.
- to satisfy auditors.

According to the Code of Safe Working Practices, a Permit-to- Work does not in itself make the job safe but contributes to measures for safe working.

- **True**
- False

Before cargo or ballast operations commence, the Master (or his representative) and the terminal representative should do what?

- **All three of the above**
- Agree in writing on the action to be taken in the event of an emergency during cargo or ballast handling operations
- Agree in writing on the transfer procedures, including the maximum loading or unloading rates
- Complete and sign the Ship/Shore Safety Checklist.

Critical Operations Check Lists are to be used for what purpose?

- **To ensure that critical equipment is operated, started or restarted correctly so that it will not result in a hazardous situation developing or compromise critical functions.**
- To ensure that all company procedures are followed at all times.
- To ensure that every item of equipment on board is operated according to the manufacturer's instructions.
- To ensure that the vessel passes external ISM audits by showing the auditor that Section 10 of the ISM Code is fully complied with.

Familiarisation of the Safety Management System (SMS) is essential in order to . . .

- **All of these answers.**
- comply with all the applicable codes, international regulations and best practices.
- ensure that the ship is operated in the safest possible way to avoid injury or damage to the environment.
- improve your skills including emergency preparedness.

Hazardous materials such as chemicals and oil will have a Material Safety Data Sheet (MSDS) available. Which of the following is NOT likely to be included in a MSDS?

- **The name of the vessel.**
- The correct storage.
- The health effects.
- The physical data (e.g. the flashpoint).

If hot work is to be undertaken above the tank deck (higher than 500mm), then cargo and slop tanks must either be cleaned and gas freed, OR emptied purged and inerted, OR filled with water, within a minimum radius around the working area. What is this min

- **30 metres.**
- 10 metres.
- 20 metres.
- 40 metres.

If hot work is to be undertaken on the tank deck or at a height of less than 500mm above the tank deck, the precautions taken should be the same as for hot work within the tank.

- **True**
- False

If the conditions under which a Hot Work Permit has been issued change, what should be done immediately?

- **Stop the hot work.**
- Amend the work permit conditions.
- Extend the validity of the permit.
- Prepare additional firefighting equipment.

No hot work should be carried out in the vicinity of bunker tanks unless which of the following is carried out?

- **The tank has been cleaned to hot work standard.**
- The tank contains fuel with a flashpoint above 60 degrees Celsius.
- The tank contains fuel with a flashpoint below 60 degrees Celsius.
- The tank is greater than 90% full.

On a Permit-to-Work for pipeline/pressure vessels, which of the following questions would NOT likely be included as part of the checks?

- **Is the lifeboat/rescue boat ready for use?**
- Has the part of the system to be worked upon been isolated?
- Has the pressure been bled off the section to be worked on?
- Have all personnel been equipped with the correct personal protective equipment?

The Code of Safe Working Practices requires that any faults relating to cargo handling equipment, testing instruments and alarm systems must be . . .

- **reported immediately.**
- repaired immediately.
- repaired in dry dock.
- reported when cargo operations are completed.

The International Shore Fire Connection is used for what purpose?

- **To enable the water fire main onboard to be connected to an external water supply from shore.**
- To enable carbon dioxide to be used from shore to fight engine room fires.
- To enable the foam line onboard to be connected to an external foam supply from shore.
- To enable the ship to assist in extinguishing a fire ashore.

What happens to a Permit-to-Work upon completion of work?

- **It is cancelled by the responsible officer.**
- It is cancelled by the person carrying out the work.
- It is left in place in case further work has to be carried out.
- It is transferred to another location where similar work is to be carried out.

What is the correct method of closing off unused cargo and bunker connections?

- **All unused connections must be closed and blanked. Blank flanges are fitted and are to be fully bolted.**
- All unused connections must be closed and blanked. Blank flanges are fitted and have every second bolt secured by a nut.
- Unused connections may be left unblanked so long as two valve separation exists between the connection and the line in use.
- Unused connections on the shore-side of the vessel must be closed and fully blanked. The offshore-side may be left unblanked.

What should happen to deck scuppers during periods of heavy rain when engaged in cargo or bunker handling?

- **One or two scuppers may be opened temporarily to allow rain water to escape but must be closed as soon as possible and in the event of an deck oil spill.**
- All deck scuppers must be kept fully closed at all times and never opened until cargo and bunker operations are completed.
- Only deck scuppers on the outboard side of the vessel are to be opened to release the rain water. When the rain stops, the scuppers can then be replugged.
- Open all deck scuppers to release the rain water. When the rain stops, the scuppers can then be replugged.

What should personnel on board barges or other craft obey when moored alongside?

- **All of these answers.**
- Prohibition of naked lights.
- Restrictions on the use of cooking appliances.
- Smoking restrictions.

What should the ship's Master do if an unauthorised craft comes alongside and attempts to secure to the vessel when at a terminal?

- **Report the craft to the terminal and if necessary stop cargo operations.**
- All of these answers.
- Sound the emergency alarm.
- Turn fire hoses onto the small craft.

What would be the minimum recommended firefighting equipment to have available when alongside a terminal?

- **Fire hoses with spray/jet nozzles run out forward, aft and adjacent to the manifold in use. Also a dry chemical extinguisher near the manifold.**
- Fire hoses with spray/jet nozzles forward and aft of the manifold and a fire blanket adjacent to the manifold in use.
- Fire hoses with spray/jet nozzles or fire extinguishers adjacent to the manifold in use.
- One fire hose and one dry chemical extinguisher near the manifold in use.

When are items on the ship/shore safety check list checked when a vessel is at a terminal?

- **Before cargo or ballast operations commence and then at regular intervals until hoses are disconnected.**
- During cargo operations.
- Immediately upon berthing and before starting cargo operations.
- Upon commencing cargo operations.

When can controlled hot work be carried out within a hazardous area on a tanker?

- **When the ship is in ballast condition.**
- When the ship is carrying out cargo operations.
- When the ship is carrying out gas freeing operations.
- When the ship is carrying out inerting operations.

When carrying out hot work with electric welding equipment, which one of the following statements is correct?

- **The ship's structure should not be used as an earthing point.**
- The cable route to the working area should be the shortest and most direct route.
- The Direct Current output source should be in excess of 70 Volts.
- The person carrying out the welding operation can work alone so long as a work permit has been issued.

When handling oil or waste oil, direct contact should be avoided with the skin by using gloves. What other method can be used to protect the skin?

- **A barrier cream.**
- A moisturiser cream.
- Talcum powder.
- Vaseline.

Where does the Code of Safe Working Practices recommend guidance on general precautions be taken from?

- **All of these answers.**
- Safety in Oil Tankers.
- The Company's Safety Management System.
- The International Safety Guide for Oil Tankers and Terminals.

Where hot work is planned to be carried out, what should the Lower Flammable Limit (LFL) reading of the atmosphere be?

- **Less than 1% LFL.**
- 1% LFL.
- Between 1% and 2% LFL.
- Greater than 1% LFL.

Where should the ship's emergency fire control plans be located?

- **In a prominently marked weather-tight enclosure outside the accommodation block.**
- At the bottom of the gangway.
- In a prominently marked wallet contained within the cargo control room.
- With the Deck Officer of the Watch at all times.

Which of the following could be considered a "designated space" where hot work is able to be carried out safely?

- **The engine room workshop.**
- The engine room.
- The forecastle deck.
- The poop deck.

Which of the following is the most important way for any company to improve their Safety Management System (SMS)?

- **Receiving constructive feedback and review from the ship's staff.**
- Receiving criticism from third parties.
- Receiving information on the introduction of new legislation.
- Receiving notification of problems by Port State Control.

Which of the following items can be described as critical machinery?

- **All of these answers.**
- Generators.
- Steering gear.
- The main engine.

Which of the following items may NOT be considered a critical operation?

- **Switching on the radar.**
- Changing over from heavy fuel to diesel.
- Lowering and raising lifeboats.
- Starting the emergency generator.

Which of the following items of equipment would be found in a chemical storage/handling area?

- **Safety goggles.**
- A dust mask.
- Ear plugs.
- Safety shoes.

Which of the following questions would NOT be part of a Permit-to-Work check list for allowing small craft to come alongside?

- **Has the ship been inerted or gas freed?**
- Are all deck and cargo openings closed?
- Has the Master of the small craft been advised of all appropriate safety regulations (e.g. emergency signals, smoking regulations etc)?
- Is adequate illumination available?

Which of the following statements apply to a storage area where chemicals are kept?

- **It must be kept well ventilated and secure.**
- It must be kept under surveillance by closed circuit television cameras.
- It must be located within the engine room workshop only.
- It must be stored in the main deck store only.

7031 – TOTS 1B-The Flammability Diagram

Version 1.0 (10)

After inerting a cargo tank, the oxygen content was found to be 7.5%, and the hydrocarbon gas content 2%. Using the Flammability Diagram, would you consider the tank to be safely inerted before gas freeing?

- **Yes, the atmosphere is below the critical dilution line.**
- No, inerting must be continued.
- No, the tank atmosphere is in the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

Define the "Lower Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration below which there is not sufficient gas to support and propagate combustion.**
- It is the gas concentration above which there is not sufficient air to support and propagate combustion.
- It is the gas concentration above which there is too much gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Define the "Upper Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration above which there is not sufficient air to support and propagate combustion.**
- It is the gas concentration above which there is too much oxygen to support and propagate combustion.
- It is the gas concentration below which there is not sufficient gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Explain why the atmosphere in a cargo tank must be sampled at different depths?

- **Due to the high relative density of hydrocarbon gas, it tends to layer, and collect in pockets.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

In practical terms, what figures can be used for the Lower and Upper Flammability Limits of oil tanker cargoes?

- **1% and 10% of gas.**
- 19% and 21% of oxygen.
- 5% and 21% of oxygen.
- 5% and 50% of gas.

The characteristics of the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the relative density of butane?

- **2.00**
- 1.00
- 2.55
- 5.50

The characteristics of the vapours given off by crude oil can be represented by those of propane. What is the relative density of propane?

- **1.55**
- 0.55
- 1.00
- 5.50

The characteristics of the vapours given off by natural gasolines can be represented by those of pentane. What is the relative density of pentane?

- **2.50**
- 0.55
- 1.00
- 5.50

The flammable limits for the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the Lower Flammable Limit of butane?

- **1.9% by volume in air.**
- 19% by volume in air.
- 5.9% by volume in air.
- 9% by volume in air.

The flammable limits for the vapours given off by crude oil can be represented by those of propane. What is the Lower Flammable Limit of propane?

- **2.2% by volume in air.**
- 10% by volume in air.
- 22% by volume in air.
- 5.2% by volume in air.

The flammable limits for the vapours given off by natural gasolines can be represented by those of pentane. What is the Lower Flammable Limit of pentane?

- **1.5% by volume in air.**
- 21% by volume in air.
- 4.5% by volume in air.
- 7.8% by volume in air.

What is the relative density of hydrocarbon gas at concentrations near the Lower Flammable Limit?

- **Approximately 1.0, the same as air.**
- Approximately 0.7, lighter than air.
- Approximately 1.5, heavier than air.
- Approximately 2.0, much heavier than air.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, what would be the effect of diluting with fresh air?

- **The hydrocarbon concentration would be reduced, and the oxygen level increased, taking the atmosphere into the Flammable Range.**
- The hydrocarbon concentration would be increased, taking the atmosphere into the Flammable Range.
- The hydrocarbon concentration would be reduced, and the oxygen level increased, resulting in a safe tank atmosphere.
- The oxygen level would be increased, resulting in a safe tank atmosphere.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, why would it be necessary to dilute the atmosphere to point H with inert gas before gas freeing with fresh air?

- **To reduce the hydrocarbon level below the critical dilution line.**
- To help prevent corrosion in the tank.
- To increase the hydrocarbon level above the Upper Flammable Limit.
- To reduce the oxygen level to below the critical dilution line.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, would you consider it to be safe to purge with inert gas with an oxygen content of 8%?

- **Yes, the atmosphere would remain outside the Flammable Range.**
- No, the tank atmosphere is in the Flammable Range.
- No, the tank must first be gas freed with fresh air.
- Yes, the tank atmosphere would remain too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point A?

- **The tank contains fresh air, with no hydrocarbon gas, and cannot support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point B?

- **The tank atmosphere is too rich to support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere is too lean to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point C?

- **The hydrocarbon gas concentration is at its Lower Flammable Limit.**
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point D?

- **The hydrocarbon gas concentration is at its Upper Flammable Limit.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too lean to support combustion.

7032 – TOTS 1B-Fixed and Portable Gas Detections Systems

Version 1.0 (3)

A fixed gas detection system for toxic gas normally uses electro-chemical detectors. What level of accuracy can be expected?

- **Approximately 5 ppm.**
- Approximately 10 % by volume.
- Approximately 30 % by volume.
- Approximately 50 ppm.

After washing and gas-freeing from a toxic cargo, what type of gas detection equipment should you use before entering a cargo tank?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

Cotton filters are sometimes used with hydrocarbon gas detectors. What is their purpose?

- **To remove water from the sample.**
- To prevent chemical contamination of the sample.
- To remove dust from the sample.
- To stop pressure surges in the sample line.

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

How will leaks in the sample pipework affect the operation of a fixed gas detection system?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

How will leaks in the sampling line affect the operation of portable gas detection equipment?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

If an Explosimeter is calibrated using span gas containing 1% butane in air, what reading may be expected?

- **Approximately 50% Lower Flammable Limit.**
- Approximately 1% Lower Flammable Limit.
- Approximately 10% Lower Flammable Limit.
- Approximately 100% Lower Flammable Limit.

Multi-toxic gas detectors normally use electro-chemical detectors. What level of accuracy can be expected?

- **Approximately 1 - 5 ppm . . .**
- Approximately 10% by volume.
- Approximately 100 - 500 ppm.
- Approximately 30% by volume.

Span gas is generally used for calibration of a fixed gas detection system. What is the composition of this gas?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure nitrogen.

What effect will inert gas have on the operation of an infrared type gas detector?

- **None. The instrument is not affected by the absence of oxygen.**
- It will Increase the reading by increasing the temperature of the chamber.
- It will reduce the reading by contaminating the sensor.
- Small particles may obscure the mirrors.

What instrument should you use to test for very low levels of oxygen before loading a sensitive cargo?

- **Chemical indicator tubes.**
- A paramagnetic oxygen analyser.
- An Explosimeter.
- An infrared gas detector.

What instruments must be provided to comply with the statutory requirements for ships fitted with an inert gas plant?

- **An oxygen analyser and an instrument to measure the percentage of hydrocarbon gas in an inerted atmosphere.**
- An Explosimeter and an oxygen analyser.
- At least two instruments to measure the percentage of hydrocarbon gas in an inerted atmosphere.
- At least two instruments to measure the percentage of oxygen in an inerted atmosphere.

What is a "Pellistor"?

- **It's a poison resistant catalytic filament.**
- It's a balancing resistor between the sensing element and a fixed compensator.
- It's a filter which prevents the spread of flame.
- It's a protective device which removes contaminants.

What is a "Sintered Filter", and what is its purpose?

- **It's a flashback arrestor used to prevent flame from escaping from the combustion chamber.**
- It's a cotton wad used to prevent water from entering the combustion chamber.
- It's a fine metal filter used to prevent contamination of the sensor.
- It's an electronic device used to smooth the signal from the sensor.

What is the composition of the span gas generally used for calibration of an Explosimeter?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure carbon dioxide.

What is the composition of the span gas generally used to zero a portable oxygen analyser?

- **High purity nitrogen or carbon dioxide.**
- Approximately 2% propane in air.
- Approximately 20% propane in air.
- Fresh air.

What is the gas concentration alarm level on a fixed gas detection system?

- **30% of Lower Flammable Limit.**
- 10% of flammable gas.
- 3% of Lower Flammable Limit.
- 30% of flammable gas.

What is the principle by which a thermal conductivity meter works?

- **To show a change in resistance of an electrically heated resistance element, due to heat loss to the gas sample by conduction.**
- To show a change in temperature of an electrically heated resistance element, due to heat loss to the gas sample by conduction.
- To show a change in the conductivity of a heated gas sample relative to that of air.
- To show change in the density of a heated gas sample relative to that of air.

What is the principle by which an infrared gas detector operates?

- **It senses the level of absorption of infrared radiation in the sample.**
- It compares the electrical resistance of the sample with fresh air.
- It compares the relative density of a sample which has been heated by an infrared source.
- It measures the resistance of a sample which has been heated by an infrared source.

What is the principle by which chemical absorption indicators operate?

- **By passing the sample through a proprietary chemical, the gas concentration is indicated by its changing colour.**
- To compare the chemical absorption rate of the sample with fresh air.
- To sense the proportion of a sample which is absorbed by a catalyst.
- To show whether a toxic gas will be absorbed by the human body.

What level of accuracy can be expected when using toxic gas detection tubes?

- **Approximately 1 ppm, depending on the type of tube in use.**
- Approximately 10%, depending on the type of tube in use.
- Approximately 100 ppm, depending on the type of tube in use.
- Approximately 30%, depending on the type of tube in use.

What operational tests should be made routinely before using a gascope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self- testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using a tankscope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using an Explosimeter?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas, and zero using atmospheric air.

What operational tests should be made routinely before using an oxygen analyser?

- **Check battery level, reference point with atmospheric air (21%), and zero using nitrogen or carbon dioxide.**
- Check battery level, reference point with span gas, and zero using nitrogen or carbon dioxide.
- Check the battery level, and self- test the alarm functions.
- Check the reference point with span gas and zero using atmospheric air.

What property of oxygen is utilised in a paramagnetic oxygen analyser?

- **Oxygen is strongly attracted to a magnetic pole.**
- Oxygen molecules will break up in a magnetic field.
- The electrical resistance of oxygen changes in a magnetic field.
- The strength of a magnetic field is reduced in the presence of oxygen.

What type of sensor is fitted in a tankscope?

- **A non-catalytic heated filament.**
- A catalytic filament resistor.
- A magnetic induction sensor.
- A PT100 temperature sensor.

What type of sensor is fitted in an Explosimeter?

- **A catalytic filament resistor.**
- A gas diffusion filter.
- A magnetic resonance chamber.
- A non-catalytic heated filament.

What will be the effect of hydrogen sulphide on the reading shown on a Combustible Gas Indicator (Explosimeter)?

- **It will reduce the reading by contaminating the sensor.**
- It will give a zero reading by destroying the sensor.
- It will have no effect because it is non-flammable.
- It will increase the reading by raising the sensitivity of the sensor.

What would you use a personal oxygen meter for?

- **For monitoring atmospheric conditions during an enclosed space entry.**
- For checking the atmosphere of a ballast tank before entry.
- For testing an enclosed space before starting hot work.
- For testing the atmosphere of an open space before starting hot work.

When changing high purity cargo grades, what instrument should be used to ensure that tanks are free of the previous product?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

When starting to test a sample using a Combustible Gas Indicator (Explosimeter), the needle immediately flicks to maximum deflection, and then returns to zero. What is indicated?

- **That the flammable gas concentration is above the Lower Flammable Limit.**
- That the battery is incorrectly fitted.
- That the instrument was not warmed up, but is now working correctly.
- That the sensor element has been contaminated by toxic gas.

Why is it necessary to use a water-retaining filter in the sample line of an oxygen analyser?

- **Because water vapour in the sample will cause damage to the measuring cell.**
- Because water in the sample will corrode the internal parts of the instrument.
- Because water reacts with the measuring cell, giving a false reading.
- Because water will build up in the sample line, reducing the flow.

Why must chemical absorption tubes only be used with pumps from the same manufacturer?

- **Because the pump and tube are matched to ensure that the correct volume of sample gas is tested.**
- Because the tubes will not fit correctly into another manufacturer's pump.
- Because using different tubes breaks the terms of the contract with the manufacturer.
- Because using the wrong tubes may lead to contamination of the pump.

Why must the span gas pressure be carefully controlled when calibrating a fixed gas detection system?

- **Excess pressure may damage the catalytic sensing element.**
- Excess pressure may cause errors in the readings.
- Excess pressure may cause leaks in the system.
- If the pressure is too high, it could cause an explosion inside the equipment.

Why must the tank atmosphere be sampled at a variety of positions when testing before entry?

- **To allow for the density of the gas and to ensure that any collected in dead spots is detected.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

7033 – TOTS 1B-Enclosed Spaces and Tank Atmosphere . . .

Version 1.0 (5)

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

For what purpose would you use a personal gas detector?

- **To monitor atmospheric conditions during an enclosed space entry.**
- To check the atmosphere of an enclosed space before entry.
- To test an enclosed space before starting hot work.
- To test the atmosphere of an open space before starting hot work.

For what purpose would you use a personal oxygen meter?

- **To monitor atmospheric conditions during an enclosed space entry.**
- Testing an enclosed space before starting hot work.
- Testing the atmosphere of an open space before starting hot work.
- To check the atmosphere of an enclosed space before entry.

How long should ventilation be continued after the atmosphere of an enclosed space has been found suitable for entry?

- **Continuously while the space is occupied, and even during temporary breaks.**
- For at least one hour after successful testing.
- It can be stopped when testing shows that the space is suitable for entry.
- It should be continued after testing until the space is actually entered.

How many different locations within a tank should be sampled to test the atmosphere before entering a tank?

- **At various depths, and through as many openings as possible.**
- As close as possible to the centre of the tank.
- At least three locations.
- In one position, at a minimum of three levels.

How many people should be allowed to enter an enclosed space to carry out routine work?

- **Only the number actually required to carry out the work.**
- Only two people.
- The number required to carry out the work, plus one extra watchman.
- The number required to carry out the work, plus the person in charge and one extra watchman.

What action must be taken if the presence of a toxic gas is suspected for which no specific testing equipment is available?

- **Treat the space as unsafe and follow the procedures according to COSWP 17.11.**
- Only allow entry after further ventilation.
- Test the space for a similar gas using any available equipment, following the procedures according to COSWP 17.6
- Test using an Explosimeter, and allow any necessary conversion factors.

What action must be taken when an enclosed space entry permit expires?

- **The space must be vacated and closed.**
- Additional crew should attend for assistance.
- The work must be completed as soon as possible.
- Work can continue until it has been completed.

What action should be taken if personnel working in an enclosed space get into difficulties?

- **The general alarm should be sounded and the rescue team should enter using breathing apparatus under the direction of the Person In Charge**
- All personnel should be instructed to leave the space immediately.
- The first person on the scene should immediately enter the space to give assistance.
- The Watchman should immediately enter the space to give assistance.

What actions must be taken to isolate a tank before entry?

- **Blank off the pipelines to the tank, close and secure all valves and inform the Officer of the Watch.**
- Request the Officer of the Watch to stop all cargo and ballast pumps.
- Stop all cargo and ballast operations.
- Stop all ventilation and inform the Officer of the Watch.

What equipment is required to be kept at the entrance of an enclosed space during entry?

- **A resuscitator, breathing apparatus, a lifeline and a harness, a torch or a lamp.**
- A resuscitator or escape breathing set.
- Enough escape breathing sets for all personnel in the space.
- Spare tools and parts for the work being undertaken.

What gas detection equipment must be used to test the atmosphere of a cargo tank before entry is permitted?

- **An oxygen meter, a combustible gas indicator and a toxic gas detector if necessary.**
- A combustible gas indicator and a toxic gas detector if necessary.
- An oxygen meter and a tankscope.
- An oxygen meter and a toxic gas detector if necessary.

What indication will be given by an Explosimeter in case of oxygen deficiency?

- **No indication.**
- Approximately 20% of the Lower Flammable Limit
- Full scale deflection.
- The indication is erratic and jumpy.

What is the maximum permitted level of toxic gas for an enclosed space entry?

- **The Occupational Exposure Limit, which varies according to the particular gas.**
- 1% of the Lower Flammable Limit.
- 10% of the Lower Flammable Limit.
- It varies according to the particular gas, but 1% is an acceptable level.

What potential hazards would be caused in a machinery space by the release of carbon dioxide for fire extinguishing purposes?

- **A lack of oxygen.**
- Carbon dioxide which is toxic to breathe.
- The carbon dioxide gas would be very cold, and could cause severe cold damage to the skin.
- Vapour, which may reduce visibility.

Who is required to standby at the entrance of an enclosed space during entry?

- **At least one watchman, equipped to communicate with the personnel inside and with the Officer of the Watch.**
- Nobody, provided the personnel inside are equipped to communicate with the Officer of the Watch.
- The Officer of the Watch and one watchman, equipped to communicate with the personnel inside.
- The Officer of the Watch, or whoever signed the Entry Permit.

Why could an unprotected steel space, which has been closed, produce a hazardous atmosphere?

- **Because oxygen in the atmosphere may be depleted by reacting with the steel (i.e. corrosion).**
- Because hydrocarbons could be liberated from rust scale in the tank.
- Because the level of nitrogen in the atmosphere would be increased by corrosion.
- Because toxic gases may be given off by the process of corrosion.

Why could an unventilated ballast tank fitted with Cathodic Protection contain a hazardous atmosphere?

- **Because hydrogen is produced by the Cathodic Protection.**
- Because hydrocarbons could be produced by the Cathodic Protection.
- Because the Cathodic Protection could produce an oxygen enriched atmosphere.
- Because the Cathodic Protection would increase the level of nitrogen in the atmosphere.

Why must the enclosed space atmosphere be sampled at a variety of positions?

- **To allow for the denisity of the gas and to check for any gas pockets which may have accumulated in blind spots.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

7034 – TOTS 1B-Familiarisation prior Cargo Operations

Version 1.0 (2)

A Ship to Shore Safety Checklist can be found in which publication?

- **ISGOTT.**
- ISPS code.
- SOPEP.
- The Mariner's Handbook.

During cargo transfer, what action would you take if the Inert Gas System fails?

- **Cease all discharge operations and advise the terminal.**
- Advise the terminal and continue discharging.
- Increase the rate of discharge.
- Review the Ship / Shore Safety Check List.

Low expansion foam is most effective extinguishing . . .

- **most hydrocarbon liquid fires.**
- ballast tank fires.
- cabin fires.
- main switchboard fires.

On what occasions will a pre-cargo transfer meeting be minuted?

- **On each and every occasion.**
- When the Captain attends.
- When there is something special to say.
- Whenever the port stay is estimated to be greater than 24 hours.

The Ship / Shore Safety Check List states that temporarily removed scupper plugs must be . . .

- **constantly monitored.**
- clearly labelled as to location.
- only removed by the person responsible for cargo operations.
- stowed in a safe place.

The principle purpose of a pre-cargo transfer meeting is to . . .

- **ensure that all involved personnel are aware of the proposed cargo operations, services needed and safety requirements.**
- arrange shore leave.
- discuss the Ship / Shore Safety Check List.
- inform the technical officers when cargo pumps will be required.

What is a "Class B" fire?

- **Fires involving flammable and combustible hydrocarbon liquids.**
- Fires involving solid material.
- Fires that are not so important.
- The type of bulkhead the fire can penetrate.

What is a schematic drawing?

- **A drawing in the form of a diagram and not to scale.**
- An approximate drawing.
- An exact technical drawing.
- The general arrangement plan.

What is the name of the diagram that is superimposed on the cargo control console?

- **The mimic diagram.**
- The cargo lines.
- The pumping plan.
- The tank plan.

When is high expansion foam most effective?

- **When rapid filling of an enclosed space is required.**
- Generally anywhere on deck.
- When it is to be used as bridge front cooling.
- When the cargo manifold is on fire.

Where would you expect to locate a mimic diagram of the cargo system?

- **In the cargo control room (CCR) operational panel.**
- Adjacent to the bunkering manifold.
- At the manifold.
- In the Chief Engineer's office.

Why is carbon dioxide a good medium to fight an electrical fire?

- **Because it is a non-conductor.**
- Because it generates static electricity.
- Because it has a cooling effect.
- Because you can smell it.

7035 – TOTS 1B-Cargo Information, Precautions . . .

Version 1.0 (5)

According to the IBC code where should in use protective equipment be stowed?

- **In special lockers in easy accessible places.**
- At the forecastle head.
- At the top of the pumproom.
- In the accommodation.

How does the IBC code describe coveralls?

- **They should be made of material resistant to chemical attack.**
- They can be made of any material but must have long sleeves.
- They must be nylon.
- They should be made of pure cotton.

On a tanker carrying Noxious Liquid Substances, what is the minimum number of protective equipment sets that must be carried?

- **At least three.**
- It can vary as it depends upon the toxicity of the cargo.
- One, provided it's the one that's required by SOLAS.
- Two.

The cargo being loaded is described as having highly toxic vapour. What must be worn on the cargo deck?

- **Breathing apparatus.**
- A dust mask.
- A face shield.
- Rubber gloves.

What can be worn in addition to or instead of goggles?

- **A face shield.**
- A smoke helmet.
- An escape set.
- Breathing apparatus.

What is cargo ingestion?

- **Cargo being swallowed.**
- Cargo having contact with the eyes.
- Cargo having contact with the skin.
- Cargo vapours being inhaled.

What is essential for an oxygen/air resuscitator to operate efficiently?

- **The casualty's airway must be clear.**
- The casualty must be unconscious.
- The casualty must not have facial hair.
- The casualty must weigh over 90 KG.

What is the common name for a Personal Escape Set?

- **EEBD (Emergency Escape Breathing Device).**
- A resuscitator.
- ELSA set.
- Self Contained Breathing Apparatus.

What is the important feature of the goggles on a tanker carrying Noxious Liquid Substances?

- **They must be tight fitting and splash proof.**
- Nothing in particular, chipping goggles are adequate
- They must be lightweight.
- They must have 2 headbands for extra safety.

What is the important feature of the lifeline and belt that form part of the safety equipment on a tanker carrying Noxious Liquid Substances?

- **The line and belt must be fireproof and resistant to the cargo being carried.**
- The equipment must be inspected at least annually.
- The length of the lifeline must be at least double the tankers moulded depth.
- They must always be orange in colour.

What is the inspection criteria for breathing apparatus on a tanker carrying Noxious Liquid Substances?

- **A monthly inspection by a responsible officer and annual inspection and testing at an approved facility.**
- A weekly inspection by the 3rd officer.
- It should be inspected by a class surveyor annually.
- The Chief Engineer inspects all the apparatus monthly.

What is the minimum air duration for an EEBD?

- **10 minutes.**
- 15 minutes.
- As per the vessels Safety Management System.
- It depends on the type of tanker.

What would you do if you looked down a cargo tank and saw a colleague apparently unconscious?

- **Raise the alarm and wait for assistance.**
- Don breathing apparatus and enter the tank.
- Immediately enter the tank to give first aid.
- Look at the permit to see who is down the tank.

When wearing the EEBD how do you support the air bottle?

- **With a strap around the neck supported on the chest**
- You carry it under your arm.
- You put the air bottle into your chemical suit pocket.
- You strap it to your back.

Where would you find information on the cargo to be loaded relating to health hazards?

- **The MSDS (Material Safety data Sheets).**
- SOLAS consolidated edition.
- The Ship Captain's Medical Guide.
- The World Health organisation.

Who has the responsibility to ensure the correct health hazard information is passed to the vessel?

- **The supplier of the cargo.**
- MARPOL inspectors.
- The ship broker.
- The ship owner.

Why is the first 0 to 4 minutes critical for the casualty?

- **After this time permanent brain damage may occur even if the casualty survives.**
- It gives the rescuers some time to get organised.
- The casualty will always die if oxygen is given after this time frame.
- The casualty will always survive if oxygen is given within this time frame.

Why would a vessel be in possession of an NLS (Noxious Liquid Substances) certificate?

- **It's required for any vessel carrying a MARPOL Annex II cargo that does not have a Certificate of Fitness for the Carriage of Chemicals.**
- The certificate is not mandatory and would usually be kept by the owner.
- The certificate is required on a petroleum product tanker.
- The certificate is required when carrying liquid petroleum gas.

7040 – TOTS 1C-Ship Handling and Characteristics

Version 1.0 (7)

At which speed could the ship's bow (or stern) thruster be mechanically damaged?

- **At a ship's speed of more than five knots.**
- At a ship's speed of more than two knots.
- When the vessel is running astern at a speed of less than two knots.
- When the vessel is stopped.

At which speed is the bow (or stern) thruster ineffective?

- **At a ship's speed of more than five knots.**
- At a ship's speed of more than two knots.
- When the vessel is running astern at a speed of less than two knots.
- When the vessel is stopped.

Describe the effect of squat on a vessel in shallow water?

- **It increases the draught, and reduces the steering ability of the vessel.**
- It decreases the draught, and increases the speed of the vessel.
- It improves the vessel's steering ability.
- Squat has no effect in shallow water.

From which direction does the wind have the greatest effect on the vessel's manoeuvring?

- **On the beam, with the vessel in ballast condition.**
- From right ahead.
- From right astern.
- On the quarter, with the vessel in loaded condition.

How could you describe "squat"?

- **The decrease in the ships under keel clearance.**
- The decrease of the draft of your ship.
- The decrease of the speed and draft of your ship in confined waters.
- The decrease of the speed of your ship in confined waters.

How long can the main engine be run on critical revolutions?

- **Long enough for the transfer from one speed setting to another.**
- As long as required.
- As long as the Chief Engineer permits.
- It should not be run on critical revolutions at all.

How should you secure the anchor cable when your vessel is anchored?

- **By using the cable stopper and windlass brake, with slight slack on the cable between the stopper and the brake.**
- By using lashing wires and the brake.
- By using the anchor windlass brake and leaving the motor in gear.
- By using the cable stopper.

If you are sailing in shallow water with 20% UKC (Under Keel Clearance), how will your ship's stopping distance be affected, relative to deep water?

- **Longer than normal.**
- About 1.5 ships longer.
- Shorter than normal.
- The same as for deep waters.

If your vessel is turning to starboard in shallow water with 20% UKC (Under Keel Clearance), what will be the diameter of the turn when compared with deep water?

- **The turn will be wider than a normal deep water turn.**
- The turn will be same as a deep water turn.
- The turn will be tighter than a normal deep water turn.
- The turn will be wider on the port side and tighter on the starboard side than a normal deep water turn.

What action should be avoided during the approach to the anchorage?

- **Crossing in front of an anchored vessel at short distance.**
- All three answers.
- Passing behind an anchored vessel at short distance and slow speed.
- Passing behind another vessel that is manoeuvring towards anchorage.

What actions should be taken immediately by the Officer of the Watch if the vessel starts dragging anchor?

- **Call the Master, inform the engine room that the main engine is required**
- Give one hours notice to the engine room that the main engine is required
- Inform the Port Authority or Coast Guard.
- Inform the vessel behind you that your vessel is dragging anchor.

What could reduce the effect of squat on your vessel?

- **Decreasing the speed of the vessel.**
- Increasing the ship's speed.
- Maintaining your ship's present speed and course.
- Trim the vessel by stern as much as possible.

What data can be found on a ship's Manoeuvring Poster?

- **Turning circles in the ballast and loaded conditions.**
- All three answers.
- Blind sectors for RADARs.
- Emergency contact numbers for CSO.

What equipment must be tested before anchoring?

- **The steering gear, anchor windlass, and main engine ahead and astern.**
- Main and emergency fire pumps.
- The anchor lights and deck lighting.
- The fire alarm system.

What factors will have an effect on the ship's squat and thus the Under Keel Clearance?

- **The ship's speed and block coefficient.**
- The nature of the sea bed, e.g. sandy, rocky, etc.
- The ship's draught.
- The ship's overall length and length/width ratio.

What information can you find from the ship's turning circles posted on the Manoeuvring Posters?

- **Information about the "wheel-over point".**
- All three answers.
- Information about the stopping distances in zig-zag manoeuvres.
- Information on the stopping distance while the vessel carries out a full turn in ballast condition.

What information should be communicated to the Master by the officer in charge when weighing anchor?

- **Cable direction and weight on the cable, number of shackles remaining in the water.**
- Apparent wind direction and force.
- Cable washers are in operation when weighing anchor.
- Quantity and description of seabed material brought up, i.e. Sand, mud, shell etc.

What is a "Williamson Turn", and when is it used?

- **It returns the vessel onto her track, on a reciprocal heading; it is used in a Man Overboard situation.**
- It is a complete 360 degree turn, used when making a lee for a pilot.
- It is an alteration of course of 180 degrees at low speed, and is used when waiting for a pilot.
- It is the use of the rudder to move the ship sideways with no fore and aft movement, used when berthing.

What is the "swinging circle"?

- **It's the arc described by the ship's stern around the anchoring point.**
- It's the arc described by the ship's bow around the anchoring point.
- It's the length of the vessel plus the scope of anchor chain.
- It's the length of the vessel.

What is the correct way of anchoring in deep water?

- **Approach the anchorage at slow speed, walk out the anchor to above the water line, stop the vessel, and then walk out the chain by the windlass to the required amount when ready.**
- Approach the anchorage at slow speed, pay out the anchor for one shackle and drop the anchor by the brake.
- Approach the anchorage at slow speed, stop the vessel and drop the anchor from the hawse pipe.
- Approach the anchorage at slow speed, stop the vessel, pay out the anchor for one shackle and drop out the anchor by the brake.

What means of position-fixing should you use during the approach to anchorage?

- **All available means.**
- GPS, because it is more accurate.
- Radar only, due to the increased workload on the bridge team.
- Visual methods, such as buoys.

What means should be used when securing the anchor for sea?

- **It should be secured by the windlass brake, cable stopper and lashing wire.**
- In good weather it can be secured by the windlass brake only.
- It should be secured by engaging the gipsy and securing the brake.
- It should be secured by the windlass brake, ready for immediate use.

What methods of position fixing should the Officer of the Watch use when at anchor?

- **The Officer Of the Watch should use all means available, including radar/visual ranges and bearings.**
- The Officer Of the Watch should rely only on GPS.
- The Officer of the Watch should verify the position against buoys and navigational aids.
- The Officer should check the position visually against other anchored vessels.

What should be done as a first response to the "Man Overboard Emergency"?

- **Raise the Emergency Alarm, drop the MOB buoy on the side of the man overboard, place wheel hard over on the side of man overboard, place engines on standby.**
- Inform the DPA.
- Inform the Master and await his arrival to the bridge.
- Mark the event on the GPS set.

What should be undertaken before you switch from sea to manoeuvring speed?

- **Give one-hour notice to the engine room before requesting them to slow down from Sea to Manoeuvring Speed.**
- Check the length of the remaining voyage before commencing to manoeuvring.
- Check the ship's ETA.
- Inform the Master and the Pilot.

What should you watch for if you are overtaking a similar size vessel in a narrow channel?

- **The bow of your vessel could be sucked towards the other vessel and the stern pushed away as you pass.**
- Make sure you maintain course because the bow of your vessel could be pushed away from the other vessel.
- That the other vessel is not decreasing its speed.
- That you keep your vessel at full sea speed during overtaking.

Where is the best place to anchor in a crowded anchorage?

- **At the edge of the anchorage.**
- In the centre of the crowded anchorage.
- In the deepest part of the crowded anchorage.
- In the fairway entrance.

Which communications links should be tested before anchoring?

- **Between the bridge and the forecastle, engine room, and steering gear**
- All GMDSS equipment.
- Between the bridge wings and the wheelhouse.
- Between the ship and the port authority or traffic information service.

Which movements of the vessel could increase the ship's draft and reduce the Under keel Clearance?

- **Squat, pitching in a seaway, heeling due to rapid alterations of course**
- Changes in the vessel's passage plan.
- Uncharted areas of seabed.
- Unmarked wrecks or seabed features.

Why should the maximum air draft be carefully calculated?

- **So that the air clearance is known when passing under bridges or electrical cables over channels or rivers.**
- Because the Officer of the Watch requires this information when using the sextant.
- Because this information is always required by port operators.
- This information is required for the GMDSS Log book.

You are sailing on a small coaster, approaching the berth port side at the speed of one knot, and have moderate wind from the starboard side and no tug. How could you assist yourself in the situation described?

- **Let go starboard anchor and pay out the cable controlling the bow as you are approaching the berth.**
- Give the rudder hard a starboard.
- Increase the speed and drop port anchor one shackle in the water.
- Let go port anchor and gradually release the chain as you are approaching the berth.

You meet another vessel port to port in a narrow channel. What interaction could you expect of both ships?

- **The bows of both ships will be pushed away from each other and the aft parts of the vessels could make contact.**
- Only the ship with the deepest draft will be affected.
- Only the ship with the lightest draft will be affected.
- The bows of both ships will be sucked together and the forward parts of the vessels could make contact.

Your vessel has a conventional right-hand propeller. How would you position your rudder before going full power astern?

- **Amidship.**
- Hard a port.
- Hard a starboard.
- Ten degrees to port.

Your vessel is anchoring with the Pilot on board. What will be your main duty as a Watchkeeping Officer?

- **Check the ship's position frequently and give the Master and Pilot the required information.**
- Only follow the Master's orders.
- Strictly follow the Pilot's orders.
- You don't have a specific duty as the Pilot is on board.

7041 – TOTS 1C-Pilotage

Version 1.0 (3)

A ship's ETA is given as local time but clarified in . . .

- **UTC (Universal Time Coordinated).**
- Daylight savings time.
- GMT (Greenwich Mean Time).
- None of these answers.

If the Captain is unable to contact the Pilot Authority directly, he may pass a message via the . . .

- **local agent.**
- local coastguard.
- local ship handler.
- Port State Control.

The Wheel-house Poster contains information regarding . . .

- **specific manoeuvring information / characteristics of the vessel.**
- boarding arrangements for pilots.
- similar information to the Pilot Card.
- the owners / managers standing orders.

The presence of a pilot on the ship does not relieve the master or officer in charge of the navigational watch from their duties and obligations for the safe conduct of the ship.

- **True**
- False

Using the Phonetic Alphabet, how would you transmit H A N D during voice communication?

- **HOTEL, ALPHA, NOVEMBER, DELTA.**
- HOTEL, ALPHA, NOVEMBER, DRAFT.
- HOTEL, ARRIVAL, NOVEMBER, DELTA.
- HUGO, ALPHA, NOVEMBER, DELTA.

What information should be passed to the Pilot when he arrives on the bridge?

- **The Master/Pilot Exchange information card.**
- ISPS procedures.
- The location of life-saving appliances.
- The VHF channels being monitored.

What is a "Pilot Card"?

- **It's a detailed description of the vessel, the main engine characteristics and operational status, and the limitations of various equipment.**
- It contains details of the Pilot's name and the maximum size of vessel he can pilot.
- It's confirmation of the Pilot's attendance on board.
- It's the card the Master signs to enable the Pilot to receive his fee.

What is an abort position?

- **The point in a passage plan where, if passed, the ship can only proceed and cannot return to the origin.**
- An anchoring position.
- The point at which the Chief Engineer can safely reduce speed.
- The position marked on the chart where the Pilots change.

What is described by the "Air draft"?

- **The height of the highest point of the vessel above the waterline.**
- The ship's draft in different terms.
- The wind force at pilot boarding position.
- The wind force on board a vessel when approaching port.

What is your understanding of mixed moorings?

- **Natural or man-made fibre ropes that are used in conjunction with wire ropes.**
- A combination wire.
- Different diameters of wire ropes.
- Some mooring lines that are secured on a winch while others are secured on the bitts.

When a helm order is given by the Pilot, the Helmsman will . . .

- **repeat and execute the order.**
- ask the Captain.
- execute the order.
- wait for the Officer of the Watch to confirm.

When discussing the UKC with the Pilot, what is your understanding of the term "controlling depth"?

- **The minimum depth of water within the limits of a channel.**
- An allowance for squat.
- The maximum draft the vessel can transit at.
- The sounding on a chart close to where you want to transit.

When is the Phonetic Alphabet used?

- **During verbal radio communications.**
- It is never used.
- Only when using email.
- Only when using telex.

When is the final "pilotage passage plan" agreed with the Pilot?

- **As soon as he arrives on the bridge.**
- By VHF prior to his arrival.
- By VTS (Vessel Traffic Services).
- The local agent will keep him informed.

When making verbal contact with the pilot station, the ship will be identified by name and . . .

- **its call sign.**
- its IMO number.
- its nationality (flag).
- its telex number.

When the vessel is under pilotage, where would you find information regarding the bridge team compliment and respective responsibilities?

- **The company's SMS (Safety Management System).**
- The Admiralty Sailing Directions.
- The Master's standing orders.
- The Pilot Card.

When would you use a combination ladder for pilot boarding?

- **When the vessel's freeboard exceeds 9 metres.**
- Only when the pilot requests it.
- When the accommodation ladder faces forward.
- When the pilot ladder is damaged.

Which volume of the Admiralty List of Radio Signals gives contact information for marine pilots?

- **Volume 6.**
- None. Information is only from Guide to Port Entry.
- The Mariner's Handbook.
- Volume 2.

Your gyro compass has failed. Would you . . .

- **inform the Pilot?**
- keep quiet in case it affects the pilot's concentration.
- switch on the radar?
- tell the agent?

7042 – TOTS 1C-Mooring Equipment and Operations

Version 1.0 (10)

At what height should the eyes of the emergency towing wires (fire wires) be positioned?

- **They should be run out and maintained at the waterline or about two metres above it.**
- They should be kept at least two meters above the minimum ballast draught level.
- They should be kept at least two metres above the maximum loaded draught level.
- They should be kept secured not less than two metres above the waterline.

During a vessel's stay on a SPM berth, it is required to continuously monitor the ship's mooring.

- **True**
- False

During ship-to-ship operations, vessels use fenders. What is the purpose of the "primary fenders"?

- **These are large fenders, which float between the ships, and are made to absorb the impact of two ships approaching during berthing and later when sailing together.**
- These fenders are fixed to the bow of the attending Tender Vessel during the transfer of the transfer hose.
- These small fenders (Manila Fender ball) are used by the crew during an emergency.
- These small fenders (tyre type) are used from the deck during an emergency.

During ship-to-ship operations, vessels use fenders. What is the purpose of the "secondary fenders"?

- **These small fenders which are suspended near the ship's bow or stern protect the vessel from ship-to-ship contact.**
- These large fenders are used on the "mother vessel" for separating the two ships.
- These small fenders (Manila Fender ball) are used by the crew during an emergency.
- These small fenders (tyre type) are used from the deck during an emergency.

How could you define "squat"?

- **The increase in the ship's draught of the point of your ship that is closest to the bottom.**
- The decrease of the draught of your ship.
- The decrease of the speed and draft of your ship in confined waters.
- The decrease of the speed of your ship in confined waters.

How often should the mooring winch brake on tankers be tested?

- **At least every 12 months or after each change of brake band lining.**
- At each repair or dry-docking.
- At fixed 12-month intervals.
- At fixed 24-month intervals.

How should you secure the anchor chain while your vessel is being anchored?

- **By using the chain stopper and windlass brake, with slight slack on the chain between the chain stopper and the brake.**
- By using lashing wires and the brake.
- By using the anchor windlass brake and leaving the motor in gear.
- By using the chain stopper.

If you are sailing in shallow water with 20% UKC (Under Keel Clearance), what will be your ship's stopping distance?

- **Longer than normal.**
- About 1.5 ships longer.
- Shorter than normal.
- The same as for deep waters.

If your vessel is turning in shallow water with 20% UKC (Under Keel Clearance), what will be the diameter of the turn when compared with deep water?

- **The turn will be wider than a normal deep water turn.**
- The turn will be same as a deep water turn.
- The turn will be tighter than a normal deep water turn.
- The turn will be wider on the port side and slighter on the starboard side than a normal deep water turn.

Is it acceptable to travel at the speed of four knots in the approach to the berth?

- **It's too fast.**
- It is up to the Pilot to decide.
- It's normal.
- It's sufficient for manoeuvring.

What could minimise the effect of squat on your vessel?

- **Decreasing the speed of the vessel as much as safe navigation permits.**
- Increasing the ship's speed if you are not already on full ahead.
- Maintaining your ship's present speed and course.
- Trim the vessel by stern as much as possible.

What important design feature should be met for construction of the forward ETA (Emergency Towing Arrangement) system?

- **The forward ETA should be capable of being deployed in harbour condition in not more than one hour.**
- The forward ETA should be able to be used for normal berthing of a vessel alongside the terminal.
- The forward ETA should be capable of being deployed in not more than one hour by two crew members.
- The forward ETA should be fitted on the centre line of the vessel.

What important design feature should be met for the construction of the aft ETA (Emergency Towing Arrangement) system?

- **The aft ETA should be pre-rigged and capable of being deployed in a controlled manner in controlled conditions in not more than 15 minutes by one person.**
- The aft ETA should be able to be remote-controlled from the stern during deployment.
- The aft ETA should have a SWL of at least 200 tonnes.
- The aft ETA should have a towing cable of at least 100 metres in length.

What information should be available on board for each mooring rope or mooring wire?

- **A certificate of quality stating the construction, the material and MBL/SWL, the position of the rope on board and the age (time of use of particular rope).**
- The MBL of mooring ropes or wires.
- The SWL of mooring ropes or wires.
- The type of construction material of the mooring ropes and wires in order to avoid using mixed mooring in the same direction.

What is an advantage of split-type mooring drums?

- **All three answers.**
- The mooring wires are not "crushed" under load.
- The use of the split drum uses the winch braking capacity to its maximum.
- There is a smaller chance that the mooring rope will be "jammed" on the storage part of reel.

What is it important to avoid during berthing operations?

- **The mooring rope fouling around the turning propeller.**
- The forward and aft winches working at the same time.
- The mooring stations remaining without communication by hand-held walkie-talkies with the wheel-house.
- The mooring winches set on the correct speed and load setting.

What is it important to discuss during the pre-berthing briefing between the Master and the ship's officers?

- **The specific berthing operation for the forthcoming berthing, the use of tugs and the position of the mooring ropes/wires to be used.**
- How many crew members should be called for berthing operations.
- The size (pulling power) of the tugs to be used during berthing.
- Which officer will be on the forecastle and which on the stern during mooring operations.

What is most important in the conversation between the wheel-house and the mooring stations during mooring operations?

- **The positive identification of stations and positive feedback.**
- To call the mooring stations at agreed intervals.
- To have working and charged walkie-talkies.
- To make conversation only in English.

What is the "Bollard Pull" of the tugboat?

- **All three answers.**
- It is the indicator of a tug's power to push or pull.
- It is the main criteria of a tug and displays the pulling force expressed in tonnes.
- It is the measure of a tug's thrust at zero speed through water with full engine RPM.

What is the CPP?

- **All three answers.**
- The CPP is a special type of propeller that can be rotated around its long axis to change its pitch.
- The CPP is the propeller that has a variable pitch in order to obtain the optimum efficiency in relation to a ship's load so that fuel can be saved.
- The CPP is when the pitch of the ship's propeller is controlled by the hydraulic system working through the propeller shaft.

What is the MBL of mooring ropes or wires?

- **Minimum Breaking Load.**
- Maximum Breaking Load.
- Minimum Braking Length
- Minimum Braking Load.

What is the correct procedure for carrying out the winch brake test?

- **The winch brake lever is tightened by the torque wrench and the torque force is increased/adjusted until the required rendering force of the winch brake is reached.**
- The MBL of the mooring rope/wire is calculated at 80% along with the required hydraulic jack force based on test equipment design characteristics.
- The MBL of the mooring ropes/wires is calculated at 60% along with the required hydraulic jack force based on test equipment design characteristics.
- The winch brake lever is tightened by hand and tested with the hydraulic brake until the required brake force is achieved.

What is the definition of "make fast"?

- **To secure the rope.**
- To act fast.
- To coil the rope around a storage drum.
- To release fast

What is the first task of the Officer in Charge of the mooring station upon arrival at the station?

- **To check the communication with the wheel-house / the Master.**
- To check that all of his crew are available at the mooring station.
- To check that all of his crew are stand-by.
- To check that all of the mooring equipment is ready for mooring.

What is the most dangerous part of mooring operations?

- **The crew staying in rope "snap-back" and "rope-tight" areas.**
- Anchoring in rivers.
- Berthing of a vessel in high tidal ports.
- Berthing with the use of a Pilot and tugs in restricted visibility conditions.

What is the principal purpose of the emergency towing wires (fire wires) on tankers?

- **To be used if there is a fire on board or at the terminal and the ship needs to be pulled off the terminal.**
- To be used for normal unberthing operation of the vessel.
- To be used for shifting the vessel alongside the jetty.
- To be used to extend the towing rope of the tugboat during unberthing.

What is the recommended procedure for securing synthetic rope on a pair of bitts?

- **To make an initial full turn around the outboard bitt and then secure the rope by using a figure eight on both bitts.**
- To make the rope fast in a figure eight on both bitts.
- To make the rope fast on the inner bitt only.
- To make the rope fast on the outboard bitt only.

What is the recommended procedure if your vessel is equipped with "self-stowing automatic tension" winches and your vessel is moored alongside the loading jetty with fixed loading arms?

- **All mooring ropes should be left secured by the manual mooring brakes and all winches should be left out of gear.**
- All winches should be kept in "auto-tension" mode.
- Head and stern line winches should be kept in "auto-tension" mode and breast and spring ropes should be kept manually secured by brakes.
- Spring line winches should be kept in "auto-tension" mode and all other ropes should be kept secured by manual brakes.

What manoeuvring information should be available in the wheel-house?

- **All three answers.**
- The Manoeuvring Booklet.
- The Manoeuvring Poster.
- The Pilot Card.

What moorings should be avoided if leading in the same direction?

- **Mixed mooring ropes of various quality and strength.**
- Mooring wires not fitted with a non-friction link between the wire and tail.
- Mooring wires not fitted with mooring tails.
- Soft mooring ropes of various ages.

What safe practices should be enforced during mooring operations?

- **All three answers.**
- The familiarisation of new personnel with mooring equipment (e.g. winches, brakes, SWL of bitts).
- The mooring station should always be manned with a sufficient number of trained and skilled Officers in Charge and crew.
- The use of Personal Protective Equipment.

What ships are required to be fitted with the aft ETA (Emergency Towing Arrangement) system?

- **Tankers over 20,000 tonnes DWT.**
- All ships over 20,000 tonnes displacement.
- All ships over 20,000 tonnes DWT.
- Tankers over 20,000 tonnes displacement.

What should be done when a vessel is approaching the Pilot Station with the Pilot on board, and the Chief Engineer advises the bridge that engine trouble could be expected during mooring operations?

- **The Master should inform the Pilot, abort mooring operations and drop anchor in a safe location, if possible.**
- The Master should be advised so he can decide if mooring operations should continue.
- The Master should call the Company DPA for advice and instructions.
- The Master should do nothing and continue with berthing as planned.

What should be the type and size (SWL) of the rope stoppers used for securing synthetic mooring ropes during mooring operations?

- **Double tail stoppers with a rope SWL of 25% of the mooring rope SWL.**
- A single tail stopper with 25% SWL of the mooring rope.
- A single tail stopper with 50% SWL of the mooring rope.
- Double tail stoppers with a rope SWL of 50% of the mooring rope SWL.

What should you watch for if you are overtaking a similar size vessel in a narrow channel?

- **The bow of your vessel could be sucked towards the other vessel and the stern pushed away as you pass.**
- That the other vessel is not decreasing its speed.
- That you keep your vessel at full sea speed during overtaking.
- There is nothing particular you need to watch for.

What signals should be displayed by ships engaged in ship-to- ship operations (while berthed together and underway)?

- **Signals for vessels restricted in their ability to manoeuvre.**
- No special signals are required during daylight.
- Signals for vessels in tow.
- Signals for vessels not under command.

What signals should be displayed by ships engaged in ship-to- ship operations while approaching one another?

- **Signals for vessels restricted in their ability to manoeuvre.**
- No special signals are required during daylight.
- Signals for vessels in tow.
- Signals for vessels not under command.

When nylon mooring tails are used in combination with mooring wire, what SWL should they have?

- **37.5% more SWL than mooring wire.**
- 10% more SWL than mooring wire.
- 25% more SWL than mooring wire.
- 50% more SWL than mooring wire.

When should the emergency towing wires (fire wires) be deployed?

- **When the vessel is berthed at the terminals and tugs are available.**
- Only on terminals where the vessel is secured alongside the jetty.
- Whenever the vessel is anchored or berthed at the terminals.
- Whenever the vessel is not in navigation.

When synthetic (other than nylon) mooring tails are used in combination with mooring wire, what SWL should they have?

- **25% more SWL than mooring wire.**
- 10% more SWL than mooring wire.
- 37.5% more SWL than mooring wire.
- 50 % more SWL than mooring wire

When will a vessel need the assistance of Tugs?

- **When it reduces speed, is no longer under total self-control and does not have sufficient sea room to manoeuvre safely without external assistance.**
- Only when it starts the mooring operation (i.e. sending mooring ropes ashore).
- Only when it starts the unberthing operation (i.e. letting go of the mooring ropes).
- Only when the Pilot recommends it to the Master.

Who has the overriding authority for decision-making on berthing over the tide?

- **The Master.**
- The Charterers.
- The Company DPA.
- The ship's Commercial Operator.

Who is responsible for selecting the number of tugs to be used during mooring operations?

- **The Master.**
- The Pilot.
- The Port Authorities.
- The vessel agents in port.

Why are the mooring equipment records important for the safe operation of a vessel?

- **Because they contain all of the information on the construction of the mooring ropes and/or wires and their records of maintenance.**
- Because they are a requirement of the Company Planned Maintenance System.
- Because they are a requirement of the ISM Code, Port State Control and USCG.
- Because they are required by Oil Major Vetting Inspectors.

Why are the non-friction type of mooring tail shackles important to use between mooring wires and tails?

- **Because they are the safest connection between the mooring wire and the soft (i.e. nylon) mooring tail and they extend the lifetime of the mooring eyes on the wires and tails.**
- Because they are made of anti-sparking material and prevent sparking when the mooring wire is paid out or pulled in on board.
- Because they are required by Oil Major Vetting Inspectors.
- They are a requirement of the PSC and USCG for large crude carriers and ULCC ships.

Why is the Master-to-Pilot exchange of information procedure important for the safety of the berthing or unberthing operation?

- **Because it is confirmation that the plan for berthing between the Pilot and Master has been discussed and agreed.**
- Because it contains the berthing plan by the Master.
- Because it contains the berthing plan by the Pilot.
- Because it contains the Pilot Card.

Why should the mooring rope (or wire) not be kept on winch brakes with an additional engaged winch gear?

- **It could cause damage to the winch, drum shaft and wire.**
- It could be broken more easily since the brake could not slip out.
- It is not a sign of good seamanship on board.
- It is not permitted by the terminal and OCIMF mooring equipment guidelines.

You are on a 300,000 tonne DWT ULCC and the terminal is asking if your vessel is provided with SPM mooring according to OCIMF requirements. What is the requirement for fittings that should be complied with?

- **Two tongue-type chain stoppers of 200 tons SWL for a 76mm chain and separate closed fairleads of at least 450 x 600 mm in size.**
- One Smith-type bracket of 150 tonnes SWL for a 76mm chain and one closed fairlead of at least 450 x 600 mm in size.
- One tongue-type chain stopper of 200 tonnes SWL for a 76mm chain and the centre line positioned closed fairlead of at least 450 x 600 mm in size.
- Two tongue-type chain stoppers of 200 tonnes SWL for a 76mm chain and one centrally positioned closed fairlead of at least 450 x 600 mm in size.

You are on a mooring station and you have released the tug rope. Who should you report this to?

- **To the ship's Master.**
- There's no need to report it as the order was received from the bridge.
- To the Pilot directly.
- To the tugboat Master.

You are sailing on a small coaster, approaching the berth port side at the speed of one knot, and have moderate wind from the starboard side and no tug. How could you assist yourself in the situation described?

- **Let go starboard anchor and pay out the cable controlling the bow as you are approaching the berth.**
- Give the rudder hard a starboard.
- Increase the speed and drop port anchor one shackle in the water.
- Let go port anchor and gradually release the chain as you are approaching the berth.

You meet another vessel port to port in a narrow channel. What interaction could you expect of both ships?

- **The bows of both ships will be pushed away from each other and aft parts could make contact.**
- Both ships will proceed on their planned courses and tracks.
- The bows of both ships will be sucked together and the ships could make contact with forward parts.
- This does not apply to tankers.

Your ULCC-sized vessel is anchoring in deep water anchorage. Who should be in charge of anchoring operations on the forecastle?

- **An experienced licensed ship officer.**
- A licensed ship officer.
- An experienced bosun.
- Any experienced crew member appointed by the Master for this operation.

Your mooring winches are fitted with a brake-holding capacity of 60 tonnes. What size mooring ropes should you request?

- **100 tonne SWL.**
- 60 tonne SWL.
- 72 tonne SWL.
- 80 tonne SWL.

Your vessel has a conventional right-hand propeller. How would you position your rudder before going full power astern?

- **Amidship.**
- Hard a port.
- Hard a starboard.
- Ten degrees to port.

Your vessel is anchoring with the Pilot on board. What will be your main duty as a Watchkeeping Officer?

- **Check the ship's position frequently and give the Master the required information.**
- No duty as the Pilot is on board.
- Only follow the Master's orders.
- Strictly follow the Pilot's orders.

Your vessel is berthing starboard side on the terminal jetty. How could you use two tugs for berthing?

- **Make the tugs fast on the ship's flat sides and use them for pushing or pulling when required.**
- Make one tug fast forward through the centre lead for pulling, while the other is positioned on the aft port quarter for pushing.
- Make the tugs fast forward and aft through the centre lead.
- Use the tugs to push the flat side of the vessel but not fast.

Your vessel is leaving the anchorage and you are the Officer in Charge of lifting anchor. Who should you report the status of the anchor to?

- **To the Master, according to his instructions.**
- There's no need to report it when the Master gives the orders.
- To the Chief Officer.
- To the Officer on Duty in the wheel-house.

7043 – TOTS 1C-Emergencies, Drills and Fire Fighting

Version 1.0 (19)

Can the SMPEP and SOPEP be combined into one manual?

- **Yes. Since the requirements are similar, they can be combined into one plan.**
- No. The SMPEP is a dry cargo ship contingency plan while the SOPEP is for tankers.
- No. The SMPEP is for a particular cargo while the SOPEP is a general plan.
- No. They address different regulations so cannot be combined.

Does the paint locker require a fixed fire fighting installation?

- **Yes, a fixed system is mandatory.**
- A portable installation is acceptable if within easy reach.
- It is only important to be able to quickly access some type of extinguishing system.
- The regulations require a portable extinguisher in the locker.

How can the fire stations and lockers be identified on board?

- **Because they are marked with internationally recognised IMO signs.**
- By following the crew reporting to their stations when the emergency alarm sounds.
- It is the duty of the ship's Safety Officer to show the crew where the specific locations are.
- One can check the general arrangement plan and search for the lockers/stations when necessary.

How is it possible to ensure thermal stresses do not affect the compressor when the BA set air bottles are filled?

- **All the answers are correct.**
- By maintaining adequate lubrication at all times.
- By making sure overheating does not take place during operation.
- By making sure the compressor is located in a well ventilated place.

How is the SMPEP different from a SOPEP?

- **It is the contingency plan for ships carrying Noxious Liquid Substances.**
- Different Flag Administrations use different terminology for the same plan.
- It is the same plan, only called by different names based on the individual company.
- SMPEP deals with Oil Pollution only.

How many foam applicators must a fixed foam extinguishing system have on deck?

- **A minimum of 4.**
- 10 in total.
- A minimum of 2.
- At least 4 on each side.

How should you ensure that the fire main does not freeze in cold weather?

- **By continuously bleeding water at extreme ends of the fire main and draining low points.**
- Sea water temperature is higher than the air temperature in cold weather so freezing should not take place.
- The fire line can be kept insulated with Denso tape so it remains warm.
- The fire mains are not likely to freeze since salt water has a lower freezing point.

Is high expansion foam suitable to fight a fire on deck?

- **It is unsuitable for use in open areas in outside locations as it is easily dispersed in light winds.**
- It depends on the expansion ratio of the foam.
- It is better to use water jets to clear the deck of the combustible substance.
- It is suitable since it gives a larger coverage area on deck.

Is it mandatory to have a fire blanket in the galley?

- **Yes, it is mandatory as stated by SOLAS.**
- It depends how committed the ship manager is to safety.
- It is up to the chief cook to decide.
- The Port State Control insists on having one in the galley.

What are SOPEP / VRP / SMPEP / PCSOPEP?

- **Oil Spill Contingency plans based on different regulations.**
- Different plans as required by different nations of the world.
- Manuals required by the ISO 9001 standards.
- Manuals segregated by the company for ease of use.

What are the different types of foam extinguishing media that may be found on board?

- **High expansion, medium expansion and/or low expansion foam**
- High Expansion Foam only.
- Low Expansion Foam only.
- Medium Expansion Foam only.

What details of the main and emergency fire pumps are described on the Fire Control Plan?

- **The capacity of the pumps are clearly indicated.**
- Only the location of the fire pumps are marked.
- The identity of the makers and model number is included.
- There is no such requirement for details.

What does the fire alarm signal sound like?

- **Normally a continuous intermittent ringing of the ship's alarm bell or whistle.**
- As decided by the Master or the ship managers.
- Intermittent long and short blasts on the ship's whistle.
- Seven short blasts followed by one long blast on the whistle.

What information is available on the ship's Fire Control Plan?

- **The firefighting arrangements and location of all firefighting equipment on board.**
- The location of all portable fire-fighting equipment on board.
- The location of all Safety Equipment (LSA & FFA) on board.
- The location of the Fixed fire fighting system fitted on board.

What is the first action that must be taken in the case of a hose burst, pipe work fracture or cargo overflow?

- **All operations must be immediately stopped and the alarm sounded.**
- Quickly repair the leakage and keep operations going to prevent the ship from being delayed.
- Sound the alarm, inform the Master and wait for his instructions.
- Stop all operations and wait for the Coast Guard and terminal people to arrive.

What is the importance of a running hours log on the BA set air bottle filling compressor?

- **To ensure periodical maintenance is carried out as required by the manufacturer.**
- It ensures the crew on board are doing their job by looking at the running hour's log.
- It indicates when a new compressor needs to be ordered.
- The running hours log is maintained as the company periodically requests to see it.

What is the importance of operating instructions posted at local positions for fire-fighting equipment?

- **Ease of operation when it's necessary to use the equipment in an emergency.**
- Instructions are provided because training standards are not uniform globally.
- It helps with the problem of language barriers between different nationalities.
- It is instructed by the Master as part of normal standard procedures.

What is the minimum frequency (for the majority of vessels) for an abandon ship drill and a fire drill for the crew, according to SOLAS Chapter lll?

- **There shall be at least one abandon ship drill and one fire drill every month.**
- The frequency of the drill is left to the discretion of the Master.
- There shall be at least one abandon ship drill and one fire drill every three months.
- There shall be at least one drill every week while the ship is at sea.

What is the minimum required service duration of an EEBD?

- **10 minutes.**
- 12 minutes.
- 15 minutes.
- 20 minutes.

What is the order of priority when dealing with contingencies?

- **Safety of life, the environment and then property.**
- Safety and protection of the environment.
- Safety of property and cargo.
- Safety of the ship and the environment.

What is the purpose of a Safety Meeting on board?

- **To review safety matters and improve the Safety Management System.**
- To discuss operational problems and the maintenance programme.
- To ensure the paperwork, as required by ISM, is completed.
- To receive and review crew complaints with regard to crew matters.

What is the requirement for galleys that have deep-fat cooking equipment?

- **All the answers are correct.**
- There must be an automatic shut off of electrical power when the fire extinguishing system is activated.
- There should be clear instructions for crew concerning manual operation of the system.
- They shall be fitted with an automatic or manual fire extinguishing system.

What is the requirement for manually operated call points on a fixed fire detection and alarm system?

- **They shall be capable of immediate operation at all times.**
- In case the automatic system fails then the manual call point is to be used.
- Manual operation can only be overidden by the automatic system from the bridge.
- No manually operated call points are required since the system is fixed.

What kind of fire alarm detectors are used on a ship?

- **They can be smoke detectors, heat detectors, flame sensors or any combination of the three.**
- A smoke detection system.
- Flame sensors.
- Heat detectors.

What kind of fixed fire-fighting installations may be fitted for the cargo pump room?

- **It should be fitted with a system suitable for machinery spaces of category A.**
- Pump rooms must be fitted with a gas fire extinguishing system only.
- Pump rooms should be fitted with a fixed high expansion foam extinguishing system only.
- The type of system is based on the type of cargo the ship is carrying.

What kind of fixed fire-fighting installations may be fitted for the machinery spaces?

- **Any of the types mentioned.**
- A fixed high pressure water spraying system.
- Fixed Gas Fire Extinguishing system.
- Fixed High Expansion Foam system.

What kind of galley equipment must be fitted with an automatic or manual fire fighting system?

- **Deep fat cooking equipment.**
- Electrical equipment.
- Galley ovens
- Galley stove hotplates

What kind of portable extinguishers are not allowed in the accommodation area?

- **Carbon Dioxide Extinguishers.**
- Dry powder Extinguishers.
- There are no restrictions.
- Water Type extinguishers.

What kind of symbols are used on a Fire Control Plan?

- **Standard IMO symbols and safety signs.**
- The symbols are issued by the respective company.
- The symbols are listed in the Maritime progress catalogue.
- The symbols are suggested by respective P & I Clubs.

What must be covered in the familiarisation of the fixed fire-fighting system on the cargo deck area?

- **All the answers are correct.**
- The fire-fighting medium storage area.
- The location and operation of the fixed fire extinguishing system.
- The remote and local activation points and procedures.

What should the Officer on Watch do in the event of a 'Man Overboard' ?

- **Raise the alarm and turn the ship towards the side of the man overboard.**
- Call the Chief officer.
- Inform The Master and wait for his arrival.
- Turn the ship away from the side of the man overboard.

What size of tanker must be fitted with a fixed deck fire-extinguishing system?

- **Tankers of 20,000 tonnes deadweight and upwards.**
- All tankers must be fitted with a fixed system.
- Tankers of 5,000 tonnes deadweight and upwards.
- Tankers of 50,000 tonnes deadweight and upwards.

What would be the immediate action taken in the event of a bunker spill?

- **Stop all transfer operations, inform, sound the alarm and initiate Emergency response.**
- Stop operations and work out a solution to cover up in collusion with the barge.
- Stop operations, check if anyone has noticed and then take any necessary action.
- Stop operations, investigate the occurrence and inform based on the quantity of the spill.

What would be the required action if a steering gear failure takes place in a busy shipping lane?

- **All the answers are correct.**
- Alert the VTS station and check if it's possible to anchor.
- SECURITE call to other ships in the vicinity so they can take evasive action.
- Switch to emergency steering and reduce speed.

When should a new crew member undergo on-board safety familiarization?

- **As soon as practicable upon joining the vessel.**
- No later than one month after joining the ship.
- No later than two months after joining the ship.
- The training schedule is subject to the time available between operations.

When should maintenance of fire-fighting equipment be carried out on board?

- **Periodically to ensure it is in continuous readiness for use in an emergency.**
- Before the annual inspection or survey to make sure there are no problems.
- Once The Master joins the vessel and then this task doesn't need to be repeated for the rest of the contract.
- When it comes within the duties assigned to the Master from the Planned Maintenance System.

When would you use the Emergency Response Service of the Class Society?

- **For damage stability and longitudinal strength calculations after a casualty.**
- It is useful for any emergency situation on board where Class is affected.
- They are called in for support if the ship is involved in a pollution incident.
- They are contacted if the ship has an accidental release of cargo in USA waters.

Where are the operating instructions for the emergency fire pump located?

- **In a clear, visible place near the emergency fire pump.**
- Adjacent to the international ship/shore connection for use by the shore fire services.
- On the bulkhead next to the ship's fire plan.
- On the ship's bridge and the Engine Control Room.

Where can the fixed fire fighting installation be located?

- **All three answers are correct.**
- It can be located in a room behind the fwd collision bulkhead, used for no other purpose.
- It can be located inside protected space.
- It can be located no more than one deck below open deck and directly accessible from open deck.

Where can you find a description of the alarm signals used on board?

- **On the ship's muster list displayed on board.**
- In the Master's standing orders.
- In the ship familiarisation checklist.
- On a notice at the top of the gangway

Where is the International Shore Connection located?

- **On either side of the accommodation block near the gangway.**
- In the forecastle store where drydock gear is kept.
- In the ship's engine control room.
- Inside the emergency headquarters.

Where is the fire alarm control panel located?

- **On the ship's navigating bridge.**
- At the general muster point.
- In the Emergency headquarters.
- In the Engine Control Room.

Where shall the isolating valve on the fire main be fitted?

- **It shall be fitted in an easily accessible and tenable position outside the machinery space.**
- It can be fitted anywhere as long as it serves the purpose of isolating the section of fire main for machinery space.
- It shall be fitted anywhere on open deck spaces; the ship's main deck being the most appropriate.
- It shall be fitted in the accommodation as the crew are always available in that area.

Where should the main control station of the deck fixed foam fire fighting system be installed?

- **Outside the cargo area, adjacent to the accommodation, readily accessible and operable in event of fire.**
- Inside the accommodation so that it is accessible and operable in event of fire.
- Inside the cargo pump room since all cargo operations are controlled from there.
- On one of the superstructure decks in an easily accessible location for use in event of fire.

Where would you find details of the fire dampers, fire flaps and ventilation on board the ship?

- **In the Fire Control Plan.**
- In the ship's Airconditioning Plan.
- In the ship's Emergency Action Plan.
- In the ship's General Arrangement Plan.

Which of the following elements should generally be included in a fire drill?

- **All of these answers**
- Checking of the fireman's outfits, personal rescue equipment and communications equipment.
- Checking the operation of watertight doors, fire doors, fire dampers and main inlets and outlets of ventilation systems in the drill area.
- The starting of a fire pump, using at least two jets of water

Which party is to be notified first in the event of a probable spill or actual spill?

- **The Coastal State Authority**
- The Company
- The local agents
- The P&I Club

Who approves the ship's Fire Control Plan?

- **The Flag Administration or Recognised Organisation.**
- The Port State Control.
- The ship's Protection and Indemnity Club.
- The ship's Technical Managers.

Who is the person identified in the Vessel Response Plan who acts on behalf of the Owner for oil spill response in USA?

- **A Qualified Individual.**
- A designated person ashore.
- An authorised person.
- The owner's representative.

Why are Company Standing Orders issued?

- **Regulations require that clear instructions should be provided by the Company on key operations.**
- For watchstanders and personnel on night duty so they know what to do.
- It is traditional age old practice followed on every ship and by every Company.
- To relieve the Company of its responsibility in the event of an unfortunate incident.

Why are Vessel Management Meetings held?

- **To promote effective team-working, planning, clarity on tasks, execution and monitoring of jobs.**
- The Management level staff are required by the regulations to hold these meetings.
- To comply with the Company Auditors who periodically check to make sure the meetings have been carried out.
- To provide an opportunity for the vessel management team to discuss the issues raised at the HSE meetings.

Why are environmental issues discussed during HSE or environmental team meetings?

- **To raise awareness among the crew on issues that affect the environment and promote pollution prevention.**
- Because there are so many rigid regulations to comply with.
- For financial reasons as the commercial viability of ship may be affected.
- To discuss tactics in the eventuality of an incident involving environmental activists such as Greenpeace.

Why are proper records required to be maintained for routine tests, checks and maintenance of fire-fighting equipment?

- **For ease of reference and to allow proper and effective planning of testing and maintenance. Also plans and records must be available for inspection when required by the Administration.**
- Because the company auditors can ask for all records of inspections.
- Because the Port State Control is likely to detain the ship if records are unavailable.
- Because the quality system requires records to be maintained to provide evidence.

Why do dry powder installations require periodical checking?

- **Solidification of the compound can occur on board over a period of time.**
- Dry powder evaporates and disappears over time.
- Other extinguishing media are more stable.
- The powder tends to change colour and loose effectiveness.

Why do foam compounds after an initial 3 year period have to be tested annually?

- **All the answers.**
- Because deterioration of the foam can reduce its properties.
- Because it is a Flag requirement.
- Because the foam compounds are likely to deteriorate over time.

Why do tankers require isolation valves to be fitted in the fire main at the poop front and on the tank deck at intervals of no more than 40 metres?

- **To preserve the integrity of the fire main system in case of fire or explosion.**
- Because additional isolation valves help to provide quick access.
- Because deck fire fighting becomes easier when different sections of fire main can be isolated.
- Because it makes it easier to renew damaged sections of the fire main while the fire pump is running.

Why does the ship require an International Shore Connection?

- **To provide a standardised connection between the ship's fire main and shore systems which may have different couplings.**
- It is a requirement of tanker terminals and a practice in the 'wet' trade.
- It is called by that name since the ship trades in International waters.
- It is used only in the drydock to keep the fire main pressurised since the ship's pumps are off.

Why is a fire control plan stored in a prominently marked weathertight enclosure outside the deckhouse?

- **It is to assist shore side fire-fighting personnel in an emergency.**
- It is kept outside in case of an accommodation fire on board.
- It is kept there because the Port State Control Inspector will look for it.
- It is stored so that crew can access the information during a drill on deck.

Why is fire fighting equipment required to be on deck prior to liquid cargo operations?

- **Because all emergencies must be anticipated and the ship must be in constant readiness.**
- It is a guideline issued by the Environment Protection Agency.
- It is a guideline issued in the Civil Liability Convention.
- It is a requirement of the International Oil Pollution Prevention Certificate.

Why is it important to follow a drill with a debriefing session?

- **To review and discuss the lessons learnt and continually improve.**
- It is a requirement of the Company Safety Policy.
- To advise the crew about their faults and assign blame to individuals.
- To motivate crew and make good use of any remaining time left over from the drill.

Why is it important to identify and locate all fire dampers and fire flaps?

- **To easily access them for quick closure in the event of fire.**
- Because the surveyor may ask members of the crew to locate them.
- It is a part of the internal Audit requirements as stated in ISM.
- This is required by The ISM familiarisation checklist.

You are responsible for LSA/FFA maintenance. 4 BA sets are used to conduct an effective fire drill and training programme. What should you do after the drill?

- **Immediately ensure the cylinders are refilled and the equipment is inspected, cleaned and returned to the location to be on standby.**
- If time is short leave the refilling as this will probably be carried out by another crew member.
- Leave the refilling and return them to their location by the next day.
- New cadets are usually responsible for this procedure as part of their training.

7044 – TOTS 1C-Passage Planning and Navigation

Version 1.0 (4)

During appraisal what sources should the Navigating Officer use to gain information?

- **From every available source.**
- From official sources approved by Company regulations.
- From official sources approved by the relevant Hydrographic Office.
- From sources agreed by the Master.

If a vessel is fitted with an unapproved electronic chart system, how can it be used in Passage Planning?

- **It should not be used for Passage Planning.**
- Any electronic chart system can be used with confidence, even if it has not been officially approved.
- It can be used fully, but positions must be plotted on a paper chart.
- It can be used in the appraisal, but this must then be checked by another method.

If, during the Appraisal it is realised that one or more up to date charts at the best possible scale are not available on board, what action should be taken?

- **The necessary charts should be requisitioned.**
- Cancelled charts can be used with caution.
- The next best scale charts should be used.
- The route should be changed to use only acceptable charts.

One of the earliest tasks in Passage Planning is to calculate the total distance. Why is this information needed at an early stage?

- **To calculate the ETA at the destination, and the consumables such as fuel and water required to reach it.**
- To assess the weather situation at the destination.
- To enable the catering department to plan food consumption in advance.
- To inform the crewing office in good time with regard to crew changes at the destination.

What actions must the vessel be able to safely take at the abort point?

- **Turn around, stop in safe water, or anchor.**
- Pick up the Pilot, providing a lee if necessary.
- Reduce speed to minimum steerage way.
- Test the main engine ahead and astern.

What are the four stages of Passage Planning?

- **Appraisal, planning, execution, monitoring.**
- Appraisal, preparation, approval, execution.
- Pilotage outwards, sea passage, anchoring, pilotage inwards.
- Planning on paper, preparing the route on the charts, discussion with the Master, correction of any errors.

What information should you record in the Bell Book and the Deck Log Book about the navigation of the vessel?

- **Sufficient information to enable the reconstruction of the ship's movement if required at a later date.**
- Only the information that is instructed by the Master to be recorded.
- Only the manoeuvring changes and status of the main engine should be recorded.
- Only the times that are important for the charterers' commercial operations (like EOSP, Pilot on board, all fast etc should be recorded).

What is achieved by carefully selecting "no-go areas" on the nautical charts required for the ship's passage?

- **It indicates all areas where the vessel cannot go with present draft.**
- It indicates all areas where the vessel cannot go on deepest draft.
- It indicates all areas where the vessel is advised not go on deepest draft.
- It indicates all areas where the vessel should not go when loaded.

What is achieved by the "appraisal" phase of passage planning?

- **Appraisal is the part of passage planning where all pertinent information is gathered and the firm foundation for the plan is built.**
- Appraisal is the part of passage plan where all the information on required charts is collected, and it is confirmed that all required charts are available on board.
- Appraisal is the part of the passage plan that needs to approved by office.
- Appraisal is the part of the passage plan where the Master decides which route the vessel will take between two ports.

What is the "tidal window"?

- **It is the times between which the tide has achieved a required height, and it is safe for the ship to transit an area.**
- It is the maximum time permitted for the vessel to reach the berth from safe anchorage in a tidal area.
- It is the time required for the vessel to reach the berth from safe anchorage in a tidal area.
- It is the time when the vessel can go alongside and start discharging cargo to reach the required safe draft.

What is the UKC (Under Keel Clearance)?

- **It is the vertical distance between the seabed and the deepest part of the ship.**
- It is an important part of the "pilot station to berth" section of the passage plan for all tankers.
- It is an important part of the "pilot station to berth" section of the passage plan for loaded tankers.
- It is an important part of the "pilot station to berth" section of the passage plan.

What is the main consideration when deciding the frequency of position fixing?

- **The vessel should be clear of dangers, and have enough time to take avoiding action between fixes.**
- Master's orders, and Company regulations.
- The availability of different methods of position fixing, and the time required to plot an accurate using them.
- The frequency should take into account the workload on the Bridge Team.

What is the meaning of the term "abort point"?

- **It is the latest position where the vessel may abandon the planned passage and either return or stop safely.**
- It is any position on the passage where the vessel can abandon the planned passage.
- It is not a phrase which should be used within the plan.
- It is the safe area, specified in the plan and marked on the chart, where the vessel can wait if the plan is abandoned.

What is the meaning of the term "margin of safety"?

- **It is a line parallel to the track, within which the vessel will always be in safe water.**
- It is another phrase meaning "no-go areas".
- It is not a phrase which should be used within the plan.
- It is the distance from the nearest danger.

What is the meaning of the term "parallel index"?

- **A radar-based constant update of a ship's cross track error.**
- A navigational instrument used for plotting bearings on a chart.
- Information on radar about the ship's position in relation to a fixed object ashore.
- Information on radar about the ship's position in relation to a ship's longitudinal track error.

What is the primary reason for the ship's passage plan to be discussed between the navigating officers and the Master before the beginning of a voyage?

- **In order to fully inform all navigation officers about the particular voyage, its specific issues and concerns.**
- In order to inform the Chief Officer and the 3rd Officer about the particular voyage and its specific issues and concerns.
- In order to inform the officers and crew about the destination port.
- In order to satisfy the Company's Navigational Procedures requirement.

What precautions must be taken when using floating marks for navigation?

- **The identity and position of the mark must be accurately established.**
- Floating marks should only be used within port limits.
- The mark should only be used on the advice of the pilot.
- Use visual bearings only, not radar information.

What type of markers can be used on nautical charts?

- **Soft 2B pencil and permanent magenta ink.**
- Ball-point pens.
- HB pencils only.
- White correction fluid.

What would you use a gnomonic chart for during Passage Planning?

- **To plot Great Circle courses and waypoints.**
- To plot Rhumb Line courses and waypoints.
- To show sections of the route for which suitable passage charts were not available.
- To show the complete passage conveniently on one chart.

When is the UKC (Under Keel Clearance) calculation required to be completed?

- **The UKC should be calculated and applied for all ships loading and discharging from ports, irrespective of any local regulations and requirements.**
- The UKC should be calculated for all tankers while arriving and departing from US ports.
- The UKC should be calculated for loaded tankers while sailing to US ports.
- The UKC should be calculated for tankers while sailing to US ports.

When should highlighters and permanent markers be used on nautical charts?

- **They should not be used.**
- For corrections according to Notices to Mariners.
- Only in areas of the chart not used directly for navigation.
- When permitted by the Master.

Which scale navigation charts should be prepared for use on the passage?

- **The largest scale available.**
- The most recently corrected charts available.
- The smallest scale available.
- Whatever scale will conveniently show the intended route.

Who is responsible for ensuring that a proper Passage Plan is drawn up?

- **The Master**
- It depends on the Company requirements.
- The Second Officer
- The ship owner

Why should the passage plan be updated on receipt of Navarea Warnings or T&P notices during the voyage?

- **Because they may contain important and urgent navigational information and warnings that were not available on board at the beginning of the voyage.**
- Because they may contain important commercial information regarding the availability of berths, tugs, pilots etc.
- Because they may contain instructions from owners or charterers regarding the voyage.
- Because they may contain urgent weather warnings.

Why would you use risk assessment techniques during Passage Planning?

- **To reduce, and where possible eliminate risks from navigational hazards.**
- There is no requirement for risk assessment to be used.
- To comply with Flag State or Company regulations.
- To reduce, and where possible eliminate risks to the Bridge Team personnel during the passage.

7045 – TOTS 1C-Navigating in Restricted Visibility

Version 1.0 (2)

A bell or gong or other device having similar sound characteristics shall produce a sound pressure level not less than how many dB at a distance of not less than 1 metre from it?

- **110dB.**
- 100dB.
- 1120dB.
- 125dB.

At what intervals should a power-driven vessel under way and making way through the water, sound one prolonged blast in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

At what intervals should a power-driven vessel underway but stopped and making no way through the water sound two prolonged blasts in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

At what intervals should a vessel being towed sound one prolonged followed by three short blasts during periods of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

At what intervals should a vessel not under command, or a vessel restricted in her ability to manoeuvre, a vessel constrained by her draft, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel, sound three blasts in succession when in or near an area of restricted visibility?

- **At intervals of not more than two minutes.**
- At intervals of not less than five minutes.
- At intervals of not less than two minutes.
- At intervals of not more than one minute.

If the vessel is fitted with two whistles that are spaced apart more than 100 metres, how should the whistles be arranged?

- **They should be arranged so that they are not sounded simultaneously.**
- They should be arranged so that only one can be used at a time.
- They should be arranged so that the forward whistle always sounds first and then the aft whistle sounds with a 10 second delay.
- They should be arranged so that they are sounded simultaneously.

In which publications can specific guidance and instructions on navigation in restricted visibility be found?

- **In the Bridge Procedures Guide, in COLREGS and in the Mariners' Handbook.**
- In the Bridge Procedures Guide, in Bridge Team Management and in the Passage Planning Guide.
- In the Deck Log Book and in the Company Navigation Manual.
- In the Master's Night Order Book.

On ships greater than 20 metres in length, the diameter of the ships bell mouth used for sound signals shall not be less than how many mm?

- **300mm.**
- 250mm.
- 275mm.
- 350mm.

What action should you take when you hear the fog signal of another vessel apparently forward of the beam, or when you cannot avoid a close quarters situation with another vessel forward of the beam?

- **Reduce speed to the minimum and if necessary take all way off and in any event navigate with extreme caution.**
- Identify the other vessel by AIS, and then establish contact with the vessel to agree the intended course of action to avoid the danger.
- Identify the other vessel by VHF using the ships call sign, and then establish the other vessels intended course of action to avoid the danger.
- Sound five short blasts on the ships whistle to indicate doubt about the others intention.

What actions required by the Collision Regulations should the Officer of the Watch comply with during periods of restricted visibility?

- **All three answers.**
- To have the ship's engines ready for immediate manoeuvres.
- To proceed at a safe speed.
- To sound the fog signals.

What additional sound signal may a pilot vessel sound when engaged in pilotage duties?

- **Four short blasts.**
- Four long blasts.
- Three short blasts.
- Two short blasts followed two long blasts.

What additional sound signal may a vessel over 100 meters in length make at anchor in an area of restricted visibility?

- **Three blasts in succession, one short one prolonged and one short.**
- Five short blasts in succession.
- Three blasts in succession, one prolonged and two short.
- Three blasts in succession, two short and one prolonged.

What is the meaning of the term "restricted visibility"?

- **Any condition in which visibility is restricted by fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.**
- Where visibility is restricted to within five nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.
- Where visibility is restricted to within three nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.
- Where visibility is restricted to within two nautical miles due to fog, falling snow, heavy rainstorms, sandstorms or any other similar weather conditions.

What sound signal should a power-driven vessel making way through the water sound during periods of restricted visibility?

- **One prolonged blast**
- one prolonged blast followed by two short blasts
- Two prolonged blast.
- Two prolonged blasts followed by one short blast.

What sound signal should a power-driven vessel underway but stopped and making no way through the water sound during periods of restricted visibility?

- **Two prolonged blasts.**
- Four prolonged blasts.
- One prolonged blast.
- Two prolonged blasts followed by two short blasts.

What sound signal should a towed vessel sound during periods of restricted visibility?

- **One prolonged blast followed by two short blasts.**
- Three prolonged blasts.
- Two prolonged blasts followed by two short blasts.
- Two prolonged blasts.

What standard sound signal must a vessel over 100 meters in length make at anchor in an area of restricted visibility?

- **Sounding the anchor bell forward immediately followed by sounding the gong aft.**
- Sounding the anchor bell aft immediately followed by sounding the gong forward.
- Sounding the gong aft followed immediately by sounding the anchor bell forward.
- Sounding the gong forward immediately followed by sounding the anchor bell aft.

When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite what sound signal should they sound when in or near an area of restricted visibility when underway and making way.

- **One prolonged blast.**
- Two prolonged blasts followed by two short blasts.
- Two prolonged blasts.
- Two short blasts

When is Rule 19 of COLREGS applicable, and to which ships?

- **This rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.**
- This rule applies only to power-driven vessels navigating in an area of restricted visibility.
- This rule applies to all power-driven vessels navigating in an area of restricted visibility.
- This rule applies to all vessels sailing in an area of restricted visibility.

When restricted visibility is encountered or is expected, who should be informed?

- **The Master.**
- The Company Operator / DPA.
- The Helmsman.
- The Lookout.

When two mast head lights are carried on a power driven vessel what height shall the after one be above the forward mast head light?

- **4.5 metres.**
- 2.5 metres.
- 3.5 metres.
- 4.0 metres.

Where is the ship-specific information about restricted visibility declared?

- **In the Master's Bridge Standing Orders.**
- In the Company's Navigation Standing Orders.
- In the Deck Log Book.
- In the Master's Night Order Book.

7046 – TOTS 1C-Heavy Weather Navigation

Version 1.0 (5)

If the vessel is in port, what is the best option if a tropical cyclone is passing within 80 nautical miles or so?

- **The best option is to sail out to sea in time to avoid the worst of the storm.**
- The best option is to stay alongside the well fendered berth that is on the lee-side of the wind.
- The best option is to stay at anchorage and make the vessel as heavy as possible by adding as much full ballast as possible.
- The best option is to stay in sheltered waters, and, if possible, on special moorings.

In relation to navigating in ice conditions, the term beset means that the ship has found itself in a position where it is surrounded by ice and is unable to move. True or False?

- **True.**
- False.

The vessel is in port and is due to sail in a few hours, however severe weather is forecast that may endanger the vessel. What should the Master do?

- **Based on risk assessment, the Master should make the decision on whether the vessel is in a "Go" or "No Go" situation and therefore determine if the ship should sail or remain in port.**
- The Master should call the company's operator for advice and further instruction.
- The Master should evaluate the weather conditions and try to solve the problem without causing a commercial delay for the vessel.
- The Master should make the decision based on available weather reports/forecasts and then wait for the company's advice.

What action should you take if you find yourself within the dangerous semi-circle of a tropical storm in the Southern hemisphere and the wind is backing?

- **You should put the wind on the port bow and proceed at full available speed, altering the course to port as the wind backs.**
- You should bring the wind on to the port quarter and proceed with all available speed turning to starboard as the wind changes direction.
- You should put the wind on the port quarter and proceed as fast as possible in a southerly direction.
- You should put the wind on the starboard quarter and proceed as fast as possible in a northerly direction.

What are the characteristics of "trade winds" in the Southern hemisphere?

- **They have a south-easterly direction and a steady average wind speed of force 4 to 5**
- They have a north-easterly direction and a steady average wind speed of force 4 to 5
- They have a north-easterly direction and mainly a variable wind speed of above force 6.
- They have a south-easterly direction and mainly a variable wind speed of above force 6.

What are the main types of "sea ice" that ships encounter?

- **Drift and fast ice.**
- Glacier ice.
- Growlers.
- Icebergs and packed ice.

What are the signs of ice in the vicinity of the vessel?

- **All three answers.**
- Abrupt moderation of the sea and swell.
- Ice blink.
- Sighting of small fragments of ice.

What can cause vessel icing on ships in freezing conditions?

- **All three answers.**
- Fog.
- Rain or wet snow.
- Sea spray or water breaking over the ship.

What impact could occur at a high speed while sailing through the ice?

- **The ship's hull could be damaged by collision with floes.**
- A vessel navigating at a high speed through the ice could have problems with icing of the sea-boxes (water suctions).
- A vessel navigating at a high speed through the ice could not keep the track through the passage opened by the icebreaker.
- The ship could "run aground" on thick ice and not be able to back up if required.

What impact could vessel icing have on the ship?

- **Vessel icing may impair the stability and safety of the ship.**
- Vessel icing may impair the manoeuvring of the vessel.
- Vessel icing may impair the security of the vessel.
- Vessel icing may impair the visibility from the bridge

What is considered "new ice"?

- **Frazil ice, Slush, Sluga and Grease ice with a thickness of up to 2.5 cm**
- First-year ice that has a thickness of between 30 to 70 cm.
- First-year ice that has less than 30 cm of thickness.
- Young ice that has less than 10 cm of thickness.

What is it important to know in order to decide the best course of action if the storm is suspected to be in the vessel's vicinity?

- **It is important to know the bearing of the centre of the storm and the path of the storm.**
- It is important to know the difference of pressure between the centre of the storm and the position of the vessel to calculate the strength of wind.
- It is important to know where the nearest port of refuge is.
- It is important to monitor the barometric pressure and wind strength and direction.

What is of vital importance when encountering a tropical cyclone in the vessel's vicinity?

- **It is imperative to avoid passing at 80 Nm or closer to the centre of the storm.**
- It is imperative to carry out an emergency drill in order to prepare for a worse case scenario.
- It is imperative to sail in the opposite direction of the storms track
- It is imperative to send out a SECURITE message to all ships in the vicinity warning them of the approcahing eye of the storm.

What is the "dangerous semicircle" of a tropical storm in the Southern hemisphere?

- **The dangerous semicircle is the section of the tropical cyclone that is on the port side of the cyclone path and its eye.**
- The dangerous semicircle is the section of the tropical cyclone that is 250 Nm fwd of the eye of the storm directly in her path.
- The dangerous semicircle is the section of the tropical cyclone that is 80 Nm fwd of the eye of the storm directly in her path.
- The dangerous semicircle is the section of the tropical cyclone that is on the starboard side of the cyclone path and its eye.

What is the "navigable semicircle" of a tropical storm in the Northern hemisphere?

- **The navigable semicircle is the section of the tropical cyclone that is on the port side of the cyclone path and its eye**
- The navigable semicircle is the section of the tropical cyclone that is 250 Nm aft of the eye of the storm directly behind her path
- The navigable semicircle is the section of the tropical cyclone that is at least 80 Nm aft of the eye of the storm behind on her path
- The navigable semicircle is the section of the tropical cyclone that is on the starboard side of the cyclone path and its eye

What is the Master required to do by SOLAS when his ship encounters dangerous ice?

- **He is required to communicate the fact to all ships in the vicinity by all means available**
- He is required to communicate the fact to his DPA
- He is required to exhibit lights and shapes declaring that his vessel is restricted in her ability to maneouvre.
- He is required to stop his ship and communicate with the nearest coastal state.

What is the difference between hurricanes and typhoons?

- **A hurricane develops in the Atlantic or East Pacific while Typhoons develop in the West Pacific.**
- A hurricane develops in the western Pacific and a typhoon in the central and eastern Pacific.
- A hurricane is a storm with winds of over 64 knots and a typhoon is between 48 and 56 knots.
- A hurricane is a tropical storm and a typhoon is a storm of the mid-latitudes (above 40 degrees).

What is the first basic ship-handling rule in ice concentrations?

- **Keep moving, even if very slowly.**
- Try to avoid icing of the ship's superstructure.
- Try to keep the vessel sailing only after the icebreakers.
- Try to keep the vessel sailing only in convoys through the ice.

What is the first principle for successful passage through the ice?

- **To maintain freedom of manoeuvre.**
- To maintain radio watch in order to receive all ice reports and information.
- To sail through the ice during the day only.
- To stay out of the ice as much as possible.

What is the general freezing point of seawater?

- **About - 2º C**
- About - 4º C
- About - 5 º C
- About 0º C

What is the indication of an approaching depression?

- **A falling barometer.**
- A rising barometer.
- The wind is rising and the temperature is falling.
- The wind is subsiding and the temperature is rising.

What is the meaning of the term "glacier ice"?

- **Ice originating from a glacier, either on land or floating on the sea as icebergs, bergy bits, growlers, or ice islands.**
- Any ice originating from fresh water melted from glaciers.
- Any ice originating from fresh water.
- Any ice originating from large icebergs made of fresh water.

What is the meaning of the term "growler"?

- **A rounded piece of glacier ice, often transparent but appearing green or black in colour, extending less than 1 metre above the sea surface and normally occupying an area of about 20 m2.**
- A smaller piece of iceberg, extending about 5 metres above the sea surface, normally occupying an area of about 100 m2.
- A special type of iceberg that is only found in the Antarctic that is about 5 metres in height and green or black in colour.
- An iceberg that grows from packed sea ice and extends less than 1 metre above sea level.

What is the meaning of the term "ramming"?

- **Attempting to break the ice by repeatedly driving the ship as far forward as possible, backing the ship out and repeating the process.**
- The noise caused by icebreakers breaking the ice.
- The unpleasant noise caused by the propeller grinding through the ice when the vessel is running ahead in the ballast condition.
- The unpleasant noise caused by the propeller grinding through the ice when the vessel is running astern.

What is the meaning of the term "sea ice"?

- **Any form of ice found at sea which originated from the freezing of seawater.**
- Any form of ice that is floating on the sea.
- Any form of ice that is found on the open sea irrespective of the origin of the ice.
- Any form of ice that is free floating away from the ice.

What is the meaning of the term the "wind is backing"?

- **It's an anticlockwise change in the direction of the wind (i.e. from northwest to west to southwest).**
- It's a change of direction in the wind from astern to ahead.
- It's a change of direction in the wind from ahead to astern.
- It's a clockwise change in the direction of the wind (i.e. from northeast to east to southeast).

What is the meaning of the term the "wind is veering"?

- **It's a clockwise change in the direction of the wind (i.e. from northeast to east to southeast).**
- It's a change of direction in the wind from astern to ahead.
- It's a change of direction in the wind from ahead to astern.
- It's an anticlockwise change in the direction of the wind (i.e. from northwest to west to southwest).

What is the most serious danger to a "beset" vessel (i.e. a vessel that has become surrounded by ice and is unable to move)?

- **Pressure from the ice that can crush the hull or nip off the bottom.**
- Danger of icebergs capsizing the vessel.
- Danger of running aground as it's carried by the moving drift ice.
- Danger of the ice causing damage to the rudder and the propeller.

What precautions should be carried out on deck before encountering heavy weather?

- **All loose objects and equipment should be secured.**
- All crew should be given heavy weather clothing, including raincoats and boots.
- All crew should be instructed not to leave their cabins.
- All equipment should be greased.

What problems can occur on board from icing from fresh water?

- **Ice can accumulate on the rigging, and radar, and radio equipment failures can occur due to ice on the aerials.**
- Fresh water supplies will stop functioning
- Heating systems will trip out on overload
- Ice can prevent the opening of all accommodation openings.

What should be the most important task when navigating in the vicinity of strong tropical storms (e.g. cyclones, hurricanes, typhoons)?

- **To stay as far as possible from the path forward of the tropical storm.**
- To frequently monitor the ship's barometer.
- To inform the crew to secure all loose objects and equipment to avoid them being lost or damaged.
- To monitor carefully all tropical storm area weather reports.

What type of ice formation is most dangerous for navigation?

- **Icebergs.**
- New ice.
- Old ice.
- Sea ice.

What warnings and instructions should be given before the vessel encounters heavy weather?

- **All three answers.**
- Instructions to rig safety lines where necessary.
- Warning and instructions should be issued to the crew on deck access.
- Warning and instructions should be issued to the engine room crew.

Where can the Master or Officer of the Watch find details regarding ice reporting schemes and the radio stations transmitting them?

- **In the Admiralty List of Radio Signals, Volume 3.**
- In the Bridge Procedures Guide.
- In the Company SMS manuals.
- In the Mariner's Handbook.

Why should the Master regulate the ship's speed in heavy weather?

- **All three answers.**
- To avoid the possibility of damage to the cargo.
- To avoid the possibility of damage to the ship.
- To prevent needless expenditure of fuel.

You are in the vicinity of a tropical storm in the Northern hemisphere and the wind is remaining in a steady direction. If the vessel is in the navigable semi-circle, you should continue the voyage in the planned direction. True or False?

- **False.**
- True.

7047 – TOTS 1C-General Tanker Section

Version 1.0 (2)

A Permit-to-Work should specify the period of validity. According to the Code of Safe Working Practices, this period should not exceed how many hours?

- **24 hours.**
- 12 hours.
- Eight hours.
- Six hours.

A single work permit is issued . . .

- **only for the work to be undertaken.**
- for a variety of tasks.
- to keep paperwork up-to-date.
- to satisfy auditors.

According to the Code of Safe Working Practices, a Permit-to- Work does not in itself make the job safe but contributes to measures for safe working.

- **True**
- False

Before cargo or ballast operations commence, the Master (or his representative) and the terminal representative should do what?

- **All three of the above**
- Agree in writing on the action to be taken in the event of an emergency during cargo or ballast handling operations
- Agree in writing on the transfer procedures, including the maximum loading or unloading rates
- Complete and sign the Ship/Shore Safety Checklist.

Critical Operations Check Lists are to be used for what purpose?

- **To ensure that critical equipment is operated, started or restarted correctly so that it will not result in a hazardous situation developing or compromise critical functions.**
- To ensure that all company procedures are followed at all times.
- To ensure that every item of equipment on board is operated according to the manufacturer's instructions.
- To ensure that the vessel passes external ISM audits by showing the auditor that Section 10 of the ISM Code is fully complied with.

Hazardous materials such as chemicals and oil will have a Material Safety Data Sheet (MSDS) available. Which of the following is NOT likely to be included in a MSDS?

- **The name of the vessel.**
- The correct storage.
- The health effects.
- The physical data (e.g. the flashpoint).

Persons can conduct safety inspections of areas outside of their direct control (e.g. the Chief Officer inspecting the engine room).

- **True**
- False

Shipboard management meetings should not be used for what purpose?

- **As a non-constructive platform for complaining to shore management about various issues.**
- As a constructive method of raising concerns with shore management about activities affecting the safe and efficient operation of the ship.
- As a method of relaying information concerning the safe and efficient operation to shore management.
- As a structured discussion between members of the shipboard management team relating to the safe and efficient operation of the ship.

The Code of Safe Working Practices requires that any faults relating to cargo handling equipment, testing instruments and alarm systems must be . . .

- **reported immediately.**
- repaired immediately.
- repaired in dry dock.
- reported when cargo operations are completed.

The International Shore Fire Connection is used for what purpose?

- **To enable the water fire main onboard to be connected to an external water supply from shore.**
- To enable carbon dioxide to be used from shore to fight engine room fires.
- To enable the foam line onboard to be connected to an external foam supply from shore.
- To enable the ship to assist in extinguishing a fire ashore.

What happens to a Permit-to-Work upon completion of work?

- **It is cancelled by the responsible officer.**
- It is cancelled by the person carrying out the work.
- It is left in place in case further work has to be carried out.
- It is transferred to another location where similar work is to be carried out.

What is the correct method of closing off unused cargo and bunker connections?

- **All unused connections must be closed and blanked. Blank flanges are fitted and are to be fully bolted.**
- All unused connections must be closed and blanked. Blank flanges are fitted and have every second bolt secured by a nut.
- Unused connections may be left unblanked so long as two valve separation exists between the connection and the line in use.
- Unused connections on the shore-side of the vessel must be closed and fully blanked. The offshore-side may be left unblanked.

What should happen to deck scuppers during periods of heavy rain when engaged in cargo or bunker handling?

- **One or two scuppers may be opened temporarily to allow rain water to escape but must be closed as soon as possible and in the event of an deck oil spill.**
- All deck scuppers must be kept fully closed at all times and never opened until cargo and bunker operations are completed.
- Only deck scuppers on the outboard side of the vessel are to be opened to release the rain water. When the rain stops, the scuppers can then be replugged.
- Open all deck scuppers to release the rain water. When the rain stops, the scuppers can then be replugged.

What should the Company do with accident and incident reports after they are sent from the ship to the office?

- **Closely examine and analyse them so as to determine the root causes or management failures, then investigate methods on how to avoid a similar accident.**
- Closely examine and analyse them so as to determine the direct causes of the accident or incident.
- Closely examine and analyse them to determine who is to blame on board the ship and take the relevant disciplinary action.
- Thank the vessel for the time taken to complete the accident/incident form and then file the report away for future reference.

What should you do if you notice a serious safety deficiency when on board?

- **Report it to a senior officer or the Safety Officer and then ensure steps are taken to rectify the problem without delay.**
- Attempt to rectify the defect on your own.
- Ignore it.
- Make a note of the problem and then inform the Safety Committee at the next Safety Meeting.

What would be the minimum recommended firefighting equipment to have available when alongside a terminal?

- **Fire hoses with spray/jet nozzles run out forward, aft and adjacent to the manifold in use. Also a dry chemical extinguisher near the manifold.**
- Fire hoses with spray/jet nozzles forward and aft of the manifold and a fire blanket adjacent to the manifold in use.
- Fire hoses with spray/jet nozzles or fire extinguishers adjacent to the manifold in use.
- One fire hose and one dry chemical extinguisher near the manifold in use.

When are items on the ship/shore safety check list checked when a vessel is at a terminal?

- **Before cargo or ballast operations commence and then at regular intervals until hoses are disconnected.**
- During cargo operations.
- Immediately upon berthing and before starting cargo operations.
- Upon commencing cargo operations.

When handling oil or waste oil, direct contact should be avoided with the skin by using gloves. What other method can be used to protect the skin?

- **A barrier cream.**
- A moisturiser cream.
- Talcum powder.
- Vaseline.

Where does the Code of Safe Working Practices recommend guidance on general precautions be taken from?

- **All of these answers.**
- Safety in Oil Tankers.
- The Company's Safety Management System.
- The International Safety Guide for Oil Tankers and Terminals.

Where should the ship's emergency fire control plans be located?

- **In a prominently marked weather-tight enclosure outside the accommodation block.**
- At the bottom of the gangway.
- In a prominently marked wallet contained within the cargo control room.
- With the Deck Officer of the Watch at all times.

Which of the following is NOT likely to be an agenda item in a safety committee meeting?

- **Vessel performance.**
- Emergency / pollution drills conducted.
- Matters outstanding from the previous meeting.
- Safety equipment under repair.

Which of the following items can be described as critical machinery?

- **All of these answers.**
- Generators.
- Steering gear.
- The main engine.

Which of the following items may NOT be considered a critical operation?

- **Switching on the radar.**
- Changing over from heavy fuel to diesel.
- Lowering and raising lifeboats.
- Starting the emergency generator.

Which of the following items of equipment would be found in a chemical storage/handling area?

- **Safety goggles.**
- A dust mask.
- Ear plugs.
- Safety shoes.

Which of the following questions could be included in the Safety Officer's Inspection Check List?

- **All of these questions.**
- Are all openings through which a person could fall suitably fenced?
- Does the crew in the area have any safety suggestions to make?
- Is protective equipment and clothing in good condition and being used correctly?

Which of the following statements apply to a storage area where chemicals are kept?

- **It must be kept well ventilated and secure.**
- It must be kept under surveillance by closed circuit television cameras.
- It must be located within the engine room workshop only.
- It must be stored in the main deck store only.

Which of the following statements reflects the importance of near-miss reporting?

- **Lessons can be learned from near misses and by analysing them, steps can be put in place to avoid accidents occurring in the future.**
- If a ship reports many near misses, the Company will consider that the ship is operating safely.
- If a ship reports very few near misses, the Company will be under the impression that the ship is operating safely.
- The Company has set targets to meet and the number of near misses sent in is essential to meet those targets.

7048 – TOTS 1C-The Flammability Diagram

Version 1.0 (15)

A gas turbine with afterburner will produce similar inert gas to an independent inert gas generator. What level of oxygen might typically be delivered?

- **1.5-2.5% by volume.**
- 5% by volume.
- 8-10% by volume.
- Approximately 12% by volume.

After inerting a cargo tank, the oxygen content was found to be 7.5%, and the hydrocarbon gas content 2%. Using the Flammability Diagram, would you consider the tank to be safely inerted before gas freeing?

- **Yes, the atmosphere is below the critical dilution line.**
- No, inerting must be continued.
- No, the tank atmosphere is in the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

At high concentrations, hydrocarbon gas has a high relative density. How will this affect the sampling procedure?

- **The atmosphere must be sampled at different depths, because the gas tends to layer and form pockets.**
- Air may be drawn into the sample line, giving a false reading.
- Gas content readings may be inaccurate due to the higher density.
- It may be impossible to obtain samples from the bottom of the tank because the gas is too heavy.

Define the "Lower Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration below which there is not sufficient gas to support and propagate combustion.**
- It is the gas concentration above which there is not sufficient air to support and propagate combustion.
- It is the gas concentration above which there is too much gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Define the "Upper Flammable Limit" of a hydrocarbon gas?

- **It is the gas concentration above which there is not sufficient air to support and propagate combustion.**
- It is the gas concentration above which there is too much oxygen to support and propagate combustion.
- It is the gas concentration below which there is not sufficient gas to support and propagate combustion.
- It is the gas concentration below which there is not sufficient oxygen to support and propagate combustion.

Explain how very cold weather conditions can cause over-pressurisation of cargo tanks?

- **Ice could form on the vents, or the P/V or high velocity vent valves could freeze.**
- Cold weather may cause malfunctioning of cargo monitoring equipment, including pressure monitors.
- Low temperatures will increase the viscosity of the cargo, and may make it unpumpable.
- The increased density of the cargo could lead to overloading.

Explain why the atmosphere in a cargo tank must be sampled at different depths?

- **Due to the high relative density of hydrocarbon gas, it tends to layer, and collect in pockets.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

Gas detection equipment showed the oxygen content of a cargo tank to be 2%, and the hydrocarbon gas content to be 2%. Using the Flammability Diagram, is it safe to gas free the tank with fresh air?

- **Yes, the atmosphere would remain outside the Flammable Range.**
- No the tank must first be purged with inert gas.
- No, the tank atmosphere is in the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

Gas detection equipment showed the oxygen content of a cargo tank to be 3%, and the hydrocarbon gas content 8%. Using the Flammability Diagram estimate whether it is safe to gas free the tank with fresh air?

- **No the tank must first be purged with inert gas.**
- No, the tank atmosphere is in the Flammable Range.
- Yes, the atmosphere would remain outside the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

If cargo tank openings are opened frequently for operations such as sampling and dipping, inert gas quality may be affected. How so?

- **Air could enter the tanks through the openings.**
- Inert gas would escape, lowering the tank pressure.
- Moisture from the atmosphere could enter the tank.
- The hydrocarbon content of the tank atmosphere would be increased.

In addition to the manufacturer's tests of the inert gas plant, what other checks must be performed before inerting operations?

- **Checks of the fixed and portable oxygen meters and gas detectors.**
- A pressure test of the cargo and inert gas pipework.
- A test of the crude oil washing system, if fitted.
- An inspection of the cargo tanks.

In practical terms, what figures can be used for the Lower and Upper Flammability Limits of oil tanker cargoes?

- **1% and 10% of gas.**
- 19% and 21% of oxygen.
- 5% and 21% of oxygen.
- 5% and 50% of gas.

Inert gas quality in slop tanks may be degraded when tank washings from non-inerted tanks are added: Why?

- **Air would be trapped in the washings, particularly if an eductor was in use.**
- Adding washings would increase the pressure in the tank, forcing inert gas out via the vent line.
- The higher temperature of the slops caused by adding washings would result in increased evolution of hydrocarbon vapours.
- The turbulence in the slops caused by adding washings would result in increased evolution of hydrocarbon vapours.

Pressure in the cargo tanks may be reduced due to low night time temperatures. How could this affect inert gas quality?

- **Air could be drawn into the system.**
- Components of the inert gas may break down at low pressure.
- The inert gas may condense into the cargo at low temperatures.
- The rate of evolution of cargo vapours could be increased.

The characteristics of the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the relative density of butane?

- **2.00**
- 1.00
- 2.55
- 5.50

The characteristics of the vapours given off by crude oil can be represented by those of propane. What is the relative density of propane?

- **1.55**
- 0.55
- 1.00
- 5.50

The characteristics of the vapours given off by natural gasolines can be represented by those of pentane. What is the relative density of pentane?

- **2.50**
- 0.55
- 1.00
- 5.50

What action must be taken if the inert gas plant fails during cargo discharge operations?

- **You should prevent air from entering the tanks, and stop discharge from inerted tanks.**
- You should carefully allow fresh air into the tanks to prevent under-pressurisation.
- You should continue discharge, monitoring oxygen and gas concentration in the cargo tanks.
- You should reduce the discharge rate until the plant is repaired.

What action should be taken if the pressure in inerted cargo tanks falls to the low pressure alarm level during a ballast passage?

- **The inert gas plant should be used to top up the tanks, and the oxygen content should be less than 5%.**
- If the tanks are not loaded, they can be opened to atmosphere to prevent under-pressurisation.
- The P/V valve will take care of fluctuations in the tank pressure.
- The tanks should be gas freed.

What adverse results may occur if cargo tanks are allowed to become over-pressurised?

- **Major deformation or failure of the tank structure could occur, leading to explosion, fire or pollution.**
- There could be a loss of cargo due to excessive venting.
- There could be difficulty continuing the loading operation due to high pressure.
- There could be leakage of cargo and vent pipelines.

What are the main components of inert gas delivered from a flue gas plant?

- **Nitrogen 83%, Carbon Dioxide 12-14%, Oxygen 2-4%**
- Carbon Dioxide 63%, Nitrogen 22-24%, Oxygen 10-12%
- Carbon Dioxide 83%, Nitrogen 12-14%, Oxygen 2-4%
- Nitrogen 60%, Carbon Dioxide 30%, Oxygen 10%

What are the main factors which could cause over-pressurisation of a cargo tank?

- **All three anwers**
- Excessively high delivery pressure from the inert gas plant or shore delivery supply
- Inadequate venting arrangements or overfilling the tank.
- Pump failure during discharge.

What inert gas monitoring instrumentation must be fitted in the cargo control room of a tanker?

- **Instruments to continuously indicate and permanently record the oxygen content and deck main pressure.**
- A device to permanently record the times of stopping and starting the inert gas plant.
- Instrumentation showing all operating parameters of the inert gas plant.
- Instrumentation showing the cargo tank temperature and pressure.

What is meant by the "Dilution" method of inerting a cargo tank?

- **The incoming inert gas is mixed with the tank atmosphere to form a homogenous mixture throughout the tank.**
- The inert gas is introduced slowly at the bottom of the tank so that a stable horizontal interface is formed.
- The inert gas is introduced slowly at the top of the tank so that a stable horizontal interface is formed.
- The inert gas is mixed with fresh air, and introduced rapidly into the tank.

What is meant by the "Displacement" method of inerting cargo tanks?

- **The inert gas is introduced slowly at the top of the tank so that a stable horizontal interface is formed.**
- The incoming inert gas is mixed with the tank atmosphere to form a homogenous mixture throughout the tank.
- The inert gas is introduced slowly at the bottom of the tank so that a stable horizontal interface is formed.
- The inert gas is mixed with fresh air, and introduced rapidly into the tank.

What is the maximum oxygen content allowed by SOLAS in the output from an inert gas plant?

- **Not more than 5%.**
- Not more than 1%.
- Not more than 21%.
- Not more than 8%.

What is the maximum oxygen content which should be maintained in a cargo tank during operations?

- **Not more than 8%, except when gas freeing.**
- Not more than 1%, except when gas freeing.
- Not more than 18%, except when loading.
- Not more than 21%, except when loading.

What is the minimum recommended pressure for cargo tanks connected with an inert gas main?

- **100 mm water gauge.**
- 0.25 bar.
- 500 mm water gauge
- One bar.

What is the operating principle of a dry type deck seal in an inert gas system?

- **A dump valve is fitted, which opens, filling the lower chamber with water to form a seal when the inert gas supply stops.**
- A large diameter non-return valve is fitted, which will close when the inert gas supply stops.
- The chamber contains a number of baffles which prevent solid materials from entering the cargo tanks, and help regulate the pressure of the inert gas main line.
- The chamber is drained when the inert gas plant is prepared for operation, but can be filled with water when the plant is not in use, forming a seal.

What is the purpose of the deck water seal in an inert gas system?

- **To prevent cargo liquid or vapour from entering the engine room.**
- To clean the inert gas before delivery to the tanks.
- To cool the inert gas before delivery to the tanks.
- To regulate the pressure of the inert gas main line, and help prevent over-pressurisation of the tanks.

What is the relative density of hydrocarbon gas at concentrations near the Lower Flammable Limit?

- **Approximately 1.0, the same as air.**
- Approximately 0.7, lighter than air.
- Approximately 1.5, heavier than air.
- Approximately 2.0, much heavier than air.

What oxygen content would normally be expected from an independent inert gas generator?

- **1.5-2.5% by volume.**
- 5% by volume.
- 8-10% by volume.
- Approximately 12% by volume.

When using the "Dilution" method of inerting a cargo tank, inert gas is introduced rapidly into the tank. Why?

- **Because the incoming inert gas must have enough velocity to reach the bottom of the tank and mix thoroughly with the tank atmosphere.**
- Because inert gas is slightly less dense than hydrocarbon vapour, and a stable horizontal interface can be formed.
- Because the incoming inert gas must have enough velocity to reach the bottom of the tank and form a stable interface.
- Because the inert gas is hotter than the tank atmosphere, and it is necessary to mix thoroughly with the tank atmosphere.

When using the "Dilution" method of inerting, it is essential that only a small number of cargo tanks are open. Why?

- **Because the inert gas must be introduced at a high enough rate to mix thoroughly with the tank atmosphere.**
- Because it is difficult to monitor the hydrocarbon levels properly if a large number of tanks are open.
- Because the incoming inert gas must form a stable interface with the tank atmosphere.
- Because the inert gas plant works more efficiently when the back pressure is higher.

When using the "Displacement" method of inerting cargo tanks, inert gas is introduced slowly at the top of the tank. Why?

- **Because inert gas is slightly less dense than hydrocarbon vapour, and a stable horizontal interface can be formed.**
- Because inert gas is denser than hydrocarbon vapour, and a stable horizontal interface can be formed.
- Because the incoming inert gas will mix with the tank atmosphere throughout the tank.
- Because the inert gas is hotter than the tank atmosphere and it is necessary to avoid violent mixing.

When using the "Displacement" method of inerting, it is common practice to inert several tanks at the same time. Why?

- **Because the inert gas must be introduced at a slow enough rate to enable the formation of a stable horizontal interface.**
- Because it is easier to compare the hydrocarbon levels in the tanks if a large number are open.
- Because the inert gas must be introduced at a high enough rate to mix thoroughly with the tank atmosphere.
- Because the inert gas plant works more efficiently when the back pressure is lower.

Why is an oxygen content of not more than 8% considered to be the safe limit in a cargo tank atmosphere?

- **Because the atmosphere is safely outside the Flammable Range.**
- Because 8% hydrocarbon gas is above the Upper Flammable Limit.
- Because 8% hydrocarbon gas is below the Lower Flammable Limit.
- Because the tank atmosphere was outside the Flammable Range at the time the readings were taken.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, what would be the effect of diluting with fresh air?

- **The hydrocarbon concentration would be reduced, and the oxygen level increased, taking the atmosphere into the Flammable Range.**
- The hydrocarbon concentration would be increased, taking the atmosphere into the Flammable Range.
- The hydrocarbon concentration would be reduced, and the oxygen level increased, resulting in a safe tank atmosphere.
- The oxygen level would be increased, resulting in a safe tank atmosphere.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, why would it be necessary to dilute the atmosphere to point H with inert gas before gas freeing with fresh air?

- **To reduce the hydrocarbon level below the critical dilution line.**
- To help prevent corrosion in the tank.
- To increase the hydrocarbon level above the Upper Flammable Limit.
- To reduce the oxygen level to below the critical dilution line.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point F, would you consider it to be safe to gas free the tank with fresh air?

- **No, the tank must first be purged with inert gas.**
- No, the tank atmosphere is in the Flammable Range.
- Yes, the atmosphere would remain outside the Flammable Range.
- Yes, the tank atmosphere would remain too rich to support combustion.

With reference to the Flammability Diagram, if a cargo tank atmosphere was at point H, would you consider it to be safe to gas free using fresh air?

- **Yes, the atmosphere would remain outside the Flammable Range.**
- No, the tank atmosphere is in the Flammable Range.
- No, the tank must first be purged with inert gas.
- Yes, the tank atmosphere would remain too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point A?

- **The tank contains fresh air, with no hydrocarbon gas, and cannot support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point B?

- **The tank atmosphere is too rich to support combustion.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere is too lean to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point C?

- **The hydrocarbon gas concentration is at its Lower Flammable Limit.**
- The hydrocarbon gas concentration is at its Upper Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too rich to support combustion.

With reference to the Flammability Diagram, what conditions are shown at point D?

- **The hydrocarbon gas concentration is at its Upper Flammable Limit.**
- The hydrocarbon gas concentration is at its Lower Flammable Limit.
- The tank atmosphere contains no oxygen, and cannot support combustion.
- The tank atmosphere is too lean to support combustion.

With reference to the Flammability Diagram, what point on the diagram shows a hydrocarbon concentration at its Lower Flammable Limit, with no inert gas?

- **Point C.**
- Point A.
- Point D.
- Point G.

With reference to the Flammability Diagram, what point on the diagram shows a hydrocarbon concentration at its Upper Flammable Limit, with no inert gas?

- **Point D.**
- Point A.
- Point C.
- Point G.

7049 – TOTS 1C-Fixed and Portable Gas Detections Systems

Version 1.0 (4)

A fixed combustible gas detection system can only detect a limited range of substances. What are these substances?

- **Hydrocarbon gases up to the Lower Flammable Limit.**
- Chemical gases up to 100%.
- Oxygen content up to 21% . . .
- Smoke and combustion products.

A fixed combustible gas detection system should not be used to assess compartments for entry purposes. Why not?

- **Because it does not indicate the oxygen content or the presence of toxic gases.**
- Because it does not give an accurate enough reading.
- Because many spaces are being monitored, and the origin of the sample is not certain.
- Since the sample takes some time to pass through the system, the reading may not be current.

After washing and gas-freeing from a toxic cargo, what type of gas detection equipment should you use before entering a cargo tank?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

An Explosimeter cannot be used in an inerted atmosphere. Why not?

- **Because adequate oxygen is required for combustion of the filament.**
- Because small particles will clog the filters.
- Because the inert gas poisons the filament.
- Because the presence of carbon dioxide will increase the readings.

Cotton filters are sometimes used with hydrocarbon gas detectors. What is their purpose?

- **To remove water from the sample.**
- To prevent chemical contamination of the sample.
- To remove dust from the sample.
- To stop pressure surges in the sample line.

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

How many different locations within a tank should be sampled to test the atmosphere before entering a tank?

- **At various depths, and through as many openings as possible.**
- As close as possible to the centre of the tank.
- At least three.
- In one position, at a minimum of three levels.

How will leaks in the sample pipework affect the operation of a fixed gas detection system?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

How will leaks in the sampling line affect the operation of portable gas detection equipment?

- **It will cause lower readings due to dilution of the sample.**
- It will cause failure due to high flow rate.
- It will cause failure due to low suction pressure.
- It will cause higher readings due to increased flow rate.

If an Explosimeter is calibrated using span gas containing 1% butane in air, what reading may be expected?

- **Approximately 50% Lower Flammable Limit.**
- Approximately 1% Lower Flammable Limit.
- Approximately 10% Lower Flammable Limit.
- Approximately 100% Lower Flammable Limit.

Many fixed gas detection systems have high and low sampling points. Why is this so?

- **So that they can be switched for cargos of different vapour density.**
- In order to ensure that all of a compartment is sampled.
- In order to ensure that suction pipe runs are of a similar length.
- So that they can be switched to account for temperature variations in the sample.

Many fixed gas detection systems monitor the atmosphere in the ballast tanks. What precautions must be taken on loaded passages?

- **Sample lines from ballast tanks must be isolated to prevent water from entering the system.**
- All sample lines must be isolated to prevent any cargo from entering the system.
- The alarm function must be switched off to prevent a continuous alarm condition.
- The system can be run as normal, but all alarms must be investigated.

Multi-toxic gas detectors normally use electro-chemical detectors. What level of accuracy can be expected?

- **Approximately 1 - 5 ppm.**
- Approximately 10% by volume.
- Approximately 100 - 500 ppm.
- Approximately 30% by volume.

Some chemical gases will not be detected by a fixed combustible gas detection system. How does this restrict the use of the system?

- **The system cannot be used to assess the level of toxic gas.**
- The flammability readings may be unreliable.
- The system cannot be used if the ship is carrying chemicals.
- The system cannot be used under very cold conditions.

Span gas is generally used for calibration of a fixed gas detection system. What is the composition of this gas?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure nitrogen.

The flammable limits for the vapours given off by clean products such as motor or aviation gasoline can be represented by those of butane. What is the Lower Flammable Limit of butane?

- **1.9% by volume in air.**
- 19% by volume in air.
- 5.9% by volume in air.
- 9% by volume in air.

The flammable limits for the vapours given off by crude oil can be represented by those of propane. What is the Lower Flammable Limit of propane?

- **2.2% by volume in air.**
- 10% by volume in air.
- 22% by volume in air.
- 5.2% by volume in air.

What effect will inert gas have on the operation of a tankscope?

- **None. The sensor does not require oxygen.**
- It will increase the reading by increasing the temperature of the filament.
- It will reduce the reading by contaminating the sensor.
- Small particles may clog the filters.

What effect will inert gas have on the operation of an Explosimeter?

- **It will inhibit operation, as the sensor requires oxygen.**
- It will reduce the reading by contaminating the sensor.
- None. The sensor does not require oxygen.
- Small particles may clog the filters.

What effect will inert gas have on the operation of an infrared type gas detector?

- **None. The instrument is not affected by the absence of oxygen.**
- It will Increase the reading by increasing the temperature of the chamber.
- It will reduce the reading by contaminating the sensor.
- Small particles may obscure the mirrors.

What equipment is required to test an enclosed space for the presence of toxic gas before entry?

- **Chemical absorption tubes.**
- A tankscope.
- An Explosimeter.
- An oxygen meter.

What indication will be given by an Explosimeter in the case of oxygen deficiency?

- **No indication.**
- Approximately 20% of the Lower Flammable Limit.
- Full scale deflection.
- The indication will be erratic and jumpy.

What instrument should you use to test for very low levels of oxygen before loading a sensitive cargo?

- **Chemical indicator tubes.**
- A paramagnetic oxygen analyser.
- An Explosimeter.
- An infrared gas detector.

What instruments must be provided to comply with the statutory requirements for ships fitted with an inert gas plant?

- **An oxygen analyser and an instrument to measure the percentage of hydrocarbon gas in an inerted atmosphere.**
- An Explosimeter and an oxygen analyser.
- At least two instruments to measure the percentage of hydrocarbon gas in an inerted atmosphere.
- At least two instruments to measure the percentage of oxygen in an inerted atmosphere.

What instruments should be used to check the atmosphere of an enclosed space before entry?

- **An Explosimeter, a portable oxygen analyser, and a toxic gas meter if required.**
- A personal gas detector and a personal toxic gas alarm if required.
- A tankscope, an oxygen analyser, and a toxic gas meter if required.
- An Explosimeter, and a toxic gas meter if required.

What is a "Pellistor"?

- **It's a poison resistant catalytic filament.**
- It's a balancing resistor between the sensing element and a fixed compensator.
- It's a filter which prevents the spread of flame.
- It's a protective device which removes contaminants.

What is a "Sintered Filter", and what is its purpose?

- **It's a flashback arrestor used to prevent flame from escaping from the combustion chamber.**
- It's a cotton wad used to prevent water from entering the combustion chamber.
- It's a fine metal filter used to prevent contamination of the sensor.
- It's an electronic device used to smooth the signal from the sensor.

What is the composition of the span gas generally used for calibration of an Explosimeter?

- **Approximately 1% butane in air.**
- A proprietary mixture of nitrogen and carbon dioxide.
- Approximately 40% propane in air.
- Pure carbon dioxide.

What is the composition of the span gas generally used to zero a portable oxygen analyser?

- **High purity nitrogen or carbon dioxide.**
- Approximately 2% propane in air.
- Approximately 20% propane in air.
- Fresh air.

What is the gas concentration alarm level on a fixed gas detection system?

- **30% of Lower Flammable Limit.**
- 10% of flammable gas.
- 3% of Lower Flammable Limit.
- 30% of flammable gas.

What is the main reason for fitting a fixed gas detection system on a ship?

- **To indicate when a flammable atmosphere exists in a compartment.**
- To indicate that ventilation systems are working correctly.
- To show that compartments are safe for entry.
- To show that compartments are safe for hot work.

What is the principle by which a thermal conductivity meter works?

- **To show a change in resistance of an electrically heated resistance element, due to heat loss to the gas sample by conduction.**
- To show a change in temperature of an electrically heated resistance element, due to heat loss to the gas sample by conduction.
- To show a change in the conductivity of a heated gas sample relative to that of air.
- To show change in the density of a heated gas sample relative to that of air.

What is the principle by which an infrared gas detector operates?

- **It senses the level of absorption of infrared radiation in the sample.**
- It compares the electrical resistance of the sample with fresh air.
- It compares the relative density of a sample which has been heated by an infrared source.
- It measures the resistance of a sample which has been heated by an infrared source.

What is the principle by which chemical absorption indicators operate?

- **By passing the sample through a proprietary chemical, the gas concentration is indicated by its changing colour.**
- To compare the chemical absorption rate of the sample with fresh air.
- To sense the proportion of a sample which is absorbed by a catalyst.
- To show whether a toxic gas will be absorbed by the human body.

What measurements can be made using a Combustible Gas Indicator (Explosimeter)?

- **Flammable gas up to the Lower Flammable Limit.**
- Flammable gas in an inert atmosphere.
- Flammable gas up to 100% concentration.
- Toxic gas up to the threshold limit value.

What operational tests should be made routinely before using a gascope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self- testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using a tankscope?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas and zero using atmospheric air.

What operational tests should be made routinely before using an Explosimeter?

- **Checking the battery level and zero using fresh air.**
- Checking the battery level and self-testing alarm functions.
- Checking the battery level and zero using nitrogen or carbon dioxide.
- Checking the reference point with span gas, and zero using atmospheric air.

What operational tests should be made routinely before using an oxygen analyser?

- **Check battery level, reference point with atmospheric air (21%), and zero using nitrogen or carbon dioxide.**
- Check battery level, reference point with span gas, and zero using nitrogen or carbon dioxide.
- Check the battery level, and self- test the alarm functions.
- Check the reference point with span gas and zero using atmospheric air.

What property of oxygen is utilised in a paramagnetic oxygen analyser?

- **Oxygen is strongly attracted to a magnetic pole.**
- Oxygen molecules will break up in a magnetic field.
- The electrical resistance of oxygen changes in a magnetic field.
- The strength of a magnetic field is reduced in the presence of oxygen.

What range of flammable gas can be detected by a catalytic resistance detector?

- **0-100% Lower Flammable Limit.**
- 0-10% gas.
- 0-10% Lower Flammable Limit.
- 0-100% gas.

What range of flammable gas can be detected by a thermal conductivity meter?

- **0-100% gas.**
- 0-10% gas.
- 0-10% Lower Flammable Limit.
- 0-100% Lower Flammable Limit.

What range of flammable gas can be detected by an infrared type detector?

- **0-100% gas.**
- 0-10% gas.
- 0-10% Lower Flammable Limit.
- 0-100% Lower Flammable Limit.

What task can a personal multi-gas detector be used for?

- **For monitoring atmospheric conditions during an enclosed space entry.**
- For checking the atmosphere of a ballast tank before entry.
- For testing an enclosed space before starting hot work.
- For testing the atmosphere of an open space before starting hot work.

What type of gas detection equipment should you use to test the atmosphere before starting hot work?

- **A Combustible Gas Indicator (Explosimeter).**
- A personal gas detector.
- A toxic gas meter.
- An oxygen analyser.

What type of sensor is fitted in a tankscope?

- **A non-catalytic heated filament.**
- A catalytic filament resistor.
- A magnetic induction sensor.
- A PT100 temperature sensor.

What type of sensor is fitted in an Explosimeter?

- **A catalytic filament resistor.**
- A gas diffusion filter.
- A magnetic resonance chamber.
- A non-catalytic heated filament.

What will be the effect of hydrogen sulphide on the reading shown on a Combustible Gas Indicator (Explosimeter)?

- **It will reduce the reading by contaminating the sensor.**
- It will give a zero reading by destroying the sensor.
- It will have no effect because it is non-flammable.
- It will increase the reading by raising the sensitivity of the sensor.

What would you use a personal oxygen meter for?

- **For monitoring atmospheric conditions during an enclosed space entry.**
- For checking the atmosphere of a ballast tank before entry.
- For testing an enclosed space before starting hot work.
- For testing the atmosphere of an open space before starting hot work.

When changing high purity cargo grades, what instrument should be used to ensure that tanks are free of the previous product?

- **Chemical indicator tubes.**
- A tankscope.
- An Explosimeter.
- An infrared gas detector.

When starting to test a sample using a Combustible Gas Indicator (Explosimeter), the needle immediately flicks to maximum deflection, then returns to zero. What is indicated?

- **That the flammable gas concentration is above the Lower Flammable Limit.**
- That the battery is incorrectly fitted.
- That the instrument was not warmed up, but is now working correctly.
- That the sensor element has been contaminated by toxic gas.

When using chemical absorption tubes, why must you use the exact number of pump strokes specified in the instructions?

- **To ensure that the correct volume of sample gas is passed through the tube.**
- Because all tubes require the same number of strokes.
- It is not necessary to use the exact number stated, it is only for general guidance.
- So that the results of all your tests are based on the same data.

Why is it necessary to use a water-retaining filter in the sample line of an oxygen analyser?

- **Because water vapour in the sample will cause damage to the measuring cell.**
- Because water in the sample will corrode the internal parts of the instrument.
- Because water reacts with the measuring cell, giving a false reading.
- Because water will build up in the sample line, reducing the flow.

Why must chemical absorption tubes only be used with pumps from the same manufacturer?

- **Because the pump and tube are matched to ensure that the correct volume of sample gas is tested.**
- Because the tubes will not fit correctly into another manufacturer's pump.
- Because using different tubes breaks the terms of the contract with the manufacturer.
- Because using the wrong tubes may lead to contamination of the pump.

Why must the tank atmosphere be sampled at a variety of positions when testing before entry?

- **To allow for the density of the gas and to ensure that any collected in dead spots is detected.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

Why should you avoid high sample pressure when using an Explosimeter?

- **The pressure may damage the catalytic filament.**
- If the flow rate is too high, the instrument does not have time to measure the gas concentration.
- If the pressure is too high, it could cause an explosion inside the instrument.
- The pressure may burst the sampling hose.

7050 – TOTS 1C-Enclosed Spaces and Tank Atmosphere . . .

Version 1.0 (8)

A Permit to Work is required before an enclosed space is entered. Who should complete this permit?

- **The Authorising Officer, normally the head of department.**
- It must be authorised by the Company office.
- The Master.
- The person who is to carry out the work.

For how long must the ventilation of a space be stopped before conducting tests of the atmosphere for an Enclosed Space Entry?

- **10 minutes.**
- 30 minutes.
- Briefly, while the tests are being carried out.
- The ventilation should be run continuously.

For what purpose would you use a personal gas detector?

- **To monitor atmospheric conditions during an enclosed space entry.**
- To check the atmosphere of an enclosed space before entry.
- To test an enclosed space before starting hot work.
- To test the atmosphere of an open space before starting hot work.

For what purpose would you use a personal oxygen meter?

- **To monitor atmospheric conditions during an enclosed space entry.**
- Testing an enclosed space before starting hot work.
- Testing the atmosphere of an open space before starting hot work.
- To check the atmosphere of an enclosed space before entry.

How long should ventilation be continued after the atmosphere of an enclosed space has been found suitable for entry?

- **Continuously while the space is occupied, and even during temporary breaks.**
- For at least one hour after successful testing.
- It can be stopped when testing shows that the space is suitable for entry.
- It should be continued after testing until the space is actually entered.

How many different locations within a tank should be sampled to test the atmosphere before entering a tank?

- **At various depths, and through as many openings as possible.**
- As close as possible to the centre of the tank.
- At least three locations.
- In one position, at a minimum of three levels.

How many people should be allowed to enter an enclosed space to carry out routine work?

- **Only the number actually required to carry out the work.**
- Only two people.
- The number required to carry out the work, plus one extra watchman.
- The number required to carry out the work, plus the person in charge and one extra watchman.

To comply with an Enclosed Space Entry Permit, when must the atmosphere of an enclosed space be tested?

- **Before entry, and at frequent intervals when personnel are in the space.**
- Before entry and after completion of work.
- Every four hours
- In the morning, before daily routine work starts.

What action should be taken if personnel working in an enclosed space feel unwell or suspect the presence of gas?

- **They should leave the space immediately and contact the Person In Charge or Officer of the Watch.**
- The Officer of the Watch, or whoever signed the entry permit, should be advised of the situation.
- They should request a crew member to bring breathing apparatus into the space.
- They should request a crew member to verify the atmosphere.

What action should be taken if personnel working in an enclosed space get into difficulties?

- **The general alarm should be sounded and the rescue team should enter using breathing apparatus under the direction of the Person In Charge**
- All personnel should be instructed to leave the space immediately.
- The first person on the scene should immediately enter the space to give assistance.
- The Watchman should immediately enter the space to give assistance.

What action should be taken if routine tests indicate the presence of gas while personnel are present in an enclosed space?

- **The personnel should leave the space immediately and contact the Person In Charge or Officer of the Watch.**
- The gas testing equipment should be checked to verify its accurate.
- The Watchman should be instructed to bring the breathing apparatus into the space as back up.
- The work must be completed as soon as possible and then vacate the space.

What actions must be taken to isolate a tank before entry?

- **Blank off the pipelines to the tank, close and secure all valves and inform the Officer of the Watch.**
- Request the Officer of the Watch to stop all cargo and ballast pumps.
- Stop all cargo and ballast operations.
- Stop all ventilation and inform the Officer of the Watch.

What documentation should be completed before carrying out welding in a double bottom tank?

- **A Risk Assessment, an Enclosed Space Entry Permit and a Hot Work Permit.**
- A Risk Assessment and a Hot Work Permit.
- A Risk Assessment and an Enclosed Space Entry Permit.
- An Enclosed Space Entry Permit and a Cold Work Permit.

What equipment is required to be kept at the entrance of an enclosed space during entry?

- **A resuscitator, breathing apparatus, a lifeline and a harness, a torch or a lamp.**
- A resuscitator or escape breathing set.
- Enough escape breathing sets for all personnel in the space.
- Spare tools and parts for the work being undertaken.

What equipment must be used to test an enclosed space for the presence of toxic gas before entry?

- **A toxic gas detector**
- A tankscope.
- An explosimeter.
- An oxygen meter.

What gas detection equipment must be used to test the atmosphere of a cargo tank before entry is permitted?

- **An oxygen meter, a combustible gas indicator and a toxic gas detector if necessary.**
- A combustible gas indicator and a toxic gas detector if necessary.
- An oxygen meter and a tankscope.
- An oxygen meter and a toxic gas detector if necessary.

What indication will be given by an Explosimeter in case of oxygen deficiency?

- **No indication.**
- Approximately 20% of the Lower Flammable Limit
- Full scale deflection.
- The indication is erratic and jumpy.

What is the maximum length of time an Enclosed Space Entry Permit can remain valid?

- **Eight hours.**
- Four hours.
- The estimated time required to complete a particular task.
- Twenty four hours

What is the maximum permitted level of hydrocarbon gas for an enclosed space entry?

- **1% of the Lower Flammable Limit.**
- 1% by volume.
- 10% of the Lower Flammable Limit.
- 10% of the Occupational Exposure Limit.

What is the maximum permitted level of toxic gas for an enclosed space entry?

- **The Occupational Exposure Limit, which varies according to the particular gas.**
- 1% of the Lower Flammable Limit.
- 10% of the Lower Flammable Limit.
- It varies according to the particular gas, but 1% is an acceptable level.

What potential hazards would be caused in a machinery space by the release of carbon dioxide for fire extinguishing purposes?

- **A lack of oxygen.**
- Carbon dioxide which is toxic to breathe.
- The carbon dioxide gas would be very cold, and could cause severe cold damage to the skin.
- Vapour, which may reduce visibility.

What use can be made of a combustible gas detector when making a Risk Assessment for tank entry?

- **Testing for hydrocarbon vapours up to the Lower Flammable Limit.**
- Testing for hydrocarbon vapours up to 100% concentration.
- Testing for oxygen content up to 21%.
- Testing for toxic gases up to any concentration.

Who is required to standby at the entrance of an enclosed space during entry?

- **At least one watchman, equipped to communicate with the personnel inside and with the Officer of the Watch.**
- Nobody, provided the personnel inside are equipped to communicate with the Officer of the Watch.
- The Officer of the Watch and one watchman, equipped to communicate with the personnel inside.
- The Officer of the Watch, or whoever signed the Entry Permit.

Who is responsible for cancelling an Enclosed Space Entry Permit when work is completed?

- **The person who authorised the permit.**
- Any other senior officer.
- The Master.
- The person who completed the work.

Why could an unprotected steel space, which has been closed, produce a hazardous atmosphere?

- **Because oxygen in the atmosphere may be depleted by reacting with the steel (i.e. corrosion).**
- Because hydrocarbons could be liberated from rust scale in the tank.
- Because the level of nitrogen in the atmosphere would be increased by corrosion.
- Because toxic gases may be given off by the process of corrosion.

Why could an unventilated ballast tank fitted with Cathodic Protection contain a hazardous atmosphere?

- **Because hydrogen is produced by the Cathodic Protection.**
- Because hydrocarbons could be produced by the Cathodic Protection.
- Because the Cathodic Protection could produce an oxygen enriched atmosphere.
- Because the Cathodic Protection would increase the level of nitrogen in the atmosphere.

Why must the enclosed space atmosphere be sampled at a variety of positions?

- **To allow for the density of the gas and to check for any gas pockets which may have accumulated in blind spots.**
- To ensure that enough gas is drawn through the equipment to avoid it remaining in the sampling line.
- To make sure that the detection equipment is working correctly.
- To obtain an average reading throughout the tank.

7051 – TOTS 1C-Familiarisation prior Cargo Operations

Version 1.0 (3)

A "Statement of Facts" is . . .

- **an accurate account of happenings in? a time based in chronological order.**
- a statement regarding the volatility of the cargo.
- an analysis of cargo samples.
- the Chief Officer's account of the cargo transfer.

A Ship to Shore Safety Checklist can be found in which publication?

- **ISGOTT.**
- ISPS code.
- SOPEP.
- The Mariner's Handbook.

A combination carrier loaded with crude oil has two tanks, each only 60% full. What affect will this have on the vessel's stability?

- **There will be a virtual loss of GM.**
- It will increase in permissible bending moments.
- It will increase the rolling period.
- The vessel will trim by the head.

A tank "sounding" can be referred to as . . .

- **tank innage.**
- tank reference point.
- tank ullage.
- trim correction.

A vessel is to maintain light ballast condition, however if logged and permissible bending moments have been exceeded, you would . . .

- **reduce ballast in Fore & After Peaks, maintaining propeller immersion and minimum draught forward?**
- ballast amidships cargo tanks?
- do nothing it is only a short voyage?
- move bunker fuel aft?

All cargo system pre-arrival testing must be recorded in the ship's . . .

- **deck logbook.**
- ESP file.
- Ship / Shore Safety Check List.
- SOPEP manual.

As part of the Safety Management System (SMS), new personnel joining the tanker must be . . .

- **familiarised with their duties and responsibilities.**
- given a minimum of 24 hours to recover after travel.
- informed of their salary.
- introduced to the Captain.

At commencement of loading, the generation of static is controlled by . . .

- **limiting the flow rate at the tank inlet.**
- earthing the manifold.
- monitoring the cargo temperature.
- only having one loading arm connected.

Before loading, you should ensure that . . .

- **the offshore manifold is closed and blanked.**
- the manifold is blanked but the valve is open.
- the pressure gauge is fitted.
- the temperature gauge is fitted between the valve and the blank.

Cargo tank temperature and pressure gauges and reliquefaction plant instruments should be calibrated . . .

- **preferably at intervals not exceeding 30 months.**
- at intervals not exceeding 12 months.
- at the Master's discretion.
- whenever a false reading is suspected.

Cargo transfer operational information will normally be available in . . .

- **the cargo control room (CCR).**
- the Chief Engineer's office.
- the SOLAS training manual located in the officers' and crew lounge.
- the wheel-house.

Damage due to sloshing is a direct result of the tank . . .

- **being partially filled (slack).**
- being filled more than 98%.
- having the tank pressure too high.
- None of these answers.

Different grades on a product tanker are normally segregated by . . .

- **Two valve separation.**
- A single valve.
- A spade blank.
- It does not matter as different densities will keep them separated.

During cargo transfer, what action would you take if the Inert Gas System fails?

- **Cease all discharge operations and advise the terminal.**
- Advise the terminal and continue discharging.
- Increase the rate of discharge.
- Review the Ship / Shore Safety Check List.

Flammability and toxicity of proposed cargo can be obtained from . . .

- **Material Safety Data Sheets (MSDS).**
- ISGOTT.
- The Approved Trim and Stability Manual.
- The SOPEP file.

How does "free surface effect" (sloshing) affect the ship's stability?

- **It reduces the metacentric height.**
- It causes an angle of loll.
- It has no effect.
- It increases the metacentric height.

How long must logbooks and other cargo records be kept on board?

- **As detailed in the owners/managers Safety & Quality Management System.**
- At the Master's discretion.
- For 12 months after discharge.
- For the life of the vessel.

If hydrocarbon gases are found in double hull ballast tanks, what action should be taken?

- **Immediately utilise the emergency inerting procedure.**
- Close the tank down.
- Read ISGOTT.
- Vent to atmosphere.

If the shore fails to deliver cargo at the requested rate, what action would the Master take?

- **He would issue a letter of protest.**
- He would inform the Class.
- He would inform the local P&I Club.
- He would take no action.

If unable to ground GMDSS equipment during cargo operations, what action may be taken?

- **To switch the equipment off.**
- To ignore any rules as you cannot comply.
- To make a logbook entry.
- To turn the volume down.

In the cargo control room (CCR), on the inert gas panel, the continuous trace of inert gas oxygen content at delivery during crude oil washing must be . . .

- **5% or less.**
- 8% is satisfactory.
- dependent upon the Chief Officer's standing orders.
- none of these. As long as the tank being washed has positive pressure, it does not matter.

In the loading limit formula, at which temperature is the pR value expressed?

- **At reference temperature.**
- At 45 degrees Centigrade.
- At ambient temperature.
- At sea temperature.

In which documentation would you expect to find duties and responsibilities relating to the cargo system?

- **In the Safety Management System (SMS).**
- In the Builders' Cargo Operations Book.
- In the Certificate of Fitness.
- In the SOLAS training manual.

In which language would you expect the operational and technical manuals to be written?

- **The official working language of the vessel.**
- English.
- It depends upon the nationality of the manufacturer.
- Several languages.

Night orders must be signed by the appropriate personnel . . .

- **every night when new orders are issued.**
- It does not matter.
- once per cargo grade only.
- once per port only.

Offshore manifold gauges need not be fitted?

- **False**
- True

On a combination carrier, where would the audible bilge alarm sound from a void space forward of the collision bulkhead?

- **The navigation bridge.**
- No alarm. It would just checked daily by soundings.
- On deck close to an accommodation entrance.
- The CCR (cargo control room).

On what occasions will a pre-cargo transfer meeting be minuted?

- **On each and every occasion.**
- When the Captain attends.
- When there is something special to say.
- Whenever the port stay is estimated to be greater than 24 hours.

Permissible bending moments are greater in which condition?

- **In harbour condition.**
- It depends upon the amount of segregated ballast.
- It depends upon the mean draught.
- There is no difference between sea and harbour condition.

Petroleum cargo pipe lines are to be tested to what pressure?

- **1.25 times their designed working pressure.**
- 1.5 times their designed working pressure.
- They are only to be tested after maintenance or when repairs have been completed.
- To normal operating pressure.

Sequential and flow through methods form part of . . .

- **the ballast exchange plan.**
- the ballast water analysis.
- the composition of a sounding book.
- the plan to clean cargo tanks.

Shore cargo hard arms are commonly known as . . .

- **chicksans.**
- cargo lines.
- hose strings.
- manifold lines.

The Officer of the Watch must regularly check the cargo tank ullages to . . .

- **ensure that the cargo is only entering the designated cargo tanks and that there is no escape of cargo into pumprooms and cofferdams.**
- determine when the cargo transfer will be completed.
- ensure the cargo temperature.
- ensure the loading rate.

The Ship / Shore Safety Check List states that temporarily removed scupper plugs must be . . .

- **constantly monitored.**
- clearly labelled as to location.
- only removed by the person responsible for cargo operations.
- stowed in a safe place.

The Ship Inspection Report Programme (SIRE) states that the loading programme must be checked by means of an approved test programme at least . . .

- **Quarterly.**
- Annually.
- As directed by the Master.
- At the Chief Officer's discretion.

The approved loading computer programme is malfunctioning, you would . . .

- **consult the Approved Trim and Stability Manual?**
- consult a general arrangement and a capacity plan?
- look for a similar cargo in the cargo record book?
- use the Chief Officer's personal programme?

The location of technical manuals relating to the cargo system . . .

- **would depend upon who is responsible as per the Plan Maintenance System.**
- are always with the Chief Engineer.
- are in the common library on the bridge.
- would depend upon where the Captain wants to put them.

The port / cargo operations log will form the basis of . . .

- **the statement of facts.**
- compliance with SOLAS.
- crew overtime.
- ISPS implementation.

The principle purpose of a pre-cargo transfer meeting is to . . .

- **ensure that all involved personnel are aware of the proposed cargo operations, services needed and safety requirements.**
- arrange shore leave.
- discuss the Ship / Shore Safety Check List.
- inform the technical officers when cargo pumps will be required.

What competent body would normally approve a loading instrument / programme?

- **The Classification Society.**
- SOLAS.
- The P&I Club.
- The ship's Master.

What does code "R" indicate in the Ship / Shore Safety Check List?

- **Re-checking at intervals not to exceed an agreed time.**
- Checks to be repeated only upon the instructions of the Chief Officer.
- Regular checks not required.
- Restricted access.

What information is used to obtain the factor for reducing the volume at the observed temperature to the volume at standard temperature?

- **API gravity and observed temperature.**
- API and the ullage of the cargo tank.
- Capacity in US barrels.
- The cargo tank ullage and the temperature.

What instrument must be used for measuring hydrocarbon gas in an inert atmosphere?

- **A Tankscope.**
- An Explosimeter.
- An oxygen meter.
- Chemical indicator tubes.

What is a schematic drawing?

- **A drawing in the form of a diagram and not to scale.**
- An approximate drawing.
- An exact technical drawing.
- The general arrangement plan.

What is meant by the operating envelope when referring to loading arms?

- **It is the maximum amount the vessel can safely range up and down the berth and generally move without damaging the loading arms.**
- It is the maximum cargo pressure the loading arm can withstand.
- It is the range of cargo temperatures the loading arm can safely withstand.
- It is the vertical reach of the loading arm.

What is the name of the diagram that is superimposed on the cargo control console?

- **The mimic diagram.**
- The cargo lines.
- The pumping plan.
- The tank plan.

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- The tank plan.

What material should manifold reducers be made of?

- **Steel.**
- Cast iron.
- GRP (glass reinforced plastic).
- None of these answers.

What medium is used to purge cargo tanks that have previously carried crude oil?

- **Inert gas.**
- A mixture of oxygen and water.
- Oxygen.
- Water.

What must be monitored before entering the pumproom during cargo operations?

- **The hydrocarbon gases and the oxygen content.**
- The amount of bilge water.
- The mechanical ventilation.
- The rescue harness.

What precaution must be taken when removing a blank flange?

- **Ensuring that the section between the blank flange and last valve does not contain oil under pressure.**
- Ensuring that the cock on the pressure gauge is closed.
- Ensuring that the correct spanner size is used to remove the bolts.
- Ensuring that the pressure gauge is fitted correctly.

When calculating the UKC, what is the "controlling depth"?

- **It's the least depth of water on the vessel's proposed track.**
- It's a figure required to calculate squat.
- It's the depth of water allowing for height of tide.
- It's the draught of the vessel when proceeding in a narrow channel.

When discharging segregated ballast, the overboard discharge must be monitored . . .

- **for contaminated ballast.**
- Monitoring is not required.
- to help calculate the rate of discharge.
- to observe when de-ballasting has been completed.

When entering cargo tank calibration tables, which components affect the resulting volume?

- **Ullage, trim and list.**
- Only innage.
- Only ullage.
- The load temperature of the cargo.

When must the Chief Officer's standing orders be followed?

- **Throughout all cargo transfer, ballasting operations and tank cleaning.**
- Only during ship to ship transfer operations.
- Only in port.
- When the Captain is ashore.

When starting a centrifugal cargo pump, the discharge valve should be initially . . .

- **Closed.**
- It does not matter.
- Open.
- Slightly open.

When would you exceed the maximum pressure ratings of the cargo system?

- **Never. All pressure ratings must be adhered to.**
- When circumstances dictated a departure from safe operating procedures.
- When you think it is safe, according to your own personal experience exceeding pressures on other occasions.
- Whenever instructed to do so by the head office.

When would you need to check the intact stability at intermediate stages of cargo transfer?

- **On each and every occasion.**
- If the Master directed you to do so.
- Seldom as it's not really required.
- When the vessel developed a list.

When would you use the Chief Officer's personally developed loading / stress monitoring computer programme?

- **Never.**
- All the time, if the results appear correct.
- Occasionally, if the approved one is non-operational.
- Once a month to check the results of the approved system.

Where would a plan indicating the location of cargo measurement instruments, sensors, pressure gauges, etc normally be found?

- **In the CCR (cargo control room).**
- At the manifold.
- In the Chief Engineer's office.
- In the reliquefaction plant.

Where would the greater ullage be when a vessel is trimmed by the head?

- **The aft end of the tank.**
- It depends upon the list.
- The forward end of the tank.
- There would be no difference.

Where would you expect cargo pump emergency stops to be located?

- **The cargo control room (CCR), the pumproom and the cargo manifold.**
- In the Master's cabin.
- In the wheel-house.
- On the poop.

Where would you expect to find the cargo pump performance curves?

- **Adjacent to the cargo / pump control console.**
- At the top of the pumproom.
- In the Chief Officer's office.
- In the engine control room (ECR).

Where would you expect to locate a mimic diagram of the cargo system?

- **In the cargo control room (CCR) operational panel.**
- Adjacent to the bunkering manifold.
- At the manifold.
- In the Chief Engineer's office.

Where would you expect to see a notice displayed regarding free surface effect on a combination carrier?

- **Next to the cargo control station.**
- At the top of the pumproom.
- In the cargo manifold.
- In the Master's office.

Where would you find approved information regarding the vessel structural strength?

- **The Approved Trim and Stability Book.**
- The Certificate of Class.
- The deck logbook.
- The OCIMF Ship Inspection Report Programme.

Where would you normally expect to find a full set of cargo operation instructions?

- **The cargo control room (CCR).**
- In the wheel-house.
- The Master's cabin.
- The SOLAS training manual in the crew's lounge.

Which competent body checks and approves a vessel's tank calibration tables?

- **The Classification Society.**
- MARPOL.
- The IMO.
- The P&I Club.

Which volume reduction petroleum tables may be used when calculating a crude oil cargo?

- **6A and / or 6B.**
- 54A or 54B.
- 84A.
- Norries tables.

Who has overall responsibility for the maintenance of the cargo system?

- **It will depend upon the owner's/manager's work instructions as detailed in the Safety Management System (SMS).**
- The Chief Engineer.
- The Chief Officer.
- The Electrical Officer.

Why are main radio transmitter aerials grounded during cargo operations?

- **To prevent a potential ignition source.**
- Because it is written in the Safety Management System (SMS).
- Because they are not needed in port.
- Because they are not required for communications in port

Why is it important to keep the cargo manifold area clear of obstructions?

- **So as not to interfere with the automatic release of a cargo hardarm.**
- For easy access in case of a pollution incident.
- Simply to keep the space tidy and ship shape.
- So ship and shore staff can work easily.

Why is it important to regularly check offshore manifold gauges during a cargo transfer?

- **To check for manifold valve leakage.**
- To check for pressure in the cargo system.
- To comply with the MSDS (Material Safety Data Sheet).
- To comply with the terminal requirements.

Why is the atmosphere of void spaces regularly checked?

- **Because it is the first indication of cargo migration due to cracks in welded seams or similar.**
- Because instructions to do so are contained in the Chief Officer's standing orders.
- In order to check for water content.
- To ensure adequate oxygen.

Why must a cargo tank atmosphere be monitored during crude oil washing?

- **To ensure that the oxygen level remains at 8% or less by volume.**
- Overall, to ensure sludge control.
- To ensure that the cleaning gun cycle is completed.
- To ensure that the washing process is adequate.

Why should you monitor void spaces, cofferdams and empty ballast tanks during a loaded passage?

- **To confirm that such spaces remain gas free.**
- To check for water content.
- To confirm that welded seams are not cracked.
- To practise the use of portable instruments.

7052 – TOTS 1C-Cargo Information, Precautions . . .

Version 1.0 (5)

According to the IBC code where should in use protective equipment be stowed?

- **In special lockers in easy accessible places.**
- At the forecastle head.
- At the top of the pumproom.
- In the accommodation.

How does one usually separate two different cargo products?

- **A two valve segregation.**
- A one valve segregation is sufficient.
- A tank valve and a crossover.
- Insert spade blanks.

How does the IBC code describe coveralls?

- **They should be made of material resistant to chemical attack.**
- They can be made of any material but must have long sleeves.
- They must be nylon.
- They should be made of pure cotton.

On a tanker carrying Noxious Liquid Substances, what is the minimum number of protective equipment sets that must be carried?

- **At least three.**
- It can vary as it depends upon the toxicity of the cargo.
- One, provided it's the one that's required by SOLAS.
- Two.

The cargo being loaded is described as having highly toxic vapour. What must be worn on the cargo deck?

- **Breathing apparatus.**
- A dust mask.
- A face shield.
- Rubber gloves.

The charter party includes information on STS transfer. What is meant by STS transfer?

- **Ship To Ship.**
- Safe Tanker System.
- Suitable Tank System.
- Suitable Tanker Stability.

Voyage orders instruct you to dose the cargo with an Anti-static additive. What is an Anti-static additive?

- **It's a substance added to a petroleum product to raise its electrical conductivity to a safe level.**
- It assists in stabilising the cargo temperature.
- It reduces static electricity when tank cleaning with portable machines and hoses.
- It's a substance that enables safer ullaging.

What action would the master take if he thought the proposed cargo was not suitable for his vessel?

- **He would confirm the contents of the charter party and then immediately inform his owners/managers and the charterers.**
- He would inform his owners.
- He would inform the charter broker.
- He would inform the QI (Qualified Individual).

What can be worn in addition to or instead of goggles?

- **A face shield.**
- A smoke helmet.
- An escape set.
- Breathing apparatus.

What is a charter party?

- **A charter party is the contract between the owner of a vessel and the charterer for the use of a vessel.**
- A group of ship brokers.
- A non-binding agreement between any two parties.
- Renting a vessel for an event.

What is cargo ingestion?

- **Cargo being swallowed.**
- Cargo having contact with the eyes.
- Cargo having contact with the skin.
- Cargo vapours being inhaled.

What is deadfreight?

- **The sum payable to the owner when the charterer has failed to load a full and complete cargo.**
- The amount of cargo not shipped on the vessel.
- The amount of cargo which is described as FOB (Free on Board).
- When the charterer is bankrupt it's the freight claimed by the owner.

What is the "pumping clause"?

- **It's the maximum time to discharge the entire cargo or a minimum average pressure at the discharge manifold.**
- It's the maximum number of cargo pumps that can be operated at one time.
- It's the maximum time the cargo pumps can be running.
- None of these answers are correct.

What is the common name for a Personal Escape Set?

- **EEBD (Emergency Escape Breathing Device).**
- A resuscitator.
- ELSA set.
- Self Contained Breathing Apparatus.

What is the important feature of the goggles on a tanker carrying Noxious Liquid Substances?

- **They must be tight fitting and splash proof.**
- Nothing in particular, chipping goggles are adequate
- They must be lightweight.
- They must have 2 headbands for extra safety.

What is the important feature of the lifeline and belt that form part of the safety equipment on a tanker carrying Noxious Liquid Substances?

- **The line and belt must be fireproof and resistant to the cargo being carried.**
- The equipment must be inspected at least annually.
- The length of the lifeline must be at least double the tankers moulded depth.
- They must always be orange in colour.

What is the inspection criteria for breathing apparatus on a tanker carrying Noxious Liquid Substances?

- **A monthly inspection by a responsible officer and annual inspection and testing at an approved facility.**
- A weekly inspection by the 3rd officer.
- It should be inspected by a class surveyor annually.
- The Chief Engineer inspects all the apparatus monthly.

What is the minimum air duration for an EEBD?

- **10 minutes.**
- 15 minutes.
- As per the vessels Safety Management System.
- It depends on the type of tanker.

When wearing the EEBD how do you support the air bottle?

- **With a strap around the neck supported on the chest**
- You carry it under your arm.
- You put the air bottle into your chemical suit pocket.
- You strap it to your back.

Where would you find information on the cargo to be loaded relating to health hazards?

- **The MSDS (Material Safety data Sheets).**
- SOLAS consolidated edition.
- The Ship Captain's Medical Guide.
- The World Health organisation.

Who has the responsibility to ensure the correct health hazard information is passed to the vessel?

- **The supplier of the cargo.**
- MARPOL inspectors.
- The ship broker.
- The ship owner.

You are concerned about the toxicity of the cargo stated in the voyage orders. What is toxicity?

- **The degree to which a substance or mixture of substances can harm humans or animals.**
- A gas below its critical temperature.
- The amount of alcohol in the blood stream due to toxicity of the cargo.
- The name given to a gas sample.

7053 – TOTS 1C-Engine room operations during manoeuvring

Version 1.0 (7)

As per SOLAS requirements, where should the quick close fuel valves' release cabinets be located?

- **Outside the engine room.**
- Inside the engine room.
- Near the engine control room.
- On the navigation bridge.

Does SOLAS legislation exist stating how many steering gear hydraulic pump units are directly connected to the emergency switchboard?

- **Yes, according to SOLAS legislation there should be a minimum of one pump.**
- According to SOLAS legislation all the pumps should be connected.
- No such requirement exists.
- SOLAS do not have the power to legislate on this requirement.

During manoeuvring what condition would you expect to consistently apply to the starting air system?

- **The starting air pressure should be kept at the maximum possible level at all times.**
- At least two air compressors should be running at all times.
- Only one air compressor should be run at all times and the second one kept on stand-by.
- The bridge should be informed of the status of the starting air capacity on a regular basis.

How many persons minimum, would you expect to be on duty in the Engine Room during manoeuvring?

- **One senior and one junior engineer.**
- All senior engineers and one junior.
- One Senior and all junior engineers.
- The Chief Engineer and one junior engineer.

If a fire has broken out in the purifier room during manoeuvring, which media should be used as a last resort?

- **Seawater.**
- Carbon Dioxide (CO2).
- Foam.
- Powder.

If one unit of the main engine is out of order due to a fracture in the liner, what action should be taken with respect to the scavenge air supply to the unit before manoeuvring?

- **The scavenge air should be isolated from the unit.**
- No action needs to be taken with respect to the scavenge air.
- The exhaust valve should be adjusted to remain open during the full cycle to prevent build up of air pressure.
- The scavenge air should be increased to improve cylinder cooling.

If the main engine remote control from the bridge and from the engine control room is defective, what should be done?

- **Transfer should be made to main engine local control.**
- Transfer should be made to main engine control from the emergency steering gear.
- Transfer should be made to main engine control from the emergency switchboard.
- Transfer should be made to main engine deck control.

If, during manoeuvring, you find that a fire had broken out in the purifier room what action should you take after raising the alarm and advising the bridge of the situation?

- **Try to extinguish the fire using portable fire fighting equipment.**
- Call a vessel in the vicinity for help.
- Call the designated person ashore for help.
- Evacuate all non-required personnel from the engine room.

In the event of a main engine scavenge fire, which of the following options is the best course of action to take immediately?

- **Reduce main engine rpm to a minimum.**
- Speed up the engine in an effort to blow the fire out.
- Start the auxiliary blowers.
- Stop the engine and keep it in one position until the fire is out.

In what configuration should the starting air system bottles be set prior to arrival manoeuvring?

- **All air bottles should be fully charged with one online and one on stand-by.**
- All air bottles should be charged and fully open.
- All air compressors should be switched to MANUAL mode.
- All air-operated deck equipment should be isolated.

Is there a requirement to increase Main Engine Cylinder Lubrication during manoeuvring?

- **No, unless special circumstances apply.**
- No, as per MARPOL requirements.
- Yes, as manoeuvring is considered a special circumstance.
- Yes, as per the lubrication oil company requirements.

Lubricating oil pressure drops if its temperature rises.

- **True**
- False

What action should be carried out FIRST if a fire has broken out in the engine room during manoeuvring?

- **Ensure the fire alarm is raised.**
- Call the bridge and ask for assistance.
- Call the Emergency Party.
- Stop the main engine by pushing the emergency stop button.

What action should be carried out if an electrical black-out occurs in the engine room during manoeuvring?

- **After the emergency diesel generator starts, run only minimum machinery for safe manoeuvring.**
- Give assistance to the Electrician and try to bring back the electrical power as soon as possible.
- Inform the bridge of the situation and try to restart the diesel generators.
- Wait for the emergency diesel generator to start and evacuate the engine room immediately.

What action should be carried out if the CO2 alarm is raised in the engine room?

- **All personnel should be evacuated from the engine room immediately.**
- All running equipment should be stopped first and then all personnel should leave the engine room.
- It should be re-checked to make sure it is not a false alarm.
- The Master should be informed immediately of the alarm.

What action should be carried out when one fuel pump becomes defective during manoeuvring?

- **The fuel pump should be hung up, the inlet valve shut and the bridge informed of the problem.**
- The main engine should be stopped immediately and the bridge informed of the problem.
- The main engine's revolution should be decreased to avoid overloading other cylinders.
- The main engine's revolution should be increased to recover the loss of power on one cylinder.

What action should be taken if the main engine's fresh water cooling pressure drops during manoeuvring?

- **Reduce the main engine revolutions.**
- Ensure both cooling water pumps are running.
- No action is required because the main engine load is low.
- Stop the main engine immediately.

What action would you take if the main engine lubricating oil pressure dropped suddenly during manoeuvring ?

- **Request permission from the bridge to stop the main engine as soon as possible and warn them that it may stop automatically.**
- The main engine revolutions should be increased to compensate.
- The main engine should be kept running while waiting for the stand-by oil pump to start.
- The main engine's revolutions should be reduced.

What action would you take if, during manoeuvring operations the main engine oil mist detector alarm was activated, the engine automatically slows down and the alarm condition clears?

- **Advise the Master and prepare to stop the engine as soon as it is practical for the manoeuvring operations.**
- Advise the Master and continue with slow speed operation, until the completion of manoeuvring.
- Advise the Master and stop the engine immediately to inspect the crankcase.
- Advise the Master of the alarm activation and continue with the normal engine operations.

What activities related to arrival manoeuvring does the International Safety Management Code (ISM) require to be logged in the Engine Log Book?

- **All pre-arrival checks and tests carried out.**
- All function tests carried out on communications (Bridge / Engine Room).
- All function tests carried out on the main propulsion system.
- All tests carried out as per arrival check lists.

What affects the distribution of electrical load between electric generators in parallel?

- **Changing individual prime mover revolutions.**
- Adjusting individual excitation current.
- Adjusting individual generator voltage.
- Removal of consumer load demand.

What condition should stand-by equipment be in prior to arrival / departure manoeuvring?

- **All stand-by equipment should be operable / available and in standby condition.**
- All stand-by equipment should be running but not delivering its intended service.
- All stand-by equipment should be switched OFF.
- All stand-by equipment should be switched ON and running.

What does MARPOL regulate in the marine industry?

- **Prevention of Marine Pollution.**
- Marine Police.
- Maritime Protocol of Loading.
- Means and Rules governing Repatriation of Leavers.

What is one of the first actions that should be carried out during manoeuvring if one or more of the main engine's turbochargers is defective?

- **Check that all auxiliary electric-powered blowers are switched on and operating correctly.**
- The main engine should be stopped and shore assistance should be called.
- The main engine's revolution should be increased for more power.
- The main engine's revolutions should be reduced.

What is the first action you should take if the main engine oil mist detector alarm sounds?

- **Slow down the main engine immediately and inform the bridge of the problem.**
- Call the Technical Superintendent for help immediately.
- Check if the alarm is genuine before initiating any action.
- Inform the bridge and re-check the main engine's bearing temperatures.

What is the main purpose of the "Arrival / Departure Checklist"?

- **To confirm that all systems necessary for manoeuvring are in working order.**
- To be used as a teaching aid to junior ranks.
- To ensure personnel changes on board are documented.
- To provide Company data required by the office.

What is the maximum allowed sulphur content in "Low Sulphur Fuel"?

- **1.5%**
- 1%
- 1.75%
- 2%

What is the normally accepted method for testing automatic start-up of stand-by equipment?

- **By manually stopping the in service unit.**
- By checking that the standby unit switch is correctly set.
- By doing a black-out simulation.
- By running both units simultaneously.

What is the purpose of a steering gear "Creep Test"?

- **To check the condition of the steering gear hydraulic system seals and valves.**
- To check if the steering gear autopilot is in good order.
- To check if the steering gear non-follow up control system is in good order.
- To check that the steering gear does not fall back from the hard-over position.

What is the usual acceptable time for the steering rudder to swing from "hard to port" to "hard to starboard"?

- **28 seconds.**
- 18 seconds.
- 38 seconds.
- 48 seconds.

What minimum starting air pressure is normally required to start a main engine?

- **12 bars.**
- 10 bars.
- 15 bars.
- 17 bars.

What should be done if the remote control of the steering system from the bridge becomes defective during manoeuvring?

- **Change over to local control and engage one person for steering.**
- Call tugboats for assistance and proceed with manoeuvring.
- Stop all operations immediately.
- Stop manoeuvring until the remote control has been repaired.

When, during manoeuvring, would you expect to changeover from heavy oil to diesel oil on a vessel where the two fuels are utilised?

- **Prior to manoeuvring.**
- At a suitable time during manoeuvring.
- It is optional dependent on whether overhaul work will be carried out on the engine.
- Just before shutting down allowing time to flush through the fuel system.

Which extinguishing media is no longer acceptable for fitting onboard vessels?

- **Halon.**
- Carbon Dioxide (CO2).
- Low expansion foam.
- Water mist.

Which of the following tests would NOT normally be carried out as part of a pre-arrival standby routine?

- **The testing of the fuel system quick-closing valves.**
- Testing communications between the bridge and engine room.
- The testing of main engine starting ahead and astern.
- The testing of the steering gear.

Which regulations cover the discharge of NOx and SOx?

- **MARPOL**
- APPS
- ISGOTT
- SOLAS

While on Main Engine stand-by during a river passage towards a port, you get a high exhaust temperature alarm on one main engine cylinder. What would you do?

- **Call the bridge and advise the Master of the problem.**
- Acknowledge the alarm and watch to see if the temperature increases.
- Acknowledge the alarm until you get into port.
- Stop the main engine.

Who should be present in the engine room during manoeuvring besides duty personnel?

- **A Senior Engineer.**
- All Engine Ratings.
- All Engine Staff.
- The Electrician and the Fitter.

Why is water not suitable for extinguishing main engine scavenge fires?

- **It will flash off and the fire may develop into a hydrogen fire.**
- It will cause the liner, piston rod and scraper box parts to rust.
- It will cool the space too quickly and cause the liner to crack.
- It will leak into the crankcase space and contaminate the lubricating oil.

Would you expect the Control Room to be manned during manoeuvring in an UMS classed vessel?

- **Yes.**
- No manning is required due to UMS conditions.
- No manning is required even if no UMS conditions apply.
- Yes, as required by the Master.

Would you expect the Engine Room itself to have personnel engaged in watchkeeping duties during manoeuvring?

- **Yes.**
- No, due to Company requirements.
- No, due to safety requirements.
- Yes, as required by the Master.

7054 – TOTS 1C-Watchkeeping in the Engine Room

Version 1.0 (5)

Crankcase Lub oil consumption of a slow speed main engine should be monitored to confirm that it is in line with the makers specific consumption . . .

- **to verify that oil is not being lost due to leakage or worn scraper boxes.**
- so that the need for regular topping up of the crankcase is not required.
- to ensure that leakage does not contaminate the bilge spaces.
- to ensure that there is no air entrainment in the system oil due to the crankcase level running low.

During cargo operations, on a vessel with steam turbine driven cargo pumps, the condenser salinity alarm sounds intermittently but resets automatically. What action should you take?

- **Test the chloride level in the boiler to check if it is rising.**
- Check the sea water pressure to the condenser to check that it is not too high.
- Ignore the alarm as it is obviously an intermittent fault.
- Slow down the turbine to reduce the load on the condenser.

During night-time under-unmanned operations, which procedure for engine entry should be followed?

- **The bridge must be informed of the entry.**
- Entry is allowed only if two people enter together.
- Entry is allowed only with the Chief Engineer's permission.
- Entry is prohibited.

Duty engineers are NOT allowed to create their own personal standing orders for watchkeeping ratings.

- **True.**
- False.

How are the conditions of the cooling water, boiler water and lube oil maintained?

- **By regular testing and treatment.**
- By maintaining the correct liquid levels.
- By shore side service technicians on a quarterly basis.
- By top up of the water level and purification of lub oils.

How do duty engineers acknowledge the Chief Engineer's standing orders and orders in the Engine Order Book?

- **By signature.**
- Acknowledgement is not required.
- By telephone confirmation to the Chief Engineer.
- By verbal verification.

How do the watch keepers ensure various equipment is maintained regularly?

- **A work list is produced from the planned maintenance system & the running hours sheet is followed.**
- Equipment to be maintained will be identified on a daily basis by the Duty officer.
- The Classification society issues a listing for each vessel of machinery to be overhauled / maintained, and time frames for doing such work.
- Through morning meetings and delegation of the 2nd engineer.

How do the watch keepers ensure various equipment is working correctly?

- **By regular performance monitoring of each piece of equipment.**
- By completing planned maintenance in accordance to maintenance schedules.
- If equipment is defective, the alarm panel will indicate that accordingly.
- Via daily log book readings (pressure / temp / etc).

How do you ensure the various equipment is maintained regularly?

- **By following a work list produced from the planned maintenance system.**
- The Classification Society issues a listing of machinery to be overhauled/maintained, and time frames for doing such work.
- The equipment to be maintained will be identified on a daily basis by the Duty officer.
- Through morning meetings and delegation of the 2nd engineer.

How do you ensure the various indicating instruments are reading correctly?

- **By calibrating the equipment for their accuracy.**
- By double checking, via portable equipment.
- If they are not reading correctly, an alarm will sound and the machinery will not operate.
- They are changed during planned maintenance.

How frequently should the various auto control devices be tested for correct operation?

- **At regular intervals as laid down in the vessel's planned maintenance system.**
- All auto control devices are checked during dry docking, and certification is issued.
- Auto control devices operate automatically, and therefore do not require intervention.
- When stand-by equipment is changed over, the auto control devices indicate any errors, via an alarm.

How many times per day is a regular equipment check to be carried out when the vessel is underway?

- **Once per watch.**
- Each hour.
- Twice a day.
- When ordered by the Chief Engineer.

How should the engine watch be kept at anchorage?

- **The vessel is on stand-by for manoeuvring.**
- No watch is required.
- The vessel is in navigation.
- The vessel is in port.

If a Rating is part of an engine watch, who is he directly responsible to?

- **To the Duty Engineer.**
- To the Chief Engineer.
- To the Master.
- To the Second Engineer.

If an alarm sounds and you are on duty during a UMS period what are the first actions you should take?

- **Cancel the alarm and investigate the cause.**
- Advise the Chief Engineer that it is necessary to return to full watchkeeping routine.
- Call the Bridge to inform them of an alarm state.
- Cancel the alarm and reduce the speed of the Main Engine.

If the vessel has unmanned machinery space, what precautions should you take before entering, when on UMS mode?

- **Inform the bridge watch keeper**
- Inform the 2nd Engineer.
- Inform the Chief Engineer.
- Inform the master.

In normal circumstances, which parameters on steam boilers are the most important?

- **The water level and steam pressure.**
- The water level and the steam temperature.
- The water pressure and the steam temperature.
- The water temperature and the steam pressure.

In port, the engine watch depends on . . .

- **cargo operations.**
- crew change operations.
- crew shore leave.
- ship supply operations.

What is a dead man alarm?

- **A switch which should be operated at regular intervals which, if not activated, will give an alarm on the bridge.**
- A direct link alarm to the office in the event of an emergency.
- An Alarm that is activated when an injured crew member is discovered.
- Terminology for an alarm which is malfunctioning or out of order.

What is the basic rule for good watchkeeping?

- **Regular visual checks.**
- To have a good engine rating as a watchkeeper.
- To have good lighting in the engine room.
- To keep the engine room clean at all times.

What is the diesel oil service tank sludge cock used for?

- **To drain any water out of the tank.**
- For draining the tank for cleaning.
- To draw off a sample for analysis.
- To draw off clean oil for cleaning equipment.

What is the normal temperature for the deep frozen meat and fish chambers?

- **Minus 18˚ Celsius (-18˚C).**
- Minus 12˚ Celsius (-12˚C).
- Minus 4˚ Celsius (-4˚C).
- Minus 8˚ Celsius (-8˚C).

What is the purpose of the Engineers Call system and when should this be activated?

- **In case of any emergency situation requiring the support of all Engine room staff.**
- During Fuel Oil and Lub Oil Bunker operations when extra assistance is required.
- The Engineers call is used in the mornings to ensure all engine staff are present in the E/R.
- When the duty officer is changing rotation, all engine officers are to be present for handover.

What is the usual required preheating temperature for 380 cSt (centistokes) of heavy fuel (HFO)?

- **130-140˚C.**
- 100-110˚C.
- 110-120˚C.
- 140-150˚C.

What should be done by the watchkeeping engineer if the relieving engineer is not ready for watch hand-over?

- **He should continue with the watch and inform the Chief Engineer.**
- He should continue with the watch.
- He should leave the engine room and inform the Chief Engineer.
- He should leave the engine room and inform the Master.

What should be done if any problem to the propulsion power or steering gear occurs during the watch?

- **Immediately inform the Officer of the Watch on the bridge.**
- Call the Chief Engineer immediately.
- Call the Master immediately.
- Immediately call for help.

What temperature should HFO be maintained at when passing through the purifier?

- **At a temperature calculated according to the density of the fuel being purified.**
- It should be maintained as close to 99 degree C as possible.
- It should be maintained at 101 to 103 degree C.
- It should be maintained at 90 to 95 degree C.

What temperature should lubricating oil be maintained at when passing through the purifier?

- **It should be maintained at 90 to 93 degree C.**
- At a temperature calculated according to the viscosity of the oil being purified.
- It should be maintained at 60 to 65 degree C.
- It should be maintained at 75 to 80 degree C.

What will happen if the "dead man alarm" is not reset on time?

- **The general alarm will sound.**
- The Chief Engineer will be notified of the alarm.
- The engineer call alarm will sound.
- The Master will be notified of the alarm.

What would rapid fluctuation of the main engine cooling water pressure indicate?

- **The possibility of a cracked cylinder liner or cylinder cover.**
- A problem with the cooling water circulating pump impeller.
- Problems with the cooling water temperature control valves.
- Speed fluctuation of the cooling water pump due to a motor problem.

When is the Engine Order Book filled in?

- **Whenever the Chief Engineer has additional orders.**
- From time to time.
- Only in port.
- When ordered by the Master.

When the engine watch is changing over from manned to unmanned (UMS) mode, which of the following equipment must be switched on?

- **The Engineer call alarm.**
- Another diesel generator.
- The second steering gear pump.
- The stand -by cooling pump.

When the engine watch is changing over from manned to unmanned (UMS) mode, who should be informed?

- **The bridge.**
- All crew.
- The Chief Engineer.
- The Master.

Where should all important requirements and activities carried out during a watch be written or recorded?

- **In the Engine Log Book.**
- In the Engine Activities Notebook.
- In the Engine Maintenance Log.
- On the blackboard in the cargo control room.

Which of the following statements is correct?

- **The watchkeeper must call the Bridge every 20 minutes to advise them if no "dead man" alarm is fitted?**
- Ear defenders must not be used in the ER as it is essential to hear all noises.
- It is expected that the maintenance of equipment will be carried out when on UMS.
- When an alarm sounds, no watchkeeper must enter the engine room under UMS until they have advised the Bridge.

While in port, all repairs on the main engine can be carried out without permission from local authorities.

- **False.**
- True.

Who is the Duty Engineer directly responsible to?

- **To the Chief Engineer.**
- To the Company.
- To the Master.
- To the Superintendent.

7055 – TOTS 1C-Operation, monitoring, machinery

Version 1.0 (7)

A general indication of good combustion in an operating engine is . . .

- **clear exhaust gases.**
- less fuel and lube oil consumption.
- less noise and less vibration.
- slightly sooty exhaust gases.

A heat exchanger where two liquids flow in opposite directions is called . . .

- **counter-flow.**
- horizontal flow.
- opposite flow.
- parallel flow.

A high vacuum reading on the suction side of a gear pump . . .

- **is indicative of partially opened suction valves or blocked strainers.**
- is indicative of damaged gears.
- means that the discharge valve is shut.
- shows normal condition.

A lub oil purifier fault alarm, if not attended to immediately . . .

- **may result in good lub oil discharging to the sludge tank.**
- may result in a breakdown.
- may result in an overload condition.
- may result in contamination of the oil.

A ship's air conditioning system is normally designed to give . . .

- **six to eight changes of air per hour.**
- ten to twelve changes of air per hour.
- twelve to sixteen changes of air per hour.
- two to three changes of air per hour.

An electrical driven ballast pump will not start because one of the main fuses has blown. Which of the following would you check, prior to trying to restart the pump motor?

- **The motor and pump rotation.**
- The control system fuses.
- The electrical power supply.
- The insulation of the motor windings.

An incinerator is normally used . . .

- **to incinerate waste oil, sludge and other waste.**
- to burn galley waste.
- to burn plastic materials.
- to burn sewage.

An overcharge or excess refrigerant in the system will be seen as . . .

- **high condenser pressure.**
- frost formation.
- high cooling effect.
- high lube oil pressure.

Any Oily Water Separation operation has to be recorded and noted in the Oily Record Book.

- **True**
- False

As the temperature of air is reduced, its capacity to carry water vapour . . .

- **is reduced.**
- is marginally increased.
- is rapidly increased.
- remains the same.

Cavitation is the result of . . .

- **the formation and collapse of air bubbles.**
- a too high discharge pressure on the pump.
- impurities in the water impacting on the impeller.
- overloading of the impeller.

Centrifugal pumps . . .

- **are not self-priming.**
- are effective for pumping high viscous fluid.
- are positive displacement pumps.
- are self priming.

Compressors must always be started in . . .

- **unloaded condition.**
- auto mode.
- loaded condition.
- manually.

Control air is normally supplied around . . .

- **7 bars.**
- 1 bar.
- 30 bars.
- 40 bars.

During discharge, problems with the inert gas plant result in the production of clean inert gas being reduced by 50%. What do you do?

- **Advise the Chief Officer so that he can reduce the discharge rate in line with the reduction of the gas supply.**
- Adjust the boiler controls to increase the fuel and reduce the air in an attempt to reduce the oxygen content of the gas.
- Allow the discharge to continue as the oxygen content in the tanks may not rise significantly during the remaining discharge period.
- Shut down the discharge until the inert gas system problem is resolved.

For any maintenance work, the sewage system should be considered as an enclosed space.

- **True**
- False

For better purification, the throughput of a purifier . . .

- **has to be maintained at minimum.**
- has no effect on purification.
- has to be maintained at maximum.
- should be changed depending upon fuel quality.

For the most efficient and practical separation of oil from water, what is required?

- **As long as possible dwell time in the separation tank.**
- A three stage separation tank.
- A two stage separation tank.
- A very high temperature in the separation tank.

How is the rudder travel stopped in the event of total telemotor failure?

- **It reaches the physical stop-limits of the steering gear.**
- No other means are provided.
- The emergency diesel alternator will start and supply power.
- The rudder cannot be stopped.

If one generator is on load and the other generator is to be taken on load (share), you need to . . .

- **synchronise phase and frequency of the incoming generator.**
- increase the speed of both generators.
- reduce the speed of both generators.
- start a third generator before load sharing.

If the auxiliary boiler feed water hotwell was found to be heavily contaminated with oil, what initial action would you take?

- **Overflow the hotwell until the water is clear then top scum the boiler.**
- Drain the hotwell and blow down the boiler, then refill with treated fresh water.
- Drain the hotwell and refill with fresh water to the correct level.
- Overflow the hotwell until clear water is seen in the sightglass.

If the safety valves fitted on an air reservoir can be isolated from the reservoir, then . . .

- **the reservoir must be fitted with a fusible plug.**
- it can be used legally without any added safety measures.
- the compressor must be fitted with a fusible plug.
- the safety valve must be fitted with a fusible plug.

In a boiler, the system is set so that when the boiler is switched on, the following sequence of operation takes place . . .

- **(1) heat up and circulate fuel; (2) purge the combustion chamber; (3) turn on the fuel and ignite the flame.**
- (1) heat and circulate the fuel(2) turn on the fuel and ignite the flame (3) purge the combustion chamber.
- (1) purge the combustion chamber; (2) heat and circulate the fuel; (3) turn on the fuel and ignite the flame.
- (1) purge the combustion chamber; (2) turn on the fuel and ignite the flame (3) heat and circulate the fuel.

In a cargo system fitted with a vertical submerged pump, the pumped fluid contained in the vertical column will fall down into the tank when the pump is stopped after discharge. The normal remedy for this is by . . .

- **using the purge connection fitted to clear the discharge pipe.**
- stripping.
- the use of manually operated valves.
- the use of non-return valves.

In a typical tubular heat exchanger, the expansion of the tubes are accommodated by . . .

- **a floating header.**
- choosing the right material.
- leaving one end of the cooler free to move axially.
- using pre-expanded tubes.

In order to test a weld repair to a lubricating oil storage tank the tank is to be tested with compressed air to check for leakage. What is the maximum pressure that should be used?

- **0.2 Bar**
- 0.02 Bar
- 1.0 Bar
- 2.5 Bar

In plate type coolers you would normally expect the plates to be made of . . .

- **titanium.**
- nickel brass.
- silver.
- stainless steel.

In ships other than those engaged in short voyages, the steering gear should be thoroughly checked and tested . . .

- **within 12 hours before departure.**
- on standby.
- when one hour's notice is given.
- within 24 hours before departure.

In the event that the sludge generated on board exceeds the incinerator capacity the excess sludge should be . . .

- **transferred to a shore facility.**
- discharged only when vessel is 12 nautical miles or more from land.
- discharged via OWS before entering port.
- transferred to fuel oil bunker tank.

It is recommended that the Oil Content Monitor of an Oily Water Separator is calibrated at least . . .

- **annually.**
- once every five years.
- once every ten years.
- once weekly.

Moisture in a refrigeration system could lead to . . .

- **ice formation and a drop in pressure on the evaporator side.**
- excessive discharge pressure.
- the dryer getting clogged.
- the loss of refrigerant.

On a tanker with 3 centrifugal cargo pumps, but with discharge reduced to 2 pumps due to shore restrictions you note that the idle pump is running backwards. What would you suspect as being the main problem?

- **The idle cargo pump non-return valve is not closing properly.**
- The non-return valve on one of the running pumps is defective and causing it to drive the idle pump . . .
- The Pumpman has not closed the idle pump suction and discharge valves.
- The Pumpman has set the valve incorrectly allowing the running pumps to drive the idle pump.

On a tanker, when on voyage with a heated cargo, the hotwell is found to be overheating due to steam returning from the cargo heating coils. What should you do?

- **Check the steam traps drains to see if the traps are functioning correctly.**
- Check that the Cargo Officer has not opened up the heating coil valves too far.
- Check the cargo heating log to see if the cargo tank temperatures are too high.
- Check the heating coil steam trap drain filters to see if they are clean.

Plate type heat exchangers can effectively deal with . . .

- **low pressure fluids.**
- high pressure fluids.
- large volumes.
- vapour and gases.

Sewage discharge regulation can be found in the . . .

- **MARPOL regulations.**
- Health & Safety regulations.
- ISGOTT regulations.
- SOLAS regulations.

The air bottle(s) drain should be regularly checked and drained . . .

- **to remove any water present.**
- because it is part of the checklist.
- because it is recommended by the makers.
- or the valves will seize and cannot be operated.

The cause of an abnormally high condenser pressure within a refrigeration system would be . . .

- **incondensable gases in the refrigerant.**
- a defective thermal expansion valve.
- a low cooling water flow through the condenser.
- an above normal level of refrigerant in the condenser.

The closed flashpoints of fuel for general use should not be less than . . .

- **60 degrees Celsius.**
- 65 degrees Celsius.
- 70 degrees Celsius.
- 75 degrees Celsius.

The control of temperature in heat exchangers when the cooling medium is seawater is normally achieved by . . .

- **by-passing a proportion or all of the hot fluid flow.**
- by-passing or limiting (controlling) the seawater flow.
- recirculating seawater discharge back to suction.
- throttling the flow of hot liquid.

The easing gear fitted to boiler safety valves is there to . . .

- **allow the valves to be opened to release pressure in the event of a boiler problem.**
- allow testing of the valve travel during safety checks.
- allow testing to make sure that the valves are not stuck shut.
- allow the valve to be eased off the seat slightly when adjusting the relief pressure.

The exhaust temperature deviation between units should be maintained within . . .

- **10%**
- 0%
- 40%
- 60%

The function of the inert gas scrubber is . . .

- **(1) to cool the gas; (2) to remove the sulphur dioxide; and (3) to remove the soot.**
- (1) to cool the gas; (2) to remove the soot.
- (1) to cool the gas; (2) to remove the sulphur dioxide.
- to remove toxic matter.

The inert gas blowers should automatically shut down in the event of . . .

- **high temperature.**
- a high motor load.
- deck seal failure.
- high pressure.

The maximum permissible oxygen content of the inert gas going into the cargo tank should be . . .

- **5% by volume.**
- 6% by volume.
- 8% by volume.
- 9% by volume.

The oil content of bilge water is normally reduced from 100 to 15 ppm and below by the use of . . .

- **coalescent filters.**
- centrifugal separation.
- chemical treatment.
- steam heating.

The water side of an intercooler is protected against overpressure by . . .

- **bursting the disc on the cylinder jacket.**
- pressure testing it with the required pressure.
- the fusible plug on the cylinder jacket.
- the relief valve on the outlet side.

To prevent damage of the compressor, the refrigerant return line . . .

- **should not contain any liquid.**
- should be a mixture of gas and liquid.
- should be condensed to a liquid state.
- should not go back to the compressor.

What are the three distinct sections of an inert gas system?

- **(1) production of the inert gas; (2) cleaning and cooling the gas; and (3) distribution of the gas.**
- (1) the boiler to produce the inert gas; (2) the scrubber to clean the gas; and (3) the deck seal.
- (1) the boiler; (2) the inert gas generator; and (3) the scrubber.
- (1) the inert gas fan; (2) the deck seal; and (3) the pressure vacuum valves.

What do you predict would be the cause, if all the auxiliary diesel engines / generators, when running in parallel operation, started shedding load and hunting?

- **It could be caused by a reduction in the flow of fuel through the main filters.**
- It could be caused by a defective droop setting on one of the automatic voltage regulators.
- It could be caused by a faulty governor speed adjustment motor on one auxiliary engine.
- It could be caused by a loss of oil from the governor control circuit.

What is considered to be the main value of using ammonia as a refrigerant gas.

- **It doesn't deplete the ozone layer.**
- It is non-corrosive and non-irritant.
- It is non-toxic and non-flammable.
- It is readily available world wide.

What is meant by hunting in a control system?

- **The system being controlled does not settle on the set point.**
- The system being controlled does not reach the set point and remains below the set point.
- The system being controlled overshoots the set point and remains above the set point.
- The system being controlled reaches the set point but then falls back and remains offset.

What is meant by the term "surge" in relation to the Main Engine turbo charger?

- **It is a stalling of the airflow across the compressor wheel.**
- It is a constant loud whistling noise.
- It is a high frequency vibration of the turbo charger rotor.
- It is a rapid increase in pressure and air flow at high velocity from the diffuser to the scavenge trunk.

What is the cause if, on a two stage main air compressor the discharge pressure is low and the 1st stage discharge temperature is higher than normal?

- **The 2nd stage suction valve is defective.**
- The 1st stage discharge valve is defective.
- The 2nd stage discharge valve is defective.
- The 2nd stage piston rings are damaged.

What is the correct sequence of events that take place in an engine cycle?

- **(1) Compression (2) Fuel injection (3) Expansion (4) Exhaust**
- (1) Compression (2) Expansion (3) Fuel Injection (4) Exhaust
- (1) Expansion (2) Exhaust (3) Fuel Injection (4) Compression
- (1) Fuel Injection (2) Exhaust (3) Expansion (4) Compression

What is the discharge regulation if the sewage is not comminuted or disinfected?

- **The ship should be on voyage and more than 12 nautical miles from the nearest land.**
- The ship should be more than 12 nautical miles from the nearest land.
- The ship should be on voyage and four nautical miles from the nearest land.
- The ship should not be in special areas and should be proceeding at not less than three knots.

What is the first action you should take when there is no water in the boiler gauge glasses?

- **Shut the fuel off from the boiler.**
- Blow down the gauge glass locally and check the water level.
- Increase the feed water supply so that the boiler doesn't run dry.
- Reduce the load on the boiler.

What is the most common cause of the lowering in performance of heat exchangers?

- **Fouling of the heat transfer surface.**
- An air lock.
- Low water flow.
- The division plate joint failure.

What is the primary function of the deck seal?

- **To prevent the backflow of cargo tank gases, even when the inert gas plant is shut down.**
- To isolate the cargo tanks from the pump room.
- To prevent cargo and toxic gas from entering into the engine room while the inert gas generator is in operation.
- To provide a non-return seal on gases from the pump room to the engine room.

What is the purpose of a refrigerant compressor shaft seal?

- **To prevent leakage of oil and refrigerant from the crankcase.**
- To lock the shaft when not in operation.
- To prevent a crankcase explosion.
- To prevent any moisture or air from entering into the compressor.

What is the purpose of boiler water treatment?

- **To prevent corrosion, scale and increase efficiency.**
- To keep the water clean.
- To maintain the correct water temperature.
- To reduce water consumption.

What would cause a medium speed main engine exhaust gas temperature to be higher and the scavenge air pressure to be lower than normal, at a fixed load?

- **A partly blocked air cooler.**
- Incorrect fuel pump timing.
- Incorrect valve timing.
- Poor fuel injection.

What would cause the reduction in viscosity and flash point in the lubricating oil of an auxiliary engine operated on marine gas oil?

- **Fuel leakage into the crankcase.**
- A build up of combustion products in the lubricating oil.
- The lubricating oil is at an operating temperature that is too high.
- The wrong grade of lubricant being added to the crankcase.

What would you expect the main source of inert gas to be in a VLCC?

- **The main and auxiliary boiler.**
- A gas turbine plant with an after burner.
- An independent inert gas generator.
- The main engine exhaust.

What would you expect to cause smoky exhaust?

- **Sticky fuel injector.**
- A sticking air valve.
- A sticking fuel pump.
- Unsuitable cylinder lubricating oil.

When charging the refrigeration system with more refrigerant gas . . .

- **the main liquid valve should be closed and the gas introduced before the regulating valve.**
- always inform the Chief Engineer.
- it is essential to weigh the refrigerant cylinder before and after the operation.
- you may choose whether to introduce the gas in the suction or discharge side.

When cooling a piston by system oil rather than water, the return temperature of the oil must be lower than it would be if using water because . . .

- **it avoids carbon deposits on metal surfaces.**
- it reduces the danger of fire.
- it reduces the danger of leaks.
- the specific heat capacity of water is less than oil.

When do you expect to have to change from low to high SW suction?

- **When the ship is entering an area where there will be restricted under keel clearance.**
- When the ship is entering a canal.
- When the ship is in heavy weather.
- When the ship loads a full cargo and is at maximum draught.

When the bunker fuel needs more purification, you should normally . . .

- **operate two purifiers in parallel.**
- change the gravity disc.
- increase the purifier throughput.
- operate two purifiers in series.

When using a cargo stripping eductor . . .

- **always open the cargo line after creating enough vacuum.**
- always open the cargo line first, and then start to build up a vacuum using seawater.
- it is always recommended to use remote operated valves.
- you do not have to wait for the vacuum to build up.

Which of the following arrangements is normally provided to facilitate complete tank draining on vessels fitted with centrifugal discharge pumps?

- **A vacuum stripping system.**
- A gravity drain.
- A tank pressurisation system.
- Centrifugal pumps connected in a series.

Which of the following is most likely to make the engine run unevenly?

- **Sticky fuel pump control racks or linkage.**
- The fuel system not being primed.
- Unsuitable fuel.
- Water in the fuel.

Which of the following proportions make up good quality inert gas, from an inert gas generator plant?

- **nitrogen 77% by volume, carbon dioxide 13%, oxygen 4%, no water content.**
- nitrogen 75% by volume, carbon dioxide 13%, oxygen 12%, some water content.
- nitrogen 80% by volume, carbon dioxide 13%, oxygen 6%, some water and sulphur content.
- nitrogen 90% by volume, carbon dioxide 1.8%, oxygen 8%, no water content.

Which of the following safety components would you expect to find in the air starting system of a main engine?

- **All three devices.**
- Flame traps.
- Non-return automatic valves.
- Relief valves.

Why is it necessary to preheat the engine before attempting to start it?

- **To reduce thermal stresses.**
- To ensure that it starts first time.
- To reduce start failure.
- To reduce the temperature gradient.

Why would you turn over a two stroke engine with starting air before commencing manoeuvring?

- **To clear the combustion space of any accumulation of liquid, and to test for potential leaks.**
- To check the propeller is clear of obstructions.
- To prime the fuel to the engine.
- To test the start air system.

You would expect a deck seal to be . . .

- **of wet or dry types.**
- gland packings.
- mechanical type seals.
- normal non-return valves.

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After the overhaul of a main bearing and after crankcase inspection, what needs to be done first before manoeuvring with the engine?

- **Turn the engine on turning gear with the lub oil pump running.**
- Blow through engine using Local / Manual control
- Blow through the engine from Control room to test controls.
- Turn the engine on turning gear.

An increase in main engine crankshaft deflection readings may be an indication of . . .

- **excessive wear of main bearing shells.**
- excessive torsional stress.
- excessive wear of thrust bearings.
- excessive weight of piston and connecting rod.

Before withdrawing the piston during the overhaul of a unit, what should the position of that piston be?

- **Top dead centre.**
- 90 degrees after top dead centre.
- 90 degrees before top dead centre.
- Bottom dead centre.

Crankshaft deflections are normally taken when the engine is . . .

- **hot.**
- at any temperature.
- cold.
- either hot or cold.

During routine rounds of the main engine, you note that the pipe leading to one of the air start valves is hot. Do you?

- **Contact the bridge and request permission to stop immediately.**
- Advise the watch keepers to monitor it and report if the temperature increases.
- Apply portable ventilators to cool the pipe down.
- Attach a portable temperature gauge to record the temperature.

Excessive crankshaft deflections are taken as an indication of . . .

- **an engine misalignment or worn main bearings.**
- a bent crank shaft.
- only an engine misalignment.
- only worn main bearings.

For accurate readings the deflection gauge should be . . .

- **at the same temperature as the engine.**
- at a higher temperature than the engine.
- at a lower temperature than the engine.
- at room temperature.

Generally which parts of an air compressor require most frequent attention?

- **The discharge valves & oil filters.**
- The cylinder head & suction valves
- The oil filter & relief valve.
- The relief valve & cylinder head.

How often would you normally expect to carry out water washing of the exhaust gas economiser?

- **Once a month.**
- Once a year.
- Once every 6 months.
- Once every two years.

How will you ensure the fire fighting and life saving equipment in the Engine room are in readiness?

- **By regular inspection, testing and maintenance.**
- By conducting drills every 3 months.
- By servicing and Certification annually.
- By servicing and Certification during dry docking.

In a large slow speed diesel engine, which of the following would you expect to be directly cooled by Sea Water?

- **Scavenging air.**
- Cylinder heads.
- Fuel valves.
- Turbocharger.

In addition to a deck mounted emergency stop button, what type of safety device would you expect to find on a grinder to protect the operator?

- **A glass shield.**
- A high voltage trip.
- A reverse current trip.
- An over current trip.

In the event of a main engine scavenge fire, which of the following options is the best course of action to take immediately?

- **Reduce main engine rpm to a minimum.**
- Speed up the engine in an effort to blow the fire out.
- Start the auxiliary blowers.
- Stop the engine and keep it in one position until the fire is out.

In which direction is the engine turned when taking crankshaft deflections?

- **Ahead.**
- Any direction.
- Astern.
- Both ahead and astern.

On a diesel engine, if the tappet clearances are greater than the manufacturer's recommend setting, the cylinder head valves will . . .

- **open late and close early.**
- open early and close early.
- open early and close late.
- open late and close late.

Prior to putting an Oily Water Separator into operation, should you?

- **Fill it completely with clean sea water.**
- First drain it completely.
- Flush the oily contents through with sea water prior to switching on the oil content monitor.
- Partially fill it and pressurize it with air.

The purpose of fine mesh screens on bunker tank vent heads is to?

- **Prevent flames from passing into the tank.**
- Prevent accidental overflow from the tank.
- Prevent dirt from getting into the fuel tank.
- Prevent the fumes from the tank affecting personnel working close by.

What action would you take to reduce chlorine levels in boiler water?

- **By controlled blow-down via the bottom blow-down valve.**
- By increasing the chemical dosage.
- By maintaining a high water level.
- By maintaining a low water level.

What actions should be taken regarding critical equipment when starting any vessel plant?

- **Ensure critical equipment is operational & critical standby equipment is ready to be deployed.**
- Ensure personnel are on standby beside all Critical Equipment.
- Ensure the critical equipment is continuously monitored and entries made in the Engine Room Log Book.
- Inform the Chief Engineer before starting any Critical Equipment.

What could be the most common reason for the hydraulic pump in a hydraulic pack to trip on high oil temperature?

- **A fouled oil cooler.**
- A broken discharge valve.
- A broken pump.
- A broken suction valve.

What do you need to refer to while calculating fuel quantities during fuel soundings?

- **The tank calibration tables.**
- The capacity plan.
- The fuel oil viscosity.
- The fuel piping diagram.

What is a piston profile gauge used for?

- **To check for burn down of the piston crown.**
- To measure the piston clearance in the liner.
- To measure the piston groove wear.
- To measure the piston rod wear.

What is the diameter of the high pressure piston in comparison with the low pressure piston?

- **It's smaller.**
- It can vary.
- It's larger.
- It's the same.

What is the effect on the plant of having a high sulphur content in fuel?

- **Corrosion.**
- A higher ignition temperature.
- A lower ignition temperature.
- Fouling of the combustion spaces.

What is the function of an accumulator in a hydraulic pack?

- **It reduces frequency of pump start/stop.**
- To act as an oil reservoir.
- To prevent over pressure.
- To remove an air lock.

What is the main reason for adding chemical to engine cooling water?

- **As a corrosion inhibitor.**
- To lower nitrates.
- To lower the boiling point.
- To reduce chloride levels.

What is the main reason for honing the liners of a medium speed engine during routine overhaul?

- **To remove the glazing of the liner and re-establish a surface which retains the lubricating oil, thus reducing. liner wear**
- To reduce any ovality or barrelling of the liner.
- To remove scratches from the liner wall.
- To remove the wear ridge at the top of the liner.

What is the normal, accepted method of keeping the turbo charger rotor air side blades clean?

- **By water washing.**
- By pneumatic cleaning.
- By soot blowing.
- By steam cleaning.

What is the possible consequence of prolonged oil carry over from a starting air compressor?

- **It could cause an air line explosion.**
- It could cause a high starting air temperature.
- It could cause lifting of the air compressor relief valve.
- It could cause low starting air pressure.

What is the recommended DC voltage output from a welding machine?

- **Less than 70V.**
- Less than 110V.
- Less than 240V.
- Less than 440V.

What is the requirement for the space where oxygen and acetylene bottles are stored on a ship?

- **They should be stored in separate well ventilated compartments on the deck.**
- They should be stored in a well ventilated compartment in the accommodation.
- They should be stored in an enclosed space.
- They should be stored in separated well ventilated compartments in the engine room workshop.

What material are welding gloves normally made of?

- **Leather.**
- Cotton.
- Rubber.
- Wool.

What measurements are taken during the overhaul of a unit of the Main engine?

- **Liner calibration, piston groove measurements and piston rod measurements.**
- Liner calibration.
- Piston groove measurements.
- Piston rod measurements.

What needs to be checked on the liner before putting back the piston during the replacement of the main engine liner?

- **The flow of cylinder lub oil.**
- The flow of camshaft lub oil.
- The flow of jacket cooling water.
- The flow of system lub oil.

What needs to be done before stopping a refrigerating plant for overhaul?

- **Collect the refrigerant in the reservoir.**
- Shut off the refrigerant to the condenser.
- Shut off the refrigerant to the evaporator.
- Shut off the water supply to the condenser.

What position of the piston are exhaust valve clearances adjusted?

- **During the compression stroke.**
- During the exhaust stroke.
- During the intake stroke.
- During the power stroke.

What type of pump would you expect to find in a hydraulic power pack?

- **A positive displacement pump.**
- A multi stage centrifugal pump.
- A single stage centrifugal pump.
- A sliding shoe pump.

What type of pump would you use for a high pressure, low volume application?

- **A positive displacement pump.**
- A multi stage centrifugal pump.
- A single stage centrifugal pump.
- A sliding shoe pump.

What type of pump would you use for a high volume, low pressure application?

- **Centrifugal.**
- Lobe.
- Reciprocating.
- Screw.

What type of seal would you find in a turbocharger?

- **A labyrinth seal.**
- A lip seal.
- A mechanical seal.
- Gland packing.

What would you normally associate a piligrim nut with?

- **The propeller.**
- The auxiliary engine.
- The boiler.
- The main engine.

What would you normally suspect if the suction of the bilge system shows an excessive vacuum?

- **A choked suction filter.**
- A hole in the suction filter mesh.
- A leaky discharge valve.
- A leaky suction valve.

Where is an oil scraper ring fitted on a piston?

- **In the bottom grooves.**
- In any one of the grooves.
- In the centre grooves.
- In the top grooves.

Where should the dial gauge be placed while taking crankshaft deflections?

- **As far as possible from the axis of the crank pin.**
- Anywhere on the crank web but as indicated in the Planned Maintenance System.
- As close as possible to the axis of the crank pin.
- The centre of the crank web.

Where would you expect to find a list of critical equipment for the vessel?

- **In the Safety Management System manual for the vessel or letter from Company.**
- In the Classification Society Rules for the vessel.
- In the Crew familiarisation booklet.
- In the Engine Room Log book.

Where would you expect to find damping wire used in a turbocharger?

- **The gas side.**
- Both gas and air sides
- The air side.
- The bearings.

Where would you look for, and expect to find, the position of various engine room double bottom tanks?

- **In the general Arrangement plan.**
- In the Engine Room Piping Schematic drawings.
- In the stability booklet.
- In the tank calibration tables.

Which of the following Main Engine spare parts would you expect to find on board?

- **A cylinder head.**
- A complete set of main bearing shells.
- A crank shaft.
- A vibration damper.

Which of the following are parameters indicated on a fuel oil test report? Sulphur, ash, water?

- **All of the above**
- The ash content.
- The sulphur content.
- The water content.

Which of the following tools are required for overhauling a turbocharger?

- **A rotor locking tool, a bearing extracting tool and a spanner for the bearing nut.**
- A bearing extracting tool.
- A rotor locking tool.
- A spanner for the bearing nut.

Which of the following tools would be required for replacement of a main engine cylinder liner?

- **A liner lifting tool, a piston lifting tool and a cylinder head lifting tool.**
- A cylinder head lifting tool.
- A liner lifting tool.
- A piston lifting tool.

Which property of fuel oil is used to select the gravity disc of the fuel oil purifier?

- **The density.**
- The sulphur content.
- The viscosity.
- The water content.

Why is an air compressor started in the unloaded condition?

- **To avoid high starting torque.**
- To avoid condensate carry over.
- To avoid damage to the suction and discharge valves.
- To avoid high discharge pressure.

7057 – TOTS 1C-Control of Operations - Care of Persons

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A Permit-to-Work delivers a risk-based approach to safety management on a ship.

- **True**
- False

A Permit-to-Work is essentially a document which . . .

- **describes the work to be done and the precautions to be taken in doing it.**
- describes the easiest way of completing a given job.
- describes the method of doing a job.
- describes the shortest way of completing work in the minimum time possible.

A Permit-to-Work should normally NOT exceed . . .

- **one working day.**
- one hour.
- one week.
- three days.

A Permit-to-Work system is . . .

- **a formal written system.**
- casual work management.
- theoretical work management.
- verbal work management.

A Port State Control Inspector advises a ship in good time on the date of a planned inspection.

- **False**
- True

A chemical from an unlabelled package should never be used unless its identity has been positively identified.

- **True**
- False

A ship is subject to Port State Control inspection in a particular region covered by MOU . . .

- **once in six months.**
- annually.
- every three months.
- every twelve months.

According to the ISPS Code, how often should security drills be conducted?

- **Every three months.**
- Every two months.
- Every two weeks.
- Monthly.

Any defect found on lifting equipment on board should be immediately reported to . . .

- **The Master or responsible person.**
- The charterers.
- The manufacturers.
- The owners.

Any failure of the oil filtering equipment shall be noted in the Oil Record Book Part 1.

- **True**
- False

As required by the ISM Code, what certificate must a company have for it to be allowed to operate ships?

- **Document of Compliance.**
- Certificate of Class.
- International Ship Security Certificate.
- Safety Management Certificate.

Bunker procedures for ships are controlled under procedures that are incorporated in the Company's Safety Management System.

- **True**
- False

Bunkering operations should be planned and executed in accordance with . . .

- **MARPOL regulations.**
- ISM Code.
- the Code of Safe Working Practices.
- the ship's safety record book.

Compressed air should never be used for cleaning work spaces.

- **True**
- False

During a vetting inspection, the oily water separator is checked for . . .

- **All of these answers.**
- alterations in the piping system.
- correct operation.
- testing of the oil sensing alarm.

During topping of operations during bunkering, what hazard is most likely to occur?

- **Air lock in the tank causing oil mist carry over from the vents.**
- Crew fatigue.
- The ship's stability might be affected.
- The temperature of bunkers might increase.

Elements of Risk include:

- **the likelihood that a hazard might occur, and the consequence of the hazardous event.**
- the completion of a job in the slowest manner possible.
- the timely completion of a given task in as little time as possible.
- the unlikelihood that a hazard might occur, and the consequence of the hazardous event not occurring.

Engine room bilge oily water pumping and disposal arrangements are checked during a vetting inspection.

- **True**
- False

For how long should the Oil Record Book be preserved?

- **Three years after the last entry.**
- Five years after the last entry.
- One year after the last entry.
- Two years after the first entry.

For which voltage should a General Electrical Work Permit be issued?

- **Under 1000 volts.**
- Under 120 volts.
- Under 230 volts.
- Under 70 volts.

How many years is a ship's IOPP Certificate normally valid for?

- **Five years.**
- Four years.
- One year.
- Two and a half years.

How often and where should the quantity of oil residues collected as a result of machinery space operations be recorded?

- **Weekly, in the ORB Part I.**
- After every operation, in the ORB Part I.
- At end of voyage, in ORB Part I.
- Daily, in the ORB Part II.

How often must a passenger lift be inspected by a competent person?

- **Every six months.**
- Every five years.
- Every two years.
- Yearly.

How often must a ship's lifting equipment be tested by a competent person?

- **Every five years.**
- Every six months.
- Every two years.
- Yearly.

How quickly should the main steering gear be capable of putting the rudder over from 35 degrees on one side to 30 degrees on the other side?

- **Not more than 28 seconds.**
- Not more than 30 seconds.
- Not more than 35 seconds.
- Not more than one minute.

Hydraulic jacks should be applied . . .

- **to safe jacking points on equipment.**
- after visual inspection of the equipment to be raised.
- to all suitable equipment for lifting purposes.
- to no equipment on board.

If the Vetting Inspector notes that the planned maintenance system being followed is not up to date and not class approved, he will . . .

- **raise an observation.**
- issue a Non-Conformance to the owners.
- request the Chief Engineer to provide a written statement.
- suspend the inspection of the engine room.

If the conditions under which a Hot Work Permit is issued should change . . .

- **hot work should be stopped immediately.**
- hot work should be continued with caution.
- hot work should be stopped for a few minutes and then started again.
- hot work would not have any impact with the change in conditions.

If work is to be performed on pressurised systems such as filters, what should be done to ensure that pressure in the system is off?

- **Open the drain cocks and air vents.**
- Place a rag over the cover when opening.
- Shut off any heating system.
- Wait for 24 hours to allow the pressure to drop.

It is a good practice to transfer E/R bilges into the cargo area slop tanks using a flexible hose.

- **False**
- True

Many of the bunkering operations overflows are due to . . .

- **human error.**
- technical failures.
- temperature of the bunkers.
- weather conditions.

On completion of the vetting inspection, the inspector passes the final report onto which database?

- **SIRE.**
- IMO.
- MSC.
- WIRE.

On completion of the vetting inspection, the inspector's initial report is . . .

- **discussed "on site" and the ship staff requested to comment.**
- handed over to port authority.
- handed over to the agent.
- kept confidential from the ship staff.

Prior to bunkering operations, the responsible officers should . . .

- **agree in writing the handling procedures and action to take in the event of an emergency.**
- delegate the responsibility of this operation to shore authorities.
- signal visually to agree the handling procedures and action to take in the event of an emergency.
- verbally agree the handling procedures and action to take in the event of an emergency.

Prior to conducting hot work in a location, the LFL should be . . .

- **Less than 1%.**
- Less than 0%.
- Less than 0.5%.
- Less than 2%.

Prior to working on electrical equipment . . .

- **all electrical power to the equipment to be isolated/safety signs posted/ work permit completed**
- all electrical power to the equipment is to be isolated and the work commenced immediately.
- make sure that the work permit is completed and filed after the completion of the work.
- the work permit used on prior occasions is to be used again to complete the job.

Rest hour regulations can be found in . . .

- **ILO 180.**
- ILO 200.
- ILO 205.
- ILO 680.

Risk Assessment is essential prior to moving around heavy weights in the machinery space.

- **True**
- False

Risk Assessments conducted on board should be . . .

- **recorded and retained within the SMS documentation of the ship.**
- destroyed on completion of the task.
- handed over to the ship owner every month.
- put up on a notice board in the mess room.

Risk Assessments should be structured in order to identify . . .

- **potential hazards.**
- a simplified method of conducting work.
- short cuts to be taken to complete a job.
- the time frame required to complete a task.

Risk management should include Risk Assessments.

- **True**
- False

Something that is a source of potential harm or damage, or a situation with potential for harm or damage is known as . . .

- **a hazard.**
- a perception.
- an act of God.
- invincible.

The Code of Safe Working Practices is applicable and addressed to . . .

- **all ranks.**
- Junior officers.
- Ratings.
- Senior officers.

The Oil Record Book Part 1 is required for every oil tanker over . . .

- **150 Gross Tonnage.**
- 250 Gross Tonnage.
- 450 Gross Tonnage.
- 500 Gross Tonnage.

The Vetting Inspector has the authority to operate ship equipment on his own during the inspection.

- **False**
- True

The flow chart for guidance on Hot Work can be found in which publication?

- **ISGOTT.**
- IAMSAR.
- MARPOL.
- SOLAS.

The main aim of Risk Assessment is to . . .

- **minimise accidents and ill health on board a ship.**
- increase profitability of the Company.
- man a ship with the minimum number of crew.
- minimise time wastage to complete a task.

The presence of a Superintendent onboard the ship during a vetting inspection . . .

- **does not relieve the ship staff of their duties during the inspection.**
- allows the ship staff to relax and seek assistance from him.
- allows the ship to keep any defects confidential.
- permits the Inspector to deal with the Superintendent directly without assistance from the ship staff.

The purpose of a Port State Control inspection is to . . .

- **ensure a ship's compliance with international regulations and conventions.**
- check the ship's service history.
- confirm that the ship has passed vetting inspections.
- make a list of additional suggestions that would benefit the ship's performance.

There are certain overriding factors that might subject a ship to re-inspection by Port State Control even though a ship was recently inspected.

- **True**
- False

What action should be taken if a Risk Assessment shows that an activity carries a very low risk?

- **Proceed with the task.**
- Do not continue with the task.
- Eliminate the risk totally.
- Redo the Risk Assessment.

What are the legal requirements for test intervals of lifting equipment?

- **Every five years.**
- Every 10 years.
- Every 15 years.
- Every six months.

What could happen if steam is admitted to a section of the steam pipe while the drains on the pipe are still closed?

- **Water hammer.**
- Excess pressure.
- Loss of feed water.
- Nothing, as this is normal procedure.

What data is necessary to correctly quantify the bunker quantity on completion of operations?

- **Specific gravity of the bunkers.**
- The chemical properties of the bunkers.
- The colour of the bunkers.
- The viscosity of the bunkers.

What is the maximum angle of list that the emergency generator and its prime mover can be capable of operating from?

- **22.5 degrees.**
- 10 degrees.
- 15 degrees.
- 6 degrees.

What is the meaning of a chemical's MSDS?

- **Material Safety Data Sheet.**
- Manual on Safety of Dangerous Substances.
- Maritime Substances Danger Statistics.
- Minimum Solubility Data Sheet.

What is the minimum capacity required for fire pumps?

- **25 m3/hr.**
- 10 m3/hr.
- 100 m3/hr.
- 50 m3/hr.

What is the minimum height requirement for guard rails fitted to all exposed parts of freeboard and superstructure decks?

- **1 metre.**
- 0.75 metres.
- 1.2 metres.
- 1.5 metres.

What is the minimum number of hours required for emergency electrical power to be available for lighting machinery spaces and machinery control position during periods of blackout in a cargo ship?

- **18 hours.**
- 24 hours.
- 36 hours.
- 8 hours.

What marking is to be clearly visible on lifting equipment designed to lift heavy weights?

- **Safe Working Load.**
- Breaking Stress of the weakest part.
- Manufacturer's details and address.
- Proof Load.

What plans should be available on board the ship fitted with derricks?

- **The Rigging Plan.**
- The Distribution Plan.
- The Isolation Plan.
- The Movement Plan.

What publication contains information on the equipment for the control of oil discharge from machinery space bilges?

- **MARPOL.**
- ILO Conventions.
- ISPS Code.
- SOLAS.

What publication provides information on hazards related to dangerous substances carried on board?

- **IMDG Code.**
- ISPS Code.
- MARPOL.
- SOLAS.

What section of the Code of Safe Working Practices deals with various working practices common to all ships?

- **Section 3.**
- Section 1.
- Section 5.
- Section 6.

What tool/system on board provides the minimum basis for safe bunkering operations as agreed through the mutual exchange of critical information?

- **The Bunker Safety Check List.**
- The Bunker Loading Master.
- The Chief Engineer.
- The Chief Officer.

What toxic hazards are to be considered when handling fuel oil bunkering?

- **The presence of H2S.**
- The presence of Halon gases.
- The presence of helium gas.
- The presence of VCM.

When completing a work permit, at least . . . should be made.

- **two copies.**
- five copies.
- one copy.
- six copies.

When may a load greater than the safe working load be lifted in a lifting appliance?

- **When a load test is required.**
- For periods less than one minute.
- In an emergency.
- Never.

Where can a description of the oily water separator be found?

- **In the IOPP Certificate Supplement.**
- In the Oil Record Book Part I.
- In the Oil Record Book Part II.
- In the SOPEP.

Where can information on machinery space chemicals be found?

- **MSDS.**
- IMTS.
- MSTS.
- RSPCA.

Where will the engineer's alarm sound if operated from the engine control room?

- **The engineer's accommodation.**
- The bridge.
- The Chief Engineer's office.
- The officer's smoke room.

Where would you find the record of construction and pollution prevention equipment for oil tankers?

- **In the supplement to the IOPP Certificate.**
- In the cargo record book.
- In the IOPP Certificate.
- In the Safety Construction Certificate.

Which among these operations is NOT required to be recorded in the Oil Record Book Part I?

- **None of these answers.**
- Ballasting of oil fuel tanks.
- Cleaning of oil fuel tanks.
- Disposal of oily residues.

Which annex of MARPOL Publication deals with regulations for the prevention of air pollution by ships?

- **Annex VI.**
- Annex I.
- Annex III.
- Annex VII.

Which chapter of the Code of Safe Working Practices describes the Risk Assessment procedures?

- **Chapter 1.**
- Chapter 11.
- Chapter 16.
- Chapter 8.

Which chapter of the Code of Safe Working Practices describes the safety system of work with reference to unmanned machinery spaces?

- **Chapter 15.**
- Chapter 2.
- Chapter 3.
- Chapter 7.

Which chapter of the Code of Safe Working Practices describes working procedures for high pressure/hydraulic/pneumatic equipment?

- **Chapter 20.**
- Chapter 14.
- Chapter 4.
- Chapter 8.

Which chapter of the Code of Safe Working Practices provides advice on maintenance of various machinery on board the ship?

- **Chapter 22.**
- Chapter 11.
- Chapter 14.
- Chapter 8.

Which chapter of the Code of Safe Working Practices provides information of lifting plants on board a ship?

- **Chapter 21.**
- Chapter 14.
- Chapter 4.
- Chapter 9.

Which component of inert gas is NOT toxic?

- **Nitrogen.**
- Carbon monoxide.
- Nitrogen dioxide.
- None.

Which document on board a ship provides vessel details to a vetting inspector during inspection?

- **VPQ.**
- OCIMF.
- PSC Report.
- SIRE.

Which of the following is NOT to be recorded in the Oil Record Book Part 1?

- **Cargo slop tank residues.**
- Accidental discharge of bilge tank residue.
- Bunkering of fuel or bulk lubricating oil.
- The condition of the oil filtering equipment.

Which of the following ozone depleting substances may be found on board a ship?

- **All of these answers.**
- CFC 114.
- Halon 1211.
- Halon 1301.

Which of the following would give a Port State Control Inspector clear grounds to detain a ship?

- **If the Master or crew member is not familiar with the essential shipboard operational procedures that relate to the safety or security of the ship and pollution prevention.**
- If the last Port State Control inspection was conducted more than three months ago.
- If the ship is registered under a flag of convenience.
- If two senior officers have gone ashore in port.

Which publication highlights the parameters a Vetting Inspector will use during his inspection?

- **VIQ.**
- MARPOL.
- The ISM Code.
- VPQ.

Which publication would assist the ship staff to prepare for a vetting inspection?

- **VIQ.**
- SOLAS.
- The ISM Code.
- VPQ.

Who is responsible for keeping the vessel's particular questionnaire updated?

- **The ship's staff.**
- The ship owner.
- The shore-based office staff.
- The vetting inspector.

Working on electrical equipment in hazardous areas of the ship requires

- **a Permit-to-Work on an electrical circuit.**
- permission from the Duty Officer.
- permission from the owners.
- the Chief Engineer to be on stand-by while the work is being done.

Would the Vetting Inspector request to sight detailed bunker transfer instructions?

- **Always.**
- Never.
- Not applicable.
- Sometimes.

7058 – TOTS 1C-Management of the Engineering Department

Version 1.0 (5)

A Class Surveyor has imposed a Condition of Class on a machinery item. What does this mean?

- **It needs to be repaired and resurveyed in a given time.**
- It has been removed from the required survey items.
- It has to be resurveyed at the next external audit.
- It needs to be repaired in a given time.

A Garbage Record Book must be kept. When plastic has been incinerated what should you do with the ash?

- **Land the ash to reception facilities.**
- Dispose of it into the sea 25 miles from the nearest land.
- Mix it with plastic garbage and dispose to shore.
- Mix it with the other ash and dispose of it into the sea.

A risk assessment should be carried out and a Hot Work Permit issued before any hot work is carried out . . .

- **outside the Engine Room workshop.**
- at night.
- during bunker operations.
- during manoeuvring.

Before entering an empty Fresh Water tank for inspection and cleaning, you should . . .

- **ventilate and test the atmosphere with a calibrated oxygen indicator.**
- open and ventilate the tank at least 1 hour before entry.
- put on a Breathing Apparatus set and enter the tank.
- ventilate and test the atmosphere with a calibrated combustible gas detector.

Having emptied, cleaned and ventilated a bunker tank for hot work, how would you ensure that the atmosphere in the tank is not combustible?

- **Test the atmosphere with an explosimeter.**
- Leave the tank lids open for 24 hours.
- Take a sample of the atmosphere in the tank and test it with an open flame.
- Test the atmosphere with a portable oxygen meter.

If a new motor is being fitted to a pump room fan how important is the direction of rotation of the motor?

- **It is very important as the fan must act as an extraction fan.**
- It is not important as the fan direction is switchable.
- It is not important as the fan is used for both supply and extraction.
- It is very important as the fan must act as a supply fan.

In order to prevent inadvertent direct overboard discharge of bilge water the main Engine Room bilge system should be . . .

- **secured.**
- blanked.
- checked regularly by a watchkeeper.
- displaying a warning notice.

Lifting and loose gear which is not part of the ship's equipment should be examined by a competent person at least once every . . .

- **12 months.**
- 24 months.
- 3 years.
- 5 years.

Oil Pollution Regulations require that any transfer of oil onboard a vessel should be recorded in . . .

- **the Oil Record Book.**
- the Bridge log book.
- the Engine Room log book.
- the Master's Official log book.

On a tanker, which of the following best describes a segregated Ballast System?

- **All ballast tanks, pipes and valves are independent of those used for cargo.**
- All ballast is discharged via an oil discharge monitor (O.D.M.E).
- Ballast and cargo tanks are separated by void spaces.
- Ballast is only taken after cargo has been flushed from the line.

One result of poorly maintained fire dampers could be . . .

- **feeding the fire with oxygen.**
- a build up of toxic fumes.
- faster cooling of the fire.
- the fixed foam system being ineffective.

Paints and thinners used in the Engine Room for cosmetic maintenance should always be . . .

- **returned to the paint locker after every use.**
- kept in a closed container to prevent contamination.
- kept in a cool place.
- stored in the Engine Room away from the Deck paint supplies.

Ships receive a Quarterly Listing. What does this contain?

- **The status of Surveys from Class.**
- A list of spares the ship has used over 3 months.
- Budget reports for the last 3 months.
- The status of Surveys by Port State Control.

The Chief Engineer's Standing Orders must contain, which of the following?

- **Duties of the watchkeeping Engineers and Ratings.**
- Duties of the Engineers in an emergency.
- Duty times of the watchkeeping Engineers and Ratings.
- Operation of the Dead Man Alarm.

The Emergency Generator is required to have clear operating instructions. Who must be familiar with the operation?

- **The ship's crew.**
- Deck Ratings.
- The Chief Engineer.
- The Training superintendent.

The Planned Maintenance System should be inspected by which of the following to confirm that it has been implemented according to Company procedures?

- **An Auditor.**
- A Class Surveyor.
- A Port State Control inspector.
- An authorised shore technician.

The Steering Gear has 2 electric motors. How is the power normally supplied?

- **Independently from the Main switchboard.**
- Both are connected via the Emergency Switchboard.
- Both are supplied from the Main Switchboard on a common power supply.
- One is supplied from the shaft generator.

The exhaust gas temperature entering the turbo-blower in relation to the main exhaust manifold temperature should be . . .

- **higher.**
- any of these.
- lower.
- the same.

Weekly testing of the Engine Room CO2 alarm should confirm that . . .

- **the Engine Room ventilation supply and exhaust fans stop automatically.**
- the Engine Room mechanical ventilation dampers close automatically.
- the Engine Room ventilation supply fans stop automatically.
- the fuel quick closing valves shut automatically.

What do the letters UMS indicate in a vessel's Class notation?

- **The vessel may be run with Unattended Machinery Spaces.**
- It defines certain Machinery Spaces on the vessel which are Unused and do not require any maintenance.
- It defines those Spaces on the vessel which must remain Unused for Machinery.
- It is a classification for Underwater Survey Maintenance.

What instrument would be used to measure the area of a power card taken from the main engine?

- **A planimeter.**
- A micrometer.
- A power meter.
- A vernier calliper.

What is the normal survey interval for machinery items?

- **Once every 5 years.**
- Once every 2.5 years.
- Once every 3 years.
- Once every 4 years.

When an Exhaust Gas Boiler is surveyed who is responsible for setting the safety valves?

- **The Chief Engineer.**
- The Class Surveyor.
- The Company Superintendent.
- The Shore Technician.

When bunkering, you note that you have some leakage from the connection manifold flange, but the Master is pressing you to depart as soon as possible to make the tide. What do you do?

- **Stop bunkering and repair the leak?**
- Notify the local authority of a potential oil spill?
- Reduce the pumping pressure to minimize the leakage?
- Spread absorbent material on the deck to soak up any spillage?

When calculating bunkers for a voyage, the amount of extra bunkers required to provide a safe margin is calculated according to . . .

- **the Company's procedures.**
- the Chief Engineer's own experience.
- the direction given by the Charterer.
- the Master's instructions.

Where would you look to find your prime responsibilities in the event of a fire?

- **The Company Management manuals.**
- The Emergency Stations (Muster List) poster.
- The Safety Equipment Certificate.
- The Shipboard Training Manual.

Which of the following would contribute to an increased risk of accidents?

- **Unsafe working conditions.**
- Good housekeeping in the Engine Room.
- Proper risk assessments.
- Regular inspections.

7059 – TOTS 1C-Pollution and Environment Protection

Version 1.0 (5)

A ship demonstrates compliance with MARPOL Annex IV by . . .

- **having a valid International Sewage Pollution Prevention Certificate.**
- having a sewage holding tank volume of 100 litres per person on board.
- having a stand-by sewage treatment plant.
- having a surveyor inspect the sewage system every six years.

After having performed a ballast operation the master must . . .

- **complete the ballast water management log.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

An environmental policy in line with ISO14001 is a commitment from . . .

- **company top management.**
- class society.
- flag administration.
- the IMO.

Discharge of sewage is prohibited when . . .

- **the vessel is 24 nautical miles from the nearest land and travelling at 3 knots.**
- the sewage is comminuted and disinfected and the vessel is 5 nautical miles from the nearest land.
- the ship is 24 nautical miles from land and has a functioning certified sewage treatment plant.
- the vessel is 24 nautical miles from the nearest land and travelling at 7 knots.

During bunker transfer operations deck scuppers should be . . .

- **plugged to prevent spilled oil flowing overboard.**
- attended by crew to collect any spilled oil that approaches.
- only fitted to the side of the vessel that bunkers are being loaded to.
- Plugged when required

During cargo or bunker operations, oil response equipment is to be . . .

- **ready for immediate use.**
- in its store room.
- in the cargo pump room.
- used elsewhere in the ship.

False entries in the Oil Record Book can lead to criminal penalties for the master and chief engineer?

- **True.**
- False.

For how long must the receipts be held with the Garbage Record Book?

- **Two years.**
- Five years.
- One year.
- Six months.

How long should the oil record book be kept for after it has been completed?

- **Three years.**
- One year.
- Six months.
- Two years.

In the event of an oil pollution incident MARPOL Annex I requires that an oil tanker of 5,000 tonnes deadweight shall have prompt access to . . .

- **computerised shore-based damage stability and residual structural strength programs.**
- an empty oil tanker of equal deadweight.
- computerised models of the ship's cargo tank system.
- the nearest port.

It is illegal to discharge oily mixtures from the ship when the vessel is stationary and outside of a special area.

- **True.**
- False.

It is legal to discharge oil outside a special area when the oily mixture originates from the cargo pump room bilges on an oil tanker.

- **False.**
- True.

No discharge into the sea shall contain chemicals or other substances in sufficient concentration so as to cause damage to the marine environment.

- **True.**
- False.

Operational malpractice of the oil filtering equipment can lead to criminal penalties for the vessel master and chief engineer?

- **True.**
- False.

Prior to exchanging ballast the master must . . .

- **perform a risk assessment.**
- inform the Port Administration.
- inform the vessel superintendent.
- perform a course change.

The cargo manifold working platform surface should be made of . . .

- **open mesh grating.**
- checquer plate.
- perforated sheet steel.
- wooden laminate.

The oil record book makes references to factors affecting quantity of oil. The accuracy of measuring quantities are affected by . . .

- **All three answers**
- Accuracy of tank measurement devices.
- Temperature variations.
- Tendency of oil to cling to the inside tank walls.

The oil record book part 2 requires which of the following to be recorded after cleaning cargo tanks?

- **All three answers.**
- The duration of cleaning.
- The method of cleaning.
- The ship's position or port.

The oil record book shall be completed on each occasion that ballast is discharged from segregated ballast tanks.

- **False.**
- True.

The oil record book shall be completed on each occasion that the cargo tanks are cleansed.

- **True.**
- False.

The oil spill removal equipment store must be . . .

- **prominently marked and easily accessible.**
- a storeroom close to the pump room.
- a storeroom in the forward area of the ship.
- just off the main engine room.

The portable pump equipment within the oil spill response kit must be . . .

- **All three answers**
- fitted with sufficient hoses.
- safe for use in hazardous areas
- Self priming

The shipboard oil pollution emergency plan shall contain details of a list of authorities to be contacted in the event of an incident.

- **True.**
- False.

The spill response station inventory should include . . .

- **personal protective equipment.**
- a supply of water.
- brushes.
- metal shovels.

To prevent inadvertent leaks, bunker manifolds not in use should . . .

- **have blank flanges fitted with gaskets and tightened with all bolts in place.**
- have blank flanges fitted and tightened with all bolts in place.
- have blank flanges fitted and tightened with some bolts in place.
- have blank flanges fitted with gaskets and tightened with some bolts in place.

To reduce cargo tank vapour release which of the following should be checked?

- **That the tightness of cargo tank hatch dogs are checked after every cargo operation.**
- That header valves are unseated.
- That the cargo pump deepwell bleed valve is open.
- That the hatch gasket is checked every year.

Uncontrolled release of volatile organic compounds (VOCs) are an environmental problem because . . .

- **they contribute to low-level ozone pollution in populated areas.**
- they contaminate the sea.
- they react with CO2.
- they react with hydrogen creating acid rain.

What can happen if an emulsion is taken up by the OWS?

- **It can lead to a false reading on the oil content monitor.**
- It can cause damage to the pump bearings.
- It can completely block the system piping.
- nothing.

What should the Master refer to In the event of an accidental discharge of oil?

- **The shipboard oil pollution emergency plan.**
- The ballast water management plan.
- The cargo tank plan.
- The ship environmental management system.

When first using a cleansing agent in the machinery space, the crew should . . .

- **confirm that it won't form an emulsion.**
- confirm that it will form an emulsion.
- use it liberally.
- use it sparingly.

When is it good practice to check the pressure-vacuum valves?

- **Prior to cargo transfer.**
- After cargo transfer operations.
- Before the next dry-docking.
- During cargo transfer operations.

When is the Master to sign the Garbage Record Book?

- **After each completed page.**
- After each discharge operation.
- After the incinerator has been used.
- At the end of the day.

Where on board might ozone depleting substances be found?

- **In refrigeration systems.**
- In compressed air systems.
- In electronic equipment.
- In sewage treatment plants.

Which of the following disposals would be permitted?

- **Food waste from a vessel outside a special area and 12 miles from the nearest land.**
- Food waste from a vessel inside a special area and 10 miles from the nearest land.
- Food waste from a vessel outside a special area and 10 miles from the nearest land.
- Paper products from a vessel inside a special area and 15 miles from the nearest land.

Which of the following is NOT a method for detecting leaks of refrigerant gases?

- **Torch.**
- Gas detector.
- Halide torch.
- Soapy water bubble test.

Which of the following is NOT an acceptable code and number identifier in the oil record book part 1?

- **M 2.**
- B 6.
- D 13.
- H 26.1.

Which of the following is NOT an option to comply with MARPOL Annex IV?

- **An approved comminuting plant.**
- A certified sewage treatment plant.
- An approved comminuting and disinfection plant.
- Holding tanks with an approved volume.

Which of the following is not part of the spill response equipment inventory?

- **The pressure washer.**
- Emulsifiers.
- Non sparking scoops, shovels and buckets.
- Sorbents.

Which of the following must be recorded in the oil record book part 1?

- **All three answers**
- Bunkering of fuel oil.
- Collection and disposal of oil residues.
- Condition of oil filtering equipment.

Which of the following should be checked prior to cargo operations?

- **All three answers**
- That pipeline drains are blanked correctly.
- That save-all drains in way of tank vents are plugged.
- That spill tank drain plugs are properly secured.

Which of the following substances is ozone-depleting?

- **Halons.**
- Ammonia.
- CO2.
- Neon.

8524 – Dropped object awareness

Version 1.1 (1150)

How often should a third party inspect all areas in height, including inaccessible areas?
Select the correct answer.

- **Yearly**
- Every second year
- Not needed
- Weekly

Select two common examples of causes to dynamic dropped objects?

- **Distractions from duty**
- **Inadequate risk assessment**
- Corrosion
- Vibration

Select two common examples of causes to static dropped objects?

- **Corrosion**
- **Vibration**
- Inadequate risk assessment
- Poor pre-lift inspections

What are examples of best practice for drop prevention equipment used when working at height for securing personnel?
Select the two correct answers.

- **Choose the type of fall protection equipment according to conditions at the work place**
- **No one should work without supervision when using fall protection**
- Do not barrier off work area below
- If your supervisor allows you to use any fall protection equipment, it does not need to be approved by the applicable regulations

What are examples of mitigation methods?
Select the correct answer.

- **Wire rope slings**
- Barrier and signals
- Bolt safety lock
- Vibration

What are the three common barriers to avoid or reduce dropped objects?
Select the correct answer.

- **Organizational, technical and operational**
- Equipment, maintenance and technical
- Manpower, knowledge and experience
- Procedures, instructions and checklists

What are your responsibilities related to dropped objects?
Select the two correct answers.

- **Continuously identify potential dropped objects**
- **Eliminate or minimize danger to yourself and others**
- Continue on work until it is completed before you act on an identified potential dropped object
- Only use observation techniques on selected areas

What energy does a dropped object have when it can cause serious injury or death?
Select the correct answer.

- **Above 40 Joule (J)**
- Above 10 Joule (J)
- Above 100 Joule (J)
- Above 20 Joule (J)

What is Fall Energy measured in? Select the correct answer.

- **Joule (J)**
- m/sec2
- Newton Meter (Nm)
- Ton/mm2

What is a static dropped object?
Select the correct answer.

- **An object that falls from its previous position under its own weight (gravity) without any applied force**
- An object that can avoid or reduce consequences of unwanted incidents
- An object that falls due to being struck by another object.
- An object that has the potential to fall but still in its static position

What is illustrated on the screen?
Select the correct answer.

- **Weak link**
- Bolt safety lock
- Heli coil
- Palnut

What is meant by a secondary barrier?
Select the correct answer.

- **A secondary barrier shall prevent, or reduce the possibility, of an unwanted incident from an object should the primary barrier fail**
- An object that moves uncontrolled from one level to another
- Bolts
- Common causes of incidents in the offshore industry

What is the definition of a Dropped Object? Select the correct answer.

- **A dropped object is an object that moves uncontrolled from one level to another**
- A dropped object is an object that cause fatality and serious injury
- A dropped object is an object that has potential to fall down
- A dropped object is an object that has sharp edges

What is the definition of a barrier related to dropped objects?
Select the correct answer.

- **A barrier is used in order to avoid or reduce consequences of unwanted incidents**
- A barrier is failure to deliver a completed Permit to Work
- A barrier is human causes that leads to an unwanted incident
- A barrier is not following procedures and improper conduct

Why are good observation techniques important?
Select the correct answer.

- It is the starting point for choosing correct fall protection equipment
- It is the starting point for minimizing the potential for dropped objects
- It is used to permanent barrier off areas
- It pleases your manager

Why is it important with good housekeeping?
Select the two most accurate answers.

- **Clear and tidy work places is less exposed to risk than the opposite**
- **Your colleagues will not be exposed to your mess**
- It will please your manager
- It will reduce every potential of dropped objects

8525 – Personal Protective Equipment

Version 1.1 (1027)

How often should you inspect your PPE to ensure it's in good condition and functioning properly?

- **Before each use**
- Once a month
- Once a week
- When needed

If you recognize failure on your PPE equipment, you will have to:

- **Inform my supervisor and get the PPE replaced**
- Repair the defect and continue use the PPE
- Use the PPE as backup equipment only
- Wear it anyway - it's unlikely that the protection has been compromised

What does the abbreviation PPE stand for?

- **Personal Protective Equipment**
- People Protecting Equipment
- Personally Protected Equipment
- Protection Providing Equipment

What is the definition of PPE?

- **Personal Protective Equipment (PPE) is equipment that will protect the user against health or safety risks at work.**
- Personal Protective Equipment (PPE) is equipment that is meant to lower the risk of an accident to occur.
- Personal Protective Equipment (PPE) is equipment that will protect the employer from lawsuits.

What is the most common PPE used to protect against hand injuries?

- **Gloves**
- Hand cream
- Hand hygiene
- Screw driver

What is the most common cause for feet injuries?

- **Not wearing safety shoes**
- Hot environment
- Not wearing socks
- Poor housekeeping

What purpose does a hard hat / helmet serve? (Select two alternatives)

- **Protection from falling objects**
- **Protection from overhead hazards**
- Hearing protection
- Keep you warm

Which of the following are NOT proper use of gloves?

- **Store gloves inside out**
- Remove any rings, watches, or bracelets that might cut or tear your gloves
- Select and use the right kind of glove for the job you are going to be performing
- Select gloves that fit

Which of these are hazards to your eye? (Select 3 alternatives)

- **Dusts, powders, fumes and mists**
- **Molten metals**
- **Splashes of water, chemicals and particles**
- Light from a reading lamp

Why is it necessary to wear proper body protection?

- **To protect you from bodily injuries**
- So the authorities can detect witch company you are working for
- So you won't be cold

Your workplace must have a hearing conservation program when employees are exposed to noise levels that are equal to or greater than . . . dBA averaged over an eight hour period:

- **85**
- 100
- 90
- 95

8527 – Fall Protection

Version 1.0 (1860)

Can a fall from less than one meter cause injurie?

- **Yes! Many injuries occur even when falling from a height of less than one meter.**
- No! This is not correct. No injuries occur when falling from a height of less than one meter.

Choose ONE option that best completes the following statement. 'Fall is among the leading causes of death in the workplace. Most falls are between . . .

- **. . . 2 and 3 meters**
- . . . 15 and 30 meters
- . . . 9 and 15 meters
- . . . .30 and 50 meters

Choose ONE option that best completes the following statement. When fall protection isn't used correctly . . .

- **Wrong use can result in serious injury or death**
- It is important to ensure the anchor point is certified to minimum seven ton
- Nothing will happen since the fall protection is made of strong material

Choose ONE option that best completes the following statement. 'Slip straps over shoulders so . . .

- **. . . the D-ring is located in middle of the back between the shoulder blades**
- . . . It is easier to reach the straps
- . . . it is easier to see if the straps are twisted
- . . . the D-ring is placed over the neck

Choose ONE option that best completes the following statement. 'The willingness to take shortcuts and risks . . .

- **. . . increase when the obvious danger level decreases**
- . . . is accepted when the obvious danger level decreases
- . . . decrease when the obvious danger level decreases
- . . . remains the same when the obvious danger level decreases

Is the following statement true or false? 'A rescue plan is only required if the worker is not able to perform a self rescue '.

- **False**
- True

Is the following statement true or false? 'People following procedures are a danger to themselves and others'.

- **False**
- True

Is the following statement true or false? 'People that don't follow the rules and regulations are a danger to themselves, and others'.

- **True**
- False

Is the following statement true or false? 'There is no need to put up a barrier below a work site if you don't have any loose items in height'.

- **False**
- True

Is the following statement true or false? 'Tools or equipment that are not secured, is a potential dropped object'.

- **True**
- False

Is the following statement true or false? 'Work at height that haven't been risk assessed adds an unacceptable risk to the job. All work at height must be risk assessed, and all precautions must be followed at all time'.

- **True**
- False

What is a fall factor? Select the ONE correct answer.

- **The force generated by the fall length in meters, before the safety system brakes the fall.**
- It is used to calculate how strong D ring you must mount on your harness before climbing at height

What is meant by self rescue?

- **Employee using his fall protection equipment to rescue him or herself**
- The emergency response team are able to rescue a person without calling for external recourses

Where can information be found, on the criteria used to condemn fall protection? Select the ONE correct answer.

- **In the manual or information supplied with the fall protection**
- From a port state inspector
- In the Code of Safe Working Practices
- On the test certificate associated with the fall protection

8532 – Ex basic refresher and assessment

Version 3.0 (4614)

"Simple device" is defined as:

- **Devices that are not able to store or generate energy**
- Certified components
- Equipment that are not allowed in EX area
- Pumps and gear boxes

An ignition source is the definition of?

- **A high temperature or a spark capable of igniting**
- A combustibles gas or liquid
- An Ex equipment

Are there any requirements for IP rating on "Ex e" equipment?

- **Yes, Minimum IP 45**
- No, it's not relevant for this concept
- Only if is installed in dust atmospheres

Are you always allowed to modify an Ex "e" junction box as you wish?

- **No. Only in accordance with the manufacturers guidelines**
- Yes, you may always mount extra gland
- Yes, you may always mount extra glands and switches

Blue cables are normally associated with?

- **Intrinsically safe circuits**
- High voltage
- Single phase Power cables

Can the extent of a zone be reduced through effective use of ventilation?

- **Yes. Both natural and mechanical ventilation may be used to limit the extent of zones.**
- No, may never have ventilation in an Ex area
- Only natural ventilation

Do you need a risk assessment in connection with zoning of a facility?

- **Yes, we must assess risk connected to explosion hazards.**
- No. A zone will always be the same regardless of the process
- Only if there is suspicion of the zone extending further than the zonal chart shows

Does there exist Ex "e" junction boxes made of GRP (Glass Reinforced Plastic)?

- **Yes**
- No, GPR is not suitable material
- No, these must resist an internal explosion and may not be made of plastic

Equipment with ATEX marking II 1 G has which level of safety?

- **Very high**
- High voltage
- Normal

How do the flame path in and Ex d enclosure work?

- **They are designed to cool down an internal explosion so that the gases that leave the enclosure will not ignite an explosive atmosphere outside**
- It is to have sufficient surface for gaskets
- Since these most often are made of a material that corrodes, it is to have sufficient sufficient surface to wear down during maintenance

How is EPL marked for equipment to be used in gas atmospheres?

- **With a small a, b or c behind the concept (E.g.: Ex eb) alternative Ga, Gb or Gc**
- With a capital A, B or C clearly marked on the label
- With a small x, y or z behind the concept (E.g.: Ex ey) alternative Gx, Gy or Gz

How many conductors are allowed under an earthing terminal on I.S. earth?

- **1**
- 2
- 3
- 4

How many earthing wires may be connected to one screw?

- **1**
- 1 to 2
- 2 to 3

Hydrogen has a temperature class of T1. Are you allowed to use equipment marked with T3 in an area with Hydrogen present?

- **Yes**
- No
- Only if it is secured with good ventilation and the ambient temperature is kept below 40 degrees Celsius

If a gas mixed with air is below the LEL for the gas in question, is it:

- **Above the lowest explosion limit**
- Above the highest explosion limit
- Below the lowest explosion limit

In which Ex-zone is it allowed to use Exe motors?

- **1 and 2**
- 0
- 0 and 1

In which Ex-zone is it allowed to use the Exn motors?

- **2**
- 0
- 1

In which zone can we use this equipment: ATEX II 2 G?

- **Zone 1**
- Zone 0
- Zone 2

In which zone is it allowed to use Exib equipment?

- **1 and 2**
- 0, 1 and 2
- Only 2

Is an Ex "ib" circuit suitable for zone 0?

- **No, only zone 1 and 2**
- Yes
- Yes, if the field device has a floating screen

Is there a need for maintenance on Ex "e" equipment?

- **Yes, in accordance to relevant, current standards**
- No, this equipment must be designed as maintenance free
- Only requirements for cleaning

May modifications inside an Ex "p" enclosure have any consequences?

- **Yes, it may cause the T-class and "purge time" to be altered, and this may only be performed by approved workshop or a notified body.**
- No, not as long as it is not placed in zone 0
- Only if one does not perform sufficient calculations

May one open Ex "d" equipment that are in use in order to inspect it?

- **No, not without isolating it from the source of supply or have a suitable procedure with gas measurement.**
- If you are careful
- Only during the night shift
- Yes

May you install uncertified equipment in an Ex "e" box?

- **No. All equipment in an Ex "e" box must be certified and covered by the certificate of the component**
- Yes, since Ex "e" has overpressure
- Yes, since Ex "e" is moulded
- Yes, since Ex 'e' is increased safety

Of the following, what is NOT defined as a potential source of release?

- **Welded pipes**
- Around process valves that could leak under normal operations
- Flanges

What are the requirements for distance between I.S. and non-I.S. circuits?

- **50mm**
- 200mm
- 3mm
- 6mm

What determines which gas groups a gas is placed in?

- **Ignition energy and explosion properties**
- Explosion properties and flashpoint
- Ignition temperature and explosion properties

What does the "T-class" on an Ex component describe?

- **The highest temperature the relevant surface of this component is able to achieve.**
- Resistance class against static influence
- The ambient temperature allowed for the area

What does the "U" in the end of a certificate reference mean. For instance PRESAFE ATEX 0989 U?

- **Only component certified**
- It has specific conditions for safe use
- Only allowed in zone 2

What does the "X" in the end of a certificate reference mean?

- **There are special conditions for safe use**
- Spelling mistake in the certificate
- This component is not ready for use and must be certified by an able 3rd party

What does the housing IP-grade show?

- **Protection against water and dust**
- Protection against gas
- Protection against power surge

What does the letter "e" mean in Ex-marking?

- **Increased safety equipment**
- Explosion-proof equipment
- Intrinsically safe equipment

What does the letter "i" mean in Ex-marking?

- **Intrinsically safe equipment**
- Explosion-proof equipment
- Increased safety equipment

What does this marking mean: EEx ib IIB T5?

- **Intrinsically safe equipment, approved for zones 1 and 2.**
- Explosion-proof equipment, approved for zones 1 and 2.
- Increased safety equipment, approved for zones 1 and 2.

What information does EPL provide?

- **Equivalent to ATEX category, indicating which zone it is suitable for**
- Classification of protection against static influence
- Impact resistance of the product

What is DSBs area of responsibility?

- **Onshore and other process plants not associated with the oil industry in Norway**
- Offshore and ship industry in Norway
- Only dust atmospheres in Norway

What is Te time on an Ex "e" motor?

- **It is the time it takes for the engine the temperature-limit for the T-class marked on the equipment, when the rotor is locked.**
- Cooling time after over-load
- The start-up time for the motor

What is particularly important with Ex d flame path?

- **That they are kept clean and not damaged**
- That they are coated with paint
- That they must be brushed down by the user when there is deep rust

What is the flash point of liquid?

- **The temperature where the surface of a liquid emitts enough vapour to be able to ignite**
- 100 degrees Celcius
- The explosion temperature

What is the maximum allowed resistance between connection point and main earthing in the intrinsically safe circuits?

- **1,0 Ohm**
- 0,1 Ohm
- 10 Ohm

What is the maximum temperature an equipment marked with T1 is allowed to reach ?

- **450 degrees Celsius**
- 100 degrees Celsius
- 200 degrees Celsius
- 300 degrees Celsius

What is the maximum temperature an equipment marked with T6 is allowed to reach?

- **85 degrees Celsius**
- 100 degrees Celsius
- 200 degrees Celsius
- 300 degrees Celsius

What is the minimum required distance between intrinsically safe end connections and non-intrinsically safe connections?

- **50 mm**
- 30 mm
- 500 mm

What is the minimum required distance between the intrinsically safe end connections and earthed metal?

- **3 mm**
- 10 mm
- 5 mm

What is the normal ambient temperature for Ex equipment?

- **-20C to +40 C**
- -10C to +40 C
- -20C to +50 C

What is the principle behind Ex "m"?

- **Components able to generate heat or sparks are encapsulated in a potting compound**
- Components able to generate heat or sparks are encapsulated in powder or sand
- Components able to generate heat or sparks are submerged in oil

What is the principle behind Ex "o"?

- **Components able to generate heat or sparks are submerged in oil**
- Components able to generate heat or sparks are encapsulated in a potting compound
- Components able to generate heat or sparks are encapsulated in powder or sand.

What is the principle behind Ex "q"?

- **Components able to generate heat or sparks are encapsulated in powder or sand**
- Components able to generate heat or sparks are submerged are encapsulated in a potting compound
- Components able to generate heat or sparks are submerged in oil

What is the principle behind Ex 'p'?

- **There is a overpressure inside the enclosure that ensures that the explosive atmosphere surrounding the enclosure does not come in contact with the sources of ignition inside.**
- It must be able to resits an internal explosion
- No components that generate sparks are allowed inside the enclosure

What is the relationship between ATEX category and zone?

- **The ATEX category represents the zone it is suitable for use in**
- The ATEX category represents the IP rating on the equipment
- The ATEX category represents the temperature class of the equipment

What is the relationship between EPL and ATEX categories?

- **ATEX categories and EPL are are providing the same information telling the Zone of use.**
- EPL is not related to EX equipment
- EPL never appears at the same time at ATEX marking and only pertains to IECEX equipment

What is the shortest distance allowed between the parallel flame paths of an Ex d enclosure and an obstruction if placed in a IIB atmosphere

- **30mm**
- 100mm
- 200mm
- 500mm

What is the standard ambient temperature for an Ex equipment if there is no marking of ambient temperature?

- **-20 to +40 degrees Celsius**
- -10 to +20 degrees Celsius
- -50 to +50 degrees Celsius

What is the subject of the ATEX directive 94/9/EC?

- **Products and protection systems for use in potentially explosive areas**
- Classification in zones
- Health and safety

What zone is equipment marked with EPL Ga intended for use in?

- **zone 0, 1 and 2**
- only zone 1
- only zone 2
- Zone 2 and less hazardous areas

When can you install equipment in a higher ambient temperature than what is standard for Ex equipment?

- **When the extended temperature range is marked on the equipment**
- It is not possible
- When the user has tested the temperature range first

Where does the FUSEX directive apply?
This is based on ATEX 94/9/EC and applies for equipment.

- **Onshore and offshore**
- Offshore
- Onshore

Where is PTIL's / PSA area of responsibility?

- **Offshore installations and associated refineries on the Norwegian sector**
- Only tankers on the Norwegian sector
- Refineries with storage capacity over 100 000 barrels on the Norwegian mainland.

Where is zone 0 most likely located?

- **Inside a tank**
- Around ventilation that could leak under normal operations
- Around welded pipes

Which ATEX area marking applies for coal mines?

- **I**
- I and II
- lI

Which Ex-equipment is allowed to use in zone 0?

- **Ex ia**
- Ex e
- Ex ib

Which colour marks the intrinsically safe circuit cables?

- **Light blue**
- Black
- Red
- Yellow

Which colour may be used to indicate intrinsically safe circuits?

- **Light blue**
- Black
- Grey
- Purple

Which criteria must you base the decision on, when selecting equipment for use in zone?

- **All of them**
- Gas group
- T-Class
- Zone

Which equipment is not allowed to use in zone 1?

- **Exn equipment**
- Exi equipment
- Zone 0 equipment

Which equipment may be used if there is a requirement for T3 equipment?

- **T3, T4, T5, T6**
- All.
- T1, T2, T3

Which equipment requires Exd glands?

- **Ex d**
- Ex e
- Ex i

Which gas group holds gasses with the quickest increase of pressure when ignited?

- **IIC**
- IIA
- IIB

Which gas group holds gasses with the slowest increase of pressure when ignited?

- **IIA**
- IIB
- IIC

Which gaskets may be used with an Ex d enclosure?

- **Only the manufacturers original gasket, provided it is allowed for the enclosure in question**
- Any type as long as it is sufficiently fastened
- Only gaskets that cover the entire flame path

Which is the marking for oil-filled devices?

- **Ex o**
- Ex d
- Ex i
- Ex p

Which is the marking for over-pressure devices?

- **Ex p**
- Ex d
- Ex i
- Ex o

Which marking do you expect to find on a flameproof enclosure with indirect entry?

- **Ex de**
- Ex qe
- Exd

Which of the following are you allowed to use in an area with potentially explosive atmospheres of IIB gas.

- **IIB and IIC equipment**
- All Ex equipment
- IIA equipment

Which of the following is an ignition source?

- **Hot surfaces**
- An Ex e light fixture
- An explosion proof enclosure

Which part of an explosion triangle is removed by the use of Ex equipment?

- **The Ignition source**
- The Gas or flammable substance
- The Oxygen

Which power-supply system is not allowed to use in the explosion-hazard areas?

- **TN-C**
- IT
- TN-S
- TT

Which protection concept is based on allowing an internal explosion while preventing ignition of the atmosphere outside the enclosure?

- **Ex 'd'**
- Ex 'i'
- Ex 'o'
- Ex 'p'

Which protection concept is based on limiting the energy in a circuit?

- **Ex 'i'**
- Ex 'd'
- Ex 'o'
- Ex 'p'

Which temperature is equipment marked with T3 guaranteed to reach?

- **The equipment is only guaranteed to not exceed its T-class, and not necessarily to reach this temperature.**
- 200 degrees Celsius
- 300 degrees Celsius
- 85 degrees Celsius

Which zone is normally associated to the inside of a tank with a flammable substance?

- **0**
- 1
- 2

Who is allowed to repair the Ex-equipment?

- **Certified workshop**
- An electrician who has taken the Ex-course
- End user

Who is responsible for placing the ATEX mark on components?

- **Manufacturers or importers of the equipment.**
- Facility owner
- User

Who may drill new holes for glands in an Ex d enclosure?

- **Only an approved workshop or the manufacturer**
- A qualified Electrician. Note: Needs Ex Basic training
- Owner
- The user

Who may not repair serious defects and do flame-path repairs?

- **End user**
- Certified workshop
- Manufacturer

Will an Ex "e" box be able to resist an internal explosion?

- **No**
- Only with IIB and IIA gases
- Yes

With which equipment is it not allowed to use Ex e glands?

- **Exd with direct cable entry**
- Exd with indirect cable entry
- Exe

With which gas group is it allowed to use the class IIB equipment?

- **IIA and IIB gases**
- All gases
- IIA gases

With which gas group is it allowed to use the class IIC equipment?

- **All gases**
- IIA and IIB gases
- IIA gases

8533 – Electrical installation in Ex-area

Version 3.0 (4428)

After 2008, are you allowed to install uncertified cable glands in Ex e equipment in Zone2?

- **No, all cable glands shall be certified**
- Yes, as long as it is made of brass

Can we modify the purge time on a Ex 'p' enclosure?

- **No, unless the documentation from the manufacturer states otherwise**
- Purge time is not applicable for Ex p enclosures. Only T-class
- Yes, adjustable depending to customer requirements

Can we use Ex 'e' cable glands as a direct entry to a Ex 'd' enclosure?

- **No**
- Depends on the Zone
- Yes

Can you install a Ex d IIA T6 enclosure in a IIC environment?

- **No, the enclosure is not designed for IIC gasses**
- Yes, but not for a IIB gas
- Yes, IIA gasses are "worse" than IIC gasses

Can you use greased tapes(like Denso tape) on flanged Ex d enclosures in a IIA environment?

- **Yes**
- No
- Only if the gap in the enclosure is below 0,01mm

Does scratches on a flame path affect the safety of Ex'd' enclosures?

- **Yes, it might not be able to cool down the escaping gasses from the internal explosion, leading to ignition of the surrounding atmosphere**
- No, as this concept is all about limiting the electrical energy, this does not affect the protection
- Yes, the IP rating might be affected. Solve this with additional grease

How is the concept Ex d protected:

- **It contains and cools down the ignited gasses prior to escaping the enclosure**
- No arcs, sparks or high temperatures allowed inside the enclosure
- Pressurized, the gas will never come in contact with internal ignition sources

How is the concept Ex e protected:

- **Prevent arcs, sparks and avoid temperatures exceeding the T-class**
- Flameproof
- Intrinsically safe
- Overpressure

How many threads is the minimum you need when installing a cable gland into a Ex 'd' enclosure?

- **5 full threads**
- 3 full threads
- 8 full threads

If no rating for the ambient temperature is stated on the component, what is the intended ambient temperature rating for the component?

- **-20°C to +40°C**
- -40°C to +40°C
- -40°C to +50°C

Into which zones are Ex 'n' equipment acceptable?

- **Only zone 2**
- Zone 0, 1, 2
- Zone 1 and 2

Into which zones are you allowed to install the following equipment: II 2 G Ex d IIC T5 Gb

- **Zone 1 and 2**
- Zone 0, 1, 2
- Zone 2

Into which zones may you use Ex n* equipment?

- **Only zone 2**
- Only with IIB gasses
- Zone 0, 1 and 2, depending on the T-class

Is it a problem with bad terminations in Ex 'e'?

- **Yes, arcs and sparks is not accepted in a Ex 'e'**
- No, the enclosure is able to withstand an internal explosion
- No, the enclosure is pressurized

May you drill new holes into Ex e enclosures?

- **Only if this is in accordance with the manufacturer's instructions**
- No, newer
- Yes, you may always modify Ex e enclosures

Name a typical example of a cylindrical flame path

- **Shaft on a Ex d motor**
- Lid on a threaded enclosure
- The threads on a compression cable gland

What are the principles behind the Ex nL concept?

- **Energy limitation**
- Explosion quenching
- Pressurization

What does the 'U' in the certificate reference mean?

- **The equipment or device is only component certified**
- The equipment can be installed without any precautions
- The equipment has got special conditions for safe use

What does the 'X' in the end of the certificate reference mean? E.g. ITS 01 ATEX 0402X

- **The equipment has got special conditions for safe use**
- The component is only equipment certified
- The equipment can be installed without any precautions

What does the roman numbers II mean in EEx e II T6?

- **The same as IIC, suitable for all gasses**
- Enclosure is for mining
- Enclosure only for dust

What equipment can we use in an environment with T4 requirements?

- **T4. T5, T6**
- Only T4
- T1, T2, T3, T4

What is the acceptable solution for spare conductors in a junction box?

- **Terminate in an appropriate terminal or connect to the same earth reference as the circuit**
- Cut wires at the base of the cable and use tape to isolate them
- Use tape to insulate the spare conductors

What is the main purpose of grease on the flame paths?

- **Protection against corrosion**
- To make the threaded enclosures easier to open
- To offer a higher IP rating than stated on the marking plate

What is the minimum IP rating for a Ex 'e' junction box with bare live parts?

- **IP 54**
- IP 20
- IP 67

What is the minimum distance between a flanged flame path of an Ex d enclosure and an obstacle in a IIC environment?

- **40mm**
- 10mm
- 30mm

What is the principles behind the Ex nR concept?

- **Restricted breathing**
- No arcs and sparks
- Simplified flameproof enclosure

What is the purpose of Ex 'p' enclosures?

- **It keeps the explosive atmosphere away from the ignition sources by keeping a positive overpressure inside the enclosure**
- Contains the internal explosion
- The sources of ignition is encapsulated in a epoxy resin

What is the purpose of the flame path on flameproof enclosures?

- **Cool down the escaping gasses from the internal explosion**
- It gives the enclosure mechanical protection
- Provide a good IP rating

What is the requirement for shutting down a Ex 'px' enclosure?

- **Loss of pressure, needs a new full purge time before start up**
- No requirements for shutdown. The equipment on the inside is certified for minimum zone 2
- Only alarm

What is the temperature represented by T6?

- **85°C**
- 280°C
- 450°C

What zones are the following equipment suitable for:
II 2 G, Ex e IIC T6 Gb

- **Zone 1 and 2**
- Zone 0, 1, 2
- Zone 2

When must you use an Ex 'd' barrier gland (compound filled) when installing directly into an Ex 'd' enclosure?

- **If the cable is not with extruded bedding**
- Always, no exception
- Never, barrier glands are typically used on Intrinsically safe circuits

8534 – Electrical Exi installation

Version 3.0 (2940)

Can a Zener barrier be polarity dependent?

- **Yes**
- No
- Only the output for the Hazardous area

Can a non-certified measuring device affect an I.S. loop?

- **Yes, it might add energy to the loop or affect the inductance and capacitance in the loop**
- No, the barrier detects a potential high energy from the field and starts reducing the voltage in the loop
- Only when connected in the field device end

Can we use a circuit for gas group IIC in an environment with IIA gas?

- **Yes**
- No
- Only if the level of protection is 'ia'

Can we use an Ex 'nL' field device in an Ex 'ic' circuit?

- **Yes**
- No
- No, only in an Ex 'ia' or 'ib' circuit.

Can we work on live I.S. circuits in a hazardous environment?

- **Yes**
- No, the enclosures of Ex 'i' shall be able to contain an internal overpressure
- No, the potential sparks may ignite a gas

Could a Zener barrier be used in a circuit for an IIC gas?

- **Yes**
- No only IIA
- Only IIA and IIB

Does a Zener barrier require an earth connection?

- **Yes**
- No
- Only when used for Zone 0

How does the Zener barrier work?

- **It has Zener diodes working as voltage suppressors and a resistor limiting the current.**
- It converts the signals from the hazardous area to 4-20mA for instrumentation circuits.
- It provides a galvanic isolation between safe area and zone.

How many faults shall Ex 'ib' circuits be safe with?

- **1**
- 2
- None, only safe during normal operation.

How many faults shall Ex 'ic' circuits be safe with?

- **None, only safe during normal operation.**
- 1
- 2
- 3

How many levels of protection are there on the Ex 'i' concept?

- **'ia', 'ib', 'ic'**
- Only 'ia'
- 'ia', 'ib'

How should the screen of an I.S. circuit be connected?

- **As described in the loop drawings**
- Only connected to earth in the field device
- To PE in both ends

If a barrier is has a level of protection [Ex ia] and a field device has Ex ib, what is the best level of protection this circuit will achieve?

- **'ib'**
- 'ia'
- 'ic'

If a barrier must be replaced and the identical spare is not on stock, can we replace this with another?

- **Yes, if the safety parameters are reassessed a nd the documentation updated**
- No, only like for like
- Yes, only consider the level of protection ('ia' for 'ia', 'ib' for 'ib')

If a color is used for I.S., this shall be?

- **Light blue**
- Light grey
- Red

If a field device suitable for gas group IIC is supplied from a barrier suitable for gas group IIB, can this be installed in an IIC environment?

- **No, the complete circuit will never be better than the weakest component**
- Yes, as long as only the field device is in the IIC environment
- Yes, the complete circuit are safe for both IIB and IIC gases, but not IIA

Intrinsically safe circuits are:

- **Low energy circuits**
- Able to contain an explosion
- Ex 'n' (Non-incendive)
- Non sparking

Is the T-class important on the field devices in an I.S. circuit?

- **Yes, all the components in zone must have a T-class**
- No, the energy is limited, so no T-class is required
- Only for 'ia' and 'ib'

Is the gas group relevant for Ex 'ia' circuits?

- **Yes**
- No
- Only for Ex 'ib' and 'ic' circutis

Of the following, what can be defined as simple apparatus in an I.S. loop?

- **Junction boxes and plugs**
- Only junction boxes
- Only plugs
- Transmitters working at maximum 24VDC and 0-25mA

On a Zener barrier located in an unclassified area, is the T-class relevant?

- **No**
- Yes, all Ex components shall have a T-class even if installed outside of zone.
- Yes, as this describes the circuits T-class

The level of protection 'ic' is suitable for Zone?

- **Zone 2**
- All Zones
- Depends on other factors, it is suitable for a IIC gas
- Zone 1 and 2

What does Ci and Li on a certified field device describe?

- **The maximum internal capacitance and inductance in the device**
- The maximum allowable capacitance and inductance in the loop
- The minimum internal capacitance and inductance in the device

What does Co and Lo on an associated apparatus (barrier) describe?

- **The maximum allowable capacitance and inductance in the connected circuit**
- The maximum internal capacitance and inductance in the barrier
- The minimum allowable capacitance and inductance in the connected circuit

What does Ui, Ii and Pi on a certified pressure transmitter describe?

- **The maximum voltage, current and power for the loop it may be used in**
- The minimum operational voltage, current and power
- The Ui, Ii and Pi are typical values for barriers, not field apparatus like pressure transmitters

What does the ATEX code II(1)G, and EPL (Ga) on a certificate label mean?

- **Associated apparatus that can feed into Zone 0**
- Device that can be located in Zone 0
- Not a valid marking.

What does the Uo, Io and Po represent?

- **The maximum output parameters from a barrier**
- The maximum input parameters to a field device
- The maximum voltage, current and power the barrier can be supplied with from E.g. a PLC

What is the Um value on a barrier?

- **The maximum voltage where the I.S. circuit is still protected**
- The maximum nominal Voltage
- The maximum output Voltage from the barrier

What is the main purpose of a galvanic isolator?

- **Limit the energy in an I.S. circuit**
- Convert the signal from the field device
- Ensure proper earth continuity

What is the maximum impedance from the I.S. earth bar in a control cabinet with Zener barriers back to the power systems earth?

- **Max 1Ω**
- Max 10Ω
- No maximum value shall be minimum 1Ω

What is the minimum clearance distance between I.S. circuits and Non-I.S. circuits?

- **50mm**
- 20mm
- 3mm
- 6mm

What is the minimum creepage distance between the metal of terminals on two separate I.S. circuits?

- **6mm**
- 20mm
- 3mm
- 50mm

What is the minimum creepage distance between the metal of a terminal on an I.S. loop and earth?

- **3mm**
- 20mm
- 50mm
- 6mm

What is the minimum requirements for cross-section for the I.S. earth back to the power systems earth?

- **One 4mm2 or two separate 1,5mm2**
- One 16mm2 or two separate 10mm2
- One 6mm2 or two separate 4mm2

What limits the energy in an I.S. circuit?

- **The associated apparatus**
- The cable
- The field device

Which of the concepts offers the highest level of protection?

- **Ex 'ia'**
- Ex 'ib'
- Ex 'ic'
- Ex 'nL'

Will the cable type and length affect an I.S. loop?

- **Yes, both with capacitance and inductance**
- No, this is classified as a simple apparatus
- Only in an I.S. loop intended for IIC gas environment

You have a label with following information:
II 3(1)G Ex nA [ia] IIB T4 Gc (Ga).
Into which zone can the equipment be installed?

- **Zone 2, and feed into Zone 0**
- Zone 0, and feed into Zone 2 only
- Zone 2, and feed into Zone 1 and 2 only

You have a label with the following information:
II 2(1)G Ex d [ia] IIB T6 Gb (Ga).
What does the concept codes Ex d [ia] represent?

- **This is an assembly with an Ex d enclosure and there is an associated apparatus for Ex ia circuits inside**
- The equipment must be supplied by an Ex 'ia' circuit
- The "clamps" on the [ia] indicates that this is only component certified

8535 – Cables, cable entry and IP-degree in Ex-area

Version 3.0 (2826)

A sheath metal Ex e junction box is located in an area exposed to vibrations and exposed to pressure washing, how would you carry out the installation?

- **By the use of a thru gland with sealing washer, serrated washer and lock nut**
- By the use of a thru gland and lock nut
- By the use of a thru gland with sealing washer and lock nut
- By the use of a thru gland with serrated washer and lock nut

Considerations when selecting a compression gland?

- **Diameter of inner sheath, braid and outer sheath of the cable**
- Diameter of the braid and outer sheath
- Diameter of the inner sheath of the cable
- Diameter of the outer sheath of the cable

During installation we discover that the equipment has been delivered with a M32 threaded hole and during planning we concluded that we needed a M25 cable gland for our cable. How can we resolve this issue?

- **By the use of a reducer**
- By the use of a blanking plug, where we seal of the existing hole and drill and tap a new one
- By the use of an adapter
- We are not allowed to modify certified equipment, we have to order a new one

Ex certified cable gland has to be according to IEC 60079-0. With the regards to Ex d, which additional standard is applicable?

- **IEC 60079-1**
- IEC 60079-14
- IEC 60079-17
- IEC 60079-7

For which protection concepts are breather and drains intended?

- **Ex d and Ex e**
- Ex d and Ex i
- Ex e and Ex i
- Only for Ex e in a IIC gas atmosphere

What IP rating will we automatically achieve when a gland is mounted perpendicular to the surface of an enclosure that is 6mm thick?

- **54**
- 44
- 45
- 56

What is the minimum required IP rating for Ex certified cable glands?

- **IP 54**
- IP 56
- IP 66
- IP 66/67

What kind of gland shall we use when connecting an Ex i circuit in an Ex d[ic] IIB T6 Gb enclosure?

- **Ex d gland**
- Ex di gland
- Ex e gland
- We don't have to use a Ex certified gland since it's a Ex i circuit

What kind of threads is allowed to use in Zone 1 area?

- **All**
- BSP
- Metric
- NPT, but only for IIB and IIA gases

What type of gland has compression on the inner sheath of cable?

- **Compression gland**
- Barrier gland
- MCT
- Thru gland

When installing an Ex e motor located in Zone 1, the cable comes directly from the MCC(Motor Control Centre) and is not substantial compact and has hygroscopic fillers. Which of the following glands would you choose?

- **Multi certified(Ex d, Ex e, Ex nR) barrier gland**
- Ex e certified compression gland
- Ex e certified thru gland
- Multi certified(Ex d, Ex e, Ex nR) thru gland

When selecting a thru gland for a Ex e enclosure, which of the following do have to consider?

- **Thread size and the outer diameter of the cable**
- Thread size and the braids diameter
- Thread size and the diameter of the inner bedding of the cable
- Thread size and the number of allowed cores through the gland

Where would you find data regarding dimensions for cables?

- **Manufacturers datasheet**
- Installation standard IEC 60079-14
- Since the IEC 60079-14 standard is additional to the low voltage installation standard these values would be given here
- These values has to be given by the certificate of the equipment into which the cable enters.

Which of the following is required to achieve IP 54 into an enclosure with a 6mm threaded entry?

- **None**
- Neoprene washer
- O-Ring
- Silicon washer

Which of the following is true regarding cables for fixed installations?

- **Sheathed with thermoplastic, thermosetting or elastomeric material, circular, compact, have extruded bedding and fillers, if any, shall be non-hygroscopic**
- Sheathed with thermoplastic, thermosetting or elastomeric material, circular, compact, have extruded bedding and fillers, if any, shall be hygroscopic
- Sheathed with thermoplastic, thermosetting or elastomeric material, rectangular, compact, have extruded bedding and fillers, if any, shall be non-hygroscopic
- Sheathed with thermoplastic, thermosetting or elastomeric material, square, compact, have extruded bedding and fillers, if any, shall be non-hygroscopic

Which of the following statements is correct?

- **Cables are not classified as equipment, thus they don't have to be ATEX and/or EPL certified**
- Cable would normally classified as Ex e IIC T6 Gb since they don't produce sparks
- Cables are classified as equipment, thus they have to ATEX and/or EPL certified
- Cables only have to be certified when they enter a Zone 0, IIC area

Which statement is correct regarding Ex certified blanking plugs?

- **You are not allowed to use a blanking plug with an adapter or reducer.**
- Ex certified blanking plugs is only available in metric threads.
- Ex certified plastic blanking plugs are not allowed above M32.
- You are allowed to use an blanking plug with and adapter or reducer, but only for Ex d

8536 – Ex inspection and maintenance

Version 3.0 (2609)

Are you allowed to bring your standard non-ex multimeter into a hazardous ares?

- **Yes, with a work permit in place and gas measurements of the actual area.**
- No, this is not permitted under any circumstances.
- No, you have to use an Ex i multimeter, with Ex e probes.
- Yes, in zone 2 without any other pre-cautions.

Are you allowed to open an energized Ex e junction box in hazardous area without any other pre-cautions?

- **No.**
- Yes, as long as you are indoors.
- Yes, in zone 1 and zone 2.
- Yes.

For which protection concepts are there inspection tables for in the standard?

- **Ex d, e, n, t, Ex i and Ex p**
- Ex d, e, n, t , Ex s and Ex q
- Ex d, e, n, t, Ex m and Ex o
- Ex d, e, n, t, Ex q and Ex o

How can you discover an un-authorized modification on a Ex i circuit board?

- **Un-original soldering compared to the rest of the circuit board.**
- By checking the product standard for Ex i, IEC 60079-11.
- It is not possible.
- This is an Ex i circuit, would not be reckoned as an un-authorized modification.

How many inspection tables are there in the standard?

- **3**
- 2
- 4
- 5

Insulation resistance should always be logged as a numeric value, why?

- **To be able to show the effect of deterioration over time.**
- It's not a requirement to check insulation resistance.
- Not, important as long as the insulation resistance is according to the standard.
- To be able to calculate the length of the cable run.

Prior to energizing an installation there shall always be carried out an inspection of the installation, which type of inspection is this?

- **Initial inspection.**
- Continuous supervision.
- Periodic inspection.
- Sample inspection.

What grade of inspection does not require you to use tools and access equipment and de-energize the equipment?

- **Visual inspection.**
- Close inspection.
- Detailed inspection.
- None of the alternatives.

What grade of inspection is typically used with a periodic inspection of transportable equipment?

- **Close inspection.**
- Detailed inspection.
- None.
- Visual inspection.

What grade of inspection requires you to de-energize the equipment to carry out your inspection?

- **Detailed inspection.**
- Close inspection.
- None of the alternatives.
- Visual inspection.

What grade of inspection requires you to use tools and access equipment but does not require you to de energize the equipment?

- **Close inspection.**
- Detailed inspection.
- None of the alternatives.
- Visual inspection.

What is a typical maintenance task?

- **Replace a terminal in an Ex e junction box.**
- Change an component on a Ex i circuit board
- Change bearings on a Ex d motor.
- Repair the flame-path of an Ex d enclosure.

What is an inspection?

- **An action taken to determine the condtion of the equipment.**
- An action taken to remove equipment from the installation.
- An action taken to repair the equipment.
- An action taken to restore the equipment to it's original state

What is the maximum interval for periodic inspections of equipment in a fixed installation?

- **36 months**
- 12 months
- 24 months
- 6 months

What is the maximum interval for periodic inspections of transportable equipment?

- **12 months**
- 24 months
- 36 months
- 6 months

What shall you always do prior to removing the IS earth of a Zener barrier?

- **Disconnect the out going circuit.**
- Calibrate your instrument.
- Check the loop impedance.
- No special requirement, this is a Ex i circuit.

What type of inspection is an annual inspection?

- **Periodic inspection.**
- Continuous supervision.
- Initial inspection.
- Sample inspection.

When checking that the "Bolts, cable entry devices (direct and indirect) and blanking elements are of the correct type and complete and tight" without de-energizing the equipment, which grade of inspection are you carrying out?

- **Close**
- Detailed
- None, this is a periodic inspection
- Visual

When checking that the "Electrical connection are tight", which grade of inspection are you carrying out?

- **Detailed**
- Close
- None, this is a periodic inspection
- Visual

When checking that the "Equipment is appropriate to the EPL/Zone requirements of the location", which grade of inspection are you carrying out?

- **Visual**
- Close
- Detailed
- None, this is a periodic inspection

Which of the following can be used to clean a flame-path?

- **Non-metallic scrapers and non-corrosive cleaning agents.**
- Metallic scrapers and corrosive cleaning agents.
- Sandpaper with grits above 40
- Sandpaper with grits under 40

Which of the following is a grade of inspection?

- **Detailed inspection.**
- Annual inspection.
- Periodic inspection.
- Sample inspection.

Which of the following is a type of inspection?

- **Periodic inspection.**
- Close inspection.
- Detailed inspection.
- Visual inspection.

Which of the following is not required in your training to do inspections?

- **Offshore certificate with helicopter evacuation.**
- General principles of area classification.
- The requirements of the standard.
- Types of protection and installation practices.

Which of the following would you need when performing an inspection?

- **Up to date documentation.**
- An agreement with an authorized workshop.
- An agreement with the manufacturer of the equipment.
- Authorization from a notified body.

Which protection concept would require you to check "Protective gas pressure and/or flow is adequate"?

- **Ex p**
- Ex d
- Ex e
- Ex n

Which protection concepts would require you to check "Electrical connection are tight"?

- **Ex e and Ex n**
- Ex d and Ex e
- Ex e and Ex o
- Ex n and Ex m

Which protection concepts would require you to check that "Circuit and/or equipment category and equipment is correct"?

- **Ex i**
- Ex d
- Ex e
- Ex m

Which standard is used for Ex inspections?

- **IEC 60079-17**
- IEC 60079-10-1
- IEC 60079-19
- IEC 60079-20-1

Who can do repairs on Ex equipment?

- **The manufacturer, his authorized representative and authorized workshops.**
- A electrician with basic Ex knowledge.
- A mechanic with basic Ex knowledge.
- A technical authority of the oil company.

8537 – Exi protection concepts (Exi a,b,c,nL)

Version 1.0 (3009)

Can a field device marked Ex nL be used in an Ex ic circuit?

- **Yes**
- No

Do we need to take into account any junction boxes and plugs in an I.S. circuit?

- **Yes, as a simple apparatus**
- No, because they are simple apparatus

If a color is used to identify an intrinsically safe circuit, what colour should this be?

- **Light blue**
- Black
- Green
- Light red

If an associated apparatus is marked with [Ex ib IIC], can this be used for applications in zone 0?

- **No**
- Yes
- Yes, if the field device has EPL Ga / ATEX cat II1G

If we install an Ex 'nL' transmitter in an I.S. loop, what level of protection will the circuit achieve at best?

- **Ex 'ic'**
- Ex 'ib'
- Ex 'nL'

Is the ambient temperature relevant for I.S. components?

- **Yes**
- Only for associated apparatus
- Only for field devices

Is the cable in an intrinsically safe circuit an important part of the system

- **Yes**
- No

Is the selection of barrier important for zone suitability?

- **Yes**
- No
- Only for zone with IIC gas

On what I.S. equipment is the T-class relevant?

- **Only for equipment installed in zone**
- All
- Only for associated apparatus

On what I.S. equipment is the gas group relevant?

- **All**
- Only for associated apparatus
- Only for equipment installed in zone

We have an associated apparatus marked "II (1) G [Ex ia] IIC (Ga)" and a field device marked with "II 2 G Ex ib IIC T4 Gb". What kind of Ex i circuit would we get?

- **Ex ib**
- Ex ia
- Ex ic
- Ex nL

What EPL / ATEX code indicates that a field device is suitable for zone 2 only?

- **Gc / II3G**
- Gb / II2G
- Gc / II1G

What is described by Uo, Io and Po?

- **The maximum output parameters from an associated apparatus**
- The maximum input parameters for a certified field device
- The nominal voltage for a certified field device
- The nominal voltage, current and power from an associated apparatus

What is the "energy limiting" component in an intrinsically safe circuit?

- **The associated apparatus**
- The cable
- The certified field device

What provides the highest level of safety?

- **Ex 'ia'**
- Ex 'ib'
- Ex 'ic'

What will be the most common use for I.S.?

- **Instrumentation**
- Motors
- Protection for flameproof enclosures

What zone can equipment marked with "II (1) G [Ex ia] IIC (Ga)" be installed in?

- **A non-hazardous area**
- Zone 0, 1 and 2
- Zone 1 and 2
- Zone 2

What zone can equipment marked with "II 2 G Ex ib IIC T4 Gb" be installed in?

- **Zone 1 and 2**
- A non-hazardous area
- Zone 0, 1 and 2
- Zone 2

Where can equipment with following marking be installed: II3(2)G Ex ec mc [ia Gb] IIC T4 Gc?

- **Zone 2 only**
- Zone 0, 1 and zone 2
- Zone 1 and zone 2

Which I.S. circuit will be suitable for zone 0?

- **Ex ia**
- Ex ib
- Ex ic
- Ex nL

Which I.S. circuit will only be suitable for zone 2?

- **Ex ic**
- Ex ia
- Ex ib

Which gas-groups have gasses with the lowest MIC?

- **IIC**
- IIA
- IIB

Which parameter shall be greatest, the barriers Co or the field devices Ci?

- **Co**
- Ci
- Not relevant

Which parameter shall be greatest, the barriers Io or the field devices Ii?

- **Ii**
- Io
- Not relevant

Which parameter shall be greatest, the barriers Lo or the field devices Li?

- **Lo**
- Li
- Not relevant

Which parameter shall be greatest, the barriers Po or the field devices Pi?

- **Pi**
- Not relevant
- Po

Which parameter shall be greatest, the barriers T-class or the field devices T-class?

- **Not relevant**
- Barriers T-class
- Field devices T-class

Which parameter shall be greatest, the barriers Uo or the field devices Ui?

- **Ui**
- Not relevant
- Uo

8538 – Exi components in Ex zone and safe zone

Version 1.0 (3741)

Are all zener barriers polarity sensitive?

- **No**
- Yes

Are there any direct electric connection between the I.S. side and the non-I.S. side of a Zener barrier?

- **Yes**
- No

Are there any requirements for simple apparatus?

- **Yes**
- No
- Only for Zone 0

Are there any requirements for simple apparatus?

- **Yes**
- No
- Only for Zone 0

Does all I.S. equipment have a T-class?

- **No, only if intended to be installed in a zone.**
- T-class not relevant for I.S.
- Yes, all Ex equipment needs a T-class

Does all I.S. equipment have polarity protection?

- **No**
- Only Zener barriers
- Yes

Does all intrinsically safe components have to be supplied from a galvanic isolator?

- **No, some equipment is "self contained" like a hand held lamp**
- Only equipment with the marking [Ex ia]
- Yes, all Ex i equipment needs a barrier

Equipment with the marking Ex 'ic' is suitable for which zone?

- **Zone 2 only**
- Zone 0, Zone 1 and Zone 2
- Zone 1 and Zone 2

For certified field devices, what may the T-class depend on?

- **Both**
- Ambient temperature
- The source of supply

How many faults shall a Zener barrier be safe with for ia?

- **2 faults**
- 1 fault
- Only safe during normal operation

How many faults shall a galvanic isolator be safe with for ib?

- **1 fault**
- 2 faults
- Only safe during normal operation

Into which zones may this apparatus be installed: II(1)G [Ex ia Ga] IIC?

- **Only in an unclassified area**
- In Zone 0, Zone 1 and Zone 2
- Only in areas with IIC gasses, Zone 0, Zone 1 and Zone 2

Into which zones may this equipment be installed: II 2(1)G Ex de [ia] IIC T4 Gb(Ga)?

- **Zone 1 and Zone 2**
- Only Zone 2
- Zone 0, Zone 1 and Zone 2

Is the nominal Voltage and Current equal to the safety parameters on barriers?

- **No**
- Only for Ex 'ic' apparatus
- Yes

Of the following which is a suitable application for Ex i?

- **Instrumentation**
- Lighting
- Motors

Typically how many zener diodes are there in a 'ia' zener barrier?

- **3**
- 2
- 4

What category and EPL will be marked for equipment made for zone 1 and zone 2?

- **ATEX cat 2 / EPL Gb**
- ATEX cat 1 / EPL Ga
- ATEX cat 1 / EPL Gc

What determines how much power the equipment may send into a hazardous environment or use in a hazardous environment?

- **The gas group it is designed for**
- The level of protection
- The T-class it is designed for

What does Um on an I.S. component describe?

- **The maximum safety voltage**
- The nominal voltage
- The voltage safety parameter

What is FISCO - Fieldbus Intrinsically Safe Concept?

- **A non-linear I.S. system**
- Not a I.S. concept
- This is the traditional barriers for Ex i

What is FISCO - Fieldbus Intrinsically Safe Concept?

- **A non-linear I.S. system**
- Not a I.S. concept
- This is the traditional barriers for Ex i

What is the minimum I.P rating if the I.S. components do not comply with the minimum requirements for creepage and clearance?

- **IP 54**
- IP 45
- IP 66

What is the minimum IP rating for I.S. equipment?

- **IP 20**
- IP 45
- IP 54

What is the minimum creepage distance between supply side and I.S. side for a Zener barrier?

- **50mm**
- 3mm
- 6mm

What is the minimum creepage distance between supply side and I.S. side for a galvanically isolated barrier?

- **50mm**
- 3mm
- 6mm

What of the following could be an 'X' condition on a certificate?

- **Both**
- The equipment has to be installed in an enclosure providing minimum IP 54.
- The equipment needs to be mechanically protected

Which parameters describe the safety parameters for a certified field device?

- **Ui, Ii, Pi**
- Un, In, Pn
- Uo, Io, Po

Which parameters describe the safety parameters for an associated apparatus?

- **Uo, Io, Po**
- Ui, Ii, Pi
- Un, In, Pn

Will a galvanic isolator normally have an earth connection?

- **No**
- Only for 'ia' and 'ib'
- Yes

Will a zener barrier normally have an earth connection?

- **Yes**
- No
- Only for 'ia' and 'ib'

8539 – Exi verification and documentation

Version 1.0 (4921)

Can barriers be connected together in parallel or series?

- **Yes, refer to documentation or standard**
- No, but one barrier may have several field devices
- No, one barrier allowed per field device

Can low energy generating components be used as 'simple apparatus'. E.g. a photocell?

- **Yes, with limitation**
- No
- Yes, without limitations

Does simple apparatus need a T-class?

- **Yes**
- No
- Only for Zone 0 and Zone 1

Does simple apparatus need a certificate label?

- **No**
- Only when applied in Zone 0 and Zone 1
- Yes

For assessing equipment, which of the values shall be greatest?

- **Ui greater or equal than Uo**
- Un greater or equal than Ui
- Uo greater or equal than Ui

For assessing equipment, which of the values shall be greatest?

- **Ii greater or equal than Io**
- In greater or equal than Ii
- Io greater or equal than Ii

For assessing equipment, which of the values shall be greatest?

- **Lo greater or equal than Li**
- Li greater or equal than Lo
- Nominal inductance greater or equal than Ii

For assessing equipment, which of the values shall be greatest?

- **Co greater or equal than Ci**
- Ci greater or equal than Li
- Nominal capacitance greater or equal than Ci

How long cable can be installed where the Cc (allowed cable inductance) is 0,080μF and the cable has a capacitance of 0,2μF/km?

- **400 meters**
- 2400 meters
- 40 meters

How much capacitance may the cable add where the circuit has the following instruments: Co – 0,11μF, Ci – 87nF?

- **0,023μF**
- 76μF
- 98nF

How will the zenerbarrier affect the circuit?

- **It's connected directly in the circuit and adds resistance to the loop**
- All zener barriers connects the return signal to earth
- It provides a segregation for the loop

If a field device has Gas group IIC stated on the equipment label, is this automatically suitable for an IIC environment?

- **No, the documentation shall reflect the achieved gas group**
- Yes, as long as the component is marked for IIC, the equipment is suitable for IIC
- Yes, I.S. circuits and components will always achieve gas group IIC

Is Ex nL equipment suitable for a 'ib' circuit?

- **No, only for an 'ic' circuit**
- No, Ex nL may not be used in an intrinsically safe circuit
- Yes

Is it a requirement to state the achieved T-class for components in an I.S. circuit?

- **Yes**
- No
- Only for Certified Apparatus

May a field device have several T-classes?

- **Yes, refer to documentation**
- No, not for certified apparatus
- Yes, but only for simple apparatus

May a simple apparatus be used in another circuit without any further verification?

- **No**
- Only for IIA, 'ic' circuits
- Yes

May we use an IIC approved field device in an IIA circuit?

- **Yes**
- Depends on the level of protection
- No

Of the following, which can be used as simple apparatus in an I.S. circuit?

- **Thermocouple**
- A transmitter
- Motor

What T-class can be achieved for many simple apparatus based on the ambient temperature and Po from the associated apparatus?

- **T4**
- T5
- T6

What T-class can normally be allocated junction boxes and plugs when used as simple apparatus below 40 C°?

- **T6**
- T1
- T4

What T-class is achieved on junction boxes, terminals and plugs in an ambient temp below 40°C?

- **T6**
- T4
- T5

What determines the T-class?

- **The field device documentation / label**
- Only the ambient temperature
- Only the power from the associated apparatus

What determines the gas group of the circuit?

- **The values used for the barrier during assessment of the circuit and the field device.**
- The gas group stated on the field device
- The values used for the barrier alone

What happens to the circuit when we use a galvanic isolated barrier?

- **The control side is segregated from the I.S. side**
- The control side is in direct electrical contact with the I.S. area
- The zero volt reference from the barrier is earthed

What happens to the safety parameters if the inductance and the capacitance is above 1% of the barrier?

- **The allowable values for Co and Lo are reduced with 50%**
- The allowable inductance are reduced with 50%
- The allowable values for all parameters are reduced with 50%

What happens with the protection level when we connect two barriers in series or parallel and assess the system?

- **Reduced from 'ia' to 'ib'**
- Nothing ,as long as both are 'ia'.
- Reduced from 'ia' to 'ic'

What is a 'DSD' for an intrinsically safe system?

- **The "Descriptive System Documentation"**
- The "Design System Dossier" (Translated from French)
- The "Documentation, System Datasheets"

What level of protection is achieved if the associated apparatus is 'ia' and the field device is 'ib'?

- **'ib'**
- This is not allowed
- 'ia'
- 'ic'

What level of protection is achieved if the associated apparatus is 'ib' and the field device is 'ia'?

- **'ib'**
- This is not allowed
- 'ia'
- 'ic'

What will the achieved gas group be for an IIC component in an IIA circuit?

- **IIA**
- IIB
- IIC, the field device is not affected by the circuit

When can we use the L/R ratio instead of assessing the inductance of the cable?

- **Only when Li is less than 1% of the Lo**
- Only when Li is more than 1% of the Lo
- Only when the capacitance is less than 1% of Co

Which standard describes the requirements for I.S. installations?

- **IEC 60079-14 and IEC 60079-25**
- IEC 60079-11
- IEC 60079-15 and IEC 60079-11

8540 – Exi installation and inspection

Version 1.0 (2796)

Are there any restrictions for measuring devices to be used in an Intrinsically safe loop?

- **Yes**
- No
- Only when working in a zone environment

From where does the clearance requirement count?

- **From the "metal" in the terminal to metal**
- From the centre of the housing of barrier or terminal
- From the edge of the housing of barrier or terminal

How many I.S. circuits may be routed through a junction box containing non-I.S. circuits without reduction of the level of protection?

- **No limitation**
- I.S. circuits may never be routed through a cabinet or junction box carrying non-I.S.
- Only 2 circuits
- Only one

How to verify how many zener diodes that are broken in a Zener barrier?

- **Unfortunatly, not possible**
- Megger test between I.S. and non-I.S. side.
- There shall allways be an indicator on the barrier side

If the cable parameters are unknown, what to do?

- **All of the options are OK.**
- Test it according to procedure in IEC 60079-14
- Use another cable
- Use the "standard" values stated in IEC 60079-14

To where may the I.S. main earth reference be connected?

- **To the installations "main earth reference"**
- From the PE locally in a cabinet
- The I.S. earth is to be left floating

What are the Ex requirements for junction boxes and plugs with a single I.S. circuit?

- **None**
- Has to be certified for the same zone as the entire circuit
- Has to be certified for the zone of installation

What are the Ex requirements for junction boxes and plugs with multiple I.S. circuits?

- **Should be certified**
- Has to be certified for the same zone as the entire circuit
- None

What dielectric voltage test shall cables for I.S. be able to withstand?

- **500VAC or 700VDC**
- 1500VAC or 2300VDC
- 150VDC or 250VDC

What grade of inspection shall be used during the initial inspection?

- **Detailed**
- Close
- Periodic
- Visual

What happens to ia circuits in a "type A" mulicore circuit?

- **They remain 'ia'**
- The level of safety is reduced to 'ib'
- The level of safety is reduced to 'ic'

What happens to ia circuits in a "type B" mulicore circuit?

- **They remain 'ia'**
- The level of safety is reduced to 'ib'
- The level of safety is reduced to 'ic'

What happens to ia circuits in a "type C" multicore circuit?

- **The level of safety is reduced to 'ic'**
- The level of safety is reduced to 'ib'
- They remain 'ia'

What is Um?

- **The maximum safety voltage on the barrier**
- The nominal operation voltage
- The output voltage

What is a "type A" cable?

- **Multicore cable with individual screen**
- Multicore cable with armouring or mechanical protection
- Multicore cable with common screen
- Unarmoured and unscreened

What is a "type B" cable?

- **Multicore cable with armouring or mechanical protection**
- Multicore cable with common screen
- Multicore cable with individual screen
- Unarmoured and unscreened

What is a “type C” cable?

- **Unarmoured and without individual screen**
- Multicore cable with armouring or mechanical protection
- Multicore cable with common screen
- Multicore cable with individual screen

What is an initial inspection?

- **A type of inspection, the commissioning verification**
- A grace of inspection, the visual inspection
- A grade of inspection, the yearly inspection

What is the maximum allowable impedance from the I.S. earth back to equipotential bonding?

- **1Ω**
- 10Ω
- 1kΩ
- 2,5$\mu\Omega$

What is the maximum allowable interval between inspections?

- **3 years**
- 1 years
- 5 years
- 6 months

What is the minimum creepage and clearance between I.S. and Non-I.S.

- **50mm**
- 3mm
- 5mm
- 6mm

What is the minimum creepage and clearance between I.S. and earth?

- **3mm**
- 50mm
- 5mm
- 6mm

What is the minimum creepage and clearance between two separate I.S. circuits?

- **6mm**
- 3mm
- 50mm
- 5mm

What is the minimum cross-section when using a single conductor for the I.S.earth to equipotential bonding?

- **4mm2**
- 10mm2
- 2,5mm2

What is the minimum cross-section when using two conductors for the I.S.earth to equipotential bonding?

- **1,5nmm2**
- 10mm2
- 2,5mm2
- 4mm2

What is the preferred colour to indicate I.S.?

- **Light blue**
- Dark red
- Light green
- Light yellow

What is the standards minimum cross-section for a bonding connection?

- **4mm2**
- 1,5mm2
- 10mm2
- 16mm2

What is the standards recommended barrier type for Zone 0?

- **Galvanically isolated**
- Only non-linear barriers
- The earthed zener barrier

What level of protection is normally suitable for areas requiring EPL Gb / ATEX cat 2?

- **ia and ib**
- ia only
- ia, ib and ic

What level of protection is normally suitable for areas requiring EPL Gc / ATEX cat 3?

- **ia, ib and ic**
- ia only
- ia, ib